(magnitude) will also be incommensurable with it.

Ἔστω δύο μεγέθη σύμμετρα τὰ A, B, τὸ δὲ ὁέτερον αὐτῶν τὸ Aὁάλλῳ τινὶ τῷ $Γ$ ἀσύμμετρονὁέστω· λέγω, ὅτι καὶ τὸ λοιπὸν τὸ B τῷ $Γ$ ἀσύμμετρόν ἐστιν.

Εἰ γάρ ἐστι σύμμετρον τὸ B τῷ $Γ$, ἀλλὰ καὶ τὸ A τῷ B σύμμετρόν ἐστιν, καὶ τὸ A ἄρα τῷ $Γ$ σύμμετρόν ἐστιν. ἀλλὰ καὶ ἀσύμμετρον· ὅπερ ἀδύνατον. οὐκ ἄρα σύμμετρόν ἐστι τὸ B τῷ $Γ$· ἀσύμμετρον ἄρα.

Ἐὰν ἄραὁῆ δύο μεγέθη σύμμετρα, καὶ τὰ ἑχῆς.

Let A and B be two commensurable magnitudes, and let one of them, A, be incommensurable with some other (magnitude), C. I say that the remaining (magnitude), B, is also incommensurable with C.

For if B is commensurable with C, but A is also commensurable with B, A is thus also commensurable with C [Prop. 10.12]. But, (it is) also incommensurable (with C). The very thing (is) impossible. Thus, B is not commensurable with C. Thus, (it is) incommensurable.

Thus, if two magnitudes are commensurable, and so on ....

Λῆμμα.

Δύο δοθεισῶν εὐθειῶν ἀνίσων εὑρεῖν, τίνι μεῖζον δύναται ἡ μείζων τῆς ἐλάσσονος.

Lemma

For two given unequal straight-lines, to find by (the square on) which (straight-line) the square on the greater (straight-line is) larger than (the square on) the lesser.[†]

Ἔστωσαν αἱ δοθεῖσαι δύοὐάνισοι εὐθεῖαι αἱ AB, $Γ$, ὧν μείζωνὁέστω ἡ AB· δεῖ δὴ εὑρεῖν, τίνι μεῖζον δύναται ἡ AB τῆς $Γ$.

Γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ $AΔB$, καὶ εἰξ αὐτὸ

Let AB and C be the two given unequal straight-lines, and let AB be the greater of them. So it is required to find by (the square on) which (straight-line) the square on AB (is) greater than (the square on) C.

ἐνηρμόσθω τῇ Γὸίση ἡ $A\Delta$, καὶ ἐπεζεύθθω ἡ ΔB. φανερὸν δή, ὅτι ὀρθή ἐστιν ἡ ὑπὸ $A\Delta B$ γωνία, καὶ ὅτι ἡ AB τῆς $A\Delta$, τουτέστι τῆς Γ, μεῖζον δύναται τῇ ΔB.

Ὁμοίως δὲ καὶ δύο δοθεισῶν εὐθειῶν ἡ δυναμένη αὐτὰξ εὑρίσκεται οὑτως.

Ἔστωσαν αἱ δοθεῖσαι δύο εὐθεῖαι αἱ $A\Delta$, ΔB, καὶ δέονὸέστω εὑρεῖν τὴν δυναμένην αὐτάξ. κείσθωσαν γάρ, ὁὡστε ὀρθὴν γωνίαν περιέθειν τὴν ὑπὸ $A\Delta$, ΔB, καὶ ἐπεζεύθθω ἡ AB· φανερὸν πάλιν, ὅτι ἡ τὰξ $A\Delta$, ΔB δυναμένη ἐστὶν ἡ AB· ὅπερ ἔδει δεῖχαι.

Let the semi-circle ADB be described on AB. And let AD, equal to C, be inserted into it [Prop. 4.1]. And let DB be joined. So (it is) clear that the angle ADB is a right-angle [Prop. 3.31], and that the square on AB (is) greater than (the square on) AD—that is to say, (the square on) C—by (the square on) DB [Prop. 1.47].

And, similarly, the square-root of (the sum of the squares on) two given straight-lines is also found likeso.

Let AD and DB be the two given straight-lines. And let it be necessary to find the square-root of (the sum of the squares on) them. For let them be laid down such as to encompass a right-angle—(namely), that (angle encompassed) by AD and DB. And let AB be joined. (It is) again clear that AB is the square-root of (the sum of the squares on) AD and DB [Prop. 1.47]. (Which is) the very thing it was required to show.

† That is, if α and β are the lengths of two given straight-lines, with α being greater than β, to find a straight-line of length γ such that $\alpha^2 = \beta^2 + \gamma^2$. Similarly, we can also find γ such that $\gamma^2 = \alpha^2 + \beta^2$.

14.

Ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον ὦσιν, δύνηται δὲ ἡ πρώτη τῆς δευτέρας μεῖζον τῷ ἀπὸ συμμέτρου ἑαυτῇ [μήκει], καὶ ἡ τρίτη τῆς τετάρτης μεῖζον δυνήσεται τῷ ἀπὸ συμμέτρου ἑαυτῇ [μήκει]. καὶ ἐὰν ἡ πρώτη τῆς δευτέρας μεῖζον δύνηται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ [μήκει], καὶ ἡ τρίτη τῆς τετάρτης μεῖζον δυνήσεται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ [μήκει].

Ἔστωσαν τέσσαρες εὐθεῖαι ἀνάλογον αἱ A, B, Γ, Δ, ὡξ ἡ A πρὸς τὴν B, οὑτως ἡ Γ πρὸς τὴν Δ, καὶ ἡ A μὲν τῆς B μεῖζον δυνάσθω τῷ ἀπὸ τῆς E, ἡ δὲ Γ τῆς Δ μεῖζον δυνάσθω τῷ ἀπὸ τῆς Z· λέγω, ὅτι, εὐίτε σύμμετρόξ ἐστιν ἡ A τῇ E, σύμμετρόξ ἐστι καὶ ἡ Γ τῇ Z, εὐίτε ἀσύμμετρόξ ἐστιν ἡ A τῇ E, ἀσύμμετρόξ ἐστι καὶ ὁ Γ τῇ Z.

Proposition 14

If four straight-lines are proportional, and the square on the first is greater than (the square on) the second by the (square) on (some straight-line) commensurable [in length] with the first, then the square on the third will also be greater than (the square on) the fourth by the (square) on (some straight-line) commensurable [in length] with the third. And if the square on the first is greater than (the square on) the second by the (square) on (some straight-line) incommensurable [in length] with the first, then the square on the third will also be greater than (the square on) the fourth by the (square) on (some straight-line) incommensurable [in length] with the third.

Let A, B, C, D be four proportional straight-lines, (such that) as A (is) to B, so C (is) to D. And let the square on A be greater than (the square on) B by the (square) on E, and let the square on C be greater than (the square on) D by the (square) on F. I say that A is either commensurable (in length) with E, and C is also commensurable with F, or A is incommensurable (in length) with E, and C is also incommensurable with F.

Ἐπεὶ γάρ ἐστιν ὡς ἡ Α πρὸς τὴν Β, οὕτως ἡ Γ πρὸς τὴν Δ, ὑέστιν ἄρα καὶ ὡς τὸ ἀπὸ τῆς Α πρὸς τὸ ἀπὸ τῆς Β, οὕτως τὸ ἀπὸ τῆς Γ πρὸς τὸ ἀπὸ τῆς Δ. ἀλλὰ τῷ μὲν ἀπὸ τῆς Α ἴσα ἐστὶ τὰ ἀπὸ τῶν Ε, Β, τῷ δὲ ἀπὸ τῆς Γ ἴσα ἐστὶ τὰ ἀπὸ τῶν Δ, Ζ. ὑέστιν ἄρα ὡς τὰ ἀπὸ τῶν Ε, Β πρὸς τὸ ἀπὸ τῆς Β, οὕτως τὰ ἀπὸ τῶν Δ, Ζ πρὸς τὸ ἀπὸ τῆς Δ· διελόντι ἄρα ἐστὶν ὡς τὸ ἀπὸ τῆς Ε πρὸς τὸ ἀπὸ τῆς Β, οὕτως τὸ ἀπὸ τῆς Ζ πρὸς τὸ ἀπὸ τῆς Δ· ὑέστιν ἄρα καὶ ὡς ἡ Ε πρὸς τὴν Β, οὕτως ἡ Ζ πρὸς τὴν Δ· ἀνάπαλιν ἄρα ἐστὶν ὡς ἡ Β πρὸς τὴν Ε, οὕτως ἡ Δ πρὸς τὴν Ζ. ὑέστι δὲ καὶ ὡς ἡ Α πρὸς τὴν Β, οὕτως ἡ Γ πρὸς τὴν Δ· διὸ ὑίσου ἄρα ἐστὶν ὡς ἡ Α πρὸς

For since as *A* is to *B*, so *C* (is) to *D*, thus as the (square) on *A* is to the (square) on *B*, so the (square) on *C* (is) to the (square) on *D* [Prop. 6.22]. But the (sum of the squares) on *E* and *B* is equal to the (square) on *A*, and the (sum of the squares) on *D* and *F* is equal to the (square) on *C*. Thus, as the (sum of the squares) on *E* and *B* is to the (square) on *B*, so the (sum of the squares) on *D* and *F* (is) to the (square) on *D*. Thus, via separation, as the (square) on *E* is to the (square) on *B*, so the (square) on *F* (is) to the (square) on *D* [Prop. 5.17]. Thus, also, as *E* is to *B*, so *F* (is) to *D* [Prop. 6.22]. Thus, inversely, as *B* is to *E*, so *D* (is) to *F* [Prop. 5.7 corr.]. But,

τὴν E, οὑτως ἡ Γ πρὸς τὴν Z. εὐίτε οὐῦν σύμμετρόξ ἐστιν ἡ A τῇ E, συμμετρόξ ἐστι καὶ ἡ Γ τῇ Z, εὐίτε ἀσύμμετρόξ ἐστιν ἡ A τῇ E, ἀσύμμετρόξ ἐστι καὶ ἡ Γ τῇ Z.

Ἐὰν ἄρα, καὶ τὰ ἑξῆς.

as A is to B, so C also (is) to D. Thus, via equality, as A is to E, so C (is) to F [Prop. 5.22]. Therefore, A is either commensurable (in length) with E, and C is also commensurable with F, or A is incommensurable (in length) with E, and C is also incommensurable with F [Prop. 10.11].

Thus, if, and so on ....

15.

Ἐὰν δύο μεγέθη σύμμετρα συντεθῇ, καὶ τὸ ὅλον ἑκατέρῳ αὐτῶν σύμμετρονὸέσται· κὸὰν τὸ ὅλον ἑνὶ αὐτῶν σύμμετρονὸῇ, καὶ τὰ ἐξ ἀρθῆς μεγέθη σύμμετραὸέσται.

Συγκείσθω γὰρ δύο μεγέθη σύμμετρα τὰ AB, $B\Gamma$· λέγω, ὅτι καὶ ὅλον τὸ $A\Gamma$ ἑκατέρῳ τῶν AB, $B\Gamma$ ἐστι σύμμετρον.

Proposition 15

If two commensurable magnitudes are added together then the whole will also be commensurable with each of them. And if the whole is commensurable with one of them then the original magnitudes will also be commensurable (with one another).

For let the two commensurable magnitudes AB and BC be laid down together. I say that the whole AC is also commensurable with each of AB and BC.

Ἐπεὶ γὰρ σύμμετρά ἐστι τὰ AB, $B\Gamma$, μετρήσει τι αὐτὰ μέγεθος. μετρείτω, καὶ ὁέστω τὸ Δ. ἐπεὶ οὐῦν τὸ Δ τὰ AB, $B\Gamma$ μετρεῖ, καὶ ὅλον τὸ $A\Gamma$ μετρήσει. μετρεῖ δὲ καὶ τὰ AB, $B\Gamma$. τὸ Δ ἄρα τὰ AB, $B\Gamma$, $A\Gamma$ μετρεῖ· σύμμετρον ἄρα ἐστὶ τὸ $A\Gamma$ ἑκατέρῳ τῶν AB, $B\Gamma$.

Ἀλλὰ δὴ τὸ $A\Gamma$ὁέστω σύμμετρον τῷ AB· λέγω δή, ὅτι καὶ τὰ AB, $B\Gamma$ σύμμετρά ἐστιν.

Ἐπεὶ γὰρ σύμμετρά ἐστι τὰ $A\Gamma$, AB, μετρήσει τι αὐτὰ μέγεθος. μετρείτω, καὶ ὁέστω τὸ Δ. ἐπεὶ οὐῦν τὸ Δ τὰ ΓA, AB μετρεῖ, καὶ λοιπὸν ἄρα τὸ $B\Gamma$ μετρήσει. μετρεῖ δὲ καὶ τὸ AB· τὸ Δ ἄρα τὰ AB, $B\Gamma$ μετρήσει· σύμμετρα ἄρα ἐστὶ τὰ AB, $B\Gamma$.

Ἐὰν ἄρα δύο μεγέθη, καὶ τὰ ἑξῆς.

For since AB and BC are commensurable, some magnitude will measure them. Let it (so) measure (them), and let it be D. Therefore, since D measures (both) AB and BC, it will also measure the whole AC. And it also measures AB and BC. Thus, D measures AB, BC, and AC. Thus, AC is commensurable with each of AB and BC [Def. 10.1].

And so let AC be commensurable with AB. I say that AB and BC are also commensurable.

For since AC and AB are commensurable, some magnitude will measure them. Let it (so) measure (them), and let it be D. Therefore, since D measures (both) CA and AB, it will thus also measure the remainder BC. And it also measures AB. Thus, D will measure (both) AB and BC. Thus, AB and BC are commensurable [Def. 10.1].

Thus, if two magnitudes, and so on ....

16.

Ἐὰν δύο μεγέθη ἀσύμμετρα συντεθῇ, καὶ τὸ ὅλον ἑκατέρῳ αὐτῶν ἀσύμμετρονὸέσται· κὸὰν τὸ ὅλον ἑνὶ αὐτῶν ἀσύμμετρονὸῇ, καὶ τὰ ἐξ ἀρθῆς μεγέθη ἀσύμμετραὸέσται.

Proposition 16

If two incommensurable magnitudes are added together then the whole will also be incommensurable with each of them. And if the whole is incommensurable with one of them then the original magnitudes will also be incommensurable (with one another).

Συγκείσθω γὰρ δύο μεγέθη ἀσύμμετρα τὰ AB, $BΓ$· λέγω, ὅτι καὶ ὅλον τὸ $AΓ$ ἑκατέρῳ τῶν AB, $BΓ$ ἀσύμμετρόν ἐστιν.

Εἰ γὰρ μή ἐστιν ἀσύμμετρα τὰ $ΓA$, AB, μετρήσει τι [αὐτὰ] μέγεθος. μετρείτω, εἰ δυνατόν, καὶ ὁέστω τὸ $Δ$. ἐπεὶ οὖν τὸ $Δ$ τὰ $ΓA$, AB μετρεῖ, καὶ λοιπὸν ἄρα τὸ $BΓ$ μετρήσει. μετρεῖ δὲ καὶ τὸ AB· τὸ $Δ$ ἄρα τὰ AB, $BΓ$ μετρεῖ. σύμμετρα ἄρα ἐστὶ τὰ AB, $BΓ$· ὑπέκειντο δὲ καὶ ἀσύμμετρα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὰ $ΓA$, AB μετρήσει τι μέγεθος· ἀσύμμετρα ἄρα ἐστὶ τὰ $ΓA$, AB. ὁμοίως δὴ δείχομεν, ὅτι καὶ τὰ $AΓ$, $ΓB$ ἀσύμμετρά ἐστιν. τὸ $AΓ$ ἄρα ἑκατέρῳ τῶν AB, $BΓ$ ἀσύμμετρόν ἐστιν.

Ἀλλὰ δὴ τὸ $AΓ$ ἑνὶ τῶν AB, $BΓ$ ἀσύμμετρονὁέστω. ὁέστω δὴ πρότερον τῷ AB· λέγω, ὅτι καὶ τὰ AB, $BΓ$ ἀσύμμετρά ἐστιν. εἰ γὰρὁέσται σύμμετρα, μετρήσει τι αὐτὰ μέγεθος. μετρείτω, καὶ ὁέστω τὸ $Δ$. ἐπεὶ οὖν τὸ $Δ$ τὰ AB, $BΓ$ μετρεῖ, καὶ ὅλον ἄρα τὸ $AΓ$ μετρήσει. μετρεῖ δὲ καὶ τὸ AB· τὸ $Δ$ ἄρα τὰ $ΓA$, AB μετρεῖ. σύμμετρα ἄρα ἐστὶ τὰ $ΓA$, AB· ὑπέκειτο δὲ καὶ ἀσύμμετρα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὰ AB, $BΓ$ μετρήσει τι μέγεθος· ἀσύμμετρα ἄρα ἐστὶ τὰ AB, $BΓ$.

Ἐὰν ἄρα δύο μεγέθη, καὶ τὰ ἑξῆς.

For let the two incommensurable magnitudes AB and BC be laid down together. I say that that the whole AC is also incommensurable with each of AB and BC.

For if CA and AB are not incommensurable then some magnitude will measure [them]. If possible, let it (so) measure (them), and let it be D. Therefore, since D measures (both) CA and AB, it will thus also measure the remainder BC. And it also measures AB. Thus, D measures (both) AB and BC. Thus, AB and BC are commensurable [Def. 10.1]. But they were also assumed (to be) incommensurable. The very thing is impossible. Thus, some magnitude cannot measure (both) CA and AB. Thus, CA and AB are incommensurable [Def. 10.1]. So, similarly, we can show that AC and CB are also incommensurable. Thus, AC is incommensurable with each of AB and BC.

And so let AC be incommensurable with one of AB and BC. So let it, first of all, be incommensurable with AB. I say that AB and BC are also incommensurable. For if they are commensurable then some magnitude will measure them. Let it (so) measure (them), and let it be D. Therefore, since D measures (both) AB and BC, it will thus also measure the whole AC. And it also measures AB. Thus, D measures (both) CA and AB. Thus, CA and AB are commensurable [Def. 10.1]. But they were also assumed (to be) incommensurable. The very thing is impossible. Thus, some magnitude cannot measure (both) AB and BC. Thus, AB and BC are incommensurable [Def. 10.1].

Thus, if two. . . magnitudes, and so on . . . .

Λῆμμα.

Ἐὰν παρά τινα εὐθεῖαν παραβληθῇ παραλληλόγραμμον ἐλλεῖπον εἰδει τετραγώνῳ, τὸ παραβληθὲν ἴσον ἐστὶ τῷ ὑπὸ τῶν ἐκ τῆς παραβολῆξ γενομένων τμ ημάτων τῆς εὐθείας.

Lemma

If a parallelogram,† falling short by a square figure, is applied to some straight-line then the applied (parallelogram) is equal (in area) to the (rectangle contained) by the pieces of the straight-line created via the application (of the parallelogram).

Παρὰ γὰρ εὐθεῖαν τὴν AB παραβεβλήσθω παραλληλόγραμμον τὸ $A\Delta$ ἐλλεῖπον εἰδει τετραγώνῳ τῷ ΔB· λέγω, ὅτι ἴσον ἐστὶ τὸ $A\Delta$ τῷ ὑπὸ τῶν $A\Gamma$, ΓB.

Καί ἐστιν αὐτόθεν φανερόν· ἐπεὶ γὰρ τετράγωνόν ἐστι τὸ ΔB, ὁίση ἐστὶν ἡ $\Delta\Gamma$ τῇ ΓB, καί ἐστι τὸ $A\Delta$ τὸ ὑπὸ τῶν $A\Gamma$, $\Gamma\Delta$, τουτέστι τὸ ὑπὸ τῶν $A\Gamma$, ΓB.

Ἐὰν ἄρα παρά τινα εὐθεῖαν, καὶ τὰ ἑξῆς.

For let the parallelogram AD, falling short by the square figure DB, be applied to the straight-line AB. I say that AD is equal to the (rectangle contained) by AC and CB.

And it is immediately obvious. For since DB is a square, DC is equal to CB. And AD is the (rectangle contained) by AC and CD—that is to say, by AC and CB.

Thus, if ... to some straight-line, and so on ....

17.

Ἐὰν ὦσι δύο εὐθεῖαι ἀνισοι, τῷ δὲ τετράτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἐλλεῖπον εἰδει τετραγώνῳ καὶ εἰξ σύμμετρα αὐτὴν διαιρῇ μήκει, ἡ μείζων τῆς ἐλάσσονος μεῖζον δυνήσεται τῷ ἀπὸ συμμέτου ἑαυτῇ [μήκει]. καὶ ἐαν ἡ μείζων τῆς ἐλάσσονος μεῖζον δύνηται τῷ ἀπὸ συμμέτρου ἑαυτῇ [μήκει], τῷ δὲ τετράρτῳ τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἐλλεῖπον εἰδει τετραγώνῳ, εἰξ σύμμετρα αὐτὴν διαιρεῖ μήκει.

Ἔστωσαν δύο εὐθεῖαι ἀνισοι αἱ A, $B\Gamma$, ὧν μείζων ἡ $B\Gamma$, τῷ δὲ τετράρτῳ μέρει τοῦ ἀπὸ ἐλάσσονος τῆς A, τουτέστι τῷ ἀπὸ τῆς ἡμισείας τῆς A, ἴσον παρὰ τὴν $B\Gamma$ παραβεβλήσθω ἐλλεῖπον εἰδει τετραγώνῳ, καὶ ἐστω τὸ ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$, σύμμετρος δὲ ἐστω ἡ $B\Delta$ τῇ $\Delta\Gamma$ μήκει· λέγω, ὅτι ἡ $B\Gamma$ τῆς A μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ.

Proposition 17[†]

If there are two unequal straight-lines, and a (rectangle) equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) commensurable in length, then the square on the greater will be larger than (the square on) the lesser by the (square) on (some straight-line) commensurable [in length] with the greater. And if the square on the greater is larger than (the square on) the lesser by the (square) on (some straight-line) commensurable [in length] with the greater, and a (rectangle) equal to the fourth (part) of the (square) on the lesser, falling short by a square figure, is applied to the greater, then it divides it into (parts which are) commensurable in length.

Let A and BC be two unequal straight-lines, of which (let) BC (be) the greater. And let a (rectangle) equal to the fourth part of the (square) on the lesser, A—that is, (equal) to the (square) on half of A—falling short by a square figure, be applied to BC. And let it be the (rectangle contained) by BD and DC [see previous lemma]. And let BD be commensurable in length with DC. I say that that the square on BC is greater than the (square on) A by (the square on some straight-line)

[†] Note that this lemma only applies to rectangular parallelograms.

Τετμήσθω γὰρ ἡ ΒΓ δίθα κατὰ τὸ Ε σημεῖον, καὶ κείσθω τῇ ΔΕὀίση ἡ ΕΖ. λοιπὴ ἄρα ἡ ΔΓὀίση ἐστὶ τῇ ΒΖ. καὶ ἐπεὶ εὐθεῖα ἡ ΒΓ τέτμηται εἰξ μὲν ἴσα κατὰ τὸ Ε, εἰξ δὲ ὀάνισα κατὰ τὸ Δ, τὸ ἄρα ὑπὸ ΒΔ, ΔΓ περιεθόμενον ὀρθογώνιον μετὰ τοῦ ἀπὸ τῆς ΕΔ τετραγώνου ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΓ τετραγώνῳ· καὶ τὰ τετραπλάσια· τὸ ἄρα τετράκις ὑπὸ τῶν ΒΔ, ΔΓ μετὰ τοῦ τετραπλασίου τοῦ ἀπὸ τῆς ΔΕ ἴσον ἐστὶ τῷ τετράκις ἀπὸ τῆς ΕΓ τετραγώνῳ. ἀλλὰ τῷ μὲν τετραπλασίῳ τοῦ ὑπὸ τῶν ΒΔ, ΔΓ ἴσον ἐστὶ τὸ ἀπὸ τῆς Α τετράγωνον, τῷ δὲ τετραπλασίῳ τοῦ ἀπὸ τῆς ΔΕ ἴσον ἐστὶ τὸ ἀπὸ τῆς ΔΖ τετράγωνον· διπλασίων γάρ ἐστιν ἡ ΔΖ τῆς ΔΕ. τῷ δὲ τετραπλασίῳ τοῦ ἀπὸ τῆς ΕΓ ἴσον ἐστὶ τὸ ἀπὸ τῆς ΒΓ τετράγωνον· διπλασίων γάρ ἐστι πάλιν ἡ ΒΓ τῆς ΓΕ. τὰ ἄρα ἀπὸ τῶν Α, ΔΖ τετράγωνα ἴσα ἐστὶ τῷ ἀπὸ τῆς ΒΓ τετράγωνῳ· ὥστε τὸ ἀπὸ τῆς ΒΓ τοῦ ἀπὸ τῆς Α μεῖζόν ἐστι τῷ ἀπὸ τῆς ΔΖ· ἡ ΒΓ ἄρα τῆς Α μεῖζον δύναται τῇ ΔΖ. δεικτέον, ὅτι καὶ σύμμετρόξ ἐστιν ἡ ΒΓ τῇ ΔΖ. ἐπεὶ γὰρ σύμμετρόξ ἐστιν ἡ ΒΔ τῇ ΔΓ μήκει, σύμμετρος ἄρα ἐστὶ καὶ ἡ ΒΓ τῇ ΓΔ μήκει. ἀλλὰ ἡ ΓΔ ταῖς ΓΔ, ΒΖ ἐστι σύμμετρος μήκει· ὀίση γάρ ἐστιν ἡ ΓΔ τῇ ΒΖ. καὶ ἡ ΒΓ ἄρα σύμμετρόξ ἐστι ταῖς ΒΖ, ΓΔ μήκει· ὥστε καὶ λοιπῇ τῇ ΖΔ σύμμετρόξ ἐστιν ἡ ΒΓ μήκει· ἡ ΒΓ ἄρα τῆς Α μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ.

Ἀλλὰ δὴ ἡ ΒΓ τῆς Α μεῖζον δυνάσθω τῷ ἀπὸ συμμέτρου ἑαυτῇ, τῷ δὲ τετράτρῳ τοῦ ἀπὸ τῆς Α ἴσον παρὰ τὴν ΒΓ παραβεβλήσθω ἐλλεῖπον εὐίδει τετραγώνῳ, καὶ ὀέστω τὸ ὑπὸ τῶν ΒΔ, ΔΓ. δεικτέον, ὅτι σύμμετρόξ ἐστιν ἡ ΒΔ τῇ ΔΓ μήκει.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, ὅτι ἡ ΒΓ τῆς Α μεῖζον δύναται τῷ ἀπὸ τῆς ΖΔ. δύναται δὲ ἡ ΒΓ τῆς Α μεῖζον τῷ ἀπὸ συμμέτρου ἑαυτῇ. σύμμετρος ἄρα ἐστὶν ἡ ΒΓ τῇ ΖΔ μήκει· ὥστε καὶ λοιπῇ συναμφοτέρῳ τῇ ΒΖ, ΔΓ σύμμετρόξ ἐστιν ἡ ΒΓ μήκει. ἀλλὰ συναμφότερος ἡ ΒΖ, ΔΓ σύμμετρόξ ἐστι τῇ ΔΓ [μήκει]. ὥστε καὶ ἡ ΒΓ τῇ ΓΔ σύμμετρόξ ἐστι μήκει· καὶ διελόντι ἄρα ἡ ΒΔ τῇ ΔΓ ἐστι

For let BC be cut in half at the point E [Prop. 1.10]. And let EF be made equal to DE [Prop. 1.3]. Thus, the remainder DC is equal to BF. And since the straight-line BC has been cut into equal (pieces) at E, and into unequal (pieces) at D, the rectangle contained by BD and DC, plus the square on ED, is thus equal to the square on EC [Prop. 2.5]. (The same) also (for) the quadruples. Thus, four times the (rectangle contained) by BD and DC, plus the quadruple of the (square) on DE, is equal to four times the square on EC. But, the square on A is equal to the quadruple of the (rectangle contained) by BD and DC, and the square on DF is equal to the quadruple of the (square) on DE. For DF is double DE. And the square on BC is equal to the quadruple of the (square) on EC. For, again, BC is double CE. Thus, the (sum of the) squares on A and DF is equal to the square on BC. Hence, the (square) on BC is greater than the (square) on A by the (square) on DF. Thus, BC is greater in square than A by DF. It must also be shown that BC is commensurable (in length) with DF. For since BD is commensurable in length with DC, BC is thus also commensurable in length with CD [Prop. 10.15]. But, CD is commensurable in length with CD plus BF. For CD is equal to BF [Prop. 10.6]. Thus, BC is also commensurable in length with BF plus CD [Prop. 10.12]. Hence, BC is also commensurable in length with the remainder FD [Prop. 10.15]. Thus, the square on BC is greater than (the square on) A by the (square) on (some straight-line) commensurable (in length) with (BC).

And so let the square on BC be greater than the (square on) A by the (square) on (some straight-line) commensurable (in length) with (BC). And let a (rectangle) equal to the fourth (part) of the (square) on A, falling short by a square figure, be applied to BC. And let it be the (rectangle contained) by BD and DC. It must be shown that BD is commensurable in length with DC.

For, similarly, by the same construction, we can show that

σύμμετρος μήκει.

Ἐὰν ἄραὸῶσι δύο εὐθεῖαιὀάνισοι, καὶ τὰ ἑχῆς.

the square on *BC* is greater than the (square on) *A* by the (square) on *FD*. And the square on *BC* is greater than the (square on) *A* by the (square) on (some straight-line) commensurable (in length) with (*BC*). Thus, *BC* is commensurable in length with *FD*. Hence, *BC* is also commensurable in length with the remaining sum of *BF* and *DC* [Prop. 10.15]. But, the sum of *BF* and *DC* is commensurable [in length] with *DC* [Prop. 10.6]. Hence, *BC* is also commensurable in length with *CD* [Prop. 10.12]. Thus, via separation, *BD* is also commensurable in length with *DC* [Prop. 10.15].

Thus, if there are two unequal straight-lines, and so on ....

† This proposition states that if $\alpha\, x - x^2 = \beta^2/4$ (where $\alpha = BC$, $x = DC$, and $\beta = A$) then α and $\sqrt{\alpha^2 - \beta^2}$ are commensurable when $\alpha - x$ are x are commensurable, and vice versa.

18.

Ἐὰνὸῶσι δύο εὐθεῖαιὀάνισοι, τῷ δὲ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἐλλεῖπον εἰὸίδει τετραγώνῳ, καὶ εἰξ ἀσυμμετρα αὐτὴν διαιρῇ [μήκει], ἡ μείζων τῆς ἐλάσσονος μεῖζον δυνήσεται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καὶ ἐαν ἡ μείζων τῆς ἐλάσσονος μεῖζον δύνηται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, τῷ δὲ τετράρτῳ τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἐλλεῖπον εἰὸίδει τετραγώνῳ, εἰξ ἀσύμμετρα αὐτὴν διαιρεῖ [μήκει].

Ἔστωσαν δύο εὐθεῖαιὀάνισοι αἱ Α, ΒΓ, ὁῶν μείζων ἡ ΒΓ, τῷ δὲ τετάρτῳ [μέρει] τοῦ ἀπὸ τῆς ἐλάσσονος τῆς Α ἴσον παρὰ τὴν ΒΓ παραβεβλήσθω ἐλλεῖπον εἰὸίδει τετραγώνῳ, καὶ ὀέστω τὸ ὑπὸ τῶν ΒΔΓ, ἀσύμμετρος δὲ ὀέστω ἡ ΒΔ τῇ ΔΓ μήκει· λέγω, ὅτι ἡ ΒΓ τῆς Α μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ.

Proposition 18†

If there are two unequal straight-lines, and a (rectangle) equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) incommensurable [in length], then the square on the greater will be larger than the (square on the) lesser by the (square) on (some straight-line) incommensurable (in length) with the greater. And if the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) incommensurable (in length) with the greater, and a (rectangle) equal to the fourth (part) of the (square) on the lesser, falling short by a square figure, is applied to the greater, then it divides it into (parts which are) incommensurable [in length].

Let *A* and *BC* be two unequal straight-lines, of which (let) *BC* (be) the greater. And let a (rectangle) equal to the fourth [part] of the (square) on the lesser, *A*, falling short by a square figure, be applied to *BC*. And let it be the (rectangle contained) by *BDC*. And let *BD* be incommensurable in length with *DC*. I say that that the square on *BC* is greater than the (square on) *A* by the (square) on (some straight-line) incommensurable (in length) with (*BC*).

Τῶν γὰρ αὐτῶν κατασκευασθέντων τῷ πρότερον ὁμοίως δείχομεν, ὅτι ἡ ΒΓ τῆς Α μεῖζον δύναται τῷ ἀπὸ τῆς ΖΔ.

For, similarly, by the same construction as before, we can show that the square on *BC* is greater than the (square on) *A* by

δεικτέον [οὐῦν], ὅτι ἀσύμμετρόξ ἐστιν ἡ ΒΓ τῇ ΔΖ μήκει. ἐπεὶ γὰρ ἀσύμμετρόξ ἐστιν ἡ ΒΔ τῇ ΔΓ μήκει, ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ ΒΓ τῇ ΓΔ μήκει. ἀλλὰ ἡ ΔΓ σύμμετρόξ ἐστι συναμφοτέραις ταῖς ΒΖ, ΔΓ· καὶ ἡ ΒΓ ἄρα ἀσύμμετρόξ ἐστι συναμφοτέραις ταῖς ΒΖ, ΔΓ. ὥστε καὶ λοιπῇ τῇ ΖΔ ἀσύμμετρόξὸέστιν ἡ ΒΓ μήκει. καὶ ἡ ΒΓ τῆς Α μεῖζον δύναται τῷ ἀπὸ τῆς ΖΔ· ἡ ΒΓ ἄρα τῆς Α μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ.

Δυνάσθω δὴ πάλιν ἡ ΒΓ τῆς Α μεῖζον τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, τῷ δὲ τετάρτῳ τοῦ ἀπὸ τῆς Α ἴσον παρὰ τὴν ΒΓ παραβεβλήσθω ἐλλεῖπον εἰόιδει τετραγώνῳ, καὶ ὀέστω τὸ ὑπὸ τῶν ΒΔ, ΔΓ. δεικτέον, ὅτι ἀσύμμετρόξ ἐστιν ἡ ΒΔ τῇ ΔΓ μήκει.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ὁμοίως δείχομεν, ὅτι ἡ ΒΓ τῆς Α μεῖζον δύναται τῷ ἀπὸ τῆς ΖΔ. ἀλλὰ ἡ ΒΓ τῆς Α μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. ἀσύμμετρος ἄρα ἐστὶν ἡ ΒΓ τῇ ΖΔ μήκει· ὥστε καὶ λοιπῇ συναμφοτέρῳ τῇ ΒΖ, ΔΓ ἀσύμμετρόξ ἐστιν ἡ ΒΓ. ἀλλὰ συναμφότερος ἡ ΒΖ, ΔΓ τῇ ΔΓ σύμμετρόξ ἐστι μήκει· καὶ ἡ ΒΓ ἄρα τῇ ΔΓ ἀσύμμετρόξ ἐστι μήκει· ὥστε καὶ διελόντι ἡ ΒΔ τῇ ΔΓ ἀσύμμετρόξ ἐστι μήκει.

Ἐὰν ἄραὸῶσι δύο εὐθεῖαι, καὶ τὰ ἑχῆς.

the (square) on FD. [Therefore] it must be shown that BC is incommensurable in length with DF. For since BD is incommensurable in length with DC, BC is thus also incommensurable in length with CD [Prop. 10.16]. But, DC is commensurable (in length) with the sum of BF and DC [Prop. 10.6]. And, thus, BC is incommensurable (in length) with the sum of BF and DC [Prop. 10.13]. Hence, BC is also incommensurable in length with the remainder FD [Prop. 10.16]. And the square on BC is greater than the (square on) A by the (square) on FD. Thus, the square on BC is greater than the (square on) A by the (square) on (some straight-line) incommensurable (in length) with (BC).

So, again, let the square on BC be greater than the (square on) A by the (square) on (some straight-line) incommensurable (in length) with (BC). And let a (rectangle) equal to the fourth [part] of the (square) on A, falling short by a square figure, be applied to BC. And let it be the (rectangle contained) by BD and DC. It must be shown that BD is incommensurable in length with DC.

For, similarly, by the same construction, we can show that the square on BC is greater than the (square) on A by the (square) on FD. But, the square on BC is greater than the (square) on A by the (square) on (some straight-line) incommensurable (in length) with (BC). Thus, BC is incommensurable in length with FD. Hence, BC is also incommensurable (in length) with the remaining sum of BF and DC [Prop. 10.16]. But, the sum of BF and DC is commensurable in length with DC [Prop. 10.6]. Thus, BC is also incommensurable in length with DC [Prop. 10.13]. Hence, via separation, BD is also incommensurable in length with DC [Prop. 10.16].

Thus, if there are two ... straight-lines, and so on ....

† This proposition states that if $\alpha\, x - x^2 = \beta^2/4$ (where $\alpha = BC$, $x = DC$, and $\beta = A$) then α and $\sqrt{\alpha^2 - \beta^2}$ are incommensurable when $\alpha - x$ are x are incommensurable, and vice versa.

19.

Τὸ ὑπὸ ὁρητῶν μήκει συμμέτρων εὐθειῶν περιεχόμενον ὀρθογώνιον ὁρητόν ἐστιν.

Ὑπὸ γὰρ ὁρητῶν μήκει συμμέτρων εὐθειῶν τῶν ΑΒ, ΒΓ ὀρθογώνιον περιεθέσθω τὸ ΑΓ· λέγω, ὅτι ὁρητόν ἐστι τὸ ΑΓ.

Proposition 19

The rectangle contained by rational straight-lines (which are) commensurable in length is rational.

For let the rectangle AC be enclosed by the rational straight-lines AB and BC (which are) commensurable in length. I say that AC is rational.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ $A\Delta$· ὁρητὸν ἄρα ἐστὶ τὸ $A\Delta$. καὶ ἐπεὶ σύμμετρός ἐστιν ἡ AB τῇ $B\Gamma$ μήκει, ὁίση δέ ἐστιν ἡ AB τῇ $B\Delta$, σύμμετρος ἄρα ἐστὶν ἡ $B\Delta$ τῇ $B\Gamma$ μήκει. καί ἐστιν ὡξ ἡ $B\Delta$ πρὸς τὴν $B\Gamma$, οὕτως τὸ ΔA πρὸς τὸ $A\Gamma$. σύμμετρον ἄρα ἐστὶ τὸ ΔA τῷ $A\Gamma$. ὁρητὸν δὲ τὸ ΔA· ὁρητὸν ἄρα ἐστὶ καὶ τὸ $A\Gamma$.

Τὸ ἄρα ὑπὸ ὁρητῶν μήκει συμμέτρων, καὶ τὰ ἑξῆς.

For let the square AD be described on AB. AD is thus rational [Def. 10.4]. And since AB is commensurable in length with BC, and AB is equal to BD, BD is thus commensurable in length with BC. And as BD is to BC, so DA (is) to AC [Prop. 6.1]. Thus, DA is commensurable with AC [Prop. 10.11]. And DA (is) rational. Thus, AC is also rational [Def. 10.4]. Thus, the ... by rational straight-lines ... commensurable, and so on ....

20.

Ἐὰν ὁρητὸν παρὰ ὁρητὴν παραβληθῇ, πλάτος ποιεῖ ὁρητὴν καὶ σύμμετρον τῇ, παρὸ ἣν παράκειται, μήκει.

Proposition 20

If a rational (area) is applied to a rational (straight-line) then it produces as breadth a (straight-line which is) rational, and commensurable in length with the (straight-line) to which it is applied.

ὁΡητὸν γὰρ τὸ ΑΓ παρὰ ὁρητὴν τὴν ΑΒ παραβεβλήσθω πλάτος ποιοῦν τὴν ΒΓ· λέγω, ὅτι ὁρητή ἐστιν ἡ ΒΓ καὶ σύμμετρος τῇ ΒΑ μήκει.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΑΒ τετράγωνον τὸ ΑΔ· ὁρητὸν ἄρα ἐστὶ τὸ ΑΔ. ὁρητὸν δὲ καὶ τὸ ΑΓ· σύμμετρον ἄρα ἐστὶ τὸ ΔΑ τῷ ΑΓ. καί ἐστιν ὡς τὸ ΔΑ πρὸς τὸ ΑΓ, οὕτως ἡ ΔΒ πρὸς τὴν ΒΓ. σύμμετρος ἄρα ἐστὶ καὶ ἡ ΔΒ τῇ ΒΓ· ὁίση δὲ ἡ ΔΒ τῇ ΒΑ· σύμμετρος ἄρα καὶ ἡ ΑΒ τῇ ΒΓ. ὁρητὴ δέ ἐστιν ἡ ΑΒ· ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΒΓ καὶ σύμμετρος τῇ ΑΒ μήκει.

Ἐὰν ἄρα ὁρητὸν παρὰ ὁρητὴν παραβληθῇ, καὶ τὰ ἑχῆς.

For let the rational (area) AC be applied to the rational (straight-line) AB, producing the (straight-line) BC as breadth. I say that BC is rational, and commensurable in length with BA.

For let the square AD be described on AB. AD is thus rational [Def. 10.4]. And AC (is) also rational. DA is thus commensurable with AC. And as DA is to AC, so DB (is) to BC [Prop. 6.1]. Thus, DB is also commensurable (in length) with BC [Prop. 10.11]. And DB (is) equal to BA. Thus, AB (is) also commensurable (in length) with BC. And AB is rational. Thus, BC is also rational, and commensurable in length with AB [Def. 10.3].

Thus, if a rational (area) is applied to a rational (straight-line), and so on ....

21.

Τὸ ὑπὸ ὁρητῶν δυνάμει μόνον συμμέτρων εὐθειῶν περιεχόμενον ὀρθογώνιονὁάλογόν ἐστιν, καὶ ἡ δυναμένη αὐτὸ ὁάλογόξ ἐστιν, καλείσθω δὲ μέση.

Proposition 21

The rectangle contained by rational straight-lines (which are) commensurable in square only is irrational, and its square-root is irrational—let it be called medial.[†]

Ὑπὸ γὰρ ὁρητῶν δυνάμει μόνον συμμέτρων εὐθειῶν τῶν AB, $BΓ$ ὀρθογώνιον περιειθέσθω τὸ $AΓ$· λέγω, ὅτι ὁ $AΓ$ ἐστιν, καὶ ἡ δυναμένη αὐτὸ ὁ ἐστιν, καλείσθω δὲ μέση.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ $AΔ$· ὁρητὸν ἄρα ἐστὶ τὸ $AΔ$. καὶ ἐπεὶ ἀσύμμετρός ἐστιν ἡ AB τῇ $BΓ$ μήκει· δυνάμει γὰρ μόνον ὑπόκεινται σύμμετροι· ὁίση δὲ ἡ AB τῇ $BΔ$, ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ $ΔB$ τῇ $BΓ$ μήκει. καί ἐστιν ὡς ἡ $ΔB$ πρὸς τὴν $BΓ$, οὕτως τὸ $AΔ$ πρὸς τὸ $AΓ$· ἀσύμμετρον ἄρα [ἐστὶ] τὸ $ΔA$ τῷ $AΓ$. ὁρητὸν δὲ τὸ $ΔA$· ὁ $AΓ$ ἄρα ἐστὶ τὸ $AΓ$· ὥστε καὶ ἡ δυναμένη τὸ $AΓ$ [τουτέστιν ἡ ἴσον αὐτῷ τετράγωνον δυναμένη] ὁ ἐστιν, καλείσθω δε μέση· ὅπερ ἔδει δεῖχαι.

For let the rectangle AC be contained by the rational straight-lines AB and BC (which are) commensurable in square only. I say that AC is irrational, and its square-root is irrational—let it be called medial.

For let the square AD be described on AB. AD is thus rational [Def. 10.4]. And since AB is incommensurable in length with BC. For they were assumed to be commensurable in square only. And AB (is) equal to BD. DB is thus also incommensurable in length with BC. And as DB is to BC, so AD (is) to AC [Prop. 6.1]. Thus, DA [is] incommensurable with AC [Prop. 10.11]. And DA (is) rational. Thus, AC is irrational [Def. 10.4]. Hence, its square-root [that is to say, the square-root of the square equal to it] is also irrational [Def. 10.4]—let it be called medial. (Which is) the very thing it was required to show.

† Thus, a medial straight-line has a length expressible as $k^{1/4}$.

Λῆμμα.

Ἐὰν ὦσι δύο εὐθεῖαι, ὁέστιν ὡς ἡ πρώτη πρὸς τὴν δευτέραν, οὕτως τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ὑπὸ τῶν δύο εὐθειῶν.

Lemma

If there are two straight-lines then as the first is to the second, so the (square) on the first (is) to the (rectangle contained) by the two straight-lines.

Ἔστωσαν δύο εὐθεῖαι αἱ ΖΕ, ΕΗ. λέγω, ὅτι ἐστὶν ὡς ἡ ΖΕ πρὸς τὴν ΕΗ, οὕτως τὸ ἀπὸ τῆς ΖΕ πρὸς τὸ ὑπὸ τῶν ΖΕ, ΕΗ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΖΕ τετράγωνον τὸ ΔΖ, καὶ συμπεπληρώσθω τὸ ΗΔ. ἐπεὶ οὐόν ἐστιν ὡς ἡ ΖΕ πρὸς τὴν ΕΗ, οὕτως τὸ ΖΔ πρὸς τὸ ΔΗ, καί ἐστι τὸ μὲν ΖΔ τὸ ἀπὸ τῆς ΖΕ, τὸ δὲ ΔΗ τὸ ὑπὸ τῶν ΔΕ, ΕΗ, τουτέστι τὸ ὑπὸ τῶν ΖΕ, ΕΗ, ὁέστιν ἄρα ὡς ἡ ΖΕ πρὸς τὴν ΕΗ, οὕτως τὸ ἀπὸ τῆς ΖΕ πρὸς τὸ ὑπὸ τῶν ΖΕ, ΕΗ. ὁμοίως δὲ καὶ ὡς τὸ ὑπὸ τῶν ΗΕ, ΕΖ πρὸς τὸ ἀπὸ τῆς ΕΖ, τουτέστιν ὡς τὸ ΗΔ πρὸς τὸ ΖΔ, οὕτως ἡ ΗΕ πρὸς τὴν ΕΖ· ὅπερ ἔδει δεῖχαι.

Let FE and EG be two straight-lines. I say that as FE is to EG, so the (square) on FE (is) to the (rectangle contained) by FE and EG.

For let the square DF be described on FE. And let GD be completed. Therefore, since as FE is to EG, so FD (is) to DG [Prop. 6.1], and FD is the (square) on FE, and DG the (rectangle contained) by DE and EG—that is to say, the (rectangle contained) by FE and EG—thus as FE is to EG, so the (square) on FE (is) to the (rectangle contained) by FE and EG. And also, similarly, as the (rectangle contained) by GE and EF is to the (square on) EF—that is to say, as GD (is) to FD—so GE (is) to EF. (Which is) the very thing it was required to show.

22.

Τὸ ἀπὸ μέσης παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ ὁρητὴν καὶ ἀσύμμετρον τῇ, παρὸ ἣν παράκειται, μήκει.

Ἔστω μέση μὲν ἡ Α, ὁρητὴ δὲ ἡ ΓΒ, καὶ τῷ ἀπὸ τῆς Α ἴσον παρὰ τὴν ΒΓ παραβεβλήσθω θωρίον ὀρθογώνιον τὸ ΒΔ πλάτος ποιοῦν τὴν ΓΔ· λέγω, ὅτι ὁρητή ἐστιν ἡ ΓΔ καὶ ἀσύμμετρος τῇ ΓΒ μήκει.

Proposition 22

The square on a medial (straight-line), being applied to a rational (straight-line), produces as breadth a (straight-line which is) rational, and incommensurable in length with the (straight-line) to which it is applied.

Let A be a medial (straight-line), and CB a rational (straight-line), and let the rectangular area BD, equal to the (square) on A, be applied to BC, producing CD as breadth. I say that CD is rational, and incommensurable in length with CB.

Ἐπεὶ γὰρ μέση ἐστὶν ἡ Α, δύναται θωρίον περιεχόμενον ὑπὸ ὀρητῶν δυνάμει μόνον συμμέτρων. δυνάσθω τὸ ΗΖ. δύναται δὲ καὶ τὸ ΒΔ· ἴσον ἄρα ἐστὶ τὸ ΒΔ τῷ ΗΖ. ὀέστι δὲ αὐτῷ καὶ ἰσογώνιον· τῶν δὲ ὁίσων τε καὶ ἰσογωνίων παραλληλογράμμων ἀντιπεπόνθασιν αἱ πλευραὶ αἱ περὶ τὰξ ἴσας γωνίας· ἀνάλογον ἄρα ἐστὶν ὡξ ἡ ΒΓ πρὸς τὴν ΕΗ, οὕτως ἡ ΕΖ πρὸς τὴν ΓΔ. ὀέστιν ἄρα καὶ ὡξ τὸ ἀπὸ τῆς ΒΓ πρὸς τὸ ἀπὸ τῆς ΕΗ, οὕτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΓΔ. σύμμετρον δέ ἐστι τὸ ἀπὸ τῆς ΓΒ τῷ ἀπὸ τῆς ΕΗ· ὀρητὴ γάρ ἐστιν ἑκατέρα αὐτῶν· σύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς ΕΖ τῷ ἀπὸ τῆς ΓΔ. ὀρητὸν δέ ἐστι τὸ ἀπὸ τῆς ΕΖ· ὀρητὸν ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς ΓΔ· ὀρητὴ ἄρα ἐστὶν ἡ ΓΔ. καὶ ἐπεὶ ἀσύμμετρόξ ἐστιν ἡ ΕΖ τῇ ΕΗ μήκει· δυνάμει γὰρ μόνον εἰσὶ σύμμετροι· ὡξ δὲ ἡ ΕΖ πρὸς τὴν ΕΗ, οὕτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ὑπὸ τῶν ΖΕ, ΕΗ, ἀσύμμετρον ἄρα [ἐστὶ] τὸ ἀπὸ τῆς ΕΖ τῷ

For since A is medial, the square on it is equal to a (rectangular) area contained by rational (straight-lines which are) commensurable in square only [Prop. 10.21]. Let the square on (A) be equal to GF. And the square on (A) is also equal to BD. Thus, BD is equal to GF. And (BD) is also equiangular with (GF). And for equal and equiangular parallelograms, the sides about the equal angles are reciprocally proportional [Prop. 6.14]. Thus, proportionally, as BC is to EG, so EF (is) to CD. And, also, as the (square) on BC is to the (square) on EG, so the (square) on EF (is) to the (square) on CD [Prop. 6.22]. And the (square) on CB is commensurable with the (square) on EG. For they are each rational. Thus, the (square) on EF is also commensurable with the (square) on CD [Prop. 10.11]. And the (square) on EF is rational. Thus, the (square) on CD is also rational [Def. 10.4]. Thus, CD is ra-

ὑπὸ τῶν ΖΕ, ΕΗ. ἀλλὰ τῷ μὲν ἀπὸ τῆς ΕΖ σύμμετρόν ἐστι τὸ ἀπὸ τῆς ΓΔ· ὀρηταὶ γάρ εἰσι δυνάμει· τῷ δὲ ὑπὸ τῶν ΖΕ, ΕΗ σύμμετρόν ἐστι τὸ ὑπὸ τῶν ΔΓ, ΓΒ· ἴσα γάρ ἐστι τῷ ἀπὸ τῆς Α· ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς ΓΔ τῷ ὑπὸ τῶν ΔΓ, ΓΒ. ὡς δὲ τὸ ἀπὸ τῆς ΓΔ πρὸς τὸ ὑπὸ τῶν ΔΓ, ΓΒ, οὕτως ἐστὶν ἡ ΔΓ πρὸς τὴν ΓΒ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΔΓ τῇ ΓΒ μήκει. ὀρητὴ ἄρα ἐστὶν ἡ ΓΔ καὶ ἀσύμμετρος τῇ ΓΒ μήκει· ὅπερ ἔδει δεῖξαι.

tional. And since EF is incommensurable in length with EG. For they are commensurable in square only. And as EF (is) to EG, so the (square) on EF (is) to the (rectangle contained) by FE and EG [see previous lemma]. The (square) on EF [is] thus incommensurable with the (rectangle contained) by FE and EG [Prop. 10.11]. But, the (square) on CD is commensurable with the (square) on EF. For they are rational in square. And the (rectangle contained) by DC and CB is commensurable with the (rectangle contained) by FE and EG. For they are (both) equal to the (square) on A. Thus, the (square) on CD is also incommensurable with the (rectangle contained) by DC and CB [Prop. 10.13]. And as the (square) on CD (is) to the (rectangle contained) by DC and CB, so DC is to CB [see previous lemma]. Thus, DC is incommensurable in length with CB [Prop. 10.11]. Thus, CD is rational, and incommensurable in length with CB. (Which is) the very thing it was required to show.

† Literally, "rational".

23.

Ἡ τῇ μέσῃ σύμμετρος μέση ἐστίν.

Ἔστω μέση ἡ Α, καὶ τῇ Α σύμμετροςὁέστω ἡ Β· λέγω, ὅτι καὶ ἡ Β μέση ἐστίν.

Ἐκκείσθω γὰρ ὀρητὴ ἡ ΓΔ, καὶ τῷ μὲν ἀπὸ τῆς Α ἴσον παρὰ τὴν ΓΔ παραβεβλήσθω θωρίον ὀρθογώνιον τὸ ΓΕ πλάτος ποιοῦν τὴν ΕΔ· ὀρητὴ ἄρα ἐστὶν ἡ ΕΔ καὶ ἀσύμμετρος τῇ ΓΔ μήκει. τῷ δὲ ἀπὸ τῆς Β ἴσον παρὰ τὴν ΓΔ παραβεβλήσθω θωρίον ὀρθογώνιον τὸ ΓΖ πλάτος ποιοῦν τὴν ΔΖ. ἐπεὶ οὐῦν σύμμετρόξ ἐστιν ἡ Α τῇ Β, σύμμετρόν ἐστι καὶ τὸ ἀπὸ τῆς Α τῷ ἀπὸ τῆς Β. ἀλλὰ τῷ μὲν ἀπὸ τῆς Α ἴσον ἐστὶ τὸ ΕΓ, τῷ δὲ ἀπὸ τῆς Β ἴσον ἐστὶ τὸ ΓΖ· σύμμετρον ἄρα ἐστὶ τὸ ΕΓ τῷ ΓΖ. καί ἐστιν ὡξ τὸ ΕΓ πρὸς τὸ ΓΖ, οὕτως ἡ ΕΔ πρὸς τὴν ΔΖ· σύμμετρος ἄρα ἐστὶν ἡ ΕΔ τῇ ΔΖ μήκει. ὀρητὴ δέ ἐστιν ἡ ΕΔ καὶ ἀσύμμετρος τῇ ΔΓ μήκει· ὀρητὴ ἄρα ἐστὶ καὶ ἡ ΔΖ καὶ ἀσύμμετρος τῇ ΔΓ μήκει· αἱ ΓΔ, ΔΖ ἄρα ὀρηταί εἰσι δυνάμει μόνον σύμμετροι. ἡ δὲ τὸ ὑπὸ ὀρητῶν δυνάμει μόνον συμμέτρων δυναμένη μέση ἐστίν. ἡ ἄρα τὸ ὑπὸ τῶν ΓΔ, ΔΖ δυναμένη μέση ἐστίν· καὶ δύναται τὸ ὑπὸ τῶν ΓΔ, ΔΖ ἡ Β· μέση ἄρα ἐστὶν ἡ Β.

Proposition 23

A (straight-line) commensurable with a medial (straight-line) is medial.

Let A be a medial (straight-line), and let B be commensurable with A. I say that B is also a medial (staight-line).

Let the rational (straight-line) CD be set out, and let the rectangular area CE, equal to the (square) on A, be applied to CD, producing ED as width. ED is thus rational, and incommensurable in length with CD [Prop. 10.22]. And let the rectangular area CF, equal to the (square) on B, be applied to CD, producing DF as width. Therefore, since A is commensurable with B, the (square) on A is also commensurable with the (square) on B. But, EC is equal to the (square) on A, and CF is equal to the (square) on B. Thus, EC is commensurable with CF. And as EC is to CF, so ED (is) to DF [Prop. 6.1]. Thus, ED is commensurable in length with DF [Prop. 10.11]. And ED is rational, and incommensurable in length with CD. DF is thus also rational [Def. 10.3], and incommensurable in length with DC [Prop. 10.13]. Thus, CD and DF are rational, and commensurable in square only. And the square-root of a (rectangle contained) by rational (straight-lines which are) commensurable in square only is medial [Prop. 10.21]. Thus, the square-root of the (rectangle contained) by CD and DF is medial. And the square on B is equal to the (rectangle contained) by CD and DF. Thus, B is a medial (straight-line).

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι τὸ τῷ μέσῳ ϑωρίῳ σύμμετρον μέσον ἐστίν.

Corollary

And (it is) clear, from this, that an (area) commensurable with a medial area† is medial.

† A medial area is equal to the square on some medial straight-line. Hence, a medial area is expressible as $k^{1/2}$.

24.

Τὸ ὑπὸ μέσων μήκει συμμέτρων εὐθειῶν περιεχόμενον ὀρθογώνιον μέσον ἐστίν.

Ὑπὸ γὰρ μέσων μήκει συμμέτρων εὐθειῶν τῶν AB, $BΓ$ περιεϑέσϑω ὀρθογώνιον τὸ $AΓ$· λέγω, ὅτι τὸ $AΓ$ μέσον ἐστίν.

Ἀναγεγράφϑω γὰρ ἀπὸ τῆς AB τετράγωνον τὸ $AΔ$· μέσον ἄρα ἐστὶ τὸ $AΔ$. καὶ ἐπεὶ σύμμετρόξ ἐστιν ἡ AB τῇ $BΓ$ μήκει, ὁίση δὲ ἡ AB τῇ $BΔ$, σύμμετρος ἄρα ἐστὶ καὶ ἡ $ΔB$ τῇ $BΓ$ μήκει· ὥστε καὶ τὸ $ΔA$ τῷ $AΓ$ σύμμετρόν ἐστιν. μέσον δὲ τὸ $ΔA$· μέσον ἄρα καὶ τὸ $AΓ$· ὅπερ ἔδει δεῖχαι.

Proposition 24

A rectangle contained by medial straight-lines (which are) commensurable in length is medial.

For let the rectangle AC be contained by the medial straight-lines AB and BC (which are) commensurable in length. I say that AC is medial.

For let the square AD be described on AB. AD is thus medial [see previous footnote]. And since AB is commensurable in length with BC, and AB (is) equal to BD, DB is thus also commensurable in length with BC. Hence, DA is also commensurable with AC [Props. 6.1, 10.11]. And DA (is) medial. Thus, AC (is) also medial [Prop. 10.23 corr.]. (Which is) the

very thing it was required to show.

25.

Τὸ ὑπὸ μέσων δυνάμει μόνον συμμέτρων εὐθειῶν περιεχόμενον ὀρθογώνιον ἤτοι ὀρητόν ἐστιν ἢ μέσον ἐστίν.

Ὑπὸ γὰρ μέσων δυνάμει μόνον συμμέτρων εὐθειῶν τῶν ΑΒ, ΒΓ ὀρθογώνιον περιεσθέσθω τὸ ΑΓ· λέγω, ὅτι τὸ ΑΓ ἤτοι ὀρητόν ἐστιν ἢ μέσον ἐστίν.

Ἀναγεγράφθω γὰρ ἀπὸ τῶν ΑΒ, ΒΓ τετράγωνα τὰ ΑΔ, ΒΕ· μέσον ἄρα ἐστὶν ἑκάτερον τῶν ΑΔ, ΒΕ. καὶ ἐκκείσθω ὀρητὴ ἡ ΖΗ, καὶ τῷ μὲν ΑΔ ἴσον παρὰ τὴν ΖΗ παραβεβλήσθω ὀρθογώνιον παραλληλόγραμμον τὸ ΗΘ πλάτος ποιοῦν τὴν ΖΘ, τῷ δὲ ΑΓ ἴσον παρὰ τὴν ΘΜ παραβεβλήσθω ὀρθογώνιον παραλληλόγραμμον τὸ ΜΚ πλάτος ποιοῦν τὴν ΘΚ, καὶ ὁέτι τῷ ΒΕ ἴσον ὁμοίως παρὰ τὴν ΚΝ παραβεβλήσθω τὸ ΝΛ πλάτος ποιοῦν τὴν ΚΛ· ἐπ΄ εὐθείας ἄρα εἰσὶν αἱ ΖΘ, ΘΚ, ΚΛ. ἐπεὶ οὐῦν μέσον ἐστὶν ἑκάτερον τῶν ΑΔ, ΒΕ, καί ἐστιν ἴσον τὸ μὲν ΑΔ τῷ ΗΘ, τὸ δὲ ΒΕ τῷ ΝΛ, μέσον ἄρα καὶ ἑκάτερον τῶν ΗΘ, ΝΛ. καὶ παρὰ ὀρητὴν τὴν ΖΗ παράκειται· ὀρητὴ ἄρα ἐστὶν ἑκατέρα τῶν ΖΘ, ΚΛ καὶ ἀσύμμετρος τῇ ΖΗ μήκει. καὶ ἐπεὶ σύμμετρόν ἐστι τὸ ΑΔ τῷ ΒΕ, σύμμετρον ἄρα ἐστὶ καὶ τὸ ΗΘ τῷ ΝΛ. καί ἐστιν ὡξ τὸ ΗΘ πρὸς τὸ ΝΛ, οὐῦτως ἡ ΖΘ πρὸς τὴν ΚΛ· σύμμετρος ἄρα ἐστὶν ἡ ΖΘ τῇ ΚΛ μήκει. αἱ ΖΘ, ΚΛ ἄρα ὀρηταί εἰσι μήκει σύμμετροι· ὀρητὸν ἄρα ἐστὶ τὸ ὑπὸ τῶν ΖΘ, ΚΛ. καὶ ἐπεὶ ὁίση ἔστὶν ἡ μὲν ΔΒ τῇ ΒΑ, ἡ δὲ ΧΒ τῇ ΒΓ, ὁέστιν ἄρα ὡξ ἡ ΔΒ πρὸς τὴν ΒΓ, οὐῦτως ἡ ΑΒ πρὸς τὴν ΒΧ. ἀλλὸ ὡξ μὲν ἡ ΔΒ πρὸς τὴν ΒΓ, οὐῦτως τὸ ΔΑ πρὸς τὸ ΑΓ· ὡξ δὲ ἡ ΑΒ πρὸς τὴν ΒΧ, οὐῦτως τὸ ΑΓ πρὸς τὸ ΓΧ· ὁέστιν ἄρα ὡξ τὸ ΔΑ πρὸς τὸ ΑΓ, οὐῦτως τὸ ΑΓ πρὸς τὸ ΓΧ. ἴσον δέ ἐστι τὸ μὲν ΑΔ τῷ ΗΘ, τὸ δὲ ΑΓ τῷ ΜΚ, τὸ δὲ ΓΧ τῷ ΝΛ· ὁέστιν ἄρα ὡξ τὸ ΗΘ πρὸς τὸ

Proposition 25

The rectangle contained by medial straight-lines (which are) commensurable in square only is either rational or medial.

For let the rectangle AC be contained by the medial straight-lines AB and BC (which are) commensurable in square only. I say that AC is either rational or medial.

For let the squares AD and BE be described on (the straight-lines) AB and BC (respectively). AD and BE are thus each medial. And let the rational (straight-line) FG be laid out. And let the rectangular parallelogram GH, equal to AD, be applied to FG, producing FH as breadth. And let the rectangular parallelogram MK, equal to AC, be applied to HM, producing HK as breadth. And, finally, let NL, equal to BE, have similarly been applied to KN, producing KL as breadth. Thus, FH, HK, and KL are in a straight-line. Therefore, since AD and BE are each medial, and AD is equal to GH, and BE to NL, GH and NL (are) thus each also medial. And they are applied to the rational (straight-line) FG. FH and KL are thus each rational, and incommensurable in length with FG [Prop. 10.22]. And since AD is commensurable with BE, GH is thus also commensurable with NL. And as GH is to NL, so FH (is) to KL [Prop. 6.1]. Thus, FH is commensurable in length with KL [Prop. 10.11]. Thus, FH and KL are rational (straight-lines which are) commensurable in length. Thus, the (rectangle contained) by FH and KL is rational [Prop. 10.19]. And since DB is equal to BA, and OB to BC, thus as DB is to BC, so AB (is) to BO. But, as DB (is) to BC, so DA (is) to AC [Props. 6.1]. And as AB (is) to BO, so AC (is) to CO [Prop. 6.1]. Thus, as DA is to AC, so AC (is) to CO. And

ΜΚ, οὑτως τὸ ΜΚ πρὸς τὸ ΝΛ· ὀέστιν ἄρα καὶ ὡξ ἡ ΖΘ πρὸς τὴν ΘΚ, οὑτως ἡ ΘΚ πρὸς τὴν ΚΛ· τὸ ἄρα ὑπὸ τῶν ΖΘ, ΚΛ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΘΚ. ὁρητὸν δὲ τὸ ὑπὸ τῶν ΖΘ, ΚΛ· ὁρητὸν ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς ΘΚ· ὁρητὴ ἄρα ἐστὶν ἡ ΘΚ. καὶ εἰ μὲν σύμμετρόξ ἐστι τῇ ΖΗ μήκει, ὁρητόν ἐστι τὸ ΘΝ· εἰ δὲ ἀσύμμετρόξ ἐστι τῇ ΖΗ μήκει, αἱ ΚΘ, ΘΜ ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· μέσον ἄρα τὸ ΘΝ. τὸ ΘΝ ἄρα ἤτοι ὁρητὸνὸή μέσον ἐστίν. ἴσον δὲ τὸ ΘΝ τῷ ΑΓ· τὸ ΑΓ ἄρα ἤτοι ὁρητὸνὸή μέσον ἐστίν.

Τὸ ἄρα ὑπὸ μέσων δυνάμει μόνον συμμέτρων, καὶ τὰ ἐχῆς.

AD is equal to GH, and AC to MK, and CO to NL. Thus, as GH is to MK, so MK (is) to NL. Thus, also, as FH is to HK, so HK (is) to KL [Props. 6.1, 5.11]. Thus, the (rectangle contained) by FH and KL is equal to the (square) on HK [Prop. 6.17]. And the (rectangle contained) by FH and KL (is) rational. Thus, the (square) on HK is also rational. Thus, HK is rational. And if it is commensurable in length with FG then HN is rational [Prop. 10.19]. And if it is incommensurable in length with FG then KH and HM are rational (straight-lines which are) commensurable in square only: thus, HN is medial [Prop. 10.21]. Thus, HN is either rational or medial. And HN (is) equal to AC. Thus, AC is either rational or medial.

Thus, the ... by medial straight-lines (which are) commensurable in square only, and so on ....

26.

Μέσον μέσου οὐϑ ὑπερέϑει ὁρητῷ.

Proposition 26

A medial (area) does not exceed a medial (area) by a rational (area).†

Εἰ γὰρ δυνατόν, μέσον τὸ ΑΒ μέσου τοῦ ΑΓ ὑπερεϑέτω ὁρητῷ τῷ ΔΒ, καὶ ἐκκείσϑω ὁρητὴ ἡ ΕΖ, καὶ τῷ ΑΒ ἴσον παρὰ τὴν ΕΖ παραβεβλήσϑω παραλληλόγραμμον ὀρϑογώνιον τὸ ΖΘ πλάτος ποιοῦν τὴν ΕΘ, τῷ δὲ ΑΓ ἴσον ἀφῃρήσϑω τὸ ΖΗ· λοιπὸν ἄρα τὸ ΒΔ λοιπῷ τῷ ΚΘ ἐστιν ἴσον. ὁρητὸν δέ ἐστι τὸ ΔΒ· ὁρητὸν ἄρα ἐστὶ καὶ τὸ ΚΘ. ἐπεὶ οὐῦν μέσον ἐστὶν ἑκάτερον τῶν ΑΒ, ΑΓ, καί ἐστι τὸ μὲν ΑΒ τῷ ΖΘ ἴσον, τὸ δὲ ΑΓ τῷ ΖΗ, μέσον ἄρα καὶ ἑκάτερον τῶν ΖΘ, ΖΗ. καὶ παρὰ ὁρητὴν τὴν ΕΖ παράκειται· ὁρητὴ ἄρα ἐστὶν ἑκατέρα τῶν ΘΕ, ΕΗ καὶ ἀσύμμετρος τῇ ΕΖ μήκει. καὶ ἐπεὶ ὁρητόν ἐστι τὸ ΔΒ καί ἐστιν ἴσον τῷ ΚΘ, ὁρητὸν ἄρα ἐστὶ καὶ τὸ

For, if possible, let the medial (area) AB exceed the medial (area) AC by the rational (area) DB. And let the rational (straight-line) EF be laid down. And let the rectangular parallelogram FH, equal to AB, be applied to to EF, producing EH as breadth. And let FG, equal to AC, be cut off (from FH). Thus, the remainder BD is equal to the remainder KH. And DB is rational. Thus, KH is also rational. Therefore, since AB and AC are each medial, and AB is equal to FH, and AC to FG, FH and FG are thus each also medial. And they are applied to the rational (straight-line) EF. Thus, HE and EG are each rational, and incommensurable in length with EF

ΚΘ. καὶ παρὰ ὁρητὴν τὴν EZ παράκειται· ὁρητὴ ἄρα ἐστὶν ἡ ΗΘ καὶ σύμμετρος τῇ EZ μήκει. ἀλλά καὶ ἡ EH ὁρητή ἐστι καὶ ἀσύμμετρος τῇ EZ μήκει· ἀσύμμετρος ἄρα ἐστὶν ἡ EH τῇ ΗΘ μήκει. καί ἐστιν ὡξ ἡ EH πρὸς τὴν ΗΘ, οὕτως τὸ ἀπὸ τῆς EH πρὸς τὸ ὑπὸ τῶν EH, ΗΘ· ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς EH τῷ ὑπὸ τῶν EH, ΗΘ. ἀλλὰ τῷ μὲν ἀπὸ τῆς EH σύμμετρά ἐστι τὰ ἀπὸ τῶν EH, ΗΘ τετράγωνα· ὁρηταὶ γὰρ ἀμφότερα· τῷ δὲ ὑπὸ τῶν EH, ΗΘ σύμμετρόν ἐστι τὸ δὶξ ὑπὸ τῶν EH, ΗΘ· διπλάσιον γάρ ἐστιν αὐτοῦ· ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν EH, ΗΘ τῷ δὶξ ὑπὸ τῶν EH, ΗΘ· καὶ συναμφότερα ἄρα τά τε ἀπὸ τῶν EH, ΗΘ καὶ τὸ δὶξ ὑπὸ τῶν EH, ΗΘ, ὅπερ ἐστὶ τὸ ἀπὸ τῆς EΘ, ἀσύμμετρόν ἐστι τοῖς ἀπὸ τῶν EH, ΗΘ. ὁρηταὶ δὲ τὰ ἀπὸ τῶν EH, ΗΘ· ὀάλογον ἄρα τὸ ἀπὸ τῆς EΘ. ὀάλογος ἄρα ἐστὶν ἡ EΘ. ἀλλὰ καὶ ὁρηρή· ὅπερ ἐστὶν ἀδύνατον.

Μέσον ἄρα μέσου οὐθ ὑπερέθει ὁρητῷ· ὅπερ ἔδει δεῖχαι.

[Prop. 10.22]. And since DB is rational, and is equal to KH, KH is thus also rational. And (KH) is applied to the rational (straight-line) EF. GH is thus rational, and commensurable in length with EF [Prop. 10.20]. But, EG is also rational, and incommensurable in length with EF. Thus, EG is incommensurable in length with GH [Prop. 10.13]. And as EG is to GH, so the (square) on EG (is) to the (rectangle contained) by EG and GH [Prop. 10.13 lem.]. Thus, the (square) on EG is incommensurable with the (rectangle contained) by EG and GH [Prop. 10.11]. But, the (sum of the) squares on EG and GH is commensurable with the (square) on EG. For (EG and GH are) both rational. And twice the (rectangle contained) by EG and GH is commensurable with the (rectangle contained) by EG and GH [Prop. 10.6]. For (the former) is double the latter. Thus, the (sum of the squares) on EG and GH is incommensurable with twice the (rectangle contained) by EG and GH [Prop. 10.13]. And thus the sum of the (squares) on EG and GH plus twice the (rectangle contained) by EG and GH, that is the (square) on EH [Prop. 2.4], is incommensurable with the (sum of the squares) on EG and GH [Prop. 10.16]. And the (sum of the squares) on EG and GH (is) rational. Thus, the (square) on EH is irrational [Def. 10.4]. Thus, EH is irrational [Def. 10.4]. But, (it is) also rational. The very thing is impossible.

Thus, a medial (area) does not exceed a medial (area) by a rational (area). (Which is) the very thing it was required to show.

† In other words, $\sqrt{k} - \sqrt{k'} \neq k''$.

27.

Μέσας εὑρεῖν δυνάμει μόνον συμμέτρους ὁρητὸν περιεχούσας.

Proposition 27

To find (two) medial (straight-lines), containing a rational (area), (which are) commensurable in square only.

Ἐκκείσθωσαν δύο ὁρηταὶ δυνάμει μόνον σύμμετροι αἱ Α, Β, καὶ εἰλήφθω τῶν Α, Β μέση ἀνάλογον ἡ Γ, καὶ γεγονέτω ὡς ἡ Α πρὸς τὴν Β, οὕτως ἡ Γ πρὸς τὴν Δ.

Καὶ ἐπεὶ αἱ Α, Β ὁρηταί εἰσι δυνάμει μόνον σύμμετροι, τὸ ἄρα ὑπὸ τῶν Α, Β, τουτέστι τὸ ἀπὸ τῆς Γ, μέσον ἐστίν. μέση ἄρα ἡ Γ. καὶ ἐπεί ἐστιν ὡς ἡ Α πρὸς τὴν Β, [οὕτως] ἡ Γ

Let the two rational (straight-lines) A and B, (which are) commensurable in square only, be laid down. And let C—the mean proportional (straight-line) to A and B—be taken [Prop. 6.13]. And let it be contrived that as A (is) to B, so C (is) to D [Prop. 6.12].

And since the rational (straight-lines) A and B are commensurable in square only, the (rectangle contained) by A and

πρὸς τὴν Δ, αἱ δὲ A, B δυνάμει μόνον [εἰσὶ] σύμμετροι, καὶ αἱ Γ, Δ ἄρα δυνάμει μόνον εἰσὶ σύμμετροι. καί ἐστι μέση ἡ Γ· μέση ἄρα καὶ ἡ Δ. αἱ Γ, Δ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. λέγω, ὅτι καὶ ὁρητὸν περιέθουσιν. ἐπεὶ γάρ ἐστιν ὡς ἡ A πρὸς τὴν B, οὕτως ἡ Γ πρὸς τὴν Δ, ἐναλλὰχ ἄρα ἐστὶν ὡς ἡ A πρὸς τὴν Γ, ἡ B πρὸς τὴν Δ. ἀλλὰ ὡς ἡ A πρὸς τὴν Γ, ἡ Γ πρὸς τὴν B· καὶ ὡς ἄρα ἡ Γ πρὸς τὴν B, οὕτως ἡ B πρὸς τὴν Δ· τὸ ἄρα ὑπὸ τῶν Γ, Δ ἴσον ἐστὶ τῷ ἀπὸ τῆς B. ὁρητὸν δὲ τὸ ἀπὸ τῆς B· ὁρητὸν ἄρα [ἐστὶ] καὶ τὸ ὑπὸ τῶν Γ, Δ.

Εὑρηνται ἄρα μέσαι δυνάμει μόνον σύμμετροι ὁρητὸν περιέθουσαι· ὅπερ ἔδει δεῖξαι.

B—that is to say, the (square) on C [Prop. 6.17]—is thus medial [Prop 10.21]. Thus, C is medial [Prop. 10.21]. And since as A is to B, [so] C (is) to D, and A and B [are] commensurable in square only, C and D are thus also commensurable in square only [Prop. 10.11]. And C is medial. Thus, D is also medial [Prop. 10.23]. Thus, C and D are medial (straight-lines which are) commensurable in square only. I say that they also contain a rational (area). For since as A is to B, so C (is) to D, thus, alternately, as A is to C, so B (is) to D [Prop. 5.16]. But, as A (is) to C, (so) C (is) to B. And thus as C (is) to B, so B (is) to D [Prop. 5.11]. Thus, the (rectangle contained) by C and D is equal to the (square) on B [Prop. 6.17]. And the (square) on B (is) rational. Thus, the (rectangle contained) by C and D [is] also rational.

Thus, (two) medial (straight-lines, C and D), containing a rational (area), (which are) commensurable in square only, be found.† (Which is) the very thing it was required to show.

† C and D have lengths $k^{1/4}$ and $k^{3/4}$ times that of A, respectively, where the length of B is $k^{1/2}$ times that of A.

28.

Μέσας εὑρεῖν δυνάμει μόνον συμμέτρους μέσον περιεθούσας.

Proposition 28

To find (two) medial (straight-lines), containing a medial (area), (which are) commensurable in square only.

Ἐκκείσθωσαν [τρεῖς] ὁρηταὶ δυνάμει μόνον σύμμετροι αἱ A, B, Γ, καὶ εἰλήφθω τῶν A, B μέση ἀνάλογον ἡ Δ, καὶ γεγονέτω ὡς ἡ B πρὸς τὴν Γ, ἡ Δ πρὸς τὴν E.

Ἐπεὶ αἱ A, B ὁρηταί εἰσι δυνάμει μόνον σύμμετροι, τὸ ἄρα ὑπὸ τῶν A, B, τουτέστι τὸ ἀπὸ τῆς Δ, μέσον ἐστίν. μέση ἄρα ἡ Δ. καὶ ἐπεὶ αἱ B, Γ δυνάμει μόνον εἰσὶ σύμμετροι, καί ἐστιν ὡς ἡ B πρὸς τὴν Γ, ἡ Δ πρὸς τὴν E, καὶ αἱ Δ, E ἄρα δυνάμει μόνον εἰσὶ σύμμετροι. μέση δὲ ἡ Δ· μέση ἄρα καὶ ἡ E· αἱ Δ, E ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. λέγω δή, ὅτι καὶ μέσον περιέθουσιν. ἐπεὶ γάρ ἐστιν ὡς ἡ B πρὸς τὴν Γ, ἡ Δ πρὸς τὴν E, ἐναλλὰχ ἄρα ὡς ἡ B πρὸς τὴν Δ, ἡ Γ πρὸς τὴν E. ὡς δὲ ἡ B πρὸς τὴν Δ, ἡ Δ πρὸς τὴν A· καὶ ὡς ὁάρα ἡ Δ πρὸς τὴν A, ἡ Γ πρὸς τὴν E· τὸ ἄρα ὑπὸ τῶν A, Γ ἴσον ἐστὶ τῷ ὑπὸ τῶν Δ, E. μέσον δὲ τὸ ὑπὸ τῶν A, Γ· μέσον ἄρα καὶ τὸ ὑπὸ τῶν Δ, E.

Ἐὑρηνται ἄρα μέσαι δυνάμει μόνον σύμμετροι μέσον περιέθουσαι· ὅπερ ἔδει δεῖξαι.

Let the [three] rational (straight-lines) A, B, and C, (which are) commensurable in square only, be laid down. And let, D, the mean proportional (straight-line) to A and B, be taken [Prop. 6.13]. And let it be contrived that as B (is) to C, (so) D (is) to E [Prop. 6.12].

Since the rational (straight-lines) A and B are commensurable in square only, the (rectangle contained) by A and B—that is to say, the (square) on D [Prop. 6.17]—is medial [Prop. 10.21]. Thus, D (is) medial [Prop. 10.21]. And since B and C are commensurable in square only, and as B is to C, (so) D (is) to E, D and E are thus commensurable in square only [Prop. 10.11]. And D (is) medial. E (is) thus also medial [Prop. 10.23]. Thus, D and E are medial (straight-lines which are) commensurable in square only. So, I say that they also enclose a medial (area). For since as B is to C, (so) D (is) to E, thus, alternately, as B (is) to D, (so) C (is) to E [Prop. 5.16]. And as B (is) to D, (so) D (is) to A. And thus as D (is) to A, (so) C (is) to E. Thus, the (rectangle contained) by A and C is equal to the (rectangle contained) by D and E

[Prop. 6.16]. And the (rectangle contained) by A and C is medial [Prop. 10.21]. Thus, the (rectangle contained) by D and E (is) also medial.

Thus, (two) medial (straight-lines, D and E), containing a medial (area), (which are) commensurable in square only, be found. (Which is) the very thing it was required to show.

† D and E have lengths $k^{1/4}$ and $k'^{1/2}/k^{1/4}$ times that of A, respectively, where the lengths of B and C are $k^{1/2}$ and $k'^{1/2}$ times that of A, respectively.

Λῆμμα 1.

Εὑρειν δύο τετραγώνους ἀριθμούξ, ὁώστε καὶ τὸν συγκειμενον ἐξ αὐτῶν εὸῖναι τετράγωνον.

Lemma I

To find two square numbers such that the sum of them is also square.

Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ AB, $BΓ$, ὀέστωσαν δὲ ἤτοιὀάρτιοιὀή περιττοί. καὶ ἐπεὶ, ἔαν τε ἀπὸ ἀρτίουὀάρτιος ἀφαιρεθῇ, ἔαν τε ἀπὸ περισσοῦ περισσός, ὁ λοιπόξὀάρτιόξ ἐστιν, ὁ λοιπόξ ἄρα ὁ $AΓ$ὀάρτιόξ ἐστιν. τετμήσθω ὁ $AΓ$ δίθα κατὰ τὸ $Δ$. ὀέστωσαν δὲ καὶ οἱ AB, $BΓ$ ἤτοι ὅμοιοι ἐπίπεδοιὀή τετράγωνοι, οὀὶ καὶ αὐτοὶ ὅμοιοί εἰσιν ἐπίπεδοι· ὁ ἄρα ἐκ τῶν AB, $BΓ$ μετὰ τοῦ ἀπὸ [τοῦ] $ΓΔ$ τετραγώνουὀίσος ἐστὶ τῷ ἀπὸ τοῦ $BΔ$ τετραγώνῳ. καί ἐστι τετράγωνος ὁ ἐκ τῶν AB, $BΓ$, ἐπειδήπερ ἐδείθθη, ὅτι, ἔαν δύο ὅμοιοι ἐπίπεδοι πολλαπλασιάσαντες ἀλλήλους ποιῶσι τινα, ὁ γενόμενος τετράγωνόξ ἐστιν. εὑρηνται ἄρα δύο τετράγωνοι ἀριθμοὶ ὅ τε ἐκ τῶν AB, $BΓ$ καὶ ὁ ἀπὸ τοῦ $ΓΔ$, οὀὶ συντεθέντες ποιοῦσι τὸν ἀπὸ τοῦ $BΔ$ τετράγωνον.

Καὶ φανερόν, ὅτι εὑρηνται πάλιν δύο τετράγωνοι ὅ τε ἀπὸ τοῦ $BΔ$ καὶ ὁ ἀπὸ τοῦ $ΓΔ$, ὁώστε τὴν ὑπεροθὴν αὐτῶν τὸν ὑπὸ AB, $BΓ$ εὸῖναι τετράγωνον, ὅταν οἱ AB, $BΓ$ ὅμοιοι ὦσιν ἐπίπεδοι. ὅταν δὲ μὴ ὦσιν ὅμοιοι ἐπίπεδοι, εὑρηνται δύο τετράγωνοι ὅ τε ἀπὸ τοῦ $BΔ$ καὶ ὁ ἀπὸ τοῦ $ΔΓ$, ὁῶν ἡ ὑπεροθὴ ὁ ὑπὸ τῶν AB, $BΓ$ οὐκὸέστι τετράγωνος· ὅπερ ἔδει δεῖχαι.

Let the two numbers AB and BC be laid down. And let them be either (both) even or (both) odd. And since, if an even (number) is subtracted from an even (number), or if an odd (number is subtracted) from an odd (number), then the remainder is even [Props. 9.24, 9.26], the remainder AC is thus even. Let AC be cut in half at D. And let AB and BC also be either similar plane (numbers), or square (numbers)—which are themselves also similar plane (numbers). Thus, the (number created) from (multiplying) AB and BC, plus the square on CD, is equal to the square on BD [Prop. 2.6]. And the (number created) from (multiplying) AB and BC is square—inasmuch as it was shown that if two similar plane (numbers) make some (number) by multiplying one another then the (number so) created is square [Prop. 9.1]. Thus, two square numbers be found—(namely,) the (number created) from (multiplying) AB and BC, and the (square) on CD—which, (when) added (together), make the square on BD.

And (it is) clear that two square (numbers) have again been found—(namely,) the (square) on BD, and the (square) on CD—such that their difference—(namely,) the (rectangle) contained by AB and BC—is square whenever AB and BC are similar plane (numbers). But, when they are not similar plane numbers, two square (numbers) be found—(namely,) the (square) on BD, and the (square) on DC—between which the difference—(namely,) the (rectangle) contained by AB and BC—is not square. (Which is) the very thing it was required to show.

Λῆμμα 2.

Εὑρεῖν δύο τετραγώνους ἀριθμούξ, ὁώστε τὸν ἐξ αὐτῶν συγκείμενον μὴ εὸῖναι τετράγωνον.

Lemma II

To find two square numbers such that the sum of them is not square.

Ἔστω γὰρ ὁ ἐκ τῶν AB, BΓ, ὡξὸέφαμεν, τετράγωνος, καὶ ὀάρτιος ὁ ΓA, καὶ τετμήσθω ὁ ΓA δίθα τῷ Δ. φανερὸν δή, ὅτι ὁ ἐκ τῶν AB, BΓ τετράγωνος μετὰ τοῦ ἀπὸ [τοῦ] ΓΔ τετραγώνουὸίσος ἐστὶ τῷ ἀπὸ [τοῦ] BΔ τετραγώνῳ. ἀφῃρήσθω μονὰξ ἡ ΔE· ὁ ἄρα ἐκ τῶν AB, BΓ μετὰ τοῦ ἀπὸ [τοῦ] ΓE ἐλάσσων ἐστὶ τοῦ ἀπὸ [τοῦ] BΔ τετραγώνου. λέγω οὐῦν, ὅτι ὁ ἐκ τῶν AB, BΓ τετράγωνος μετὰ τοῦ ἀπὸ [τοῦ] ΓE οὐκὸέσται τετράγωνος.

Εἰ γὰρὸέσται τετράγωνος, ἤτοιὸίσος ἐστὶ τῷ ἀπὸ [τοῦ] BEὸὴ ἐλάσσων τοῦ ἀπὸ [τοῦ] BE, οὐκέτι δὲ καὶ μείζων, ὅινα μὴ τμ ηθῇ ἡ μονάξ. ὀέστω, εἰ δυνατόν, πρότερον ὁ ἐκ τῶν AB, BΓ μετὰ τοῦ ἀπὸ ΓEὸίσος τῷ ἀπὸ BE, καὶ ὀέστω τῆς ΔE μονάδος διπλασίων ὁ HA. ἐπεὶ οὐῦν ὅλος ὁ AΓ ὅλου τοῦ ΓΔ ἐστι διπλασίων, ὁῶν ὁ AH τοῦ ΔE ἐστι διπλασίων, καὶ λοιπὸξ ἄρα ὁ HΓ λοιποῦ τοῦ EΓ ἐστι διπλασίων· δίθα ἄρα τέτμ ηται ὁ HΓ τῷ E. ὁ ἄρα ἐκ τῶν HB, BΓ μετὰ τοῦ ἀπὸ ΓEὸίσος ἐστὶ τῷ ἀπὸ BE τετραγώνῳ. ἀλλὰ καὶ ὁ ἐκ τῶν AB, BΓ μετὰ τοῦ ἀπὸ ΓEὸίσος ὑπόκειται τῷ ἀπὸ [τοῦ] BE τετραγώνῳ· ὁ ἄρα ἐκ τῶν HB, BΓ μετὰ τοῦ ἀπὸ ΓEὸίσος ἐστὶ τῷ ἐκ τῶν AB, BΓ μετὰ τοῦ ἀπὸ ΓE. καὶ κοινοῦ ἀφαιρεθέντος τοῦ ἀπὸ ΓE συνάγεται ὁ ABὸίσος τῷ HB· ὅπερὸάτοπον. οὐκ ἄρα ὁ ἐκ τῶν AB, BΓ μετὰ τοῦ ἀπὸ [τοῦ] ΓEὸίσος ἐστὶ τῷ ἀπὸ BE. λέγω δή, ὅτι οὐδὲ ἐλάσσων τοῦ ἀπὸ BE. εἰ γὰρ δυνατόν, ὀέστω τῷ ἀπὸ BZὸίσος, καὶ τοῦ ΔZ διπλασίων ὁ ΘA. καὶ συναθθήσεται πάλιν διπλασίων ὁ ΘΓ τοῦ ΓZ· ὁώστε καὶ τὸν ΓΘ δίθα τετμῆσθαι κατὰ τὸ Z, καὶ διὰ τοῦτο τὸν ἐκ τῶν ΘB, BΓ μετὰ τοῦ ἀπὸ ZΓ ἴσον γίνεσθαι τῷ ἀπὸ BZ. ὑπόκειται δὲ καὶ ὁ ἐκ τῶν AB, BΓ μετὰ τοῦ ἀπὸ ΓEὸίσος τῷ ἀπὸ BZ. ὁώστε καὶ ὁ ἐκ τῶν ΘB, BΓ μετὰ τοῦ ἀπὸ ΓZὸίσοςὸέσται τῷ ἐκ τῶν AB, BΓ μετὰ τοῦ ἀπὸ ΓE· ὅπερὸάτοπον. οὐκ ἄρα ὁ ἐκ τῶν AB, BΓ μετὰ τοῦ ἀπὸ ΓEὸίσος ἐστὶ [τῷ] ἐλάσσονι τοῦ ἀπὸ BE. ἐδείθθη δέ, ὅτι οὐδὲ [αὐτῷ] τῷ ἀπὸ BE. οὐκ ἄρα ὁ ἐκ τῶν AB, BΓ μετὰ τοῦ ἀπὸ ΓE τετράγωνόξ ἐστιν. ὅπερ ἔδει δεῖχαι.

For let the (number created from (multiplying) AB and BC, as we said, be square. And (let) CA (be) even. And let CA be cut in half at D. So it is clear that the square (number created) from (multiplying) AB and BC, plus the square on CD, is equal to the square on BD [see previous lemma]. Let the unit DE be subtracted (from BD). Thus, the (number created) from (multiplying) AB and BC, plus the (square) on CE, is less than the square on BD. I say, therefore, that the square (number created) from (multiplying) AB and BC, plus the (square) on CE, is not square.

For if it is square, it is either equal to the (square) on BE, or less than the (square) on BE, but cannot any more be greater (than the square on BE), lest the unit be divided. First of all, if possible, let the (number created) from (multiplying) AB and BC, plus the (square) on CE, be equal to the (square) on BE. And let GA be double the unit DE. Therefore, since the whole of AC is double the whole of CD, of which AG is double DE, the remainder GC is thus double the remainder EC. Thus, GC has been cut in half at E. Thus, the (number created) from (multiplying) GB and BC, plus the (square) on CE, is equal to the square on BE [Prop. 2.6]. But, the (number created) from (multiplying) AB and BC, plus the (square) on CE, was also assumed (to be) equal to the square on BE. Thus, the (number created) from (multiplying) GB and BC, plus the (square) on CE, is equal to the (number created) from (multiplying) AB and BC, plus the (square) on CE. And subtracting the (square) on CE from both, AB is inferred (to be) equal to GB. The very thing is absurd. Thus, the (number created) from (multiplying) AB and BC, plus the (square) on CE, is not equal to the (square) on BE. So I say that (it is) not less than the (square) on BE either. For, if possible, let it be equal to the (square) on BF. And (let) HA (be) double DF. And it can again be inferred that HC (is) double CF. Hence, CH has also been cut in half at F. And, on account of this, the (number created) from (multiplying) HB and BC, plus the (square) on FC, becomes equal to the (square) on BF [Prop. 2.6]. And the (number created) from (multiplying) AB and BC, plus the (square) on CE, was also assumed (to be) equal to the (square) on BF. Hence, the (number created) from (multiplying) HB and BC, plus the (square) on CF, will also be equal to the (number created) from (multiplying) AB and BC, plus the (square) on CE. The very thing is absurd. Thus, the (number created) from (multiplying) AB and BC, plus the (square) on CE, is not equal to less than the (square) on BE. And it was shown that (is it) not equal to the (square) on BE either. Thus, the (number created) from (multiplying) AB and BC, plus the square on CE, is not

29.

Εὑρεῖν δύο ὁρητὰξ δυνάμει μόνον συμμέτρους, ὁώστε τὴν μείζονα τῆς ἐλάσσονος μεῖζον δύνασθαι τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει.

Ἐκκείσθω γάρ τις ὁρητὴ ἡ AB καὶ δύο τετράγωνοι ἀριθμοὶ οἱ $ΓΔ$, $ΔE$, ὁώστε τὴν ὑπεροθὴν αὐτῶν τὸν $ΓE$ μὴ εὶῖναι τετράγωνον, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ AZB, καὶ πεποιήσθω ὡξ ὁ $ΔΓ$ πρὸς τὸν $ΓE$, οὁύτως τὸ ἀπὸ τῆς BA τετράγωνον πρὸς τὸ ἀπὸ τῆς AZ τετράγωνον, καὶ ἐπεζεύθθω ἡ ZB.

Ἐπεὶ [οὁῦν] ἐστιν ὡξ τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ, οὁύτως ὁ $ΔΓ$ πρὸς τὸν $ΓE$, τὸ ἀπὸ τῆς BA ἄρα πρὸς τὸ ἀπὸ τῆς AZ λόγον ἔχει, ὅν ἀριθμὸξ ὁ $ΔΓ$ πρὸς ἀριθμὸν τὸν $ΓE$· σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς BA τῷ ἀπὸ τῆς AZ. ὁρητὸν δὲ τὸ ἀπὸ τῆς AB· ὁρητὸν ἄρα καὶ τὸ ἀπὸ τῆς AZ· ὁρητὴ ἄρα καὶ ἡ AZ. καὶ ἐπεὶ ὁ $ΔΓ$ πρὸς τὸν $ΓE$ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς BA ἄρα πρὸς τὸ ἀπὸ τῆς AZ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ AB τῇ AZ μήκει· αἱ BA, AZ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι. καὶ ἐπεί [ἐστιν] ὡξ ὁ $ΔΓ$ πρὸς τὸν $ΓE$, οὁύτως τὸ

square. (Which is) the very thing it was required to show.

Proposition 29

To find two rational (straight-lines which are) commensurable in square only, such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line which is) commensurable in length with the greater.

For let some rational (straight-line) AB be laid down, and two square numbers, CD and DE, such that the difference between them, CE, is not square [Prop. 10.28 lem. I]. And let the semi-circle AFB be drawn on AB. And let it be contrived that as DC (is) to CE, so the square on BA (is) to the square on AF [Prop. 10.6 corr.]. And let FB be joined.

[Therefore,] since as the (square) on BA is to the (square) on AF, so DC (is) to CE, the (square) on BA thus has to the (square) on AF the ratio which the number DC (has) to the number CE. Thus, the (square) on BA is commensurable with the (square) on AF [Prop. 10.6]. And the (square) on AB (is) rational [Def. 10.4]. Thus, the (square) on AF (is) also rational. Thus, AF (is) also rational. And since DC does not have to CE the ratio which (some) square number (has) to (some) square number, the (square) on BA thus does not have to the (square) on AF the ratio which (some) square number has to (some) square number either. Thus, AB is incommensurable in

ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ, ἀναστρέψαντι ἄρα ὡς ὁ $ΓΔ$ πρὸς τὸν $ΔE$, οὕτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ. ὁ δὲ $ΓΔ$ πρὸς τὸν $ΔE$ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ τὸ ἀπὸ τῆς AB ἄρα πρὸς τὸ ἀπὸ τῆς BZ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· σύμμετρος ἄρα ἐστὶν ἡ AB τῇ BZ μήκει. καί ἐστι τὸ ἀπὸ τῆς AB ἴσον τοῖς ἀπὸ τῶν AZ, ZB· ἡ AB ἄρα τῆς AZ μεῖζον δύναται τῇ BZ συμμέτρῳ ἑαυτῇ.

Εὑρηνται ἄρα δύο ὀρηταὶ δυνάμει μόνον σύμμετροι αἱ BA, AZ, ὡστε τὴν μεῖζονα τὴν AB τῆς ἐλάσσονος τῆς AZ μεῖζον δύνασθαι τῷ ἀπὸ τῆς BZ συμμέτρου ἑαυτῇ μήκει· ὅπερ ἔδει δεῖξαι.

length with AF [Prop. 10.9]. Thus, the rational (straight-lines) BA and AF are commensurable in square only. And since as DC [is] to CE, so the (square) on BA (is) to the (square) on AF, thus, via conversion, as CD (is) to DE, so the (square) on AB (is) to the (square) on BF [Props. 5.19 corr., 3.31, 1.47]. And CD has to DE the ratio which (some) square number (has) to (some) square number. Thus, the (square) on AB also has to the (square) on BF the ratio which (some) square number has to (some) square number. AB is thus commensurable in length with BF [Prop. 10.9]. And the (square) on AB is equal to the (sum of the squares) on AF and FB [Prop. 1.47]. Thus, the square on AB is greater than (the square on) AF by (the square on) BF, (which is) commensurable (in length) with (AB).

Thus, two rational (straight-lines), BA and AF, commensurable in square only, be found such that the square on the greater, AB, is larger than (the square on) the lesser, AF, by the (square) on BF, (which is) commensurable in length with (AB).[†] (Which is) the very thing it was required to show.

[†] BA and AF have lengths 1 and $\sqrt{1-k^2}$ times that of AB, respectively, where $k = \sqrt{DE/CD}$.

30.

Εὑρεῖν δύο ὀρητὰξ δυνάμει μόνον συμμέτρους, ὡστε τὴν μείζονα τῆς ἐλάσσονος μεῖζον δύνασθαι τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ μήκει.

Proposition 30

To find two rational (straight-lines which are) commensurable in square only, such that the square on the greater is larger than the (the square on) lesser by the (square) on (some straight-line which is) incommensurable in length with the greater.

Ἐκκείσθω ὁρητὴ ἡ AB καὶ δύο τετράγωνοι ἀριθμοὶ οἱ $ΓE$, $EΔ$, ὁώστε τὸν συγκείμενον ἐξ αὐτῶν τὸν $ΓΔ$ μὴ εἶναι τετράγωνον, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ AZB, καὶ πεποιήσθω ὡξ ὁ $ΔΓ$ πρὸς τὸν $ΓE$, οὕτως τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ, καὶ ἐπεζεύθθω ἡ ZB.

Ὁμοίως δὴ δείχομεν τῷ πρὸ τούτου, ὅτι αἱ BA, AZ ὁρηταί εἰσι δυνάμει μόνον σύμμετροι. καὶ ἐπεί ἐστιν ὡξ ὁ $ΔΓ$ πρὸς τὸν $ΓE$, οὕτως τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ, ἀναστρέψαντι ἄρα ὡξ ὁ $ΓΔ$ πρὸς τὸν $ΔE$, οὕτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ. ὁ δὲ $ΓΔ$ πρὸς τὸν $ΔE$ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· οὐδὸ ἄρα τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ AB τῇ BZ μήκει. καὶ δύναται ἡ AB τῆς AZ μεῖζον τῷ ἀπὸ τῆς ZB ἀσυμμέτρου ἑαυτῇ.

Αἱ AB, AZ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ AB τῆς AZ μεῖζον δύναται τῷ ἀπὸ τῆς ZB ἀσυμμέτρου ἑαυτῇ μήκει· ὅπερ ἔδει δεῖχαι.

Let the rational (straight-line) AB be laid out, and the two square numbers, CE and ED, such that the sum of them, CD, is not square [Prop. 10.28 lem. II]. And let the semi-circle AFB be drawn on AB. And let it be contrived that as DC (is) to CE, so the (square) on BA (is) to the (square) on AF [Prop. 10.6 corr]. And let FB be joined.

So, similarly to the (proposition) before this, we can show that BA and AF are rational (straight-lines which are) commensurable in square only. And since as DC is to CE, so the (square) on BA (is) to the (square) on AF, thus, via conversion, as CD (is) to DE, so the (square) on AB (is) to the (square) on BF [Props. 5.19 corr., 3.31, 1.47]. And CD does not have to DE the ratio which (some) square number (has) to (some) square number. Thus, the (square) on AB does not have to the (square) on BF the ratio which (some) square number has to (some) square number either. Thus, AB is incommensurable in length with BF [Prop. 10.9]. And the square on AB is greater than the (square on) AF by the (square) on FB [Prop. 1.47], (which is) incommensurable (in length) with (AB).

Thus, AB and AF are rational (straight-lines which are) commensurable in square only, and the square on AB is greater than (the square on) AF by the (square) on FB, (which is) incommensurable (in length) with (AB).[†] (Which is) the very thing it was required to show.

† AB and AF have lengths 1 and $1/\sqrt{1+k^2}$ times that of AB, respectively, where $k = \sqrt{DE/CE}$.

31.

Εὑρεῖν δύο μέσας δυνάμει μόνον συμμέτρους ὁρητὸν περιεχούσας, ὥστε τὴν μείζονα τῆς ἐλάσσονος μεῖζον δύνασθαι τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει.

Proposition 31

To find two medial (straight-lines), commensurable in square only, (and) containing a rational (area), such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) commensurable in length with the greater.

Ἐκκείσθωσαν δύο ὁρηταὶ δυνάμει μόνον σύμμετροι αἱ A, B, ὥστε τὴν A μείζονα οὖσαν τῆς ἐλάσσονος τῆς B μεῖζον δύνασθαι τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει. καὶ τῷ ὑπὸ τῶν A, B ἴσον ὀέστω τὸ ἀπὸ τῆς $Γ$. μέσον δὲ τὸ ὑπὸ τῶν A, B· μέσον ἄρα καὶ τὸ ἀπὸ τῆς $Γ$· μέση ἄρα καὶ ἡ $Γ$. τῷ δὲ ἀπὸ τῆς B ἴσον ὀέστω τὸ ὑπὸ τῶν $Γ$, $Δ$· ὁρητὸν δὲ τὸ ἀπὸ τῆς B· ὁρητὸν ἄρα καὶ τὸ ὑπὸ τῶν $Γ$, $Δ$. καὶ ἐπεί ἐστιν ὡξ ἡ A πρὸς τὴν

Let two rational (straight-lines), A and B, commensurable in square only, be laid out, such that the square on the greater A is larger than the (square on the) lesser B by the (square) on (some straight-line) commensurable in length with (A) [Prop. 10.29]. And let the (square) on C be equal to the (rectangle contained) by A and B. And the (rectangle contained by) A and B (is) medial [Prop. 10.21]. Thus, the (square) on C (is)

Β, οὑτως τὸ ὑπὸ τῶν Α, Β πρὸς τὸ ἀπὸ τῆς Β, ἀλλὰ τῷ μὲν ὑπὸ τῶν Α, Β ἴσον ἐστὶ τὸ ἀπὸ τῆς Γ, τῷ δὲ ἀπὸ τῆς Β ἴσον τὸ ὑπὸ τῶν Γ, Δ, ὡξ ἄρα ἡ Α πρὸς τὴν Β, οὑτως τὸ ἀπὸ τῆς Γ πρὸς τὸ ὑπὸ τῶν Γ, Δ. ὡξ δὲ τὸ ἀπὸ τῆς Γ πρὸς τὸ ὑπὸ τῶν Γ, Δ, οὑτως ἡ Γ πρὸς τὴν Δ· καὶ ὡξ ἄρα ἡ Α πρὸς τὴν Β, οὑτως ἡ Γ πρὸς τὴν Δ. σύμμετρος δὲ ἡ Α τῇ Β δυνάμει μόνον· σύμμετρος ἄρα καὶ ἡ Γ τῇ Δ δυνάμει μόνον. καί ἐστι μέση ἡ Γ· μέση ἄρα καὶ ἡ Δ. καὶ ἐπεί ἐστιν ὡξ ἡ Α πρὸς τὴν Β, ἡ Γ πρὸς τὴν Δ, ἡ δὲ Α τῆς Β μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ ἡ Γ ἄρα τῆς Δ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ.

Εὑρηνται ἄρα δύο μέσαι δυνάμει μόνον σύμμετροι αἱ Γ, Δ ὁρητὸν περιέθουσαι, καὶ ἡ Γ τῆς Δ μεῖζον δυνάται τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει.

Ὁμοίως δὴ δειθθήσεται καὶ τῷ ἀπὸ ἀσυμμέτρου, ὅταν ἡ Α τῆς Β μεῖζον δύνηται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ.

also medial. Thus, C (is) also medial [Prop. 10.21]. And let the (rectangle contained) by C and D be equal to the (square) on B. And the (square) on B (is) rational. Thus, the (rectangle contained) by C and D (is) also rational. And since as A is to B, so the (rectangle contained) by A and B (is) to the (square) on B [Prop. 10.21 lem.], but the (square) on C is equal to the (rectangle contained) by A and B, and the (rectangle contained) by C and D to the (square) on B, thus as A (is) to B, so the (square) on C (is) to the (rectangle contained) by C and D. And as the (square) on C (is) to the (rectangle contained) by C and D, so C (is) to D [Prop. 10.21 lem.]. And thus as A (is) to B, so C (is) to D. And A is commensurable in square only with B. Thus, C (is) also commensurable in square only with D [Prop. 10.11]. And C is medial. Thus, D (is) also medial [Prop. 10.23]. And since as A is to B, (so) C (is) to D, and the square on A is greater than (the square on) B by the (square) on (some straight-line) commensurable (in length) with (A), the square on C is thus also greater than (the square on) D by the (square) on (some straight-line) commensurable (in length) with (C) [Prop. 10.14].

Thus, two medial (straight-lines), C and D, commensurable in square only, (and) containing a rational (area), be found. And the square on C is greater than (the square on) D by the (square) on (some straight-line) commensurable in length with (C).[†]

So, similarly, (the proposition) can also be demonstrated for (some straight-line) incommensurable (in length with C), provided that the square on A is greater than (the square on B) by the (square) on (some straight-line) incommensurable (in length) with (A) [Prop. 10.30].[‡]

[†] C and D have lengths $(1 - k^2)^{1/4}$ and $(1 - k^2)^{3/4}$ times that of A, respectively, where k is defined in the footnote to Prop. 10.29.

[‡] C and D would have lengths $1/(1 + k^2)^{1/4}$ and $1/(1 + k^2)^{3/4}$ times that of A, respectively, where k is defined in the footnote to Prop. 10.30.

32.

Εὑρεῖν δύο μέσας δυνάμει μόνον συμμέτρους μέσον περιεχούσας, ὁώστε τὴν μείζονα τῆς ἐλάσσονος μεῖζον δύνασθαι τῷ ἀπὸ συμμέτρου ἑαυτῇ.

Proposition 32

To find two medial (straight-lines), commensurable in square only, (and) containing a medial (area), such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) commensurable (in length) with the greater.

Ἐκκείσθωσαν τρεῖς ὁρηταὶ δυνάμει μόνον σύμμετροι αἱ Α, Β, Γ, ὁώστε τὴν Α τῆς Γ μεῖζον δύνασθαι τῷ ἀπὸ συμμέτρου

Let three rational (straight-lines), A, B and C, commensurable in square only, be laid out such that the square

ἑαυτῇ, καὶ τῷ μὲν ὑπὸ τῶν A, B ἴσον ὑπόκειται τὸ ἀπὸ τῆς Δ. μέσον ἄρα τὸ ἀπὸ τῆς Δ· καὶ ἡ Δ ἄρα μέση ἐστίν. τῷ δὲ ὑπὸ τῶν B, Γ ἴσον ὑποκείσθω τὸ ὑπὸ τῶν Δ, E. καὶ ἐπεί ἐστιν ὡς τὸ ὑπὸ τῶν A, B πρὸς τὸ ὑπὸ τῶν B, Γ, οὕτως ἡ A πρὸς τὴν Γ, ἀλλὰ τῷ μὲν ὑπὸ τῶν A, B ἴσον ἐστὶ τὸ ἀπὸ τῆς Δ, τῷ δὲ ὑπὸ τῶν B, Γ ἴσον τὸ ὑπὸ τῶν Δ, E, ὐστιν ἄρα ὡς ἡ A πρὸς τὴν Γ, οὕτως τὸ ἀπὸ τῆς Δ πρὸς τὸ ὑπὸ τῶν Δ, E. ὡς δὲ τὸ ἀπὸ τῆς Δ πρὸς τὸ ὑπὸ τῶν Δ, E, οὕτως ἡ Δ πρὸς τὴν E· καὶ ὡς ἄρα ἡ A πρὸς τὴν Γ, οὕτως ἡ Δ πρὸς τὴν E. σύμμετρος δὲ ἡ A τῇ Γ δυνάμει [μόνον]. σύμμετρος ἄρα καὶ ἡ Δ τῇ E δυνάμει μόνον. μέση δὲ ἡ Δ· μέση ἄρα καὶ ἡ E. καὶ ἐπεί ἐστιν ὡς ἡ A πρὸς τὴν Γ, ἡ Δ πρὸς τὴν E, ἡ δὲ A τῆς Γ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ ἡ Δ ἄρα τῆς E μεῖζον δυνήσεται τῷ ἀπὸ συμμέτρου ἑαυτῇ. λέγω δή, ὅτι καὶ μέσον ἐστὶ τὸ ὑπὸ τῶν Δ, E. ἐπεὶ γὰρ ἴσον ἐστὶ τὸ ὑπὸ τῶν B, Γ τῷ ὑπὸ τῶν Δ, E, μέσον δὲ τὸ ὑπὸ τῶν B, Γ [αἱ γὰρ B, Γ ὁρηταί εἰσι δυνάμει μόνον σύμμετροι], μέσον ἄρα καὶ τὸ ὑπὸ τῶν Δ, E.

Εὑρηνται ἄρα δύο μέσαι δυνάμει μόνον σύμμετροι αἱ Δ, E μέσον περιέθουσαι, ὥστε τὴν μείζονα τῆς ἐλάσσονος μεῖζον δύνασθαι τῷ ἀπὸ συμμέτρου ἑαυτῇ.

Ὁμοίως δὴ πάλιν διεθθήσεται καὶ τῷ ἀπὸ ἀσυμμέτρου, ὅταν ἡ A τῆς Γ μεῖζον δύνηται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῃ.

on A is greater than (the square on C) by the (square) on (some straight-line) commensurable (in length) with (A) [Prop. 10.29]. And let the (square) on D be equal to the (rectangle contained) by A and B. Thus, the (square) on D (is) medial. Thus, D is also medial [Prop. 10.21]. And let the (rectangle contained) by D and E be equal to the (rectangle contained) by B and C. And since as the (rectangle contained) by A and B is to the (rectangle contained) by B and C, so A (is) to C [Prop. 10.21 lem.], but the (square) on D is equal to the (rectangle contained) by A and B, and the (rectangle contained) by D and E to the (rectangle contained) by B and C, thus as A is to C, so the (square) on D (is) to the (rectangle contained) by D and E. And as the (square) on D (is) to the (rectangle contained) by D and E, so D (is) to E [Prop. 10.21 lem.]. And thus as A (is) to C, so D (is) to E. And A (is) commensurable in square [only] with C. Thus, D (is) also commensurable in square only with E [Prop. 10.11]. And D (is) medial. Thus, E (is) also medial [Prop. 10.23]. And since as A is to C, (so) D (is) to E, and the square on A is greater than (the square on) C by the (square) on (some straight-line) commensurable (in length) with (A), the square on D will thus also be greater than (the square on) E by the (square) on (some straight-line) commensurable (in length) with (D) [Prop. 10.14]. So, I also say that the (rectangle contained) by D and E is medial. For since the (rectangle contained) by B and C is equal to the (rectangle contained) by D and E, and the (rectangle contained) by B and C (is) medial [for B and C are rational (straight-lines which are) commensurable in square only] [Prop. 10.21], the (rectangle contained) by D and E (is) thus also medial.

Thus, two medial (straight-lines), D and E, commensurable in square only, (and) containing a medial (area), be found such that the square on the greater is larger than the (square on the) lesser by the (square) on (some straight-line) commensurable (in length) with the greater.[†].

So, similarly, (the proposition) can again also be demonstrated for (some straight-line) incommensurable (in length with the greater), provided that the square on A is greater than (the square on) C by the (square) on (some straight-line) incommensurable (in length) with (A) [Prop. 10.30].[‡]

[†] D and E have lengths $k'^{1/4}$ and $k'^{1/4}\sqrt{1-k^2}$ times that of A, respectively, where the length of B is $k'^{1/2}$ times that of A, and k is defined in the footnote to Prop. 10.29.

[‡] D and E would have lengths $k'^{1/4}$ and $k'^{1/4}/\sqrt{1+k^2}$ times that of A, respectively, where the length of B is $k'^{1/2}$ times that of A, and k is defined in the footnote to Prop. 10.30.

Λῆμμα.

Ἔστω τρίγωνον ὀρθογώνιον τὸ $AB\Gamma$ ὀρθὴν ἔχον τὴν A, καὶ ἤχθω κάθετος ἡ $A\Delta$· λέγω, ὅτι τὸ μὲν ὑπὸ τῶν $\Gamma B\Delta$ ἴσον ἐστὶ τῷ ἀπὸ τῆς BA, τὸ δὲ ὑπὸ τῶν $B\Gamma A$ ἴσον τῷ ἀπὸ τῆς ΓA, καὶ τὸ ὑπὸ τῶν $B\Delta$, $\Delta\Gamma$ ἴσον τῷ ἀπὸ τῆς $A\Delta$, καὶ

Lemma

Let ABC be a right-angled triangle having the (angle) A a right-angle. And let the perpendicular AD be drawn. I say that the (rectangle contained) by CBD is equal to the (square) on BA, and the (rectangle contained) by BCD (is) equal to the

ὁέτι τὸ ὑπὸ τῶν ΒΓ, ΑΔ ἴσον [ἐστὶ] τῷ ὑπὸ τῶν ΒΑ, ΑΓ.

Καὶ πρῶτον, ὅτι τὸ ὑπὸ τῶν ΓΒΔ ἴσον [ἐστὶ] τῷ ἀπὸ τῆς ΒΑ.

(square) on *CA*, and the (rectangle contained) by *BD* and *DC* (is) equal to the (square) on *AD*, and, further, the (rectangle contained) by *BC* and *AD* [is] equal to the (rectangle contained) by *BA* and *AC*.

And, first of all, (let us prove) that the (rectangle contained) by *CBD* [is] equal to the (square) on *BA*.

Ἐπεὶ γὰρ ἐν ὀρθογωνίῳ τριγώνῳ ἀπὸ τῆς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετος ὀῆκται ἡ ΑΔ, τὰ ΑΒΔ, ΑΔΓ ἄρα τρίγωνα ὅμοιά ἐστι τῷ τε ὅλῳ τῷ ΑΒΓ καὶ ἀλλήλοις. καὶ ἐπεὶ ὅμοιόν ἐστι τὸ ΑΒΓ τρίγωνον τῷ ΑΒΔ τριγώνῳ, ὁέστιν ἄρα ὡξ ἡ ΓΒ πρὸς τὴν ΒΑ, οὁύτως ἡ ΒΑ πρὸς τὴν ΒΔ· τὸ ἄρα ὑπὸ τῶν ΓΒΔ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΒ.

Διὰ τὰ αὐτὰ δὴ καὶ τὸ ὑπὸ τῶν ΒΓΔ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΓ.

Καὶ ἐπεί, ἐαν ἐν ὀρθογωνίῳ τριγώνῳ ἀπὸ τῆς ὀρθῆς γωνίας ἐπὶ τὴν βάσιν κάθετος ἀθθῇ, ἡ ἀθθεῖσα τῶν τῆς βάσεως τμ ημάτων μέση ἀνάλογόν ἐστιν, ὁέστιν ἄρα ὡξ ἡ ΒΑ πρὸς τὴν ΔΑ, οὁύτως ἡ ΑΔ πρὸς τὴν ΔΓ· τὸ ἄρα ὑπὸ τῶν ΒΔ, ΔΓ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΔΑ.

Λέγω, ὅτι καὶ τὸ ὑπὸ τῶν ΒΓ, ΑΔ ἴσον ἐστὶ τῷ ὑπὸ τῶν ΒΑ, ΑΓ. ἐπεὶ γὰρ, ὡξὁέφαμεν, ὅμοιόν ἐστι τὸ ΑΒΓ τῷ ΑΒΔ, ὁέστιν ἄρα ὡξ ἡ ΒΓ πρὸς τὴν ΓΑ, οὁύτως ἡ ΒΑ πρὸς τὴν ΑΔ. τὸ ἄρα ὑπὸ τῶν ΒΓ, ΑΔ ἴσον ἐστὶ τῷ ὑπὸ τῶν ΒΑ, ΑΓ·

For since *AD* has been drawn from the right-angle in a right-angled triangle, perpendicular to the base, *ABD* and *ADC* are thus triangles (which are) similar to the whole, *ABC*, and to one another [Prop. 6.8]. And since triangle *ABC* is similar to triangle *ABD*, thus as *CB* is to *BA*, so *BA* (is) to *BD* [Prop. 6.4]. Thus, the (rectangle contained) by *CBD* is equal to the (square) on *AB* [Prop. 6.17].

So, for the same (reasons), the (rectangle contained) by *BCD* is also equal to the (square) on *AC*.

And since if a (straight-line) is drawn from the right-angle in a right-angled triangle, perpendicular to the base, the (straight-line so) drawn is the mean proportional to the pieces of the base [Prop. 6.8 corr.], thus as *BD* is to *DA*, so *AD* (is) to *DC*. Thus, the (rectangle contained) by *BD* and *DC* is equal to the (square) on *DA* [Prop. 6.17].

I also say that the (rectangle contained) by *BC* and *AD* is equal to the (rectangle contained) by *BA* and *AC*. For since,

ὅπερ ἔδει δεῖξαι.

33.

Εὑρεῖν δύο εὐθείας δυνάμει ἀσυμμέτρους ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων ὁρητόν, τὸ δὸ ὑπὸ αὐτῶν μέσον.

Ἐκκείσθωσαν δύο ὁρηταὶ δυνάμει μόνον σύμμετροι αἱ AB, $BΓ$, ὥστε τὴν μείζονα τὴν AB τῆς ἐλάσσονος τῆς $BΓ$ μείζον δύνασθαι τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, καὶ τετμήσθω ἡ $BΓ$ δίθα κατὰ τὸ $Δ$, καὶ τῷ ἀφ' ὁποτέρας τῶν $BΔ$, $ΔΓ$ ἴσον παρὰ τὴν AB παραβεβλήσθω παραλληλόγραμμον ἐλλεῖπον εἰδει τετραγώνῳ, καὶ ὀέστω τὸ ὑπὸ τῶν AEB, καὶ γεγράφθω ἐπὶ τῆς AB ημικύκλιον τὸ AZB, καὶ ἤθθω τῇ AB πρὸς ὀρθὰξ ἡ EZ, καὶ ἐπεζεύθθωσαν αἱ AZ, ZB.

Let the two rational (straight-lines) AB and BC, (which are) commensurable in square only, be laid out such that the square on the greater, AB, is larger than (the square on) the lesser, BC, by the (square) on (some straight-line which is) incommensurable (in length) with (AB) [Prop. 10.30]. And let BC be cut in half at D. And let a parallelogram equal to the (square) on either of BD or DC, (and) falling short by a square figure, be applied to AB [Prop. 6.28], and let it be the (rectangle contained) by AEB. And let the semi-circle AFB be drawn on AB. And let EF be drawn at right-angles to AB. And let AF and FB be joined.

Καὶ ἐπεὶ [δύο] εὐθεῖαιὀάνισοί εἰσιν αἱ AB, $BΓ$, καὶ ἡ AB τῆς $BΓ$ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, τῷ δὲ τετάρτῳ τοῦ ἀπὸ τῆς $BΓ$, τουτέστι τῷ ἀπὸ τῆς ἡμισείας αὐτῆς, ἴσον παρὰ τὴν AB παραβέβληται παραλληλόγραμμον ἐλλεῖπον εἰδει τετραγώνῳ καὶ ποιεῖ τὸ ὑπὸ τῶν AEB, ἀσύμμετρος ἄρα ἐστὶν ἡ AE τῇ EB. καί ἐστιν ὡξ ἡ AE πρὸς EB, οὕτως τὸ ὑπὸ τῶν BA, AE πρὸς τὸ ὑπὸ τῶν AB, BE, ἴσον δὲ τὸ μὲν ὑπὸ τῶν BA, AE τῷ ἀπὸ τῆς AZ, τὸ δὲ ὑπὸ τῶν AB, BE τῷ ἀπὸ τῆς BZ· ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς AZ τῷ ἀπὸ τῆς ZB· αἱ AZ, ZB ἄρα δυνάμει εἰσὶν ἀσύμμετροι. καὶ ἐπεὶ ἡ AB ὁρητή ἐστιν, ὁρητὸν ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς AB· ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AZ, ZB ὁρητόν ἐστιν. καὶ ἐπεὶ πάλιν τὸ ὑπὸ τῶν AE, EB ἴσον ἐστὶ τῷ ἀπὸ τῆς EZ, ὑπόκειται δὲ τὸ ὑπὸ τῶν AE, EB καὶ τῷ ἀπὸ τῆς $BΔ$ ἴσον, ὀίση ἄρα ἐστὶν ἡ ZE τῇ $BΔ$· διπλῆ ἄρα ἡ $BΓ$ τὴξ ZE· ὥστε καὶ τὸ ὑπὸ τῶν AB, $BΓ$ σύμμετρόν ἐστι τῷ ὑπὸ τῶν AB, EZ. μέσον δὲ τὸ ὑπὸ τῶν AB, $BΓ$· μέσον ἄρα καὶ τὸ ὑπὸ τῶν AB, EZ. ἴσον δὲ τὸ ὑπὸ τῶν AB, EZ τῷ ὑπὸ τῶν AZ, ZB· μέσον ἄρα καὶ τὸ ὑπὸ τῶν AZ, ZB. ἐδείθθη δὲ καὶ ὁρητὸν τὸ

as we said, ABC is similar to ABD, thus as BC is to CA, so BA (is) to AD [Prop. 6.4]. Thus, the (rectangle contained) by BC and AD is equal to the (rectangle contained) by BA and AC [Prop. 6.16]. (Which is) the very thing it was required to show.

Proposition 33

To find two straight-lines (which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial.

Let the two rational (straight-lines) AB and BC, (which are) commensurable in square only, be laid out such that the square on the greater, AB, is larger than (the square on) the lesser, BC, by the (square) on (some straight-line which is) incommensurable (in length) with (AB) [Prop. 10.30]. And let BC be cut in half at D. And let a parallelogram equal to the (square) on either of BD or DC, (and) falling short by a square figure, be applied to AB [Prop. 6.28], and let it be the (rectangle contained) by AEB. And let the semi-circle AFB be drawn on AB. And let EF be drawn at right-angles to AB. And let AF and FB be joined.

And since AB and BC are [two] unequal straight-lines, and the square on AB is greater than (the square on) BC by the (square) on (some straight-line which is) incommensurable (in length) with (AB). And a parallelogram, equal to one quarter of the (square) on BC—that is to say, (equal) to the (square) on half of it—(and) falling short by a square figure, has been applied to AB, and makes the (rectangle contained) by AEB. AE is thus incommensurable (in length) with EB [Prop. 10.18]. And as AE is to EB, so the (rectangle contained) by BA and AE (is) to the (rectangle contained) by AB and BE. And the (rectangle contained) by BA and AE (is) equal to the (square) on AF, and the (rectangle contained) by AB and BE to the (square) on BF [Prop. 10.32 lem.]. The (square) on AF is thus incommensurable with the (square) on FB [Prop. 10.11]. Thus, AF and FB are incommensurable in square. And since AB is rational, the (square) on AB is also rational. Hence, the sum of the (squares) on AF and FB is also rational [Prop. 1.47]. And, again, since the (rectangle contained) by AE and EB is equal to the (square) on EF, and

συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων.

Εὕρηνται ἄρα δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ AZ, ZB ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων ὁρητόν, τὸ δὲ ὑπὸ αὐτῶν μέσον· ὅπερ ἔδει δεῖξαι.

the (rectangle contained) by AE and EB was assumed (to be) equal to the (square) on BD, FE is thus equal to BD. Thus, BC is double FE. And hence the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and EF [Prop. 10.6]. And the (rectangle contained) by AB and BC (is) medial [Prop. 10.21]. Thus, the (rectangle contained) by AB and EF (is) also medial [Prop. 10.23 corr.]. And the (rectangle contained) by AB and EF (is) equal to the (rectangle contained) by AF and FB [Prop. 10.32 lem.]. Thus, the (rectangle contained) by AF and FB (is) also medial. And the sum of the squares on them was also shown (to be) rational.

Thus, the two straight-lines, AF and FB, (which are) incommensurable in square, be found, making the sum of the squares on them rational, and the (rectangle contained) by them medial. (Which is) the very thing it was required to show.

† AF and FB have lengths $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2}$ and $\sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$ times that of AB, respectively, where k is defined in the footnote to Prop. 10.30.

34.

Εὑρεῖν δύο εὐθείας δυνάμει ἀσυμμέτρους ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον, τὸ δὸ ὑπὸ αὐτῶν ὁρητόν.

Proposition 34

To find two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational.

Ἐκκείσθωσαν δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB, BΓ ὁρητὸν περιέθουσαι τὸ ὑπὸ αὐτῶν, ὁώστε τὴν AB τῆς BΓ μεῖζον δύνασθαι τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, καὶ γεγράφθω ἐπὶ τῆς AB τὸ AΔB ἡμικύκλιον, καὶ τετμήσθω ἡ BΓ δίθα κατὰ τὸ E, καὶ παραβεβλήσθω παρὰ τὴν AB τῷ ἀπὸ τῆς BE ἴσον παραλληλόγραμμον ἐλλεῖπον εὐίδει τετραγώνῳ τὸ ὑπὸ τῶν AZB· ἀσύμμετρος ἄρα [ἐστὶν] ἡ AZ τῇ ZB μήκει. καὶ ἤθθω ἀπὸ τοῦ Z τῇ AB πρὸς ὀρθὰξ ἡ ZΔ, καὶ ἐπεζεύθθωσαν αἱ AΔ, ΔB.

Ἐπεὶ ἀσύμμετρόξ ἐστιν ἡ AZ τῇ ZB, ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ὑπὸ τῶν BA, AZ τῷ ὑπὸ τῶν AB, BZ. ἴσον δὲ τὸ μὲν ὑπὸ τῶν BA, AZ τῷ ἀπὸ τῆς AΔ, τὸ δὲ ὑπὸ τῶν AB, BZ τῷ ἀπὸ τῆς ΔB· ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς AΔ τῷ ἀπὸ τῆς ΔB. καὶ ἐπεὶ μέσον ἐστὶ τὸ ἀπὸ τῆς AB, μέσον ἄρα καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AΔ, ΔB. καὶ ἐπεὶ διπλῆ ἐστιν ἡ BΓ τῆς ΔZ, διπλάσιον ἄρα καὶ τὸ ὑπὸ τῶν AB, BΓ τοῦ ὑπὸ τῶν AB, ZΔ. ὁρητὸν δὲ τὸ ὑπὸ τῶν AB, BΓ·

Let the two medial (straight-lines) AB and BC, (which are) commensurable in square only, be laid out having the (rectangle contained) by them rational, (and) such that the square on AB is greater than (the square on) BC by the (square) on (some straight-line) incommensurable (in length) with (AB) [Prop. 10.31]. And let the semi-circle ADB be drawn on AB. And let BC be cut in half at E. And let a (rectangular) parallelogram equal to the (square) on BE, (and) falling short by a square figure, be applied to AB, (and let it be) the (rectangle contained by) AFB [Prop. 6.28]. Thus, AF [is] incommensurable in length with FB [Prop. 10.18]. And let FD be drawn from F at right-angles to AB. And let AD and DB be joined.

Since AF is incommensurable (in length) with FB, the (rectangle contained) by BA and AF is thus also incommensurable with the (rectangle contained) by AB and BF [Prop. 10.11]. And the (rectangle contained) by BA and AF (is) equal to the (square) on AD, and the (rectangle contained)

ὁρητὸν ἄρα καὶ τὸ ὑπὸ τῶν AB, $Z\Delta$. τὸ δὲ ὑπὸ τῶν AB, $Z\Delta$ ἴσον τῷ ὑπὸ τῶν $A\Delta$, ΔB· ὥστε καὶ τὸ ὑπὸ τῶν $A\Delta$, ΔB ὁρητόν ἐστιν.

Εὑρηνται ἄρα δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ $A\Delta$, ΔB ποιοῦσαι τὸ [μὲν] συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον, τὸ δὸ ὑπὸ αὐτῶν ὁρητόν· ὅπερ ἔδει δεῖχαι.

by AB and BF to the (square) on DB [Prop. 10.32 lem.]. Thus, the (square) on AD is also incommensurable with the (square) on DB. And since the (square) on AB is medial, the sum of the (squares) on AD and DB (is) thus also medial [Props. 3.31, 1.47]. And since BC is double DF [see previous proposition], the (rectangle contained) by AB and BC (is) thus also double the (rectangle contained) by AB and FD. And the (rectangle contained) by AB and BC (is) rational. Thus, the (rectangle contained) by AB and FD (is) also rational [Prop. 10.6, Def. 10.4]. And the (rectangle contained) by AB and FD (is) equal to the (rectangle contained) by AD and DB [Prop. 10.32 lem.]. And hence the (rectangle contained) by AD and DB is rational.

Thus, two straight-lines, AD and DB, (which are) incommensurable in square, be found, making the sum of the squares on them medial, and the (rectangle contained) by them rational.† (Which is) the very thing it was required to show.

† AD and DB have lengths $\sqrt{[(1+k^2)^{1/2}+k]/[2\,(1+k^2)]}$ and $\sqrt{[(1+k^2)^{1/2}-k]/[2\,(1+k^2)]}$ times that of AB, respectively, where k is defined in the footnote to Prop. 10.29.

35.

Εὑρεῖν δύο εὐθείας δυνάμει ἀσυμμέτρους ποιούσας τό τε συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπὸ αὐτῶν μέσον καὶ ὀέτι ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπὸ αὐτῶν τετραγώνῳ.

Proposition 35

To find two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the squares on them.

Ἐκκείσθωσαν δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB, $B\Gamma$ μέσον περιέθουσαι, ὥστε τὴν AB τῆς $B\Gamma$ μεῖζον δύνασθαι τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ $A\Delta B$, καὶ τὰ λοιπὰ γεγονέτω τοῖς ἐπάνω ὁμοίως.

Καὶ ἐπεὶ ἀσύμμετρόξ ἐστιν ἡ AZ τῇ ZB μήκει, ἀσύμμετρόξ ἐστι καὶ ἡ $A\Delta$ τῇ ΔB δυνάμει. καὶ ἐπεὶ μέσον ἐστὶ τὸ ἀπὸ τῆς AB, μέσον ἄρα καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $A\Delta$, ΔB. καὶ ἐπεὶ τὸ ὑπὸ τῶν AZ, ZB ἴσον ἐστὶ τῷ ἀφ' ἑκατέρας τῶν BE, ΔZ, ὁίση ἄρα ἐστὶν ἡ BE τῇ ΔZ· διπλῆ ἄρα ἡ $B\Gamma$ τῆς $Z\Delta$· ὥστε καὶ τὸ ὑπὸ τῶν AB, $B\Gamma$ διπλάσιόν ἐστι τοῦ ὑπὸ τῶν AB, $Z\Delta$. μέσον δὲ τὸ ὑπὸ τῶν AB, $B\Gamma$· μέσον ἄρα καὶ τὸ ὑπὸ τῶν AB, $Z\Delta$. καί ἐστιν ἴσον τῷ ὑπὸ

Let the two medial (straight-lines) AB and BC, (which are) commensurable in square only, be laid out containing a medial (area), such that the square on AB is greater than (the square on) BC by the (square) on (some straight-line) incommensurable (in length) with (AB) [Prop. 10.32]. And let the semicircle ADB be drawn on AB. And let the remainder (of the figure) be generated similarly to the above (proposition).

And since AF is incommensurable in length with FB [Prop. 10.18], AD is also incommensurable in square with DB [Prop. 10.11]. And since the (square) on AB is medial, the sum of the (squares) on AD and DB (is) thus also medial [Props. 3.31, 1.47]. And since the (rectangle contained) by AF and FB is equal to the (square) on each of BE and DF, BE is

τῶν $A\Delta$, ΔB· μέσον ἄρα καὶ τὸ ὑπὸ τῶν $A\Delta$, ΔB. καὶ ἐπεὶ ἀσύμμετρόξ ἐστιν ἡ AB τῇ $B\Gamma$ μήκει, σύμμετρος δὲ ἡ ΓB τῇ BE, ἀσύμμετρος ἄρα καὶ ἡ AB τῇ BE μήκει· ὥστε καὶ τὸ ἀπὸ τῆς AB τῷ ὑπὸ τῶν AB, BE ἀσύμμετρόν ἐστιν. ἀλλὰ τῷ μὲν ἀπὸ τῆς AB ἴσα ἐστὶ τὰ ἀπὸ τῶν $A\Delta$, ΔB, τῷ δὲ ὑπὸ τῶν AB, BE ἴσον ἐστὶ τὸ ὑπὸ τῶν AB, $Z\Delta$, τουτέστι τὸ ὑπὸ τῶν $A\Delta$, ΔB· ἀσύμμετρον ἄρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $A\Delta$, ΔB τῷ ὑπὸ τῶν $A\Delta$, ΔB.

Εὑρηνται ἄρα δύο εὐθεῖαι αἱ $A\Delta$, ΔB δυνάμει ἀσύμμετροι ποιοῦσαι τό τε συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν μέσον καὶ τὸ ὑπὸ αὐτῶν μέσον καὶ ὀέτι ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων· ὅπερ ἔδει δεῖχαι.

thus equal to DF. Thus, BC (is) double FD. And hence the (rectangle contained) by AB and BC is double the (rectangle) contained by AB and FD. And the (rectangle contained) by AB and BC (is) medial. Thus, the (rectangle contained) by AB and FD (is) also medial. And it is equal to the (rectangle contained) by AD and DB [Prop. 10.32 lem.]. Thus, the (rectangle contained) by AD and DB (is) also medial. And since AB is incommensurable in length with BC, and CB (is) commensurable (in length) with BE, AB (is) thus also incommensurable in length with BE [Prop. 10.13]. And hence the (square) on AB is also incommensurable with the (rectangle contained) by AB and BE [Prop. 10.11]. But the (sum of the squares) on AD and DB is equal to the (square) on AB [Prop. 1.47]. And the (rectangle contained) by AB and FD—that is to say, the (rectangle contained) by AD and DB—is equal to the (rectangle contained) by AB and BE. Thus, the sum of the (squares) on AD and DB is incommensurable with the (rectangle contained) by AD and DB.

Thus, two straight-lines, AD and DB, (which are) incommensurable in square, be found, making the sum of the (squares) on them medial, and the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the squares on them.† (Which is) the very thing it was required to show.

† AD and DB have lengths $k'^{1/4}\sqrt{[1+k/(1+k^2)^{1/2}]/2}$ and $k'^{1/4}\sqrt{[1-k/(1+k^2)^{1/2}]/2}$ times that of AB, respectively, where k and k' are defined in the footnote to Prop. 10.32.

36.

Ἐὰν δύο ὁρηταὶ δυνάμει μόνον σύμμετροι συντεθῶσιν, ἡ ὅληὀάλογόξ ἐστιν, καλείσθω δὲ ἐκ δύο ὀνομάτων.

Proposition 36

If two rational (straight-lines which are) commensurable in square only are added together then the whole (straight-line) is irrational—let it be called a binomial (straight-line).†

Συγκείσθωσαν γὰρ δύο ὁρηταὶ δυνάμει μόνον σύμμετροι αἱ AB, $B\Gamma$· λέγω, ὅτι ὅλη ἡ $A\Gamma$ὀάλογόξ ἐστιν.

Ἐπεὶ γὰρ ἀσύμμετρόξ ἐστιν ἡ AB τῇ $B\Gamma$ μήκει· δυνάμει γὰρ μόνον εἰσὶ σύμμετροι· ὡξ δὲ ἡ AB πρὸς τὴν $B\Gamma$, οὕτως τὸ ὑπὸ τῶν $AB\Gamma$ πρὸς τὸ ἀπὸ τῆς $B\Gamma$, ἀσύμμετρον ἄρα ἐστὶ τὸ ὑπὸ τῶν AB, $B\Gamma$ τῷ ἀπὸ τῆς $B\Gamma$. ἀλλὰ τῷ μὲν ὑπὸ τῶν AB, $B\Gamma$ σύμμετρόν ἐστι τὸ δὶξ ὑπὸ τῶν AB, $B\Gamma$, τῷ δὲ ἀπὸ τῆς $B\Gamma$ σύμμετρά ἐστι τὰ ἀπὸ τῶν AB, $B\Gamma$· αἱ γὰρ AB, $B\Gamma$ ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀσύμμετρον ἄρα ἐστὶ τὸ δὶξ ὑπὸ τῶν AB, $B\Gamma$ τοῖς ἀπὸ τῶν AB, $B\Gamma$. καὶ συνθέντι τὸ δὶξ ὑπὸ τῶν AB, $B\Gamma$ μετὰ τῶν ἀπὸ τῶν AB, $B\Gamma$, τουτέστι τὸ ἀπὸ τῆς $A\Gamma$, ἀσύμμετρόν ἐστι τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν AB, $B\Gamma$· ὁρητὸν δὲ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB, $B\Gamma$· ὀάλογον ἄρα [ἐστὶ] τὸ ἀπὸ τῆς $A\Gamma$· ὥστε καὶ ἡ $A\Gamma$ὀάλογόξ

For let the two rational (straight-lines), AB and BC, (which are) commensurable in square only, be laid down together. I say that the whole (straight-line), AC, is irrational. For since AB is incommensurable in length with BC—for they are commensurable in square only—and as AB (is) to BC, so the (rectangle contained) by ABC (is) to the (square) on BC, the (rectangle contained) by AB and BC is thus incommensurable with the (square) on BC [Prop. 10.11]. But, twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. And (the sum of) the (squares) on AB and BC is commensurable with the (square) on BC—for the rational (straight-lines) AB and BC are commensurable in square only [Prop. 10.15]. Thus, twice the (rectangle contained) by AB and BC is incommensurable

ἐστιν, καλείσθω δὲ ἐκ δύο ὀνομάτων· ὅπερ ἔδει δεῖξαι.

with (the sum of) the (squares) on AB and BC [Prop. 10.13]. And, via composition, twice the (rectangle contained) by AB and BC, plus (the sum of) the (squares) on AB and BC—that is to say, the (square) on AC [Prop. 2.4]—is incommensurable with the sum of the (squares) on AB and BC [Prop. 10.16]. And the sum of the (squares) on AB and BC (is) rational. Thus, the (square) on AC [is] irrational [Def. 10.4]. Hence, AC is also irrational [Def. 10.4]—let it be called a binomial (straight-line).‡ (Which is) the very thing it was required to show.

† Literally, "from two names".

‡ Thus, a binomial straight-line has a length expressible as $1 + k^{1/2}$ [or, more generally, $\rho\,(1 + k^{1/2})$, where ρ is rational—the same proviso applies to the definitions in the following propositions]. The binomial and the corresponding apotome, whose length is expressible as $1 - k^{1/2}$ (see Prop. 10.73), are the positive roots of the quartic $x^4 - 2\,(1 + k)\,x^2 + (1 - k)^2 = 0$.

37.

Ἐὰν δύο μέσαι δυνάμει μόνον σύμμετροι συντεθῶσι ὁρητὸν περιέθουσαι, ἡ ὅλη ἀλογός ἐστιν, καλείσθω δὲ ἐκ δύο μέσων πρώτη.

Proposition 37

If two medial (straight-lines), commensurable in square only, which contain a rational (area), are added together then the whole (straight-line) is irrational—let it be called a first bimedial (straight-line).†

Συγκείσθωσαν γὰρ δύο μέσαι δυνάμει μόνον σύμμετροι αἱ AB, $BΓ$ ὁρητὸν περιέθουσαι· λέγω, ὅτι ὅλη ἡ $AΓ$ ἀλογός ἐστιν.

Ἐπεὶ γὰρ ἀσύμμετρός ἐστιν ἡ AB τῇ $BΓ$ μήκει, καὶ τὰ ἀπὸ τῶν AB, $BΓ$ ἄρα ἀσύμμετρά ἐστι τῷ δὶς ὑπὸ τῶν AB, $BΓ$· καὶ συνθέντι τὰ ἀπὸ τῶν AB, $BΓ$ μετὰ τοῦ δὶς ὑπὸ τῶν AB, $BΓ$, ὅπερ ἐστὶ τὸ ἀπὸ τῆς $AΓ$, ἀσύμμετρόν ἐστι τῷ ὑπὸ τῶν AB, $BΓ$. ὁρητὸν δὲ τὸ ὑπὸ τῶν AB, $BΓ$· ὑπόκεινται γὰρ αἱ AB, $BΓ$ ὁρητὸν περιέθουσαι· ἀλογον ἄρα τὸ ἀπὸ τῆς $AΓ$· ἀλογος ἄρα ἡ $AΓ$, καλείσθω δὲ ἐκ δύο μέσων πρώτη· ὅπερ ἔδει δεῖξαι.

For let the two medial (straight-lines), AB and BC, commensurable in square only, (and) containing a rational (area), be laid down together. I say that the whole (straight-line), AC, is irrational.

For since AB is incommensurable in length with BC, (the sum of) the (squares) on AB and BC is thus also incommensurable with twice the (rectangle contained) by AB and BC [see previous proposition]. And, via composition, (the sum of) the (squares) on AB and BC, plus twice the (rectangle contained) by AB and BC—that is, the (square) on AC [Prop. 2.4]—is incommensurable with the (rectangle contained) by AB and BC [Prop. 10.16]. And the (rectangle contained) by AB and BC (is) rational—for AB and BC were assumed to enclose a rational (area). Thus, the (square) on AC (is) irrational. Thus, AC (is) irrational [Def. 10.4]—let it be called a first bimedial (straight-line).‡ (Which is) the very thing it was required to show.

† Literally, "first from two medials".

‡ Thus, a first bimedial straight-line has a length expressible as $k^{1/4} + k^{3/4}$. The first bimedial and the corresponding first apotome of a medial, whose length is expressible as $k^{1/4} - k^{3/4}$ (see Prop. 10.74), are the positive roots of the quartic $x^4 - 2\,\sqrt{k}\,(1 + k)\,x^2 + k\,(1 - k)^2 = 0$.

38.

Ἐὰν δύο μέσαι δυνάμει μόνον σύμμετροι συντεθῶσι μέσον περιέθουσαι, ἡ ὅλη ἀλογός ἐστιν, καλείσθω δὲ ἐκ δύο μέσων

Proposition 38

If two medial (straight-lines), commensurable in square only, which contain a medial (area), are added together then the

δυετέρα.

Συγκείσθωσαν γὰρ δύο μέσαι δυνάμει μόνον σύμμετροι αἱ ΑΒ, ΒΓ μέσον περιέθουσαι· λέγω, ὅτιὁάλογόξ ἐστιν ἡ ΑΓ.

Ἐκκείσθω γὰρ ὁρητὴ ἡ ΔΕ, καὶ τῷ ἀπὸ τῆς ΑΓ ἴσον παρὰ τὴν ΔΕ παραβεβλήσθω τὸ ΔΖ πλάτος ποιοῦν τὴν ΔΗ. καὶ ἐπεὶ τὸ ἀπὸ τῆς ΑΓ ἴσον ἐστὶ τοῖς τε ἀπὸ τῶν ΑΒ, ΒΓ καὶ τῷ δὶξ ὑπὸ τῶν ΑΒ, ΒΓ, παραβεβλήσθω δὴ τοῖς ἀπὸ τῶν ΑΒ, ΒΓ παρὰ τὴν ΔΕ ἴσον τὸ ΕΘ· λοιπὸν ἄρα τὸ ΘΖ ἴσον ἐστὶ τῷ δὶξ ὑπὸ τῶν ΑΒ, ΒΓ. καὶ ἐπεὶ μέση ἐστὶν ἑκατέρα τῶν ΑΒ, ΒΓ, μέσα ἄρα ἐστὶ καὶ τὰ ἀπὸ τῶν ΑΒ, ΒΓ. μέσον δὲ ὑπόκειται καὶ τὸ δὶξ ὑπὸ τῶν ΑΒ, ΒΓ. καί ἐστι τοῖς μὲν ἀπὸ τῶν ΑΒ, ΒΓ ἴσον τὸ ΕΘ, τῷ δὲ δὶξ ὑπὸ τῶν ΑΒ, ΒΓ ἴσον τὸ ΖΘ· μέσον ἄρα ἑκάτερον τῶν ΕΘ, ΘΖ. καὶ παρὰ ὁρητὴν τὴν ΔΕ παράκειται· ὁρητὴ ἄρα ἐστὶν ἑκατέρα τῶν ΔΘ, ΘΗ καὶ ἀσύμμετρος τῇ ΔΕ μήκει. ἐπεὶ οὐῦν ἀσύμμετρόξ ἐστιν ἡ ΑΒ τῇ ΒΓ μήκει, καί ἐστιν ὡξ ἡ ΑΒ πρὸς τὴν ΒΓ, οὕτως τὸ ἀπὸ τῆς ΑΒ πρὸς τὸ ὑπὸ τῶν ΑΒ, ΒΓ, ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΒ τῷ ὑπὸ τῶν ΑΒ, ΒΓ. ἀλλὰ τῷ μὲν ἀπὸ τῆς ΑΒ σύμμετρόν ἐστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΒ, ΒΓ τετραγώνων, τῷ δὲ ὑπὸ τῶν ΑΒ, ΒΓ σύμμετρόν ἐστι τὸ δὶξ ὑπὸ τῶν ΑΒ, ΒΓ. ἀσύμμετρον ἄρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΒ, ΒΓ τῷ δὶξ ὑπὸ τῶν ΑΒ, ΒΓ. ἀλλὰ τοῖς μὲν ἀπὸ τῶν ΑΒ, ΒΓ ἴσον ἐστὶ τὸ ΕΘ, τῷ δὲ δὶξ ὑπὸ τῶν ΑΒ, ΒΓ ἴσον ἐστὶ τὸ ΘΖ. ἀσύμμετρον ἄρα ἐστὶ τὸ ΕΘ τῷ ΘΖ· ὡστε καὶ ἡ ΔΘ τῇ ΘΗ ἐστιν ἀσύμμετρος μήκει. αἱ ΔΘ, ΘΗ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι. ὡστε ἡ ΔΗὁάλογόξ ἐστιν. ὁρητὴ δὲ ἡ ΔΕ· τὸ δὲ ὑπὸ ἀλόγου καὶ

For let the two medial (straight-lines), AB and BC, commensurable in square only, (and) containing a medial (area), be laid down together [Prop. 10.28]. I say that AC is irrational.

For let the rational (straight-line) DE be laid down, and let (the rectangle) DF, equal to the (square) on AC, be applied to DE, making DG as breadth [Prop. 1.44]. And since the (square) on AC is equal to (the sum of) the (squares) on AB and BC, plus twice the (rectangle contained) by AB and BC [Prop. 2.4], so let (the rectangle) EH, equal to (the sum of) the squares on AB and BC, be applied to DE. The remainder HF is thus equal to twice the (rectangle contained) by AB and BC. And since AB and BC are each medial, (the sum of) the squares on AB and BC is thus also medial.‡ And twice the (rectangle contained) by AB and BC was also assumed (to be) medial. And EH is equal to (the sum of) the squares on AB and BC, and FH (is) equal to twice the (rectangle contained) by AB and BC. Thus, EH and HF (are) each medial. And they were applied to the rational (straight-line) DE. Thus, DH and HG are each rational, and incommensurable in length with DE [Prop. 10.22]. Therefore, since AB is incommensurable in length with BC, and as AB is to BC, so the (square) on AB (is) to the (rectangle contained) by AB and BC [Prop. 10.21 lem.], the (square) on AB is thus incommensurable with the (rectangle contained) by AB and BC [Prop. 10.11]. But, the sum of the squares on AB and BC is commensurable with the (square) on AB [Prop. 10.15], and twice the (rectangle contained) by

δυετέρα.

whole (straight-line) is irrational—let it be called a second bimedial (straight-line).

ὁρητῆς περιεχόμενον ὀρθογώνιονὁάλογόν ἐστιν· ὁάλογον ἄρα ἐστὶ τὸ ΔΖ θωρίον, καὶ ἡ δυναμένη [αὐτὸ] ὁάλογόξ ἐστιν. δύναται δὲ τὸ ΔΖ ἡ ΑΓ· ὁάλογος ἄρα ἐστὶν ἡ ΑΓ, καλείσθω δὲ ἐκ δύο μέσων δευτέρα. ὅπερ ἔδει δεῖχαι.

AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. Thus, the sum of the (squares) on AB and BC is incommensurable with twice the (rectangle contained) by AB and BC [Prop. 10.13]. But, EH is equal to (the sum of) the squares on AB and BC, and HF is equal to twice the (rectangle) contained by AB and BC. Thus, EH is incommensurable with HF. Hence, DH is also incommensurable in length with HG [Props. 6.1, 10.11]. Thus, DH and HG are rational (straight-lines which are) commensurable in square only. Hence, DG is irrational [Prop. 10.36]. And DE (is) rational. And the rectangle contained by irrational and rational (straight-lines) is irrational [Prop. 10.20]. The area DF is thus irrational, and (so) the square-root [of it] is irrational [Def. 10.4]. And AC is the square-root of DF. AC is thus irrational—let it be called a second bimedial (straight-line).[§] (Which is) the very thing it was required to show.

[†] Literally, "second from two medials".

[‡] Since, by hypothesis, the squares on AB and BC are commensurable—see Props. 10.15, 10.23.

[§] Thus, a second bimedial straight-line has a length expressible as $k^{1/4} + k'^{1/2}/k^{1/4}$. The second bimedial and the corresponding second apotome of a medial, whose length is expressible as $k^{1/4} - k'^{1/2}/k^{1/4}$ (see Prop. 10.75), are the positive roots of the quartic $x^4 - 2\,[(k + k')/\sqrt{k}\,]\,x^2 + [(k - k')^2/k] = 0$.

39.

Ἐὰν δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων ὁρητόν, τὸ δὸ ὑπὸ αὐτῶν μέσον, ἡ ὅλη εὐθεῖαὁάλογόξ ἐστιν, καλείσθω δὲ μείζων.

Proposition 39

If two straight-lines (which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial, are added together then the whole straight-line is irrational—let it be called a major (straight-line).

Συγκείσθωσαν γὰρ δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ ΑΒ, ΒΓ ποιοῦσαι τὰ προκείμενα· λέγω, ὅτιὁάλογόξ ἐστιν ἡ ΑΓ.

Ἐπεὶ γὰρ τὸ ὑπὸ τῶν ΑΒ, ΒΓ μέσον ἐστίν, καὶ τὸ δὶξ [ἄρα] ὑπὸ τῶν ΑΒ, ΒΓ μέσον ἐστίν. τὸ δὲ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΒ, ΒΓ ὁρητόν· ἀσύμμετρον ἄρα ἐστὶ τὸ δὶξ ὑπὸ τῶν ΑΒ, ΒΓ τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν ΑΒ, ΒΓ· ὥστε καὶ τὰ ἀπὸ τῶν ΑΒ, ΒΓ μετὰ τοῦ δὶξ ὑπὸ τῶν ΑΒ, ΒΓ, ὅπερ ἐστὶ τὸ ἀπὸ τῆς ΑΓ, ἀσύμμετρόν ἐστι τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν ΑΒ, ΒΓ [ὁρητὸν δὲ τὸ συγμείμενον ἐκ τῶν ἀπὸ τῶν ΑΒ, ΒΓ]· ὁάλογον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΓ. ὥστε καὶ ἡ ΑΓὁάλογόξ ἐστιν, καλείσθω δὲ μείζων. ὅπερ ἔδει δεῖχαι.

For let the two straight-lines, AB and BC, incommensurable in square, and fulfilling the prescribed (conditions), be laid down together [Prop. 10.33]. I say that AC is irrational.

For since the (rectangle contained) by AB and BC is medial, twice the (rectangle contained) by AB and BC is [thus] also medial [Props. 10.6, 10.23 corr.]. And the sum of the (squares) on AB and BC (is) rational. Thus, twice the (rectangle contained) by AB and BC is incommensurable with the sum of the (squares) on AB and BC [Def. 10.4]. Hence, (the sum of) the squares on AB and BC, plus twice the (rectangle contained) by AB and BC—that is, the (square) on AC [Prop. 2.4]—is also incommensurable with the sum of the (squares) on AB and BC [Prop. 10.16] [and the sum of the (squares) on AB and BC (is) rational]. Thus, the (square) on AC is irrational. Hence, AC is also irrational [Def. 10.4]—let it be called a major (straight-line).[†] (Which is) the very thing it was required to show.

† Thus, a major straight-line has a length expressible as $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + \sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$. The major and the corresponding minor, whose length is expressible as $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} - \sqrt{[1 - k/(1 + k^2)^{1/2}]/2}$ (see Prop. 10.76), are the positive roots of the quartic $x^4 - 2\,x^2 + k^2/(1 + k^2) = 0$.

40.

Ἐὰν δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον, τὸ δὸ ὑπὸ αὐτῶν ὁρητόν, ἡ ὅλη εὐθεῖαὁάλογόξ ἐστιν, καλείσθω δὲ ὁρητὸν καὶ μέσον δυναμένη.

Proposition 40

If two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational, are added together then the whole straight-line is irrational—let it be called the square-root of a rational plus a medial (area).

Συγκείσθωσαν γὰρ δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ AB, BΓ ποιοῦσαι τὰ προκείμενα· λέγω, ὅτιὁάλογόξ ἐστιν ἡ AΓ.

Ἐπεὶ γὰρ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB, BΓ μέσον ἐστίν, τὸ δὲ δὶξ ὑπὸ τῶν AB, BΓ ὁρητόν, ἀσύμμετρον ἄρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB, BΓ τῷ δὶξ ὑπὸ τῶν AB, BΓ· ὁώστε καὶ τὸ ἀπὸ τῆς AΓ ἀσύμμετρόν ἐστι τῷ δὶξ ὑπὸ τῶν AB, BΓ. ὁρητὸν δὲ τὸ δὶξ ὑπὸ τῶν AB, BΓ· ὁάλογον ἄρα τὸ ἀπὸ τῆς AΓ. ὁάλογος ἄρα ἡ AΓ, καλείσθω δὲ ὁρητὸν καὶ μέσον δυναμένη. ὅπερ ἔδει δεῖχαι.

For let the two straight-lines, AB and BC, incommensurable in square, (and) fulfilling the prescribed (conditions), be laid down together [Prop. 10.34]. I say that AC is irrational.

For since the sum of the (squares) on AB and BC is medial, and twice the (rectangle contained) by AB and BC (is) rational, the sum of the (squares) on AB and BC is thus incommensurable with twice the (rectangle contained) by AB and BC. Hence, the (square) on AC is also incommensurable with twice the (rectangle contained) by AB and BC [Prop. 10.16]. And twice the (rectangle contained) by AB and BC (is) rational. The (square) on AC (is) thus irrational. Thus, AC (is) irrational [Def. 10.4]—let it be called the square-root of a rational plus a medial (area).† (Which is) the very thing it was required to show.

† Thus, the square-root of a rational plus a medial (area) has a length expressible as $\sqrt{[(1 + k^2)^{1/2} + k]/[2\,(1 + k^2)]} + \sqrt{[(1 + k^2)^{1/2} - k]/[2\,(1 + k^2)]}$. This and the corresponding irrational with a minus sign, whose length is expressible as $\sqrt{[(1 + k^2)^{1/2} + k]/[2\,(1 + k^2)]} - \sqrt{[(1 + k^2)^{1/2} - k]/[2\,(1 + k^2)]}$ (see Prop. 10.77), are the positive roots of the quartic $x^4 - (2/\sqrt{1 + k^2})\,x^2 + k^2/(1 + k^2)^2 = 0$.

41.

Ἐὰν δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιοῦσαι τό τε συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπὸ αὐτῶν μέσον καὶ ὀέτι ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων, ἡ ὅλη εὐθεῖαὁάλογόξ ἐστιν, καλείσθω δὲ δύο μέσα δυναμένη.

Proposition 41

If two straight-lines (which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the squares on them, are added together then the whole straight-line is irrational—let it be called the square-root of (the sum of) two medial (areas).

Συγκείσθωσαν γὰρ δύο εὐθεῖαι δυνάμει ἀσύμμετροι αἱ AB, BΓ ποιοῦσαι τὰ προκείμενα· λέγω, ὅτι ἡ AΓὀάλογόξ ἐστιν.

Ἐκκείσθω ὁρητὴ ἡ ΔE, καὶ παραβεβλήσθω παρὰ τὴν ΔE τοῖς μὲν ἀπὸ τῶν AB, BΓ ἴσον τὸ ΔZ, τῷ δὲ δἰξ ὑπὸ τῶν AB, BΓ ἴσον τὸ HΘ· ὅλον ἄρα τὸ ΔΘ ἴσον ἐστὶ τῷ ἀπὸ τῆς AΓ τετραγώνῳ. καὶ ἐπεὶ μέσον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB, BΓ, καί ἐστιν ἴσον τῷ ΔZ, μέσον ἄρα ἐστὶ καὶ τὸ ΔZ. καὶ παρὰ ὁρητὴν τὴν ΔE παράκειται· ὁρητὴ ἄρα ἐστὶν ἡ ΔH καὶ ἀσύμμετρος τῇ ΔE μήκει. διὰ τὰ αὐτὰ δὴ καὶ ἡ HK ὁρητή ἐστι καὶ ἀσύμμετρος τῇ HZ, τουτέστι τῇ ΔE, μήκει. καὶ ἐπεὶ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν AB, BΓ τῷ δἰξ ὑπὸ τῶν AB, BΓ, ἀσύμμετρόν ἐστι τὸ ΔZ τῷ HΘ· ὥστε καὶ ἡ ΔH τῇ HK ἀσύμμετρόξ ἐστιν. καὶ εἰσι ὁρηταί· αἱ ΔH,

For let the two straight-lines, AB and BC, incommensurable in square, (and) fulfilling the prescribed (conditions), be laid down together [Prop. 10.35]. I say that AC is irrational.

Let the rational (straight-line) DE be laid out, and let (the rectangle) DF, equal to (the sum of) the (squares) on AB and BC, and (the rectangle) GH, equal to twice the (rectangle contained) by AB and BC, be applied to DE. Thus, the whole of DH is equal to the square on AC [Prop. 2.4]. And since the sum of the (squares) on AB and BC is medial, and is equal to DF, DF is thus also medial. And it is applied to the rational (straight-line) DE. Thus, DG is rational, and incommensurable in length with DE [Prop. 10.22]. So, for the same (reasons), GK is also rational, and incommensurable in length with GF—that is to say, DE. And since (the sum of) the

ΗΚ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ὁάλογος ἄρα ἐστὶν ἡ ΔΚ ἡ καλουμένη ἐκ δύο ὀνομάτων. ὁρητὴ δὲ ἡ ΔΕ· ὁάλογον ἄρα ἐστὶ τὸ ΔΘ καὶ ἡ δυναμένη αὐτὸ ὁάλογός ἐστιν. δύναται δὲ τὸ ΘΔ ἡ ΑΓ· ὁάλογος ἄρα ἐστὶν ἡ ΑΓ, καλείσθω δὲ δύο μέσα δυναμένη. ὅπερ ἔδει δεῖχαι.

(squares) on AB and BC is incommensurable with twice the (rectangle contained) by AB and BC, DF is incommensurable with GH. Hence, DG is also incommensurable (in length) with GK [Props. 6.1, 10.11]. And they are rational. Thus, DG and GK are rational (straight-lines which are) commensurable in square only. Thus, DK is irrational, and that (straight-line which is) called binomial [Prop. 10.36]. And DE (is) rational. Thus, DH is irrational, and its square-root is irrational [Def. 10.4]. And AC (is) the square-root of HD. Thus, AC is irrational—let it be called the square-root of (the sum of) two medial (areas).[†] (Which is) the very thing it was required to show.

[†] Thus, the square-root of (the sum of) two medial (areas) has a length expressible as $k'^{1/4}\left(\sqrt{[1+k/(1+k^2)^{1/2}]/2}+\sqrt{[1-k/(1+k^2)^{1/2}]/2}\right)$. This and the corresponding irrational with a minus sign, whose length is expressible as $k'^{1/4}\left(\sqrt{[1+k/(1+k^2)^{1/2}]/2}-\sqrt{[1-k/(1+k^2)^{1/2}]/2}\right)$ (see Prop. 10.78), are the positive roots of the quartic $x^4-2\,k'^{1/2}\,x^2+k'\,k^2/(1+k^2)=0$.

Λῆμμα.

Ὅτι δὲ αἱ εἰρημέναιὁάλογοι μοναθῶξ διαιροῦνται εἰξ τὰξ εὐθείας, ἐξ ὧν σύγκεινται ποιουσῶν τὰ προκείμενα εὁίδη, δείχομεν ἤδη προεκθέμενοι λημμάτιον τοιοῦτον·

Lemma

We will now demonstrate that the aforementioned irrational (straight-lines) are uniquely divided into the straight-lines of which they are the sum, and which produce the prescribed types, (after) setting forth the following lemma.

Ἐκκείσθω εὐθεῖα ἡ ΑΒ καὶ τετμήσθω ἡ ὅλη εἰξὁάνισα καθὸ ἑκάτερον τῶν Γ, Δ, ὑποκείσθω δὲ μείζων ἡ ΑΓ τῆς ΔΒ· λέγω, ὅτι τὰ ἀπὸ τῶν ΑΓ, ΓΒ μείζονά ἐστι τῶν ἀπὸ τῶν ΑΔ, ΔΒ.

Τετμήσθω γὰρ ἡ ΑΒ δίθα κατὰ τὸ Ε. καὶ ἐπεὶ μείζων ἐστὶν ἡ ΑΓ τῆς ΔΒ, κοινὴ ἀφῃρήσθω ἡ ΔΓ· λοιπὴ ἄρα ἡ ΑΔ λοιπῆς τῆς ΓΒ μείζων ἐστίν. ὁίση δὲ ἡ ΑΕ τῇ ΕΒ· ἐλάττων ἄρα ἡ ΔΕ τῆς ΕΓ· τὰ Γ, Δ ἄρα σημεῖα οὐκ ἴσον ἀπέθουσι τῆς διθοτομίας. καὶ ἐπεὶ τὸ ὑπὸ τῶν ΑΓ, ΓΒ μετὰ τοῦ ἀπὸ τῆς ΕΓ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΒ, ἀλλὰ μὴν καὶ τὸ ὑπὸ τῶν ΑΔ, ΔΒ μετὰ τοῦ ἀπὸ ΔΕ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΒ, τὸ ἄρα ὑπὸ τῶν ΑΓ, ΓΒ μετὰ τοῦ ἀπὸ τῆς ΕΓ ἴσον ἐστὶ τῷ ὑπὸ τῶν ΑΔ, ΔΒ μετὰ τοῦ ἀπὸ τῆς ΔΕ· ὧν τὸ ἀπὸ τῆς ΔΕὁέλασσόν ἐστι τοῦ ἀπὸ τῆς ΕΓ· καὶ λοιπὸν ἄρα τὸ ὑπὸ τῶν ΑΓ, ΓΒὁέλασσόν ἐστι τοῦ ὑπὸ τῶν ΑΔ, ΔΒ. ὥστε καὶ τὸ δὶξ ὑπὸ τῶν ΑΓ, ΓΒὁέλασσόν ἐστι τοῦ δὶξ ὑπὸ τῶν ΑΔ, ΔΒ. καὶ λοιπὸν ἄρα τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓΒ μεῖζόν ἐστι τοῦ συγκειμένου ἐκ τῶν ἀπὸ τῶν ΑΔ, ΔΒ. ὅπερ ἔδει δεῖχαι.

Let the straight-line AB be laid out, and let the whole (straight-line) be cut into unequal parts at each of the (points) C and D. And let AC be assumed (to be) greater than DB. I say that (the sum of) the (squares) on AC and CB is greater than (the sum of) the (squares) on AD and DB.

For let AB be cut in half at E. And since AC is greater than DB, let DC be subtracted from both. Thus, the remainder AD is greater than the remainder CB. And AE (is) equal to EB. Thus, DE (is) less than EC. Thus, points C and D are not equally far from the point of bisection. And since the (rectangle contained) by AC and CB, plus the (square) on EC, is equal to the (square) on EB [Prop. 2.5], but, moreover, the (rectangle contained) by AD and DB, plus the (square) on DE, is also equal to the (square) on EB [Prop. 2.5], the (rectangle contained) by AC and CB, plus the (square) on EC, is thus equal to the (rectangle contained) by AD and DB, plus the (square) on DE. And, of these, the (square) on DE is less than the (square) on EC. And, thus, the remaining (rectangle contained) by AC and CB is less than the (rectangle contained) by AD and DB. And, hence, twice the (rectangle contained) by AC and CB is less than twice the (rectangle contained) by AD and DB. And thus the remaining sum of the (squares) on AC and CB is greater than the sum of the (squares) on AD and DB.[†] (Which is) the very thing it was required to show.

† Since, $AC^2 + CB^2 + 2\,AC\,CB = AD^2 + DB^2 + 2\,AD\,DB = AB^2$.

42.

Ἡ ἐκ δύο ὀνομάτων κατὰ ὁὲν μόνον σημεῖον διαιρεῖται εἰξ τὰ ὀνόματα.

Ἔστω ἐκ δύο ὀνομάτων ἡ AB διηρημένη εἰξ τὰ ὀνόματα κατὰ τὸ Γ· αἱ AΓ, ΓB ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι. λέγω, ὅτι ἡ AB κατὸ ὀάλλο σημεῖον οὐ διαιρεῖται εἰξ δύο ὁρητὰξ δυνάμει μόνον συμμέτρους.

Εἰ γὰρ δυνατόν, διηρήσθω καὶ κατὰ τὸ Δ, ὁώστε καὶ τὰξ AΔ, ΔB ὁρητὰξ εὸῖναι δυνάμει μόνον συμμέτρους. φανερὸν δή, ὅτι ἡ AΓ τῇ ΔB οὐκὸέστιν ἡ αὐτή. εἰ γὰρ δυνατόν, ὀέστω. ὀέσται δὴ καὶ ἡ AΔ τῇ ΓB ἡ αὐτή· καὶ ὀέσται ὡξ ἡ AΓ πρὸς τὴν ΓB, οὁύτως ἡ BΔ πρὸς τὴν ΔA, καὶ ὀέσται ἡ AB κατὰ τὸ αὐτὸ τῇ κατὰ τὸ Γ διαιρέσει διαιρεθεῖσα καὶ κατὰ τὸ Δ· ὅπερ οὐθ ὑπόκειται. οὐκ ἄρα ἡ AΓ τῇ ΔB ἐστιν ἡ αὐτή. διὰ δὴ τοῦτο καὶ τὰ Γ, Δ σημεῖα οὐκ ἴσον ἀπέθουσι τῆς διθοτομίας. ὁῷ ἄρα διαφέρει τὰ ἀπὸ τῶν AΓ, ΓB τῶν ἀπὸ τῶν AΔ, ΔB, τούτῳ διαφέρει καὶ τὸ δὶξ ὑπὸ τῶν AΔ, ΔB τοῦ δὶξ ὑπὸ τῶν AΓ, ΓB διὰ τὸ καὶ τὰ ἀπὸ τῶν AΓ, ΓB μετὰ τοῦ δὶξ ὑπὸ τῶν AΓ, ΓB καὶ τὰ ἀπὸ τῶν AΔ, ΔB μετὰ τοῦ δὶξ ὑπὸ τῶν AΔ, ΔB ἴσα εὸῖναι τῷ ἀπὸ τῆς AB. ἀλλὰ τὰ ἀπὸ τῶν AΓ, ΓB τῶν ἀπὸ τῶν AΔ, ΔB διαφέρει ὁρητῷ· ὁρητὰ γὰρ ἀμφότερα· καὶ τὸ δὶξ ἄρα ὑπὸ τῶν AΔ, ΔB τοῦ δὶξ ὑπὸ τῶν AΓ, ΓB διαφέρει ὁρητῷ μέσα ὄντα· ὅπερὸάτοπον· μέσον γὰρ μέσου οὐθ ὑπερέθει ὁρητῷ.

Οὐθ ἄρα ἡ ἐκ δύο ὀνομάτων κατὸ ὀάλλο καὶ ὀάλλο σημεῖον διαιρεῖται· καθὸ ὁὲν ἄρα μόνον· ὅπερ ἔδει δεῖχαι.

Proposition 42

A binomial (straight-line) can be divided into its (component) terms at one point only.†

Let AB be a binomial (straight-line) which has been divided into its (component) terms at C. AC and CB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.36]. I say that AB cannot be divided at another point into two rational (straight-lines which are) commensurable in square only.

For, if possible, let it also be divided at D, such that AD and DB are also rational (straight-lines which are) commensurable in square only. So, (it is) clear that AC is not the same as DB. For, if possible, let it be (the same). So, AD will also be the same as CB. And as AC will be to CB, so BD (will be) to DA. And AB will (thus) also be divided at D in the same (manner) as the division at C. The very opposite was assumed. Thus, AC is not the same as DB. So, on account of this, points C and D are not equally far from the point of bisection. Thus, by whatever (amount the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB, twice the (rectangle contained) by AD and DB also differs from twice the (rectangle contained) by AC and CB by this (same amount)—on account of both (the sum of) the (squares) on AC and CB, plus twice the (rectangle contained) by AC and CB, and (the sum of) the (squares) on AD and DB, plus twice the (rectangle contained) by AD and DB, being equal to the (square) on AB [Prop. 2.4]. But, (the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB by a rational (area). For (they are) both rational (areas). Thus, twice the (rectangle contained) by AD and DB also differs from twice the (rectangle contained) by AC and CB by a rational (area, despite both) being medial (areas) [Prop. 10.21]. The very thing is absurd. For a medial (area) cannot exceed a medial (area) by a rational (area) [Prop. 10.26].

Thus, a binomial (straight-line) cannot be divided (into its component terms) at different points. Thus, (it can be so divided) at one point only. (Which is) the very thing it was required to show.

† In other words, $k + k'^{1/2} = k'' + k'''^{1/2}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$. Likewise, $k^{1/2} + k'^{1/2} = k''^{1/2} + k'''^{1/2}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$ (or, equivalently, $k'' = k'$ and $k''' = k$).

43.

Proposition 43

Ἡ ἐκ δύο μέσων πρώτη καθ' ὁὲν μόνον σημεῖον διαιρεῖται.

A first bimedial (straight-line) can be divided (into its component terms) at one point only.†

Ἔστω ἐκ δύο μέσων πρώτη ἡ AB διῃρημένη κατὰ τὸ $Γ$, ὁώστε τὰξ $AΓ$, $ΓB$ μέσας εὁῖναι δυνάμει μόνον συμμέτρους ὁρητὸν περιεχούσας· λέγω, ὅτι ἡ AB κατὰ ὁάλλο σημεῖον οὐ διαιρεῖται.

Εἰ γὰρ δυνατόν διῃρήσθω καὶ κατὰ τὸ $Δ$, ὁώστε καὶ τὰξ $AΔ$, $ΔB$ μέσας εὁῖναι δυνάμει μόνον συμμέτρους ὁρητὸν περιεχούσας. ἐπεὶ οὁῦν, ὁῷ διαφέρει τὸ δὶξ ὑπὸ τῶν $AΔ$, $ΔB$ τοῦ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$, τούτῳ διαφέρει τὰ ἀπὸ τῶν $AΓ$, $ΓB$ τῶν ἀπὸ τῶν $AΔ$, $ΔB$, ὁρητῷ δὲ διαφέρει τὸ δὶξ ὑπὸ τῶν $AΔ$, $ΔB$ τοῦ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$· ὁρητὰ γὰρ ἀμφότερα· ὁρητῷ ἄρα διαφέρει καὶ τὰ ἀπὸ τῶν $AΓ$, $ΓB$ τῶν ἀπὸ τῶν $AΔ$, $ΔB$ μέσα ὄντα· ὅπερ ὁάτοπον.

Let AB be a first bimedial (straight-line) which has been divided at C, such that AC and CB are medial (straight-lines), commensurable in square only, (and) containing a rational (area) [Prop. 10.37]. I say that AB cannot be (so) divided at another point.

For, if possible, let it also be divided at D, such that AD and DB are also medial (straight-lines), commensurable in square only, (and) containing a rational (area). Since, therefore, by whatever (amount) twice the (rectangle contained) by AD and DB differs from twice the (rectangle contained) by AC and CB, (the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB by this (same amount) [Prop. 10.41 lem.]. And twice the (rectangle contained) by AD and DB differs from twice the (rectangle contained) by AC and CB by a rational (area). For (they are) both rational (areas). (The sum of) the (squares) on AC and CB thus differs from (the sum of) the (squares) on AD and DB by a rational (area, despite both) being medial (areas). The very thing is absurd [Prop. 10.26].

Οὐκ ἄρα ἡ ἐκ δύο μέσων πρώτη κατὸ ὁάλλο καὶ ὁάλλο σημεῖον διαιρεῖται εἰξ τὰ ὀνόματα· καθ' ὁὲν ἄρα μόνον· ὅπερ ἔδει δεῖχαι.

Thus, a first bimedial (straight-line) cannot be divided into its (component) terms at different points. Thus, (it can be so divided) at one point only. (Which is) the very thing it was required to show.

† In other words, $k^{1/4} + k^{3/4} = k'^{1/4} + k'^{3/4}$ has only one solution: i.e., $k' = k$.

44.

Proposition 44

Ἡ ἐκ δύο μέσων δευτέρα καθ' ὁὲν μόνον σημεῖον διαιρεῖται.

A second bimedial (straight-line) can be divided (into its component terms) at one point only.†

Ἔστω ἐκ δύο μέσων δευτέρα ἡ AB διῃρημένη κατὰ τὸ $Γ$, ὁώστε τὰξ $AΓ$, $ΓB$ μέσας εὁῖναι δυνάμει μόνον συμμέτρους μέσον περιεχούσας· φανερὸν ὁή, ὅτι τὸ $Γ$ οὐκὁέστι κατὰ τῆς διθοτομίας, ὅτι οὐκ εἰσὶ μήκει σύμμετροι. λέγω, ὅτι ἡ AB κατὸ ὁάλλο σημεῖον οὐ διαιρεῖται.

Let AB be a second bimedial (straight-line) which has been divided at C, so that AC and BC are medial (straight-lines), commensurable in square only, (and) containing a medial (area) [Prop. 10.38]. So, (it is) clear that C is not (located) at the point of bisection, since (AC and BC) are not commensurable in length. I say that AB cannot be (so) divided at another point.

Εἰ γὰρ δυνατόν, διῃρήσθω καὶ κατὰ τὸ Δ, ὥστε τὴν ΑΓ τῇ ΔΒ μὴ εἶναι τὴν αὐτήν, ἀλλὰ μείζονα καθὸ ὑπόθεσιν τὴν ΑΓ· δῆλον δή, ὅτι καὶ τὰ ἀπὸ τῶν ΑΔ, ΔΒ, ὧν ἐπάνω ἐδείχαμεν, ἐλάσσονα τῶν ἀπὸ τῶν ΑΓ, ΓΒ· καὶ τὰ ξ ΑΔ, ΔΒ μέσας εἶναι δυνάμει μόνον συμμέτρους μέσον περιεχούσας. καὶ ἐκκείσθω ὁρητὴ ἡ ΕΖ, καὶ τῷ μὲν ἀπὸ τῆς ΑΒ ἴσον παρὰ τὴν ΕΖ παραλληλόγραμμον ὀρθογώνιον παραβεβλήσθω τὸ ΕΚ, τοῖς δὲ ἀπὸ τῶν ΑΓ, ΓΒ ἴσον ἀφῃρήσθω τὸ ΕΗ· λοιπὸν ἄρα τὸ ΘΚ ἴσον ἐστὶ τῷ δὶς ὑπὸ τῶν ΑΓ, ΓΒ. πάλιν δὴ τοῖς ἀπὸ τῶν ΑΔ, ΔΒ, ὅπερ ἐλάσσονα ἐδείθθη τῶν ἀπὸ τῶν ΑΓ, ΓΒ, ἴσον ἀφῃρήσθω τὸ ΕΛ· καὶ λοιπὸν ἄρα τὸ ΜΚ ἴσον τῷ δὶς ὑπὸ τῶν

For, if possible, let it also have been (so) divided at D, so that AC is not the same as DB, but AC (is), by hypothesis, greater. So, (it is) clear that (the sum of) the (squares) on AD and DB is also less than (the sum of) the (squares) on AC and CB, as we showed above [Prop. 10.41 lem.]. And AD and DB are medial (straight-lines), commensurable in square only, (and) containing a medial (area). And let the rational (straight-line) EF be laid down. And let the rectangular parallelogram EK, equal to the (square) on AB, be applied to EF. And let EG, equal to (the sum of) the (squares) on AC and CB, be cut off (from EK). Thus, the remainder, HK, is equal to twice the (rectangle contained) by AC and CB [Prop. 2.4].

ΑΔ, ΔΒ. καὶ ἐπεὶ μέσα ἐστὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ, μέσον ἄρα [καὶ] τὸ ΕΗ. καὶ παρὰ ὁρητὴν τὴν ΕΖ παράκειται· ὁρητὴ ἄρα ἐστὶν ἡ ΕΘ καὶ ἀσύμμετρος τῇ ΕΖ μήκει. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΘΝ ὁρητή ἐστι καὶ ἀσύμμετρος τῇ ΕΖ μήκει. καὶ ἐπεὶ αἱ ΑΓ, ΓΒ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι, ἀσύμμετρος ἄρα ἐστὶν ἡ ΑΓ τῇ ΓΒ μήκει. ὡς δὲ ἡ ΑΓ πρὸς τὴν ΓΒ, οὕτως τὸ ἀπὸ τῆς ΑΓ πρὸς τὸ ὑπὸ τῶν ΑΓ, ΓΒ· ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΓ τῷ ὑπὸ τῶν ΑΓ, ΓΒ. ἀλλὰ τῷ μὲν ἀπὸ τῆς ΑΓ σύμμετρά ἐστι τὰ ἀπὸ τῶν ΑΓ, ΓΒ· δυνάμει γάρ εἰσι σύμμετροι αἱ ΑΓ, ΓΒ. τῷ δὲ ὑπὸ τῶν ΑΓ, ΓΒ σύμμετρόν ἐστι τὸ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ. καὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ ἄρα ἀσύμμετρά ἐστι τῷ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ. ἀλλὰ τοῖς μὲν ἀπὸ τῶν ΑΓ, ΓΒ ἴσον ἐστὶ τὸ ΕΗ, τῷ δὲ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ ἴσον τὸ ΘΚ· ἀσύμμετρον ἄρα ἐστὶ τὸ ΕΗ τῷ ΘΚ· ὥστε καὶ ἡ ΕΘ τῇ ΘΝ ἀσύμμετρός ἐστι μήκει. καί εἰσι ὁρηταί· αἱ ΕΘ, ΘΝ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι. ἐὰν δὲ δύο ὁρηταὶ δυνάμει μόνον σύμμετροι συντεθῶσιν, ἡ ὅληὀάλογός ἐστιν ἡ καλουμένη ἐκ δύο ὀνομάτων· ἡ ΕΝ ἄρα ἐκ δύο ὀνομάτων ἐστὶ διῃρημένη κατὰ τὸ Θ. κατὰ τὰ αὐτὰ δὴ δειθθήσονται καὶ αἱ ΕΜ, ΜΝ ὁρηταὶ δυνάμει μόνον σύμμετροι· καὶ ὀέσται ἡ ΕΝ ἐκ δύο ὀνομάτων κατὸ ὀάλλο καὶ ὀάλλο διῃρημένη τό τε Θ καὶ τὸ Μ, καὶ οὐκὀέστιν ἡ ΕΘ τῇ ΜΝ ἡ αὐτή, ὅτι τὰ ἀπὸ τῶν ΑΓ, ΓΒ μείζονά ἐστι τῶν ἀπὸ τῶν ΑΔ, ΔΒ. ἀλλὰ τὰ ἀπὸ τῶν ΑΔ, ΔΒ μείζονά ἐστι τοῦ δὶξ ὑπὸ ΑΔ, ΔΒ· πολλῷ ἄρα καὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ, τουτέστι τὸ ΕΗ, μείζόν ἐστι τοῦ δὶξ ὑπὸ τῶν ΑΔ, ΔΒ, τουτέστι τοῦ ΜΚ· ὥστε καὶ ἡ ΕΘ τῆς ΜΝ μείζων ἐστίν. ἡ ἄρα ΕΘ τῇ ΜΝ οὐκὀέστιν ἡ αὐτή· ὅπερ ἔδει δεῖχαι.

So, again, let EL, equal to (the sum of) the (squares) on AD and DB—which was shown (to be) less than (the sum of) the (squares) on AC and CB—be cut off (from EK). And, thus, the remainder, MK, (is) equal to twice the (rectangle contained) by AD and DB. And since (the sum of) the (squares) on AC and CB is medial, EG (is) thus [also] medial. And it is applied to the rational (straight-line) EF. Thus, EH is rational, and incommensurable in length with EF [Prop. 10.22]. So, for the same (reasons), HN is also rational, and incommensurable in length with EF. And since AC and CB are medial (straight-lines which are) commensurable in square only, AC is thus incommensurable in length with CB. And as AC (is) to CB, so the (square) on AC (is) to the (rectangle contained) by AC and CB [Prop. 10.21 lem.]. Thus, the (square) on AC is incommensurable with the (rectangle contained) by AC and CB [Prop. 10.11]. But, (the sum of) the (squares) on AC and CB is commensurable with the (square) on AC. For, AC and CB are commensurable in square [Prop. 10.15]. And twice the (rectangle contained) by AC and CB is commensurable with the (rectangle contained) by AC and CB [Prop. 10.6]. And thus (the sum of) the squares on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB [Prop. 10.13]. But, EG is equal to (the sum of) the (squares) on AC and CB, and HK equal to twice the (rectangle contained) by AC and CB. Thus, EG is incommensurable with HK. Hence, EH is also incommensurable in length with HN [Props. 6.1, 10.11]. And (they are) rational (straight-lines). Thus, EH and HN are rational (straight-lines which are) commensurable in square only. And if two rational (straight-lines which are) commensurable in square only are added together then the whole (straight-line) is that irrational called binomial [Prop. 10.36]. Thus, EN is a binomial (straight-line) which has been divided (into its component terms) at H. So, according to the same (reasoning), EM and MN can be shown (to be) rational (straight-lines which are) commensurable in square only. And EN will (thus) be a binomial (straight-line) which has been divided (into its component terms) at the different (points) H and M (which is absurd [Prop. 10.42]). And EH is not the same as MN, since (the sum of) the (squares) on AC and CB is greater than (the sum of) the (squares) on AD and DB. But, (the sum of) the (squares) on AD and DB is greater than twice the (rectangle contained) by AD and DB [Prop. 10.59 lem.]. Thus, (the sum of) the (squares) on AC and CB—that is to say, EG—is also much greater than twice the (rectangle contained) by AD and DB—that is to say, MK. Hence, EH is also greater than MN [Prop. 6.1]. Thus, EH is not the same as MN. (Which is) the very thing it was required to show.

† In other words, $k^{1/4} + k'^{1/2}/k^{1/4} = k''^{1/4} + k'''^{1/2}/k''^{1/4}$ has only one solution: i.e., $k'' = k$ and $k''' = k'$.

45.

Ἡ μείζων κατὰ τὸ αὐτὸ μόνον σημεῖον διαιρεῖται.

Ἔστω μείζων ἡ AB διῃρημένη κατὰ τὸ $Γ$, ὥστε τὰξ $AΓ$, $ΓB$ δυνάμει ἀσυμμέτρους εἶναι ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν $AΓ$, $ΓB$ τετραγώνων ὁρητόν, τὸ δὸ ὑπὸ τῶν $AΓ$, $ΓB$ μέσον· λέγω, ὅτι ἡ AB κατὸ ὀάλλο σημεῖον οὐ διαιρεῖται.

Εἰ γὰρ δυνατόν, διῃρήσθω καὶ κατὰ τὸ $Δ$, ὥστε καὶ τὰξ $AΔ$, $ΔB$ δυνάμει ἀσυμμέτρους εἶναι ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν $AΔ$, $ΔB$ ὁρητόν, τὸ δὸ ὑπὸ αὐτῶν μέσον. καὶ ἐπεί, ὅῳ διαφέρει τὰ ἀπὸ τῶν $AΓ$, $ΓB$ τῶν ἀπὸ τῶν $AΔ$, $ΔB$, τούτῳ διαφέρει καὶ τὸ δἰξ ὑπὸ τῶν $AΔ$, $ΔB$ τοῦ δἰξ ὑπὸ τῶν $AΓ$, $ΓB$, ἀλλὰ τὰ ἀπὸ τῶν $AΓ$, $ΓB$ τῶν ἀπὸ τῶν $AΔ$, $ΔB$ ὑπερέθει ὁρητῷ· ὁρητὰ γὰρ ἀμφότερα· καὶ τὸ δἰξ ὑπὸ τῶν $AΔ$, $ΔB$ ἄρα τοῦ δἰξ ὑπὸ τῶν $AΓ$, $ΓB$ ὑπερέθει ὁρητῷ μέσα ὄντα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ μείζων κατὸ ὀάλλο καὶ ὀάλλο σημεῖον διαιρεῖται· κατὰ τὸ αὐτὸ ἄρα μόνον διαιρεῖται· ὅπερ ἔδει δεῖχαι.

Proposition 45

A major (straight-line) can only be divided (into its component terms) at the same point.†

Let AB be a major (straight-line) which has been divided at C, so that AC and CB are incommensurable in square, making the sum of the squares on AC and CB rational, and the (rectangle contained) by AC and CD medial [Prop. 10.39]. I say that AB cannot be (so) divided at another point.

For, if possible, let it also be divided at D, such that AD and DB are also incommensurable in square, making the sum of the (squares) on AD and DB rational, and the (rectangle contained) by them medial. And since, by whatever (amount the sum of) the (squares) on AC and CB differs from (the sum of) the (squares) on AD and DB, twice the (rectangle contained) by AD and DB also differs from twice the (rectangle contained) by AC and CB by this (same amount). But, (the sum of) the (squares) on AC and CB exceeds (the sum of) the (squares) on AD and DB by a rational (area). For (they are) both rational (areas). Thus, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by a rational (area), (despite both) being medial (areas). The very thing is impossible [Prop. 10.26]. Thus, a major (straight-line) cannot be divided (into its component terms) at different points. Thus, it can only be (so) divided at the same (point). (Which is) the very thing it was required to show.

† In other words, $\sqrt{[1 + k/(1 + k^2)^{1/2}]/2} + \sqrt{[1 - k/(1 + k^2)^{1/2}]/2} = \sqrt{[1 + k'/(1 + k'^2)^{1/2}]/2} + \sqrt{[1 - k'/(1 + k'^2)^{1/2}]/2}$ has only one solution: i.e., $k' = k$.

46.

Ἡ ὁρητὸν καὶ μέσον δυναμένη καθὸ ὁὲν μόνον σημεῖον διαιρεῖται.

Ἔστω ὁρητὸν καὶ μέσον δυναμένη ἡ AB διῃρημένη κατὰ τὸ $Γ$, ὥστε τὰξ $AΓ$, $ΓB$ δυνάμει ἀσυμμέτρους εἶναι ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν $AΓ$, $ΓB$ μέσον, τὸ δὲ δἰξ ὑπὸ τῶν $AΓ$, $ΓB$ ὁρητόν· λέγω, ὅτι ἡ AB κατὸ ὀάλλο σημεῖον οὐ διαιρεῖται.

Εἰ γὰρ δυνατόν, διῃρήσθω καὶ κατὰ τὸ $Δ$, ὥστε καὶ τὰξ $AΔ$, $ΔB$ δυνάμει ἀσυμμέτρους εἶναι ποιούσας τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν $AΔ$, $ΔB$ μέσον, τὸ δὲ δἰξ ὑπὸ τῶν $AΔ$, $ΔB$ ὁρητόν. ἐπεὶ οὐῦν, ὅῳ διαφέρει τὸ δἰξ ὑπὸ τῶν

Proposition 46

The square-root of a rational plus a medial (area) can be divided (into its component terms) at one point only.†

Let AB be the square-root of a rational plus a medial (area) which has been divided at C, so that AC and CB are incommensurable in square, making the sum of the (squares) on AC and CB medial, and twice the (rectangle contained) by AC and CB rational [Prop. 10.40]. I say that AB cannot be (so) divided at another point.

For, if possible, let it also be divided at D, so that AD and DB are also incommensurable in square, making the sum of the (squares) on AD and DB medial, and twice the (rectangle

ΑΓ, ΓΒ τοῦ δὶξ ὑπὸ τῶν ΑΔ, ΔΒ, τούτῳ διαφέρει καὶ τὰ ἀπὸ τῶν ΑΔ, ΔΒ τῶν ἀπὸ τῶν ΑΓ, ΓΒ, τὸ δὲ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ τοῦ δὶξ ὑπὸ τῶν ΑΔ, ΔΒ ὑπερέθει ὁρητῷ, καὶ τὰ ἀπὸ τῶν ΑΔ, ΔΒ ἄρα τῶν ἀπὸ τῶν ΑΓ, ΓΒ ὑπερέθει ὁρητῷ μέσα ὄντα· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ ὁρητὸν καὶ μέσον δυναμένη κατὰ ὀάλλο καὶ ὀάλλο σημεῖον διαιρεῖται. κατὰ ὁὲν ἄρα σημεῖον διαιρεῖται· ὅπερ ἔδει δεῖχαι.

contained) by AD and DB rational. Therefore, since by whatever (amount) twice the (rectangle contained) by AC and CB differs from twice the (rectangle contained) by AD and DB, (the sum of) the (squares) on AD and DB also differs from (the sum of) the (squares) on AC and CB by this (same amount). And twice the (rectangle contained) by AC and CB exceeds twice the (rectangle contained) by AD and DB by a rational (area). (The sum of) the (squares) on AD and DB thus also exceeds (the sum of) the (squares) on AC and CB by a rational (area), (despite both) being medial (areas). The very thing is impossible [Prop. 10.26]. Thus, the square-root of a rational plus a medial (area) cannot be divided (into its component terms) at different points. Thus, it can be (so) divided at one point (only). (Which is) the very thing it was required to show.

† In other words, $\sqrt{[(1+k^2)^{1/2}+k]/[2\,(1+k^2)]} + \sqrt{[(1+k^2)^{1/2}-k]/[2\,(1+k^2)]} = \sqrt{[(1+k'^2)^{1/2}+k']/[2\,(1+k'^2)]} + \sqrt{[(1+k'^2)^{1/2}-k']/[2\,(1+k'^2)]}$ has only one solution: i.e., $k' = k$.

47.

Ἡ δύο μέσα δυναμένη καθὸ ὁὲν μόνον σημεῖον διαιρεῖται.

Proposition 47

The square-root of (the sum of) two medial (areas) can be divided (into its component terms) at one point only.†

Ἔστω [δύο μέσα δυναμένη] ἡ ΑΒ διῃρημένη κατὰ τὸ Γ,

Let AB be [the square-root of (the sum of) two medial (ar-

ὁώστε τὰξ ΑΓ, ΓΒ δυνάμει ἀσυμμέτρους εὐῖναι ποιούσας τό τε συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓΒ μέσον καὶ τὸ ὑπὸ τῶν ΑΓ, ΓΒ μέσον καὶ ὀέτι ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπὸ αὐτῶν. λέγω, ὅτι ἡ ΑΒ κατὸ ὀάλλο σημεῖον οὐ διαιρεῖται ποιοῦσα τὰ προκείμενα.

Εἰ γὰρ δυνατόν, διῃρήσθω κατὰ τὸ Δ, ὁώστε πάλιν δηλονότι τὴν ΑΓ τῇ ΔΒ μὴ εὐῖναι τὴν αὐτήν, ἀλλὰ μείζονα καθὸ ὑπόθεσιν τὴν ΑΓ, καὶ ἐκκείσθω ὁρητὴ ἡ ΕΖ, καὶ παραβεβλήσθω παρὰ τὴν ΕΖ τοῖς μὲν ἀπὸ τῶν ΑΓ, ΓΒ ἴσον τὸ ΕΗ, τῷ δὲ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ ἴσον τὸ ΘΚ· ὅλον ἄρα τὸ ΕΚ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΒ τετραγώνῳ. πάλιν δὴ παραβεβλήσθω παρὰ τὴν ΕΖ τοῖς ἀπὸ τῶν ΑΔ, ΔΒ ἴσον τὸ ΕΛ· λοιπὸν ἄρα τὸ δὶξ ὑπὸ τῶν ΑΔ, ΔΒ λοιπῷ τῷ ΜΚ ἴσον ἐστίν. καὶ ἐπεὶ μέσον ὑπόκειται τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓΒ, μέσον ἄρα ἐστὶ καὶ τὸ ΕΗ. καὶ παρὰ ὁρητὴν τὴν ΕΖ παράκειται· ὁρητὴ ἄρα ἐστὶν ἡ ΘΕ καὶ ἀσύμμετρος τῇ ΕΖ μήκει. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΘΝ ὁρητή ἐστι καὶ ἀσύμμετρος τῇ ΕΖ μήκει. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓΒ τῷ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ, καὶ τὸ ΕΗ ἄρα τῷ ΗΝ ἀσύμμετρόν ἐστιν· ὁώστε καὶ ἡ ΕΘ τῇ ΘΝ ἀσύμμετρόξ ἐστιν. καί εἰσι ὁρηταί· αἱ ΕΘ, ΘΝ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἡ ΕΝ ἄρα ἐκ δύο ὀνομάτων ἐστὶ διῃρημένη κατὰ τὸ Θ. ὁμοίως δὴ δείχομεν, ὅτι καὶ κατὰ τὸ Μ διῄρηται. καὶ οὐκὸέστιν ἡ ΕΘ τῇ ΜΝ ἡ αὐτή· ἡ ἄρα ἐκ δύο ὀνομάτων κατὸ ὀάλλο καὶ ὀάλλο σημεῖον διῃρηται· ὅπερ ἐστίνὸάτοπον. οὐκ ἄρα ἡ δύο μέσα δυναμένη κατὸ ἄλλο καὶ ὀάλλο σημεῖον διαιρεῖται· καθὸ ὁὲν ἄρα μόνον [σημεῖον] διαιρεῖται.

eas)] which has been divided at C, such that AC and CB are incommensurable in square, making the sum of the (squares) on AC and CB medial, and the (rectangle contained) by AC and CB medial, and, moreover, incommensurable with the sum of the (squares) on (AC and CB) [Prop. 10.41]. I say that AB cannot be divided at another point fulfilling the prescribed (conditions).

For, if possible, let it be divided at D, such that AC is again manifestly not the same as DB, but AC (is), by hypothesis, greater. And let the rational (straight-line) EF be laid down. And let EG, equal to (the sum of) the (squares) on AC and CB, and HK, equal to twice the (rectangle contained) by AC and CB, be applied to EF. Thus, the whole of EK is equal to the square on AB [Prop. 2.4]. So, again, let EL, equal to (the sum of) the (squares) on AD and DB, be applied to EF. Thus, the remainder—twice the (rectangle contained) by AD and DB—is equal to the remainder, MK. And since the sum of the (squares) on AC and CB was assumed (to be) medial, EG is also medial. And it is applied to the rational (straight-line) EF. HE is thus rational, and incommensurable in length with EF [Prop. 10.22]. So, for the same (reasons), HN is also rational, and incommensurable in length with EF. And since the sum of the (squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB, EG is thus also incommensurable with GN. Hence, EH is also incommensurable with HN [Props. 6.1, 10.11]. And they are (both) rational (straight-lines). Thus, EH and HN are rational (straight-lines which are) commensurable in square only. Thus, EN is a binomial (straight-line) which has been divided (into its component terms) at H [Prop. 10.36]. So, similarly, we can show that it has also been (so) divided at M. And EH is not the same as MN. Thus, a binomial (straight-line) has been divided (into its component terms) at different points. The very thing is absurd [Prop. 10.42]. Thus, the square-root of (the sum of) two medial (areas) cannot be divided (into its component terms) at different points. Thus, it can be (so) divided at one [point] only.

† In other words, $k'^{1/4}\sqrt{[1+k/(1+k^2)^{1/2}]/2}+k'^{1/4}\sqrt{[1-k/(1+k^2)^{1/2}]/2}=k'''^{1/4}\sqrt{[1+k''/(1+k''^2)^{1/2}]/2}$ $+k'''^{1/4}\sqrt{[1-k''/(1+k''^2)^{1/2}]/2}$ has only one solution: i.e., $k''=k$ and $k'''=k'$.

Ὅροι δεύτεροι.

5. Ὑποκειμένης ὁρητῆς καὶ τῆς ἐκ δύο ὀνομάτων διῃρημένης εἰξ τὰ ὀνόματα, ἧξ τὸ μεῖζον ὄνομα τοῦ ἐλάσσονος μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει, ἂν μὲν τὸ μεῖζον ὄνομα σύμμετρονὸῇ μήκει τῇ ἐκκειμένῃ ὁρητῇ, καλείσθω [ἡ ὅλη] ἐκ δύο ὀνομάτων πρώτη.

6. Ἐὰν δὲ τὸ ἐλάσσον ὄνομα σύμμετρονὸῇ μήκει τῇ ἐκκειμένῃ ὁρητῇ, καλείσθω ἐκ δύο ὀνομάτων δευτέρα.

7. Ἐὰν δὲ μ ηδέτερον τῶν ὀνομάτων σύμμετρονὸῇ μήκει τῇ ἐκκειμένῃ ὁρητῇ, καλείσθω ἐκ δύο ὀνομάτων τρίτη.

Definitions II

5. Given a rational (straight-line), and a binomial (straight-line) which has been divided into its (component) terms, of which the square on the greater term is larger than (the square on) the lesser by the (square) on (some straight-line) commensurable in length with (the greater) then, if the greater term is commensurable in length with the rational (straight-line previously) laid out, let [the whole] (straight-line) be called a first binomial (straight-line).

6. And if the lesser term is commensurable in length with

8. Πάλιν δὴ ἐαν τὸ μεῖζον ὄνομα [τοῦ ἐλάσσονος] μεῖζον δύνηται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ μήκει, ἐαν μὲν τὸ μεῖζον ὄνομα σύμμετρονὸῇ μήκει τῇ ἐκκειμένῃ ὁρητῇ, καλείσθω ἐκ δύο ὀνομάτων τετάρτη.

9. Ἐὰν δὲ τὸ ὀέλασσον, πέμπτη.

10. Ἐὰν δὲ μ ηδέτερον, ὁέκτη.

8. And if neither of the terms is commensurable in length with the rational (straight-line previously) laid out then let (the whole straight-line) be called a second binomial (straight-line).

7. And if neither of the terms is commensurable in length with the rational (straight-line previously) laid out then let (the whole straight-line) be called a third binomial (straight-line).

8. So, again, if the square on the greater term is larger than (the square on) [the lesser] by the (square) on (some straight-line) incommensurable in length with (the greater) then, if the greater term is commensurable in length with the rational (straight-line previously) laid out, let (the whole straight-line) be called a fourth binomial (straight-line).

9. And if the lesser (term is commensurable), a fifth (binomial straight-line).

10. And if neither (term is commensurable), a sixth (binomial straight-line).

48.

Εὑρεῖν τὴν ἐκ δύο ὀνομάτων πρώτην.

Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΑΓ, ΓΒ, ὁώστε τὸν συγκείμενον ἐξ αὐτῶν τὸν ΑΒ πρὸς μὲν τὸν ΒΓ λόγον ἔχειν, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, πρὸς δὲ τὸν ΓΑ λόγον μὴ ἔχειν, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, καὶ ἐκκείσθω τις ὁρητὴ ἡ Δ, καὶ τῇ Δ σύμμετροσὸέστω μήκει ἡ ΕΖ. ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΕΖ. καὶ γεγονέτω ὡξ ὁ ΒΑ ἀριθμὸξ πρὸς τὸν ΑΓ, οὁύτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ. ὁ δὲ ΑΒ πρὸς τὸν ΑΓ λόγον ἔχει, ὃν ἀριθμὸξ πρὸς ἀριθμόν· καὶ τὸ ἀπὸ τῆς ΕΖ ἄρα πρὸς τὸ ἀπὸ τῆς ΖΗ λόγον ἔχει, ὃν ἀριθμὸξ πρὸς ἀριθμόν· ὁώστε σύμμετρόν ἐστι τὸ ἀπὸ τῆς ΕΖ τῷ ἀπὸ τῆς ΖΗ. καὶ ἐστι ὁρητὴ ἡ ΕΖ· ὁρητὴ ἄρα καὶ ἡ ΖΗ. καὶ ἐπεὶ ὁ ΒΑ πρὸς τὸν ΑΓ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς ΕΖ ἄρα πρὸς τὸ ἀπὸ τῆς ΖΗ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΕΖ τῇ ΖΗ μήκει. αἱ ΕΖ, ΖΗ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΕΗ. λέγω, ὅτι καὶ πρώτη.

Proposition 48

To find a first binomial (straight-line).

Let two numbers AC and CB be laid down such that their sum AB has to BC the ratio which (some) square number (has) to (some) square number, and does not have to CA the ratio which (some) square number (has) to (some) square number [Prop. 10.28 lem. I]. And let some rational (straight-line) D be laid down. And let EF be commensurable in length with D. EF is thus also rational [Def. 10.3]. And let it be contrived that as the number BA (is) to AC, so the (square) on EF (is) to the (square) on FG [Prop. 10.6 corr.]. And AB has to AC the ratio which (some) number (has) to (some) number. Thus, the (square) on EF also has to the (square) on FG the ratio which (some) number (has) to (some) number. Hence, the (square) on EF is commensurable with the (square) on FG [Prop. 10.6]. And EF is rational. Thus, FG (is) also rational. And since BA does not have to AC the ratio which (some) square number (has) to (some) square number, thus the (square) on EF does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, EF is incommensurable in length with FG [Prop 10.9]. EF and FG are thus rational (straight-lines which are) commensurable in square only. Thus, EG is a binomial (straight-line) [Prop. 10.36]. I say that (it is) also a first (binomial straight-line).

Ἐπεὶ γάρ ἐστιν ὡς ὁ ΒΑ ἀριθμὸς πρὸς τὸν ΑΓ, οὕτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ, μείζων δὲ ὁ ΒΑ τοῦ ΑΓ, μεῖζον ἄρα καὶ τὸ ἀπὸ τῆς ΕΖ τοῦ ἀπὸ τῆς ΖΗ. ὑπόκειται οὖν τῷ ἀπὸ τῆς ΕΖ ἴσα τὰ ἀπὸ τῶν ΖΗ, Θ. καὶ ἐπεί ἐστιν ὡς ὁ ΒΑ πρὸς τὸν ΑΓ, οὕτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ, ἀναστρέψαντι ἄρα ἐστὶν ὡς ὁ ΑΒ πρὸς τὸν ΒΓ, οὕτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς Θ. ὁ δὲ ΑΒ πρὸς τὸν ΒΓ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ τὸ ἀπὸ τῆς ΕΖ ἄρα πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. σύμμετρος ἄρα ἐστὶν ἡ ΕΖ τῇ Θ μήκει· ἡ ΕΖ ἄρα τῆς ΖΗ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καί εἰσι ὁρηταὶ αἱ ΕΖ, ΖΗ, καὶ σύμμετρος ἡ ΕΖ τῇ Δ μήκει.

Ἡ ΕΗ ἄρα ἐκ δύο ὀνομάτων ἐστὶ πρώτη· ὅπερ ἔδει δεῖξαι.

For since as the number BA is to AC, so the (square) on EF (is) to the (square) on FG, and BA (is) greater than AC, the (square) on EF (is) thus also greater than the (square) on FG [Prop. 5.14]. Therefore, let (the sum of) the (squares) on FG and H be equal to the (square) on EF. And since as BA is to AC, so the (square) on EF (is) to the (square) on FG, thus, via conversion, as AB is to BC, so the (square) on EF (is) to the (square) on H [Prop. 5.19 corr.]. And AB has to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on EF also has to the (square) on H the ratio which (some) square number (has) to (some) square number. Thus, EF is commensurable in length with H [Prop. 10.9]. Thus, the square on EF is greater than (the square on) FG by the (square) on (some straight-line) commensurable (in length) with (EF). And EF and FG are rational (straight-lines). And EF (is) commensurable in length with D.

Thus, EG is a first binomial (straight-line) [Def. 10.5].[†] (Which is) the very thing it was required to show.

[†]If the rational straight-line has unit length then the length of a first binomial straight-line is $k + k\sqrt{1 - k'^2}$. This, and the first apotome, whose length is $k - k\sqrt{1 - k'^2}$ [Prop. 10.85], are the roots of $x^2 - 2kx + k^2k'^2 = 0$.

49.

Εὑρεῖν τὴν ἐκ δύο ὀνομάτων δευτέραν.

Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΑΓ, ΓΒ, ὥστε τὸν συγκείμενον ἐξ αὐτῶν τὸν ΑΒ πρὸς μὲν τὸν ΒΓ λόγον ἔχειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, πρὸς δὲ τὸν ΑΓ λόγον μὴ ἔχειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, καὶ ἐκκείσθω ὁρητὴ ἡ Δ, καὶ τῇ Δ σύμμετροςὁέστω ἡ ΕΖ μήκει· ὁρητὴ ἄρα ἐστὶν ἡ ΕΖ. γεγονέτω δὴ καὶ ὡς ὁ ΓΑ ἀριθμὸς πρὸς τὸν ΑΒ, οὕτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ· σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΕΖ τῷ ἀπὸ τῆς ΖΗ. ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΖΗ. καὶ ἐπεὶ ὁ ΓΑ ἀριθμὸς πρὸς τὸν ΑΒ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. ἀσύμμε-

Proposition 49

To find a second binomial (straight-line).

Let the two numbers AC and CB be laid down such that their sum AB has to BC the ratio which (some) square number (has) to (some) square number, and does not have to AC the ratio which (some) square number (has) to (some) square number [Prop. 10.28 lem. I]. And let the rational (straight-line) D be laid down. And let EF be commensurable in length with D. EF is thus a rational (straight-line). So, let it also be contrived that as the number CA (is) to AB, so the (square) on EF (is) to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square) on EF is commensurable with the (square) on FG [Prop. 10.6]. Thus, FG is also a rational (straight-line). And since the number CA does not have to AB the ratio which (some) square

τρος ἄρα ἐστὶν ἡ EZ τῇ ZH μήκει· αἱ EZ, ZH ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ EH. δεικτέον δή, ὅτι καὶ δευτέρα.

number (has) to (some) square number, the (square) on EF does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, EF is incommensurable in length with FG [Prop. 10.9]. EF and FG are thus rational (straight-lines which are) commensurable in square only. Thus, EG is a binomial (straight-line) [Prop. 10.36]. So, we must show that (it is) also a second (binomial straight-line).

Ἐπεὶ γὰρ ἀνάπαλίν ἐστιν ὡς ὁ BA ἀριθμὸξ πρὸς τὸν $AΓ$, οὑτως τὸ ἀπὸ τῆς HZ πρὸς τὸ ἀπὸ τῆς ZE, μείζων δὲ ὁ BA τοῦ $AΓ$, μεῖζον ἄρα [καὶ] τὸ ἀπὸ τῆς HZ τοῦ ἀπὸ τῆς ZE. ὀέστω τῷ ἀπὸ τῆς HZ ἴσα τὰ ἀπὸ τῶν EZ, $Θ$· ἀναστρέψαντι ἄρα ἐστὶν ὡξ ὁ AB πρὸς τὸν $BΓ$, οὑτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς $Θ$. ἀλλὸ ὁ AB πρὸς τὸν $BΓ$ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· καὶ τὸ ἀπὸ τῆς ZH ἄρα πρὸς τὸ ἀπὸ τῆς $Θ$ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν. σύμμετρος ἄρα ἐστὶν ἡ ZH τῇ $Θ$ μήκει· ὁὡστε ἡ ZH τῆς ZE μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καί εἰσι ὁρηταὶ αἱ ZH, ZE δυνάμει μόνον σύμμετροι, καὶ τὸ EZὸέλασσον ὄνομα τῇ ἐκκειμένῃ ὁρητῇ σύμμετρόν ἐστι τῇ $Δ$ μήκει.

Ἡ EH ἄρα ἐκ δύο ὀνομάτων ἐστὶ δευτέρα· ὅπερ ἔδει δεῖχαι.

For since, inversely, as the number BA is to AC, so the (square) on GF (is) to the (square) on FE [Prop. 5.7 corr.], and BA (is) greater than AC, the (square) on GF (is) thus [also] greater than the (square) on FE [Prop. 5.14]. Let (the sum of) the (squares) on EF and H be equal to the (square) on GF. Thus, via conversion, as AB is to BC, so the (square) on FG (is) to the (square) on H [Prop. 5.19 corr.]. But, AB has to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG also has to the (square) on H the ratio which (some) square number (has) to (some) square number. Thus, FG is commensurable in length with H [Prop. 10.9]. Hence, the square on FG is greater than (the square on) FE by the (square) on (some straight-line) commensurable in length with (FG). And FG and FE are rational (straight-lines which are) commensurable in square only. And the lesser term EF is commensurable in length with the rational (straight-line) D (previously) laid down.

Thus, EG is a second binomial (straight-line) [Def. 10.6].[†] (Which is) the very thing it was required to show.

[†] If the rational straight-line has unit length then the length of a second binomial straight-line is $k/\sqrt{1-k'^2}+k$. This, and the second apotome, whose length is $k/\sqrt{1-k'^2}-k$ [Prop. 10.86], are the roots of $x^2-(2\,k/\sqrt{1-k'^2})\,x+k^2\,[k'^2/(1-k'^2)]=0$.

50.

Proposition 50

Εὑρεῖν τὴν ἐκ δύο ὀνομάτων τρίτην.

Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ $AΓ$, $ΓB$, ὁὡστε τὸν συγκείμενον ἐξ αὐτῶν τὸν AB πρὸς μὲν τὸν $BΓ$ λόγον ἔχειν, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, πρὸς δὲ τὸν $AΓ$ λόγον μὴ ἔχειν, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν. ἐκκείσθω δέ τις καὶ ὁἄλλος μὴ τετράγωνος ἀριθμὸξ ὁ $Δ$, καὶ πρὸς ἑκάτερον τῶν BA, $AΓ$ λόγον μὴ ἐθέτω, ὃν

To find a third binomial (straight-line).

Let the two numbers AC and CB be laid down such that their sum AB has to BC the ratio which (some) square number (has) to (some) square number, and does not have to AC the ratio which (some) square number (has) to (some) square number. And let some other non-square number D also be laid down, and let it not have to each of BA and AC the ratio

τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· καὶ ἐκκείσθω τις ὁρητὴ εὐθεῖα ἡ Ε, καὶ γεγονέτω ὡξ ὁ Δ πρὸς τὸν ΑΒ, οὕτως τὸ ἀπὸ τῆς Ε πρὸς τὸ ἀπὸ τῆς ΖΗ· σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς Ε τῷ ἀπὸ τῆς ΖΗ. καί ἐστι ὁρητὴ ἡ Ε· ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΖΗ. καὶ ἐπεὶ ὁ Δ πρὸς τὸν ΑΒ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς Ε πρὸς τὸ ἀπὸ τῆς ΖΗ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ Ε τῇ ΖΗ μήκει. γεγονέτω δὴ πάλιν ὡξ ἡ ΒΑ ἀριθμὸξ πρὸς τὸν ΑΓ, οὕτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ· σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΖΗ τῷ ἀπὸ τῆς ΗΘ. ὁρητὴ δὲ ἡ ΖΗ· ὁρητὴ ἄρα καὶ ἡ ΗΘ. καὶ ἐπεὶ ὁ ΒΑ πρὸς τὸν ΑΓ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΘΗ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΖΗ τῇ ΗΘ μήκει. αἱ ΖΗ, ΗΘ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἡ ΖΘ ἄρα ἐκ δύο ὀνομάτων ἐστίν. λέγω δή, ὅτι καὶ τρίτη.

which (some) square number (has) to (some) square number. And let some rational straight-line E be laid down, and let it be contrived that as D (is) to AB, so the (square) on E (is) to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square) on E is commensurable with the (square) on FG [Prop. 10.6]. And E is a rational (straight-line). Thus, FG is also a rational (straight-line). And since D does not have to AB the ratio which (some) square number has to (some) square number, the (square) on E does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. E is thus incommensurable in length with FG [Prop. 10.9]. So, again, let it be contrived that as the number BA (is) to AC, so the (square) on FG (is) to the (square) on GH [Prop. 10.6 corr.]. Thus, the (square) on FG is commensurable with the (square) on GH [Prop. 10.6]. And FG (is) a rational (straight-line). Thus, GH (is) also a rational (straight-line). And since BA does not have to AC the ratio which (some) square number (has) to (some) square number, the (square) on FG does not have to the (square) on HG the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. FG and GH are thus rational (straight-lines which are) commensurable in square only. Thus, FH is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a third (binomial straight-line).

Ἐπεὶ γάρ ἐστιν ὡξ ὁ Δ πρὸς τὸν ΑΒ, οὕτως τὸ ἀπὸ τῆς Ε πρὸς τὸ ἀπὸ τῆς ΖΗ, ὡξ δὲ ὁ ΒΑ πρὸς τὸν ΑΓ, οὕτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ, διὸ ὐίσου ἄρα ἐστὶν ὡξ ὁ Δ πρὸς τὸν ΑΓ, οὕτως τὸ ἀπὸ τῆς Ε πρὸς τὸ ἀπὸ τῆς ΗΘ. ὁ δὲ Δ πρὸς τὸν ΑΓ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· οὐδὲ τὸ ἀπὸ τῆς Ε ἄρα πρὸς τὸ ἀπὸ τῆς ΗΘ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ Ε τῇ ΗΘ μήκει. καὶ ἐπεί ἐστιν ὡξ ὁ ΒΑ πρὸς τὸν ΑΓ, οὕτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ, μεῖζον ἄρα τὸ ἀπὸ τῆς ΖΗ τοῦ ἀπὸ τῆς ΗΘ. ὐέστω οὐῦν τῷ ἀπὸ τῆς ΖΗ ἴσα τὰ ἀπὸ τῶν ΗΘ, Κ·

For since as D is to AB, so the (square) on E (is) to the (square) on FG, and as BA (is) to AC, so the (square) on FG (is) to the (square) on GH, thus, via equality, as D (is) to AC, so the (square) on E (is) to the (square) on GH [Prop. 5.22]. And D does not have to AC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on E does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. Thus, E is incommensurable in length with GH [Prop. 10.9]. And since as BA is to AC, so the (square) on FG (is) to the (square) on GH, the (square) on FG (is) thus greater than the

ἀναστρέψαντι ἄρα [ἐστὶν] ὡς ὁ ΑΒ πρὸς τὸν ΒΓ, οὕτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς Κ. ὁ δὲ ΑΒ πρὸς τὸν ΒΓ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ τὸ ἀπὸ τῆς ΖΗ ἄρα πρὸς τὸ ἀπὸ τῆς Κ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· σύμμετρος ἄρα [ἐστὶν] ἡ ΖΗ τῇ Κ μήκει. ἡ ΖΗ ἄρα τῆς ΗΘ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καί εἰσιν αἱ ΖΗ, ΗΘ ὁρηταὶ δυνάμει μόνον σύμμετροι, καὶ οὐδετέρα αὐτῶν σύμμετρός ἐστι τῇ Ε μήκει.

Ἡ ΖΘ ἄρα ἐκ δύο ὀνομάτων ἐστὶ τρίτη· ὅπερ ἔδει δεῖξαι.

(square) on GH [Prop. 5.14]. Therefore, let (the sum of) the (squares) on GH and K be equal to the (square) on FG. Thus, via conversion, as AB [is] to BC, so the (square) on FG (is) to the (square) on K [Prop. 5.19 corr.]. And AB has to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG also has to the (square) on K the ratio which (some) square number (has) to (some) square number. Thus, FG [is] commensurable in length with K [Prop. 10.9]. Thus, the square on FG is greater than (the square on) GH by the (square) on (some straight-line) commensurable (in length) with (FG). And FG and GH are rational (straight-lines which are) commensurable in square only, and neither of them is commensurable in length with E.

Thus, FH is a third binomial (straight-line) [Def. 10.7].[†] (Which is) the very thing it was required to show.

[†] If the rational straight-line has unit length then the length of a third binomial straight-line is $k^{1/2}\,(1+\sqrt{1-k'^2})$. This, and the third apotome, whose length is $k^{1/2}\,(1-\sqrt{1-k'^2})$ [Prop. 10.87], are the roots of $x^2-2\,k^{1/2}\,x+k\,k'^2=0$.

51.

Εὑρεῖν τὴν ἐκ δύο ὀνομάτων τετάρτην.

Proposition 51

To find a fourth binomial (straight-line).

Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΑΓ, ΓΒ, ὥστε τὸν ΑΒ πρὸς τὸν ΒΓ λόγον μὴ ἔχειν μήτε μὴν πρὸς τὸν ΑΓ, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. καὶ ἐκκείσθω ὁρητὴ ἡ Δ, καὶ τῇ Δ σύμμετροςὁέστω μήκει ἡ ΕΖ· ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΕΖ. καὶ γεγονέτω ὡς ὁ ΒΑ ἀριθμὸς πρὸς τὸν ΑΓ, οὕτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ· σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΕΖ τῷ ἀπὸ τῆς ΖΗ· ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΖΗ. καὶ ἐπεὶ ὁ ΒΑ πρὸς τὸν ΑΓ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΕΖ τῇ ΖΗ μήκει. αἱ ΕΖ, ΖΗ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ὥστε ἡ ΕΗ ἐκ

Let the two numbers AC and CB be laid down such that AB does not have to BC, or to AC either, the ratio which (some) square number (has) to (some) square number [Prop. 10.28 lem. I]. And let the rational (straight-line) D be laid down. And let EF be commensurable in length with D. Thus, EF is also a rational (straight-line). And let it be contrived that as the number BA (is) to AC, so the (square) on EF (is) to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square) on EF is commensurable with the (square) on FG [Prop. 10.6]. Thus, FG is also a rational (straight-line). And since BA does not have to AC the ratio which (some) square number (has) to (some) square number, the (square) on EF

δύο ὀνομάτων ἐστίν. λέγω δή, ὅτι καὶ τετάρτη.

Ἐπεὶ γάρ ἐστιν ὡς ὁ ΒΑ πρὸς τὸν ΑΓ, οὕτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς ΖΗ [μείζων δὲ ὁ ΒΑ τοῦ ΑΓ], μεῖζον ἄρα τὸ ἀπὸ τῆς ΕΖ τοῦ ἀπὸ τῆς ΖΗ. ὑέστω οὐὖν τῷ ἀπὸ τῆς ΕΖ ἴσα τὰ ἀπὸ τῶν ΖΗ, Θ· ἀναστρέψαντι ἄρα ὡς ὁ ΑΒ ἀριθμὸξ πρὸς τὸν ΒΓ, οὕτως τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς Θ. ὁ δὲ ΑΒ πρὸς τὸν ΒΓ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· οὐδὸ ἄρα τὸ ἀπὸ τῆς ΕΖ πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν. ἀσύμμετρος ἄρα ἐστὶν ἡ ΕΖ τῇ Θ μήκει· ἡ ΕΖ ἄρα τῆς ΗΖ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καί εἰσιν αἱ ΕΖ, ΖΗ ὁρηταὶ δυνάμει μόνον σύμμετροι, καὶ ἡ ΕΖ τῇ Δ σύμμετρόξ ἐστι μήκει.

Ἡ ΕΗ ἄρα ἐκ δύο ὀνομάτων ἐστὶ τετάρτη· ὅπερ ἔδει δεῖχαι.

does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, EF is incommensurable in length with FG [Prop. 10.9]. Thus, EF and FG are rational (straight-lines which are) commensurable in square only. Hence, EG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a fourth (binomial straight-line).

For since as BA is to AC, so the (square) on EF (is) to the (square) on FG [and BA (is) greater than AC], the (square) on EF (is) thus greater than the (square) on FG [Prop. 5.14]. Therefore, let (the sum of) the squares on FG and H be equal to the (square) on EF. Thus, via conversion, as the number AB (is) to BC, so the (square) on EF (is) to the (square) on H [Prop. 5.19 corr.]. And AB does not have to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on EF does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, EF is incommensurable in length with H [Prop. 10.9]. Thus, the square on EF is greater than (the square on) GF by the (square) on (some straight-line) incommensurable (in length) with (EF). And EF and FG are rational (straight-lines which are) commensurable in square only. And EF is commensurable in length with D.

Thus, EG is a fourth binomial (straight-line) [Def. 10.8].[†] (Which is) the very thing it was required to show.

[†] If the rational straight-line has unit length then the length of a fourth binomial straight-line is $k\,(1 + 1/\sqrt{1 + k'})$. This, and the fourth apotome, whose length is $k\,(1 - 1/\sqrt{1 + k'})$ [Prop. 10.88], are the roots of $x^2 - 2\,k\,x + k^2\,k'/(1 + k') = 0$.

52.

Εὑρεῖν τὴν ἐκ δύο ὀνομάτων πέμπτην.

Proposition 52

To find a fifth binomial straight-line.

Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΑΓ, ΓΒ, ὥστε τὸν ΑΒ πρὸς ἑκάτερον αὐτῶν λόγον μὴ ἔχειν, ὃν τετράγωνος ἀριθμὸξ πρὸς

Let the two numbers AC and CB be laid down such that AB does not have to either of them the ratio which (some) square

τετράγωνον ἀριθμόν, καὶ ἐκκείσθω ὁρητή τις εὐθεῖα ἡ Δ, καὶ τῇ Δ σύμμετροςὸέστω [μήκει] ἡ EZ· ὁρητὴ ἄρα ἡ EZ. καὶ γεγονέτω ὡς ὁ ΓΑ πρὸς τὸν AB, οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς ZH. ὁ δὲ ΓΑ πρὸς τὸν AB λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδὲ τὸ ἀπὸ τῆς EZ ἄρα πρὸς τὸ ἀπὸ τῆς ZH λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. αἱ EZ, ZH ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ EH. λέγω δή, ὅτι καὶ πέμπτη.

Ἐπεὶ γάρ ἐστιν ὡς ὁ ΓΑ πρὸς τὸν AB, οὕτως τὸ ἀπὸ τῆς EZ πρὸς τὸ ἀπὸ τῆς ZH, ἀνάπαλιν ὡς ὁ BA πρὸς τὸν AΓ, οὕτως τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς ZE· μεῖζον ἄρα τὸ ἀπὸ τῆς HZ τοῦ ἀπὸ τῆς ZE. ὀέστω οὐῦν τῷ ἀπὸ τῆς HZ ἴσα τὰ ἀπὸ τῶν EZ, Θ· ἀναστρέψαντι ἄρα ἐστὶν ὡς ὁ AB ἀριθμὸς πρὸς τὸν BΓ, οὕτως τὸ ἀπὸ τῆς HZ πρὸς τὸ ἀπὸ τῆς Θ. ὁ δὲ AB πρὸς τὸν BΓ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδὸ ἄρα τὸ ἀπὸ τῆς ZH πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν. ἀσύμμετρος ἄρα ἐστὶν ἡ ZH τῇ Θ μήκει· ὥστε ἡ ZH τῆς ZE μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καί εἰσιν αἱ HZ, ZE ὁρηταὶ δυνάμει μόνον σύμμετροι, καὶ τὸ EZὀέλαττον ὄνομα σύμμετρόν ἐστι τῇ ἐκκειμένῃ ὁρητῇ τῇ Δ μήκει.

Ἡ EH ἄρα ἐκ δύο ὀνομάτων ἐστὶ πέμπτη· ὅπερ ἔδει δεῖχαι.

number (has) to (some) square number [Prop. 10.38 lem.]. And let some rational straight-line D be laid down. And let EF be commensurable [in length] with D. Thus, EF (is) a rational (straight-line). And let it be contrived that as CA (is) to AB, so the (square) on EF (is) to the (square) on FG [Prop. 10.6 corr.]. And CA does not have to AB the ratio which (some) square number (has) to (some) square number. Thus, the (square) on EF does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, EF and FG are rational (straight-lines which are) commensurable in square only [Prop. 10.9]. Thus, EG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a fifth (binomial straight-line).

For since as CA is to AB, so the (square) on EF (is) to the (square) on FG, inversely, as BA (is) to AC, so the (square) on FG (is) to the (square) on FE [Prop. 5.7 corr.]. Thus, the (square) on GF (is) greater than the (square) on FE [Prop. 5.14]. Therefore, let (the sum of) the (squares) on EF and H be equal to the (square) on GF. Thus, via conversion, as the number AB is to BC, so the (square) on GF (is) to the (square) on H [Prop. 5.19 corr.]. And AB does not have to BC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with H [Prop. 10.9]. Hence, the square on FG is greater than (the square on) FE by the (square) on (some straight-line) incommensurable (in length) with (FG). And GF and FE are rational (straight-lines which are) commensurable in square only. And the lesser term EF is commensurable in length with the rational (straight-line previously) laid down, D.

Thus, EG is a fifth binomial (straight-line).† (Which is) the very thing it was required to show.

† If the rational straight-line has unit length then the length of a fifth binomial straight-line is $k\,(\sqrt{1+k'}+1)$. This, and the fifth apotome, whose length is $k\,(\sqrt{1+k'}-1)$ [Prop. 10.89], are the roots of $x^2 - 2k\sqrt{1+k'}\,x + k^2k' = 0$.

53.

Εὑρεῖν τὴν ἐκ δύο ὀνομάτων ὀέκτην.

Proposition 53

To find a sixth binomial (straight-line).

Ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΑΓ, ΓΒ, ὥστε τὸν ΑΒ πρὸς ἑκάτερον αὐτῶν λόγον μὴ ἔχειν, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ὑπόκειται δὲ καὶ ὁπότερος ἀριθμὸς ὁ Δ μὴ τετράγωνος ὢν μηδὲ πρὸς ἑκάτερον τῶν ΒΑ, ΑΓ λόγον ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· καὶ ἐκκείσθω τις ὁρητὴ εὐθεῖα ἡ Ε, καὶ γεγονέτω ὡς ὁ Δ πρὸς τὸν ΑΒ, οὕτως τὸ ἀπὸ τῆς Ε πρὸς τὸ ἀπὸ τῆς ΖΗ· σύμμετρον ἄρα τὸ ἀπὸ τῆς Ε τῷ ἀπὸ τῆς ΖΗ. καί ἐστι ὁρητὴ ἡ Ε· ὁρητὴ ἄρα καὶ ἡ ΖΗ. καὶ ἐπεὶ οὐκ ἔχει ὁ Δ πρὸς τὸν ΑΒ λόγον, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς Ε ἄρα πρὸς τὸ ἀπὸ τῆς ΖΗ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἡ Ε τῇ ΖΗ μήκει. γεγονέτω δὴ πάλιν ὡς ὁ ΒΑ πρὸς τὸν ΑΓ, οὕτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ. σύμμετρον ἄρα τὸ ἀπὸ τῆς ΖΗ τῷ ἀπὸ τῆς ΘΗ. ὁρητὸν ἄρα τὸ ἀπὸ τῆς ΘΗ· ὁρητὴ ἄρα ἡ ΘΗ. καὶ ἐπεὶ ὁ ΒΑ πρὸς τὸν ΑΓ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὲ τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΖΗ τῇ ΗΘ μήκει. αἱ ΖΗ, ΗΘ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΖΘ. δεικτέον δή, ὅτι καὶ ὁέκτη.

Ἐπεὶ γάρ ἐστιν ὡς ὁ Δ πρὸς τὸν ΑΒ, οὕτως τὸ ἀπὸ τῆς Ε πρὸς τὸ ἀπὸ τῆς ΖΗ, ὅτι δὲ καὶ ὡς ὁ ΒΑ πρὸς τὸν ΑΓ, οὕτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ, διὸ ὅτι ἄρα ἐστὶν ὡς ὁ Δ πρὸς τὸν ΑΓ, οὕτως τὸ ἀπὸ τῆς Ε πρὸς τὸ ἀπὸ τῆς ΗΘ. ὁ δὲ Δ πρὸς τὸν ΑΓ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδὲ τὸ ἀπὸ τῆς Ε ἄρα πρὸς τὸ ἀπὸ τῆς ΗΘ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ Ε τῇ ΗΘ μήκει. ἐδείχθη δὲ καὶ τῇ ΖΗ ἀσύμμετρος· ἑκατέρα ἄρα τῶν ΖΗ, ΗΘ ἀσύμμετρός ἐστι τῇ Ε μήκει. καὶ ἐπεί ἐστιν ὡς ὁ ΒΑ πρὸς τὸν ΑΓ, οὕτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ, μεῖζον ἄρα τὸ ἀπὸ τῆς ΖΗ τοῦ ἀπὸ τῆς ΗΘ. ὑπόκειται οὐκοῦν τῷ ἀπὸ [τῆς] ΖΗ ἴσα τὰ ἀπὸ τῶν ΗΘ, Κ· ἀναστρέψαντι ἄρα ὡς ὁ ΑΒ πρὸς ΒΓ, οὕτως τὸ ἀπὸ ΖΗ πρὸς τὸ ἀπὸ τῆς Κ. ὁ δὲ ΑΒ πρὸς τὸν ΒΓ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ὥστε οὐδὲ τὸ ἀπὸ ΖΗ πρὸς τὸ ἀπὸ τῆς Κ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον

Let the two numbers AC and CB be laid down such that AB does not have to each of them the ratio which (some) square number (has) to (some) square number. And let D also be another number, which is not square, and does not have to each of BA and AC the ratio which (some) square number (has) to (some) square number either [Prop. 10.28 lem. I]. And let some rational straight-line E be laid down. And let it be contrived that as D (is) to AB, so the (square) on E (is) to the (square) on FG [Prop. 10.6 corr.]. Thus, the (square) on E (is) commensurable with the (square) on FG [Prop. 10.6]. And E is rational. Thus, FG (is) also rational. And since D does not have to AB the ratio which (some) square number (has) to (some) square number, the (square) on E thus does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, E (is) incommensurable in length with FG [Prop. 10.9]. So, again, let it have be contrived that as BA (is) to AC, so the (square) on FG (is) to the (square) on GH [Prop. 10.6 corr.]. The (square) on FG (is) thus commensurable with the (square) on HG [Prop. 10.6]. The (square) on HG (is) thus rational. Thus, HG (is) rational. And since BA does not have to AC the ratio which (some) square number (has) to (some) square number, the (square) on FG does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. Thus, FG and GH are rational (straight-lines which are) commensurable in square only. Thus, FH is a binomial (straight-line) [Prop. 10.36]. So, we must show that (it is) also a sixth (binomial straight-line).

For since as D is to AB, so the (square) on E (is) to the (square) on FG, and also as BA is to AC, so the (square) on FG (is) to the (square) on GH, thus, via equality, as D is to AC, so the (square) on E (is) to the (square) on GH [Prop. 5.22]. And D does not have to AC the ratio which (some) square number (has) to (some) square number. Thus, the (square) on E does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. E is thus incommensurable in length with GH [Prop. 10.9]. And (E) was also shown (to be) incommensurable (in length) with FG.

ἀριθμόν. ἀσύμμετρος ἄρα ἐστὶν ἡ ΖΗ τῇ Κ μήκει· ἡ ΖΗ ἄρα τῆς ΗΘ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καί εἰσιν αἱ ΖΗ, ΗΘ ὁρηταὶ δυνάμει μόνον σύμμετροι, καὶ οὐδετέρα αὐτῶν σύμμετρός ἐστι μήκει τῇ ἐκκειμένῃ ὁρητῇ τῇ Ε.

Ἡ ΖΘ ἄρα ἐκ δύο ὀνομάτων ἐστὶν ὀκτη· ὅπερ ἔδει δεῖξαι.

Thus, FG and GH are each incommensurable in length with E. And since as BA is to AC, so the (square) on FG (is) to the (square) on GH, the (square) on FG (is) thus greater than the (square) on GH [Prop. 5.14]. Therefore, let (the sum of) the (squares) on GH and K be equal to the (square) on FG. Thus, via conversion, as AB (is) to BC, so the (square) on FG (is) to the (square) on K [Prop. 5.19 corr.]. And AB does not have to BC the ratio which (some) square number (has) to (some) square number. Hence, the (square) on FG does not have to the (square) on K the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with K [Prop. 10.9]. The square on FG is thus greater than (the square on) GH by the (square) on (some straight-line which is) incommensurable (in length) with (FG). And FG and GH are rational (straight-lines which are) commensurable in square only, and neither of them is commensurable in length with the rational (straight-line) E (previously) laid down.

Thus, FH is a sixth binomial (straight-line) [Def. 10.10].† (Which is) the very thing it was required to show.

† If the rational straight-line has unit length then the length of a sixth binomial straight-line is $\sqrt{k}+\sqrt{k'}$. This, and the sixth apotome, whose length is $\sqrt{k}-\sqrt{k'}$ [Prop. 10.90], are the roots of $x^2 - 2\sqrt{k}\,x + (k-k') = 0$.

Λῆμμα.

Ἔστω δύο τετράγωνα τὰ ΑΒ, ΒΓ καὶ κείσθωσαν ὥστε ἐπ᾽ εὐθείας εἶναι τὴν ΔΒ τῇ ΒΕ· ἐπ᾽ εὐθείας ἄρα ἐστὶ καὶ ἡ ΖΒ τῇ ΒΗ. καὶ συμπεπληρώσθω τὸ ΑΓ παραλληλόγραμμον· λέγω, ὅτι τετράγωνόν ἐστι τὸ ΑΓ, καὶ ὅτι τῶν ΑΒ, ΒΓ μέσον ἀνάλογόν ἐστι τὸ ΔΗ, καὶ ὁέτι τῶν ΑΓ, ΓΒ μέσον ἀνάλογόν ἐστι τὸ ΔΓ.

Ἐπεὶ γὰρόίση ἐστὶν ἡ μὲν ΔΒ τῇ ΒΖ, ἡ δὲ ΒΕ τῇ ΒΗ, ὅλη ἄρα ἡ ΔΕ ὅλῃ τῇ ΖΗ ἐστινὸίση. ἀλλὸ ἡ μὲν ΔΕ ἑκατέρᾳ τῶν ΑΘ, ΚΓ ἐστινὸίση, ἡ δὲ ΖΗ ἑκατέρᾳ τῶν ΑΚ, ΘΓ ἐστινὸίση· καὶ ἑκατέρα ἄρα τῶν ΑΘ, ΚΓ ἑκατέρᾳ τῶν ΑΚ, ΘΓ ἐστινὸίση. ἰσόπλευρον ἄρα ἐστὶ τὸ ΑΓ παραλληλόγραμμον· ὁέστι δὲ καὶ ὀρθογώνιον· τετράγωνον ἄρα ἐστὶ τὸ ΑΓ.

Καὶ ἐπεὶ ἐστιν ὡς ἡ ΖΒ πρὸς τὴν ΒΗ, οὕτως ἡ ΔΒ πρὸς τὴν ΒΕ, ἀλλὸ ὡς μὲν ἡ ΖΒ πρὸς τὴν ΒΗ, οὕτως τὸ ΑΒ πρὸς τὸ ΔΗ, ὡς δὲ ἡ ΔΒ πρὸς τὴν ΒΕ, οὕτως τὸ ΔΗ πρὸς τὸ ΒΓ, καὶ ὡς ἄρα τὸ ΑΒ πρὸς τὸ ΔΗ, οὕτως τὸ ΔΗ πρὸς τὸ ΒΓ. τῶν ΑΒ, ΒΓ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΔΗ.

Λέγω δή, ὅτι καὶ τῶν ΑΓ, ΓΒ μέσον ἀνάλογόν [ἐστι] τὸ ΔΓ.

Lemma

Let AB and BC be two squares, and let them be laid down such that DB is straight-on to BE. FB is, thus, also straight-on to BG. And let the parallelogram AC be completed. I say that AC is a square, and that DG is the mean proportional to AB and BC, and, moreover, DC is the mean proportional to AC and CB.

For since DB is equal to BF, and BE to BG, the whole of DE is thus equal to the whole of FG. But DE is equal to each of AH and KC, and FG is equal to each of AK and HC [Prop. 1.34]. Thus, AH and KC are also equal to AK and HC, respectively. Thus, the parallelogram AC is equilateral. And (it is) also right-angled. Thus, AC is a square.

And since as FB is to BG, so DB (is) to BE, but as FB (is) to BG, so AB (is) to DG, and as DB (is) to BE, so DG (is) to BC [Prop. 6.1], thus also as AB (is) to DG, so DG (is) to BC [Prop. 5.11]. Thus, DG is the mean proportional to AB and BC.

So I also say that DC [is] the mean proportional to AC and CB.

Ἐπεὶ γάρ ἐστιν ὡξ ἡ ΑΔ πρὸς τὴν ΔΚ, οὑτως ἡ ΚΗ πρὸς τὴν ΗΓ· ὁίση γάρ [ἐστιν] ἑκατέρα ἑκατέρᾳ· καὶ συνθέντι ὡξ ἡ ΑΚ πρὸς ΚΔ, οὑτως ἡ ΚΓ πρὸς ΓΗ, ἀλλὰ ὡξ μὲν ἡ ΑΚ πρὸς ΚΔ, οὑτως τὸ ΑΓ πρὸς τὸ ΓΔ, ὡξ δὲ ἡ ΚΓ πρὸς ΓΗ, οὑτως τὸ ΔΓ πρὸς ΓΒ, καὶ ὡξ ἄρα τὸ ΑΓ πρὸς ΔΓ, οὑτως τὸ ΔΓ πρὸς τὸ ΒΓ. τῶν ΑΓ, ΓΒ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΔΓ· ὁἀ προέκειτο δεῖχαι.

For since as *AD* is to *DK*, so *KG* (is) to *GC*. For [they are] respectively equal. And, via composition, as *AK* (is) to *KD*, so *KC* (is) to *CG* [Prop. 5.18]. But as *AK* (is) to *KD*, so *AC* (is) to *CD*, and as *KC* (is) to *CG*, so *DC* (is) to *CB* [Prop. 6.1]. Thus, also, as *AC* (is) to *DC*, so *DC* (is) to *BC* [Prop. 5.11]. Thus, *DC* is the mean proportional to *AC* and *CB*. Which (is the very thing) it was prescribed to show.

54.

Ἐὰν θωρίον περιέθηται ὑπὸ ὁρητῆς καὶ τῆς ἐκ δύο ὀνομάτων πρώτης, ἡ τὸ θωρίον δυναμένηὁάλογόξ ἐστιν ἡ καλουμένη ἐκ δύο ὀνομάτων.

Proposition 54

If an area is contained by a rational (straight-line) and a first binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called binomial.[†]

Θωρίον γὰρ τὸ ΑΓ περιεθέσθω ὑπὸ ὁρητῆς τῆς ΑΒ καὶ τῆς ἐκ δύο ὀνομάτων πρώτης τῆς ΑΔ· λέγω, ὅτι ἡ τὸ ΑΓ θωρίον δυναμένηὀάλογόξ ἐστιν ἡ καλουμένη ἐκ δύο ὀνομάτων.

Ἐπεὶ γὰρ ἐκ δύο ὀνομάτων ἐστὶ πρώτη ἡ ΑΔ, διῃρήσθω εἰξ τὰ ὀνόματα κατὰ τὸ Ε, καὶ ὀέστω τὸ μεῖζον ὄνομα τὸ ΑΕ. φανερὸν δή, ὅτι αἱ ΑΕ, ΕΔ ὁρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΑΕ τῆς ΕΔ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῃ, καὶ ἡ ΑΕ σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῇ τῇ ΑΒ μήκει. τετμήσθω δὴ ἡ ΕΔ δίθα κατὰ τὸ Ζ σημεῖον. καὶ ἐπεὶ ἡ ΑΕ τῆς ΕΔ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος, τουτέστι τῷ ἀπὸ τῆς ΕΖ, ἴσον παρὰ τὴν μείζονα τὴν ΑΕ παραβληθῇ ἐλλεῖπον εοίδει τετραγώνῳ, εἰξ σύμμετρα αὐτὴν διαιρεῖ. παραβεβλήσθω ο῾ν παρὰ τὴν ΑΕ τῷ ἀπὸ τῆς ΕΖ ἴσον τὸ ὑπὸ ΑΗ, ΗΕ· σύμμετρος ἄρα ἐστὶν ἡ ΑΗ τῇ ΕΗ μήκει. καὶ ἤχθωσαν ἀπὸ τῶν Η, Ε, Ζ ὁποτέρᾳ τῶν ΑΒ, ΓΔ παράλληλοι αἱ ΗΘ, ΕΚ, ΖΛ· καὶ τῷ μὲν ΑΘ παραλληλογράμμῳ ἴσον τετράγωνον συνεστάτω τὸ ΣΝ, τῷ δὲ ΗΚ ἴσον τὸ ΝΠ, καὶ κείσθω ὁώστε ἐπ΄ εὐθείας εοῖναι τὴν ΜΝ τῇ ΝΧ· ἐπ΄ εὐθείας ἄρα ἐστὶ καὶ ἡ ΡΝ τῇ ΝΟ. καὶ συμπεπληρώσθω τὸ ΣΠ παραλληλόγραμμον· τετράγωνον ἄρα ἐστὶ τὸ ΣΠ. καὶ ἐπεὶ τὸ ὑπὸ τῶν ΑΗ, ΗΕ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΖ, ὀέστιν ἄρα ὡξ ἡ ΑΗ πρὸς ΕΖ, οὁύτως ἡ ΖΕ πρὸς ΕΗ· καὶ ὡξ ἄρα τὸ ΑΘ πρὸς ΕΛ, τὸ ΕΛ πρὸς ΚΗ· τῶν ΑΘ, ΗΚ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΕΛ. ἀλλὰ τὸ μὲν ΑΘ ἴσον ἐστὶ τῷ ΣΝ, τὸ δὲ ΗΚ ἴσον τῷ ΝΠ· τῶν ΣΝ, ΝΠ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΕΛ. ὀέστι δὲ τῶν αὐτῶν τῶν ΣΝ, ΝΠ μέσον ἀνάλογον καὶ τὸ ΜΡ· ἴσον ἄρα ἐστὶ τὸ ΕΛ τῷ ΜΡ· ὁώστε καὶ τῷ ΟΧ ἴσον ἐστίν. ὀέστι δὲ καὶ τὰ ΑΘ, ΗΚ τοῖς ΣΝ, ΝΠ ἴσα· ὅλον ἄρα τὸ ΑΓ ἴσον ἐστὶν ὅλῳ τῷ ΣΠ, τουτέστι τῷ ἀπὸ τῆς ΜΧ τετραγώνῳ· τὸ ΑΓ ἄρα δύναται ἡ ΜΧ. λέγω, ὅτι ἡ ΜΧ ἐκ δύο ὀνομάτων ἐστίν.

Ἐπεὶ γὰρ σύμμετρόξ ἐστιν ἡ ΑΗ τῇ ΗΕ, σύμμετρόξ ἐστι καὶ ἡ ΑΕ ἑκατέρᾳ τῶν ΑΗ, ΗΕ. ὑπόκειται δὲ καὶ ἡ ΑΕ τῇ ΑΒ σύμμετρος· καὶ αἱ ΑΗ, ΗΕ ἄρα τῇ ΑΒ σύμμετροί εἰσιν. καί ἐστι ὁρητὴ ἡ ΑΒ· ὁρητὴ ἄρα ἐστὶ καὶ ἑκατέρα τῶν ΑΗ, ΗΕ· ὁρητὸν ἄρα ἐστὶν ἑκάτερον τῶν ΑΘ, ΗΚ, καί ἐστι σύμμετρον τὸ ΑΘ τῷ ΗΚ. ἀλλὰ τὸ μὲν ΑΘ τῷ ΣΝ ἴσον ἐστίν, τὸ δὲ ΗΚ τῷ ΝΠ· καὶ τὸ ΣΝ ἄρα τῷ ΝΠ σύμμετρόν ἐστιν. ὡξ δὲ τὸ ΣΝ πρὸς ΝΠ, οὁύτως ἡ ΜΝ πρὸς ΝΧ· καὶ ὡξ ἄρα ἡ ΜΝ πρὸς ΝΧ, οὁύτως ἡ ΜΝ πρὸς ΡΝ· σύμμετρόξ ἐστιν ἡ ΜΝ τῇ ΡΝ μήκει. καί ἐστιν ἡ ΜΝ τῇ ΝΟ μήκει ἐκκειμένῃ ὁρητῇ τῇ ΜΟ μήκει. καὶ ἡ ΡΝ ἄρα τῇ ΝΟ μήκει ἐκκειμένῃ ὁρητῇ τῇ ΜΟ μήκει σύμμετρόξ ἐστιν. ὁρητὴ δὲ ἑκατέρα τῶν ΜΝ, ΝΟ· ὁρηταὶ δὲ καὶ δυνάμει μόνον σύμμετροι εἰσὶν αἱ ΜΝ, ΝΟ, ἐπειδήπερ ἡ ΜΝ τῆς ΝΟ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ ἡ ΜΝ σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῇ τῇ ΜΟ μήκει· ἡ ΜΟ ἄρα ἐκ δύο ὀνομάτων ἐστὶ πρώτη. ἔστω αὐτῆς διαίρεσις τὸ Τ, καὶ ὀέστω τὸ μεῖζον ὄνομα τὸ ΜΤ· αἱ ΜΤ, ΤΟ ἄρα ὁηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΜΤ τῆς ΤΟ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ ἡ ΜΤ σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῇ τῇ ΜΟ μήκει. τετμήσθω δὴ ἡ ΤΟ δίθα κατὰ τὸ Υ σημεῖον. καὶ ἐπεὶ ἡ ΜΤ τῆς ΤΟ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος, τουτέστι τῷ ἀπὸ τῆς ΤΥ, ἴσον παρὰ τὴν μείζονα τὴν ΜΤ παραβληθῇ ἐλλεῖπον εοίδει τετραγώνῳ, εἰξ σύμμετρα αὐτὴν διαιρεῖ. παραβεβλήσθω ο῾ν παρὰ τὴν ΜΤ τῷ ἀπὸ τῆς ΤΥ ἴσον τὸ ὑπὸ ΜΦ, ΦΤ· σύμμετρος ἄρα ἐστὶν ἡ ΜΦ τῇ ΦΤ μήκει. καὶ ἤχθωσαν ἀπὸ τῶν Φ, Τ, Υ ὁποτέρᾳ τῶν ΜΝ, ΝΟ παράλληλοι αἱ ΦΧ, ΤΨ, ΥΩ· καὶ τῷ μὲν ΜΧ παραλληλογράμμῳ ἴσον τετράγωνον συνεστάτω τὸ ΣΝ, τῷ δὲ ΦΨ ἴσον τὸ ΝΠ, καὶ κείσθω ὁώστε ἐπ΄ εὐθείας εοῖναι τὴν ΜΝ τῇ ΝΧ· ἐπ΄ εὐθείας ἄρα ἐστὶ καὶ ἡ ΡΝ τῇ ΝΟ. καὶ συμπεπληρώσθω τὸ ΣΠ παραλληλόγραμμον· τετράγωνον ἄρα ἐστὶ τὸ ΣΠ. καὶ ἐπεὶ τὸ ὑπὸ τῶν ΜΦ, ΦΤ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΤΥ, ὀέστιν ἄρα ὡξ ἡ ΜΦ πρὸς ΤΥ, οὁύτως ἡ ΤΥ πρὸς ΦΤ· καὶ ὡξ ἄρα τὸ ΜΧ πρὸς ΤΩ, τὸ ΤΩ πρὸς ΦΨ· τῶν ΜΧ, ΦΨ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΤΩ. ἀλλὰ τὸ μὲν ΜΧ ἴσον ἐστὶ τῷ ΣΝ, τὸ δὲ ΦΨ ἴσον τῷ ΝΠ· τῶν ΣΝ, ΝΠ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΤΩ. ὀέστι δὲ τῶν αὐτῶν τῶν ΣΝ, ΝΠ μέσον ἀνάλογον καὶ τὸ ΜΡ· ἴσον ἄρα ἐστὶ τὸ ΤΩ τῷ ΜΡ· ὁώστε καὶ τῷ ΟΧ ἴσον ἐστίν. ὀέστι δὲ καὶ τὰ ΜΧ, ΦΨ τοῖς ΣΝ, ΝΠ ἴσα· ὅλον ἄρα τὸ ΜΤ ἴσον ἐστὶν ὅλῳ τῷ ΣΠ, τουτέστι τῷ ἀπὸ τῆς ΜΧ τετραγώνῳ· τὸ ΜΤ ἄρα δύναται ἡ ΜΧ. λέγω, ὅτι ἡ ΜΧ ἐκ δύο ὀνομάτων ἐστὶν ἑτέρως ἢ ἡ ΜΟ, τουτέστιν οὐκ ἔστιν ἡ ΜΟ ἐν ἑνὶ μόνῳ σημείῳ διαιρεθεῖσα εἰξ τὰ ὀνόματα εἰξ ἃ καὶ ἡ ΜΧ διαιρεθήσεται. εἰ γὰρ δυνατόν, ἔστω αὐτῆς διαίρεσις τὸ Ξ, καὶ ὀέστω τὸ μεῖζον ὄνομα τὸ ΜΞ· αἱ ΜΞ, ΞΟ ἄρα ὁηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΜΞ τῆς ΞΟ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ ἡ ΜΞ σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῇ τῇ ΜΟ μήκει. τετμήσθω δὴ ἡ ΞΟ δίθα κατὰ τὸ Ψ σημεῖον. καὶ ἐπεὶ ἡ ΜΞ τῆς ΞΟ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος, τουτέστι τῷ ἀπὸ τῆς ΞΨ, ἴσον παρὰ τὴν μείζονα τὴν ΜΞ παραβληθῇ ἐλλεῖπον εοίδει τετραγώνῳ, εἰξ σύμμετρα αὐτὴν διαιρεῖ. παραβεβλήσθω ο῾ν παρὰ τὴν ΜΞ τῷ ἀπὸ τῆς ΞΨ ἴσον τὸ ὑπὸ ΜΖ, ΖΞ· σύμμετρος ἄρα ἐστὶν ἡ ΜΖ τῇ ΖΞ μήκει. καὶ ἤχθωσαν ἀπὸ τῶν Ζ, Ξ, Ψ ὁποτέρᾳ τῶν ΜΝ, ΝΟ παράλληλοι αἱ ΖΓ, ΞΔ, ΨΒ· καὶ τῷ μὲν ΜΓ παραλληλογράμμῳ ἴσον τετράγωνον συνεστάτω τὸ ΣΝ, τῷ δὲ ΖΔ ἴσον τὸ ΝΠ, καὶ κείσθω ὁώστε ἐπ΄ εὐθείας εοῖναι τὴν ΜΝ τῇ ΝΓ· ἐπ΄ εὐθείας ἄρα ἐστὶ καὶ ἡ ΡΝ τῇ ΝΟ. καὶ συμπεπληρώσθω τὸ ΣΠ παραλληλόγραμμον· τετράγωνον ἄρα ἐστὶ τὸ ΣΠ. καὶ ἐπεὶ τὸ ὑπὸ τῶν ΜΖ, ΖΞ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΞΨ, ὀέστιν ἄρα ὡξ ἡ ΜΖ πρὸς ΞΨ, οὁύτως ἡ ΞΨ πρὸς ΖΞ· καὶ ὡξ ἄρα τὸ ΜΓ πρὸς ΞΒ, τὸ ΞΒ πρὸς ΖΔ· τῶν ΜΓ, ΖΔ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΞΒ. ἀλλὰ τὸ μὲν ΜΓ ἴσον ἐστὶ τῷ ΣΝ, τὸ δὲ ΖΔ ἴσον τῷ ΝΠ· τῶν ΣΝ, ΝΠ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΞΒ. ὀέστι δὲ τῶν αὐτῶν τῶν ΣΝ, ΝΠ μέσον ἀνάλογον καὶ τὸ ΜΡ· ἴσον ἄρα ἐστὶ τὸ ΞΒ τῷ ΜΡ· ὁώστε καὶ τῷ ΟΓ ἴσον ἐστίν. ὀέστι δὲ καὶ τὰ ΜΓ, ΖΔ τοῖς ΣΝ, ΝΠ ἴσα· ὅλον ἄρα τὸ ΜΞ ἴσον ἐστὶν ὅλῳ τῷ ΣΠ, τουτέστι τῷ ἀπὸ τῆς ΜΓ τετραγώνῳ· τὸ ΜΞ ἄρα δύναται ἡ ΜΓ. ἐδείχθη δὲ καὶ τὸ ΜΤ τὸ αὐτὸ τῷ ἀπὸ τῆς ΜΧ τετραγώνῳ· τὸ ΜΤ ἄρα δύναται ἡ ΜΧ. ἐδείχθη δὲ καὶ τὸ ΜΞ τὸ αὐτὸ

τῷ ΝΠ· καὶ τὰ ΣΝ, ΝΠ ἄρα, τουτέστι τὰ ἀπὸ τῶν ΜΝ, ΝΧ, ὁρητά ἐστι καὶ σύμμετρα. καὶ ἐπεὶ ἀσύμμετρόξ ἐστιν ἡ ΑΕ τῇ ΕΔ μήκει, ἀλλὰ ἡ μὲν ΑΕ τῇ ΑΗ ἐστι σύμμετρος, ἡ δὲ ΔΕ τῇ ΕΖ σύμμετρος, ἀσύμμετρος ἄρα καὶ ἡ ΑΗ τῇ ΕΖ· ὥστε καὶ τὸ ΑΘ τῷ ΕΛ ἀσύμμετρόν ἐστιν. ἀλλὰ τὸ μὲν ΑΘ τῷ ΣΝ ἐστιν ἴσον, τὸ δὲ ΕΛ τῷ ΜΡ· καὶ τὸ ΣΝ ἄρα τῷ ΜΡ ἀσύμμετρόν ἐστιν. ἀλλὰ ὡξ τὸ ΣΝ πρὸς ΜΡ, ἡ ΟΝ πρὸς τὴν ΝΡ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΟΝ τῇ ΝΡ. ὁίση δὲ ἡ μὲν ΟΝ τῇ ΜΝ, ἡ δὲ ΝΡ τῇ ΝΧ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΜΝ τῇ ΝΧ. καί ἐστι τὸ ἀπὸ τῆς ΜΝ σύμμετρον τῷ ἀπὸ τῆς ΝΧ, καὶ ὁρητὸν ἑκάτερον· αἱ ΜΝ, ΝΧ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι.

Ἡ ΜΧ ἄρα ἐκ δύο ὀνομάτων ἐστὶ καὶ δύναται τὸ ΑΓ· ὅπερ ἔδει δεῖχαι.

AH plus GK is equal to SN plus NQ. Thus, the whole of AC is equal to the whole of SQ—that is to say, to the square on MO. Thus, MO (is) the square-root of (area) AC. I say that MO is a binomial (straight-line).

For since AG is commensurable (in length) with GE, AE is also commensurable (in length) with each of AG and GE [Prop. 10.15]. And AE was also assumed (to be) commensurable (in length) with AB. Thus, AG and GE are also commensurable (in length) with AB [Prop. 10.12]. And AB is rational. AG and GE are thus each also rational. Thus, AH and GK are each rational (areas), and AH is commensurable with GK [Prop. 10.19]. But, AH is equal to SN, and GK to NQ. SN and NQ—that is to say, the (squares) on MN and NO (respectively)—are thus also rational and commensurable. And since AE is incommensurable in length with ED, but AE is commensurable (in length) with AG, and DE (is) commensurable (in length) with EF, AG (is) thus also incommensurable (in length) with EF [Prop. 10.13]. Hence, AH is also incommensurable with EL [Props. 6.1, 10.11]. But, AH is equal to SN, and EL to MR. Thus, SN is also incommensurable with MR. But, as SN (is) to MR, (so) PN (is) to NR [Prop. 6.1]. PN is thus incommensurable (in length) with NR [Prop. 10.11]. And PN (is) equal to MN, and NR to NO. Thus, MN is incommensurable (in length) with NO. And the (square) on MN is commensurable with the (square) on NO, and each (is) rational. MN and NO are thus rational (straight-lines which are) commensurable in square only.

Thus, MO is (both) a binomial (straight-line) [Prop. 10.36], and the square-root of AC. (Which is) the very thing it was required to show.

† If the rational straight-line has unit length then this proposition states that the square-root of a first binomial straight-line is a binomial straight-line: i.e., a first binomial straight-line has a length $k + k\sqrt{1-k'^2}$ whose square-root can be written $\rho(1+\sqrt{k''})$, where $\rho = \sqrt{k(1+k')/2}$ and $k'' = (1-k')/(1+k')$. This is the length of a binomial straight-line (see Prop. 10.36), since ρ is rational.

55.

Ἐὰν θωρίον περιέθηται ὑπὸ ὁρητῆς καὶ τῆς ἐκ δύο ὀνομάτων δευτέρας, ἡ τὸ θωρίον δυναμένηὀάλογόξ ἐστιν ἡ καλουμένη ἐκ δύο μέσων πρώτη.

Proposition 55

If an area is contained by a rational (straight-line) and a second binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called first bimedial.†

Περιειθέσθω γὰρ θωρίον τὸ ΑΒΓΔ ὑπὸ ὁρητῆς τῆς ΑΒ καὶ τῆς ἐκ δύο ὀνομάτων δυετέρας τῆς ΑΔ· λέγω, ὅτι ἡ τὸ ΑΓ θωρίον δυναμένη ἐκ δύο μέσων πρώτη ἐστίν.

Ἐπεὶ γὰρ ἐκ δύο ὀνομάτων δευτέρα ἐστὶν ἡ ΑΔ, διῃρήσθω εἰξ τὰ ὀνόματα κατὰ τὸ Ε, ὥστε τὸ μεῖζον ὄνομα εὐῖναι τὸ ΑΕ· αἱ ΑΕ, ΕΔ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΑΕ τῆς ΕΔ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ τὸ ὀέλαττον ὄνομα ἡ ΕΔ σύμμετρόν ἐστι τῇ ΑΒ μήκει. τετμήσθω ἡ ΕΔ δίθα κατὰ τὸ Ζ, καὶ τῷ ἀπὸ τῆς ΕΖ ἴσον παρὰ τὴν ΑΕ παραβεβλήσθω ἐλλεῖπον εὐίδει τετραγώνῳ τὸ ὑπὸ τῶν ΑΗΕ· σύμμετρος ἄρα ἡ ΑΗ τῇ ΗΕ μήκει. καὶ διὰ τῶν Η, Ε, Ζ παράλληλοι ἤθθωσαν ταῖς ΑΒ, ΓΔ αἱ ΗΘ, ΕΚ, ΖΛ, καὶ τῷ μὲν ΑΘ παραλληλογράμμῳ ἴσον τετράγωνον συνεστάτω τὸ ΣΝ, τῷ δὲ ΗΚ ἴσον τετράγωνον τὸ ΝΠ, καὶ κείσθω ὥστε ἐπ΄ εὐθείας εὐῖναι τὴν ΜΝ τῇ ΝΧ· ἐπ΄ εὐθείας ἄρα [ἐστὶ] καὶ ἡ ΡΝ τῇ ΝΟ. καὶ συμπεπληρώσθω τὸ ΣΠ τετράγωνον· φανερὸν δὴ ἐκ τοῦ προδεδειγμένου, ὅτι τὸ ΜΡ μέσον ἀνάλογόν ἐστι τῶν ΣΝ, ΝΠ, καὶ ἴσον τῷ ΕΛ, καὶ ὅτι τὸ ΑΓ θωρίον δύναται ἡ ΜΧ. δεικτέον δή, ὅτι ἡ ΜΧ ἐκ δύο μέσων ἐστὶ πρώτη.

Ἐπεὶ ἀσύμμετρόξ ἐστιν ἡ ΑΕ τῇ ΕΔ μήκει, σύμμετρος δὲ ἡ ΕΔ τῇ ΑΒ, ἀσύμμετρος ἄρα ἡ ΑΕ τῇ ΑΒ. καὶ ἐπεὶ σύμμετρόξ ἐστιν ἡ ΑΗ τῇ ΕΗ, σύμμετρόξ ἐστι καὶ ἡ ΑΕ ἑκατέρᾳ τῶν ΑΗ, ΗΕ. ἀλλὰ ἡ ΑΕ ἀσύμμετρος τῇ ΑΒ μήκει· καὶ αἱ ΑΗ, ΗΕ ἄρα ἀσύμμετροί εἰσι τῇ ΑΒ. αἱ ΒΑ, ΑΗ, ΗΕ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ὥστε μέσον ἐστὶν ἑκάτερον τῶν ΑΘ, ΗΚ. ὥστε καὶ ἑκάτερον τῶν ΣΝ, ΝΠ μέσον ἐστίν. καὶ αἱ ΜΝ, ΝΧ ἄρα μέσαι εἰσίν. καὶ ἐπεὶ σύμμετρος ἡ ΑΗ τῇ ΗΕ μήκει, σύμμετρόν ἐστι καὶ τὸ ΑΘ τῷ ΗΚ, τουτέστι τὸ ΣΝ τῷ ΝΠ, τουτέστι τὸ ἀπὸ τῆς ΜΝ τῷ ἀπὸ τῆς ΝΧ [ὥστε δυνάμει εἰσὶ σύμμετροι αἱ ΜΝ, ΝΧ]. καὶ ἐπεὶ ἀσύμμετρόξ ἐστιν ἡ ΑΕ τῇ ΕΔ μήκει, ἀλλὸ ἡ μὲν ΑΕ σύμμετρόξ ἐστι τῇ ΑΗ, ἡ δὲ ΕΔ τῇ ΕΖ σύμμετρος, ἀσύμμετρος ἄρα ἡ ΑΗ τῇ ΕΖ· ὥστε καὶ τὸ ΑΘ τῷ ΕΛ ἀσύμμετρόν ἐστιν, τουτέστι τὸ ΣΝ τῷ ΜΡ, τουτέστιν ὁ ΟΝ τῇ ΝΡ, τουτέστιν ἡ ΜΝ τῇ ΝΧ ἀσύμμετρόξ ἐστι μήκει. ἐδείθθησαν δὲ αἱ ΜΝ, ΝΧ καὶ μέσαι οὐῦσαι καὶ δυνάμει σύμμετροι· αἱ ΜΝ,

For let the area $ABCD$ be contained by the rational (straight-line) AB and by the second binomial (straight-line) AD. I say that the square-root of area AC is a first bimedial (straight-line).

For since AD is a second binomial (straight-line), let it be divided into its (component) terms at E, such that AE is the greater term. Thus, AE and ED are rational (straight-lines which are) commensurable in square only, and the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE), and the lesser term ED is commensurable in length with AB [Def. 10.6]. Let ED be cut in half at F. And let the (rectangle contained) by AGE, equal to the (square) on EF, be applied to AE, falling short by a square figure. AG (is) thus commensurable in length with GE [Prop. 10.17]. And let GH, EK, and FL be drawn through (points) G, E, and F (respectively), parallel to AB and CD. And let the square SN, equal to the parallelogram AH, be constructed, and the square NQ, equal to GK. And let MN be laid down so as to be straight-on to NO. Thus, RN [is] also straight-on to NP. And let the square SQ be completed. So, (it is) clear from what has been previously demonstrated [Prop. 10.53 lem.] that MR is the mean proportional to SN and NQ, and (is) equal to EL, and that MO is the square-root of the area AC. So, we must show that MO is a first bimedial (straight-line).

Since AE is incommensurable in length with ED, and ED (is) commensurable (in length) with AB, AE (is) thus incommensurable (in length) with AB [Prop. 10.13]. And since AG is commensurable (in length) with EG, AE is also commensurable (in length) with each of AG and GE [Prop. 10.15]. But, AE is incommensurable in length with AB. Thus, AG and GE are also (both) incommensurable (in length) with AB [Prop. 10.13]. Thus, BA, AG, and (BA, and) GE are (pairs of) rational (straight-lines which are) commensurable in square only. And, hence, each of AH and GK is a medial (area) [Prop. 10.21]. Hence, each of SN and NQ is also a medial (area). Thus, MN and NO are medial (straight-lines). And

ΝΧ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. λέγω δή, ὅτι καὶ ὁρητὸν περιέθουσιν. ἐπεὶ γὰρ ἡ ΔΕ ὑπόκειται ἑκατέρᾳ τῶν ΑΒ, ΕΖ σύμμετρος, σύμμετρος ἄρα καὶ ἡ ΕΖ τῇ ΕΚ. καὶ ὁρητὴ ἑκατέρα αὐτῶν· ὁρητὸν ἄρα τὸ ΕΛ, τουτέστι τὸ ΜΡ· τὸ δὲ ΜΡ ἐστι τὸ ὑπὸ τῶν ΜΝΧ. ἐὰν δὲ δύο μέσαι δυνάμει μόνον σύμμετροι συντεθῶσι ὁρητὸν περιέθουσαι, ἡ ὅληὀάλογόξ ἐστιν, καλεῖται δὲ ἐκ δύο μέσων πρώτη.

Ἡ ἄρα ΜΧ ἐκ δύο μέσων ἐστὶ πρώτη· ὅπερ ἔδει δεῖξαι.

since AG (is) commensurable in length with GE, AH is also commensurable with GK—that is to say, SN with NQ—that is to say, the (square) on MN with the (square) on NO [hence, MN and NO are commensurable in square] [Props. 6.1, 10.11]. And since AE is incommensurable in length with ED, but AE is commensurable (in length) with AG, and ED commensurable (in length) with EF, AG (is) thus incommensurable (in length) with EF [Prop. 10.13]. Hence, AH is also incommensurable with EL—that is to say, SN with MR—that is to say, PN with NR—that is to say, MN is incommensurable in length with NO [Props. 6.1, 10.11]. But MN and NO have also been shown to be medial (straight-lines) which are commensurable in square. Thus, MN and NO are medial (straight-lines which are) commensurable in square only. So, I say that they also contain a rational (area). For since DE was assumed (to be) commensurable (in length) with each of AB and EF, EF (is) thus also commensurable with EK [Prop. 10.12]. And they (are) each rational. Thus, EL—that is to say, MR—(is) rational [Prop. 10.19]. And MR is the (rectangle contained) by MNO. And if two medial (straight-lines), commensurable in square only, which contain a rational (area), are added together, then the whole is (that) irrational (straight-line which is) called first bimedial [Prop. 10.37].

Thus, MO is a first bimedial (straight-line). (Which is) the very thing it was required to show.

† If the rational straight-line has unit length then this proposition states that the square-root of a second binomial straight-line is a first bimedial straight-line: i.e., a second binomial straight-line has a length $k/\sqrt{1-k'^2}+k$ whose square-root can be written $\rho\,(k''^{1/4}+k''^{3/4})$, where $\rho=\sqrt{(k/2)\,(1+k')/(1-k')}$ and $k''=(1-k')/(1+k')$. This is the length of a first bimedial straight-line (see Prop. 10.37), since ρ is rational.

56.

Ἐὰν ϑωρίον περιέϑηται ὑπὸ ὁρητῆς καὶ τῆς ἐκ δύο ὀνομάτων τρίτης, ἡ τὸ ϑωρίον δυναμένηὀάλογόξ ἐστιν ἡ καλουμένη ἐκ δύο μέσων δευτέρα.

Θωρίον γὰρ τὸ ΑΒΓΔ περιεϑέσϑω ὑπὸ ὁρητῆς τῆς ΑΒ καὶ τῆς ἐκ δύο ὀνομάτων τρίτης τῆς ΑΔ διῃρημένης εἰξ τὰ ὀνόματα κατὰ τὸ Ε, ὧν μεῖζόν ἐστι τὸ ΑΕ· λέγω, ὅτι ἡ τὸ ΑΓ ϑωρίον δυναμένηὀάλογόξ ἐστιν ἡ καλουμένη ἐκ δύο μέσων δευτέρα.

Proposition 56

If an area is contained by a rational (straight-line) and a third binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called second bimedial.†

For let the area $ABCD$ be contained by the rational (straight-line) AB and by the third binomial (straight-line) AD, which has been divided into its (component) terms at E, of which AE is the greater. I say that the square-root of area AC is the irrational (straight-line which is) called second bimedial.

Κατεσκευάσθω γὰρ τὰ αὐτὰ τοῖς πρότερον. καὶ ἐπεὶ ἐκ δύο ὀνομάτων ἐστὶ τρίτη ἡ $A\Delta$, αἱ AE, $E\Delta$ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ AE τῆς $E\Delta$ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ οὐδετέρα τῶν AE, $E\Delta$ σύμμετρός [ἐστι] τῇ AB μήκει. ὁμοίως δὴ τοῖς προδεδειγμένοις δείχομεν, ὅτι ἡ MX ἐστιν ἡ τὸ $A\Gamma$ θωρίον δυναμένη, καὶ αἱ MN, NX μέσαι εἰσὶ δυνάμει μόνον σύμμετροι· ὥστε ἡ MX ἐκ δύο μέσων ἐστίν. δεικτέον δή, ὅτι καὶ δευτέρα.

[Καὶ] ἐπεὶ ἀσύμμετρός ἐστιν ἡ ΔE τῇ AB μήκει, τουτέστι τῇ EK, σύμμετρος δὲ ἡ ΔE τῇ EZ, ἀσύμμετρος ἄρα ἐστὶν ἡ EZ τῇ EK μήκει. καί εἰσι ῥηταί· αἱ ZE, EK ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι. μέσον ἄρα [ἐστὶ] τὸ $E\Lambda$, τουτέστι τὸ MP· καὶ περιέχεται ὑπὸ τῶν $MN\Xi$· μέσον ἄρα ἐστὶ τὸ ὑπὸ τῶν $MN\Xi$.

Ἡ MX ἄρα ἐκ δύο μέσων ἐστὶ δευτέρα· ὅπερ ἔδει δεῖξαι.

For let the same construction be made as previously. And since AD is a third binomial (straight-line), AE and ED are thus rational (straight-lines which are) commensurable in square only, and the square on AE is greater than (the square on) ED by the (square) on (some straight-line) commensurable (in length) with (AE), and neither of AE and ED [is] commensurable in length with AB [Def. 10.7]. So, similarly to that which has been previously demonstrated, we can show that MO is the square-root of area AC, and MN and NO are medial (straight-lines which are) commensurable in square only. Hence, MO is bimedial. So, we must show that (it is) also second (bimedial).

[And] since DE is incommensurable in length with AB—that is to say, with EK—and DE (is) commensurable (in length) with EF, EF is thus incommensurable in length with EK [Prop. 10.13]. And they are (both) rational (straight-lines). Thus, FE and EK are rational (straight-lines which are) commensurable in square only. EL—that is to say, MR—[is] thus medial [Prop. 10.21]. And it is contained by MNO. Thus, the (rectangle contained by) MNO is medial.

Thus, MO is a second bimedial (straight-line) [Prop. 10.38]. (Which is) the very thing it was required to show.

† If the rational straight-line has unit length then this proposition states that the square-root of a third binomial straight-line is a second bimedial straight-line: i.e., a third binomial straight-line has a length $k^{1/2}\,(1 + \sqrt{1 - k'^2})$ whose square-root can be written $\rho\,(k^{1/4} + k''^{1/2}/k^{1/4})$, where $\rho = \sqrt{(1 + k')/2}$ and $k'' = k\,(1 - k')/(1 + k')$. This is the length of a second bimedial straight-line (see Prop. 10.38), since ρ is rational.

57.

Ἐὰν θωρίον περιέθηται ὑπὸ ὁρητῆς καὶ τῆς ἐκ δύο ὀνομάτων τετάρτης, ἡ τὸ θωρίον δυναμένηὀάλογόξ ἐστιν ἡ καλουμένη μείζων.

Proposition 57

If an area is contained by a rational (straight-line) and a fourth binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called major.†

Θωρίον γὰρ τὸ ΑΓ περιειθέσθω ὑπὸ ὁρητῆς τῆς ΑΒ καὶ τῆς ἐκ δύο ὀνομάτων τετάρτης τῆς ΑΔ διῃρημένης εἰξ τὰ ὀνόματα κατὰ τὸ Ε, ὧν μεῖζον ὸέστω τὸ ΑΕ· λέγω, ὅτι ἡ τὸ ΑΓ θωρίον δυναμένηὸάλογός ἐστιν ἡ καλουμένη μείζων.

Ἐπεὶ γὰρ ἡ ΑΔ ἐκ δύο ὀνομάτων ἐστὶ τετάρτη, αἱ ΑΕ, ΕΔ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΑΕ τῆς ΕΔ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, καὶ ἡ ΑΕ τῇ ΑΒ σύμμετρός [ἐστι] μήκει. τετμήσθω ἡ ΔΕ δίθα κατὰ τὸ Ζ, καὶ τῷ ἀπὸ τῆς ΕΖ ἴσον παρὰ τὴν ΑΕ παραβεβλήσθω παραλληλόγραμμον τὸ ὑπὸ ΑΗ, ΗΕ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΑΗ τῇ ΗΕ μήκει. ἤχθωσαν παράλληλοι τῇ ΑΒ αἱ ΗΘ, ΕΚ, ΖΛ, καὶ τὰ λοιπὰ τὰ αὐτὰ τοῖς πρὸ τούτου γεγονέτω· φανερὸν δή, ὅτι ἡ τὸ ΑΓ θωρίον δυναμένη ἐστὶν ἡ ΜΧ. δεικτέον δή, ὅτι ἡ ΜΧὸάλογός ἐστιν ἡ καλουμένη μείζων.

Ἐπεὶ ἀσύμμετρός ἐστιν ἡ ΑΗ τῇ ΕΗ μήκει, ἀσύμμετρόν ἐστι καὶ τὸ ΑΘ τῷ ΗΚ, τουτέστι τὸ ΣΝ τῷ ΝΠ· αἱ ΜΝ, ΝΧ ἄρα δυνάμει εἰσὶν ἀσύμμετροι. καὶ ἐπεὶ σύμμετρός ἐστιν ἡ ΑΕ τῇ ΑΒ μήκει, ὁρητόν ἐστι τὸ ΑΚ· καί ἐστιν ἴσον τοῖς ἀπὸ τῶν ΜΝ, ΝΧ· ὁρητὸν ἄρα [ἐστὶ] καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΜΝ, ΝΧ. καὶ ἐπεὶ ἀσύμμετρός [ἐστιν] ἡ ΔΕ τῇ ΑΒ μήκει, τουτέστι τῇ ΕΚ, ἀλλὰ ἡ ΔΕ σύμμετρός ἐστι τῇ ΕΖ, ἀσύμμετρος ἄρα ἡ ΕΖ τῇ ΕΚ μήκει. αἱ ΕΚ, ΕΖ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· μέσον ἄρα τὸ ΛΕ, τουτέστι τὸ ΜΡ. καὶ περιέθεται ὑπὸ τῶν ΜΝ, ΝΧ· μέσον ἄρα ἐστὶ τὸ ὑπὸ τῶν ΜΝ, ΝΧ. καὶ ὁρητὸν τὸ [συγκείμενον] ἐκ τῶν ἀπὸ τῶν ΜΝ, ΝΧ, καί εἰσιν ἀσύμμετροι αἱ ΜΝ, ΝΧ δυνάμει. ἐὰν δὲ δύο εὐθεῖαι δυνάμει ἀσύμμετροι συντεθῶσι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων ὁρητόν, τὸ δὸ ὑπὸ αὐτῶν μέσον, ἡ ὅληὸάλογός ἐστιν, καλεῖται δὲ μείζων.

Ἡ ΜΧ ἄραὸάλογός ἐστιν ἡ καλουμένη μείζων, καὶ δύναται τὸ ΑΓ θωρίον· ὅπερ ἔδει δεῖχαι.

For let the area AC be contained by the rational (straight-line) AB and the fourth binomial (straight-line) AD, which has been divided into its (component) terms at E, of which let AE be the greater. I say that the square-root of AC is the irrational (straight-line which is) called major.

For since AD is a fourth binomial (straight-line), AE and ED are thus rational (straight-lines which are) commensurable in square only, and the square on AE is greater than (the square on) ED by the (square) on (some straight-line) incommensurable (in length) with (AE), and AE [is] commensurable in length with AB [Def. 10.8]. Let DE be cut in half at F, and let the parallelogram (contained by) AG and GE, equal to the (square) on EF, (and falling short by a square figure) be applied to AE. AG is thus incommensurable in length with GE [Prop. 10.18]. Let GH, EK, and FL be drawn parallel to AB, and let the rest (of the construction) be made the same as the (proposition) before this. So, it is clear that MO is the square-root of area AC. So, we must show that MO is the irrational (straight-line which is) called major.

Since AG is incommensurable in length with EG, AH is also incommensurable with GK—that is to say, SN with NQ [Props. 6.1, 10.11]. Thus, MN and NO are incommensurable in square. And since AE is commensurable in length with AB, AK is rational [Prop. 10.19]. And it is equal to the (sum of the squares) on MN and NO. Thus, the sum of the (squares) on MN and NO [is] also rational. And since DE [is] incommensurable in length with AB [Prop. 10.13]—that is to say, with EK—but DE is commensurable (in length) with EF, EF (is) thus incommensurable in length with EK [Prop. 10.13]. Thus, EK and EF are rational (straight-lines which are) commensurable in square only. LE—that is to say, MR—(is) thus medial [Prop. 10.21]. And it is contained by MN and NO. The (rectangle contained) by MN and NO is thus medial. And the [sum] of the (squares) on MN and NO (is) rational, and MN and NO are incommensurable in square. And if two straight-lines (which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained)

by them medial, are added together, then the whole is the irrational (straight-line which is) called major [Prop. 10.39].

Thus, MO is the irrational (straight-line which is) called major. And (it is) the square-root of area AC. (Which is) the very thing it was required to show.

† If the rational straight-line has unit length then this proposition states that the square-root of a fourth binomial straight-line is a major straight-line: i.e., a fourth binomial straight-line has a length $k\,(1 + 1/\sqrt{1 + k'})$ whose square-root can be written $\rho\,\sqrt{[1 + k''/(1 + k''^2)^{1/2}]/2} + \rho\,\sqrt{[1 - k''/(1 + k''^2)^{1/2}]/2}$, where $\rho = \sqrt{k}$ and $k''^2 = k'$. This is the length of a major straight-line (see Prop. 10.39), since ρ is rational.

58.

Ἐὰν ϑωρίον περιέϑηται ὑπὸ ῥητῆς καὶ τῆς ἐκ δύο ὀνομάτων πέμπτης, ἡ τὸ ϑωρίον δυναμένη ἀλογόν ἐστιν ἡ καλουμένη ῥητὸν καὶ μέσον δυναμένη.

Θωρίον γὰρ τὸ ΑΓ περιεϑέσθω ὑπὸ ῥητῆς τῆς ΑΒ καὶ τῆς ἐκ δύο ὀνομάτων πέμπτης τῆς ΑΔ διῃρημένης εἰς τὰ ὀνόματα κατὰ τὸ Ε, ὥστε τὸ μεῖζον ὄνομα εἶναι τὸ ΑΕ· λέγω [δή], ὅτι ἡ τὸ ΑΓ ϑωρίον δυναμένη ἀλογόν ἐστιν ἡ καλουμένη ῥητὸν καὶ μέσον δυναμένη.

Κατεσκευάσθω γὰρ τὰ αὐτὰ τοῖς πρότερον δεδειγμένοις· φανερὸν δή, ὅτι ἡ τὸ ΑΓ ϑωρίον δυναμένη ἐστὶν ἡ ΜΧ. δεικτέον δή, ὅτι ἡ ΜΧ ἐστιν ἡ ῥητὸν καὶ μέσον δυναμένη.

Proposition 58

If an area is contained by a rational (straight-line) and a fifth binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called the square-root of a rational plus a medial (area).†

For let the area AC be contained by the rational (straight-line) AB and the fifth binomial (straight-line) AD, which has been divided into its (component) terms at E, such that AE is the greater term. [So] I say that the square-root of area AC is the irrational (straight-line which is) called the square-root of a rational plus a medial (area).

For let the same construction be made as that shown previously. So, (it is) clear that MO is the square-root of area AC. So, we must show that MO is the square-root of a rational plus a medial (area).

Ἐπεὶ γὰρ ἀσύμμετρός ἐστιν ἡ ΑΗ τῇ ΗΕ, ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ΑΘ τῷ ΘΕ, τουτέστι τὸ ἀπὸ τῆς ΜΝ τῷ ἀπὸ τῆς ΝΧ· αἱ ΜΝ, ΝΧ ἄρα δυνάμει εἰσὶν ἀσύμμετροι. καὶ ἐπεὶ ἡ ΑΔ ἐκ δύο ὀνομάτων ἐστὶ πέμπτη, καί [ἐστιν] ὀέλασσον αὐτῆς τμῆμα τὸ ΕΔ, σύμμετρος ἄρα ἡ ΕΔ τῇ ΑΒ μήκει. ἀλλὰ ἡ ΑΕ τῇ ΕΔ ἐστιν ἀσύμμετρος· καὶ ἡ ΑΒ ἄρα τῇ ΑΕ ἐστιν ἀσύμμετρος μήκει [αἱ ΒΑ, ΑΕ ὀρηταί εἰσι δυνάμει μόνον σύμμετροι]· μέσον ἄρα ἐστὶ τὸ ΑΚ, τουτέστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΜΝ, ΝΧ. καὶ ἐπεὶ σύμμετρός ἐστιν ἡ ΔΕ τῇ ΑΒ μήκει, τουτέστι τῇ ΕΚ, ἀλλὰ ἡ ΔΕ τῇ ΕΖ σύμμετρός ἐστιν, καὶ ἡ ΕΖ ἄρα τῇ ΕΚ σύμμετρός ἐστιν. καὶ ὀρητὴ ἡ

For since AG is incommensurable (in length) with GE [Prop. 10.18], AH is thus also incommensurable with HE—that is to say, the (square) on MN with the (square) on NO [Props. 6.1, 10.11]. Thus, MN and NO are incommensurable in square. And since AD is a fifth binomial (straight-line), and ED [is] its lesser segment, ED (is) thus commensurable in length with AB [Def. 10.9]. But, AE is incommensurable (in length) with ED. Thus, AB is also incommensurable in length with AE [BA and AE are rational (straight-lines which are) commensurable in square only] [Prop. 10.13]. Thus, AK—that is to say, the sum of the (squares) on MN and NO—is medial

ΕΚ· ὁρητὸν ἄρα καὶ τὸ ΕΛ, τουτέστι τὸ ΜΡ, τουτέστι τὸ ὑπὸ ΜΝΧ· αἱ ΜΝ, ΝΧ ἄρα δυνάμει ἀσύμμετροί εἰσι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον, τὸ δὲ ὑπὸ αὐτῶν ὁρητόν.

Ἡ ΜΧ ἄρα ὁρητὸν καὶ μέσον δυναμένη ἐστὶ καὶ δύναται τὸ ΑΓ θωρίον· ὅπερ ἔδει δεῖξαι.

[Prop. 10.21]. And since DE is commensurable in length with AB—that is to say, with EK—but, DE is commensurable (in length) with EF, EF is thus also commensurable (in length) with EK [Prop. 10.12]. And EK (is) rational. Thus, EL—that is to say, MR—that is to say, the (rectangle contained) by MNO—(is) also rational [Prop. 10.19]. MN and NO are thus (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational.

Thus, MO is the square-root of a rational plus a medial (area) [Prop. 10.40]. And (it is) the square-root of area AC. (Which is) the very thing it was required to show.

† If the rational straight-line has unit length then this proposition states that the square-root of a fifth binomial straight-line is the square root of a rational plus a medial area: i.e., a fifth binomial straight-line has a length $k\,(\sqrt{1+k'}+1)$ whose square-root can be written $\rho\,\sqrt{[(1+k''^2)^{1/2}+k'']/[2\,(1+k''^2)]}+\rho\,\sqrt{[(1+k''^2)^{1/2}-k'']/[2\,(1+k''^2)]}$, where $\rho=\sqrt{k\,(1+k''^2)}$ and $k''^2=k'$. This is the length of the square root of a rational plus a medial area (see Prop. 10.40), since ρ is rational.

59.

Ἐὰν θωρίον περιέθηται ὑπὸ ὁρητῆς καὶ τῆς ἐκ δύο ὀνομάτων ὀέκτης, ἡ τὸ θωρίον δυναμένηὀάλογός ἐστιν ἡ καλουμένη δύο μέσα δυναμένη.

Proposition 59

If an area is contained by a rational (straight-line) and a sixth binomial (straight-line) then the square-root of the area is the irrational (straight-line which is) called the square-root of (the sum of) two medial (areas).†

Θωρίον γὰρ τὸ ΑΒΓΔ περιεθέσθω ὑπὸ ὁρητῆς τῆς ΑΒ καὶ τῆς ἐκ δύο ὀνομάτων ὀέκτης τῆς ΑΔ διῃρημένης εἰξ τὰ ὀνόματα κατὰ τὸ Ε, ὥστε τὸ μεῖζον ὄνομα εὀῖναι τὸ ΑΕ· λέγω, ὅτι ἡ τὸ ΑΓ δυναμένη ἡ δύο μέσα δυναμένη ἐστίν.

Κατεσκευάσθω [γὰρ] τὰ αὐτὰ τοῖς προδεδειγμένοις. φανερὸν δή, ὅτι [ἡ] τὸ ΑΓ δυναμένη ἐστὶν ἡ ΜΧ, καὶ ὅτι ἀσύμμετρόξ ἐστιν ἡ ΜΝ τῇ ΝΧ δυνάμει. καὶ ἐπεὶ ἀσύμμετρόξ ἐστιν ἡ ΕΑ τῇ ΑΒ μήκει, αἱ ΕΑ, ΑΒ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· μέσον ἄρα ἐστὶ τὸ ΑΚ, τουτέστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΜΝ, ΝΧ. πάλιν, ἐπεὶ ἀσύμμετρόξ ἐστιν ἡ ΕΔ τῇ ΑΒ μήκει, ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ ΖΕ τῇ ΕΚ· αἱ ΖΕ, ΕΚ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· μέσον ἄρα ἐστὶ τὸ ΕΛ, τουτέστι τὸ ΜΡ, τουτέστι τὸ ὑπὸ τῶν ΜΝΧ. καὶ

For let the area $ABCD$ be contained by the rational (straight-line) AB and the sixth binomial (straight-line) AD, which has been divided into its (component) terms at E, such that AE is the greater term. So, I say that the square-root of AC is the square-root of (the sum of) two medial (areas).

[For] let the same construction be made as that shown previously. So, (it is) clear that MO is the square-root of AC, and that MN is incommensurable in square with NO. And since EA is incommensurable in length with AB [Def. 10.10], EA and AB are thus rational (straight-lines which are) commensurable in square only. Thus, AK—that is to say, the sum of the (squares) on MN and NO—is medial [Prop. 10.21]. Again, since ED is incommensurable in length with AB

ἐπεὶ ἀσύμμετρος ἡ AE τῇ EZ, καὶ τὸ AK τῷ $EΛ$ ἀσύμμετρόν ἐστιν. ἀλλὰ τὸ μὲν AK ἐστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν MN, NX, τὸ δὲ $EΛ$ ἐστι τὸ ὑπὸ τῶν MNX· ἀσύμμετρον ἄρα ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν MNX τῷ ὑπὸ τῶν MNX. καί ἐστι μέσον ἑκάτερον αὐτῶν, καὶ αἱ MN, NX δυνάμει εἰσὶν ἀσύμμετροι.

Ἡ MX ἄρα δύο μέσα δυναμένη ἐστὶ καὶ δύναται τὸ $AΓ$· ὅπερ ἔδει δεῖξαι.

[Def. 10.10], FE is thus also incommensurable (in length) with EK [Prop. 10.13]. Thus, FE and EK are rational (straight-lines which are) commensurable in square only. Thus, EL—that is to say, MR—that is to say, the (rectangle contained) by MNO—is medial [Prop. 10.21]. And since AE is incommensurable (in length) with EF, AK is also incommensurable with EL [Props. 6.1, 10.11]. But, AK is the sum of the (squares) on MN and NO, and EL is the (rectangle contained) by MNO. Thus, the sum of the (squares) on MNO is incommensurable with the (rectangle contained) by MNO. And each of them is medial. And MN and NO are incommensurable in square.

Thus, MO is the square-root of (the sum of) two medial (areas) [Prop. 10.41]. And (it is) the square-root of AC. (Which is) the very thing it was required to show.

† If the rational straight-line has unit length then this proposition states that the square-root of a sixth binomial straight-line is the square root of the sum of two medial areas: i.e., a sixth binomial straight-line has a length $\sqrt{k} + \sqrt{k'}$ whose square-root can be written $k^{1/4}\left(\sqrt{[1 + k''/(1 + k''^2)^{1/2}]/2} + \sqrt{[1 - k''/(1 + k''^2)^{1/2}]/2}\right)$, where $k''^2 = (k - k')/k'$. This is the length of the square-root of the sum of two medial areas (see Prop. 10.41).

Λῆμμα.

Ἐὰν εὐθεῖα γραμμὴ τμηθῇ εἰξὸάνισα, τὰ ἀπὸ τῶν ἀνίσων τετράγωνα μείζονά ἐστι τοῦ δὶξ ὑπὸ τῶν ἀνίσων περιεχομένου ὀρθογωνίου.

Lemma

If a straight-line is cut unequally then (the sum of) the squares on the unequal (parts) is greater than twice the rectangle contained by the unequal (parts).

Ἔστω εὐθεῖα ἡ AB καὶ τετμήσθω εἰξὸάνισα κατὰ τὸ $Γ$, καὶ ὀέστω μείζων ἡ $AΓ$· λέγω, ὅτι τὰ ἀπὸ τῶν $AΓ$, $ΓB$ μείζονά ἐστι τοῦ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$.

Τετμήσθω γὰρ ἡ AB δίθα κατὰ τὸ $Δ$. ἐπεὶ οὐῦν εὐθεῖα γραμμὴ τέτμ ηται εἶξ μὲν ἴσα κατὰ τὸ $Δ$, εἶξ δὲ ὀάνισα κατὰ τὸ $Γ$, τὸ ἄρα ὑπὸ τῶν $AΓ$, $ΓB$ μετὰ τοῦ ἀπὸ $ΓΔ$ ἴσον ἐστὶ τῷ ἀπὸ $AΔ$· ὥστε τὸ ὑπὸ τῶν $AΓ$, $ΓB$ὀέλαττόν ἐστι τοῦ ἀπὸ $AΔ$· τὸ ἄρα δὶξ ὑπὸ τῶν $AΓ$, $ΓB$ὀέλαττονὸὴ διπλάσιόν ἐστι τοῦ ἀπὸ $AΔ$. ἀλλὰ τὰ ἀπὸ τῶν $AΓ$, $ΓB$ διπλάσιά [ἐστι] τῶν ἀπὸ τῶν $AΔ$, $ΔΓ$· τὰ ἄρα ἀπὸ τῶν $AΓ$, $ΓB$ μείζονά ἐστι τοῦ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$· ὅπερ ἔδει δεῖξαι.

Let AB be a straight-line, and let it be cut unequally at C, and let AC be greater (than CB). I say that (the sum of) the (squares) on AC and CB is greater than twice the (rectangle contained) by AC and CB.

For let AB be cut in half at D. Therefore, since a straight-line has been cut into equal (parts) at D, and into unequal (parts) at C, the (rectangle contained) by AC and CB, plus the (square) on CD, is thus equal to the (square) on AD [Prop. 2.5]. Hence, the (rectangle contained) by AC and CB is less than the (square) on AD. Thus, twice the (rectangle contained) by AC and CB is less than double the (square) on AD. But, (the sum of) the (squares) on AC and CB [is] double (the sum of) the (squares) on AD and DC [Prop. 2.9]. Thus, (the sum of) the (squares) on AC and CB is greater than twice the (rectangle contained) by AC and CB. (Which is) the very thing it was required to show.

60.

Τὸ ἀπὸ τῆς ἐκ δύο ὀνομάτων παρὰ ὀρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πρώτην.

Proposition 60

The square on a binomial (straight-line) applied to a rational (straight-line) produces as breadth a first binomial (straight-line).†

Ἔστω ἐκ δύο ὀνομάτων ἡ AB διῃρημένη εἰξ τὰ ὀνόματα κατὰ τὸ $Γ$, ὥστε τὸ μεῖζον ὄνομα εἶναι τὸ $AΓ$, καὶ ἐκκείσθω ὁρητὴ ἡ $ΔE$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν $ΔE$ παραβεβλήσθω τὸ $ΔEZH$ πλάτος ποιοῦν τὴν $ΔH$· λέγω, ὅτι ἡ $ΔH$ ἐκ δύο ὀνομάτων ἐστὶ πρώτη.

Παραβεβλήσθω γὰρ παρὰ τὴν $ΔE$ τῷ μὲν ἀπὸ τῆς $AΓ$ ἴσον τὸ $ΔΘ$, τῷ δὲ ἀπὸ τῆς $BΓ$ ἴσον τὸ $KΛ$· λοιπὸν ἄρα τὸ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$ ἴσον ἐστὶ τῷ MZ. τετμήσθω ἡ MH δίθα κατὰ τὸ N, καὶ παράλληλος ἤχθω ἡ NX [ἑκατέρᾳ τῶν $MΛ$, HZ]. ἑκάτερον ἄρα τῶν MX, NZ ἴσον ἐστὶ τῷ ἁπαχ ὑπὸ τῶν $AΓB$. καὶ ἐπεὶ ἐκ δύο ὀνομάτων ἐστὶν ἡ AB διῃρημένη εἰξ τὰ ὀνόματα κατὰ τὸ $Γ$, αἱ $AΓ$, $ΓB$ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· τὰ ἄρα ἀπὸ τῶν $AΓ$, $ΓB$ ὁρητά ἐστι καὶ σύμμετρα ἀλλήλοις· ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $AΓ$, $ΓB$. καί ἐστιν ἴσον τῷ $ΔΛ$· ὁρητὸν ἄρα ἐστὶ τὸ $ΔΛ$. καὶ παρὰ ὁρητὴν τὴν $ΔE$ παράκειται· ὁρητὴ ἄρα ἐστὶν ἡ $ΔM$ καὶ σύμμετρος τῇ $ΔE$ μήκει. πάλιν, ἐπεὶ αἱ $AΓ$, $ΓB$ ὁρηταί εἰσι δυνάμει μόνον σύμμετροι, μέσον ἄρα ἐστὶ τὸ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$, τουτέστι τὸ MZ. καὶ παρὰ ὁρητὴν τὴν $MΛ$ παράκειται· ὁρητὴ ἄρα καὶ ἡ MH καὶ ἀσύμμετρος τῇ $MΛ$, τουτέστι τῇ $ΔE$, μήκει. ὐέστι δὲ καὶ ἡ $MΔ$ ὁρητὴ καὶ τῇ $ΔE$ μήκει σύμμετρος· ἀσύμμετρος ἄρα ἐστὶν ἡ $ΔM$ τῇ MH μήκει. καί εἰσι ὁρηταί· αἱ $ΔM$, MH ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ $ΔH$. δεικτέον δή, ὅτι καὶ πρώτη.

Ἐπεὶ τῶν ἀπὸ τῶν $AΓ$, $ΓB$ μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν $AΓB$, καὶ τῶν $ΔΘ$, $KΛ$ ἄρα μέσον ἀνάλογόν ἐστι τὸ MX.

Let AB be a binomial (straight-line), having been divided into its (component) terms at C, such that AC is the greater term. And let the rational (straight-line) DE be laid down. And let the (rectangle) $DEFG$, equal to the (square) on AB, be applied to DE, producing DG as breadth. I say that DG is a first binomial (straight-line).

For let DH, equal to the (square) on AC, and KL, equal to the (square) on BC, be applied to DE. Thus, the remaining twice the (rectangle contained) by AC and CB is equal to MF [Prop. 2.4]. Let MG be cut in half at N, and let NO be drawn parallel [to each of ML and GF]. MO and NF are thus each equal to once the (rectangle contained) by ACB. And since AB is a binomial (straight-line), having been divided into its (component) terms at C, AC and CB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.36]. Thus, the (squares) on AC and CB are rational, and commensurable with one another. And hence the sum of the (squares) on AC and CB (is rational) [Prop. 10.15], and is equal to DL. Thus, DL is rational. And it is applied to the rational (straight-line) DE. DM is thus rational, and commensurable in length with DE [Prop. 10.20]. Again, since AC and CB are rational (straight-lines which are) commensurable in square only, twice the (rectangle contained) by AC and CB—that is to say, MF—is thus medial [Prop. 10.21]. And it is applied to the rational (straight-line) ML. MG is thus also rational, and incommensurable in length with ML—that is to say, with DE [Prop. 10.22]. And MD is also rational, and com-

ὀέστιν ἄρα ὡξ τὸ ΔΘ πρὸς τὸ ΜΧ, οὐτως τὸ ΜΧ πρὸς τὸ ΚΛ, τουτέστιν ὡξ ἡ ΔΚ πρὸς τὴν ΜΝ, ἡ ΜΝ πρὸς τὴν ΜΚ· τὸ ἄρα ὑπὸ τῶν ΔΚ, ΚΜ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΜΝ. καὶ ἐπεὶ σύμμετρόν ἐστι τὸ ἀπὸ τῆς ΑΓ τῷ ἀπὸ τῆς ΓΒ, σύμμετρόν ἐστι καὶ τὸ ΔΘ τῷ ΚΛ· ὁώστε καὶ ἡ ΔΚ τῇ ΚΜ σύμμετρόξ ἐστιν. καὶ ἐπεὶ μείζονά ἐστι τὰ ἀπὸ τῶν ΑΓ, ΓΒ τοῦ διξ ὑπὸ τῶν ΑΓ, ΓΒ, μεῖζον ἄρα καὶ τὸ ΔΛ τοῦ ΜΖ· ὁώστε καὶ ἡ ΔΜ τῆς ΜΗ μείζων ἐστίν. καί ἐστιν ἴσον τὸ ὑπὸ τῶν ΔΚ, ΚΜ τῷ ἀπὸ τῆς ΜΝ, τουτέστι τῷ τετάρτῳ τοῦ ἀπὸ τῆς ΜΗ -.7πτ η, καὶ σύμμετρος ἡ ΔΚ τῇ ΚΜ. ἔαν δὲ ὠσι δύο εὐθεῖαιὀάνισοι, τῷ δὲ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παρὰ τὴν μείζονα παραβληθῇ ἐλλεῖπον εὀίδει τετραγώνῳ καὶ εἰξ σύμμετρα αὐτὴν διαιρῇ, ἡ μείζων τῆς ἐλάσσονος μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ· ἡ ΔΜ ἄρα τῆς ΜΗ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καί εἰσι ὀρηταὶ αἱ ΔΜ, ΜΗ , καὶ ἡ ΔΜ μεῖζον ὄνομα οὀῦσα σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὀρητῇ τῇ ΔΕ μήκει.

Ἡ ΔΗ ἄρα ἐκ δύο ὀνομάτων ἐστὶ πρώτη· ὅπερ ἔδει δεῖχαι.

mensurable in length with DE. Thus, DM is incommensurable in length with MG [Prop. 10.13]. And they are rational. DM and MG are thus rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line) [Prop. 10.36]. So, we must show that (it is) also a first (binomial straight-line).

Since the (rectangle contained) by ACB is the mean proportional to the squares on AC and CB [Prop. 10.53 lem.], MO is thus also the mean proportional to DH and KL. Thus, as DH is to MO, so MO (is) to KL—that is to say, as DK (is) to MN, (so) MN (is) to MK [Prop. 6.1]. Thus, the (rectangle contained) by DK and KM is equal to the (square) on MN [Prop. 6.17]. And since the (square) on AC is commensurable with the (square) on CB, DH is also commensurable with KL. Hence, DK is also commensurable with KM [Props. 6.1, 10.11]. And since (the sum of) the squares on AC and CB is greater than twice the (rectangle contained) by AC and CB [Prop. 10.59 lem.], DL (is) thus also greater than MF. Hence, DM is also greater than MG [Props. 6.1, 5.14]. And the (rectangle contained) by DK and KM is equal to the (square) on MN—that is to say, to one quarter the (square) on MG. And DK (is) commensurable (in length) with KM. And if there are two unequal straight-lines, and a (rectangle) equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) commensurable (in length), then the square on the greater is larger than (the square on) the lesser by the (square) on (some straight-line) commensurable (in length) with the greater [Prop. 10.17]. Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) commensurable (in length) with (DM). And DM and MG are rational. And DM, which is the greater term, is commensurable in length with the (previously) laid down rational (straight-line) DE.

Thus, DG is a first binomial (straight-line) [Def. 10.5]. (Which is) the very thing it was required to show.

† In other words, the square of a binomial is a first binomial. See Prop. 10.54.

61.

Τὸ ἀπὸ τῆς ἐκ δύο μέσων πρώτης παρὰ ὀρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων δευτέραν.

Ἔστω ἐκ δύο μέσων πρώτη ἡ ΑΒ διῃρημένη εἰξ τὰξ μέσας κατὰ τὸ Γ, ὁῶν μείζων ἡ ΑΓ, καὶ ἐκκείσθω ὀρητὴ ἡ ΔΕ, καὶ παραβεβλήσθω παρὰ τὴν ΔΕ τῷ ἀπὸ τῆς ΑΒ ἴσον παραλληλόγραμμον τὸ ΔΖ πλάτος ποιοῦν τὴν ΔΗ· λέγω, ὅτι ἡ ΔΗ ἐκ δύο ὀνομάτων ἐστὶ δευτέρα.

Κατεσκευάσθω γὰρ τὰ αὐτὰ τοῖς πρὸ τούτου. καὶ ἐπεὶ ἡ ΑΒ ἐκ δύο μέσων ἐστὶ πρώτη διῃρημένη κατὰ τὸ Γ, αἱ ΑΓ, ΓΒ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι ὀρητὸν περιέθου-

Proposition 61

The square on a first bimedial (straight-line) applied to a rational (straight-line) produces as breadth a second binomial (straight-line).†

Let AB be a first bimedial (straight-line) having been divided into its (component) medial (straight-lines) at C, of which AC (is) the greater. And let the rational (straight-line) DE be laid down. And let the parallelogram DF, equal to the (square) on AB, be applied to DE, producing DG as breadth. I say that DG is a second binomial (straight-line).

For let the same construction be made as in the (proposi-

σαι· ὥστε καὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ μέσα ἐστίν. μέσον ἄρα ἐστὶ τὸ ΔΛ. καὶ παρὰ ῥητὴν τὴν ΔΕ παραβέβληται· ῥητὴ ἄρα ἐστίν ἡ ΜΔ καὶ ἀσύμμετρος τῇ ΔΕ μήκει. πάλιν, ἐπεὶ ῥητόν ἐστι τὸ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ, ῥητόν ἐστι καὶ τὸ ΜΖ. καὶ παρὰ ῥητὴν τὴν ΜΛ παράκειται· ῥητὴ ἄρα [ἐστὶ] καὶ ἡ ΜΗ καὶ μήκει σύμμετρος τῇ ΜΛ, τουτέστι τῇ ΔΕ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΔΜ τῇ ΜΗ μήκει. καί εἰσι ῥηταί· αἱ ΔΜ, ΜΗ ἄρα ῥηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΔΗ. δεικτέον δή, ὅτι καὶ δευτέρα.

tion) before this. And since AB is a first bimedial (straight-line), having been divided at C, AC and CB are thus medial (straight-lines) commensurable in square only, and containing a rational (area) [Prop. 10.37]. Hence, the (squares) on AC and CB are also medial [Prop. 10.21]. Thus, DL is medial [Props. 10.15, 10.23 corr.]. And it has been applied to the rational (straight-line) DE. MD is thus rational, and incommensurable in length with DE [Prop. 10.22]. Again, since twice the (rectangle contained) by AC and CB is rational, MF is also rational. And it is applied to the rational (straight-line) ML. Thus, MG [is] also rational, and commensurable in length with ML—that is to say, with DE [Prop. 10.20]. DM is thus incommensurable in length with MG [Prop. 10.13]. And they are rational. DM and MG are thus rational, and commensurable in square only. DG is thus a binomial (straight-line) [Prop. 10.36]. So, we must show that (it is) also a second (binomial straight-line).

Ἐπεὶ γὰρ τὰ ἀπὸ τῶν ΑΓ, ΓΒ μείζονά ἐστι τοῦ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ, μεῖζον ἄρα καὶ τὸ ΔΛ τοῦ ΜΖ· ὥστε καὶ ἡ ΔΜ τῆς ΜΗ . καὶ ἐπεὶ σύμμετρόν ἐστι τὸ ἀπὸ τῆς ΑΓ τῷ ἀπὸ τῆς ΓΒ, σύμμετρόν ἐστι καὶ τὸ ΔΘ τῷ ΚΛ· ὥστε καὶ ἡ ΔΚ τῇ ΚΜ σύμμετρός ἐστιν. καί ἐστι τὸ ὑπὸ τῶν ΔΚΜ ἴσον τῷ ἀπὸ τῆς ΜΝ· ἡ ΔΜ ἄρα τῆς ΜΗ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καί ἐστιν ἡ ΜΗ σύμμετρος τῇ ΔΕ μήκει.

Ἡ ΔΗ ἄρα ἐκ δύο ὀνομάτων ἐστὶ δευτέρα.

For since (the sum of) the squares on AC and CB is greater than twice the (rectangle contained) by AC and CB [Prop. 10.59], DL (is) thus also greater than MF. Hence, DM (is) also (greater) than MG [Prop. 6.1]. And since the (square) on AC is commensurable with the (square) on CB, DH is also commensurable with KL. Hence, DK is also commensurable (in length) with KM [Props. 6.1, 10.11]. And the (rectangle contained) by DKM is equal to the (square) on MN. Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) commensurable (in

length) with (*DM*) [Prop. 10.17]. And *MG* is commensurable in length with *DE*.

Thus, *DG* is a second binomial (straight-line) [Def. 10.6].

[†] In other words, the square of a first bimedial is a second binomial. See Prop. 10.55.

62.

Τὸ ἀπὸ τῆς ἐκ δύο μέσων δευτέρας παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τρίτην.

Ἔστω ἐκ δύο μέσων δευτέρα ἡ ΑΒ διῃρημένη εἰξ τὰξ μέσας κατὰ τὸ Γ, ὥστε τὸ μεῖζον τμῆμα εὐῖναι τὸ ΑΓ, ὁρητὴ δέ τις ὑποκείσθω ἡ ΔΕ, καὶ παρὰ τὴν ΔΕ τῷ ἀπὸ τῆς ΑΒ ἴσον παραλληλόγραμμον παραβεβλήσθω τὸ ΔΖ πλάτος ποιοῦν τὴν ΔΗ· λέγω, ὅτι ἡ ΔΗ ἐκ δύο ὀνομάτων ἐστὶ τρίτη.

Κατεσκευάσθω τὰ αὐτὰ τοῖς προδεδειγμένοις. καὶ ἐπεὶ ἐκ δύο μέσων δευτέρα ἐστὶν ἡ ΑΒ διῃρημένη κατὰ τὸ Γ, αἱ ΑΓ, ΓΒ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι· ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓΒ μέσον ἐστίν. καί ἐστιν ἴσον τῷ ΔΛ· μέσον ἄρα καὶ τὸ ΔΛ. καὶ παράκειται παρὰ ὁρητὴν τὴν ΔΕ· ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΜΔ καὶ ἀσύμμετρος τῇ ΔΕ μήκει. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΜΗ ὁρητή ἐστι καὶ ἀσύμμετρος τῇ ΜΛ, τουτέστι τῇ ΔΕ, μήκει· ὁρητὴ ἄρα ἐστὶν ἑκατέρα τῶν ΔΜ, ΜΗ καὶ ἀσύμμετρος τῇ ΔΕ μήκει. καὶ ἐπεὶ ἀσύμμετρόξ ἐστιν ἡ ΑΓ τῇ ΓΒ μήκει, ὡξ δὲ ἡ ΑΓ πρὸς τὴν ΓΒ, οὕτως τὸ ἀπὸ τῆς ΑΓ πρὸς τὸ ὑπὸ τῶν ΑΓΒ, ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς ΑΓ τῷ ὑπὸ τῶν ΑΓΒ. ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓΒ τῷ δὶξ ὑπὸ τῶν ΑΓΒ ἀσύμμετρόν ἐστιν, τουτέστι τὸ ΔΛ τῷ ΜΖ· ὥστε καὶ ἡ ΔΜ τῷ ΜΗ ἀσύμμετρόξ ἐστιν. καί εἰσι ὁρηταί· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΔΗ. δεικτέον [δή], ὅτι καὶ τρίτη.

Proposition 62

The square on a second bimedial (straight-line) applied to a rational (straight-line) produces as breadth a third binomial (straight-line).[†]

Let *AB* be a second bimedial (straight-line) having been divided into its (component) medial (straight-lines) at *C*, such that *AC* is the greater segment. And let *DE* be some rational (straight-line). And let the parallelogram *DF*, equal to the (square) on *AB*, be applied to *DE*, producing *DG* as breadth. I say that *DG* is a third binomial (straight-line).

Let the same construction be made as that shown previously. And since *AB* is a second bimedial (straight-line), having been divided at *C*, *AC* and *CB* are thus medial (straight-lines) commensurable in square only, and containing a medial (area) [Prop. 10.38]. Hence, the sum of the (squares) on *AC* and *CB* is also medial [Props. 10.15, 10.23 corr.]. And it is equal to *DL*. Thus, *DL* (is) also medial. And it is applied to the rational (straight-line) *DE*. *MD* is thus also rational, and incommensurable in length with *DE* [Prop. 10.22]. So, for the same (reasons), *MG* is also rational, and incommensurable in length with *ML*—that is to say, with *DE*. Thus, *DM* and *MG* are each rational, and incommensurable in length with *DE*. And since *AC* is incommensurable in length with *CB*, and as *AC* (is) to *CB*, so the (square) on *AC* (is) to the (rectangle contained) by *ACB* [Prop. 10.21 lem.], the (square) on *AC* (is) also incommensurable with the (rectangle contained) by *ACB* [Prop. 10.11]. And hence the sum of the (squares) on *AC* and *CB* is incommensurable with twice the (rectangle contained) by *ACB*—that is to say, *DL* with *MF* [Props. 10.12, 10.13]. Hence, *DM* is also incommensurable (in length) with *MG* [Props. 6.1, 10.11]. And they are rational. *DG* is thus a binomial (straight-line) [Prop. 10.36]. [So] we must show that (it is) also a third (binomial straight-line).

Ὁμοίως δὴ τοῖς προτέροις ἐπιλογιούμεθα, ὅτι μείζων ἐστὶν ἡ ΔM τῆς MH , καὶ σύμμετρος ἡ ΔK τῇ KM. καί ἐστι τὸ ὑπὸ τῶν ΔKM ἴσον τῷ ἀπὸ τῆς MN· ἡ ΔM ἄρα τῆς MH

So, similarly to the previous (propositions), we can conclude that DM is greater than MG, and DK (is) commensurable (in length) with KM. And the (rectangle contained) by DKM is equal to the (square) on MN. Thus, the square

μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καὶ οὐδετέρα τῶν ΔΜ, ΜΗ σύμμετρόξ ἐστι τῇ ΔΕ μήκει.

Ἡ ΔΗ ἄρα ἐκ δύο ὀνομάτων ἐστὶ τρίτη· ὅπερ ἔδει δεῖχαι.

on DM is greater than (the square on) MG by the (square) on (some straight-line) commensurable (in length) with (DM) [Prop. 10.17]. And neither of DM and MG is commensurable in length with DE.

Thus, DG is a third binomial (straight-line) [Def. 10.7]. (Which is) the very thing it was required to show.

† In other words, the square of a second bimedial is a third binomial. See Prop. 10.56.

63.

Τὸ ἀπὸ τῆς μείζονος παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τετάρτην.

Ἔστω μείζων ἡ ΑΒ διῃρημένη κατὰ τὸ Γ, ὁώστε μείζονα εὐῖναι τὴν ΑΓ τῆς ΓΒ, ὁρητὴ δὲ ἡ ΔΕ, καὶ τῷ ἀπὸ τῆς ΑΒ ἴσον παρὰ τὴν ΔΕ παραβεβλήσθω τὸ ΔΖ παραλληλόγραμμον πλάτος ποιοῦν τὴν ΔΗ· λέγω, ὅτι ἡ ΔΗ ἐκ δύο ὀνομάτων ἐστὶ τετάρτη.

Proposition 63

The square on a major (straight-line) applied to a rational (straight-line) produces as breadth a fourth binomial (straight-line).†

Let AB be a major (straight-line) having been divided at C, such that AC is greater than CB, and (let) DE (be) a rational (straight-line). And let the parallelogram DF, equal to the (square) on AB, be applied to DE, producing DG as breadth. I say that DG is a fourth binomial (straight-line).

Κατεσκευάσθω τὰ αὐτὰ τοῖς προδεδειγμένοις. καὶ ἐπεὶ μείζων ἐστὶν ἡ ΑΒ διῃρημένη κατὰ τὸ Γ, αἱ ΑΓ, ΓΒ δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων ὁρητόν, τὸ δὲ ὑπὸ αὐτῶν μέσον. ἐπεὶ οὐῦν ὁρητόν ἐστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓΒ, ὁρητὸν ἄρα ἐστὶ τὸ ΔΛ· ὁρητὴ ἄρα καὶ ἡ ΔΜ καὶ σύμμετρος τῇ ΔΕ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ, τουτέστι τὸ ΜΖ, καὶ παρὰ ὁρητὴν ἐστι τὴν ΜΛ, ὁρητὴ ἄρα

Let the same construction be made as that shown previously. And since AB is a major (straight-line), having been divided at C, AC and CB are incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial [Prop. 10.39]. Therefore, since the sum of the (squares) on AC and CB is rational, DL is thus rational. Thus, DM (is) also rational, and commensurable in length with DE [Prop. 10.20]. Again, since

ἐστὶ καὶ ἡ ΜΗ καὶ ἀσύμμετρος τῇ ΔΕ μήκει· ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ ΔΜ τῇ ΜΗ μήκει. αἱ ΔΜ, ΜΗ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΔΗ. δεικτέον [δή], ὅτι καὶ τετάρτη.

Ὁμοίως δὴ δείχομεν τοῖς πρότερον, ὅτι μείζων ἐστὶν ἡ ΔΜ τῆς ΜΗ , καὶ ὅτι τὸ ὑπὸ ΔΚΜ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΜΝ. ἐπεὶ οὐκοῦν ἀσύμμετρόν ἐστι τὸ ἀπὸ τῆς ΑΓ τῷ ἀπὸ τῆς ΓΒ, ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ΔΘ τῷ ΚΛ· ὥστε ἀσύμμετρος καὶ ἡ ΔΚ τῇ ΚΜ ἐστιν. ἐὰν δὲ ὦσι δύο εὐθεῖαι ἀνισοι, τῷ δὲ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ἐλάσσονος ἴσον παραλληλόγραμμον παρὰ τὴν μείζονα παραβληθῇ ἐλλεῖπον εἰδει τετραγώνῳ καὶ εἰς ἀσύμμετρα αὐτὴν διαιρῇ, ἡ μείζων τῆς ἐλάσσονος μεῖζον δυνήσεται τῷ ἀπὸ ἀσύμμέτρου ἑαυτῇ μήκει· ἡ ΔΜ ἄρα τῆς ΜΗ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καί εἰσιν αἱ ΔΜ, ΜΗ ὁρηταὶ δυνάμει μόνον σύμμετροι, καὶ ἡ ΔΜ σύμμετρός ἐστι τῇ ἐκκειμένῃ ὁρητῇ τῇ ΔΕ.

Ἡ ΔΗ ἄρα ἐκ δύο ὀνομάτων ἐστὶ τετάρτη· ὅπερ ἔδει δεῖχαι.

twice the (rectangle contained) by AC and CB—that is to say, MF—is medial, and is (applied to) the rational (straight-line) ML, MG is thus also rational, and incommensurable in length with DE [Prop. 10.22]. DM is thus also incommensurable in length with MG [Prop. 10.13]. DM and MG are thus rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line) [Prop. 10.36]. [So] we must show that (it is) also a fourth (binomial straight-line).

So, similarly to the previous (propositions), we can show that DM is greater than MG, and that the (rectangle contained) by DKM is equal to the (square) on MN. Therefore, since the (square) on AC is incommensurable with the (square) on CB, DH is also incommensurable with KL. Hence, DK is also incommensurable with KM [Props. 6.1, 10.11]. And if there are two unequal straight-lines, and a parallelogram equal to the fourth part of the (square) on the lesser, falling short by a square figure, is applied to the greater, and divides it into (parts which are) incommensurable (in length), then the square on the greater will be larger than (the square on) the lesser by the (square) on (some straight-line) incommensurable in length with the greater [Prop. 10.18]. Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) incommensurable (in length) with (DM). And DM and MG are rational (straight-lines which are) commensurable in square only. And DM is commensurable (in length) with the (previously) laid down rational (straight-line) DE.

Thus, DG is a fourth binomial (straight-line) [Def. 10.8]. (Which is) the very thing it was required to show.

† In other words, the square of a major is a fourth binomial. See Prop. 10.57.

64.

Τὸ ἀπὸ τῆς ὁρητὸν καὶ μέσον δυναμένης παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πέμπτην.

Ἔστω ὁρητὸν καὶ μέσον δυναμένη ἡ ΑΒ διῃρημένη εἰς τὰξ εὐθείας κατὰ τὸ Γ, ὥστε μείζονα εἶναι τὴν ΑΓ, καὶ ἐκκείσθω ὁρητὴ ἡ ΔΕ, καὶ τῷ ἀπὸ τῆς ΑΒ ἴσον παρὰ τὴν ΔΕ παραβεβλήσθω τὸ ΔΖ πλάτος ποιοῦν τὴν ΔΗ· λέγω, ὅτι ἡ ΔΗ ἐκ δύο ὀνομάτων ἐστὶ πέμπτη.

Proposition 64

The square on the square-root of a rational plus a medial (area) applied to a rational (straight-line) produces as breadth a fifth binomial (straight-line).†

Let AB be the square-root of a rational plus a medial (area) having been divided into its (component) straight-lines at C, such that AC is greater. And let the rational (straight-line) DE be laid down. And let the (parallelogram) DF, equal to the (square) on AB, be applied to DE, producing DG as breadth. I say that DG is a fifth binomial straight-line.

Κατεσκευάσθω τὰ αὐτα τοῖς πρὸ τούτου. ἐπεὶ οὐῦν ὁρητὸν καὶ μέσον δυναμένη ἐστὶν ἡ AB διῃρημένη κατὰ τὸ Γ, αἱ AΓ, ΓB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον, τὸ δὸ ὑπὸ αὐτῶν ὁρητόν. ἐπεὶ οὐῦν μέσον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AΓ, ΓB, μέσον ἄρα ἐστὶ τὸ ΔΛ· ὥστε ὁρητή ἐστιν ἡ ΔM καὶ μήκει ἀσύμμετρος τῇ ΔE. πάλιν, ἐπεὶ ὁρητόν ἐστι τὸ δὶξ ὑπὸ τῶν AΓB, τουτέστι τὸ MZ, ὁρητὴ ἄρα ἡ MH καὶ σύμμετρος τῇ ΔE. ἀσύμμετρος ἄρα ἡ ΔM τῇ MH · αἱ ΔM, MH ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΔH. λέγω δή, ὅτι καὶ πέμπτη.

Ὁμοίως γὰρ διεθθήσεται, ὅτι τὸ ὑπὸ τῶν ΔKM ἴσον ἐστὶ τῷ ἀπὸ τῆς MN, καὶ ἀσύμμετρος ἡ ΔK τῇ KM μήκει· ἡ ΔM ἄρα τῆς MH μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καί εἰσιν αἱ ΔM, MH [ὁρηταὶ] δυνάμει μόνον σύμμετροι, καὶ ἡ ἐλάσσων ἡ MH σύμμετρος τῇ ΔE μήκει.

Ἡ ΔH ἄρα ἐκ δύο ὀνομάτων ἐστὶ πέμπτη· ὅπερ ἔδει δεῖχαι.

Let the same construction be made as in the (propositions) before this. Therefore, since AB is the square-root of a rational plus a medial (area), having been divided at C, AC and CB are thus incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational [Prop. 10.40]. Therefore, since the sum of the (squares) on AC and CB is medial, DL is thus medial. Hence, DM is rational and incommensurable in length with DE [Prop. 10.22]. Again, since twice the (rectangle contained) by ACB—that is to say, MF—is rational, MG (is) thus rational and commensurable (in length) with DE [Prop. 10.20]. DM (is) thus incommensurable (in length) with MG [Prop. 10.13]. Thus, DM and MG are rational (straight-lines which are) commensurable in square only. Thus, DG is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a fifth (binomial straight-line).

For, similarly (to the previous propositions), it can be shown that the (rectangle contained) by DKM is equal to the (square) on MN, and DK (is) incommensurable in length with KM. Thus, the square on DM is greater than (the square on) MG by the (square) on (some straight-line) incommensurable (in length) with (DM) [Prop. 10.18]. And DM and MG are [rational] (straight-lines which are) commensurable in square only, and the lesser MG is commensurable in length with DE.

Thus, DG is a fifth binomial (straight-line) [Def. 10.9]. (Which is) the very thing it was required to show.

† In other words, the square of the square-root of a rational plus medial is a fifth binomial. See Prop. 10.58.

65.

Τὸ ἀπὸ τῆς δύο μέσα δυναμένης παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων ὁέκτην.

Ἔστω δύο μέσα δυναμένη ἡ AB διῃρημένη κατὰ τὸ Γ, ὁρητὴ δὲ ὑέστω ἡ ΔE, καὶ παρὰ τὴν ΔE τῷ ἀπὸ τῆς AB ἴσον παραβεβλήσθω τὸ ΔZ πλάτος ποιοῦν τὴν ΔH· λέγω, ὅτι ἡ ΔH ἐκ δύο ὀνομάτων ἐστὶν ὁέκτη.

Proposition 65

The square on the square-root of (the sum of) two medial (areas) applied to a rational (straight-line) produces as breadth a sixth binomial (straight-line).†

Let *AB* be the square-root of (the sum of) two medial (areas), having been divided at *C*. And let *DE* be a rational (straight-line). And let the (parallelogram) *DF*, equal to the (square) on *AB*, be applied to *DE*, producing *DG* as breadth. I say that *DG* is a sixth binomial (straight-line).

Κατεσκευάσθω γὰρ τὰ αὐτὰ τοῖς πρότερον. καὶ ἐπεὶ ἡ AB δύο μέσα δυναμένη ἐστὶ διῃρημένη κατὰ τὸ Γ, αἱ ΑΓ, ΓB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τό τε συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπὸ αὐτῶν μέσον καὶ ὀέτι ἀσύμμετρον τὸ ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων συγκείμενον τῷ ὑπὸ αὐτῶν· ὁώστε κατὰ τὰ προδεδειγμένα μέσον ἐστὶν ἑκάτερον τῶν ΔΛ, MZ. καὶ παρὰ ὁρητὴν τὴν ΔE παράκειται· ὁρητὴ ἄρα ἐστὶν ἑκατέρα τῶν ΔM, MH καὶ ἀσύμμετρος τῇ ΔE μήκει. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΓ, ΓB τῷ δἰξ ὑπὸ τῶν ΑΓ, ΓB, ἀσύμμετρον ἄρα ἐστὶ τὸ ΔΛ τῷ MZ. ἀσύμμετρος ἄρα καὶ ἡ ΔM τῇ MH · αἱ ΔM, MH ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΔH. λέγω ὁή, ὅτι καὶ ὁέκτη.

Ὁμοίως ὁὴ πάλιν ὁεῖχομεν, ὅτι τὸ ὑπὸ τῶν ΔKM ἴσον ἐστὶ τῷ ἀπὸ τῆς MN, καὶ ὅτι ἡ ΔK τῇ KM μήκει ἐστὶν

For let the same construction be made as in the previous (propositions). And since *AB* is the square-root of (the sum of) two medial (areas), having been divided at *C*, *AC* and *CB* are thus incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, moreover, the sum of the squares on them incommensurable with the (rectangle contained) by them [Prop. 10.41]. Hence, according to what has been previously demonstrated, *DL* and *MF* are each medial. And they are applied to the rational (straight-line) *DE*. Thus, *DM* and *MG* are each rational, and incommensurable in length with *DE* [Prop. 10.22]. And since the sum of the (squares) on *AC* and *CB* is incommensurable with twice the (rectangle contained) by *AC* and *CB*, *DL* is thus incommensurable with *MF*. Thus, *DM* (is) also incommensurable (in length) with *MG*

ἀσύμμετρος· καὶ διὰ τὰ αὐτὰ δὴ ἡ ΔΜ τῆς ΜΗ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ μήκει. καὶ οὐδετέρα τῶν ΔΜ, ΜΗ σύμμετρός ἐστι τῇ ἐκκειμένῃ ὁρητῇ τῇ ΔΕ μήκει.

Ἡ ΔΗ ἄρα ἐκ δύο ὀνομάτων ἐστὶν ὀκτη· ὅπερ ἔδει δεῖξαι.

[Props. 6.1, 10.11]. *DM* and *MG* are thus rational (straight-lines which are) commensurable in square only. Thus, *DG* is a binomial (straight-line) [Prop. 10.36]. So, I say that (it is) also a sixth (binomial straight-line).

So, similarly (to the previous propositions), we can again show that the (rectangle contained) by *DKM* is equal to the (square) on *MN*, and that *DK* is incommensurable in length with *KM*. And so, for the same (reasons), the square on *DM* is greater than (the square on) *MG* by the (square) on (some straight-line) incommensurable in length with (*DM*) [Prop. 10.18]. And neither of *DM* and *MG* is commensurable in length with the (previously) laid down rational (straight-line) *DE*.

Thus, *DG* is a sixth binomial (straight-line) [Def. 10.10]. (Which is) the very thing it was required to show.

† In other words, the square of the square-root of two medials is a sixth binomial. See Prop. 10.59.

66.

Ἡ τῇ ἐκ δύο ὀνομάτων μήκει σύμμετρος καὶ αὐτὴ ἐκ δύο ὀνομάτων ἐστὶ καὶ τῇ τάχει ἡ αὐτή.

Ἔστω ἐκ δύο ὀνομάτων ἡ ΑΒ, καὶ τῇ ΑΒ μήκει σύμμετροςὁέστω ἡ ΓΔ· λέγω, ὅτι ἡ ΓΔ ἐκ δύο ὀνομάτων ἐστὶ καὶ τῇ τάχει ἡ αὐτὴ τῇ ΑΒ.

Proposition 66

A (straight-line) commensurable in length with a binomial (straight-line) is itself also binomial, and the same in order.

Let *AB* be a binomial (straight-line), and let *CD* be commensurable in length with *AB*. I say that *CD* is a binomial (straight-line), and (is) the same in order as *AB*.

Ἐπεὶ γὰρ ἐκ δύο ὀνομάτων ἐστὶν ἡ ΑΒ, διῃρήσθω εἰξ τὰ ὀνόματα κατὰ τὸ Ε, καὶ ὀέστω μεῖζον ὄνομα τὸ ΑΕ· αἱ ΑΕ, ΕΒ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι. γεγονέτω ὡξ ἡ ΑΒ πρὸς τὴν ΓΔ, οὐτως ἡ ΑΕ πρὸς τὴν ΓΖ· καὶ λοιπὴ ἄρα ἡ ΕΒ πρὸς λοιπὴν τὴν ΖΔ ἐστιν, ὡξ ἡ ΑΒ πρὸς τὴν ΓΔ. σύμμετρος δὲ ἡ ΑΒ τῇ ΓΔ μήκει· σύμμετρος ἄρα ἐστὶ καὶ ἡ μὲν ΑΕ τῇ ΓΖ, ἡ δὲ ΕΒ τῇ ΖΔ. καί εἰσι ὁρηταὶ αἱ ΑΕ, ΕΒ· ὁρηταὶ ἄρα εἰσὶ καὶ αἱ ΓΖ, ΖΔ. καὶ ἐστιν ὡξ ἡ ΑΕ πρὸς ΓΖ, ἡ ΕΒ πρὸς ΖΔ. ἐναλλὰχ ἄρα ἐστὶν ὡξ ἡ ΑΕ πρὸς ΕΒ, ἡ ΓΖ πρὸς ΖΔ. αἱ δὲ ΑΕ, ΕΒ δυνάμει μόνον [εἰσὶ] σύμμετροι· καὶ αἱ ΓΖ, ΖΔ ἄρα δυνάμει μόνον εἰσὶ σύμμετροι. καί εἰσι ὁρηταί· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΓΔ. λέγω δή, ὅτι τῇ τάχει ἐστὶν ἡ αὐτὴ τῇ ΑΒ.

Ἡ γὰρ ΑΕ τῆς ΕΒ μεῖζον δύναται ἤτοι τῷ ἀπὸ συμμέτρου ἑαυτῇ ὀὴ τῷ ἀπὸ ἀσυμμέτρου. εἰ μὲν οὐῦν ἡ ΑΕ τῆς ΕΒ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ ἡ ΓΖ τῆς ΖΔ μεῖζον

For since *AB* is a binomial (straight-line), let it be divided into its (component) terms at *E*, and let *AE* be the greater term. *AE* and *EB* are thus rational (straight-lines which are) commensurable in square only [Prop. 10.36]. Let it be contrived that as *AB* (is) to *CD*, so *AE* (is) to *CF* [Prop. 6.12]. Thus, the remainder *EB* is also to the remainder *FD*, as *AB* (is) to *CD* [Props. 6.16, 5.19 corr.]. And *AB* (is) commensurable in length with *CD*. Thus, *AE* is also commensurable (in length) with *CF*, and *EB* with *FD* [Prop. 10.11]. And *AE* and *EB* are rational. Thus, *CF* and *FD* are also rational. And as *AE* is to *CF*, (so) *EB* (is) to *FD* [Prop. 5.11]. Thus, alternately, as *AE* is to *EB*, (so) *CF* (is) to *FD* [Prop. 5.16]. And *AE* and *EB* [are] commensurable in square only. Thus, *CF* and *FD* are also commensurable in square only [Prop. 10.11]. And they are rational. *CD* is thus a binomial (straight-line) [Prop. 10.36]. So, I say that it is the same in order as *AB*.

δυνήσεται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καὶ εἰ μὲν σύμμετρόξ ἐστιν ἡ AE τῇ ἐκκειμένῃ ὁρητῇ, καὶ ἡ $ΓZ$ σύμμετρος αὐτῇ ὀέσται, καὶ διὰ τοῦτο ἑκατέρα τῶν AB, $ΓΔ$ ἐκ δύο ὀνομάτων ἐστὶ πρώτη, τουτέστι τῇ τάχει ἡ αὐτή. εἰ δὲ ἡ EB σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῇ, καὶ ἡ $ZΔ$ σύμμετρόξ ἐστιν αὐτῇ, καὶ διὰ τοῦτο πάλιν τῇ τάχει ἡ αὐτὴ ὀέσται τῇ AB· ἑκατέρα γὰρ αὐτῶνὀέσται ἐκ δύο ὀνομάτων δευτέρα. εἰ δὲ οὐδετέρα τῶν AE, EB σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῇ, οὐδετέρα τῶν $ΓZ$, $ZΔ$ σύμμετρος αὐτῇ ὀέσται, καί ἐστιν ἑκατέρα τρίτη. εἰ δὲ ἡ AE τῆς EB μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, καὶ ἡ $ΓZ$ τὴξ $ZΔ$ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καὶ εἰ μὲν ἡ AE σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῃ, καὶ ἡ $ΓZ$ σύμμετρόξ ἐστιν αὐτῇ, καὶ ἐστιν ἑκατέρα τετάρτη. εἰ δὲ ἡ EB, καὶ ἡ $ZΔ$, καὶ ὀέσται ἑκατέρα πέμπτη. εἰ δὲ οὐδετέρα τῶν AE, EB, καὶ τῶν $ΓZ$, $ZΔ$ οὐδετέρα σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῇ, καὶ ὀέσται ἑκατέρα ὁέκτη.

Ὅ̓Ωστε ἡ τῇ ἐκ δύο ὀνομάτων μήκει σύμμετρος ἐκ δύο ὀνομάτων ἐστὶ καὶ τῇ τάχει ἡ αὐτή· ὅπερ ἔδει δεῖχαι.

For the square on AE is greater than (the square on) EB by the (square) on (some straight-line) either commensurable or incommensurable (in length) with (AE). Therefore, if the square on AE is greater than (the square on) EB by the (square) on (some straight-line) commensurable (in length) with (AE) then the square on CF will also be greater than (the square on) FD by the (square) on (some straight-line) commensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable (in length) with (some previously) laid down rational (straight-line) then CF will also be commensurable (in length) with it [Prop. 10.12]. And, on account of this, AB and CD are each first binomial (straight-lines) [Def. 10.5]—that is to say, the same in order. And if EB is commensurable (in length) with the (previously) laid down rational (straight-line) then FD is also commensurable (in length) with it [Prop. 10.12], and, again, on account of this, (CD) will be the same in order as AB. For each of them will be second binomial (straight-lines) [Def. 10.6]. And if neither of AE and EB is commensurable (in length) with the (previously) laid down rational (straight-line) then neither of CF and FD will be commensurable (in length) with it [Prop. 10.13], and each (of AB and CD) is a third (binomial straight-line) [Def. 10.7]. And if the square on AE is greater than (the square on) EB by the (square) on (some straight-line) incommensurable (in length) with (AE) then the square on CF is also greater than (the square on) FD by the (square) on (some straight-line) incommensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable (in length) with the (previously) laid down rational (straight-line) then CF is also commensurable (in length) with it [Prop. 10.12], and each (of AB and CD) is a fourth (binomial straight-line) [Def. 10.8]. And if EB (is commensurable in length with the previously laid down rational straight-line) then FD (is) also (commensurable in length with it), and each (of AB and CD) will be a fifth (binomial straight-line) [Def. 10.9]. And if neither of AE and EB (is commensurable in length with the previously laid down rational straight-line) then also neither of CF and FD is commensurable (in length) with the laid down rational (straight-line), and each (of AB and CD) will be a sixth (binomial straight-line) [Def. 10.10].

Hence, a (straight-line) commensurable in length with a binomial (straight-line) is a binomial (straight-line), and the same in order. (Which is) the very thing it was required to show.

67.

Ἡ τῇ ἐκ δύο μέσων μήκει σύμμετρος καὶ αὐτὴ ἐκ δύο μέσων ἐστὶ καὶ τῇ τάχει ἡ αὐτή.

Proposition 67

A (straight-line) commensurable in length with a bimedial (straight-line) is itself also bimedial, and the same in order.

Ἔστω ἐκ δύο μέσων ἡ AB, καὶ τῇ AB σύμμετροςὁέστω μήκει ἡ ΓΔ· λέγω, ὅτι ἡ ΓΔ ἐκ δύο μέσων ἐστὶ καὶ τῇ τάχει ἡ αὐτὴ τῇ AB.

Ἐπεὶ γὰρ ἐκ δύο μέσων ἐστὶν ἡ AB, διῃρήσθω εἰξ τὰξ μέσας κατὰ τὸ E· αἱ AE, EB ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. καὶ γεγονέτω ὡξ ἡ AB πρὸς ΓΔ, ἡ AE πρὸς ΓZ· καὶ λοιπὴ ἄρα ἡ EB πρὸς λοιπὴν τὴν ZΔ ἐστιν, ὡξ ἡ AB πρὸς ΓΔ. σύμμετρος δὲ ἡ AB τῇ ΓΔ μήκει· σύμμετρος ἄρα καὶ ἑκατέρα τῶν AE, EB ἑκατέρᾳ τῶν ΓZ, ZΔ. μέσαι δὲ αἱ AE, EB· μέσαι ἄρα καὶ αἱ ΓZ, ZΔ. καὶ ἐπεί ἐστιν ὡξ ἡ AE πρὸς EB, ἡ ΓZ πρὸς ZΔ, αἱ δὲ AE, EB δυνάμει μόνον σύμμετροί εἰσιν, καὶ αἱ ΓZ, ZΔ [ἄρα] δυνάμει μόνον σύμμετροί εἰσιν, ἐδείθθησαν δὲ καὶ μέσαι· ἡ ΓΔ ἄρα ἐκ δύο μέσων ἐστίν. λέγω δή, ὅτι καὶ τῇ τάχει ἡ αὐτή ἐστι τῇ AB.

Ἐπεὶ γάρ ἐστιν ὡξ ἡ AE πρὸς EB, ἡ ΓZ πρὸς ZΔ, καὶ ὡξ ἄρα τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AEB, οὑτως τὸ ἀπὸ τῆς ΓZ πρὸς τὸ ὑπὸ τῶν ΓZΔ· ἐναλλὰχ ὡξ τὸ ἀπὸ τῆς AE πρὸς τὸ ἀπὸ τῆς ΓZ, οὑτως τὸ ὑπὸ τῶν AEB πρὸς τὸ ὑπὸ τῶν ΓZΔ. σύμμετρον δὲ τὸ ἀπὸ τῆς AE τῷ ἀπὸ τῆς ΓZ· σύμμετρον ἄρα καὶ τὸ ὑπὸ τῶν AEB τῷ ὑπὸ τῶν ΓZΔ. εὐίτε οὐῦν ὁρητόν ἐστι τὸ ὑπὸ τῶν AEB, καὶ τὸ ὑπὸ τῶν ΓZΔ ὁρητόν ἐστιν [καὶ διὰ τοῦτό ἐστιν ἐκ δύο μέσων πρώτη]. εὐίτε μέσον, μέσον, καί ἐστιν ἑκατέρα δευτέρα.

Καὶ διὰ τοῦτοὐέσται ἡ ΓΔ τῇ AB τῇ τάχει ἡ αὐτή· ὅπερ ἔδει δεῖχαι.

Let *AB* be a bimedial (straight-line), and let *CD* be commensurable in length with *AB*. I say that *CD* is bimedial, and the same in order as *AB*.

For since *AB* is a bimedial (straight-line), let it be divided into its (component) medial (straight-lines) at *E*. Thus, *AE* and *EB* are medial (straight-lines which are) commensurable in square only [Props. 10.37, 10.38]. And let it be contrived that as *AB* (is) to *CD*, (so) *AE* (is) to *CF* [Prop. 6.12]. And thus as the remainder *EB* is to the remainder *FD*, so *AB* (is) to *CD* [Props. 5.19 corr., 6.16]. And *AB* (is) commensurable in length with *CD*. Thus, *AE* and *EB* are also commensurable (in length) with *CF* and *FD*, respectively [Prop. 10.11]. And *AE* and *EB* (are) medial. Thus, *CF* and *FD* (are) also medial [Prop. 10.23]. And since as *AE* is to *EB*, (so) *CF* (is) to *FD*, and *AE* and *EB* are commensurable in square only, *CF* and *FD* are [thus] also commensurable in square only [Prop. 10.11]. And they were also shown (to be) medial. Thus, *CD* is a bimedial (straight-line). So, I say that it is also the same in order as *AB*.

For since as *AE* is to *EB*, (so) *CF* (is) to *FD*, thus also as the (square) on *AE* (is) to the (rectangle contained) by *AEB*, so the (square) on *CF* (is) to the (rectangle contained) by *CFD* [Prop. 10.21 lem.]. Alternately, as the (square) on *AE* (is) to the (square) on *CF*, so the (rectangle contained) by *AEB* (is) to the (rectangle contained) by *CFD* [Prop. 5.16]. And the (square) on *AE* (is) commensurable with the (square) on *CF*. Thus, the (rectangle contained) by *AEB* (is) also commensurable with the (rectangle contained) by *CFD* [Prop. 10.11]. Therefore, either the (rectangle contained) by *AEB* is rational, and the (rectangle contained) by *CFD* is rational [and, on account of this, (*AE* and *CD*) are first bimedial (straight-lines)], or (the rectangle contained by *AEB* is) medial, and (the rectangle contained by *CFD* is) medial, and (*AB* and *CD*) are each second (bimedial straight-lines) [Props. 10.23, 10.37, 10.38].

And, on account of this, *CD* will be the same in order as *AB*. (Which is) the very thing it was required to show.

68.

Ἡ τῇ μείζονι σύμμετρος καὶ αὐτὴ μείζων ἐστίν.

Proposition 68

A (straight-line) commensurable (in length) with a major (straight-line) is itself also major.

Ἔστω μείζων ἡ AB, καὶ τῇ AB σύμμετροςὁέστω ἡ $ΓΔ$· λέγω, ὅτι ἡ $ΓΔ$ μείζων ἐστίν.

Διῃρήσθω ἡ AB κατὰ τὸ E· αἱ AE, EB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων ὁρητόν, τὸ δὸ ὑπὸ αὐτῶν μέσον· καὶ γεγονέτω τὰ αὐτὰ τοῖς πρότερον. καὶ ἐπεί ἐστιν ὡξ ἡ AB πρὸς τὴν $ΓΔ$, οὑτως ἥ τε AE πρὸς τὴν $ΓZ$ καὶ ἡ EB πρὸς τὴν $ZΔ$, καὶ ὡξ ἄρα ἡ AE πρὸς τὴν $ΓZ$, οὑτως ἡ EB πρὸς τὴν $ZΔ$. σύμμετρος δὲ ἡ AB τῇ $ΓΔ$· σύμμετρος ἄρα καὶ ἑκατέρα τῶν AE, EB ἑκατέρᾳ τῶν $ΓZ$, $ZΔ$. καὶ ἐπεί ἐστιν ὡξ ἡ AE πρὸς τὴν $ΓZ$, οὑτως ἡ EB πρὸς τὴν $ZΔ$, καὶ ἐναλλὰχ ὡξ ἡ AE πρὸς EB, οὑτως ἡ $ΓZ$ πρὸς $ZΔ$, καὶ συνθέντι ἄρα ἐστὶν ὡξ ἡ AB πρὸς τὴν BE, οὑτως ἡ $ΓΔ$ πρὸς τὴν $ΔZ$· καὶ ὡξ ἄρα τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BE, οὑτως τὸ ἀπὸ τῆς $ΓΔ$ πρὸς τὸ ἀπὸ τῆς $ΔZ$. ὁμοίως δὴ δείχομεν, ὅτι καὶ ὡξ τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς AE, οὑτως τὸ ἀπὸ τῆς $ΓΔ$ πρὸς τὸ ἀπὸ τῆς $ΓZ$. καὶ ὡξ ἄρα τὸ ἀπὸ τῆς AB πρὸς τὰ ἀπὸ τῶν AE, EB, οὑτως τὸ ἀπὸ τῆς $ΓΔ$ πρὸς τὰ ἀπὸ τῶν $ΓZ$, $ZΔ$· καὶ ἐναλλὰχ ἄρα ἐστὶν ὡξ τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς $ΓΔ$, οὑτως τὰ ἀπὸ τῶν AE, EB πρὸς τὰ ἀπὸ τῶν $ΓZ$, $ZΔ$. σύμμετρον δὲ τὸ ἀπὸ τῆς AB τῷ ἀπὸ τῆς $ΓΔ$· σύμμετρα ἄρα καὶ τὰ ἀπὸ τῶν AE, EB τοῖς ἀπὸ τῶν $ΓZ$, $ZΔ$. καί ἐστι τὰ ἀπὸ τῶν AE, EB ὁάμα ὁρητόν, καὶ τὰ ἀπὸ τῶν $ΓZ$, $ZΔ$ ὁάμα ὁρητόν ἐστιν. ὁμοίως δὲ καὶ τὸ δὶξ ὑπὸ τῶν AE, EB σύμμετρόν ἐστι τῷ δὶξ ὑπὸ τῶν $ΓZ$, $ZΔ$. καί ἐστι μέσον τὸ δὶξ ὑπὸ τῶν AE, EB· μέσον ἄρα καὶ τὸ δὶξ ὑπὸ τῶν $ΓZ$, $ZΔ$. αἱ $ΓZ$, $ZΔ$ ἄρα δυνάμει ἀσύμμετροί εἰσι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων ὁάμα ὁρητόν, τὸ δὲ δὶξ ὑπὸ αὐτῶν μέσον· ὅλη ἄρα ἡ $ΓΔ$ὁάλογόξ ἐστιν ἡ καλουμένη μείζων.

Ἡ ἄρα τῇ μείζονι σύμμετρος μείζων ἐστίν· ὅπερ ἔδει δεῖχαι.

Let AB be a major (straight-line), and let CD be commensurable (in length) with AB. I say that CD is a major (straight-line).

Let AB be divided (into its component terms) at E. AE and EB are thus incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial [Prop. 10.39]. And let (the) same (things) be contrived as in the previous (propositions). And since as AB is to CD, so AE (is) to CF and EB to FD, thus also as AE (is) to CF, so EB (is) to FD [Prop. 5.11]. And AB (is) commensurable (in length) with CD. Thus, AE and EB (are) also commensurable (in length) with CF and FD, respectively [Prop. 10.11]. And since as AE is to CF, so EB (is) to FD, also, alternately, as AE (is) to EB, so CF (is) to FD [Prop. 5.16], and thus, via composition, as AB is to BE, so CD (is) to DF [Prop. 5.18]. And thus as the (square) on AB (is) to the (square) on BE, so the (square) on CD (is) to the (square) on DF [Prop. 6.20]. So, similarly, we can also show that as the (square) on AB (is) to the (square) on AE, so the (square) on CD (is) to the (square) on CF. And thus as the (square) on AB (is) to (the sum of) the (squares) on AE and EB, so the (square) on CD (is) to (the sum of) the (squares) on CF and FD. And thus, alternately, as the (square) on AB is to the (square) on CD, so (the sum of) the (squares) on AE and EB (is) to (the sum of) the (squares) on CF and FD [Prop. 5.16]. And the (square) on AB (is) commensurable with the (square) on CD. Thus, (the sum of) the (squares) on AE and EB (is) also commensurable with (the sum of) the (squares) on CF and FD [Prop. 10.11]. And the (squares) on AE and EB (added) together are rational. The (squares) on CF and FD (added) together (are) thus also rational. So, similarly, twice the (rectangle contained) by AE and EB is also commensurable with twice the (rectangle contained) by CF and FD. And twice the (rectangle contained) by AE and EB is medial. Therefore, twice the (rectangle contained) by CF and FD (is) also medial [Prop. 10.23 corr.]. CF and FD are thus (straight-lines which are) incommensurable in square [Prop 10.13], simultaneously making the sum of the squares on them rational, and twice the (rectangle contained) by them medial. The whole, CD, is thus that irrational (straight-line) called major [Prop. 10.39].

Thus, a (straight-line) commensurable (in length) with a major (straight-line) is major. (Which is) the very thing it was

required to show.

69.

Ἡ τῇ ὁρητὸν καὶ μέσον δυναμένῃ σύμμετρος [καὶ αὐτὴ] ὁρητὸν καὶ μέσον δυναμένη ἐστίν.

Ἔστω ὁρητὸν καὶ μέσον δυναμένη ἡ AB, καὶ τῇ AB σύμμετροςὁέστω ἡ $ΓΔ$· δεικτέον, ὅτι καὶ ἡ $ΓΔ$ ὁρητὸν καὶ μέσον δυναμένη ἐστίν.

Διῃρήσθω ἡ AB εἰξ τὰξ εὐθείας κατὰ τὸ E· αἱ AE, EB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον, τὸ δὸ ὑπὸ αὐτῶν ὁρητόν· καὶ τὰ αὐτὰ κατεσκευάσθω τοῖς πρότερον. ὁμοίως δὴ δείχομεν, ὅτι καὶ αἱ $ΓZ$, $ZΔ$ δυνάμει εἰσὶν ἀσύμμετροι, καὶ σύμμετρον τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AE, EB τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν $ΓZ$, $ZΔ$, τὸ δὲ ὑπὸ AE, EB τῷ ὑπὸ $ΓZ$, $ZΔ$· ὁώστε καὶ τὸ [μὲν] συγκείμενον ἐκ τῶν ἀπὸ τῶν $ΓZ$, $ZΔ$ τετραγώνων ἐστὶ μέσον, τὸ δὸ ὑπὸ τῶν $ΓZ$, $ZΔ$ ὁρητόν.

ὁΡητὸν ἄρα καὶ μέσον δυναμένη ἐστὶν ἡ $ΓΔ$· ὅπερ ἔδει δεῖχαι.

Let AB be the square-root of a rational plus a medial (area), and let CD be commensurable (in length) with AB. We must show that CD is also the square-root of a rational plus a medial (area).

Let AB be divided into its (component) straight-lines at E. AE and EB are thus incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them rational [Prop. 10.40]. And let the same construction be made as in the previous (propositions). So, similarly, we can show that CF and FD are also incommensurable in square, and that the sum of the (squares) on AE and EB (is) commensurable with the sum of the (squares) on CF and FD, and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD. And hence the sum of the squares on CF and FD is medial, and the (rectangle contained) by CF and FD (is) rational.

Thus, CD is the square-root of a rational plus a medial (area) [Prop. 10.40]. (Which is) the very thing it was required to show.

Proposition 69

A (straight-line) commensurable (in length) with the square-root of a rational plus a medial (area) is [itself also] the square-root of a rational plus a medial (area).

70.

Ἡ τῇ δύο μέσα δυναμένῃ σύμμετρος δύο μέσα δυναμένη ἐστίν.

Ἔστω δύο μέσα δυναμένη ἡ AB, καὶ τῇ AB σύμμετρος ἡ $ΓΔ$· δεικτέον, ὅτι καὶ ἡ $ΓΔ$ δύο μέσα δυναμένη ἐστίν.

Proposition 70

A (straight-line) commensurable (in length) with the square-root of (the sum of) two medial (areas) is (itself also) the square-root of (the sum of) two medial (areas).

Let AB be the square-root of (the sum of) two medial (areas), and (let) CD (be) commensurable (in length) with AB. We must show that CD is also the square-root of (the sum of) two medial (areas).

Ἐπεὶ γὰρ δύο μέσα δυναμένη ἐστὶν ἡ AB, διῃρήσθω εἰξ τὰξ εὐθείας κατὰ τὸ E· αἱ AE, EB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τό τε συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν [τετραγώνων] μέσον καὶ τὸ ὑπὸ αὐτῶν μέσον καὶ ὀέτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE, EB τετραγώνων τῷ ὑπὸ τῶν AE, EB· καὶ κατεσκευάσθω τὰ αὐτὰ τοῖς πρότερον. ὁμοίως δὴ δείχομεν, ὅτι καὶ αἱ $ΓZ$, $ZΔ$ δυνάμει εἰσὶν ἀσύμμετροι καὶ σύμμετρον τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AE, EB τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν $ΓZ$, $ZΔ$, τὸ δὲ ὑπὸ τῶν AE, EB τῷ ὑπὸ τῶν $ΓZ$, $ZΔ$· ὥστε καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $ΓZ$, $ZΔ$ τετραγώνων μέσον ἐστὶ καὶ τὸ ὑπὸ τῶν $ΓZ$, $ZΔ$ μέσον καὶ ὀέτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $ΓZ$, $ZΔ$ τετραγώνων τῷ ὑπὸ τῶν $ΓZ$, $ZΔ$.

Ἡ ἄρα $ΓΔ$ δύο μέσα δυναμένη ἐστίν· ὅπερ ἔδει δεῖχαι.

For since AB is the square-root of (the sum of) two medial (areas), let it be divided into its (component) straight-lines at E. Thus, AE and EB are incommensurable in square, making the sum of the [squares] on them medial, and the (rectangle contained) by them medial, and, moreover, the sum of the (squares) on AE and EB incommensurable with the (rectangle) contained by AE and EB [Prop. 10.41]. And let the same construction be made as in the previous (propositions). So, similarly, we can show that CF and FD are also incommensurable in square, and (that) the sum of the (squares) on AE and EB (is) commensurable with the sum of the (squares) on CF and FD, and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD. Hence, the sum of the squares on CF and FD is also medial, and the (rectangle contained) by CF and FD (is) medial, and, moreover, the sum of the squares on CF and FD (is) incommensurable with the (rectangle contained) by CF and FD.

Thus, CD is the square-root of (the sum of) two medial (areas) [Prop. 10.41]. (Which is) the very thing it was required to show.

71.

ὁΡητοῦ καὶ μέσου συντιθεμένου τέσσαρεςὀάλογοι γίγνονται ἤτοι ἐκ δύο ὀνομάτωνὁὴ ἐκ δύο μέσων πρώτηὁὴ μείζωνὁὴ ὁρητὸν καὶ μέσον δυναμένη.

Ἔστω ὁρητὸν μὲν τὸ AB, μέσον δὲ τὸ $ΓΔ$· λέγω, ὅτι ἡ τὸ $AΔ$ θωρίον δυναμένη ἤτοι ἐκ δύο ὀνομάτων ἐστὶνὁὴ ἐκ δύο μέσων πρώτηὁὴ μείζωνὁὴ ὁρητὸν καὶ μέσον δυναμένη.

Τὸ γὰρ AB τοῦ $ΓΔ$ ἤτοι μεῖζόν ἐστινὁὴ ὀέλασσον. ὀέστω πρότερον μεῖζον· καὶ ἐκκείσθω ὁρητὴ ἡ EZ, καὶ παραβεβλήσθω παρὰ τὴν EZ τῷ AB ἴσον τὸ EH πλάτος ποιοῦν τὴν $EΘ$· τῷ δὲ $ΔΓ$ ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ $ΘI$ πλάτος ποιοῦν τὴν $ΘK$. καὶ ἐπεὶ ὁρητόν ἐστι τὸ AB καί ἐστιν ἴσον τῷ EH, ὁρητὸν ἄρα καὶ τὸ EH. καὶ παρὰ [ὁρητὴν] τὴν EZ παραβέβληται πλάτος ποιοῦν τὴν $EΘ$· ἡ $EΘ$ ἄρα ὁρητή ἐστι καὶ σύμμετρος τῇ EZ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ $ΓΔ$ καί ἐστιν ἴσον τῷ $ΘI$, μέσον ἄρα ἐστὶ καὶ τὸ $ΘI$. καὶ παρὰ ὁρητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν $ΘK$· ὁρητὴ ἄρα ἐστὶν ἡ $ΘK$ καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ μέσον ἐστὶ τὸ $ΓΔ$, ὁρητὸν δὲ τὸ AB, ἀσύμμετρον ἄρα ἐστὶ τὸ AB τῷ $ΓΔ$· ὥστε καὶ τὸ EH ἀσύμμετρόν ἐστι τῷ

Proposition 71

When a rational and a medial (area) are added together, four irrational (straight-lines) arise (as the square-roots of the total area)—either a binomial, or a first bimedial, or a major, or the square-root of a rational plus a medial (area).

Let AB be a rational (area), and CD a medial (area). I say that the square-root of area AD is either binomial, or first bimedial, or major, or the square-root of a rational plus a medial (area).

For AB is either greater or less than CD. Let it, first of all, be greater. And let the rational (straight-line) EF be laid down. And let (the rectangle) EG, equal to AB, be applied to EF, producing EH as breadth. And let (the recatangle) HI, equal to DC, be applied to EF, producing HK as breadth. And since AB is rational, and is equal to EG, EG is thus also rational. And it has been applied to the [rational] (straight-line) EF, producing EH as breadth. EH is thus rational, and commensurable in length with EF [Prop. 10.20]. Again, since CD is medial, and is equal to HI, HI is thus also medial. And it is applied to the rational (straight-line) EF, producing HK as

ΘΙ. ὡξ δὲ τὸ ΕΗ πρὸς τὸ ΘΙ, οὕτως ἐστὶν ἡ ΕΘ πρὸς τὴν ΘΚ· ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ ΕΘ τῇ ΘΚ μήκει. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ ΕΘ, ΘΚ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΕΚ διῃρημένη κατὰ τὸ Θ. καὶ ἐπεὶ μεῖζόν ἐστι τὸ ΑΒ τοῦ ΓΔ, ἴσον δὲ τὸ μὲν ΑΒ τῷ ΕΗ, τὸ δὲ ΓΔ τῷ ΘΙ, μεῖζον ἄρα καὶ τὸ ΕΗ τοῦ ΘΙ· καὶ ἡ ΕΘ ἄρα μείζων ἐστὶ τῆς ΘΚ. ἤτοι οὐκοῦν ἡ ΕΘ τῆς ΘΚ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκειὁὴ τῷ ἀπὸ ἀσυμμέτρου. δυνάσθω πρότερον τῷ ἀπὸ συμμέτρου ἑαυτῇ· καί ἐστιν ἡ μείζων ἡ ΘΕ σύμμετρος τῇ ἐκκειμένῃ ὁρητῇ τῇ ΕΖ· ἡ ἄρα ΕΚ ἐκ δύο ὀνομάτων ἐστὶ πρώτη. ὁρητὴ δὲ ἡ ΕΖ· ἐαν δὲ θωρίον περιέθηται ὑπὸ ὁρητῆς καὶ τῆς ἐκ δύο ὀνομάτων πρώτης, ἡ τὸ θωρίον δυναμένη ἐκ δύο ὀνομάτων ἐστίν. ἡ ἄρα τὸ ΕΙ δυναμένη ἐκ δύο ὀνομάτων ἐστίν· ὁὥστε καὶ ἡ τὸ ΑΔ δυναμένη ἐκ δύο ὀνομάτων ἐστίν. ἀλλὰ δὴ δυνάσθω ἡ ΕΘ τῆς ΘΚ μεῖζον τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ· καί ἐστιν ἡ μείζων ἡ ΕΘ σύμμετρος τῇ ἐκκειμένῃ ὁρητῇ τῇ ΕΖ μήκει· ἡ ἄρα ΕΚ ἐκ δύο ὀνομάτων ἐστὶ τετάρτη. ὁρητὴ δὲ ἡ ΕΖ· ἐαν δὲ θωρίον περιέθηται ὑπὸ ὁρητῆς καὶ τῆς ἐκ δύο ὀνομάτων τετάρτης, ἡ τὸ θωρίον δυναμένηὀάλογόξ ἐστιν ἡ καλουμένη μείζων. ἡ ἄρα τὸ ΕΙ θωρίον δυναμένη μείζων ἐστίν· ὁὥστε καὶ ἡ τὸ ΑΔ δυναμένη μείζων ἐστίν.

Ἀλλὰ δὴ ὀέστωὀέλασσον τὸ ΑΒ τοῦ ΓΔ· καὶ τὸ ΕΗ ἄραὀέλασσόν ἐστι τοῦ ΘΙ· ὁὥστε καὶ ἡ ΕΘ ἐλάσσων ἐστὶ τῆς ΘΚ. ἤτοι δὲ ἡ ΘΚ τῆς ΕΘ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ ὀὴ τῷ ἀπὸ ἀσυμμέτρου. δυνάσθω πρότερον τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει· καί ἐστιν ἡ ἐλάσσων ἡ ΕΘ σύμμετρος τῇ ἐκκειμένῃ ὁρητῇ τῇ ΕΖ μήκει· ἡ ἄρα ΕΚ ἐκ δύο ὀνομάτων ἐστὶ δευτέρα. ὁρητὴ δὲ ἡ ΕΖ· ἐαν δὲ θωρίον περιέθηται ὑπὸ ὁρητῆς καὶ τῆς ἐκ δύο ὀνομάτων δευτέρας, ἡ τὸ θωρίον δυναμένη ἐκ δύο μέσων ἐστὶ πρώτη. ἡ ἄρα τὸ ΕΙ θωρίον δυναμένη ἐκ δύο μέσων ἐστὶ πρώτη· ὁὥστε καὶ ἡ τὸ ΑΔ δυναμένη ἐκ δύο μέσων ἐστὶ πρώτη. ἀλλὰ δὴ ἡ ΘΚ τῆς ΘΕ μεῖζον δυνάσθω τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καί ἐστιν ἡ ἐλάσσων ἡ ΕΘ σύμμετρος τῇ ἐκκειμένῃ ὁρητῇ τῇ ΕΖ· ἡ ἄρα ΕΚ ἐκ δύο ὀνομάτων ἐστὶ πέμπτη. ὁρητὴ δὲ ἡ ΕΖ· ἐαν δὲ θωρίον περιέθηται ὑπὸ ὁρητῆς καὶ τῆς ἐκ δύο ὀνομάτων πέμπτης, ἡ τὸ θωρίον δυναμένη ὁρητὸν καὶ μέσον δυναμένη ἐστίν. ἡ ἄρα τὸ ΕΙ θωρίον δυναμένη ὁρητὸν καὶ μέσον δυναμένη ἐστίν· ὁὥστε καὶ ἡ τὸ ΑΔ θωρίον δυναμένη ὁρητὸν καὶ μέσον δυναμένη ἐστίν.

breadth. HK is thus rational, and incommensurable in length with EF [Prop. 10.22]. And since CD is medial, and AB rational, AB is thus incommensurable with CD. Hence, EG is also incommensurable with HI. And as EG (is) to HI, so EH is to HK [Prop. 6.1]. Thus, EH is also incommensurable in length with HK [Prop. 10.11]. And they are both rational. Thus, EH and HK are rational (straight-lines which are) commensurable in square only. EK is thus a binomial (straight-line), having been divided (into its component terms) at H [Prop. 10.36]. And since AB is greater than CD, and AB (is) equal to EG, and CD to HI, EG (is) thus also greater than HI. Thus, EH is also greater than HK [Prop. 5.14]. Therefore, the square on EH is greater than (the square on) HK either by the (square) on (some straight-line) commensurable in length with (EH), or by the (square) on (some straight-line) incommensurable (in length with EH). Let it, first of all, be greater by the (square) on (some straight-line) commensurable (in length with EH). And the greater (of the two components of EK) HE is commensurable (in length) with the (previously) laid down (straight-line) EF. EK is thus a first binomial (straight-line) [Def. 10.5]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a first binomial (straight-line) then the square-root of the area is a binomial (straight-line) [Prop. 10.54]. Thus, the square-root of EI is a binomial (straight-line). Hence the square-root of AD is also a binomial (straight-line). And, so, let the square on EH be greater than (the square on) HK by the (square) on (some straight-line) incommensurable (in length) with (EH). And the greater (of the two components of EK) EH is commensurable in length with the (previously) laid down rational (straight-line) EF. Thus, EK is a fourth binomial (straight-line) [Def. 10.8]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a fourth binomial (straight-line) then the square-root of the area is the irrational (straight-line) called major [Prop. 10.57]. Thus, the square-root of area EI is a major (straight-line). Hence, the square-root of AD is also major.

And so, let AB be less than CD. Thus, EG is also less than HI. Hence, EH is also less than HK [Props. 6.1, 5.14]. And the square on HK is greater than (the square on) EH either by the (square) on (some straight-line) commensurable (in length) with (HK), or by the (square) on (some straight-line) incommensurable (in length) with (HK). Let it, first of all, be greater by the square on (some straight-line) commensurable in length with (HK). And the lesser (of the two components of EK) EH is commensurable in length with the (previously) laid down rational (straight-line) EF. Thus, EK is a second binomial (straight-line) [Def. 10.6]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a second binomial (straight-line) then the square-root of the area is a first bimedial (straight-line) [Prop. 10.55]. Thus, the square-

root of area EI is a first bimedial (straight-line). Hence, the square-root of AD is also a first bimedial (straight-line). And so, let the square on HK be greater than (the square on) HE by the (square) on (some straight-line) incommensurable (in length) with (HK). And the lesser (of the two components of EK) EH is commensurable (in length) with the (previously) laid down rational (straight-line) EF. Thus, EK is a fifth binomial (straight-line) [Def. 10.9]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a fifth binomial (straight-line) then the square-root of the area is the square-root of a rational plus a medial (area) [Prop. 10.58]. Thus, the square-root of area EI is the square-root of a rational plus a medial (area). Hence, the square-root of area AD is also the square-root of a rational plus a medial (area).

Thus, when a rational and a medial area are added together,

ὁΡητοῦ ἄρα καὶ μέσου συντιθεμένου τέσσαρες ὀάλογοι γίγνονται ἤτοι ἐκ δύο ὀνομάτων ἡ ἐκ δύο μέσων πρώτη ἡ μείζων ἡ ὁρητὸν καὶ μέσον δυναμένη· ὅπερ ἔδει δεῖξαι.

62.

Δύο μέσων ἀσυμμέτρων ἀλλήλοις συντιθεμένων αἱ λοιπαὶ δύο ὀάλογοι γίγνονται ἤτοι ἐκ δύο μέσων δευτέρα ἡ [ἤ] δύο μέσα δυναμένη.

Συγκείσθω γὰρ δύο μέσα ἀσύμμετρα ἀλλήλοις τὰ AB, ΓΔ· λέγω, ὅτι ἡ τὸ AΔ ὡρίον δυναμένη ἤτοι ἐκ δύο μέσων ἐστὶ δευτέρα ἡ δύο μέσα δυναμένη.

Τὸ γὰρ AB τοῦ ΓΔ ἤτοι μεῖζόν ἐστιν ἢ ὐέλασσον. ὐέστω, εἰ τύθον, πρότερον μεῖζον τὸ AB τοῦ ΓΔ· καὶ ἐκκείσθω ὁρητὴ ἡ EZ, καὶ τῷ μὲν AB ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ EH πλάτος ποιοῦν τὴν EΘ, τῷ δὲ ΓΔ ἴσον τὸ ΘI πλάτος ποιοῦν τὴν ΘK. καὶ ἐπεὶ μέσον ἐστὶν ἑκάτερον τῶν AB, ΓΔ, μέσον ἄρα καὶ ἑκάτερον τῶν EH, ΘI. καὶ παρὰ ὁρητὴν τὴν ZE παράκειται πλάτος ποιοῦν τὰξ EΘ, ΘK· ἑκατέρα ἄρα τῶν EΘ, ΘK ὁρητή ἐστι καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ AB τῷ ΓΔ, καί ἐστιν ἴσον τὸ μὲν AB τῷ EH, τὸ δὲ ΓΔ τῷ ΘI, ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ EH τῷ ΘI. ὡξ δὲ τὸ EH πρὸς τὸ ΘI, οὕτως ἐστὶν ἡ EΘ πρὸς ΘK· ἀσύμμετρος ἄρα ἐστὶν ἡ EΘ τῇ ΘK μήκει. αἱ EΘ, ΘK ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ EK. ἤτοι δὲ ἡ EΘ τῆς ΘK μεῖζον δύναται τῷ ἀπὸ four irrational (straight-lines) arise (as the square-roots of the total area)—either a binomial, or a first bimedial, or a major, or the square-root of a rational plus a medial (area). (Which is) the very thing it was required to show.

Proposition 72

When two medial (areas which are) incommensurable with one another are added together, the remaining two irrational (straight-lines) arise (as the square-roots of the total area)—either a second bimedial, or the square-root of (the sum of) two medial (areas).

For let the two medial (areas) AB and CD, (which are) incommensurable with one another, be added together. I say that the square-root of area AD is either a second bimedial, or the square-root of (the sum of) two medial (areas).

For AB is either greater than or less than CD. By chance, let AB, first of all, be greater than CD. And let the rational (straight-line) EF be laid down. And let EG, equal to AB, be applied to EF, producing EH as breadth, and HI, equal to CD, producing HK as breadth. And since AB and CD are each medial, EG and HI (are) thus also each medial. And they are applied to the rational straight-line FE, producing EH and HK (respectively) as breadth. Thus, EH and HK are each rational (straight-lines which are) incommensurable in length with EF [Prop. 10.22]. And since AB is incommensurable with CD, and AB is equal to EG, and CD to HI, EG is thus also in-

συμμέτρου ἑαυτῇ ὀὴ τῷ ἀπὸ ἀσυμμέτρου. δυνάσθω πρότερον τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει· καὶ οὐδετέρα τῶν ΕΘ, ΘΚ σύμμετρός ἐστι τῇ ἐκκειμένῃ ὁρητῇ τῇ EZ μήκει· ἡ EK ἄρα ἐκ δύο ὀνομάτων ἐστὶ τρίτη. ὁρητὴ δὲ ἡ EZ· ἐαν δὲ θωρίον περιέθηται ὑπὸ ὁρητῆς καὶ τῆς ἐκ δύο ὀνομάτων τρίτης, ἡ τὸ θωρίον δυναμένη ἐκ δύο μέσων ἐστὶ δευτέρα· ἡ ἄρα τὸ EI, τουτέστι τὸ ΑΔ, δυναμένη ἐκ δύο μέσων ἐστὶ δευτέρα. ἀλλα δὴ ἡ ΕΘ τῆς ΘΚ μεῖζον δυνάσθω τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ μήκει· καὶ ἀσύμμετρός ἐστιν ἑκατέρα τῶν ΕΘ, ΘΚ τῇ EZ μήκει· ἡ ἄρα EK ἐκ δύο ὀνομάτων ἐστὶν ὀέκτη. ἐαν δὲ θωρίον περιέθηται ὑπὸ ὁρητῆς καὶ τῆς ἐκ δύο ὀνομάτων ὀέκτης, ἡ τὸ θωρίον δυναμένη ἡ δύο μέσα δυναμένη ἐστίν· ὡστε καὶ ἡ τὸ ΑΔ θωρίον δυναμένη ἡ δύο μέσα δυναμένη ἐστίν.

commensurable with HI. And as EG (is) to HI, so EH is to HK [Prop. 6.1]. EH is thus incommensurable in length with HK [Prop. 10.11]. Thus, EH and HK are rational (straight-lines which are) commensurable in square only. EK is thus a binomial (straight-line) [Prop. 10.36]. And the square on EH is greater than (the square on) HK either by the (square) on (some straight-line) commensurable (in length) with (EH), or by the (square) on (some straight-line) incommensurable (in length with EH). Let it, first of all, be greater by the square on (some straight-line) commensurable in length with (EH). And neither of EH or HK is commensurable in length with the (previously) laid down rational (straight-line) EF. Thus, EK is a third binomial (straight-line) [Def. 10.7]. And EF (is) rational. And if an area is contained by a rational (straight-line) and a third binomial (straight-line) then the square-root of the area is a second bimedial (straight-line) [Prop. 10.56]. Thus, the square-root of EI—that is to say, of AD—is a second bimedial. And so, let the square on EH be greater than (the square) on HK by the (square) on (some straight-line) incommensurable in length with (EH). And EH and HK are each incommensurable in length with EF. Thus, EK is a sixth binomial (straight-line) [Def. 10.10]. And if an area is contained by a rational (straight-line) and a sixth binomial (straight-line) then the square-root of the area is the square-root of (the sum of) two medial (areas) [Prop. 10.59]. Hence, the square-root of area AD is also the square-root of (the sum of) two medial (areas).

[Ὁμοίως δὴ δείξομεν, ὅτι κἂν ἔλαττον ᾖ τὸ AB τοῦ $ΓΔ$, ἡ τὸ $AΔ$ χωρίον δυναμένη ἢ ἐκ δύο μέσων δευτέρα ἐστὶν ἤτοι δύο μέσα δυναμένη].

Δύο ἄρα μέσων ἀσυμμέτρων ἀλλήλοις συντιθεμένων αἱ λοιπαὶ δύο ἀλόγοι γίγνονται ἤτοι ἐκ δύο μέσων δευτέρα ἢ δύο μέσα δυναμένη.

Ἡ ἐκ δύο ὀνομάτων καὶ αἱ μετὰ αὐτὴν ἄλογοι οὔτε τῇ μέσῃ οὔτε ἀλλήλαις εἰσὶν αἱ αὐταί. τὸ μὲν γὰρ ἀπὸ μέσης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ ῥητὴν καὶ ἀσύμμετρον τῇ παρ' ἣν παράκειται μήκει. τὸ δὲ ἀπὸ τῆς ἐκ δύο ὀνομάτων παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πρώτην. τὸ δὲ ἀπὸ τῆς ἐκ δύο μέσων πρώτης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων δευτέραν. τὸ δὲ ἀπὸ τῆς ἐκ δύο μέσων δευτέρας παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τρίτην. τὸ δὲ ἀπὸ τῆς μείζονος παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων τετάρτην. τὸ δὲ ἀπὸ τῆς ῥητὸν καὶ μέσον δυναμένης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων πέμπτην. τὸ δὲ ἀπὸ τῆς δύο μέσα δυναμένης παρὰ ῥητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων ἕκτην. τὰ δὸ εἰρημένα πλάτη διαφέρει τοῦ τε πρώτου καὶ ἀλλήλων, τοῦ μὲν πρώτου, ὅτι ῥητή ἐστιν, ἀλλήλων δέ, ὅτι τῇ τάχει οὐκ εἰσὶν αἱ αὐταί· ὥστε καὶ αὐταὶ αἱ ἀλογοι διαφέρουσιν ἀλλήλων.

[So, similarly, we can show that, even if AB is less than CD, the square-root of area AD is either a second bimedial or the square-root of (the sum of) two medial (areas).]

Thus, when two medial (areas which are) incommensurable with one another are added together, the remaining two irrational (straight-lines) arise (as the square-roots of the total area)—either a second bimedial, or the square-root of (the sum of) two medial (areas).

A binomial (straight-line), and the (other) irrational (straight-lines) after it, are neither the same as a medial (straight-line) nor (the same) as one another. For the (square) on a medial (straight-line), applied to a rational (straight-line), produces as breadth a rational (straight-line which is) also incommensurable in length with (the straight-line) to which it is applied [Prop. 10.22]. And the (square) on a binomial (straight-line), applied to a rational (straight-line), produces as breadth a first binomial [Prop. 10.60]. And the (square) on a first bimedial (straight-line), applied to a rational (straight-line), produces as breadth a second binomial [Prop. 10.61]. And the (square) on a second bimedial (straight-line), applied to a rational (straight-line), produces as breadth a third binomial [Prop. 10.62]. And the (square) on a major (straight-line), applied to a rational (straight-line), produces as breadth a fourth binomial [Prop. 10.63]. And the (square) on the square-root of a ratio-

nal plus a medial (area), applied to a rational (straight-line), produces as breadth a fifth binomial [Prop. 10.64]. And the (square) on the square-root of (the sum of) two medial (areas), applied to a rational (straight-line), produces as breadth a sixth binomial [Prop. 10.65]. And the aforementioned breadths differ from the first (breadth), and from one another—from the first, because it is rational—and from one another, because they are not the same in order. Hence, the (previously mentioned) irrational (straight-lines) themselves also differ from one another.

73.

Ἐὰν ἀπὸ ὁρητῆς ὁρητὴ ἀφαιρεθῇ δυνάμει μόνον σύμμετρος οὐσῦσα τῇ ὅλῃ, ἡ λοιπὴ ἀλογός ἐστιν· καλείσθω δὲ ἀποτομή.

Proposition 73

If a rational (straight-line), which is commensurable in square only with the whole, is subtracted from a(nother) rational (straight-line) then the remainder is an irrational (straight-line). Let it be called an apotome.

Ἀπὸ γὰρ ὁρητῆς τῆς AB ὁρητὴ ἀφῃρήσθω ἡ $BΓ$ δυνάμει μόνον σύμμετρος οὐσῦσα τῇ ὅλῃ· λέγω, ὅτι ἡ λοιπὴ ἡ $AΓ$ ἀλογός ἐστιν ἡ καλουμένη ἀποτομή.

Ἐπεὶ γὰρ ἀσύμμετρός ἐστιν ἡ AB τῇ $BΓ$ μήκει, καί ἐστιν ὡς ἡ AB πρὸς τὴν $BΓ$, οὑτως τὸ ἀπὸ τῆς AB πρὸς τὸ ὑπὸ τῶν AB, $BΓ$, ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς AB τῷ ὑπὸ τῶν AB, $BΓ$. ἀλλὰ τῷ μὲν ἀπὸ τῆς AB σύμμετρά ἐστι τὰ ἀπὸ τῶν AB, $BΓ$ τετράγωνα, τῷ δὲ ὑπὸ τῶν AB, $BΓ$ σύμμετρόν ἐστι τὸ δὶξ ὑπὸ τῶν AB, $BΓ$. καὶ ἐπειδήπερ τὰ ἀπὸ τῶν AB, $BΓ$ ἴσα ἐστὶ τῷ δὶξ ὑπὸ τῶν AB, $BΓ$ μετὰ τοῦ ἀπὸ $ΓA$, καὶ λοιπῷ ἄρα τῷ ἀπὸ τῆς $AΓ$ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν AB, $BΓ$. ὁρητὰ δὲ τὰ ἀπὸ τῶν AB, $BΓ$· ἀλογος ἄρα ἐστὶν ἡ $AΓ$· καλείσθω δὲ ἀποτομή. ὅπερ ἔδει δεῖχαι.

For let the rational (straight-line) BC, which commensurable in square only with the whole, be subtracted from the rational (straight-line) AB. I say that the remainder AC is that irrational (straight-line) called an apotome.

For since AB is incommensurable in length with BC, and as AB is to BC, so the (square) on AB (is) to the (rectangle contained) by AB and BC [Prop. 10.21 lem.], the (square) on AB is thus incommensurable with the (rectangle contained) by AB and BC [Prop. 10.11]. But, the (sum of the) squares on AB and BC is commensurable with the (square) on AB [Prop. 10.15], and twice the (rectangle contained) by AB and BC is commensurable with the (rectangle contained) by AB and BC [Prop. 10.6]. And, inasmuch as the (sum of the squares) on AB and BC is equal to twice the (rectangle contained) by AB and BC plus the (square) on CA [Prop. 2.7], the (sum of the squares) on AB and BC is thus also incommensurable with the remaining (square) on AC [Props. 10.13, 10.16]. And the (sum of the squares) on AB and BC is rational. AC is thus an irrational (straight-line) [Def. 10.4]. And let it be called an apotome.[†] (Which is) the very thing it was required to show.

[†] See footnote to Prop. 10.36.

74.

Proposition 74

Ἐὰν ἀπὸ μέσης μέση ἀφαιρεθῇ δυνάμει μόνον σύμμετρος οὐσῦσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ὁρητὸν περιέθουσα, ἡ λοιπὴ ἀλογός ἐστιν· καλείσθω δὲ μέσης ἀποτομὴ πρώτη.

If a medial (straight-line), which is commensurable in square only with the whole, and which contains a rational (area) with the whole, is subtracted from a(nother) medial (straight-line) then the remainder is an irrational (straight-line). Let it be called a first apotome of a medial (straight-line).

Ἀπὸ γὰρ μέσης τῆς ΑΒ μέση ἀφῃρήσθω ἡ ΒΓ δυνάμει μόνον σύμμετρος οὐσῦσα τῇ ΑΒ, μετὰ δὲ τῆς ΑΒ ὁρητὸν ποιοῦσα τὸ ὑπὸ τῶν ΑΒ, ΒΓ· λέγω, ὅτι ἡ λοιπὴ ἡ ΑΓἀλογός ἐστιν· καλείσθω δὲ μέσης ἀποτομὴ πρώτη.

Ἐπεὶ γὰρ αἱ ΑΒ, ΒΓ μέσαι εἰσίν, μέσα ἐστὶ καὶ τὰ ἀπὸ τῶν ΑΒ, ΒΓ. ὁρητὸν δὲ τὸ δἰξ ὑπὸ τῶν ΑΒ, ΒΓ· ἀσύμμετρα ἄρα τὰ ἀπὸ τῶν ΑΒ, ΒΓ τῷ δἰξ ὑπὸ τῶν ΑΒ, ΒΓ· καὶ λοιπῷ ἄρα τῷ ἀπὸ τῆς ΑΓ ἀσύμμετρόν ἐστι τὸ δἰξ ὑπὸ τῶν ΑΒ, ΒΓ, ἐπεὶ κὰν τὸ ὅλον ἑνὶ αὐτῶν ἀσύμμετρονὁῇ, καὶ τὰ ἐξ ἀρθῆς μεγέθη ἀσύμμετραὁέσται. ὁρητὸν δὲ τὸ δἰξ ὑπὸ τῶν ΑΒ, ΒΓ· ὀάλογον ἄρα τὸ ἀπὸ τῆς ΑΓ· ὀάλογος ἄρα ἐστὶν ἡ ΑΓ· καλείσθω δὲ μέσης ἀποτομὴ πρώτη.

For let the medial (straight-line) *BC*, which is commensurable in square only with *AB*, and which makes with *AB* the rational (rectangle contained) by *AB* and *BC*, be subtracted from the medial (straight-line) *AB* [Prop. 10.27]. I say that the remainder *AC* is an irrational (straight-line). Let it be called the first apotome of a medial (straight-line).

For since *AB* and *BC* are medial (straight-lines), the (sum of the squares) on *AB* and *BC* is also medial. And twice the (rectangle contained) by *AB* and *BC* (is) rational. The (sum of the squares) on *AB* and *BC* (is) thus incommensurable with twice the (rectangle contained) by *AB* and *BC*. Thus, twice the (rectangle contained) by *AB* and *BC* is also incommensurable with the remaining (square) on *AC* [Prop. 2.7], since if the whole is incommensurable with one of the (constituent magnitudes) then the original magnitudes will also be incommensurable (with one another) [Prop. 10.16]. And twice the (rectangle contained) by *AB* and *BC* (is) rational. Thus, the (square) on *AC* is irrational. Thus, *AC* is an irrational (straight-line) [Def. 10.4]. Let it be called a first apotome of a medial (straight-line).†

† See footnote to Prop. 10.37.

75.

Ἐὰν ἀπὸ μέσης μέση ἀφαιρεθῇ δυνάμει μόνον σύμμετρος οὐσῦσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης μέσον περιέθουσα, ἡ λοιπὴ ἀλογός ἐστιν· καλείσθω δὲ μέσης ἀποτομὴ δευτέρα.

Ἀπὸ γὰρ μέσης τῆς ΑΒ μέση ἀφῃρήσθω ἡ ΓΒ δυνάμει μόνον σύμμετρος οὐσῦσα τῇ ὅλῃ τῇ ΑΒ, μετὰ δὲ τῆς ὅλης τῆς ΑΒ μέσον περιέθουσα τὸ ὑπὸ τῶν ΑΒ, ΒΓ· λέγω, ὅτι ἡ λοιπὴ ἡ ΑΓἀλογός ἐστιν· καλείσθω δὲ μέσης ἀποτομὴ δευτέρα.

Proposition 75

If a medial (straight-line), which is commensurable in square only with the whole, and which contains a medial (area) with the whole, is subtracted from a(nother) medial (straight-line) then the remainder is an irrational (straight-line). Let it be called a second apotome of a medial (straight-line).

For let the medial (straight-line) *CB*, which is commensurable in square only with the whole, *AB*, and which contains with the whole, *AB*, the medial (rectangle contained) by *AB* and *BC*, be subtracted from the medial (straight-line) *AB* [Prop. 10.28]. I say that the remainder *AC* is an irrational (straight-line). Let it be called a second apotome of a medial

(straight-line).

Ἐκκείσθω γὰρ ὁρητὴ ἡ ΔΙ, καὶ τοῖς μὲν ἀπὸ τῶν ΑΒ, ΒΓ ἴσον παρὰ τὴν ΔΙ παραβεβλήσθω τὸ ΔΕ πλάτος ποιοῦν τὴν ΔΗ, τῷ δὲ δὶξ ὑπὸ τῶν ΑΒ, ΒΓ ἴσον παρὰ τὴν ΔΙ παραβεβλήσθω τὸ ΔΘ πλάτος ποιοῦν τὴν ΔΖ· λοιπὸν ἄρα τὸ ΖΕ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΓ. καὶ ἐπεὶ μέσα καὶ σύμμετρά ἐστι τὰ ἀπὸ τῶν ΑΒ, ΒΓ, μέσον ἄρα καὶ τὸ ΔΕ. καὶ παρὰ ὁρητὴν τὴν ΔΙ παράκειται πλάτος ποιοῦν τὴν ΔΗ· ὁρητὴ ἄρα ἐστὶν ἡ ΔΗ καὶ ἀσύμμετρος τῇ ΔΙ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ ὑπὸ τῶν ΑΒ, ΒΓ, καὶ τὸ δὶξ ἄρα ὑπὸ τῶν ΑΒ, ΒΓ μέσον ἐστίν. καί ἐστιν ἴσον τῷ ΔΘ· καὶ τὸ ΔΘ ἄρα μέσον ἐστίν. καὶ παρὰ ὁρητὴν τὴν ΔΙ παραβέβληται πλάτος ποιοῦν τὴν ΔΖ· ὁρητὴ ἄρα ἐστὶν ἡ ΔΖ καὶ ἀσύμμετρος τῇ ΔΙ μήκει. καὶ ἐπεὶ αἱ ΑΒ, ΒΓ δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετρος ἄρα ἐστὶν ἡ ΑΒ τῇ ΒΓ μήκει· ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς

For let the rational (straight-line) DI be laid down. And let DE, equal to the (sum of the squares) on AB and BC, be applied to DI, producing DG as breadth. And let DH, equal to twice the (rectangle contained) by AB and BC, be applied to DI, producing DF as breadth. The remainder FE is thus equal to the (square) on AC [Prop. 2.7]. And since the (squares) on AB and BC are medial and commensurable (with one another), DE (is) thus also medial [Props. 10.15, 10.23 corr.]. And it is applied to the rational (straight-line) DI, producing DG as breadth. Thus, DG is rational, and incommensurable in length with DI [Prop. 10.22]. Again, since the (rectangle contained) by AB and BC is medial, twice the (rectangle contained) by AB and BC is thus also medial [Prop. 10.23 corr.]. And it is equal to DH. Thus, DH is also medial. And it has been applied

AB τετράγωνον τῷ ὑπὸ τῶν AB, BΓ. ἀλλὰ τῷ μὲν ἀπὸ τῆς AB σύμμετρά ἐστι τὰ ἀπὸ τῶν AB, BΓ, τῷ δὲ ὑπὸ τῶν AB, BΓ σύμμετρόν ἐστι τὸ δἰξ ὑπὸ τῶν AB, BΓ· ἀσύμμετρον ἄρα ἐστὶ τὸ δἰξ ὑπὸ τῶν AB, BΓ τοῖς ἀπὸ τῶν AB, BΓ. ἴσον δὲ τοῖς μὲν ἀπὸ τῶν AB, BΓ τὸ ΔE, τῷ δὲ δἰξ ὑπὸ τῶν AB, BΓ τὸ ΔΘ· ἀσύμμετρον ἄρα [ἐστὶ] τὸ ΔE τῷ ΔΘ. ὡξ δὲ τὸ ΔE πρὸς τὸ ΔΘ, οὕτως ἡ HΔ πρὸς τὴν ΔZ· ἀσύμμετρος ἄρα ἐστὶν ἡ HΔ τῇ ΔZ. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ ἄρα HΔ, ΔZ ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἡ ZH ἄρα ἀποτομή ἐστιν. ὁρητὴ δὲ ἡ ΔI· τὸ δὲ ὑπὸ ὁρητῆς καὶ ἀλόγου περιεχόμενονὁάλογόν ἐστιν, καί ἡ δυναμένη αὐτὸ ὁάλογόξ ἐστιν. καὶ δύναται τὸ ZE ἡ AΓ· ἡ AΓ ἄραὁάλογόξ ἐστιν· καλείσθω δὲ μέσης ἀποτομὴ δευτέρα. ὅπερ ἔδει δεῖχαι.

to the rational (straight-line) *DI*, producing *DF* as breadth. *DF* is thus rational, and incommensurable in length with *DI* [Prop. 10.22]. And since *AB* and *BC* are commensurable in square only, *AB* is thus incommensurable in length with *BC*. Thus, the square on *AB* (is) also incommensurable with the (rectangle contained) by *AB* and *BC* [Props. 10.21 lem., 10.11]. But, the (sum of the squares) on *AB* and *BC* is commensurable with the (square) on *AB* [Prop. 10.15], and twice the (rectangle contained) by *AB* and *BC* is commensurable with the (rectangle contained) by *AB* and *BC* [Prop. 10.6]. Thus, twice the (rectangle contained) by *AB* and *BC* is incommensurable with the (sum of the squares) on *AB* and *BC* [Prop. 10.13]. And *DE* is equal to the (sum of the squares) on *AB* and *BC*, and *DH* to twice the (rectangle contained) by *AB* and *BC*. Thus, *DE* [is] incommensurable with *DH*. And as *DE* (is) to *DH*, so *GD* (is) to *DF* [Prop. 6.1]. Thus, *GD* is incommensurable with *DF* [Prop. 10.11]. And they are both rational (straight-lines). Thus, *GD* and *DF* are rational (straight-lines which are) commensurable in square only. Thus, *FG* is an apotome [Prop. 10.73]. And *DI* (is) rational. And the (area) contained by a rational and an irrational (straight-line) is irrational [Prop. 10.20], and its square-root is irrational. And *AC* is the square-root of *FE*. Thus, *AC* is an irrational (straight-line) [Def. 10.4]. And let it be called the second apotome of a medial (straight-line).† (Which is) the very thing it was required to show.

† See footnote to Prop. 10.38.

76.

Ἐὰν ἀπὸ εὐθείας εὐθεῖα ἀφαιρεθῇ δυνάμει ἀσύμμετρος οὐσῦσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιοῦσα τὰ μὲν ἀπὸ αὐτῶν ὁάμα ὁρητόν, τὸ δὸ ὑπὸ αὐτῶν μέσον, ἡ λοιπὴ ὁάλογόξ ἐστιν· καλείσθω δὲ ἐλάσσων.

Proposition 76

If a straight-line, which is incommensurable in square with the whole, and with the whole makes the (squares) on them (added) together rational, and the (rectangle contained) by them medial, is subtracted from a(nother) straight-line then the remainder is an irrational (straight-line). Let it be called a minor (straight-line).

Ἀπὸ γὰρ εὐθείας τῆς AB εὐθεῖα ἀφῃρήσθω ἡ BΓ δυνάμει ἀσύμμετρος οὐσῦσα τῇ ὅλῃ ποιοῦσα τὰ προκείμενα. λέγω, ὅτι ἡ λοιπὴ ἡ AΓὁάλογόξ ἐστιν ἡ καλουμένη ἐλάσσων.

Ἐπεὶ γὰρ τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AB, BΓ τετραγώνων ὁρητόν ἐστιν, τὸ δὲ δἰξ ὑπὸ τῶν AB, BΓ μέσον, ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν AB, BΓ τῷ δἰξ ὑπὸ τῶν AB, BΓ· καὶ ἀναστρέψαντι λοιπῷ τῷ ἀπὸ τῆς AΓ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν AB, BΓ. ὁρηταὶ δὲ τὰ ἀπὸ τῶν AB, BΓ· ὁάλογον ἄρα τὸ ἀπὸ τῆς AΓ· ὁάλογος ἄρα ἡ AΓ· καλείσθω

For let the straight-line *BC*, which is incommensurable in square with the whole, and fulfils the (other) prescribed (conditions), be subtracted from the straight-line *AB* [Prop. 10.33]. I say that the remainder *AC* is that irrational (straight-line) called minor.

For since the sum of the squares on *AB* and *BC* is rational, and twice the (rectangle contained) by *AB* and *BC* (is) medial, the (sum of the squares) on *AB* and *BC* is thus incommensurable with twice the (rectangle contained) by *AB* and

δὲ ἐλάσσων. ὅπερ ἔδει δεῖξαι.

BC. And, via conversion, the (sum of the squares) on AB and BC is incommensurable with the remaining (square) on AC [Props. 2.7, 10.16]. And the (sum of the squares) on AB and BC (is) rational. The (square) on AC (is) thus irrational. Thus, AC (is) an irrational (straight-line) [Def. 10.4]. Let it be called a minor (straight-line).† (Which is) the very thing it was required to show.

† See footnote to Prop. 10.39.

77.

Ἐὰν ἀπὸ εὐθείας εὐθεῖα ἀφαιρεθῇ δυνάμει ἀσύμμετρος οὐσῦσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιοῦσα τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον, τὸ δὲ δὶξ ὑπὸ αὐτῶν ὁρητόν, ἡ λοιπὴ ὀάλογόξ ἐστιν· καλείσθω δὲ ἡ μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσα.

Proposition 77

If a straight-line, which is incommensurable in square with the whole, and with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them rational, is subtracted from a(nother) straight-line then the remainder is an irrational (straight-line). Let it be called that which makes with a rational (area) a medial whole.

Ἀπὸ γὰρ εὐθείας τῆς AB εὐθεῖα ἀφῃρήσθω ἡ BΓ δυνάμει ἀσύμμετος οὐσῦσα τῇ AB ποιοῦσα τὰ προκείμενα· λέγω, ὅτι ἡ λοιπὴ ἡ AΓὀάλογόξ ἐστιν ἡ προειρημένη.

Ἐπεὶ γὰρ τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AB, BΓ τετραγώνων μέσον ἐστίν, τὸ δὲ δὶξ ὑπὸ τῶν AB, BΓ ὁρητόν, ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν AB, BΓ τῷ δὶξ ὑπὸ τῶν AB, BΓ· καὶ λοιπὸν ἄρα τὸ ἀπὸ τῆς AΓ ἀσύμμετρόν ἐστι τῷ δὶξ ὑπὸ τῶν AB, BΓ. καί ἐστι τὸ δὶξ ὑπὸ τῶν AB, BΓ ὁρητόν· τὸ ἄρα ἀπὸ τῆς AΓὀάλογόν ἐστιν· ὀάλογος ἄρα ἐστὶν ἡ AΓ· καλείσθω δὲ ἡ μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσα. ὅπερ ἔδει δεῖξαι.

For let the straight-line BC, which is incommensurable in square with AB, and fulfils the (other) prescribed (conditions), be subtracted from the straight-line AB [Prop. 10.34]. I say that the remainder AC is the aforementioned irrational (straight-line).

For since the sum of the squares on AB and BC is medial, and twice the (rectangle contained) by AB and BC rational, the (sum of the squares) on AB and BC is thus incommensurable with twice the (rectangle contained) by AB and BC. Thus, the remaining (square) on AC is also incommensurable with twice the (rectangle contained) by AB and BC [Props. 2.7, 10.16]. And twice the (rectangle contained) by AB and BC is rational. Thus, the (square) on AC is irrational. Thus, AC is an irrational (straight-line) [Def. 10.4]. And let it be called that which makes with a rational (area) a medial whole.† (Which is) the very thing it was required to show.

† See footnote to Prop. 10.40.

78.

Ἐὰν ἀπὸ εὐθείας εὐθεῖα ἀφαιρεθῇ δυνάμει ἀσύμμετρος οὐσῦσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιοῦσα τό τε συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον τό τε δὶξ ὑπὸ αὐτῶν μέσον καὶ ὀέτι τὰ ἀπὸ αὐτῶν τετράγωνα ἀσύμμετρα τῷ δὶξ ὑπὸ αὐτῶν, ἡ λοιπὴ ὀάλογόξ ἐστιν· καλείσθω δὲ ἡ μετὰ μέσου μέσον τὸ ὅλον ποιοῦσα.

Proposition 78

If a straight-line, which is incommensurable in square with the whole, and with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them medial, and, moreover, the (sum of the) squares on them incommensurable with twice the (rectangle contained) by them, is subtracted from a(nother) straight-line then the remainder is an irrational (straight-line). Let it be called that which makes with a medial (area) a medial whole.

Ἀπὸ γὰρ εὐθείας τῆς AB εὐθεῖα ἀφῃρήσθω ἡ $BΓ$ δυνάμει ἀσύμμετρος οὐσα τῇ AB ποιοῦσα τὰ προκείμενα· λέγω, ὅτι ἡ λοιπὴ ἡ $AΓ$ὀάλογόξ ἐστιν ἡ καλουμένη ἡ μετὰ μέσου μέσον τὸ ὅλον ποιοῦσα.

Ἐκκείσθω γὰρ ὁρητὴ ἡ $ΔI$, καὶ τοῖς μὲν ἀπὸ τῶν AB, $BΓ$ ἴσον παρὰ τὴν $ΔI$ παραβεβλήσθω τὸ $ΔE$ πλάτος ποιοῦν τὴν $ΔH$, τῷ δὲ δὶξ ὑπὸ τῶν AB, $BΓ$ ἴσον ἀφῃρήσθω τὸ $ΔΘ$ [πλάτος ποιοῦν τὴν $ΔZ$]. λοιπὸν ἄρα τὸ ZE ἴσον ἐστὶ τῷ ἀπὸ τῆς $AΓ$· ὥστε ἡ $AΓ$ δύναται τὸ ZE. καὶ ἐπεὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AB, $BΓ$ τετραγώνων μέσον ἐστὶ καί ἐστιν ἴσον τῷ $ΔE$, μέσον ἄρα [ἐστὶ] τὸ $ΔE$. καὶ παρὰ ὁρητὴν τὴν $ΔI$ παράκειται πλάτος ποιοῦν τὴν $ΔH$· ὁρητὴ ἄρα ἐστὶν ἡ $ΔH$ καὶ ἀσύμμετρος τῇ $ΔI$ μήκει. πάλιν, ἐπεὶ τὸ δὶξ ὑπὸ τῶν AB, $BΓ$ μέσον ἐστὶ καί ἐστιν ἴσον τῷ $ΔΘ$, τὸ ἄρα $ΔΘ$ μέσον ἐστίν. καὶ παρὰ ὁρητὴν τὴν $ΔI$ παράκειται πλάτος ποιοῦν τὴν $ΔZ$· ὁρητὴ ἄρα ἐστὶ καὶ ἡ $ΔZ$ καὶ ἀσύμμετρος τῇ $ΔI$ μήκει. καὶ ἐπεὶ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν AB, $BΓ$ τῷ δὶξ ὑπὸ τῶν AB, $BΓ$, ἀσύμμετρον ἄρα καὶ τὸ $ΔE$ τῷ $ΔΘ$. ὡξ δὲ τὸ $ΔE$ πρὸς τὸ $ΔΘ$, οὕτως ἐστὶ καὶ ἡ $ΔH$ πρὸς τὴν $ΔZ$· ἀσύμμετρος ἄρα ἡ $ΔH$ τῇ $ΔZ$. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ $HΔ$, $ΔZ$ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι. ἀποτομὴ ἄρα ἐστίν ἡ ZH· ὁρητὴ δὲ ἡ $ZΘ$. τὸ δὲ ὑπὸ ὁρητῆς καὶ ἀποτομῆς περιεχόμενον [ὀρθογώνιον] ὀάλογόν ἐστιν, καὶ ἡ δυναμένη αὐτὸ ὀάλογόξ ἐστιν· καὶ δύναται τὸ ZE ἡ $AΓ$· ἡ $AΓ$ ἄραὀάλογόξ ἐστιν· καλείσθω δὲ ἡ μετὰ μέσου μέσον τὸ ὅλον

For let the straight-line BC, which is incommensurable in square AB, and fulfils the (other) prescribed (conditions), be subtracted from the (straight-line) AB [Prop. 10.35]. I say that the remainder AC is the irrational (straight-line) called that which makes with a medial (area) a medial whole.

For let the rational (straight-line) DI be laid down. And let DE, equal to the (sum of the squares) on AB and BC, be applied to DI, producing DG as breadth. And let DH, equal to twice the (rectangle contained) by AB and BC, be subtracted (from DE) [producing DF as breadth]. Thus, the remainder FE is equal to the (square) on AC [Prop. 2.7]. Hence, AC is the square-root of FE. And since the sum of the squares on AB and BC is medial, and is equal to DE, DE [is] thus medial. And it is applied to the rational (straight-line) DI, producing DG as breadth. Thus, DG is rational, and incommensurable in length with DI [Prop 10.22]. Again, since twice the (rectangle contained) by AB and BC is medial, and is equal to DH, DH is thus medial. And it is applied to the rational (straight-line) DI, producing DF as breadth. Thus, DF is also rational, and incommensurable in length with DI [Prop. 10.22]. And since the (sum of the squares) on AB and BC is incommensurable with twice the (rectangle contained) by AB and BC, DE (is) also incommensurable with DH. And as DE (is) to DH, so DG also is to DF [Prop. 6.1]. Thus, DG (is) incommensurable (in length) with DF [Prop. 10.11]. And they are both

ποιοῦσα. ὅπερ ἔδει δεῖχαι.

rational. Thus, GD and DF are rational (straight-lines which are) commensurable in square only. Thus, FG is an apotome [Prop. 10.73]. And FH (is) rational. And the [rectangle] contained by a rational (straight-line) and an apotome is irrational [Prop. 10.20], and its square-root is irrational. And AC is the square-root of FE. Thus, AC is irrational. Let it be called that which makes with a medial (area) a medial whole.† (Which is) the very thing it was required to show.

† See footnote to Prop. 10.41.

79.

Τῇ ἀποτομῇ μία [μόνον] προσαρμόζει εὐθεῖα ὁρητὴ δυνάμει μόνον σύμμετρος οὐσῦα τῇ ὅλῃ.

Proposition 79

[Only] one rational straight-line, which is commensurable in square only with the whole, can be attached to an apotome.†

Ἔστω ἀποτομὴ ἡ AB, προσαρμόζουσα δὲ αὐτῇ ἡ $BΓ$· αἱ $AΓ$, $ΓB$ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· λέγω, ὅτι τῇ AB ἑτέρα οὐ προσαρμόζει ὁρητὴ δυνάμει μόνον σύμμετρος οὐσῦα τῇ ὅλῃ.

Εἰ γὰρ δυνατόν, προσαρμοζέτω ἡ $BΔ$· καὶ αἱ $AΔ$, $ΔB$ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι. καὶ ἐπεί, ὧ ὑπερέθει τὰ ἀπὸ τῶν $AΔ$, $ΔB$ τοῦ δὶξ ὑπὸ τῶν $AΔ$, $ΔB$, τούτῳ ὑπερέθει καὶ τὰ ἀπὸ τῶν $AΓ$, $ΓB$ τοῦ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$· τῷ γὰρ αὐτῷ τῷ ἀπὸ τῆς AB ἀμφότερα ὑπερέθει· ἐναλλὰχ ἄρα, ὧ ὑπερέθει τὰ ἀπὸ τῶν $AΔ$, $ΔB$ τῶν ἀπὸ τῶν $AΓ$, $ΓB$, τούτῳ ὑπερέθει [καὶ] τὸ δὶξ ὑπὸ τῶν $AΔ$, $ΔB$ τοῦ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$. τὰ δὲ ἀπὸ τῶν $AΔ$, $ΔB$ τῶν ἀπὸ τῶν $AΓ$, $ΓB$ ὑπερέθει ὁρητῷ· ὁρητὰ γὰρ ἀμφότερα. καὶ τὸ δὶξ ἄρα ὑπὸ τῶν $AΔ$, $ΔB$ τοῦ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$ ὑπερέθει ὁρητῷ· ὅπερ ἐστὶν ἀδύνατον· μέσα γὰρ ἀμφότερα, μέσον δὲ μέσου οὐθ ὑπερέθει ὁρητῷ. τῇ ἄρα AB ἑτέρα οὐ προσαρμόζει ὁρητὴ δυνάμει μόνον σύμμετρος οὐσῦα τῇ ὅλῃ.

Μία ἄρα μόνη τῇ ἀποτομῇ προσαρμόζει ὁρητὴ δυνάμει μόνον σύμμετρος οὐσῦα τῇ ὅλῃ· ὅπερ ἔδει δεῖχαι.

Let AB be an apotome, with BC (so) attached to it. AC and CB are thus rational (straight-lines which are) commensurable in square only [Prop. 10.73]. I say that another rational (straight-line), which is commensurable in square only with the whole, cannot be attached to AB.

For, if possible, let BD be (so) attached (to AB). Thus, AD and DB are also rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And since by whatever (area) the (sum of the squares) on AD and DB exceeds twice the (rectangle contained) by AD and DB, the (sum of the squares) on AC and CB also exceeds twice the (rectangle contained) by AC and CB by this (same area). For both exceed by the same (area)—(namely), the (square) on AB [Prop. 2.7]. Thus, alternately, by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB, twice the (rectangle contained) by AD and DB [also] exceeds twice the (rectangle contained) by AC and CB by this (same area). And the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB by a rational (area). For both (are) rational (areas). Thus, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Prop. 10.21], and a medial (area) cannot exceed a(nother) medial (area) by a rational (area) [Prop. 10.26]. Thus, another rational (straight-line), which is commensurable in square only with the whole, cannot be attached to AB.

Thus, only one rational (straight-line), which is commensurable in square only with the whole, can be attached to an apotome. (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.42, with minus signs instead of plus signs.

80.

Τῇ μέσης ἀποτομῇ πρώτῃ μία μόνον προσαρμόζει εὐθεῖα μέση δυνάμει μόνον σύμμετρος οὐσῦσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ὁρητὸν περιέθουσα.

Ἔστω γὰρ μέσης ἀποτομὴ πρώτη ἡ AB, καὶ τῇ AB προσαρμοζέτω ἡ $BΓ$· αἱ $AΓ$, $ΓB$ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι ὁρητὸν περιέθουσαι τὸ ὑπὸ τῶν $AΓ$, $ΓB$· λέγω, ὅτι τῇ AB ἑτέρα οὐ προσαρμόζει μέση δυνάμει μόνον σύμμετρος οὐσῦσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ὁρητὸν περιέθουσα.

Εἰ γὰρ δυνατόν, προσαρμοζέτω καὶ ἡ $ΔB$· αἱ ἄρα $AΔ$, $ΔB$ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι ὁρητὸν περιέθουσαι τὸ ὑπὸ τῶν $AΔ$, $ΔB$. καὶ ἐπεί, ὁῷ ὑπερέθει τὰ ἀπὸ τῶν $AΔ$, $ΔB$ τοῦ δὶξ ὑπὸ τῶν $AΔ$, $ΔB$, τούτῳ ὑπερέθει καὶ τὰ ἀπὸ τῶν $AΓ$, $ΓB$ τοῦ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$· τῷ γὰρ αὐτῷ [πάλιν] ὑπερέθουσι τῷ ἀπὸ τῆς AB· ἐναλλὰχ ἄρα, ὁῷ ὑπερέθει τὰ ἀπὸ τῶν $AΔ$, $ΔB$ τῶν ἀπὸ τῶν $AΓ$, $ΓB$, τούτῳ ὑπερέθει καὶ τὸ δὶξ ὑπὸ τῶν $AΔ$, $ΔB$ τοῦ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$. τὸ δὲ δὶξ ὑπὸ τῶν $AΔ$, $ΔB$ τοῦ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$ ὑπερέθει ὁρητῷ· ὁρητὰ γὰρ ἀμφότερα. καὶ τὰ ἀπὸ τῶν $AΔ$, $ΔB$ ἄρα τῶν ἀπὸ τῶν $AΓ$, $ΓB$ [τετραγώνων] ὑπερέθει ὁρητῷ· ὅπερ ἐστὶν ἀδύνατον· μέσα γάρ ἐστιν ἀμφότερα, μέσον δὲ μέσου οὐθ ὑπερέθει ὁρητῷ.

Τῇ ἄρα μέσης ἀποτομῇ πρώτῃ μία μόνον προσαρμόζει εὐθεῖα μέση δυνάμει μόνον σύμμετρος οὐσῦσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ὁρητὸν περιέθουσα· ὅπερ ἔδει δεῖχαι.

Proposition 80

Only one medial straight-line, which is commensurable in square only with the whole, and contains a rational (area) with the whole, can be attached to a first apotome of a medial (straight-line).†

For let AB be a first apotome of a medial (straight-line), and let BC be (so) attached to AB. Thus, AC and CB are medial (straight-lines which are) commensurable in square only, containing a rational (area)—(namely, that contained) by AC and CB [Prop. 10.74]. I say that a(nother) medial (straight-line), which is commensurable in square only with the whole, and contains a rational (area) with the whole, cannot be attached to AB.

For, if possible, let DB also be (so) attached to AB. Thus, AD and DB are medial (straight-lines which are) commensurable in square only, containing a rational (area)—(namely, that) contained by AD and DB [Prop. 10.74]. And since by whatever (area) the (sum of the squares) on AD and DB exceeds twice the (rectangle contained) by AD and DB, the (sum of the squares) on AC and CB also exceeds twice the (rectangle contained) by AC and CB by this (same area). For [again] both exceed by the same (area)—(namely), the (square) on AB [Prop. 2.7]. Thus, alternately, by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by this (same area). And twice the (rectangle contained) by AD and DB exceeds twice the (rectangle contained) by AC and CB by a rational (area). For both (are) rational (areas). Thus, the (sum of the squares) on AD and DB also exceeds the (sum of the) [squares] on AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Props. 10.15, 10.23 corr.], and a medial (area) cannot exceed a(nother) medial (area) by a rational (area) [Prop. 10.26].

Thus, only one medial (straight-line), which is commensurable in square only with the whole, and contains a rational (area) with the whole, can be attached to a first apotome of a medial (straight-line). (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.43, with minus signs instead of plus signs.

81.

Τῇ μέσης ἀποτομῇ δευτέρᾳ μία μόνον προσαρμόζει εὐθεῖα

Proposition 81

Only one medial straight-line, which is commensurable in

μέση δυνάμει μόνον σύμμετρος τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης μέσον περιέθουσα.

square only with the whole, and contains a medial (area) with the whole, can be attached to a second apotome of a medial (straight-line).[†]

Ἔστω μέσης ἀποτομὴ δευτέρα ἡ ΑΒ καὶ τῇ ΑΒ προσαρμόζουσα ἡ ΒΓ· αἱ ἄρα ΑΓ, ΓΒ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι τὸ ὑπὸ τῶν ΑΓ, ΓΒ· λέγω, ὅτι τῇ ΑΒ ἑτέρα οὐ προσαρμόσει εὐθεῖα μέση δυνάμει μόνον σύμμετρος οὐοῦσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης μέσον περιέθουσα.

Εἰ γὰρ δυνατόν, προσαρμοζέτω ἡ ΒΔ· καὶ αἱ ΑΔ, ΔΒ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι τὸ ὑπὸ τῶν ΑΔ, ΔΒ. καὶ ἐκκείσθω ὁρητὴ ἡ ΕΖ, καὶ τοῖς μὲν ἀπὸ τῶν ΑΓ, ΓΒ ἴσον παρὰ τὴν ΕΖ παραβεβλήσθω τὸ ΕΗ πλάτος ποιοῦν τὴν ΕΜ· τῷ δὲ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ ἴσον ἀφῃρήσθω τὸ ΘΗ πλάτος ποιοῦν τὴν ΘΜ· λοιπὸν ἄρα τὸ ΕΛ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΒ· ὥστε ἡ ΑΒ δύναται τὸ ΕΛ. πάλιν δὴ τοῖς ἀπὸ τῶν ΑΔ, ΔΒ ἴσον παρὰ τὴν ΕΖ παραβεβλήσθω τὸ ΕΙ πλάτος ποιοῦν τὴν ΕΝ· ὁέστι δὲ καὶ τὸ ΕΛ ἴσον τῷ ἀπὸ τῆς ΑΒ τετραγώνῳ· λοιπὸν ἄρα τὸ ΘΙ ἴσον ἐστὶ τῷ δὶξ ὑπὸ τῶν ΑΔ, ΔΒ. καὶ ἐπεὶ μέσαι εἰσὶν αἱ ΑΓ, ΓΒ, μέσα ἄρα ἐστὶ καὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ. καί ἐστιν ἴσα τῷ ΕΗ· μέσον ἄρα καὶ τὸ ΕΗ. καὶ παρὰ ὁρητὴν τὴν ΕΖ παράκειται πλάτος ποιοῦν τὴν ΕΜ· ὁρητὴ ἄρα ἐστὶν ἡ ΕΜ καὶ ἀσύμμετρος τῇ ΕΖ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ ὑπὸ τῶν ΑΓ, ΓΒ, καὶ τὸ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ μέσον ἐστίν. καί ἐστιν ἴσον τῷ ΘΗ· καὶ τὸ ΘΗ ἄρα μέσον ἐστίν. καὶ παρὰ ὁρητὴν τὴν ΕΖ παράκειται πλάτος ποιοῦν τὴν ΘΜ· ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΘΜ καὶ ἀσύμμετρος τῇ ΕΖ μήκει. καὶ ἐπεὶ αἱ ΑΓ, ΓΒ δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετρος ἄρα ἐστὶν ἡ ΑΓ τῇ ΓΒ μήκει. ὡξ δὲ ἡ ΑΓ πρὸς τὴν ΓΒ, οὕτως ἐστὶ τὸ ἀπὸ τῆς ΑΓ πρὸς τὸ ὑπὸ

Let *AB* be a second apotome of a medial (straight-line), with *BC* (so) attached to *AB*. Thus, *AC* and *CB* are medial (straight-lines which are) commensurable in square only, containing a medial (area)—(namely, that contained) by *AC* and *CB* [Prop. 10.75]. I say that a(nother) medial straight-line, which is commensurable in square only with the whole, and contains a medial (area) with the whole, cannot be attached to *AB*.

For, if possible, let *BD* be (so) attached. Thus, *AD* and *DB* are also medial (straight-lines which are) commensurable in square only, containing a medial (area)—(namely, that contained) by *AD* and *DB* [Prop. 10.75]. And let the rational (straight-line) *EF* be laid down. And let *EG*, equal to the (sum of the squares) on *AC* and *CB*, be applied to *EF*, producing *EM* as breadth. And let *HG*, equal to twice the (rectangle contained) by *AC* and *CB*, be subtracted (from *EG*), producing *HM* as breadth. The remainder *EL* is thus equal to the (square) on *AB* [Prop. 2.7]. Hence, *AB* is the square-root of *EL*. So, again, let *EI*, equal to the (sum of the squares) on *AD* and *DB* be applied to *EF*, producing *EN* as breadth. And *EL* is also equal to the square on *AB*. Thus, the remainder *HI* is equal to twice the (rectangle contained) by *AD* and *DB* [Prop. 2.7]. And since *AC* and *CB* are (both) medial (straight-lines), the (sum of the squares) on *AC* and *CB* is also medial. And it is equal to *EG*. Thus, *EG* is also medial [Props. 10.15, 10.23 corr.]. And it is applied to the rational

τῶν ΑΓ, ΓΒ· ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΓ τῷ ὑπὸ τῶν ΑΓ, ΓΒ. ἀλλὰ τῷ μὲν ἀπὸ τῆς ΑΓ σύμμετρά ἐστι τὰ ἀπὸ τῶν ΑΓ, ΓΒ, τῷ δὲ ὑπὸ τῶν ΑΓ, ΓΒ σύμμετρόν ἐστι τὸ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ· ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν ΑΓ, ΓΒ τῷ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ. καί ἐστι τοῖς μὲν ἀπὸ τῶν ΑΓ, ΓΒ ἴσον τὸ ΕΗ, τῷ δὲ δὶξ ὑπὸ τῶν ΑΓ, ΓΒ ἴσον τὸ ΗΘ· ἀσύμμετρον ἄρα ἐστὶ τὸ ΕΗ τῷ ΘΗ. ὡς δὲ τὸ ΕΗ πρὸς τὸ ΘΗ, οὕτως ἐστὶν ἡ ΕΜ πρὸς τὴν ΘΜ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΕΜ τῇ ΜΘ μήκει. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ ΕΜ, ΜΘ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ ΕΘ, προσαρμόζουσα δὲ αὐτῇ ἡ ΘΜ. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ ΘΝ αὐτῇ προσαρμόζει· τῇ ἄρα ἀποτομῇ ὀάλλη καὶ ὀάλλη προσαρμόζει εὐθεῖα δυνάμει μόνον σύμμετρος οὐῦσα τῇ ὅλῃ· ὅπερ ἐστὶν ἀδύνατον.

Τῇ ἄρα μέσης ἀποτομῇ δευτέρᾳ μία μόνον προσαρμόζει εὐθεῖα μέση δυνάμει μόνον σύμμετρος οὐῦσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης μέσον περιέθουσα· ὅπερ ἔδει δεῖχαι.

(straight-line) EF, producing EM as breadth. Thus, EM is rational, and incommensurable in length with EF [Prop. 10.22]. Again, since the (rectangle contained) by AC and CB is medial, twice the (rectangle contained) by AC and CB is also medial [Prop. 10.23 corr.]. And it is equal to HG. Thus, HG is also medial. And it is applied to the rational (straight-line) EF, producing HM as breadth. Thus, HM is also rational, and incommensurable in length with EF [Prop. 10.22]. And since AC and CB are commensurable in square only, AC is thus incommensurable in length with CB. And as AC (is) to CB, so the (square) on AC is to the (rectangle contained) by AC and CB [Prop. 10.21 corr.]. Thus, the (square) on AC is incommensurable with the (rectangle contained) by AC and CB [Prop. 10.11]. But, the (sum of the squares) on AC and CB is commensurable with the (square) on AC, and twice the (rectangle contained) by AC and CB is commensurable with the (rectangle contained) by AC and CB [Prop. 10.6]. Thus, the (sum of the squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB [Prop. 10.13]. And EG is equal to the (sum of the squares) on AC and CB. And GH is equal to twice the (rectangle contained) by AC and CB. Thus, EG is incommensurable with HG. And as EG (is) to HG, so EM is to HM [Prop. 6.1]. Thus, EM is incommensurable in length with MH [Prop. 10.11]. And they are both rational (straight-lines). Thus, EM and MH are rational (straight-lines which are) commensurable in square only. Thus, EH is an apotome [Prop. 10.73], and HM (is) attached to it. So, similarly, we can show that HN (is) also (commensurable in square only with EN and is) attached to (EH). Thus, different straight-lines, which are commensurable in square only with the whole, are attached to an apotome. The very thing is impossible [Prop. 10.79].

Thus, only one medial straight-line, which is commensurable in square only with the whole, and contains a medial (area) with the whole, can be attached to a second apotome of a medial (straight-line). (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.44, with minus signs instead of plus signs.

82.

Τῇ ἐλάσσονι μία μόνον προσαρμόζει εὐθεῖα δυνάμει ἀσύμμετρος οὐῦσα τῇ ὅλῃ ποιοῦσα μετὰ τῆς ὅλης τὸ μὲν ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων ὁρητόν, τὸ δὲ δὶξ ὑπὸ αὐτῶν μέσον.

Proposition 82

Only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the (sum of the) squares on them rational, and twice the (rectangle contained) by them medial, can be attached to a minor

(straight-line).

Ἔστω ἡ ἐλάσσων ἡ AB, καὶ τῇ AB προσαρμόζουσαὐέστω ἡ $BΓ$· αἱ ἄρα $AΓ$, $ΓB$ δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων ὁρητόν, τὸ δὲ δὶξ ὑπὸ αὐτῶν μέσον· λέγω, ὅτι τῇ AB ἑτέρα εὐθεῖα οὐ προσαρμόσει τὰ αὐτὰ ποιοῦσα.

Εἰ γὰρ δυνατόν, προσαρμοζέτω ἡ $BΔ$· καὶ αἱ $AΔ$, $ΔB$ ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὰ προειρημένα. καὶ ἐπεί, ὁῷ ὑπερέθει τὰ ἀπὸ τῶν $AΔ$, $ΔB$ τῶν ἀπὸ τῶν $AΓ$, $ΓB$, τούτῳ ὑπερέθει καὶ τὸ δὶξ ὑπὸ τῶν $AΔ$, $ΔB$ τοῦ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$, τὰ δὲ ἀπὸ τῶν $AΔ$, $ΔB$ τετράγωνα τῶν ἀπὸ τῶν $AΓ$, $ΓB$ τετραγώνων ὑπερέθει ὁρητῷ· ὁρητὰ γάρ ἐστιν ἀμφότερα· καὶ τὸ δὶξ ὑπὸ τῶν $AΔ$, $ΔB$ ἄρα τοῦ δὶξ ὑπὸ τῶν $AΓ$, $ΓB$ ὑπερέθει ὁρητῷ· ὅπερ ἐστὶν ἀδύνατον· μέσα γάρ ἐστιν ἀμφότερα.

Τῇ ἄρα ἐλάσσονι μία μόνον προσαρμόζει εὐθεῖα δυνάμει ἀσύμμετρος οὐῦσα τῇ ὅλῃ καὶ ποιοῦσα τὰ μὲν ἀπὸ αὐτῶν τετράγωνα ὁάμα ὁρητόν, τὸ δὲ δὶξ ὑπὸ αὐτῶν μέσον· ὅπερ ἔδει δεῖχαι.

Let AB be a minor (straight-line), and let BC be (so) attached to AB. Thus, AC and CB are (straight-lines which are) incommensurable in square, making the sum of the squares on them rational, and twice the (rectangle contained) by them medial [Prop. 10.76]. I say that another another straight-line fulfilling the same (conditions) cannot be attached to AB.

For, if possible, let BD be (so) attached (to AB). Thus, AD and DB are also (straight-lines which are) incommensurable in square, fulfilling the (other) aforementioned (conditions) [Prop. 10.76]. And since by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by this (same area) [Prop. 2.7]. And the (sum of the) squares on AD and DB exceeds the (sum of the) squares on AC and CB by a rational (area). For both are rational (areas). Thus, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Prop. 10.26].

Thus, only one straight-line, which is incommensurable in square with the whole, and (with the whole) makes the squares on them (added) together rational, and twice the (rectangle contained) by them medial, can be attached to a minor (straight-line). (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.45, with minus signs instead of plus signs.

83.

Τῇ μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιούσῃ μία μόνον προσαρμόζει εὐθεῖα δυνάμει ἀσύμμετρος οὐῦσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιοῦσα τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον, τὸ δὲ δὶξ ὑπὸ αὐτῶν ὁρητόν.

Proposition 83

Only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them rational, can be attached to that (straight-line) which with a rational (area) makes a medial whole.†

Ἔστω ἡ μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσα ἡ AB, καὶ τῇ AB προσαρμοζέτω ἡ $BΓ$· αἱ ἄρα $AΓ$, $ΓB$ δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὰ προκείμενα· λέγω, ὅτι τῇ AB ἑτέρα οὐ προσαρμόσει τὰ αὐτὰ ποιοῦσα.

Εἰ γὰρ δυνατόν, προσαρμοζέτω ἡ $BΔ$· καὶ αἱ $AΔ$, $ΔB$ ἄρα εὐθεῖαι δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὰ προκείμενα. ἐπεὶ οὖν, ὁῷ ὑπερέθει τὰ ἀπὸ τῶν $AΔ$, $ΔB$ τῶν ἀπὸ τῶν $AΓ$,

Let AB be a (straight-line) which with a rational (area) makes a medial whole, and let BC be (so) attached to AB. Thus, AC and CB are (straight-lines which are) incommensurable in square, fulfilling the (other) proscribed (conditions) [Prop. 10.77]. I say that another (straight-line) fulfilling the same (conditions) cannot be attached to AB.

For, if possible, let BD be (so) attached (to AB). Thus,

ΓΒ, τούτῳ ὑπερέθει καὶ τὸ δἰξ ὑπὸ τῶν ΑΔ, ΔΒ τοῦ δἰξ ὑπὸ τῶν ΑΓ, ΓΒ ἀκολούθως τοῖς πρὸ αὐτοῦ, τὸ δὲ δἰξ ὑπὸ τῶν ΑΔ, ΔΒ τοῦ δἰξ ὑπὸ τῶν ΑΓ, ΓΒ ὑπερέθει ὁρητῷ· ὁρητὰ γάρ ἐστιν ἀμφότερα· καὶ τὰ ἀπὸ τῶν ΑΔ, ΔΒ ἄρα τῶν ἀπὸ τῶν ΑΓ, ΓΒ ὑπερέθει ὁρητῷ· ὅπερ ἐστὶν ἀδύνατον· μέσα γάρ ἐστιν ἀμφότερα.

Οὐκ ἄρα τῇ ΑΒ ἑτέρα προσαρμόσει εὐθεῖα δυνάμει ἀσύμμετρος οὐσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιοῦσα τὰ προειρημένα· μία ἄρα μόνον προσαρμόσει· ὅπερ ἔδει δεῖχαι.

AD and DB are also straight-lines (which are) incommensurable in square, fulfilling the (other) prescribed (conditions) [Prop. 10.77]. Therefore, analogously to the (propositions) before this, since by whatever (area) the (sum of the squares) on AD and DB exceeds the (sum of the squares) on AC and CB, twice the (rectangle contained) by AD and DB also exceeds twice the (rectangle contained) by AC and CB by this (same area). And twice the (rectangle contained) by AD and DB exceeds twice the (rectangle contained) by AC and CB by a rational (area). For they are (both) rational (areas). Thus, the (sum of the squares) on AD and DB also exceeds the (sum of the squares) on AC and CB by a rational (area). The very thing is impossible. For both are medial (areas) [Prop. 10.26].

Thus, another straight-line cannot be attached to AB, which is incommensurable in square with the whole, and fulfills the (other) aforementioned (conditions) with the whole. Thus, only one (such straight-line) can be (so) attached. (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.46, with minus signs instead of plus signs.

84.

Τῇ μετὰ μέσου μέσον τὸ ὅλον ποιούσῃ μία μόνη προσαρμόζει εὐθεῖα δυνάμει ἀσύμμετρος οὐσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιοῦσα τό τε συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον τό τε δἰξ ὑπὸ αὐτῶν μέσον καὶ ὁέτι ἀσύμμετρον τῷ συγκειμένῳ ἐκ τῶν ἀπὸ αὐτῶν.

Ἔστω ἡ μετὰ μέσου μέσον τὸ ὅλον ποιοῦσα ἡ ΑΒ, προσαρμόζουσα δὲ αὐτῇ ἡ ΒΓ· αἱ ἄρα ΑΓ, ΓΒ δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὰ προειρημένα. λέγω, ὅτι τῇ ΑΒ ἑτέρα οὐ προσαρμόσει ποιοῦσα προειρημένα.

Proposition 84

Only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the sum of the squares on them medial, and twice the (rectangle contained) by them medial, and, moreover, incommensurable with the sum of the (squares) on them, can be attached to that (straight-line) which with a medial (area) makes a medial whole.†

Let AB be a (straight-line) which with a medial (area) makes a medial whole, BC being (so) attached to it. Thus, AC and CB are incommensurable in square, fulfilling the (other) aforementioned (conditions) [Prop. 10.78]. I say that a(nother) (straight-line) fulfilling the aforementioned (conditions) cannot be attached to AB.

Εἰ γὰρ δυνατόν, προσαρμοζέτω ἡ $B\Delta$, ὥστε καὶ τὰξ $A\Delta$, ΔB δυνάμει ἀσυμμέτρους εὐῖναι ποιούσας τά τε ἀπὸ τῶν $A\Delta$, ΔB τετράγωνα ὁάμα μέσον καὶ τὸ δὶξ ὑπὸ τῶν $A\Delta$, ΔB μέσον καὶ ὀέτι τὰ ἀπὸ τῶν $A\Delta$, ΔB ἀσύμμετρα τῷ δὶξ ὑπὸ τῶν $A\Delta$, ΔB· καὶ ἐκκείσθω ὁρητὴ ἡ EZ, καὶ τοῖς μὲν ἀπὸ τῶν $A\Gamma$, ΓB ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ EH πλάτος ποιοῦν τὴν EM, τῷ δὲ δὶξ ὑπὸ τῶν $A\Gamma$, ΓB ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ ΘH πλάτος ποιοῦν τὴν ΘM· λοιπὸν ἄρα τὸ ἀπὸ τῆς AB ἴσον ἐστὶ τῷ $E\Lambda$· ἡ ἄρα AB δύναται τὸ $E\Lambda$. πάλιν τοῖς ἀπὸ τῶν $A\Delta$, ΔB ἴσον παρὰ τὴν EZ παραβεβλήσθω τὸ EI πλάτος ποιοῦν τὴν EN. ὀέστι δὲ καὶ τὸ ἀπὸ τῆς AB ἴσον τῷ $E\Lambda$· λοιπὸν ἄρα τὸ δὶξ ὑπὸ τῶν $A\Delta$, ΔB ἴσον [ἐστὶ] τῷ ΘI. καὶ ἐπεὶ μέσον ἐστὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $A\Gamma$, ΓB καί ἐστιν ἴσον τῷ EH, μέσον ἄρα ἐστὶ καὶ τὸ EH. καὶ παρὰ ὁρητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν EM· ὁρητὴ ἄρα ἐστὶν ἡ EM καὶ ἀσύμμετρος τῇ EZ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ δὶξ ὑπὸ τῶν $A\Gamma$, ΓB καί ἐστιν ἴσον τῷ ΘH, μέσον ἄρα καὶ τὸ ΘH. καὶ παρὰ ὁρητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν ΘM· ὁρητὴ ἄρα ἐστὶν ἡ ΘM καὶ ἀσύμμετρος τῇ EZ μήκει. καὶ ἐπεὶ ἀσύμμετρά ἐστι τὰ ἀπὸ τῶν $A\Gamma$, ΓB τῷ δὶξ ὑπὸ τῶν $A\Gamma$, ΓB, ἀσύμμετρόν ἐστι καὶ τὸ EH τῷ ΘH· ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ EM τῇ $M\Theta$ μήκει. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ ἄρα EM, $M\Theta$ ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ $E\Theta$, προσαρμόζουσα δὲ αὐτῇ ἡ ΘM. ὁμοίως δὴ δείχομεν, ὅτι ἡ $E\Theta$ πάλιν ἀποτομή ἐστιν, προσαρμόζουσα δὲ αὐτῇ ἡ ΘN. τῇ ἄρα ἀποτομῇ ὀάλλη καὶ ὀάλλη προσαρμόζει ὁρητὴ δυνάμει μόνον σύμμετρος οὐῦσα τῇ ὅλῃ· ὅπερ ἐδείθθη ἀδύνατον. οὐκ ἄρα τῇ AB ἑτέρα προσαρμόσει εὐθεῖα.

Τῇ ἄρα AB μία μόνον προσαρμόζει εὐθεῖα δυνάμει ἀσύμμετρος οὐῦσα τῇ ὅλῃ, μετὰ δὲ τῆς ὅλης ποιοῦσα τά τε ἀπὸ

For, if possible, let BD be (so) attached. Hence, AD and DB are also (straight-lines which are) incommensurable in square, making the squares on AD and DB (added) together medial, and twice the (rectangle contained) by AD and DB medial, and, moreover, the (sum of the squares) on AD and DB incommensurable with twice the (rectangle contained) by AD and DB [Prop. 10.78]. And let the rational (straight-line) EF be laid down. And let EG, equal to the (sum of the squares) on AC and CB, be applied to EF, producing EM as breadth. And let HG, equal to twice the (rectangle contained) by AC and CB, be applied to EF, producing HM as breadth. Thus, the remaining (square) on AB is equal to EL [Prop. 2.7]. Thus, AB is the square-root of EL. Again, let EI, equal to the (sum of the squares) on AD and DB, be applied to EF, producing EN as breadth. And the (square) on AB is also equal to EL. Thus, the remaining twice the (rectangle contained) by AD and DB [is] equal to HI [Prop. 2.7]. And since the sum of the (squares) on AC and CB is medial, and is equal to EG, EG is thus also medial. And it is applied to the rational (straight-line) EF, producing EM as breadth. EM is thus rational, and incommensurable in length with EF [Prop. 10.22]. Again, since twice the (rectangle contained) by AC and CB is medial, and is equal to HG, HG is thus also medial. And it is applied to the rational (straight-line) EF, producing HM as breadth. HM is thus rational, and incommensurable in length with EF [Prop. 10.22]. And since the (sum of the squares) on AC and CB is incommensurable with twice the (rectangle contained) by AC and CB, EG is also incommensurable with HG. Thus, EM is also incommensurable in length with MH [Props. 6.1, 10.11]. And they are both rational (straight-lines). Thus, EM

αὐτῶν τετράγωνα ὁάμα μέσον καὶ τὸ δὶξ ὑπὸ αὐτῶν μέσον καὶ ὀέτι τὰ ἀπὸ αὐτῶν τετράγωνα ἀσύμμετρα τῷ δὶξ ὑπὸ αὐτῶν· ὅπερ ἔδει δεῖχαι.

and MH are rational (straight-lines which are) commensurable in square only. Thus, EH is an apotome [Prop. 10.73], with HM attached to it. So, similarly, we can show that EH is again an apotome, with HN attached to it. Thus, different rational (straight-lines), which are commensurable in square only with the whole, are attached to an apotome. The very thing was shown (to be) impossible [Prop. 10.79]. Thus, another straight-line cannot be (so) attached to AB.

Thus, only one straight-line, which is incommensurable in square with the whole, and (together) with the whole makes the squares on them (added) together medial, and twice the (rectangle contained) by them medial, and, moreover, the (sum of the) squares on them incommensurable with the (rectangle contained) by them, can be attached to AB. (Which is) the very thing it was required to show.

† This proposition is equivalent to Prop. 10.47, with minus signs instead of plus signs.

Ὅροι τρίτοι.

11. Ὑποκειμένης ὀρητῆς καὶ ἀποτομῆς, ἐαν μὲν ἡ ὅλη τῆς προσαρμοζούσης μεῖζον δύνηται τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει, καὶ ἡ ὅλη σύμμετροσὸῇ τῇ ἐκκειμένῃ ὀρητῇ μήκει, καλείσθω ἀποτομὴ πρώτη.

12. Ἐὰν δὲ ἡ προσαρμόζουσα σύμμετροσὸῇ τῇ ἐκκειμένῃ ὀρητῇ μήκει, καὶ ἡ ὅλη τῆς προσαρμοζούσης μεῖζον δύνηται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καλείσθω ἀποτομὴ δευτέρα.

13. Ἐὰν δὲ μ ηδετέρα σύμμετροσὸῇ τῇ ἐκκειμένῃ ὀρητῇ μήκει, ἡ δὲ ὅλη τῆς προσαρμοζούσης μεῖζον δύνηται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καλείσθω ἀποτομὴ τρίτη.

14. Πάλιν, ἐαν ἡ ὅλη τῆς προσαρμοζούσης μεῖζον δύνηται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ [μήκει], ἐαν μὲν ἡ ὅλη σύμμετροσὸῇ τῇ ἐκκειμένῃ ὀρητῇ μήκει, καλείσθω ἀποτομὴ τετάρτη.

15. Ἐὰν δὲ ἡ προσαρμόζουσα, πέμπτη.

16. Ἐὰν δὲ μ ηδετέρα, ὀέκτη.

Definitions III

11. Given a rational (straight-line) and an apotome, if the square on the whole is greater than the (square on a straight-line) attached (to the apotome) by the (square) on (some straight-line) commensurable in length with (the whole), and the whole is commensurable in length with the (previously) laid down rational (straight-line), then let the (apotome) be called a first apotome.

12. And if the attached (straight-line) is commensurable in length with the (previously) laid down rational (straight-line), and the square on the whole is greater than (the square on) the attached (straight-line) by the (square) on (some straight-line) commensurable (in length) with (the whole), then let the (apotome) be called a second apotome.

13. And if neither of (the whole or the attached straight-line) is commensurable in length with the (previously) laid down rational (straight-line), and the square on the whole is greater than (the square on) the attached (straight-line) by the (square) on (some straight-line) commensurable (in length) with (the whole), then let the (apotome) be called a third apotome.

14. Again, if the square on the whole is greater than (the square on) the attached (straight-line) by the (square) on (some straight-line) incommensurable [in length] with (the whole), and the whole is commensurable in length with the (previously) laid down rational (straight-line), then let the (apotome) be called a fourth apotome.

15. And if the attached (straight-line is commensurable), a fifth (apotome).

16. And if neither (the whole nor the attached straight-line is commensurable), a sixth (apotome).

85.

Proposition 85

Εὑρεῖν τὴν πρώτην ἀποτομήν.

To find a first apotome.

Ἐκκείσθω ὁρητὴ ἡ Α, καὶ τῇ Α μήκει σύμμετροςὸέστω ἡ ΒΗ· ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΒΗ. καὶ ἐκκείσθωσαν δύο τετράγωνοι ἀριθμοὶ οἱ ΔΕ, ΕΖ, ὧν ἡ ὑπεροθὴ ὁ ΖΔ μὴ ὀέστω τετράγωνος· οὐδὸ ἄρα ὁ ΕΔ πρὸς τὸν ΔΖ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν. καὶ πεποιήσθω ὡξ ὁ ΕΔ πρὸς τὸν ΔΖ, οὑτως τὸ ἀπὸ τῆς ΒΗ τετράγωνον πρὸς τὸ ἀπὸ τὴξ ΗΓ τετράγωνον· σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΒΗ τῷ ἀπὸ τῆς ΗΓ. ὁρητὸν δὲ τὸ ἀπὸ τῆς ΒΗ· ὁρητὸν ἄρα καὶ τὸ ἀπὸ τῆς ΗΓ· ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΗΓ. καὶ ἐπεὶ ὁ ΕΔ πρὸς τὸν ΔΖ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, οὐδὸ ἄρα τὸ ἀπὸ τῆς ΒΗ πρὸς τὸ ἀπὸ τῆς ΗΓ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΒΗ τῇ ΗΓ μήκει. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ ΒΗ, ΗΓ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἡ ἄρα ΒΓ ἀποτομή ἐστιν. λέγω δή, ὅτι καὶ πρώτη.

ὁ῀ι γὰρ μεῖζόν ἐστι τὸ ἀπὸ τῆς ΒΗ τοῦ ἀπὸ τῆς ΗΓ, ὀέστω τὸ ἀπὸ τῆς Θ. καὶ ἐπεί ἐστιν ὡξ ὁ ΕΔ πρὸς τὸν ΖΔ, οὑτως τὸ ἀπὸ τῆς ΒΗ πρὸς τὸ ἀπὸ τῆς ΗΓ, καὶ ἀναστρέψαντι ἄρα ἐστὶν ὡξ ὁ ΔΕ πρὸς τὸν ΕΖ, οὑτως τὸ ἀπὸ τῆς ΗΒ πρὸς τὸ ἀπὸ τῆς Θ. ὁ δὲ ΔΕ πρὸς τὸν ΕΖ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἑκάτερος γὰρ τετράγωνόξ ἐστιν· καὶ τὸ ἀπὸ τῆς ΗΒ ἄρα πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· σύμμετρος ἄρα ἐστὶν ἡ ΒΗ τῇ Θ μήκει. καὶ δύναται ἡ ΒΗ τῆς ΗΓ μεῖζον τῷ ἀπὸ τῆς Θ· ἡ ΒΗ ἄρα τῆς ΗΓ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει. καί ἐστιν ἡ ὅλη ἡ ΒΗ σύμμετρος τῇ ἐκκειμένῃ ὁρητῇ μήκει τῇ Α. ἡ ΒΓ ἄρα ἀποτομή ἐστι πρώτη.

Εὑρηται ἄρα ἡ πρώτη ἀποτομὴ ἡ ΒΓ· ὅπερ ἔδει εὑρεῖν.

Let the rational (straight-line) *A* be laid down. And let *BG* be commensurable in length with *A*. *BG* is thus also a rational (straight-line). And let two square numbers *DE* and *EF* be laid down, and let their difference *FD* be not square [Prop. 10.28 lem. I]. Thus, *ED* does not have to *DF* the ratio which (some) square number (has) to (some) square number. And let it be contrived that as *ED* (is) to *DF*, so the square on *BG* (is) to the square on *GC* [Prop. 10.6. corr.]. Thus, the (square) on *BG* is commensurable with the (square) on *GC* [Prop. 10.6]. And the (square) on *BG* (is) rational. Thus, the (square) on *GC* (is) also rational. Thus, *GC* is also rational. And since *ED* does not have to *DF* the ratio which (some) square number (has) to (some) square number, the (square) on *BG* thus does not have to the (square) on *GC* the ratio which (some) square number (has) to (some) square number either. Thus, *BG* is incommensurable in length with *GC* [Prop. 10.9]. And they are both rational (straight-lines). Thus, *BG* and *GC* are rational (straight-lines which are) commensurable in square only. Thus, *BC* is an apotome [Prop. 10.73]. So, I say that (it is) also a first (apotome).

Let the (square) on *H* be that (area) by which the (square) on *BG* is greater than the (square) on *GC* [Prop. 10.13 lem.]. And since as *ED* is to *FD*, so the (square) on *BG* (is) to the (square) on *GC*, thus, via conversion, as *DE* is to *EF*, so the (square) on *GB* (is) to the (square) on *H* [Prop. 5.19 corr.]. And *DE* has to *EF* the ratio which (some) square-number (has) to (some) square-number. For each is a square (number). Thus, the (square) on *GB* also has to the (square) on *H* the ratio which (some) square number (has) to (some) square number. Thus, *BG* is commensurable in length with *H* [Prop. 10.9]. And the square on *BG* is greater than (the square on) *GC* by the (square) on *H*. Thus, the square on *BG* is greater than (the square on) *GC* by the (square) on (some straight-line) commensurable in length with (*BG*). And the whole, *BG*, is commensurable in length with the (previously) laid down rational (straight-line) *A*. Thus, *BC* is a first apotome [Def. 10.11].

Thus, the first apotome *BC* has been found. (Which is) the very thing it was required to find.

† See footnote to Prop. 10.48.

Εὑρεῖν τὴν δευτέραν ἀποτομήν.

Ἐκκείσθω ὁρητὴ ἡ Α καὶ τῇ Α σύμμετρος μήκει ἡ ΗΓ. ὁρητὴ ἄρα ἐστὶν ἡ ΗΓ. καὶ ἐκκείσθωσαν δύο τετράγωνοι ἀριθμοὶ οἱ ΔΕ, ΕΖ, ὧν ἡ ὑπεροθὴ ὁ ΔΖ μὴ ὁέστω τετράγωνος. καὶ πεποιήσθω ὡξ ὁ ΖΔ πρὸς τὸν ΔΕ, οὑτως τὸ ἀπὸ τῆς ΓΗ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΗΒ τετράγωνον. σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΓΗ τετράγωνον τῷ ἀπὸ τῆς ΗΒ τετραγώνῳ. ὁρητὸν δὲ τὸ ἀπὸ τῆς ΓΗ. ὁρητὸν ἄρα [ἐστὶ] καὶ τὸ ἀπὸ τῆς ΗΒ· ὁρητὴ ἄρα ἐστὶν ἡ ΒΗ. καὶ ἐπεὶ τὸ ἀπὸ τῆς ΗΓ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΗΒ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, ἀσύμμετρόξ ἐστιν ἡ ΓΗ τῇ ΗΒ μήκει. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ ΓΗ, ΗΒ ἄρα ρηταί εἰσι δυνάμει μόνον σύμμετροι· ἡ ΒΓ ἄρα ἀποτομή ἐστιν. λέγω δή, ὅτι καὶ δευτέρα.

To find a second apotome.

Let the rational (straight-line) A, and GC (which is) commensurable in length with A, be laid down. Thus, GC is a rational (straight-line). And let the two square numbers DE and EF be laid down, and let their difference DF be not square [Prop. 10.28 lem. I]. And let it be contrived that as FD (is) to DE, so the square on CG (is) to the square on GB [Prop. 10.6 corr.]. Thus, the square on CG is commensurable with the square on GB [Prop. 10.6]. And the (square) on CG (is) rational. Thus, the (square) on GB [is] also rational. Thus, BG is a rational (straight-line). And since the square on GC does not have to the (square) on GB the ratio which (some) square number (has) to (some) square number, CG is incommensurable in length with GB [Prop. 10.9]. And they are both rational (straight-lines). Thus, CG and GB are rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. So, I say that it is also a second (apotome).

ὅτι γὰρ μεῖζόν ἐστι τὸ ἀπὸ τῆς ΒΗ τοῦ ἀπὸ τῆς ΗΓ, ὁέστω τὸ ἀπὸ τῆς Θ. ἐπεὶ οὐῦν ἐστιν ὡξ τὸ ἀπὸ τῆς ΒΗ πρὸς τὸ ἀπὸ τῆς ΗΓ, οὑτως ὁ ΕΔ ἀριθμὸξ πρὸς τὸν ΔΖ ἀριθμόν, ἀναστρέψαντι ἄρα ἐστὶν ὡξ τὸ ἀπὸ τῆς ΒΗ πρὸς τὸ ἀπὸ τῆς Θ, οὑτως ὁ ΔΕ πρὸς τὸν ΕΖ. καί ἐστιν ἑκάτερος τῶν ΔΕ, ΕΖ τετράγωνος· τὸ ἄρα ἀπὸ τῆς ΒΗ πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· σύμμετρος ἄρα ἐστὶν ἡ ΒΗ τῇ Θ μήκει. καὶ δύναται ἡ ΒΗ τῆς ΗΓ μεῖζον τῷ ἀπὸ τῆς Θ· ἡ ΒΗ ἄρα τῆς ΗΓ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει. καί ἐστιν ἡ προσαρμόζουσα ἡ ΓΗ τῇ ἐκκειμένῃ ὁρητῇ σύμμετρος τῇ Α. ἡ ΒΓ ἄρα ἀποτομή ἐστι δευτέτα.

Εὑρηται ἄρα δευτέρα ἀποτομὴ ἡ ΒΓ· ὅπερ ἔδει δεῖχαι.

For let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.1 lem.]. Therefore, since as the (square) on BG is to the (square) on GC, so the number ED (is) to the number DF, thus, also, via conversion, as the (square) on BG is to the (square) on H, so DE (is) to EF [Prop. 5.19 corr.]. And DE and EF are each square (numbers). Thus, the (square) on BG has to the (square) on H the ratio which (some) square number (has) to (some) square number. Thus, BG is commensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H. Thus, the square on BG is greater than (the square on) GC by the (square) on (some straight-line) commensurable in length with (BG). And the attachment CG is commensurable (in length) with the (previously) laid down rational (straight-line) A. Thus, BC is a second apotome [Def. 10.12].†

Thus, the second apotome BC has been found. (Which is) the very thing it was required to show.

† See footnote to Prop. 10.49.

87.

Εὑρεῖν τὴν τρίτην ἀποτομήν.

Proposition 87

To find a third apotome.

Ἐκκείσθω ὁρητὴ ἡ Α, καὶ ἐκκείσθωσαν τρεῖς ἀριθμοὶ οἱ Ε, ΒΓ, ΓΔ λόγον μὴ ἔχοντες πρὸς ἀλλήλους, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, ὁ δὲ ΓΒ πρὸς τὸν ΒΔ λόγον ἐθέτω, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, καὶ πεποιήσθω ὡξ μὲν ὁ Ε πρὸς τὸν ΒΓ, οὕτως τὸ ἀπὸ τῆς Α τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΗ τετράγωνον, ὡξ δὲ ὁ ΒΓ πρὸς τὸν ΓΔ, οὕτως τὸ ἀπὸ τῆς ΖΗ τετράγωνον πρὸς τὸ ἀπὸ τηξ ΗΘ. ἐπεὶ οὐῦν ἐστιν ὡξ ὁ Ε πρὸς τὸν ΒΓ, οὕτως τὸ ἀπὸ τῆς Α τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΗ τετράγωνον, σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς Α τετράγωνον τῷ ἀπὸ τῆς ΖΗ τετραγώνῳ. ὁρητὸν δὲ τὸ ἀπὸ τῆς Α τετράγωνον. ὁρητὸν ἄρα καὶ τὸ ἀπὸ τῆς ΖΗ· ὁρητὴ ἄρα ἐστὶν ἡ ΖΗ. καὶ ἐπεὶ ὁ Ε πρὸς τὸν ΒΓ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, οὐδὸ ἄρα τὸ ἀπὸ τῆς Α τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΗ [τετράγωνον] λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ Α τῇ ΖΗ μήκει. πάλιν, ἐπεί ἐστιν ὡξ ὁ ΒΓ πρὸς τὸν ΓΔ, οὕτως τὸ ἀπὸ τῆς ΖΗ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΗΘ, σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΖΗ τῷ ἀπὸ τῆς ΗΘ. ὁρητὸν δὲ τὸ ἀπὸ τῆς ΖΗ· ὁρητὸν ἄρα καὶ τὸ ἀπὸ τῆς ΗΘ· ὁρητὴ ἄρα ἐστὶν ἡ ΗΘ. καὶ ἐπεὶ ὁ ΒΓ πρὸς τὸν ΓΔ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, οὐδὸ ἄρα τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΖΗ τῇ ΗΘ μήκει. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ ΖΗ, ΗΘ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ ΖΘ. λέγω δή, ὅτι καὶ τρίτη.

Ἐπεὶ γάρ ἐστιν ὡξ μὲν ὁ Ε πρὸς τὸν ΒΓ, οὕτως τὸ ἀπὸ τῆς Α τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΗ, ὡξ δὲ ὁ ΒΓ πρὸς τὸν ΓΔ, οὕτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΘΗ, διὸ ὁίσου ἄρα ἐστὶν ὡξ ὁ Ε πρὸς τὸν ΓΔ, οὕτως τὸ ἀπὸ τῆς Α πρὸς τὸ ἀπὸ τῆς ΘΗ. ὁ δὲ Ε πρὸς τὸν ΓΔ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· οὐδὸ ἄρα τὸ ἀπὸ τῆς Α πρὸς τὸ ἀπὸ τῆς ΗΘ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἡ Α τῇ ΗΘ μήκει. οὐδετέρα ἄρα τῶν ΖΗ, ΗΘ σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῇ τῇ Α μήκει. ὅῷ οὐῦν μεῖζόν ἐστι τὸ ἀπὸ τῆς ΖΗ τοῦ ἀπὸ τῆς ΗΘ, ὀέστω τὸ ἀπὸ τῆς Κ. ἐπεὶ οὐῦν ἐστιν

Let the rational (straight-line) A be laid down. And let the three numbers, E, BC, and CD, not having to one another the ratio which (some) square number (has) to (some) square number, be laid down. And let CB have to BD the ratio which (some) square number (has) to (some) square number. And let it be contrived that as E (is) to BC, so the square on A (is) to the square on FG, and as BC (is) to CD, so the square on FG (is) to the (square) on GH [Prop. 10.6 corr.]. Therefore, since as E is to BC, so the square on A (is) to the square on FG, the square on A is thus commensurable with the square on FG [Prop. 10.6]. And the square on A (is) rational. Thus, the (square) on FG (is) also rational. Thus, FG is a rational (straight-line). And since E does not have to BC the ratio which (some) square number (has) to (some) square number, the square on A thus does not have to the [square] on FG the ratio which (some) square number (has) to (some) square number either. Thus, A is incommensurable in length with FG [Prop. 10.9]. Again, since as BC is to CD, so the square on FG is to the (square) on GH, the square on FG is thus commensurable with the (square) on GH [Prop. 10.6]. And the (square) on FG (is) rational. Thus, the (square) on GH (is) also rational. Thus, GH is a rational (straight-line). And since BC does not have to CD the ratio which (some) square number (has) to (some) square number, the (square) on FG thus does not have to the (square) on GH the ratio which (some) square number (has) to (some) square number either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. And both are rational (straight-lines). FG and GH are thus rational (straight-lines which are) commensurable in square only. Thus, FH is an apotome [Prop. 10.73]. So, I say that (it is) also a third (apotome).

For since as E is to BC, so the square on A (is) to the (square) on FG, and as BC (is) to CD, so the (square) on FG (is) to the (square) on HG, thus, via equality, as E is to CD, so the (square) on A (is) to the (square) on HG [Prop. 5.22]. And E does not have to CD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on A does not have to the (square) on GH the ratio which (some) square

ὢξ ὁ ΒΓ πρὸς τὸν ΓΔ, οὕτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ, ἀναστρέψαντι ἄρα ἐστὶν ὢξ ὁ ΒΓ πρὸς τὸν ΒΔ, οὕτως τὸ ἀπὸ τῆς ΖΗ τετράγωνον πρὸς τὸ ἀπὸ τῆς Κ. ὁ δὲ ΒΓ πρὸς τὸν ΒΔ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν. καὶ τὸ ἀπὸ τῆς ΖΗ ἄρα πρὸς τὸ ἀπὸ τῆς Κ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν. σύμμετρόξ ἄρα ἐστὶν ἡ ΖΗ τῇ Κ μήκει, καὶ δύναται ἡ ΖΗ τῆς ΗΘ μεῖζον τῷ ἀπὸ συμμέτρου ἑαυτῇ. καὶ οὐδετέρα τῶν ΖΗ, ΗΘ σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῇ τῇ Α μήκει· ἡ ΖΘ ἄρα ἀποτομή ἐστι τρίτη.

Εὑρηται ἄρα ἡ τρίτη ἀποτομὴ ἡ ΖΘ· ὅπερ ἔδει δεῖξαι.

number (has) to (some) square number either. *A* (is) thus incommensurable in length with *GH* [Prop. 10.9]. Thus, neither of *FG* and *GH* is commensurable in length with the (previously) laid down rational (straight-line) *A*. Therefore, let the (square) on *K* be that (area) by which the (square) on *FG* is greater than the (square) on *GH* [Prop. 10.13 lem.]. Therefore, since as *BC* is to *CD*, so the (square) on *FG* (is) to the (square) on *GH*, thus, via conversion, as *BC* is to *BD*, so the square on *FG* (is) to the square on *K* [Prop. 5.19 corr.]. And *BC* has to *BD* the ratio which (some) square number (has) to (some) square number. Thus, the (square) on *FG* also has to the (square) on *K* the ratio which (some) square number (has) to (some) square number. *FG* is thus commensurable in length with *K* [Prop. 10.9]. And the square on *FG* is (thus) greater than (the square on) *GH* by the (square) on (some straight-line) commensurable (in length) with (*FG*). And neither of *FG* and *GH* is commensurable in length with the (previously) laid down rational (straight-line) *A*. Thus, *FH* is a third apotome [Def. 10.13].

Thus, the third apotome *FH* has been found. (Which is) very thing it was required to show.

† See footnote to Prop. 10.50.

88.

Εὑρεῖν τὴν τετάρτην ἀποτομήν.

Proposition 88

To find a fourth apotome.

Ἐκκείσθω ὁρητὴ ἡ Α καὶ τῇ Α μήκει σύμμετρος ἡ ΒΗ· ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΒΗ. καὶ ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΔΖ, ΖΕ, ὥστε τὸν ΔΕ ὅλον πρὸς ἑκάτερον τῶν ΔΖ, ΕΖ λόγον μὴ ἔχειν, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν. καὶ πεποιήσθω ὢξ ὁ ΔΕ πρὸς τὸν ΕΖ, οὕτως τὸ ἀπὸ τῆς ΒΗ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΗΓ· σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΒΗ τῷ ἀπὸ τῆς ΗΓ. ὁρητὸν δὲ τὸ ἀπὸ τῆς ΒΗ· ὁρητὸν ἄρα καὶ τὸ ἀπὸ τῆς ΗΓ· ὁρητὴ ἄρα ἐστὶν ἡ ΗΓ. καὶ ἐπεὶ ὁ ΔΕ πρὸς τὸν ΕΖ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, οὐδὸ ἄρα τὸ ἀπὸ τῆς ΒΗ πρὸς τὸ ἀπὸ τῆς ΗΓ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΒΗ τῇ ΗΓ μήκει. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ ΒΗ, ΗΓ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ ΒΓ. [λέγω δή, ὅτι καὶ τετάρτη.]

ὅ῎ι οὐῦν μεῖζόν ἐστι τὸ ἀπὸ τῆς ΒΗ τοῦ ἀπὸ τῆς ΗΓ, ὀέστω

Let the rational (straight-line) *A*, and *BG* (which is) commensurable in length with *A*, be laid down. Thus, *BG* is also a rational (straight-line). And let the two numbers *DF* and *FE* be laid down such that the whole, *DE*, does not have to each of *DF* and *EF* the ratio which (some) square number (has) to (some) square number. And let it be contrived that as *DE* (is) to *EF*, so the square on *BG* (is) to the (square) on *GC* [Prop. 10.6 corr.]. The (square) on *BG* is thus commensurable with the (square) on *GC* [Prop. 10.6]. And the (square) on *BG* (is) rational. Thus, the (square) on *GC* (is) also rational. Thus, *GC* (is) a rational (straight-line). And since *DE* does not have to *EF* the ratio which (some) square number (has) to (some) square number, the (square) on *BG* thus does not have to the (square) on *GC* the ratio which (some) square number (has) to (some) square number either. Thus, *BG* is incommensurable in length with *GC* [Prop. 10.9]. And they are both rational

τὸ ἀπὸ τῆς Θ. ἐπεὶ οὐῦν ἐστιν ὡξ ὁ ΔΕ πρὸς τὸν ΕΖ, οὑτως τὸ ἀπὸ τῆς ΒΗ πρὸς τὸ ἀπὸ τῆς ΗΓ, καὶ ἀναστρέψαντι ἄρα ἐστὶν ὡξ ὁ ΕΔ πρὸς τὸν ΔΖ, οὑτως τὸ ἀπὸ τῆς ΗΒ πρὸς τὸ ἀπὸ τῆς Θ. ὁ δὲ ΕΔ πρὸς τὸν ΔΖ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· οὐδὸ ἄρα τὸ ἀπὸ τῆς ΗΒ πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΒΗ τῇ Θ μήκει. καὶ δύναται ἡ ΒΗ τῆς ΗΓ μεῖζον τῷ ἀπὸ τῆς Θ· ἡ ἄρα ΒΗ τῆς ΗΓ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καὶ ἐστιν ὅλη ἡ ΒΗ σύμμετρος τῇ ἐκκειμένῃ ὁρητῇ μήκει τῇ Α. ἡ ἄρα ΒΓ ἀποτομή ἐστι τετάρτη.

Εὑρηται ἄρα ἡ τετάρτη ἀποτομή· ὅπερ ἔδει δεῖχαι.

(straight-lines). Thus, BG and GC are rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. [So, I say that (it is) also a fourth (apotome).]

Now, let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.13 lem.]. Therefore, since as DE is to EF, so the (square) on BG (is) to the (square) on GC, thus, also, via conversion, as ED is to DF, so the (square) on GB (is) to the (square) on H [Prop. 5.19 corr.]. And ED does not have to DF the ratio which (some) square number (has) to (some) square number. Thus, the (square) on GB does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H. Thus, the square on BG is greater than (the square) on GC by the (square) on (some straight-line) incommensurable (in length) with (BG). And the whole, BG, is commensurable in length with the the (previously) laid down rational (straight-line) A. Thus, BC is a fourth apotome [Def. 10.14].†

Thus, a fourth apotome has been found. (Which is) the very thing it was required to show.

† See footnote to Prop. 10.51.

89.

Εὑρεῖν τὴν πέμπτην ἀποτομήν.

Proposition 89

To find a fifth apotome.

Ἐκκείσθω ὁρητὴ ἡ Α, καὶ τῇ Α μήκει σύμμετροςὸέστω ἡ ΓΗ· ὁρητὴ ἄρα [ἐστὶν] ἡ ΓΗ. καὶ ἐκκείσθωσαν δύο ἀριθμοὶ οἱ ΔΖ, ΖΕ, ὁώστε τὸν ΔΕ πρὸς ἑκάτερον τῶν ΔΖ, ΖΕ λόγον πάλιν μὴ ἔχειν, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· καὶ πεποιήσθω ὡξ ὁ ΖΕ πρὸς τὸν ΕΔ, οὑτως τὸ ἀπὸ τῆς ΓΗ πρὸς τὸ ἀπὸ τῆς ΗΒ. ὁρητὸν ἄρα καὶ τὸ ἀπὸ τῆς ΗΒ· ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΒΗ. καὶ ἐπεί ἐστιν ὡξ ὁ ΔΕ πρὸς τὸν ΕΖ, οὑτως τὸ ἀπὸ τῆς ΒΗ πρὸς τὸ ἀπὸ τῆς ΗΓ, ὁ δὲ ΔΕ πρὸς τὸν ΕΖ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, οὐδὸ ἄρα τὸ ἀπὸ τῆς ΒΗ πρὸς τὸ ἀπὸ τῆς ΗΓ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΒΗ τῇ ΗΓ μήκει. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ ΒΗ, ΗΓ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἡ ΒΓ ἄρα ἀποτομή ἐστιν. λέγω δή, ὅτι καὶ πέμπτη.

Let the rational (straight-line) A be laid down, and let CG be commensurable in length with A. Thus, CG [is] a rational (straight-line). And let the two numbers DF and FE be laid down such that DE again does not have to each of DF and FE the ratio which (some) square number (has) to (some) square number. And let it be contrived that as FE (is) to ED, so the (square) on CG (is) to the (square) on GB. Thus, the (square) on GB (is) also rational [Prop. 10.6]. Thus, BG is also rational. And since as DE is to EF, so the (square) on BG (is) to the (square) on GC. And DE does not have to EF the ratio which (some) square number (has) to (some) square number. The (square) on BG thus does not have to the (square) on GC the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with GC [Prop. 10.9]. And they are both rational (straight-lines). BG

ὅ῾ι γὰρ μεῖζόν ἐστι τὸ ἀπὸ τῆς ΒΗ τοῦ ἀπὸ τῆς ΗΓ, ὀέστω τὸ ἀπὸ τῆς Θ. ἐπεὶ οὐῦν ἐστιν ὡξ τὸ ἀπὸ τὴξ ΒΗ πρὸς τὸ ἀπὸ τῆς ΗΓ, οὑτως ὁ ΔΕ πρὸς τὸν ΕΖ, ἀναστρέψαντι ἄρα ἐστὶν ὡξ ὁ ΕΔ πρὸς τὸν ΔΖ, οὑτως τὸ ἀπὸ τῆς ΒΗ πρὸς τὸ ἀπὸ τῆς Θ, ὁ δὲ ΕΔ πρὸς τὸν ΔΖ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· οὐδὸ ἄρα τὸ ἀπὸ τῆς ΒΗ πρὸς τὸ ἀπὸ τῆς Θ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΒΗ τῇ Θ μήκει. καὶ δύναται ἡ ΒΗ τῆς ΗΓ μεῖζον τῷ ἀπὸ τῆς Θ· ἡ ΗΒ ἄρα τῆς ΗΓ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ μήκει. καί ἐστιν ἡ προσαρμόζουσα ἡ ΓΗ σύμμετρος τῇ ἐκκειμένῃ ὁρητῇ τῇ Α μήκει· ἡ ἄρα ΒΓ ἀποτομή ἐστι πέμπτη.

Εὑρηται ἄρα ἡ πέμπτη ἀποτομὴ ἡ ΒΓ· ὅπερ ἔδει δεῖχαι.

and GC are thus rational (straight-lines which are) commensurable in square only. Thus, BC is an apotome [Prop. 10.73]. So, I say that (it is) also a fifth (apotome).

For, let the (square) on H be that (area) by which the (square) on BG is greater than the (square) on GC [Prop. 10.13 lem.]. Therefore, since as the (square) on BG (is) to the (square) on GC, so DE (is) to EF, thus, via conversion, as ED is to DF, so the (square) on BG (is) to the (square) on H [Prop. 5.19 corr.]. And ED does not have to DF the ratio which (some) square number (has) to (some) square number. Thus, the (square) on BG does not have to the (square) on H the ratio which (some) square number (has) to (some) square number either. Thus, BG is incommensurable in length with H [Prop. 10.9]. And the square on BG is greater than (the square on) GC by the (square) on H. Thus, the square on GB is greater than (the square on) GC by the (square) on (some straight-line) incommensurable in length with (GB). And the attachment CG is commensurable in length with the (previously) laid down rational (straight-line) A. Thus, BC is a fifth apotome [Def. 10.15].†

Thus, the fifth apotome BC has been found. (Which is) the very thing it was required to show.

† See footnote to Prop. 10.52.

90.

Εὑρεῖν τὴν ὁέκτην ἀποτομήν.

Ἐκκείσθω ὁρητὴ ἡ Α καὶ τρεῖς ἀριθμοὶ οἱ Ε, ΒΓ, ΓΔ λόγον μὴ ἔχοντες πρὸς ἀλλήλους, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ὁέτι δὲ καὶ ὁ ΓΒ πρὸς τὸν ΒΔ λόγον μὴ ἐθετώ, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· καὶ πεποιήσθω ὡξ μὲν ὁ Ε πρὸς τὸν ΒΓ, οὑτως τὸ ἀπὸ τῆς Α πρὸς τὸ ἀπὸ τῆς ΖΗ, ὡξ δὲ ὁ ΒΓ πρὸς τὸν ΓΔ, οὑτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ.

Proposition 90

To find a sixth apotome.

Let the rational (straight-line) A, and the three numbers E, BC, and CD, not having to one another the ratio which (some) square number (has) to (some) square number, be laid down. Furthermore, let CB also not have to BD the ratio which (some) square number (has) to (some) square number. And let it be contrived that as E (is) to BC, so the (square) on A (is) to the (square) on FG, and as BC (is) to CD, so the (square) on FG (is) to the (square) on GH [Prop. 10.6 corr.].

Ἐπεὶ οὐῦν ἐστιν ὡξ ὁ Ε πρὸς τὸν ΒΓ, οὑτως τὸ ἀπὸ τῆς Α πρὸς τὸ ἀπὸ τῆς ΖΗ, σύμμετρον ἄρα τὸ ἀπὸ τῆς Α τῷ ἀπὸ τῆς ΖΗ. ὁρητὸν δὲ τὸ ἀπὸ τῆς Α· ὁρητὸν ἄρα καὶ

Therefore, since as E is to BC, so the (square) on A (is) to the (square) on FG, the (square) on A (is) thus commensurable with the (square) on FG [Prop. 10.6]. And the (square)

τὸ ἀπὸ τῆς ΖΗ· ὀρητὴ ἄρα ἐστὶ καὶ ἡ ΖΗ. καὶ ἐπεὶ ὁ Ε πρὸς τὸν ΒΓ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὸ ἄρα τὸ ἀπὸ τῆς Α πρὸς τὸ ἀπὸ τῆς ΖΗ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ Α τῇ ΖΗ μήκει. πάλιν, ἐπεί ἐστιν ὡς ὁ ΒΓ πρὸς τὸν ΓΔ, οὕτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ, σύμμετρον ἄρα τὸ ἀπὸ τῆς ΖΗ τῷ ἀπὸ τῆς ΗΘ. ὀρητὸν δὲ τὸ ἀπὸ τῆς ΖΗ· ὀρητὸν ἄρα καὶ τὸ ἀπὸ τῆς ΗΘ· ὀρητὴ ἄρα καὶ ἡ ΗΘ. καὶ ἐπεὶ ὁ ΒΓ πρὸς τὸν ΓΔ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν, οὐδὸ ἄρα τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΖΗ τῇ ΗΘ μήκει. καί εἰσιν ἀμφότεραι ὀρηταί· αἱ ΖΗ, ΗΘ ἄρα ὀρηταί εἰσι δυνάμει μόνον σύμμετροι· ἡ ἄρα ΖΘ ἀποτομή ἐστιν. λέγω δή, ὅτι καὶ ὀέκτη.

Ἐπεὶ γάρ ἐστιν ὡς μὲν ὁ Ε πρὸς τὸν ΒΓ, οὕτως τὸ ἀπὸ τῆς Α πρὸς τὸ ἀπὸ τῆς ΖΗ, ὡς δὲ ὁ ΒΓ πρὸς τὸν ΓΔ, οὕτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ, διὸ ὀίσου ἄρα ἐστὶν ὡς ὁ Ε πρὸς τὸν ΓΔ, οὕτως τὸ ἀπὸ τῆς Α πρὸς τὸ ἀπὸ τῆς ΗΘ. ὁ δὲ Ε πρὸς τὸν ΓΔ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδὸ ἄρα τὸ ἀπὸ τῆς Α πρὸς τὸ ἀπὸ τῆς ΗΘ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ Α τῇ ΗΘ μήκει· οὐδετέρα ἄρα τῶν ΖΗ, ΗΘ σύμμετρός ἐστι τῇ Α ὀρητῇ μήκει. ὀῷ οὐῦν μεῖζόν ἐστι τὸ ἀπὸ τῆς ΖΗ τοῦ ἀπὸ τῆς ΗΘ, ὀέστω τὸ ἀπὸ τῆς Κ. ἐπεὶ οὐῦν ἐστιν ὡς ὁ ΒΓ πρὸς τὸν ΓΔ, οὕτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς ΗΘ, ἀναστρέψαντι ἄρα ἐστὶν ὡς ὁ ΓΒ πρὸς τὸν ΒΔ, οὕτως τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς Κ. ὁ δὲ ΓΒ πρὸς τὸν ΒΔ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· οὐδὸ ἄρα τὸ ἀπὸ τῆς ΖΗ πρὸς τὸ ἀπὸ τῆς Κ λόγον ἔχει, ὃν τετράγωνος ἀριθμὸς πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΖΗ τῇ Κ μήκει. καὶ δύναται ἡ ΖΗ τῆς ΗΘ μεῖζον τῷ ἀπὸ τῆς Κ· ἡ ΖΗ ἄρα τῆς ΗΘ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ μήκει. καὶ οὐδετέρα τῶν ΖΗ, ΗΘ σύμμετρός ἐστι τῇ ἐκκειμένῃ ὀρητῇ μήκει τῇ Α. ἡ ἄρα ΖΘ ἀποτομή ἐστιν ὀέκτη.

Εὑρηται ἄρα ἡ ὀέκτη ἀποτομὴ ἡ ΖΘ· ὅπερ ἔδει δεῖξαι.

on A (is) rational. Thus, the (square) on FG (is) also rational. Thus, FG is also a rational (straight-line). And since E does not have to BC the ratio which (some) square number (has) to (some) square number, the (square) on A thus does not have to the (square) on FG the ratio which (some) square number (has) to (some) square number either. Thus, A is incommensurable in length with FG [Prop. 10.9]. Again, since as BC is to CD, so the (square) on FG (is) to the (square) on GH, the (square) on FG (is) thus commensurable with the (square) on GH [Prop. 10.6]. And the (square) on FG (is) rational. Thus, the (square) on GH (is) also rational. Thus, GH (is) also rational. And since BC does not have to CD the ratio which (some) square number (has) to (some) square number, the (square) on FG thus does not have to the (square) on GH the ratio which (some) square (number) has to (some) square (number) either. Thus, FG is incommensurable in length with GH [Prop. 10.9]. And both are rational (straight-lines). Thus, FG and GH are rational (straight-lines which are) commensurable in square only. Thus, FH is an apotome [Prop. 10.73]. So, I say that (it is) also a sixth (apotome).

For since as E is to BC, so the (square) on A (is) to the (square) on FG, and as BC (is) to CD, so the (square) on FG (is) to the (square) on GH, thus, via equality, as E is to CD, so the (square) on A (is) to the (square) on GH [Prop. 5.22]. And E does not have to CD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on A does not have to the (square) GH the ratio which (some) square number (has) to (some) square number either. A is thus incommensurable in length with GH [Prop. 10.9]. Thus, neither of FG and GH is commensurable in length with the rational (straight-line) A. Therefore, let the (square) on K be that (area) by which the (square) on FG is greater than the (square) on GH [Prop. 10.13 lem.]. Therefore, since as BC is to CD, so the (square) on FG (is) to the (square) on GH, thus, via conversion, as CB is to BD, so the (square) on FG (is) to the (square) on K [Prop. 5.19 corr.]. And CB does not have to BD the ratio which (some) square number (has) to (some) square number. Thus, the (square) on FG does not have to the (square) on K the ratio which (some) square number (has) to (some) square number either. FG is thus incommensurable in length with K [Prop. 10.9]. And the square on FG is greater than (the square on) GH by the (square) on K. Thus, the square on FG is greater than (the square on) GH by the (square) on (some straight-line) incommensurable in length with (FG). And neither of FG and GH is commensurable in length with the (previously) laid down rational (straight-line) A. Thus, FH is a sixth apotome [Def. 10.16].

Thus, the sixth apotome FH has been found. (Which is) the very thing it was required to show.

† See footnote to Prop. 10.53.

91.

Ἐὰν θωρίον περιέθηται ὑπὸ ὁρητῆς καὶ ἀποτομῆς πρώτης, ἡ τὸ θωρίον δυναμένη ἀπορομή ἐστιν.

Περιεθέσθω γὰρ θωρίον τὸ ΑΒ ὑπὸ ὁρητῆς τῆς ΑΓ καὶ ἀποτομῆς πρώτης τῆς ΑΔ· λέγω, ὅτι ἡ τὸ ΑΒ θωρίον δυναμένη ἀποτομή ἐστιν.

Ἐπεὶ γὰρ ἀποτομή ἐστι πρώτη ἡ ΑΔ, ὀέστω αὐτῇ προσαρμόζουσα ἡ ΔΗ· αἱ ΑΗ, ΗΔ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι. καὶ ὅλη ἡ ΑΗ σύμμετρός ἐστι τῇ ἐκκειμένῃ ὁρητῇ τῇ ΑΓ, καὶ ἡ ΑΗ τῆς ΗΔ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει· ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΔΗ ἴσον παρὰ τὴν ΑΗ παραβληθῇ ἐλλεῖπον εἰόδει τετραγώνῳ, εἰξ σύμμετρα αὐτὴν διαιρεῖ. τετμήσθω ἡ ΔΗ δίθα κατὰ τὸ Ε, καὶ τῷ ἀπὸ τῆς ΕΗ ἴσον παρὰ τὴν ΑΗ παραβεβλήσθω ἐλλεῖπον εἰόδει τετραγώνῳ, καὶ ὀέστω τὸ ὑπὸ τῶν ΑΖ, ΖΗ· σύμμετρος ἄρα ἐστὶν ἡ ΑΖ τῇ ΖΗ. καὶ διὰ τῶν Ε, Ζ, Η σημείων τῇ ΑΓ παράλληλοι ἤχθωσαν αἱ ΕΘ, ΖΙ, ΗΚ.

Proposition 91

If an area is contained by a rational (straight-line) and a first apotome then the square-root of the area is an apotome.

For let the area AB be contained by the rational (straight-line) AC and the first apotome AD. I say that the square-root of area AB is an apotome.

For since AD is a first apotome, let DG be its attachment. Thus, AG and DG are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And the whole, AG, is commensurable (in length) with the (previously) laid down rational (straight-line) AC, and the square on AG is greater than (the square on) GD by the (square) on (some straight-line) commensurable in length with (AG) [Def. 10.11]. Thus, if (an area) equal to the fourth part of the (square) on DG is applied to AG, falling short by a square figure, then it divides (AG) into (parts which are) commensurable (in length) [Prop. 10.17]. Let DG be cut in half at E. And let (an area) equal to the (square) on EG be applied to AG, falling short by a square figure. And let it be the (rectangle contained) by AF and FG. AF is thus commensurable (in length) with FG. And let EH, FI, and GK be drawn through points E, F, and G (respectively), parallel to AC.

Καὶ ἐπεὶ σύμμετρός ἐστιν ἡ ΑΖ τῇ ΖΗ μήκει, καὶ ἡ ΑΗ ἄρα ἑκατέρᾳ τῶν ΑΖ, ΖΗ σύμμετρός ἐστι μήκει. ἀλλὰ ἡ ΑΗ σύμμετρός ἐστι τῇ ΑΓ· καὶ ἑκατέρα ἄρα τῶν ΑΖ, ΖΗ σύμμετρός ἐστι τῇ ΑΓ μήκει. καί ἐστι ὁρητὴ ἡ ΑΓ· ὁρητὴ ἄρα καὶ ἑκατέρα τῶν ΑΖ, ΖΗ· ὥστε καὶ ἑκάτερον τῶν ΑΙ, ΖΚ ὁρητόν ἐστιν. καὶ ἐπεὶ σύμμετρός ἐστιν ἡ ΔΕ τῇ ΕΗ μήκει, καὶ ἡ ΔΗ ἄρα ἑκατέρᾳ τῶν ΔΕ, ΕΗ σύμμετρός ἐστι μήκει. ὁρητὴ δὲ ἡ ΔΗ καὶ ἀσύμμετρος τῇ ΑΓ μήκει· ὁρητὴ ἄρα καὶ ἑκατέρα τῶν ΔΕ, ΕΗ καὶ ἀσύμμετρος τῇ ΑΓ μήκει· ἑκάτερον ἄρα τῶν ΔΘ, ΕΚ μέσον ἐστίν.

Κείσθω δὴ τῷ μὲν ΑΙ ἴσον τετράγωνον τὸ ΛΜ, τῷ δὲ ΖΚ ἴσον τετράγωνον ἀφῃρήσθω κοινὴν γωνίαν ἔχον αὐτῷ τὴν

And since AF is commensurable in length with FG, AG is thus also commensurable in length with each of AF and FG [Prop. 10.15]. But AG is commensurable (in length) with AC. Thus, each of AF and FG is also commensurable in length with AC [Prop. 10.12]. And AC is a rational (straight-line). Thus, AF and FG (are) each also rational (straight-lines). Hence, AI and FK are also each rational (areas) [Prop. 10.19]. And since DE is commensurable in length with EG, DG is thus also commensurable in length with each of DE and EG [Prop. 10.15]. And DG (is) rational, and incommensurable in length with AC. DE and EG (are) thus each rational, and incommensurable in length with AC [Prop. 10.13]. Thus, DH

ὑπὸ ΛΟΜ τὸ ΝΧ· περὶ τὴν αὐτὴν ἄρα διάμετρόν ἐστι τὰ ΛΜ, ΝΧ τετράγωνα. ὀέστω αὐτῶν διάμετρος ἡ ΟΡ, καὶ καταγεγράφθω τὸ σχῆμα. ἐπεὶ οὐῦν ἴσον ἐστὶ τὸ ὑπὸ τῶν ΑΖ, ΖΗ περιεχόμενον ὀρθογώνιον τῷ ἀπὸ τῆς ΕΗ τετραγώνῳ, ὀέστιν ἄρα ὡξ ἡ ΑΖ πρὸς τὴν ΕΗ, οὐύτως ἡ ΕΗ πρὸς τὴν ΖΗ. ἀλλὰ ὡξ μὲν ἡ ΑΖ πρὸς τὴν ΕΗ, οὐύτως τὸ ΑΙ πρὸς τὸ ΕΚ, ὡξ δὲ ἡ ΕΗ πρὸς τὴν ΖΗ, οὐύτως ἐστὶ τὸ ΕΚ πρὸς τὸ ΚΖ· τῶν ἄρα ΑΙ, ΚΖ μέσον ἀνάλογόν ἐστι τὸ ΕΚ. ὀέστι δὲ καὶ τῶν ΛΜ, ΝΧ μέσον ἀνάλογον τὸ ΜΝ, ὡξ ἐν τοῖςὀέμπροσθεν ἐδείθθη, καί ἐστι τὸ [μὲν] ΑΙ τῷ ΛΜ τετραγώνῳ ἴσον, τὸ δὲ ΚΖ τῷ ΝΧ· καὶ τὸ ΜΝ ἄρα τῷ ΕΚ ἴσον ἐστίν. ἀλλὰ τὸ μὲν ΕΚ τῷ ΔΘ ἐστιν ἴσον, τὸ δὲ ΜΝ τῷ ΛΧ· τὸ ἄρα ΔΚ ἴσον ἐστὶ τῷ ΥΦΘ γνώμονι καὶ τῷ ΝΧ. ὀέστι δὲ καὶ τὸ ΑΚ ἴσον τοῖς ΛΜ, ΝΧ τετραγώνοις· λοιπὸν ἄρα τὸ ΑΒ ἴσον ἐστὶ τῷ ΣΤ. τὸ δὲ ΣΤ τὸ ἀπὸ τῆς ΛΝ ἐστι τετράγωνον· τὸ ἄρα ἀπὸ τῆς ΛΝ τετράγωνον ἴσον ἐστὶ τῷ ΑΒ· ἡ ΛΝ ἄρα δύναται τὸ ΑΒ. λέγω δή, ὅτι ἡ ΛΝ ἀποτομή ἐστιν.

Ἐπεὶ γὰρ ὀρητόν ἐστιν ἑκάτερον τῶν ΑΙ, ΖΚ, καί ἐστιν ἴσον τοῖς ΛΜ, ΝΧ, καὶ ἑκάτερον ἄρα τῶν ΛΜ, ΝΧ ὀρητόν ἐστιν, τουτέστι τὸ ἀπὸ ἑκατέρας τῶν ΛΟ, ΟΝ· καὶ ἑκατέρα ἄρα τῶν ΛΟ, ΟΝ ὀρητή ἐστιν. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ ΔΘ καί ἐστιν ἴσον τῷ ΛΧ, μέσον ἄρα ἐστὶ καὶ τὸ ΛΧ. ἐπεὶ οὐῦν τὸ μὲν ΛΧ μέσον ἐστίν, τὸ δὲ ΝΧ ὀρητόν, ἀσύμμετρον ἄρα ἐστὶ τὸ ΛΧ τῷ ΝΧ· ὡξ δὲ τὸ ΛΧ πρὸς τὸ ΝΧ, οὐύτως ἐστὶν ἡ ΛΟ πρὸς τὴν ΟΝ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΛΟ τῇ ΟΝ μήκει. καί εἰσιν ἀμφότεραι ὀρηταί· αἱ ΛΟ, ΟΝ ἄρα ὀρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ ΛΝ. καὶ δύναται τὸ ΑΒ θωρίον· ἡ ἄρα τὸ ΑΒ θωρίον δυναμένη ἀποτομή ἐστιν.

Ἐὰν ἄρα θωρίον περιέθηται ὑπὸ ὀρητῆς καὶ τὰ ἑχῆς.

and *EK* are each medial (areas) [Prop. 10.21].

So let the square *LM*, equal to *AI*, be laid down. And let the square *NO*, equal to *FK*, be subtracted (from *LM*), having with it the common angle *LPM*. Thus, the squares *LM* and *NO* are about the same diagonal [Prop. 6.26]. Let *PR* be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since the rectangle contained by *AF* and *FG* is equal to the square *EG*, thus as *AF* is to *EG*, so *EG* (is) to *FG* [Prop. 6.17]. But, as *AF* (is) to *EG*, so *AI* (is) to *EK*, and as *EG* (is) to *FG*, so *EK* is to *KF* [Prop. 6.1]. Thus, *EK* is the mean proportional to *AI* and *KF* [Prop. 5.11]. And *MN* is also the mean proportional to *LM* and *NO*, as shown before [Prop. 10.53 lem.]. And *AI* is equal to the square *LM*, and *KF* to *NO*. Thus, *MN* is also equal to *EK*. But, *EK* is equal to *DH*, and *MN* to *LO* [Prop. 1.43]. Thus, *DK* is equal to the gnomon *UVW* and *NO*. And *AK* is also equal to (the sum of) the squares *LM* and *NO*. Thus, the remainder *AB* is equal to *ST*. And *ST* is the square on *LN*. Thus, the square on *LN* is equal to *AB*. Thus, *LN* is the square-root of *AB*. So, I say that *LN* is an apotome.

For since *AI* and *FK* are each rational (areas), and are equal to *LM* and *NO* (respectively), thus *LM* and *NO*—that is to say, the (squares) on each of *LP* and *PN* (respectively)—are also each rational (areas). Thus, *LP* and *PN* are also each rational (straight-lines). Again, since *DH* is a medial (area), and is equal to *LO*, *LO* is thus also a medial (area). Therefore, since *LO* is medial, and *NO* rational, *LO* is thus incommensurable with *NO*. And as *LO* (is) to *NO*, so *LP* is to *PN* [Prop. 6.1]. *LP* is thus incommensurable in length with *PN* [Prop. 10.11]. And they are both rational (straight-lines). Thus, *LP* and *PN* are rational (straight-lines which are) commensurable in square only. Thus, *LN* is an apotome [Prop. 10.73]. And it is the square-root of area *AB*. Thus, the square-root of area *AB* is an apotome.

Thus, if an area is contained by a rational (straight-line), and so on ....

92.

Ἐὰν θωρίον περιέθηται ὑπὸ ὀρητῆς καὶ ἀποτομῆς δευτέρας, ἡ τὸ θωρίον δυναμένη μέσης ἀποτομή ἐστι πρώτη.

Proposition 92

If an area is contained by a rational (straight-line) and a second apotome then the square-root of the area is a first apotome of a medial (straight-line).

Θωρίον γὰρ τὸ AB περιειθέσθω ὑπὸ ὀρητῆς τῆς $AΓ$ καὶ ἀποτομῆς δευτέρας τῆς $AΔ$· λέγω, ὅτι ἡ τὸ AB θωρίον δυναμένη μέσης ἀποτομή ἐστι πρώτη.

Ἔστω γὰρ τῇ $AΔ$ προσαρμόζουσα ἡ $ΔH$· αἱ ἄρα AH, $HΔ$ ὀρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ προσαρμόζουσα ἡ $ΔH$ σύμμετρός ἐστι τῇ ἐκκειμένῃ ὀρητῇ τῇ $AΓ$, ἡ δὲ ὅλη ἡ AH τῆς προσαρμοζούσης τῆς $HΔ$ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει. ἐπεὶ οὖν ἡ AH τῆς $HΔ$ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, ἐὰν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς $HΔ$ ἴσον παρὰ τὴν AH παραβληθῇ ἐλλεῖπον εἰδίδει τετραγώνῳ, εἰξ σύμμετρα αὐτὴν διαιρεῖ. τετμήσθω οὖν ἡ $ΔH$ δίθα κατὰ τὸ E· καὶ τῷ ἀπὸ τῆς EH ἴσον παρὰ τὴν AH παραβεβλήσθω ἐλλεῖπον εἰδίδει τετραγώνῳ, καὶ ἐστω τὸ ὑπὸ τῶν AZ, ZH· σύμμετρος ἄρα ἐστὶν ἡ AZ τῇ ZH μήκει. καὶ ἡ AH ἄρα ἑκατέρᾳ τῶν AZ, ZH σύμμετρός ἐστι μήκει. ὀρητὴ δὲ ἡ AH καὶ ἀσύμμετρος τῇ $AΓ$ μήκει· καὶ ἑκατέρα ἄρα τῶν AZ, ZH ὀρητή ἐστι καὶ ἀσύμμετρος τῇ $AΓ$ μήκει· ἑκάτερον ἄρα τῶν AI, ZK μέσον ἐστίν. πάλιν, ἐπεὶ σύμμετρός ἐστιν ἡ $ΔE$ τῇ EH, καὶ ἡ $ΔH$ ἄρα ἑκατέρᾳ τῶν $ΔE$, EH σύμμετρός ἐστιν. ἀλλὰ ἡ $ΔH$ σύμμετρός ἐστι τῇ $AΓ$ μήκει [ὀρητὴ ἄρα καὶ ἑκατέρα τῶν $ΔE$, EH καὶ σύμμετρος τῇ $AΓ$ μήκει]. ἑκάτερον ἄρα τῶν $ΔΘ$, EK ὀρητόν ἐστιν.

Συνεστάτω οὖν τῷ μὲν AI ἴσον τετράγωνον τὸ $ΛM$, τῷ δὲ ZK ἴσον ἀφῃρήσθω τὸ NX περὶ τὴν αὐτὴν γωνίαν ὂν τῷ $ΛM$ τὴν ὑπὸ τῶν $ΛOM$· περὶ τὴν αὐτὴν ἄρα ἐστὶ διάμετρον τὰ $ΛM$, NX τετράγωνα. ὀέστω αὐτῶν διάμετρος ἡ OP, καὶ καταγεγράφθω τὸ σχῆμα. ἐπεὶ οὖν τὰ AI, ZK μέσα ἐστὶ καὶ ἐστιν ἴσα τοῖς ἀπὸ τῶν $ΛO$, ON, καὶ τὰ ἀπὸ τῶν $ΛO$, ON [ἄρα] μέσα ἐστίν· καὶ αἱ $ΛO$, ON ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. καὶ ἐπεὶ τὸ ὑπὸ τῶν AZ, ZH ἴσον ἐστὶ τῷ ἀπὸ τῆς EH, ὀέστιν ἄρα ὡξ ἡ AZ πρὸς τὴν EH, οὐτως ἡ EH πρὸς τὴν ZH· ἀλλὸ ὡξ μὲν ἡ AZ πρὸς τὴν EH, οὐτως τὸ AI πρὸς τὸ EK· ὡξ δὲ ἡ EH πρὸς τὴν ZH, οὐτως [ἐστὶ] τὸ EK πρὸς τὸ ZK· τῶν ἄρα AI, ZK μέσον ἀνάλογόν ἐστι τὸ EK. ὀέστι δὲ καὶ τῶν $ΛM$, NX τετραγώνων μέσον ἀνάλογον τὸ MN· καί ἐστιν ἴσον τὸ μὲν AI τῷ $ΛM$, τὸ δὲ ZK τῷ NX· καὶ τὸ MN ἄρα ἴσον ἐστὶ τῷ EK. ἀλλὰ τῷ μὲν EK ἴσον [ἐστὶ]

For let the area AB be contained by the rational (straight-line) AC and the second apotome AD. I say that the square-root of area AB is the first apotome of a medial (straight-line).

For let DG be an attachment to AD. Thus, AG and GD are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and the attachment DG is commensurable (in length) with the (previously) laid down rational (straight-line) AC, and the square on the whole, AG, is greater than (the square on) the attachment, GD, by the (square) on (some straight-line) commensurable in length with (AG) [Def. 10.12]. Therefore, since the square on AG is greater than (the square on) GD by the (square) on (some straight-line) commensurable (in length) with (AG), thus if (an area) equal to the fourth part of the (square) on GD is applied to AG, falling short by a square figure, then it divides (AG) into (parts which are) commensurable (in length) [Prop. 10.17]. Therefore, let DG be cut in half at E. And let (an area) equal to the (square) on EG be applied to AG, falling short by a square figure. And let it be the (rectangle contained) by AF and FG. Thus, AF is commensurable in length with FG. AG is thus also commensurable in length with each of AF and FG [Prop. 10.15]. And AG (is) a rational (straight-line), and incommensurable in length with AC. AF and FG are thus also each rational (straight-lines), and incommensurable in length with AC [Prop. 10.13]. Thus, AI and FK are each medial (areas) [Prop. 10.21]. Again, since DE is commensurable (in length) with EG, thus DG is also commensurable (in length) with each of DE and EG [Prop. 10.15]. But, DG is commensurable in length with AC [thus, DE and EG are also each rational, and commensurable in length with AC]. Thus, DH and EK are each rational (areas) [Prop. 10.19].

Therefore, let the square LM, equal to AI, be constructed. And let NO, equal to FK, which is about the same angle LPM as LM, be subtracted (from LM). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since AI and FK are medial (areas), and are

τὸ ΔΘ, τῷ δὲ ΜΝ ἴσον τὸ ΛΧ· ὅλον ἄρα τὸ ΔΚ ἴσον ἐστὶ τῷ ΥΦΘ γνώμονι καὶ τῷ ΝΧ. ἐπεὶ οὐκοῦν ὅλον τὸ ΑΚ ἴσον ἐστὶ τοῖς ΛΜ, ΝΧ, ὧν τὸ ΔΚ ἴσον ἐστὶ τῷ ΥΦΘ γνώμονι καὶ τῷ ΝΧ, λοιπὸν ἄρα τὸ ΑΒ ἴσον ἐστὶ τῷ ΤΣ. τὸ δὲ ΤΣ ἐστι τὸ ἀπὸ τῆς ΛΝ· τὸ ἀπὸ τῆς ΛΝ ἄρα ἴσον ἐστὶ τῷ ΑΒ θωρίῳ· ἡ ΛΝ ἄρα δύναται τὸ ΑΒ θωρίον. λέγω [δή], ὅτι ἡ ΛΝ μέσης ἀποτομή ἐστι πρώτη.

Ἐπεὶ γὰρ ὀρητόν ἐστι τὸ ΕΚ καί ἐστιν ἴσον τῷ ΛΧ, ὀρητὸν ἄρα ἐστὶ τὸ ΛΧ, τουτέστι τὸ ὑπὸ τῶν ΛΟ, ΟΝ. μέσον δὲ ἐδείθθη τὸ ΝΧ· ἀσύμμετρον ἄρα ἐστὶ τὸ ΛΧ τῷ ΝΧ· ὡς δὲ τὸ ΛΧ πρὸς τὸ ΝΧ, οὕτως ἐστὶν ἡ ΛΟ πρὸς ΟΝ· αἱ ΛΟ, ΟΝ ἄρα ἀσύμμετροί εἰσι μήκει. αἱ ἄρα ΛΟ, ΟΝ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι ὀρητὸν περιέθουσαι· ἡ ΛΝ ἄρα μέσης ἀποτομή ἐστι πρώτη· καὶ δύναται τὸ ΑΒ θωρίον.

Ἡ ἄρα τὸ ΑΒ θωρίον δυναμένη μέσης ἀποτομή ἐστι πρώτη· ὅπερ ἔδει δεῖχαι.

equal to the (squares) on LP and PN (respectively), [thus] the (squares) on LP and PN are also medial. Thus, LP and PN are also medial (straight-lines which are) commensurable in square only.† And since the (rectangle contained) by AF and FG is equal to the (square) on EG, thus as AF is to EG, so EG (is) to FG [Prop. 10.17]. But, as AF (is) to EG, so AI (is) to EK. And as EG (is) to FG, so EK [is] to FK [Prop. 6.1]. Thus, EK is the mean proportional to AI and FK [Prop. 5.11]. And MN is also the mean proportional to the squares LM and NO [Prop. 10.53 lem.]. And AI is equal to LM, and FK to NO. Thus, MN is also equal to EK. But, DH [is] equal to EK, and LO equal to MN [Prop. 1.43]. Thus, the whole (of) DK is equal to the gnomon UVW and NO. Therefore, since the whole (of) AK is equal to LM and NO, of which DK is equal to the gnomon UVW and NO, the remainder AB is thus equal to TS. And TS is the (square) on LN. Thus, the (square) on LN is equal to the area AB. LN is thus the square-root of area AB. [So], I say that LN is the first apotome of a medial (straight-line).

For since EK is a rational (area), and is equal to LO, LO—that is to say, the (rectangle contained) by LP and PN—is thus a rational (area). And NO was shown (to be) a medial (area). Thus, LO is incommensurable with NO. And as LO (is) to NO, so LP is to PN [Prop. 6.1]. Thus, LP and PN are incommensurable in length [Prop. 10.11]. LP and PN are thus medial (straight-lines which are) commensurable in square only, and which contain a rational (area). Thus, LN is the first apotome of a medial (straight-line) [Prop. 10.74]. And it is the square-root of area AB.

Thus, the square root of area AB is the first apotome of a medial (straight-line). (Which is) the very thing it was required to show.

† There is an error in the argument here. It should just say that LP and PN are commensurable in square, rather than in square only, since LP and PN are only shown to be incommensurable in length later on.

93.

Ἐὰν θωρίον περιέθηται ὑπὸ ὀρητῆς καὶ ἀποτομῆς τρίτης, ἡ τὸ θωρίον δυναμένη μέσης ἀποτομή ἐστι δευτέρα.

Θωρίον γὰρ τὸ ΑΒ περιεθέσθω ὑπὸ ὀρητῆς τῆς ΑΓ καὶ ἀποτομῆς τρίτης τῆς ΑΔ· λέγω, ὅτι ἡ τὸ ΑΒ θωρίον δυναμένη μέσης ἀποτομή ἐστι δευτέρα.

Proposition 93

If an area is contained by a rational (straight-line) and a third apotome then the square-root of the area is a second apotome of a medial (straight-line).

For let the area AB be contained by the rational (straight-line) AC and the third apotome AD. I say that the square-root of area AB is the second apotome of a medial (straight-line).

Ἔστω γὰρ τῇ ΑΔ προσαρμόζουσα ἡ ΔΗ· αἱ ΑΗ, ΗΔ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ οὐδετέρα τῶν ΑΗ, ΗΔ σύμμετρός ἐστι μήκει τῇ ἐκκειμένῃ ὁρητῇ τῇ ΑΓ, ἡ δὲ ὅλη ἡ ΑΗ τὴξ προσαρμοζούσης τῆς ΔΗ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ. ἐπεὶ οὐκῦν ἡ ΑΗ τῆς ΗΔ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΔΗ ἴσον παρὰ τὴν ΑΗ παραβληθῇ ἐλλεῖπον εἰδίδει τετραγώνῳ, εἰξ σύμμετρα αὐτὴν διελεῖ. τετμήσθω οὐκῦν ἡ ΔΗ δίθα κατὰ τὸ Ε, καὶ τῷ ἀπὸ τῆς ΕΗ ἴσον παρὰ τὴν ΑΗ παραβεβλήσθω ἐλλεῖπον εἰδίδει τετραγώνῳ, καὶ ἐστω τὸ ὑπὸ τῶν ΑΖ, ΖΗ. καὶ ἤθθωσαν διὰ τῶν Ε, Ζ, Η σημείων τῇ ΑΓ παράλληλοι αἱ ΕΘ, ΖΙ, ΗΚ· σύμμετροι ἄρα εἰσὶν αἱ ΑΖ, ΖΗ· σύμμετρον ἄρα καὶ τὸ ΑΙ τῷ ΖΚ. καὶ ἐπεὶ αἱ ΑΖ, ΖΗ σύμμετροί εἰσι μήκει, καὶ ἡ ΑΗ ἄρα ἑκατέρᾳ τῶν ΑΖ, ΖΗ σύμμετρός ἐστι μήκει. ὁρητὴ δὲ ἡ ΑΗ καὶ ἀσύμμετρος τῇ ΑΓ μήκει· ὥστε καὶ αἱ ΑΖ, ΖΗ. ἑκάτερον ἄρα τῶν ΑΙ, ΖΚ μέσον ἐστίν. πάλιν, ἐπεὶ σύμμετρός ἐστιν ἡ ΔΕ τῇ ΕΗ μήκει, καὶ ἡ ΔΗ ἄρα ἑκατέρᾳ τῶν ΔΕ, ΕΗ σύμμετρός ἐστι μήκει. ὁρητὴ δὲ ἡ ΗΔ καὶ ἀσύμμετρος τῇ ΑΓ μήκει· ὁρητὴ ἄρα καὶ ἑκατέρα τῶν ΔΕ, ΕΗ καὶ ἀσύμμετρος τῇ ΑΓ μήκει· ἑκάτερον ἄρα τῶν ΔΘ, ΕΚ μέσον ἐστίν. καὶ ἐπεὶ αἱ ΑΗ, ΗΔ δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετρος ἄρα ἐστὶ μήκει ἡ ΑΗ τῇ ΗΔ. ἀλλὸ ἡ μὲν ΑΗ τῇ ΑΖ σύμμετρός ἐστι μήκει ἡ δὲ ΔΗ τῇ ΕΗ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΑΖ τῇ ΕΗ μήκει. ὡξ δὲ ἡ ΑΖ πρὸς τὴν ΕΗ, οὐτως ἐστὶ τὸ ΑΙ πρὸς τὸ ΕΚ· ἀσύμμετρον ἄρα ἐστὶ τὸ ΑΙ τῷ ΕΚ.

Συνεστάτω οὐκῦν τῷ μὲν ΑΙ ἴσον τετράγωνον τὸ ΛΜ, τῷ δὲ ΖΚ ἴσον ἀφῇρήσθω τὸ ΝΧ περὶ τὴν αὐτὴν γωνίαν ὃν τῷ ΛΜ· περὶ τὴν αὐτὴν ἄρα διάμετρόν ἐστι τὰ ΛΜ, ΝΧ. ὀέστω αὐτῶν διάμετρος ἡ ΟΡ, καὶ καταγεγράφθω τὸ σχῆμα. ἐπεὶ οὐκῦν τὸ ὑπὸ τῶν ΑΖ, ΖΗ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΗ, ὀέστιν ἄρα ὡξ ἡ ΑΖ πρὸς τὴν ΕΗ, οὐτως ἡ ΕΗ πρὸς τὴν ΖΗ. ἀλλὸ ὡξ μὲν ἡ ΑΖ πρὸς τὴν ΕΗ, οὐτως ἐστὶ τὸ ΑΙ πρὸς τὸ ΕΚ· ὡξ δὲ ἡ ΕΗ πρὸς τὴν ΖΗ, οὐτως ἐστὶ τὸ ΕΚ πρὸς τὸ ΖΚ· καὶ ὡξ ἄρα τὸ ΑΙ πρὸς τὸ ΕΚ, οὐτως τὸ ΕΚ πρὸς τὸ ΖΚ· τῶν ἄρα ΑΙ, ΖΚ μέσον ἀνάλογόν ἐστι τὸ ΕΚ. ὀέστι δὲ καὶ τῶν ΛΜ, ΝΧ τετραγώνων μέσον ἀνάλογον τὸ ΜΝ· καί ἐστιν ἴσον

For let DG be an attachment to AD. Thus, AG and GD are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and neither of AG and GD is commensurable in length with the (previously) laid down rational (straight-line) AC, and the square on the whole, AG, is greater than (the square on) the attachment, DG, by the (square) on (some straight-line) commensurable (in length) with (AG) [Def. 10.13]. Therefore, since the square on AG is greater than (the square on) GD by the (square) on (some straight-line) commensurable (in length) with (AG), thus if (an area) equal to the fourth part of the square on DG is applied to AG, falling short by a square figure, then it divides (AG) into (parts which are) commensurable (in length) [Prop. 10.17]. Therefore, let DG be cut in half at E. And let (an area) equal to the (square) on EG be applied to AG, falling short by a square figure. And let it be the (rectangle contained) by AF and FG. And let EH, FI, and GK be drawn through points E, F, and G (respectively), parallel to AC. Thus, AF and FG are commensurable (in length). AI (is) thus also commensurable with FK [Props. 6.1, 10.11]. And since AF and FG are commensurable in length, AG is thus also commensurable in length with each of AF and FG [Prop. 10.15]. And AG (is) rational, and incommensurable in length with AC. Hence, AF and FG (are) also (rational, and incommensurable in length with AC) [Prop. 10.13]. Thus, AI and FK are each medial (areas) [Prop. 10.21]. Again, since DE is commensurable in length with EG, DG is also commensurable in length with each of DE and EG [Prop. 10.15]. And GD (is) rational, and incommensurable in length with AC. Thus, DE and EG (are) each also rational, and incommensurable in length with AC [Prop. 10.13]. DH and EK are thus each medial (areas) [Prop. 10.21]. And since AG and GD are commensurable in square only, AG is thus incommensurable in length with GD. But, AG is commensurable in length with AF, and DG with EG. Thus, AF is incommensurable in length with EG [Prop. 10.13]. And as AF (is) to EG, so AI is to EK [Prop. 6.1]. Thus, AI is incommensurable with EK [Prop. 10.11].

τὸ μὲν AI τῷ ΛM, τὸ δὲ ZK τῷ NX· καὶ τὸ EK ἄρα ἴσον ἐστὶ τῷ MN. ἀλλὰ τὸ μὲν MN ἴσον ἐστὶ τῷ ΛX, τὸ δὲ EK ἴσον [ἐστὶ] τῷ $\Delta\Theta$· καὶ ὅλον ἄρα τὸ ΔK ἴσον ἐστὶ τῷ $\Upsilon\Phi\Theta$ γνώμονι καὶ τῷ NX. ὐέστι δὲ καὶ τὸ AK ἴσον τοῖς ΛM, NX· λοιπὸν ἄρα τὸ AB ἴσον ἐστὶ τῷ ΣT, τουτέστι τῷ ἀπὸ τῆς ΛN τετραγώνῳ· ἡ ΛN ἄρα δύναται τὸ AB θωρίον. λέγω, ὅτι ἡ ΛN μέσης ἀποτομή ἐστι δευτέρα.

Ἐπεὶ γὰρ μέσα ἐδείχθη τὰ AI, ZK καί ἐστιν ἴσα τοῖς ἀπὸ τῶν ΛO, ON, μέσον ἄρα καὶ ἑκάτερον τῶν ἀπὸ τῶν ΛO, ON· μέση ἄρα ἑκατέρα τῶν ΛO, ON. καὶ ἐπεὶ σύμμετρόν ἐστι τὸ AI τῷ ZK, σύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς ΛO τῷ ἀπὸ τῆς ON. πάλιν, ἐπεὶ ἀσύμμετρον ἐδείχθη τὸ AI τῷ EK, ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ΛM τῷ MN, τουτέστι τὸ ἀπὸ τῆς ΛO τῷ ὑπὸ τῶν ΛO, ON· ὥστε καὶ ἡ ΛO ἀσύμμετρός ἐστι μήκει τῇ ON· αἱ ΛO, ON ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. λέγω δή, ὅτι καὶ μέσον περιέθουσιν.

Ἐπεὶ γὰρ μέσον ἐδείχθη τὸ EK καί ἐστιν ἴσον τῷ ὑπὸ τῶν ΛO, ON, μέσον ἄρα ἐστὶ καὶ τὸ ὑπὸ τῶν ΛO, ON· ὥστε αἱ ΛO, ON μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι. ἡ ΛN ἄρα μέσης ἀποτομή ἐστι δευτέρα· καὶ δύναται τὸ AB θωρίον.

Ἡ ἄρα τὸ AB θωρίον δυναμένη μέσης ἀποτομή ἐστι δευτέρα· ὅπερ ἔδει δεῖξαι.

Therefore, let the square LM, equal to AI, be constructed. And let NO, equal to FK, which is about the same angle as LM, be subtracted (from LM). Thus, LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since the (rectangle contained) by AF and FG is equal to the (square) on EG, thus as AF is to EG, so EG (is) to FG [Prop. 6.17]. But, as AF (is) to EG, so AI is to EK [Prop. 6.1]. And as EG (is) to FG, so EK is to FK [Prop. 6.1]. And thus as AI (is) to EK, so EK (is) to FK [Prop. 5.11]. Thus, EK is the mean proportional to AI and FK. And MN is also the mean proportional to the squares LM and NO [Prop. 10.53 lem.]. And AI is equal to LM, and FK to NO. Thus, EK is also equal to MN. But, MN is equal to LO, and EK [is] equal to DH [Prop. 1.43]. And thus the whole of DK is equal to the gnomon UVW and NO. And AK (is) also equal to LM and NO. Thus, the remainder AB is equal to ST—that is to say, to the square on LN. Thus, LN is the square-root of area AB. I say that LN is the second apotome of a medial (straight-line).

For since AI and FK were shown (to be) medial (areas), and are equal to the (squares) on LP and PN (respectively), the (squares) on each of LP and PN (are) thus also medial. Thus, LP and PN (are) each medial (straight-lines). And since AI is commensurable with FK [Props. 6.1, 10.11], the (square) on LP (is) thus also commensurable with the (square) on PN. Again, since AI was shown (to be) incommensurable with EK, LM is thus also incommensurable with MN—that is to say, the (square) on LP with the (rectangle contained) by LP and PN. Hence, LP is also incommensurable in length with PN [Props. 6.1, 10.11]. Thus, LP and PN are medial (straight-lines which are) commensurable in square only. So, I say that they also contain a medial (area).

For since EK was shown (to be) a medial (area), and is equal to the (rectangle contained) by LP and PN, the (rectangle contained) by LP and PN is thus also medial. Hence, LP and PN are medial (straight-lines which are) commensurable in square only, and which contain a medial (area). Thus, LN is the second apotome of a medial (straight-line) [Prop. 10.75]. And it is the square-root of area AB.

Thus, the square-root of area AB is the second apotome of a medial (straight-line). (Which is) the very thing it was required to show.

94.

Ἐὰν θωρίον περιέθηται ὑπὸ ὀρητῆς καὶ ἀποτομῆς τετάρτης, ἡ τὸ θωρίον δυναμένη ἐλάσσων ἐστίν.

Θωρίον γὰρ τὸ AB περιεθέσθω ὑπὸ ὀρητῆς τῆς $A\Gamma$ καὶ ἀποτομῆς τετάρτης τῆς $A\Delta$· λέγω, ὅτι ἡ τὸ AB θωρίον δυναμένη ἐλάσσων ἐστίν.

Ἔστω γὰρ τῇ $A\Delta$ προσαρμόζουσα ἡ ΔH· αἱ ἄρα AH,

Proposition 94

If an area is contained by a rational (straight-line) and a fourth apotome then the square-root of the area is a minor (straight-line).

For let the area AB be contained by the rational (straight-line) AC and the fourth apotome AD. I say that the square-root of area AB is a minor (straight-line).

ΗΔ ὀρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΑΗ σύμμετρός ἐστι τῇ ἐκκειμένῃ ὀρητῇ τῇ ΑΓ μήκει, ἡ δὲ ὅλη ἡ ΑΗ τῆς προσαρμοζούσης τῆς ΔΗ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ μήκει. ἐπεὶ οὐκοῦν ἡ ΑΗ τῆς ΗΔ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ μήκει, ἐὰν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΔΗ ἴσον παρὰ τὴν ΑΗ παραβληθῇ ἐλλεῖπον εἰδίδει τετραγώνῳ, εἰξ ἀσύμμετρα αὐτὴν διελεῖ. τετμήσθω οὐκοῦν ἡ ΔΗ δίθα κατὰ τὸ Ε, καὶ τῷ ἀπὸ τῆς ΕΗ ἴσον παρὰ τὴν ΑΗ παραβεβλήσθω ἐλλεῖπον εἰδίδει τετραγώνῳ, καὶ ἐστω τὸ ὑπὸ τῶν ΑΖ, ΖΗ· ἀσύμμετρος ἄρα ἐστὶ μήκει ἡ ΑΖ τῇ ΖΗ. ἤχθωσαν οὐκοῦν διὰ τῶν Ε, Ζ, Η παράλληλοι ταῖς ΑΓ, ΒΔ αἱ ΕΘ, ΖΙ, ΗΚ. ἐπεὶ οὐκοῦν ὀρητή ἐστιν ἡ ΑΗ καὶ σύμμετρος τῇ ΑΓ μήκει, ὀρητὸν ἄρα ἐστὶν ὅλον τὸ ΑΚ. πάλιν, ἐπεὶ ἀσύμμετρός ἐστιν ἡ ΔΗ τῇ ΑΓ μήκει, καί εἰσιν ἀμφότεραι ὀρηταί, μέσον ἄρα ἐστὶ τὸ ΔΚ. πάλιν, ἐπεὶ ἀσύμμετρός ἐστὶν ἡ ΑΖ τῇ ΖΗ μήκει, ἀσύμμετρον ἄρα καὶ τὸ ΑΙ τῷ ΖΚ.

For let *DG* be an attachment to *AD*. Thus, *AG* and *DG* are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and *AG* is commensurable in length with the (previously) laid down rational (straight-line) *AC*, and the square on the whole, *AG*, is greater than (the square on) the attachment, *DG*, by the square on (some straight-line) incommensurable in length with (*AG*) [Def. 10.14]. Therefore, since the square on *AG* is greater than (the square on) *GD* by the (square) on (some straight-line) incommensurable in length with (*AG*), thus if (some area), equal to the fourth part of the (square) on *DG*, is applied to *AG*, falling short by a square figure, then it divides (*AG*) into (parts which are) incommensurable (in length) [Prop. 10.18]. Therefore, let *DG* be cut in half at *E*, and let (some area), equal to the (square) on *EG*, be applied to *AG*, falling short by a square figure, and let it be the (rectangle contained) by *AF* and *FG*. Thus, *AF* is incommensurable in length with *FG*. Therefore, let *EH*, *FI*, and *GK* be drawn through *E*, *F*, and *G* (respectively), parallel to *AC* and *BD*. Therefore, since *AG* is rational, and commensurable in length with *AC*, the whole (area) *AK* is thus rational [Prop. 10.19]. Again, since *DG* is incommensurable in length with *AC*, and both are rational (straight-lines), *DK* is thus a medial (area) [Prop. 10.21]. Again, since *AF* is incommensurable in length with *FG*, *AI* (is) thus also incommensurable with *FK* [Props. 6.1, 10.11].

Συνεστάτω οὐκοῦν τῷ μὲν ΑΙ ἴσον τετράγωνον τὸ ΛΜ, τῷ δὲ ΖΚ ἴσον ἀφῃρήσθω περὶ τὴν αὐτὴν γωνίαν τὴν ὑπὸ τῶν ΛΟΜ τὸ ΝΧ. περὶ τὴν αὐτὴν ἄρα διάμετόν ἐστι τὰ ΛΜ, ΝΧ τετράγωνα. ὀέστω αὐτῶν διάμετος ἡ ΟΡ, καὶ καταγεγράφθω τὸ σχῆμα. ἐπεὶ οὐκοῦν τὸ ὑπὸ τῶν ΑΖ, ΖΗ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΗ, ἀνάλογον ἄρα ἐστὶν ὡξ ἡ ΑΖ πρὸς τὴν ΕΗ, οὑτως ἡ ΕΗ πρὸς τὴν ΖΗ. ἀλλὰ ὡξ μὲν ἡ ΑΖ πρὸς τὴν ΕΗ, οὑτως ἐστὶ τὸ ΑΙ πρὸς τὸ ΕΚ, ὡξ δὲ ἡ ΕΗ πρὸς τὴν ΖΗ, οὑτως ἐστὶ τὸ ΕΚ πρὸς τὸ ΖΚ· τῶν ἄρα ΑΙ, ΖΚ μέσον ἀνάλογόν ἐστι τὸ ΕΚ. ὀέστι δὲ καὶ τῶν ΛΜ, ΝΧ τετραγώνων μέσον ἀνάλογον τὸ ΜΝ, καί ἐστιν ἴσον τὸ μὲν ΑΙ τῷ ΛΜ, τὸ δὲ ΖΚ τῷ ΝΧ· καὶ τὸ ΕΚ ἄρα ἴσον ἐστὶ τῷ ΜΝ. ἀλλὰ τῷ μὲν ΕΚ

Therefore, let the square *LM*, equal to *AI*, be constructed. And let *NO*, equal to *FK*, (and) about the same angle, *LPM*, be subtracted (from *LM*). Thus, the squares *LM* and *NO* are about the same diagonal [Prop. 6.26]. Let *PR* be their (common) diagonal, and let the (rest of the) figure be drawn. Therefore, since the (rectangle contained) by *AF* and *FG* is equal to the (square) on *EG*, thus, proportionally, as *AF* is to *EG*, so *EG* (is) to *FG* [Prop. 6.17]. But, as *AF* (is) to *EG*, so *AI* is to *EK*, and as *EG* (is) to *FG*, so *EK* is to *FK* [Prop. 6.1]. Thus, *EK* is the mean proportional to *AI* and *FK* [Prop. 5.11]. And *MN* is also the mean proportional to the squares *LM* and *NO* [Prop. 10.13 lem.], and *AI* is equal to *LM*, and *FK* to *NO*. *EK*

ἴσον ἐστὶ τὸ ΔΘ, τῷ δὲ ΜΝ ἴσον ἐστὶ τὸ ΛΧ· ὅλον ἄρα τὸ ΔΚ ἴσον ἐστὶ τῷ ΥΦΘ γνώμονι καὶ τῷ ΝΧ. ἐπεὶ οὖν ὅλον τὸ ΑΚ ἴσον ἐστὶ τοῖς ΛΜ, ΝΧ τετραγώνοις, ὧν τὸ ΔΚ ἴσον ἐστὶ τῷ ΥΦΘ γνώμονι καὶ τῷ ΝΧ τετραγώνῳ, λοιπὸν ἄρα τὸ ΑΒ ἴσον ἐστὶ τῷ ΣΤ, τουτέστι τῷ ἀπὸ τῆς ΛΝ τετραγώνῳ· ἡ ΛΝ ἄρα δύναται τὸ ΑΒ θωρίον. λέγω, ὅτι ἡ ΛΝὀάλογός ἐστιν ἡ καλουμένη ἐλάσσων.

Ἐπεὶ γὰρ ὀρητόν ἐστι τὸ ΑΚ καί ἐστιν ἴσον τοῖς ἀπὸ τῶν ΛΟ, ΟΝ τετράγωνοις, τὸ ἄρα συγκείμενον ἐκ τῶν ἀπὸ τῶν ΛΟ, ΟΝ ὀρητόν ἐστιν. πάλιν, ἐπεὶ τὸ ΔΚ μέσον ἐστίν, καί ἐστιν ἴσον τὸ ΔΚ τῷ δὶξ ὑπὸ τῶν ΛΟ, ΟΝ, τὸ ἄρα δὶξ ὑπὸ τῶν ΛΟ, ΟΝ μέσον ἐστίν. καὶ ἐπεὶ ἀσύμμετρον ἐδείθθη τὸ ΑΙ τῷ ΖΚ, ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς ΛΟ τετράγωνον τῷ ἀπὸ τῆς ΟΝ τετραγώνῳ. αἱ ΛΟ, ΟΝ ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων ὀρητόν, τὸ δὲ δὶξ ὑπὸ αὐτῶν μέσον. ἡ ΛΝ ἄραὀάλογός ἐστιν ἡ καλουμένη ἐλάσσων· καὶ δύναται τὸ ΑΒ θωρίον.

Ἡ ἄρα τὸ ΑΒ θωρίον δυναμένη ἐλάσσων ἐστίν· ὅπερ ἔδει δεῖχαι.

is thus also equal to MN. But, DH is equal to EK, and LO is equal to MN [Prop. 1.43]. Thus, the whole of DK is equal to the gnomon UVW and NO. Therefore, since the whole of AK is equal to the (sum of the) squares LM and NO, of which DK is equal to the gnomon UVW and the square NO, the remainder AB is thus equal to ST—that is to say, to the square on LN. Thus, LN is the square-root of area AB. I say that LN is the irrational (straight-line which is) called minor.

For since AK is rational, and is equal to the (sum of the) squares LP and PN, the sum of the (squares) on LP and PN is thus rational. Again, since DK is medial, and DK is equal to twice the (rectangle contained) by LP and PN, thus twice the (rectangle contained) by LP and PN is medial. And since AI was shown (to be) incommensurable with FK, the square on LP (is) thus also incommensurable with the square on PN. Thus, LP and PN are (straight-lines which are) incommensurable in square, making the sum of the squares on them rational, and twice the (rectangle contained) by them medial. LN is thus the irrational (straight-line) called minor [Prop. 10.76]. And it is the square-root of area AB.

Thus, the square-root of area AB is a minor (straight-line). (Which is) the very thing it was required to show.

95.

Ἐὰν θωρίον περιέθηται ὑπὸ ὀρητῆς καὶ ἀποτομῆς πέμπτης, ἡ τὸ θωρίον δυναμένη [ἡ] μετὰ ὀρητοῦ μέσον τὸ ὅλον ποιοῦσά ἐστιν.

Proposition 95

If an area is contained by a rational (straight-line) and a fifth apotome then the square-root of the area is that (straight-line) which with a rational (area) makes a medial whole.

Θωρίον γὰρ τὸ ΑΒ περιεθέσθω ὑπὸ ὀρητῆς τῆς ΑΓ καὶ ἀποτομῆς πέμπτης τῆς ΑΔ· λέγω, ὅτι ἡ τὸ ΑΒ θωρίον δυναμένη [ἡ] μετὰ ὀρητοῦ μέσον τὸ ὅλον ποιοῦσά ἐστιν.

Ἔστω γὰρ τῇ ΑΔ προσαρμόζουσα ἡ ΔΗ· αἱ ἄρα ΑΗ, ΗΔ ὀρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ προσαρμόζουσα ἡ ΗΔ σύμμετρός ἐστι μήκει τῇ ἐκκειμένῃ ὀρητῇ τῇ ΑΓ, ἡ δὲ ὅλη ἡ ΑΗ τῆς προσαρμοζούσης τῆς ΔΗ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΔΗ ἴσον παρὰ τὴν ΑΗ παραβληθῇ ἐλλεῖπον εἰδει τετρα-

For let the area AB be contained by the rational (straight-line) AC and the fifth apotome AD. I say that the square-root of area AB is that (straight-line) which with a rational (area) makes a medial whole.

For let DG be an attachment to AD. Thus, AG and DG are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and the attachment GD is commensurable in length the the (previously) laid down rational (straight-line) AC, and the square on the whole, AG, is greater

γώνῳ, εἰξ ἀσύμμετρα αὐτὴν διελεῖ. τετμήσθω οὐῶν ἡ ΔΗ δίθα κατὰ τὸ Ε σημεῖον, καὶ τῷ ἀπὸ τῆς ΕΗ ἴσον παρὰ τὴν ΑΗ παραβεβλήσθω ἐλλεῖπον εἰδει τετραγώνῳ καὶ ὑέστω τὸ ὑπὸ τῶν ΑΖ, ΖΗ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΑΖ τῇ ΖΗ μήκει. καὶ ἐπεὶ ἀσύμμετρόξ ἐστὶν ἡ ΑΗ τῇ ΓΑ μήκει, καί εἰσιν ἀμφότεραι ὁρηταί, μέσον ἄρα ἐστὶ τὸ ΑΚ. πάλιν, ἐπεὶ ὁρητή ἐστιν ἡ ΔΗ καὶ σύμμετρος τῇ ΑΓ μήκει, ὁρητόν ἐστι τὸ ΔΚ.

Συνεστάτω οὐῶν τῷ μὲν ΑΙ ἴσον τετράγωνον τὸ ΛΜ, τῷ δὲ ΖΚ ἴσον τετράγωνον ἀφῃρήσθω τὸ ΝΧ περὶ τὴν αὐτὴν γωνίαν τὴν ὑπὸ ΛΟΜ· περὶ τὴν αὐτὴν ἄρα διάμετρόν ἐστι τὰ ΛΜ, ΝΧ τετράγωνα. ὑέστω αὐτῶν διάμετρος ἡ ΟΡ, καὶ καταγεγράφθω τὸ σχῆμα. ὁμοίως δὴ δείχομεν, ὅτι ἡ ΛΝ δύναται τὸ ΑΒ θωρίον. λέγω, ὅτι ἡ ΛΝ ἡ μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσά ἐστιν.

Ἐπεὶ γὰρ μέσον ἐδείθθη τὸ ΑΚ καί ἐστιν ἴσον τοῖς ἀπὸ τῶν ΛΟ, ΟΝ, τὸ ἄρα συγκείμενον ἐκ τῶν ἀπὸ τῶν ΛΟ, ΟΝ μέσον ἐστίν. πάλιν, ἐπεὶ ὁρητόν ἐστι τὸ ΔΚ καί ἐστιν ἴσον τῷ δὶξ ὑπὸ τῶν ΛΟ, ΟΝ, καὶ αὐτὸ ὁρητόν ἐστιν. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ ΑΙ τῷ ΖΚ, ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς ΛΟ τῷ ἀπὸ τῆς ΟΝ· αἱ ΛΟ, ΟΝ ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον, τὸ δὲ δὶξ ὑπὸ αὐτῶν ὁρητόν. ἡ λοιπὴ ἄρα ἡ ΛΝὀάλογόξ ἐστιν ἡ καλουμένη μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσα· καὶ δύναται τὸ ΑΒ θωρίον.

Ἡ τὸ ΑΒ ἄρα θωρίον δυναμένη μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσά ἐστιν· ὅπερ ἔδει δεῖχαι.

than (the square on) the attachment, *DG*, by the (square) on (some straight-line) incommensurable (in length) with (*AG*) [Def. 10.15]. Thus, if (some area), equal to the fourth part of the (square) on *DG*, is applied to *AG*, falling short by a square figure, then it divides (*AG*) into (parts which are) incommensurable (in length) [Prop. 10.18]. Therefore, let *DG* be divided in half at point *E*, and let (some area), equal to the (square) on *EG*, be applied to *AG*, falling short by a square figure, and let it be the (rectangle contained) by *AF* and *FG*. Thus, *AF* is incommensurable in length with *FG*. And since *AG* is incommensurable in length with *CA*, and both are rational (straight-lines), *AK* is thus a medial (area) [Prop. 10.21]. Again, since *DG* is rational, and commensurable in length with *AC*, *DK* is a rational (area) [Prop. 10.19].

Therefore, let the square *LM*, equal to *AI*, be constructed. And let the square *NO*, equal to *FK*, (and) about the same angle, *LPM*, be subtracted (from *NO*). Thus, the squares *LM* and *NO* are about the same diagonal [Prop. 6.26]. Let *PR* be their (common) diagonal, and let (the rest of) the figure be drawn. So, similarly (to the previous propositions), we can show that *LN* is the square-root of area *AB*. I say that *LN* is that (straight-line) which with a rational (area) makes a medial whole.

For since *AK* was shown (to be) a medial (area), and is equal to (the sum of) the squares on *LP* and *PN*, the sum of the (squares) on *LP* and *PN* is thus medial. Again, since *DK* is rational, and is equal to twice the (rectangle contained) by *LP* and *PN*, (the latter) is also rational. And since *AI* is incommensurable with *FK*, the (square) on *LP* is thus also incommensurable with the (square) on *PN*. Thus, *LP* and *PN* are (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by them rational. Thus, the remainder *LN* is the irrational (straight-line) called that which with a rational (area) makes a medial whole [Prop. 10.77]. And it is the square-root of area *AB*.

Thus, the square-root of area *AB* is that (straight-line) which with a rational (area) makes a medial whole. (Which is) the very thing it was required to show.

96.

Ἐὰν θωρίον περιέθηται ὑπὸ ὁρητῆς καὶ ἀποτομῆς ὁέκτης, ἡ τὸ θωρίον δυναμένη μετὰ μέσου μέσον τὸ ὅλον ποιοῦσά ἐστιν.

Proposition 96

If an area is contained by a rational (straight-line) and a sixth apotome then the square-root of the area is that (straight-line) which with a medial (area) makes a medial whole.

Θωρίον γὰρ τὸ AB περιειθέσθω ὑπὸ ὀρητῆς τῆς $AΓ$ καὶ ἀποτομῆς ὀέκτης τῆς $AΔ$· λέγω, ὅτι ἡ τὸ AB θωρίον δυναμένη [ἡ] μετὰ μέσου μέσον τὸ ὅλον ποιοῦσά ἐστιν.

Ἔστω γὰρ τῇ $AΔ$ προσαρμόζουσα ἡ $ΔH$· αἱ ἄρα AH, $HΔ$ ὀρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ οὐδετέρα αὐτῶν σύμμετρός ἐστι τῇ ἐκκειμένῃ ὀρητῇ τῇ $AΓ$ μήκει, ἡ δὲ ὅλη ἡ AH τῆς προσαρμοζούσης τῆς $ΔH$ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ μήκει. ἐπεὶ οὐῦν ἡ AH τῆς $HΔ$ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ μήκει, ἔαν ἄρα τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς $ΔH$ ἴσον παρὰ τὴν AH παραβληθῇ ἐλλεῖπον εἰδει τετραγώνῳ, εἰξ ἀσύμμετρα αὐτὴν διελεῖ. τετμήσθω οὐῦν ἡ $ΔH$ δίθα κατὰ τὸ E [σημεῖον], καὶ τῷ ἀπὸ τῆς EH ἴσον παρὰ τὴν AH παραβεβλήσθω ἐλλεῖπον εἰδει τετραγώνῳ, καὶ ὀέστω τὸ ὑπὸ τῶν AZ, ZH· ἀσύμμετρος ἄρα ἐστὶν ἡ AZ τῇ ZH μήκει. ὡξ δὲ ἡ AZ πρὸς τὴν ZH, οὑτως ἐστὶ τὸ AI πρὸς τὸ ZK· ἀσύμμετρον ἄρα ἐστὶ τὸ AI τῷ ZK. καὶ ἐπεὶ αἱ AH, $AΓ$ ὀρηταί εἰσι δυνάμει μόνον σύμμετροι, μέσον ἐστὶ τὸ AK. πάλιν, ἐπεὶ αἱ $AΓ$, $ΔH$ ὀρηταί εἰσι καὶ ἀσύμμετροι μήκει, μέσον ἐστὶ καὶ τὸ $ΔK$. ἐπεὶ οὐῦν αἱ AH, $HΔ$ δυνάμει μόνον σύμμετροί εἰσιν, ἀσύμμετρος ἄρα ἐστὶν ἡ AH τῇ $HΔ$ μήκει. ὡξ δὲ ἡ AH πρὸς τὴν $HΔ$, οὑτως ἐστὶ τὸ AK πρὸς τὸ $KΔ$· ἀσύμμετρον ἄρα ἐστὶ τὸ AK τῷ $KΔ$.

Συνεστάτω οὐῦν τῷ μὲν AI ἴσον τετράγωνον τὸ $ΛM$, τῷ δὲ ZK ἴσον ἀφῃρήσθω περὶ τὴν αὐτὴν γωνίαν τὸ NX· περὶ τὴν αὐτὴν ἄρα διάμετρόν ἐστι τὰ $ΛM$, NX τετράγωνα. ὀέστω αὐτῶν διάμετρος ἡ OP, καὶ καταγεγράφθω τὸ σχῆμα. ὁμοίως δὴ τοῖς ἐπάνω δείχομεν, ὅτι ἡ $ΛN$ δύναται τὸ AB θωρίον. λέγω, ὅτι ἡ $ΛN$ [ἡ] μετὰ μέσου μέσον τὸ ὅλον ποιοῦσά ἐστιν.

Ἐπεὶ γὰρ μέσον ἐδείθθη τὸ AK καί ἐστιν ἴσον τοῖς ἀπὸ τῶν $ΛO$, ON, τὸ ἄρα συγκείμενον ἐκ τῶν ἀπὸ τῶν $ΛO$, ON μέσον ἐστίν. πάλιν, ἐπεὶ μέσον ἐδείθθη τὸ $ΔK$ καί ἐστιν ἴσον τῷ δὶξ ὑπὸ τῶν $ΛO$, ON, καὶ τὸ δὶξ ὑπὸ τῶν $ΛO$, ON μέσον ἐστίν. καὶ ἐπεὶ ἀσύμμετρον ἐδείθθη τὸ AK τῷ $ΔK$, ἀσύμμετρα [ἄρα] ἐστὶ καὶ τὰ ἀπὸ τῶν $ΛO$, ON τετράγωνα τῷ δὶξ ὑπὸ τῶν $ΛO$, ON. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ AI τῷ ZK, ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς $ΛO$ τῷ ἀπὸ τῆς ON· αἱ $ΛO$, ON ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τό τε συγκείμενον ἐκ τῶν

For let the area AB be contained by the rational (straight-line) AC and the sixth apotome AD. I say that the square-root of area AB is that (straight-line) which with a medial (area) makes a medial whole.

For let DG be an attachment to AD. Thus, AG and GD are rational (straight-lines which are) commensurable in square only [Prop. 10.73], and neither of them is commensurable in length with the (previously) laid down rational (straight-line) AC, and the square on the whole, AG, is greater than (the square on) the attachment, DG, by the (square) on (some straight-line) incommensurable in length with (AG) [Def. 10.16]. Therefore, since the square on AG is greater than (the square on) GD by the (square) on (some straight-line) incommensurable in length with (AG), thus if (some area), equal to the fourth part of square on DG, is applied to AG, falling short by a square figure, then it divides (AG) into (parts which are) incommensurable (in length) [Prop. 10.18]. Therefore, let DG be cut in half at [point] E. And let (some area), equal to the (square) on EG, be applied to AG, falling short by a square figure. And let it be the (rectangle contained) by AF and FG. AF is thus incommensurable in length with FG. And as AF (is) to FG, so AI is to FK [Prop. 6.1]. Thus, AI is incommensurable with FK [Prop. 10.11]. And since AG and AC are rational (straight-lines which are) commensurable in square only, AK is a medial (area) [Prop. 10.21]. Again, since AC and DG are rational (straight-lines which are) incommensurable in length, DK is also a medial (area) [Prop. 10.21]. Therefore, since AG and GD are commensurable in square only, AG is thus incommensurable in length with GD. And as AG (is) to GD, so AK is to KD [Prop. 6.1]. Thus, AK is incommensurable with KD [Prop. 10.11].

Therefore, let the square LM, equal to AI, be constructed. And let NO, equal to FK, (and) about the same angle, be subtracted (from LM). Thus, the squares LM and NO are about the same diagonal [Prop. 6.26]. Let PR be their (common) diagonal, and let (the rest of) the figure be drawn. So, similarly to the above, we can show that LN is the square-root of area

ἀπὸ αὐτῶν τετραγώνων μέσον καὶ τὸ δὶξ ὑπὸ αὐτῶν μέσονὸέτι τε τὰ ἀπὸ αὐτῶν τετράγωνα ἀσύμμετρα τῷ δὶξ ὑπὸ αὐτῶν. ἡ ἄρα ΛΝὸάλογόξ ἐστιν ἡ καλουμέμ η μετὰ μέσου μέσον τὸ ὅλον ποιοῦσα· καὶ δύναται τὸ ΑΒ θωρίον.

Ἡ ἄρα τὸ θωρίον δυναμένη μετὰ μέσου μέσον τὸ ὅλον ποιοῦσά ἐστιν· ὅπερ ἔδει δεῖχαι.

AB. I say that LN is that (straight-line) which with a medial (area) makes a medial whole.

For since AK was shown (to be) a medial (area), and is equal to the (sum of the) squares on LP and PN, the sum of the (squares) on LP and PN is medial. Again, since DK was shown (to be) a medial (area), and is equal to twice the (rectangle contained) by LP and PN, twice the (rectangle contained) by LP and PN is also medial. And since AK was shown (to be) incommensurable with DK, [thus] the (sum of the) squares on LP and PN is also incommensurable with twice the (rectangle contained) by LP and PN. And since AI is incommensurable with FK, the (square) on LP (is) thus also incommensurable with the (square) on PN. Thus, LP and PN are (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by medial, and, furthermore, the (sum of the) squares on them incommensurable with twice the (rectangle contained) by them. Thus, LN is the irrational (straight-line) called that which with a medial (area) makes a medial whole [Prop. 10.78]. And it is the square-root of area AB.

Thus, the square-root of area (AB) is that (straight-line) which with a medial (area) makes a medial whole. (Which is) the very thing it was required to show.

97.

Τὸ ἀπὸ ἀποτομῆς παρὰ ὀρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πρώτην.

Ἔστω ἀποτομὴ ἡ ΑΒ, ὀρητὴ δὲ ἡ ΓΔ, καὶ τῷ ἀπὸ τῆς ΑΒ ἴσον παρὰ τὴν ΓΔ παραβεβλήσθω τὸ ΓΕ πλάτος ποιοῦν τὴν ΓΖ· λέγω, ὅτι ἡ ΓΖ ἀποτομή ἐστι πρώτη.

Proposition 97

The (square) on an apotome, applied to a rational (straight-line), produces a first apotome as breadth.

Let AB be an apotome, and CD a rational (straight-line). And let CE, equal to the (square) on AB, be applied to CD, producing CF as breadth. I say that CF is a first apotome.

Ἔστω γὰρ τῇ AB προσαρμόζουσα ἡ BH· αἱ ἄρα AH, HB ὁρηταί εἰσι δυνάμει μόνον σύμμετροι. καὶ τῷ μὲν ἀπὸ τῆς AH ἴσον παρὰ τὴν $\Gamma\Delta$ παραβεβλήσθω τὸ $\Gamma\Theta$, τῷ δὲ ἀπὸ τῆς BH τὸ $K\Lambda$. ὅλον ἄρα τὸ $\Gamma\Lambda$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH, HB· ὧν τὸ ΓE ἴσον ἐστὶ τῷ ἀπὸ τῆς AB· λοιπὸν ἄρα τὸ $Z\Lambda$ ἴσον ἐστὶ τῷ δὶς ὑπὸ τῶν AH, HB. τετμήσθω ἡ ZM δίχα κατὰ τὸ N σημεῖον, καὶ ἤχθω διὰ τοῦ N τῇ $\Gamma\Delta$ παράλληλος ἡ NX· ἑκάτερον ἄρα τῶν ZX, ΛN ἴσον ἐστὶ τῷ ὑπὸ τῶν AH, HB. καὶ ἐπεὶ τὰ ἀπὸ τῶν AH, HB ὁρητά ἐστιν, καὶ ἐστι τοῖς ἀπὸ τῶν AH, HB ἴσον τὸ ΔM, ὁρητὸν ἄρα ἐστὶ τὸ ΔM. καὶ παρὰ ὁρητὴν τὴν $\Gamma\Delta$ παραβέβληται πλάτος ποιοῦν τὴν ΓM· ὁρητὴ ἄρα ἐστὶν ἡ ΓM καὶ σύμμετρος τῇ $\Gamma\Delta$ μήκει. πάλιν, ἐπεὶ μέσον ἐστὶ τὸ δὶς ὑπὸ τῶν AH, HB, καὶ τῷ δὶς ὑπὸ τῶν AH, HB ἴσον τὸ $Z\Lambda$, μέσον ἄρα τὸ $Z\Lambda$. καὶ παρὰ ὁρητὴν τὴν $\Gamma\Delta$ παράκειται πλάτος ποιοῦν τὴν ZM· ὁρητὴ ἄρα ἐστὶν ἡ ZM καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει. καὶ ἐπεὶ τὰ μὲν ἀπὸ τῶν AH, HB ὁρητά ἐστιν, τὸ δὲ δὶς ὑπὸ τῶν AH, HB μέσον, ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν AH, HB τῷ δὶς ὑπὸ τῶν AH, HB. καὶ τοῖς μὲν ἀπὸ τῶν AH, HB ἴσον ἐστὶ τὸ $\Gamma\Lambda$, τῷ δὲ δὶς ὑπὸ τῶν AH, HB τὸ $Z\Lambda$· ἀσύμμετρον ἄρα ἐστὶ τὸ ΔM τῷ $Z\Lambda$. ὡς δὲ τὸ ΔM πρὸς τὸ $Z\Lambda$, οὕτως ἐστὶν ἡ ΓM πρὸς τὴν ZM. ἀσύμμετρος ἄρα ἐστὶν ἡ ΓM τῇ ZM μήκει. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ ἄρα ΓM, MZ ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἡ ΓZ ἄρα ἀποτομή ἐστιν. λέγω δή, ὅτι καὶ πρώτη.

Ἐπεὶ γὰρ τῶν ἀπὸ τῶν AH, HB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH, HB, καί ἐστι τῷ μὲν ἀπὸ τῆς AH ἴσον τὸ $\Gamma\Theta$, τῷ δὲ ἀπὸ τῆς BH ἴσον τὸ $K\Lambda$, τῷ δὲ ὑπὸ τῶν AH, HB τὸ $N\Lambda$, καὶ τῶν $\Gamma\Theta$, $K\Lambda$ ἄρα μέσον ἀνάλογόν ἐστι τὸ $N\Lambda$· ὀέστιν ἄρα ὡς τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, οὕτως τὸ $N\Lambda$ πρὸς τὸ $K\Lambda$. ἀλλὰ

For let BG be an attachment to AB. Thus, AG and GB are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And let CH, equal to the (square) on AG, and KL, (equal) to the (square) on BG, be applied to CD. Thus, the whole of CL is equal to the (sum of the squares) on AG and GB, of which CE is equal to the (square) on AB. The remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Let FM be cut in half at point N. And let NO be drawn through N, parallel to CD. Thus, FO and LN are each equal to the (rectangle contained) by AG and GB. And since the (sum of the squares) on AG and GB is rational, and DM is equal to the (sum of the squares) on AG and GB, DM is thus rational. And it has been applied to the rational (straight-line) CD, producing CM as breadth. Thus, CM is rational, and commensurable in length with CD [Prop. 10.20]. Again, since twice the (rectangle contained) by AG and GB is medial, and FL (is) equal to twice the (rectangle contained) by AG and GB, FL (is) thus a medial (area). And it is applied to the rational (straight-line) CD, producing FM as breadth. FM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since the (sum of the squares) on AG and GB is rational, and twice the (rectangle contained) by AG and GB medial, the (sum of the squares) on AG and GB is thus incommensurable with twice the (rectangle contained) by AG and GB. And CL is equal to the (sum of the squares) on AG and GB, and FL to twice the (rectangle contained) by AG and GB. DM is thus incommensurable with FL. And as DM (is) to FL, so CM is to FM [Prop. 6.1]. CM is thus incommensurable in length with FM [Prop. 10.11]. And both are rational

ὧξ μὲν τὸ ΓΘ πρὸς τὸ ΝΛ, οὕτως ἐστὶν ἡ ΓΚ πρὸς τὴν ΝΜ· ὧξ δὲ τὸ ΝΛ πρὸς τὸ ΚΛ, οὕτως ἐστὶν ἡ ΝΜ πρὸς τὴν ΚΜ· τὸ ἄρα ὑπὸ τῶν ΓΚ, ΚΜ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΝΜ, τουτέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΖΜ. καὶ ἐπεὶ σύμμετρόν ἐστι τὸ ἀπὸ τῆς ΑΗ τῷ ἀπὸ τῆς ΗΒ, σύμμετρόν [ἐστι] καὶ τὸ ΓΘ τῷ ΚΛ. ὧξ δὲ τὸ ΓΘ πρὸς τὸ ΚΛ, οὕτως ἡ ΓΚ πρὸς τὴν ΚΜ· σύμμετρος ἄρα ἐστὶν ἡ ΓΚ τῇ ΚΜ. ἐπεὶ οὖν δύο εὐθεῖαι ἀνισοί εἰσιν αἱ ΓΜ, ΜΖ, καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΖΜ ἴσον παρὰ τὴν ΓΜ παραβέβληται ἐλλεῖπον εἰδέι τετραγώνῳ τὸ ὑπὸ τῶν ΓΚ, ΚΜ, καὶ ἐστι σύμμετρος ἡ ΓΚ τῇ ΚΜ, ἡ ἄρα ΓΜ τῆς ΜΖ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει. καί ἐστιν ἡ ΓΜ σύμμετρος τῇ ἐκκειμένῃ ὁρητῇ τῇ ΓΔ μήκει· ἡ ἄρα ΓΖ ἀποτομή ἐστι πρώτη.

Τὸ ἄρα ἀπὸ ἀποτομῆς παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πρώτην· ὅπερ ἔδει δεῖξαι.

(straight-lines). Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a first (apotome).

For since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and CH is equal to the (square) on AG, and KL equal to the (square) on BG, and NL to the (rectangle contained) by AG and GB, NL is thus also the mean proportional to CH and KL. Thus, as CH is to NL, so NL (is) to KL. But, as CH (is) to NL, so CK is to NM, and as NL (is) to KL, so NM is to KM [Prop. 6.1]. Thus, the (rectangle contained) by CK and KM is equal to the (square) on NM—that is to say, to the fourth part of the (square) on FM [Prop. 6.17]. And since the (square) on AG is commensurable with the (square) on GB, CH [is] also commensurable with KL. And as CH (is) to KL, so CK (is) to KM [Prop. 6.1]. CK is thus commensurable (in length) with KM [Prop. 10.11]. Therefore, since CM and MF are two unequal straight-lines, and the (rectangle contained) by CK and KM, equal to the fourth part of the (square) on FM, has been applied to CM, falling short by a square figure, and CK is commensurable (in length) with KM, the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) commensurable in length with (CM) [Prop. 10.17]. And CM is commensurable in length with the (previously) laid down rational (straight-line) CD. Thus, CF is a first apotome [Def. 10.15].

Thus, the (square) on an apotome, applied to a rational (straight-line), produces a first apotome as breadth. (Which is) the very thing it was required to show.

98.

Τὸ ἀπὸ μέσης ἀποτομῆς πρώτης παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν δευτέραν.

Ἔστω μέσης ἀποτομὴ πρώτη ἡ ΑΒ, ὁρητὴ δὲ ἡ ΓΔ, καὶ τῷ ἀπὸ τῆς ΑΒ ἴσον παρὰ τὴν ΓΔ παραβεβλήσθω τὸ ΓΕ πλάτος ποιοῦν τὴν ΓΖ· λέγω, ὅτι ἡ ΓΖ ἀποτομή ἐστι δευτέρα.

Ἔστω γὰρ τῇ ΑΒ προσαρμόζουσα ἡ ΒΗ· αἱ ἄρα ΑΗ, ΗΒ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι ὁρητὸν περιέθουσαι. καὶ τῷ μὲν ἀπὸ τῆς ΑΗ ἴσον παρὰ τὴν ΓΔ παραβεβλήσθω τὸ ΓΘ πλάτος ποιοῦν τὴν ΓΚ, τῷ δὲ ἀπὸ τῆς ΗΒ ἴσον τὸ ΚΛ πλάτος ποιοῦν τὴν ΚΜ· ὅλον ἄρα τὸ ΓΛ ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΑΗ, ΗΒ· μέσον ἄρα καὶ τὸ ΓΛ. καὶ παρὰ ὁρητὴν τὴν ΓΔ παράκειται πλάτος ποιοῦν τὴν ΓΜ· ὁρητὴ ἄρα ἐστὶν ἡ ΓΜ καὶ ἀσύμμετρος τῇ ΓΔ μήκει. καὶ ἐπεὶ τὸ ΓΛ ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΑΗ, ΗΒ, ὧν τὸ ἀπὸ τῆς ΑΒ ἴσον ἐστὶ τῷ ΓΕ, λοιπὸν ἄρα τὸ δὶξ ὑπὸ τῶν ΑΗ, ΗΒ ἴσον ἐστὶ τῷ ΖΛ. ὁρητὸν δέ [ἐστι] τὸ δὶξ ὑπὸ τῶν ΑΗ, ΗΒ· ὁρητὸν ἄρα τὸ ΖΛ. καὶ παρὰ ὁρητὴν τὴν ΖΕ παράκειται πλάτος ποιοῦν τὴν ΖΜ· ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΖΜ καὶ σύμμετρος τῇ ΓΔ μήκει. ἐπεὶ οὖν τὰ μὲν ἀπὸ τῶν ΑΗ, ΗΒ, τουτέστι τὸ ΓΛ, μέσον ἐστίν, τὸ

Proposition 98

The (square) on a first apotome of a medial (straight-line), applied to a rational (straight-line), produces a second apotome as breadth.

Let AB be a first apotome of a medial (straight-line), and CD a rational (straight-line). And let CE, equal to the (square) on AB, be applied to CD, producing CF as breadth. I say that CF is a second apotome.

For let BG be an attachment to AB. Thus, AG and GB are medial (straight-lines which are) commensurable in square only, containing a rational (area) [Prop. 10.74]. And let CH, equal to the (square) on AG, be applied to CD, producing CK as breadth, and KL, equal to the (square) on GB, producing KM as breadth. Thus, the whole of CL is equal to the (sum of the squares) on AG and GB. Thus, CL (is) also a medial (area) [Props. 10.15, 10.23 corr.]. And it is applied to the rational (straight-line) CD, producing CM as breadth. CM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since CL is equal to the (sum of the squares) on AG and GB, of which the (square) on AB is equal to CE, the remainder,

δὲ δὶξ ὑπὸ τῶν ΑΗ, ΗΒ, τουτέστι τὸ ΖΛ, ὁρητόν ἀσύμμετρον ἄρα ἐστὶ τὸ ΓΛ τῷ ΖΛ. ὡς δὲ τὸ ΓΛ πρὸς τὸ ΖΛ, οὕτως ἐστὶν ἡ ΓΜ πρὸς τὴν ΖΜ· ἀσύμμετρος ἄρα ἡ ΓΜ τῇ ΖΜ μήκει. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ ἄρα ΓΜ, ΜΖ ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἡ ΓΖ ἄρα ἀποτομή ἐστιν. λέγω δή, ὅτι καὶ δευτέρα.

twice the (rectangle contained) by AG and GB, is thus equal to FL [Prop. 2.7]. And twice the (rectangle contained) by AG and GB [is] rational. Thus, FL (is) rational. And it is applied to the rational (straight-line) FE, producing FM as breadth. FM is thus also rational, and commensurable in length with CD [Prop. 10.20]. Therefore, since the (sum of the squares) on AG and GB—that is to say, CL—is medial, and twice the (rectangle contained) by AG and GB—that is to say, FL—(is) rational, CL is thus incommensurable with FL. And as CL (is) to FL, so CM is to FM [Prop. 6.1]. Thus, CM (is) incommensurable in length with FM [Prop. 10.11]. And they are both rational (straight-lines). Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a second (apotome).

Τετμήσθω γὰρ ἡ ΖΜ δίθα κατὰ τὸ Ν, καὶ ἤχθω διὰ τοῦ Ν τῇ ΓΔ παράλληλος ἡ ΝΧ· ἑκάτερον ἄρα τῶν ΖΧ, ΝΛ ἴσον ἐστὶ τῷ ὑπὸ τῶν ΑΗ, ΗΒ. καὶ ἐπεὶ τῶν ἀπὸ τῶν ΑΗ, ΗΒ τετραγώνων μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν ΑΗ, ΗΒ, καί ἐστιν ἴσον τὸ μὲν ἀπὸ τῆς ΑΗ τῷ ΓΘ, τὸ δὲ ὑπὸ τῶν ΑΗ, ΗΒ τῷ ΝΛ, τὸ δὲ ἀπὸ τῆς ΒΗ τῷ ΚΛ, καὶ τῶν ΓΘ, ΚΛ ἄρα μέσον ἀνάλογόν ἐστι τὸ ΝΛ· ὐέστιν ἄρα ὡς τὸ ΓΘ πρὸς τὸ ΝΛ, οὕτως τὸ ΝΛ πρὸς τὸ ΚΛ. ἀλλὰ ὡς μὲν τὸ ΓΘ πρὸς τὸ ΝΛ, οὕτως ἐστὶν ἡ ΓΚ πρὸς τὴν ΝΜ, ὡς δὲ τὸ ΝΛ πρὸς τὸ ΚΛ, οὕτως ἐστὶν ἡ ΝΜ πρὸς τὴν ΜΚ· ὡς ἄρα ἡ ΓΚ πρὸς τὴν ΝΜ, οὕτως ἐστὶν ἡ ΝΜ πρὸς τὴν ΚΜ· τὸ ἄρα ὑπὸ τῶν ΓΚ, ΚΜ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΝΜ, τουτέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΖΜ [καὶ ἐπεὶ σύμμετρόν ἐστι τὸ ἀπὸ τῆς ΑΗ τῷ

For let FM be cut in half at N. And let NO be drawn through (point) N, parallel to CD. Thus, FO and NL are each equal to the (rectangle contained) by AG and GB. And since the (rectangle contained) by AG and GB is the mean proportional to the squares on AG and GB [Prop. 10.21 lem.], and the (square) on AG is equal to CH, and the (rectangle contained) by AG and GB to NL, and the (square) on BG to KL, NL is thus also the mean proportional to CH and KL. Thus, as CH is to NL, so NL (is) to KL [Prop. 5.11]. But, as CH (is) to NL, so CK is to NM, and as NL (is) to KL, so NM is to MK [Prop. 6.1]. Thus, as CK (is) to NM, so NM is to KM [Prop. 5.11]. The (rectangle contained) by CK and KM is thus equal to the (square) on NM [Prop. 6.17]—that is to

ἀπὸ τῆς ΒΗ, σύμμετρόν ἐστι καὶ τὸ ΓΘ τῷ ΚΛ, τουτέστιν ἡ ΓΚ τῇ ΚΜ]. ἐπεὶ οὖν δύο εὐθεῖαι ἀνισοί εἰσιν αἱ ΓΜ, ΜΖ, καὶ τῷ τετάρῳ μέρει τοῦ ἀπὸ τῆς ΜΖ ἴσον παρὰ τὴν μείζονα τὴν ΓΜ παραβέβληται ἐλλεῖπον εἰδει τετραγώνῳ τὸ ὑπὸ τῶν ΓΚ, ΚΜ καὶ εἰξ σύμμετρα αὐτὴν διαιρεῖ, ἡ ἄρα ΓΜ τῆς ΜΖ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ μήκει. καί ἐστιν ἡ προσαρμόζουσα ἡ ΖΜ σύμμετρος μήκει τῇ ἐκκειμένῃ ὁρητῇ τῇ ΓΔ· ἡ ἄρα ΓΖ ἀποτομή ἐστι δευτέρα.

Τὸ ἄρα ἀπὸ μέσης ἀποτομῆς πρώτης παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν δευτέραν· ὅπερ ἔδει δεῖχαι.

say, to the fourth part of the (square) on FM [and since the (square) on AG is commensurable with the (square) on BG, CH is also commensurable with KL—that is to say, CK with KM]. Therefore, since CM and MF are two unequal straight-lines, and the (rectangle contained) by CK and KM, equal to the fourth part of the (square) on MF, has been applied to the greater CM, falling short by a square figure, and divides it into commensurable (parts), the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) commensurable in length with (CM) [Prop. 10.17]. The attachment FM is also commensurable in length with the (previously) laid down rational (straight-line) CD. CF is thus a second apotome [Def. 10.16].

Thus, the (square) on a first apotome of a medial (straight-line), applied to a rational (straight-line), produces a second apotome as breadth. (Which is) the very thing it was required to show.

99.

Τὸ ἀπὸ μέσης ἀποτομῆς δευτέρας παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τρίτην.

Ἔστω μέσης ἀποτομὴ δευτέρα ἡ ΑΒ, ὁρητὴ δὲ ἡ ΓΔ, καὶ τῷ ἀπὸ τῆς ΑΒ ἴσον παρὰ τὴν ΓΔ παραβεβλήσθω τὸ ΓΕ πλάτος ποιοῦν τὴν ΓΖ· λέγω, ὅτι ἡ ΓΖ ἀποτομή ἐστι τρίτη.

Proposition 99

The (square) on a second apotome of a medial (straight-line), applied to a rational (straight-line), produces a third apotome as breadth.

Let AB be the second apotome of a medial (straight-line), and CD a rational (straight-line). And let CE, equal to the (square) on AB, be applied to CD, producing CF as breadth. I say that CF is a third apotome.

Ἔστω γὰρ τῇ ΑΒ προσαρμόζουσα ἡ ΒΗ· αἱ ἄρα ΑΗ, ΗΒ μέσαι εἰσὶ δυνάμει μόνον σύμμετροι μέσον περιέθουσαι.

For let BG be an attachment to AB. Thus, AG and GB are medial (straight-lines which are) commensurable in square

καὶ τῷ μὲν ἀπὸ τῆς AH ἴσον παρὰ τὴν $\Gamma\Delta$ παραβεβλήσθω τὸ $\Gamma\Theta$ πλάτος ποιοῦν τὴν ΓK, τῷ δὲ ἀπὸ τῆς BH ἴσον παρὰ τὴν $K\Theta$ παραβεβλήσθω τὸ $K\Lambda$ πλάτος ποιοῦν τὴν KM· ὅλον ἄρα τὸ $\Gamma\Lambda$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH, HB [καί ἐστι μέσα τὰ ἀπὸ τῶν AH, HB]· μέσον ἄρα καὶ τὸ $\Gamma\Lambda$. καὶ παρὰ ὁρητὴν τὴν $\Gamma\Delta$ παραβέβληται πλάτος ποιοῦν τὴν ΓM· ὁρητὴ ἄρα ἐστὶν ἡ ΓM καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει. καὶ ἐπεὶ ὅλον τὸ $\Gamma\Lambda$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH, HB, ὧν τὸ ΓE ἴσον ἐστὶ τῷ ἀπὸ τῆς AB, λοιπὸν ἄρα τὸ ΛZ ἴσον ἐστὶ τῷ δὶξ ὑπὸ τῶν AH, HB. τετμήσθω οὐῦν ἡ ZM δίθα κατὰ τὸ N σημεῖον, καὶ τῇ $\Gamma\Delta$ παράλληλος ἤχθω ἡ NX· ἑκάτερον ἄρα τῶν ZX, $N\Lambda$ ἴσον ἐστὶ τῷ ὑπὸ τῶν AH, HB. μέσον δὲ τὸ ὑπὸ τῶν AH, HB· μέσον ἄρα ἐστὶ καὶ τὸ $Z\Lambda$. καὶ παρὰ ὁρητὴν τὴν EZ παράκειται πλάτος ποιοῦν τὴν ZM· ὁρητὴ ἄρα καὶ ἡ ZM καὶ ἀσύμμετρος τῇ $\Gamma\Delta$ μήκει. καὶ ἐπεὶ αἱ AH, HB δυνάμει μόνον εἰσὶ σύμμετροι, ἀσύμμετρος ἄρα [ἐστὶ] μήκει ἡ AH τῇ HB· ἀσύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς AH τῷ ὑπὸ τῶν AH, HB. ἀλλὰ τῷ μὲν ἀπὸ τῆς AH σύμμετρά ἐστι τὰ ἀπὸ τῶν AH, HB, τῷ δὲ ὑπὸ τῶν AH, HB τὸ δὶξ ὑπὸ τῶν AH, HB· ἀσύμμετρα ἄρα ἐστὶ τὰ ἀπὸ τῶν AH, HB τῷ δὶξ ὑπὸ τῶν AH, HB. ἀλλὰ τοῖς μὲν ἀπὸ τῶν AH, HB ἴσον ἐστὶ τὸ $\Gamma\Lambda$, τῷ δὲ δὶξ ὑπὸ τῶν AH, HB ἴσον ἐστὶ τὸ $Z\Lambda$· ἀσύμμετρον ἄρα ἐστὶ τὸ $\Gamma\Lambda$ τῷ $Z\Lambda$. ὡς δὲ τὸ $\Gamma\Lambda$ πρὸς τὸ $Z\Lambda$, οὑτως ἐστὶν ἡ ΓM πρὸς τὴν ZM· ἀσύμμετρος ἄρα ἐστὶν ἡ ΓM τῇ ZM μήκει. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ ἄρα ΓM, MZ ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ ΓZ. λέγω δή, ὅτι καὶ τρίτη.

Ἐπεὶ γὰρ σύμμετρόν ἐστι τὸ ἀπὸ τῆς AH τῷ ἀπὸ τῆς HB, σύμμετρον ἄρα καὶ τὸ $\Gamma\Theta$ τῷ $K\Lambda$· ὡστε καὶ ἡ ΓK τῇ KM. καὶ ἐπεὶ τῶν ἀπὸ τῶν AH, HB μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν AH, HB, καί ἐστι τῷ μὲν ἀπὸ τῆς AH ἴσον τὸ $\Gamma\Theta$, τῷ δὲ ἀπὸ τῆς HB ἴσον τὸ $K\Lambda$, τῷ δὲ ὑπὸ τῶν AH, HB ἴσον τὸ $N\Lambda$, καὶ τῶν $\Gamma\Theta$, $K\Lambda$ ἄρα μέσον ἀνάλογόν ἐστι τὸ $N\Lambda$· ὁέστιν ἄρα ὡς τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, οὑτως τὸ $N\Lambda$ πρὸς τὸ $K\Lambda$. ἀλλὰ ὡς μὲν τὸ $\Gamma\Theta$ πρὸς τὸ $N\Lambda$, οὑτως ἐστὶν ἡ ΓK πρὸς τὴν NM, ὡς δὲ τὸ $N\Lambda$ πρὸς τὸ $K\Lambda$, οὑτως ἐστὶν ἡ NM πρὸς τὴν KM· ὡς ἄρα ἡ ΓK πρὸς τὴν MN, οὑτως ἐστὶν ἡ MN πρὸς τὴν KM· τὸ ἄρα ὑπὸ τῶν ΓK, KM ἴσον ἐστὶ τῷ [ἀπὸ τῆς MN, τουτέστι τῷ] τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM. ἐπεὶ οὐῦν δύο εὐθεῖαιὁάνισοί εἰσιν αἱ ΓM, MZ, καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ZM ἴσον παρὰ τὴν ΓM παραβέβληται ἐλλεῖπον εὁίδει τετραγώνῳ καὶ εἰξ σύμμετρα αὐτὴν διαιρεῖ, ἡ ΓM ἄρα τῆς MZ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καὶ οὐδετέρα τῶν ΓM, MZ σύμμετρόξ ἐστι μήκει τῇ ἐκκειμένῃ ὁρητῇ τῇ $\Gamma\Delta$· ἡ ἄρα ΓZ ἀποτομή ἐστι τρίτη.

Τὸ ἄρα ἀπὸ μέσης ἀποτομῆς δευτέρας παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τρίτην· ὅπερ ἔδει δεῖχαι.

only, containing a medial (area) [Prop. 10.75]. And let CH, equal to the (square) on AG, be applied to CD, producing CK as breadth. And let KL, equal to the (square) on BG, be applied to KH, producing KM as breadth. Thus, the whole of CL is equal to the (sum of the squares) on AG and GB [and the (sum of the squares) on AG and GB is medial]. CL (is) thus also medial [Props. 10.15, 10.23 corr.]. And it has been applied to the rational (straight-line) CD, producing CM as breadth. Thus, CM is rational, and incommensurable in length with CD [Prop. 10.22]. And since the whole of CL is equal to the (sum of the squares) on AG and GB, of which CE is equal to the (square) on AB, the remainder LF is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Therefore, let FM be cut in half at point N. And let NO be drawn parallel to CD. Thus, FO and NL are each equal to the (rectangle contained) by AG and GB. And the (rectangle contained) by AG and GB (is) medial. Thus, FL is also medial. And it is applied to the rational (straight-line) EF, producing FM as breadth. FM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since AG and GB are commensurable in square only, AG [is] thus incommensurable in length with GB. Thus, the (square) on AG is also incommensurable with the (rectangle contained) by AG and GB [Props. 6.1, 10.11]. But, the (sum of the squares) on AG and GB is commensurable with the (square) on AG, and twice the (rectangle contained) by AG and GB with the (rectangle contained) by AG and GB. The (sum of the squares) on AG and GB is thus incommensurable with twice the (rectangle contained) by AG and GB [Prop. 10.13]. But, CL is equal to the (sum of the squares) on AG and GB, and FL is equal to twice the (rectangle contained) by AG and GB. Thus, CL is incommensurable with FL. And as CL (is) to FL, so CM is to FM [Prop. 6.1]. CM is thus incommensurable in length with FM [Prop. 10.11]. And they are both rational (straight-lines). Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a third (apotome).

For since the (square) on AG is commensurable with the (square) on GB, CH (is) thus also commensurable with KL. Hence, CK (is) also (commensurable in length) with KM [Props. 6.1, 10.11]. And since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and CH is equal to the (square) on AG, and KL equal to the (square) on GB, and NL equal to the (rectangle contained) by AG and GB, NL is thus also the mean proportional to CH and KL. Thus, as CH is to NL, so NL (is) to KL. But, as CH (is) to NL, so CK is to NM, and as NL (is) to KL, so NM (is) to KM [Prop. 6.1]. Thus, as CK (is) to MN, so MN is to KM [Prop. 5.11]. Thus, the (rectangle contained) by CK and KM is equal to the [(square) on MN—that is to say, to the] fourth part of the (square) on

100.

Τὸ ἀπὸ ἐλάσσονος παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τετάρτην.

Ἔστω ἐλάσσων ἡ AB, ὁρητὴ δὲ ἡ $ΓΔ$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ ὁρητὴν τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΕ$ πλάτος ποιοῦν τὴν $ΓΖ$· λέγω, ὅτι ἡ $ΓΖ$ ἀποτομή ἐστι τετάρτη.

Ἔστω γὰρ τῇ AB προσαρμόζουσα ἡ BH· αἱ ἄρα AH, HB δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AH, HB τετραγώνων ὁρητόν, τὸ δὲ δὶξ ὑπὸ τῶν AH, HB μέσον. καὶ τῷ μὲν ἀπὸ τῆς AH ἴσον παρὰ τὴν $ΓΔ$ παραβεβλήσθω τὸ $ΓΘ$ πλάτος ποιοῦν τὴν $ΓΚ$, τῷ δὲ ἀπὸ τῆς BH ἴσον τὸ $ΚΛ$ πλάτος ποιοῦν τὴν $ΚΜ$· ὅλον ἄρα τὸ $ΓΛ$ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH, HB. καί ἐστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AH, HB ὁρητόν· ὁρητὸν ἄρα ἐστὶ καὶ τὸ $ΓΛ$.

Proposition 100

The (square) on a minor (straight-line), applied to a rational (straight-line), produces a fourth apotome as breadth.

Let AB be a minor (straight-line), and CD a rational (straight-line). And let CE, equal to the (square) on AB, be applied to the rational (straight-line) CD, producing CF as breadth. I say that CF is a fourth apotome.

For let BG be an attachment to AB. Thus, AG and GB are incommensurable in square, making the sum of the squares on AG and GB rational, and twice the (rectangle contained) by AG and GB medial [Prop. 10.76]. And let CH, equal to the (square) on AG, be applied to CD, producing CK as breadth, and KL, equal to the (square) on BG, producing KM as breadth. Thus, the whole of CL is equal to the (sum of FM [Prop. 6.17]. Therefore, since CM and MF are two unequal straight-lines, and (some area), equal to the fourth part of the (square) on FM, has been applied to CM, falling short by a square figure, and divides it into commensurable (parts), the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) commensurable (in length) with (CM) [Prop. 10.17]. And neither of CM and MF is commensurable in length with the (previously) laid down rational (straight-line) CD. CF is thus a third apotome [Def. 10.13].

Thus, the (square) on a second apotome of a medial (straight-line), applied to a rational (straight-line), produces a third apotome as breadth. (Which is) the very thing it was required to show.

καὶ παρὰ ὁρητὴν τὴν ΓΔ παράκειται πλάτος ποιοῦν τὴν ΓΜ· ὁρητὴ ἄρα καὶ ἡ ΓΜ καὶ σύμμετρος τῇ ΓΔ μήκει. καὶ ἐπεὶ ὅλον τὸ ΓΛ ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΑΗ, ΗΒ, ὧν τὸ ΓΕ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΒ, λοιπὸν ἄρα τὸ ΖΛ ἴσον ἐστὶ τῷ δὶξ ὑπὸ τῶν ΑΗ, ΗΒ. τετμήσθω οὐῦν ἡ ΖΜ δίθα κατὰ τὸ Ν σημεῖον, καὶ ἤθθω δια τοῦ Ν ὁποτέρᾳ τῶν ΓΔ, ΜΛ παράλληλος ἡ ΝΧ· ἑκάτερον ἄρα τῶν ΖΧ, ΝΛ ἴσον ἐστὶ τῷ ὑπὸ τῶν ΑΗ, ΗΒ. καὶ ἐπεὶ τὸ δὶξ ὑπὸ τῶν ΑΗ, ΗΒ μέσον ἐστὶ καί ἐστιν ἴσον τῷ ΖΛ, καὶ τὸ ΖΛ ἄρα μέσον ἐστίν. καὶ παρὰ ὁρητὴν τὴν ΖΕ παράκειται πλάτος ποιοῦν τὴν ΖΜ· ὁρητὴ ἄρα ἐστὶν ἡ ΖΜ καὶ ἀσύμμετρος τῇ ΓΔ μήκει. καὶ ἐπεὶ τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν ΑΗ, ΗΒ ὁρητόν ἐστιν, τὸ δὲ δὶξ ὑπὸ τῶν ΑΗ, ΗΒ μέσον, ἀσύμμετρα [ἄρα] ἐστὶ τὰ ἀπὸ τῶν ΑΗ, ΗΒ τῷ δὶξ ὑπὸ τῶν ΑΗ, ΗΒ. ἴσον δέ [ἐστι] τὸ ΓΛ τοῖς ἀπὸ τῶν ΑΗ, ΗΒ, τῷ δὲ δὶξ ὑπὸ τῶν ΑΗ, ΗΒ ἴσον τὸ ΖΛ· ἀσύμμετρον ἄρα [ἐστὶ] τὸ ΓΛ τῷ ΖΛ. ὡξ δὲ τὸ ΓΛ πρὸς τὸ ΖΛ, οὐύτως ἐστὶν ἡ ΓΜ πρὸς τὴν ΜΖ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΓΜ τῇ ΜΖ μήκει. καί εἰσιν ἀμφότεραι ὁρηταί· αἱ ἄρα ΓΜ, ΜΖ ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ ΓΖ. λέγω [δή], ὅτι καὶ τετάρτη.

Ἐπεὶ γὰρ αἱ ΑΗ, ΗΒ δυνάμει εἰσὶν ἀσύμμετροι, ἀσύμμετρον ἄρα καὶ τὸ ἀπὸ τῆς ΑΗ τῷ ἀπὸ τῆς ΗΒ. καί ἐστι τῷ μὲν ἀπὸ τῆς ΑΗ ἴσον τὸ ΓΘ, τῷ δὲ ἀπὸ τῆς ΗΒ ἴσον τὸ ΚΛ· ἀσύμμετρον ἄρα ἐστὶ τὸ ΓΘ τῷ ΚΛ. ὡξ δὲ τὸ ΓΘ πρὸς τὸ ΚΛ, οὐύτως ἐστὶν ἡ ΓΚ πρὸς τὴν ΚΜ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΓΚ τῇ ΚΜ μήκει. καὶ ἐπεὶ τῶν ἀπὸ τῶν ΑΗ, ΗΒ μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν ΑΗ, ΗΒ, καί ἐστιν ἴσον τὸ μὲν ἀπὸ τῆς ΑΗ τῷ ΓΘ, τὸ δὲ ἀπὸ τῆς ΗΒ τῷ ΚΛ, τὸ δὲ ὑπὸ τῶν ΑΗ, ΗΒ τῷ ΝΛ, τῶν ἄρα ΓΘ, ΚΛ μέσον ἀνάλογόν ἐστι τὸ ΝΛ· ὀέστιν ἄρα ὡξ τὸ ΓΘ πρὸς τὸ ΝΛ, οὐύτως τὸ ΝΛ πρὸς τὸ ΚΛ. ἀλλὸ ὡξ μὲν τὸ ΓΘ πρὸς τὸ ΝΛ, οὐύτως ἐστὶν ἡ ΓΚ πρὸς τὴν ΝΜ, ὡξ δὲ τὸ ΝΛ πρὸς τὸ ΚΛ, οὐύτως ἐστὶν ἡ ΝΜ πρὸς τὴν ΚΜ· ὡξ ἄρα ἡ ΓΚ πρὸς τὴν ΜΝ, οὐύτως ἐστὶν ἡ ΜΝ πρὸς τὴν ΚΜ· τὸ ἄρα ὑπὸ τῶν ΓΚ, ΚΜ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΜΝ, τουτέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΖΜ. ἐπεὶ οὐῦν δύο εὐθεῖαιὀάνισοί εἰσιν αἱ ΓΜ, ΜΖ, καὶ τῷ τετράρτῳ μέρει τοῦ ἀπὸ τῆς ΜΖ ἴσον παρὰ τὴν ΓΜ παραβέβληται ἐλλεῖπον εὐίδει τετραγώνῳ τὸ ὑπὸ τῶν ΓΚ, ΚΜ καὶ εἰξ ἀσύμμετρα αὐτὴν διαιρεῖ, ἡ ἄρα ΓΜ τῆς ΜΖ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καί ἐστιν ὅλη ἡ ΓΜ σύμμετρος μήκει τῇ ἐκκειμένῃ ὁρητῇ τῇ ΓΔ· ἡ ἄρα ΓΖ ἀποτομή ἐστι τετάρτη.

Τὸ ἄρα ἀπὸ ἐλάσσονος καὶ τὰ ἑχῆς.

the squares) on AG and GB. And the sum of the (squares) on AG and GB is rational. CL is thus also rational. And it is applied to the rational (straight-line) CD, producing CM as breadth. Thus, CM (is) also rational, and commensurable in length with CD [Prop. 10.20]. And since the whole of CL is equal to the (sum of the squares) on AG and GB, of which CE is equal to the (square) on AB, the remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Therefore, let FM be cut in half at point N. And let NO be drawn through N, parallel to either of CD or ML. Thus, FO and NL are each equal to the (rectangle contained) by AG and GB. And since twice the (rectangle contained) by AG and GB is medial, and is equal to FL, FL is thus also medial. And it is applied to the rational (straight-line) FE, producing FM as breadth. Thus, FM is rational, and incommensurable in length with CD [Prop. 10.22]. And since the sum of the (squares) on AG and GB is rational, and twice the (rectangle contained) by AG and GB medial, the (sum of the squares) on AG and GB is [thus] incommensurable with twice the (rectangle contained) by AG and GB. And CL (is) equal to the (sum of the squares) on AG and GB, and FL equal to twice the (rectangle contained) by AG and GB. CL [is] thus incommensurable with FL. And as CL (is) to FL, so CM is to MF [Prop. 6.1]. CM is thus incommensurable in length with MF [Prop. 10.11]. And both are rational (straight-lines). Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. [So], I say that (it is) also a fourth (apotome).

For since AG and GB are incommensurable in square, the (square) on AG (is) thus also incommensurable with the (square) on GB. And CH is equal to the (square) on AG, and KL equal to the (square) on GB. Thus, CH is incommensurable with KL. And as CH (is) to KL, so CK is to KM [Prop. 6.1]. CK is thus incommensurable in length with KM [Prop. 10.11]. And since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and the (square) on AG is equal to CH, and the (square) on GB to KL, and the (rectangle contained) by AG and GB to NL, NL is thus the mean proportional to CH and KL. Thus, as CH is to NL, so NL (is) to KL. But, as CH (is) to NL, so CK is to NM, and as NL (is) to KL, so NM is to KM [Prop. 6.1]. Thus, as CK (is) to MN, so MN is to KM [Prop. 5.11]. The (rectangle contained) by CK and KM is thus equal to the (square) on MN—that is to say, to the fourth part of the (square) on FM [Prop. 6.17]. Therefore, since CM and MF are two unequal straight-lines, and the (rectangle contained) by CK and KM, equal to the fourth part of the (square) on MF, has been applied to CM, falling short by a square figure, and divides it into incommensurable (parts), the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) incommensurable (in length) with (CM)

101.

Τὸ ἀπὸ τῆς μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιούσης παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πέμπτην.

Ἔστω ἡ μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσα ἡ AB, ὁρητὴ δὲ ἡ ΓΔ, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν ΓΔ παραβεβλήσθω τὸ ΓΕ πλάτος ποιοῦν τὴν ΓΖ· λέγω, ὅτι ἡ ΓΖ ἀποτομή ἐστι πέμπτη.

Ἔστω γὰρ τῇ AB προσαρμόζουσα ἡ BH· αἱ ἄρα AH, HB εὐθεῖαι δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον, τὸ δὲ δὶξ ὑπὸ αὐτῶν ὁρητόν, καὶ τῷ μὲν ἀπὸ τῆς AH ἴσον παρὰ τὴν ΓΔ παραβεβλήσθω τὸ ΓΘ, τῷ δὲ ἀπὸ τῆς HB ὅίσον τὸ ΚΛ· ὅλον ἄρα τὸ ΓΛ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH, HB. τὸ δὲ συγκείμενον ἐκ τῶν ἀπὸ τῶν AH, HB ὁάμα μέσον ἐστίν· μέσον ἄρα ἐστὶ τὸ ΓΛ. καὶ παρὰ ὁρητὴν τὴν ΓΔ παράκειται πλάτος ποιοῦν τὴν ΓΜ· ὁρητὴ ἄρα ἐστὶν ἡ ΓΜ καὶ ἀσύμμετρος τῇ ΓΔ. καὶ ἐπεὶ ὅλον τὸ ΓΛ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH, HB, ὁῶν τὸ ΓΕ ἴσον ἐστὶ τῷ ἀπὸ τῆς AB, λοιπὸν ἄρα τὸ ΖΛ ἴσον ἐστὶ τῷ δὶξ ὑπὸ τῶν AH, HB. τετμήσθω οὐῦν ἡ ZM δίθα κατὰ τὸ N, καὶ ἤθθω διὰ τοῦ N ὁποτέρᾳ τῶν ΓΔ, ΜΛ παράλληλος ἡ NX· ἑκάτερον ἄρα τῶν ZX, NΛ ἴσον ἐστὶ τῷ ὑπὸ τῶν AH, HB, καὶ ἐπεὶ τὸ δὶξ ὑπὸ τῶν AH, HB ὁρητόν ἐστι καί [ἐστιν] ἴσον τῷ ZΛ,

Proposition 101

The (square) on that (straight-line) which with a rational (area) makes a medial whole, applied to a rational (straight-line), produces a fifth apotome as breadth.

Let AB be that (straight-line) which with a rational (area) makes a medial whole, and CD a rational (straight-line). And let CE, equal to the (square) on AB, be applied to CD, producing CF as breadth. I say that CF is a fifth apotome.

Let BG be an attachment to AB. Thus, the straight-lines AG and GB are incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by them rational [Prop. 10.77]. And let CH, equal to the (square) on AG, be applied to CD, and KL, equal to the (square) on GB. The whole of CL is thus equal to the (sum of the squares) on AG and GB. And the sum of the (squares) on AG and GB together is medial. Thus, CL is medial. And it has been applied to the rational (straight-line) CD, producing CM as breadth. CM is thus rational, and incommensurable (in length) with CD [Prop. 10.22]. And since the whole of CL is equal to the (sum of the squares) on AG and GB, of which CE is equal to the (square) on AB, the remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. Therefore, let FM be cut in half at N. And let NO be drawn

ὀρητὸν ἄρα ἐστὶ τὸ ΖΛ. καὶ παρὰ ὀρητὴν τὴν ΕΖ παράκειται πλάτος ποιοῦν τὴν ΖΜ· ὀρητὴ ἄρα ἐστὶν ἡ ΖΜ καὶ σύμμετρος τῇ ΓΔ μήκει. καὶ ἐπεὶ τὸ μὲν ΓΛ μέσον ἐστίν, τὸ δὲ ΖΛ ὀρητόν, ἀσύμμετρον ἄρα ἐστὶ τὸ ΓΛ τῷ ΖΛ. ὡς δὲ τὸ ΓΛ πρὸς τὸ ΖΛ, οὕτως ἡ ΓΜ πρὸς τὴν ΜΖ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΓΜ τῇ ΜΖ μήκει. καί εἰσιν ἀμφότεραι ὀρηταί· αἱ ἄρα ΓΜ, ΜΖ ὀρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ ΓΖ. λέγω δή, ὅτι καὶ πέμπτη.

Ὁμοίως γὰρ δείχομεν, ὅτι τὸ ὑπὸ τῶν ΓΚΜ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΝΜ, τουτέστι τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΖΜ. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ ἀπὸ τῆς ΑΗ τῷ ἀπὸ τῆς ΗΒ, ἴσον δὲ τὸ μὲν ἀπὸ τῆς ΑΗ τῷ ΓΘ, τὸ δὲ ἀπὸ τῆς ΗΒ τῷ ΚΛ, ἀσύμμετρον ἄρα τὸ ΓΘ τῷ ΚΛ. ὡς δὲ τὸ ΓΘ πρὸς τὸ ΚΛ, οὕτως ἡ ΓΚ πρὸς τὴν ΚΜ· ἀσύμμετρος ἄρα ἡ ΓΚ τῇ ΚΜ μήκει. ἐπεὶ οὖν δύο εὐθεῖαι ἀνίσοί εἰσιν αἱ ΓΜ, ΜΖ, καὶ τῷ τετάρτῳ μέρει τοῦ ἀπὸ τῆς ΖΜ ἴσον παρὰ τὴν ΓΜ παραβέβληται ἐλλεῖπον εἰδέι τετραγώνῳ καὶ εἰξ ἀσύμμετρα αὐτὴν διαιρεῖ, ἡ ἄρα ΓΜ τῆς ΜΖ μεῖζον δύναται τῷ ἀπὸ ἀσύμμέτρου ἑαυτῇ. καί ἐστιν ἡ προσαρμόζουσα ἡ ΖΜ σύμμετρος τῇ ἐκκειμένῃ ὀρητῇ τῇ ΓΔ· ἡ ἄρα ΓΖ ἀποτομή ἐστι πέμπτη· ὅπερ ἔδει δεῖχαι.

through N, parallel to either of CD or ML. Thus, FO and NL are each equal to the (rectangle contained) by AG and GB. And since twice the (rectangle contained) by AG and GB is rational, and [is] equal to FL, FL is thus rational. And it is applied to the rational (straight-line) EF, producing FM as breadth. Thus, FM is rational, and commensurable in length with CD [Prop. 10.20]. And since CL is medial, and FL rational, CL is thus incommensurable with FL. And as CL (is) to FL, so CM (is) to MF [Prop. 6.1]. CM is thus incommensurable in length with MF [Prop. 10.11]. And both are rational. Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a fifth (apotome).

For, similarly (to the previous propositions), we can show that the (rectangle contained by) CKM is equal to the (square) on NM—that is to say, to the fourth part of the (square) on FM. And since the (square) on AG is incommensurable with the (square) on GB, and the (square) on AG (is) equal to CH, and the (square) on GB to KL, CH (is) thus incommensurable with KL. And as CH (is) to KL, so CK (is) to KM [Prop. 6.1]. Thus, CK (is) incommensurable in length with KM [Prop. 10.11]. Therefore, since CM and MF are two unequal straight-lines, and (some area), equal to the fourth part of the (square) on FM, has been applied to CM, falling short by a square figure, and divides it into incommensurable (parts), the square on CM is thus greater than (the square on) MF by the (square) on (some straight-line) incommensurable (in length) with (CM) [Prop. 10.18]. And the attachment FM is commensurable with the (previously) laid down rational (straight-line) CD. Thus, CF is a fifth apotome [Def. 10.15]. (Which is) the very thing it was required to show.

102.

Τὸ ἀπὸ τῆς μετὰ μέσου μέσον τὸ ὅλον ποιούσης παρὰ ὀρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν ὀέκτην.

Proposition 102

The (square) on that (straight-line) which with a medial (area) makes a medial whole, applied to a rational (straight-line), produces a sixth apotome as breadth.

Ἔστω ἡ μετὰ μέσου μέσον τὸ ὅλον ποιοῦσα ἡ AB, ὁρητὴ δὲ ἡ ΓΔ, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν ΓΔ παραβεβλήσθω τὸ ΓΕ πλάτος ποιοῦν τὴν ΓΖ· λέγω, ὅτι ἡ ΓΖ ἀποτομή ἐστιν ὁέκτη.

Ἔστω γὰρ τῇ AB προσαρμόζουσα ἡ BH· αἱ ἄρα AH, HB δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τό τε συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον καὶ τὸ δὶξ ὑπὸ τῶν AH, HB μέσον καὶ ἀσύμμετρον τὰ ἀπὸ τῶν AH, HB τῷ δὶξ ὑπὸ τῶν AH, HB. παραβεβλήσθω οὐῦν παρὰ τὴν ΓΔ τῷ μὲν ἀπὸ τῆς AH ἴσον τὸ ΓΘ πλάτος ποιοῦν τὴν ΓΚ, τῷ δὲ ἀπὸ τῆς BH τὸ ΚΛ· ὅλον ἄρα τὸ ΓΛ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH, HB· μέσον ἄρα [ἐστὶ] καὶ τὸ ΓΛ. καὶ παρὰ ὁρητὴν τὴν ΓΔ παράκειται πλάτος ποιοῦν τὴν ΓΜ· ὁρητὴ ἄρα ἐστὶν ἡ ΓΜ καὶ ἀσύμμετρος τῇ ΓΔ μήκει. ἐπεὶ οὐῦν τὸ ΓΛ ἴσον ἐστὶ τοῖς ἀπὸ τῶν AH, HB, ὁῶν τὸ ΓΕ ἴσον τῷ ἀπὸ τῆς AB, λοιπὸν ἄρα τὸ ΖΛ ἴσον ἐστὶ τῷ δὶξ ὑπὸ τῶν AH, HB. καὶ ἐστι τὸ δὶξ ὑπὸ τῶν AH, HB μέσον· καὶ τὸ ΖΛ ἄρα μέσον ἐστίν. καὶ παρὰ ὁρητὴν τὴν ΖΕ παράκειται πλάτος ποιοῦν τὴν ΖΜ· ὁρητὴ ἄρα ἐστὶν ἡ ΖΜ καὶ ἀσύμμετρος τῇ ΓΔ μήκει. καὶ ἐπεὶ τὰ ἀπὸ τῶν AH, HB ἀσύμμετρά ἐστι τῷ δὶξ ὑπὸ τῶν AH, HB, καί ἐστι τοῖς μὲν ἀπὸ τῶν AH, HB ἴσον τὸ ΓΛ, τῷ δὲ δὶξ ὑπὸ τῶν AH, HB ἴσον τὸ ΖΛ, ἀσύμμετρος ἄρα [ἐστὶ] τὸ ΓΛ τῷ ΖΛ. ὡξ δὲ τὸ ΓΛ πρὸς τὸ ΖΛ, οὐύτως ἐστὶν ἡ ΓΜ πρὸς τὴν ΜΖ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΓΜ τῇ ΜΖ μήκει. καί εἰσιν ἀμφότεραι ὁρηταί. αἱ ΓΜ, ΜΖ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ ΓΖ. λέγω δή, ὅτι καὶ ὁέκτη.

Ἐπεὶ γὰρ τὸ ΖΛ ἴσον ἐστὶ τῷ δὶξ ὑπὸ τῶν AH, HB, τετμήσθω δίθα ἡ ΖΜ κατὰ τὸ N, καὶ ᾖθθω διὰ τοῦ N τῇ ΓΔ παράλληλος ἡ ΝΧ· ἑκάτερον ἄρα τῶν ΖΧ, ΝΛ ἴσον ἐστὶ τῷ

Let AB be that (straight-line) which with a medial (area) makes a medial whole, and CD a rational (straight-line). And let CE, equal to the (square) on AB, be applied to CD, producing CF as breadth. I say that CF is a sixth apotome.

For let BG be an attachment to AB. Thus, AG and GB are incommensurable in square, making the sum of the squares on them medial, and twice the (rectangle contained) by AG and GB medial, and the (sum of the squares) on AG and GB incommensurable with twice the (rectangle contained) by AG and GB [Prop. 10.78]. Therefore, let CH, equal to the (square) on AG, be applied to CD, producing CK as breadth, and KL, equal to the (square) on BG. Thus, the whole of CL is equal to the (sum of the squares) on AG and GB. CL [is] thus also medial. And it is applied to the rational (straight-line) CD, producing CM as breadth. Thus, CM is rational, and incommensurable in length with CD [Prop. 10.22]. Therefore, since CL is equal to the (sum of the squares) on AG and GB, of which CE (is) equal to the (square) on AB, the remainder FL is thus equal to twice the (rectangle contained) by AG and GB [Prop. 2.7]. And twice the (rectangle contained) by AG and GB (is) medial. Thus, FL is also medial. And it is applied to the rational (straight-line) FE, producing FM as breadth. FM is thus rational, and incommensurable in length with CD [Prop. 10.22]. And since the (sum of the squares) on AG and GB is incommensurable with twice the (rectangle contained) by AG and GB, and CL equal to the (sum of the squares) on AG and GB, and FL equal to twice the (rectangle contained) by AG and GB, CL [is] thus incommensurable with FL. And as CL (is) to FL, so CM is to MF [Prop. 6.1]. Thus, CM is

ὑπὸ τῶν ΑΗ, ΗΒ. καὶ ἐπεὶ αἱ ΑΗ, ΗΒ δυνάμει εἰσὶν ἀσύμμετροι, ἀσύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΗ τῷ ἀπὸ τῆς ΗΒ. ἀλλὰ τῷ μὲν ἀπὸ τῆς ΑΗ ἴσον ἐστὶ τὸ ΓΘ, τῷ δὲ ἀπὸ τῆς ΗΒ ἴσον ἐστὶ τὸ ΚΛ· ἀσύμμετρον ἄρα ἐστὶ τὸ ΓΘ τῷ ΚΛ. ὡς δὲ τὸ ΓΘ πρὸς τὸ ΚΛ, οὕτως ἐστὶν ἡ ΓΚ πρὸς τὴν ΚΜ· ἀσύμμετρος ἄρα ἐστὶν ἡ ΓΚ τῇ ΚΜ. καὶ ἐπεὶ τῶν ἀπὸ τῶν ΑΗ, ΗΒ μέσον ἀνάλογόν ἐστι τὸ ὑπὸ τῶν ΑΗ, ΗΒ, καὶ ἐστι τῷ μὲν ἀπὸ τῆς ΑΗ ἴσον τὸ ΓΘ, τῷ δὲ ἀπὸ τῆς ΗΒ ἴσον τὸ ΚΛ, τῷ δὲ ὑπὸ τῶν ΑΗ, ΗΒ ἴσον τὸ ΝΛ, καὶ τῶν ἄρα ΓΘ, ΚΛ μέσον ἀνάλογόν ἐστι τὸ ΝΛ· ὀέστιν ἄρα ὡς τὸ ΓΘ πρὸς τὸ ΝΛ, οὕτως τὸ ΝΛ πρὸς τὸ ΚΛ. καὶ διὰ τὰ αὐτὰ ἡ ΓΜ τῆς ΜΖ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καὶ οὐδετέρα αὐτῶν σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῇ τῇ ΓΔ· ἡ ΓΖ ἄρα ἀποτομή ἐστιν ὀέκτη· ὅπερ ἔδει δεῖχαι.

incommensurable in length with MF [Prop. 10.11]. And they are both rational. Thus, CM and MF are rational (straight-lines which are) commensurable in square only. CF is thus an apotome [Prop. 10.73]. So, I say that (it is) also a sixth (apotome).

For since FL is equal to twice the (rectangle contained) by AG and GB, let FM be cut in half at N, and let NO be drawn through N, parallel to CD. Thus, FO and NL are each equal to the (rectangle contained) by AG and GB. And since AG and GB are incommensurable in square, the (square) on AG is thus incommensurable with the (square) on GB. But, CH is equal to the (square) on AG, and KL is equal to the (square) on GB. Thus, CH is incommensurable with KL. And as CH (is) to KL, so CK is to KM [Prop. 6.1]. Thus, CK is incommensurable (in length) with KM [Prop. 10.11]. And since the (rectangle contained) by AG and GB is the mean proportional to the (squares) on AG and GB [Prop. 10.21 lem.], and CH is equal to the (square) on AG, and KL equal to the (square) on GB, and NL equal to the (rectangle contained) by AG and GB, NL is thus also the mean proportional to CH and KL. Thus, as CH is to NL, so NL (is) to KL. And for the same (reasons as the preceding propositions), the square on CM is greater than (the square on) MF by the (square) on (some straight-line) incommensurable (in length) with (CM) [Prop. 10.18]. And neither of them is commensurable with the (previously) laid down rational (straight-line) CD. Thus, CF is a sixth apotome [Def. 10.16]. (Which is) the very thing it was required to show.

103.

Ἡ τῇ ἀποτομῇ μήκει σύμμετρος ἀποτομή ἐστι καὶ τῇ τάχει ἡ αὐτή.

Ἔστω ἀποτομὴ ἡ ΑΒ, καὶ τῇ ΑΒ μήκει σύμμετροςὀέστω ἡ ΓΔ· λέγω, ὅτι καὶ ἡ ΓΔ ἀποτομή ἐστι καὶ τῇ τάχει ἡ αὐτὴ τῇ ΑΒ.

Proposition 103

A (straight-line) commensurable in length with an apotome is an apotome, and (is) the same in order.

Let AB be an apotome, and let CD be commensurable in length with AB. I say that CD is also an apotome, and (is) the same in order as AB.

Ἐπεὶ γὰρ ἀποτομή ἐστιν ἡ ΑΒ, ὀέστω αὐτῇ προσαρμόζουσα ἡ ΒΕ· αἱ ΑΕ, ΕΒ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι. καὶ τῷ τῆς ΑΒ πρὸς τὴν ΓΔ λόγῳ ὁ αὐτὸξ γεγονέτω ὁ τῆς

For since AB is an apotome, let BE be an attachment to it. Thus, AE and EB are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And let it be

BE πρὸς τὴν ΔZ· καὶ ὡς ὁὲν ἄρα πρὸς ὁέν, πάντα [ἐστὶ] πρὸς πάντα· ὀέστιν ἄρα καὶ ὡς ὅλη ἡ AE πρὸς ὅλην τὴν ΓZ, οὑτως ἡ AB πρὸς τὴν ΓΔ. σύμμετρος δὲ ἡ AB τῇ ΓΔ μήκει· σύμμετρος ἄρα καὶ ἡ AE μὲν τῇ ΓZ, ἡ δὲ BE τῇ ΔZ. καὶ αἱ AE, EB ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· καὶ αἱ ΓZ, ZΔ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι [ἀποτομὴ ἄρα ἐστὶν ἡ ΓΔ. λέγω δή, ὅτι καὶ τῇ τάχει ἡ αὐτὴ τῇ AB].

Ἐπεὶ οὐῦν ἐστιν ὡς ἡ AE πρὸς τὴν ΓZ, οὑτως ἡ BE πρὸς τὴν ΔZ, ἐναλλὰχ ἄρα ἐστὶν ὡς ἡ AE πρὸς τὴν EB, οὑτως ἡ ΓZ πρὸς τὴν ZΔ. ἦτοι δὴ ἡ AE τῆς EB μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ ὀὴ τῷ ἀπὸ ἀσυμμέτρου. εἰ μὲν οὐῦν ἡ AE τῆς EB μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ ἡ ΓZ τῆς ZΔ μεῖζον δυνήσεται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καὶ εἰ μὲν σύμμετρόξ ἐστιν ἡ AE τῇ ἐκκειμένῃ ὁρητῇ μήκει, καὶ ἡ ΓZ, εἰ δὲ ἡ BE, καὶ ἡ ΔZ, εἰ δὲ οὐδετέρα τῶν AE, EB, καὶ οὐδετέρα τῶν ΓZ, ZΔ. εἰ δὲ ἡ AE [τῆς EB] μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, καὶ ἡ ΓZ τῆς ZΔ μεῖζον δυνήσεται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καὶ εἰ μὲν σύμμετρόξ ἐστιν ἡ AE τῇ ἐκκειμένῃ ὁρητῇ μήκει, καὶ ἡ ΓZ, εἰ δὲ ἡ BE, καὶ ἡ ΔZ, εἰ δὲ οὐδετέρα τῶν AE, EB, οὐδετέρα τῶν ΓZ, ZΔ.

Ἀποτομὴ ἄρα ἐστὶν ἡ ΓΔ καὶ τῇ τάχει ἡ αὐτὴ τῇ AB· ὅπερ ἔδει δεῖχαι.

contrived that the (ratio) of BE to DF is the same as the ratio of AB to CD [Prop. 6.12]. Thus, also, as one is to one, (so) all [are] to all [Prop. 5.12]. And thus as the whole AE is to the whole CF, so AB (is) to CD. And AB (is) commensurable in length with CD. AE (is) thus also commensurable (in length) with CF, and BE with DF [Prop. 10.11]. And AE and BE are rational (straight-lines which are) commensurable in square only. Thus, CF and FD are also rational (straight-lines which are) commensurable in square only [Prop. 10.13]. [CD is thus an apotome. So, I say that (it is) also the same in order as AB.]

Therefore, since as AE is to CF, so BE (is) to DF, thus, alternately, as AE is to EB, so CF (is) to FD [Prop. 5.16]. So, the square on AE is greater than (the square on) EB either by the (square) on (some straight-line) commensurable, or by the (square) on (some straight-line) incommensurable, (in length) with (AE). Therefore, if the (square) on AE is greater than (the square on) EB by the (square) on (some straight-line) commensurable (in length) with (AE) then the square on CF will also be greater than (the square on) FD by the (square) on (some straight-line) commensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable in length with a (previously) laid down rational (straight-line) then so (is) CF [Prop. 10.12], and if BE (is commensurable), so (is) DF, and if neither of AE or EB (are commensurable), neither (are) either of CF or FD [Prop. 10.13]. And if the (square) on AE is greater [than (the square on) EB] by the (square) on (some straight-line) incommensurable (in length) with (AE) then the (square) on CF will also be greater than (the square on) FD by the (square) on (some straight-line) incommensurable (in length) with (CF) [Prop. 10.14]. And if AE is commensurable in length with a (previously) laid down rational (straight-line), so (is) CF [Prop. 10.12], and if BE (is commensurable), so (is) DF, and if neither of AE or EB (are commensurable), neither (are) either of CF or FD [Prop. 10.13].

Thus, CD is an apotome, and (is) the same in order as AB [Defs. 10.11—10.16]. (Which is) the very thing it was required to show.

104.

Ἡ τῇ μέσης ἀποτομῇ σύμμετρος μέσης ἀποτομή ἐστι καὶ τῇ τάχει ἡ αὐτή.

Proposition 104

A (straight-line) commensurable (in length) with an apotome of a medial (straight-line) is an apotome of a medial (straight-line), and (is) the same in order.

Ἔστω μέσης ἀποτομὴ ἡ AB, καὶ τῇ AB μήκει σύμμετρος ὁέστω ἡ $ΓΔ$· λέγω, ὅτι καὶ ἡ $ΓΔ$ μέσης ἀποτομή ἐστι καὶ τῇ τάχει ἡ αὐτὴ τῇ AB.

Ἐπεὶ γὰρ μέσης ἀποτομή ἐστιν ἡ AB, ὁέστω αὐτῇ προσαρμόζουσα ἡ EB. αἱ AE, EB ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι. καὶ γεγονέτω ὡς ἡ AB πρὸς τὴν $ΓΔ$, οὕτως ἡ BE πρὸς τὴν $ΔZ$· σύμμετρος ἄρα [ἐστὶ] καὶ ἡ AE τῇ $ΓZ$, ἡ δὲ BE τῇ $ΔZ$. αἱ δὲ AE, EB μέσαι εἰσὶ δυνάμει μόνον σύμμετροι· καὶ αἱ $ΓZ$, $ZΔ$ ἄρα μέσαι εἰσὶ δυνάμει μόνον σύμμετροι· μέσης ἄρα ἀποτομή ἐστιν ἡ $ΓΔ$. λέγω δή, ὅτι καὶ τῇ τάχει ἐστὶν ἡ αὐτὴ τῇ AB.

Ἐπεὶ [γάρ] ἐστιν ὡς ἡ AE πρὸς τὴν EB, οὕτως ἡ $ΓZ$ πρὸς τὴν $ZΔ$ [ἀλλὰ ὡς μὲν ἡ AE πρὸς τὴν EB, οὕτως τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AE, EB, ὡς δὲ ἡ $ΓZ$ πρὸς τὴν $ZΔ$, οὕτως τὸ ἀπὸ τῆς $ΓZ$ πρὸς τὸ ὑπὸ τῶν $ΓZ$, $ZΔ$], ὐέστιν ἄρα καὶ ὡς τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AE, EB, οὕτως τὸ ἀπὸ τῆς $ΓZ$ πρὸς τὸ ὑπὸ τῶν $ΓZ$, $ZΔ$ [καὶ ἐναλλὰχ ὡς τὸ ἀπὸ τῆς AE πρὸς τὸ ἀπὸ τῆς $ΓZ$, οὕτως τὸ ὑπὸ τῶν AE, EB πρὸς τὸ ὑπὸ τῶν $ΓZ$, $ZΔ$]. σύμμετρον δὲ τὸ ἀπὸ τῆς AE τῷ ἀπὸ τῆς $ΓZ$· σύμμετρον ἄρα ἐστὶ καὶ τὸ ὑπὸ τῶν AE, EB τῷ ὑπὸ τῶν $ΓZ$, $ZΔ$. εὐίτε οὐῦν ὀρητόν ἐστι τὸ ὑπὸ τῶν AE, EB, ὀρητὸνὀέσται καὶ τὸ ὑπὸ τῶν $ΓZ$, $ZΔ$, εὐίτε μέσον [ἐστὶ] τὸ ὑπὸ τῶν AE, EB, μέσον [ἐστὶ] καὶ τὸ ὑπὸ τῶν $ΓZ$, $ZΔ$.

Μέσης ἄρα ἀποτομή ἐστιν ἡ $ΓΔ$ καὶ τῇ τάχει ἡ αὐτὴ τῇ AB· ὅπερ ἔδει δεῖχαι.

Let AB be an apotome of a medial (straight-line), and let CD be commensurable in length with AB. I say that CD is also an apotome of a medial (straight-line), and (is) the same in order as AB.

For since AB is an apotome of a medial (straight-line), let EB be an attachment to it. Thus, AE and EB are medial (straight-lines which are) commensurable in square only [Props. 10.74, 10.75]. And let it be contrived that as AB is to CD, so BE (is) to DF [Prop. 6.12]. Thus, AE [is] also commensurable (in length) with CF, and BE with DF [Props. 5.12, 10.11]. And AE and EB are medial (straight-lines which are) commensurable in square only. CF and FD are thus also medial (straight-lines which are) commensurable in square only [Props. 10.23, 10.13]. Thus, CD is an apotome of a medial (straight-line) [Props. 10.74, 10.75]. So, I say that it is also the same in order as AB.

[For] since as AE is to EB, so CF (is) to FD [Props. 5.12, 5.16] [but as AE (is) to EB, so the (square) on AE (is) to the (rectangle contained) by AE and EB, and as CF (is) to FD, so the (square) on CF (is) to the (rectangle contained) by CF and FD], thus as the (square) on AE is to the (rectangle contained) by AE and EB, so the (square) on CF also (is) to the (rectangle contained) by CF and FD [Prop. 10.21 lem.] [and, alternately, as the (square) on AE (is) to the (square) on CF, so the (rectangle contained) by AE and EB (is) to the (rectangle contained) by CF and FD]. And the (square) on AE (is) commensurable with the (square) on CF. Thus, the (rectangle contained) by AE and EB is also commensurable with the (rectangle contained) by CF and FD [Props. 5.16, 10.11]. Therefore, either the (rectangle contained) by AE and EB is rational, and the (rectangle contained) by CF and FD will also be rational [Def. 10.4], or the (rectangle contained) by AE and EB [is] medial, and the (rectangle contained) by CF and FD [is] also medial [Prop. 10.23 corr.].

Therefore, CD is the apotome of a medial (straight-line), and is the same in order as AB [Props. 10.74, 10.75]. (Which is) the very thing it was required to show.

Ἡ τῇ ἐλάσσονι σύμμετρος ἐλάσσων ἐστίν.

Ἔστω γὰρ ἐλάσσων ἡ AB καὶ τῇ AB σύμμετρος ἡ $ΓΔ$· λέγω, ὅτι καὶ ἡ $ΓΔ$ ἐλάσσων ἐστίν.

A (straight-line) commensurable (in length) with a minor (straight-line) is a minor (straight-line).

For let AB be a minor (straight-line), and (let) CD (be) commensurable (in length) with AB. I say that CD is also a minor (straight-line).

Γεγονέτω γὰρ τὰ αὐτά· καὶ ἐπεὶ αἱ AE, EB δυνάμει εἰσὶν ἀσύμμετροι, καὶ αἱ $ΓZ$, $ZΔ$ ἄρα δυνάμει εἰσὶν ἀσύμμετροι. ἐπεὶ οὐκοῦν ἐστιν ὡς ἡ AE πρὸς τὴν EB, οὕτως ἡ $ΓZ$ πρὸς τὴν $ZΔ$, ὐέστιν ἄρα καὶ ὡς τὸ ἀπὸ τῆς AE πρὸς τὸ ἀπὸ τῆς EB, οὕτως τὸ ἀπὸ τῆς $ΓZ$ πρὸς τὸ ἀπὸ τῆς $ZΔ$. συνθέντι ἄρα ἐστὶν ὡς τὰ ἀπὸ τῶν AE, EB πρὸς τὸ ἀπὸ τῆς EB, οὕτως τὰ ἀπὸ τῶν $ΓZ$, $ZΔ$ πρὸς τὸ ἀπὸ τῆς $ZΔ$ [καὶ ἐναλλάχ]· σύμμετρον δέ ἐστι τὸ ἀπὸ τῆς BE τῷ ἀπὸ τῆς $ΔZ$· σύμμετρον ἄρα καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE, EB τετραγώνων τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν $ΓZ$, $ZΔ$ τετραγώνων. ὁρητὸν δέ ἐστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE, EB τετραγώνων· ὁρητὸν ἄρα ἐστὶ καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν $ΓZ$, $ZΔ$ τετραγώνων. πάλιν, ἐπεί ἐστιν ὡς τὸ ἀπὸ τῆς AE πρὸς τὸ ὑπὸ τῶν AE, EB, οὕτως τὸ ἀπὸ τῆς $ΓZ$ πρὸς τὸ ὑπὸ τῶν $ΓZ$, $ZΔ$, σύμμετρον δὲ τὸ ἀπὸ τῆς AE τετράγωνον τῷ ἀπὸ τῆς $ΓZ$ τετραγώνῳ, σύμμετρον ἄρα ἐστὶ καὶ τὸ ὑπὸ τῶν AE, EB τῷ ὑπὸ τῶν $ΓZ$, $ZΔ$. μέσον δὲ τὸ ὑπὸ τῶν AE, EB· μέσον ἄρα καὶ τὸ ὑπὸ τῶν $ΓZ$, $ZΔ$· αἱ $ΓZ$, $ZΔ$ ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων ὁρητόν, τὸ δὸ ὑπὸ αὐτῶν μέσον.

Ἐλάσσων ἄρα ἐστὶν ἡ $ΓΔ$· ὅπερ ἔδει δεῖχαι.

For let the same things be contrived (as in the former proposition). And since AE and EB are (straight-lines which are) incommensurable in square [Prop. 10.76], CF and FD are thus also (straight-lines which are) incommensurable in square [Prop. 10.13]. Therefore, since as AE is to EB, so CF (is) to FD [Props. 5.12, 5.16], thus also as the (square) on AE is to the (square) on EB, so the (square) on CF (is) to the (square) on FD [Prop. 6.22]. Thus, via composition, as the (sum of the squares) on AE and EB is to the (square) on EB, so the (sum of the squares) on CF and FD (is) to the (square) on FD [Prop. 5.18], [also alternately]. And the (square) on BE is commensurable with the (square) on DF [Prop. 10.104]. The sum of the squares on AE and EB (is) thus also commensurable with the sum of the squares on CF and FD [Prop. 5.16, 10.11]. And the sum of the (squares) on AE and EB is rational [Prop. 10.76]. Thus, the sum of the (squares) on CF and FD is also rational [Def. 10.4]. Again, since as the (square) on AE is to the (rectangle contained) by AE and EB, so the (square) on CF (is) to the (rectangle contained) by CF and FD [Prop. 10.21 lem.], and the square on AE (is) commensurable with the square on CF, the (rectangle contained) by AE and EB is thus also commensurable with the (rectangle contained) by CF and FD. And the (rectangle contained) by AE and EB (is) medial [Prop. 10.76]. Thus, the (rectangle contained) by CF and FD (is) also medial [Prop. 10.23 corr.]. CF and FD are thus (straight-lines which are) incommensurable in square, making the sum of the squares on them rational, and the (rectangle contained) by them medial.

Thus, CD is a minor (straight-line) [Prop. 10.76]. (Which is) the very thing it was required to show.

106.

Ἡ τῇ μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιούσῃ σύμμετρος μετὰ

Proposition 106

A (straight-line) commensurable (in length) with a (straight-

ὁρητοῦ μέσον τὸ ὅλον ποιοῦσά ἐστιν.

Ἔστω μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσα ἡ AB καὶ τῇ AB σύμμετρος ἡ $ΓΔ$· λέγω, ὅτι καὶ ἡ $ΓΔ$ μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσά ἐστιν.

Ἔστω γὰρ τῇ AB προσαρμόζουσα ἡ BE· αἱ AE, EB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν AE, EB τετραγώνων μέσον, τὸ δὸ ὑπὸ αὐτῶν ὁρητόν. καὶ τὰ αὐτὰ κατεσκευάσθω. ὁμοίως δὴ δείχομεν τοῖς πρότερον, ὅτι αἱ $ΓZ$, $ZΔ$ ἐν τῷ αὐτῷ λόγῳ εἰσὶ ταῖς AE, EB, καὶ σύμμετρόν ἐστι τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE, EB τετραγώνων τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν $ΓZ$, $ZΔ$ τετραγώνων, τὸ δὲ ὑπὸ τῶν AE, EB τῷ ὑπὸ τῶν $ΓZ$, $ZΔ$· ὥστε καὶ αἱ $ΓZ$, $ZΔ$ δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τὸ μὲν συγκείμενον ἐκ τῶν ἀπὸ τῶν $ΓZ$, $ZΔ$ τετραγώνων μέσον, τὸ δὸ ὑπὸ αὐτῶν ὁρητόν.

line) which with a rational (area) makes a medial whole is a (straight-line) which with a rational (area) makes a medial whole.

Let AB be a (straight-line) which with a rational (area) makes a medial whole, and (let) CD (be) commensurable (in length) with AB. I say that CD is also a (straight-line) which with a rational (area) makes a medial (whole).

For let BE be an attachment to AB. Thus, AE and EB are (straight-lines which are) incommensurable in square, making the sum of the squares on AE and EB medial, and the (rectangle contained) by them rational [Prop. 10.77]. And let the same construction be made (as in the previous propositions). So, similarly to the previous (propositions), we can show that CF and FD are in the same ratio as AE and EB, and the sum of the squares on AE and EB is commensurable with the sum of the squares on CF and FD, and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD. Hence, CF and FD are also (straight-lines which are) incommensurable in square, making the sum of the squares on CF and FD medial, and the (rectangle contained) by them rational.

Ἡ $ΓΔ$ ἄρα μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσά ἐστιν· ὅπερ ἔδει δεῖξαι.

CD is thus a (straight-line) which with a rational (area) makes a medial whole [Prop. 10.77]. (Which is) the very thing it was required to show.

107.

Ἡ τῇ μετὰ μέσου μέσον τὸ ὅλον ποιούσῃ σύμμετρος καὶ αὐτὴ μετὰ μέσου μέσον τὸ ὅλον ποιοῦσά ἐστιν.

Proposition 107

A (straight-line) commensurable (in length) with a (straight-line) which with a medial (area) makes a medial whole is itself also a (straight-line) which with a medial (area) makes a medial whole.

Ἔστω μετὰ μέσου μέσον τὸ ὅλον ποιοῦσα ἡ AB, καὶ τῇ ABὸέστω σύμμετρος ἡ $ΓΔ$· λέγω, ὅτι καὶ ἡ $ΓΔ$ μετὰ μέσου μέσον τὸ ὅλον ποιοῦσά ἐστιν.

Ἔστω γὰρ τῇ AB προσαρμόζουσα ἡ BE, καὶ τὰ αὐτὰ κατεσκευάσθω· αἱ AE, EB ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τό τε συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπὸ αὐτῶν μέσον καὶ ὀέτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων τῷ ὑπὸ αὐτῶν. καί εἰσιν, ὡξ ἐδείθθη, αἱ AE, EB σύμμετροι ταῖς $ΓZ$, $ZΔ$, καὶ τὸ συγκείμενον ἐκ τῶν ἀπὸ τῶν AE, EB τετραγώνων τῷ συγκειμένῳ ἐκ τῶν ἀπὸ τῶν $ΓZ$, $ZΔ$, τὸ δὲ ὑπὸ τῶν AE, EB τῷ ὑπὸ τῶν $ΓZ$, $ZΔ$· καὶ αἱ $ΓZ$, $ZΔ$ ἄρα δυνάμει εἰσὶν ἀσύμμετροι ποιοῦσαι τό τε συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν τετραγώνων μέσον καὶ τὸ ὑπὸ ἃ ὐτῶν μέσον καὶ ὀέτι ἀσύμμετρον τὸ συγκείμενον ἐκ τῶν ἀπὸ αὐτῶν [τετραγώνων] τῷ ὑπὸ αὐτῶν.

Ἡ $ΓΔ$ ἄρα μετὰ μέσου μέσον τὸ ὅλον ποιοῦσά ἐστιν· ὅπερ ἔδει δεῖχαι.

Let AB be a (straight-line) which with a medial (area) makes a medial whole, and let CD be commensurable (in length) with AB. I say that CD is also a (straight-line) which with a medial (area) makes a medial whole.

For let BE be an attachment to AB. And let the same construction be made (as in the previous propositions). Thus, AE and EB are (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, further, the sum of the squares on them incommensurable with the (rectangle contained) by them [Prop. 10.78]. And, as was shown (previously), AE and EB are commensurable (in length) with CF and FD (respectively), and the sum of the squares on AE and EB with the sum of the squares on CF and FD, and the (rectangle contained) by AE and EB with the (rectangle contained) by CF and FD. Thus, CF and FD are also (straight-lines which are) incommensurable in square, making the sum of the squares on them medial, and the (rectangle contained) by them medial, and, further, the sum of the [squares] on them incommensurable with the (rectangle contained) by them.

Thus, CD is a (straight-line) which with a medial (area) makes a medial whole [Prop. 10.78]. (Which is) the very thing it was required to show.

108.

Ἀπὸ ὀρητοῦ μέσου ἀφαιρουμένου ἡ τὸ λοιπὸν θωρίον δυναμένη μία δύο ἀλόγων γίνεται ἤτοι ἀποτομὴ ὀὴ ἐλάσσων.

Proposition 108

A medial (area) being subtracted from a rational (area), one of two irrational (straight-lines) arise (as) the square-root of the remaining area—either an apotome, or a minor (straight-line).

Ἀπὸ γὰρ ὁρητοῦ τοῦ ΒΓ μέσον ἀφῃρήσθω τὸ ΒΔ· λέγω, ὅτι ἡ τὸ λοιπὸν δυναμένη τὸ ΕΓ μία δύο ἀλόγων γίνεται ἤτοι ἀποτομὴ ὀὴ ἐλάσσων.

Ἐκκείσθω γὰρ ὁρητὴ ἡ ΖΗ, καὶ τῷ μὲν ΒΓ ἴσον παρὰ τὴν ΖΗ παραβεβλήσθω ὀρθογώνιον παραλληλόγραμμον τὸ ΗΘ, τῷ δὲ ΔΒ ἴσον ἀφῃρήσθω τὸ ΗΚ· λοιπὸν ἄρα τὸ ΕΓ ἴσον ἐστὶ τῷ ΛΘ. ἐπεὶ οὐῦν ὁρητόν μέν ἐστι τὸ ΒΓ, μέσον δὲ τὸ ΒΔ, ἴσον δὲ τὸ μὲν ΒΓ τῷ ΗΘ, τὸ δὲ ΒΔ τῷ ΗΚ, ὁρητὸν μὲν ἄρα ἐστὶ τὸ ΗΘ, μέσον δὲ τὸ ΗΚ. καὶ παρὰ ὁρητὴν τὴν ΖΗ παράκειται· ὁρητὴ μὲν ἄρα ἡ ΖΘ καὶ σύμμετρος τῇ ΖΗ μήκει, ὁρητὴ δὲ ἡ ΖΚ καὶ ἀσύμμετρος τῇ ΖΗ μήκει· ἀσύμμετρος ἄρα ἐστὶν ἡ ΖΘ τῇ ΖΚ μήκει. αἱ ΖΘ, ΖΚ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ ΚΘ, προσαρμόζουσα δὲ αὐτῇ ἡ ΚΖ. ἤτοι δὴ ἡ ΘΖ τῆς ΖΚ μεῖζον δύναται τῷ ἀπὸ συμμέτρουὸὴ οὐύ.

Δυνάσθω πρότερον τῷ ἀπὸ συμμέτρου. καί ἐστιν ὅλη ἡ ΘΖ σύμμετρος τῇ ἐκκειμένῃ ὁρητῇ μήκει τῇ ΖΗ· ἀποτομὴ ἄρα πρώτη ἐστὶν ἡ ΚΘ. τὸ δὸ ὑπὸ ὁρητῆς καὶ ἀποτομῆς πρώτης περιεχόμενον ἡ δυναμένη ἀποτομή ἐστιν. ἡ ἄρα τὸ ΛΘ, τουτέστι τὸ ΕΓ, δυναμένη ἀποτομή ἐστιν.

Εἰ δὲ ἡ ΘΖ τῆς ΖΚ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, καί ἐστιν ὅλη ἡ ΖΘ σύμμετρος τῇ ἐκκειμένῃ ὁρητῇ μήκει τῇ ΖΗ, ἀποτομὴ τετάρτη ἐστὶν ἡ ΚΘ. τὸ δὸ ὑπὸ ὁρητῆς καὶ ἀποτομῆς τετάρτης περιεχόμενον ἡ δυναμένη ἐλάσσων ἐστίν· ὅπερ ἔδει δεῖχαι.

For let the medial (area) BD be subtracted from the rational (area) BC. I say that one of two irrational (straight-lines) arise (as) the square-root of the remaining (area), EC—either an apotome, or a minor (straight-line).

For let the rational (straight-line) FG be laid out, and let the right-angled parallelogram GH, equal to BC, be applied to FG, and let GK, equal to DB, be subtracted (from GH). Thus, the remainder EC is equal to LH. Therefore, since BC is a rational (area), and BD a medial (area), and BC (is) equal to GH, and BD to GK, GH is thus a rational (area), and GK a medial (area). And they are applied to the rational (straight-line) FG. Thus, FH (is) rational, and commensurable in length with FG [Prop. 10.20], and FK (is) also rational, and incommensurable in length with FG [Prop. 10.22]. Thus, FH is incommensurable in length with FK [Prop. 10.13]. FH and FK are thus rational (straight-lines which are) commensurable in square only. Thus, KH is an apotome [Prop. 10.73], and KF an attachment to it. So, the square on HF is greater than (the square on) FK by the (square) on (some straight-line which is) either commensurable, or not (commensurable), (in length with HF).

First, let the square (on it) be (greater) by the (square) on (some straight-line which is) commensurable (in length with HF). And the whole of HF is commensurable in length with the (previously) laid down rational (straight-line) FG. Thus, KH is a first apotome [Def. 10.1]. And the square-root of an (area) contained by a rational (straight-line) and a first apotome is an apotome [Prop. 10.91]. Thus, the square-root of LH—that is to say, (of) EC—is an apotome.

And if the square on HF is greater than (the square on) FK by the (square) on (some straight-line which is) incommensurable (in length) with (HF), and (since) the whole of FH is commensurable in length with the (previously) laid down rational (straight-line) FG, KH is a fourth apotome [Prop. 10.14]. And the square-root of an (area) contained

109.

Ἀπὸ μέσου ὁρητοῦ ἀφαιρουμένουὀάλλαι δύοὀάλογοι γίνονται ἤτοι μέσης ἀποτομὴ πρώτηὀὴ μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσα.

Ἀπὸ γὰρ μέσου τοῦ ΒΓ ὁρητὸν ἀφῃρήσθω τὸ ΒΔ. λέγω, ὅτι ἡ τὸ λοιπὸν τὸ ΕΓ δυναμένη μία δύο ἀλόγων γίνεται ἤτοι μέσης ἀποτομὴ πρώτηὀὴ μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσα.

Ἐκκείσθω γὰρ ὁρητὴ ἡ ΖΗ, καὶ παραβεβλήσθω ὁμοίως τὰ θωρία. ὀέστι δὴ ἀκολούθως ὁρητὴ μὲν ἡ ΖΘ καὶ ἀσύμμετρος τῇ ΖΗ μήκει, ὁρητὴ δὲ ἡ ΚΖ καὶ σύμμετρος τῇ ΖΗ μήκει· αἱ ΖΘ, ΖΚ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ ΚΘ, προσαρμόζουσα δὲ ταύτῃ ἡ ΖΚ. ἤτοι δὴ ἡ ΘΖ τῆς ΖΚ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ ὀὴ τῷ ἀπὸ ἀσυμμέτρου.

by a rational (straight-line) and a fourth apotome is a minor (straight-line) [Prop. 10.94]. (Which is) the very thing it was required to show.

Proposition 109

A rational (area) being subtracted from a medial (area), two other irrational (straight-lines) arise (as the square-root of the remaining area)—either a first apotome of a medial (straight-line), or that (straight-line) which with a rational (area) makes a medial whole.

For let the rational (area) BD be subtracted from the medial (area) BC. I say that one of two irrational (straight-lines) arise (as) the square-root of the remaining (area), EC—either a first apotome of a medial (straight-line), or that (straight-line) which with a rational (area) makes a medial whole.

For let the rational (straight-line) FG be laid down, and let similar areas (to the preceding proposition) have been applied (to it). So, accordingly, FH is rational, and incommensurable in length with FG, and KF (is) also rational, and commensurable in length with FG. Thus, FH and FK are rational (straight-lines which are) commensurable in square only [Prop. 10.13]. KH is thus an apotome [Prop. 10.73], and FK an attachment to it. So, the square on HF is greater than (the square on) FK either by the (square) on (some straight-line) commensurable (in length) with (HF), or by the (square) on (some straight-line) incommensurable (in length with HF).

Εἰ μὲν οὐῦν ἡ ΘΖ τῆς ΖΚ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καί ἐστιν ἡ προσαρμόζουσα ἡ ΖΚ σύμμετρος τῇ ἐκκειμένῃ ὁρητῇ μήκει τῇ ΖΗ, ἀποτομὴ δευτέρα ἐστὶν ἡ ΚΘ. ὁρητὴ δὲ ἡ ΖΗ· ὥστε ἡ τὸ ΛΘ, τουτέστι τὸ ΕΓ, δυναμένη μέσης ἀποτομὴ πρώτη ἐστίν.

Εἰ δὲ ἡ ΘΖ τῆς ΖΚ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου, καί ἐστιν ἡ προσαρμόζουσα ἡ ΖΚ σύμμετρος τῇ ἐκκειμένῃ ὁρητῇ μήκει τῇ ΖΗ, ἀποτομὴ πέμπτη ἐστὶν ἡ ΚΘ· ὥστε ἡ τὸ ΕΓ δυναμένη μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσά ἐστιν· ὅπερ ἔδει δεῖξαι.

Therefore, if the square on HF is greater than (the square on) FK by the (square) on (some straight-line) commensurable (in length) with (HF), and (since) the attachment FK is commensurable in length with the (previously) laid down rational (straight-line) FG, KH is a second apotome [Def. 10.12]. And FG (is) rational. Hence, the square-root of LH—that is to say, (of) EC—is a first apotome of a medial (straight-line) [Prop. 10.92].

And if the square on HF is greater than (the square on) FK by the (square) on (some straight-line) incommensurable (in length with HF), and (since) the attachment FK is commensurable in length with the (previously) laid down rational (straight-line) FG, KH is a fifth apotome [Def. 10.15]. Hence, the square-root of EC is that (straight-line) which with a rational (area) makes a medial whole [Prop. 10.95]. (Which is) the very thing it was required to show.

110.

Ἀπὸ μέσου μέσου ἀφαιρουμένου ἀσυμμέτρου τῷ ὅλῳ α-

Proposition 110

A medial (area), incommensurable with the whole, being

αἱ λοιπαὶ δύο ἀλογοι γίνονται ἤτοι μέσης ἀποτομὴ δευτέρα ἢ μετὰ μέσου μέσον τὸ ὅλον ποιοῦσα.

Ἀφῃρήσθω γὰρ ὡς ἐπὶ τῶν προκειμένων καταγραφῶν ἀπὸ μέσου τοῦ ΒΓ μέσον τὸ ΒΔ ἀσύμμετρον τῷ ὅλῳ· λέγω, ὅτι ἡ τὸ ΕΓ δυναμένη μία ἐστὶ δύο ἀλόγων ἤτοι μέσης ἀποτομὴ δευτέρα ἢ μετὰ μέσου μέσον τὸ ὅλον ποιοῦσα.

subtracted from a medial (area), the two remaining irrational (straight-lines) arise (as) the (square-root of the area)—either a second apotome of a medial (straight-line), or that (straight-line) which with a medial (area) makes a medial whole.

For, as in the previous figures, let the medial (area) *BD*, incommensurable with the whole, be subtracted from the medial (area) *BC*. I say that the square-root of *EC* is one of two irrational (straight-lines)—either a second apotome of a medial (straight-line), or that (straight-line) which with a medial (area) makes a medial whole.

Ἐπεὶ γὰρ μέσον ἐστὶν ἑκάτερον τῶν ΒΓ, ΒΔ, καὶ ἀσύμμετρον τὸ ΒΓ τῷ ΒΔ, ὀέσται ἀκολούθως ὁρητὴ ἑκατέρα τῶν ΖΘ, ΖΚ καὶ ἀσύμμετρος τῇ ΖΗ μήκει. καὶ ἐπεὶ ἀσύμμετρόν ἐστι τὸ ΒΓ τῷ ΒΔ, τουτέστι τὸ ΗΘ τῷ ΗΚ, ἀσύμμετρος καὶ ἡ ΘΖ τῇ ΖΚ· αἱ ΖΘ, ΖΚ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ ΚΘ [προσαρμόζουσα δὲ ἡ ΖΚ. ἤτοι δὴ ἡ ΖΘ τῆς ΖΚ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἢ τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ].

Εἰ μὲν δὴ ἡ ΖΘ τῆς ΖΚ μεῖζον δύναται τῷ ἀπὸ συμμέτρου

For since *BC* and *BD* are each medial (areas), and *BC* (is) incommensurable with *BD*, accordingly, FH and FK will each be rational (straight-lines), and incommensurable in length with FG [Prop. 10.22]. And since *BC* is incommensurable with *BD*—that is to say, GH with GK—HF (is) also incommensurable (in length) with FK [Props. 6.1, 10.11]. Thus, FH and FK are rational (straight-lines which are) commensurable in square only. KH is thus as apotome [Prop. 10.73], [and FK an attachment (to it). So, the square on FH is greater than (the

ἑαυτῇ, καὶ οὐθετέρα τῶν ΖΘ, ΖΚ σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῇ μήκει τῇ ΖΗ, ἀποτομὴ τρίτη ἐστὶν ἡ ΚΘ. ὁρητὴ δὲ ἡ ΚΛ, τὸ δὸ ὑπὸ ὁρητῆς καὶ ἀποτομῆς τρίτης περιεχόμενον ὀρθογώνιονὸάλογόν ἐστιν, καὶ ἡ δυναμένη αὐτὸ ὀάλογόξ ἐστιν, καλεῖται δὲ μέσης ἀποτομὴ δευτέρα· ὥστε ἡ τὸ ΛΘ, τουτέστι τὸ ΕΓ, δυναμένη μέσης ἀποτομή ἐστι δευτερά.

Εἰ δὲ ἡ ΖΘ τῆς ΖΚ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ [μήκει], καὶ οὐθετέρα τῶν ΘΖ, ΖΚ σύμμετρόξ ἐστι τῇ ΖΗ μήκει, ἀποτομὴ ὁέκτη ἐστὶν ἡ ΚΘ. τὸ δὸ ὑπὸ ὁρητῆς καὶ ἀποτομῆς ὁέκτης ἡ δυναμένη ἐστὶ μετὰ μέσου μέσον τὸ ὅλον ποιοῦσα. ἡ τὸ ΛΘ ἄρα, τουτέστι τὸ ΕΓ, δυναμένη μετὰ μέσου μέσον τὸ ὅλον ποιοῦσά ἐστιν· ὅπερ ἔδει δεῖχαι.

square on) FK either by the (square) on (some straight-line) commensurable, or by the (square) on (some straight-line) incommensurable, (in length) with (FH).]

So, if the square on FH is greater than (the square on) FK by the (square) on (some straight-line) commensurable (in length) with (FH), and (since) neither of FH and FK is commensurable in length with the (previously) laid down rational (straight-line) FG, KH is a third apotome [Def. 10.3]. And KL (is) rational. And the rectangle contained by a rational (straight-line) and a third apotome is irrational, and the square-root of it is that irrational (straight-line) called a second apotome of a medial (straight-line) [Prop. 10.93]. Hence, the square-root of LH—that is to say, (of) EC—is a second apotome of a medial (straight-line).

And if the square on FH is greater than (the square on) FK by the (square) on (some straight-line) incommensurable [in length] with (FH), and (since) neither of HF and FK is commensurable in length with FG, KH is a sixth apotome [Def. 10.16]. And the square-root of the (rectangle contained) by a rational (straight-line) and a sixth apotome is that (straight-line) which with a medial (area) makes a medial whole [Prop. 10.96]. Thus, the square-root of LH—that is to say, (of) EC—is that (straight-line) which with a medial (area) makes a medial whole. (Which is) the very thing it was required to show.

111.

Ἡ ἀποτομὴ οὐκὸέστιν ἡ αὐτὴ τῇ ἐκ δύο ὀνομάτων.

Proposition 111

An apotome is not the same as a binomial.

Ἔστω ἀποτομὴ ἡ AB· λέγω, ὅτι ἡ AB οὐκὸέστιν ἡ αὐτὴ τῇ ἐκ δύο ὀνομάτων.

Εἰ γὰρ δυνατόν, ὀέστω· καὶ ἐκκείσθω ὁρητὴ ἡ $ΔΓ$, καὶ τῷ ἀπὸ τῆς AB ἴσον παρὰ τὴν $ΓΔ$ παραβεβλήσθω ὀρθογώνιον τὸ $ΓΕ$ πλάτος ποιοῦν τὴν $ΔΕ$. ἐπεὶ οὐῦν ἀποτομή ἐστιν ἡ AB, ἀποτομὴ πρώτη ἐστὶν ἡ $ΔΕ$. ὀέστω αὐτῇ προσαρμόζουσα ἡ EZ· αἱ $ΔZ$, ZE ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ $ΔZ$ τῆς ZE μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ ἡ $ΔZ$ σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῇ μήκει τῇ $ΔΓ$.

Let AB be an apotome. I say that AB is not the same as a binomial.

For, if possible, let it be (the same). And let a rational (straight-line) DC be laid down. And let the rectangle CE, equal to the (square) on AB, be applied to CD, producing DE as breadth. Therefore, since AB is an apotome, DE is a first apotome [Prop. 10.97]. Let EF be an attachment to it. Thus, DF and FE are rational (straight-lines which are) commensurable in square only, and the square on DF is greater than (the

πάλιν, ἐπεὶ ἐκ δύο ὀνομάτων ἐστὶν ἡ AB, ἐκ δύο ἄρα ὀνομάτων πρώτη ἐστὶν ἡ ΔE. διῃρήσθω εἰς τὰ ὀνόματα κατὰ τὸ H, καὶ ὀέστω μεῖζον ὄνομα τὸ ΔH· αἱ ΔH, HE ἄρα ὀρηταί εἰσι δυνάμει μόνον σύμμετροι, καὶ ἡ ΔH τῆς HE μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ τὸ μεῖζον ἡ ΔH σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὀρητῇ μήκει τῇ $\Delta\Gamma$. καὶ ἡ ΔZ ἄρα τῇ ΔH σύμμετρόξ ἐστι μήκει· καὶ λοιπὴ ἄρα ἡ HZ σύμμετρόξ ἐστι τῇ ΔZ μήκει. [ἐπεὶ οὖν σύμμετρόξ ἐστιν ἡ ΔZ τῇ HZ, ὀρητὴ δέ ἐστιν ἡ ΔZ, ὀρητὴ ἄρα ἐστὶ καὶ ἡ HZ. ἐπεὶ οὐῦν σύμμετρόξ ἐστιν ἡ ΔZ τῇ HZ μήκει] ἀσύμμετρος δὲ ἡ ΔZ τῇ EZ μήκει. ἀσύμμετρος ἄρα ἐστὶ καὶ ἡ ZH τῇ EZ μήκει. αἱ HZ, ZE ἄρα ὀρηταί [εἰσι] δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ EH. ἀλλὰ καὶ ὀρητή· ὅπερ ἐστὶν ἀδύνατον.

Ἡ ἄρα ἀποτομὴ οὐκὸέστιν ἡ αὐτὴ τῇ ἐκ δύο ὀνομάτων· ὅπερ ἔδει δεῖχαι.

square on) FE by the (square) on (some straight-line) commensurable (in length) with (DF), and DF is commensurable in length with the (previously) laid down rational (straight-line) DC [Def. 10.10]. Again, since AB is a binomial, DE is thus a first binomial [Prop. 10.60]. Let (DE) be divided into its (component) terms at G, and let DG be the greater term. Thus, DG and GE are rational (straight-lines which are) commensurable in square only, and the square on DG is greater than (the square on) GE by the (square) on (some straight-line) commensurable (in length) with (DG), and the greater (term) DG is commensurable in length with the (previously) laid down rational (straight-line) DC [Def. 10.5]. Thus, DF is also commensurable in length with DG [Prop. 10.12]. The remainder GF is thus commensurable in length with DF [Prop. 10.15]. [Therefore, since DF is commensurable with GF, and DF is rational, GF is thus also rational. Therefore, since DF is commensurable in length with GF,] DF (is) incommensurable in length with EF. Thus, FG is also incommensurable in length with EF [Prop. 10.13]. GF and FE [are] thus rational (straight-lines which are) commensurable in square only. Thus, EG is an apotome [Prop. 10.73]. But, (it is) also rational. The very thing is impossible.

Thus, an apotome is not the same as a binomial. (Which is) the very thing it was required to show.

[Πόρισμα.]

Ἡ ἀποτομὴ καὶ αἱ μετὸ αὐτὴνὀάλογοι οοῦτε τῇ μέσῃ οοῦτε ἀλλήλαις εἰσὶν αἱ αὐταί.

Τὸ μὲν γὰρ ἀπὸ μέσης παρὰ ὀρητὴν παραβαλλόμενον πλάτος ποιεῖ ὀρητὴν καὶ ἀσύμμετρον τῇ, παρὸ ἣν παράκειται, μήκει, τὸ δὲ ἀπὸ ἀποτομῆς παρὰ ὀρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πρώτην, τὸ δὲ ἀπὸ μέσης ἀποτομῆς πρώτης παρὰ ὀρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν δευτέραν, τὸ δὲ ἀπὸ μέσης ἀποτομῆς δευτέρας παρὰ ὀρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τρίτην, τὸ δὲ ἀπὸ ἐλάσσονος παρὰ ὀρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν τετάρτην, τὸ δὲ ἀπὸ τῆς μετὰ ὀρητοῦ μέσον τὸ ὅλον ποιούσης παρὰ ὀρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πέμπτην, τὸ δὲ ἀπὸ τῆς μετὰ μέσου μέσον τὸ ὅλον ποιούσης παρὰ ὀρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν ὀέκτην. ἐπεὶ οὐῦν τὰ εἰρημένα πλάτη διαφέρει τοῦ τε πρώτου καὶ ἀλλήλων, τοῦ μὲν πρώτου, ὅτι ὀρητή ἐστιν, ἀλλήλων δὲ, ἐπεὶ τῇ τάχει οὐκ εἰσὶν αἱ αὐταί, δῆλον, ὡξ καὶ αὐταὶ αἱ ὀάλογοι διαφέρουσιν ἀλλήλων. καὶ ἐπεὶ δέδεικται ἡ ἀποτομὴ οὐκ οὐῦσα ἡ αὐτὴ τῇ ἐκ δύο ὀνομάτων, ποιοῦσι δὲ πλάτη παρὰ ὀρητὴν παραβαλλόμεναι αἱ μετὰ τὴν ἀποτομὴν ἀποτομὰξ ἀκολούθως ἑκάστη τῇ τάχει τῇ καθὸ αὐτήν, αἱ δὲ μετὰ τὴν ἐκ δύο ὀνομάτων τὰξ ἐκ δύο ὀνομάτων καὶ αὐταὶ τῇ τάχει ἀκολούθως, ὀέτεραι ἄρα εἰσὶν αἱ μετὰ τὴν ἀποτομὴν καὶ ὀέτεραι αἱ μετὰ τὴν ἐκ δύο ὀνομάτων, ὡξ εοῖναι τῇ τάχει πάσας ἀλόγους ιγ,

[Corollary]

The apotome and the irrational (straight-lines) after it are neither the same as a medial (straight-line) nor (the same) as one another.

For the (square) on a medial (straight-line), applied to a rational (straight-line), produces as breadth a rational (straight-line which is) incommensurable in length with the (straight-line) to which (the area) is applied [Prop. 10.22]. And the (square) on an apotome, applied to a rational (straight-line), produces as breadth a first apotome [Prop. 10.97]. And the (square) on a first apotome of a medial (straight-line), applied to a rational (straight-line), produces as breadth a second apotome [Prop. 10.98]. And the (square) on a second apotome of a medial (straight-line), applied to a rational (straight-line), produces as breadth a third apotome [Prop. 10.99]. And (square) on a minor (straight-line), applied to a rational (straight-line), produces as breadth a fourth apotome [Prop. 10.100]. And (square) on that (straight-line) which with a rational (area) produces a medial whole, applied to a rational (straight-line), produces as breadth a fifth apotome [Prop. 10.101]. And (square) on that (straight-line) which with a medial (area) produces a medial whole, applied to a rational (straight-line), produces as breadth a sixth apotome [Prop. 10.102]. Therefore, since the aforementioned breadths differ from the first (breadth), and from one another—from the first, because it is rational, and

from one another since they are not the same in order—clearly, the irrational (straight-lines) themselves also differ from one another. And since it has been shown that an apotome is not the same as a binomial [Prop. 10.111], and (that) the (irrational straight-lines) after the apotome, being applied to a rational (straight-line), produce as breadth, each according to its own (order), apotomes, and (that) the (irrational straight-lines) after the binomial themselves also (produce as breadth), according (to their) order, binomials, the (irrational straight-lines) after the apotome are thus different, and the (irrational straight-lines) after the binomial (are also) different, so that there are, in order, 13 irrational (straight-lines) in all:

Μέσην, Medial,

Ἐκ δύο ὀνομάτων, Binomial,

Ἐκ δύο μέσων πρώτην, First bimedial,

Ἐκ δύο μέσων δευτέραν, Second bimedial,

Μείζονα, Major,

μὴ ὁΡητὸν καὶ μέσον δυναμένην, Square-root of a rational plus a medial (area),

Δύο μέσα δυναμένην, Square-root of (the sum of) two medial (areas),

Ἀποτομήν, Apotome,

Μέσης ἀποτομὴν πρώτην, First apotome of a medial,

Μέσης ἀποτομὴν δευτέραν, Second apotome of a medial,

Ἐλάσσονα, Minor,

Μετὰ ὁρητοῦ μέσον τὸ ὅλον ποιοῦσαν, That which with a rational (area) produces a medial whole,

Μετὰ μέσου μέσον τὸ ὅλον ποιοῦσαν. That which with a medial (area) produces a medial whole.

112.

Τὸ ἀπὸ ὁρητῆς παρὰ τὴν ἐκ δύο ὀνομάτων παραβαλλόμενον πλάτος ποιεῖ ἀποτομήν, ἧς τὰ ὀνόματα σύμμετρά ἐστι τοῖς τῆς ἐκ δύο ὀνομάτων ὀνόμασι καὶ ὁέτι ἐν τῷ αὐτῷ λόγῳ, καὶ ὁέτι ἡ γινομένη ἀποτομὴ τὴν αὐτὴν ὁέχει τάχιν τῇ ἐκ δύο ὀνομάτων.

Proposition 112[†]

The (square) on a rational (straight-line), applied to a binomial (straight-line), produces as breadth an apotome whose terms are commensurable (in length) with the terms of the binomial, and, furthermore, in the same ratio. Moreover, the created apotome will have the same order as the binomial.

Ἔστω ὁρητὴ μὲν ἡ A, ἐκ δύο ὀνομάτων δὲ ἡ $BΓ$, ἧξ μεῖζον ὄνομα ὑέστω ἡ $ΔΓ$, καὶ τῷ ἀπὸ τῆς A ἴσον ὑέστω τὸ ὑπὸ τῶν $BΓ$, EZ· λέγω, ὅτι ἡ EZ ἀποτομή ἐστιν, ἧξ τὰ ὀνόματα σύμμετρά ἐστι τοῖς $ΓΔ$, $ΔB$, καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ ὑέτι ἡ EZ τὴν αὐτὴν ὑέχει τάχιν τῇ $BΓ$.

Ἔστω γὰρ πάλιν τῷ ἀπὸ τῆς A ἴσον τὸ ὑπὸ τῶν $BΔ$, H. ἐπεὶ οὐῦν τὸ ὑπὸ τῶν $BΓ$, EZ ἴσον ἐστὶ τῷ ὑπὸ τῶν $BΔ$, H, ὑέστιν ἄρα ὡξ ἡ $ΓB$ πρὸς τὴν $BΔ$, οὑύτως ἡ H πρὸς τὴν EZ. μείζων δὲ ἡ $ΓB$ τῆς $BΔ$· μείζων ἄρα ἐστὶ καὶ ἡ H τῆς EZ. ὑέστω τῇ Hὑίσῃ ἡ $EΘ$· ὑέστιν ἄρα ὡξ ἡ $ΓB$ πρὸς τὴν $BΔ$, οὑύτως ἡ $ΘE$ πρὸς τὴν EZ· διελόντι ἄρα ἐστὶν ὡξ ἡ $ΓΔ$ πρὸς τὴν $BΔ$, οὑύτως ἡ $ΘZ$ πρὸς τὴν ZE. γεγονέτω ὡξ ἡ $ΘZ$ πρὸς τὴν ZE, οὑύτως ἡ ZE πρὸς τὴν KE· καὶ ὅλη ἄρα ἡ $ΘK$ πρὸς ὅλην τὴν KZ ἐστιν, ὡξ ἡ ZK πρὸς KE· ὡξ γὰρ ὁὲν τῶν ἡγουμένων πρὸς ὁὲν τῶν ἑπομένων, οὑύτως ὁάπαντα τὰ ἡγούμενα πρὸς ὁάπαντα τὰ ἑπόμενα. ὡξ δὲ ἡ ZK πρὸς KE, οὑύτως ἐστὶν ἡ $ΓΔ$ πρὸς τὴν $ΔB$· καὶ ὡξ ἄρα ἡ $ΘK$ πρὸς KZ, οὑύτως ἡ $ΓΔ$ πρὸς τὴν $ΔB$. σύμμετρον δὲ τὸ ἀπὸ τῆς $ΓΔ$ τῷ ἀπὸ τῆς $ΔB$· σύμμετρον ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς $ΘK$ τῷ ἀπὸ τῆς KZ. καί ἐστιν ὡξ τὸ ἀπὸ τῆς $ΘK$ πρὸς τὸ ἀπὸ τῆς KZ, οὑύτως ἡ $ΘK$ πρὸς τὴν KE, ἐπεὶ αἱ τρεῖς αἱ $ΘK$, KZ, KE ἀνάλογόν εἰσιν. σύμμετρος ἄρα ἡ $ΘK$ τῇ KE μήκει. ὁώστε καὶ ἡ $ΘE$ τῇ EK σύμμετρόξ ἐστι μήκει. καὶ ἐπεὶ τὸ ἀπὸ τῆς A ἴσον ἐστὶ τῷ ὑπὸ τῶν $EΘ$, $BΔ$, ὁρητὸν δέ ἐστι τὸ ἀπὸ τῆς A, ὁρητὸν ἄρα ἐστὶ καὶ τὸ ὑπὸ τῶν $EΘ$, $BΔ$. καὶ παρὰ ὁρητὴν τὴν $BΔ$ παράκειται· ὁρητὴ ἄρα ἐστὶν ἡ $EΘ$ καὶ σύμμετρος τῇ $BΔ$ μήκει· ὁώστε καὶ ἡ σύμμετρος αὐτῇ ἡ EK ὁρητή ἐστι καὶ σύμμετρος τῇ $BΔ$ μήκει. ἐπεὶ οὐῦν ἐστιν ὡξ ἡ $ΓΔ$ πρὸς $ΔB$, οὑύτως ἡ ZK πρὸς KE, αἱ δὲ $ΓΔ$, $ΔB$ δυνάμει μόνον εἰσὶ σύμμετροι, καὶ αἱ ZK, KE δυνάμει μόνον εἰσὶ σύμμετροι. ὁρητὴ δέ ἐστιν ἡ KE· ὁρητὴ ἄρα ἐστὶ καὶ ἡ ZK. αἱ ZK, KE ἄρα ὁρηταὶ δυνάμει μόνον εἰσὶ σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ EZ.

Ὁ Ἤτοι δὲ ἡ $ΓΔ$ τῆς $ΔB$ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ ὁὴ τῷ ἀπὸ ἀσυμμέτρου.

Εἰ μὲν οὐῦν ἡ $ΓΔ$ τῆς $ΔB$ μεῖζον δύναται τῷ ἀπὸ συμμέτρου [ἑαυτῇ], καὶ ἡ ZK τῆς KE μεῖζον δυνήσεται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καὶ εἰ μὲν σύμμετρόξ ἐστιν ἡ $ΓΔ$ τῇ ἐκκειμένῃ ὁρητῇ μήκει, καὶ ἡ ZK· εἰ δὲ ἡ $BΔ$, καὶ ἡ KE· εἰ δὲ οὐὁετέρα τῶν $ΓΔ$, $ΔB$, καὶ οὐὁετέρα τῶν ZK, KE.

Εἰ δὲ ἡ $ΓΔ$ τῆς $ΔB$ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου

Let A be a rational (straight-line), and BC a binomial (straight-line), of which let DC be the greater term. And let the (rectangle contained) by BC and EF be equal to the (square) on A. I say that EF is an apotome whose terms are commensurable (in length) with CD and DB, and in the same ratio, and, moreover, that EF will have the same order as BC.

For, again, let the (rectangle contained) by BD and G be equal to the (square) on A. Therefore, since the (rectangle contained) by BC and EF is equal to the (rectangle contained) by BD and G, thus as CB is to BD, so G (is) to EF [Prop. 6.16]. And CB (is) greater than BD. Thus, G is also greater than EF [Props. 5.16, 5.14]. Let EH be equal to G. Thus, as CB is to BD, so HE (is) to EF. Thus, via separation, as CD is to BD, so HF (is) to FE [Prop. 5.17]. Let it be contrived that as HF (is) to FE, so FK (is) to KE. And, thus, the whole HK is to the whole KF, as FK (is) to KE. For as one of the leading (proportional magnitudes is) to one of the following, so all of the leading (magnitudes) are to all of the following [Prop. 5.12]. And as FK (is) to KE, so CD is to DB [Prop. 5.11]. And, thus, as HK (is) to KF, so CD is to DB [Prop. 5.11]. And the (square) on CD (is) commensurable with the (square) on DB [Prop. 10.36]. The (square) on HK is thus also commensurable with the (square) on KF [Props. 6.22, 10.11]. And as the (square) on HK is to the (square) on KF, so HK (is) to KE, since the three (straight-lines) HK, KF, and KE are proportional [Def. 5.9]. HK is thus commensurable in length with KE [Prop. 10.11]. Hence, HE is also commensurable in length with EK [Prop. 10.15]. And since the (square) on A is equal to the (rectangle contained) by EH and BD, and the (square) on A is rational, the (rectangle contained) by EH and BD is thus also rational. And it is applied to the rational (straight-line) BD. Thus, EH is rational, and commensurable in length with BD [Prop. 10.20]. And, hence, the (straight-line) commensurable (in length) with it, EK, is also rational [Def. 10.3], and commensurable in length with BD [Prop. 10.12]. Therefore, since as CD is to DB, so FK (is) to KE, and CD and DB are (straight-lines which are) commensurable in square only, FK and KE are also commensurable in square only [Prop. 10.11]. And KE is rational. Thus, FK is also rational. FK and KE are thus rational (straight-lines which are) commensurable in square only. Thus, EF is an apotome [Prop. 10.73].

And the square on CD is greater than (the square on) DB

ἑαυτῇ, καὶ ἡ ZK τῆς KE μεῖζον δυνήσεται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καὶ εἰ μὲν ἡ ΓΔ σύμμετρόξ ἐστι τῇ ἐκκειμένῃ ὁρητῇ μήκει, καὶ ἡ ZK· εἰ δὲ ἡ BΔ, καὶ ἡ KE· εἰ δὲ οὐδετέρα τῶν ΓΔ, ΔB, καὶ οὐδετέρα τῶν ZK, KE· ὁώστε ἀποτομή ἐστιν ἡ ZE, ἧξ τὰ ὀνόματα τὰ ZK, KE σύμμετρά ἐστι τοῖς τῆς ἐκ δύο ὀνομάτων ὀνόμασι τοῖς ΓΔ, ΔB καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ τὴν αὐτῆν τάχιν ἔχει τῇ BΓ· ὅπερ ἔδει δεῖχαι.

either by the (square) on (some straight-line) commensurable, or by the (square) on (some straight-line) incommensurable, (in length) with (CD).

Therefore, if the square on CD is greater than (the square on) DB by the (square) on (some straight-line) commensurable (in length) with [CD] then the square on FK will also be greater than (the square on) KE by the (square) on (some straight-line) commensurable (in length) with (FK) [Prop. 10.14]. And if CD is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) FK [Props. 10.11, 10.12]. And if BD (is commensurable), (so) also (is) KE [Prop. 10.12]. And if neither of CD or DB (is commensurable), neither also (are) either of FK or KE.

And if the square on CD is greater than (the square on) DB by the (square) on (some straight-line) incommensurable (in length) with (CD) then the square on FK will also be greater than (the square on) KE by the (square) on (some straight-line) incommensurable (in length) with (FK) [Prop. 10.14]. And if CD is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) FK [Props. 10.11, 10.12]. And if BD (is commensurable), (so) also (is) KE [Prop. 10.12]. And if neither of CD or DB (is commensurable), neither also (are) either of FK or KE. Hence, FE is an apotome whose terms, FK and KE, are commensurable (in length) with the terms, CD and DB, of the binomial, and in the same ratio. And (FE) has the same order as BC [Defs. 10.5—10.10]. (Which is) the very thing it was required to show.

† Heiberg considers this proposition, and the succeeding ones, to be relatively early interpolations into the original text.

113.

Τὸ ἀπὸ ὁρητῆς παρὰ ἀποτομὴν παραβαλλόμενον πλάτος ποιεῖ τὴν ἐκ δύο ὀνομάτων, ἧξ τὰ ὀνόματα σύμμετρά ἐστι τοῖς τῆς ἀποτομῆς ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, ὁέτι δὲ ἡ γινομένη ἐκ δύο ὀνομάτων τὴν αὐτὴν τάχιν ἔχει τῇ ἀποτομῇ.

Proposition 113

The (square) on a rational (straight-line), applied to an apotome, produces as breadth a binomial whose terms are commensurable with the terms of the apotome, and in the same ratio. Moreover, the created binomial has the same order as the apotome.

Ἔστω ὁρητὴ μὲν ἡ A, ἀποτομὴ δὲ ἡ BΔ, καὶ τῷ ἀπὸ τῆς A ἴσονὁέστω τὸ ὑπὸ τῶν BΔ, KΘ, ὁώστε τὸ ἀπὸ τῆς A ὁρητῆς παρὰ τὴν BΔ ἀποτομὴν παραβαλλόμενον πλάτος

Let A be a rational (straight-line), and BD an apotome. And let the (rectangle contained) by BD and KH be equal to the (square) on A, such that the square on the rational (straight-

ποιεῖ τὴν ΚΘ· λέγω, ὅτι ἐκ δύο ὀνομάτων ἐστὶν ἡ ΚΘ, ἧξ τὰ ὀνόματα σύμμετρά ἐστι τοῖς τῆς ΒΔ ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ ὀέτι ἡ ΚΘ τὴν αὐτὴν ἔχει τάχιν τῇ ΒΔ.

Ἔστω γὰρ τῇ ΒΔ προσαρμόζουσα ἡ ΔΓ· αἱ ΒΓ, ΓΔ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι. καὶ τῷ ἀπὸ τῆς Α ἴσονὸέστω καὶ τὸ ὑπὸ τῶν ΒΓ, Η. ὁρητὸν δὲ τὸ ἀπὸ τῆς Α· ὁρητὸν ἄρα καὶ τὸ ὑπὸ τῶν ΒΓ, Η. καὶ παρὰ ὁρητὴν τὴν ΒΓ παραβέβληται· ὁρητὴ ἄρα ἐστὶν ἡ Η καὶ σύμμετρος τῇ ΒΓ μήκει. ἐπεὶ οὐῦν τὸ ὑπὸ τῶν ΒΓ, Η ἴσον ἐστὶ τῷ ὑπὸ τῶν ΒΔ, ΚΘ, ἀνάλογον ἄρα ἐστὶν ὡξ ἡ ΓΒ πρὸς ΒΔ, οὕτως ἡ ΚΘ πρὸς Η. μείζων δὲ ἡ ΒΓ τῆς ΒΔ· μείζων ἄρα καὶ ἡ ΚΘ τῆς Η. κείσθω τῇ Ηὀίση ἡ ΚΕ· σύμμετρος ἄρα ἐστὶν ἡ ΚΕ τῇ ΒΓ μήκει. καὶ ἐπεί ἐστιν ὡξ ἡ ΓΒ πρὸς ΒΔ, οὕτως ἡ ΘΚ πρὸς ΚΕ, ἀναστρέψαντι ἄρα ἐστὶν ὡξ ἡ ΒΓ πρὸς τὴν ΓΔ, οὕτως ἡ ΚΘ πρὸς ΘΕ. γεγονέτω ὡξ ἡ ΚΘ πρὸς ΘΕ, οὕτως ἡ ΘΖ πρὸς ΖΕ· καὶ λοιπὴ ἄρα ἡ ΚΖ πρὸς ΖΘ ἐστιν, ὡξ ἡ ΚΘ πρὸς ΘΕ, τουτέστιν [ὡξ] ἡ ΒΓ πρὸς ΓΔ. αἱ δὲ ΒΓ, ΓΔ δυνάμει μόνον [εἰσί] σύμμετροι· καὶ αἱ ΚΖ, ΖΘ ἄρα δυνάμει μόνον εἰσὶ σύμμετροι· καὶ ἐπεί ἐστιν ὡξ ἡ ΚΘ πρὸς ΘΕ, ἡ ΚΖ πρὸς ΖΘ, ἀλλὸ ὡξ ἡ ΚΘ πρὸς ΘΕ, ἡ ΘΖ πρὸς ΖΕ, καὶ ὡξ ἄρα ἡ ΚΖ πρὸς ΖΘ, ἡ ΘΖ πρὸς ΖΕ· ὁὡστε καὶ ὡξ ἡ πρώτη πρὸς τὴν τρίτην, τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας· καὶ ὡξ ἄρα ἡ ΚΖ πρὸς ΖΕ, οὕτως τὸ ἀπὸ τῆς ΚΖ πρὸς τὸ ἀπὸ τῆς ΖΘ. σύμμετρον δέ ἐστι τὸ ἀπὸ τῆς ΚΖ τῷ ἀπὸ τῆς ΖΘ· αἱ γὰρ ΚΖ, ΖΘ δυνάμει εἰσὶ σύμμετροι· σύμμετρος ἄρα ἐστὶ καὶ ἡ ΚΖ τῇ ΖΕ μήκει· ὁὡστε ἡ ΚΖ καὶ τῇ ΚΕ σύμμετρός [ἐστι] μήκει. ὁρητὴ δέ ἐστιν ἡ ΚΕ καὶ σύμμετρος τῇ ΒΓ μήκει. ὁρητὴ ἄρα καὶ ἡ ΚΖ καὶ σύμμετρος τῇ ΒΓ μήκει. καὶ ἐπεί ἐστιν ὡξ ἡ ΒΓ πρὸς ΓΔ, οὕτως ἡ ΚΖ πρὸς ΖΘ, ἐναλλὰχ ὡξ ἡ ΒΓ πρὸς ΚΖ, οὕτως ἡ ΔΓ πρὸς ΖΘ. σύμμετρος δὲ ἡ ΒΓ τῇ ΚΖ· σύμμετρος ἄρα καὶ ἡ ΖΘ τῇ ΓΔ μήκει. αἱ ΒΓ, ΓΔ δὲ ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· καὶ αἱ ΚΖ, ΖΘ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἐκ δύο ὀνομάτων ἐστὶν ἄρα ἡ ΚΘ.

Εἰ μὲν οὐῦν ἡ ΒΓ τῆς ΓΔ μεῖζον δύναται τῷ ἀπὸ συμμέτρου ἑαυτῇ, καὶ ἡ ΚΖ τῆς ΖΘ μεῖζον δυνήσεται τῷ ἀπὸ συμμέτρου ἑαυτῇ. καὶ εἰ μὲν σύμμετρός ἐστιν ἡ ΒΓ τῇ ἐκκειμένῃ ὁρητῇ μήκει, καὶ ἡ ΚΖ, εἰ δὲ ἡ ΓΔ σύμμετρός ἐστι τῇ ἐκκειμένῃ ὁρητῇ μήκει, καὶ ἡ ΖΘ, εἰ δὲ οὐδετέρα τῶν ΒΓ, ΓΔ, οὐδετέρα τῶν ΚΖ, ΖΘ.

Εἰ δὲ ἡ ΒΓ τῆς ΓΔ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, καὶ ἡ ΚΖ τῆς ΖΘ μεῖζον δυνήσεται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. καὶ εἰ μὲν σύμμετρός ἐστιν ἡ ΒΓ τῇ ἐκκειμένῃ ὁρητῇ μήκει, καὶ ἡ ΚΖ, εἰ δὲ ἡ ΓΔ, καὶ ἡ ΖΘ, εἰ δὲ οὐδετέρα τῶν ΒΓ, ΓΔ, οὐδετέρα τῶν ΚΖ, ΖΘ.

Ἐκ δύο ἄρα ὀνομάτων ἐστὶν ἡ ΚΘ, ἧξ τὰ ὀνόματα τὰ ΚΖ, ΖΘ σύμμετρά [ἐστι] τοῖς τῆς ἀποτομῆς ὀνόμασι τοῖς ΒΓ, ΓΔ καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ ὀέτι ἡ ΚΘ τῇ ΒΓ τὴν αὐτὴν ὁέχει τάχιν· ὅπερ ἔδει δεῖχαι.

line) A, applied to the apotome BD, produces KH as breadth. I say that KH is a binomial whose terms are commensurable with the terms of BD, and in the same ratio, and, moreover, that KH has the same order as BD.

For let DC be an attachment to BD. Thus, BC and CD are rational (straight-lines which are) commensurable in square only [Prop. 10.73]. And let the (rectangle contained) by BC and G also be equal to the (square) on A. And the (square) on A (is) rational. The (rectangle contained) by BC and G (is) thus also rational. And it has been applied to the rational (straight-line) BC. Thus, G is rational, and commensurable in length with BC [Prop. 10.20]. Therefore, since the (rectangle contained) by BC and G is equal to the (rectangle contained) by BD and KH, thus, proportionally, as CB is to BD, so KH (is) to G [Prop. 6.16]. And BC (is) greater than BD. Thus, KH (is) also greater than G [Prop. 5.16, 5.14]. Let KE be made equal to G. KE is thus commensurable in length with BC. And since as CB is to BD, so HK (is) to KE, thus, via conversion, as BC (is) to CD, so KH (is) to HE [Prop. 5.19 corr.]. Let it be contrived that as KH (is) to HE, so HF (is) to FE. And thus the remainder KF is to FH, as KH (is) to HE—that is to say, [as] BC (is) to CD [Prop. 5.19]. And BC and CD [are] commensurable in square only. KF and FH are thus also commensurable in square only [Prop. 10.11]. And since as KH is to HE, (so) KF (is) to FH, but as KH (is) to HE, (so) HF (is) to FE, thus, also as KF (is) to FH, (so) HF (is) to FE [Prop. 5.11]. And hence as the first (is) to the third, so the (square) on the first (is) to the (square) on the second [Def. 5.9]. And thus as KF (is) to FE, so the (square) on KF (is) to the (square) on FH. And the (square) on KF is commensurable with the (square) on FH. For KF and FH are commensurable in square. Thus, KF is also commensurable in length with FE [Prop. 10.11]. Hence, KF [is] also commensurable in length with KE [Prop. 10.15]. And KE is rational, and commensurable in length with BC. Thus, KF (is) also rational, and commensurable in length with BC [Prop. 10.12]. And since as BC is to CD, (so) KF (is) to FH, alternately, as BC (is) to KF, so DC (is) to FH [Prop. 5.16]. And BC (is) commensurable (in length) with KF. Thus, FH (is) also commensurable in length with CD [Prop. 10.11]. And BC and CD are rational (straight-lines which are) commensurable in square only. KF and FH are thus also rational (straight-lines which are) commensurable in square only [Def. 10.3, Prop. 10.13]. Thus, KH is a binomial [Prop. 10.36].

Therefore, if the square on BC is greater than (the square on) CD by the (square) on (some straight-line) commensurable (in length) with (BC), then the square on KF will also be greater than (the square on) FH by the (square) on (some straight-line) commensurable (in length) with (KF) [Prop. 10.14]. And if BC is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is)

KF [Prop. 10.12]. And if *CD* is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) *FH* [Prop. 10.12]. And if neither of *BC* or *CD* (are commensurable), neither also (are) either of *KF* or *FH* [Prop. 10.13].

And if the square on *BC* is greater than (the square on) *CD* by the (square) on (some straight-line) incommensurable (in length) with (*BC*) then the square on *KF* will also be greater than (the square on) *FH* by the (square) on (some straight-line) incommensurable (in length) with (*KF*) [Prop. 10.14]. And if *BC* is commensurable in length with a (previously) laid down rational (straight-line), (so) also (is) *KF* [Prop. 10.12]. And if *CD* is commensurable, (so) also (is) *FH* [Prop. 10.12]. And if neither of *BC* or *CD* (are commensurable), neither also (are) either of *KF* or *FH* [Prop. 10.13].

KH is thus a binomial whose terms, *KF* and *FH*, [are] commensurable (in length) with the terms, *BC* and *CD*, of the apotome, and in the same ratio. Moreover, *KH* will have the same order as *BC* [Defs. 10.5—10.10]. (Which is) the very thing it was required to show.

114.

Ἐὰν θωρίον περιέθηται ὑπὸ ἀποτομῆς καὶ τῆς ἐκ δύο ὀνομάτων, ἧξ τὰ ὀνόματα σύμμετρά τέ ἐστι τοῖς τῆς ἀποτομῆς ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, ἡ τὸ θωρίον δυναμένη ὀρητή ἐστιν.

Περιεθέσθω γὰρ θωρίον τὸ ὑπὸ τῶν AB, ΓΔ ὑπὸ ἀποτομῆς τῆς AB καὶ τῆς ἐκ δύο ὀνομάτων τῆς ΓΔ, ἧξ μεῖζον ὄνομα ἐστω τὸ ΓΕ, καὶ ἐστω τὰ ὀνόματα τῆς ἐκ δύο ὀνομάτων τὰ ΓΕ, ΕΔ σύμμετρά τε τοῖς τῆς ἀποτομῆς ὀνόμασι τοῖς AZ, ZB καὶ ἐν τῷ αὐτῷ λόγῳ, καὶ ἐστω ἡ τὸ ὑπὸ τῶν AB, ΓΔ δυναμένη ἡ H· λέγω, ὅτι ὀρητή ἐστιν ἡ H.

Ἐκκείσθω γὰρ ὀρητὴ ἡ Θ, καὶ τῷ ἀπὸ τῆς Θ ἴσον παρὰ τὴν ΓΔ παραβεβλήσθω πλάτος ποιοῦν τὴν ΚΛ· ἀποτομὴ ἄρα ἐστὶν ἡ ΚΛ, ἧξ τὰ ὀνόματα ἐστω τὰ ΚΜ, ΜΛ σύμμετρα τοῖς τῆς ἐκ δύο ὀνομάτων ὀνόμασι τοῖς ΓΕ, ΕΔ καὶ ἐν τῷ αὐτῷ λόγῳ. ἀλλὰ καὶ αἱ ΓΕ, ΕΔ σύμμετροί τέ εἰσι ταῖς AZ, ZB καὶ ἐν τῷ αὐτῷ λόγῳ· ὀέστιν ἄρα ὡξ ἡ AZ πρὸς τὴν ZB, οὕτως ἡ ΚΜ πρὸς ΜΛ. ἐναλλὰχ ἄρα ἐστὶν ὡξ ἡ AZ πρὸς τὴν ΚΜ, οὕτως ἡ BZ πρὸς τὴν ΛΜ· καὶ λοιπὴ ἄρα ἡ AB πρὸς λοιπὴν τὴν ΚΛ ἐστιν ὡξ ἡ AZ πρὸς ΚΜ. σύμμετρος δὲ ἡ AZ τῇ ΚΜ· σύμμετρος ἄρα ἐστὶ καὶ ἡ AB τῇ ΚΛ. καί ἐστιν ὡξ ἡ AB πρὸς ΚΛ, οὕτως τὸ ὑπὸ τῶν ΓΔ, AB πρὸς τὸ ὑπὸ τῶν ΓΔ, ΚΛ· σύμμετρον ἄρα ἐστὶ καὶ τὸ ὑπὸ τῶν ΓΔ, AB τῷ ὑπὸ τῶν ΓΔ, ΚΛ. ἴσον δὲ τὸ ὑπὸ τῶν ΓΔ, ΚΛ τῷ ἀπὸ τῆς Θ· σύμμετρον ἄρα ἐστὶ τὸ ὑπὸ τῶν ΓΔ, AB τῷ ἀπὸ τῆς Θ. τῷ δὲ ὑπὸ τῶν ΓΔ, AB ἴσον ἐστὶ τὸ ἀπὸ τῆς H· σύμμετρον ἄρα ἐστὶ τὸ ἀπὸ τῆς H τῷ ἀπὸ τῆς Θ. ὀρητὸν δὲ τὸ ἀπὸ τῆς Θ· ὀρητὸν ἄρα ἐστὶ καὶ τὸ ἀπὸ τῆς H· ὀρητὴ ἄρα ἐστὶν ἡ H. καὶ δύναται τὸ ὑπὸ τῶν ΓΔ, AB.

Proposition 114

If an area is contained by an apotome, and a binomial whose terms are commensurable with, and in the same ratio as, the terms of the apotome then the square-root of the area is a rational (straight-line).

For let an area, the (rectangle contained) by *AB* and *CD*, be contained by the apotome *AB*, and the binomial *CD*, of which let the greater term be *CE*. And let the terms of the binomial, *CE* and *ED*, be commensurable with the terms of the apotome, *AF* and *FB* (respectively), and in the same ratio. And let the square-root of the (rectangle contained) by *AB* and *CD* be *G*. I say that *G* is a rational (straight-line).

For let the rational (straight-line) *H* be laid down. And let (some rectangle), equal to the (square) on *H*, be applied to *CD*, producing *KL* as breadth. Thus, *KL* is an apotome, of which let the terms, *KM* and *ML*, be commensurable with the terms of the binomial, *CE* and *ED* (respectively), and in the same ratio [Prop. 10.112]. But, *CE* and *ED* are also commensurable with *AF* and *FB* (respectively), and in the same ratio. Thus, as *AF* is to *FB*, so *KM* (is) to *ML*. Thus, alternately, as *AF* is to *KM*, so *BF* (is) to *LM* [Prop. 5.16]. Thus, the remainder *AB* is also to the remainder *KL* as *AF* (is) to *KM* [Prop. 5.19]. And *AF* (is) commensurable with *KM* [Prop. 10.12]. *AB* is thus also commensurable with *KL* [Prop. 10.11]. And as *AB* is to *KL*, so the (rectangle contained) by *CD* and *AB* (is) to the (rectangle contained) by *CD* and *KL* [Prop. 6.1]. Thus, the (rectangle contained) by *CD* and *AB* is also commensurable with the (rectangle contained) by *CD* and *KL* [Prop. 10.11]. And the (rectangle contained) by *CD* and *KL* (is) equal to the (square) on *H*. Thus, the (rectangle contained) by *CD* and *AB*

is commensurable with the (square) on H. And the (square) on G is equal to the (rectangle contained) by CD and AB. The (square) on G is thus commensurable with the (square) on H. And the (square) on H (is) rational. Thus, the (square) on G is also rational. G is thus rational. And it is the square-root of the (rectangle contained) by CD and AB.

Ἐὰν ἄρα θωρίον περιέθηται ὑπὸ ἀποτομῆς καὶ τῆς ἐκ δύο ὀνομάτων, ἧξ τὰ ὀνόματα σύμμετρά ἐστι τοῖς τῆς ἀποτομῆς ὀνόμασι καὶ ἐν τῷ αὐτῷ λόγῳ, ἡ τὸ θωρίον δυναμένη ὀρητή ἐστιν.

Thus, if an area is contained by an apotome, and a binomial whose terms are commensurable with, and in the same ratio as, the terms of the apotome, then the square-root of the area is a rational (straight-line).

Πόρισμα.

Καὶ γέγονεν ἡμῖν καὶ διὰ τούτου φανερόν, ὅτι δυνατόν ἐστι ὀρητὸν θωρίον ὑπὸ ἀλόγων εὐθειῶν περιέθεσθαι. ὅπερ ἔδει δεῖχαι.

Corollary

And it has also been made clear to us, through this, that it is possible for a rational area to be contained by irrational straight-lines. (Which is) the very thing it was required to show.

115.

Proposition 115

Ἀπὸ μέσης ὀἀπειροιὀάλογοι γίνονται, καὶ οὐδεμία οὐδεμιᾷ τῶν πρότερον ἡ αὐτή.

Ἔστω μέση ἡ A· λέγω, ὅτι ἀπὸ τῆς A ὀἀπειροιὀάλογοι γίνονται, καὶ οὐδεμία οὐδεμιᾷ τῶν πρότερον ἡ αὐτή.

An infinite (series) of irrational (straight-lines) can be created from a medial (straight-line), and none of them is the same as any of the preceding (straight-lines).

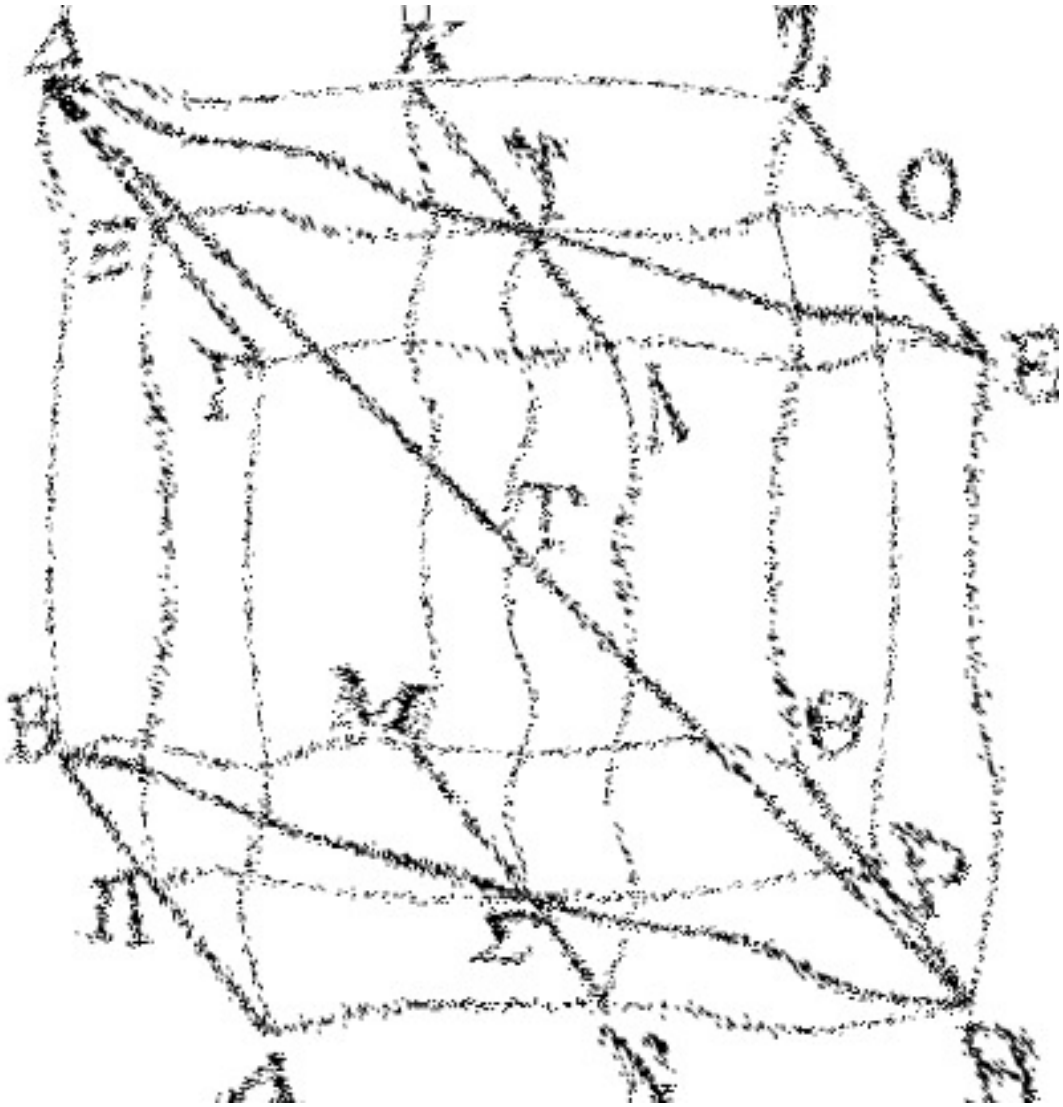
Let A be a medial (straight-line). I say that an infinite (series) of irrational (straight-lines) can be created from A, and that none of them is the same as any of the preceding (straight-lines).

Ἐκκείσθω ὁρητὴ ἡ B, καὶ τῷ ὑπὸ τῶν B, A ἴσον ὀέστω τὸ ἀπὸ τῆς Γ· ὀάλογος ἄρα ἐστὶν ἡ Γ· τὸ γὰρ ὑπὸ ἀλόγου καὶ ὁρητῆς ὀάλογόν ἐστιν. καὶ οὐδεμιᾷ τῶν πρότερον ἡ αὐτή· τὸ γὰρ ἀπὸ οὐδεμιᾶς τῶν πρότερον παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ μέσην. πάλιν δὴ τῷ ὑπὸ τῶν B, Γ ἴσον ὀέστω τὸ ἀπὸ τῆς Δ· ὀάλογον ἄρα ἐστὶ τὸ ἀπὸ τῆς Δ. ὀάλογος ἄρα ἐστὶν ἡ Δ· καὶ οὐδεμιᾷ τῶν πρότερον ἡ αὐτή· τὸ γὰρ ἀπὸ οὐδεμιᾶς τῶν πρότερον παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ τὴν Γ. ὁμοίως δὴ τῆς τοιαύτης τάχεως ἐπ᾽ ἄπειρον προβαινούσης φανερόν, ὅτι ἀπὸ τῆς μέσης ὀἀπειροιὀάλογοι γίνονται, καὶ οὐδεμία οὐδεμιᾷ τῶν πρότερον ἡ αὐτή· ὅπερ ἔδει δεῖξαι.

Let the rational (straight-line) B be laid down. And let the (square) on C be equal to the (rectangle contained) by B and A. Thus, C is irrational [Def. 10.4]. For an (area contained) by an irrational and a rational (straight-line) is irrational [Prop. 10.20]. And (C is) not the same as any of the preceding (straight-lines). For the (square) on none of the preceding (straight-lines), applied to a rational (straight-line), produces a medial (straight-line) as breadth. So, again, let the (square) on D be equal to the (rectangle contained) by B and C. Thus, the (square) on D is irrational [Prop. 10.20]. D is thus irrational [Def. 10.4]. And (D is) not the same as any of the preceding (straight-lines). For the (square) on none of the preceding (straight-lines), applied to a rational (straight-line), produces C as breadth. So, similarly, this arrangement being advanced to infinity, it is clear that an infinite (series) of irrational (straight-lines) can be created from a medial (straight-line), and that none of them is the same as any of the preceding (straight-lines). (Which is) the very thing it was required to show.

ELEMENTS BOOK 11

Elementary Stereometry



Ὅροι.

1. Στερεόν ἐστι τὸ μῆκος καὶ πλάτος καὶ βάθος ἔχον.

2. Στερεοῦ δὲ πέρας ἐπιφάνεια.

3. Εὐθεῖα πρὸς ἐπίπεδον ὀρθή ἐστιν, ὅταν πρὸς πάσας τὰς ἁπτομένας αὐτῆς εὐθείας καὶ οὐσας ἐν τῷ [ὑποκειμένῳ] ἐπιπέδῳ ὀρθὰς ποιῇ γωνίας.

4. Ἐπίπεδον πρὸς ἐπίπεδον ὀρθόν ἐστιν, ὅταν αἱ τῇ κοινῇ τομῇ τῶν ἐπιπέδων πρὸς ὀρθὰς ἀγόμεναι εὐθεῖαι ἐν ἑνὶ τῶν ἐπιπέδων τῷ λοιπῷ ἐπιπέδῳ πρὸς ὀρθὰς ὦσιν.

5. Εὐθείας πρὸς ἐπίπεδον κλίσις ἐστίν, ὅταν ἀπὸ τοῦ μετεώρου πέρατος τῆς εὐθείας ἐπὶ τὸ ἐπίπεδον κάθετος ἀθθῇ, καὶ ἀπὸ τοῦ γενομένου σημείου ἐπὶ τὸ ἐν τῷ ἐπιπέδῳ πέρας τῆς εὐθείας εὐθεῖα ἐπιζευθθῇ, ἡ περιεθομένη γωνία ὑπὸ τῆς ἀθθείσης καὶ τῆς ἐφεστώσης.

6. Ἐπιπέδου πρὸς ἐπίπεδον κλίσις ἐστὶν ἡ περιεθομένη ὀξεῖα γωνία ὑπὸ τῶν πρὸς ὀρθὰς τῇ κοινῇ τομῇ ἀγομένων πρὸς τῷ αὐτῷ σημείῳ ἐν ἑκατέρῳ τῶν ἐπιπέδων.

7. Ἐπίπεδον πρὸς ἐπίπεδον ὁμοίως κεκλίσθαι λέγεται καὶ ὁέτερον πρὸς ὁέτερον, ὅταν αἱ εἰρημέναι τῶν κλίσεων γωνίαι ἴσαι ἀλλήλαις ὦσιν.

8. Παράλληλα ἐπίπεδά ἐστι τὰ ἀσύμπτωτα.

9. Ὅμοια στερεὰ σθήματά ἐστι τὰ ὑπὸ ὁμοίων ἐπιπέδων περιεχόμεναοίσων τὸ πλῆθος.

10. Ὁἴσα δὲ καὶ ὅμοια στερεὰ σθήματά ἐστι τὰ ὑπὸ ὁμοίων ἐπιπέδων περιεχόμεναοίσων τῷ πλήθει καὶ τῷ μεγέθει.

11. Στερεὰ γωνία ἐστὶν ἡ ὑπὸ πλειόνωνὸὴ δύο γραμμῶν ἁπτομένων ἀλλήλων καὶ μὴ ἐν τῇ αὐτῇ ἐπιφανείᾳ οὐσῶν πρὸς πάσαις ταῖς γραμμαῖξ κλίσις. ὁάλλως· στερεὰ γωνία ἐστὶν ἡ ὑπὸ πλειόνωνὸὴ δύο γωνιῶν ἐπιπέδων περιεθομένη μὴ οὐσῶν ἐν τῷ αὐτῷ ἐπιπέδῳ πρὸς ἑνὶ σημείῳ συνισταμένων.

12. Πυραμίξ ἐστι σχῆμα στερεὸν ἐπιπέδοις περιθόμενον ἀπὸ ἑνὸς ἐπιπέδου πρὸς ἑνὶ σημείῳ συνεστώξ.

13. Πρίσμα ἐστὶ σχῆμα στερεὸν ἐπιπέδοις περιεχόμενον, ὁῶν δύο τὰ ἀπεναντίον ἴσα τε καὶ ὅμοιά ἐστι καὶ παράλληλα, τὰ δὲ λοιπὰ παραλληλόγραμμα.

14. Σφαῖρά ἐστιν, ὅταν ἡμικυκλίου μενούσης τῆς διαμέτρου περιενεθθὲν τὸ ἡμικύκλιον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, τὸ περιληφθὲν σχῆμα.

15. Ὁἄχων δὲ τῆς σφαίρας ἐστὶν ἡ μένουσα εὐθεῖα, περὶ ἣν τὸ ἡμικύκλιον στρέφεται.

16. Κέντρον δὲ τῆς σφαίρας ἐστὶ τὸ αὐτό, ὃ καὶ τοῦ ἡμικυκλίου.

17. Διάμετρος δὲ τῆς σφαίρας ἐστὶν εὐθεῖά τις διὰ τοῦ κέντρου ἠγμένη καὶ περατουμένη ἐφ᾽ ἑκάτερα τὰ μέρη ὑπὸ τῆς ἐπιφανείας τῆς σφαίρας.

18. Κῶνόξ ἐστιν, ὅταν ὀρθογωνίου τριγώνου μενούσης μιᾶς πλευρᾶξ τῶν περὶ τὴν ὀρθὴν γωνίαν περιενεθθὲν τὸ τρίγωνον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, τὸ περιληφθὲν σχῆμα. κὀὰν μὲν ἡ μένουσα εὐθεῖαὁίσηὸῇ τῇ λοιπῇ

[τῇ] περὶ τὴν ὀρθὴν περιφερομένῃ, ὀρθογώνιος ὀέσται ὁ κῶνος, ἐαν δὲ ἐλάττων, ἀμβλυγώνιος, ἐαν δὲ μείζων, ὀχυγώνιος.

19. Ὁ Ἄχων δὲ τοῦ κώνου ἐστὶν ἡ μένουσα εὐθεῖα, περὶ ἣν τὸ τρίγωνον στρέφεται.

20. Βάσις δὲ ὁ κύκλος ὁ ὑπὸ τῆς περιφερομένης εὐθείας γραφόμενος.

21. Κύλινδρός ἐστιν, ὅταν ὀρθογωνίου παραλληλογράμμου μενούσης μιᾶς πλευρᾶξ τῶν περὶ τὴν ὀρθὴν γωνίαν περιενεθθὲν τὸ παραλληλόγραμμον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, τὸ περιληφθὲν σχῆμα.

22. Ὁ Ἄχων δὲ τοῦ κυλίνδρου ἐστὶν ἡ μένουσα εὐθεῖα, περὶ ἣν τὸ παραλληλόγραμμον στρέφεται.

23. Βάσεις δὲ οἱ κύκλοι οἱ ὑπὸ τῶν ἀπεναντίον περιαγομένων δύο πλευρῶν γραφόμενοι.

24. Ὅμοιοι κῶνοι καὶ κύλινδροί εἰσιν, ὧν οὶ τεὸάχονες καὶ αἱ διάμετροι τῶν βάσεων ἀνάλογόν εἰσιν.

25. Κύβος ἐστὶ σχῆμα στερεὸν ὑπὸ ὸἓχ τετραγώνωνὸίσων περιεχόμενον.

26. Ὀκτάεδρόν ἐστὶ σχῆμα στερεὸν ὑπὸ ὀκτὼ τριγώνωνὸίσων καὶ ἰσοπλεύρων περιεχόμενον.

27. Εἰκοσάεδρόν ἐστι σχῆμα στερεὸν ὑπὸ εὸίκοσι τριγώνωνὸίσων καὶ ἰσοπλεύρων περιεχόμενον.

28. Δωδεκάεδρόν ἐστι σχῆμα στερεὸν ὑπὸ δώδεκα πενταγώνωνὸίσων καὶ ἰσοπλεύρων καὶ ἰσογωνίων περιεχόμενον.

Definitions

1. A solid is a (figure) having length and breadth and depth.

2. The extremity of a solid (is) a surface.

3. A straight-line is at right-angles to a plane when it makes right-angles with all of the straight-lines joined to it which are also in the plane.

4. A plane is at right-angles to a(nother) plane when (all of) the straight-lines drawn in one of the planes, at right-angles to the common section of the planes, are at right-angles to the remaining plane.

5. The inclination of a straight-line to a plane is the angle contained by the drawn and standing (straight-lines), when a perpendicular is lead to the plane from the end of the (standing) straight-line raised (out of the plane), and a straight-line is (then) joined from the point (so) generated to the end of the (standing) straight-line (lying) in the plane.

6. The inclination of a plane to a(nother) plane is the acute angle contained by the (straight-lines), (one) in each of the planes, drawn at right-angles to the common segment (of the planes), at the same point.

7. A plane is said to be similarly inclined to a plane, as another to another, when the aforementioned angles of inclination are equal to one another.

8. Parallel planes are those which do not meet (one another).

9. Similar solid figures are those contained by equal numbers of similar planes (which are similarly arranged).

10. But equal and similar solid figures are those contained by similar planes equal in number and in magnitude (which are similarly arranged).

11. A solid angle is the inclination (constituted) by more than two lines joining one another (at the same point), and not being in the same surface, to all of the lines. Otherwise, a solid angle is that contained by more than two plane angles, not being in the same plane, and constructed at one point.

12. A pyramid is a solid figure, contained by planes, (which is) constructed from one plane to one point.

13. A prism is a solid figure, contained by planes, of which the two opposite (planes) are equal, similar, and parallel, and the remaining (planes are) parallelograms.

14. A sphere is the figure enclosed when, the diameter of a semicircle remaining (fixed), the semicircle is carried around, and again established at the same (position) from which it began to be moved.

15. And the axis of the sphere is the fixed straight-line about which the semicircle is turned.

16. And the center of the sphere is the same as that of the semicircle.

17. And the diameter of the sphere is any straight-line which is drawn through the center and terminated in both directions by the surface of the sphere.

18. A cone is the figure enclosed when, one of the sides of a right-angled triangle about the right-angle remaining (fixed), the triangle is carried around, and again established at the same (position) from which it began to be moved. And if the fixed straight-line is equal to the remaining (straight-line) about the right-angle, (which is) carried around, then the cone will be right-angled, and if less, obtuse-angled, and if greater, acute-angled.

19. And the axis of the cone is the fixed straight-line about which the triangle is turned.

20. And the base (of the cone is) the circle described by the (remaining) straight-line (about the right-angle which is) carried around (the axis).

21. A cylinder is the figure enclosed when, one of the sides of a right-angled parallelogram about the right-angle remaining (fixed), the parallelogram is carried around, and again established at the same (position) from which it began to be moved.

22. And the axis of the cylinder is the stationary straight-line about which the parallelogram is turned.

23. And the bases (of the cylinder are) the circles described by the two opposite sides (which are) carried around.

24. Similar cones and cylinders are those for which the axes and the diameters of the bases are proportional.

25. A cube is a solid figure contained by six equal squares.

26. An octahedron is a solid figure contained by eight equal and equilateral triangles.

27. An icosahedron is a solid figure contained by twenty equal and equilateral triangles.

28. A dodecahedron is a solid figure contained by twelve equal, equilateral, and equiangular pentagons.

1.

Εὐθείας γραμμῆς μέρος μέν τι οὐκόέστιν ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, μέρος δέ τι ἐν μετεωροτέρῳ.

Εἰ γὰρ δυνατόν, εὐθείας γραμμῆς τῆς ΑΒΓ μέρος μέν τι τὸ ΑΒὸέστω ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, μέρος δέ τι τὸ ΒΓ ἐν μετεωροτέρῳ.

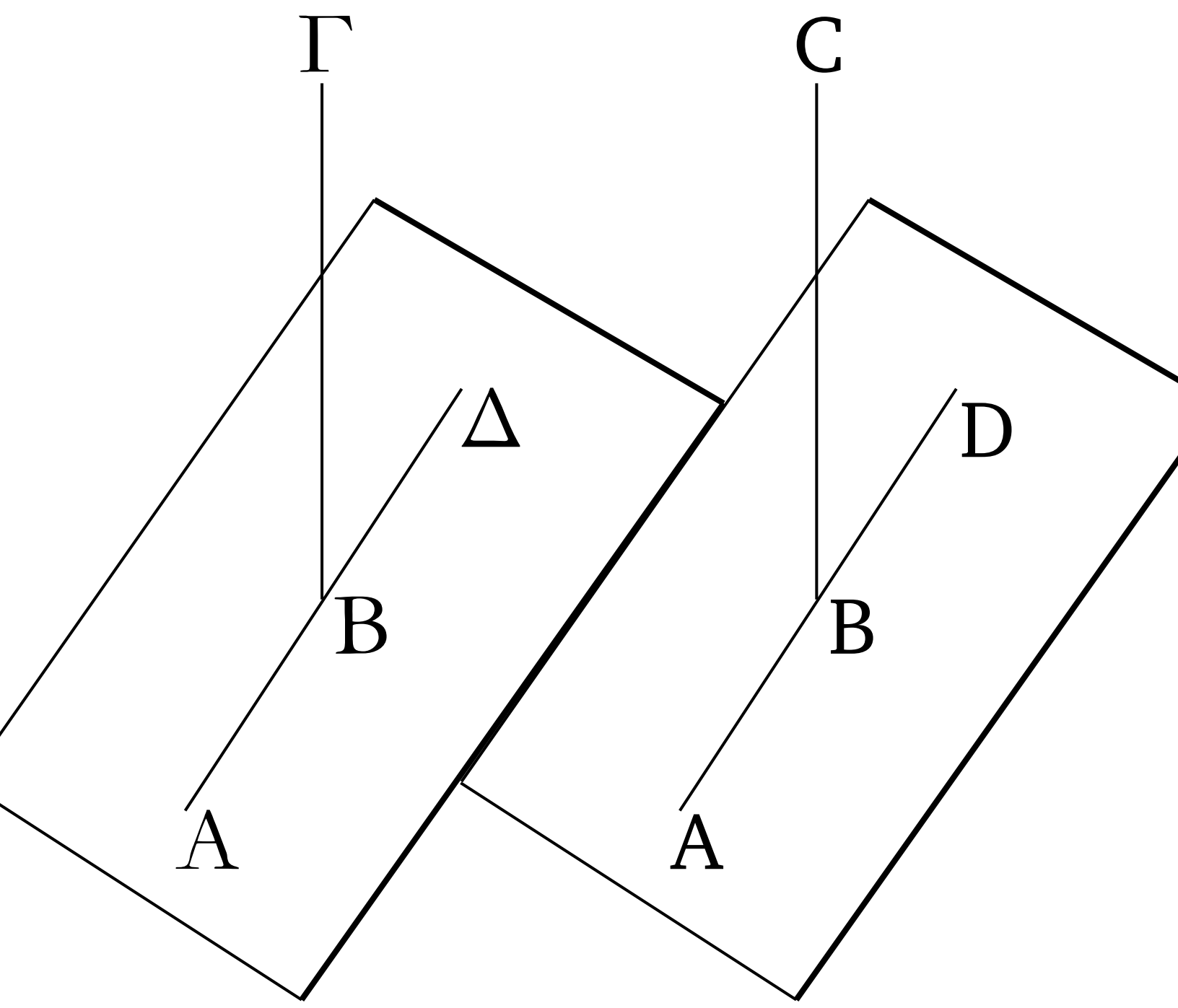
ὸΕσται δή τις τῇ ΑΒ συννεθὴξ εὐθεῖα ἐπ΄ εὐθείας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ. ὸέστω ἡ ΒΔ· δύο ἄρα εὐθειῶν τῶν ΑΒΓ, ΑΒΔ κοινὸν τμῆμά ἐστιν ἡ ΑΒ· ὅπερ ἐστὶν ἀδύνατον, ἐπειδήπερ ἓαν κέντρῳ τῷ Β καὶ διαστήματι τῷ ΑΒ κύκλον γράψωμεν, αἱ διάμετροι ἀνίσους ἀπολήψονται τοῦ κύκλου περιφερείας.

Proposition 1[†]

Some part of a straight-line cannot be in a reference plane, and some part in a more elevated (plane).

For, if possible, let some part, *AB*, of the straight-line *ABC* be in a reference plane, and some part, *BC*, in a more elevated (plane).

In the reference plane, there will be some straight-line continuous with, and straight-on to, *AB*.[‡] Let it be *BD*. Thus, *AB* is a common segment of the two (different) straight-lines *ABC* and *ABD*. The very thing is impossible, inasmuch as if we draw a circle with center *B* and radius *AB* then the diameters (*ABD* and *ABC*) will cut off unequal circumferences of the circle.



Εὐθείας ἄρα γραμμῆς μέρος μέν τι οὐκὸέστιν ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ ἐν μετεωροτέρῳ· ὅπερ ἔδει δεῖχαι.

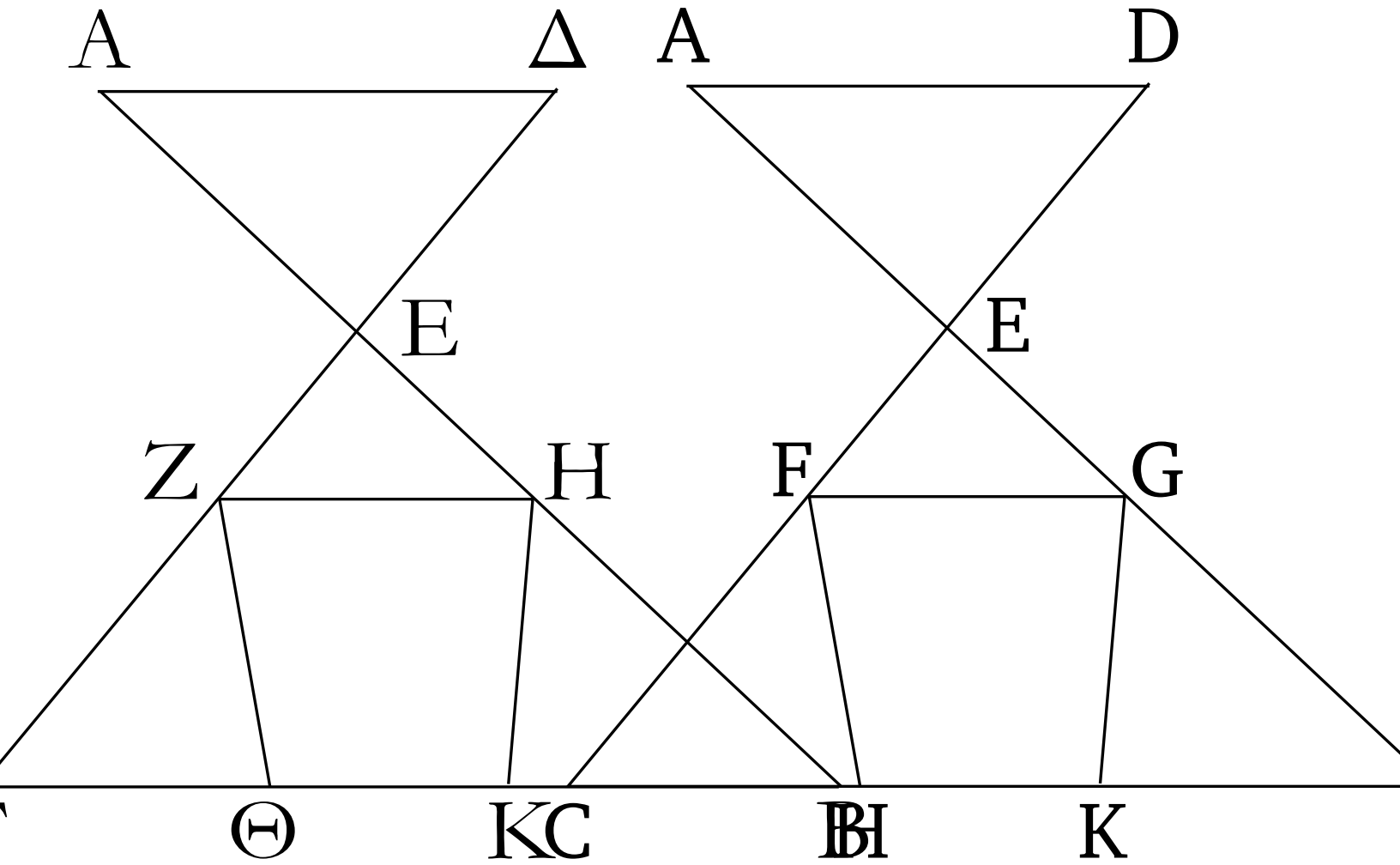
Thus, some part of a straight-line cannot be in a reference plane, and (some part) in a more elevated (plane). (Which is) the very thing it was required to show.

† The proofs of the first three propositions in this book are not at all rigorous. Hence, these three propositions should properly be regarded as additional axioms.

‡ This assumption essentially presupposes the validity of the proposition under discussion.

Ἐὰν δύο εὐθεῖαι τέμνωσιν ἀλλήλας, ἐν ἑνί εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἑνί ἐστιν ἐπιπέδῳ.

If two straight-lines cut one another then they are in one plane, and every triangle (formed using segments of both lines) is in one plane.



Δύο γὰρ εὐθεῖαι αἱ ΑΒ, ΓΔ τεμνέτωσαν ἀλλήλας κατὰ τὸ Ε σημεῖον. λέγω, ὅτι αἱ ΑΒ, ΓΔ ἐν ἑνί εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἑνί ἐστιν ἐπιπέδῳ.

Εἰλήφθω γὰρ ἐπὶ τῶν ΕΓ, ΕΒ τυθόντα σημεῖα τὰ Ζ, Η, καὶ ἐπεζεύθθωσαν αἱ ΓΒ, ΖΗ, καὶ διήθθωσαν αἱ ΖΘ, ΗΚ· λέγω πρῶτον, ὅτι τὸ ΕΓΒ τρίγωνον ἐν ἑνί ἐστιν ἐπιπέδῳ. εἰ γάρ ἐστι τοῦ ΕΓΒ τριγώνου μέρος ἤτοι τὸ ΖΘΓὸἠ τὸ ΗΒΚ ἐν τῷ ὑποκειμένῳ [ἐπιπέδῳ], τὸ δὲ λοιπὸν ἐνὸάλλῳ, ὸέσται καὶ μιᾶς τῶν ΕΓ, ΕΒ εὐθειῶν μέρος μέν τι ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ ἐν αλλῳ. εἰ δὲ τοῦ ΕΓΒ τριγώνου τὸ ΖΓΒΗ μέροςὸῇ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ λοιπὸν ἐνὸάλλῳ, ὸέσται καὶ ἀμφοτέρων τῶν ΕΓ, ΕΒ εὐθειῶν μέρος μέν τι ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ ἐνὸάλλω· ὅπερὸάτοπον ἐδείθθη. τὸ ἄρα ΕΓΒ τρίγωνον ἐν ἑνί ἐστιν ἐπιπέδῳ. ἐν ὁῷ δέ ἐστι τὸ ΕΓΒ τρίγωνον, ἐν τούτῳ καὶ ἑκατέρα τῶν ΕΓ, ΕΒ, ἐν ὁῷ δὲ ἑκατέρα τῶν ΕΓ, ΕΒ, ἐν τούτῳ καὶ αἱ ΑΒ, ΓΔ. αἱ ΑΒ, ΓΔ

For let the two straight-lines *AB* and *CD* have cut one another at point *E*. I say that *AB* and *CD* are in one plane, and that every triangle (formed using segments of both lines) is in one plane.

For let the random points *F* and *G* be taken on *EC* and *EB* (respectively). And let *CB* and *FG* be joined, and let *FH* and *GK* be drawn across. I say, first of all, that triangle *ECB* is in one (reference) plane. For if part of triangle *ECB*, either *FHC* or *GBK*, is in the reference [plane], and the remainder in a different (plane) then a part of one the straight-lines *EC* and *EB* will also be in the reference plane, and (a part) in a different (plane). And if the part *FCBG* of triangle *ECB* is in the reference plane, and the remainder in a different (plane) then parts of both of the straight-lines *EC* and *EB* will also be in the reference plane, and (parts) in a different (plane). The very thing was shown to be absurb [Prop. 11.1]. Thus, triangle

ἄρα εὐθεῖαι ἐν ἑνί εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἑνί ἐστιν ἐπιπέδῳ· ὅπερ ἔδει δεῖξαι.

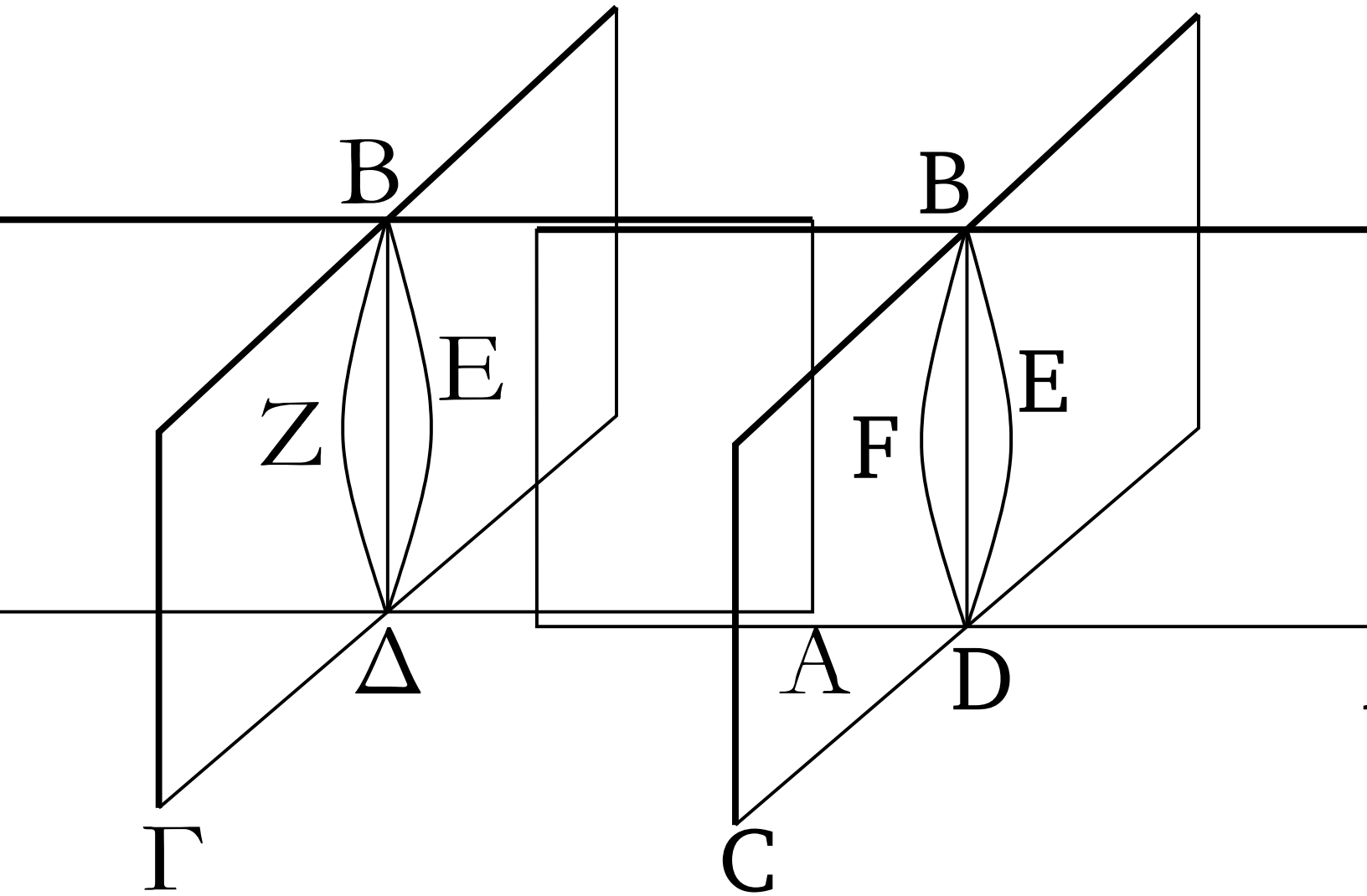
ECB is in one plane. And in whichever (plane) triangle *ECB* is (found), in that (plane) *EC* and *EB* (will) each also (be found). And in whichever (plane) *EC* and *EB* (are) each (found), in that (plane) *AB* and *CD* (will) also (be found) [Prop. 11.1]. Thus, the straight-lines *AB* and *CD* are in one plane, and every triangle (formed using segments of both lines) is in one plane. (Which is) the very thing it was required to show.

3.

Ἐὰν δύο ἐπίπεδα τεμνῇ ὀάλληλα, ἡ κοινὴ αὐτῶν τομὴ εὐθεῖά ἐστιν.

Proposition 3

If two planes cut one another then their common section is a straight-line.



Δύο γὰρ ἐπίπεδα τὰ AB, BΓ τεμνέτωὀάλληλα, κοινὴ δὲ αὐτῶν τομὴ ὀέστω ἡ ΔB γραμμή· λέγω, ὅτι ἡ ΔB γραμμὴ εὐθεῖά ἐστιν.

Εἰ γὰρ μή, ἐπεζεύθθω ἀπὸ τοῦ Δ ἐπὶ τὸ B ἐν μὲν τῷ AB ἐπιπέδῳ εὐθεῖα ἡ ΔEB, ἐν δὲ τῷ BΓ ἐπιπέδῳ εὐθεῖα ἡ ΔZB.

For let the two planes *AB* and *BC* cut one another, and let their common section be the line *DB*. I say that the line *DB* is straight.

For, if not, let the straight-line *DEB* be joined from *D* to *B* in the plane *AB*, and the straight-line *DFB* in the plane *BC*.

ὀέσται δὴ δύο εὐθειῶν τῶν ΔΕΒ, ΔΖΒ τὰ αὐτὰ πέρατα, καὶ περιέχουσι δηλαδὴ θωρίον· ὅπερ ὀάτοπον. οὐκ ἄρα αἱ ΔΕΒ, ΔΖΒ εὐθεῖαί εἰσιν. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ὀάλλη τις ἀπὸ τοῦ Δ ἐπὶ τὸ Β ἐπιζευγνυμένη εὐθεῖαὀέσται πλὴν τῆς ΔΒ κοινῆς τομῆς τῶν ΑΒ, ΒΓ ἐπιπέδων.

Ἐὰν ἄρα δύο ἐπίπεδα τέμνῃ ὀάλληλα, ἡ κοινὴ αὐτῶν τομὴ εὐθεῖά ἐστιν· ὅπερ ἔδει δεῖχαι.

So two straight-lines, DEB and DFB, will have the same ends, and they will clearly enclose an area. The very thing (is) absurd. Thus, DEB and DFB are not straight-lines. So, similarly, we can show than no other straight-line can be joined from D to B except DB, the common section of the planes AB and BC.

Thus, if two planes cut one another then their common section is a straight-line. (Which is) the very thing it was required to show.

4.

Ἐὰν εὐθεῖα δύο εὐθείαις τεμνούσαις ἀλλήλας πρὸς ὀρθὰξ ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, καὶ τῷ διὸ αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰξὀέσται.

Εὐθεῖα γάρ τις ἡ ΕΖ δύο εὐθείαις ταῖς ΑΒ, ΓΔ τεμνούσαις ἀλλήλας κατὰ τὸ Ε σημεῖον ἀπὸ τοῦ Ε πρὸς ὀρθὰξ ἐφεστάτω· λέγω, ὅτι ἡ ΕΖ καὶ τῷ διὰ τῶν ΑΒ, ΓΔ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν.

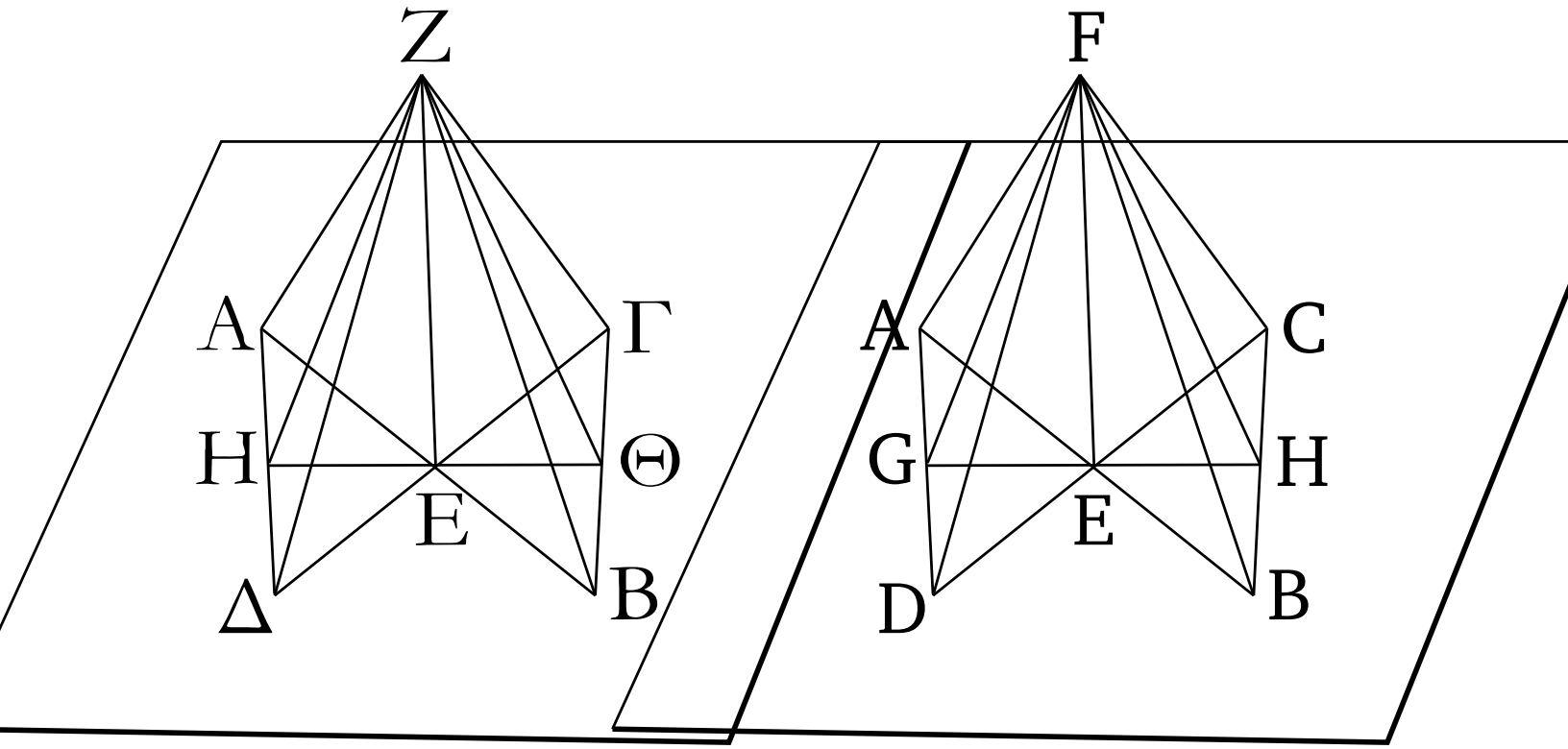
Ἀπειλήφθωσαν γὰρ αἱ ΑΕ, ΕΒ, ΓΕ, ΕΔ ἴσαι ἀλλήλαις, καὶ διήχθω τις διὰ τοῦ Ε, ὡξὀέτυθεν, ἡ ΗΕΘ, καὶ ἐπεζεύχθωσαν αἱ ΑΔ, ΓΒ, καὶ ὀέτι ἀπὸ τυχόντος τοῦ Ζ ἐπεζεύχθωσαν αἱ ΖΑ, ΖΗ, ΖΔ, ΖΓ, ΖΘ, ΖΒ.

Proposition 4

If a straight-line is set up at right-angles to two straight-lines cutting one another, at the common point of section, then it will also be at right-angles to the plane (passing) through them (both).

For let some straight-line EF have (been) set up at right-angles to two straight-lines, AB and CD, cutting one another at point E, at E. I say that EF is also at right-angles to the plane (passing) through AB and CD.

For let AE, EB, CE and ED be cut off from (the two straight-lines so as to be) equal to one another. And let GEH be drawn, at random, through E (in the plane passing through AB and CD). And let AD and CB be joined. And, furthermore, let FA, FG, FD, FC, FH, and FB be joined from the random (point) F (on EF).



Καὶ ἐπεὶ δύο αἱ ΑΕ, ΕΔ δυσὶ ταῖς ΓΕ, ΕΒ ἴσαι εἰσὶ καὶ

For since the two (straight-lines) AE and ED are equal to

γωνίας ἴσας περιέθουσιν, βάσις ἄρα ἡ $A\Delta$ βάσει τῇ ΓBὁίση ἐστίν, καὶ τὸ $AE\Delta$ τρίγωνον τῷ ΓEB τριγώνῳ ἴσονὁέσται· ὁώστε καὶ γωνία ἡ ὑπὸ ΔAE γωνίᾳ τῇ ὑπὸ $EB\Gamma$ὁίση [ἐστίν]. ὁέστι δὲ καὶ ἡ ὑπὸ AEH γωνία τῇ ὑπὸ $BE\Theta$ὁίση. δύο δὴ τρίγωνά ἐστι τὰ AHE, $BE\Theta$ τὰξ δύο γωνίας δυσὶ γωνίαις ἴσας ἔχοντα ἑκατέραν ἑκατέρᾳ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην τὴν πρὸς ταῖς ἴσαις γωνίαις τὴν AE τῇ EB· καὶ τὰξ λοιπὰξ ἄρα πλευρὰξ ταῖς λοιπαῖξ πλευραῖξ ἴσας ὁέχουσιν. ὁίση ἄρα ἡ μὲν HE τῇ $E\Theta$, ἡ δὲ AH τῇ $B\Theta$. καὶ ἐπεὶ ὁίση ἐστὶν ἡ AE τῇ EB, κοινὴ δὲ καὶ πρὸς ὀρθὰξ ἡ ZE, βάσις ἄρα ἡ ZA βάσει τῇ ZB ἐστινὁίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ $Z\Gamma$ τῇ $Z\Delta$ ἐστινὁίση. καὶ ἐπεὶ ὁίση ἐστὶν ἡ $A\Delta$ τῇ ΓB, ὁέστι δὲ καὶ ἡ ZA τῇ ZBὁίση, δύο δὴ αἱ ZA, $A\Delta$ δυσὶ ταῖς ZB, $B\Gamma$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ βάσις ἡ $Z\Delta$ βάσει τῇ $Z\Gamma$ ἐδείθηὁίση· καὶ γωνία ἄρα ἡ ὑπὸ $ZA\Delta$ γωνίᾳ τῇ ὑπὸ $ZB\Gamma$ὁίση ἐστίν. καὶ ἐπεὶ πάλιν ἐδείθη ἡ AH τῇ $B\Theta$ὁίση, ἀλλὰ μὴν καὶ ἡ ZA τῇ ZBὁίση, δύο δὴ αἱ ZA, AH δυσὶ ταῖς ZB, $B\Theta$ ἴσαι εἰσίν. καὶ γωνία ἡ ὑπὸ ZAH ἐδείθηὁίση τῇ ὑπὸ $ZB\Theta$· βάσις ἄρα ἡ ZH βάσει τῇ $Z\Theta$ ἐστινὁίση. καὶ ἐπεὶ πάλινὁίση ἐδείθη ἡ HE τῇ $E\Theta$, κοινὴ δὲ ἡ EZ, δύο δὴ αἱ HE, EZ δυσὶ ταῖς ΘE, EZ ἴσαι εἰσίν· καὶ βάσις ἡ ZH βάσει τῇ $Z\Theta$ὁίση· γωνία ἄρα ἡ ὑπὸ HEZ γωνίᾳ τῇ ὑπὸ ΘEZὁίση ἐστίν. ὀρθὴ ἄρα ἑκατέρα τῶν ὑπὸ HEZ, ΘEZ γωνιῶν. ἡ ZE ἄρα πρὸς τὴν $H\Theta$ τυθόντως διὰ τοῦ E ἀθθεῖσαν ὀρθὴ ἐστιν. ὁμοίως δὴ δείχομεν, ὅτι ἡ ZE καὶ πρὸς πάσας τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐόυσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας. εὐθεῖα δὲ πρὸς ἐπίπεδον ὀρθή ἐστιν, ὅταν πρὸς πάσας τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐόυσας ἐν τῷ αὐτῷ ἐπιπέδῳ ὀρθὰξ ποιῇ γωνίας· ἡ ZE ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. τὸ δὲ ὑποκείμενον ἐπίπεδόν ἐστι τὸ διὰ τῶν AB, $\Gamma\Delta$ εὐθειῶν. ἡ ZE ἄρα πρὸς ὀρθάξ ἐστι τῷ διὰ τῶν AB, $\Gamma\Delta$ ἐπιπέδῳ.

Ἐὰν ἄρα εὐθεῖα δύο εὐθείαις τεμνούσαις ἀλλήλας πρὸς ὀρθὰξ ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, καὶ τῷ διὰ αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰξὁέσται· ὅπερ ἔδει δεῖξαι.

the two (straight-lines) CE and EB, and they enclose equal angles [Prop. 1.15], the base AD is thus equal to the base CB, and triangle AED will be equal to triangle CEB [Prop. 1.4]. Hence, the angle DAE [is] equal to the angle EBC. And the angle AEG (is) also equal to the angle BEH [Prop. 1.15]. So AGE and BEH are two triangles having two angles equal to two angles, respectively, and one side equal to one side—(namely), those by the equal angles, AE and EB. Thus, they will also have the remaining sides equal to the remaining sides [Prop. 1.26]. Thus, GE (is) equal to EH, and AG to BH. And since AE is equal to EB, and FE is common and at right-angles, the base FA is thus equal to the base FB [Prop. 1.4]. So, for the same (reasons), FC is also equal to FD. And since AD is equal to CB, and FA is also equal to FB, the two (straight-lines) FA and AD are equal to the two (straight-lines) FB and BC, respectively. And the base FD was shown (to be) equal to the base FC. Thus, the angle FAD is also equal to the angle FBC [Prop. 1.8]. And, again, since AG was shown (to be) equal to BH, but FA (is) also equal to FB, the two (straight-lines) FA and AG are equal to the two (straight-lines) FB and BH (respectively). And the angle FAG was shown (to be) equal to the angle FBH. Thus, the base FG is equal to the base FH [Prop. 1.4]. And, again, since GE was shown (to be) equal to EH, and EF (is) common, the two (straight-lines) GE and EF are equal to the two (straight-lines) HE and EF (respectively). And the base FG (is) equal to the base FH. Thus, the angle GEF is equal to the angle HEF [Prop. 1.8]. Each of the angles GEF and HEF (are) thus right-angles [Def. 1.10]. Thus, FE is at right-angles to GH, which was drawn at random through E (in the reference plane passing though AB and AC). So, similarly, we can show that FE will make right-angles with all straight-lines joined to it which are in the reference plane. And a straight-line is at right-angles to a plane when it makes right-angles with all straight-lines joined to it which are in the plane [Def. 11.3]. Thus, FE is at right-angles to the reference plane. And the reference plane is that (passing) through the straight-lines AB and CD. Thus, FE is at right-angles to the plane (passing) through AB and CD.

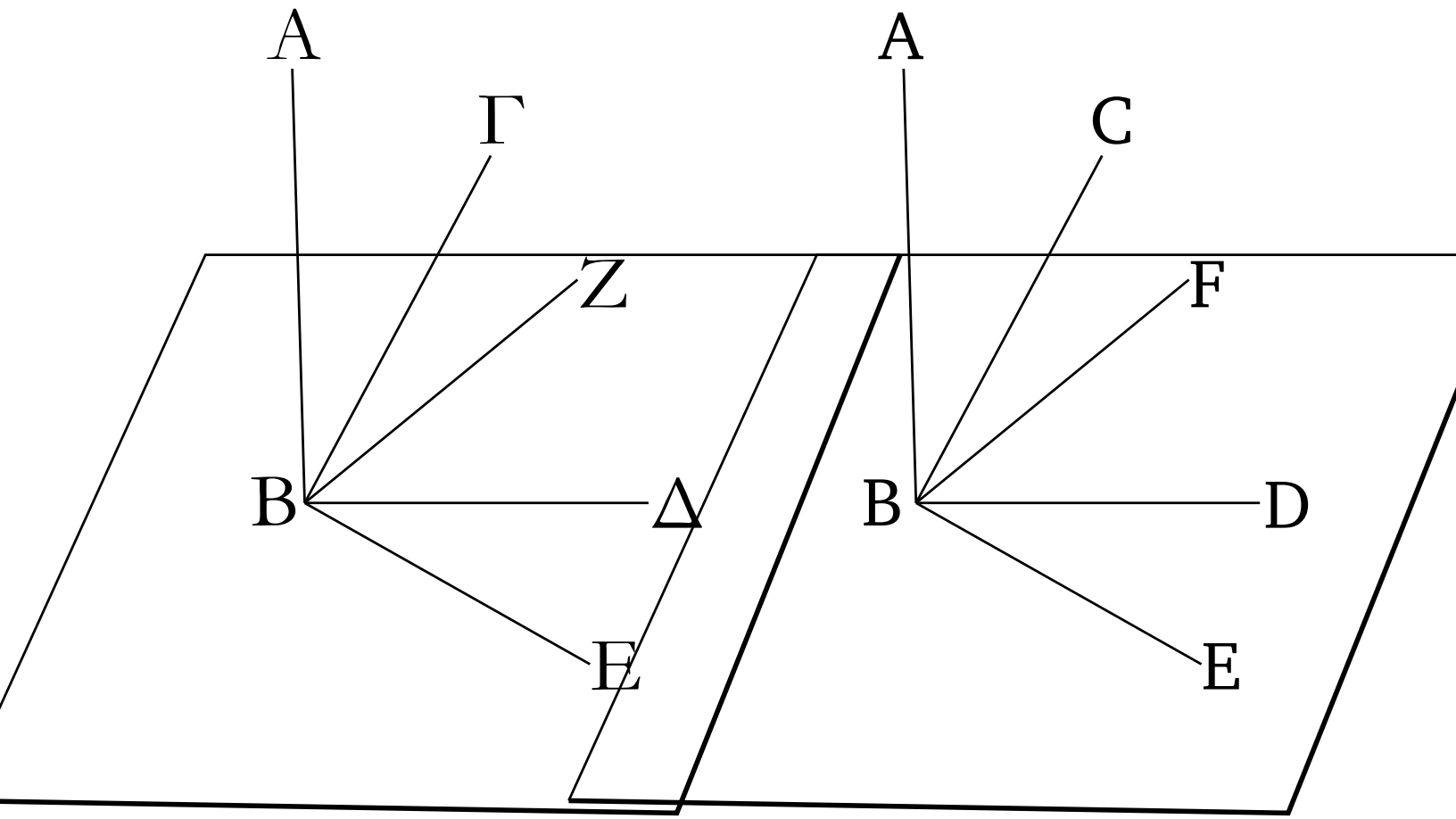
Thus, if a straight-line is set up at right-angles to two straight-lines cutting one another, at the common point of section, then it will also be at right-angles to the plane (passing) through them (both). (Which is) the very thing it was required to show.

5.

Ἐὰν εὐθεῖα τρισὶν εὐθείαις ἁπτομέναις ἀλλήλων πρὸς ὀρθὰξ ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, αἱ τρεῖς εὐθεῖαι ἐν ἑνί εἰσιν ἐπιπέδῳ.

Proposition 5

If a straight-line is set up at right-angles to three straight-lines cutting one another, at the common point of section, then the three straight-lines are in one plane.



Εὐθεῖα γάρ τις ἡ ΑΒ τρισὶν εὐθείαις ταῖς ΒΓ, ΒΔ, ΒΕ πρὸς ὀρθὰξ ἐπὶ τῆς κατὰ τὸ Β ἁφῆς ἐφεστάτω· λέγω, ὅτι αἱ ΒΓ, ΒΔ, ΒΕ ἐν ἑνί εἰσιν ἐπιπέδῳ.

μὴ γάρ, ἀλλὰ εἰ δυνατόν, ὀέστωσαν αἱ μὲν ΒΔ, ΒΕ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, ἡ δὲ ΒΓ ἐν μετεωροτέρῳ, καὶ ἐκβεβλήσθω τὸ διὰ τῶν ΑΒ, ΒΓ ἐπίπεδον· κοινὴν δὴ τομὴν ποιήσει ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιείτω τὴν ΒΖ. ἐν ἑνὶ ἄρα εἰσὶν ἐπιπέδῳ τῷ διηγμένῳ διὰ τῶν ΑΒ, ΒΓ αἱ τρεῖς εὐθεῖαι αἱ ΑΒ, ΒΓ, ΒΖ. καὶ ἐπεὶ ἡ ΑΒ ὀρθή ἐστι πρὸς ἑκατέραν τῶν ΒΔ, ΒΕ, καὶ τῷ διὰ τῶν ΒΔ, ΒΕ ἄρα ἐπιπέδῳ ὀρθή ἐστιν ἡ ΑΒ. τὸ δὲ διὰ τῶν ΒΔ, ΒΕ ἐπίπεδον τὸ ὑποκείμενόν ἐστιν· ἡ ΑΒ ἄρα ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον. ὥστε καὶ πρὸς πάσας τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας ἡ ΑΒ. ὁάπτεται δὲ αὐτῆς ἡ ΒΖ οˆσα ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ· ἡ ἄρα ὑπὸ ΑΒΖ γωνία ὀρθή ἐστιν. ὑπόκειται δὲ καὶ ἡ ὑπὸ ΑΒΓ ὀρθή· ὁίση ἄρα ἡ ὑπὸ ΑΒΖ γωνία τῇ ὑπὸ ΑΒΓ. καί εἰσιν ἐν ἑνὶ ἐπιπέδῳ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ ΒΓ εὐθεῖα ἐν μετεωροτέρῳ ἐστὶν ἐπιπέδῳ· αἱ τρεῖς ἄρα εὐθεῖαι αἱ ΒΓ, ΒΔ, ΒΕ ἐν ἑνί εἰσιν ἐπιπέδῳ.

Ἐὰν ἄρα εὐθεῖα τρισὶν εὐθείαις ἁπτομέναις ἀλλήλων ἐπὶ τῆς ἁφῆς πρὸς ὀρθὰξ ἐπισταθῇ, αἱ τρεῖς εὐθεῖαι ἐν ἑνί εἰσιν

For let some straight-line *AB* be set up at right-angles to three straight-lines *BC*, *BD*, and *BE*, at the (common) point of section *B*. I say that *BC*, *BD*, and *BE* are in one plane.

For (if) not, and if possible, let *BD* and *BE* be in the reference plane, and *BC* in a more elevated (plane). And let the plane through *AB* and *BC* be produced. So it will make a straight-line as a common section with the reference plane [Def. 11.3]. Let it make *BF*. Thus, the three straight-lines *AB*, *BC*, and *BF* are in one plane—(namely), that drawn through *AB* and *BC*. And since *AB* is at right-angles to each of *BD* and *BE*, *AB* is thus also at right-angles to the plane (passing) through *BD* and *BE* [Prop. 11.4]. And the plane (passing) through *BD* and *BE* is the reference plane. Thus, *AB* is at right-angles to the reference plane. Hence, *AB* will also make right-angles with all straight-lines joined to it which are also in the reference plane [Def. 11.3]. And *BF*, which is in the reference plane, is joined to it. Thus, the angle *ABF* is a right-angle. And *ABC* was also assumed to be a right-angle. Thus, angle *ABF* (is) equal to *ABC*. And they are in one plane. The very thing is impossible. Thus, *BC* is not in a more elevated plane. Thus, the three straight-lines *BC*, *BD*, and *BE* are in one plane.

ἐπιπέδῳ· ὅπερ ἔδει δεῖχαι.

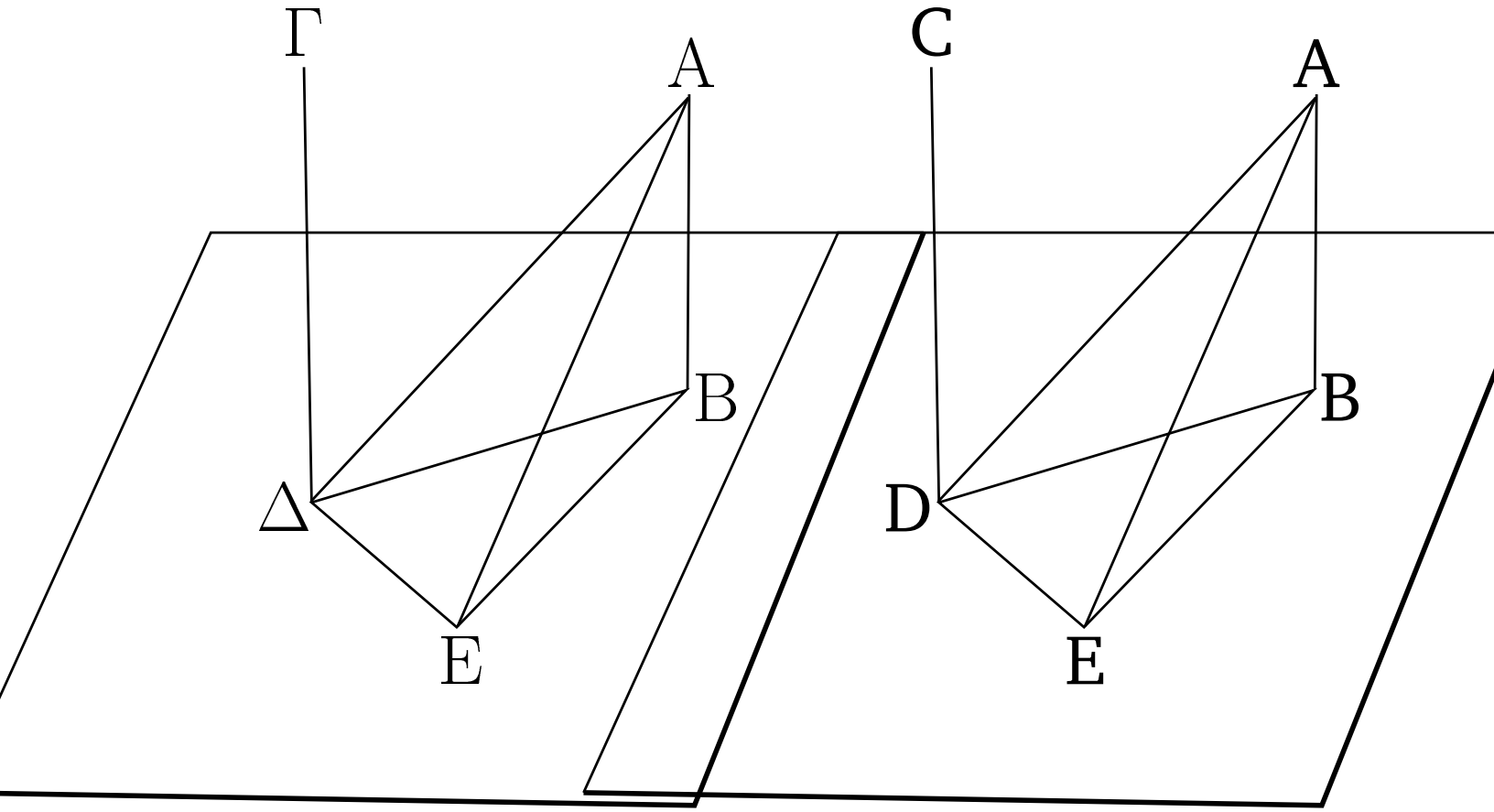
Thus, if a straight-line is set up at right-angles to three straight-lines cutting one another, at the (common) point of section, then the three straight-lines are in one plane. (Which is) the very thing it was required to show.

6.

Ἐὰν δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξ ὦσιν, παράλληλοιὀέσονται αἱ εὐθεῖαι.

Proposition 6

If two straight-lines are at right-angles to the same plane then the straight-lines will be parallel.†



Δύο γὰρ εὐθεῖαι αἱ ΑΒ, ΓΔ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξὀέστωσαν· λέγω, ὅτι παράλληλόξ ἐστιν ἡ ΑΒ τῇ ΓΔ.

Συμβαλλέτωσαν γὰρ τῷ ὑποκειμένῳ ἐπιπέδῳ κατὰ τὰ Β, Δ σημεῖα, καὶ ἐπεζεύθθω ἡ ΒΔ εὐθεῖα, καὶ ἤθθω τῇ ΒΔ πρὸς ὀρθὰξ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ἡ ΔΕ, καὶ κείσθω τῇ ΑΒὀίση ἡ ΔΕ, καὶ ἐπεζεύθθωσαν αἱ ΒΕ, ΑΕ, ΑΔ.

Καὶ ἐπεὶ ἡ ΑΒ ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσας [ἄρα] τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας. ὁάπτεται δὲ τῆς ΑΒ ἑκατέρα τῶν ΒΔ, ΒΕ οὐῦσα ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ· ὀρθὴ ἄρα ἐστὶν ἑκατέρα τῶν ὑπὸ ΑΒΔ, ΑΒΕ γωνιῶν. διὰ τὰ αὐτὰ δὴ καὶ ἑκατέρα τῶν ὑπὸ ΓΔΒ, ΓΔΕ ὀρθή ἐστιν. καὶ ἐπεὶ ὀίση ἐστὶν ἡ ΑΒ τῇ ΔΕ, κοινὴ δὲ ἡ ΒΔ, δύο δὴ αἱ ΑΒ, ΒΔ δυσὶ ταῖς ΕΔ, ΔΒ ἴσαι εἰσίν·

For let the two straight-lines *AB* and *CD* be at right-angles to a reference plane. I say that *AB* is parallel to *CD*.

For let them meet the reference plane at points *B* and *D* (respectively). And let the straight-line *BD* be joined. And let *DE* be drawn at right-angles to *BD* in the reference plane. And let *DE* be made equal to *AB*. And let *BE*, *AE*, and *AD* be joined.

And since *AB* is at right-angles to the reference plane, it will [thus] also make right-angles with all straight-lines joined to it which are in the reference plane [Def. 11.3]. And *BD* and *BE*, which are in the reference plane, are each joined to *AB*. Thus, each of the angles *ABD* and *ABE* are right-angles. So, for the same (reasons), each of the angles *CDB* and *CDE* are also right-angles. And since *AB* is equal to *DE*, and *BD*

καὶ γωνίας ὀρθὰξ περιέθουσιν· βάσις ἄρα ἡ $A\Delta$ βάσει τῇ BE ἐστινὀίση. καὶ ἐπεὶ ὀίση ἐστὶν ἡ AB τῇ ΔE, ἀλλὰ καὶ ἡ $A\Delta$ τῇ BE, δύο δὴ αἱ AB, BE δυσὶ ταῖς $E\Delta$, ΔA ἴσαι εἰσίν· καὶ βάσις αὐτῶν κοινὴ ἡ AE· γωνία ἄρα ἡ ὑπὸ ABE γωνιᾷ τῇ ὑπὸ $E\Delta A$ ἐστινὀίση. ὀρθὴ δὲ ἡ ὑπὸ ABE· ὀρθὴ ἄρα καὶ ἡ ὑπὸ $E\Delta A$· ἡ $E\Delta$ ἄρα πρὸς τὴν ΔA ὀρθή ἐστιν. ὀέστι δὲ καὶ πρὸς ἑκατέραν τῶν $B\Delta$, $\Delta\Gamma$ ὀρθή. ἡ $E\Delta$ ἄρα τρισὶν εὐθείαις ταῖς $B\Delta$, ΔA, $\Delta\Gamma$ πρὸς ὀρθὰξ ἐπὶ τῆς ἁφῆς ἐφέστηκεν· αἱ τρεῖς ἄρα εὐθεῖαι αἱ $B\Delta$, ΔA, $\Delta\Gamma$ ἐν ἑνί εἰσιν ἐπιπέδῳ. ἐν ὁῷ δὲ αἱ ΔB, ΔA, ἐν τούτῳ καὶ ἡ AB· πᾶν γὰρ τρίγωνον ἐν ἑνί ἐστιν ἐπιπέδῳ· αἱ ἄρα AB, $B\Delta$, $\Delta\Gamma$ εὐθεῖαι ἐν ἑνί εἰσιν ἐπιπέδῳ. καί ἐστιν ὀρθὴ ἑκατέρα τῶν ὑπὸ $AB\Delta$, $B\Delta\Gamma$ γωνιῶν· παράλληλος ἄρα ἐστὶν ἡ AB τῇ $\Gamma\Delta$.

Ἐὰν ἄρα δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξ ὦσιν, παράλληλοιὀέσονται αἱ εὐθεῖαι· ὅπερ ἔδει δεῖχαι.

(is) common, the two (straight-lines) AB and BD are equal to the two (straight-lines) ED and DB (respectively). And they contain right-angles. Thus, the base AD is equal to the base BE [Prop. 1.4]. And since AB is equal to DE, and AD (is) also (equal) to BE, the two (straight-lines) AB and BE are thus equal to the two (straight-lines) ED and DA (respectively). And their base AE (is) common. Thus, angle ABE is equal to angle EDA [Prop. 1.8]. And ABE (is) a right-angle. Thus, EDA (is) also a right-angle. ED is thus at right-angles to DA. And it is also at right-angles to each of BD and DC. Thus, ED is standing at right-angles to the three straight-lines BD, DA, and DC at the (common) point of section. Thus, the three straight-lines BD, DA, and DC are in one plane [Prop. 11.5]. And in which(ever) plane DB and DA (are found), in that (plane) AB (will) also (be found). For every triangle is in one plane [Prop. 11.2]. And each of the angles ABD and BDC is a right-angle. Thus, AB is parallel to CD [Prop. 1.28].

Thus, if two straight-lines are at right-angles to the same plane then the straight-lines will be parallel. (Which is) the very thing it was required to show.

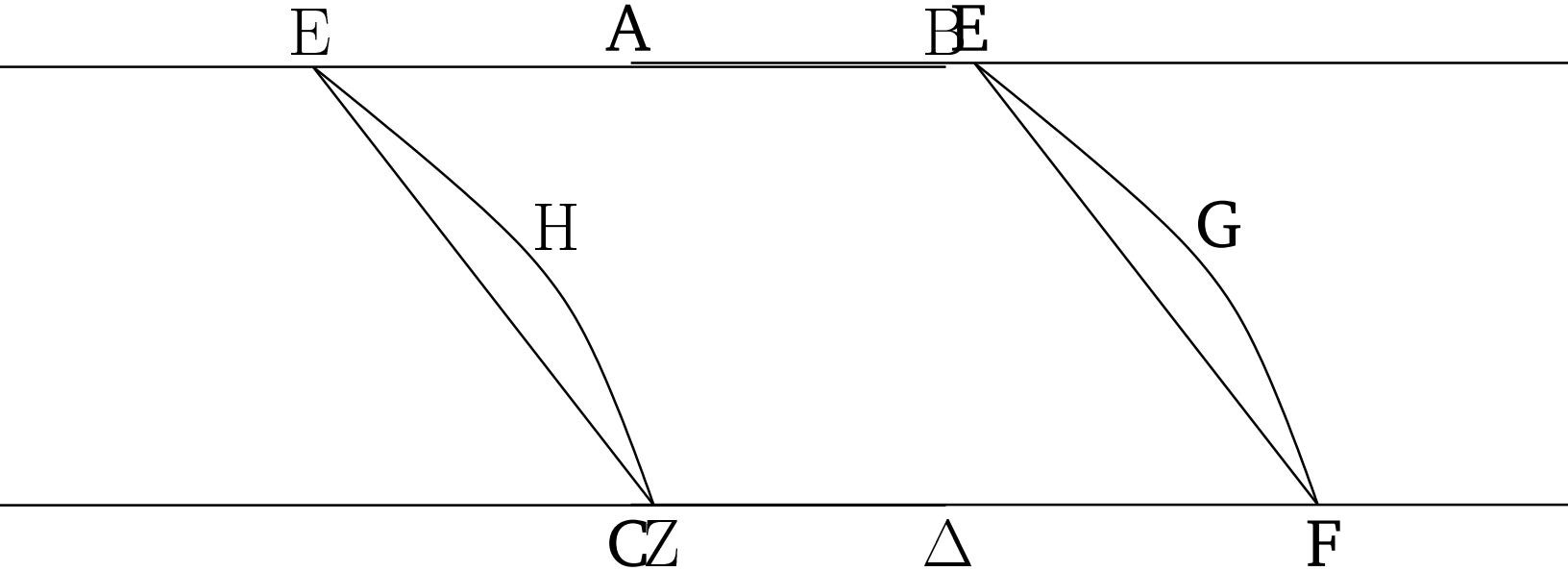
† In other words, the two straight-lines lie in the same plane, and never meet when produced in either direction.

7.

Ἐὰνὀῶσι δύο εὐθεῖαι παράλληλοι, ληφθῇ δὲ ἐφ' ἑκατέρας αὐτῶν τυθόντα σημεῖα, ἡ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις.

Proposition 7

If there are two parallel straight-lines, and random points are taken on each of them, then the straight-line joining the two points is in the same plane as the parallel (straight-lines).



Ἔστωσαν δύο εὐθεῖαι παράλληλοι αἱ AB, $\Gamma\Delta$, καὶ εἰλήφθω ἐφ' ἑκατέρας αὐτῶν τυνθόντα σημεῖα τὰ E, Z· λέγω, ὅτι ἡ ἐπὶ τὰ E, Z σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις.

μὴ γάρ, ἀλλὰ εἰ δυνατόν, ὀέστω ἐν μετεωροτέρῳ ὡξ ἡ E-

Let AB and CD be two parallel straight-lines, and let the random points E and F be taken on each of them (respectively). I say that the straight-line joining points E and F is in the same (reference) plane as the parallel (straight-lines).

For (if) not, and if possible, let it be in a more elevated

HZ, καὶ διήθθω διὰ τῆς EHZ ἐπίπεδον· τομὴν δὴ ποιήσει ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιείτω ὡξ τὴν EZ· δύο ἄρα εὐθεῖαι αἱ EHZ, EZ θωρίον περιέχουσιν· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ ἀπὸ τοῦ E ἐπὶ τὸ Z ἐπιζευγνυμένη εὐθεῖαι ἐν μετεωροτέρῳ ἐστὶν ἐπιπέδῳ· ἐν τῷ διὰ τῶν AB, ΓB ἄρα παραλλήλων ἐστὶν ἐπιπέδῳ ἡ ἀπὸ τοῦ E ἐπὶ τὸ Z ἐπιζευγνυμένη εὐθεῖα.

Ἐὰν ἄραὸῶσι δύο εὐθεῖαι παράλληλοι, ληφθῇ δὲ ἐφ΄ ἑκατέρας αὐτῶν τυθόντα σημεῖα, ἡ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις· ὅπερ ἔδει δεῖχαι.

8.

Ἐὰνὸῶσι δύο εὐθεῖαι παράλληλοι, ἡ δὲ ἑτέρα αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰξὸῇ, καὶ ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὸέσται.

Ἔστωσαν δύο εὐθεῖαι παράλληλοι αἱ AB, ΓΔ, ἡ δὲ ἑτέρα αὐτῶν ἡ AB τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξὸέστω· λέγω, ὅτι καὶ ἡ λοιπὴ ἡ ΓΔ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὸέσται.

Συμβαλλέτωσαν γὰρ αἱ AB, ΓΔ τῷ ὑποκειμένῳ ἐπιπέδῳ κατὰ τὰ B, Δ σημεῖα, καὶ ἐπεζεύθθω ἡ BΔ· αἱ AB, ΓΔ, BΔ ἄρα ἐν ἑνί εἰσιν ἐπιπέδῳ. ἤθθω τῇ BA πρὸς ὀρθὰξ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ἡ ΔE, καὶ κείσθω τῇ ABὸίση ἡ ΔE, καὶ ἐπεζεύθθωσαν αἱ BE, AE, AΔ.

(plane), such as EGF. And let a plane be drawn through EGF. So it will make a straight cutting in the reference plane [Prop. 11.3]. Let it make EF. Thus, two straight-lines (with the same end-points), EGF and EF, will enclose an area. The very thing is impossible. Thus, the straight-line joining *E* to *F* is not in a more elevated plane. The straight-line joining *E* to *F* is thus in the plane through the parallel (straight-lines) *AB* and *CD*.

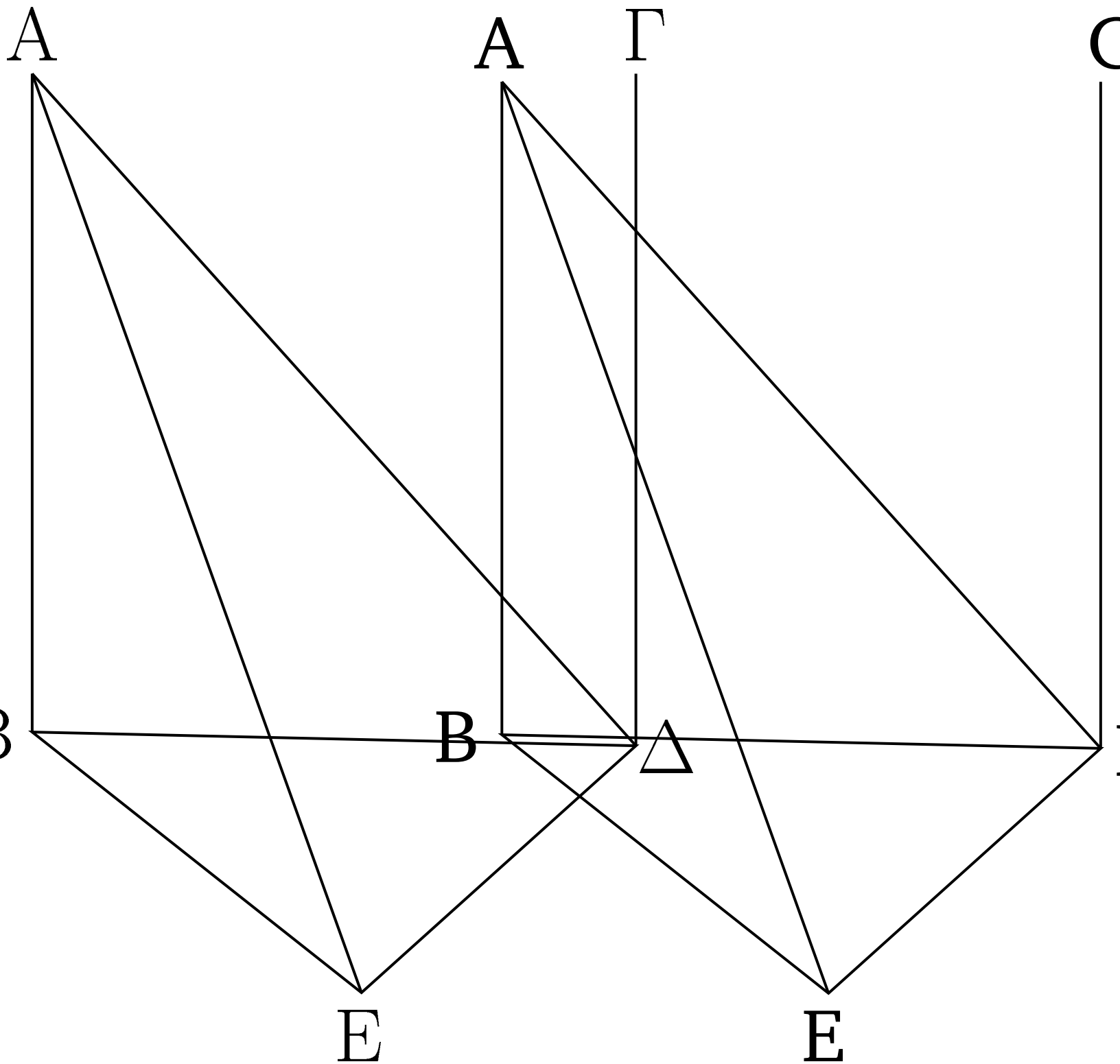
Thus, if there are two parallel straight-lines, and random points are taken on each of them, then the straight-line joining the two points is in the same plane as the parallel (straight-lines). (Which is) the very thing it was required to show.

Proposition 8

If two straight-lines are parallel, and one of them is at right-angles to some plane, then the remaining (one) will also be at right-angles to the same plane.

Let *AB* and *CD* be two parallel straight-lines, and let one of them, *AB*, be at right-angles to a reference plane. I say that the remaining (one), *CD*, will also be at right-angles to the same plane.

For let *AB* and *CD* meet the reference plane at points *B* and *D* (respectively). And let *BD* be joined. *AB*, *CD*, and *BD* are thus in one plane [Prop. 11.7]. Let *DE* be drawn at right-angles to *BD* in the reference plane, and let *DE* be made equal to *AB*, and let *BE*, *AE*, and *AD* be joined.



Καὶ ἐπεὶ ἡ AB ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας

And since *AB* is at right-angles to the reference plane, *AB* is thus also at right-angles to all of the straight-lines joined

ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν ἡ AB· ὀρθὴ ἄρα [ἐστὶν] ἑκατέρα τῶν ὑπὸ $AB\Delta$, ABE γωνιῶν. καὶ ἐπεὶ εἰξ παραλλήλους τὰξ AB, $\Gamma\Delta$ εὐθεῖα ἐμπέπτωκεν ἡ $B\Delta$, αἱ ἄρα ὑπὸ $AB\Delta$, $\Gamma\Delta B$ γωνίαι δυσὶν ὀρθαῖξ ἴσαι εἰσίν. ὀρθὴ δὲ ἡ ὑπὸ $AB\Delta$· ὀρθὴ ἄρα καὶ ἡ ὑπὸ $\Gamma\Delta B$· ἡ $\Gamma\Delta$ ἄρα πρὸς τὴν $B\Delta$ ὀρθή ἐστιν. καὶ ἐπεὶ ὁίση ἐστὶν ἡ AB τῇ ΔE, κοινὴ δὲ ἡ $B\Delta$, δύο δὴ αἱ AB, $B\Delta$ δυσὶ ταῖς $E\Delta$, ΔB ἴσαι εἰσίν· καὶ γωνία ἡ ὑπὸ $AB\Delta$ γωνίᾳ τῇ ὑπὸ $E\Delta B$ὁίση· ὀρθὴ γὰρ ἑκατέρα· βάσις ἄρα ἡ $A\Delta$ βάσει τῇ BEὁίση. καὶ ἐπεὶ ὁίση ἐστὶν ἡ μὲν AB τῇ ΔE, ἡ δὲ BE τῇ $A\Delta$, δύο δὴ αἱ AB, BE δυσὶ ταῖς $E\Delta$, ΔA ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ. καὶ βάσις αὐτῶν κοινὴ ἡ AE· γωνία ἄρα ἡ ὑπὸ ABE γωνίᾳ τῇ ὑπὸ $E\Delta A$ ἐστινὁίση. ὀρθὴ δὲ ἡ ὑπὸ ABE· ὀρθὴ ἄρα καὶ ἡ ὑπὸ $E\Delta A$· ἡ $E\Delta$ ἄρα πρὸς τὴν $A\Delta$ ὀρθή ἐστιν. ὀέστι δὲ καὶ πρὸς τὴν ΔB ὀρθή· ἡ $E\Delta$ ἄρα καὶ τῷ διὰ τῶν $B\Delta$, ΔA ἐπιπέδῳ ὀρθή ἐστιν. καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ διὰ τῶν $B\Delta A$ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας ἡ $E\Delta$. ἐν δὲ τῷ διὰ τῶν $B\Delta A$ ἐπιπέδῳ ἐστὶν ἡ $\Delta\Gamma$, ἐπειδήπερ ἐν τῷ διὰ τῶν $B\Delta A$ ἐπιπέδῳ ἐστὶν αἱ AB, $B\Delta$, ἐν ὁῷ δὲ αἱ AB, $B\Delta$, ἐν τούτῳ ἐστὶ καὶ ἡ $\Delta\Gamma$. ἡ $E\Delta$ ἄρα τῇ $\Delta\Gamma$ πρὸς ὀρθάξ ἐστιν· ὁώστε καὶ ἡ $\Gamma\Delta$ τῇ ΔE πρὸς ὀρθάξ ἐστιν. ὀέστι δὲ καὶ ἡ $\Gamma\Delta$ τῇ $B\Delta$ πρὸς ὀρθάξ. ἡ $\Gamma\Delta$ ἄρα δύο εὐθείαις τεμνούσαις ἀλλήλας ταῖς ΔE, ΔB ἀπὸ τῆς κατὰ τὸ Δ τομῆς πρὸς ὀρθὰξ ἐφέστηκεν· ὁώστε ἡ $\Gamma\Delta$ καὶ τῷ διὰ τῶν ΔE, ΔB ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. τὸ δὲ διὰ τῶν ΔE, ΔB ἐπίπεδον τὸ ὑποκείμενόν ἐστιν· ἡ $\Gamma\Delta$ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν.

Ἐὰν ἄραὁῶσι δύο εὐθεῖαι παράλληλοι, ἡ δὲ μία αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰξὁῇ, καὶ ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὁέσται· ὅπερ ἔδει δεῖχαι.

to it which are in the reference plane [Def. 11.3]. Thus, the angles ABD and ABE [are] each right-angles. And since the straight-line BD has met the parallel (straight-lines) AB and CD, the (sum of the) angles ABD and CDB is thus equal to two right-angles [Prop. 1.29]. And ABD (is) a right-angle. Thus, CDB (is) also a right-angle. CD is thus at right-angles to BD. And since AB is equal to DE, and BD (is) common, the two (straight-lines) AB and BD are equal to the two (straight-lines) ED and DB (respectively). And angle ABD (is) equal to angle EDB. For each (is) a right-angle. Thus, the base AD (is) equal to the base BE [Prop. 1.4]. And since AB is equal to DE, and BE to AD, the two (sides) AB, BE are equal to the two (sides) ED, DA, respectively. And their base AE is common. Thus, angle ABE is equal to angle EDA [Prop. 1.8]. And ABE (is) a right-angle. EDA (is) thus also a right-angle. Thus, ED is at right-angles to AD. And it is also at right-angles to DB. Thus, ED is also at right-angles to the plane through BD and DA [Prop. 11.4]. And ED will thus make right-angles with all of the straight-lines joined to it which are also in the plane through BDA. And DC is in the plane through BDA, inasmuch as AB and BD are in the plane through BDA [Prop. 11.2], and in which(ever plane) AB and BD (are found), DC is also (found). Thus, ED is at right-angles to DC. Hence, CD is also at right-angles to DE. And CD is also at right-angles to BD. Thus, CD is standing at right-angles to two straight-lines, DE and DB, which meet one another, at the (point) of section, D. Hence, CD is also at right-angles to the plane through DE and DB [Prop. 11.4]. And the plane through DE and DB is the reference (plane). CD is thus at right-angles to the reference plane.

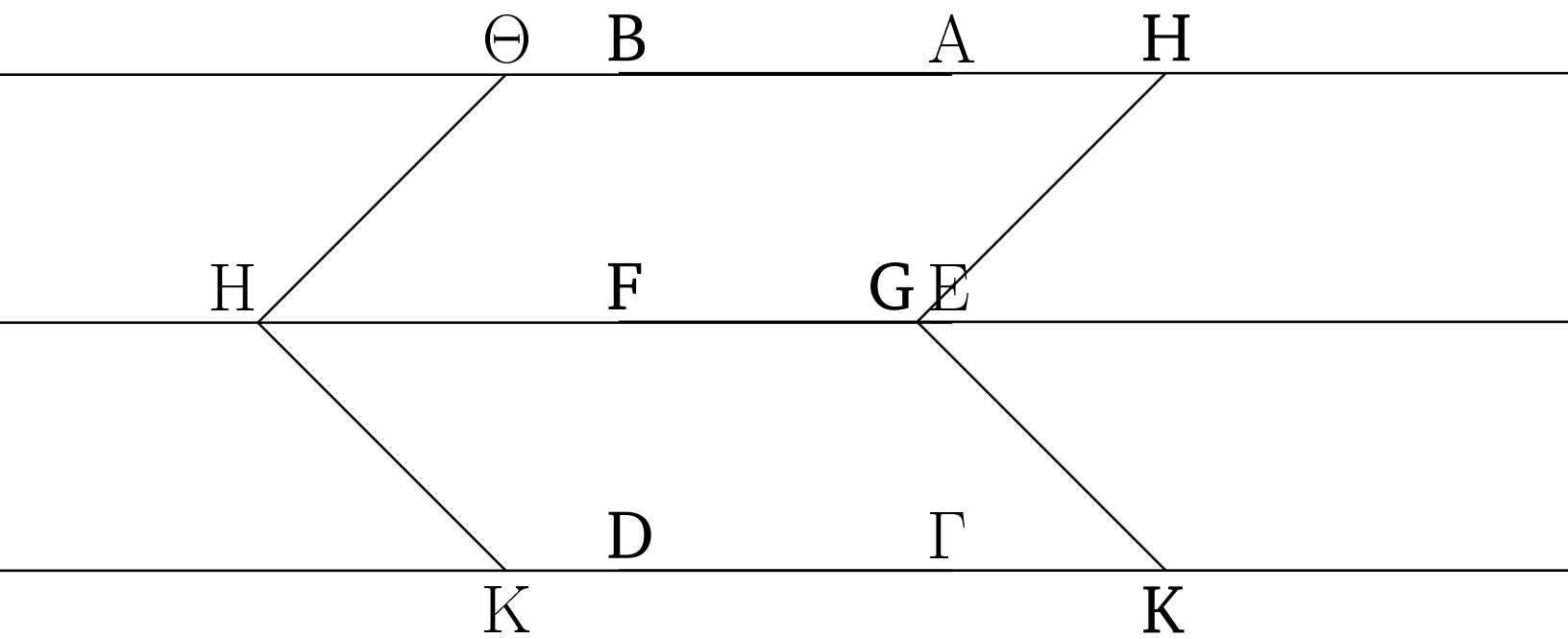
Thus, if two straight-lines are parallel, and one of them is at right-angles to some plane, then the remaining (one) will also be at right-angles to the same plane. (Which is) the very thing it was required to show.

9.

Αἱ τῇ αὐτῇ εὐθείᾳ παράλληλοι καὶ μὴ οὐῦσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ καὶ ἀλλήλαις εἰσὶ παράλληλοι.

Proposition 9

(Straight-lines) parallel to the same straight-line, and which are not in the same plane as it, are also parallel to one another.



Ἔστω γὰρ ἑκατέρα τῶν AB, ΓΔ τῇ EZ παράλληλος μὴ οὐσῦσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ· λέγω, ὅτι παράλληλόξ ἐστιν ἡ AB τῇ ΓΔ.

Εἰλήφθω γὰρ ἐπὶ τῆς EZ τυθὸν σημεῖον τὸ H, καὶ ἀπὸ αὐτοῦ τῇ EZ ἐν μὲν τῷ διὰ τῶν EZ, AB ἐπιπέδῳ πρὸς ὀρθὰξ ἤθθω ἡ HΘ, ἐν δὲ τῷ διὰ τῶν ZE, ΓΔ τῇ EZ πάλιν πρὸς ὀρθὰξ ἤθθω ἡ HK.

Καὶ ἐπεὶ ἡ EZ πρὸς ἑκατέραν τῶν HΘ, HK ὀρθή ἐστιν, ἡ EZ ἄρα καὶ τῷ διὰ τῶν HΘ, HK ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. καί ἐστιν ἡ EZ τῇ AB παράλληλος· καὶ ἡ AB ἄρα τῷ διὰ τῶν ΘHK ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΓΔ τῷ διὰ τῶν ΘHK ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν· ἑκατέρα ἄρα τῶν AB, ΓΔ τῷ διὰ τῶν ΘHK ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. ἔαν δὲ δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθάξ ὦσιν, παράλληλοί εἰσιν αἱ εὐθεῖαι· παράλληλος ἄρα ἐστὶν ἡ AB τῇ ΓΔ· ὅπερ ἔδει δεῖχαι.

For let *AB* and *CD* each be parallel to *EF*, not being in the same plane as it. I say that *AB* is parallel to *CD*.

For let some point *G* be taken at random on *EF*. And from it let *GH* be drawn at right-angles to *EF* in the plane through *EF* and *AB*. And let *GK* be drawn, again at right-angles to *EF*, in the plane through *FE* and *CD*.

And since *EF* is at right-angles to each of *GH* and *GK*, *EF* is thus also at right-angles to the plane through *GH* and *GK* [Prop. 11.4]. And *EF* is parallel to *AB*. Thus, *AB* is also at right-angles to the plane through *HGK* [Prop. 11.8]. So, for the same (reasons), *CD* is also at right-angles to the plane through *HGK*. Thus, *AB* and *CD* are each at right-angles to the plane through *HGK*. And if two straight-lines are at right–angles to the same plane then the straight-lines are parallel [Prop. 11.6]. Thus, *AB* is parallel to *CD*. (Which is) the very thing it was required to show.

10.

Ἐὰν δύο εὐθεῖαι ἁπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἁπτομένας ἀλλήλωνὀῶσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ, ὁίσας γωνίας περιέχουσιν.

Δύο γὰρ εὐθεῖαι αἱ AB, BΓ ἁπτόμεναι ἀλλήλων παρὰ δύο εὐθείας τὰξ ΔE, EZ ἁπτομένας ἀλλήλωνὀέστωσαν μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ· λέγω, ὅτιὀίση ἐστὶν ἡ ὑπὸ ABΓ γωνία τῇ ὑπὸ ΔEZ.

Ἀπειλήφθωσαν γὰρ αἱ BA, BΓ, EΔ, EZ ἴσαι ἀλλήλαις, καὶ ἐπεζεύθθωσαν αἱ AΔ, ΓZ, BE, AΓ, ΔZ.

Καὶ ἐπεὶ ἡ BA τῇ EΔὀίση ἐστὶ καὶ παράλληλος, καὶ ἡ AΔ ἄρα τῇ BEὀίση ἐστὶ καὶ παράλληλος. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΓZ τῇ BEὀίση ἐστὶ καὶ παράλληλος· ἑκατέρα ἄρα τῶν

Proposition 10

If two straight-lines joined to one another are (respectively) parallel to two straight-lines joined to one another, (but are) not in the same plane, then they will contain equal angles.

For let the two straight-lines joined to one another, *AB* and *BC*, be (respectively) parallel to the two straight-lines joined to one another, *DE* and *EF*, (but) not in the same plane. I say that angle *ABC* is equal to (angle) *DEF*.

For let *BA*, *BC*, *ED*, and *EF* be cut off (so as to be, respectively) equal to one another. And let *AD*, *CF*, *BE*, *AC*, and *DF* be joined.

And since *BA* is equal and parallel to *ED*, *AD* is thus also equal and parallel to *BE* [Prop. 1.33]. So, for the same reasons,

ΑΔ, ΓΖ τῇ ΒΕ ἰσάκις ἐστὶ καὶ παράλληλος. αἱ δὲ τῇ αὐτῇ εὐθείᾳ παράλληλοι καὶ μὴ οὐσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ καὶ ἀλλήλαις εἰσὶ παράλληλοι· παράλληλος ἄρα ἐστὶν ἡ ΑΔ τῇ ΓΖ καὶ ἴση. καὶ ἐπιζευγνύουσιν αὐτὰξ αἱ ΑΓ, ΔΖ· καὶ ἡ ΑΓ ἄρα τῇ ΔΖ ἰσάκις ἐστὶ καὶ παράλληλος. καὶ ἐπεὶ δύο αἱ ΑΒ, ΒΓ δυσὶ ταῖς ΔΕ, ΕΖ ἴσαι εἰσίν, καὶ βάσις ἡ ΑΓ βάσει τῇ ΔΖ ἴση, γωνία ἄρα ἡ ὑπὸ ΑΒΓ γωνίᾳ τῇ ὑπὸ ΔΕΖ ἐστιν ἴση.

CF is also equal and parallel to *BE*. Thus, *AD* and *CF* are each equal and parallel to *BE*. And straight-lines parallel to the same straight-line, and which are not in the same plane as it, are also parallel to one another [Prop. 11.9]. Thus, *AD* is parallel and equal to *CF*. And *AC* and *DF* join them. Thus, *AC* is also equal and parallel to *DF* [Prop. 1.33]. And since the two (straight-lines) *AB* and *BC* are equal to the two (straight-lines) *DE* and *EF* (respectively), and the base *AC* (is) equal to the base *DF*, the angle *ABC* is thus equal to the (angle) *DEF* [Prop. 1.8].

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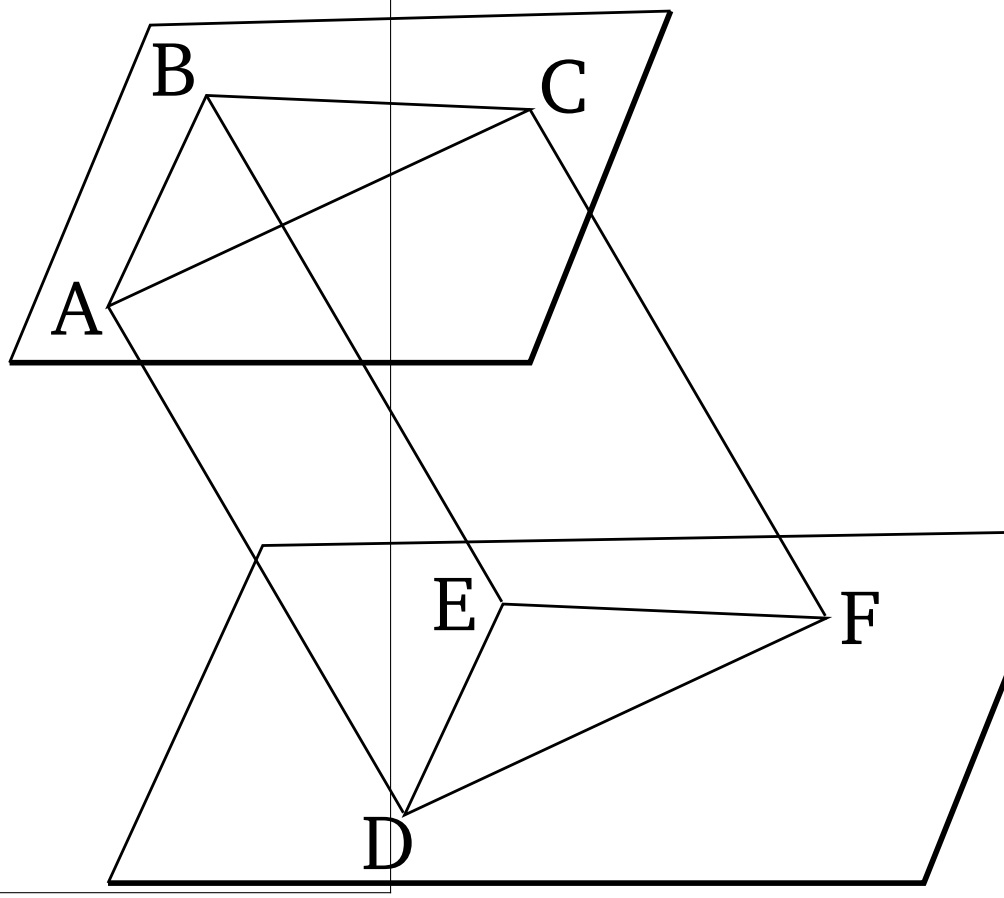
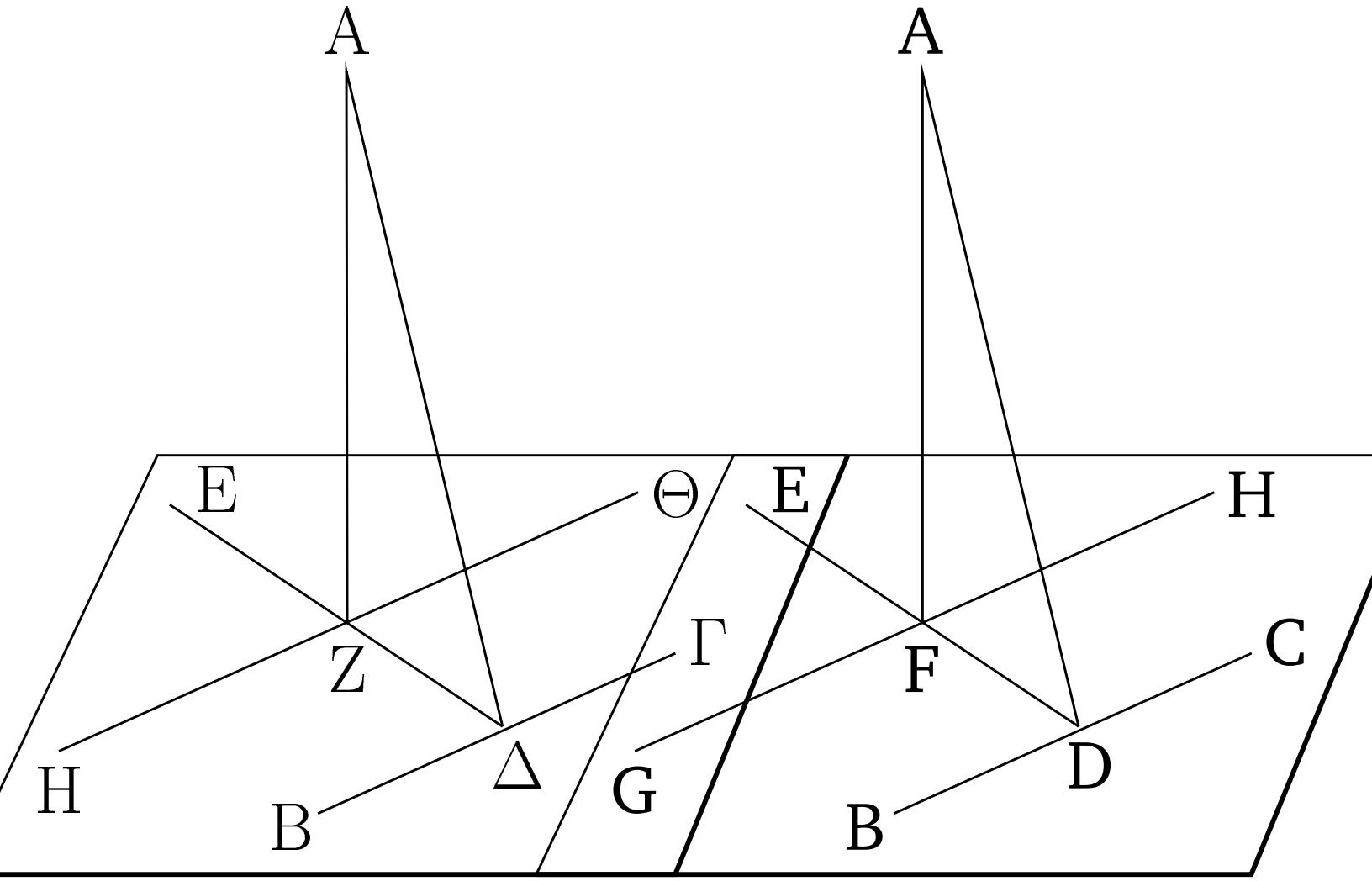


Ἐὰν ἄρα δύο εὐθεῖαι ἅπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἁπτομένας ἀλλήλων ὁσῶσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ, ἴσας γωνίας περιέχουσιν· ὅπερ ἔδει δεῖξαι.

Thus, if two straight-lines joined to one another are (respectively) parallel to two straight-lines joined to one another, (but are) not in the same plane, then they will contain equal angles. (Which is) the very thing it was required to show.

Ἀπὸ τοῦ δοθέντος σημείου μετεώρου ἐπὶ τὸ δοθὲν ἐπίπεδον κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

To draw a perpendicular straight-line from a given raised point to a given plane.



Ἔστω τὸ μὲν δοθὲν σημεῖον μετέωρον τὸ Α, τὸ δὲ δοθὲν ἐπίπεδον τὸ ὑποκείμενον· δεῖ δὴ ἀπὸ τοῦ Α σημείου ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Διήχθω γάρ τις ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖα, ὡξὸέτυθεν, ἡ ΒΓ, καὶ ἤχθω ἀπὸ τοῦ Α σημείου ἐπὶ τὴν ΒΓ κάθετος ἡ ΑΔ. εἰ μὲν οὖν ἡ ΑΔ κάθετός ἐστι καὶ ἐπὶ τὸ ὑποκείμενον ἐπίπεδον, γεγονὸξὸὰν εὸίη τὸ ἐπιταχθέν. εἰ δὲ οὐύ, ἤχθω ἀπὸ τοῦ Δ σημείου τῇ ΒΓ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξ ἡ ΔΕ, καὶ ἤχθω ἀπὸ τοῦ Α ἐπὶ τὴν ΔΕ κάθετος ἡ ΑΖ, καὶ διὰ τοῦ Ζ σημείου τῇ ΒΓ παράλληλος ἤχθω ἡ ΗΘ.

Καὶ ἐπεὶ ἡ ΒΓ ἑκατέρᾳ τῶν ΔΑ, ΔΕ πρὸς ὀρθάξ ἐστιν, ἡ ΒΓ ἄρα καὶ τῷ διὰ τῶν ΕΔΑ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. καί ἐστιν αὐτῇ παράλληλος ἡ ΗΘ· ἐαν δὲ ὀῶσι δύο εὐθεῖαι παράλληλοι, ἡ δὲ μία αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰξὸῇ, καὶ

Let *A* be the given raised point, and the given plane the reference (plane). So, it is required to draw a perpendicular straight-line from point *A* to the reference plane.

Let some random straight-line *BC* be drawn across in the reference plane, and let the (straight-line) *AD* be drawn from point *A* perpendicular to *BC* [Prop. 1.12]. If, therefore, *AD* is also perpendicular to the reference plane then that which was prescribed will have occurred. And, if not, let *DE* be drawn in the reference plane from point *D* at right-angles to *BC* [Prop. 1.11], and let the (straight-line) *AF* be drawn from *A* perpendicular to *DE* [Prop. 1.12], and let *GH* be drawn through point *F*, parallel to *BC* [Prop. 1.31].

And since *BC* is at right-angles to each of *DA* and *DE*, *BC* is thus also at right-angles to the plane through *EDA* [Prop. 11.4]. And *GH* is parallel to it. And if two straight-lines

ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὸέσται· καὶ ἡ ΗΘ ἄρα τῷ διὰ τῶν ΕΔ, ΔΑ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐύσας ἐν τῷ διὰ τῶν ΕΔ, ΔΑ ἐπιπέδῳ ὀρθή ἐστιν ἡ ΗΘ. ὁάπτεται δὲ αὐτῆς ἡ ΑΖ οὐῦσα ἐν τῷ διὰ τῶν ΕΔ, ΔΑ ἐπιπέδῳ· ἡ ΗΘ ἄρα ὀρθή ἐστι πρὸς τὴν ΖΑ· ὁώστε καὶ ἡ ΖΑ ὀρθή ἐστι πρὸς τὴν ΘΗ. ὀέστι δὲ ἡ ΑΖ καὶ πρὸς τὴν ΔΕ ὀρθή· ἡ ΑΖ ἄρα πρὸς ἑκατέραν τῶν ΗΘ, ΔΕ ὀρθή ἐστιν. ἔαν δὲ εὐθεῖα δυσὶν εὐθείαις τεμνούσαις ἀλλήλας ἐπὶ τῆς τομῆς πρὸς ὀρθὰξ ἐπισταθῇ, καὶ τῷ διὸ αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰξὸέσται· ἡ ΖΑ ἄρα τῷ διὰ τῶν ΕΔ, ΗΘ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. τὸ δὲ διὰ τῶν ΕΔ, ΗΘ ἐπίπεδόν ἐστι τὸ ὑποκείμενον· ἡ ΑΖ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν.

Ἀπὸ τοῦ ἄρα δοθέντος σημείου μετεώρου τοῦ Α ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετος εὐθεῖα γραμμὴ ὀῆκται ἡ ΑΖ· ὅπερ ἔδει ποιῆσαι.

12.

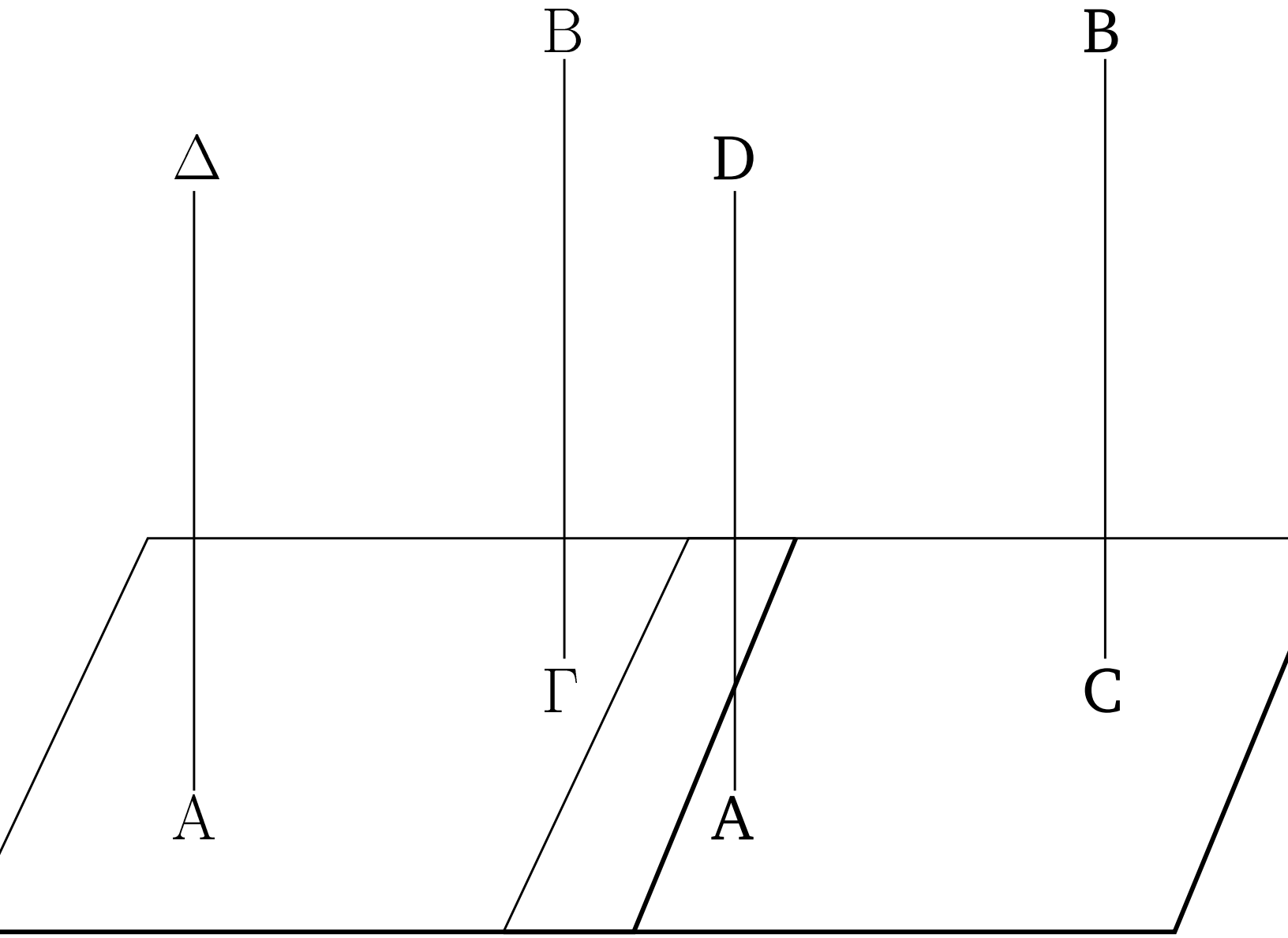
Τῷ δοθέντι ἐπιπέδῳ ἀπὸ τοῦ πρὸς αὐτῷ δοθέντος σημείου πρὸς ὀρθὰξ εὐθεῖαν γραμμὴν ἀναστῆσαι.

are parallel, and one of them is at right-angles to some plane, then the remaining (straight-line) will also be at right-angles to the same plane [Prop. 11.8]. Thus, GH is also at right-angles to the plane through ED and DA. And GH is thus at right-angles to all of the straight-lines joined to it which are also in the plane through ED and AD [Def. 11.3]. And AF, which is in the plane through ED and DA, is joined to it. Thus, GH is at right-angles to FA. Hence, FA is also at right-angles to HG. And AF is also at right-angles to DE. Thus, AF is at right-angles to each of GH and DE. And if a straight-line is set up at right-angles to two straight-lines cutting one another, at the point of section, then it will also be at right-angles to the plane through them [Prop. 11.4]. Thus, FA is at right-angles to the plane through ED and GH. And the plane through ED and GH is the reference (plane). Thus, AF is at right-angles to the reference plane.

Thus, the straight-line AF has been drawn from the given raised point A perpendicular to the reference plane. (Which is) the very thing it was required to do.

Proposition 12

To set up a straight-line at right-angles to a given plane from a given point in it.



Ἔστω τὸ μὲν δοθὲν ἐπίπεδον τὸ ὑποκείμενον, τὸ δὲ πρὸς αὐτῷ σημεῖον τὸ A· δεῖ δὴ ἀπὸ τοῦ A σημείου τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξ εὐθεῖαν γραμμὴν ἀναστῆσαι.

Νενοήσθω τι σημεῖον μετέωρον τὸ B, καὶ ἀπὸ τοῦ B ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετος ἤχθω ἡ $BΓ$, καὶ διὰ τοῦ A σημείου τῇ $BΓ$ παράλληλος ἤχθω ἡ $AΔ$.

Ἐπεὶ οὖν δύο εὐθεῖαι παράλληλοί εἰσιν αἱ $AΔ$, $ΓB$, ἡ δὲ μία αὐτῶν ἡ $BΓ$ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν, καὶ ἡ λοιπὴ ἄρα ἡ $AΔ$ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξ ἐστιν.

Τῷ ἄρα δοθέντι ἐπιπέδῳ ἀπὸ τοῦ πρὸς αὐτῷ σημείου τοῦ A πρὸς ὀρθὰξ ἀνέσταται ἡ $AΔ$· ὅπερ ἔδει ποιῆσαι.

Let the given plane be the reference (plane), and A a point in it. So, it is required to set up a straight-line at right-angles to the reference plane at point A.

Let some raised point B be assumed, and let the perpendicular (straight-line) BC be drawn from B to the reference plane [Prop. 11.11]. And let AD be drawn from point A parallel to BC [Prop. 1.31].

Therefore, since AD and CB are two parallel straight-lines, and one of them, BC, is at right-angles to the reference plane, the remaining (one) AD is thus also at right-angles to the reference plane [Prop. 11.8].

Thus, AD has been set up at right-angles to the given plane, from the point in it, A. (Which is) the very thing it was required

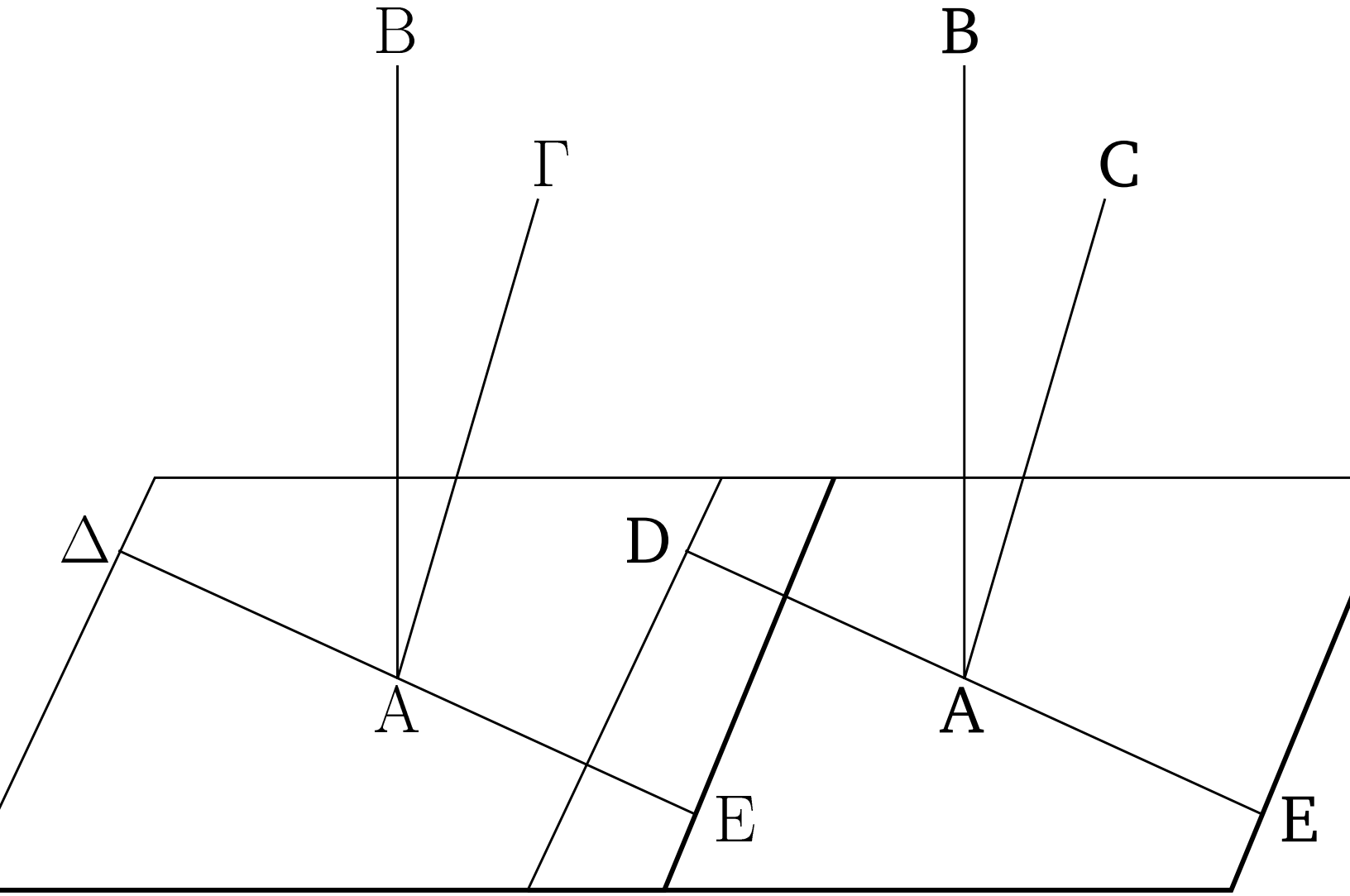
13.

Ἀπὸ τοῦ αὐτοῦ σημείου τῷ αὐτῷ ἐπιπέδῳ δύο εὐθεῖαι πρὸς ὀρθὰξ οὐκ ἀναστήσονται ἐπὶ τὰ αὐτὰ μέρη.

to do.

Proposition 13

Two (different) straight-lines cannot be set up at the same point at right-angles to the same plane, on the same side.



Εἰ γὰρ δυνατόν, ἀπὸ τοῦ αὐτοῦ σημείου τοῦ Α τῷ ὑποκειμένῳ ἐπιπέδῳ δύο εὐθεῖαι αἱ ΑΒ, ΒΓ πρὸς ὀρθὰξ ἀνεστάτωσαν ἐπὶ τὰ αὐτὰ μέρη, καὶ διήχθω τὸ διὰ τῶν ΒΑ, ΑΓ ἐπίπεδον· τομὴν δὴ ποιήσει διὰ τοῦ Α ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιείτω τὴν ΔΑΕ· αἱ ἄρα ΑΒ, ΑΓ, ΔΑΕ εὐθεῖαι ἐν ἑνὶ εἰσιν ἐπιπέδῳ. καὶ ἐπεὶ ἡ ΓΑ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας. ὁάπτεται δὲ αὐτῆς ἡ ΔΑΕ οὐῦσα ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ· ἡ ἄρα ὑπὸ ΓΑΕ γωνία ὀρθή ἐστιν. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΒΑΕ ὀρθή ἐστιν· ὁίση ἄρα ἡ ὑπὸ ΓΑΕ τῇ ὑπὸ

For, if possible, let the two straight-lines *AB* and *AC* be set up at the same point *A* at right-angles to the reference plane, on the same side. And let the plane through *BA* and *AC* be drawn. So it will make a straight cutting (passing) through (point) *A* in the reference plane [Prop. 11.3]. Let it have made *DAE*. Thus, *AB*, *AC*, and *DAE* are straight-lines in one plane. And since *CA* is at right-angles to the reference plane, it will thus also make right-angles with all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. And *DAE*, which is in the reference plane, is joined to it. Thus, angle *CAE* is a right-angle. So, for the same (reasons), *BAE* is also

BAE καί εἰσιν ἐν ἑνὶ ἐπιπέδῳ· ὅπερ ἐστὶν ἀδύνατον.

Οὐκ ἄρα ἀπὸ τοῦ αὐτοῦ σημείου τῷ αὐτῷ ἐπιπέδῳ δύο εὐθεῖαι πρὸς ὀρθὰξ ἀνασταθήσονται ἐπὶ τὰ αὐτὰ μέρη· ὅπερ ἔδει δεῖχαι.

a right-angle. Thus, *CAE* (is) equal to *BAE*. And they are in one plane. The very thing is impossible.

Thus, two (different) straight-lines cannot be set up at the same point at right-angles to the same plane, on the same side. (Which is) the very thing it was required to show.

14.

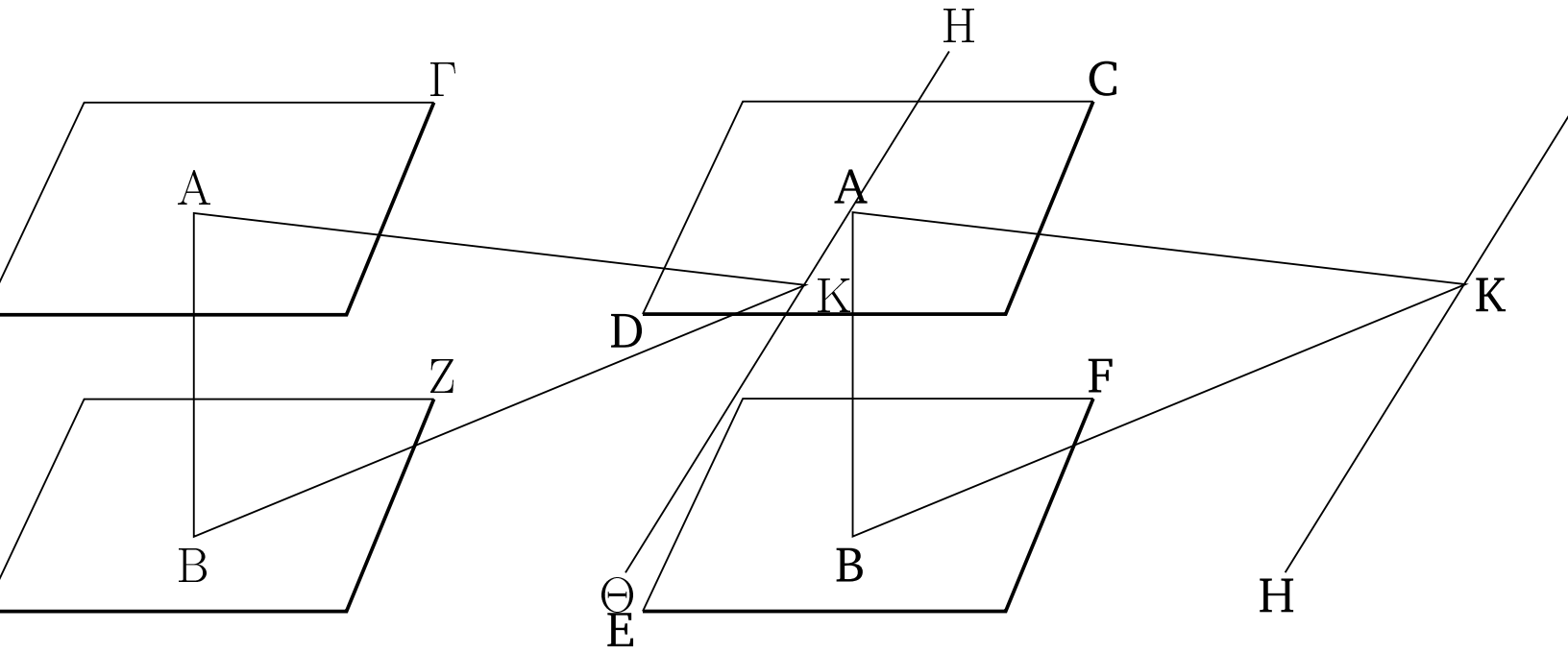
Πρὸξ ὰ ἐπίπεδα ἡ αὐτὴ εὐθεῖα ὀρθή ἐστιν, παράλληλα-ὀέσται τὰ ἐπίπεδα.

Εὐθεῖα γάρ τις ἡ AB πρὸς ἑκάτερον τῶν ΓΔ, EZ ἐπιπέδων πρὸς ὀρθὰξὸέστω· λέγω, ὅτι παράλληλά ἐστι τὰ ἐπίπεδα.

Proposition 14

Planes to which the same straight-line is at right-angles will be parallel planes.

For let some straight-line *AB* be at right-angles to each of the planes *CD* and *EF*. I say that the planes are parallel.



Εἰ γὰρ μή, ἐκβαλλόμενα συμπεσοῦνται. συμπιπτέτωσαν· ποιήσουσι δὴ κοινὴν τομὴν εὐθεῖαν. ποιείτωσαν τὴν HΘ, καὶ εἰλήφθω ἐπὶ τῆς HΘ τυθὸν σημεῖον τὸ K, καὶ ἐπεζεύθθωσαν αἱ AK, BK.

Καὶ ἐπεὶ ἡ AB ὀρθή ἐστι πρὸς τὸ EZ ἐπίπεδον, καὶ πρὸς τὴν BK ἄρα εὐθεῖαν οὐὖσαν ἐν τῷ EZ ἐκβληθέντι ἐπιπέδῳ ὀρθή ἐστιν ἡ AB· ἡ ἄρα ὑπὸ ABK γωνία ὀρθή ἐστιν. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ BAK ὀρθή ἐστιν. τριγώνου δὴ τοῦ ABK αἱ δύο γωνίαι αἱ ὑπὸ ABK, BAK δυσὶν ὀρθαῖξ εἰσιν ἴσαι· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὰ ΓΔ, EZ ἐπίπεδα ἐκβαλλόμενα συμπεσοῦνται· παράλληλα ἄρα ἐστὶ τὰ ΓΔ, EZ ἐπίπεδα.

Πρὸξ ὰ ἐπίπεδα ἄρα ἡ αὐτὴ εὐθεῖα ὀρθή ἐστιν, παράλληλά ἐστι τὰ ἐπίπεδα· ὅπερ ἔδει δεῖχαι.

For, if not, being produced, they will meet. Let them have met. So they will make a straight-line as a common section [Prop. 11.3]. Let them have made *GH*. And let some random point *K* be taken on *GH*. And let *AK* and *BK* be joined.

And since *AB* is at right-angles to the plane *EF*, *AB* is thus also at right-angles to *BK*, which is a straight-line in the produced plane *EF* [Def. 11.3]. Thus, angle *ABK* is a right-angle. So, for the same (reasons), *BAK* is also a right-angle. So the (sum of the) two angles *ABK* and *BAK* in the triangle *ABK* is equal to two right-angles. The very thing is impossible [Prop. 1.17]. Thus, planes *CD* and *EF*, being produced, will not meet. Planes *CD* and *EF* are thus parallel [Def. 11.8].

Thus, planes to which the same straight-line is at right-angles are parallel planes. (Which is) the very thing it was

15.

Ἐὰν δύο εὐθεῖαι ἁπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἁπτομένας ἀλλήλωνὁῶσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ οὁῦσαι, παράλληλά ἐστι τὰ διὸ αὐτῶν ἐπίπεδα.

Δύο γὰρ εὐθεῖαι ἁπτόμεναι ἀλλήλων αἱ ΑΒ, ΒΓ παρὰ δύο εὐθείας ἁπτομένας ἀλλήλων τὰξ ΔΕ, ΕΖὁέστωσαν μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ οὁῦσαι· λέγω, ὅτι ἐκβαλλόμενα τὰ διὰ τῶν ΑΒ, ΒΓ, ΔΕ, ΕΖ ἐπίπεδα οὐ συμπεσεῖται ἀλλήλοις.

ὁΗθθω γὰρ ἀπὸ τοῦ Β σημείου ἐπὶ τὸ διὰ τῶν ΔΕ, ΕΖ ἐπίπεδον κάθετος ἡ ΒΗ καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ Η σημεῖον, καὶ διὰ τοῦ Η τῇ μὲν ΕΔ παράλληλος ἤθθω ἡ ΗΘ, τῇ δὲ ΕΖ ἡ ΗΚ.

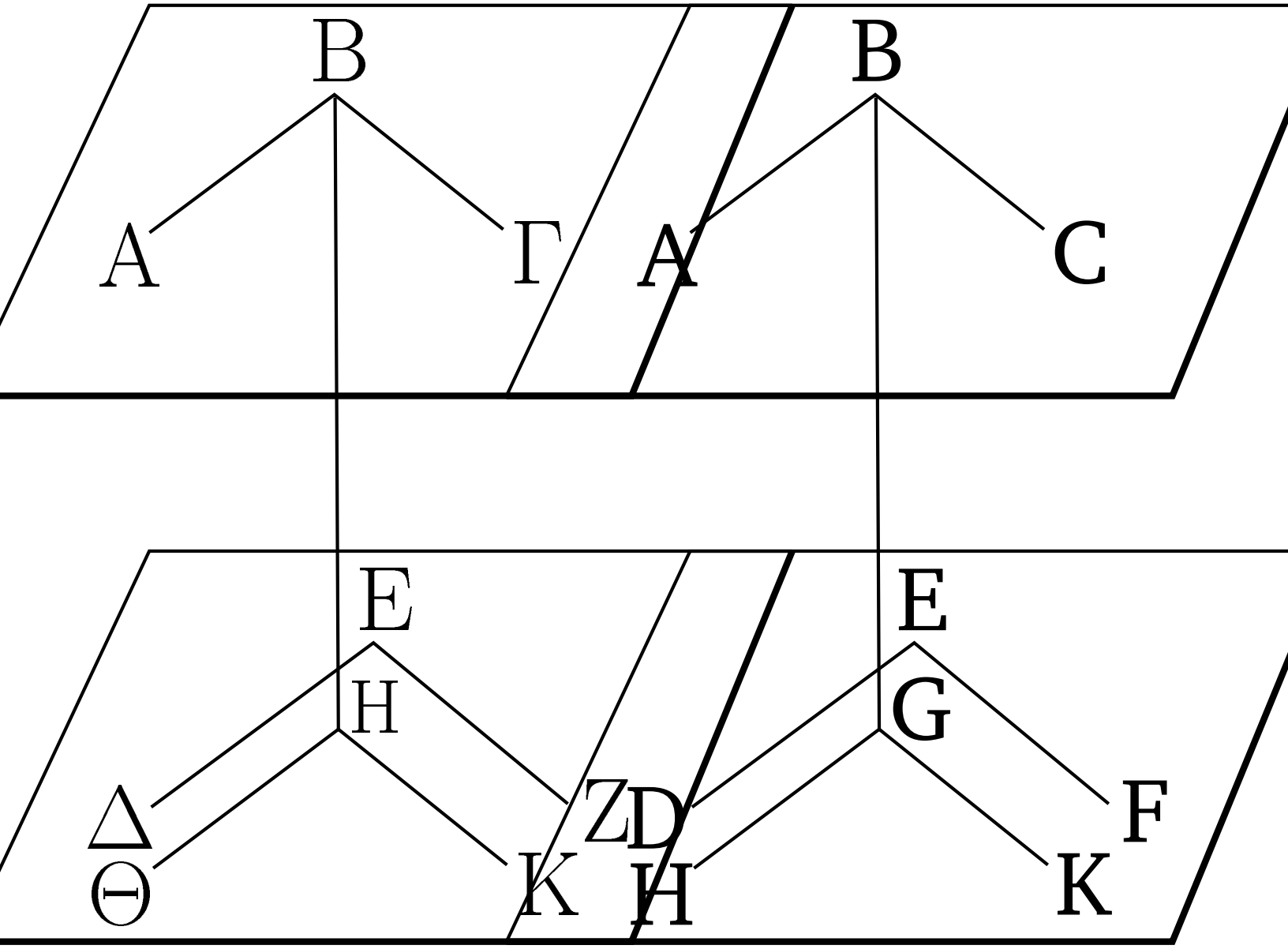
required to show.

Proposition 15

If two straight-lines joined to one another are parallel (respectively) to two straight-lines joined to one another, which are not in the same plane, then the planes through them are parallel (to one another).

For let the two straight-lines joined to one another, AB and BC, be parallel to the two straight-lines joined to one another, DE and EF (respectively), not being in the same plane. I say that the planes through AB, BC and DE, EF will not meet one another (when) produced.

For let BG be drawn from point B perpendicular to the plane through DE and EF [Prop. 11.11], and let it meet the plane at point G. And let GH be drawn through G parallel to ED, and GK (parallel) to EF [Prop. 1.31].



Καὶ ἐπεὶ ἡ ΒΗ ὀρθή ἐστι πρὸς τὸ διὰ τῶν ΔΕ, ΕΖ ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ διὰ τῶν ΔΕ, ΕΖ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας. ὅάπτεται δὲ αὐτῆς ἑκατέρα τῶν ΗΘ, ΗΚ οὐῦσα ἐν τῷ διὰ τῶν ΔΕ, ΕΖ ἐπιπέδῳ· ὀρθὴ ἄρα ἐστὶν ἑκατέρα τῶν ὑπὸ ΒΗΘ, ΒΗΚ γωνιῶν. καὶ ἐπεὶ παράλληλόξ ἐστιν ἡ ΒΑ τῇ ΗΘ, αἱ ἄρα ὑπὸ ΗΒΑ, ΒΗΘ γωνίαι δυσὶν ὀρθαῖξ ἴσαι εἰσίν. ὀρθὴ δὲ ἡ ὑπὸ ΒΗΘ· ὀρθὴ ἄρα καὶ ἡ ὑπὸ ΗΒΑ· ἡ ΗΒ ἄρα τῇ ΒΑ πρὸς ὀρθάξ ἐστιν. διὰ τὰ αὐτὰ δὴ ἡ ΗΒ καὶ τῇ ΒΓ ἐστι πρὸς ὀρθάξ. ἐπεὶ οὐῦν εὐθεῖα ἡ ΗΒ δυσὶν εὐθείαις ταῖς ΒΑ, ΒΓ τεμνούσαις ἀλλήλας πρὸς ὀρθὰξ ἐφέστηκεν, ἡ ΗΒ ἄρα καὶ τῷ διὰ τῶν ΒΑ, ΒΓ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. [διὰ τὰ αὐτὰ δὴ ἡ ΒΗ καὶ τῷ διὰ τῶν ΗΘ, ΗΚ ἐπιπέδῳ πρὸς

And since *BG* is at right-angles to the plane through *DE* and *EF*, it will thus also make right-angles with all of the straight-lines joined to it, which are also in the plane through *DE* and *EF* [Def. 11.3]. And each of *GH* and *GK*, which are in the plane through *DE* and *EF*, are joined to it. Thus, each of the angles *BGH* and *BGK* are right-angles. And since *BA* is parallel to *GH* [Prop. 11.9], the (sum of the) angles *GBA* and *BGH* is equal to two right-angles [Prop. 1.29]. And *BGH* (is) a right-angle. *GBA* (is) thus also a right-angle. Thus, *GB* is at right-angles to *BA*. So, for the same (reasons), *GB* is also at right-angles to *BC*. Therefore, since the straight-line *GB* has been set up at right-angles to two straight-lines, *BA* and *BC*, cutting one another, *GB* is thus at right-angles to the plane

ὀρθάξ ἐστιν. τὸ δὲ διὰ τῶν ΗΘ, ΗΚ ἐπίπεδόν ἐστι τὸ διὰ τῶν ΔΕ, ΕΖ· ἡ ΒΗ ἄρα τῷ διὰ τῶν ΔΕ, ΕΖ ἐπιπέδῳ ἐστὶ πρὸς ὀρθάξ. ἐδείχθη δὲ ἡ ΗΒ καὶ τῷ διὰ τῶν ΑΒ, ΒΓ ἐπιπέδῳ πρὸς ὀρθάξ]. πρὸς ὰ δὲ ἐπίπεδα ἡ αὐτὴ εὐθεῖα ὀρθή ἐστιν, παράλληλά ἐστι τὰ ἐπίπεδα· παράλληλον ἄρα ἐστὶ τὸ διὰ τῶν ΑΒ, ΒΓ ἐπίπεδον τῷ διὰ τῶν ΔΕ, ΕΖ.

Ἐὰν ἄρα δύο εὐθεῖαι ἁπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἁπτομένας ἀλλήλωνὸῶσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ, παράλληλά ἐστι τὰ διὸ αὐτῶν ἐπίπεδα· ὅπερ ἔδει δεῖχαι.

through BA and BC [Prop. 11.4]. [So, for the same (reasons), BG is also at right-angles to the plane through GH and GK. And the plane through GH and GK is the (plane) through DE and EF. And it was also shown that GB is at right-angles to the plane through AB and BC.] And planes to which the same straight-line is at right-angles are parallel planes [Prop. 11.14]. Thus, the plane through AB and BC is parallel to the (plane) through DE and EF.

Thus, if two straight-lines joined to one another are parallel (respectively) to two straight-lines joined to one another, which are not in the same plane, then the planes through them are parallel (to one another). (Which is) the very thing it was required to show.

16.

Ἐὰν δύο ἐπίπεδα παράλληλα ὑπό ἐπιπέδου τινὸξ τέμνηται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν.

Δύο γὰρ ἐπίπεδα παράλληλα τὰ ΑΒ, ΓΔ ὑπὸ ἐπιπέδου τοῦ ΕΖΗΘ τεμνέσθω, κοιναὶ δὲ αὐτῶν τομαὶ ὀέστωσαν αἱ ΕΖ, ΗΘ· λέγω, ὅτι παράλληλόξ ἐστιν ἡ ΕΖ τῇ ΗΘ.

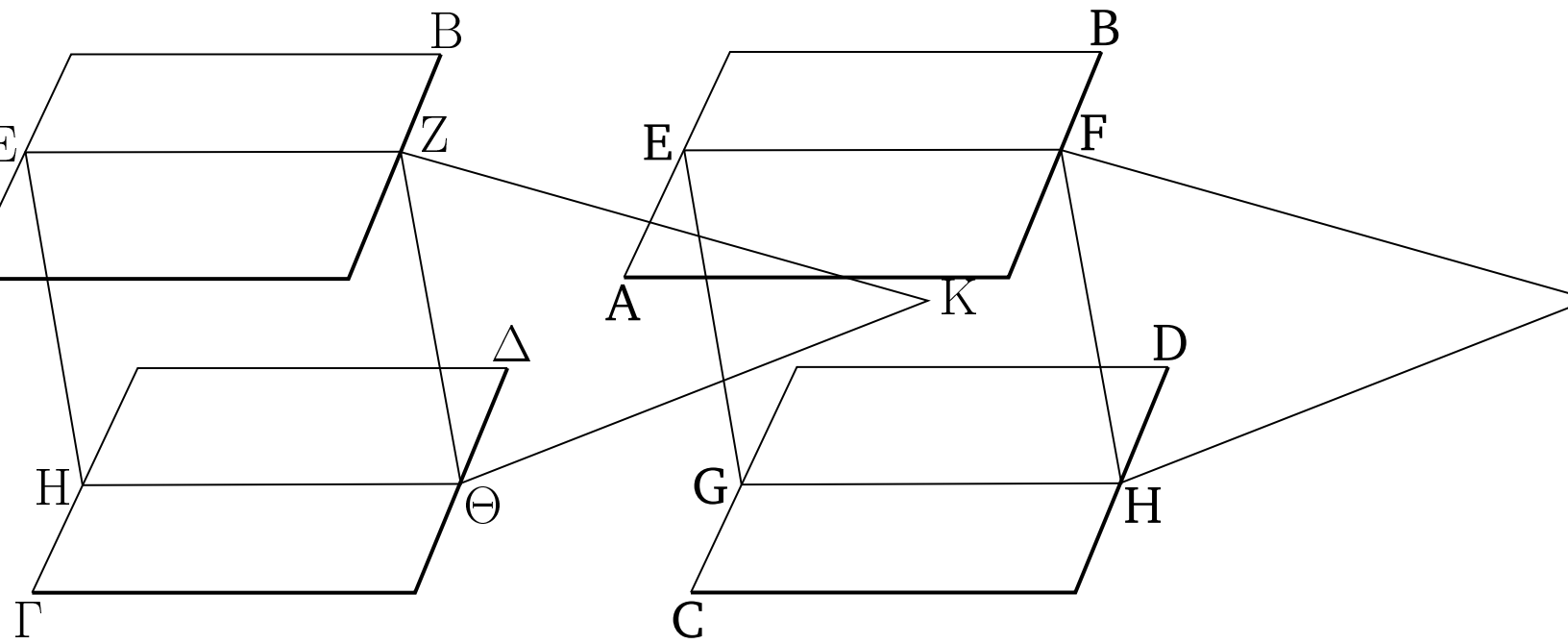
Εἰ γὰρ μή, ἐκβαλλόμεναι αἱ ΕΖ, ΗΘ ἤτοι ἐπὶ τὰ Ζ, Θ μέρηὸἡ ἐπὶ τὰ Ε, Η συμπεσοῦνται. ἐκβεβλήσθωσαν ὡξ ἐπὶ τὰ Ζ, Θ μέρη καὶ συμπιπτέτωσαν πρότερον κατὰ τὸ Κ. καὶ ἐπεὶ ἡ ΕΖΚ ἐν τῷ ΑΒ ἐστιν ἐπιπέδῳ, καὶ πάντα ἄρα τὰ ἐπὶ τῆς ΕΖΚ σημεῖα ἐν τῷ ΑΒ ἐστιν ἐπιπέδῳ. ὃὲν δὲ τῶν ἐπὶ τῆς ΕΖΚ εὐθείας σημείων ἐστὶ τὸ Κ· τὸ Κ ἄρα ἐν τῷ ΑΒ ἐστιν ἐπιπέδῳ. διὰ τὰ αὐτὰ δὴ τὸ Κ καὶ ἐν τῷ ΓΔ ἐστιν ἐπιπέδῳ· τὰ ΑΒ, ΓΔ ἄρα ἐπίπεδα ἐκβαλλόμενα συμπεσοῦνται. οὐ συμπίπτουσι δὲ διὰ τὸ παράλληλα ὑποκεῖσθαι· οὐκ ἄρα αἱ ΕΖ, ΗΘ εὐθεῖαι ἐκβαλλόμεναι ἐπὶ τὰ Ζ, Θ μέρη συμπεσοῦνται. ὁμοίως δὴ δείχομεν, ὅτι αἱ ΕΖ, ΗΘ εὐθεῖαι οὐδέ ἐπὶ τὰ Ε, Η μέρη ἐκβαλλόμεναι συμπεσοῦνται. αἱ δὲ ἐπὶ μ ηδέτερα τὰ μέρη συμπίπτουσαι παράλληλοί εἰσιν. παράλληλος ἄρα ἐστὶν ἡ ΕΖ τῇ ΗΘ.

Proposition 16

If two parallel planes are cut by some plane then their common sections are parallel.

For let the two parallel planes AB and CD be cut by the plane $EFGH$. And let EF and GH be their common sections. I say that EF is parallel to GH.

For, if not, being produced, EF and GH will meet either in the direction of F, H, or of E, G. Let them be produced, as in the direction of F, H, and let them, first of all, have met at K. And since EFK is in the plane AB, all of the points on EFK are thus also in the plane AB [Prop. 11.1]. And K is one of the points on EFK. Thus, K is in the plane AB. So, for the same (reasons), K is also in the plane CD. Thus, the planes AB and CD, being produced, will meet. But they do not meet, on account of being (initially) assumed (to be mutually) parallel. Thus, the straight-lines EF and GH, being produced in the direction of F, H, will not meet. So, similarly, we can show that the straight-lines EF and GH, being produced in the direction of E, G, will not meet either. And (straight-lines in one plane which), being produced, do not meet in either direction are parallel [Def. 1.23]. EF is thus parallel to GH.



Ἐὰν ἄρα δύο ἐπίπεδα παράλληλα ὑπό ἐπιπέδου τινὸξ τέμνηται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν· ὅπερ ἔδει δεῖχαι.

Thus, if two parallel planes are cut by some plane then their common sections are parallel. (Which is) the very thing it was required to show.

17.

Ἐὰν δύο εὐθεῖαι ὑπό παραλλήλων ἐπιπεδων τέμνωνται, εἰξ τοὐξ αὐτοὐξ λόγους τμ ηθήσονται.

Δύο γὰρ εὐθεῖαι αἱ AB, ΓΔ ὑπό παραλλήλων ἐπιπέδων τῶν HΘ, KΛ, MN τεμνέσθωσαν κατὰ τὰ A, E, B, Γ, Z, Δ σημεῖα· λέγω, ὅτι ἐστὶν ὡξ ἡ AE εὐθεῖα πρὸς τὴν EB, οὑτως ἡ ΓZ πρὸς τὴν ZΔ.

Ἐπεζεύθθωσαν γὰρ αἱ AΓ, BΔ, AΔ, καὶ συμβαλλέτω ἡ AΔ τῷ KΛ ἐπιπέδῳ κατὰ τὸ X σημεῖον, καὶ ἐπεζεύθθωσαν αἱ EX, XZ.

Καὶ ἐπεὶ δύο ἐπίπεδα παράλληλα τὰ KΛ, MN ὑπό ἐπιπέδου τοῦ EBΔX τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ αἱ EX, BΔ παράλληλοί εἰσιν. διὰ τὰ αὐτὰ δὴ ἐπεὶ δύο ἐπίπεδα παράλληλα τὰ HΘ, KΛ ὑπό ἐπιπέδου τοῦ AXZΓ τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ αἱ AΓ, XZ παράλληλοί εἰσιν. καὶ ἐπεὶ τριγώνου τοῦ ABΔ παρὰ μίαν τῶν πλευρῶν τὴν BΔ εὐθεῖαὀῆκται ἡ EX, ἀνάλογον ἄρα ἐστὶν ὡξ ἡ AE πρὸς EB, οὑτως ἡ AX πρὸς XΔ. πάλιν ἐπεὶ τριγώνου τοῦ AΔΓ παρὰ μίαν τῶν πλευρῶν τὴν AΓ εὐθεῖαὀῆκται ἡ XZ, ἀνάλογόν ἐστιν ὡξ ἡ AX πρὸς XΔ, οὑτως ἡ ΓZ πρὸς ZΔ. ἐδείθθη δὲ καὶ ὡξ ἡ AX πρὸς XΔ, οὑτως ἡ AE πρὸς EB· καὶ ὡξ ἄρα ἡ AE πρὸς EB, οὑτως ἡ ΓZ πρὸς ZΔ.

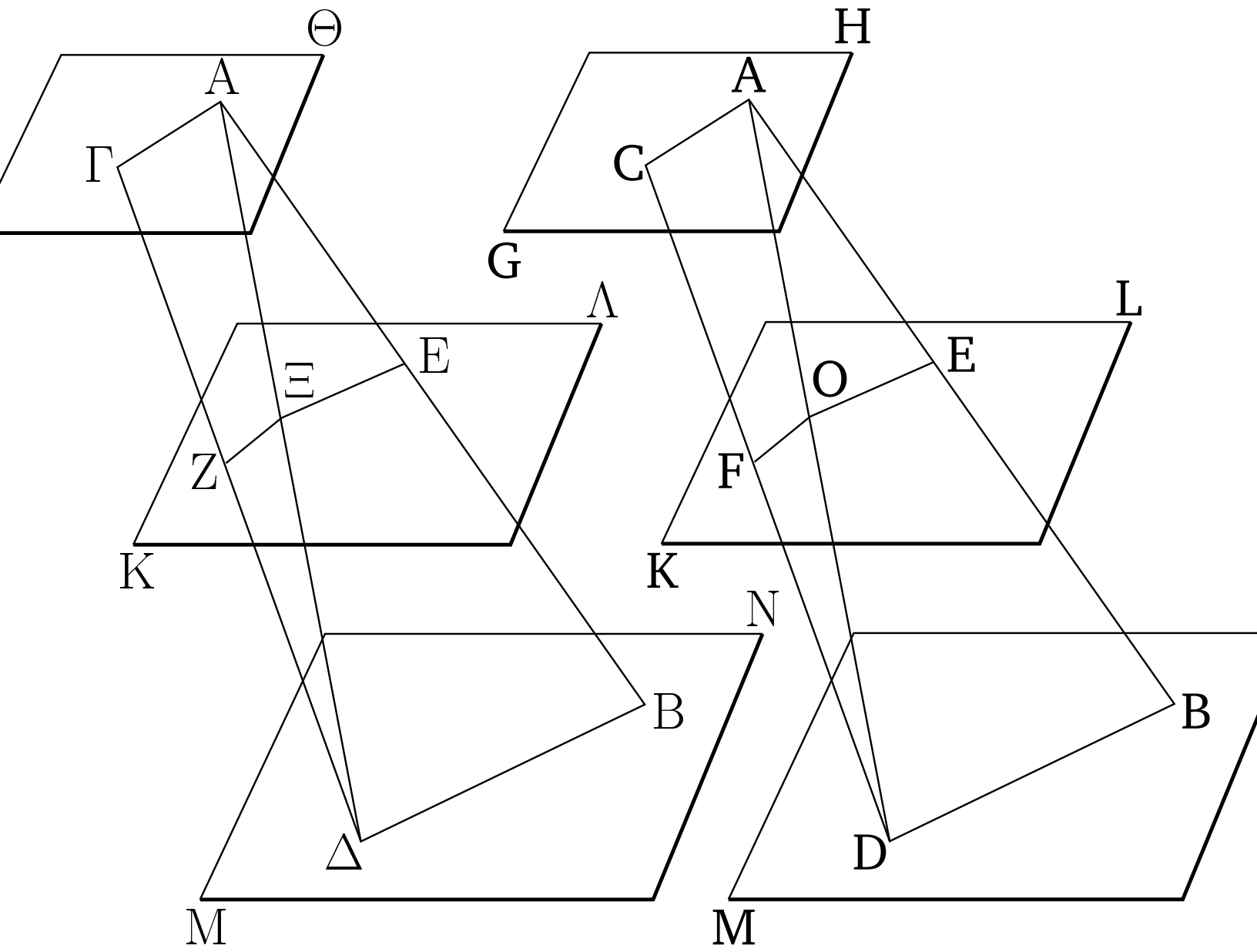
Proposition 17

If two straight-lines are cut by parallel planes then they will be cut in the same ratios.

For let the two straight-lines AB and CD be cut by the parallel planes GH, KL, and MN at the points A, E, B, and C, F, D (respectively). I say that as the straight-line AE is to EB, so CF (is) to FD.

For let AC, BD, and AD be joined, and let AD meet the plane KL at point O, and let EO and OF be joined.

And since two parallel planes KL and MN are cut by the plane $EBDO$, their common sections EO and BD are parallel [Prop. 11.16]. So, for the same (reasons), since two parallel planes GH and KL are cut by the plane $AOFC$, their common sections AC and OF are parallel [Prop. 11.16]. And since the straight-line EO has been drawn parallel to one of the sides BD of triangle ABD, thus, proportionally, as AE is to EB, so AO (is) to OD [Prop. 6.2]. Again, since the straight-line OF has been drawn parallel to one of the sides AC of triangle ADC, proportionally, as AO is to OD, so CF (is) to FD [Prop. 6.2]. And it was also shown that as AO (is) to OD, so AE (is) to EB. And thus as AE (is) to EB, so CF (is) to FD [Prop. 5.11].



Ἐὰν ἄρα δύο εὐθεῖαι ὑπὸ παραλλήλων ἐπιπέδων τέμνωνται, εἰξ τοὐξ αὐτοὐξ λόγους τμ ηθήσονται· ὅπερ ἔδει δειχαι.

Thus, if two straight-lines are cut by parallel planes then they will be cut in the same ratios. (Which is) the very thing it was required to show.

18.

Ἐὰν εὐθεῖα ἐπιπέδῳ τινὶ πρὸς ὀρθὰξὸῇ, καὶ πάντα τὰ διὸ αὐτῆς ἐπίπεδα τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὸέσται.

Εὐθεῖα γάρ τις ἡ AB τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξὸέστω· λέγω, ὅτι καὶ πάντα τὰ διὰ τῆς AB ἐπίπεδα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν.

Ἐκβεβλήσθω γὰρ διὰ τῆς AB ἐπίπεδον τὸ ΔE, καὶ ὀέστω κοινὴ τομὴ τοῦ ΔE ἐπιπέδου καὶ τοῦ ὑποκειμένου ἡ ΓE, καὶ

Proposition 18

If a straight-line is at right-angles to some plane then all of the planes (passing) through it will also be at right-angles to the same plane.

For let some straight-line *AB* be at right-angles to a reference plane. I say that all of the planes (passing) through *AB* are also at right-angles to the reference plane.

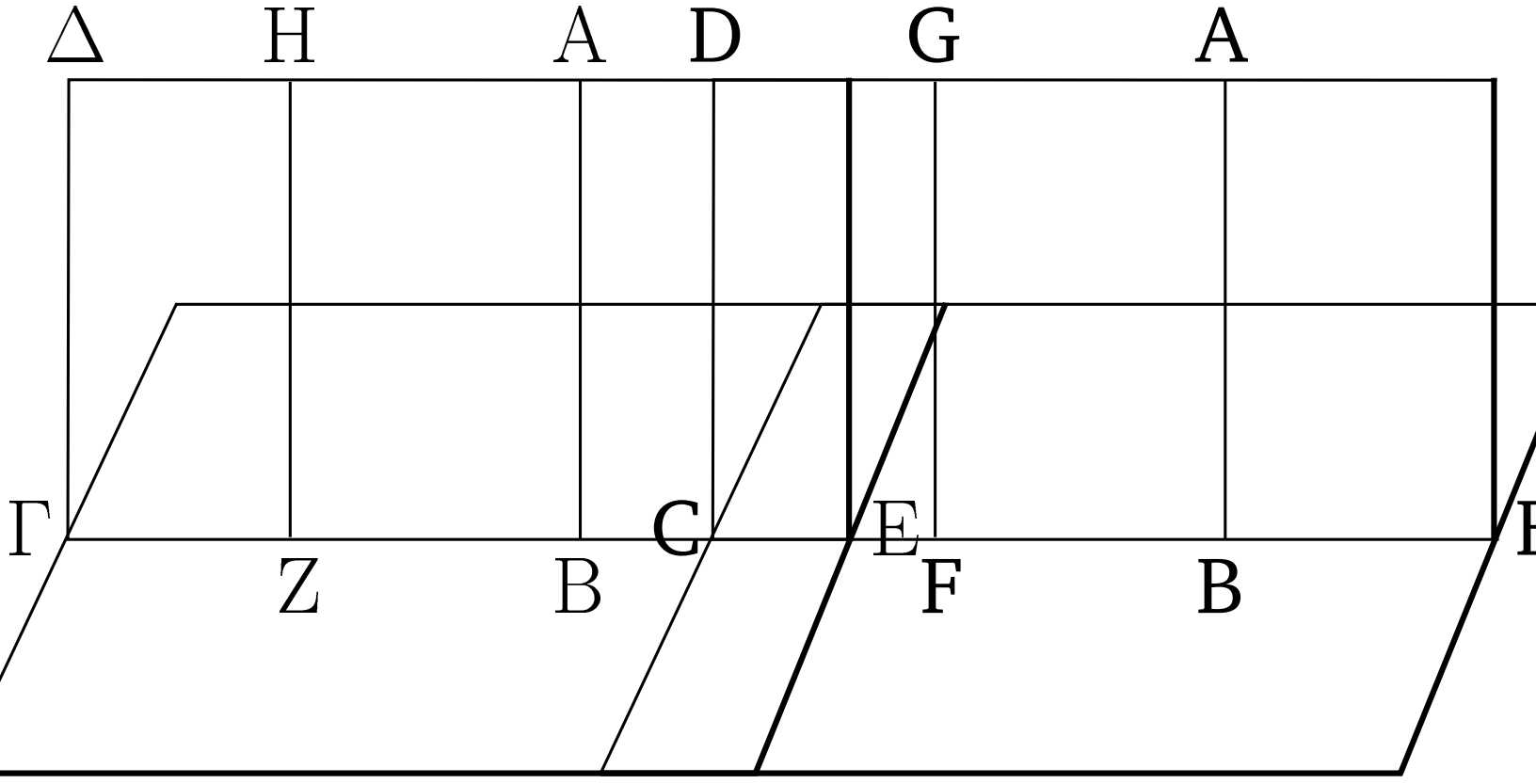
For let the plane *DE* be produced through *AB*. And let

εἰλήφθω ἐπὶ τῆς ΓΕ τυθὸν σημεῖον τὸ Ζ, καὶ ἀπὸ τοῦ Ζ τῇ ΓΕ πρὸς ὀρθὰξ ἤθθω ἐν τῷ ΔΕ ἐπιπέδῳ ἡ ΖΗ.

Καὶ ἐπεὶ ἡ ΑΒ πρὸς τὸ ὑποκείμενον ἐπίπεδον ὀρθή ἐστιν, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθή ἐστιν ἡ ΑΒ· ὥστε καὶ πρὸς τὴν ΓΕ ὀρθή ἐστιν· ἡ ἄρα ὑπὸ ΑΒΖ γωνία ὀρθή ἐστιν. ὀέστι δὲ καὶ ἡ ὑπὸ ΗΖΒ ὀρθὴ· παράλληλος ἄρα ἐστὶν ἡ ΑΒ τῇ ΖΗ. ἡ δὲ ΑΒ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν· καὶ ἡ ΖΗ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. καὶ ἐπίπεδον πρὸς ἐπίπεδον ὀρθόν ἐστιν, ὅταν αἱ τῇ κοινῇ τομῇ τῶν ἐπιπέδων πρὸς ὀρθὰξ ἀγόμεναι εὐθεῖαι ἐν ἑνὶ τῶν ἐπιπέδων τῷ λοιπῷ ἐπιπέδῳ πρὸς ὀρθὰξ ὦσιν. καὶ τῇ κοινῇ τομῇ τῶν ἐπιπέδων τῇ ΓΕ ἐν ἑνὶ τῶν ἐπιπέδων τῷ ΔΕ πρὸς ὀρθὰξ ἀθθεῖσα ἡ ΖΗ ἐδείθθη τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ· τὸ ἄρα ΔΕ ἐπίπεδον ὀρθόν ἐστι πρὸς τὸ ὑποκείμενον. ὁμοίως δὴ δειθθήσεται καὶ πάντα τὰ διὰ τῆς ΑΒ ἐπίπεδα ὀρθὰ τυγθανοντα πρὸς τὸ ὑποκείμενον ἐπίπεδον.

CE be the common section of the plane *DE* and the reference (plane). And let some random point *F* be taken on *CE*. And let *FG* be drawn from *F*, at right-angles to *CE*, in the plane *DE* [Prop. 1.11].

And since *AB* is at right-angles to the reference plane, *AB* is thus also at right-angles to all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. Hence, it is also at right-angles to *CE*. Thus, angle *ABF* is a right-angle. And *GFB* is also a right-angle. Thus, *AB* is parallel to *FG* [Prop. 1.28]. And *AB* is at right-angles to the reference plane. Thus, *FG* is also at right-angles to the reference plane [Prop. 11.8]. And a plane is at right-angles to a(nother) plane when the straight-lines drawn at right-angles to the common section of the planes, (and lying) in one of the planes, are at right-angles to the remaining plane [Def. 11.4]. And *FG*, (which was) drawn at right-angles to the common section of the planes, *CE*, in one of the planes, *DE*, was shown to be at right-angles to the reference plane. Thus, plane *DE* is at right-angles to the reference (plane). So, similarly, it can be shown that all of the planes (passing) at random through *AB* (are) at right-angles to the reference plane.



Ἐὰν ἄρα εὐθεῖα ἐπιπέδῳ τινὶ πρὸς ὀρθὰξὸῇ, καὶ πάντα τὰ διὸ αὐτῆς ἐπίπεδα τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὸέσται· ὅπερ ἔδει δεῖχαι.

Thus, if a straight-line is at right-angles to some plane then all of the planes (passing) through it will also be at right-angles to the same plane. (Which is) the very thing it was required to

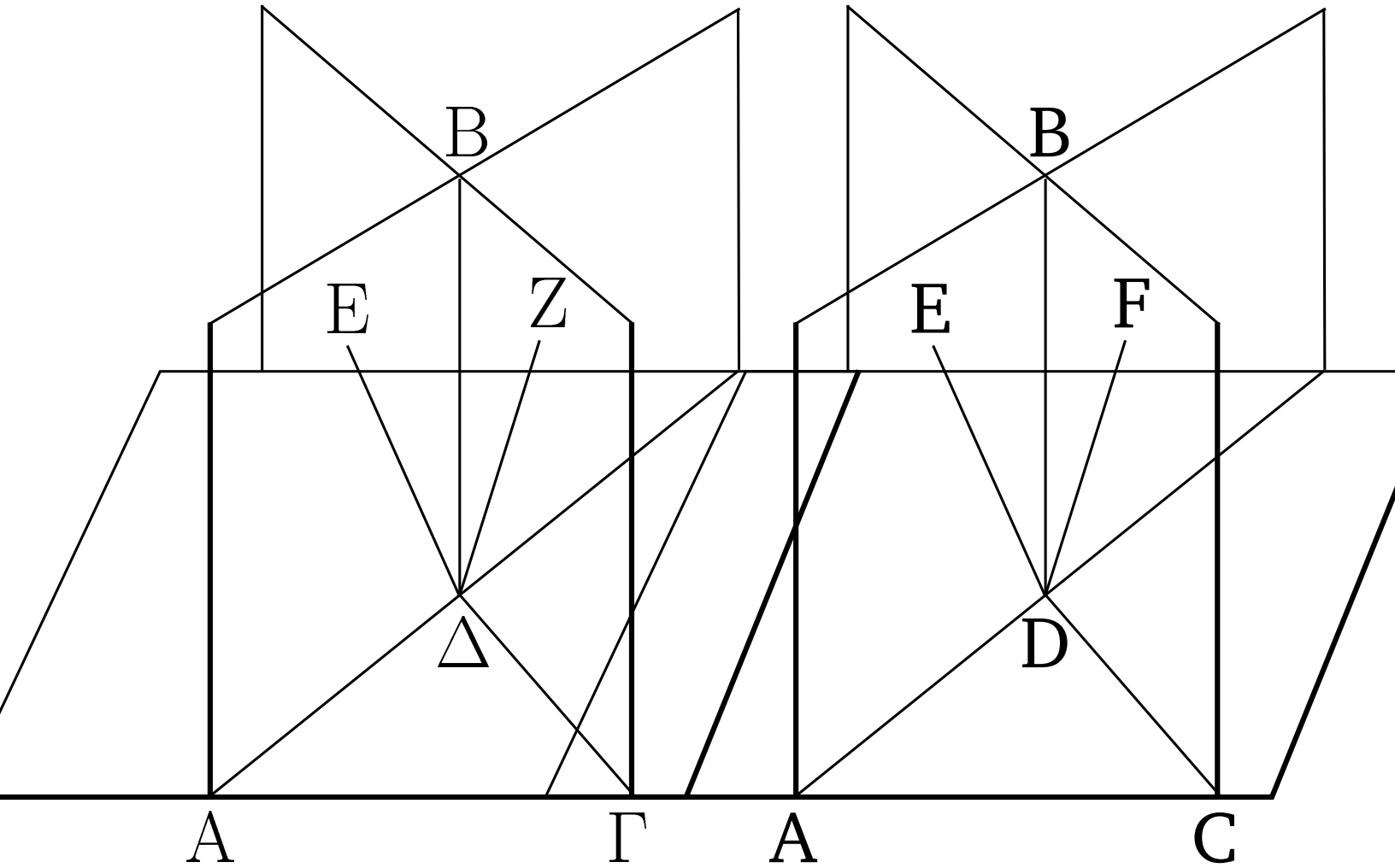
19.

Ἐὰν δύο ἐπίπεδα τέμνοντα ὀάλληλα ἐπιπέδῳ τινὶ πρὸς ὀρθὰξὸῇ, καὶ ἡ κοινὴ αὐτῶν τομὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὸέσται.

show.

Proposition 19

If two planes cutting one another are at right-angles to some plane then their common section will also be at right-angles to the same plane.



Δύο γὰρ ἐπίπεδα τὰ AB, BΓ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξὸέστω, κοινὴ δὲ αὐτῶν τομὴ ὀέστω ἡ BΔ· λέγω, ὅτι ἡ BΔ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν.

μὴ γάρ, καὶ ἤθθωσαν ἀπὸ τοῦ Δ σημείου ἐν μὲν τῷ AB ἐπιπέδῳ τῇ AΔ εὐθείᾳ πρὸς ὀρθὰξ ἡ ΔE, ἐν δὲ τῷ BΓ ἐπιπέδῳ τῇ ΓΔ πρὸς ὀρθὰξ ἡ ΔZ.

Καὶ ἐπεὶ τὸ AB ἐπίπεδον ὀρθόν ἐστι πρὸς τὸ ὑποκείμενον, καὶ τῇ κοινῇ αὐτῶν τομῇ τῇ AΔ πρὸς ὀρθὰξ ἐν τῷ AB ἐπιπέδῳ ὀῆκται ἡ ΔE, ἡ ΔE ἄρα ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ ΔZ ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον. ἀπὸ τοῦ αὐτοῦ ἄρα σημείου τοῦ Δ τῷ ὑποκειμένῳ ἐπιπέδῳ δύο εὐθεῖα πρὸς ὀρθὰξ ἀνεσταμέναι

For let the two planes *AB* and *BC* be at right-angles to a reference plane, and let their common section be *BD*. I say that *BD* is at right-angles to the reference plane.

For (if) not, let *DE* also be drawn from point *D*, in the plane *AB*, at right-angles to the straight-line *AD*, and *DF*, in the plane *BC*, at right-angles to *CD*.

And since the plane *AB* is at right-angles to the reference (plane), and *DE* has been drawn at right-angles to their common section *AD*, in the plane *AB*, *DE* is thus at right-angles to the reference plane [Def. 11.4]. So, similarly, we can show that *DF* is also at right-angles to the reference plane. Thus, two (different) straight-lines are set up, at the same point *D*, at

εἰσὶν ἐπὶ τὰ αὐτὰ μέρη· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ ἀπὸ τοῦ Δ σημείου ἀνασταθήσεται πρὸς ὀρθὰξ πλὴν τῆς ΔΒ κοινῆς τομῆς τῶν ΑΒ, ΒΓ ἐπιπέδων.

Ἐὰν ἄρα δύο ἐπίπεδα τέμνοντα ἀλλήλα ἐπιπέδῳ τινὶ πρὸς ὀρθὰξ ᾖ, καὶ ἡ κοινὴ αὐτῶν τομὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξ ἔσται· ὅπερ ἔδει δεῖξαι.

right-angles to the reference plane, on the same side. The very thing is impossible [Prop. 11.13]. Thus, no (other straight-line) except the common section *DB* of the planes *AB* and *BC* can be set up at point *D*, at right-angles to the reference plane.

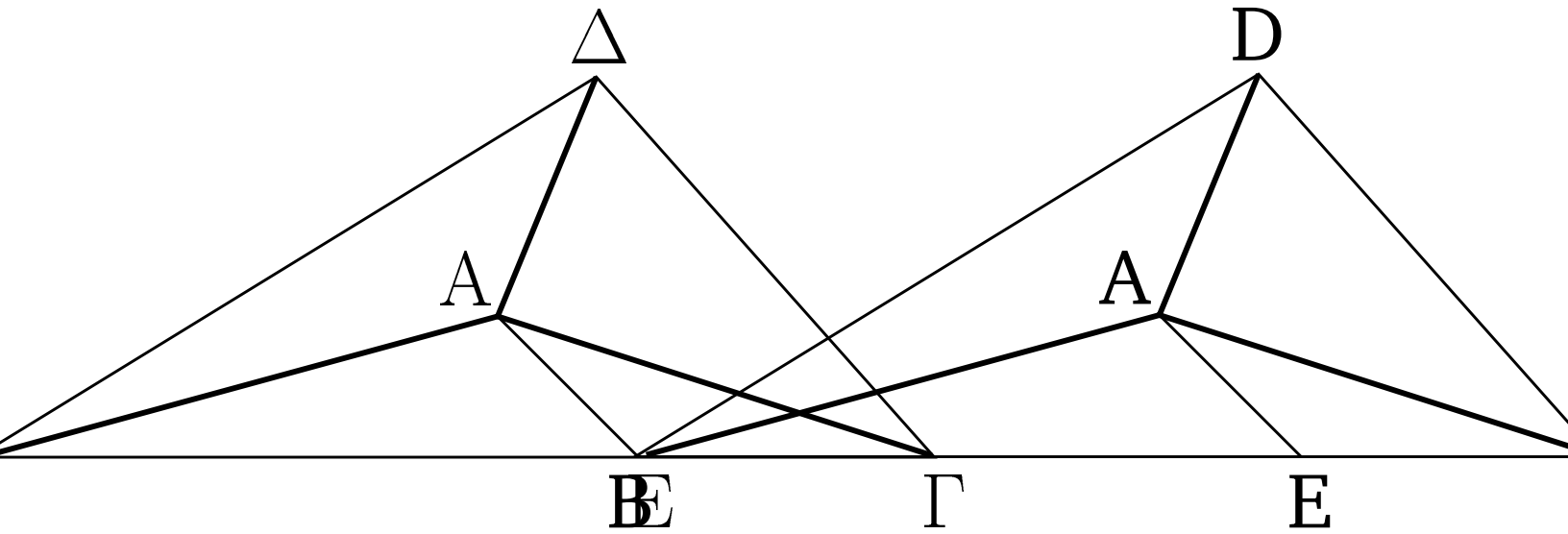
Thus, if two planes cutting one another are at right-angles to some plane then their common section will also be at right-angles to the same plane. (Which is) the very thing it was required to show.

20.

Ἐὰν στερεὰ γωνία ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέθηται, δύο ὁποιαιοῦν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι.

Proposition 20

If a solid angle is contained by three plane angles then (the sum of) any two (angles) is greater than the remaining (one), (the angles) being taken up in any (possible way).



Στερεὰ γὰρ γωνία ἡ πρὸς τῷ Α ὑπὸ τριῶν γωνιῶν ἐπιπέδων τῶν ὑπὸ ΒΑΓ, ΓΑΔ, ΔΑΒ περιεθέσθω· λέγω, ὅτι τῶν ὑπὸ ΒΑΓ, ΓΑΔ, ΔΑΒ γωνιῶν δύο ὁποιαιοῦν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι.

Εἰ μὲν οὖν αἱ ὑπὸ ΒΑΓ, ΓΑΔ, ΔΑΒ γωνίαι ἴσαι ἀλλήλαις εἰσίν, φανερόν, ὅτι δύο ὁποιαιοῦν τῆς λοιπῆς μείζονές εἰσιν. εἰ δὲ οὐ, ἔστω μείζων ἡ ὑπὸ ΒΑΓ, καὶ συνεστάτω πρὸς τῇ ΑΒ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Α τῇ ὑπὸ ΔΑΒ γωνίᾳ ἐν τῷ διὰ τῶν ΒΑΓ ἐπιπέδῳ ἴση ἡ ὑπὸ ΒΑΕ, καὶ κείσθω τῇ ΑΔ ἴση ἡ ΑΕ, καὶ διὰ τοῦ Ε σημείου διαθεῖσα ἡ ΒΕΓ τεμνέτω τὰξ ΑΒ, ΑΓ εὐθείας κατὰ τὰ Β, Γ σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΔΒ, ΔΓ.

Καὶ ἐπεὶ ἴση ἐστὶν ἡ ΔΑ τῇ ΑΕ, κοινὴ δὲ ἡ ΑΒ, δύο δυσὶν ἴσαι· καὶ γωνία ἡ ὑπὸ ΔΑΒ γωνίᾳ τῇ ὑπὸ ΒΑΕ ἴση· βάσις ἄρα ἡ ΔΒ βάσει τῇ ΒΕ ἐστιν ἴση. καὶ ἐπεὶ δύο αἱ ΒΔ, ΔΓ τῆς ΒΓ μείζονές εἰσιν, ὧν ἡ ΔΒ τῇ ΒΕ ἐδείθθη ἴση, λοιπὴ ἄρα ἡ ΔΓ λοιπῆς τῆς ΕΓ μείζων ἐστίν. καὶ ἐπεὶ ἴση ἐστὶν ἡ ΔΑ τῇ ΑΕ, κοινὴ δὲ ἡ ΑΓ, καὶ βάσις ἡ ΔΓ βάσεως τῆς ΕΓ μείζων ἐστίν, γωνία ἄρα ἡ ὑπὸ ΔΑΓ γωνίας τῆς ὑπὸ ΕΑΓ μείζων ἐστίν. ἐδείθθη δὲ καὶ ἡ ὑπὸ ΔΑΒ τῇ ὑπὸ ΒΑΕ ἴση· αἱ

For let the solid angle *A* be contained by the three plane angles *BAC*, *CAD*, and *DAB*. I say that (the sum of) any two of the angles *BAC*, *CAD*, and *DAB* is greater than the remaining (one), (the angles) being taken up in any (possible way).

For if the angles *BAC*, *CAD*, and *DAB* are equal to one another then (it is) clear that (the sum of) any two is greater than the remaining (one). But, if not, let *BAC* be greater (than *CAD* or *DAB*). And let (angle) *BAE*, equal to the angle *DAB*, be constructed in the plane through *BAC*, on the straight-line *AB*, at the point *A* on it. And let *AE* be made equal to *AD*. And *BEC* being drawn across through point *E*, let it cut the straight-lines *AB* and *AC* at points *B* and *C* (respectively). And let *DB* and *DC* be joined.

And since *DA* is equal to *AE*, and *AB* (is) common, the two (straight-lines *AD* and *AB* are) equal to the two (straight-lines *EA* and *AB*, respectively). And angle *DAB* (is) equal to angle *BAE*. Thus, the base *DB* is equal to the base *BE* [Prop. 1.4]. And since the (sum of the) two (straight-lines) *BD* and *DC* is greater than *BC* [Prop. 1.20], of which *DB* was shown (to be) equal to *BE*, the remainder *DC* is thus greater

ἄρα ὑπὸ ΔΑΒ, ΔΑΓ τῆς ὑπὸ ΒΑΓ μείζονές εἰσιν. ὁμοίως δὴ δείχομεν, ὅτι καὶ αἱ λοιπαὶ σύνδυο λαμβανόμεναι τῆς λοιπῆς μείζονές εἰσιν.

Ἐὰν ἄρα στερεὰ γωνία ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέθηται, δύο ὁποιαιοῦν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι· ὅπερ ἔδει δεῖξαι.

than the remainder *EC*. And since *DA* is equal to *AE*, but *AC* (is) common, and the base *DC* is greater than the base *EC*, the angle *DAC* is thus greater than the angle *EAC* [Prop. 1.25]. And *DAB* was also shown (to be) equal to *BAE*. Thus, (the sum of) *DAB* and *DAC* is greater than *BAC*. So, similarly, we can also show that the remaining (angles), being taken in pairs, are greater than the remaining (one).

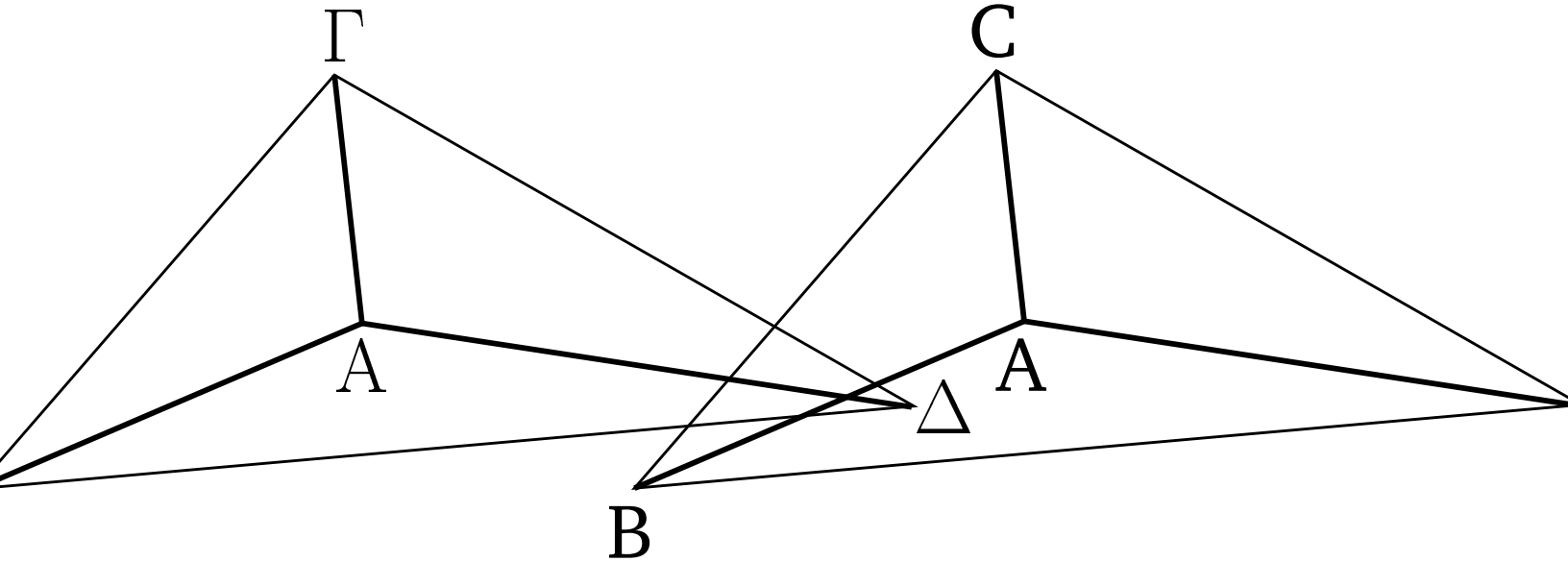
Thus, if a solid angle is contained by three plane angles then (the sum of) any two (angles) is greater than the remaining (one), (the angles) being taken up in any (possible way). (Which is) the very thing it was required to show.

21.

Ὅπασα στερεὰ γωνία ὑπὸ ἐλασσόνων [ἢ] τεσσάρων ὀρθῶν γωνιῶν ἐπιπέδων περιέθεται.

Proposition 21

Any solid angle is contained by plane angles (whose sum is) less [than] four right-angles.[†]



Ἔστω στερεὰ γωνία ἡ πρὸς τῷ Α περιεθομένη ὑπὸ ἐπιπέδων γωνιῶν τῶν ὑπὸ ΒΑΓ, ΓΑΔ, ΔΑΒ· λέγω, ὅτι αἱ ὑπὸ ΒΑΓ, ΓΑΔ, ΔΑΒ τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν.

Εἰλήφθω γὰρ ἐφ' ἑκάστης τῶν ΑΒ, ΑΓ, ΑΔ τυθόντα σημεῖα τὰ Β, Γ, Δ, καὶ ἐπεζεύχθωσαν αἱ ΒΓ, ΓΔ, ΔΒ. καὶ ἐπεὶ στερεὰ γωνία ἡ πρὸς τῷ Β ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέθεται τῶν ὑπὸ ΓΒΑ, ΑΒΔ, ΓΒΔ, δύο ὁποιαιοῦν τῆς λοιπῆς μείζονές εἰσιν· αἱ ἄρα ὑπὸ ΓΒΑ, ΑΒΔ τῆς ὑπὸ ΓΒΔ μείζονές εἰσιν. διὰ τὰ αὐτὰ δὴ καὶ αἱ μὲν ὑπὸ ΒΓΑ, ΑΓΔ τῆς ὑπὸ ΒΓΔ μείζονές εἰσιν, αἱ δὲ ὑπὸ ΓΔΑ, ΑΔΒ τῆς ὑπὸ ΓΔΒ μείζονές εἰσιν· αἱ ὀὲχ ἄρα γωνίαι αἱ ὑπὸ ΓΒΑ, ΑΒΔ, ΒΓΑ, ΑΓΔ, ΓΔΑ, ΑΔΒ τριῶν τῶν ὑπὸ ΓΒΔ, ΒΓΑ, ΓΔΒ μείζονές εἰσιν. ἀλλὰ αἱ τρεῖς αἱ ὑπὸ ΓΒΔ, ΒΔΓ, ΒΓΔ δυσὶν ὀρθαῖξ ἴσαι εἰσίν· αἱ ὀὲχ ἄρα αἱ ὑπὸ ΓΒΑ, ΑΒΔ, ΒΓΑ, ΑΓΔ, ΓΔΑ, ΑΔΒ δύο ὀρθῶν μείζονές εἰσιν. καὶ ἐπεὶ ἑκάστου τῶν ΑΒΓ, ΑΓΔ, ΑΔΒ τριγώνων αἱ τρεῖς γωνίαι δυσὶν ὀρθαῖξ ἴσαι εἰσίν,

Let the solid angle *A* be contained by the plane angles *BAC*, *CAD*, and *DAB*. I say that (the sum of) *BAC*, *CAD*, and *DAB* is less than four right-angles.

For let the random points *B*, *C*, and *D* be taken on each of (the straight-lines) *AB*, *AC*, and *AD* (respectively). And let *BC*, *CD*, and *DB* be joined. And since the solid angle at *B* is contained by the three plane angles *CBA*, *ABD*, and *CBD*, (the sum of) any two is greater than the remaining (one) [Prop. 11.20]. Thus, (the sum of) *CBA* and *ABD* is greater than *CBD*. So, for the same (reasons), (the sum of) *BCA* and *ACD* is also greater than *BCD*, and (the sum of) *CDA* and *ADB* is greater than *CDB*. Thus, the (sum of the) six angles *CBA*, *ABD*, *BCA*, *ACD*, *CDA*, and *ADB* is greater than the (sum of the) three (angles) *CBD*, *BCD*, and *CDB*. But, the (sum of the) three (angles) *CBD*, *BDC*, and *BCD* is equal to two right-angles [Prop. 1.32]. Thus, the (sum of the) six angles

αἱ ἄρα τῶν τριῶν τριγώνων ἐννέα γωνίαι αἱ ὑπὸ ΓΒΑ, ΑΓΒ, ΒΑΓ, ΑΓΔ, ΓΔΑ, ΓΑΔ, ΑΔΒ, ΔΒΑ, ΒΑΔ ὀξὲχ ὀρθαῖξ ἴσαι εἰσίν, ὁῶν αἱ ὑπὸ ΑΒΓ, ΒΓΑ, ΑΓΔ, ΓΔΑ, ΑΔΒ, ΔΒΑ ὀξὲχ γωνίαι δύο ὀρθῶν εἰσι μείζονες· λοιπαὶ ἄρα αἱ ὑπὸ ΒΑΓ, ΓΑΔ, ΔΑΒ τρεῖξ [γωνίαι] περιέθουσαι τὴν στερεὰν γωνίαν τεσσάρων ὀρθῶν ἐλάσσονέξ εἰσιν.

Ὅλπασα ἄρα στερεὰ γωνία ὑπὸ ἐλασσόνων [ἢ] τεσσάρων ὀρθῶν γωνιῶν ἐπιπέδων περιέθεται· ὅπερ ἔδει δεῖχαι.

CBA, *ABD*, *BCA*, *ACD*, *CDA*, and *ADB* is greater than two right-angles. And since the (sum of the) three angles of each of the triangles *ABC*, *ACD*, and *ADB* is equal to two right-angles, the (sum of the) nine angles *CBA*, *ACB*, *BAC*, *ACD*, *CDA*, *CAD*, *ADB*, *DBA*, and *BAD* of the three triangles is equal to six right-angles, of which the (sum of the) six angles *ABC*, *BCA*, *ACD*, *CDA*, *ADB*, and *DBA* is greater than two right-angles. Thus, the (sum of the) remaining three [angles] *BAC*, *CAD*, and *DAB*, containing the solid angle, is less than four right-angles.

Thus, any solid angle is contained by plane angles (whose sum is) less [than] four right-angles. (Which is) the very thing it was required to show.

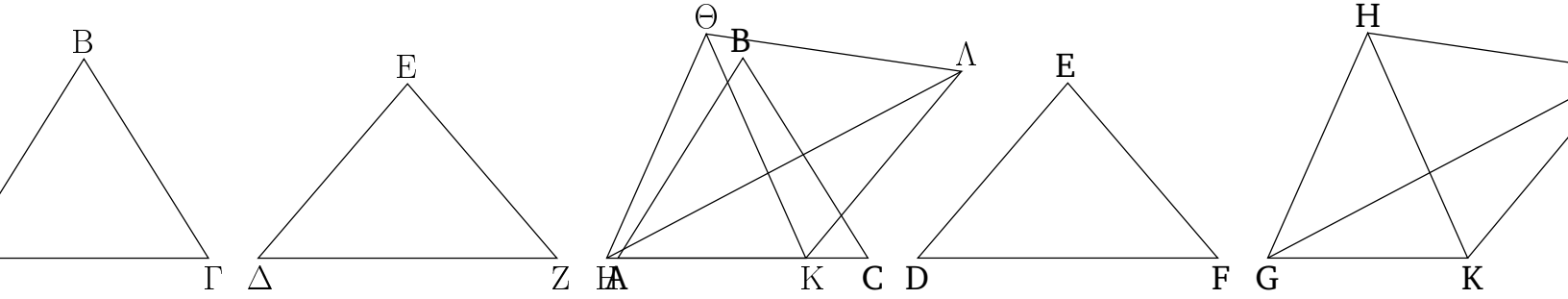
† This proposition is only proved for the case of a solid angle contained by three plane angles. However, the generalization to a solid angle contained by more than three plane angles is straightforward.

22.

Ἐὰν ὁῶσι τρεῖξ γωνίαι ἐπίπεδοι, ὁῶν αἱ δύο τῆξ λοιπῆξ μείζονέξ εἰσι πάντῃ μεταλαμβανόμεναι, περιέθωσι δὲ αὐτὰξ ἴσαι εὐθεῖαι, δυνατόν ἐστιν ἐκ τῶν ἐπιζευγνυουσῶν τὰξ ἴσαξ εὐθείαξ τρίγωνον συστήσασθαι.

Proposition 22

If there are three plane angles, of which (the sum of any) two is greater than the remaining (one), (the angles) being taken up in any (possible way), and if equal straight-lines contain them, then it is possible to construct a triangle from (the straight-lines created by) joining the (ends of the) equal straight-lines.



Ἔστωσαν τρεῖξ γωνίαι ἐπίπεδοι αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ, ὁῶν αἱ δύο τῆξ λοιπῆξ μείζονέξ εἰσι πάντῃ μεταλαμβανόμεναι, αἱ μὲν ὑπὸ ΑΒΓ, ΔΕΖ τῆξ ὑπὸ ΗΘΚ, αἱ δὲ ὑπὸ ΔΕΖ, ΗΘΚ τῆξ ὑπὸ ΑΒΓ, καὶ ὀέτι αἱ ὑπὸ ΗΘΚ, ΑΒΓ τῆξ ὑπὸ ΔΕΖ, καὶ ὀέστωσαν ἴσαι αἱ ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ εὐθεῖαι, καὶ ἐπεζεύθθωσαν αἱ ΑΓ, ΔΖ, ΗΚ· λέγω, ὅτι δυνατόν ἐστιν ἐκ τῶνὸίσων ταῖξ ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι, τουτέστιν ὅτι τῶν ΑΓ, ΔΖ, ΗΚ δύο ὁποιαιοῦν τῆξ λοιπῆξ μείζονέξ εἰσιν.

Εἰ μὲν οὐῦν αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ γωνίαι ἴσαι ἀλλήλαις εἰσίν, φανερόν, ὅτι καὶ τῶν ΑΓ, ΔΖ, ΗΚὸίσων γινομένων δυνατόν ἐστιν ἐκ τῶνὸίσων ταῖξ ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι. εἰ δὲ οὐύ, ὀέστωσανὸάνισοι, καὶ συνεστάτω πρὸξ τῇ ΘΚ εὐθείᾳ καὶ τῷ πρὸξ αὐτῇ σημείῳ τῷ Θ τῇ ὑπὸ ΑΒΓ γωνίᾳ ὁίση ἡ ὑπὸ ΚΘΛ· καὶ κείσθω μιᾷ τῶν ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚὁίση ἡ ΘΛ, καὶ ἐπεζεύθθωσαν αἱ ΚΛ, ΗΛ. καὶ ἐπεὶ δύο αἱ ΑΒ, ΒΓ δυσὶ ταῖξ ΚΘ, ΘΛ ἴσαι εἰσίν, καὶ γωνία ἡ πρὸξ τῷ Β γωνίᾳ τῇ ὑπὸ ΚΘΛὁίση, βάσιξ ἄρα ἡ ΑΓ βάσει τῇ ΚΛὁίση.

Let *ABC*, *DEF*, and *GHK* be three plane angles, of which the sum of any) two is greater than the remaining (one), (the angles) being taken up in any (possible way)—(that is), *ABC* and *DEF* (greater) than *GHK*, *DEF* and *GHK* (greater) than *ABC*, and, further, *GHK* and *ABC* (greater) than *DEF*. And let *AB*, *BC*, *DE*, *EF*, *GH*, and *HK* be equal straight-lines. And let *AC*, *DF*, and *GK* be joined. I say that that it is possible to construct a triangle out of (straight-lines) equal to *AC*, *DF*, and *GK*—that is to say, that (the sum of) any two of *AC*, *DF*, and *GK* is greater than the remaining (one).

Now, if the angles *ABC*, *DEF*, and *GHK* are equal to one another then (it is) clear that, (with) *AC*, *DF*, and *GK* also becoming equal, it is possible to construct a triangle from (straight-lines) equal to *AC*, *DF*, and *GK*. And if not, let them be unequal, and let *KHL*, equal to angle *ABC*, be constructed on the straight-line *HK*, at the point *H* on it. And let *HL* be made equal to one of *AB*, *BC*, *DE*, *EF*, *GH*, and *HK*. And let

καὶ ἐπεὶ αἱ ὑπὸ ΑΒΓ, ΗΘΚ τῆς ὑπὸ ΔΕΖ μείζονές εἰσιν, ὁίση δὲ ἡ ὑπὸ ΑΒΓ τῇ ὑπὸ ΚΘΛ, ἡ ἄρα ὑπὸ ΗΘΛ τῆς ὑπὸ ΔΕΖ μείζων ἐστίν. καὶ ἐπεὶ δύο αἱ ΗΘ, ΘΛ δύο ταῖς ΔΕ, ΕΖ ἴσαι εἰσίν, καὶ γωνία ἡ ὑπὸ ΗΘΛ γωνίας τῆς ὑπὸ ΔΕΖ μείζων, βάσις ἄρα ἡ ΗΛ βάσεως τῆς ΔΖ μείζων ἐστίν. ἀλλὰ αἱ ΗΚ, ΚΛ τῆς ΗΛ μείζονές εἰσιν. πολλῷ ἄρα αἱ ΗΚ, ΚΛ τῆς ΔΖ μείζονές εἰσιν. ὁίση δὲ ἡ ΚΛ τῇ ΑΓ· αἱ ΑΓ, ΗΚ ἄρα τῆς λοιπῆς τῆς ΔΖ μείζονές εἰσιν. ὁμοίως δὴ δείχομεν, ὅτι καὶ αἱ μὲν ΑΓ, ΔΖ τῆς ΗΚ μείζονές εἰσιν, καὶ ὀέτι αἱ ΔΖ, ΗΚ τῆς ΑΓ μείζονές εἰσιν. δυνατὸν ἄρα ἐστὶν ἐκ τῶνὁίσων ταῖς ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι· ὅπερ ἔδει δεῖχαι.

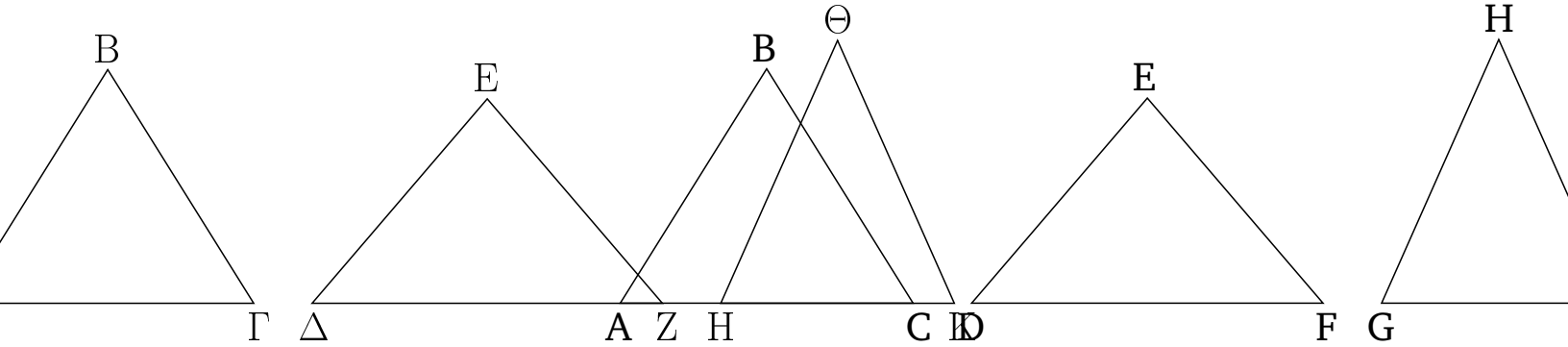
KL and GL be joined. And since the two (straight-lines) AB and BC are equal to the two (straight-lines) KH and HL (respectively), and the angle at B (is) equal to KHL, the base AC is thus equal to the base KL [Prop. 1.4]. And since (the sum of) ABC and GHK is greater than DEF, and ABC equal to KHL, GHL is thus greater than DEF. And since the two (straight-lines) GH and HL are equal to the two (straight-lines) DE and EF (respectively), and angle GHL (is) greater than DEF, the base GL is thus greater than the base DF [Prop. 1.24]. But, (the sum of) GK and KL is greater than GL [Prop. 1.20]. Thus, (the sum of) GK and KL is much greater than DF. And KL (is) equal to AC. Thus, (the sum of) AC and GK is greater than the remaining (straight-line) DF. So, similarly, we can show that (the sum of) AC and DF is greater than GK, and, further, that (the sum of) DF and GK is greater than AC. Thus, it is possible to construct a triangle from (straight-lines) equal to AC, DF, and GK. (Which is) the very thing it was required to show.

23.

Ἐκ τριῶν γωνιῶν ἐπιπέδων, ὧν αἱ δύο τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι, στερεὰν γωνίαν συστήσασθαι· δεῖ δὴ τὰξ τρεῖς τεσσάρων ὀρθῶν ἐλάσσονας εὐῖναι.

Proposition 23

To construct a solid angle from three (given) plane angles, (the sum of) two of which is greater than the remaining (one, the angles) being taken up in any (possible way). So, it is necessary for the (sum of the) three (angles) to be less than four right-angles [Prop. 11.21].



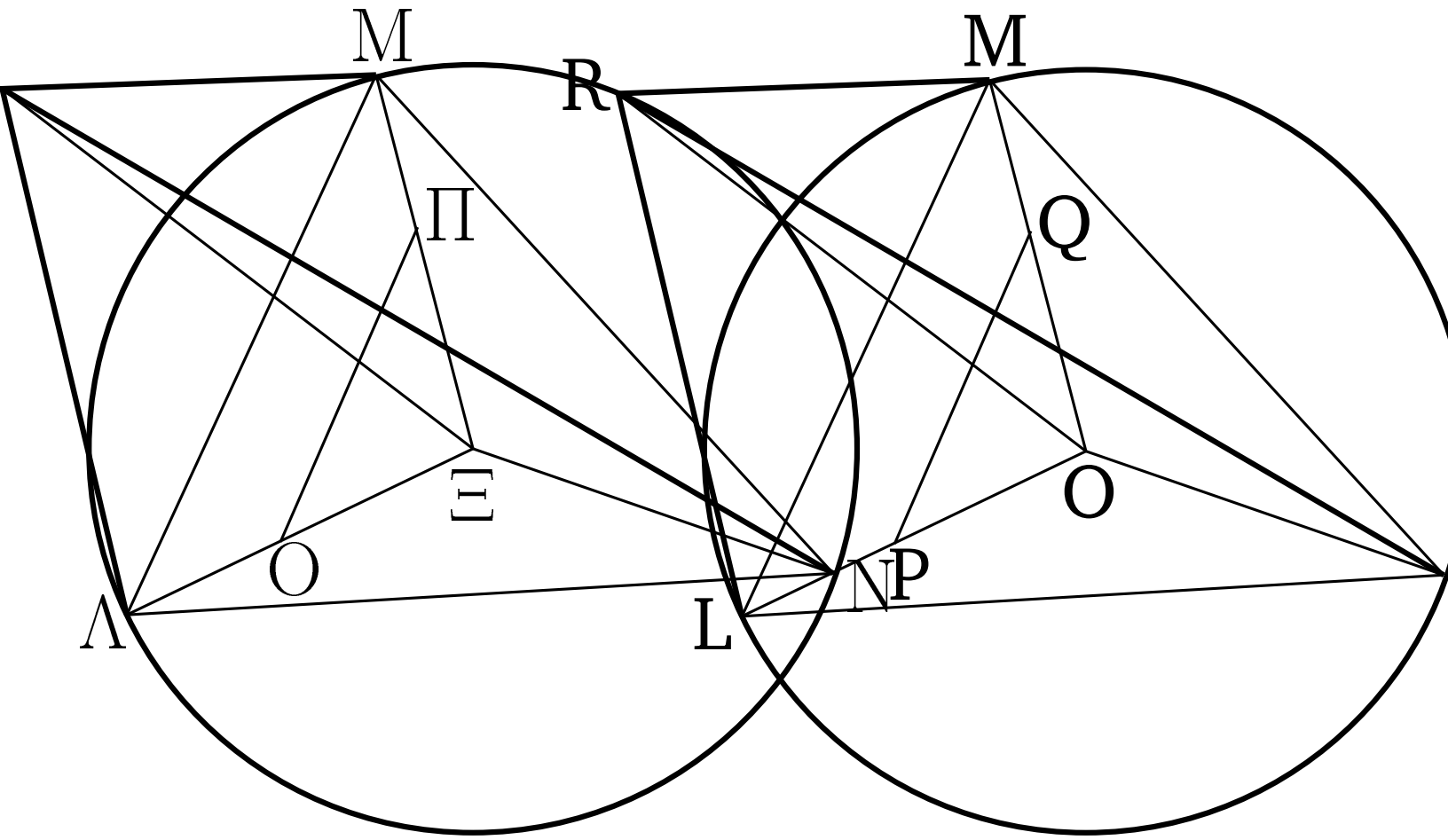
Ἔστωσαν αἱ δοθεῖσαι τρεῖς γωνίαι ἐπίπεδοι αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ, ὧν αἱ δύο τῆς λοιπῆς μείζονεςὀέστωσαν πάντῃ μεταλαμβανόμεναι, ὀέτι δὲ αἱ τρεῖς τεσσάρων ὀρθῶν ἐλάσσονες· δεῖ δὴ ἐκ τῶνὁίσων ταῖς ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ στερεὰν γωνίαν συστήσασθαι.

Ἀπειλήφθωσαν ἴσαι αἱ ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ, καὶ ἐπεζεύθθωσαν αἱ ΑΓ, ΔΖ, ΗΚ· δυνατὸν ἄρα ἐστὶν ἐκ τῶνὁίσων ταῖς ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι. συνεστάτω τὸ ΛΜΝ, ὁώστε ἴσην εὐῖναι τὴν μὲν ΑΓ τῇ ΛΜ, τὴν δὲ ΔΖ τῇ ΜΝ, καὶ ὀέτι τὴν ΗΚ τῇ ΝΛ, καὶ περιγεγράφθω περὶ τὸ ΛΜΝ τρίγωνον κύκλος ὁ ΛΜΝ, καὶ εἰλήφθω αὐτοῦ τὸ κέντρον καὶ ὀέστω τὸ Χ, καὶ ἐπεζεύθθωσαν αἱ ΛΧ, ΜΧ, ΝΧ·

Let ABC, DEF, and GHK be the three given plane angles, of which let (the sum of) two be greater than the remaining (one, the angles) being taken up in any (possible way), and, further, (let) the (sum of the) three (be) less than four right-angles. So, it is necessary to construct a solid angle from (plane angles) equal to ABC, DEF, and GHK.

Let AB, BC, DE, EF, GH, and HK be cut off (so as to be) equal (to one another). And let AC, DF, and GK be joined. It is, thus, possible to construct a triangle from (straight-lines) equal to AC, DF, and GK [Prop. 11.22]. Let (such a triangle), LMN, have be constructed, such that AC is equal to LM, DF to MN, and, further, GK to NL. And let the circle LMN be circumscribed about triangle LMN [Prop. 4.5]. And let its center be found, and let it be (at) O. And let LO, MO, and NO be

joined.



Λέγω, ὅτι ἡ ΑΒ μείζων ἐστὶ τῆς ΛΧ. εἰ γὰρ μή, ἤτοιὀίση ἐστὶν ἡ ΑΒ τῇ ΛΧὀὴ ἐλάττων. ὀέστω πρότερονὀίση. καὶ ἐπεὶ ὀίση ἐστὶν ἡ ΑΒ τῇ ΛΧ, ἀλλὰ ἡ μὲν ΑΒ τῇ ΒΓ ἐστινὀίση, ἡ δὲ ΧΛ τῇ ΧΜ, δύο δὴ αἱ ΑΒ, ΒΓ δύο ταῖς ΛΧ, ΧΜ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ βάσις ἡ ΑΓ βάσει τῇ ΛΜ ὑπόκειταιὀίση· γωνία ἄρα ἡ ὑπὸ ΑΒΓ γωνίᾳ τῇ ὑπὸ ΛΧΜ ἐστινὀίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ μὲν ὑπὸ ΔΕΖ τῇ ὑπὸ ΜΧΝ ἐστινὀίση, καὶ ὀέτι ἡ ὑπὸ ΗΘΚ τῇ ὑπὸ ΝΧΛ· αἱ ἄρα τρεῖς αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ γωνίαι τρισὶ ταῖς ὑπὸ ΛΧΜ, ΜΧΝ, ΝΧΛ εἰσιν ἴσαι. ἀλλὰ αἱ τρεῖς αἱ ὑπὸ ΛΧΜ, ΜΧΝ, ΝΧΛ τέτταρσιν ὀρθαῖξ εἰσιν ἴσαι· καὶ αἱ τρεῖς ἄρα αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ τέτταρσιν ὀρθαῖξ ἴσαι εἰσίν. ὑπόκεινται δὲ καὶ τεσσάρων ὀρθῶν ἐλάσσονες· ὅπερὀάτοπον. οὐκ ἄρα ἡ ΑΒ τῇ ΛΧὀίση ἐστίν. λέγω δή, ὅτι οὐδὲ ἐλάττων ἐστὶν ἡ ΑΒ τῆς ΛΧ. εἰ γὰρ δυνατόν, ὀέστω· καὶ κείσθω τῇ μὲν ΑΒὀίση ἡ ΧΟ, τῇ δὲ ΒΓὀίση ἡ ΧΠ, καὶ ἐπεζεύθθω ἡ ΟΠ. καὶ ἐπεὶ ὀίση ἐστὶν ἡ ΑΒ τῇ ΒΓ, ὀίση ἐστὶ καὶ ἡ ΧΟ τῇ ΧΠ· ὥστε καὶ λοιπὴ ἡ ΛΟ τῇ ΠΜ ἐστινὀίση. παράλληλος ἄρα ἐστὶν ἡ ΛΜ τῇ ΟΠ, καὶ ἰσογώνιον τὸ ΛΜΧ τῷ ΟΠΧ· ὀέστιν ἄρα ὡξ ἡ ΧΛ πρὸς ΛΜ, οὕτως ἡ ΧΟ πρὸς ΟΠ· ἐναλλὰχ ὡξ ἡ ΛΧ πρὸς ΧΟ,

I say that AB is greater than LO. For, if not, AB is either equal to, or less than, LO. Let it, first of all, be equal. And since AB is equal to LO, but AB is equal to BC, and OL to OM, so the two (straight-lines) AB and BC are equal to the two (straight-lines) LO and OM, respectively. And the base AC was assumed (to be) equal to the base LM. Thus, angle ABC is equal to angle LOM [Prop. 1.8]. So, for the same (reasons), DEF is also equal to MON, and, further, GHK to NOL. Thus, the three angles ABC, DEF, and GHK are equal to the three angles LOM, MON, and NOL, respectively. But, the (sum of the) three angles LOM, MON, and NOL is equal to four right-angles. Thus, the (sum of the) three angles ABC, DEF, and GHK is also equal to four right-angles. And it was also assumed (to be) less than four right-angles. The very thing (is) absurd. Thus, AB is not equal to LO. So, I say that AB is not less than LO either. For, if possible, let it be (less). And let OP be made equal to AB, and OQ equal to BC, and let PQ be joined. And since AB is equal to BC, OP is also equal to OQ. Hence, the remainder LP is also equal to (the remainder) QM. LM is thus parallel to PQ [Prop. 6.2], and (triangle) LMO (is)

οὑτως ἡ ΛΜ πρὸς ΟΠ. μείζων δὲ ἡ ΛΧ τῆς ΧΟ· μείζων ἄρα καὶ ἡ ΛΜ τῆς ΟΠ. ἀλλὰ ἡ ΛΜ κεῖται τῇ ΑΓὁίση· καὶ ἡ ΑΓ ἄρα τῆς ΟΠ μείζων ἐστίν. ἐπεὶ οὐν δύο αἱ ΑΒ, ΒΓ δυσὶ ταῖς ΟΧ, ΧΠ ἴσαι εἰσίν, καὶ βάσις ἡ ΑΓ βάσεως τῆς ΟΠ μείζων ἐστίν, γωνία ἄρα ἡ ὑπὸ ΑΒΓ γωνίας τῆς ὑπὸ ΟΧΠ μεῖζων ἐστίν. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ μὲν ὑπὸ ΔΕΖ τῆς ὑπὸ ΜΧΝ μείζων ἐστίν, ἡ δὲ ὑπὸ ΗΘΚ τῆς ὑπὸ ΝΧΛ. αἱ ἄρα τρεῖς γωνίαι αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ τριῶν τῶν ὑπὸ ΛΧΜ, ΜΧΝ, ΝΧΛ μείζονές εἰσιν. ἀλλὰ αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ τεσσάρων ὀρθῶν ἐλάσσονες ὑπόκεινται· πολλῷ ἄρα αἱ ὑπὸ ΛΧΜ, ΜΧΝ, ΝΧΛ τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν. ἀλλὰ καὶ ἴσαι· ὅπερ ἐστὶνὁάτοπον. οὐκ ἄρα ἡ ΑΒ ἐλάσσων ἐστὶ τῆς ΛΧ. ἐδείθθη δέ, ὅτι οὐδὲ ὁίση· μείζων ἄρα ἡ ΑΒ τῆς ΛΧ.

Ἀνεστάτω δὴ ἀπὸ τοῦ Χ σημείου τῷ τοῦ ΛΜΝ κύκλου ἐπιπέδῳ πρὸς ὀρθὰξ ἡ ΧΡ, καὶ ὁῷ μεῖζόν ἐστι τὸ ἀπὸ τῆς ΑΒ τετράγωνον τοῦ ἀπὸ τῆς ΛΧ, ἐκείνῳ ἴσονὁέστω τὸ ἀπὸ τῆς ΧΡ, καὶ ἐπεζεύθθωσαν αἱ ΡΛ, ΡΜ, ΡΝ.

Καὶ ἐπεὶ ἡ ΡΧ ὀρθὴ ἐστι πρὸς τὸ τοῦ ΛΜΝ κύκλου ἐπίπεδον, καὶ πρὸς ἑκάστην ἄρα τῶν ΛΧ, ΜΧ, ΝΧ ὀρθή ἐστιν ἡ ΡΧ. καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΛΧ τῇ ΧΜ, κοινὴ δὲ καὶ πρὸς ὀρθὰξ ἡ ΧΡ, βάσις ἄρα ἡ ΡΛ βάσει τῇ ΡΜ ἐστινὁίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΡΝ ἑκατέρᾳ τῶν ΡΛ, ΡΜ ἐστινὁίση· αἱ τρεῖς ἄρα αἱ ΡΛ, ΡΜ, ΡΝ ἴσαι ἀλλήλαις εἰσίν. καὶ ἐπεὶ ὁῷ μεῖζόν ἐστι τὸ ἀπὸ τῆς ΑΒ τοῦ ἀπὸ τῆς ΛΧ, ἐκείνῳ ἴσον ὑπόκειται τὸ ἀπὸ τῆς ΧΡ, τὸ ἄρα ἀπὸ τῆς ΑΒ ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΛΧ, ΧΡ. τοῖς δὲ ἀπὸ τῶν ΛΧ, ΧΡ ἴσον ἐστὶ τὸ ἀπὸ τῆς ΛΡ· ὀρθὴ γὰρ ἡ ὑπὸ ΛΧΡ· τὸ ἄρα ἀπὸ τῆς ΑΒ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΡΛ· ὁίση ἄρα ἡ ΑΒ τῇ ΡΛ. ἀλλὰ τῇ μὲν ΑΒὁίση ἐστὶν ἑκάστη τῶν ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ, τῇ δὲ ΡΛὁίση ἑκατέρα τῶν ΡΜ, ΡΝ· ἑκάστη ἄρα τῶν ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ ἑκάστῃ τῶν ΡΛ, ΡΜ, ΡΝὁίση ἐστίν. καὶ ἐπεὶ δύο αἱ ΛΡ, ΡΜ δυσὶ ταῖς ΑΒ, ΒΓ ἴσαι εἰσίν, καὶ βάσις ἡ ΛΜ βάσει τῇ ΑΓ ὑπόκειταιὁίση, γωνία ἄρα ἡ ὑπὸ ΛΡΜ γωνίᾳ τῇ ὑπὸ ΑΒΓ ἐστινὁίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ μὲν ὑπὸ ΜΡΝ τῇ ὑπὸ ΔΕΖ ἐστινὁίση, ἡ δὲ ὑπὸ ΛΡΝ τῇ ὑπὸ ΗΘΚ.

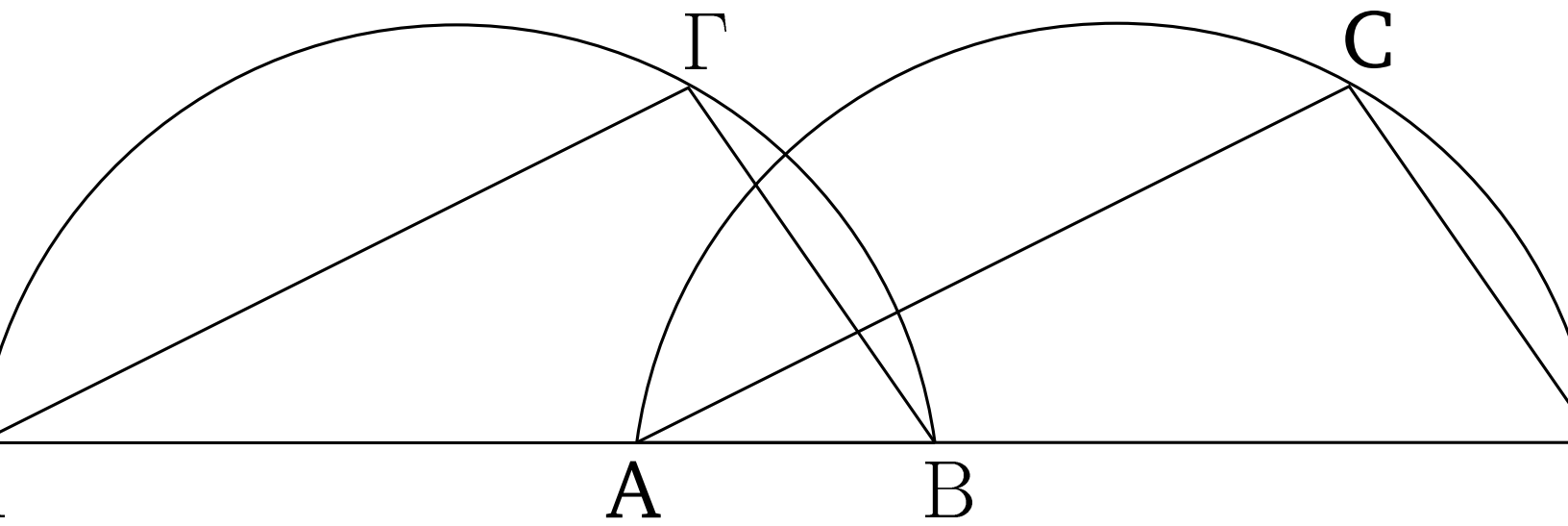
Ἐκ τριῶν ἄρα γωνιῶν ἐπιπέδων τῶν ὑπὸ ΛΡΜ, ΜΡΝ, ΛΡΝ, αὁί εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις ταῖς ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ, στερεὰ γωνία συνέσταται ἡ πρὸς τῷ Ρ περιεθομένη ὑπὸ τῶν ΛΡΜ, ΜΡΝ, ΛΡΝ γωνιῶν· ὅπερ ἔδει ποιῆσαι.

equiangular with (triangle) *PQO* [Prop. 1.29]. Thus, as *OL* is to *LM*, so *OP* (is) to *PQ* [Prop. 6.4]. Alternately, as *LO* (is) to *OP*, so *LM* (is) to *PQ* [Prop. 5.16]. And *LO* (is) greater than *OP*. Thus, *LM* (is) also greater than *PQ* [Prop. 5.14]. But *LM* was made equal to *AC*. Thus, *AC* is also greater than *PQ*. Therefore, since the two (straight-lines) *AB* and *BC* are equal to the two (straight-lines) *PO* and *OQ* (respectively), and the base *AC* is greater than the base *PQ*, the angle *ABC* is thus greater than the angle *POQ* [Prop. 1.25]. So, similarly, we can show that *DEF* is also greater than *MON*, and *GHK* than *NOL*. Thus, the (sum of the) three angles *ABC*, *DEF*, and *GHK* is greater than the (sum of the) three angles *LOM*, *MON*, and *NOL*. But, (the sum of) *ABC*, *DEF*, and *GHK* was assumed (to be) less than four right-angles. Thus, (the sum of) *LOM*, *MON*, and *NOL* is much less than four right-angles. But, (it is) also equal (to four right-angles). The very thing is absurd. Thus, *AB* is not less than *LO*. And it was shown (to be) not equal either. Thus, *AB* (is) greater than *LO*.

So let *OR* be set up at point *O* at right-angles to the plane of circle *LMN* [Prop. 11.12]. And let the (square) on *OR* be equal to that (area) by which the square on *AB* is greater than the (square) on *LO* [Prop. 11.23 lem.]. And let *RL*, *RM*, and *RN* be joined.

And since *RO* is at right-angles to the plane of circle *LMN*, *RO* is thus also at right-angles to each of *LO*, *MO*, and *NO*. And since *LO* is equal to *OM*, and *OR* is common and at right-angles, the base *RL* is thus equal to the base *RM* [Prop. 1.4]. So, for the same (reasons), *RN* is also equal to each of *RL* and *RM*. Thus, the three (straight-lines) *RL*, *RM*, and *RN* are equal to one another. And since the (square) on *OR* was assumed to be equal to that (area) by which the (square) on *AB* is greater than the (square) on *LO*, the (square) on *AB* is thus equal to the (sum of the squares) on *LO* and *OR*. And the (square) on *LR* is equal to the (sum of the squares) on *LO* and *OR*. For *LOR* (is) a right-angle [Prop. 1.47]. Thus, the (square) on *AB* is equal to the (square) on *RL*. Thus, *AB* (is) equal to *RL*. But, each of *BC*, *DE*, *EF*, *GH*, and *HK* is equal to *AB*, and each of *RM* and *RN* equal to *RL*. Thus, each of *AB*, *BC*, *DE*, *EF*, *GH*, and *HK* is equal to each of *RL*, *RM*, and *RN*. And since the two (straight-lines) *LR* and *RM* are equal to the two (straight-lines) *AB* and *BC* (respectively), and the base *LM* was assumed (to be) equal to the base *AC*, the angle *LRM* is thus equal to the angle *ABC* [Prop. 1.8]. So, for the same (reasons), *MRN* is also equal to *DEF*, and *LRN* to *GHK*.

Thus, the solid angle *R*, contained by the angles *LRM*, *MRN*, and *LRN*, has been constructed out of the three plane angles *LRM*, *MRN*, and *LRN*, which are equal to the three given (plane angles) *ABC*, *DEF*, and *GHK* (respectively). (Which is) the very thing it was required to do.



Λῆμμα.

ὁ Ὸν δὲ τρόπον, ὁῷ μεῖζόν ἐστι τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς ΛX, ἐκείνῳ ἴσον λαβεῖνὸέστι τὸ ἀπὸ τῆς XP, δείχομεν οὁύτως. ἐκκείσθωσαν αἱ AB, ΛX εὐθεῖαι, καὶ ὀέστω μείζων ἡ AB, καὶ γεγράφθω ἐπ᾽ αὐτῆς ἡμικύκλιον τὸ ABΓ, καὶ εἰξ τὸ ABΓ ἡμικύκλιον ἐνηρμόσθω τῇ ΛX εὐθείᾳ μὴ μείζονι οὁύσῃ τῆς AB διαμέτρουὸίση ἡ AΓ, καὶ ἐπεζεύθθω ἡ ΓB. ἐπεὶ οὁῦν ἐν ἡμικυκλίῳ τῷ AΓB γωνία ἐστὶν ἡ ὑπὸ AΓB, ὀρθὴ ἄρα ἐστὶν ἡ ὑπὸ AΓB. τὸ ἄρα ἀπὸ τῆς AB ἴσον ἐστὶ τοῖς ἀπὸ τῶν AΓ, ΓB. ὁώστε τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς AΓ μεῖζόν ἐστι τῷ ἀπὸ τῆς ΓB. ὁίση δὲ ἡ AΓ τῇ ΛX. τὸ ἄρα ἀπὸ τῆς AB τοῦ ἀπὸ τῆς ΛX μεῖζόν ἐστι τῷ ἀπὸ τῆς ΓB. ἐαν οὁῦν τῇ BΓ ἴσην τὴν XP ἀπολάβωμεν, ὀέσται τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς ΛX μεῖζον τῷ ἀπὸ τῆς XP· ὅπερ προέκειτο ποιῆσαι.

Lemma

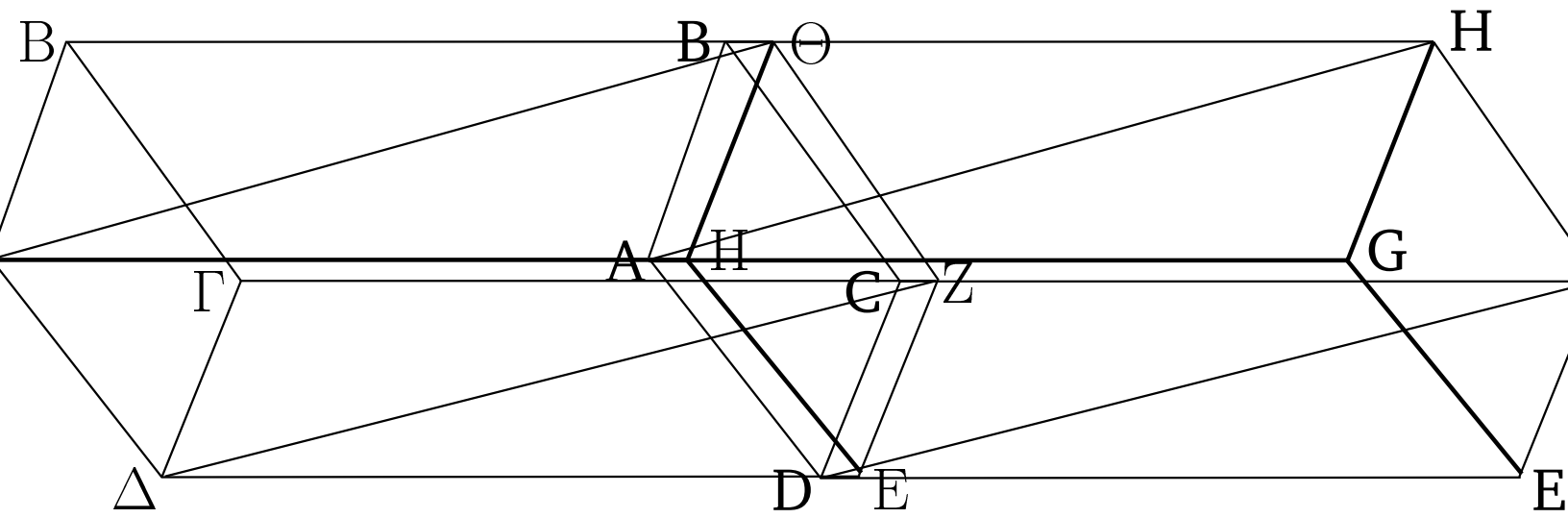
And we can demonstrate, thusly, in which manner to take the (square) on *OR* equal to that (area) by which the (square) on *AB* is greater than the (square) on *LO*. Let the straight-lines *AB* and *LO* be set out, and let *AB* be greater, and let the semicircle *ABC* be drawn around it. And let *AC*, equal to the straight-line *LO*, which is not greater than the diameter *AB*, be inserted into the semicircle *ABC* [Prop. 4.1]. And let *CB* be joined. Therefore, since the angle *ACB* is in the semicircle *ACB*, *ACB* is thus a right-angle [Prop. 3.31]. Thus, the (square) on *AB* is equal to the (sum of the) squares on *AC* and *CB* [Prop. 1.47]. Hence, the (square) on *AB* is greater than the (square) on *AC* by the (square) on *CB*. And *AC* (is) equal to *LO*. Thus, the (square) on *AB* is greater than the (square) on *LO* by the (square) on *CB*. Therefore, if we take *OR* equal to *BC* then the (square) on *AB* will be greater than the (square) on *LO* by the (square) on *OR*. (Which is) the very thing it was prescribed to do.

24.

Ἐὰν στερεὸν ὑπὸ παραλλήλων ἐπιπέδων περιέθηται, τὰ ἀπεναντίον αὐτοῦ ἐπίπεδα ἴσα τε καὶ παραλληλόγραμμά ἐστιν.

Proposition 24

If a solid (figure) is contained by (six) parallel planes then its opposite planes are both equal and parallelogrammic.



Στερεὸν γὰρ τὸ ΓΔΘΗ ὑπὸ παραλλήλων ἐπιπέδων περιεθέσθω τῶν ΑΓ, ΗΖ, ΑΘ, ΔΖ, ΒΖ, ΑΕ· λέγω, ὅτι τὰ ἀπεναντίον αὐτοῦ ἐπίπεδα ἴσα τε καὶ παραλληλόγραμμά ἐστιν.

Ἐπεὶ γὰρ δύο ἐπίπεδα παράλληλα τὰ ΒΗ, ΓΕ ὑπὸ ἐπιπέδου τοῦ ΑΓ τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν. παράλληλος ἄρα ἐστὶν ἡ ΑΒ τῇ ΔΓ. πάλιν, ἐπεὶ δύο ἐπίπεδα παράλληλα τὰ ΒΖ, ΑΕ ὑπὸ ἐπιπέδου τοῦ ΑΓ τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν. παράλληλος ἄρα ἐστὶν ἡ ΒΓ τῇ ΑΔ. ἐδείχθη δὲ καὶ ἡ ΑΒ τῇ ΔΓ παράλληλος· παραλληλόγραμμον ἄρα ἐστὶ τὸ ΑΓ. ὁμοίως δὴ δείξομεν, ὅτι καὶ ὁἑκαστον τῶν ΔΖ, ΖΗ, ΗΒ, ΒΖ, ΑΕ παραλληλόγραμμόν ἐστιν.

Ἐπεζεύθθωσαν αἱ ΑΘ, ΔΖ. καὶ ἐπεὶ παράλληλόξ ἐστιν ἡ μὲν ΑΒ τῇ ΔΓ, ἡ δὲ ΒΘ τῇ ΓΖ, δύο δὴ αἱ ΑΒ, ΒΘ ἁπτόμεναι ἀλλήλων παρὰ δύο εὐθείας τὰξ ΔΓ, ΓΖ ἁπτομένας ἀλλήλων εἰσὶν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ· ἴσας ἄρα γωνίας περιέχουσιν· ὁίση ἄρα ἡ ὑπὸ ΑΒΘ γωνία τῇ ὑπὸ ΔΓΖ. καὶ ἐπεὶ δύο αἱ ΑΒ, ΒΘ δυσὶ ταῖς ΔΓ, ΓΖ ἴσαι εἰσίν, καὶ γωνία ἡ ὑπὸ ΑΒΘ γωνίᾳ τῇ ὑπὸ ΔΓΖ ἐστινὁίση, βάσις ἄρα ἡ ΑΘ βάσει τῇ ΔΖ ἐστινὁίση, καὶ τὸ ΑΒΘ τρίγωνον τῷ ΔΓΖ τριγώνῳ ἴσον ἐστίν. καί ἐστι τοῦ μὲν ΑΒΘ διπλάσιον τὸ ΒΗ παραλληλόγραμμον, τοῦ δὲ ΔΓΖ διπλάσιον τὸ ΓΕ παραλληλόγραμμον· ἴσον ἄρα τὸ ΒΗ παραλληλόγραμμον τῷ ΓΕ παραλληλογράμμῳ· ὁμοίως δὴ δείχομεν, ὅτι καὶ τὸ μὲν ΑΓ τῷ ΗΖ ἐστιν ἴσον, τὸ δὲ ΑΕ τῷ ΒΖ.

Ἐὰν ἄρα στερεὸν ὑπὸ παραλλήλων ἐπιπέδων περιέθηται, τὰ ἀπεναντίον αὐτοῦ ἐπίπεδα ἴσα τε καὶ παραλληλόγραμμά ἐστιν· ὅπερ ἔδει δεῖχαι.

For let the solid (figure) $CDHG$ be contained by the parallel planes AC, GF, and AH, DF, and BF, AE. I say that its opposite planes are both equal and parallelogrammic.

For since the two parallel planes BG and CE are cut by the plane AC, their common sections are parallel [Prop. 11.16]. Thus, AB is parallel to DC. Again, since the two parallel planes BF and AE are cut by the plane AC, their common sections are parallel [Prop. 11.16]. Thus, BC is parallel to AD. And AB was also shown (to be) parallel to DC. Thus, AC is a parallelogram. So, similarly, we can also show that DF, FG, GB, BF, and AE are each parallelograms.

Let AH and DF be joined. And since AB is parallel to DC, and BH to CF, so the two (straight-lines) joining one another, AB and BH, are parallel to the two straight-lines joining one another, DC and CF (respectively), not (being) in the same plane. Thus, they will contain equal angles [Prop. 11.10]. Thus, angle ABH (is) equal to (angle) DCF. And since the two (straight-lines) AB and BH are equal to the two (straight-lines) DC and CF (respectively) [Prop. 1.34], and angle ABH is equal to angle DCF, the base AH is thus equal to the base DF, and triangle ABH is equal to triangle DCF [Prop. 1.4]. And parallelogram BG is double (triangle) ABH, and parallelogram CE double (triangle) DCF [Prop. 1.34]. Thus, parallelogram BG (is) equal to parallelogram CE. So, similarly, we can show that AC is also equal to GF, and AE to BF.

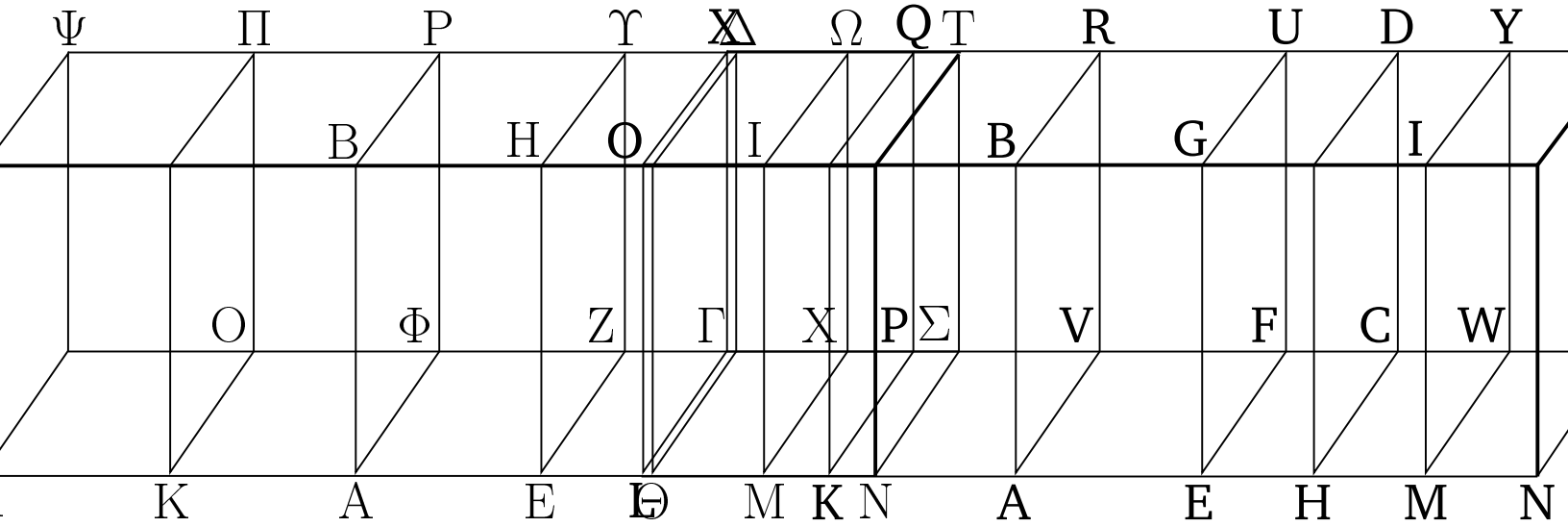
Thus, if a solid (figure) is contained by (six) parallel planes then its opposite planes are both equal and parallelogrammic. (Which is) the very thing it was required to show.

25.

Ἐὰν στερεὸν παραλληλεπίπεδον ἐπιπέδῳ τμ ηθῇ παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ὁέσται ὡξ ἡ βάσις πρὸς τὴν βάσιν, οὁύτως τὸ στερεὸν πρὸς τὸ στερεόν.

Proposition 25

If a parallelipiped solid is cut by a plane which is parallel to the opposite planes (of the parallelipiped) then as the base (is) to the base, so the solid will be to the solid.



Στερεὸν γὰρ παραλληλεπίπεδον τὸ ΑΒΓΔ ἐπιπέδῳ τῷ ΖΗ τετμήσθω παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις τοῖς ΡΑ, ΔΘ· λέγω, ὅτι ἐστὶν ὡς ἡ ΑΕΖΦ βάσις πρὸς τὴν ΕΘΓΖ βάσιν, οὕτως τὸ ΑΒΖΥ στερεὸν πρὸς τὸ ΕΗΓΔ στερεόν.

Ἐκβεβλήσθω γὰρ ἡ ΑΘ ἐφ' ἑκάτερα τὰ μέρη, καὶ κείσθωσαν τῇ μὲν ΑΕ ἴσαι ὁσαιδηποτοῦν αἱ ΑΚ, ΚΛ, τῇ δὲ ΕΘ ἴσαι ὁσαιδηποτοῦν αἱ ΘΜ, ΜΝ, καὶ συμπεπληρώσθω τὰ ΛΟ, ΚΦ, ΘΘ, ΜΣ παραλληλόγραμμα καὶ τὰ ΛΠ, ΚΡ, ΔΜ, ΜΤ στερεά.

Καὶ ἐπεὶ ἴσαι εἰσὶν αἱ ΛΚ, ΚΑ, ΑΕ εὐθεῖαι ἀλλήλαις, ἴσα ἐστὶ καὶ τὰ μὲν ΛΟ, ΚΦ, ΑΖ παραλληλόγραμμα ἀλλήλοις, τὰ δὲ ΚΧ, ΚΒ, ΑΗ ἀλλήλοις καὶ ὀέτι τὰ ΛΨ, ΚΠ, ΑΡ ἀλλήλοις· ἀπεναντίον γάρ. διὰ τὰ αὐτὰ δὴ καὶ τὰ μὲν ΕΓ, ΘΘ, ΜΣ παραλληλόγραμμα ἴσα εἰσὶν ἀλλήλοις, τὰ δὲ ΘΗ, ΘΙ, ΙΝ ἴσα εἰσὶν ἀλλήλοις, καὶ ὀέτι τὰ ΔΘ, ΜΩ, ΝΤ· τρία ἄρα ἐπίπεδα τῶν ΛΠ, ΚΡ, ΑΥ στερεῶν τρισὶν ἐπιπέδοις ἐστὶν ἴσα. ἀλλὰ τὰ τρία τρισὶ τοῖς ἀπεναντίον ἐστὶν ἴσα· τὰ ἄρα τρία στερεὰ τὰ ΛΠ, ΚΡ, ΑΥ ἴσα ἀλλήλοις ἐστίν. διὰ τὰ αὐτὰ δὴ καὶ τὰ τρία στερεὰ τὰ ΕΔ, ΔΜ, ΜΤ ἴσα ἀλλήλοις ἐστίν· ὁσαπλασίων ἄρα ἐστὶν ἡ ΛΖ βάσις τῆς ΑΖ βάσεως, τοσαυταπλάσιόν ἐστι καὶ τὸ ΛΥ στερεὸν τοῦ ΑΥ στερεοῦ. διὰ τὰ αὐτὰ δὴ ὁσαπλασίων ἐστὶν ἡ ΝΖ βάσις τῆς ΖΘ βάσεως, τοσαυταπλάσιόν ἐστι καὶ τὸ ΝΥ στερεὸν τοῦ ΘΥ στερεοῦ. καὶ εἰ ὁίση ἐστὶν ἡ ΛΖ βάσις τῇ ΝΖ βάσει, ἴσον ἐστὶ καὶ τὸ ΛΥ στερεὸν τῷ ΝΥ στερεῷ, καὶ εἰ ὑπερέθει ἡ ΛΖ βάσις τῆς ΝΖ βάσεως, ὑπερέθει καὶ τὸ ΛΥ στερεὸν τοῦ ΝΥ στερεοῦ, καὶ εἰ ἐλλείπει, ἐλλείπει. τεσσάρων δὴ ὄντων μεγεθῶν, δύο μὲν βάσεων τῶν ΑΖ, ΖΘ, δύο δὲ στερεῶν τῶν ΑΥ, ΥΘ, εὐίληπται ἰσάκις πολλαπλάσια τῆς μὲν ΑΖ βάσεως καὶ τοῦ ΑΥ στερεοῦ ἥ τε ΛΖ βάσις καὶ τὸ ΛΥ στερεόν, τῆς δὲ ΘΖ βάσεως καὶ τοῦ ΘΥ στερεοῦ ἥ τε ΝΖ βάσις καὶ τὸ ΝΥ στερεόν, καὶ δέδεικται, ὅτι εἰ ὑπερέθει ἡ ΛΖ βάσις τῆς ΖΝ βάσεως, ὑπερέθει καὶ τὸ ΛΥ στερεὸν τοῦ ΝΥ [στερεοῦ], καὶ εἰ ὁίση, ἴσον, καὶ εἰ ἐλλείπει, ἐλλείπει. ὀέστιν ἄρα ὡς ἡ

For let the parallelipiped solid $ABCD$ be cut by the plane FG which is parallel to the opposite planes RA and DH. I say that as the base $AEFV$ (is) to the base $EHCF$, so the solid $ABFU$ (is) to the solid $EGCD$.

For let AH be produced in each direction. And let any number whatsoever (of lengths), AK and KL, be made equal to AE, and any number whatsoever (of lengths), HM and MN, equal to EH. And let the parallelograms LP, KV, HW, and MS be completed, and the solids LQ, KR, DM, and MT.

And since the straight-lines LK, KA, and AE are equal to one another, the parallelograms LP, KV, and AF are also equal to one another, and KO, KB, and AG (are equal) to one another, and, further, LX, KQ, and AR (are equal) to one another. For (they are) opposite [Prop. 11.24]. So, for the same (reasons), the parallelograms EC, HW, and MS are also equal to one another, and HG, HI, and IN are equal to one another, and, further, DH, MY, and NT (are equal to one another). Thus, three planes of (one of) the solids LQ, KR, and AU are equal to the (corresponding) three planes (of the others). But, the three planes (in one of the soilds) are equal to the three opposite planes [Prop. 11.24]. Thus, the three solids LQ, KR, and AU are equal to one another [Def. 11.10]. So, for the same (reasons), the three solids ED, DM, and MT are also equal to one another. Thus, as many multiples as the base LF is of the base AF, so many multiples is the solid LU also of the the solid AU. So, for the same (reasons), as many multiples as the base NF is of the base FH, so many multiples is the solid NU also of the solid HU. And if the base LF is equal to the base NF then the solid LU is also equal to the solid NU.† And if the base LF exceeds the base NF then the solid LU also exceeds the solid NU. And if (LF) is less than (NF) then (LU) is (also) less than (NU). So, there are four magnitudes, the two bases AF and FH, and the two solids AU and UH, and equal mul-

ΑΖ βάσις πρὸς τὴν ΖΘ βάσιν, οὕτως τὸ ΑΥ στερεὸν πρὸς τὸ ΥΘ στερεόν· ὅπερ ἔδει δεῖξαι.

tiples be taken of the base AF and the solid AU— (namely), the base LF and the solid LU—and of the base HF and the solid HU—(namely), the base NF and the solid NU. And it has been shown that if the base LF exceeds the base FN then the solid LU also exceeds the [solid] NU, and if (LF is) equal (to FN) then (LU is) equal (to NU), and if (LF is) less than (FN) then (LU is) less than (NU). Thus, as the base AF is to the base FH, so the solid AU (is) to the solid UH [Def. 5.5]. (Which is) the very thing it was required to show.

† Here, Euclid assumes that $LF \gtreqless NF$ implies $LU \gtreqless NU$. This is easily demonstrated.

26.

Πρὸξ τῇ δοθείσῃ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῇ δοθείσῃ στερεᾷ γωνίᾳ ἴσην στερεὰν γωνίαν συστήσασθαι.

Ἔστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ ΑΒ, τὸ δὲ πρὸς αὐτῇ δοθὲν σημεῖον τὸ Α, ἡ δὲ δοθεῖσα στερεὰ γωνία ἡ πρὸς τῷ Δ περιεθομένη ὑπὸ τῶν ὑπὸ ΕΔΓ, ΕΔΖ, ΖΔΓ γωνιῶν ἐπιπέδων· δεῖ δὴ πρὸς τῇ ΑΒ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Α τῇ πρὸς τῷ Δ στερεᾷ γωνίᾳ ἴσην στερεὰν γωνίαν συστήσασθαι.

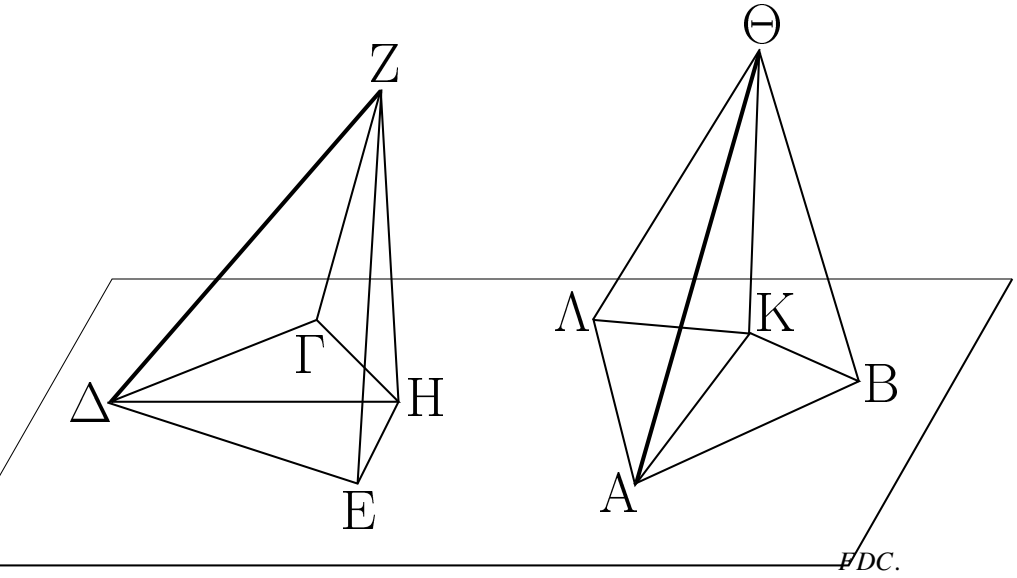
Εἰλήφθω γὰρ ἐπὶ τῆς ΔΖ τυθὸν σημεῖον τὸ Ζ, καὶ ἤχθω ἀπὸ τοῦ Ζ ἐπὶ τὸ διὰ τῶν ΕΔ, ΔΓ ἐπίπεδον κάθετος ἡ ΖΗ, καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ Η, καὶ ἐπεζεύχθω ἡ ΔΗ, καὶ συνεστάτω πρὸς τῇ ΑΒ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Α τῇ μὲν ὑπὸ ΕΔΓ γωνίᾳ ὁίση ἡ ὑπὸ ΒΑΛ, τῇ δὲ ὑπὸ ΕΔΗοίση ἡ ὑπὸ ΒΑΚ, καὶ κείσθω τῇ ΔΗοίση ἡ ΑΚ, καὶ ἀνεστάτω ἀπὸ τοῦ Κ σημείου τῷ διὰ τῶν ΒΑΛ ἐπιπέδῳ πρὸς ὀρθὰξ ἡ ΚΘ, καὶ κείσθωοίση τῇ ΗΖ ἡ ΚΘ, καὶ ἐπεζεύχθω ἡ ΘΑ· λέγω, ὅτι ἡ πρὸς τῷ Α στερεὰ γωνία περιεθομένη ὑπὸ τῶν ΒΑΛ, ΒΑΘ, ΘΑΛ γωνιῶνοίση ἐστὶ τῇ πρὸς τῷ Δ στερεᾷ γωνίᾳ τῇ περιεθομένῃ ὑπὸ τῶν ΕΔΓ, ΕΔΖ, ΖΔΓ γωνιῶν.

Proposition 26

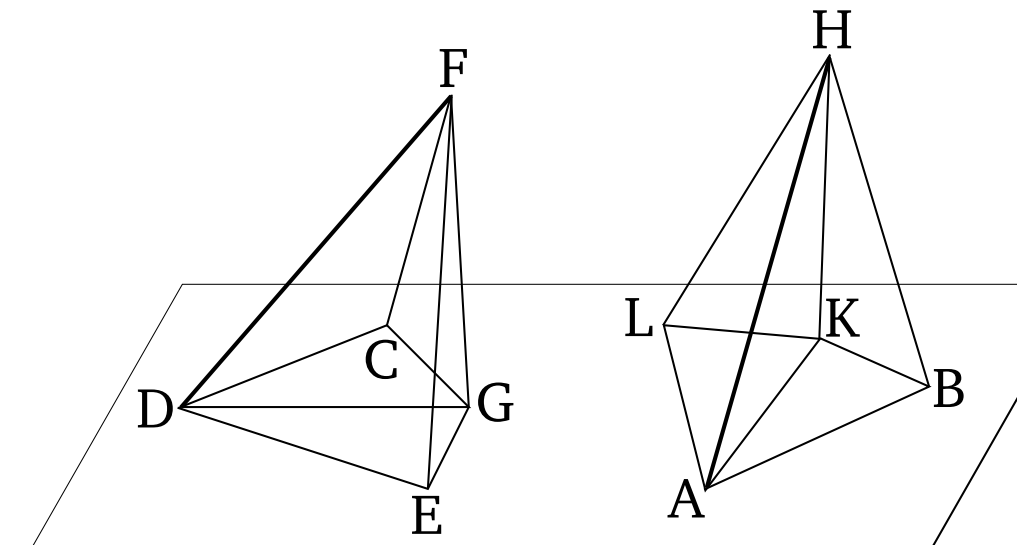
To construct a solid angle equal to a given solid angle on a given straight-line, and at a given point on it.

Let AB be the given straight-line, and A the given point on it, and D the given solid angle, contained by the plane angles EDC, EDF, and FDC. So, it is necessary to construct a solid angle equal to the solid angle D on the straight-line AB, and at the point A on it.

For let some random point F be taken on DF, and let FG be drawn from F perpendicular to the plane through ED and DC [Prop. 11.11], and let it meet the plane at G, and let DG be joined. And let BAL, equal to the angle EDC, and BAK, equal to EDG, be constructed on the straight-line AB at the point A on it [Prop. 1.23]. And let AK be made equal to DG. And let KH be set up at the point K at right-angles to the plane through BAL [Prop. 11.12]. And let KH be made equal to GF. And let HA be joined. I say that the solid angle at A, contained by the (plane) angles BAL, BAH, and HAL, is equal to the solid angle at D, contained by the (plane) angles EDC, EDF, and



FDC.



Ἀπειλήφθωσαν γὰρ ἴσαι αἱ ΑΒ, ΔΕ, καὶ ἐπεζεύχθωσαν αἱ ΘΒ, ΚΒ, ΖΕ, ΗΕ. καὶ ἐπεὶ ἡ ΖΗ ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας· ὀρθὴ ἄρα ἐστὶν ἑκατέρα τῶν ὑπὸ ΖΗΔ, ΖΗΕ γωνιῶν. διὰ τὰ αὐτὰ δὴ καὶ ἑκατέρα τῶν ὑπὸ ΘΚΑ, ΘΚΒ γωνιῶν ὀρθή ἐστιν. καὶ ἐπεὶ δύο αἱ ΚΑ, ΑΒ δύο ταῖς ΗΔ, ΔΕ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, καὶ γωνίας ἴσας περιέθουσιν, βάσις ἄρα ἡ ΚΒ βάσει τῇ ΗΕοίση ἐστίν. ὀέστι δὲ καὶ ἡ ΚΘ τῇ ΗΖοίση· καὶ γωνίας ὀρθὰξ περιέθουσιν· ὀίση ἄρα καὶ ἡ ΘΒ τῇ ΖΕ. πάλιν ἐπεὶ δύο αἱ ΑΚ, ΚΘ δυσὶ ταῖς ΔΗ, ΗΖ ἴσαι εἰσίν, καὶ γωνίας ὀρθὰξ περιέθουσιν, βάσις ἄρα ἡ ΑΘ βάσει τῇ ΖΔοίση ἐστίν. ὀέστι δὲ καὶ ἡ ΑΒ τῇ ΔΕοίση· δύο δὴ αἱ ΘΑ, ΑΒ δύο ταῖς ΔΖ, ΔΕ ἴσαι εἰσίν. καὶ βάσις ἡ ΘΒ βάσει τῇ ΖΕοίση· γωνία ἄρα ἡ ὑπὸ ΒΑΘ γωνίᾳ τῇ ὑπὸ ΕΔΖ ἐστινὀίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΘΑΛ τῇ ὑπὸ ΖΔΓ ἐστινὀίση. ὀέστι δὲ καὶ ἡ ὑπὸ ΒΑΛ τῇ ὑπὸ ΕΔΓοίση.

Πρὸξ ἄρα τῇ δοθείσῃ εὐθείᾳ τῇ ΑΒ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Α τῇ δοθείσῃ στερεᾷ γωνίᾳ τῇ πρὸς τῷ Δοίση συνέσταται· ὅπερ ἔδει ποιῆσαι.

For let AB and DE be cut off (so as to be) equal, and let HB, KB, FE, and GE be joined. And since FG is at right-angles to the reference plane (EDC), it will also make right-angles with all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. Thus, the angles FGD and FGE are right-angles. So, for the same (reasons), the angles HKA and HKB are also right-angles. And since the two (straight-lines) KA and AB are equal to the two (straight-lines) GD and DE, respectively, and they contain equal angles, the base KB is thus equal to the base GE [Prop. 1.4]. And KH is also equal to GF. And they contain right-angles (with the respective bases). Thus, HB (is) also equal to FE [Prop. 1.4]. Again, since the two (straight-lines) AK and KH are equal to the two (straight-lines) DG and GF (respectively), and they contain right-angles, the base AH is thus equal to the base FD [Prop. 1.4]. And AB (is) also equal to DE. So, the two (straight-lines) HA and AB are equal to the two (straight-lines) DF and DE (respectively). And the base HB (is) equal to the base FE. Thus, the angle BAH is equal to the angle EDF [Prop. 1.8]. So, for the same (reasons), HAL is also equal to FDC. And BAL is also equal to EDC.

Thus, (a solid angle) has been constructed, equal to the given solid angle at D, on the given straight-line AB, at the given point A on it. (Which is) the very thing it was required to do.

27.

Ἀπὸ τῆς δοθείσης εὐθείας τῷ δοθέντι στερεῷ παραλληλεπιπέδῳ ὅμοιόν τε καὶ ὁμοίως κείμενον στερεὸν παραλληλεπίπεδον ἀναγράψαι.

Ἔστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ ΑΒ, τὸ δὲ δοθὲν στερεὸν παραλληλεπίπεδον τὸ ΓΔ· δεῖ δὴ ἀπὸ τῆς δοθείσης εὐθείας

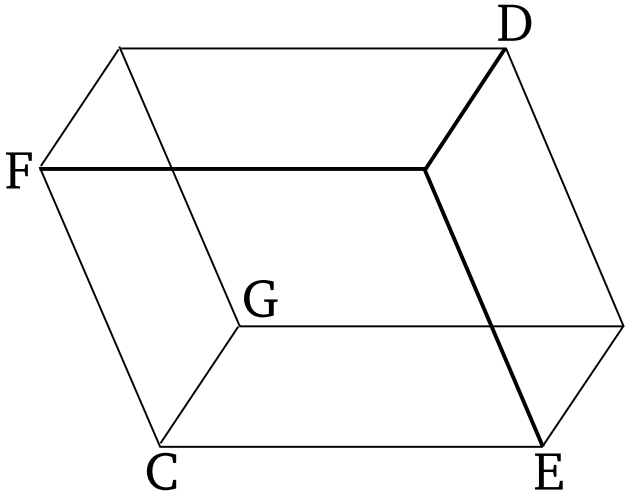
Proposition 27

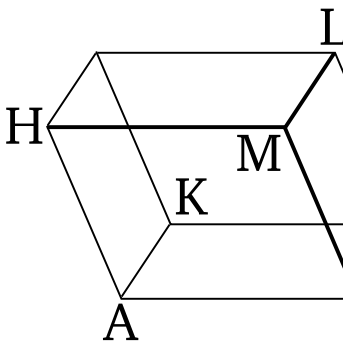
To describe a parallelepiped solid similar, and similarly laid out, to a given parallelepiped solid on a given straight-line.

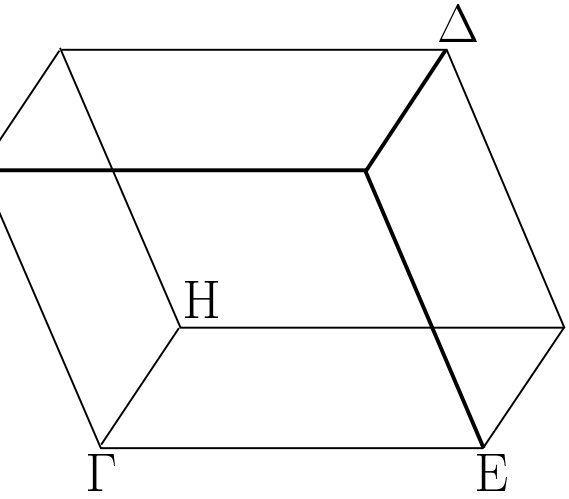
Let the given straight-line be AB, and the given parallelepiped solid CD. So, it is necessary to describe a paral-

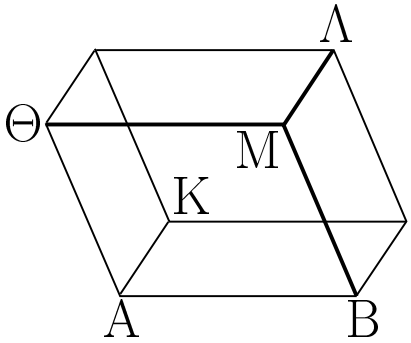
τῆς AB τῷ δοθέντι στερεῷ παραλληλεπιπέδῳ τῷ $ΓΔ$ ὁμοιόν τε καὶ ὁμοίως κείμενον στερεὸν παραλληλεπίπεδον ἀναγράψαι.

Συνεστάτω γὰρ πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ πρὸς τῷ $Γ$ στερεᾷ γωνίᾳ ὁίση ἡ περιεθομένη ὑπὸ τῶν $BAΘ$, $ΘAK$, KAB, δώστε ἴσην εὶναι τὴν μὲν ὑπὸ $BAΘ$ γωνίαν τῇ ὑπὸ $EΓZ$, τὴν δὲ ὑπὸ BAK τῇ ὑπὸ $EΓH$, τὴν δὲ ὑπὸ $KAΘ$ τῇ ὑπὸ $HΓZ$· καὶ γεγονέτω ὡς μὲν ἡ $EΓ$ πρὸς τὴν $ΓH$, οὑτως ἡ BA πρὸς τὴν AK, ὡς δὲ ἡ $HΓ$ πρὸς τὴν $ΓZ$, οὑτως ἡ KA πρὸς τὴν $AΘ$. καὶ διὸ ὁίσου ἄρα ἐστὶν ὡς ἡ $EΓ$ πρὸς τὴν $ΓZ$, οὑτως ἡ BA πρὸς τὴν $AΘ$. καὶ συμπεπληρώσθω τὸ $ΘB$ παραλληλόγραμμον καὶ τὸ $AΛ$ στερεόν.









Καὶ ἐπεί ἐστιν ὡς ἡ $EΓ$ πρὸς τὴν $ΓH$, οὑτως ἡ BA πρὸς τὴν AK, καὶ περὶ ἴσας γωνίας τὰξ ὑπὸ $EΓH$, BAK αἱ πλευραὶ ἀνάλογόν εἰσιν, ὅμοιον ἄρα ἐστὶ τὸ HE παραλληλόγραμμον τῷ KB παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν $KΘ$ παραλληλόγραμμον τῷ HZ παραλληλογράμμῳ ὅμοιόν ἐστι καὶ ὀέτι τὸ ZE τῷ $ΘB$· τρία ἄρα παραλληλόγραμμα τοῦ $ΓΔ$ στερεοῦ τρισὶ παραλληλογράμμοις τοῦ $AΛ$ στερεοῦ ὅμοιά ἐστιν. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖς ἀπεναντίον ἴσα τέ ἐστι καὶ ὅμοια,

And since as EC is to CG, so BA (is) to AK, and the sides about the equal angles ECG and BAK are (thus) proportional, the parallelogram GE is thus similar to the parallelogram KB. So, for the same (reasons), the parallelogram KH is also similar to the parallelogram GF, and, further, FE (is similar) to HB. Thus, three of the parallelograms of solid CD are similar to three of the parallelograms of solid AL. But, the (former) three are equal and similar to the three opposite, and the (latter) three are equal and similar to the three opposite. Thus, the

τὰ δὲ τρία τρισὶ τοῖς ἀπεναντίον ἴσα τέ ἐστι καὶ ὅμοια· ὅλον ἄρα τὸ ΓΔ στερεὸν ὅλῳ τῷ ΑΛ στερεῷ ὅμοιόν ἐστιν.

Ἀπὸ τῆς δοθείσης ἄρα εὐθείας τῆς ΑΒ τῷ δοθέντι στερεῷ παραλληλεπιπέδῳ τῷ ΓΔ ὁμοιόν τε καὶ ὁμοίως κείμενον ἀναγέγραπται τὸ ΑΛ· ὅπερ ἔδει ποιῆσαι.

whole solid CD is similar to the whole solid AL [Def. 11.9].

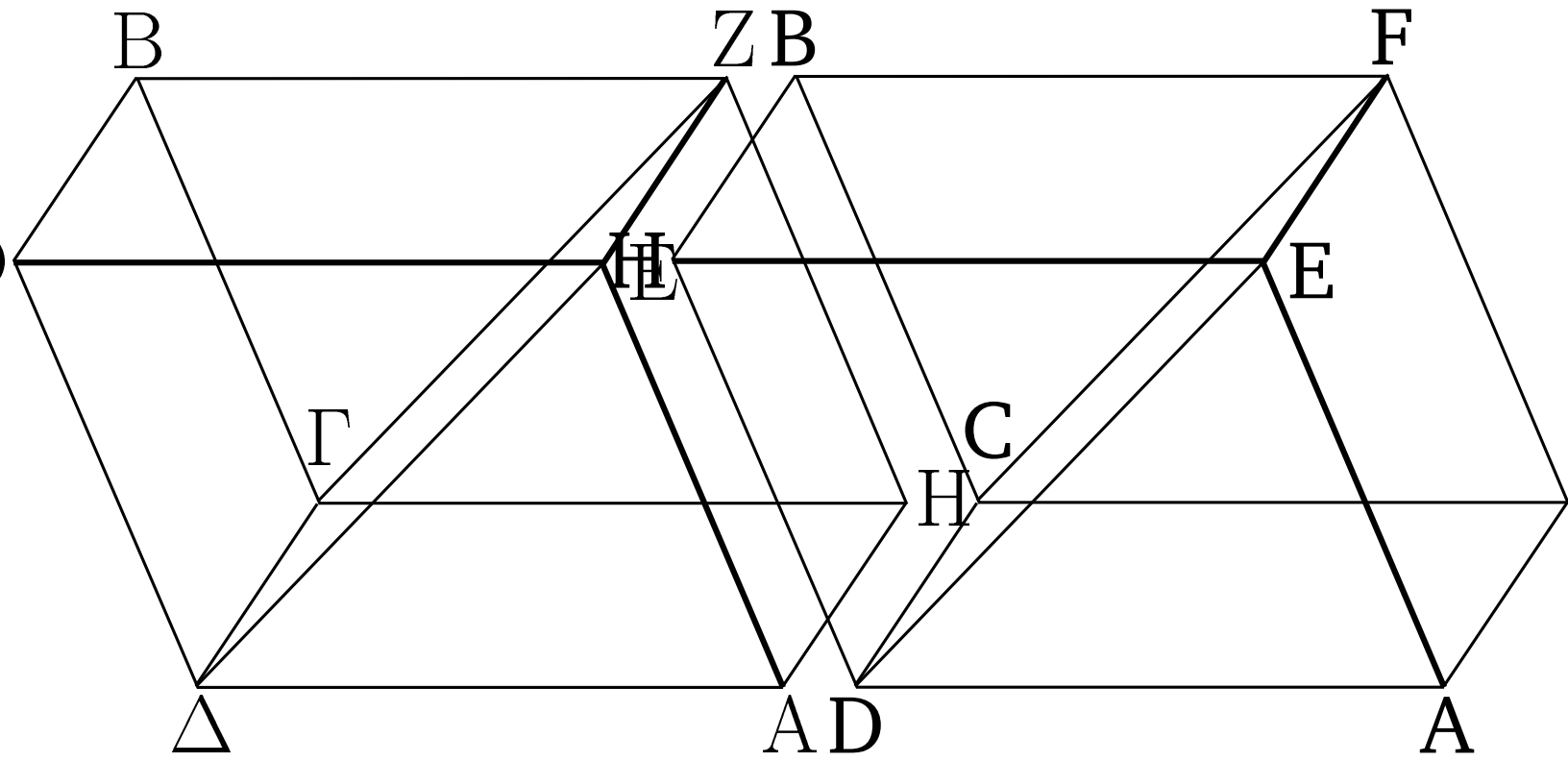
Thus, AL, similar, and similarly laid out, to the given parallelepiped solid CD, has been described on the given straight-lines AB. (Which is) the very thing it was required to do.

28.

Ἐὰν στερεὸν παραλληλεπίπεδον ἐπιπέδῳ τμ ηθῇ κατὰ τὰξ διαγωνίους τῶν ἀπεναντίον ἐπιπέδων, δίθα τμ ηθήσεται τὸ στερεὸν ὑπὸ τοῦ ἐπιπέδου.

Proposition 28

If a parallelepiped solid is cut by a plane (passing) through the diagonals of (a pair of) opposite planes then the solid will be cut in half by the plane.



Στερεὸν γὰρ παραλληλεπίπεδον τὸ ΑΒ ἐπιπέδῳ τῷ ΓΔΕΖ τετμήσθω κατὰ τὰξ διαγωνίους τῶν ἀπεναντίον ἐπιπέδων τὰξ ΓΖ, ΔΕ· λέγω, ὅτι δίθα τμ ηθήσεται τὸ ΑΒ στερεὸν ὑπὸ τοῦ ΓΔΕΖ ἐπιπέδου.

Ἐπεὶ γὰρ ἴσον ἐστὶ τὸ μὲν ΓΗΖ τρίγωνον τῷ ΓΖΒ τριγώνῳ, τὸ δὲ ΑΔΕ τῷ ΔΕΘ, ὀέστι δὲ καὶ τὸ μὲν ΓΑ παραλληλόγραμμον τῷ ΕΒ ἴσον· ἀπεναντίον γάρ· τὸ δὲ ΗΕ τῷ ΓΘ, καὶ τὸ πρίσμα ἄρα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν ΓΗΖ, ΑΔΕ, τριῶν δὲ παραλληλογράμμων τῶν ΗΕ, ΑΓ, ΓΕ ἴσον ἐστὶ τῷ πρίσματι τῷ περιεθομένῳ ὑπὸ δύο μὲν τριγώνων τῶν ΓΖΒ, ΔΕΘ, τριῶν δὲ παραλληλογράμμων τῶν ΓΘ, ΒΕ, ΓΕ· ὑπὸ γὰρὸίσων ἐπιπέδων περιέθονται τῷ τε πλήθει καὶ τῷ μεγέθει. ὁώστε ὅλον τὸ ΑΒ στερεὸν δίθα τέτμ ηται ὑπὸ τοῦ ΓΔΕΖ ἐπιπέδου· ὅπερ ἔδει δεῖχαι.

For let the parallelepiped solid AB be cut by the plane $CDEF$ (passing) through the diagonals of the opposite planes CF and DE.[†] I say that the solid AB will be cut in half by the plane $CDEF$.

For since triangle CGF is equal to triangle CFB, and ADE (is equal) to DEH [Prop. 1.34], and parallelogram CA is also equal to EB—for (they are) opposite [Prop. 11.24]—and GE (equal) to CH, thus the prism contained by the two triangles CGF and ADE, and the three parallelograms GE, AC, and CE, is also equal to the prism contained by the two triangles CFB and DEH, and the three parallelograms CH, BE, and CE. For they are contained by planes (which are) equal in number and in magnitude [Def. 11.10].[‡] Thus, the whole of solid AB is cut in half by the plane $CDEF$. (Which is) the very thing it was required to show.

† Here, it is assumed that the two diagonals lie in the same plane. The proof is easily supplied.

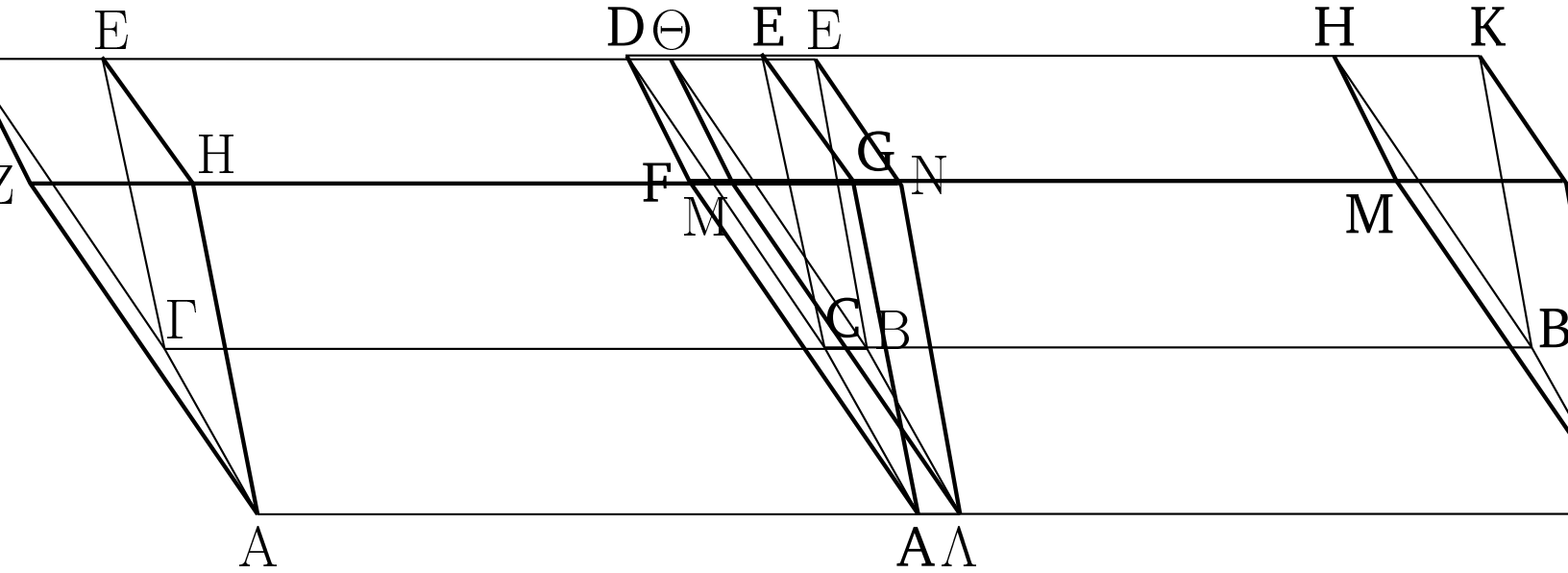
‡ However, strictly speaking, the prisms are not similarly arranged, being mirror images of one another.

29.

Τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν, ἴσα ἀλλήλοις ἐστίν.

Proposition 29

Parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up are on the same straight-lines, are equal to one another.



Ἔστω ἐπὶ τῆς αὐτῆς βάσεως τῆς AB στερεὰ παραλληλεπίπεδα τὰ $ΓΜ$, $ΓΝ$ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ AH, AZ, $ΛΜ$, $ΛΝ$, $ΓΔ$, $ΓΕ$, $ΒΘ$, $ΒΚ$ ἐπὶ τῶν αὐτῶν εὐθειῶν ὑέστωσαν τῶν ZN, $ΔΚ$· λέγω, ὅτι ἴσον ἐστὶ τὸ $ΓΜ$ στερεὸν τῷ $ΓΝ$ στερεῷ.

Ἐπεὶ γὰρ παραλληλόγραμμόν ἐστιν ἑκάτερον τῶν $ΓΘ$, $ΓΚ$, ὁίση ἐστὶν ἡ $ΓΒ$ ἑκατέρᾳ τῶν $ΔΘ$, $ΕΚ$· ὥστε καὶ ἡ $ΔΘ$ τῇ $ΕΚ$ ἐστινὁίση. κοινὴ ἀφῃρήσθω ἡ $ΕΘ$· λοιπὴ ἄρα ἡ $ΔΕ$ λοιπῇ τῇ $ΘΚ$ ἐστινὁίση. ὥστε καὶ τὸ μὲν $ΔΓΕ$ τρίγωνον τῷ $ΘΒΚ$ τριγώνῳ ἴσον ἐστίν, τὸ δὲ $ΔΗ$ παραλληλόγραμμον τῷ $ΘΝ$ παραλληογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $ΑΖΗ$ τρίγωνον τῷ $ΜΛΝ$ τριγώνῳ ἴσον ἐστίν. ὀέστι δὲ καὶ τὸ μὲν $ΓΖ$ παραλληλόγραμμον τῷ $ΒΜ$ παραλληλογράμμῳ ἴσον, τὸ δὲ $ΓΗ$ τῷ $ΒΝ$· ἀπεναντίον γάρ· καὶ τὸ πρίσμα ἄρα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν $ΑΖΗ$, $ΔΓΕ$, τριῶν δὲ παραλληλογράμμων τῶν $ΑΔ$, $ΔΗ$, $ΓΗ$ ἴσον ἐστὶ τῷ πρίσματι τῷ περιεθομένῳ ὑπὸ δύο μὲν τριγώνων τῶν $ΜΛΝ$, $ΘΒΚ$, τριῶν δὲ παραλληλογράμμων τῶν $ΒΜ$, $ΘΝ$, $ΒΝ$. κοινὸν προσκείσθω τὸ στερεόν, οὗ βάσις μὲν τὸ $ΑΒ$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $ΗΕΘΜ$· ὅλον ἄρα τὸ $ΓΜ$ στερεὸν παραλληλεπίπεδον ὅλῳ τῷ $ΓΝ$ στερεῷ παραλληλεπιπέδῳ ἴσον ἐστίν.

Τὰ ἄρα ἐπὶ τῆς αὐτῆς βάσεως ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι ἐπὶ τῶν

For let the parallelepiped solids CM and CN be on the same base AB, and (have) the same height, and let the (ends of the straight-lines) standing up in them, AG, AF, LM, LN, CD, CE, BH, and BK, be on the same straight-lines, FN and DK. I say that solid CM is equal to solid CN.

For since CH and CK are each parallelograms, CB is equal to each of DH and EK [Prop. 1.34]. Hence, DH is also equal to EK. Let EH be subtracted from both. Thus, the remainder DE is equal to the remainder HK. Hence, triangle DCE is also equal to triangle HBK [Props. 1.4, 1.8], and parallelogram DG to parallelogram HN [Prop. 1.36]. So, for the same (reasons), traingle AFG is also equal to triangle MLN. And parallelogram CF is also equal to parallelogram BM, and CG to BN [Prop. 11.24]. For they are opposite. Thus, the prism contained by the two triangles AFG and DCE, and the three parallelograms AD, DG, and CG, is equal to the prism contained by the two triangles MLN and HBK, and the three parallelograms BM, HN, and BN. Let the solid whose base (is) parallelogram AB, and (whose) opposite (face is) $GEHM$, be added to both (prisms). Thus, the whole parallelepiped solid CM is equal to the whole parallelepiped solid CN.

Thus, parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the

αὐτῶν εἰσιν εὐθειῶν, ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

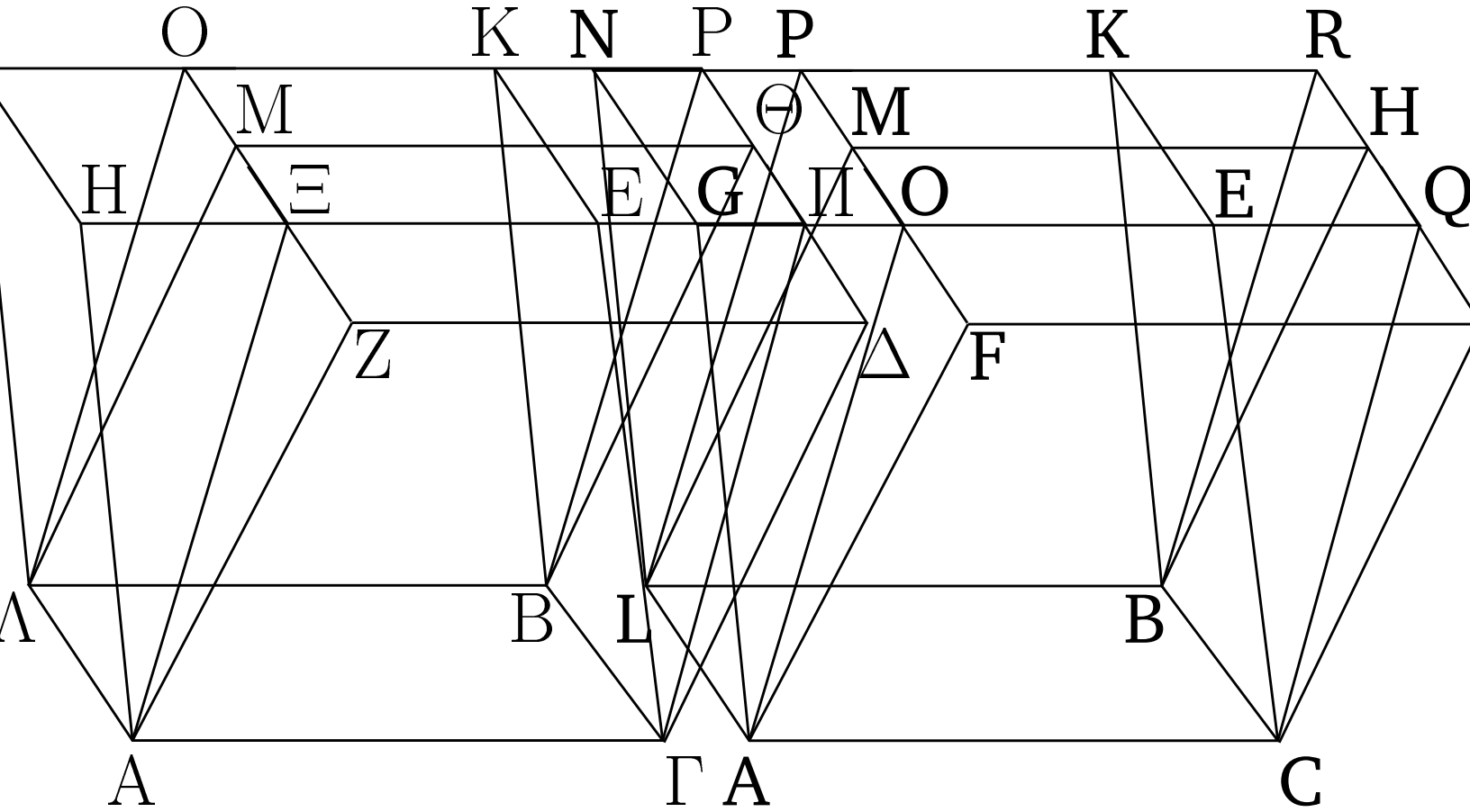
straight-lines) standing up (are) on the same straight-lines, are equal to one another. (Which is) the very thing it was required to show.

30.

Τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι οὐκ εἰσὶν ἐπὶ τῶν αὐτῶν εὐθειῶν, ἴσα ἀλλήλοις ἐστίν.

Proposition 30

Parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up are not on the same straight-lines, are equal to one another.



Ἔστω ἐπὶ τῆς αὐτῆς βάσεως τῆς ΑΒ στερεὰ παραλληλεπίπεδα τὰ ΓΜ, ΓΝ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ ΑΖ, ΑΗ, ΛΜ, ΛΝ, ΓΔ, ΓΕ, ΒΘ, ΒΚ μὴ ὀέστωσαν ἐπὶ τῶν αὐτῶν εὐθειῶν· λέγω, ὅτι ἴσον ἐστὶ τὸ ΓΜ στερεὸν τῷ ΓΝ στερεῷ.

Ἐκβεβλήσθωσαν γὰρ αἱ ΝΚ, ΔΘ καὶ συμπιπτέτωσαν ἀλλήλαις κατὰ τὸ Ρ, καὶ ὀέτι ἐκβεβλήσθωσαν αἱ ΖΜ, ΗΕ ἐπὶ τὰ Ο, Π, καὶ ἐπεζεύθθωσαν αἱ ΑΧ, ΛΟ, ΓΠ, ΒΡ. ἴσον δή ἐστι τὸ ΓΜ στερεόν, οὗ βάσις μὲν τὸ ΑΓΒΛ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ ΖΔΘΜ, τῷ ΓΟ στερεῷ, οὗ βάσις μὲν τὸ ΑΓΒΛ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ ΧΠΡΟ· ἐπί τε γὰρ τῆς αὐτῆς βάσεώξ εἰσι τῆς ΑΓΒΛ καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ ΑΖ, ΑΧ, ΛΜ, ΛΟ, ΓΔ, ΓΠ, ΒΘ, ΒΡ ἐπὶ τῶν

Let the parallelepiped solids CM and CN be on the same base, AB, and (have) the same height, and let the (ends of the straight-lines) standing up in them, AF, AG, LM, LN, CD, CE, BH, and BK, not be on the same straight-lines. I say that the solid CM is equal to the solid CN.

For let NK and DH be produced, and let them have joined one another at R. And, further, let FM and GE be produced to P and Q (respectively). And let AO, LP, CQ, and BR be joined. So, solid CM, whose base (is) parallelogram $ACBL$, and opposite (face) $FDHM$, is equal to solid CP, whose base (is) parallelogram $ACBL$, and opposite (face) $OQRP$. For they are on the same base, $ACBL$, and (have) the same height, and the (ends of the straight-lines) standing up in them, AF, AO,

αὐτῶν εἰσιν εὐθειῶν τῶν ΖΟ, ΔΡ. ἀλλὰ τὸ ΓΟ στερεόν, οὗ βάσις μέν ἐστι τὸ ΑΓΒΛ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ ΧΠΡΟ, ἴσον ἐστὶ τῷ ΓΝ στερεῷ, οὗ βάσις μὲν τὸ ΑΓΒΛ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ ΗΕΚΝ· ἐπί τε γὰρ πάλιν τῆς αὐτῆς βάσεώξ εἰσι τῆς ΑΓΒΛ καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ ΑΗ, ΑΧ, ΓΕ, ΓΠ, ΛΝ, ΛΟ, ΒΚ, ΒΡ ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν τῶν ΗΠ, ΝΡ. ὥστε καὶ τὸ ΓΜ στερεὸν ἴσον ἐστὶ τῷ ΓΝ στερεῷ.

Τὰ ἄρα ἐπὶ τῆς αὐτῆς βάσεως στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι οὐκ εἰσὶν ἐπὶ τῶν αὐτῶν εὐθειῶν, ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

LM, *LP*, *CD*, *CQ*, *BH*, and *BR*, are on the same straight-lines, *FP* and *DR* [Prop. 11.29]. But, solid *CP*, whose base is parallelogram *ACBL*, and opposite (face) *OQRP*, is equal to solid *CN*, whose base (is) parallelogram *ACBL*, and opposite (face) *GEKN*. For, again, they are on the same base, *ACBL*, and (have) the same height, and the (ends of the straight-lines) standing up in them, *AG*, *AO*, *CE*, *CQ*, *LN*, *LP*, *BK*, and *BR*, are on the same straight-lines, *GQ* and *NR* [Prop. 11.29]. Hence, solid *CM* is also equal to solid *CN*.

Thus, parallelepiped solids (which are) on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up are not on the same straight-lines, are equal to one another. (Which is) the very thing it was required to show.

31.

Τὰ ἐπὶ ὁίσων βάσεων ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοις ἐστίν.

Ἔστω ἐπὶ ὁίσων βάσεων τῶν ΑΒ, ΓΔ στερεὰ παραλληλεπίπεδα τὰ ΑΕ, ΓΖ ὑπὸ τὸ αὐτὸ ὕψος. λέγω, ὅτι ἴσον ἐστὶ τὸ ΑΕ στερεὸν τῷ ΓΖ στερεῷ.

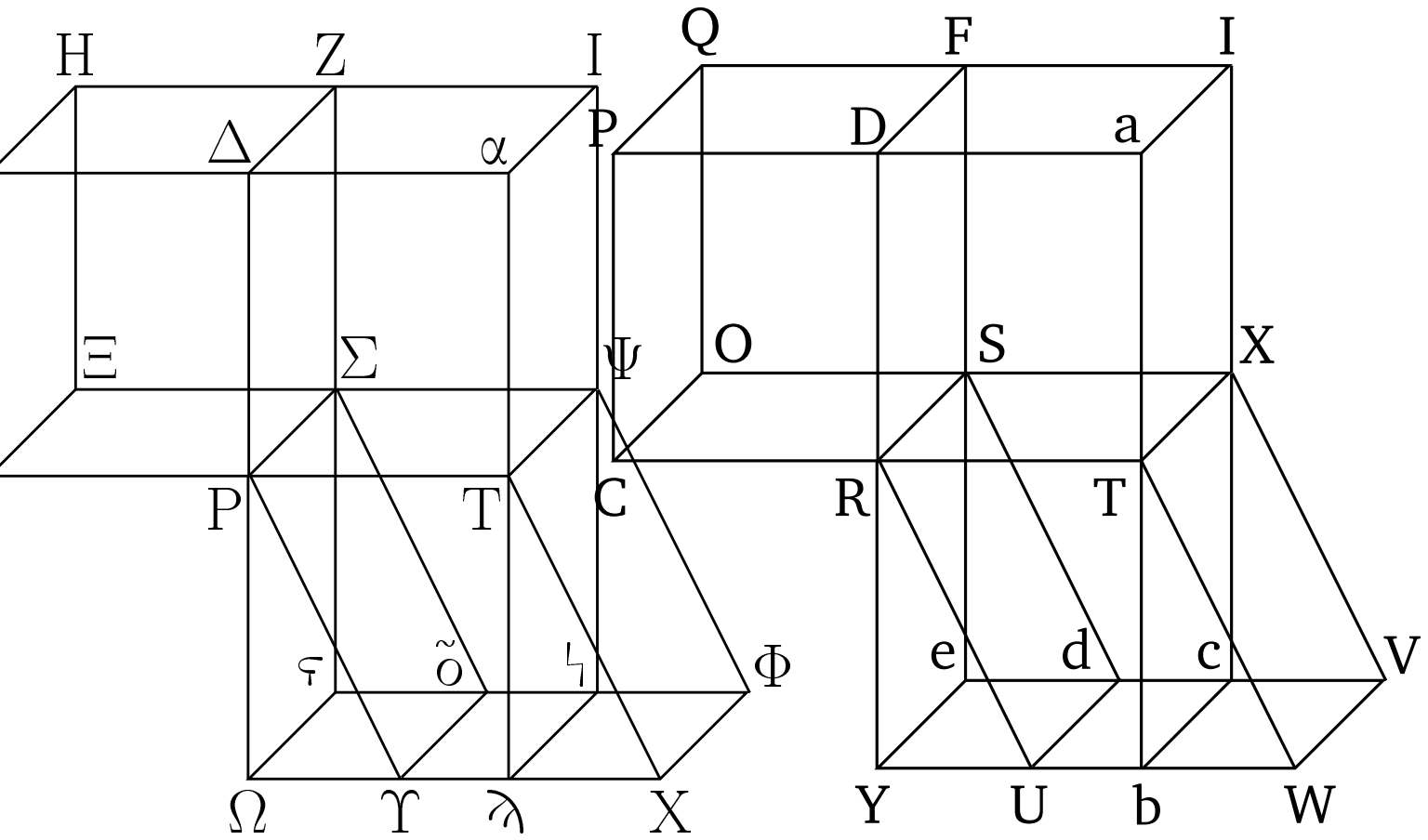
Ἔστωσαν δὴ πρότερον αἱ ἐφεστηκυῖαι αἱ ΘΚ, ΒΕ, ΑΗ, ΛΜ, ΟΠ, ΔΖ, ΓΧ, ΡΣ πρὸς ὀρθὰξ ταῖς ΑΒ, ΓΔ βάσεσιν, καὶ ἐκβεβλήσθω ἐπ΄ εὐθείας τῇ ΓΡ εὐθεῖα ἡ ΡΤ, καὶ συνεστάτω πρὸς τῇ ΡΤ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Ρ τῇ ὑπὸ ΑΛΒ γωνίᾳ ὁίση ἡ ὑπὸ ΤΡΥ, καὶ κείσθω τῇ μὲν ΑΛὁίση ἡ ΡΤ, τῇ δὲ ΛΒὁίση ἡ ΡΥ, καὶ συμπεπληρώσθω ἥ τε ΡΘ βάσις καὶ τὸ ΨΥ στερεόν.

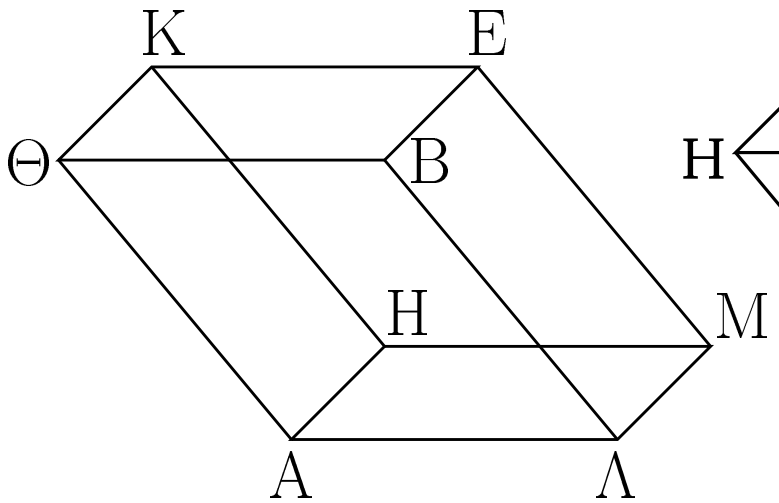
Proposition 31

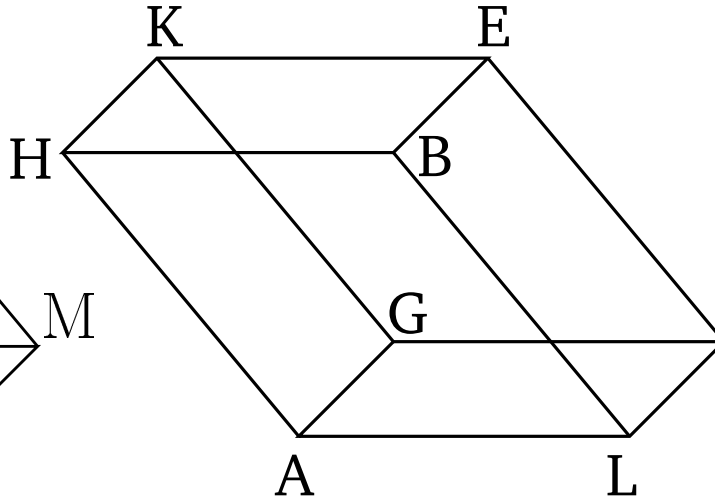
Parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another.

Let the parallelepiped solids *AE* and *CF* be on the equal bases *AB* and *CD* (respectively), and (have) the same height. I say that solid *AE* is equal to solid *CF*.

So, let the (straight-lines) standing up, *HK*, *BE*, *AG*, *LM*, *PQ*, *DF*, *CO*, and *RS*, first of all, be at right-angles to the bases *AB* and *CD*. And let *RT* be produced in a straight-line with *CR*. And let (angle) *TRU*, equal to angle *ALB*, be constructed on the straight-line *RT*, at the point *R* on it [Prop. 1.23]. And let *RT* be made equal to *AL*, and *RU* to *LB*. And let the base *RW*, and the solid *XU*, be completed.





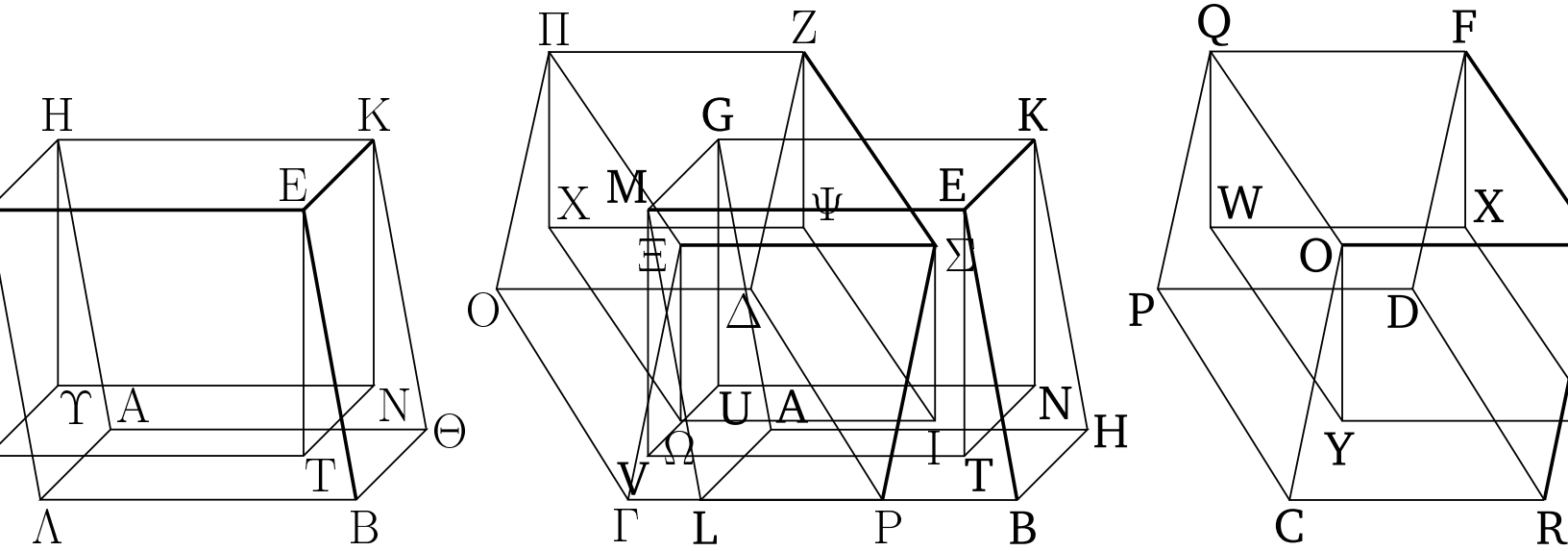


Καὶ ἐπεὶ δύο αἱ ΤΡ, ΡΥ δυσὶ ταῖς ΑΛ, ΛΒ ἴσαι εἰσίν, καὶ γωνίας ἴσας περιέθουσιν, ἴσον ἄρα καὶ ὅμοιον τὸ ΡΘ παραλληλόγραμμον τῷ ΘΛ παραλληλογράμμῳ. καὶ ἐπεὶ πάλινὁίση μὲν ἡ ΑΛ τῇ ΡΤ, ἡ δὲ ΛΜ τῇ ΡΣ, καὶ γωνίας ὀρθὰξ περιέθουσιν, ἴσον ἄρα καὶ ὅμοιόν ἐστι τὸ ΡΨ παραλληλόγραμμον

And since the two (straight-lines) *TR* and *RU* are equal to the two (straight-lines) *AL* and *LB* (respectively), and they contain equal angles, parallelogram *RW* is thus equal and similar to parallelogram *HL* [Prop. 6.14]. And, again, since *AL* is equal to *RT*, and *LM* to *RS*, and they contain right-angles,

τῷ AM παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΛE τῷ ΣY ἴσον τέ ἐστι καὶ ὅμοιον· τρία ἄρα παραλληλόγραμμα τοῦ AE στερεοῦ τρισὶ παραλληλογράμμοις τοῦ ΨY στερεοῦ ἴσα τέ ἐστι καὶ ὅμοια. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖς ἀπεναντίον ἴσα τέ ἐστι καὶ ὅμοια, τὰ δὲ τρία τρισὶ τοῖς ἀπεναντίον· ὅλον ἄρα τὸ AE στερεὸν παραλληλεπίπεδον ὅλῳ τῷ ΨY στερεῷ παραλληλεπιπέδῳ ἴσον ἐστίν. διήθθωσαν αἱ ΔP, ΘY καὶ συμπιπτέτωσαν ἀλλήλαις κατὰ τὸ Ω, καὶ διὰ τοῦ T τῇ $\Delta\Omega$ παράλληλος ἤθθω ἡ aT, καὶ ἐκβεβλήσθω ἡ $O\Delta$ κατὰ τὸ a, καὶ συμπεπληρώσθω τὰ $\Omega\Psi$, PI στερεά. ἴσον δή ἐστι τὸ $\Psi\Omega$ στερεόν, οὗ βάσις μέν ἐστι τὸ $P\Psi$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ Ω, τῷ ΨY στερεῷ, οὗ βάσις μὲν τὸ $P\Psi$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $Y\Phi$· ἐπί τε γὰρ τῆς αὐτῆς βάσεώξ εἰσι τῆς $P\Psi$ καὶ ὑπὸ τὸ αὐτὸ ὑψος, ὧν αἱ ἐφεστῶσαι αἱ $P\Omega$, PY, T, $T\Theta$, Σ, Σo, $\Psi\rho$, $\Psi\Phi$ ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν τῶν ΩX, $\varsigma\Phi$. ἀλλὰ τὸ ΨY στερεὸν τῷ AE ἐστιν ἴσον· καὶ τὸ $\Psi\Omega$ ἄρα στερεὸν τῷ AE στερεῷ ἐστιν ἴσον. καὶ ἐπεὶ ἴσον ἐστὶ τὸ $PYXT$ παραλληλόγραμμον τῷ ΩT παραλληλογράμμῳ· ἐπί τε γὰρ τῆς αὐτῆς βάσεώς εἰσι τῆς PT καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς PT, ΩX· ἀλλὰ τὸ $PYXT$ τῷ $\Gamma\Delta$ ἐστιν ἴσον, ἐπεὶ καὶ τῷ AB, καὶ τὸ ΩT ἄρα παραλληλόγραμμον τῷ $\Gamma\Delta$ ἐστιν ἴσον. ἄλλο δὲ τὸ ΔT· ἔστιν ἄρα ὡς ἡ $\Gamma\Delta$ βάσις πρὸς τὴν ΔT, οὕτως ἡ ΩT πρὸς τὴν ΔT. καὶ ἐπεὶ στερεὸν παραλληλεπίπεδον τὸ ΓI ἐπιπέδῳ τῷ PZ τέτμηται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἔστιν ὡς ἡ $\Gamma\Delta$ βάσις πρὸς τὴν ΔT βάσιν, οὕτως τὸ ΓZ στερεὸν πρὸς τὸ PI στερεόν. διὰ τὰ αὐτὰ δή, ἐπεὶ στερεὸν παραλληλεπίπεδον τὸ ΩI ἐπιπέδῳ τῷ $P\Psi$ τέτμηται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἔστιν ὡς ἡ ΩT βάσις πρὸς τὴν $T\Lambda$ βάσιν, οὕτως τὸ $\Omega\Psi$ στερεὸν πρὸς τὸ PI. ἀλλ᾽ ὡς ἡ $\Gamma\Delta$ βάσις πρὸς τὴν ΔT, οὕτως ἡ ΩT πρὸς τὴν ΔT· καὶ ὡς ἄρα τὸ ΓZ στερεὸν πρὸς τὸ PI στερεόν, οὕτως τὸ $\Omega\Psi$ στερεὸν πρὸς τὸ PI. ἑκάτερον ἄρα τῶν ΓZ, $\Omega\Psi$ στερεῶν πρὸς τὸ PI τὸν αὐτὸν ἔχει λόγον· ἴσον ἄρα ἐστὶ τὸ ΓZ στερεὸν τῷ $\Omega\Psi$ στερεῷ. ἀλλὰ τὸ $\Omega\Psi$ τῷ AE ἐδείχθη ἴσον· καὶ τὸ AE ἄρα τῷ ΓZ ἐστιν ἴσον.

parallelogram RX is thus equal and similar to parallelogram AM [Prop. 6.14]. So, for the same (reasons), LE is also equal and similar to SU. Thus, three parallelograms of solid AE are equal and similar to three parallelograms of solid XU. But, the three (faces of the former solid) are equal and similar to the three opposite (faces), and the three (faces of the latter solid) to the three opposite (faces) [Prop. 11.24]. Thus, the whole parallelepiped solid AE is equal to the whole parallelepiped solid XU [Def. 11.10]. Let DR and WU be drawn across, and let them have met one another at Y. And let aTb be drawn through T parallel to DY. And let PD be produced to a. And let the solids YX and RI be completed. So, solid XY, whose base is parallelogram RX, and opposite (face) Yc, is equal to solid XU, whose base (is) parallelogram RX, and opposite (face) UV. For they are on the same base RX, and (have) the same height, and the (ends of the straight-lines) standing up in them, RY, RU, Tb, TW, Se, Sd, Xc and XV, are on the same straight-lines, YW and eV [Prop. 11.29]. But, solid XU is equal to AE. Thus, solid XY is also equal to solid AE. And since parallelogram $RUWT$ is equal to parallelogram YT. For they are on the same base RT, and between the same parallels RT and YW [Prop. 1.35]. But, $RUWT$ is equal to CD, since (it is) also (equal) to AB. Parallelogram YT is thus also equal to CD. And DT is another (parallelogram). Thus, as base CD is to DT, so YT (is) to DT [Prop. 5.7]. And since the parallelepiped solid CI has been cut by the plane RF, which is parallel to the opposite planes (of CI), as base CD is to base DT, so solid CF (is) to solid RI [Prop. 11.25]. So, for the same (reasons), since the parallelepiped solid YI has been cut by the plane RX, which is parallel to the opposite planes (of YI), as base YT is to base TD, so solid YX (is) to solid RI [Prop. 11.25]. But, as base CD (is) to DT, so YT (is) to DT. And, thus, as solid CF (is) to solid RI, so solid YX (is) to solid RI. Thus, solids CF and YX each have the same ratio to RI [Prop. 5.11]. Thus, solid CF is equal to solid YX [Prop. 5.9]. But, YX was show (to be) equal to AE. Thus, AE is also equal to CF.



μὴ ὀέστωσαν δὴ αἱ ἐφεστηκυῖαι αἱ ΑΗ, ΘΚ, ΒΕ, ΛΜ, ΓΧ, ΟΠ, ΔΖ, ΡΣ πρὸς ὀρθὰξ ταῖς ΑΒ, ΓΔ βάσεσιν· λέγω πάλιν, ὅτι ὅισον τὸ ΑΕ στερεὸν τῷ ΓΖ στερεῷ. ἤθθωσαν γὰρ ἀπὸ τῶν Κ, Ε, Η, Μ, Π, Ζ, Χ, Σ σημείων ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετοι αἱ ΚΝ, ΕΤ, ΗΥ, ΜΦ, ΠΘ, ΖΨ, ΧΩ, ΣΙ, καὶ συμβαλλέτωσαν τῷ ἐπιπέδῳ κατὰ τὰ Ν, Τ, Υ, Φ, Θ, Ψ, Ω, Ι σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΝΤ, ΝΥ, ΥΦ, ΤΦ, ΘΨ, ΘΩ, ΩΙ, ΙΨ. ἴσον δή ἐστι τὸ ΚΦ στερεὸν τῷ ΠΙ στερεῷ· ἐπί τε γὰρὸίσων βάσεών εἰσι τῶν ΚΜ, ΠΣ καὶ ὑπὸ τὸ αὐτὸ ὁύψος, ὁῶν αἱ ἐφεστῶσαι πρὸς ὀρθάξ εἰσι ταῖς βάσεσιν. ἀλλὰ τὸ μὲν ΚΦ στερεὸν τῷ ΑΕ στερεῷ ἐστιν ἴσον, τὸ δὲ ΠΙ τῷ ΓΖ· ἐπί τε γὰρ τῆς αὐτῆς βάσεώξ εἰσι καὶ ὑπὸ τὸ αὐτὸ ὁύψος, ὁῶν αἱ ἐφεστῶσαι οὐύκ εἰσιν ἐπὶ τῶν αὐτῶν εὐθειῶν. καὶ τὸ ΑΕ ἄρα στερεὸν τῷ ΓΖ στερεῷ ἐστιν ἴσον.

Τὰ ἄρα ἐπὶ ὁίσων βάσεων ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὁύψος ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖχαι.

And so let the (straight-lines) standing up, AG, HK, BE, LM, CO, PQ, DF, and RS, not be at right-angles to the bases AB and CD. Again, I say that solid AE (is) equal to solid CF. For let KN, ET, GU, MV, QW, FX, OY, and SI be drawn from points K, E, G, M, Q, F, O, and S (respectively) perpendicular to the reference plane (i.e., the plane of the bases AB and CD), and let them have met the plane at points N, T, U, V, W, X, Y, and I (respectively). And let NT, NU, UV, TV, WX, WY, YI, and IX be joined. So solid KV is equal to solid QI. For they are on the equal bases KM and QS, and (have) the same height, and the (straight-lines) standing up in them are at right-angles to their bases (see first part of proposition). But, solid KV is equal to solid AE, and QI to CF. For they are on the same base, and (have) the same height, and the (straight-lines) standing up in them are not on the same straight-lines [Prop. 11.30]. Thus, solid AE is also equal to solid CF.

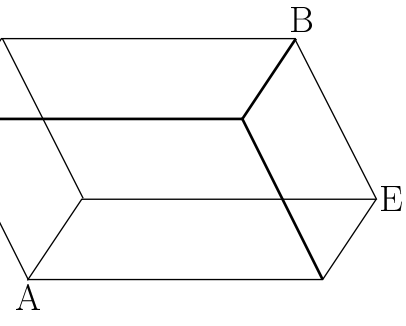
Thus, parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another. (Which is) the very thing it was required to show.

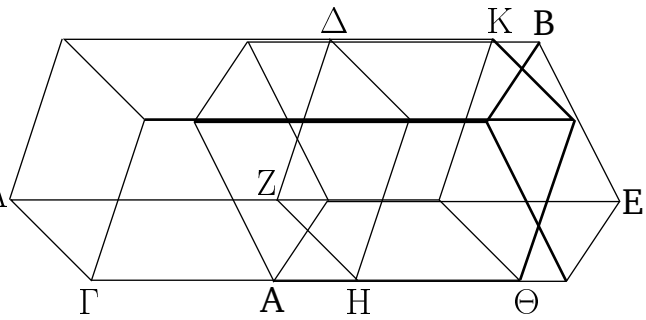
32.

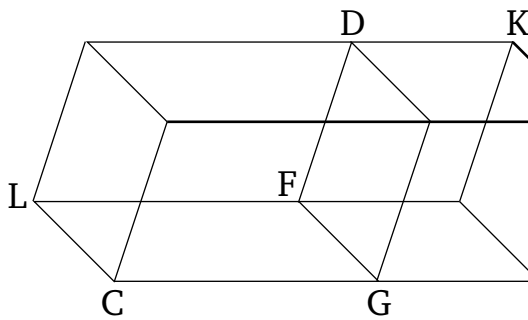
Τὰ ὑπὸ τὸ αὐτὸ ὁύψος ὄντα στερεὰ παραλληλεπίπεδα πρὸςὀάλληλά ἐστιν ὡξ αἱ βάσεις.

Proposition 32

Parallelepiped solids which (have) the same height are to one another as their bases.







Ἔστω ὑπὸ τὸ αὐτὸ ὕψος στερεὰ παραλληλεπίπεδα τὰ AB, ΓΔ· λέγω, ὅτι τὰ AB, ΓΔ στερεὰ παραλληλεπίπεδα πρὸς ἀλλήλα ἐστὶν ὡς αἱ βάσεις, τουτέστιν ὅτι ἐστὶν ὡς ἡ AE βάσις πρὸς τὴν ΓZ βάσιν, οὕτως τὸ AB στερεὸν πρὸς τὸ ΓΔ στερεόν.

Παραβεβλήσθω γὰρ παρὰ τὴν ZH τῷ AE ἴσον τὸ ZΘ, καὶ ἀπὸ βάσεως μὲν τῆς ZΘ, ὕψους δὲ τοῦ αὐτοῦ τῷ ΓΔ στερεὸν παραλληλεπίπεδον συμπεπληρώσθω τὸ HK. ἴσον δή ἐστι τὸ AB στερεὸν τῷ HK στερεῷ· ἐπί τε γὰρ ἴσων βάσεών εἰσι τῶν AE, ZΘ καὶ ὑπὸ τὸ αὐτὸ ὕψος. καὶ ἐπεὶ στερεὸν παραλληλεπίπεδον τὸ ΓK ἐπιπέδῳ τῷ ΔH τέτμηται παραλλήλῳ ὅντι τοῖς ἀπεναντίον ἐπιπέδοις, ὡς ἄρα ἡ βάσις ἡ ΓZ πρὸς τὴν ZΘ βάσιν, οὕτως τὸ ΓΔ στερεὸν πρὸς τὸ ΔΘ στερεόν. ἀλλ' ὡς μὲν ἡ ZΘ βάσις πρὸς τὴν AE βάσιν, οὕτως τὸ HK στερεὸν πρὸς τὸ AB στερεόν· ὡς ἄρα καὶ ἡ βάσις πρὸς τὴν ΓZ βάσιν, οὕτως τὸ AB στερεὸν πρὸς τὸ ΓΔ στερεόν.

Τὰ ἄρα ὑπὸ τὸ αὐτὸ ὕψος ὄντα στερεὰ παραλληλεπίπεδα πρὸς ἀλλήλα ἐστὶν ὡς αἱ βάσεις· ὅπερ ἔδει δεῖξαι.

Let *AB* and *CD* be parallelepiped solids (having) the same height. I say that the parallelepiped solids *AB* and *CD* are to one another as their bases. That is to say, as base *AE* is to base *CF*, so solid *AB* (is) to solid *CD*.

For let *FH*, equal to *AE*, be applied to *FG* (in the angle *FGH* equal to angle *LCG*) [Prop. 1.45]. And let the parallelepiped solid *GK*, (having) the same height as *CD*, be completed on the base *FH*. So solid *AB* is equal to solid *GK*. For they are on the equal bases *AE* and *FH*, and (have) the same height [Prop. 11.31]. And since the parallelepiped solid *CK* has been cut by the plane *DG*, which is parallel to the opposite planes (of *CK*), thus as the base *CF* is to the base *FH*, so the solid *CD* (is) to the solid *DH* [Prop. 11.25]. And base *FH* (is) equal to base *AE*, and solid *GK* to solid *AB*. And thus as base *AE* is to base *CF*, so solid *AB* (is) to solid *CD*.

Thus, parallelepiped solids which (have) the same height are to one another as their bases. (Which is) the very thing it was required to show.

33.

Τὰ ὅμοια στερεὰ παραλληλεπίπεδα πρὸς ἀλλήλα ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν.

Ἔστω ὅμοια στερεὰ παραλληλεπίπεδα τὰ AB, ΓΔ, ὁμόλογος δὲ ἐστω ἡ AE τῇ ΓZ· λέγω, ὅτι τὸ AB στερεὸν πρὸς τὸ ΓΔ στερεὸν τριπλασίονα λόγον ἔχει, ἤπερ ἡ AE πρὸς τὴν ΓZ.

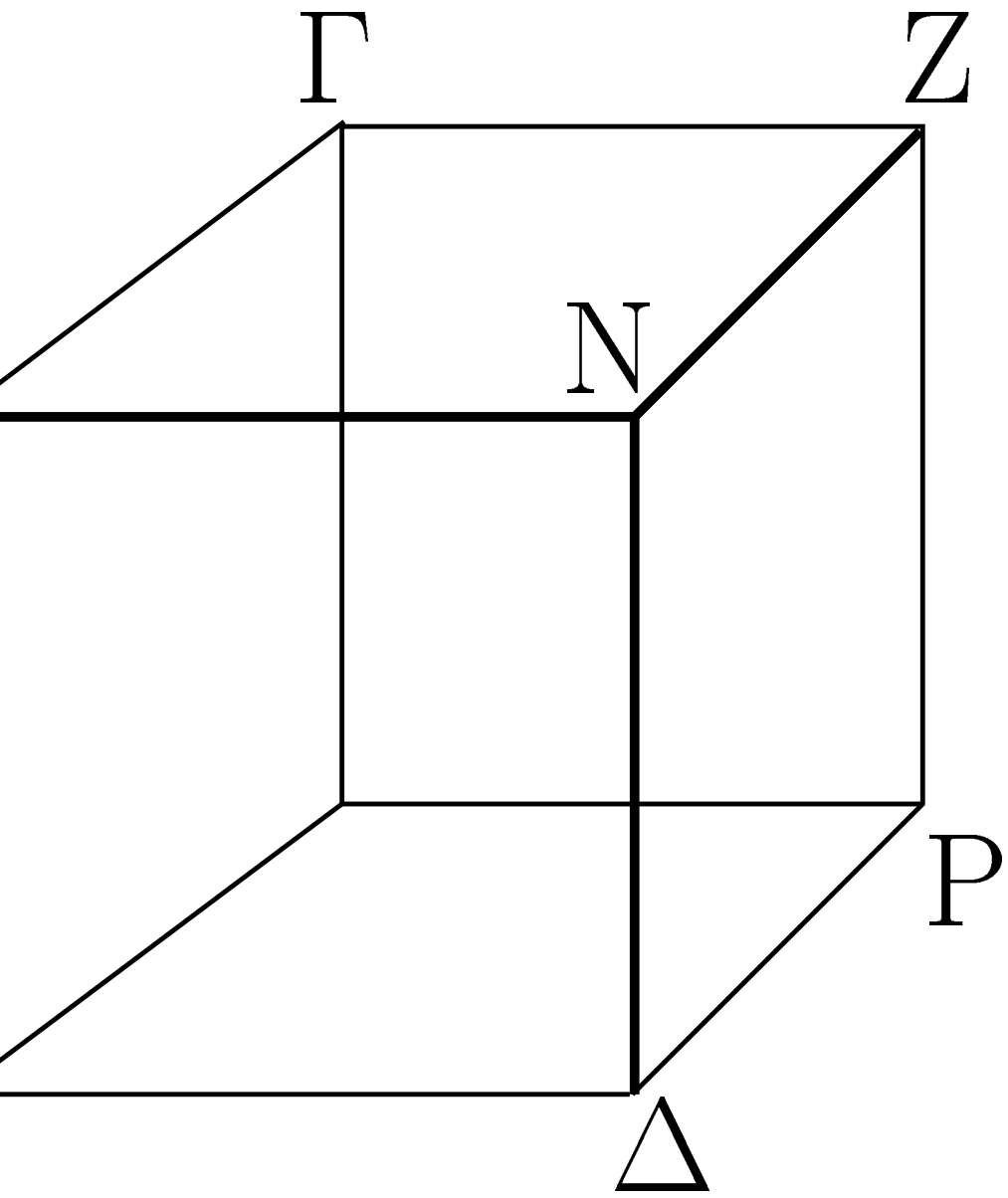
Ἐκβεβλήσθωσαν γὰρ ἐπ' εὐθείας ταῖς AE, HE, ΘE αἱ EK, EΛ, EM, καὶ κείσθω τῇ μὲν ΓZ ἴση ἡ EK, τῇ δὲ ZN ἴση ἡ EΛ, καὶ ἔτι τῇ ZP ἴση ἡ EM, καὶ συμπεπληρώσθω τὸ KΛ παραλληλόγραμμον καὶ τὸ KO στερεόν.

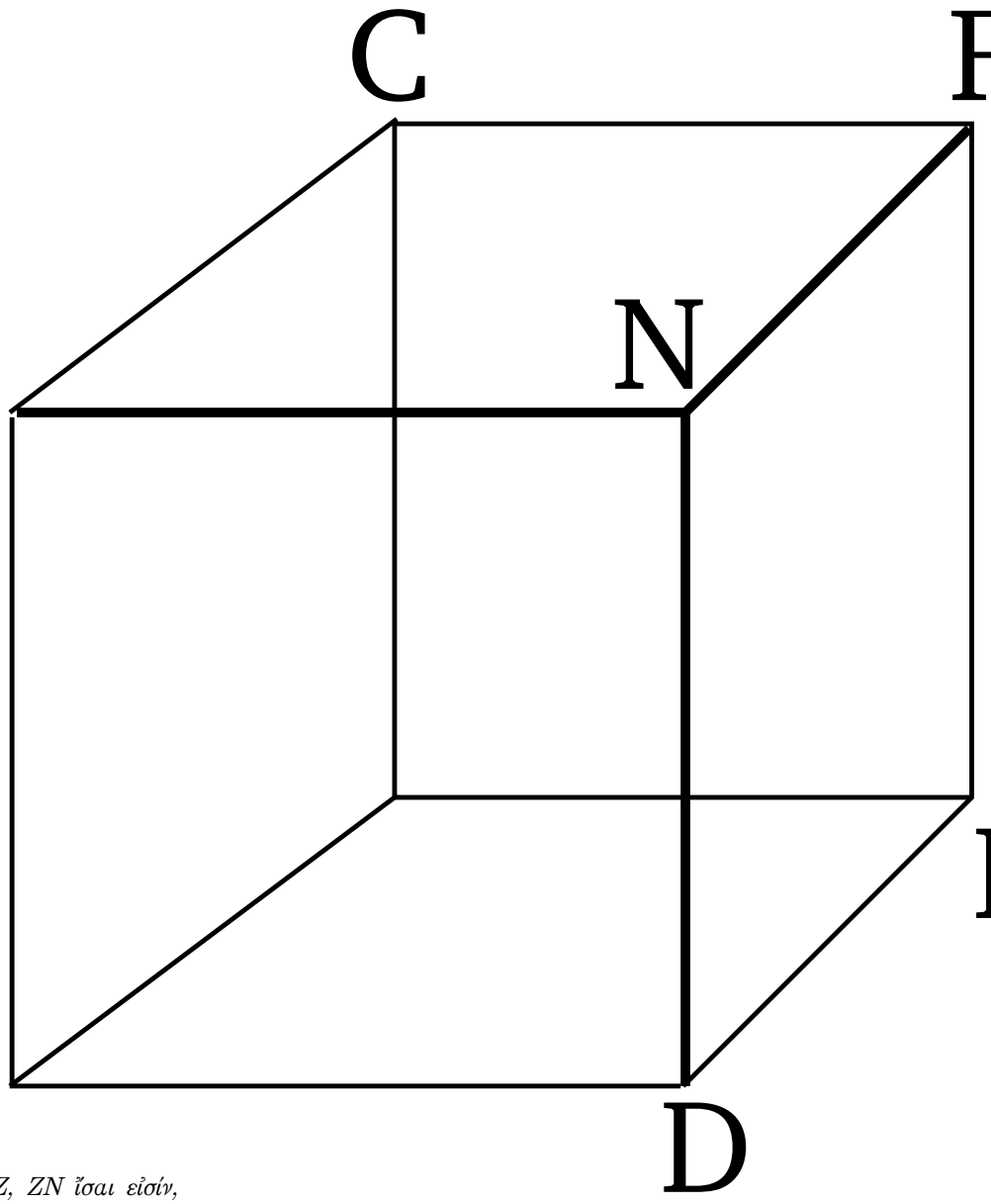
Proposition 33

Similar parallelepiped solids are to one another as the cubed ratio of their corresponding sides.

Let *AB* and *CD* be similar parallelepiped solids, and let *AE* correspond to *CF*. I say that solid *AB* has to solid *CD* the cubed ratio that *AE* (has) to *CF*.

For let *EK*, *EL*, and *EM* be produced in a straight-line with *AE*, *GE*, and *HE* (respectively). And let *EK* be made equal to *CF*, and *EL* equal to *FN*, and, further, *EM* equal to *FR*. And let the parallelogram *KL* be completed, and the solid *KP*.


Γ
Z
N
P
Δ

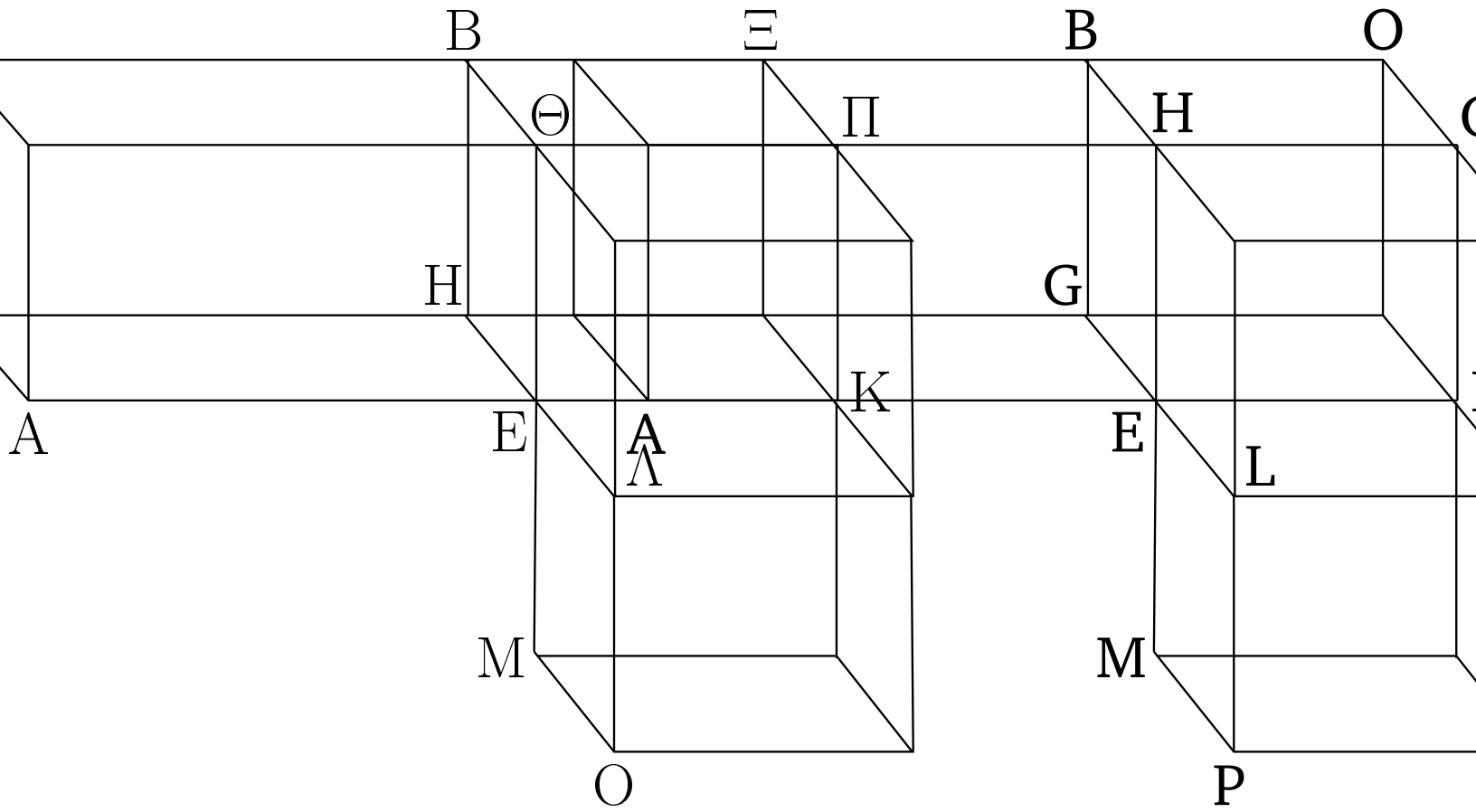


Καὶ ἐπεὶ δύο αἱ ΚΕ, ΕΛ δυσὶ ταῖς ΓΖ, ΖΝ ἴσαι εἰσίν, ἀλλὰ καὶ γωνία ἡ ὑπὸ ΚΕΛ γωνίᾳ τῇ ὑπὸ ΓΖΝ ἐστινὁίση, ἐπειδήπερ καὶ ἡ ὑπὸ ΑΕΗ τῇ ὑπὸ ΓΖΝ ἐστινὁίση διὰ τὴν ὁμοιότητα τῶν ΑΒ, ΓΔ στερεῶν, ἴσον ἄρα ἐστὶ [καὶ ὅμοιον] τὸ ΚΛ παραλληλόγραμμον τῷ ΓΝ παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν ΚΜ παραλληλόγραμμον ἴσον ἐστὶ καὶ ὅμοιον τῷ ΓΡ [παραλληλογράμμῳ] καὶ ὀέτι τὸ ΕΟ τῷ ΔΖ· τρία ἄρα παραλληλόγραμμα τοῦ ΚΟ στερεοῦ τρισὶ παραλληλογράμμοις τοῦ ΓΔ στερεοῦ ἴσα ἐστὶ καὶ ὅμοια. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖς ἀπεναντίον ἴσα ἐστὶ καὶ ὅμοια, τὰ δὲ τρία τρισὶ τοῖς ἀπεναντίον ἴσα ἐστὶ καὶ ὅμοια· ὅλον ἄρα τὸ ΚΟ στερεὸν ὅλῳ τῷ ΓΔ στερεῷ ἴσον ἐστὶ καὶ ὅμοιον. συμπεπληρώσθω τὸ ΗΚ παραλληλόγραμμον, καὶ ἀπὸ βάσεων μὲν τῶν ΗΚ, ΚΛ παραλληλόγραμμων, ὑψους δὲ τοῦ αὐτοῦ τῷ ΑΒ στερεὰ

And since the two (straight-lines) KE and EL are equal to the two (straight-lines) CF and FN, but angle KEL is also equal to angle CFN, inasmuch as AEG is also equal to CFN, on account of the similarity of the solids AB and CD, parallelogram KL is thus equal [and similar] to parallelogram CN. So, for the same (reasons), parallelogram KM is also equal and similar to [parallelogram] CR, and, further, EP to DF. Thus, three parallelograms of solid KP are equal and similar to three parallelograms of solid CD. But the three (former parallelograms) are equal and similar to the three opposite (parallelograms), and the three (latter parallelograms) are equal and similar to the three opposite (parallelograms) [Prop. 11.24]. Thus, the whole of solid KP is equal and sim-

συμπεπληρώσθω τὰ ΕΧ, ΛΠ. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν ΑΒ, ΓΔ στερεῶν ἐστιν ὡς ἡ ΑΕ πρὸς τὴν ΓΖ, οὕτως ἡ ΕΗ πρὸς τὴν ΖΝ, καὶ ἡ ΕΘ πρὸς τὴν ΖΡ, ὁίση δὲ ἡ μὲν ΓΖ τῇ ΕΚ, ἡ δὲ ΖΝ τῇ ΕΛ, ἡ δὲ ΖΡ τῇ ΕΜ, ὀέστιν ἄρα ὡξ ἡ ΑΕ πρὸς τὴν ΕΚ, οὕτως ἡ ΗΕ πρὸς τὴν ΕΛ καὶ ἡ ΘΕ πρὸς τὴν ΕΜ. ἀλλὸ ὡξ μὲν ἡ ΑΕ πρὸς τὴν ΕΚ, οὕτως τὸ ΑΗ [παραλληλόγραμμον] πρὸς τὸ ΗΚ παραλληλόγραμμον, ὡξ δὲ ἡ ΗΕ πρὸς τὴν ΕΛ, οὕτως τὸ ΗΚ πρὸς τὸ ΚΛ, ὡξ δὲ ἡ ΘΕ πρὸς ΕΜ, οὕτως τὸ ΠΕ πρὸς τὸ ΚΜ· καὶ ὡξ ἄρα τὸ ΑΗ παραλληλόγραμμον πρὸς τὸ ΗΚ, οὕτως τὸ ΗΚ πρὸς τὸ ΚΛ καὶ τὸ ΠΕ πρὸς τὸ ΚΜ. ἀλλὸ ὡξ μὲν τὸ ΑΗ πρὸς τὸ ΗΚ, οὕτως τὸ ΑΒ στερεὸν πρὸς τὸ ΕΧ στερεόν, ὡξ δὲ τὸ ΗΚ πρὸς τὸ ΚΛ, οὕτως τὸ ΧΕ στερεὸν πρὸς τὸ ΠΛ στερεόν, ὡξ δὲ τὸ ΠΕ πρὸς τὸ ΚΜ, οὕτως τὸ ΠΛ στερεὸν πρὸς τὸ ΚΟ στερεόν· καὶ ὡξ ἄρα τὸ ΑΒ στερεὸν πρὸς τὸ ΕΧ, οὕτως τὸ ΕΧ πρὸς τὸ ΠΛ καὶ τὸ ΠΛ πρὸς τὸ ΚΟ. ἒαν δὲ τέσσαρα μεγέθη κατὰ τὸ συνεθὲξ ἀνάλογονὸῇ, τὸ πρῶτον πρὸς τὸ τέταρτον τριπλασίονα λόγον ἔχει ἤπερ πρὸς τὸ δεύτερον· τὸ ΑΒ ἄρα στερεὸν πρὸς τὸ ΚΟ τριπλασίονα λόγον ἔχει ἤπερ τὸ ΑΒ πρὸς τὸ ΕΧ. ἀλλὸ ὡξ τὸ ΑΒ πρὸς τὸ ΕΧ, οὕτως τὸ ΑΗ παραλληλόγραμμον πρὸς τὸ ΗΚ καὶ ἡ ΑΕ εὐθεῖα πρὸς τὴν ΕΚ· ὡστε καὶ τὸ ΑΒ στερεὸν πρὸς τὸ ΚΟ τριπλασίονα λόγον ἔχει ἤπερ ἡ ΑΕ πρὸς τὴν ΕΚ. ἴσον δὲ τὸ [μὲν] ΚΟ στερεὸν τῷ ΓΔ στερεῷ, ἡ δὲ ΕΚ εὐθεῖα τῇ ΓΖ· καὶ τὸ ΑΒ ἄρα στερεὸν πρὸς τὸ ΓΔ στερεὸν τριπλασίονα λόγον ἔχει ἤπερ ἡ ὁμόλογος αὐτοῦ πλευρὰ ἡ ΑΕ πρὸς τὴν ὁμόλογον πλευρὰν τὴν ΓΖ.

ilar to the whole of solid CD [Def. 11.10]. Let parallelogram GK be completed. And let the the solids EO and LQ, with bases the parallelograms GK and KL (respectively), and with the same height as AB, be completed. And since, on account of the similarity of solids AB and CD, as AE is to CF, so EG (is) to FN, and EH to FR [Defs. 6.1, 11.9], and CF (is) equal to EK, and FN to EL, and FR to EM, thus as AE is to EK, so GE (is) to EL, and HE to EM. But, as AE (is) to EK, so [parallelogram] AG (is) to parallelogram GK, and as GE (is) to EL, so GK (is) to KL, and as HE (is) to EM, so QE (is) to KM [Prop. 6.1]. And thus as parallelogram AG (is) to GK, so GK (is) to KL, and QE (is) to KM. But, as AG (is) to GK, so solid AB (is) to solid EO, and as GK (is) to KL, so solid OE (is) to solid QL, and as QE (is) to KM, so solid QL (is) to solid KP [Prop. 11.32]. And, thus, as solid AB is to EO, so EO (is) to QL, and QL to KP. And if four magnitudes are continuously proportional then the first has to the fourth the cubed ratio that (it has) to the second [Def. 5.10]. Thus, solid AB has to KP the cubed ratio which AB (has) to EO. But, as AB (is) to EO, so parallelogram AG (is) to GK, and the straight-line AE to EK [Prop. 6.1]. Hence, solid AB also has to KP the cubed ratio that AE (has) to EK. And solid KP (is) equal to solid CD, and straight-line EK to CF. Thus, solid AB also has to solid CD the cubed ratio which its corresponding side AE (has) to the corresponding side CF.



Τὰ ἄρα ὅμοια στερεὰ παραλληλεπίπεδα ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν· ὅπερ ἔδει δεῖξαι.

Thus, similar parallelepiped solids are to one another as the cubed ratio of their corresponding sides. (Which is) the very thing it was required to show.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον ὦσιν, ὡς ἡ πρώτη πρὸς τὴν τετάρτην, οὕτω τὸ ἀπὸ τῆς πρώτης στερεὸν παραλληλεπίπεδον πρὸς τὸ ἀπὸ τῆς δευτέρας τὸ ὅμοιον καὶ ὁμοίως ἀναγραφόμενον, ἐπείπερ καὶ ἡ πρώτη πρὸς τὴν τετάρτην τριπλασίονα λόγον ἔχει ἤπερ πρὸς τὴν δευτέραν.

Corollary

So, (it is) clear, from this, that if four straight-lines are (continuously) proportional then as the first is to the fourth, so the parallelepiped solid on the first will be to the similar, and similarly described, parallelepiped solid on the second, since the first also has to the fourth the cubed ratio that (it has) to the second.

34.

Τῶν ἴσων στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν· καὶ ὧν στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, ἴσα ἐστὶν ἐκεῖνα.

Ἔστω ἴσα στερεὰ παραλληλεπίπεδα τὰ AB, $ΓΔ$· λέγω, ὅτι τῶν AB, $ΓΔ$ στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, καί ἐστιν ὡς ἡ $EΘ$ βάσις πρὸς τὴν $NΠ$ βάσιν, οὕτως τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος.

Ἔστωσαν γὰρ πρότερον αἱ ἐφεστηκυῖαι αἱ AH, EZ, $ΛB$,

Proposition 34†

The bases of equal parallelepiped solids are reciprocally proportional to their heights. And those parallelepiped solids whose bases are reciprocally proportional to their heights are equal.

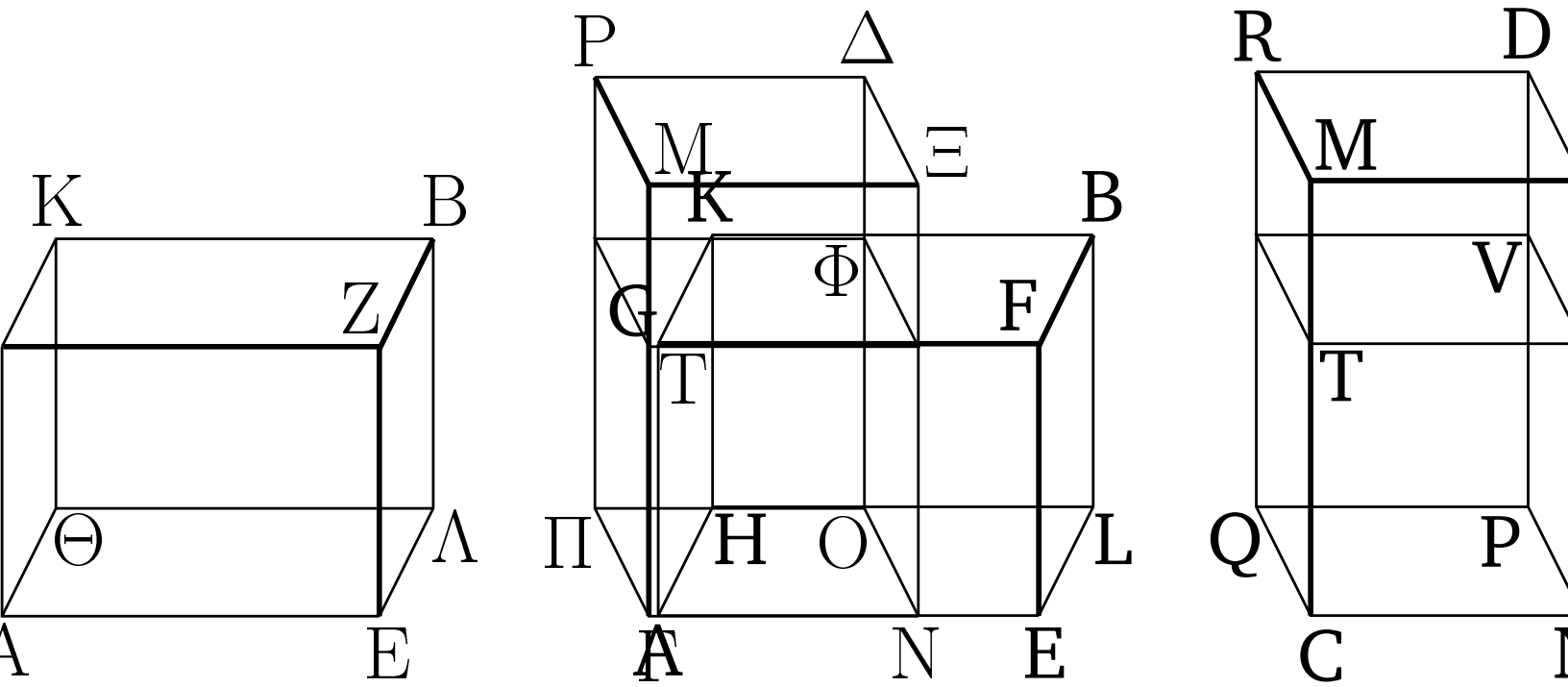
Let AB and CD be equal parallelepiped solids. I say that the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights, and (so) as base EH is to base NQ, so the height of solid CD (is) to the height of solid AB.

ΘΚ, ΓΜ, ΝΧ, ΟΔ, ΠΡ πρὸς ὀρθὰξ ταῖς βάσεσιν αὐτῶν· λέγω, ὅτι ἐστὶν ὡξ ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὕτως ἡ ΓΜ πρὸς τὴν ΑΗ.

Εἰ μὲν οὐῦνὀίση ἐστὶν ἡ ΕΘ βάσιν τῇ ΝΠ βάσει, ὀέστι δὲ καὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ ἴσον, ὀέσται καὶ ἡ ΓΜ τῇ ΑΗὀίση. τὰ γὰρ ὑπὸ τὸ αὐτὸ ὕψος στερεὰ παραλληλεπίπεδα πρὸςὀάλληλά ἐστιν ὡξ αἱ βάσεις. καὶ ὀέσται ὡξ ἡ ΕΘ βάσις πρὸς τὴν ΝΠ, οὕτως ἡ ΓΜ πρὸς τὴν ΑΗ, καὶ φανερόν, ὅτι τῶν ΑΒ, ΓΔ στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

For, first of all, let the (straight-lines) standing up, AG, EF, LB, HK, CM, NO, PD, and QR, be at right-angles to their bases. I say that as base EH is to base NQ, so CM (is) to AG.

Therefore, if base EH is equal to base NQ, and solid AB is also equal to solid CD, CM will also be equal to AG. For parallelepiped solids of the same height are to one another as their bases [Prop. 11.32]. And as base EH (is) to NQ, so CM will be to AG. And (so it is) clear that the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights.



μὴ ὀέστω δὴ ὀίση ἡ ΕΘ βάσις τῇ ΝΠ βάσει, ἀλλὸ ὀέστω μείζων ἡ ΕΘ. ὀέστι δὲ καὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ ἴσον· μείζων ἄρα ἐστὶ καὶ ἡ ΓΜ τῆς ΑΗ. κείσθω οὐῦν τῇ ΑΗὀίση ἡ ΓΤ, καὶ συμπεπληρώσθω ἀπὸ βάσεως μὲν τῆς ΝΠ, ὕψους δὲ τοῦ ΓΤ, στερεὸν παραλληλεπίπεδον τὸ ΦΓ. καὶ ἐπεὶ ἴσον ἐστὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ, ὀέχωθεν δὲ τὸ ΓΦ, τὰ δὲ ἴσα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔχει λόγον, ὀέστιν ἄρα ὡξ τὸ ΑΒ στερεὸν πρὸς τὸ ΓΦ στερεόν, οὕτως τὸ ΓΔ στερεὸν πρὸς τὸ ΓΦ στερεόν. ἀλλὸ ὡξ μὲν τὸ ΑΒ στερεὸν πρὸς τὸ ΓΦ στερεόν, οὕτως ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν· ἰσοϋψῆ γὰρ τὰ ΑΒ, ΓΦ στερεά· ὡξ δὲ τὸ ΓΔ στερεὸν πρὸς τὸ ΓΦ στερεόν, οὕτως ἡ ΜΠ βάσις πρὸς τὴν ΤΠ βάσιν καὶ ἡ ΓΜ πρὸς τὴν ΓΤ· καὶ ὡξ ἄρα ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὕτως ἡ ΜΓ πρὸς τὴν ΓΤ. ὀίση δὲ ἡ ΓΤ τῇ ΑΗ· καὶ ὡξ ἄρα ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὕτως ἡ ΜΓ πρὸς τὴν ΑΗ. τῶν ΑΒ, ΓΔ ἄρα στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

So let base EH not be equal to base NQ, but let EH be greater. And solid AB is also equal to solid CD. Thus, CM is also greater than AG. Therefore, let CT be made equal to AG. And let the parallelepiped solid VC be completed on the base NQ, with height CT. And since solid AB is equal to solid CD, and CV (is) extrinsic (to them), and equal (magnitudes) have the same ratio to the same (magnitude) [Prop. 5.7], thus as solid AB is to solid CV, so solid CD (is) to solid CV. But, as solid AB (is) to solid CV, so base EH (is) to base NQ. For the solids AB and CV (are) of equal height [Prop. 11.32]. And as solid CD (is) to solid CV, so base MQ (is) to base TQ [Prop. 11.25], and CM to CT [Prop. 6.1]. And, thus, as base EH is to base NQ, so MC (is) to AG. And CT (is) equal to AG. And thus as base EH (is) to base NQ, so MC (is) to AG. Thus, the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights.

Πάλιν δὴ τῶν ΑΒ, ΓΔ στερεῶν παραλληλεπιπέδων ἀντι-

So, again, let the bases of the parallelepipid solids AB and CD be reciprocally proportional to their heights, and let base

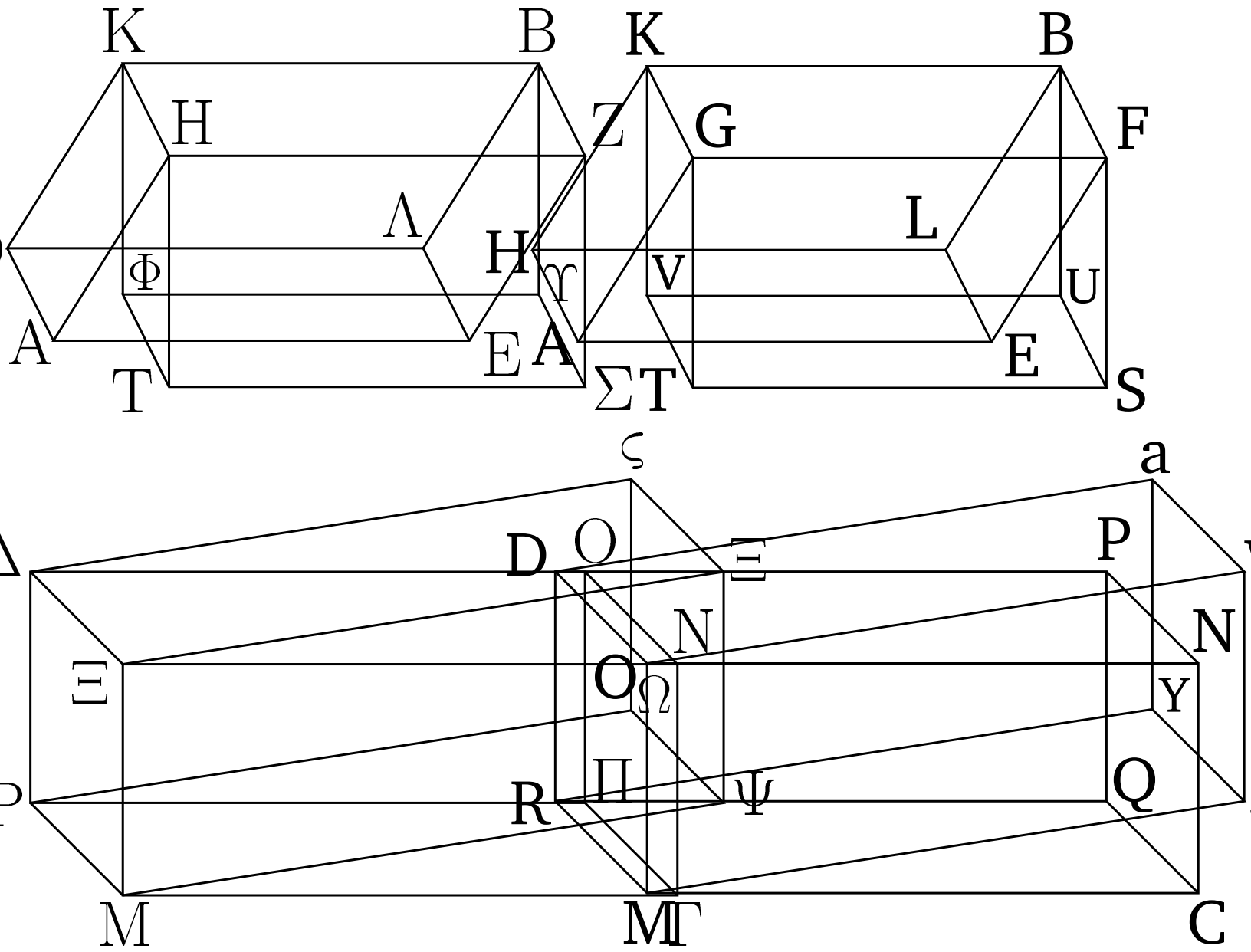
πεπονθέτωσαν αἱ βάσεις τοῖς ὑψεσιν, καὶ ὀέστω ὡς ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὑτως τὸ τοῦ ΓΔ στερεοῦ ὑψος πρὸς τὸ τοῦ ΑΒ στερεοῦ ὑψος· λέγω, ὅτι ἴσον ἐστὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ.

Ἔστωσαν [γὰρ] πάλιν αἱ ἐφεστηκυῖαι πρὸς ὀρθὰξ ταῖς βάσεσιν. καὶ εἰ μὲνὀίση ἐστὶν ἡ ΕΘ βάσις τῇ ΝΠ βάσει, καί ἐστιν ὡξ ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὑτως τὸ τοῦ ΓΔ στερεοῦ ὑψος πρὸς τὸ τοῦ ΑΒ στερεοῦ ὑψος, ἴσον ἄρα ἐστὶ καὶ τὸ τοῦ ΓΔ στερεοῦ ὑψος τῷ τοῦ ΑΒ στερεοῦ ὑψει. τὰ δὲ ἐπὶ ὁίσων βάσεων στερεά παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὑψος ἴσα ἀλλήλοις ἐστίν· ἴσον ἄρα ἐστὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ.

μὴ ὀέστω δὴ ἡ ΕΘ βάσις τῇ ΝΠ [βάσει] ὁίση, ἀλλὸ ὀέστω μείζων ἡ ΕΘ· μεῖζον ἄρα ἐστὶ καὶ τὸ τοῦ ΓΔ στερεοῦ ὑψος τοῦ τοῦ ΑΒ στερεοῦ ὑψους, τουτέστιν ἡ ΓΜ τῆς ΑΗ. κείσθω τῇ ΑΗὁίση πάλιν ἡ ΓΤ, καὶ συμπεπληρώσθω ὁμοίως τὸ ΓΦ στερεόν. ἐπεί ἐστιν ὡξ ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὑτως ἡ ΜΓ πρὸς τὴν ΑΗ, ὁίση δὲ ἡ ΑΗ τῇ ΓΤ, ὀέστιν ἄρα ὡξ ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὑτως ἡ ΓΜ πρὸς τὴν ΓΤ. ἀλλὸ ὡξ μὲν ἡ ΕΘ [βάσις] πρὸς τὴν ΝΠ βάσιν, οὑτως τὸ ΑΒ στερεὸν πρὸς τὸ ΓΦ στερεόν· ἰσοϋψῆ γάρ ἐστι τὰ ΑΒ, ΓΦ στερεά· ὡξ δὲ ἡ ΓΜ πρὸς τὴν ΓΤ, οὑτως ἥ τε ΜΠ βάσις πρὸς τὴν ΠΤ βάσιν καὶ τὸ ΓΔ στερεὸν πρὸς τὸ ΓΦ στερεόν. καὶ ὡξ ἄρα τὸ ΑΒ στερεὸν πρὸς τὸ ΓΦ στερεόν, οὑτως τὸ ΓΔ στερεὸν πρὸς τὸ ΓΦ στερεόν· ἑκάτερον ἄρα τῶν ΑΒ, ΓΔ πρὸς τὸ ΓΦ τὸν αὐτὸν ἔχει λόγον. ἴσον ἄρα ἐστὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ.

EH be to base NQ, as the height of solid CD (is) to the height of solid AB. I say that solid AB is equal to solid CD. [For] let the (straight-lines) standing up again be at right-angles to the bases. And if base EH is equal to base NQ, and as base EH is to base NQ, so the height of solid CD (is) to the height of solid AB, the height of solid CD is thus also equal to the height of solid AB. And parallelepiped solids on equal bases, and also with the same height, are equal to one another [Prop. 11.31]. Thus, solid AB is equal to solid CD.

So, let base EH not be equal to [base] NQ, but let EH be greater. Thus, the height of solid CD is also greater than the height of solid AB, that is to say CM (greater) than AG. Let CT again be made equal to AG, and let the solid CV be similarly completed. Since as base EH is to base NQ, so MC (is) to AG, and AG (is) equal to CT, thus as base EH (is) to base NQ, so CM (is) to CT. But, as [base] EH (is) to base NQ, so solid AB (is) to solid CV. For solids AB and CV are of equal heights [Prop. 11.32]. And as CM (is) to CT, so (is) base MQ to base QT [Prop. 6.1], and solid CD to solid CV [Prop. 11.25]. And thus as solid AB (is) to solid CV, so solid CD (is) to solid CV. Thus, AB and CD each have the same ratio to CV. Thus, solid AB is equal to solid CD [Prop. 5.9].



μὴ ὀέστωσαν δὴ αἱ ἐφεστηκυῖαι αἱ ΖΕ, ΒΛ, ΗΑ, ΚΘ, ΧΝ, ΔΟ, ΜΓ, ΡΠ πρὸς ὀρθὰξ ταῖς βάσεσιν αὐτῶν, καὶ ἤθθωσαν ἀπὸ τῶν Ζ, Η, Β, Κ, Χ, Μ, Ρ, Δ σημείων ἐπὶ τὰ διὰ τῶν ΕΘ, ΝΠ ἐπίπεδα κάθετοι καὶ συμβαλλέτωσαν τοῖς ἐπιπέδοις κατὰ τὰ Σ, Τ, Υ, Φ, Θ, Ψ, Ω, ξ, καὶ συμπεπληρώσθω τὰ ΖΦ, ΧΩ στερεά· λέγω, ὅτι καὶ οὑτωςὁίσων ὄντων τῶν ΑΒ, ΓΔ στερεῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν, καί ἐστιν ὡξ ἡ ΕΘ βᾶσιν πρὸς τὴν ΝΠ βάσιν, οὑτως τὸ τοῦ ΓΔ στερεοῦ ὑψος πρὸς τὸ τοῦ ΑΒ στερεοῦ ὑψος.

Ἐπεὶ ἴσον ἐστὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ, ἀλλὰ τὸ μὲν ΑΒ τῷ ΒΤ ἐστιν ἴσον· ἐπί τε γὰρ τῆς αὐτῆς βάσεώξ εἰσι τῆς ΖΚ καὶ ὑπὸ τὸ αὐτὸ ὑψος· τὸ δὲ ΓΔ στερεὸν τῷ ΔΨ ἐστιν

So, let the (straight-lines) standing up, FE, BL, GA, KH, ON, DP, MC, and RQ, not be at right-angles to their bases. And let perpendiculars be drawn to the planes through EH and NQ from points F, G, B, K, O, M, R, and D, and let them have joined the planes at (points) S, T, U, V, W, X, Y, and a (respectively). And let the solids FV and OY be completed. In this case, also, I say that the solids AB and CD being equal, their bases are reciprocally proportional to their heights, and (so) as base EH is to base NQ, so the height of solid CD (is) to the height of solid AB.

Since solid AB is equal to solid CD, but AB is equal to BT. For they are on the same base FK, and (have) the same

ἴσον· ἐπί τε γὰρ πάλιν τῆς αὐτῆς βάσεώξ εἰσι τῆς PX καὶ ὑπὸ τὸ αὐτὸ ὕψος· καὶ τὸ BT ἄρα στερεὸν τῷ $\Delta\Psi$ στερεῷ ἴσον ἐστίν. ὀέστιν ἄρα ὡξ ἡ ZK βάσις πρὸς τὴν XP βάσιν, οὕτως τὸ τοῦ $\Delta\Psi$ στερεοῦ ὕψος πρὸς τὸ τοῦ BT στερεοῦ ὕψος. ὁίση δὲ ἡ μὲν ZK βάσις τῇ $E\Theta$ βάσει, ἡ δὲ XP βάσις τῇ $N\Pi$ βάσει· ὀέστιν ἄρα ὡξ ἡ $E\Theta$ βάσις πρὸς τὴν $N\Pi$ βάσιν, οὕτως τὸ τοῦ $\Delta\Psi$ στερεοῦ ὕψος πρὸς τὸ τοῦ BT στερεοῦ ὕψος. τὰ δὸ αὐτὰ ὕψη ἐστὶ τῶν $\Delta\Psi$, BT στερεῶν καὶ τῶν $\Delta\Gamma$, BA· ὀέστιν ἄρα ὡξ ἡ $E\Theta$ βάσις πρὸς τὴν $N\Pi$ βάσιν, οὕτως τὸ τοῦ $\Delta\Gamma$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος. τῶν AB, $\Gamma\Delta$ ἄρα στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

Πάλιν δὴ τῶν AB, $\Gamma\Delta$ στερεῶν παραλληλεπιπέδων ἀντιπεπονθέτωσαν αἱ βάσεις τοῖς ὕψεσιν, καὶ ὀέστω ὡξ ἡ $E\Theta$ βάσις πρὸς τὴν $N\Pi$ βάσιν, οὕτως τὸ τοῦ $\Gamma\Delta$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος· λέγω, ὅτι ἴσον ἐστὶ τὸ AB στερεὸν τῷ $\Gamma\Delta$ στερεῷ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεί ἐστιν ὡξ ἡ $E\Theta$ βάσις πρὸς τὴν $N\Pi$ βάσιν, οὕτως τὸ τοῦ $\Gamma\Delta$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος, ὁίση δὲ ἡ μὲν $E\Theta$ βάσις τῇ ZK βάσει, ἡ δὲ $N\Pi$ τῇ XP, ὀέστιν ἄρα ὡξ ἡ ZK βάσις πρὸς τὴν XP βάσιν, οὕτως τὸ τοῦ $\Gamma\Delta$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος. τὰ δὸ αὐτὰ ὕψη ἐστὶ τῶν AB, $\Gamma\Delta$ στερεῶν καὶ τῶν BT, $\Delta\Psi$· ὀέστιν ἄρα ὡξ ἡ ZK βάσις πρὸς τὴν XP βάσιν, οὕτως τὸ τοῦ $\Delta\Psi$ στερεοῦ ὕψος πρὸς τὸ τοῦ BT στερεοῦ ὕψος. τῶν BT, $\Delta\Psi$ ἄρα στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν· ἴσον ἄρα ἐστὶ τὸ BT στερεὸν τῷ $\Delta\Psi$ στερεῷ. ἀλλὰ τὸ μὲν BT τῷ BA ἴσον ἐστίν· ἐπί τε γὰρ τῆς αὐτῆς βάσεως [εἰσι] τῆς ZK καὶ ὑπὸ τὸ αὐτὸ ὕψος. τὸ δὲ $\Delta\Psi$ στερεὸν τῷ $\Delta\Gamma$ στερεῷ ἴσον ἐστίν. καὶ τὸ AB ἄρα στερεὸν τῷ $\Gamma\Delta$ στερεῷ ἐστιν ἴσον· ὅπερ ἔδει δεῖχαι.

height [Props. 11.29, 11.30]. And solid CD is equal is equal to DX. For, again, they are on the same base RO, and (have) the same height [Props. 11.29, 11.30]. Solid BT is thus also equal to solid DX. Thus, as base FK (is) to base OR, so the height of solid DX (is) to the height of solid BT (see first part of proposition). And base FK (is) equal to base EH, and base OR to NQ. Thus, as base EH is to base NQ, so the height of solid DX (is) to the height of solid BT. And solids DX, BT are the same height as (solids) DC, BA (respectively). Thus, as base EH is to base NQ, so the height of solid DC (is) to the height of solid AB. Thus, the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights.

So, again, let the bases of the parallelepiped solids AB and CD be reciprocally proportional to their heights, and (so) let base EH be to base NQ, as the height of solid CD (is) to the height of solid AB. I say that solid AB is equal to solid CD.

For, with the same construction (as before), since as base EH is to base NQ, so the height of solid CD (is) to the height of solid AB, and base EH (is) equal to base FK, and NQ to OR, thus as base FK is to base OR, so the height of solid CD (is) to the height of solid AB. And solids AB, CD are the same height as (solids) BT, DX (respectively). Thus, as base FK is to base OR, so the height of solid DX (is) to the height of solid BT. Thus, the bases of the parallelepiped solids BT and DX are reciprocally proportional to their heights. Thus, solid BT is equal to solid DX (see first part of proposition). But, BT is equal to BA. For [they are] on the same base FK, and (have) the same height [Props. 11.29, 11.30]. And solid DX is equal to solid DC [Props. 11.29, 11.30]. Thus, solid AB is also equal to solid CD. (Which is) the very thing it was required to show.

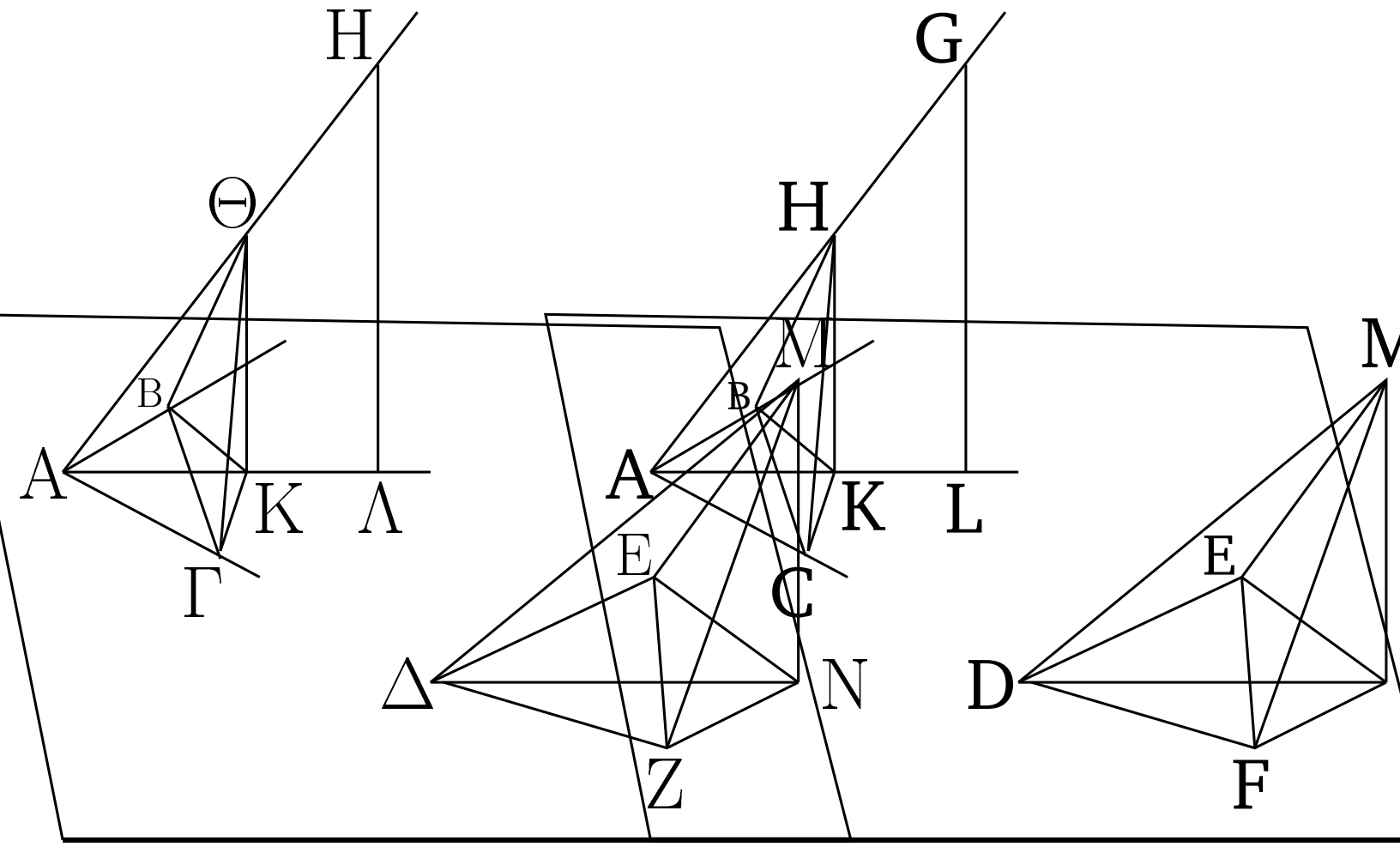
† This proposition assumes that (a) if two parallelepipeds are equal, and have equal bases, then their heights are equal, and (b) if the bases of two equal parallelepipeds are unequal, then that solid which has the lesser base has the greater height.

35.

Ἐὰνὸῶσι δύο γωνίαι ἐπίπεδοι ἴσαι, ἐπὶ δὲ τῶν κορυφῶν αὐτῶν μετέωροι εὐθεῖαι ἐπισταθῶσιν ἴσας γωνίας περιέθουσαι μετὰ τῶν ἐξ ἀρθῆς εὐθειῶν ἑκατέραν ἑκατέρᾳ, ἐπὶ δὲ τῶν μετεώρων ληφθῇ τυθόντα σημεῖα, καὶ ἀπὸ αὐτῶν ἐπὶ τὰ ἐπίπεδα, ἐν οὶῖξ εἰσιν αἱ ἐξ ἀρθῆς γωνίαι, κάθετοι ἀθθῶσιν, ἀπὸ δὲ τῶν γενομένων σημείων ἐν τοῖς ἐπιπέδοις ἐπὶ τὰξ ἐξ ἀρθῆς γωνίας ἐπιζευθθῶσιν εὐθεῖαι, ἴσας γωνίας περιέχουσι μετὰ τῶν μετεώρων.

Proposition 35

If there are two equal plane angles, and raised straight-lines are stood on the apexes of them, containing equal angles respectively with the original straight-lines (forming the angles), and random points are taken on the raised (straight-lines), and perpendiculars are drawn from them to the planes in which the original angles are, and straight-lines are joined from the points created in the planes to the (vertices of the) original angles, then they will enclose equal angles with the raised (straight-lines).



Ἔστωσαν δύο γωνίαι εὐθύγραμμοι ἴσαι αἱ ὑπὸ ΒΑΓ, ΕΔΖ, ἀπὸ δὲ τῶν Α, Δ σημείων μετέωροι εὐθεῖαι ἐφεστάτωσαν αἱ ΑΗ, ΔΜ ἴσας γωνίας περιέθουσιν μετὰ τῶν ἐξ ἀρθῆς εὐθειῶν ἑκατέραν ἑκατέρᾳ, τὴν μὲν ὑπὸ ΜΔΕ τῇ ὑπὸ ΗΑΒ, τὴν δὲ ὑπὸ ΜΔΖ τῇ ὑπὸ ΗΑΓ, καὶ εἰλήφθω ἐπὶ τῶν ΑΗ, ΔΜ τυθόντα σημεῖα τὰ Η, Μ, καὶ ἤχθωσαν ἀπὸ τῶν Η, Μ σημείων ἐπὶ τὰ διὰ τῶν ΒΑΓ, ΕΔΖ ἐπίπεδα κάθετοι αἱ ΗΛ, ΜΝ, καὶ συμβαλλέτωσαν τοῖς ἐπιπέδοις κατὰ τὰ Λ, Ν, καὶ ἐπεζεύθθωσαν αἱ ΛΑ, ΝΔ· λέγω, ὅτιὁίση ἐστὶν ἡ ὑπὸ ΗΑΛ γωνία τῇ ὑπὸ ΜΔΝ γωνίᾳ.

Κείσθω τῇ ΔΜὁίση ἡ ΑΘ, καὶ ἤχθω διὰ τοῦ Θ σημείου τῇ ΗΛ παράλληλος ἡ ΘΚ. ἡ δὲ ΗΛ κάθετόξ ἐστιν ἐπὶ τὸ διὰ τῶν ΒΑΓ ἐπίπεδον· καὶ ἡ ΘΚ ἄρα κάθετόξ ἐστιν ἐπὶ τὸ διὰ τῶν ΒΑΓ ἐπίπεδον. ἤχθωσαν ἀπὸ τῶν Κ, Ν σημείων ἐπὶ τὰξ ΑΓ, ΔΖ, ΑΒ, ΔΕ εὐθείας κάθετοι αἱ ΚΓ, ΝΖ, ΚΒ, ΝΕ, καὶ ἐπεζεύθθωσαν αἱ ΘΓ, ΓΒ, ΜΖ, ΖΕ. ἐπεὶ τὸ ἀπὸ τῆς ΘΑ ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΘΚ, ΚΑ, τῷ δὲ ἀπὸ τῆς ΚΑ ἴσα ἐστὶ τὰ ἀπὸ τῶν ΚΓ, ΓΑ, καὶ τὸ ἀπὸ τῆς ΘΑ ἄρα ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΘΚ, ΚΓ, ΓΑ. τοῖς δὲ ἀπὸ τῶν ΘΚ, ΚΓ ἴσον ἐστὶ τὸ ἀπὸ τῆς ΘΓ· τὸ ἄρα ἀπὸ τῆς ΘΑ ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΘΓ, ΓΑ.

Let *BAC* and *EDF* be two equal rectilinear angles. And let the raised straight-lines *AG* and *DM* be stood on points *A* and *D*, containing equal angles respectively with the original straight-lines. (That is) *MDE* (equal) to *GAB*, and *MDF* (to) *GAC*. And let the random points *G* and *M* be taken on *AG* and *DM* (respectively). And let the *GL* and *MN* be drawn from points *G* and *M* perpendicular to the planes through *BAC* and *EDF* (respectively). And let them have joined the planes at points *L* and *N* (respectively). And let *LA* and *ND* be joined. I say that angle *GAL* is equal to angle *MDN*.

Let *AH* be made equal to *DM*. And let *HK* be drawn through point *H* parallel to *GL*. And *GL* is perpendicular to the plane through *BAC*. Thus, *HK* is also perpendicular to the plane through *BAC* [Prop. 11.8]. And let *KC*, *NF*, *KB*, and *NE* be drawn from points *K* and *N* perpendicular to the straight-lines *AC*, *DF*, *AB*, and *DE*. And let *HC*, *CB*, *MF*, and *FE* be joined. Since the (square) on *HA* is equal to the (sum of the squares) on *HK* and *KA* [Prop. 1.47], and the (sum of the squares) on *KC* and *CA* is equal to the (square) on *KA* [Prop. 1.47], thus the (square) on *HA* is equal to the (sum of

ὀρθὴ ἄρα ἐστὶν ἡ ὑπὸ ΘΓΑ γωνία. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΔΖΜ γωνία ὀρθή ἐστιν. ὁίση ἄρα ἐστὶν ἡ ὑπὸ ΑΓΘ γωνία τῇ ὑπὸ ΔΖΜ. ὀέστι δὲ καὶ ἡ ὑπὸ ΘΑΓ τῇ ὑπὸ ΜΔΖὁίση. δύο δὴ τρίγωνά ἐστι τὰ ΜΔΖ, ΘΑΓ δύο γωνίας δυσὶ γωνίαις ἴσας ἔχοντα ἑκατέραν ἑκατέρᾳ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην τὴν ὑποτείνουσαν ὑπὸ μίαν τῶνὁίσων γωνιῶν τὴν ΘΑ τῇ ΜΔ· καὶ τὰξ λοιπὰξ ἄρα πλευρὰξ ταῖς λοιπαῖξ πλευραῖξ ἴσας ὁέχει ἑκατέραν ἑκαρέρᾳ. ὁίση ἄρα ἐστὶν ἡ ΑΓ τῇ ΔΖ. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ ΑΒ τῇ ΔΕ ἐστινὁίση. ἐπεὶ οὁῦνὁίση ἐστὶν ἡ μὲν ΑΓ τῇ ΔΖ, ἡ δὲ ΑΒ τῇ ΔΕ, δύο δὴ αἱ ΓΑ, ΑΒ δυσὶ ταῖς ΖΔ, ΔΕ ἴσαι εἰσίν. ἀλλὰ καὶ γωνία ἡ ὑπὸ ΓΑΒ γωνίᾳ τῇ ὑπὸ ΖΔΕ ἐστινὁίση· βάσις ἄρα ἡ ΒΓ βάσει τῇ ΕΖὁίση ἐστὶ καὶ τὸ τρίγωνον τῷ τριγώνῳ καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖξ γωνίαις· ὁίση ἄρα ἡ ὑπὸ ΑΓΒ γωνία τῇ ὑπὸ ΔΖΕ. ὀέστι δὲ καὶ ὀρθὴ ἡ ὑπὸ ΑΓΚ ὀρθῇ τῇ ὑπὸ ΔΖΝὁίση· καὶ λοιπὴ ἄρα ἡ ὑπὸ ΒΓΚ λοιπῇ τῇ ὑπὸ ΕΖΝ ἐστινὁίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΓΒΚ τῇ ὑπὸ ΖΕΝ ἐστινὁίση. δύο δὴ τρίγωνά ἐστι τὰ ΒΓΚ, ΕΖΝ [τὰξ] δύο γωνίας δυσὶ γωνίαις ἴσας ἔχοντα ἑκατέραν ἑκατέρᾳ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην τὴν πρὸς ταῖς ἴσαις γωνίαις τὴν ΒΓ τῇ ΕΖ· καὶ τὰξ λοιπὰξ ἄρα πλευρὰξ ταῖς λοιπαῖξ πλευραῖξ ἴσας ὁέχουσιν. ὁίση ἄρα ἐστὶν ἡ ΓΚ τῇ ΖΝ. ὀέστι δὲ καὶ ἡ ΑΓ τῇ ΔΖὁίση· δύο δὴ αἱ ΑΓ, ΓΚ δυσὶ ταῖς ΔΖ, ΖΝ ἴσαι εἰσίν· καὶ ὀρθὰξ γωνίας περιέθουσιν. βάσις ἄρα ἡ ΑΚ βάσει τῇ ΔΝὁίση ἐστίν. καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΑΘ τῇ ΔΜ, ἴσον ἐστὶ καὶ τὸ ἀπὸ τῆς ΑΘ τῷ ἀπὸ τῆς ΔΜ. ἀλλὰ τῷ μὲν ἀπὸ τῆς ΑΘ ἴσα ἐστὶ τὰ ἀπὸ τῶν ΑΚ, ΚΘ· ὀρθὴ γὰρ ἡ ὑπὸ ΑΚΘ· τῷ δὲ ἀπὸ τῆς ΔΜ ἴσα τὰ ἀπὸ τῶν ΔΝ, ΝΜ· ὀρθὴ γὰρ ἡ ὑπὸ ΔΝΜ· τὰ ἄρα ἀπὸ τῶν ΑΚ, ΚΘ ἴσα ἐστὶ τοῖς ἀπὸ τῶν ΔΝ, ΝΜ, ὁῶν τὸ ἀπὸ τῆς ΑΚ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΔΝ· λοιπὸν ἄρα τὸ ἀπὸ τῆς ΚΘ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΝΜ· ὁίση ἄρα ἡ ΘΚ τῇ ΜΝ. καὶ ἐπεὶ δύο αἱ ΘΑ, ΑΚ δυσὶ ταῖς ΜΔ, ΔΝ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, καὶ βάσις ἡ ΘΚ βάσει τῇ ΜΝ ἐδείθηὁίση, γωνία ἄρα ἡ ὑπὸ ΘΑΚ γωνίᾳ τῇ ὑπὸ ΜΔΝ ἐστινὁίση.

Ἐὰν ἄραὁῶσι δύο γωνίαι ἐπίπεδοι ἴσαι καὶ τὰ ἑχῆς τῆς προτάσεως [ὅπερ ἔδει δεῖχαι].

the squares) on HK, KC, and CA. And the (square) on HC is equal to the (sum of the squares) on HK and KC [Prop. 1.47]. Thus, the (square) on HA is equal to the (sum of the squares) on HC and CA. Thus, angle HCA is a right-angle [Prop. 1.48]. So, for the same (reasons), angle DFM is also a right-angle. Thus, angle ACH is equal to (angle) DFM. And HAC is also equal to MDF. So, MDF and HAC are two triangles having two angles equal to two angles, respectively, and one side equal to one side—(namely), that subtending one of the equal angles —(that is), HA (equal) to MD. Thus, they will also have the remaining sides equal to the remaining sides, respectively [Prop. 1.26]. Thus, AC is equal to DF. So, similarly, we can show that AB is also equal to DE. Therefore, since AC is equal to DF, and AB to DE, so the two (straight-lines) CA and AB are equal to the two (straight-lines) FD and DE (respectively). But, angle CAB is also equal to angle FDE. Thus, base BC is equal to base EF, and triangle (ACB) to triangle (DFE), and the remaining angles to the remaining angles (respectively) [Prop. 1.4]. Thus, angle ACB (is) equal to DFE. And the right-angle ACK is also equal to the right-angle DFN. Thus, the remainder BCK is equal to the remainder EFN. So, for the same (reasons), CBK is also equal to FEN. So, BCK and EFN are two triangles having two angles equal to two angles, respectively, and one side equal to one side—(namely), that by the equal angles—(that is), BC (equal) to EF. Thus, they will also have the remaining sides equal to the remaining sides (respectively) [Prop. 1.26]. Thus, CK is equal to FN. And AC (is) also equal to DF. So, the two (straight-lines) AC and CK are equal to the two (straight-lines) DF and FN (respectively). And they enclose right-angles. Thus, base AK is equal to base DN [Prop. 1.4]. And since AH is equal to DM, the (square) on AH is also equal to the (square) on DM. But, the the (sum of the squares) on AK and KH is equal to the (square) on AH. For angle AKH (is) a right-angle [Prop. 1.47]. And the (sum of the squares) on DN and NM (is) equal to the square on DM. For angle DNM (is) a right-angle [Prop. 1.47]. Thus, the (sum of the squares) on AK and KH is equal to the (sum of the squares) on DN and NM, of which the (square) on AK is equal to the (square) on DN. Thus, the remaining (square) on KH is equal to the (square) on NM. Thus, HK (is) equal to MN. And since the two (straight-lines) HA and AK are equal to the two (straight-lines) MD and DN, respectively, and base HK was shown (to be) equal to base MN, angle HAK is thus equal to angle MDN [Prop. 1.8].

Thus, if there are two equal plane angles, and so on of the proposition. [(Which is) the very thing it was required to show].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι, ἐανὁῶσι δύο γωνίαι ἐπίπε-

Corollary

So, it is clear, from this, that if there are two equal plane an-

δοι ἴσαι, ἐπισταθῶσι δὲ ἐπ᾽ αὐτῶν μετέωροι εὐθεῖαι ἴσαι ἴσας γωνίας περιέθουσαι μετὰ τῶν ἐξ ἀρθῆς εὐθειῶν ἑκατέραν ἑκατέρᾳ, αἱ ἀπὸ αὐτῶν κάθετοι ἀγόμεναι ἐπὶ τὰ ἐπίπεδα, ἐν οἷς εἰσιν αἱ ἐξ ἀρθῆς γωνίαι, ἴσαι ἀλλήλαις εἰσίν. ὅπερ ἔδει δεῖξαι.

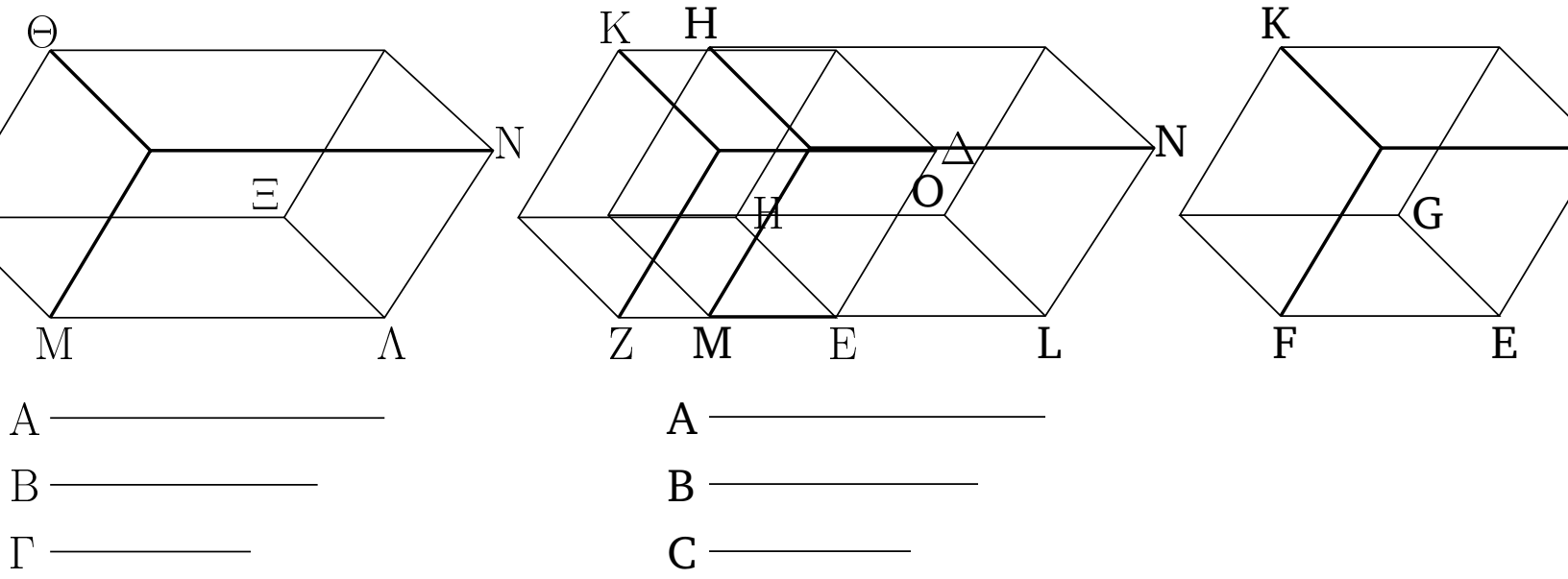
gles, and equal raised straight-lines are stood on them (at their apexes), containing equal angles respectively with the original straight-lines (forming the angles), then the perpendiculars drawn from (the raised ends of) them to the planes in which the original angles lie are equal to one another. (Which is) the very thing it was required to show.

36.

Ἐὰν τρεῖς εὐθεῖαι ἀνάλογον ὦσιν, τὸ ἐκ τῶν τριῶν στερεὸν παραλληλεπίπεδον ἴσον ἐστὶ τῷ ἀπὸ τῆς μέσης στερεῷ παραλληλεπιπέδῳ ἰσοπλεύρῳ μέν, ἰσογωνίῳ δὲ τῷ προειρημένῳ.

Proposition 36

If three straight-lines are (continuously) proportional then the parallelepiped solid (formed) from the three (straight-lines) is equal to the equilateral parallelepiped solid on the middle (straight-line which is) equiangular to the aforementioned (parallelepiped solid).



Ἔστωσαν τρεῖς εὐθεῖαι ἀνάλογον αἱ Α, Β, Γ, ὡς ἡ Α πρὸς τὴν Β, οὕτως ἡ Β πρὸς τὴν Γ· λέγω, ὅτι τὸ ἐκ τῶν Α, Β, Γ στερεὸν ἴσον ἐστὶ τῷ ἀπὸ τῆς Β στερεῷ ἰσοπλεύρῳ μέν, ἰσογωνίῳ δὲ τῷ προειρημένῳ.

Ἐκκείσθω στερεὰ γωνία ἡ πρὸς τῷ Ε περιεθομένη ὑπὸ τῶν ὑπὸ ΔΕΗ, ΗΕΖ, ΖΕΔ, καὶ κείσθω τῇ μὲν Βὁίση ἑκάστη τῶν ΔΕ, ΗΕ, ΕΖ, καὶ συμπεπληρώσθω τὸ ΕΚ στερεὸν παραλληλεπίπεδον, τῇ δὲ Αὁίση ἡ ΛΜ, καὶ συνεστάτω πρὸς τῇ ΛΜ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Λ τῇ πρὸς τῷ Ε στερεᾷ γωνίᾳ ὁίση στερεὰ γωνία ἡ περιεθομένη ὑπὸ τῶν ΝΛΧ, ΧΛΜ, ΜΛΝ, καὶ κείσθω τῇ μὲν Βὁίση ἡ ΛΧ, τῇ δὲ Γὁίση ἡ ΛΝ. καὶ ἐπεί ἐστιν ὡς ἡ Α πρὸς τὴν Β, οὕτως ἡ Β πρὸς τὴν Γ, ὁίση δὲ ἡ μὲν Α τῇ ΛΜ, ἡ δὲ Β ἑκατέρᾳ τῶν ΛΧ, ΕΔ, ἡ δὲ Γ τῇ ΛΝ, ὲστιν ἄρα ὡς ἡ ΛΜ πρὸς τὴν ΕΖ, οὕτως ἡ ΔΕ πρὸς τὴν ΛΝ. καὶ περὶ ἴσας γωνίας τὰξ ὑπὸ ΝΛΜ, ΔΕΖ αἱ πλευραὶ ἀντιπεπόνθασιν· ἴσον ἄρα ἐστὶ τὸ ΜΝ παραλληλόγραμμον τῷ ΔΖ παραλληλογραμμῳ. καὶ ἐπεὶ δύο γωνίαι ἐπίπεδοι εὐθύγραμμοι ἴσαι εἰσὶν αἱ ὑπὸ ΔΕΖ, ΝΛΜ,

Let *A*, *B*, and *C* be three (continuously) proportional straight-lines, (such that) as *A* (is) to *B*, so *B* (is) to *C*. I say that the (parallelepiped) solid (formed) from *A*, *B*, and *C* is equal to the equilateral solid on *B* (which is) equiangular with the aforementioned (solid).

Let the solid angle at *E*, contained by *DEG*, *GEF*, and *FED*, be set out. And let *DE*, *GE*, and *EF* each be made equal to *B*. And let the parallelepiped solid *EK* be completed. And (let) *LM* (be made) equal to *A*. And let the solid angle contained by *NLO*, *OLM*, and *MLN* be constructed on the straight-line *LM*, and at the point *L* on it, (so as to be) equal to the solid angle *E* [Prop. 11.23]. And let *LO* be made equal to *B*, and *LN* equal to *C*. And since as *A* (is) to *B*, so *B* (is) to *C*, and *A* (is) equal to *LM*, and *B* to each of *LO* and *ED*, and *C* to *LN*, thus as *LM* (is) to *EF*, so *DE* (is) to *LN*. And (so) the sides around the equal angles *NLM* and *DEF* are reciprocally proportional. Thus, parallelogram *MN* is equal to parallelogram *DF* [Prop. 6.14]. And since the two plane rectilinear

καὶ ἐπ΄ αὐτῶν μετέωροι εὐθεῖαι ἐφεστᾶσιν αἱ ΛΧ, ΕΗ ἴσαι τε ἀλλήλαις καὶ ἴσας γωνίας περιέθουσαι μετὰ τῶν ἐξ ἀρθῆς εὐθειῶν ἑκατέραν ἑκατέρᾳ, αἱ ἄρα ἀπὸ τῶν Η, Χ σημείων κάθετοι ἀγόμεναι ἐπὶ τὰ διὰ τῶν ΝΛΜ, ΔΕΖ ἐπίπεδα ἴσαι ἀλλήλαις εἰσίν· ὥστε τὰ ΛΘ, ΕΚ στερεὰ ὑπὸ τὸ αὐτὸ ὕψος ἐστίν. τὰ δὲ ἐπὶ ὁίσων βάσεων στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοις ἐστίν· ἴσον ἄρα ἐστὶ τὸ ΘΛ στερεὸν τῷ ΕΚ στερεῷ. καί ἐστι τὸ μὲν ΛΘ τὸ ἐκ τῶν Α, Β, Γ στερεόν, τὸ δὲ ΕΚ τὸ ἀπὸ τῆς Β στερεόν· τὸ ἄρα ἐκ τῶν Α, Β, Γ στερεὸν παραλληλεπίπεδον ἴσον ἐστὶ τῷ ἀπὸ τῆς Β στερεῷ ἰσοπλεύρῳ μέν, ἰσογωνίῳ δὲ τῷ προειρημένῳ· ὅπερ ἔδει δεῖξαι.

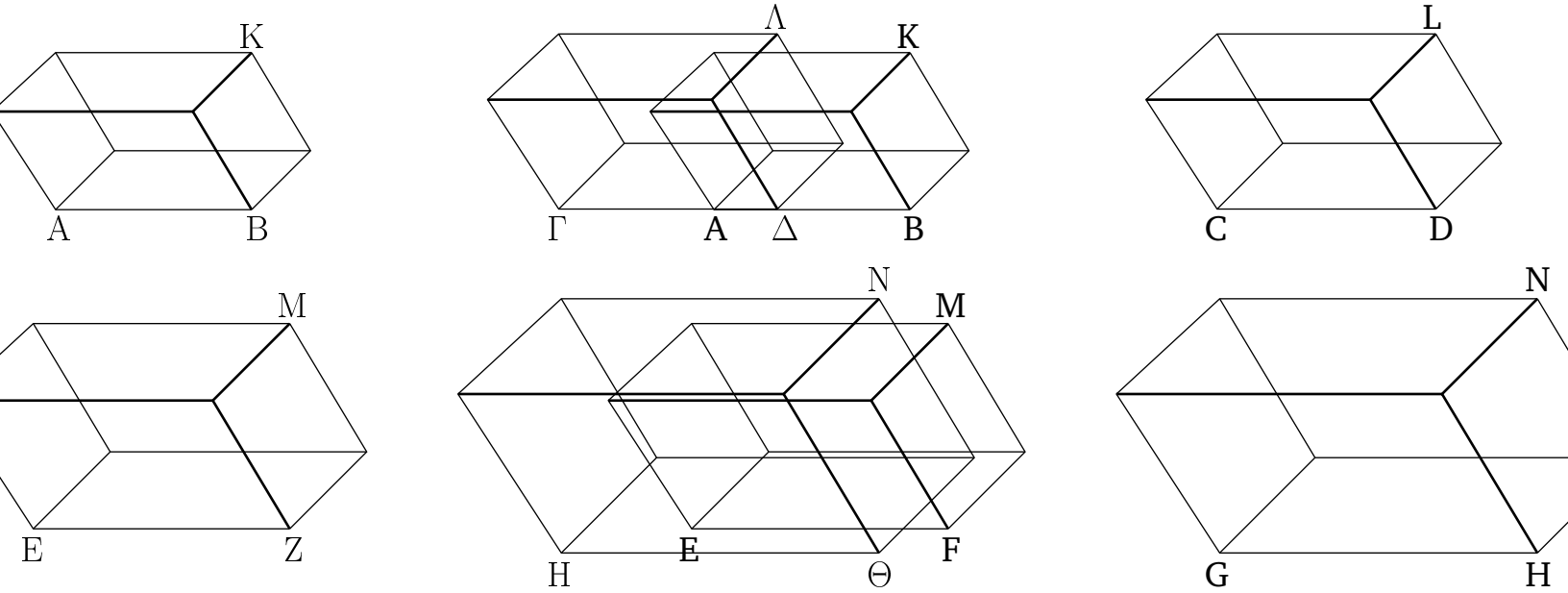
angles *DEF* and *NLM* are equal, and the raised straight-lines stood on them (at their apexes), *LO* and *EG*, are equal to one another, and contain equal angles respectively with the original straight-lines (forming the angles), the perpendiculars drawn from points *G* and *O* to the planes through *NLM* and *DEF* (respectively) are thus equal to one another [Prop. 11.35 corr.]. Thus, the solids *LH* and *EK* (have) the same height. And parallelepiped solids on equal bases, and with the same height, are equal to one another [Prop. 11.31]. Thus, solid *HL* is equal to solid *EK*. And *LH* is the solid (formed) from *A*, *B*, and *C*, and *EK* the solid on *B*. Thus, the parallelepiped solid (formed) from *A*, *B*, and *C* is equal to the equilateral solid on *B* (which is) equiangular with the aforementioned (solid). (Which is) the very thing it was required to show.

37.

Ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον ὦσιν, καὶ τὰ ἀπὸ αὐτῶν στερεὰ παραλληλεπίπεδα ὅμοιά τε καὶ ὁμοίως ἀναγραφόμενα ἀνάλογονὸέσται· καὶ ἐαν τὰ ἀπὸ αὐτῶν στερεὰ παραλληλεπίπεδα ὅμοιά τε καὶ ὁμοίως ἀναγραφόμενα ἀνάλογονὸῇ, καὶ αὐταὶ αἱ εὐθεῖαι ἀνάλογονὸέσονται.

Proposition 37†

If four straight-lines are proportional then the similar, and similarly described, parallelepiped solids on them will also be proportional. And if the similar, and similarly described, parallelepiped solids on them are proportional then the straight-lines themselves will be proportional.



Ἔστωσαν τέσσαρες εὐθεῖαι ἀνάλογον αἱ ΑΒ, ΓΔ, ΕΖ, ΗΘ, ὡς ἡ ΑΒ πρὸς τὴν ΓΔ, οὕτως ἡ ΕΖ πρὸς τὴν ΗΘ, καὶ ἀναγεγράφθωσαν ἀπὸ τῶν ΑΒ, ΓΔ, ΕΖ, ΗΘ ὅμοιά τε καὶ ὁμοίως κείμενα στερεὰ παραλληλεπίπεδα τὰ ΚΑ, ΛΓ, ΜΕ, ΝΗ· λέγω, ὅτι ἐστὶν ὡς τὸ ΚΑ πρὸς τὸ ΛΓ, οὕτως τὸ ΜΕ πρὸς τὸ ΝΗ.

Ἐπεὶ γὰρ ὅμοιόν ἐστι τὸ ΚΑ στερεὸν παραλληλεπίπεδον τῷ ΛΓ, τὸ ΚΑ ἄρα πρὸς τὸ ΛΓ τριπλασίονα λόγον ἔχει ἤπερ ἡ ΑΒ πρὸς τὴν ΓΔ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΜΕ πρὸς τὸ ΝΗ

Let *AB*, *CD*, *EF*, and *GH*, be four proportional straight-lines, (such that) as *AB* (is) to *CD*, so *EF* (is) to *GH*. And let the similar, and similarly laid out, parallelepiped solids *KA*, *LC*, *ME* and *NG* be described on *AB*, *CD*, *EF*, and *GH* (respectively). I say that as *KA* is to *LC*, so *ME* (is) to *NG*.

For since the parallelepiped solid *KA* is similar to *LC*, *KA* thus has to *LC* the cubed ratio that *AB* (has) to *CD* [Prop. 11.33]. So, for the same (reasons), *ME* also has to *NG* the cubed ratio that *EF* (has) to *GH* [Prop. 11.33]. And since

τριπλασίονα λόγον ἔχει ἥπερ ἡ EZ πρὸς τὴν HΘ. καί ἐστιν ὡς ἡ AB πρὸς τὴν ΓΔ, οὕτως ἡ EZ πρὸς τὴν HΘ. καὶ ὡς ἄρα τὸ AK πρὸς τὸ ΛΓ, οὕτως τὸ ME πρὸς τὸ NH.

Ἀλλὰ δὴ ὀέστω ὡς τὸ AK στερεὸν πρὸς τὸ ΛΓ στερεόν, οὕτως τὸ ME στερεὸν πρὸς τὸ NH· λέγω, ὅτι ἐστὶν ὡς ἡ AB εὐθεῖα πρὸς τὴν ΓΔ, οὕτως ἡ EZ πρὸς τὴν HΘ.

Ἐπεὶ γὰρ πάλιν τὸ KA πρὸς τὸ ΛΓ τριπλασίονα λόγον ἔχει ἥπερ ἡ AB πρὸς τὴν ΓΔ, ἔχει δὲ καὶ τὸ ME πρὸς τὸ NH τριπλασίονα λόγον ἥπερ ἡ EZ πρὸς τὴν HΘ, καί ἐστιν ὡς τὸ KA πρὸς τὸ ΛΓ, οὕτως τὸ ME πρὸς τὸ NH, καὶ ὡς ἄρα ἡ AB πρὸς τὴν ΓΔ, οὕτως ἡ EZ πρὸς τὴν HΘ.

Ἐὰν ἄρα τέσσαρες εὐθεῖαι ἀνάλογονὸῶσι καὶ τὰ ἐχῆς τῆς προτάσεως· ὅπερ ἔδει δεῖχαι.

as *AB* is to *CD*, so *EF* (is) to *GH*, thus, also, as *AK* (is) to *LC*, so *ME* (is) to *NG*.

And so let solid *AK* be to solid *LC*, as solid *ME* (is) to *NG*. I say that as straight-line *AB* is to *CD*, so *EF* (is) to *GH*.

For, again, since *KA* has to *LC* the cubed ratio that *AB* (has) to *CD* [Prop. 11.33], and *ME* also has to *NG* the cubed ratio that *EF* (has) to *GH* [Prop. 11.33], and as *KA* is to *LC*, so *ME* (is) to *NG*, thus, also, as *AB* (is) to *CD*, so *EF* (is) to *GH*.

Thus, if four straight-lines are proportional, and so on of the proposition. (Which is) the very thing it was required to show.

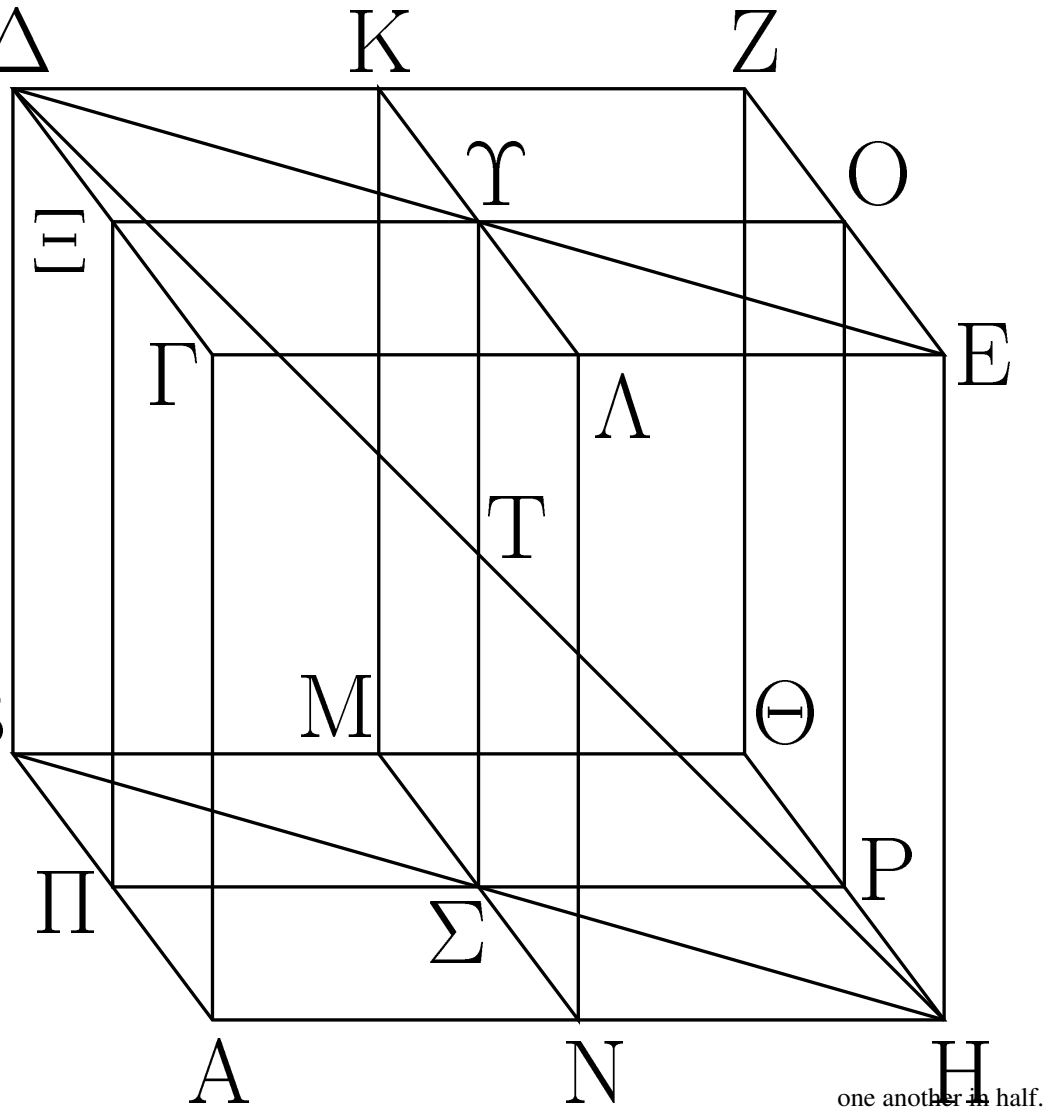
† This proposition assumes that if two ratios are equal then the cube of the former is also equal to the cube of the latter, and vice versa.

38.

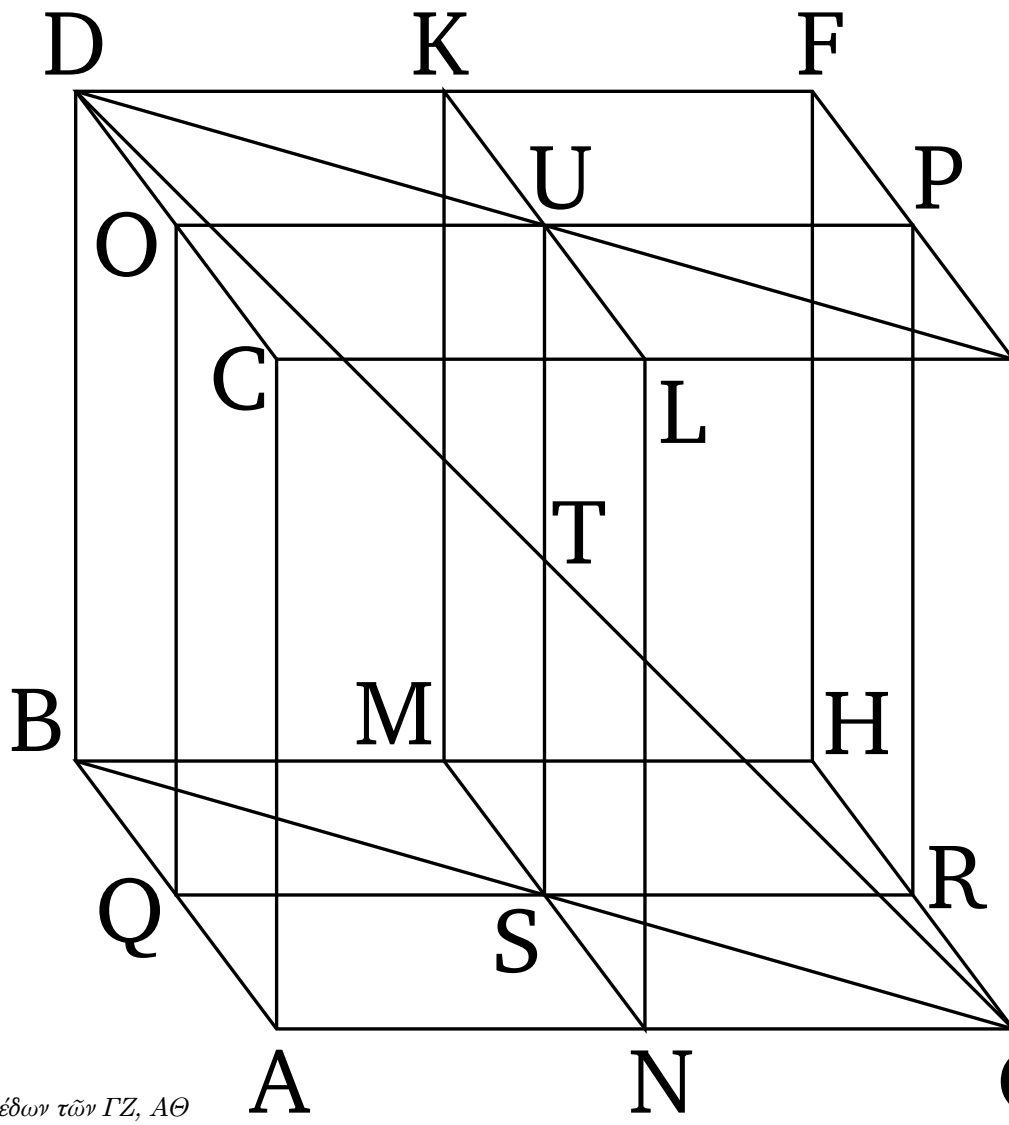
Ἐὰν κύβου τῶν ἀπεναντίον ἐπιπέδων αἱ πλευραὶ δίθα τμ ηθῶσιν, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβληθῇ, ἡ κοινὴ τομὴ τῶν ἐπιπέδων καὶ ἡ τοῦ κύβου διάμετρος δίθα τέμνουσιν ἀλλήλας.

Proposition 38

If the sides of the opposite planes of a cube are cut in half, and planes are produced through the pieces, then the common section of the (latter) planes and the diameter of the cube cut



one another in half.



Κύβου γὰρ τοῦ ΑΖ τῶν ἀπεναντίον ἐπιπέδων τῶν ΓΖ, ΑΘ αἱ πλευραὶ δίθα τετμήσθωσαν κατὰ τὰ Κ, Λ, Μ, Ν, Χ, Π, Ο, Ρ σημεῖα, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβεβλήσθω τὰ ΚΝ, ΧΡ, κοινὴ δὲ τομὴ τῶν ἐπιπέδωνὁέστω ἡ ΥΣ, τοῦ δὲ ΑΖ κύβου διαγώνιος ἡ ΔΗ. λέγω, ὅτιὁίση ἐστὶν ἡ μὲν ΥΤ τῇ ΤΣ, ἡ δὲ ΔΤ τῇ ΤΗ.

Ἐπεζεύθθωσαν γὰρ αἱ ΔΥ, ΥΕ, ΒΣ, ΣΗ. καὶ ἐπεὶ παράλληλόξ ἐστιν ἡ ΔΧ τῇ ΟΕ, αἱ ἐναλλὰχ γωνίαι αἱ ὑπὸ ΔΧΥ, ΥΟΕ ἴσαι ἀλλήλαις εἰσίν. καὶ ἐπεὶ ὁίση ἐστὶν ἡ μὲν ΔΧ τῇ ΟΕ, ἡ δὲ ΧΥ τῇ ΥΟ, καὶ γωνίας ἴσας περιέθουσιν, βάσις ἄρα ἡ ΔΥ τῇ ΥΕ ἐστινὁίση, καὶ τὸ ΔΧΥ τρίγωνον τῷ ΟΥΕ τριγώνῳ ἐστὶν ἴσον καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖξ γωνίαις ἴσαι· ὁίση ἄρα ἡ ὑπὸ ΧΥΔ γωνία τῇ ὑπὸ ΟΥΕ γωνίᾳ. διὰ δὴ τοῦτο εὐθεῖά ἐστιν ἡ ΔΥΕ. διὰ τὰ αὐτὰ δὴ καὶ ΒΣΗ εὐθεῖά ἐστιν, καὶ ὁίση ἡ ΒΣ τῇ ΣΗ. καὶ ἐπεὶ ἡ ΓΑ τῇ ΔΒὁίση ἐστὶ καὶ παράλληλος, ἀλλὰ ἡ ΓΑ καὶ τῇ ΕΗὁίση τέ ἐστι καὶ παράλληλος, καὶ ἡ ΔΒ ἄρα τῇ ΕΗὁίση τέ ἐστι καὶ παράλληλος.

For let the opposite planes CF and AH of the cube AF be cut in half at the points K, L, M, N, O, Q, P, and R. And let the planes KN and OR be produced through the pieces. And let US be the common section of the planes, and DG the diameter of cube AF. I say that UT is equal to TS, and DT to TG.

For let DU, UE, BS, and SG be joined. And since DO is parallel to PE, the alternate angles DOU and UPE are equal to one another [Prop. 1.29]. And since DO is equal to PE, and OU to UP, and they contain equal angles, base DU is thus equal to base UE, and triangle DOU is equal to triangle PUE, and the remaining angles (are) equal to the remaining angles [Prop. 1.4]. Thus, angle OUD (is) equal to angle PUE. So, for this (reason), DUE is a straight-line [Prop. 1.14]. So, for the same (reason), BSG is also a straight-line, and BS equal to SG. And since CA is equal and parallel to DB, but CA is also equal and parallel to EG, DB is thus also equal and parallel

καὶ ἐπιζευγνύουσιν αὐτὰξ εὐθεῖαι αἱ ΔΕ, ΒΗ· παράλληλος ἄρα ἐστὶν ἡ ΔΕ τῇ ΒΗ. ὁίση ἄρα ἡ μὲν ὑπὸ ΕΔΤ γωνία τῇ ὑπὸ ΒΗΤ· ἐναλλὰχ γάρ· ἡ δὲ ὑπὸ ΔΤΥ τῇ ὑπὸ ΗΤΣ. δύο δὴ τρίγωνά ἐστι τὰ ΔΤΥ, ΗΤΣ τὰξ δύο γωνίας ταῖς δυσὶ γωνίαις ἴσας ἔχοντα καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην τὴν ὑποτείνουσαν ὑπὸ μίαν τῶνὁίσων γωνιῶν τὴν ΔΥ τῇ ΗΣ· ἡμίσειαι γάρ εἰσι τῶν ΔΕ, ΒΗ· καὶ τὰξ λοιπὰξ πλευρὰξ ταῖς λοιπαῖξ πλευραῖξ ἴσας ὁέχει. ὁίση ἄρα ἡ μὲν ΔΤ τῇ ΤΗ, ἡ δὲ ΥΤ τῇ ΤΣ.

Ἐὰν ἄρα κύβου τῶν ἀπεναντίον ἐπιπέδων αἱ πλευραὶ δίθα τμ ηθῶσιν, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβληθῇ, ἡ κοινὴ τομὴ τῶν ἐπιπέδων καὶ ἡ τοῦ κύβου διάμετρος δίθα τέμνουσιν ἀλλήλας· ὅπερ ἔδει δεῖχαι.

to *EG* [Prop. 11.9]. And the straight-lines *DE* and *BG* join them. *DE* is thus parallel to *BG* [Prop. 1.33]. Thus, angle *EDT* (is) equal to *BGT*. For (they are) alternate [Prop. 1.29]. And (angle) *DTU* (is equal) to *GTS* [Prop. 1.15]. So, *DTU* and *GTS* are two triangles having two angles equal to two angles, and one side equal to one side—(namely), that subtended by one of the equal angles—(that is), *DU* (equal) to *GS*. For they are halves of *DE* and *BG* (respectively). (Thus), they will also have the remaining sides equal to the remaining sides [Prop. 1.26]. Thus, *DT* (is) equal to *TG*, and *UT* to *TS*.

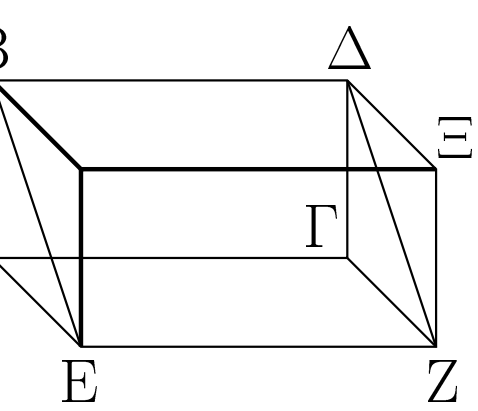
Thus, if the sides of the opposite planes of a cube are cut in half, and planes are produced through the pieces, then the common section of the (latter) planes and the diameter of the cube cut one another in half. (Which is) the very thing it was required to show.

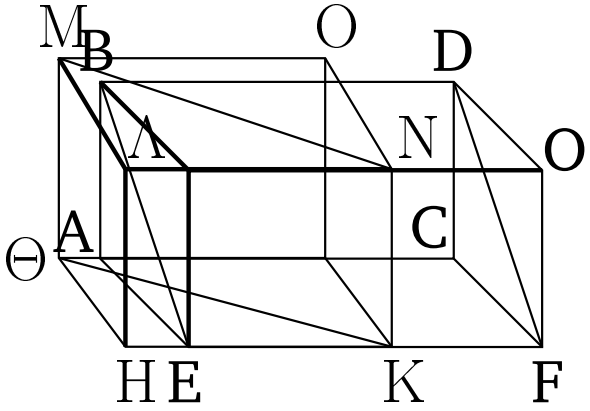
39.

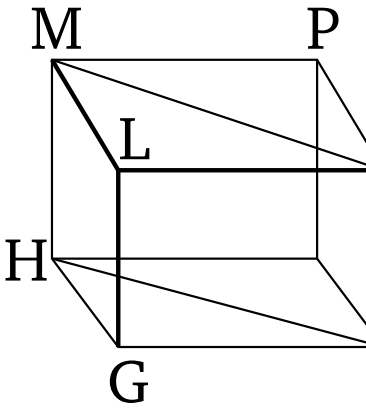
Proposition 39

Ἐὰνὸῇ δύο πρίσματα ἰσοϋψῆ, καὶ τὸ μὲνὸέθῇ βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ ὁῇ τὸ παραλληλόγραμμον τοῦ τριγώνου, ἴσαὸέσται τὰ πρίσματα.

If there are two equal height prisms, and one has a parallelogram, and the other a triangle, (as a) base, and the parallelogram is double the triangle, then the prisms will be equal.







Ἔστω δύο πρίσματα ἰσοϋψῆ τὰ ΑΒΓΔΕΖ, ΗΘΚΛΜΝ, καὶ τὸ μὲν ἐθέτω βάσιν τὸ ΑΖ παραλληλόγραμμον, τὸ δὲ τὸ ΗΘΚ τρίγωνον, διπλάσιον δὲ ὁέστω τὸ ΑΖ παραλληλόγραμμον τοῦ ΗΘΚ τριγώνου· λέγω, ὅτι ἴσον ἐστὶ τὸ ΑΒΓΔΕΖ πρίσμα τῷ ΗΘΚΛΜΝ πρίσματι.

Συμπεπληρώσθω γὰρ τὰ ΑΧ, ΗΟ στερεά. ἐπεὶ διπλάσιόν ἐστι τὸ ΑΖ παραλληλόγραμμον τοῦ ΗΘΚ τριγώνου, ὁέστι δὲ καὶ τὸ ΘΚ παραλληλόγραμμον διπλάσιον τοῦ ΗΘΚ τριγώνου, ἴσον ἄρα ἐστὶ τὸ ΑΖ παραλληλόγραμμον τῷ ΘΚ παραλληλογράμμῳ. τὰ δὲ ἐπὶ ὁίσων βάσεων ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὁύψος ἴσα ἀλλήλοις ἐστίν· ἴσον ἄρα ἐστὶ τὸ ΑΧ στερεὸν τῷ ΗΟ στερεῷ. καί ἐστι τοῦ μὲν ΑΧ στερεοῦ ἥμισυ τὸ ΑΒΓΔΕΖ πρίσμα, τοῦ δὲ ΗΟ στερεοῦ ἥμισυ τὸ ΗΘΚΛΜΝ πρίσμα· ἴσον ἄρα ἐστὶ τὸ ΑΒΓΔΕΖ πρίσμα τῷ ΗΘΚΛΜΝ πρίσματι.

Ἐὰν ἄραὸῇ δύο πρίσματα ἰσοϋψῆ, καὶ τὸ μὲνὸέθῇ βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ ὁῇ τὸ πα-

Let *ABCDEF* and *GHKLMN* be two equal height prisms, and let the former have the parallelogram *AF*, and the latter the triangle *GHK*, as a base. And let parallelogram *AF* be twice triangle *GHK*. I say that prism *ABCDEF* is equal to prism *GHKLMN*.

For let the solids *AO* and *GP* be completed. Since parallelogram *AF* is double triangle *GHK*, and parallelogram *HK* is also double triangle *GHK* [Prop. 1.34], parallelogram *AF* is thus equal to parallelogram *HK*. And parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another [Prop. 11.31]. Thus, solid *AO* is equal to solid *GP*. And prism *ABCDEF* is half of solid *AO*, and prism *GHKLMN* half of solid *GP* [Prop. 11.28]. Prism *ABCDEF* is thus equal to prism *GHKLMN*.

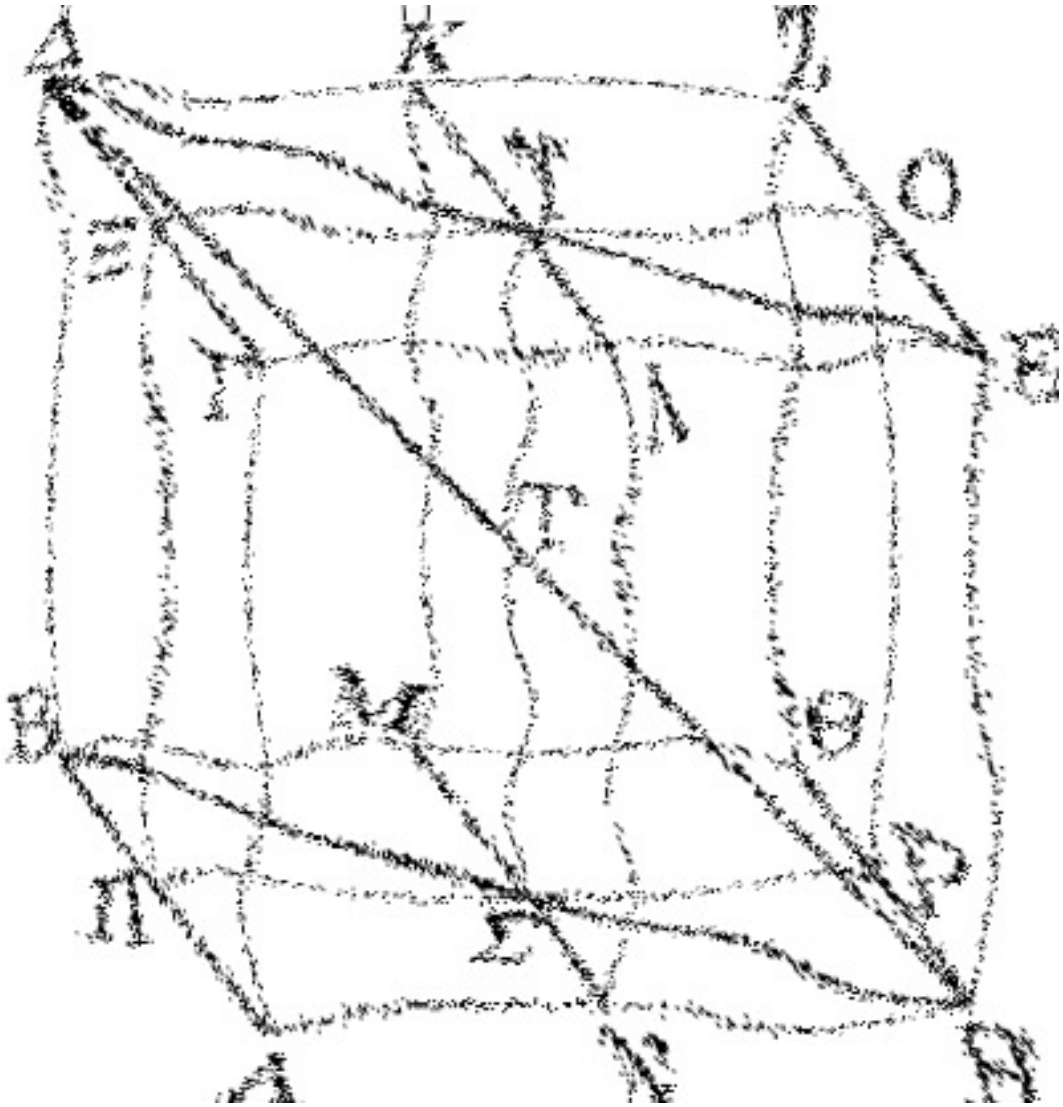
Thus, if there are two equal height prisms, and one has a parallelogram, and the other a triangle, (as a) base, and the parallelogram is double the triangle, then the prisms are equal.

ραλληλόγραμμον τοῦ τριγώνου, ἴσα δ' ἐστὶ τὰ πρίσματα· ὅπερ ἔδει δεῖξαι.

(Which is) the very thing it was required to show.

ELEMENTS BOOK 11

Elementary Stereometry



Ὅροι.

1. Στερεόν ἐστι τὸ μῆκος καὶ πλάτος καὶ βάθος ἔχον.

2. Στερεοῦ δὲ πέρας ἐπιφάνεια.

3. Εὐθεῖα πρὸς ἐπίπεδον ὀρθή ἐστιν, ὅταν πρὸς πάσας τὰς ἁπτομένας αὐτῆς εὐθείας καὶ οὐσας ἐν τῷ [ὑποκειμένῳ] ἐπιπέδῳ ὀρθὰς ποιῇ γωνίας.

4. Ἐπίπεδον πρὸς ἐπίπεδον ὀρθόν ἐστιν, ὅταν αἱ τῇ κοινῇ τομῇ τῶν ἐπιπέδων πρὸς ὀρθὰς ἀγόμεναι εὐθεῖαι ἐν ἑνὶ τῶν ἐπιπέδων τῷ λοιπῷ ἐπιπέδῳ πρὸς ὀρθὰς ὦσιν.

5. Εὐθείας πρὸς ἐπίπεδον κλίσις ἐστίν, ὅταν ἀπὸ τοῦ μετεώρου πέρατος τῆς εὐθείας ἐπὶ τὸ ἐπίπεδον κάθετος ἀθθῇ, καὶ ἀπὸ τοῦ γενομένου σημείου ἐπὶ τὸ ἐν τῷ ἐπιπέδῳ πέρας τῆς εὐθείας εὐθεῖα ἐπιζευθθῇ, ἡ περιεθομένη γωνία ὑπὸ τῆς ἀθθείσης καὶ τῆς ἐφεστώσης.

6. Ἐπιπέδου πρὸς ἐπίπεδον κλίσις ἐστὶν ἡ περιεθομένη ὀξεῖα γωνία ὑπὸ τῶν πρὸς ὀρθὰς τῇ κοινῇ τομῇ ἀγομένων πρὸς τῷ αὐτῷ σημείῳ ἐν ἑκατέρῳ τῶν ἐπιπέδων.

7. Ἐπίπεδον πρὸς ἐπίπεδον ὁμοίως κεκλίσθαι λέγεται καὶ ὁέτερον πρὸς ὁέτερον, ὅταν αἱ εἰρημέναι τῶν κλίσεων γωνίαι ἴσαι ἀλλήλαις ὦσιν.

8. Παράλληλα ἐπίπεδά ἐστι τὰ ἀσύμπτωτα.

9. Ὅμοια στερεὰ σθήματά ἐστι τὰ ὑπὸ ὁμοίων ἐπιπέδων περιεχόμεναοίσων τὸ πλῆθος.

10. Ὁἴσα δὲ καὶ ὅμοια στερεὰ σθήματά ἐστι τὰ ὑπὸ ὁμοίων ἐπιπέδων περιεχόμεναοίσων τῷ πλήθει καὶ τῷ μεγέθει.

11. Στερεὰ γωνία ἐστὶν ἡ ὑπὸ πλειόνωνὸὴ δύο γραμμῶν ἁπτομένων ἀλλήλων καὶ μὴ ἐν τῇ αὐτῇ ἐπιφανείᾳ οὐσῶν πρὸς πάσαις ταῖς γραμμαῖξ κλίσις. ὁάλλως· στερεὰ γωνία ἐστὶν ἡ ὑπὸ πλειόνωνὸὴ δύο γωνιῶν ἐπιπέδων περιεθομένη μὴ οὐσῶν ἐν τῷ αὐτῷ ἐπιπέδῳ πρὸς ἑνὶ σημείῳ συνισταμένων.

12. Πυραμίξ ἐστι σχῆμα στερεὸν ἐπιπέδοις περιθόμενον ἀπὸ ἑνὸς ἐπιπέδου πρὸς ἑνὶ σημείῳ συνεστώξ.

13. Πρίσμα ἐστὶ σχῆμα στερεὸν ἐπιπέδοις περιεχόμενον, ὁῶν δύο τὰ ἀπεναντίον ἴσα τε καὶ ὅμοιά ἐστι καὶ παράλληλα, τὰ δὲ λοιπὰ παραλληλόγραμμα.

14. Σφαῖρά ἐστιν, ὅταν ἡμικυκλίου μενούσης τῆς διαμέτρου περιενεθθὲν τὸ ἡμικύκλιον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, τὸ περιληφθὲν σχῆμα.

15. Ὁἄχων δὲ τῆς σφαίρας ἐστὶν ἡ μένουσα εὐθεῖα, περὶ ἣν τὸ ἡμικύκλιον στρέφεται.

16. Κέντρον δὲ τῆς σφαίρας ἐστὶ τὸ αὐτό, ὃ καὶ τοῦ ἡμικυκλίου.

17. Διάμετρος δὲ τῆς σφαίρας ἐστὶν εὐθεῖά τις διὰ τοῦ κέντρου ἠγμένη καὶ περατουμένη ἐφ᾽ ἑκάτερα τὰ μέρη ὑπὸ τῆς ἐπιφανείας τῆς σφαίρας.

18. Κῶνόξ ἐστιν, ὅταν ὀρθογωνίου τριγώνου μενούσης μιᾶς πλευρᾶξ τῶν περὶ τὴν ὀρθὴν γωνίαν περιενεθθὲν τὸ τρίγωνον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, τὸ περιληφθὲν σχῆμα. κὀὰν μὲν ἡ μένουσα εὐθεῖαὁίσηὸῇ τῇ λοιπῇ

[τῇ] περὶ τὴν ὀρθὴν περιφερομένῃ, ὀρθογώνιος ὀέσται ὁ κῶνος, ἐαν δὲ ἐλάττων, ἀμβλυγώνιος, ἐαν δὲ μείζων, ὀχυγώνιος.

19. Ὁ Ἄχων δὲ τοῦ κώνου ἐστὶν ἡ μένουσα εὐθεῖα, περὶ ἣν τὸ τρίγωνον στρέφεται.

20. Βάσις δὲ ὁ κύκλος ὁ ὑπὸ τῆς περιφερομένης εὐθείας γραφόμενος.

21. Κύλινδρός ἐστιν, ὅταν ὀρθογωνίου παραλληλογράμμου μενούσης μιᾶς πλευρᾶξ τῶν περὶ τὴν ὀρθὴν γωνίαν περιενεθθὲν τὸ παραλληλόγραμμον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, τὸ περιληφθὲν σχῆμα.

22. Ὁ Ἄχων δὲ τοῦ κυλίνδρου ἐστὶν ἡ μένουσα εὐθεῖα, περὶ ἣν τὸ παραλληλόγραμμον στρέφεται.

23. Βάσεις δὲ οἱ κύκλοι οἱ ὑπὸ τῶν ἀπεναντίον περιαγομένων δύο πλευρῶν γραφόμενοι.

24. Ὅμοιοι κῶνοι καὶ κύλινδροί εἰσιν, ὧν οὶ τεὸάχονες καὶ αἱ διάμετροι τῶν βάσεων ἀνάλογόν εἰσιν.

25. Κύβος ἐστὶ σχῆμα στερεὸν ὑπὸ ὸἓχ τετραγώνωνὸίσων περιεχόμενον.

26. Ὀκτάεδρόν ἐστὶ σχῆμα στερεὸν ὑπὸ ὀκτὼ τριγώνωνὸίσων καὶ ἰσοπλεύρων περιεχόμενον.

27. Εἰκοσάεδρόν ἐστι σχῆμα στερεὸν ὑπὸ εὸίκοσι τριγώνωνὸίσων καὶ ἰσοπλεύρων περιεχόμενον.

28. Δωδεκάεδρόν ἐστι σχῆμα στερεὸν ὑπὸ δώδεκα πενταγώνωνὸίσων καὶ ἰσοπλεύρων καὶ ἰσογωνίων περιεχόμενον.

Definitions

1. A solid is a (figure) having length and breadth and depth.

2. The extremity of a solid (is) a surface.

3. A straight-line is at right-angles to a plane when it makes right-angles with all of the straight-lines joined to it which are also in the plane.

4. A plane is at right-angles to a(nother) plane when (all of) the straight-lines drawn in one of the planes, at right-angles to the common section of the planes, are at right-angles to the remaining plane.

5. The inclination of a straight-line to a plane is the angle contained by the drawn and standing (straight-lines), when a perpendicular is lead to the plane from the end of the (standing) straight-line raised (out of the plane), and a straight-line is (then) joined from the point (so) generated to the end of the (standing) straight-line (lying) in the plane.

6. The inclination of a plane to a(nother) plane is the acute angle contained by the (straight-lines), (one) in each of the planes, drawn at right-angles to the common segment (of the planes), at the same point.

7. A plane is said to be similarly inclined to a plane, as another to another, when the aforementioned angles of inclination are equal to one another.

8. Parallel planes are those which do not meet (one another).

9. Similar solid figures are those contained by equal numbers of similar planes (which are similarly arranged).

10. But equal and similar solid figures are those contained by similar planes equal in number and in magnitude (which are similarly arranged).

11. A solid angle is the inclination (constituted) by more than two lines joining one another (at the same point), and not being in the same surface, to all of the lines. Otherwise, a solid angle is that contained by more than two plane angles, not being in the same plane, and constructed at one point.

12. A pyramid is a solid figure, contained by planes, (which is) constructed from one plane to one point.

13. A prism is a solid figure, contained by planes, of which the two opposite (planes) are equal, similar, and parallel, and the remaining (planes are) parallelograms.

14. A sphere is the figure enclosed when, the diameter of a semicircle remaining (fixed), the semicircle is carried around, and again established at the same (position) from which it began to be moved.

15. And the axis of the sphere is the fixed straight-line about which the semicircle is turned.

16. And the center of the sphere is the same as that of the semicircle.

17. And the diameter of the sphere is any straight-line which is drawn through the center and terminated in both directions by the surface of the sphere.

18. A cone is the figure enclosed when, one of the sides of a right-angled triangle about the right-angle remaining (fixed), the triangle is carried around, and again established at the same (position) from which it began to be moved. And if the fixed straight-line is equal to the remaining (straight-line) about the right-angle, (which is) carried around, then the cone will be right-angled, and if less, obtuse-angled, and if greater, acute-angled.

19. And the axis of the cone is the fixed straight-line about which the triangle is turned.

20. And the base (of the cone is) the circle described by the (remaining) straight-line (about the right-angle which is) carried around (the axis).

21. A cylinder is the figure enclosed when, one of the sides of a right-angled parallelogram about the right-angle remaining (fixed), the parallelogram is carried around, and again established at the same (position) from which it began to be moved.

22. And the axis of the cylinder is the stationary straight-line about which the parallelogram is turned.

23. And the bases (of the cylinder are) the circles described by the two opposite sides (which are) carried around.

24. Similar cones and cylinders are those for which the axes and the diameters of the bases are proportional.

25. A cube is a solid figure contained by six equal squares.

26. An octahedron is a solid figure contained by eight equal and equilateral triangles.

27. An icosahedron is a solid figure contained by twenty equal and equilateral triangles.

28. A dodecahedron is a solid figure contained by twelve equal, equilateral, and equiangular pentagons.

1.

Εὐθείας γραμμῆς μέρος μέν τι οὐκόέστιν ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, μέρος δέ τι ἐν μετεωροτέρῳ.

Εἰ γὰρ δυνατόν, εὐθείας γραμμῆς τῆς ΑΒΓ μέρος μέν τι τὸ ΑΒὸέστω ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, μέρος δέ τι τὸ ΒΓ ἐν μετεωροτέρῳ.

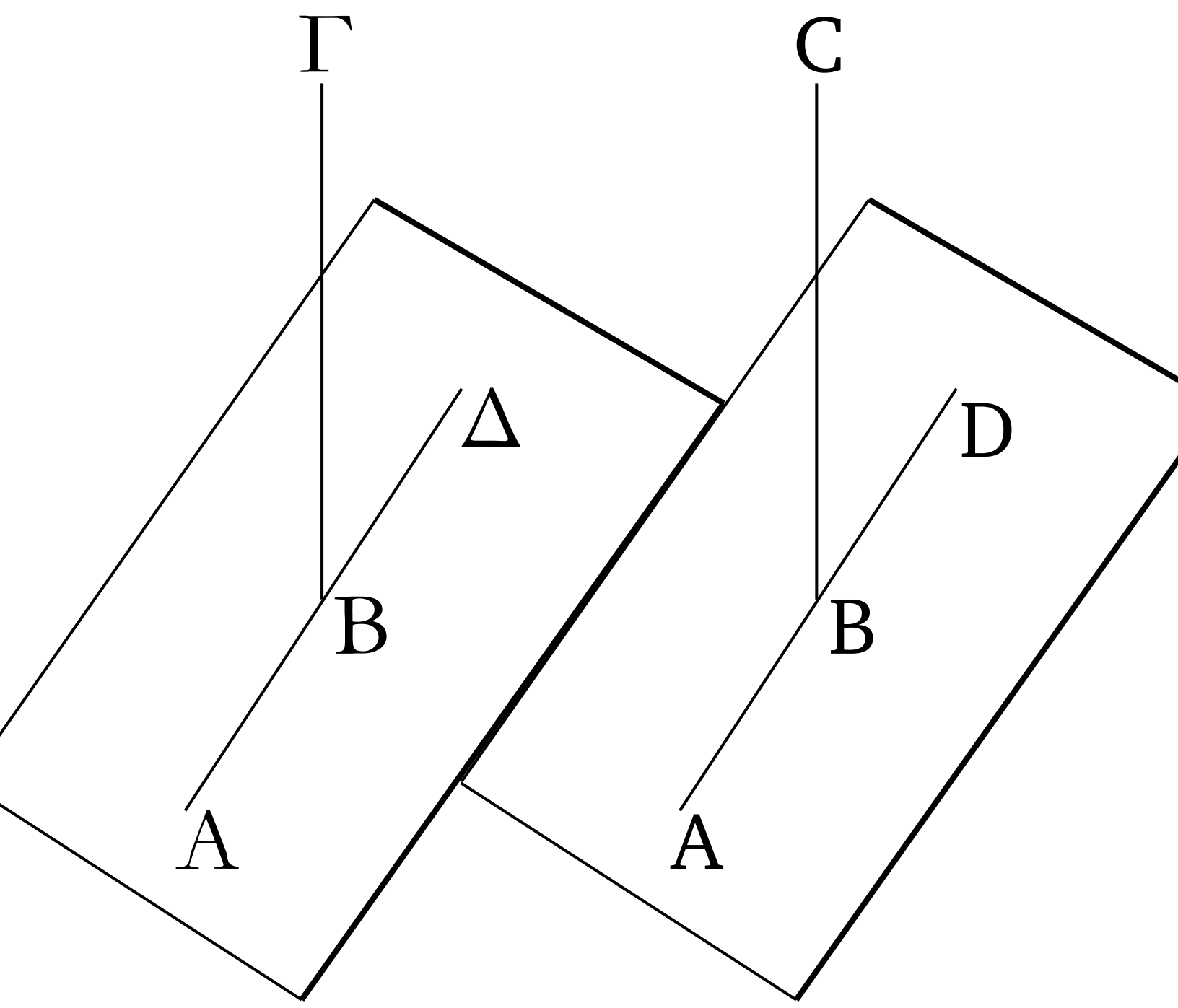
ὸΕσται δή τις τῇ ΑΒ συννεθὴξ εὐθεῖα ἐπ΄ εὐθείας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ. ὸέστω ἡ ΒΔ· δύο ἄρα εὐθειῶν τῶν ΑΒΓ, ΑΒΔ κοινὸν τμῆμά ἐστιν ἡ ΑΒ· ὅπερ ἐστὶν ἀδύνατον, ἐπειδήπερ ἓαν κέντρῳ τῷ Β καὶ διαστήματι τῷ ΑΒ κύκλον γράψωμεν, αἱ διάμετροι ἀνίσους ἀπολήψονται τοῦ κύκλου περιφερείας.

Proposition 1[†]

Some part of a straight-line cannot be in a reference plane, and some part in a more elevated (plane).

For, if possible, let some part, *AB*, of the straight-line *ABC* be in a reference plane, and some part, *BC*, in a more elevated (plane).

In the reference plane, there will be some straight-line continuous with, and straight-on to, *AB*.[‡] Let it be *BD*. Thus, *AB* is a common segment of the two (different) straight-lines *ABC* and *ABD*. The very thing is impossible, inasmuch as if we draw a circle with center *B* and radius *AB* then the diameters (*ABD* and *ABC*) will cut off unequal circumferences of the circle.



Εὐθείας ἄρα γραμμῆς μέρος μέν τι οὐκὸέστιν ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ ἐν μετεωροτέρῳ· ὅπερ ἔδει δεῖχαι.

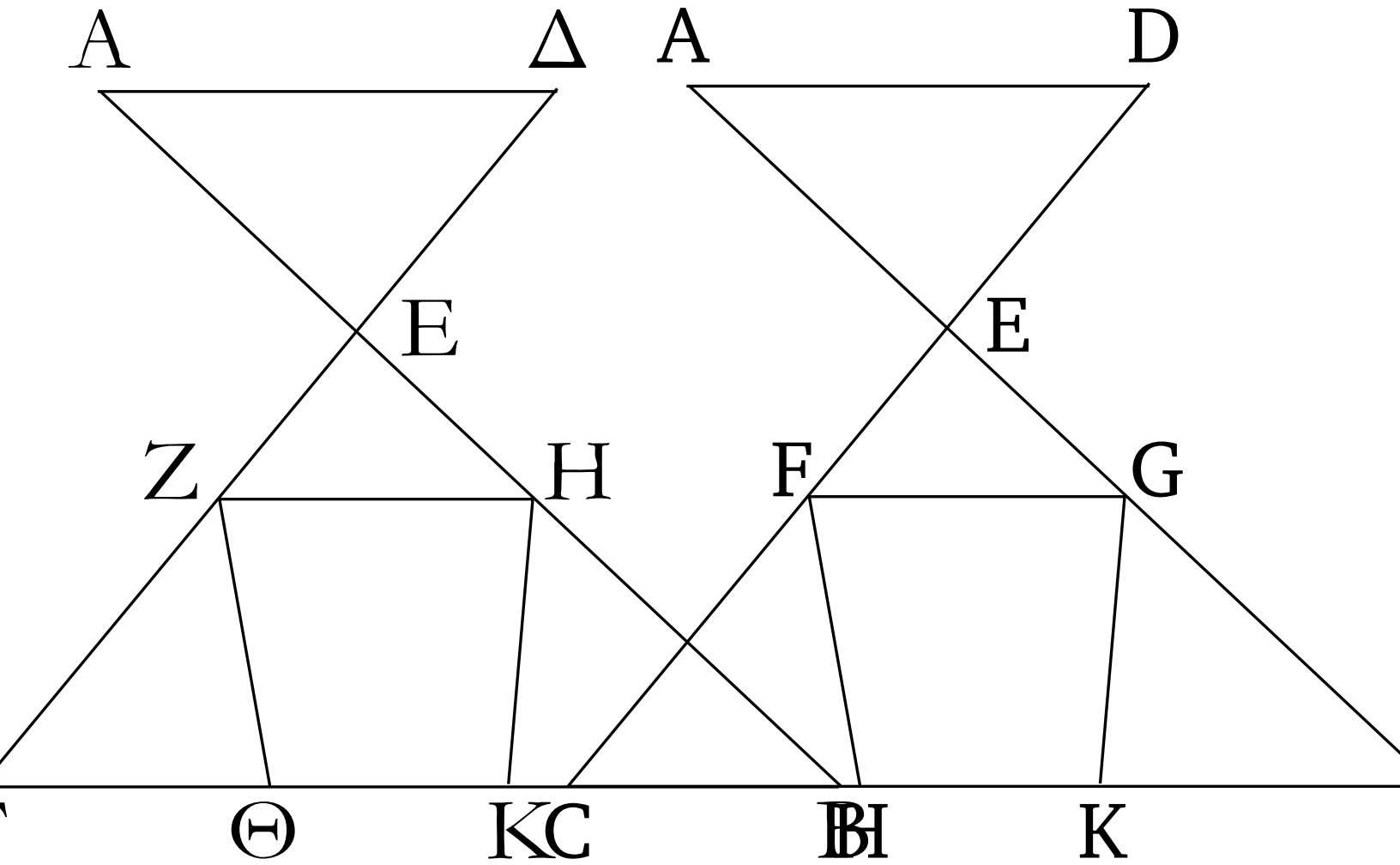
Thus, some part of a straight-line cannot be in a reference plane, and (some part) in a more elevated (plane). (Which is) the very thing it was required to show.

† The proofs of the first three propositions in this book are not at all rigorous. Hence, these three propositions should properly be regarded as additional axioms.

‡ This assumption essentially presupposes the validity of the proposition under discussion.

Ἐὰν δύο εὐθεῖαι τέμνωσιν ἀλλήλας, ἐν ἑνί εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἑνί ἐστιν ἐπιπέδῳ.

If two straight-lines cut one another then they are in one plane, and every triangle (formed using segments of both lines) is in one plane.



Δύο γὰρ εὐθεῖαι αἱ ΑΒ, ΓΔ τεμνέτωσαν ἀλλήλας κατὰ τὸ Ε σημεῖον. λέγω, ὅτι αἱ ΑΒ, ΓΔ ἐν ἑνί εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἑνί ἐστιν ἐπιπέδῳ.

Εἰλήφθω γὰρ ἐπὶ τῶν ΕΓ, ΕΒ τυθόντα σημεῖα τὰ Ζ, Η, καὶ ἐπεζεύθθωσαν αἱ ΓΒ, ΖΗ, καὶ διήθθωσαν αἱ ΖΘ, ΗΚ· λέγω πρῶτον, ὅτι τὸ ΕΓΒ τρίγωνον ἐν ἑνί ἐστιν ἐπιπέδῳ. εἰ γάρ ἐστι τοῦ ΕΓΒ τριγώνου μέρος ἤτοι τὸ ΖΘΓὸἠ τὸ ΗΒΚ ἐν τῷ ὑποκειμένῳ [ἐπιπέδῳ], τὸ δὲ λοιπὸν ἐνὸάλλῳ, ὸέσται καὶ μιᾶς τῶν ΕΓ, ΕΒ εὐθειῶν μέρος μέν τι ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ ἐν αλλῳ. εἰ δὲ τοῦ ΕΓΒ τριγώνου τὸ ΖΓΒΗ μέροςὸῇ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ λοιπὸν ἐνὸάλλῳ, ὸέσται καὶ ἀμφοτέρων τῶν ΕΓ, ΕΒ εὐθειῶν μέρος μέν τι ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, τὸ δὲ ἐνὸάλλω· ὅπερὸάτοπον ἐδείθθη. τὸ ἄρα ΕΓΒ τρίγωνον ἐν ἑνί ἐστιν ἐπιπέδῳ. ἐν ὁῷ δέ ἐστι τὸ ΕΓΒ τρίγωνον, ἐν τούτῳ καὶ ἑκατέρα τῶν ΕΓ, ΕΒ, ἐν ὁῷ δὲ ἑκατέρα τῶν ΕΓ, ΕΒ, ἐν τούτῳ καὶ αἱ ΑΒ, ΓΔ. αἱ ΑΒ, ΓΔ

For let the two straight-lines *AB* and *CD* have cut one another at point *E*. I say that *AB* and *CD* are in one plane, and that every triangle (formed using segments of both lines) is in one plane.

For let the random points *F* and *G* be taken on *EC* and *EB* (respectively). And let *CB* and *FG* be joined, and let *FH* and *GK* be drawn across. I say, first of all, that triangle *ECB* is in one (reference) plane. For if part of triangle *ECB*, either *FHC* or *GBK*, is in the reference [plane], and the remainder in a different (plane) then a part of one the straight-lines *EC* and *EB* will also be in the reference plane, and (a part) in a different (plane). And if the part *FCBG* of triangle *ECB* is in the reference plane, and the remainder in a different (plane) then parts of both of the straight-lines *EC* and *EB* will also be in the reference plane, and (parts) in a different (plane). The very thing was shown to be absurb [Prop. 11.1]. Thus, triangle

ἄρα εὐθεῖαι ἐν ἑνί εἰσιν ἐπιπέδῳ, καὶ πᾶν τρίγωνον ἐν ἑνί ἐστιν ἐπιπέδῳ· ὅπερ ἔδει δεῖξαι.

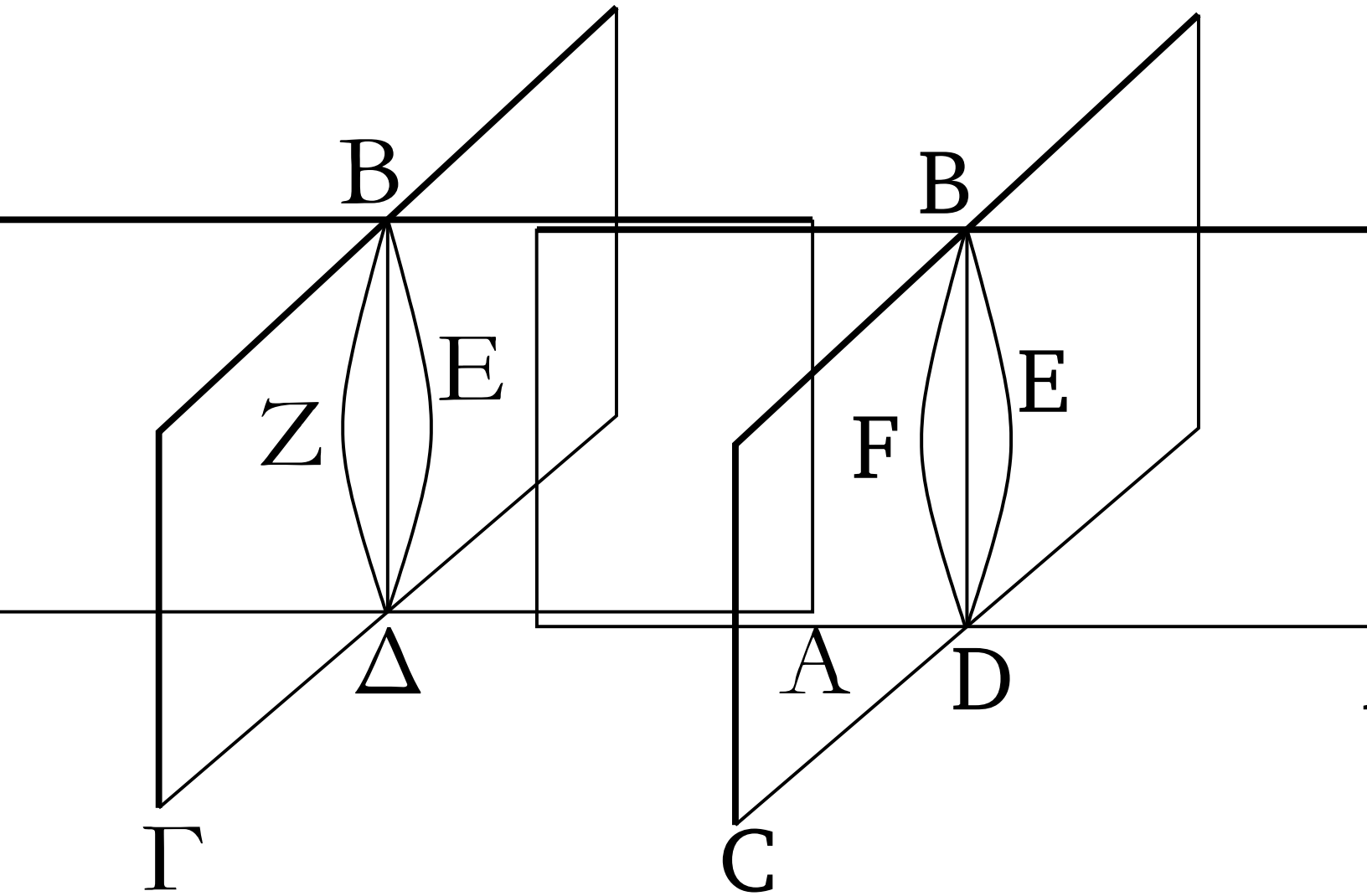
ECB is in one plane. And in whichever (plane) triangle *ECB* is (found), in that (plane) *EC* and *EB* (will) each also (be found). And in whichever (plane) *EC* and *EB* (are) each (found), in that (plane) *AB* and *CD* (will) also (be found) [Prop. 11.1]. Thus, the straight-lines *AB* and *CD* are in one plane, and every triangle (formed using segments of both lines) is in one plane. (Which is) the very thing it was required to show.

3.

Ἐὰν δύο ἐπίπεδα τεμνῇ ὀάλληλα, ἡ κοινὴ αὐτῶν τομὴ εὐθεῖά ἐστιν.

Proposition 3

If two planes cut one another then their common section is a straight-line.



Δύο γὰρ ἐπίπεδα τὰ AB, BΓ τεμνέτωὀάλληλα, κοινὴ δὲ αὐτῶν τομὴ ὀέστω ἡ ΔB γραμμή· λέγω, ὅτι ἡ ΔB γραμμὴ εὐθεῖά ἐστιν.

Εἰ γὰρ μή, ἐπεζεύθθω ἀπὸ τοῦ Δ ἐπὶ τὸ B ἐν μὲν τῷ AB ἐπιπέδῳ εὐθεῖα ἡ ΔEB, ἐν δὲ τῷ BΓ ἐπιπέδῳ εὐθεῖα ἡ ΔZB.

For let the two planes *AB* and *BC* cut one another, and let their common section be the line *DB*. I say that the line *DB* is straight.

For, if not, let the straight-line *DEB* be joined from *D* to *B* in the plane *AB*, and the straight-line *DFB* in the plane *BC*.

ὀέσται δὴ δύο εὐθειῶν τῶν ΔΕΒ, ΔΖΒ τὰ αὐτὰ πέρατα, καὶ περιέχουσι δηλαδὴ θωρίον· ὅπερ ὀάτοπον. οὐκ ἄρα αἱ ΔΕΒ, ΔΖΒ εὐθεῖαί εἰσιν. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ὀάλλη τις ἀπὸ τοῦ Δ ἐπὶ τὸ Β ἐπιζευγνυμένη εὐθεῖαὀέσται πλὴν τῆς ΔΒ κοινῆς τομῆς τῶν ΑΒ, ΒΓ ἐπιπέδων.

Ἐὰν ἄρα δύο ἐπίπεδα τέμνῃ ὀάλληλα, ἡ κοινὴ αὐτῶν τομὴ εὐθεῖά ἐστιν· ὅπερ ἔδει δεῖχαι.

So two straight-lines, DEB and DFB, will have the same ends, and they will clearly enclose an area. The very thing (is) absurd. Thus, DEB and DFB are not straight-lines. So, similarly, we can show than no other straight-line can be joined from D to B except DB, the common section of the planes AB and BC.

Thus, if two planes cut one another then their common section is a straight-line. (Which is) the very thing it was required to show.

4.

Ἐὰν εὐθεῖα δύο εὐθείαις τεμνούσαις ἀλλήλας πρὸς ὀρθὰξ ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, καὶ τῷ διὸ αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰξὀέσται.

Εὐθεῖα γάρ τις ἡ ΕΖ δύο εὐθείαις ταῖς ΑΒ, ΓΔ τεμνούσαις ἀλλήλας κατὰ τὸ Ε σημεῖον ἀπὸ τοῦ Ε πρὸς ὀρθὰξ ἐφεστάτω· λέγω, ὅτι ἡ ΕΖ καὶ τῷ διὰ τῶν ΑΒ, ΓΔ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν.

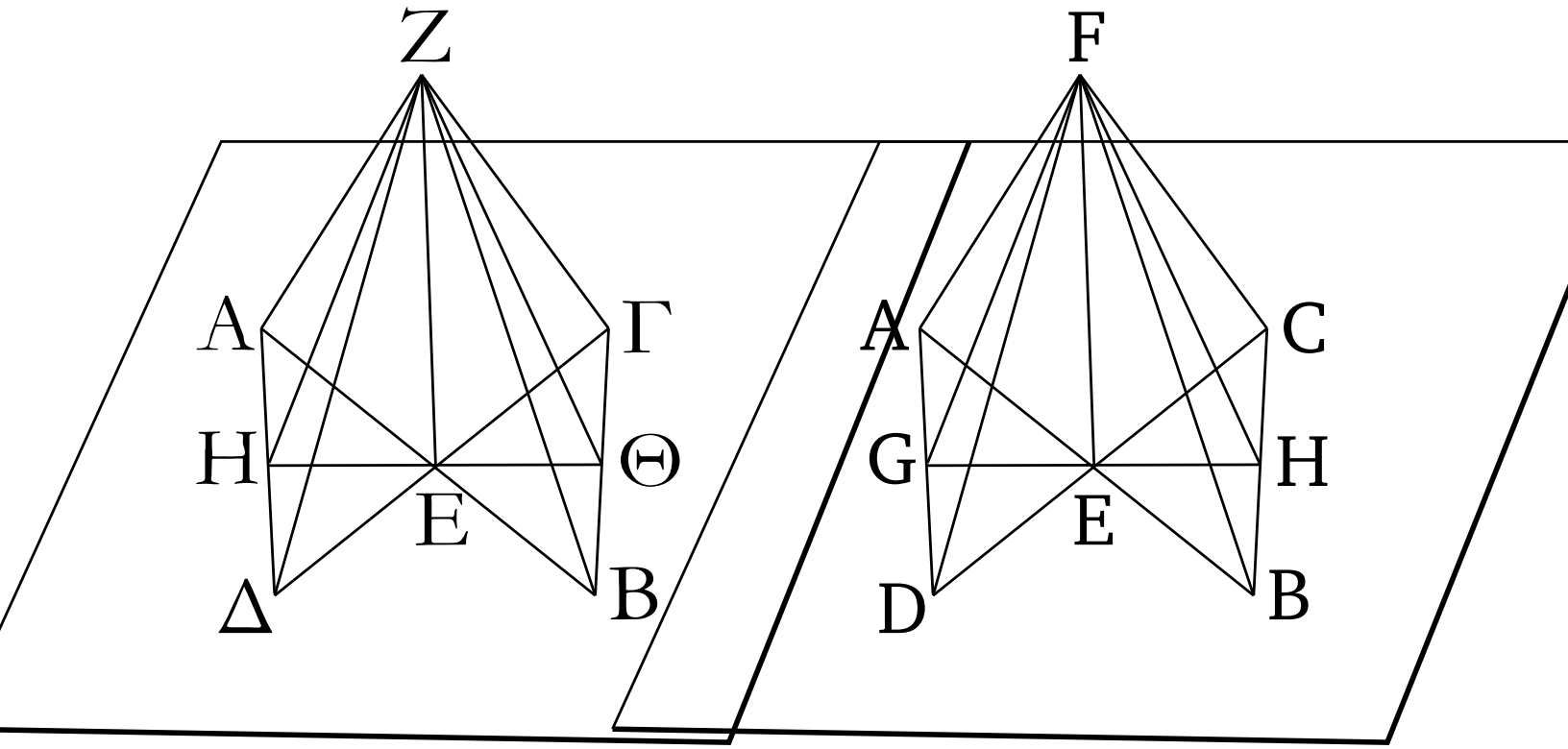
Ἀπειλήφθωσαν γὰρ αἱ ΑΕ, ΕΒ, ΓΕ, ΕΔ ἴσαι ἀλλήλαις, καὶ διήχθω τις διὰ τοῦ Ε, ὡξὀέτυθεν, ἡ ΗΕΘ, καὶ ἐπεζεύχθωσαν αἱ ΑΔ, ΓΒ, καὶ ὀέτι ἀπὸ τυχόντος τοῦ Ζ ἐπεζεύχθωσαν αἱ ΖΑ, ΖΗ, ΖΔ, ΖΓ, ΖΘ, ΖΒ.

Proposition 4

If a straight-line is set up at right-angles to two straight-lines cutting one another, at the common point of section, then it will also be at right-angles to the plane (passing) through them (both).

For let some straight-line EF have (been) set up at right-angles to two straight-lines, AB and CD, cutting one another at point E, at E. I say that EF is also at right-angles to the plane (passing) through AB and CD.

For let AE, EB, CE and ED be cut off from (the two straight-lines so as to be) equal to one another. And let GEH be drawn, at random, through E (in the plane passing through AB and CD). And let AD and CB be joined. And, furthermore, let FA, FG, FD, FC, FH, and FB be joined from the random (point) F (on EF).



Καὶ ἐπεὶ δύο αἱ ΑΕ, ΕΔ δυσὶ ταῖς ΓΕ, ΕΒ ἴσαι εἰσὶ καὶ

For since the two (straight-lines) AE and ED are equal to

γωνίας ἴσας περιέθουσιν, βάσις ἄρα ἡ $A\Delta$ βάσει τῇ ΓBὁίση ἐστίν, καὶ τὸ $AE\Delta$ τρίγωνον τῷ ΓEB τριγώνῳ ἴσονὁέσται· ὁώστε καὶ γωνία ἡ ὑπὸ ΔAE γωνίᾳ τῇ ὑπὸ $EB\Gamma$ὁίση [ἐστίν]. ὁέστι δὲ καὶ ἡ ὑπὸ AEH γωνία τῇ ὑπὸ $BE\Theta$ὁίση. δύο δὴ τρίγωνά ἐστι τὰ AHE, $BE\Theta$ τὰξ δύο γωνίας δυσὶ γωνίαις ἴσας ἔχοντα ἑκατέραν ἑκατέρᾳ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην τὴν πρὸς ταῖς ἴσαις γωνίαις τὴν AE τῇ EB· καὶ τὰξ λοιπὰξ ἄρα πλευρὰξ ταῖς λοιπαῖξ πλευραῖξ ἴσας ὁέχουσιν. ὁίση ἄρα ἡ μὲν HE τῇ $E\Theta$, ἡ δὲ AH τῇ $B\Theta$. καὶ ἐπεὶ ὁίση ἐστὶν ἡ AE τῇ EB, κοινὴ δὲ καὶ πρὸς ὀρθὰξ ἡ ZE, βάσις ἄρα ἡ ZA βάσει τῇ ZB ἐστινὁίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ $Z\Gamma$ τῇ $Z\Delta$ ἐστινὁίση. καὶ ἐπεὶ ὁίση ἐστὶν ἡ $A\Delta$ τῇ ΓB, ὁέστι δὲ καὶ ἡ ZA τῇ ZBὁίση, δύο δὴ αἱ ZA, $A\Delta$ δυσὶ ταῖς ZB, $B\Gamma$ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ βάσις ἡ $Z\Delta$ βάσει τῇ $Z\Gamma$ ἐδείθηὁίση· καὶ γωνία ἄρα ἡ ὑπὸ $ZA\Delta$ γωνίᾳ τῇ ὑπὸ $ZB\Gamma$ὁίση ἐστίν. καὶ ἐπεὶ πάλιν ἐδείθη ἡ AH τῇ $B\Theta$ὁίση, ἀλλὰ μὴν καὶ ἡ ZA τῇ ZBὁίση, δύο δὴ αἱ ZA, AH δυσὶ ταῖς ZB, $B\Theta$ ἴσαι εἰσίν. καὶ γωνία ἡ ὑπὸ ZAH ἐδείθηὁίση τῇ ὑπὸ $ZB\Theta$· βάσις ἄρα ἡ ZH βάσει τῇ $Z\Theta$ ἐστινὁίση. καὶ ἐπεὶ πάλινὁίση ἐδείθη ἡ HE τῇ $E\Theta$, κοινὴ δὲ ἡ EZ, δύο δὴ αἱ HE, EZ δυσὶ ταῖς ΘE, EZ ἴσαι εἰσίν· καὶ βάσις ἡ ZH βάσει τῇ $Z\Theta$ὁίση· γωνία ἄρα ἡ ὑπὸ HEZ γωνίᾳ τῇ ὑπὸ ΘEZὁίση ἐστίν. ὀρθὴ ἄρα ἑκατέρα τῶν ὑπὸ HEZ, ΘEZ γωνιῶν. ἡ ZE ἄρα πρὸς τὴν $H\Theta$ τυθόντως διὰ τοῦ E ἀθθεῖσαν ὀρθὴ ἐστιν. ὁμοίως δὴ δείχομεν, ὅτι ἡ ZE καὶ πρὸς πάσας τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐόυσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας. εὐθεῖα δὲ πρὸς ἐπίπεδον ὀρθή ἐστιν, ὅταν πρὸς πάσας τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐόυσας ἐν τῷ αὐτῷ ἐπιπέδῳ ὀρθὰξ ποιῇ γωνίας· ἡ ZE ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. τὸ δὲ ὑποκείμενον ἐπίπεδόν ἐστι τὸ διὰ τῶν AB, $\Gamma\Delta$ εὐθειῶν. ἡ ZE ἄρα πρὸς ὀρθάξ ἐστι τῷ διὰ τῶν AB, $\Gamma\Delta$ ἐπιπέδῳ.

Ἐὰν ἄρα εὐθεῖα δύο εὐθείαις τεμνούσαις ἀλλήλας πρὸς ὀρθὰξ ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, καὶ τῷ διὰ αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰξὁέσται· ὅπερ ἔδει δεῖξαι.

the two (straight-lines) CE and EB, and they enclose equal angles [Prop. 1.15], the base AD is thus equal to the base CB, and triangle AED will be equal to triangle CEB [Prop. 1.4]. Hence, the angle DAE [is] equal to the angle EBC. And the angle AEG (is) also equal to the angle BEH [Prop. 1.15]. So AGE and BEH are two triangles having two angles equal to two angles, respectively, and one side equal to one side—(namely), those by the equal angles, AE and EB. Thus, they will also have the remaining sides equal to the remaining sides [Prop. 1.26]. Thus, GE (is) equal to EH, and AG to BH. And since AE is equal to EB, and FE is common and at right-angles, the base FA is thus equal to the base FB [Prop. 1.4]. So, for the same (reasons), FC is also equal to FD. And since AD is equal to CB, and FA is also equal to FB, the two (straight-lines) FA and AD are equal to the two (straight-lines) FB and BC, respectively. And the base FD was shown (to be) equal to the base FC. Thus, the angle FAD is also equal to the angle FBC [Prop. 1.8]. And, again, since AG was shown (to be) equal to BH, but FA (is) also equal to FB, the two (straight-lines) FA and AG are equal to the two (straight-lines) FB and BH (respectively). And the angle FAG was shown (to be) equal to the angle FBH. Thus, the base FG is equal to the base FH [Prop. 1.4]. And, again, since GE was shown (to be) equal to EH, and EF (is) common, the two (straight-lines) GE and EF are equal to the two (straight-lines) HE and EF (respectively). And the base FG (is) equal to the base FH. Thus, the angle GEF is equal to the angle HEF [Prop. 1.8]. Each of the angles GEF and HEF (are) thus right-angles [Def. 1.10]. Thus, FE is at right-angles to GH, which was drawn at random through E (in the reference plane passing though AB and AC). So, similarly, we can show that FE will make right-angles with all straight-lines joined to it which are in the reference plane. And a straight-line is at right-angles to a plane when it makes right-angles with all straight-lines joined to it which are in the plane [Def. 11.3]. Thus, FE is at right-angles to the reference plane. And the reference plane is that (passing) through the straight-lines AB and CD. Thus, FE is at right-angles to the plane (passing) through AB and CD.

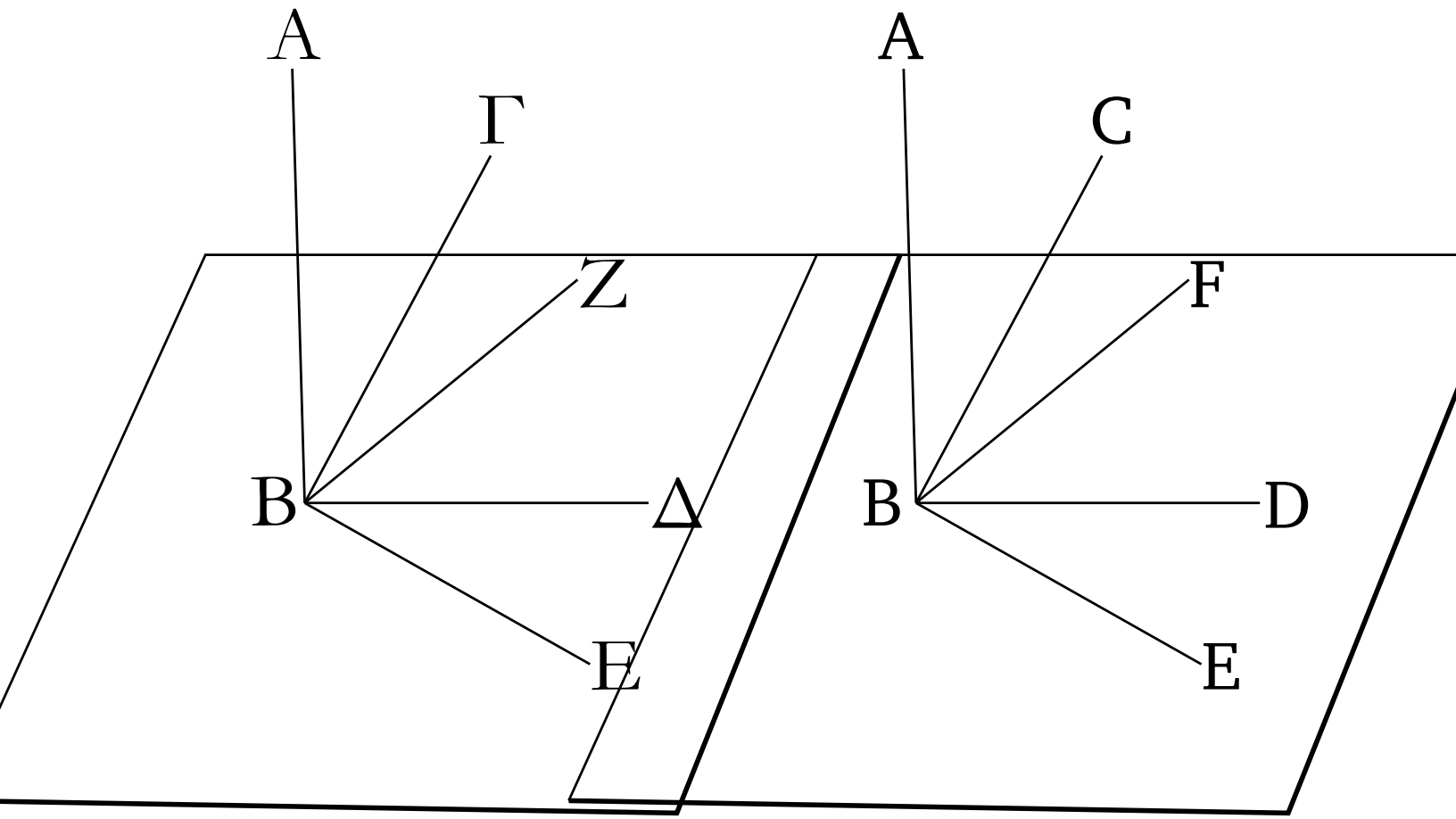
Thus, if a straight-line is set up at right-angles to two straight-lines cutting one another, at the common point of section, then it will also be at right-angles to the plane (passing) through them (both). (Which is) the very thing it was required to show.

5.

Ἐὰν εὐθεῖα τρισὶν εὐθείαις ἁπτομέναις ἀλλήλων πρὸς ὀρθὰξ ἐπὶ τῆς κοινῆς τομῆς ἐπισταθῇ, αἱ τρεῖς εὐθεῖαι ἐν ἑνί εἰσιν ἐπιπέδῳ.

Proposition 5

If a straight-line is set up at right-angles to three straight-lines cutting one another, at the common point of section, then the three straight-lines are in one plane.



Εὐθεῖα γάρ τις ἡ ΑΒ τρισὶν εὐθείαις ταῖς ΒΓ, ΒΔ, ΒΕ πρὸς ὀρθὰξ ἐπὶ τῆς κατὰ τὸ Β ἁφῆς ἐφεστάτω· λέγω, ὅτι αἱ ΒΓ, ΒΔ, ΒΕ ἐν ἑνί εἰσιν ἐπιπέδῳ.

μὴ γάρ, ἀλλὰ εἰ δυνατόν, ὀέστωσαν αἱ μὲν ΒΔ, ΒΕ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ, ἡ δὲ ΒΓ ἐν μετεωροτέρῳ, καὶ ἐκβεβλήσθω τὸ διὰ τῶν ΑΒ, ΒΓ ἐπίπεδον· κοινὴν δὴ τομὴν ποιήσει ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιείτω τὴν ΒΖ. ἐν ἑνὶ ἄρα εἰσὶν ἐπιπέδῳ τῷ διηγμένῳ διὰ τῶν ΑΒ, ΒΓ αἱ τρεῖς εὐθεῖαι αἱ ΑΒ, ΒΓ, ΒΖ. καὶ ἐπεὶ ἡ ΑΒ ὀρθή ἐστι πρὸς ἑκατέραν τῶν ΒΔ, ΒΕ, καὶ τῷ διὰ τῶν ΒΔ, ΒΕ ἄρα ἐπιπέδῳ ὀρθή ἐστιν ἡ ΑΒ. τὸ δὲ διὰ τῶν ΒΔ, ΒΕ ἐπίπεδον τὸ ὑποκείμενόν ἐστιν· ἡ ΑΒ ἄρα ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον. ὥστε καὶ πρὸς πάσας τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας ἡ ΑΒ. ὁάπτεται δὲ αὐτῆς ἡ ΒΖ οˆσα ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ· ἡ ἄρα ὑπὸ ΑΒΖ γωνία ὀρθή ἐστιν. ὑπόκειται δὲ καὶ ἡ ὑπὸ ΑΒΓ ὀρθή· ὁίση ἄρα ἡ ὑπὸ ΑΒΖ γωνία τῇ ὑπὸ ΑΒΓ. καί εἰσιν ἐν ἑνὶ ἐπιπέδῳ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ ΒΓ εὐθεῖα ἐν μετεωροτέρῳ ἐστὶν ἐπιπέδῳ· αἱ τρεῖς ἄρα εὐθεῖαι αἱ ΒΓ, ΒΔ, ΒΕ ἐν ἑνί εἰσιν ἐπιπέδῳ.

Ἐὰν ἄρα εὐθεῖα τρισὶν εὐθείαις ἁπτομέναις ἀλλήλων ἐπὶ τῆς ἁφῆς πρὸς ὀρθὰξ ἐπισταθῇ, αἱ τρεῖς εὐθεῖαι ἐν ἑνί εἰσιν

For let some straight-line *AB* be set up at right-angles to three straight-lines *BC*, *BD*, and *BE*, at the (common) point of section *B*. I say that *BC*, *BD*, and *BE* are in one plane.

For (if) not, and if possible, let *BD* and *BE* be in the reference plane, and *BC* in a more elevated (plane). And let the plane through *AB* and *BC* be produced. So it will make a straight-line as a common section with the reference plane [Def. 11.3]. Let it make *BF*. Thus, the three straight-lines *AB*, *BC*, and *BF* are in one plane—(namely), that drawn through *AB* and *BC*. And since *AB* is at right-angles to each of *BD* and *BE*, *AB* is thus also at right-angles to the plane (passing) through *BD* and *BE* [Prop. 11.4]. And the plane (passing) through *BD* and *BE* is the reference plane. Thus, *AB* is at right-angles to the reference plane. Hence, *AB* will also make right-angles with all straight-lines joined to it which are also in the reference plane [Def. 11.3]. And *BF*, which is in the reference plane, is joined to it. Thus, the angle *ABF* is a right-angle. And *ABC* was also assumed to be a right-angle. Thus, angle *ABF* (is) equal to *ABC*. And they are in one plane. The very thing is impossible. Thus, *BC* is not in a more elevated plane. Thus, the three straight-lines *BC*, *BD*, and *BE* are in one plane.

ἐπιπέδῳ· ὅπερ ἔδει δεῖχαι.

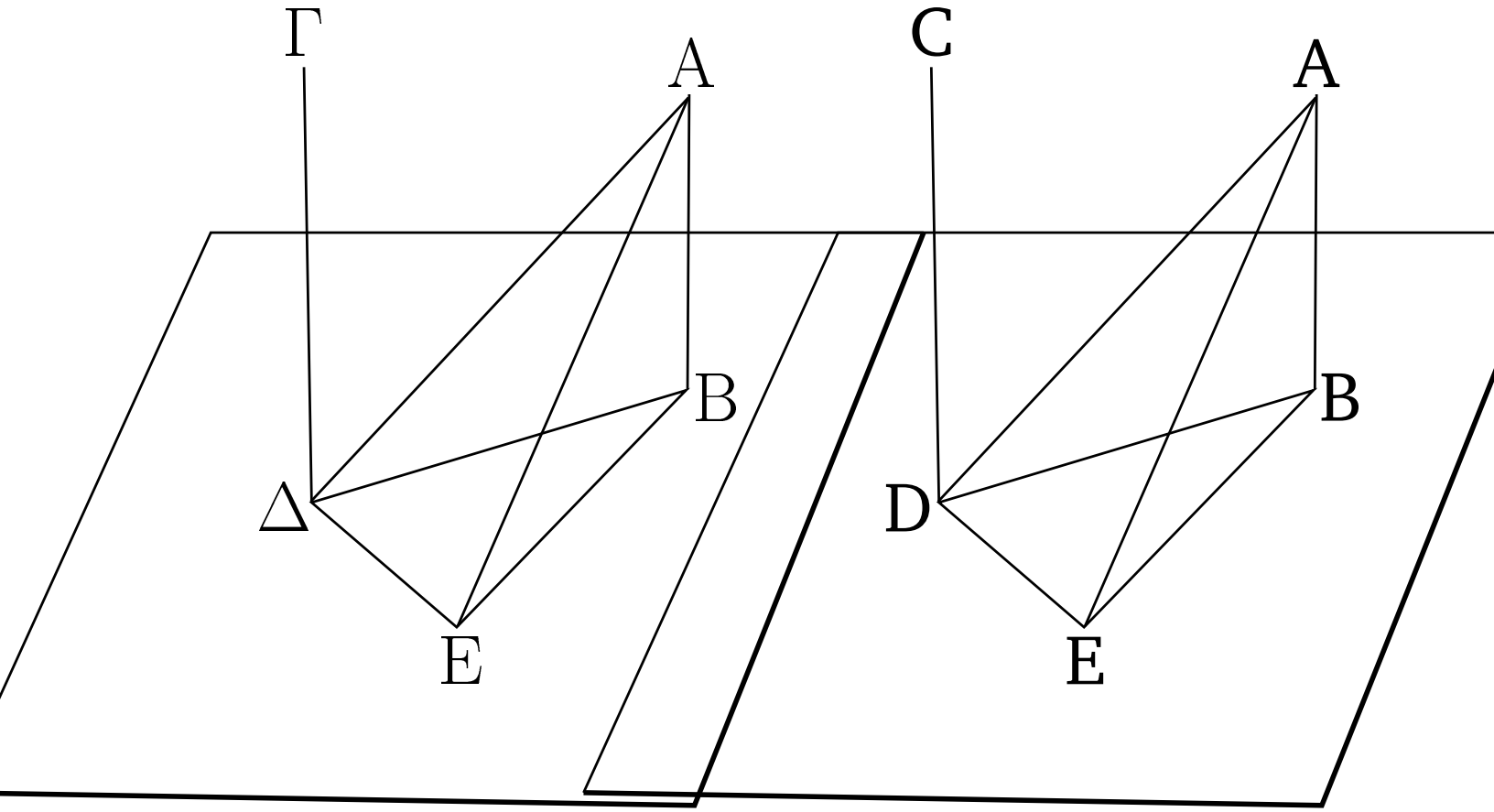
Thus, if a straight-line is set up at right-angles to three straight-lines cutting one another, at the (common) point of section, then the three straight-lines are in one plane. (Which is) the very thing it was required to show.

6.

Ἐὰν δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξ ὦσιν, παράλληλοιὀέσονται αἱ εὐθεῖαι.

Proposition 6

If two straight-lines are at right-angles to the same plane then the straight-lines will be parallel.†



Δύο γὰρ εὐθεῖαι αἱ ΑΒ, ΓΔ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξὀέστωσαν· λέγω, ὅτι παράλληλόξ ἐστιν ἡ ΑΒ τῇ ΓΔ.

Συμβαλλέτωσαν γὰρ τῷ ὑποκειμένῳ ἐπιπέδῳ κατὰ τὰ Β, Δ σημεῖα, καὶ ἐπεζεύθθω ἡ ΒΔ εὐθεῖα, καὶ ἤθθω τῇ ΒΔ πρὸς ὀρθὰξ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ἡ ΔΕ, καὶ κείσθω τῇ ΑΒὀίση ἡ ΔΕ, καὶ ἐπεζεύθθωσαν αἱ ΒΕ, ΑΕ, ΑΔ.

Καὶ ἐπεὶ ἡ ΑΒ ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσας [ἄρα] τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας. ὁάπτεται δὲ τῆς ΑΒ ἑκατέρα τῶν ΒΔ, ΒΕ οὐῦσα ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ· ὀρθὴ ἄρα ἐστὶν ἑκατέρα τῶν ὑπὸ ΑΒΔ, ΑΒΕ γωνιῶν. διὰ τὰ αὐτὰ δὴ καὶ ἑκατέρα τῶν ὑπὸ ΓΔΒ, ΓΔΕ ὀρθή ἐστιν. καὶ ἐπεὶ ὀίση ἐστὶν ἡ ΑΒ τῇ ΔΕ, κοινὴ δὲ ἡ ΒΔ, δύο δὴ αἱ ΑΒ, ΒΔ δυσὶ ταῖς ΕΔ, ΔΒ ἴσαι εἰσίν·

For let the two straight-lines *AB* and *CD* be at right-angles to a reference plane. I say that *AB* is parallel to *CD*.

For let them meet the reference plane at points *B* and *D* (respectively). And let the straight-line *BD* be joined. And let *DE* be drawn at right-angles to *BD* in the reference plane. And let *DE* be made equal to *AB*. And let *BE*, *AE*, and *AD* be joined.

And since *AB* is at right-angles to the reference plane, it will [thus] also make right-angles with all straight-lines joined to it which are in the reference plane [Def. 11.3]. And *BD* and *BE*, which are in the reference plane, are each joined to *AB*. Thus, each of the angles *ABD* and *ABE* are right-angles. So, for the same (reasons), each of the angles *CDB* and *CDE* are also right-angles. And since *AB* is equal to *DE*, and *BD*

καὶ γωνίας ὀρθὰξ περιέθουσιν· βάσις ἄρα ἡ $A\Delta$ βάσει τῇ BE ἐστινὀίση. καὶ ἐπεὶ ὀίση ἐστὶν ἡ AB τῇ ΔE, ἀλλὰ καὶ ἡ $A\Delta$ τῇ BE, δύο δὴ αἱ AB, BE δυσὶ ταῖς $E\Delta$, ΔA ἴσαι εἰσίν· καὶ βάσις αὐτῶν κοινὴ ἡ AE· γωνία ἄρα ἡ ὑπὸ ABE γωνιᾷ τῇ ὑπὸ $E\Delta A$ ἐστινὀίση. ὀρθὴ δὲ ἡ ὑπὸ ABE· ὀρθὴ ἄρα καὶ ἡ ὑπὸ $E\Delta A$· ἡ $E\Delta$ ἄρα πρὸς τὴν ΔA ὀρθή ἐστιν. ὀέστι δὲ καὶ πρὸς ἑκατέραν τῶν $B\Delta$, $\Delta\Gamma$ ὀρθή. ἡ $E\Delta$ ἄρα τρισὶν εὐθείαις ταῖς $B\Delta$, ΔA, $\Delta\Gamma$ πρὸς ὀρθὰξ ἐπὶ τῆς ἁφῆς ἐφέστηκεν· αἱ τρεῖς ἄρα εὐθεῖαι αἱ $B\Delta$, ΔA, $\Delta\Gamma$ ἐν ἑνί εἰσιν ἐπιπέδῳ. ἐν ὁῷ δὲ αἱ ΔB, ΔA, ἐν τούτῳ καὶ ἡ AB· πᾶν γὰρ τρίγωνον ἐν ἑνί ἐστιν ἐπιπέδῳ· αἱ ἄρα AB, $B\Delta$, $\Delta\Gamma$ εὐθεῖαι ἐν ἑνί εἰσιν ἐπιπέδῳ. καί ἐστιν ὀρθὴ ἑκατέρα τῶν ὑπὸ $AB\Delta$, $B\Delta\Gamma$ γωνιῶν· παράλληλος ἄρα ἐστὶν ἡ AB τῇ $\Gamma\Delta$.

Ἐὰν ἄρα δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξ ὦσιν, παράλληλοιὀέσονται αἱ εὐθεῖαι· ὅπερ ἔδει δεῖχαι.

(is) common, the two (straight-lines) AB and BD are equal to the two (straight-lines) ED and DB (respectively). And they contain right-angles. Thus, the base AD is equal to the base BE [Prop. 1.4]. And since AB is equal to DE, and AD (is) also (equal) to BE, the two (straight-lines) AB and BE are thus equal to the two (straight-lines) ED and DA (respectively). And their base AE (is) common. Thus, angle ABE is equal to angle EDA [Prop. 1.8]. And ABE (is) a right-angle. Thus, EDA (is) also a right-angle. ED is thus at right-angles to DA. And it is also at right-angles to each of BD and DC. Thus, ED is standing at right-angles to the three straight-lines BD, DA, and DC at the (common) point of section. Thus, the three straight-lines BD, DA, and DC are in one plane [Prop. 11.5]. And in which(ever) plane DB and DA (are found), in that (plane) AB (will) also (be found). For every triangle is in one plane [Prop. 11.2]. And each of the angles ABD and BDC is a right-angle. Thus, AB is parallel to CD [Prop. 1.28].

Thus, if two straight-lines are at right-angles to the same plane then the straight-lines will be parallel. (Which is) the very thing it was required to show.

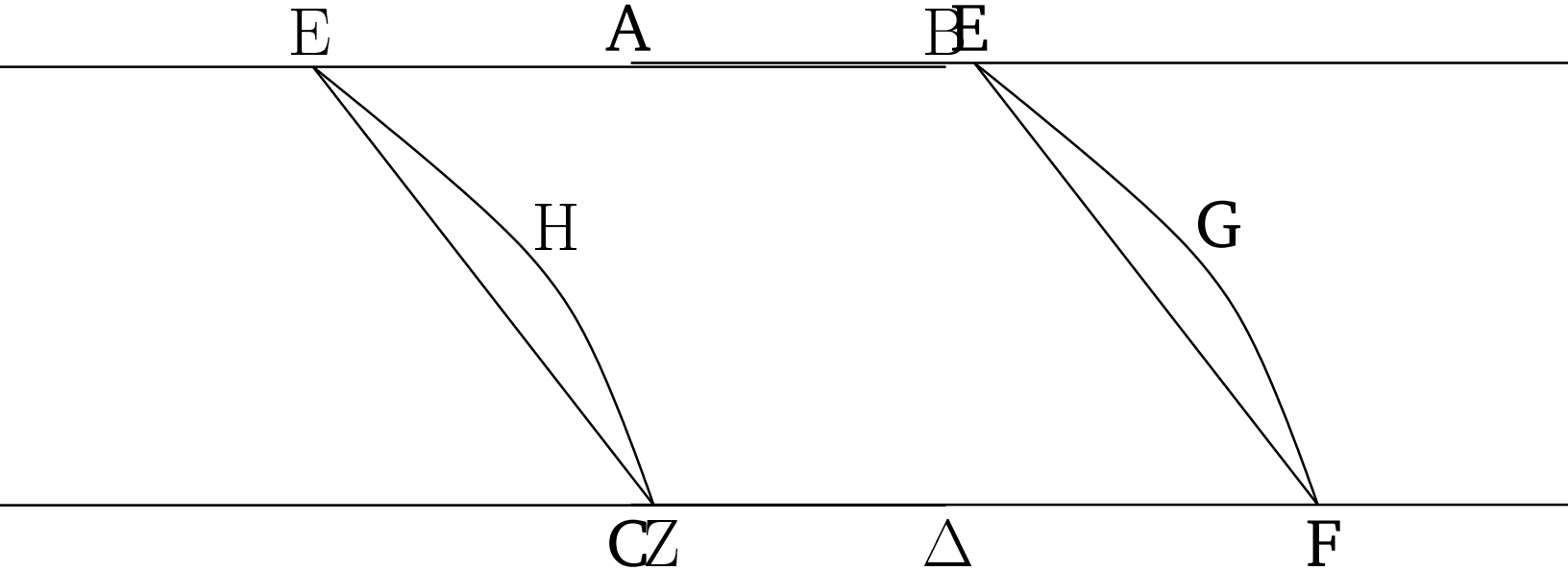
† In other words, the two straight-lines lie in the same plane, and never meet when produced in either direction.

7.

Ἐὰνὀῶσι δύο εὐθεῖαι παράλληλοι, ληφθῇ δὲ ἐφ' ἑκατέρας αὐτῶν τυθόντα σημεῖα, ἡ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις.

Proposition 7

If there are two parallel straight-lines, and random points are taken on each of them, then the straight-line joining the two points is in the same plane as the parallel (straight-lines).



Ἔστωσαν δύο εὐθεῖαι παράλληλοι αἱ AB, $\Gamma\Delta$, καὶ εἰλήφθω ἐφ' ἑκατέρας αὐτῶν τυνθόντα σημεῖα τὰ E, Z· λέγω, ὅτι ἡ ἐπὶ τὰ E, Z σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις.

μὴ γάρ, ἀλλὰ εἰ δυνατόν, ὀέστω ἐν μετεωροτέρῳ ὡξ ἡ E-

Let AB and CD be two parallel straight-lines, and let the random points E and F be taken on each of them (respectively). I say that the straight-line joining points E and F is in the same (reference) plane as the parallel (straight-lines).

For (if) not, and if possible, let it be in a more elevated

HZ, καὶ διήθθω διὰ τῆς EHZ ἐπίπεδον· τομὴν δὴ ποιήσει ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιείτω ὡξ τὴν EZ· δύο ἄρα εὐθεῖαι αἱ EHZ, EZ θωρίον περιέχουσιν· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἡ ἀπὸ τοῦ E ἐπὶ τὸ Z ἐπιζευγνυμένη εὐθεῖαι ἐν μετεωροτέρῳ ἐστὶν ἐπιπέδῳ· ἐν τῷ διὰ τῶν AB, ΓB ἄρα παραλλήλων ἐστὶν ἐπιπέδῳ ἡ ἀπὸ τοῦ E ἐπὶ τὸ Z ἐπιζευγνυμένη εὐθεῖα.

Ἐὰν ἄραὸῶσι δύο εὐθεῖαι παράλληλοι, ληφθῇ δὲ ἐφ΄ ἑκατέρας αὐτῶν τυθόντα σημεῖα, ἡ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις· ὅπερ ἔδει δεῖχαι.

8.

Ἐὰνὸῶσι δύο εὐθεῖαι παράλληλοι, ἡ δὲ ἑτέρα αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰξὸῇ, καὶ ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὸέσται.

Ἔστωσαν δύο εὐθεῖαι παράλληλοι αἱ AB, ΓΔ, ἡ δὲ ἑτέρα αὐτῶν ἡ AB τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξὸέστω· λέγω, ὅτι καὶ ἡ λοιπὴ ἡ ΓΔ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὸέσται.

Συμβαλλέτωσαν γὰρ αἱ AB, ΓΔ τῷ ὑποκειμένῳ ἐπιπέδῳ κατὰ τὰ B, Δ σημεῖα, καὶ ἐπεζεύθθω ἡ BΔ· αἱ AB, ΓΔ, BΔ ἄρα ἐν ἑνί εἰσιν ἐπιπέδῳ. ἤθθω τῇ BA πρὸς ὀρθὰξ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ἡ ΔE, καὶ κείσθω τῇ ABὸίση ἡ ΔE, καὶ ἐπεζεύθθωσαν αἱ BE, AE, AΔ.

(plane), such as EGF. And let a plane be drawn through EGF. So it will make a straight cutting in the reference plane [Prop. 11.3]. Let it make EF. Thus, two straight-lines (with the same end-points), EGF and EF, will enclose an area. The very thing is impossible. Thus, the straight-line joining *E* to *F* is not in a more elevated plane. The straight-line joining *E* to *F* is thus in the plane through the parallel (straight-lines) *AB* and *CD*.

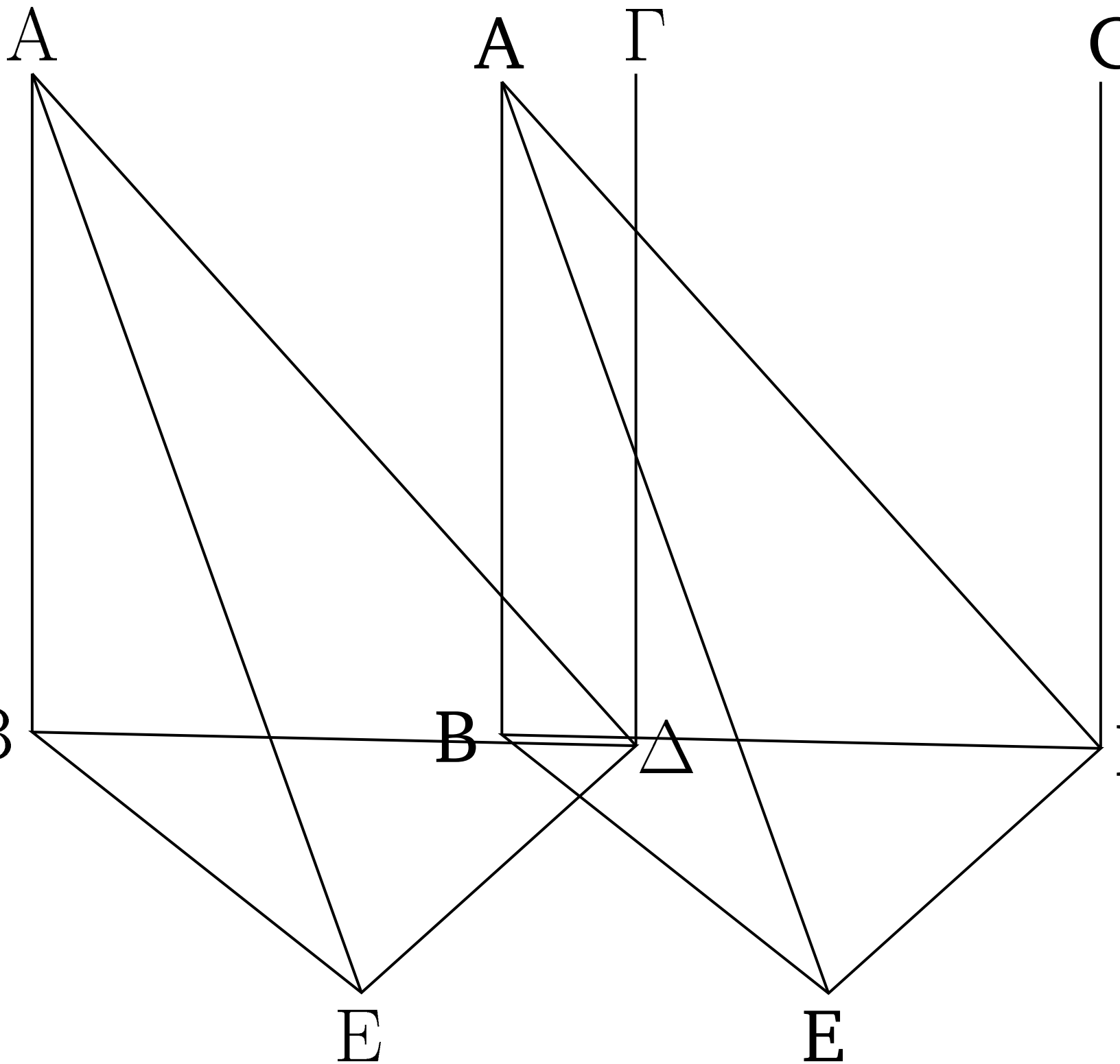
Thus, if there are two parallel straight-lines, and random points are taken on each of them, then the straight-line joining the two points is in the same plane as the parallel (straight-lines). (Which is) the very thing it was required to show.

Proposition 8

If two straight-lines are parallel, and one of them is at right-angles to some plane, then the remaining (one) will also be at right-angles to the same plane.

Let *AB* and *CD* be two parallel straight-lines, and let one of them, *AB*, be at right-angles to a reference plane. I say that the remaining (one), *CD*, will also be at right-angles to the same plane.

For let *AB* and *CD* meet the reference plane at points *B* and *D* (respectively). And let *BD* be joined. *AB*, *CD*, and *BD* are thus in one plane [Prop. 11.7]. Let *DE* be drawn at right-angles to *BD* in the reference plane, and let *DE* be made equal to *AB*, and let *BE*, *AE*, and *AD* be joined.



Καὶ ἐπεὶ ἡ AB ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας

And since *AB* is at right-angles to the reference plane, *AB* is thus also at right-angles to all of the straight-lines joined

ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν ἡ AB· ὀρθὴ ἄρα [ἐστὶν] ἑκατέρα τῶν ὑπὸ $AB\Delta$, ABE γωνιῶν. καὶ ἐπεὶ εἰξ παραλλήλους τὰξ AB, $\Gamma\Delta$ εὐθεῖα ἐμπέπτωκεν ἡ $B\Delta$, αἱ ἄρα ὑπὸ $AB\Delta$, $\Gamma\Delta B$ γωνίαι δυσὶν ὀρθαῖξ ἴσαι εἰσίν. ὀρθὴ δὲ ἡ ὑπὸ $AB\Delta$· ὀρθὴ ἄρα καὶ ἡ ὑπὸ $\Gamma\Delta B$· ἡ $\Gamma\Delta$ ἄρα πρὸς τὴν $B\Delta$ ὀρθή ἐστιν. καὶ ἐπεὶ ὁίση ἐστὶν ἡ AB τῇ ΔE, κοινὴ δὲ ἡ $B\Delta$, δύο δὴ αἱ AB, $B\Delta$ δυσὶ ταῖς $E\Delta$, ΔB ἴσαι εἰσίν· καὶ γωνία ἡ ὑπὸ $AB\Delta$ γωνίᾳ τῇ ὑπὸ $E\Delta B$ὁίση· ὀρθὴ γὰρ ἑκατέρα· βάσις ἄρα ἡ $A\Delta$ βάσει τῇ BEὁίση. καὶ ἐπεὶ ὁίση ἐστὶν ἡ μὲν AB τῇ ΔE, ἡ δὲ BE τῇ $A\Delta$, δύο δὴ αἱ AB, BE δυσὶ ταῖς $E\Delta$, ΔA ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ. καὶ βάσις αὐτῶν κοινὴ ἡ AE· γωνία ἄρα ἡ ὑπὸ ABE γωνίᾳ τῇ ὑπὸ $E\Delta A$ ἐστινὁίση. ὀρθὴ δὲ ἡ ὑπὸ ABE· ὀρθὴ ἄρα καὶ ἡ ὑπὸ $E\Delta A$· ἡ $E\Delta$ ἄρα πρὸς τὴν $A\Delta$ ὀρθή ἐστιν. ὀέστι δὲ καὶ πρὸς τὴν ΔB ὀρθή· ἡ $E\Delta$ ἄρα καὶ τῷ διὰ τῶν $B\Delta$, ΔA ἐπιπέδῳ ὀρθή ἐστιν. καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ διὰ τῶν $B\Delta A$ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας ἡ $E\Delta$. ἐν δὲ τῷ διὰ τῶν $B\Delta A$ ἐπιπέδῳ ἐστὶν ἡ $\Delta\Gamma$, ἐπειδήπερ ἐν τῷ διὰ τῶν $B\Delta A$ ἐπιπέδῳ ἐστὶν αἱ AB, $B\Delta$, ἐν ὁῷ δὲ αἱ AB, $B\Delta$, ἐν τούτῳ ἐστὶ καὶ ἡ $\Delta\Gamma$. ἡ $E\Delta$ ἄρα τῇ $\Delta\Gamma$ πρὸς ὀρθάξ ἐστιν· ὁώστε καὶ ἡ $\Gamma\Delta$ τῇ ΔE πρὸς ὀρθάξ ἐστιν. ὀέστι δὲ καὶ ἡ $\Gamma\Delta$ τῇ $B\Delta$ πρὸς ὀρθάξ. ἡ $\Gamma\Delta$ ἄρα δύο εὐθείαις τεμνούσαις ἀλλήλας ταῖς ΔE, ΔB ἀπὸ τῆς κατὰ τὸ Δ τομῆς πρὸς ὀρθὰξ ἐφέστηκεν· ὁώστε ἡ $\Gamma\Delta$ καὶ τῷ διὰ τῶν ΔE, ΔB ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. τὸ δὲ διὰ τῶν ΔE, ΔB ἐπίπεδον τὸ ὑποκείμενόν ἐστιν· ἡ $\Gamma\Delta$ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν.

Ἐὰν ἄραὁῶσι δύο εὐθεῖαι παράλληλοι, ἡ δὲ μία αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰξὁῇ, καὶ ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὁέσται· ὅπερ ἔδει δεῖχαι.

to it which are in the reference plane [Def. 11.3]. Thus, the angles ABD and ABE [are] each right-angles. And since the straight-line BD has met the parallel (straight-lines) AB and CD, the (sum of the) angles ABD and CDB is thus equal to two right-angles [Prop. 1.29]. And ABD (is) a right-angle. Thus, CDB (is) also a right-angle. CD is thus at right-angles to BD. And since AB is equal to DE, and BD (is) common, the two (straight-lines) AB and BD are equal to the two (straight-lines) ED and DB (respectively). And angle ABD (is) equal to angle EDB. For each (is) a right-angle. Thus, the base AD (is) equal to the base BE [Prop. 1.4]. And since AB is equal to DE, and BE to AD, the two (sides) AB, BE are equal to the two (sides) ED, DA, respectively. And their base AE is common. Thus, angle ABE is equal to angle EDA [Prop. 1.8]. And ABE (is) a right-angle. EDA (is) thus also a right-angle. Thus, ED is at right-angles to AD. And it is also at right-angles to DB. Thus, ED is also at right-angles to the plane through BD and DA [Prop. 11.4]. And ED will thus make right-angles with all of the straight-lines joined to it which are also in the plane through BDA. And DC is in the plane through BDA, inasmuch as AB and BD are in the plane through BDA [Prop. 11.2], and in which(ever plane) AB and BD (are found), DC is also (found). Thus, ED is at right-angles to DC. Hence, CD is also at right-angles to DE. And CD is also at right-angles to BD. Thus, CD is standing at right-angles to two straight-lines, DE and DB, which meet one another, at the (point) of section, D. Hence, CD is also at right-angles to the plane through DE and DB [Prop. 11.4]. And the plane through DE and DB is the reference (plane). CD is thus at right-angles to the reference plane.

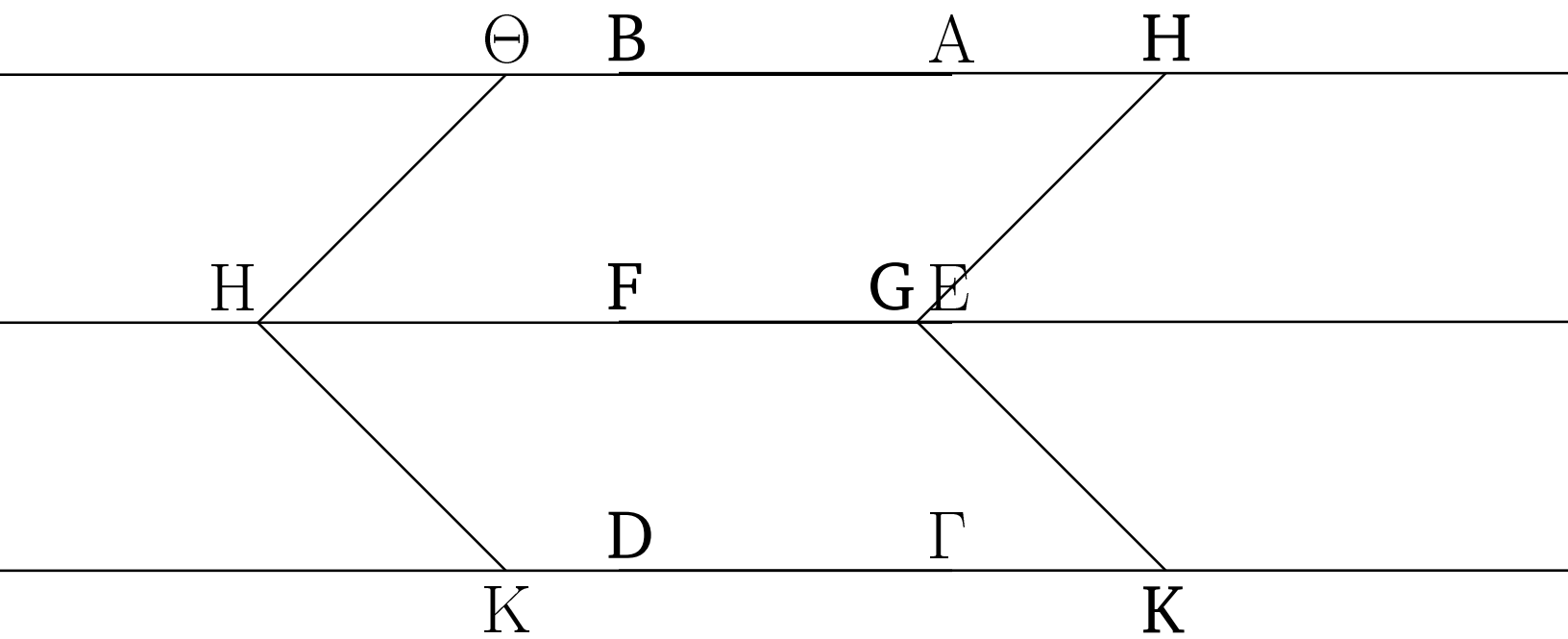
Thus, if two straight-lines are parallel, and one of them is at right-angles to some plane, then the remaining (one) will also be at right-angles to the same plane. (Which is) the very thing it was required to show.

9.

Αἱ τῇ αὐτῇ εὐθείᾳ παράλληλοι καὶ μὴ οὐῦσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ καὶ ἀλλήλαις εἰσὶ παράλληλοι.

Proposition 9

(Straight-lines) parallel to the same straight-line, and which are not in the same plane as it, are also parallel to one another.



Ἔστω γὰρ ἑκατέρα τῶν AB, ΓΔ τῇ EZ παράλληλος μὴ οὐσῦσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ· λέγω, ὅτι παράλληλόξ ἐστιν ἡ AB τῇ ΓΔ.

Εἰλήφθω γὰρ ἐπὶ τῆς EZ τυθὸν σημεῖον τὸ H, καὶ ἀπὸ αὐτοῦ τῇ EZ ἐν μὲν τῷ διὰ τῶν EZ, AB ἐπιπέδῳ πρὸς ὀρθὰξ ἤθθω ἡ HΘ, ἐν δὲ τῷ διὰ τῶν ZE, ΓΔ τῇ EZ πάλιν πρὸς ὀρθὰξ ἤθθω ἡ HK.

Καὶ ἐπεὶ ἡ EZ πρὸς ἑκατέραν τῶν HΘ, HK ὀρθή ἐστιν, ἡ EZ ἄρα καὶ τῷ διὰ τῶν HΘ, HK ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. καί ἐστιν ἡ EZ τῇ AB παράλληλος· καὶ ἡ AB ἄρα τῷ διὰ τῶν ΘHK ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΓΔ τῷ διὰ τῶν ΘHK ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν· ἑκατέρα ἄρα τῶν AB, ΓΔ τῷ διὰ τῶν ΘHK ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. ἔαν δὲ δύο εὐθεῖαι τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθάξ ὦσιν, παράλληλοί εἰσιν αἱ εὐθεῖαι· παράλληλος ἄρα ἐστὶν ἡ AB τῇ ΓΔ· ὅπερ ἔδει δεῖχαι.

For let *AB* and *CD* each be parallel to *EF*, not being in the same plane as it. I say that *AB* is parallel to *CD*.

For let some point *G* be taken at random on *EF*. And from it let *GH* be drawn at right-angles to *EF* in the plane through *EF* and *AB*. And let *GK* be drawn, again at right-angles to *EF*, in the plane through *FE* and *CD*.

And since *EF* is at right-angles to each of *GH* and *GK*, *EF* is thus also at right-angles to the plane through *GH* and *GK* [Prop. 11.4]. And *EF* is parallel to *AB*. Thus, *AB* is also at right-angles to the plane through *HGK* [Prop. 11.8]. So, for the same (reasons), *CD* is also at right-angles to the plane through *HGK*. Thus, *AB* and *CD* are each at right-angles to the plane through *HGK*. And if two straight-lines are at right–angles to the same plane then the straight-lines are parallel [Prop. 11.6]. Thus, *AB* is parallel to *CD*. (Which is) the very thing it was required to show.

10.

Ἐὰν δύο εὐθεῖαι ἁπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἁπτομένας ἀλλήλωνὀῶσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ, ὁίσας γωνίας περιέχουσιν.

Δύο γὰρ εὐθεῖαι αἱ AB, BΓ ἁπτόμεναι ἀλλήλων παρὰ δύο εὐθείας τὰξ ΔE, EZ ἁπτομένας ἀλλήλωνὀέστωσαν μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ· λέγω, ὅτιὀίση ἐστὶν ἡ ὑπὸ ABΓ γωνία τῇ ὑπὸ ΔEZ.

Ἀπειλήφθωσαν γὰρ αἱ BA, BΓ, EΔ, EZ ἴσαι ἀλλήλαις, καὶ ἐπεζεύθθωσαν αἱ AΔ, ΓZ, BE, AΓ, ΔZ.

Καὶ ἐπεὶ ἡ BA τῇ EΔὀίση ἐστὶ καὶ παράλληλος, καὶ ἡ AΔ ἄρα τῇ BEὀίση ἐστὶ καὶ παράλληλος. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΓZ τῇ BEὀίση ἐστὶ καὶ παράλληλος· ἑκατέρα ἄρα τῶν

Proposition 10

If two straight-lines joined to one another are (respectively) parallel to two straight-lines joined to one another, (but are) not in the same plane, then they will contain equal angles.

For let the two straight-lines joined to one another, *AB* and *BC*, be (respectively) parallel to the two straight-lines joined to one another, *DE* and *EF*, (but) not in the same plane. I say that angle *ABC* is equal to (angle) *DEF*.

For let *BA*, *BC*, *ED*, and *EF* be cut off (so as to be, respectively) equal to one another. And let *AD*, *CF*, *BE*, *AC*, and *DF* be joined.

And since *BA* is equal and parallel to *ED*, *AD* is thus also equal and parallel to *BE* [Prop. 1.33]. So, for the same reasons,

ΑΔ, ΓΖ τῇ ΒΕ ἰσάκις ἐστὶ καὶ παράλληλος. αἱ δὲ τῇ αὐτῇ εὐθείᾳ παράλληλοι καὶ μὴ οὐσαι αὐτῇ ἐν τῷ αὐτῷ ἐπιπέδῳ καὶ ἀλλήλαις εἰσὶ παράλληλοι· παράλληλος ἄρα ἐστὶν ἡ ΑΔ τῇ ΓΖ καὶ ἴση. καὶ ἐπιζευγνύουσιν αὐτὰξ αἱ ΑΓ, ΔΖ· καὶ ἡ ΑΓ ἄρα τῇ ΔΖ ἰσάκις ἐστὶ καὶ παράλληλος. καὶ ἐπεὶ δύο αἱ ΑΒ, ΒΓ δυσὶ ταῖς ΔΕ, ΕΖ ἴσαι εἰσίν, καὶ βάσις ἡ ΑΓ βάσει τῇ ΔΖ ἴση, γωνία ἄρα ἡ ὑπὸ ΑΒΓ γωνίᾳ τῇ ὑπὸ ΔΕΖ ἐστιν ἴση.

CF is also equal and parallel to *BE*. Thus, *AD* and *CF* are each equal and parallel to *BE*. And straight-lines parallel to the same straight-line, and which are not in the same plane as it, are also parallel to one another [Prop. 11.9]. Thus, *AD* is parallel and equal to *CF*. And *AC* and *DF* join them. Thus, *AC* is also equal and parallel to *DF* [Prop. 1.33]. And since the two (straight-lines) *AB* and *BC* are equal to the two (straight-lines) *DE* and *EF* (respectively), and the base *AC* (is) equal to the base *DF*, the angle *ABC* is thus equal to the (angle) *DEF* [Prop. 1.8].

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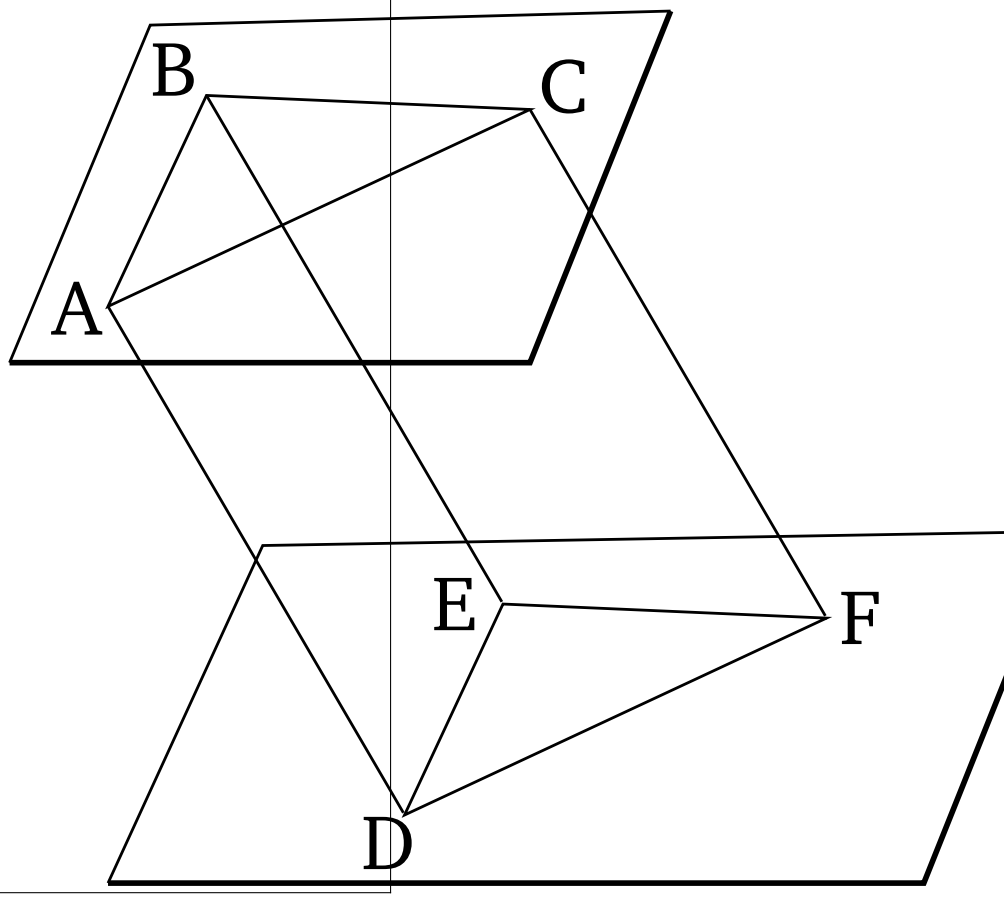
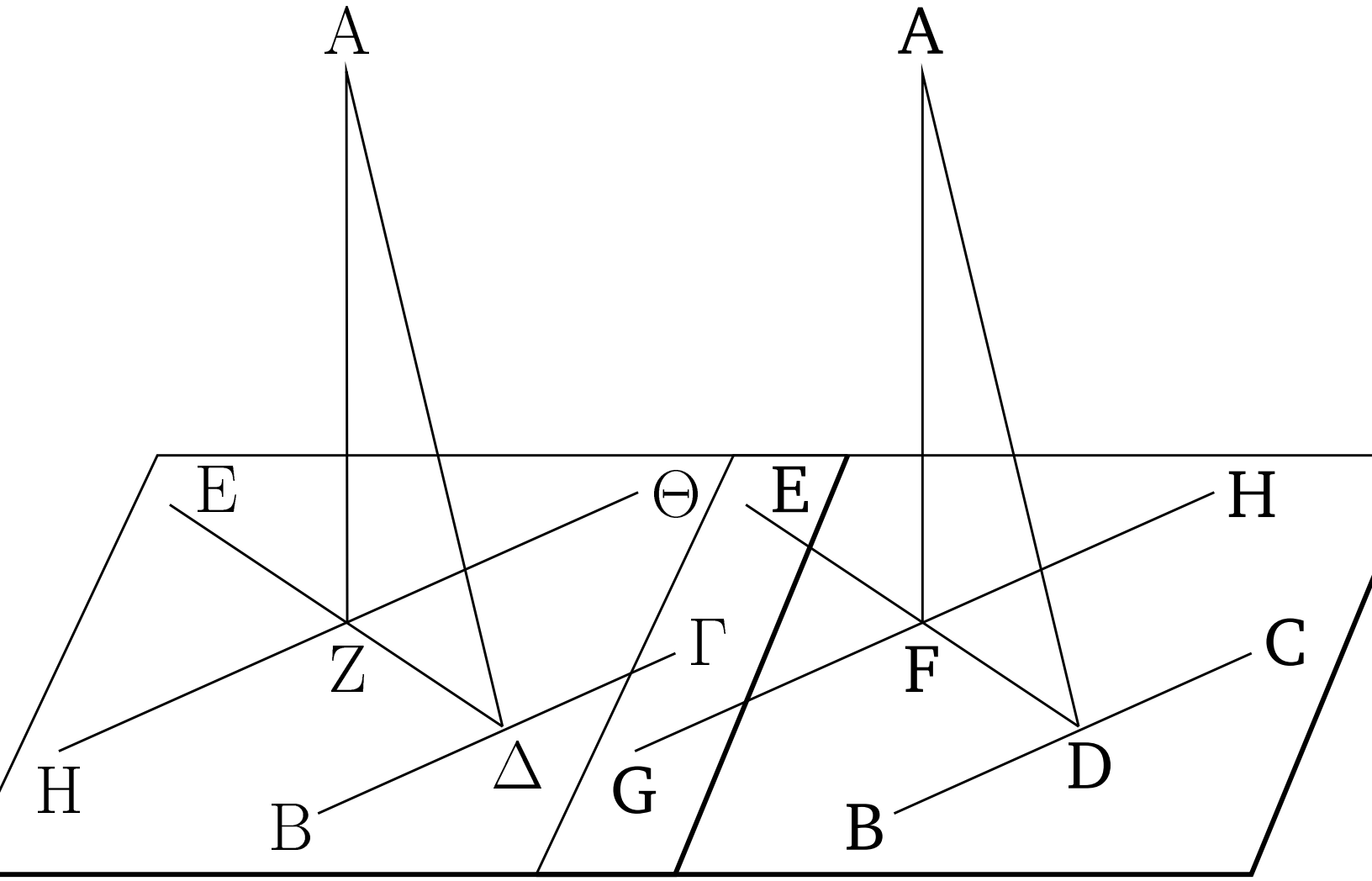


Ἐὰν ἄρα δύο εὐθεῖαι ἅπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἁπτομένας ἀλλήλων ὁσῶσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ, ἴσας γωνίας περιέχουσιν· ὅπερ ἔδει δεῖξαι.

Thus, if two straight-lines joined to one another are (respectively) parallel to two straight-lines joined to one another, (but are) not in the same plane, then they will contain equal angles. (Which is) the very thing it was required to show.

Ἀπὸ τοῦ δοθέντος σημείου μετεώρου ἐπὶ τὸ δοθὲν ἐπίπεδον κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

To draw a perpendicular straight-line from a given raised point to a given plane.



Ἔστω τὸ μὲν δοθὲν σημεῖον μετέωρον τὸ Α, τὸ δὲ δοθὲν ἐπίπεδον τὸ ὑποκείμενον· δεῖ δὴ ἀπὸ τοῦ Α σημείου ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετον εὐθεῖαν γραμμὴν ἀγαγεῖν.

Διήχθω γάρ τις ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖα, ὡξὸέτυθεν, ἡ ΒΓ, καὶ ἤχθω ἀπὸ τοῦ Α σημείου ἐπὶ τὴν ΒΓ κάθετος ἡ ΑΔ. εἰ μὲν οὖν ἡ ΑΔ κάθετός ἐστι καὶ ἐπὶ τὸ ὑποκείμενον ἐπίπεδον, γεγονὸξὸὰν εὸίη τὸ ἐπιταχθέν. εἰ δὲ οὐύ, ἤχθω ἀπὸ τοῦ Δ σημείου τῇ ΒΓ ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξ ἡ ΔΕ, καὶ ἤχθω ἀπὸ τοῦ Α ἐπὶ τὴν ΔΕ κάθετος ἡ ΑΖ, καὶ διὰ τοῦ Ζ σημείου τῇ ΒΓ παράλληλος ἤχθω ἡ ΗΘ.

Καὶ ἐπεὶ ἡ ΒΓ ἑκατέρᾳ τῶν ΔΑ, ΔΕ πρὸς ὀρθάξ ἐστιν, ἡ ΒΓ ἄρα καὶ τῷ διὰ τῶν ΕΔΑ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. καί ἐστιν αὐτῇ παράλληλος ἡ ΗΘ· ἐαν δὲ ὀῶσι δύο εὐθεῖαι παράλληλοι, ἡ δὲ μία αὐτῶν ἐπιπέδῳ τινὶ πρὸς ὀρθὰξὸῇ, καὶ

Let *A* be the given raised point, and the given plane the reference (plane). So, it is required to draw a perpendicular straight-line from point *A* to the reference plane.

Let some random straight-line *BC* be drawn across in the reference plane, and let the (straight-line) *AD* be drawn from point *A* perpendicular to *BC* [Prop. 1.12]. If, therefore, *AD* is also perpendicular to the reference plane then that which was prescribed will have occurred. And, if not, let *DE* be drawn in the reference plane from point *D* at right-angles to *BC* [Prop. 1.11], and let the (straight-line) *AF* be drawn from *A* perpendicular to *DE* [Prop. 1.12], and let *GH* be drawn through point *F*, parallel to *BC* [Prop. 1.31].

And since *BC* is at right-angles to each of *DA* and *DE*, *BC* is thus also at right-angles to the plane through *EDA* [Prop. 11.4]. And *GH* is parallel to it. And if two straight-lines

ἡ λοιπὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὸέσται· καὶ ἡ ΗΘ ἄρα τῷ διὰ τῶν ΕΔ, ΔΑ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐύσας ἐν τῷ διὰ τῶν ΕΔ, ΔΑ ἐπιπέδῳ ὀρθή ἐστιν ἡ ΗΘ. ὁάπτεται δὲ αὐτῆς ἡ ΑΖ οὐῦσα ἐν τῷ διὰ τῶν ΕΔ, ΔΑ ἐπιπέδῳ· ἡ ΗΘ ἄρα ὀρθή ἐστι πρὸς τὴν ΖΑ· ὁώστε καὶ ἡ ΖΑ ὀρθή ἐστι πρὸς τὴν ΘΗ. ὀέστι δὲ ἡ ΑΖ καὶ πρὸς τὴν ΔΕ ὀρθή· ἡ ΑΖ ἄρα πρὸς ἑκατέραν τῶν ΗΘ, ΔΕ ὀρθή ἐστιν. ἔαν δὲ εὐθεῖα δυσὶν εὐθείαις τεμνούσαις ἀλλήλας ἐπὶ τῆς τομῆς πρὸς ὀρθὰξ ἐπισταθῇ, καὶ τῷ διὸ αὐτῶν ἐπιπέδῳ πρὸς ὀρθὰξὸέσται· ἡ ΖΑ ἄρα τῷ διὰ τῶν ΕΔ, ΗΘ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. τὸ δὲ διὰ τῶν ΕΔ, ΗΘ ἐπίπεδόν ἐστι τὸ ὑποκείμενον· ἡ ΑΖ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν.

Ἀπὸ τοῦ ἄρα δοθέντος σημείου μετεώρου τοῦ Α ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετος εὐθεῖα γραμμὴ ὀῆκται ἡ ΑΖ· ὅπερ ἔδει ποιῆσαι.

12.

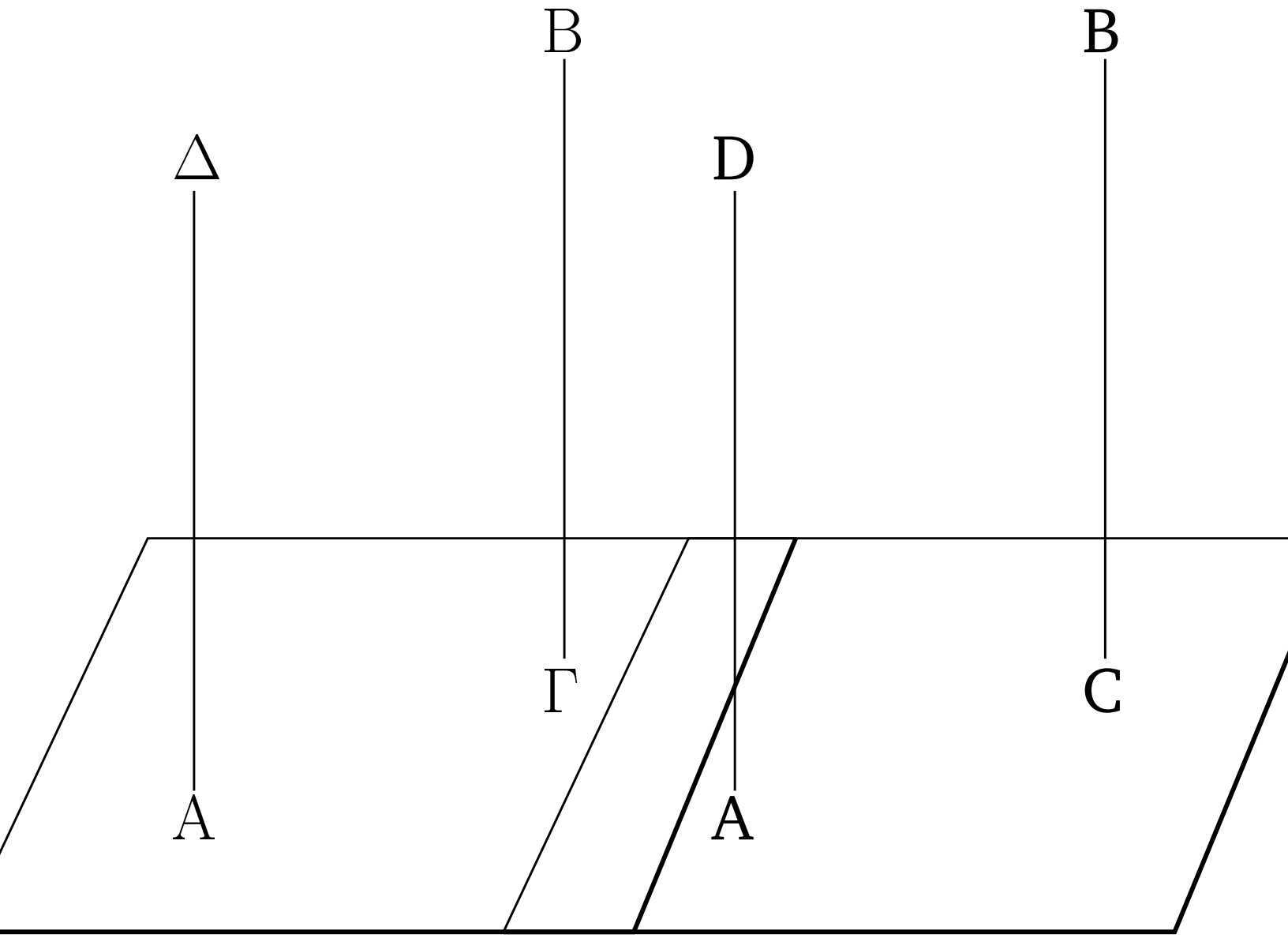
Τῷ δοθέντι ἐπιπέδῳ ἀπὸ τοῦ πρὸς αὐτῷ δοθέντος σημείου πρὸς ὀρθὰξ εὐθεῖαν γραμμὴν ἀναστῆσαι.

are parallel, and one of them is at right-angles to some plane, then the remaining (straight-line) will also be at right-angles to the same plane [Prop. 11.8]. Thus, GH is also at right-angles to the plane through ED and DA. And GH is thus at right-angles to all of the straight-lines joined to it which are also in the plane through ED and AD [Def. 11.3]. And AF, which is in the plane through ED and DA, is joined to it. Thus, GH is at right-angles to FA. Hence, FA is also at right-angles to HG. And AF is also at right-angles to DE. Thus, AF is at right-angles to each of GH and DE. And if a straight-line is set up at right-angles to two straight-lines cutting one another, at the point of section, then it will also be at right-angles to the plane through them [Prop. 11.4]. Thus, FA is at right-angles to the plane through ED and GH. And the plane through ED and GH is the reference (plane). Thus, AF is at right-angles to the reference plane.

Thus, the straight-line AF has been drawn from the given raised point A perpendicular to the reference plane. (Which is) the very thing it was required to do.

Proposition 12

To set up a straight-line at right-angles to a given plane from a given point in it.



Ἔστω τὸ μὲν δοθὲν ἐπίπεδον τὸ ὑποκείμενον, τὸ δὲ πρὸς αὐτῷ σημεῖον τὸ A· δεῖ δὴ ἀπὸ τοῦ A σημείου τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξ εὐθεῖαν γραμμὴν ἀναστῆσαι.

Νενοήσθω τι σημεῖον μετέωρον τὸ B, καὶ ἀπὸ τοῦ B ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετος ἤχθω ἡ $BΓ$, καὶ διὰ τοῦ A σημείου τῇ $BΓ$ παράλληλος ἤχθω ἡ $AΔ$.

Ἐπεὶ οὖν δύο εὐθεῖαι παράλληλοί εἰσιν αἱ $AΔ$, $ΓB$, ἡ δὲ μία αὐτῶν ἡ $BΓ$ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν, καὶ ἡ λοιπὴ ἄρα ἡ $AΔ$ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξ ἐστιν.

Τῷ ἄρα δοθέντι ἐπιπέδῳ ἀπὸ τοῦ πρὸς αὐτῷ σημείου τοῦ A πρὸς ὀρθὰξ ἀνέσταται ἡ $AΔ$· ὅπερ ἔδει ποιῆσαι.

Let the given plane be the reference (plane), and A a point in it. So, it is required to set up a straight-line at right-angles to the reference plane at point A.

Let some raised point B be assumed, and let the perpendicular (straight-line) BC be drawn from B to the reference plane [Prop. 11.11]. And let AD be drawn from point A parallel to BC [Prop. 1.31].

Therefore, since AD and CB are two parallel straight-lines, and one of them, BC, is at right-angles to the reference plane, the remaining (one) AD is thus also at right-angles to the reference plane [Prop. 11.8].

Thus, AD has been set up at right-angles to the given plane, from the point in it, A. (Which is) the very thing it was required

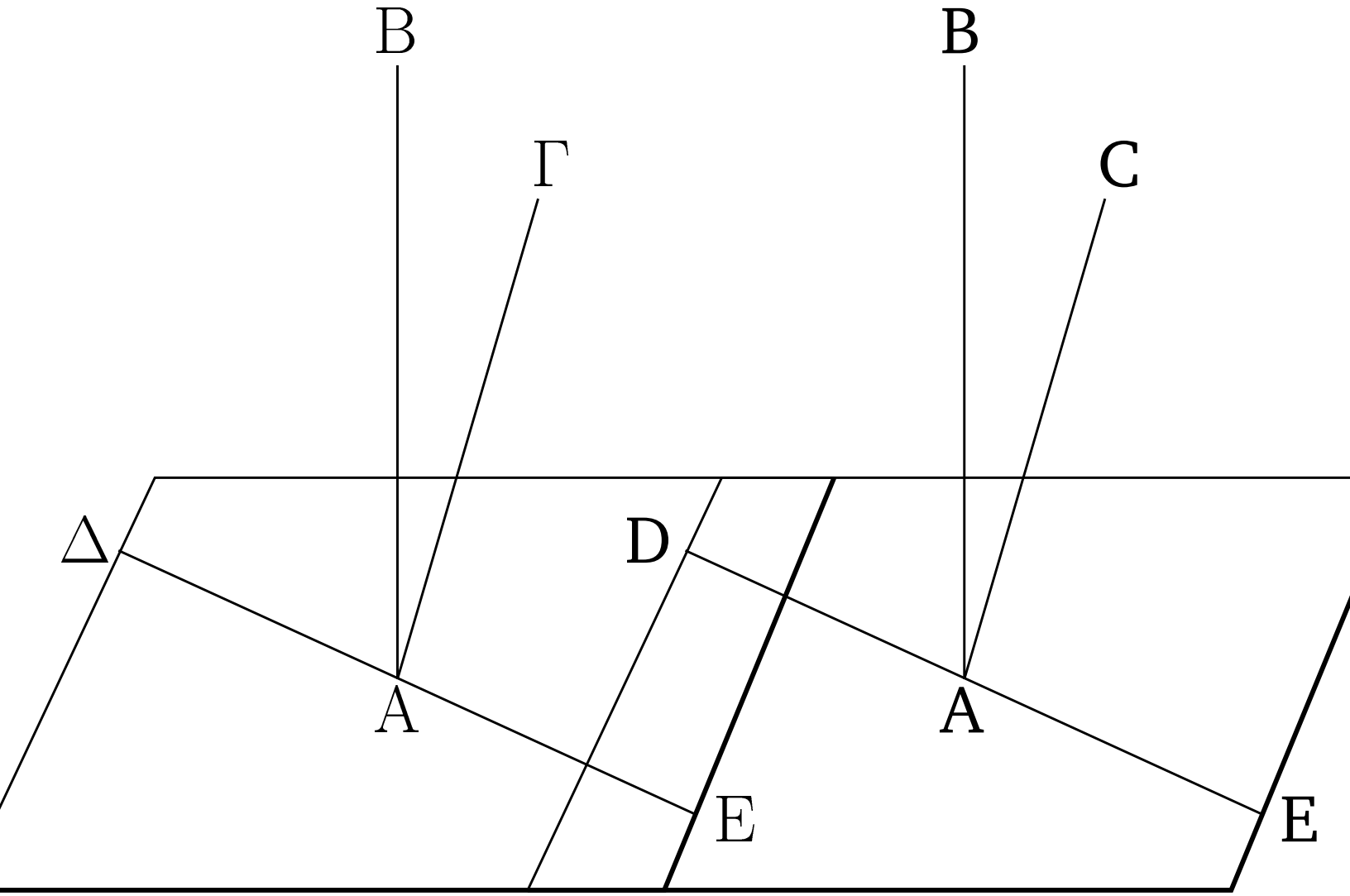
13.

Ἀπὸ τοῦ αὐτοῦ σημείου τῷ αὐτῷ ἐπιπέδῳ δύο εὐθεῖαι πρὸς ὀρθὰξ οὐκ ἀναστήσονται ἐπὶ τὰ αὐτὰ μέρη.

to do.

Proposition 13

Two (different) straight-lines cannot be set up at the same point at right-angles to the same plane, on the same side.



Εἰ γὰρ δυνατόν, ἀπὸ τοῦ αὐτοῦ σημείου τοῦ Α τῷ ὑποκειμένῳ ἐπιπέδῳ δύο εὐθεῖαι αἱ ΑΒ, ΒΓ πρὸς ὀρθὰξ ἀνεστάτωσαν ἐπὶ τὰ αὐτὰ μέρη, καὶ διήχθω τὸ διὰ τῶν ΒΑ, ΑΓ ἐπίπεδον· τομὴν δὴ ποιήσει διὰ τοῦ Α ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ εὐθεῖαν. ποιείτω τὴν ΔΑΕ· αἱ ἄρα ΑΒ, ΑΓ, ΔΑΕ εὐθεῖαι ἐν ἑνὶ εἰσιν ἐπιπέδῳ. καὶ ἐπεὶ ἡ ΓΑ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας. ὁάπτεται δὲ αὐτῆς ἡ ΔΑΕ οὐῦσα ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ· ἡ ἄρα ὑπὸ ΓΑΕ γωνία ὀρθή ἐστιν. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΒΑΕ ὀρθή ἐστιν· ὁίση ἄρα ἡ ὑπὸ ΓΑΕ τῇ ὑπὸ

For, if possible, let the two straight-lines *AB* and *AC* be set up at the same point *A* at right-angles to the reference plane, on the same side. And let the plane through *BA* and *AC* be drawn. So it will make a straight cutting (passing) through (point) *A* in the reference plane [Prop. 11.3]. Let it have made *DAE*. Thus, *AB*, *AC*, and *DAE* are straight-lines in one plane. And since *CA* is at right-angles to the reference plane, it will thus also make right-angles with all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. And *DAE*, which is in the reference plane, is joined to it. Thus, angle *CAE* is a right-angle. So, for the same (reasons), *BAE* is also

BAE καί εἰσιν ἐν ἑνὶ ἐπιπέδῳ· ὅπερ ἐστὶν ἀδύνατον.

Οὐκ ἄρα ἀπὸ τοῦ αὐτοῦ σημείου τῷ αὐτῷ ἐπιπέδῳ δύο εὐθεῖαι πρὸς ὀρθὰξ ἀνασταθήσονται ἐπὶ τὰ αὐτὰ μέρη· ὅπερ ἔδει δεῖχαι.

a right-angle. Thus, *CAE* (is) equal to *BAE*. And they are in one plane. The very thing is impossible.

Thus, two (different) straight-lines cannot be set up at the same point at right-angles to the same plane, on the same side. (Which is) the very thing it was required to show.

14.

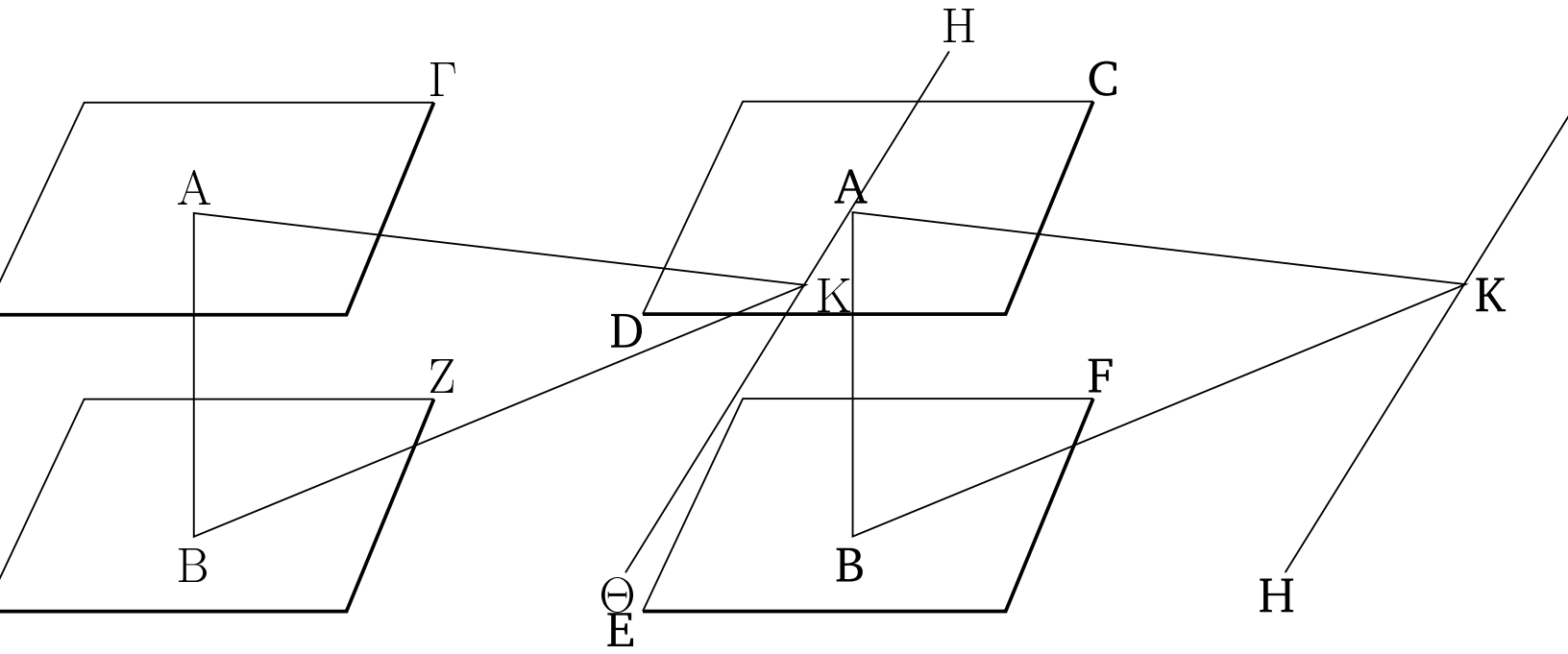
Πρὸξ ὰ ἐπίπεδα ἡ αὐτὴ εὐθεῖα ὀρθή ἐστιν, παράλληλα-ὀέσται τὰ ἐπίπεδα.

Εὐθεῖα γάρ τις ἡ AB πρὸς ἑκάτερον τῶν ΓΔ, EZ ἐπιπέδων πρὸς ὀρθὰξὸέστω· λέγω, ὅτι παράλληλά ἐστι τὰ ἐπίπεδα.

Proposition 14

Planes to which the same straight-line is at right-angles will be parallel planes.

For let some straight-line *AB* be at right-angles to each of the planes *CD* and *EF*. I say that the planes are parallel.



Εἰ γὰρ μή, ἐκβαλλόμενα συμπεσοῦνται. συμπιπτέτωσαν· ποιήσουσι δὴ κοινὴν τομὴν εὐθεῖαν. ποιείτωσαν τὴν HΘ, καὶ εἰλήφθω ἐπὶ τῆς HΘ τυθὸν σημεῖον τὸ K, καὶ ἐπεζεύθθωσαν αἱ AK, BK.

Καὶ ἐπεὶ ἡ AB ὀρθή ἐστι πρὸς τὸ EZ ἐπίπεδον, καὶ πρὸς τὴν BK ἄρα εὐθεῖαν οὐὖσαν ἐν τῷ EZ ἐκβληθέντι ἐπιπέδῳ ὀρθή ἐστιν ἡ AB· ἡ ἄρα ὑπὸ ABK γωνία ὀρθή ἐστιν. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ BAK ὀρθή ἐστιν. τριγώνου δὴ τοῦ ABK αἱ δύο γωνίαι αἱ ὑπὸ ABK, BAK δυσὶν ὀρθαῖξ εἰσιν ἴσαι· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τὰ ΓΔ, EZ ἐπίπεδα ἐκβαλλόμενα συμπεσοῦνται· παράλληλα ἄρα ἐστὶ τὰ ΓΔ, EZ ἐπίπεδα.

Πρὸξ ὰ ἐπίπεδα ἄρα ἡ αὐτὴ εὐθεῖα ὀρθή ἐστιν, παράλληλά ἐστι τὰ ἐπίπεδα· ὅπερ ἔδει δεῖχαι.

For, if not, being produced, they will meet. Let them have met. So they will make a straight-line as a common section [Prop. 11.3]. Let them have made *GH*. And let some random point *K* be taken on *GH*. And let *AK* and *BK* be joined.

And since *AB* is at right-angles to the plane *EF*, *AB* is thus also at right-angles to *BK*, which is a straight-line in the produced plane *EF* [Def. 11.3]. Thus, angle *ABK* is a right-angle. So, for the same (reasons), *BAK* is also a right-angle. So the (sum of the) two angles *ABK* and *BAK* in the triangle *ABK* is equal to two right-angles. The very thing is impossible [Prop. 1.17]. Thus, planes *CD* and *EF*, being produced, will not meet. Planes *CD* and *EF* are thus parallel [Def. 11.8].

Thus, planes to which the same straight-line is at right-angles are parallel planes. (Which is) the very thing it was

15.

Ἐὰν δύο εὐθεῖαι ἁπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἁπτομένας ἀλλήλωνὁῶσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ οὁῦσαι, παράλληλά ἐστι τὰ διὸ αὐτῶν ἐπίπεδα.

Δύο γὰρ εὐθεῖαι ἁπτόμεναι ἀλλήλων αἱ ΑΒ, ΒΓ παρὰ δύο εὐθείας ἁπτομένας ἀλλήλων τὰξ ΔΕ, ΕΖὁέστωσαν μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ οὁῦσαι· λέγω, ὅτι ἐκβαλλόμενα τὰ διὰ τῶν ΑΒ, ΒΓ, ΔΕ, ΕΖ ἐπίπεδα οὐ συμπεσεῖται ἀλλήλοις.

ὁΗθθω γὰρ ἀπὸ τοῦ Β σημείου ἐπὶ τὸ διὰ τῶν ΔΕ, ΕΖ ἐπίπεδον κάθετος ἡ ΒΗ καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ Η σημεῖον, καὶ διὰ τοῦ Η τῇ μὲν ΕΔ παράλληλος ἤθθω ἡ ΗΘ, τῇ δὲ ΕΖ ἡ ΗΚ.

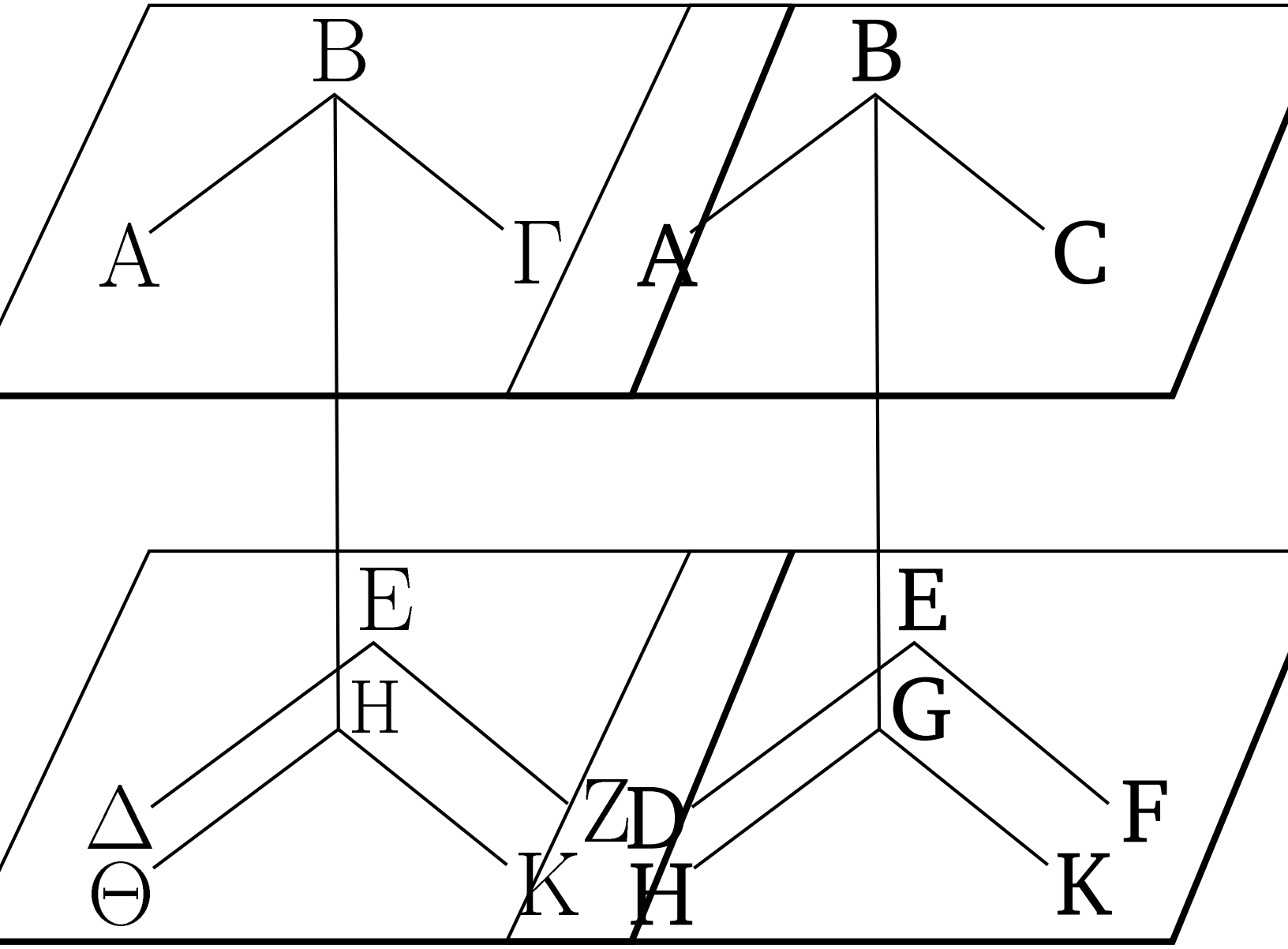
required to show.

Proposition 15

If two straight-lines joined to one another are parallel (respectively) to two straight-lines joined to one another, which are not in the same plane, then the planes through them are parallel (to one another).

For let the two straight-lines joined to one another, AB and BC, be parallel to the two straight-lines joined to one another, DE and EF (respectively), not being in the same plane. I say that the planes through AB, BC and DE, EF will not meet one another (when) produced.

For let BG be drawn from point B perpendicular to the plane through DE and EF [Prop. 11.11], and let it meet the plane at point G. And let GH be drawn through G parallel to ED, and GK (parallel) to EF [Prop. 1.31].



Καὶ ἐπεὶ ἡ ΒΗ ὀρθή ἐστι πρὸς τὸ διὰ τῶν ΔΕ, ΕΖ ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ διὰ τῶν ΔΕ, ΕΖ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας. ὅάπτεται δὲ αὐτῆς ἑκατέρα τῶν ΗΘ, ΗΚ οὐῦσα ἐν τῷ διὰ τῶν ΔΕ, ΕΖ ἐπιπέδῳ· ὀρθὴ ἄρα ἐστὶν ἑκατέρα τῶν ὑπὸ ΒΗΘ, ΒΗΚ γωνιῶν. καὶ ἐπεὶ παράλληλόξ ἐστιν ἡ ΒΑ τῇ ΗΘ, αἱ ἄρα ὑπὸ ΗΒΑ, ΒΗΘ γωνίαι δυσὶν ὀρθαῖξ ἴσαι εἰσίν. ὀρθὴ δὲ ἡ ὑπὸ ΒΗΘ· ὀρθὴ ἄρα καὶ ἡ ὑπὸ ΗΒΑ· ἡ ΗΒ ἄρα τῇ ΒΑ πρὸς ὀρθάξ ἐστιν. διὰ τὰ αὐτὰ δὴ ἡ ΗΒ καὶ τῇ ΒΓ ἐστι πρὸς ὀρθάξ. ἐπεὶ οὐῦν εὐθεῖα ἡ ΗΒ δυσὶν εὐθείαις ταῖς ΒΑ, ΒΓ τεμνούσαις ἀλλήλας πρὸς ὀρθὰξ ἐφέστηκεν, ἡ ΗΒ ἄρα καὶ τῷ διὰ τῶν ΒΑ, ΒΓ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. [διὰ τὰ αὐτὰ δὴ ἡ ΒΗ καὶ τῷ διὰ τῶν ΗΘ, ΗΚ ἐπιπέδῳ πρὸς

And since *BG* is at right-angles to the plane through *DE* and *EF*, it will thus also make right-angles with all of the straight-lines joined to it, which are also in the plane through *DE* and *EF* [Def. 11.3]. And each of *GH* and *GK*, which are in the plane through *DE* and *EF*, are joined to it. Thus, each of the angles *BGH* and *BGK* are right-angles. And since *BA* is parallel to *GH* [Prop. 11.9], the (sum of the) angles *GBA* and *BGH* is equal to two right-angles [Prop. 1.29]. And *BGH* (is) a right-angle. *GBA* (is) thus also a right-angle. Thus, *GB* is at right-angles to *BA*. So, for the same (reasons), *GB* is also at right-angles to *BC*. Therefore, since the straight-line *GB* has been set up at right-angles to two straight-lines, *BA* and *BC*, cutting one another, *GB* is thus at right-angles to the plane

ὀρθάξ ἐστιν. τὸ δὲ διὰ τῶν ΗΘ, ΗΚ ἐπίπεδόν ἐστι τὸ διὰ τῶν ΔΕ, ΕΖ· ἡ ΒΗ ἄρα τῷ διὰ τῶν ΔΕ, ΕΖ ἐπιπέδῳ ἐστὶ πρὸς ὀρθάξ. ἐδείχθη δὲ ἡ ΗΒ καὶ τῷ διὰ τῶν ΑΒ, ΒΓ ἐπιπέδῳ πρὸς ὀρθάξ]. πρὸς ὰ δὲ ἐπίπεδα ἡ αὐτὴ εὐθεῖα ὀρθή ἐστιν, παράλληλά ἐστι τὰ ἐπίπεδα· παράλληλον ἄρα ἐστὶ τὸ διὰ τῶν ΑΒ, ΒΓ ἐπίπεδον τῷ διὰ τῶν ΔΕ, ΕΖ.

Ἐὰν ἄρα δύο εὐθεῖαι ἁπτόμεναι ἀλλήλων παρὰ δύο εὐθείας ἁπτομένας ἀλλήλωνὸῶσι μὴ ἐν τῷ αὐτῷ ἐπιπέδῳ, παράλληλά ἐστι τὰ διὸ αὐτῶν ἐπίπεδα· ὅπερ ἔδει δεῖχαι.

through BA and BC [Prop. 11.4]. [So, for the same (reasons), BG is also at right-angles to the plane through GH and GK. And the plane through GH and GK is the (plane) through DE and EF. And it was also shown that GB is at right-angles to the plane through AB and BC.] And planes to which the same straight-line is at right-angles are parallel planes [Prop. 11.14]. Thus, the plane through AB and BC is parallel to the (plane) through DE and EF.

Thus, if two straight-lines joined to one another are parallel (respectively) to two straight-lines joined to one another, which are not in the same plane, then the planes through them are parallel (to one another). (Which is) the very thing it was required to show.

16.

Ἐὰν δύο ἐπίπεδα παράλληλα ὑπό ἐπιπέδου τινὸξ τέμνηται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν.

Δύο γὰρ ἐπίπεδα παράλληλα τὰ ΑΒ, ΓΔ ὑπὸ ἐπιπέδου τοῦ ΕΖΗΘ τεμνέσθω, κοιναὶ δὲ αὐτῶν τομαὶ ὀέστωσαν αἱ ΕΖ, ΗΘ· λέγω, ὅτι παράλληλόξ ἐστιν ἡ ΕΖ τῇ ΗΘ.

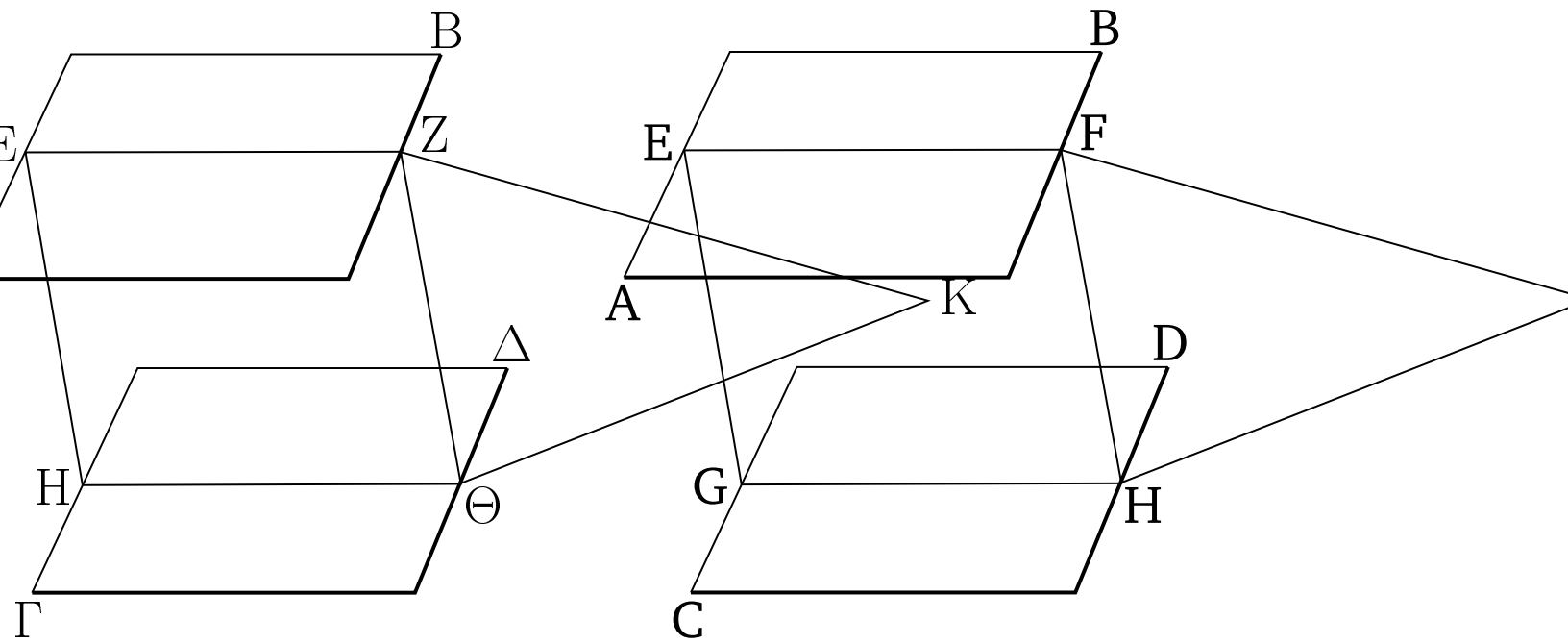
Εἰ γὰρ μή, ἐκβαλλόμεναι αἱ ΕΖ, ΗΘ ἤτοι ἐπὶ τὰ Ζ, Θ μέρηὸἡ ἐπὶ τὰ Ε, Η συμπεσοῦνται. ἐκβεβλήσθωσαν ὡξ ἐπὶ τὰ Ζ, Θ μέρη καὶ συμπιπτέτωσαν πρότερον κατὰ τὸ Κ. καὶ ἐπεὶ ἡ ΕΖΚ ἐν τῷ ΑΒ ἐστιν ἐπιπέδῳ, καὶ πάντα ἄρα τὰ ἐπὶ τῆς ΕΖΚ σημεῖα ἐν τῷ ΑΒ ἐστιν ἐπιπέδῳ. ὃὲν δὲ τῶν ἐπὶ τῆς ΕΖΚ εὐθείας σημείων ἐστὶ τὸ Κ· τὸ Κ ἄρα ἐν τῷ ΑΒ ἐστιν ἐπιπέδῳ. διὰ τὰ αὐτὰ δὴ τὸ Κ καὶ ἐν τῷ ΓΔ ἐστιν ἐπιπέδῳ· τὰ ΑΒ, ΓΔ ἄρα ἐπίπεδα ἐκβαλλόμενα συμπεσοῦνται. οὐ συμπίπτουσι δὲ διὰ τὸ παράλληλα ὑποκεῖσθαι· οὐκ ἄρα αἱ ΕΖ, ΗΘ εὐθεῖαι ἐκβαλλόμεναι ἐπὶ τὰ Ζ, Θ μέρη συμπεσοῦνται. ὁμοίως δὴ δείχομεν, ὅτι αἱ ΕΖ, ΗΘ εὐθεῖαι οὐδέ ἐπὶ τὰ Ε, Η μέρη ἐκβαλλόμεναι συμπεσοῦνται. αἱ δὲ ἐπὶ μ ηδέτερα τὰ μέρη συμπίπτουσαι παράλληλοί εἰσιν. παράλληλος ἄρα ἐστὶν ἡ ΕΖ τῇ ΗΘ.

Proposition 16

If two parallel planes are cut by some plane then their common sections are parallel.

For let the two parallel planes AB and CD be cut by the plane $EFGH$. And let EF and GH be their common sections. I say that EF is parallel to GH.

For, if not, being produced, EF and GH will meet either in the direction of F, H, or of E, G. Let them be produced, as in the direction of F, H, and let them, first of all, have met at K. And since EFK is in the plane AB, all of the points on EFK are thus also in the plane AB [Prop. 11.1]. And K is one of the points on EFK. Thus, K is in the plane AB. So, for the same (reasons), K is also in the plane CD. Thus, the planes AB and CD, being produced, will meet. But they do not meet, on account of being (initially) assumed (to be mutually) parallel. Thus, the straight-lines EF and GH, being produced in the direction of F, H, will not meet. So, similarly, we can show that the straight-lines EF and GH, being produced in the direction of E, G, will not meet either. And (straight-lines in one plane which), being produced, do not meet in either direction are parallel [Def. 1.23]. EF is thus parallel to GH.



Ἐὰν ἄρα δύο ἐπίπεδα παράλληλα ὑπό ἐπιπέδου τινὸξ τέμνηται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν· ὅπερ ἔδει δεῖχαι.

Thus, if two parallel planes are cut by some plane then their common sections are parallel. (Which is) the very thing it was required to show.

17.

Ἐὰν δύο εὐθεῖαι ὑπό παραλλήλων ἐπιπεδων τέμνωνται, εἰξ τοὐξ αὐτοὐξ λόγους τμ ηθήσονται.

Δύο γὰρ εὐθεῖαι αἱ AB, ΓΔ ὑπό παραλλήλων ἐπιπέδων τῶν HΘ, KΛ, MN τεμνέσθωσαν κατὰ τὰ A, E, B, Γ, Z, Δ σημεῖα· λέγω, ὅτι ἐστὶν ὡξ ἡ AE εὐθεῖα πρὸς τὴν EB, οὑτως ἡ ΓZ πρὸς τὴν ZΔ.

Ἐπεζεύθθωσαν γὰρ αἱ AΓ, BΔ, AΔ, καὶ συμβαλλέτω ἡ AΔ τῷ KΛ ἐπιπέδῳ κατὰ τὸ X σημεῖον, καὶ ἐπεζεύθθωσαν αἱ EX, XZ.

Καὶ ἐπεὶ δύο ἐπίπεδα παράλληλα τὰ KΛ, MN ὑπό ἐπιπέδου τοῦ EBΔX τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ αἱ EX, BΔ παράλληλοί εἰσιν. διὰ τὰ αὐτὰ δὴ ἐπεὶ δύο ἐπίπεδα παράλληλα τὰ HΘ, KΛ ὑπό ἐπιπέδου τοῦ AXZΓ τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ αἱ AΓ, XZ παράλληλοί εἰσιν. καὶ ἐπεὶ τριγώνου τοῦ ABΔ παρὰ μίαν τῶν πλευρῶν τὴν BΔ εὐθεῖαὀῆκται ἡ EX, ἀνάλογον ἄρα ἐστὶν ὡξ ἡ AE πρὸς EB, οὑτως ἡ AX πρὸς XΔ. πάλιν ἐπεὶ τριγώνου τοῦ AΔΓ παρὰ μίαν τῶν πλευρῶν τὴν AΓ εὐθεῖαὀῆκται ἡ XZ, ἀνάλογόν ἐστιν ὡξ ἡ AX πρὸς XΔ, οὑτως ἡ ΓZ πρὸς ZΔ. ἐδείθθη δὲ καὶ ὡξ ἡ AX πρὸς XΔ, οὑτως ἡ AE πρὸς EB· καὶ ὡξ ἄρα ἡ AE πρὸς EB, οὑτως ἡ ΓZ πρὸς ZΔ.

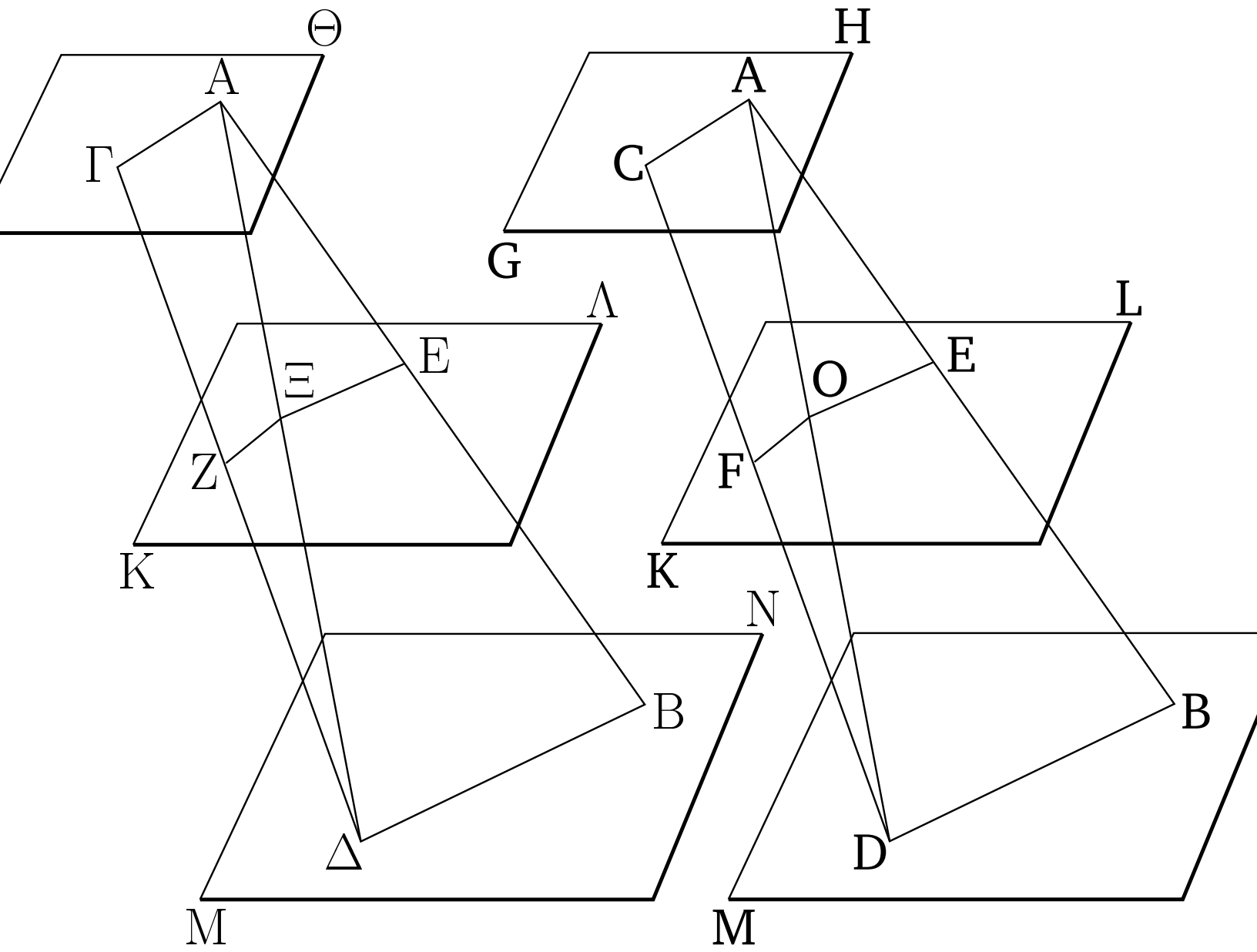
Proposition 17

If two straight-lines are cut by parallel planes then they will be cut in the same ratios.

For let the two straight-lines AB and CD be cut by the parallel planes GH, KL, and MN at the points A, E, B, and C, F, D (respectively). I say that as the straight-line AE is to EB, so CF (is) to FD.

For let AC, BD, and AD be joined, and let AD meet the plane KL at point O, and let EO and OF be joined.

And since two parallel planes KL and MN are cut by the plane $EBDO$, their common sections EO and BD are parallel [Prop. 11.16]. So, for the same (reasons), since two parallel planes GH and KL are cut by the plane $AOFC$, their common sections AC and OF are parallel [Prop. 11.16]. And since the straight-line EO has been drawn parallel to one of the sides BD of triangle ABD, thus, proportionally, as AE is to EB, so AO (is) to OD [Prop. 6.2]. Again, since the straight-line OF has been drawn parallel to one of the sides AC of triangle ADC, proportionally, as AO is to OD, so CF (is) to FD [Prop. 6.2]. And it was also shown that as AO (is) to OD, so AE (is) to EB. And thus as AE (is) to EB, so CF (is) to FD [Prop. 5.11].



Ἐὰν ἄρα δύο εὐθεῖαι ὑπὸ παραλλήλων ἐπιπέδων τέμνωνται, εἰξ τοὐξ αὐτοὐξ λόγους τμ ηθήσονται· ὅπερ ἔδει δειχαι.

Thus, if two straight-lines are cut by parallel planes then they will be cut in the same ratios. (Which is) the very thing it was required to show.

18.

Ἐὰν εὐθεῖα ἐπιπέδῳ τινὶ πρὸς ὀρθὰξὸῇ, καὶ πάντα τὰ διὸ αὐτῆς ἐπίπεδα τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὸέσται.

Εὐθεῖα γάρ τις ἡ AB τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξὸέστω· λέγω, ὅτι καὶ πάντα τὰ διὰ τῆς AB ἐπίπεδα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν.

Ἐκβεβλήσθω γὰρ διὰ τῆς AB ἐπίπεδον τὸ ΔE, καὶ ὀέστω κοινὴ τομὴ τοῦ ΔE ἐπιπέδου καὶ τοῦ ὑποκειμένου ἡ ΓE, καὶ

Proposition 18

If a straight-line is at right-angles to some plane then all of the planes (passing) through it will also be at right-angles to the same plane.

For let some straight-line *AB* be at right-angles to a reference plane. I say that all of the planes (passing) through *AB* are also at right-angles to the reference plane.

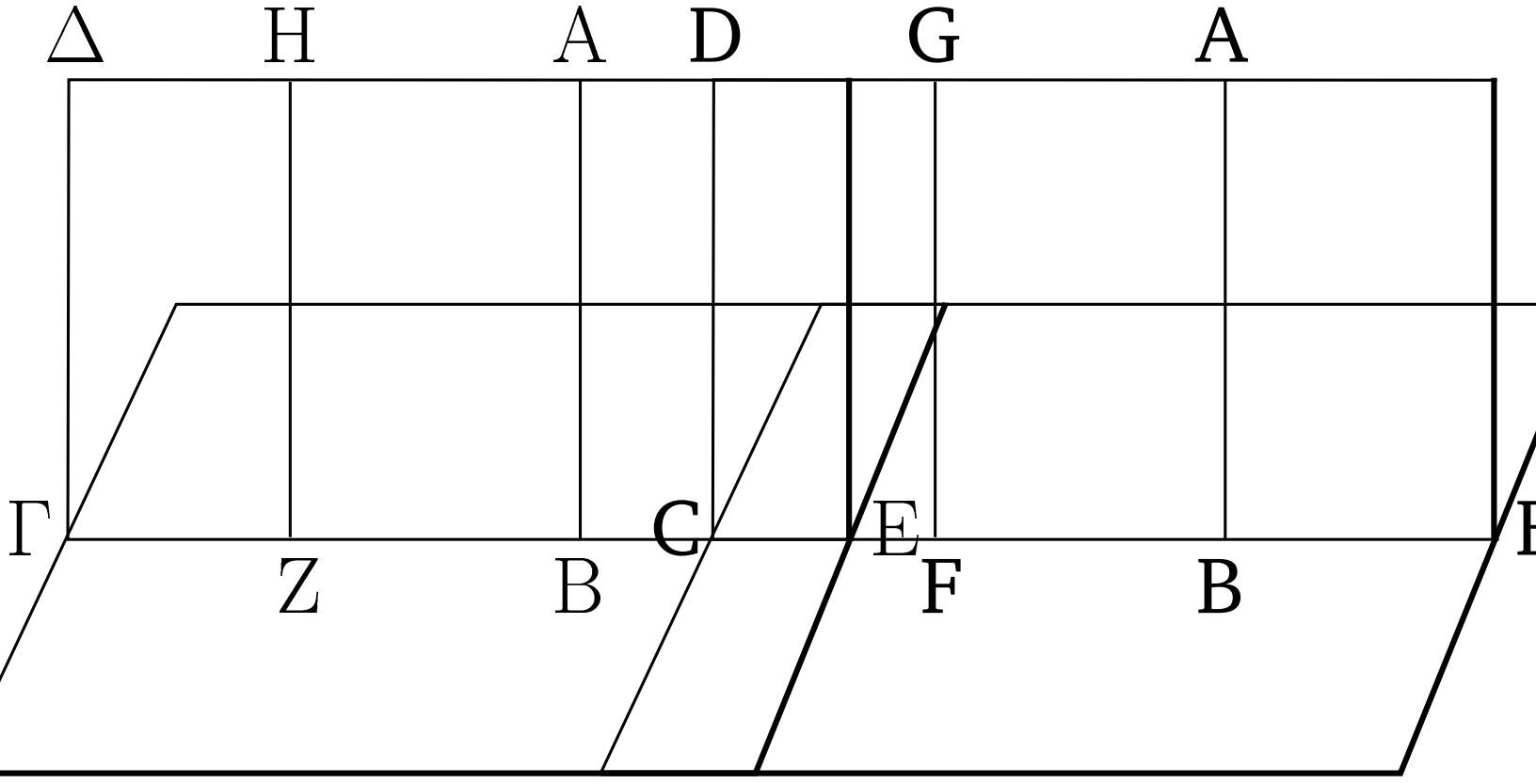
For let the plane *DE* be produced through *AB*. And let

εἰλήφθω ἐπὶ τῆς ΓΕ τυθὸν σημεῖον τὸ Ζ, καὶ ἀπὸ τοῦ Ζ τῇ ΓΕ πρὸς ὀρθὰξ ἤθθω ἐν τῷ ΔΕ ἐπιπέδῳ ἡ ΖΗ.

Καὶ ἐπεὶ ἡ ΑΒ πρὸς τὸ ὑποκείμενον ἐπίπεδον ὀρθή ἐστιν, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθή ἐστιν ἡ ΑΒ· ὥστε καὶ πρὸς τὴν ΓΕ ὀρθή ἐστιν· ἡ ἄρα ὑπὸ ΑΒΖ γωνία ὀρθή ἐστιν. ὀέστι δὲ καὶ ἡ ὑπὸ ΗΖΒ ὀρθὴ· παράλληλος ἄρα ἐστὶν ἡ ΑΒ τῇ ΖΗ. ἡ δὲ ΑΒ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν· καὶ ἡ ΖΗ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. καὶ ἐπίπεδον πρὸς ἐπίπεδον ὀρθόν ἐστιν, ὅταν αἱ τῇ κοινῇ τομῇ τῶν ἐπιπέδων πρὸς ὀρθὰξ ἀγόμεναι εὐθεῖαι ἐν ἑνὶ τῶν ἐπιπέδων τῷ λοιπῷ ἐπιπέδῳ πρὸς ὀρθὰξ ὦσιν. καὶ τῇ κοινῇ τομῇ τῶν ἐπιπέδων τῇ ΓΕ ἐν ἑνὶ τῶν ἐπιπέδων τῷ ΔΕ πρὸς ὀρθὰξ ἀθθεῖσα ἡ ΖΗ ἐδείθθη τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ· τὸ ἄρα ΔΕ ἐπίπεδον ὀρθόν ἐστι πρὸς τὸ ὑποκείμενον. ὁμοίως δὴ δειθθήσεται καὶ πάντα τὰ διὰ τῆς ΑΒ ἐπίπεδα ὀρθὰ τυγθανοντα πρὸς τὸ ὑποκείμενον ἐπίπεδον.

CE be the common section of the plane *DE* and the reference (plane). And let some random point *F* be taken on *CE*. And let *FG* be drawn from *F*, at right-angles to *CE*, in the plane *DE* [Prop. 1.11].

And since *AB* is at right-angles to the reference plane, *AB* is thus also at right-angles to all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. Hence, it is also at right-angles to *CE*. Thus, angle *ABF* is a right-angle. And *GFB* is also a right-angle. Thus, *AB* is parallel to *FG* [Prop. 1.28]. And *AB* is at right-angles to the reference plane. Thus, *FG* is also at right-angles to the reference plane [Prop. 11.8]. And a plane is at right-angles to a(nother) plane when the straight-lines drawn at right-angles to the common section of the planes, (and lying) in one of the planes, are at right-angles to the remaining plane [Def. 11.4]. And *FG*, (which was) drawn at right-angles to the common section of the planes, *CE*, in one of the planes, *DE*, was shown to be at right-angles to the reference plane. Thus, plane *DE* is at right-angles to the reference (plane). So, similarly, it can be shown that all of the planes (passing) at random through *AB* (are) at right-angles to the reference plane.



Ἐὰν ἄρα εὐθεῖα ἐπιπέδῳ τινὶ πρὸς ὀρθὰξὸῇ, καὶ πάντα τὰ διὸ αὐτῆς ἐπίπεδα τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὸέσται· ὅπερ ἔδει δεῖχαι.

Thus, if a straight-line is at right-angles to some plane then all of the planes (passing) through it will also be at right-angles to the same plane. (Which is) the very thing it was required to

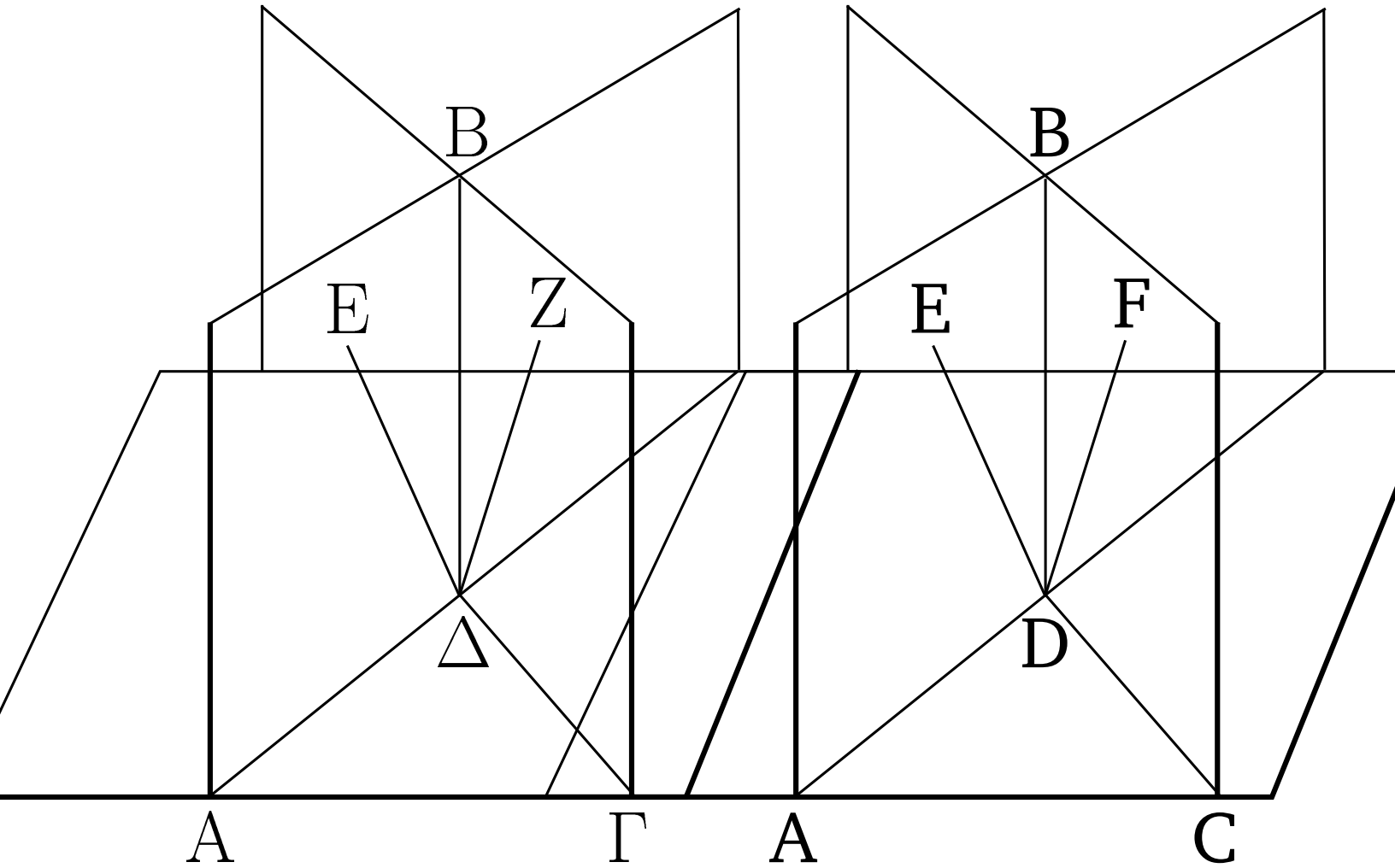
19.

Ἐὰν δύο ἐπίπεδα τέμνοντα ὀάλληλα ἐπιπέδῳ τινὶ πρὸς ὀρθὰξὸῇ, καὶ ἡ κοινὴ αὐτῶν τομὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξὸέσται.

show.

Proposition 19

If two planes cutting one another are at right-angles to some plane then their common section will also be at right-angles to the same plane.



Δύο γὰρ ἐπίπεδα τὰ AB, BΓ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθὰξὸέστω, κοινὴ δὲ αὐτῶν τομὴ ὀέστω ἡ BΔ· λέγω, ὅτι ἡ BΔ τῷ ὑποκειμένῳ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν.

μὴ γάρ, καὶ ἤθθωσαν ἀπὸ τοῦ Δ σημείου ἐν μὲν τῷ AB ἐπιπέδῳ τῇ AΔ εὐθείᾳ πρὸς ὀρθὰξ ἡ ΔE, ἐν δὲ τῷ BΓ ἐπιπέδῳ τῇ ΓΔ πρὸς ὀρθὰξ ἡ ΔZ.

Καὶ ἐπεὶ τὸ AB ἐπίπεδον ὀρθόν ἐστι πρὸς τὸ ὑποκείμενον, καὶ τῇ κοινῇ αὐτῶν τομῇ τῇ AΔ πρὸς ὀρθὰξ ἐν τῷ AB ἐπιπέδῳ ὀῆκται ἡ ΔE, ἡ ΔE ἄρα ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ ΔZ ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον. ἀπὸ τοῦ αὐτοῦ ἄρα σημείου τοῦ Δ τῷ ὑποκειμένῳ ἐπιπέδῳ δύο εὐθεῖα πρὸς ὀρθὰξ ἀνεσταμέναι

For let the two planes *AB* and *BC* be at right-angles to a reference plane, and let their common section be *BD*. I say that *BD* is at right-angles to the reference plane.

For (if) not, let *DE* also be drawn from point *D*, in the plane *AB*, at right-angles to the straight-line *AD*, and *DF*, in the plane *BC*, at right-angles to *CD*.

And since the plane *AB* is at right-angles to the reference (plane), and *DE* has been drawn at right-angles to their common section *AD*, in the plane *AB*, *DE* is thus at right-angles to the reference plane [Def. 11.4]. So, similarly, we can show that *DF* is also at right-angles to the reference plane. Thus, two (different) straight-lines are set up, at the same point *D*, at

εἰσὶν ἐπὶ τὰ αὐτὰ μέρη· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα τῷ ὑποκειμένῳ ἐπιπέδῳ ἀπὸ τοῦ Δ σημείου ἀνασταθήσεται πρὸς ὀρθὰξ πλὴν τῆς ΔΒ κοινῆς τομῆς τῶν ΑΒ, ΒΓ ἐπιπέδων.

Ἐὰν ἄρα δύο ἐπίπεδα τέμνοντα ἀλλήλα ἐπιπέδῳ τινὶ πρὸς ὀρθὰξ ᾖ, καὶ ἡ κοινὴ αὐτῶν τομὴ τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθὰξ ἔσται· ὅπερ ἔδει δεῖξαι.

right-angles to the reference plane, on the same side. The very thing is impossible [Prop. 11.13]. Thus, no (other straight-line) except the common section *DB* of the planes *AB* and *BC* can be set up at point *D*, at right-angles to the reference plane.

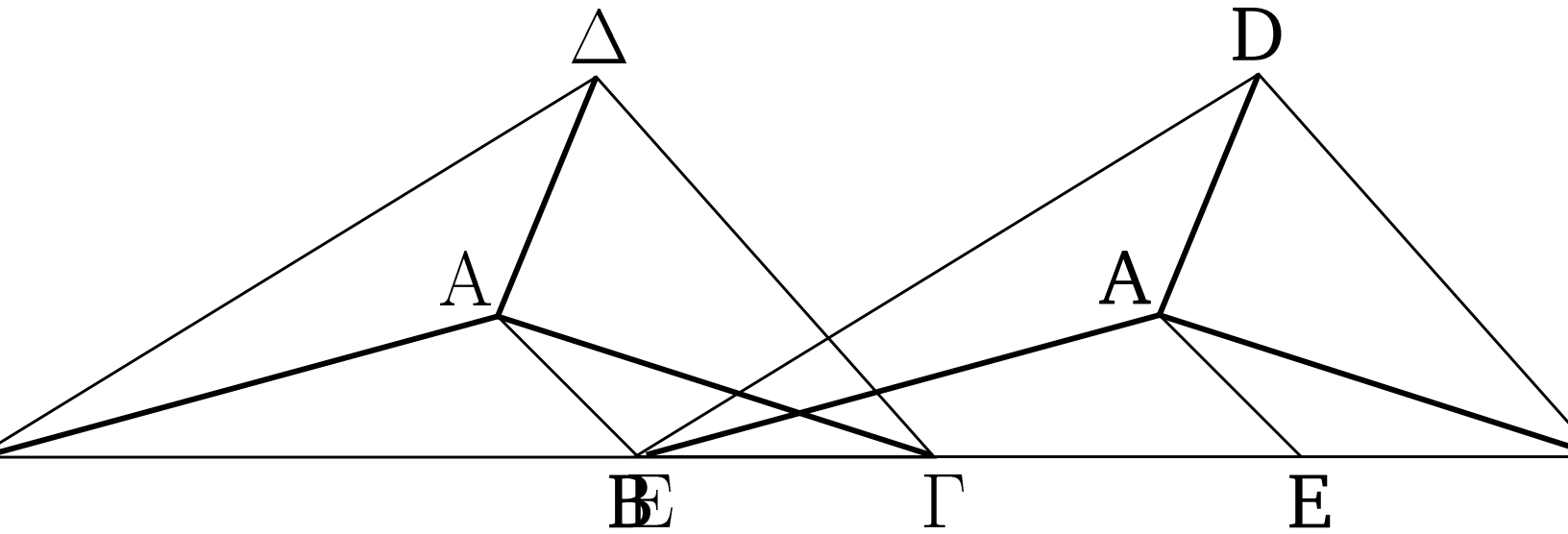
Thus, if two planes cutting one another are at right-angles to some plane then their common section will also be at right-angles to the same plane. (Which is) the very thing it was required to show.

20.

Ἐὰν στερεὰ γωνία ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέθηται, δύο ὁποιαιοῦν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι.

Proposition 20

If a solid angle is contained by three plane angles then (the sum of) any two (angles) is greater than the remaining (one), (the angles) being taken up in any (possible way).



Στερεὰ γὰρ γωνία ἡ πρὸς τῷ Α ὑπὸ τριῶν γωνιῶν ἐπιπέδων τῶν ὑπὸ ΒΑΓ, ΓΑΔ, ΔΑΒ περιεθέσθω· λέγω, ὅτι τῶν ὑπὸ ΒΑΓ, ΓΑΔ, ΔΑΒ γωνιῶν δύο ὁποιαιοῦν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι.

Εἰ μὲν οὖν αἱ ὑπὸ ΒΑΓ, ΓΑΔ, ΔΑΒ γωνίαι ἴσαι ἀλλήλαις εἰσίν, φανερόν, ὅτι δύο ὁποιαιοῦν τῆς λοιπῆς μείζονές εἰσιν. εἰ δὲ οὐ, ἔστω μείζων ἡ ὑπὸ ΒΑΓ, καὶ συνεστάτω πρὸς τῇ ΑΒ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Α τῇ ὑπὸ ΔΑΒ γωνίᾳ ἐν τῷ διὰ τῶν ΒΑΓ ἐπιπέδῳ ἴση ἡ ὑπὸ ΒΑΕ, καὶ κείσθω τῇ ΑΔ ἴση ἡ ΑΕ, καὶ διὰ τοῦ Ε σημείου διαθεῖσα ἡ ΒΕΓ τεμνέτω τὰξ ΑΒ, ΑΓ εὐθείας κατὰ τὰ Β, Γ σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΔΒ, ΔΓ.

Καὶ ἐπεὶ ἴση ἐστὶν ἡ ΔΑ τῇ ΑΕ, κοινὴ δὲ ἡ ΑΒ, δύο δυσὶν ἴσαι· καὶ γωνία ἡ ὑπὸ ΔΑΒ γωνίᾳ τῇ ὑπὸ ΒΑΕ ἴση· βάσις ἄρα ἡ ΔΒ βάσει τῇ ΒΕ ἐστιν ἴση. καὶ ἐπεὶ δύο αἱ ΒΔ, ΔΓ τῆς ΒΓ μείζονές εἰσιν, ὧν ἡ ΔΒ τῇ ΒΕ ἐδείθθη ἴση, λοιπὴ ἄρα ἡ ΔΓ λοιπῆς τῆς ΕΓ μείζων ἐστίν. καὶ ἐπεὶ ἴση ἐστὶν ἡ ΔΑ τῇ ΑΕ, κοινὴ δὲ ἡ ΑΓ, καὶ βάσις ἡ ΔΓ βάσεως τῆς ΕΓ μείζων ἐστίν, γωνία ἄρα ἡ ὑπὸ ΔΑΓ γωνίας τῆς ὑπὸ ΕΑΓ μείζων ἐστίν. ἐδείθθη δὲ καὶ ἡ ὑπὸ ΔΑΒ τῇ ὑπὸ ΒΑΕ ἴση· αἱ

For let the solid angle *A* be contained by the three plane angles *BAC*, *CAD*, and *DAB*. I say that (the sum of) any two of the angles *BAC*, *CAD*, and *DAB* is greater than the remaining (one), (the angles) being taken up in any (possible way).

For if the angles *BAC*, *CAD*, and *DAB* are equal to one another then (it is) clear that (the sum of) any two is greater than the remaining (one). But, if not, let *BAC* be greater (than *CAD* or *DAB*). And let (angle) *BAE*, equal to the angle *DAB*, be constructed in the plane through *BAC*, on the straight-line *AB*, at the point *A* on it. And let *AE* be made equal to *AD*. And *BEC* being drawn across through point *E*, let it cut the straight-lines *AB* and *AC* at points *B* and *C* (respectively). And let *DB* and *DC* be joined.

And since *DA* is equal to *AE*, and *AB* (is) common, the two (straight-lines *AD* and *AB* are) equal to the two (straight-lines *EA* and *AB*, respectively). And angle *DAB* (is) equal to angle *BAE*. Thus, the base *DB* is equal to the base *BE* [Prop. 1.4]. And since the (sum of the) two (straight-lines) *BD* and *DC* is greater than *BC* [Prop. 1.20], of which *DB* was shown (to be) equal to *BE*, the remainder *DC* is thus greater

ἄρα ὑπὸ ΔΑΒ, ΔΑΓ τῆς ὑπὸ ΒΑΓ μείζονές εἰσιν. ὁμοίως δὴ δείχομεν, ὅτι καὶ αἱ λοιπαὶ σύνδυο λαμβανόμεναι τῆς λοιπῆς μείζονές εἰσιν.

Ἐὰν ἄρα στερεὰ γωνία ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέθηται, δύο ὁποιαιοῦν τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι· ὅπερ ἔδει δεῖξαι.

than the remainder *EC*. And since *DA* is equal to *AE*, but *AC* (is) common, and the base *DC* is greater than the base *EC*, the angle *DAC* is thus greater than the angle *EAC* [Prop. 1.25]. And *DAB* was also shown (to be) equal to *BAE*. Thus, (the sum of) *DAB* and *DAC* is greater than *BAC*. So, similarly, we can also show that the remaining (angles), being taken in pairs, are greater than the remaining (one).

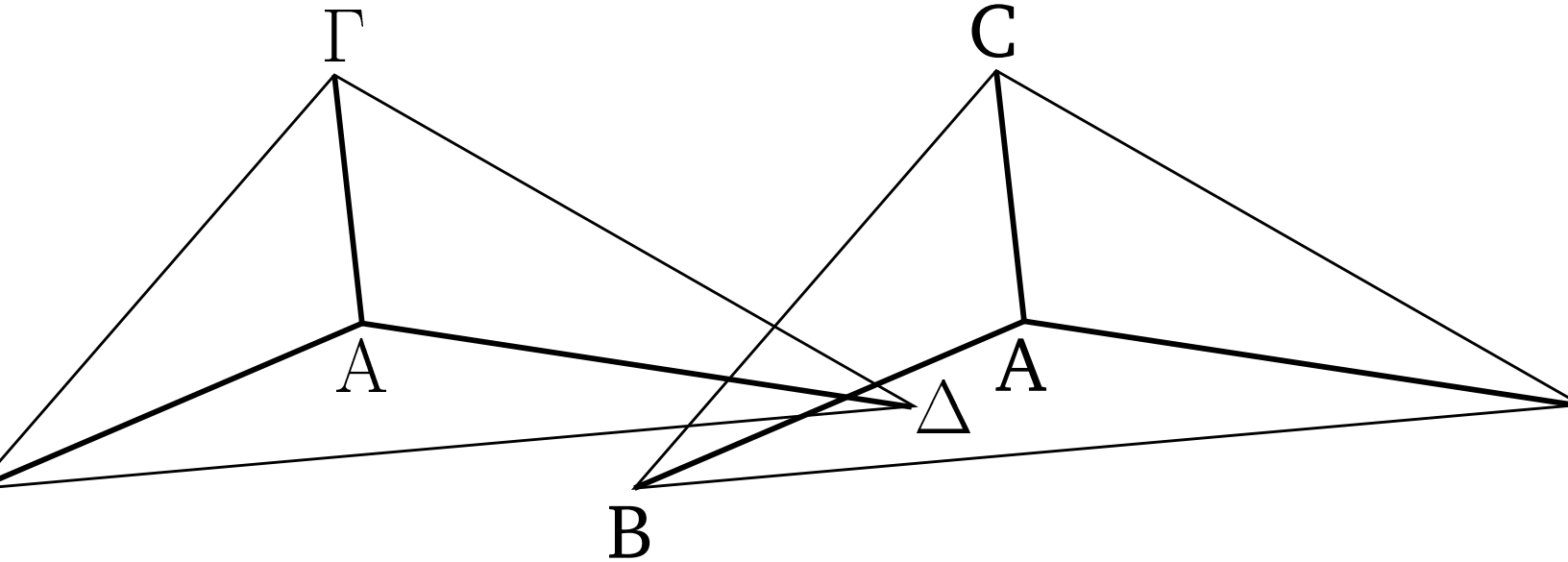
Thus, if a solid angle is contained by three plane angles then (the sum of) any two (angles) is greater than the remaining (one), (the angles) being taken up in any (possible way). (Which is) the very thing it was required to show.

21.

Ὅπασα στερεὰ γωνία ὑπὸ ἐλασσόνων [ἢ] τεσσάρων ὀρθῶν γωνιῶν ἐπιπέδων περιέθεται.

Proposition 21

Any solid angle is contained by plane angles (whose sum is) less [than] four right-angles.[†]



Ἔστω στερεὰ γωνία ἡ πρὸς τῷ Α περιεθομένη ὑπὸ ἐπιπέδων γωνιῶν τῶν ὑπὸ ΒΑΓ, ΓΑΔ, ΔΑΒ· λέγω, ὅτι αἱ ὑπὸ ΒΑΓ, ΓΑΔ, ΔΑΒ τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν.

Εἰλήφθω γὰρ ἐφ' ἑκάστης τῶν ΑΒ, ΑΓ, ΑΔ τυθόντα σημεῖα τὰ Β, Γ, Δ, καὶ ἐπεζεύχθωσαν αἱ ΒΓ, ΓΔ, ΔΒ. καὶ ἐπεὶ στερεὰ γωνία ἡ πρὸς τῷ Β ὑπὸ τριῶν γωνιῶν ἐπιπέδων περιέθεται τῶν ὑπὸ ΓΒΑ, ΑΒΔ, ΓΒΔ, δύο ὁποιαιοῦν τῆς λοιπῆς μείζονές εἰσιν· αἱ ἄρα ὑπὸ ΓΒΑ, ΑΒΔ τῆς ὑπὸ ΓΒΔ μείζονές εἰσιν. διὰ τὰ αὐτὰ δὴ καὶ αἱ μὲν ὑπὸ ΒΓΑ, ΑΓΔ τῆς ὑπὸ ΒΓΔ μείζονές εἰσιν, αἱ δὲ ὑπὸ ΓΔΑ, ΑΔΒ τῆς ὑπὸ ΓΔΒ μείζονές εἰσιν· αἱ ὀὲχ ἄρα γωνίαι αἱ ὑπὸ ΓΒΑ, ΑΒΔ, ΒΓΑ, ΑΓΔ, ΓΔΑ, ΑΔΒ τριῶν τῶν ὑπὸ ΓΒΔ, ΒΓΑ, ΓΔΒ μείζονές εἰσιν. ἀλλὰ αἱ τρεῖς αἱ ὑπὸ ΓΒΔ, ΒΔΓ, ΒΓΔ δυσὶν ὀρθαῖξ ἴσαι εἰσίν· αἱ ὀὲχ ἄρα αἱ ὑπὸ ΓΒΑ, ΑΒΔ, ΒΓΑ, ΑΓΔ, ΓΔΑ, ΑΔΒ δύο ὀρθῶν μείζονές εἰσιν. καὶ ἐπεὶ ἑκάστου τῶν ΑΒΓ, ΑΓΔ, ΑΔΒ τριγώνων αἱ τρεῖς γωνίαι δυσὶν ὀρθαῖξ ἴσαι εἰσίν,

Let the solid angle *A* be contained by the plane angles *BAC*, *CAD*, and *DAB*. I say that (the sum of) *BAC*, *CAD*, and *DAB* is less than four right-angles.

For let the random points *B*, *C*, and *D* be taken on each of (the straight-lines) *AB*, *AC*, and *AD* (respectively). And let *BC*, *CD*, and *DB* be joined. And since the solid angle at *B* is contained by the three plane angles *CBA*, *ABD*, and *CBD*, (the sum of) any two is greater than the remaining (one) [Prop. 11.20]. Thus, (the sum of) *CBA* and *ABD* is greater than *CBD*. So, for the same (reasons), (the sum of) *BCA* and *ACD* is also greater than *BCD*, and (the sum of) *CDA* and *ADB* is greater than *CDB*. Thus, the (sum of the) six angles *CBA*, *ABD*, *BCA*, *ACD*, *CDA*, and *ADB* is greater than the (sum of the) three (angles) *CBD*, *BCD*, and *CDB*. But, the (sum of the) three (angles) *CBD*, *BDC*, and *BCD* is equal to two right-angles [Prop. 1.32]. Thus, the (sum of the) six angles

αἱ ἄρα τῶν τριῶν τριγώνων ἐννέα γωνίαι αἱ ὑπὸ ΓΒΑ, ΑΓΒ, ΒΑΓ, ΑΓΔ, ΓΔΑ, ΓΑΔ, ΑΔΒ, ΔΒΑ, ΒΑΔ ὀξὲχ ὀρθαῖξ ἴσαι εἰσίν, ὁῶν αἱ ὑπὸ ΑΒΓ, ΒΓΑ, ΑΓΔ, ΓΔΑ, ΑΔΒ, ΔΒΑ ὀξὲχ γωνίαι δύο ὀρθῶν εἰσι μείζονες· λοιπαὶ ἄρα αἱ ὑπὸ ΒΑΓ, ΓΑΔ, ΔΑΒ τρεῖξ [γωνίαι] περιέθουσαι τὴν στερεὰν γωνίαν τεσσάρων ὀρθῶν ἐλάσσονέξ εἰσιν.

Ὅλπασα ἄρα στερεὰ γωνία ὑπὸ ἐλασσόνων [ἢ] τεσσάρων ὀρθῶν γωνιῶν ἐπιπέδων περιέθεται· ὅπερ ἔδει δεῖχαι.

CBA, *ABD*, *BCA*, *ACD*, *CDA*, and *ADB* is greater than two right-angles. And since the (sum of the) three angles of each of the triangles *ABC*, *ACD*, and *ADB* is equal to two right-angles, the (sum of the) nine angles *CBA*, *ACB*, *BAC*, *ACD*, *CDA*, *CAD*, *ADB*, *DBA*, and *BAD* of the three triangles is equal to six right-angles, of which the (sum of the) six angles *ABC*, *BCA*, *ACD*, *CDA*, *ADB*, and *DBA* is greater than two right-angles. Thus, the (sum of the) remaining three [angles] *BAC*, *CAD*, and *DAB*, containing the solid angle, is less than four right-angles.

Thus, any solid angle is contained by plane angles (whose sum is) less [than] four right-angles. (Which is) the very thing it was required to show.

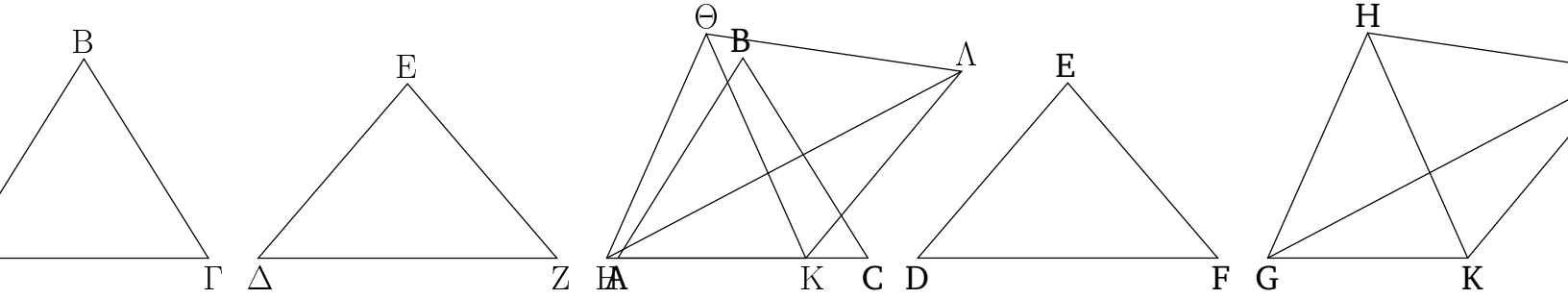
† This proposition is only proved for the case of a solid angle contained by three plane angles. However, the generalization to a solid angle contained by more than three plane angles is straightforward.

22.

Ἐὰν ὁῶσι τρεῖξ γωνίαι ἐπίπεδοι, ὁῶν αἱ δύο τῆξ λοιπῆξ μείζονέξ εἰσι πάντῃ μεταλαμβανόμεναι, περιέθωσι δὲ αὐτὰξ ἴσαι εὐθεῖαι, δυνατόν ἐστιν ἐκ τῶν ἐπιζευγνυουσῶν τὰξ ἴσαξ εὐθείαξ τρίγωνον συστήσασθαι.

Proposition 22

If there are three plane angles, of which (the sum of any) two is greater than the remaining (one), (the angles) being taken up in any (possible way), and if equal straight-lines contain them, then it is possible to construct a triangle from (the straight-lines created by) joining the (ends of the) equal straight-lines.



Ἔστωσαν τρεῖξ γωνίαι ἐπίπεδοι αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ, ὁῶν αἱ δύο τῆξ λοιπῆξ μείζονέξ εἰσι πάντῃ μεταλαμβανόμεναι, αἱ μὲν ὑπὸ ΑΒΓ, ΔΕΖ τῆξ ὑπὸ ΗΘΚ, αἱ δὲ ὑπὸ ΔΕΖ, ΗΘΚ τῆξ ὑπὸ ΑΒΓ, καὶ ὀέτι αἱ ὑπὸ ΗΘΚ, ΑΒΓ τῆξ ὑπὸ ΔΕΖ, καὶ ὀέστωσαν ἴσαι αἱ ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ εὐθεῖαι, καὶ ἐπεζεύθθωσαν αἱ ΑΓ, ΔΖ, ΗΚ· λέγω, ὅτι δυνατόν ἐστιν ἐκ τῶνὸίσων ταῖξ ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι, τουτέστιν ὅτι τῶν ΑΓ, ΔΖ, ΗΚ δύο ὁποιαιοῦν τῆξ λοιπῆξ μείζονέξ εἰσιν.

Εἰ μὲν οὐῦν αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ γωνίαι ἴσαι ἀλλήλαις εἰσίν, φανερόν, ὅτι καὶ τῶν ΑΓ, ΔΖ, ΗΚὸίσων γινομένων δυνατόν ἐστιν ἐκ τῶνὸίσων ταῖξ ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι. εἰ δὲ οὐύ, ὀέστωσανὸάνισοι, καὶ συνεστάτω πρὸξ τῇ ΘΚ εὐθείᾳ καὶ τῷ πρὸξ αὐτῇ σημείῳ τῷ Θ τῇ ὑπὸ ΑΒΓ γωνίᾳ ὁίση ἡ ὑπὸ ΚΘΛ· καὶ κείσθω μιᾷ τῶν ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚὁίση ἡ ΘΛ, καὶ ἐπεζεύθθωσαν αἱ ΚΛ, ΗΛ. καὶ ἐπεὶ δύο αἱ ΑΒ, ΒΓ δυσὶ ταῖξ ΚΘ, ΘΛ ἴσαι εἰσίν, καὶ γωνία ἡ πρὸξ τῷ Β γωνίᾳ τῇ ὑπὸ ΚΘΛὁίση, βάσιξ ἄρα ἡ ΑΓ βάσει τῇ ΚΛὁίση.

Let *ABC*, *DEF*, and *GHK* be three plane angles, of which the sum of any) two is greater than the remaining (one), (the angles) being taken up in any (possible way)—(that is), *ABC* and *DEF* (greater) than *GHK*, *DEF* and *GHK* (greater) than *ABC*, and, further, *GHK* and *ABC* (greater) than *DEF*. And let *AB*, *BC*, *DE*, *EF*, *GH*, and *HK* be equal straight-lines. And let *AC*, *DF*, and *GK* be joined. I say that that it is possible to construct a triangle out of (straight-lines) equal to *AC*, *DF*, and *GK*—that is to say, that (the sum of) any two of *AC*, *DF*, and *GK* is greater than the remaining (one).

Now, if the angles *ABC*, *DEF*, and *GHK* are equal to one another then (it is) clear that, (with) *AC*, *DF*, and *GK* also becoming equal, it is possible to construct a triangle from (straight-lines) equal to *AC*, *DF*, and *GK*. And if not, let them be unequal, and let *KHL*, equal to angle *ABC*, be constructed on the straight-line *HK*, at the point *H* on it. And let *HL* be made equal to one of *AB*, *BC*, *DE*, *EF*, *GH*, and *HK*. And let

καὶ ἐπεὶ αἱ ὑπὸ ΑΒΓ, ΗΘΚ τῆς ὑπὸ ΔΕΖ μείζονές εἰσιν, ὁίση δὲ ἡ ὑπὸ ΑΒΓ τῇ ὑπὸ ΚΘΛ, ἡ ἄρα ὑπὸ ΗΘΛ τῆς ὑπὸ ΔΕΖ μείζων ἐστίν. καὶ ἐπεὶ δύο αἱ ΗΘ, ΘΛ δύο ταῖς ΔΕ, ΕΖ ἴσαι εἰσίν, καὶ γωνία ἡ ὑπὸ ΗΘΛ γωνίας τῆς ὑπὸ ΔΕΖ μείζων, βάσις ἄρα ἡ ΗΛ βάσεως τῆς ΔΖ μείζων ἐστίν. ἀλλὰ αἱ ΗΚ, ΚΛ τῆς ΗΛ μείζονές εἰσιν. πολλῷ ἄρα αἱ ΗΚ, ΚΛ τῆς ΔΖ μείζονές εἰσιν. ὁίση δὲ ἡ ΚΛ τῇ ΑΓ· αἱ ΑΓ, ΗΚ ἄρα τῆς λοιπῆς τῆς ΔΖ μείζονές εἰσιν. ὁμοίως δὴ δείχομεν, ὅτι καὶ αἱ μὲν ΑΓ, ΔΖ τῆς ΗΚ μείζονές εἰσιν, καὶ ὀέτι αἱ ΔΖ, ΗΚ τῆς ΑΓ μείζονές εἰσιν. δυνατὸν ἄρα ἐστὶν ἐκ τῶνὁίσων ταῖς ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι· ὅπερ ἔδει δεῖχαι.

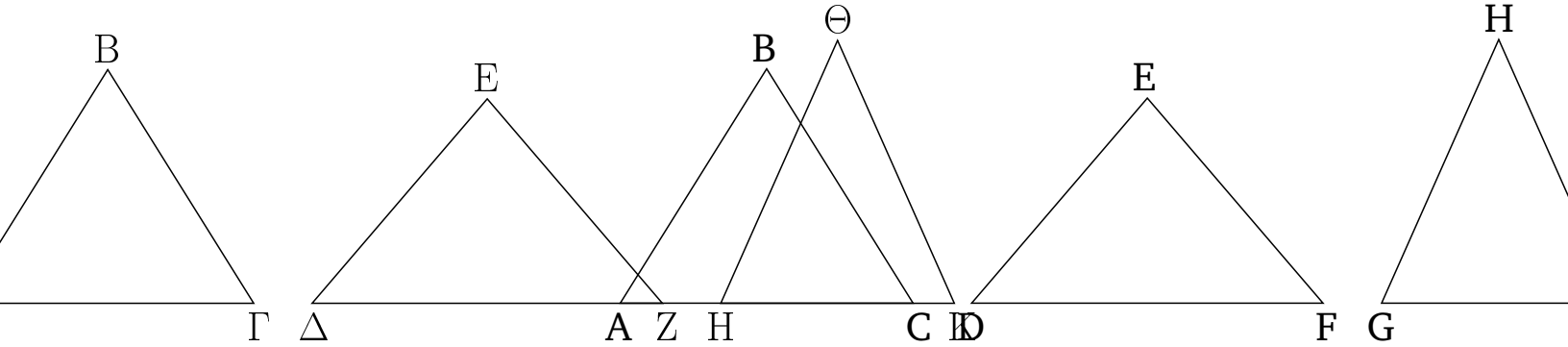
KL and GL be joined. And since the two (straight-lines) AB and BC are equal to the two (straight-lines) KH and HL (respectively), and the angle at B (is) equal to KHL, the base AC is thus equal to the base KL [Prop. 1.4]. And since (the sum of) ABC and GHK is greater than DEF, and ABC equal to KHL, GHL is thus greater than DEF. And since the two (straight-lines) GH and HL are equal to the two (straight-lines) DE and EF (respectively), and angle GHL (is) greater than DEF, the base GL is thus greater than the base DF [Prop. 1.24]. But, (the sum of) GK and KL is greater than GL [Prop. 1.20]. Thus, (the sum of) GK and KL is much greater than DF. And KL (is) equal to AC. Thus, (the sum of) AC and GK is greater than the remaining (straight-line) DF. So, similarly, we can show that (the sum of) AC and DF is greater than GK, and, further, that (the sum of) DF and GK is greater than AC. Thus, it is possible to construct a triangle from (straight-lines) equal to AC, DF, and GK. (Which is) the very thing it was required to show.

23.

Ἐκ τριῶν γωνιῶν ἐπιπέδων, ὧν αἱ δύο τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι, στερεὰν γωνίαν συστήσασθαι· δεῖ δὴ τὰξ τρεῖς τεσσάρων ὀρθῶν ἐλάσσονας εὐῖναι.

Proposition 23

To construct a solid angle from three (given) plane angles, (the sum of) two of which is greater than the remaining (one, the angles) being taken up in any (possible way). So, it is necessary for the (sum of the) three (angles) to be less than four right-angles [Prop. 11.21].



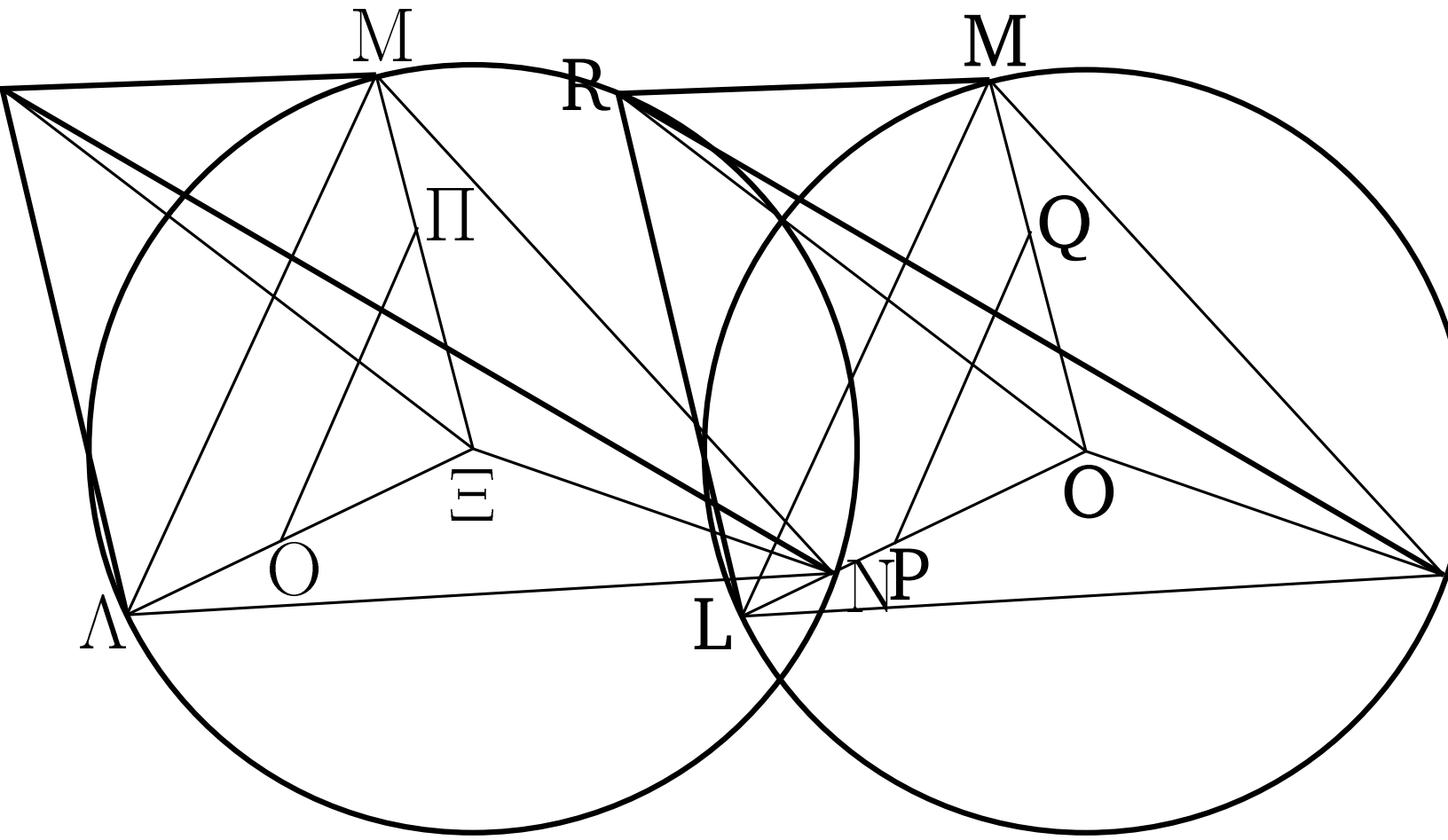
Ἔστωσαν αἱ δοθεῖσαι τρεῖς γωνίαι ἐπίπεδοι αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ, ὧν αἱ δύο τῆς λοιπῆς μείζονεςὀέστωσαν πάντῃ μεταλαμβανόμεναι, ὀέτι δὲ αἱ τρεῖς τεσσάρων ὀρθῶν ἐλάσσονες· δεῖ δὴ ἐκ τῶνὁίσων ταῖς ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ στερεὰν γωνίαν συστήσασθαι.

Ἀπειλήφθωσαν ἴσαι αἱ ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ, καὶ ἐπεζεύθθωσαν αἱ ΑΓ, ΔΖ, ΗΚ· δυνατὸν ἄρα ἐστὶν ἐκ τῶνὁίσων ταῖς ΑΓ, ΔΖ, ΗΚ τρίγωνον συστήσασθαι. συνεστάτω τὸ ΛΜΝ, ὁώστε ἴσην εὐῖναι τὴν μὲν ΑΓ τῇ ΛΜ, τὴν δὲ ΔΖ τῇ ΜΝ, καὶ ὀέτι τὴν ΗΚ τῇ ΝΛ, καὶ περιγεγράφθω περὶ τὸ ΛΜΝ τρίγωνον κύκλος ὁ ΛΜΝ, καὶ εἰλήφθω αὐτοῦ τὸ κέντρον καὶ ὀέστω τὸ Χ, καὶ ἐπεζεύθθωσαν αἱ ΛΧ, ΜΧ, ΝΧ·

Let ABC, DEF, and GHK be the three given plane angles, of which let (the sum of) two be greater than the remaining (one, the angles) being taken up in any (possible way), and, further, (let) the (sum of the) three (be) less than four right-angles. So, it is necessary to construct a solid angle from (plane angles) equal to ABC, DEF, and GHK.

Let AB, BC, DE, EF, GH, and HK be cut off (so as to be) equal (to one another). And let AC, DF, and GK be joined. It is, thus, possible to construct a triangle from (straight-lines) equal to AC, DF, and GK [Prop. 11.22]. Let (such a triangle), LMN, have be constructed, such that AC is equal to LM, DF to MN, and, further, GK to NL. And let the circle LMN be circumscribed about triangle LMN [Prop. 4.5]. And let its center be found, and let it be (at) O. And let LO, MO, and NO be

joined.



Λέγω, ὅτι ἡ ΑΒ μείζων ἐστὶ τῆς ΛΧ. εἰ γὰρ μή, ἤτοιὀίση ἐστὶν ἡ ΑΒ τῇ ΛΧὀὴ ἐλάττων. ὀέστω πρότερονὀίση. καὶ ἐπεὶ ὀίση ἐστὶν ἡ ΑΒ τῇ ΛΧ, ἀλλὰ ἡ μὲν ΑΒ τῇ ΒΓ ἐστινὀίση, ἡ δὲ ΧΛ τῇ ΧΜ, δύο δὴ αἱ ΑΒ, ΒΓ δύο ταῖς ΛΧ, ΧΜ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ βάσις ἡ ΑΓ βάσει τῇ ΛΜ ὑπόκειταιὀίση· γωνία ἄρα ἡ ὑπὸ ΑΒΓ γωνίᾳ τῇ ὑπὸ ΛΧΜ ἐστινὀίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ μὲν ὑπὸ ΔΕΖ τῇ ὑπὸ ΜΧΝ ἐστινὀίση, καὶ ὀέτι ἡ ὑπὸ ΗΘΚ τῇ ὑπὸ ΝΧΛ· αἱ ἄρα τρεῖς αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ γωνίαι τρισὶ ταῖς ὑπὸ ΛΧΜ, ΜΧΝ, ΝΧΛ εἰσιν ἴσαι. ἀλλὰ αἱ τρεῖς αἱ ὑπὸ ΛΧΜ, ΜΧΝ, ΝΧΛ τέτταρσιν ὀρθαῖξ εἰσιν ἴσαι· καὶ αἱ τρεῖς ἄρα αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ τέτταρσιν ὀρθαῖξ ἴσαι εἰσίν. ὑπόκεινται δὲ καὶ τεσσάρων ὀρθῶν ἐλάσσονες· ὅπερὀάτοπον. οὐκ ἄρα ἡ ΑΒ τῇ ΛΧὀίση ἐστίν. λέγω δή, ὅτι οὐδὲ ἐλάττων ἐστὶν ἡ ΑΒ τῆς ΛΧ. εἰ γὰρ δυνατόν, ὀέστω· καὶ κείσθω τῇ μὲν ΑΒὀίση ἡ ΧΟ, τῇ δὲ ΒΓὀίση ἡ ΧΠ, καὶ ἐπεζεύθθω ἡ ΟΠ. καὶ ἐπεὶ ὀίση ἐστὶν ἡ ΑΒ τῇ ΒΓ, ὀίση ἐστὶ καὶ ἡ ΧΟ τῇ ΧΠ· ὥστε καὶ λοιπὴ ἡ ΛΟ τῇ ΠΜ ἐστινὀίση. παράλληλος ἄρα ἐστὶν ἡ ΛΜ τῇ ΟΠ, καὶ ἰσογώνιον τὸ ΛΜΧ τῷ ΟΠΧ· ὀέστιν ἄρα ὡξ ἡ ΧΛ πρὸς ΛΜ, οὕτως ἡ ΧΟ πρὸς ΟΠ· ἐναλλὰχ ὡξ ἡ ΛΧ πρὸς ΧΟ,

I say that AB is greater than LO. For, if not, AB is either equal to, or less than, LO. Let it, first of all, be equal. And since AB is equal to LO, but AB is equal to BC, and OL to OM, so the two (straight-lines) AB and BC are equal to the two (straight-lines) LO and OM, respectively. And the base AC was assumed (to be) equal to the base LM. Thus, angle ABC is equal to angle LOM [Prop. 1.8]. So, for the same (reasons), DEF is also equal to MON, and, further, GHK to NOL. Thus, the three angles ABC, DEF, and GHK are equal to the three angles LOM, MON, and NOL, respectively. But, the (sum of the) three angles LOM, MON, and NOL is equal to four right-angles. Thus, the (sum of the) three angles ABC, DEF, and GHK is also equal to four right-angles. And it was also assumed (to be) less than four right-angles. The very thing (is) absurd. Thus, AB is not equal to LO. So, I say that AB is not less than LO either. For, if possible, let it be (less). And let OP be made equal to AB, and OQ equal to BC, and let PQ be joined. And since AB is equal to BC, OP is also equal to OQ. Hence, the remainder LP is also equal to (the remainder) QM. LM is thus parallel to PQ [Prop. 6.2], and (triangle) LMO (is)

οὑτως ἡ ΛΜ πρὸς ΟΠ. μείζων δὲ ἡ ΛΧ τῆς ΧΟ· μείζων ἄρα καὶ ἡ ΛΜ τῆς ΟΠ. ἀλλὰ ἡ ΛΜ κεῖται τῇ ΑΓὁίση· καὶ ἡ ΑΓ ἄρα τῆς ΟΠ μείζων ἐστίν. ἐπεὶ οὐν δύο αἱ ΑΒ, ΒΓ δυσὶ ταῖς ΟΧ, ΧΠ ἴσαι εἰσίν, καὶ βάσις ἡ ΑΓ βάσεως τῆς ΟΠ μείζων ἐστίν, γωνία ἄρα ἡ ὑπὸ ΑΒΓ γωνίας τῆς ὑπὸ ΟΧΠ μεῖζων ἐστίν. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ μὲν ὑπὸ ΔΕΖ τῆς ὑπὸ ΜΧΝ μείζων ἐστίν, ἡ δὲ ὑπὸ ΗΘΚ τῆς ὑπὸ ΝΧΛ. αἱ ἄρα τρεῖς γωνίαι αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ τριῶν τῶν ὑπὸ ΛΧΜ, ΜΧΝ, ΝΧΛ μείζονές εἰσιν. ἀλλὰ αἱ ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ τεσσάρων ὀρθῶν ἐλάσσονες ὑπόκεινται· πολλῷ ἄρα αἱ ὑπὸ ΛΧΜ, ΜΧΝ, ΝΧΛ τεσσάρων ὀρθῶν ἐλάσσονές εἰσιν. ἀλλὰ καὶ ἴσαι· ὅπερ ἐστὶνὁάτοπον. οὐκ ἄρα ἡ ΑΒ ἐλάσσων ἐστὶ τῆς ΛΧ. ἐδείθθη δέ, ὅτι οὐδὲ ὁίση· μείζων ἄρα ἡ ΑΒ τῆς ΛΧ.

Ἀνεστάτω δὴ ἀπὸ τοῦ Χ σημείου τῷ τοῦ ΛΜΝ κύκλου ἐπιπέδῳ πρὸς ὀρθὰξ ἡ ΧΡ, καὶ ὁῷ μεῖζόν ἐστι τὸ ἀπὸ τῆς ΑΒ τετράγωνον τοῦ ἀπὸ τῆς ΛΧ, ἐκείνῳ ἴσονὁέστω τὸ ἀπὸ τῆς ΧΡ, καὶ ἐπεζεύθθωσαν αἱ ΡΛ, ΡΜ, ΡΝ.

Καὶ ἐπεὶ ἡ ΡΧ ὀρθὴ ἐστι πρὸς τὸ τοῦ ΛΜΝ κύκλου ἐπίπεδον, καὶ πρὸς ἑκάστην ἄρα τῶν ΛΧ, ΜΧ, ΝΧ ὀρθή ἐστιν ἡ ΡΧ. καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΛΧ τῇ ΧΜ, κοινὴ δὲ καὶ πρὸς ὀρθὰξ ἡ ΧΡ, βάσις ἄρα ἡ ΡΛ βάσει τῇ ΡΜ ἐστινὁίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΡΝ ἑκατέρᾳ τῶν ΡΛ, ΡΜ ἐστινὁίση· αἱ τρεῖς ἄρα αἱ ΡΛ, ΡΜ, ΡΝ ἴσαι ἀλλήλαις εἰσίν. καὶ ἐπεὶ ὁῷ μεῖζόν ἐστι τὸ ἀπὸ τῆς ΑΒ τοῦ ἀπὸ τῆς ΛΧ, ἐκείνῳ ἴσον ὑπόκειται τὸ ἀπὸ τῆς ΧΡ, τὸ ἄρα ἀπὸ τῆς ΑΒ ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΛΧ, ΧΡ. τοῖς δὲ ἀπὸ τῶν ΛΧ, ΧΡ ἴσον ἐστὶ τὸ ἀπὸ τῆς ΛΡ· ὀρθὴ γὰρ ἡ ὑπὸ ΛΧΡ· τὸ ἄρα ἀπὸ τῆς ΑΒ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΡΛ· ὁίση ἄρα ἡ ΑΒ τῇ ΡΛ. ἀλλὰ τῇ μὲν ΑΒὁίση ἐστὶν ἑκάστη τῶν ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ, τῇ δὲ ΡΛὁίση ἑκατέρα τῶν ΡΜ, ΡΝ· ἑκάστη ἄρα τῶν ΑΒ, ΒΓ, ΔΕ, ΕΖ, ΗΘ, ΘΚ ἑκάστῃ τῶν ΡΛ, ΡΜ, ΡΝὁίση ἐστίν. καὶ ἐπεὶ δύο αἱ ΛΡ, ΡΜ δυσὶ ταῖς ΑΒ, ΒΓ ἴσαι εἰσίν, καὶ βάσις ἡ ΛΜ βάσει τῇ ΑΓ ὑπόκειταιὁίση, γωνία ἄρα ἡ ὑπὸ ΛΡΜ γωνίᾳ τῇ ὑπὸ ΑΒΓ ἐστινὁίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ μὲν ὑπὸ ΜΡΝ τῇ ὑπὸ ΔΕΖ ἐστινὁίση, ἡ δὲ ὑπὸ ΛΡΝ τῇ ὑπὸ ΗΘΚ.

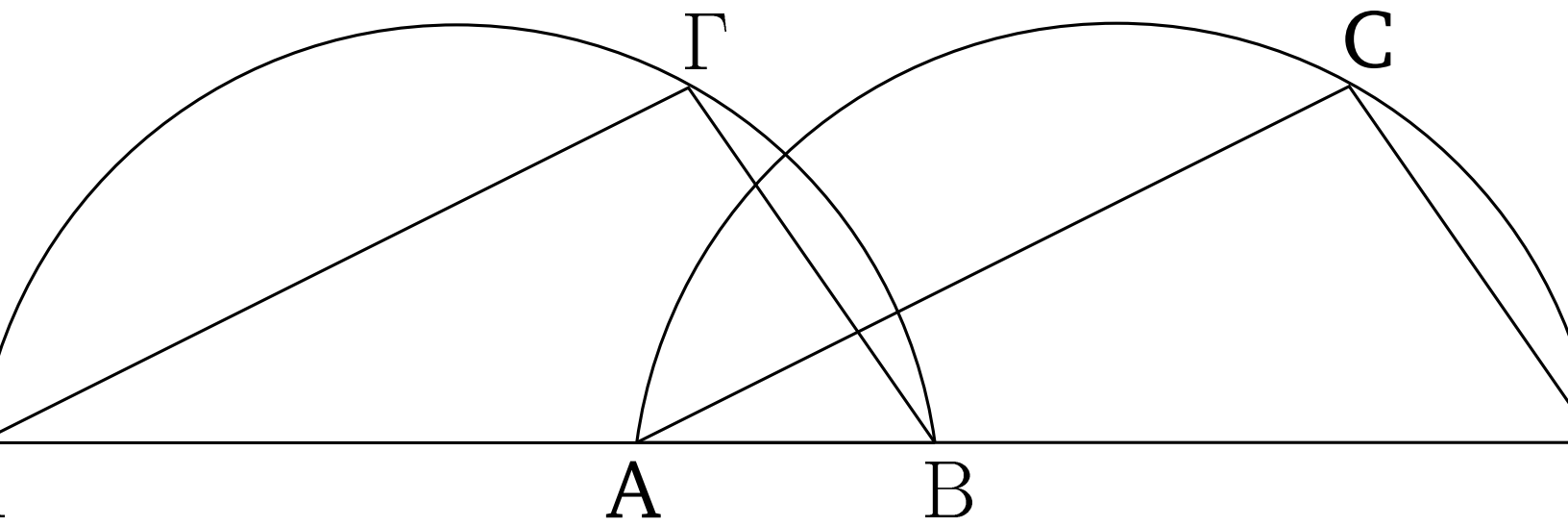
Ἐκ τριῶν ἄρα γωνιῶν ἐπιπέδων τῶν ὑπὸ ΛΡΜ, ΜΡΝ, ΛΡΝ, αὁί εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις ταῖς ὑπὸ ΑΒΓ, ΔΕΖ, ΗΘΚ, στερεὰ γωνία συνέσταται ἡ πρὸς τῷ Ρ περιεθομένη ὑπὸ τῶν ΛΡΜ, ΜΡΝ, ΛΡΝ γωνιῶν· ὅπερ ἔδει ποιῆσαι.

equiangular with (triangle) *PQO* [Prop. 1.29]. Thus, as *OL* is to *LM*, so *OP* (is) to *PQ* [Prop. 6.4]. Alternately, as *LO* (is) to *OP*, so *LM* (is) to *PQ* [Prop. 5.16]. And *LO* (is) greater than *OP*. Thus, *LM* (is) also greater than *PQ* [Prop. 5.14]. But *LM* was made equal to *AC*. Thus, *AC* is also greater than *PQ*. Therefore, since the two (straight-lines) *AB* and *BC* are equal to the two (straight-lines) *PO* and *OQ* (respectively), and the base *AC* is greater than the base *PQ*, the angle *ABC* is thus greater than the angle *POQ* [Prop. 1.25]. So, similarly, we can show that *DEF* is also greater than *MON*, and *GHK* than *NOL*. Thus, the (sum of the) three angles *ABC*, *DEF*, and *GHK* is greater than the (sum of the) three angles *LOM*, *MON*, and *NOL*. But, (the sum of) *ABC*, *DEF*, and *GHK* was assumed (to be) less than four right-angles. Thus, (the sum of) *LOM*, *MON*, and *NOL* is much less than four right-angles. But, (it is) also equal (to four right-angles). The very thing is absurd. Thus, *AB* is not less than *LO*. And it was shown (to be) not equal either. Thus, *AB* (is) greater than *LO*.

So let *OR* be set up at point *O* at right-angles to the plane of circle *LMN* [Prop. 11.12]. And let the (square) on *OR* be equal to that (area) by which the square on *AB* is greater than the (square) on *LO* [Prop. 11.23 lem.]. And let *RL*, *RM*, and *RN* be joined.

And since *RO* is at right-angles to the plane of circle *LMN*, *RO* is thus also at right-angles to each of *LO*, *MO*, and *NO*. And since *LO* is equal to *OM*, and *OR* is common and at right-angles, the base *RL* is thus equal to the base *RM* [Prop. 1.4]. So, for the same (reasons), *RN* is also equal to each of *RL* and *RM*. Thus, the three (straight-lines) *RL*, *RM*, and *RN* are equal to one another. And since the (square) on *OR* was assumed to be equal to that (area) by which the (square) on *AB* is greater than the (square) on *LO*, the (square) on *AB* is thus equal to the (sum of the squares) on *LO* and *OR*. And the (square) on *LR* is equal to the (sum of the squares) on *LO* and *OR*. For *LOR* (is) a right-angle [Prop. 1.47]. Thus, the (square) on *AB* is equal to the (square) on *RL*. Thus, *AB* (is) equal to *RL*. But, each of *BC*, *DE*, *EF*, *GH*, and *HK* is equal to *AB*, and each of *RM* and *RN* equal to *RL*. Thus, each of *AB*, *BC*, *DE*, *EF*, *GH*, and *HK* is equal to each of *RL*, *RM*, and *RN*. And since the two (straight-lines) *LR* and *RM* are equal to the two (straight-lines) *AB* and *BC* (respectively), and the base *LM* was assumed (to be) equal to the base *AC*, the angle *LRM* is thus equal to the angle *ABC* [Prop. 1.8]. So, for the same (reasons), *MRN* is also equal to *DEF*, and *LRN* to *GHK*.

Thus, the solid angle *R*, contained by the angles *LRM*, *MRN*, and *LRN*, has been constructed out of the three plane angles *LRM*, *MRN*, and *LRN*, which are equal to the three given (plane angles) *ABC*, *DEF*, and *GHK* (respectively). (Which is) the very thing it was required to do.



Λῆμμα.

ὁ Ὸν δὲ τρόπον, ὁῷ μεῖζόν ἐστι τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς ΛX, ἐκείνῳ ἴσον λαβεῖνὸέστι τὸ ἀπὸ τῆς XP, δείχομεν οὁύτως. ἐκκείσθωσαν αἱ AB, ΛX εὐθεῖαι, καὶ ὀέστω μείζων ἡ AB, καὶ γεγράφθω ἐπ᾽ αὐτῆς ἡμικύκλιον τὸ ABΓ, καὶ εἰξ τὸ ABΓ ἡμικύκλιον ἐνηρμόσθω τῇ ΛX εὐθείᾳ μὴ μείζονι οὁύσῃ τῆς AB διαμέτρουὸίση ἡ AΓ, καὶ ἐπεζεύθθω ἡ ΓB. ἐπεὶ οὁῦν ἐν ἡμικυκλίῳ τῷ AΓB γωνία ἐστὶν ἡ ὑπὸ AΓB, ὀρθὴ ἄρα ἐστὶν ἡ ὑπὸ AΓB. τὸ ἄρα ἀπὸ τῆς AB ἴσον ἐστὶ τοῖς ἀπὸ τῶν AΓ, ΓB. ὁώστε τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς AΓ μεῖζόν ἐστι τῷ ἀπὸ τῆς ΓB. ὁίση δὲ ἡ AΓ τῇ ΛX. τὸ ἄρα ἀπὸ τῆς AB τοῦ ἀπὸ τῆς ΛX μεῖζόν ἐστι τῷ ἀπὸ τῆς ΓB. ἐαν οὁῦν τῇ BΓ ἴσην τὴν XP ἀπολάβωμεν, ὀέσται τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς ΛX μεῖζον τῷ ἀπὸ τῆς XP· ὅπερ προέκειτο ποιῆσαι.

Lemma

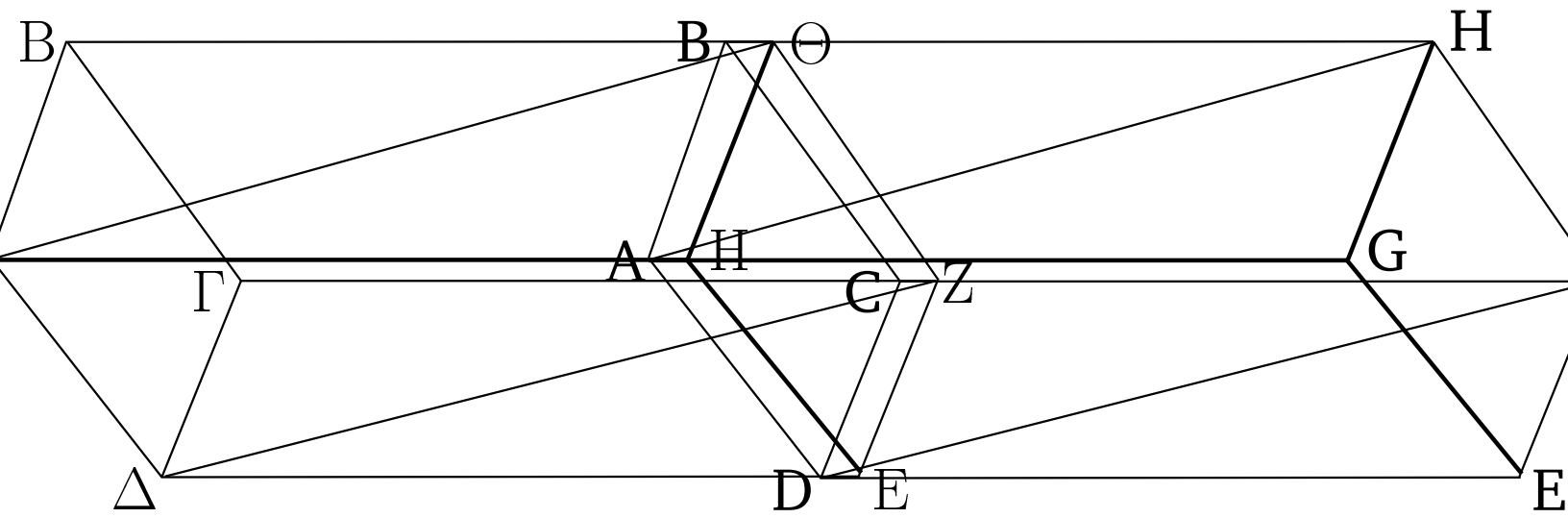
And we can demonstrate, thusly, in which manner to take the (square) on *OR* equal to that (area) by which the (square) on *AB* is greater than the (square) on *LO*. Let the straight-lines *AB* and *LO* be set out, and let *AB* be greater, and let the semicircle *ABC* be drawn around it. And let *AC*, equal to the straight-line *LO*, which is not greater than the diameter *AB*, be inserted into the semicircle *ABC* [Prop. 4.1]. And let *CB* be joined. Therefore, since the angle *ACB* is in the semicircle *ACB*, *ACB* is thus a right-angle [Prop. 3.31]. Thus, the (square) on *AB* is equal to the (sum of the) squares on *AC* and *CB* [Prop. 1.47]. Hence, the (square) on *AB* is greater than the (square) on *AC* by the (square) on *CB*. And *AC* (is) equal to *LO*. Thus, the (square) on *AB* is greater than the (square) on *LO* by the (square) on *CB*. Therefore, if we take *OR* equal to *BC* then the (square) on *AB* will be greater than the (square) on *LO* by the (square) on *OR*. (Which is) the very thing it was prescribed to do.

24.

Ἐὰν στερεὸν ὑπὸ παραλλήλων ἐπιπέδων περιέθηται, τὰ ἀπεναντίον αὐτοῦ ἐπίπεδα ἴσα τε καὶ παραλληλόγραμμά ἐστιν.

Proposition 24

If a solid (figure) is contained by (six) parallel planes then its opposite planes are both equal and parallelogrammic.



Στερεὸν γὰρ τὸ ΓΔΘΗ ὑπὸ παραλλήλων ἐπιπέδων περιεθέσθω τῶν ΑΓ, ΗΖ, ΑΘ, ΔΖ, ΒΖ, ΑΕ· λέγω, ὅτι τὰ ἀπεναντίον αὐτοῦ ἐπίπεδα ἴσα τε καὶ παραλληλόγραμμά ἐστιν.

Ἐπεὶ γὰρ δύο ἐπίπεδα παράλληλα τὰ ΒΗ, ΓΕ ὑπὸ ἐπιπέδου τοῦ ΑΓ τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν. παράλληλος ἄρα ἐστὶν ἡ ΑΒ τῇ ΔΓ. πάλιν, ἐπεὶ δύο ἐπίπεδα παράλληλα τὰ ΒΖ, ΑΕ ὑπὸ ἐπιπέδου τοῦ ΑΓ τέμνεται, αἱ κοιναὶ αὐτῶν τομαὶ παράλληλοί εἰσιν. παράλληλος ἄρα ἐστὶν ἡ ΒΓ τῇ ΑΔ. ἐδείχθη δὲ καὶ ἡ ΑΒ τῇ ΔΓ παράλληλος· παραλληλόγραμμον ἄρα ἐστὶ τὸ ΑΓ. ὁμοίως δὴ δείξομεν, ὅτι καὶ ὁἑκαστον τῶν ΔΖ, ΖΗ, ΗΒ, ΒΖ, ΑΕ παραλληλόγραμμόν ἐστιν.

Ἐπεζεύθθωσαν αἱ ΑΘ, ΔΖ. καὶ ἐπεὶ παράλληλόξ ἐστιν ἡ μὲν ΑΒ τῇ ΔΓ, ἡ δὲ ΒΘ τῇ ΓΖ, δύο δὴ αἱ ΑΒ, ΒΘ ἁπτόμεναι ἀλλήλων παρὰ δύο εὐθείας τὰξ ΔΓ, ΓΖ ἁπτομένας ἀλλήλων εἰσὶν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ· ἴσας ἄρα γωνίας περιέχουσιν· ὁίση ἄρα ἡ ὑπὸ ΑΒΘ γωνία τῇ ὑπὸ ΔΓΖ. καὶ ἐπεὶ δύο αἱ ΑΒ, ΒΘ δυσὶ ταῖς ΔΓ, ΓΖ ἴσαι εἰσίν, καὶ γωνία ἡ ὑπὸ ΑΒΘ γωνίᾳ τῇ ὑπὸ ΔΓΖ ἐστινὁίση, βάσις ἄρα ἡ ΑΘ βάσει τῇ ΔΖ ἐστινὁίση, καὶ τὸ ΑΒΘ τρίγωνον τῷ ΔΓΖ τριγώνῳ ἴσον ἐστίν. καί ἐστι τοῦ μὲν ΑΒΘ διπλάσιον τὸ ΒΗ παραλληλόγραμμον, τοῦ δὲ ΔΓΖ διπλάσιον τὸ ΓΕ παραλληλόγραμμον· ἴσον ἄρα τὸ ΒΗ παραλληλόγραμμον τῷ ΓΕ παραλληλογράμμῳ· ὁμοίως δὴ δείχομεν, ὅτι καὶ τὸ μὲν ΑΓ τῷ ΗΖ ἐστιν ἴσον, τὸ δὲ ΑΕ τῷ ΒΖ.

Ἐὰν ἄρα στερεὸν ὑπὸ παραλλήλων ἐπιπέδων περιέθηται, τὰ ἀπεναντίον αὐτοῦ ἐπίπεδα ἴσα τε καὶ παραλληλόγραμμά ἐστιν· ὅπερ ἔδει δεῖχαι.

For let the solid (figure) $CDHG$ be contained by the parallel planes AC, GF, and AH, DF, and BF, AE. I say that its opposite planes are both equal and parallelogrammic.

For since the two parallel planes BG and CE are cut by the plane AC, their common sections are parallel [Prop. 11.16]. Thus, AB is parallel to DC. Again, since the two parallel planes BF and AE are cut by the plane AC, their common sections are parallel [Prop. 11.16]. Thus, BC is parallel to AD. And AB was also shown (to be) parallel to DC. Thus, AC is a parallelogram. So, similarly, we can also show that DF, FG, GB, BF, and AE are each parallelograms.

Let AH and DF be joined. And since AB is parallel to DC, and BH to CF, so the two (straight-lines) joining one another, AB and BH, are parallel to the two straight-lines joining one another, DC and CF (respectively), not (being) in the same plane. Thus, they will contain equal angles [Prop. 11.10]. Thus, angle ABH (is) equal to (angle) DCF. And since the two (straight-lines) AB and BH are equal to the two (straight-lines) DC and CF (respectively) [Prop. 1.34], and angle ABH is equal to angle DCF, the base AH is thus equal to the base DF, and triangle ABH is equal to triangle DCF [Prop. 1.4]. And parallelogram BG is double (triangle) ABH, and parallelogram CE double (triangle) DCF [Prop. 1.34]. Thus, parallelogram BG (is) equal to parallelogram CE. So, similarly, we can show that AC is also equal to GF, and AE to BF.

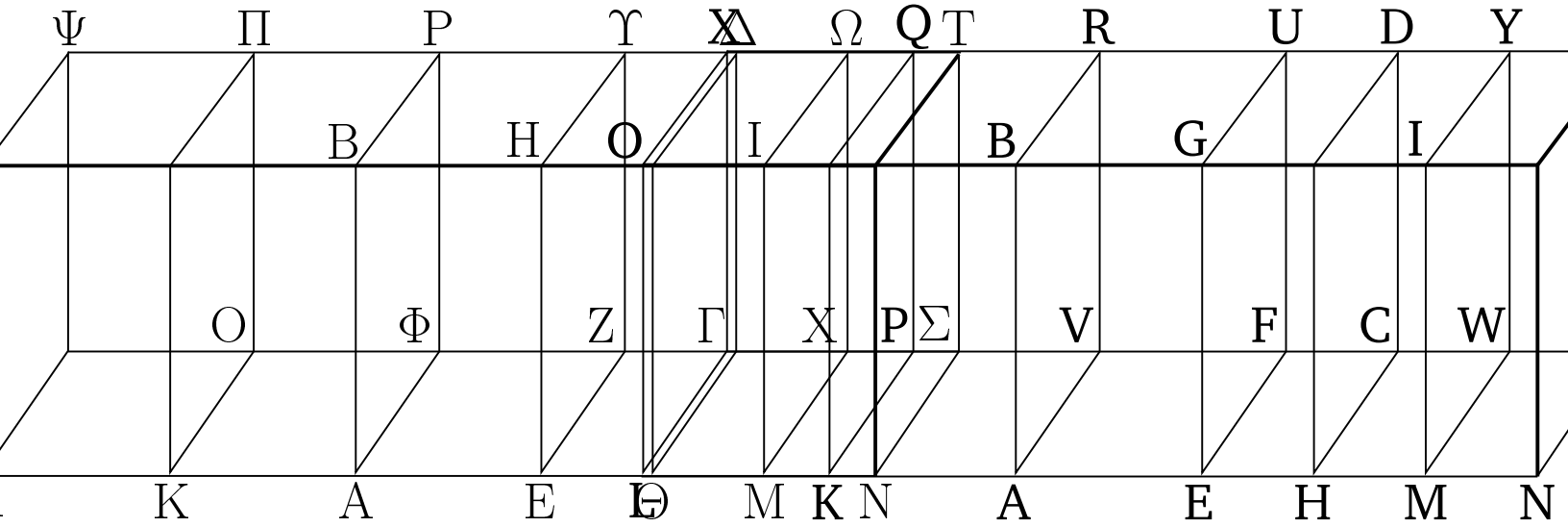
Thus, if a solid (figure) is contained by (six) parallel planes then its opposite planes are both equal and parallelogrammic. (Which is) the very thing it was required to show.

25.

Ἐὰν στερεὸν παραλληλεπίπεδον ἐπιπέδῳ τμ ηθῇ παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ὁέσται ὡξ ἡ βάσις πρὸς τὴν βάσιν, οὁύτως τὸ στερεὸν πρὸς τὸ στερεόν.

Proposition 25

If a parallelipiped solid is cut by a plane which is parallel to the opposite planes (of the parallelipiped) then as the base (is) to the base, so the solid will be to the solid.



Στερεὸν γὰρ παραλληλεπίπεδον τὸ ΑΒΓΔ ἐπιπέδῳ τῷ ΖΗ τετμήσθω παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις τοῖς ΡΑ, ΔΘ· λέγω, ὅτι ἐστὶν ὡς ἡ ΑΕΖΦ βάσις πρὸς τὴν ΕΘΓΖ βάσιν, οὕτως τὸ ΑΒΖΥ στερεὸν πρὸς τὸ ΕΗΓΔ στερεόν.

Ἐκβεβλήσθω γὰρ ἡ ΑΘ ἐφ' ἑκάτερα τὰ μέρη, καὶ κείσθωσαν τῇ μὲν ΑΕ ἴσαι ὁσαιδηποτοῦν αἱ ΑΚ, ΚΛ, τῇ δὲ ΕΘ ἴσαι ὁσαιδηποτοῦν αἱ ΘΜ, ΜΝ, καὶ συμπεπληρώσθω τὰ ΛΟ, ΚΦ, ΘΘ, ΜΣ παραλληλόγραμμα καὶ τὰ ΛΠ, ΚΡ, ΔΜ, ΜΤ στερεά.

Καὶ ἐπεὶ ἴσαι εἰσὶν αἱ ΛΚ, ΚΑ, ΑΕ εὐθεῖαι ἀλλήλαις, ἴσα ἐστὶ καὶ τὰ μὲν ΛΟ, ΚΦ, ΑΖ παραλληλόγραμμα ἀλλήλοις, τὰ δὲ ΚΧ, ΚΒ, ΑΗ ἀλλήλοις καὶ ὀέτι τὰ ΛΨ, ΚΠ, ΑΡ ἀλλήλοις· ἀπεναντίον γάρ. διὰ τὰ αὐτὰ δὴ καὶ τὰ μὲν ΕΓ, ΘΘ, ΜΣ παραλληλόγραμμα ἴσα εἰσὶν ἀλλήλοις, τὰ δὲ ΘΗ, ΘΙ, ΙΝ ἴσα εἰσὶν ἀλλήλοις, καὶ ὀέτι τὰ ΔΘ, ΜΩ, ΝΤ· τρία ἄρα ἐπίπεδα τῶν ΛΠ, ΚΡ, ΑΥ στερεῶν τρισὶν ἐπιπέδοις ἐστὶν ἴσα. ἀλλὰ τὰ τρία τρισὶ τοῖς ἀπεναντίον ἐστὶν ἴσα· τὰ ἄρα τρία στερεὰ τὰ ΛΠ, ΚΡ, ΑΥ ἴσα ἀλλήλοις ἐστίν. διὰ τὰ αὐτὰ δὴ καὶ τὰ τρία στερεὰ τὰ ΕΔ, ΔΜ, ΜΤ ἴσα ἀλλήλοις ἐστίν· ὁσαπλασίων ἄρα ἐστὶν ἡ ΛΖ βάσις τῆς ΑΖ βάσεως, τοσαυταπλάσιόν ἐστι καὶ τὸ ΛΥ στερεὸν τοῦ ΑΥ στερεοῦ. διὰ τὰ αὐτὰ δὴ ὁσαπλασίων ἐστὶν ἡ ΝΖ βάσις τῆς ΖΘ βάσεως, τοσαυταπλάσιόν ἐστι καὶ τὸ ΝΥ στερεὸν τοῦ ΘΥ στερεοῦ. καὶ εἰ ὁίση ἐστὶν ἡ ΛΖ βάσις τῇ ΝΖ βάσει, ἴσον ἐστὶ καὶ τὸ ΛΥ στερεὸν τῷ ΝΥ στερεῷ, καὶ εἰ ὑπερέθει ἡ ΛΖ βάσις τῆς ΝΖ βάσεως, ὑπερέθει καὶ τὸ ΛΥ στερεὸν τοῦ ΝΥ στερεοῦ, καὶ εἰ ἐλλείπει, ἐλλείπει. τεσσάρων δὴ ὄντων μεγεθῶν, δύο μὲν βάσεων τῶν ΑΖ, ΖΘ, δύο δὲ στερεῶν τῶν ΑΥ, ΥΘ, εὐίληπται ἰσάκις πολλαπλάσια τῆς μὲν ΑΖ βάσεως καὶ τοῦ ΑΥ στερεοῦ ἥ τε ΛΖ βάσις καὶ τὸ ΛΥ στερεόν, τῆς δὲ ΘΖ βάσεως καὶ τοῦ ΘΥ στερεοῦ ἥ τε ΝΖ βάσις καὶ τὸ ΝΥ στερεόν, καὶ δέδεικται, ὅτι εἰ ὑπερέθει ἡ ΛΖ βάσις τῆς ΖΝ βάσεως, ὑπερέθει καὶ τὸ ΛΥ στερεὸν τοῦ ΝΥ [στερεοῦ], καὶ εἰ ὁίση, ἴσον, καὶ εἰ ἐλλείπει, ἐλλείπει. ὀέστιν ἄρα ὡς ἡ

For let the parallelipiped solid $ABCD$ be cut by the plane FG which is parallel to the opposite planes RA and DH. I say that as the base $AEFV$ (is) to the base $EHCF$, so the solid $ABFU$ (is) to the solid $EGCD$.

For let AH be produced in each direction. And let any number whatsoever (of lengths), AK and KL, be made equal to AE, and any number whatsoever (of lengths), HM and MN, equal to EH. And let the parallelograms LP, KV, HW, and MS be completed, and the solids LQ, KR, DM, and MT.

And since the straight-lines LK, KA, and AE are equal to one another, the parallelograms LP, KV, and AF are also equal to one another, and KO, KB, and AG (are equal) to one another, and, further, LX, KQ, and AR (are equal) to one another. For (they are) opposite [Prop. 11.24]. So, for the same (reasons), the parallelograms EC, HW, and MS are also equal to one another, and HG, HI, and IN are equal to one another, and, further, DH, MY, and NT (are equal to one another). Thus, three planes of (one of) the solids LQ, KR, and AU are equal to the (corresponding) three planes (of the others). But, the three planes (in one of the soilds) are equal to the three opposite planes [Prop. 11.24]. Thus, the three solids LQ, KR, and AU are equal to one another [Def. 11.10]. So, for the same (reasons), the three solids ED, DM, and MT are also equal to one another. Thus, as many multiples as the base LF is of the base AF, so many multiples is the solid LU also of the the solid AU. So, for the same (reasons), as many multiples as the base NF is of the base FH, so many multiples is the solid NU also of the solid HU. And if the base LF is equal to the base NF then the solid LU is also equal to the solid NU.† And if the base LF exceeds the base NF then the solid LU also exceeds the solid NU. And if (LF) is less than (NF) then (LU) is (also) less than (NU). So, there are four magnitudes, the two bases AF and FH, and the two solids AU and UH, and equal mul-

ΑΖ βάσις πρὸς τὴν ΖΘ βάσιν, οὕτως τὸ ΑΥ στερεὸν πρὸς τὸ ΥΘ στερεόν· ὅπερ ἔδει δεῖξαι.

tiples be taken of the base AF and the solid AU— (namely), the base LF and the solid LU—and of the base HF and the solid HU—(namely), the base NF and the solid NU. And it has been shown that if the base LF exceeds the base FN then the solid LU also exceeds the [solid] NU, and if (LF is) equal (to FN) then (LU is) equal (to NU), and if (LF is) less than (FN) then (LU is) less than (NU). Thus, as the base AF is to the base FH, so the solid AU (is) to the solid UH [Def. 5.5]. (Which is) the very thing it was required to show.

† Here, Euclid assumes that $LF \gtreqless NF$ implies $LU \gtreqless NU$. This is easily demonstrated.

26.

Πρὸξ τῇ δοθείσῃ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῇ δοθείσῃ στερεᾷ γωνίᾳ ἴσην στερεὰν γωνίαν συστήσασθαι.

Ἔστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ ΑΒ, τὸ δὲ πρὸς αὐτῇ δοθὲν σημεῖον τὸ Α, ἡ δὲ δοθεῖσα στερεὰ γωνία ἡ πρὸς τῷ Δ περιεθομένη ὑπὸ τῶν ὑπὸ ΕΔΓ, ΕΔΖ, ΖΔΓ γωνιῶν ἐπιπέδων· δεῖ δὴ πρὸς τῇ ΑΒ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Α τῇ πρὸς τῷ Δ στερεᾷ γωνίᾳ ἴσην στερεὰν γωνίαν συστήσασθαι.

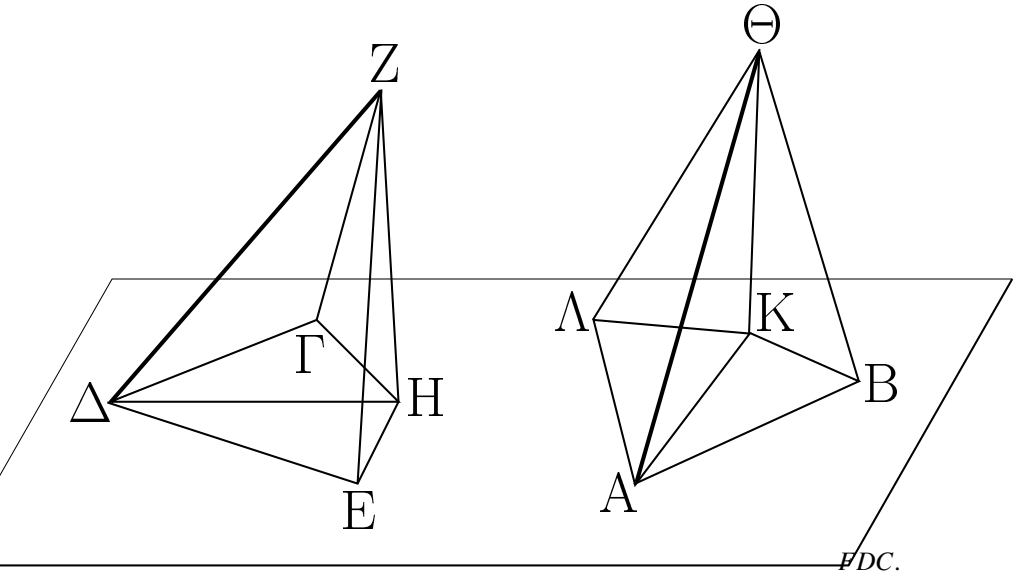
Εἰλήφθω γὰρ ἐπὶ τῆς ΔΖ τυθὸν σημεῖον τὸ Ζ, καὶ ἤχθω ἀπὸ τοῦ Ζ ἐπὶ τὸ διὰ τῶν ΕΔ, ΔΓ ἐπίπεδον κάθετος ἡ ΖΗ, καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ Η, καὶ ἐπεζεύχθω ἡ ΔΗ, καὶ συνεστάτω πρὸς τῇ ΑΒ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Α τῇ μὲν ὑπὸ ΕΔΓ γωνίᾳ ὁίση ἡ ὑπὸ ΒΑΛ, τῇ δὲ ὑπὸ ΕΔΗοίση ἡ ὑπὸ ΒΑΚ, καὶ κείσθω τῇ ΔΗοίση ἡ ΑΚ, καὶ ἀνεστάτω ἀπὸ τοῦ Κ σημείου τῷ διὰ τῶν ΒΑΛ ἐπιπέδῳ πρὸς ὀρθὰξ ἡ ΚΘ, καὶ κείσθωοίση τῇ ΗΖ ἡ ΚΘ, καὶ ἐπεζεύχθω ἡ ΘΑ· λέγω, ὅτι ἡ πρὸς τῷ Α στερεὰ γωνία περιεθομένη ὑπὸ τῶν ΒΑΛ, ΒΑΘ, ΘΑΛ γωνιῶνοίση ἐστὶ τῇ πρὸς τῷ Δ στερεᾷ γωνίᾳ τῇ περιεθομένῃ ὑπὸ τῶν ΕΔΓ, ΕΔΖ, ΖΔΓ γωνιῶν.

Proposition 26

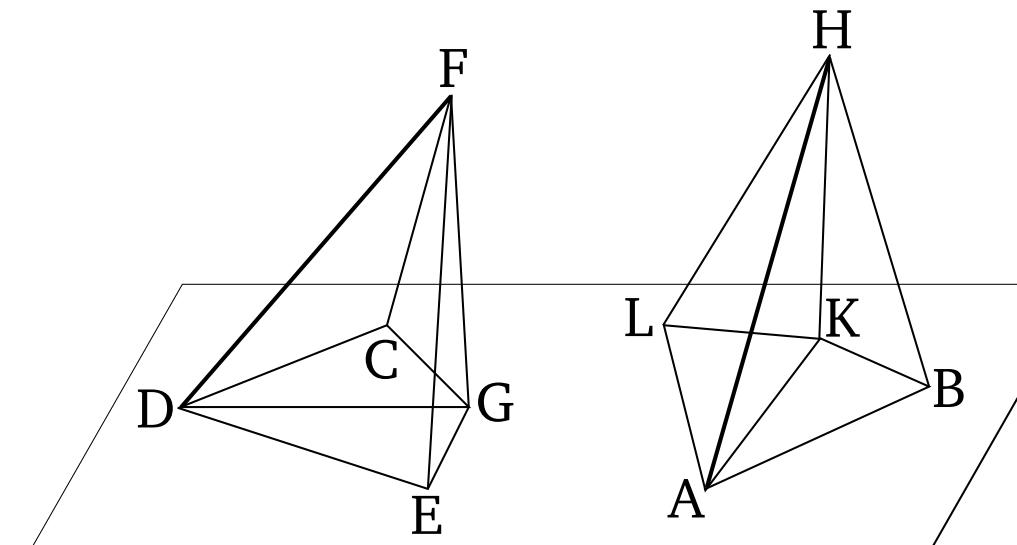
To construct a solid angle equal to a given solid angle on a given straight-line, and at a given point on it.

Let AB be the given straight-line, and A the given point on it, and D the given solid angle, contained by the plane angles EDC, EDF, and FDC. So, it is necessary to construct a solid angle equal to the solid angle D on the straight-line AB, and at the point A on it.

For let some random point F be taken on DF, and let FG be drawn from F perpendicular to the plane through ED and DC [Prop. 11.11], and let it meet the plane at G, and let DG be joined. And let BAL, equal to the angle EDC, and BAK, equal to EDG, be constructed on the straight-line AB at the point A on it [Prop. 1.23]. And let AK be made equal to DG. And let KH be set up at the point K at right-angles to the plane through BAL [Prop. 11.12]. And let KH be made equal to GF. And let HA be joined. I say that the solid angle at A, contained by the (plane) angles BAL, BAH, and HAL, is equal to the solid angle at D, contained by the (plane) angles EDC, EDF, and



FDC.



Ἀπειλήφθωσαν γὰρ ἴσαι αἱ ΑΒ, ΔΕ, καὶ ἐπεζεύχθωσαν αἱ ΘΒ, ΚΒ, ΖΕ, ΗΕ. καὶ ἐπεὶ ἡ ΖΗ ὀρθή ἐστι πρὸς τὸ ὑποκείμενον ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ ὑποκειμένῳ ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας· ὀρθὴ ἄρα ἐστὶν ἑκατέρα τῶν ὑπὸ ΖΗΔ, ΖΗΕ γωνιῶν. διὰ τὰ αὐτὰ δὴ καὶ ἑκατέρα τῶν ὑπὸ ΘΚΑ, ΘΚΒ γωνιῶν ὀρθή ἐστιν. καὶ ἐπεὶ δύο αἱ ΚΑ, ΑΒ δύο ταῖς ΗΔ, ΔΕ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, καὶ γωνίας ἴσας περιέθουσιν, βάσις ἄρα ἡ ΚΒ βάσει τῇ ΗΕοίση ἐστίν. ὀέστι δὲ καὶ ἡ ΚΘ τῇ ΗΖοίση· καὶ γωνίας ὀρθὰξ περιέθουσιν· ὀίση ἄρα καὶ ἡ ΘΒ τῇ ΖΕ. πάλιν ἐπεὶ δύο αἱ ΑΚ, ΚΘ δυσὶ ταῖς ΔΗ, ΗΖ ἴσαι εἰσίν, καὶ γωνίας ὀρθὰξ περιέθουσιν, βάσις ἄρα ἡ ΑΘ βάσει τῇ ΖΔοίση ἐστίν. ὀέστι δὲ καὶ ἡ ΑΒ τῇ ΔΕοίση· δύο δὴ αἱ ΘΑ, ΑΒ δύο ταῖς ΔΖ, ΔΕ ἴσαι εἰσίν. καὶ βάσις ἡ ΘΒ βάσει τῇ ΖΕοίση· γωνία ἄρα ἡ ὑπὸ ΒΑΘ γωνίᾳ τῇ ὑπὸ ΕΔΖ ἐστινὀίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΘΑΛ τῇ ὑπὸ ΖΔΓ ἐστινὀίση. ὀέστι δὲ καὶ ἡ ὑπὸ ΒΑΛ τῇ ὑπὸ ΕΔΓοίση.

Πρὸξ ἄρα τῇ δοθείσῃ εὐθείᾳ τῇ ΑΒ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Α τῇ δοθείσῃ στερεᾷ γωνίᾳ τῇ πρὸς τῷ Δοίση συνέσταται· ὅπερ ἔδει ποιῆσαι.

For let AB and DE be cut off (so as to be) equal, and let HB, KB, FE, and GE be joined. And since FG is at right-angles to the reference plane (EDC), it will also make right-angles with all of the straight-lines joined to it which are also in the reference plane [Def. 11.3]. Thus, the angles FGD and FGE are right-angles. So, for the same (reasons), the angles HKA and HKB are also right-angles. And since the two (straight-lines) KA and AB are equal to the two (straight-lines) GD and DE, respectively, and they contain equal angles, the base KB is thus equal to the base GE [Prop. 1.4]. And KH is also equal to GF. And they contain right-angles (with the respective bases). Thus, HB (is) also equal to FE [Prop. 1.4]. Again, since the two (straight-lines) AK and KH are equal to the two (straight-lines) DG and GF (respectively), and they contain right-angles, the base AH is thus equal to the base FD [Prop. 1.4]. And AB (is) also equal to DE. So, the two (straight-lines) HA and AB are equal to the two (straight-lines) DF and DE (respectively). And the base HB (is) equal to the base FE. Thus, the angle BAH is equal to the angle EDF [Prop. 1.8]. So, for the same (reasons), HAL is also equal to FDC. And BAL is also equal to EDC.

Thus, (a solid angle) has been constructed, equal to the given solid angle at D, on the given straight-line AB, at the given point A on it. (Which is) the very thing it was required to do.

27.

Ἀπὸ τῆς δοθείσης εὐθείας τῷ δοθέντι στερεῷ παραλληλεπιπέδῳ ὅμοιόν τε καὶ ὁμοίως κείμενον στερεὸν παραλληλεπίπεδον ἀναγράψαι.

Ἔστω ἡ μὲν δοθεῖσα εὐθεῖα ἡ ΑΒ, τὸ δὲ δοθὲν στερεὸν παραλληλεπίπεδον τὸ ΓΔ· δεῖ δὴ ἀπὸ τῆς δοθείσης εὐθείας

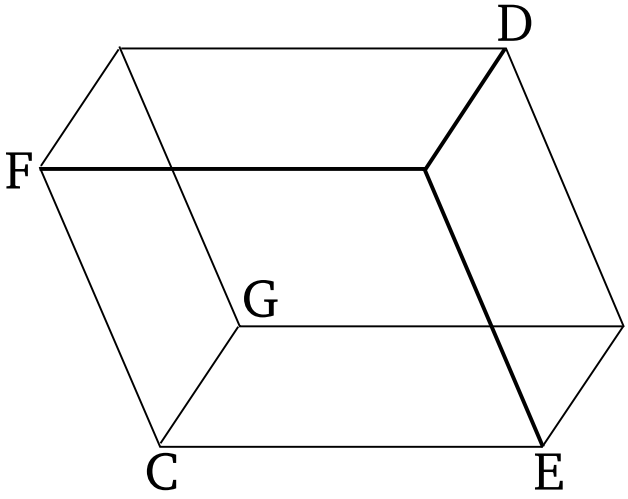
Proposition 27

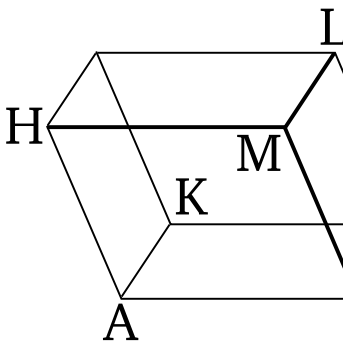
To describe a parallelepiped solid similar, and similarly laid out, to a given parallelepiped solid on a given straight-line.

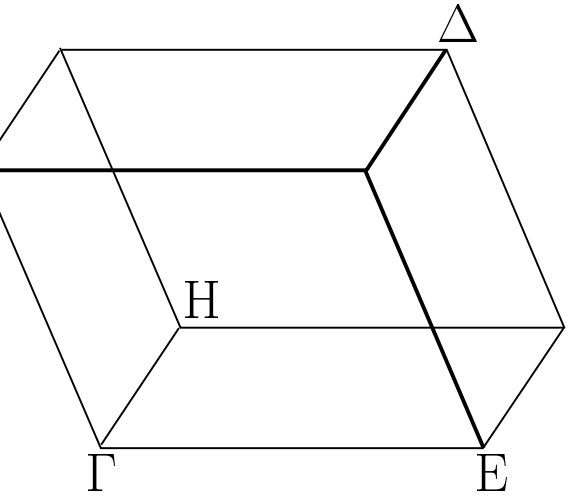
Let the given straight-line be AB, and the given parallelepiped solid CD. So, it is necessary to describe a paral-

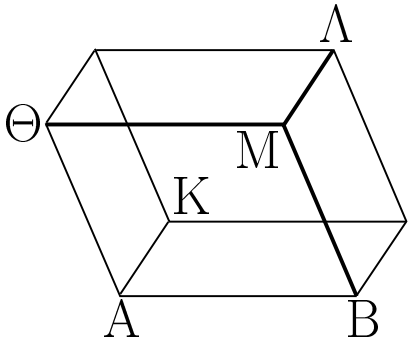
τῆς AB τῷ δοθέντι στερεῷ παραλληλεπιπέδῳ τῷ $ΓΔ$ ὁμοιόν τε καὶ ὁμοίως κείμενον στερεὸν παραλληλεπίπεδον ἀναγράψαι.

Συνεστάτω γὰρ πρὸς τῇ AB εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ A τῇ πρὸς τῷ $Γ$ στερεᾷ γωνίᾳ ὁίση ἡ περιεθομένη ὑπὸ τῶν $BAΘ$, $ΘAK$, KAB, δώστε ἴσην εὶναι τὴν μὲν ὑπὸ $BAΘ$ γωνίαν τῇ ὑπὸ $EΓZ$, τὴν δὲ ὑπὸ BAK τῇ ὑπὸ $EΓH$, τὴν δὲ ὑπὸ $KAΘ$ τῇ ὑπὸ $HΓZ$· καὶ γεγονέτω ὡς μὲν ἡ $EΓ$ πρὸς τὴν $ΓH$, οὑτως ἡ BA πρὸς τὴν AK, ὡς δὲ ἡ $HΓ$ πρὸς τὴν $ΓZ$, οὑτως ἡ KA πρὸς τὴν $AΘ$. καὶ διὸ ὁίσου ἄρα ἐστὶν ὡς ἡ $EΓ$ πρὸς τὴν $ΓZ$, οὑτως ἡ BA πρὸς τὴν $AΘ$. καὶ συμπεπληρώσθω τὸ $ΘB$ παραλληλόγραμμον καὶ τὸ $AΛ$ στερεόν.









Καὶ ἐπεί ἐστιν ὡς ἡ $EΓ$ πρὸς τὴν $ΓH$, οὑτως ἡ BA πρὸς τὴν AK, καὶ περὶ ἴσας γωνίας τὰξ ὑπὸ $EΓH$, BAK αἱ πλευραὶ ἀνάλογόν εἰσιν, ὅμοιον ἄρα ἐστὶ τὸ HE παραλληλόγραμμον τῷ KB παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν $KΘ$ παραλληλόγραμμον τῷ HZ παραλληλογράμμῳ ὅμοιόν ἐστι καὶ ὀέτι τὸ ZE τῷ $ΘB$· τρία ἄρα παραλληλόγραμμα τοῦ $ΓΔ$ στερεοῦ τρισὶ παραλληλογράμμοις τοῦ $AΛ$ στερεοῦ ὅμοιά ἐστιν. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖς ἀπεναντίον ἴσα τέ ἐστι καὶ ὅμοια,

And since as EC is to CG, so BA (is) to AK, and the sides about the equal angles ECG and BAK are (thus) proportional, the parallelogram GE is thus similar to the parallelogram KB. So, for the same (reasons), the parallelogram KH is also similar to the parallelogram GF, and, further, FE (is similar) to HB. Thus, three of the parallelograms of solid CD are similar to three of the parallelograms of solid AL. But, the (former) three are equal and similar to the three opposite, and the (latter) three are equal and similar to the three opposite. Thus, the

τὰ δὲ τρία τρισὶ τοῖς ἀπεναντίον ἴσα τέ ἐστι καὶ ὅμοια· ὅλον ἄρα τὸ ΓΔ στερεὸν ὅλῳ τῷ ΑΛ στερεῷ ὅμοιόν ἐστιν.

Ἀπὸ τῆς δοθείσης ἄρα εὐθείας τῆς ΑΒ τῷ δοθέντι στερεῷ παραλληλεπιπέδῳ τῷ ΓΔ ὁμοιόν τε καὶ ὁμοίως κείμενον ἀναγέγραπται τὸ ΑΛ· ὅπερ ἔδει ποιῆσαι.

whole solid CD is similar to the whole solid AL [Def. 11.9].

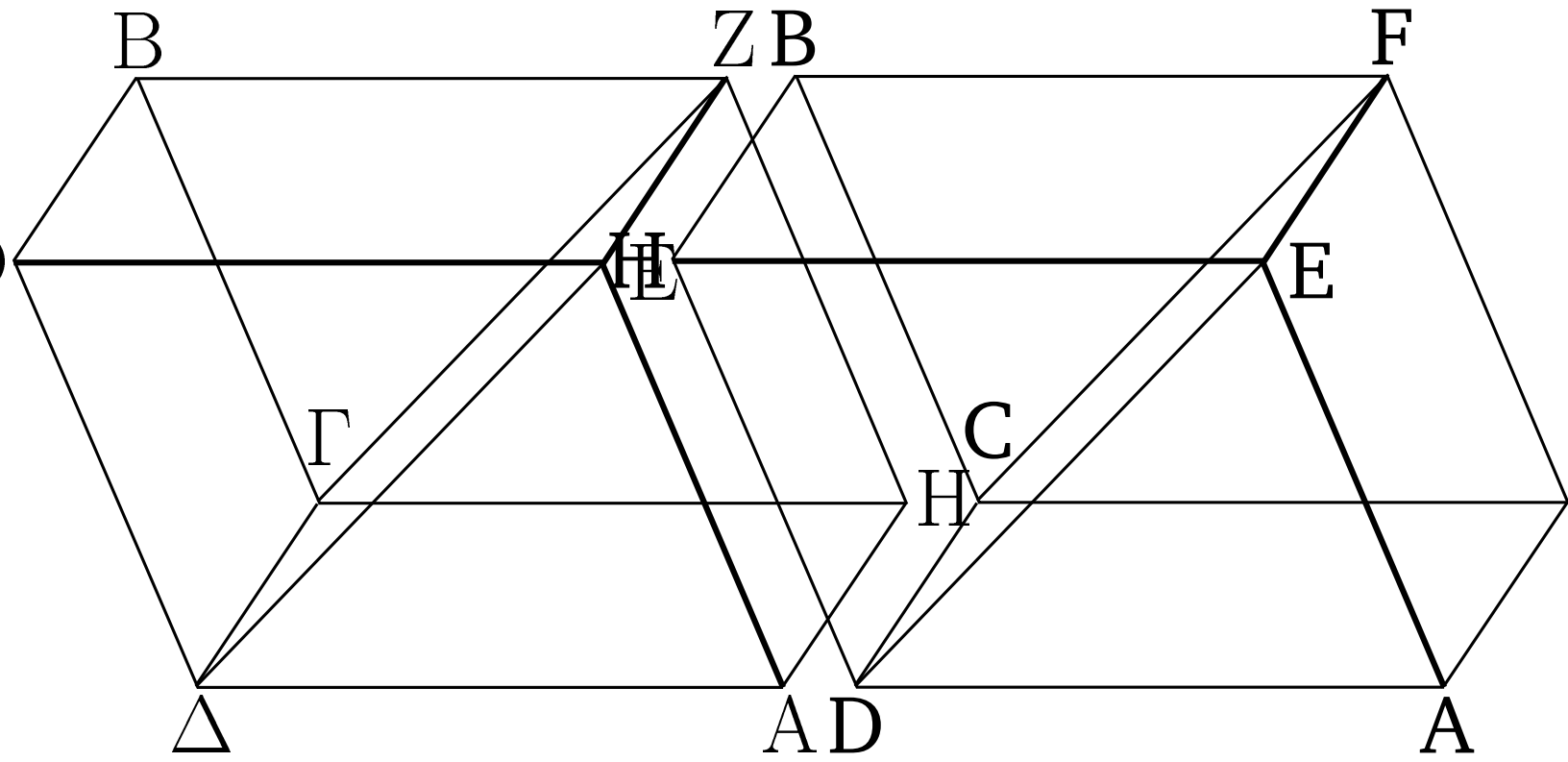
Thus, AL, similar, and similarly laid out, to the given parallelepiped solid CD, has been described on the given straight-lines AB. (Which is) the very thing it was required to do.

28.

Ἐὰν στερεὸν παραλληλεπίπεδον ἐπιπέδῳ τμ ηθῇ κατὰ τὰξ διαγωνίους τῶν ἀπεναντίον ἐπιπέδων, δίθα τμ ηθήσεται τὸ στερεὸν ὑπὸ τοῦ ἐπιπέδου.

Proposition 28

If a parallelepiped solid is cut by a plane (passing) through the diagonals of (a pair of) opposite planes then the solid will be cut in half by the plane.



Στερεὸν γὰρ παραλληλεπίπεδον τὸ ΑΒ ἐπιπέδῳ τῷ ΓΔΕΖ τετμήσθω κατὰ τὰξ διαγωνίους τῶν ἀπεναντίον ἐπιπέδων τὰξ ΓΖ, ΔΕ· λέγω, ὅτι δίθα τμ ηθήσεται τὸ ΑΒ στερεὸν ὑπὸ τοῦ ΓΔΕΖ ἐπιπέδου.

Ἐπεὶ γὰρ ἴσον ἐστὶ τὸ μὲν ΓΗΖ τρίγωνον τῷ ΓΖΒ τριγώνῳ, τὸ δὲ ΑΔΕ τῷ ΔΕΘ, ὀέστι δὲ καὶ τὸ μὲν ΓΑ παραλληλόγραμμον τῷ ΕΒ ἴσον· ἀπεναντίον γάρ· τὸ δὲ ΗΕ τῷ ΓΘ, καὶ τὸ πρίσμα ἄρα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν ΓΗΖ, ΑΔΕ, τριῶν δὲ παραλληλογράμμων τῶν ΗΕ, ΑΓ, ΓΕ ἴσον ἐστὶ τῷ πρίσματι τῷ περιεθομένῳ ὑπὸ δύο μὲν τριγώνων τῶν ΓΖΒ, ΔΕΘ, τριῶν δὲ παραλληλογράμμων τῶν ΓΘ, ΒΕ, ΓΕ· ὑπὸ γὰρὸίσων ἐπιπέδων περιέθονται τῷ τε πλήθει καὶ τῷ μεγέθει. ὁώστε ὅλον τὸ ΑΒ στερεὸν δίθα τέτμ ηται ὑπὸ τοῦ ΓΔΕΖ ἐπιπέδου· ὅπερ ἔδει δεῖχαι.

For let the parallelepiped solid AB be cut by the plane $CDEF$ (passing) through the diagonals of the opposite planes CF and DE.[†] I say that the solid AB will be cut in half by the plane $CDEF$.

For since triangle CGF is equal to triangle CFB, and ADE (is equal) to DEH [Prop. 1.34], and parallelogram CA is also equal to EB—for (they are) opposite [Prop. 11.24]—and GE (equal) to CH, thus the prism contained by the two triangles CGF and ADE, and the three parallelograms GE, AC, and CE, is also equal to the prism contained by the two triangles CFB and DEH, and the three parallelograms CH, BE, and CE. For they are contained by planes (which are) equal in number and in magnitude [Def. 11.10].[‡] Thus, the whole of solid AB is cut in half by the plane $CDEF$. (Which is) the very thing it was required to show.

† Here, it is assumed that the two diagonals lie in the same plane. The proof is easily supplied.

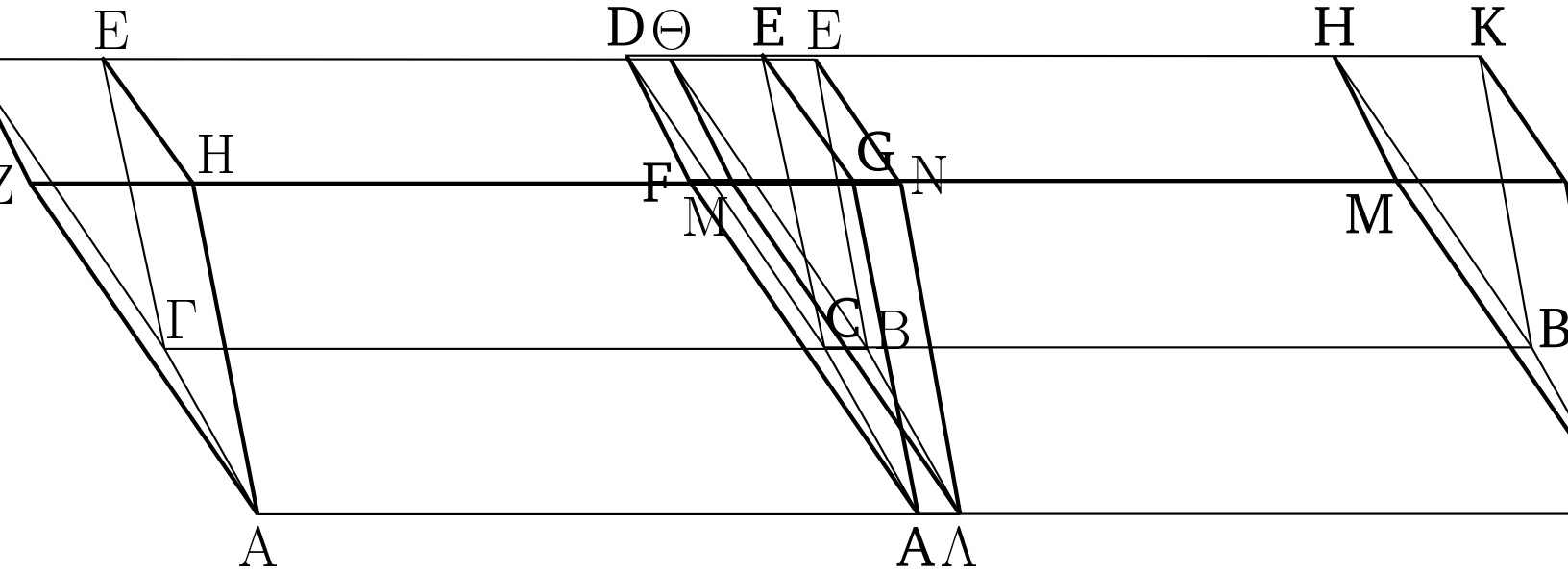
‡ However, strictly speaking, the prisms are not similarly arranged, being mirror images of one another.

29.

Τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν, ἴσα ἀλλήλοις ἐστίν.

Proposition 29

Parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up are on the same straight-lines, are equal to one another.



Ἔστω ἐπὶ τῆς αὐτῆς βάσεως τῆς AB στερεὰ παραλληλεπίπεδα τὰ $ΓΜ$, $ΓΝ$ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ AH, AZ, $ΛΜ$, $ΛΝ$, $ΓΔ$, $ΓΕ$, $ΒΘ$, $ΒΚ$ ἐπὶ τῶν αὐτῶν εὐθειῶν ὑέστωσαν τῶν ZN, $ΔΚ$· λέγω, ὅτι ἴσον ἐστὶ τὸ $ΓΜ$ στερεὸν τῷ $ΓΝ$ στερεῷ.

Ἐπεὶ γὰρ παραλληλόγραμμόν ἐστιν ἑκάτερον τῶν $ΓΘ$, $ΓΚ$, ὁίση ἐστὶν ἡ $ΓΒ$ ἑκατέρᾳ τῶν $ΔΘ$, $ΕΚ$· ὥστε καὶ ἡ $ΔΘ$ τῇ $ΕΚ$ ἐστινὁίση. κοινὴ ἀφῃρήσθω ἡ $ΕΘ$· λοιπὴ ἄρα ἡ $ΔΕ$ λοιπῇ τῇ $ΘΚ$ ἐστινὁίση. ὥστε καὶ τὸ μὲν $ΔΓΕ$ τρίγωνον τῷ $ΘΒΚ$ τριγώνῳ ἴσον ἐστίν, τὸ δὲ $ΔΗ$ παραλληλόγραμμον τῷ $ΘΝ$ παραλληογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ $ΑΖΗ$ τρίγωνον τῷ $ΜΛΝ$ τριγώνῳ ἴσον ἐστίν. ὀέστι δὲ καὶ τὸ μὲν $ΓΖ$ παραλληλόγραμμον τῷ $ΒΜ$ παραλληλογράμμῳ ἴσον, τὸ δὲ $ΓΗ$ τῷ $ΒΝ$· ἀπεναντίον γάρ· καὶ τὸ πρίσμα ἄρα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν $ΑΖΗ$, $ΔΓΕ$, τριῶν δὲ παραλληλογράμμων τῶν $ΑΔ$, $ΔΗ$, $ΓΗ$ ἴσον ἐστὶ τῷ πρίσματι τῷ περιεθομένῳ ὑπὸ δύο μὲν τριγώνων τῶν $ΜΛΝ$, $ΘΒΚ$, τριῶν δὲ παραλληλογράμμων τῶν $ΒΜ$, $ΘΝ$, $ΒΝ$. κοινὸν προσκείσθω τὸ στερεόν, οὗ βάσις μὲν τὸ $ΑΒ$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $ΗΕΘΜ$· ὅλον ἄρα τὸ $ΓΜ$ στερεὸν παραλληλεπίπεδον ὅλῳ τῷ $ΓΝ$ στερεῷ παραλληλεπιπέδῳ ἴσον ἐστίν.

Τὰ ἄρα ἐπὶ τῆς αὐτῆς βάσεως ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι ἐπὶ τῶν

For let the parallelepiped solids CM and CN be on the same base AB, and (have) the same height, and let the (ends of the straight-lines) standing up in them, AG, AF, LM, LN, CD, CE, BH, and BK, be on the same straight-lines, FN and DK. I say that solid CM is equal to solid CN.

For since CH and CK are each parallelograms, CB is equal to each of DH and EK [Prop. 1.34]. Hence, DH is also equal to EK. Let EH be subtracted from both. Thus, the remainder DE is equal to the remainder HK. Hence, triangle DCE is also equal to triangle HBK [Props. 1.4, 1.8], and parallelogram DG to parallelogram HN [Prop. 1.36]. So, for the same (reasons), traingle AFG is also equal to triangle MLN. And parallelogram CF is also equal to parallelogram BM, and CG to BN [Prop. 11.24]. For they are opposite. Thus, the prism contained by the two triangles AFG and DCE, and the three parallelograms AD, DG, and CG, is equal to the prism contained by the two triangles MLN and HBK, and the three parallelograms BM, HN, and BN. Let the solid whose base (is) parallelogram AB, and (whose) opposite (face is) $GEHM$, be added to both (prisms). Thus, the whole parallelepiped solid CM is equal to the whole parallelepiped solid CN.

Thus, parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the

αὐτῶν εἰσιν εὐθειῶν, ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

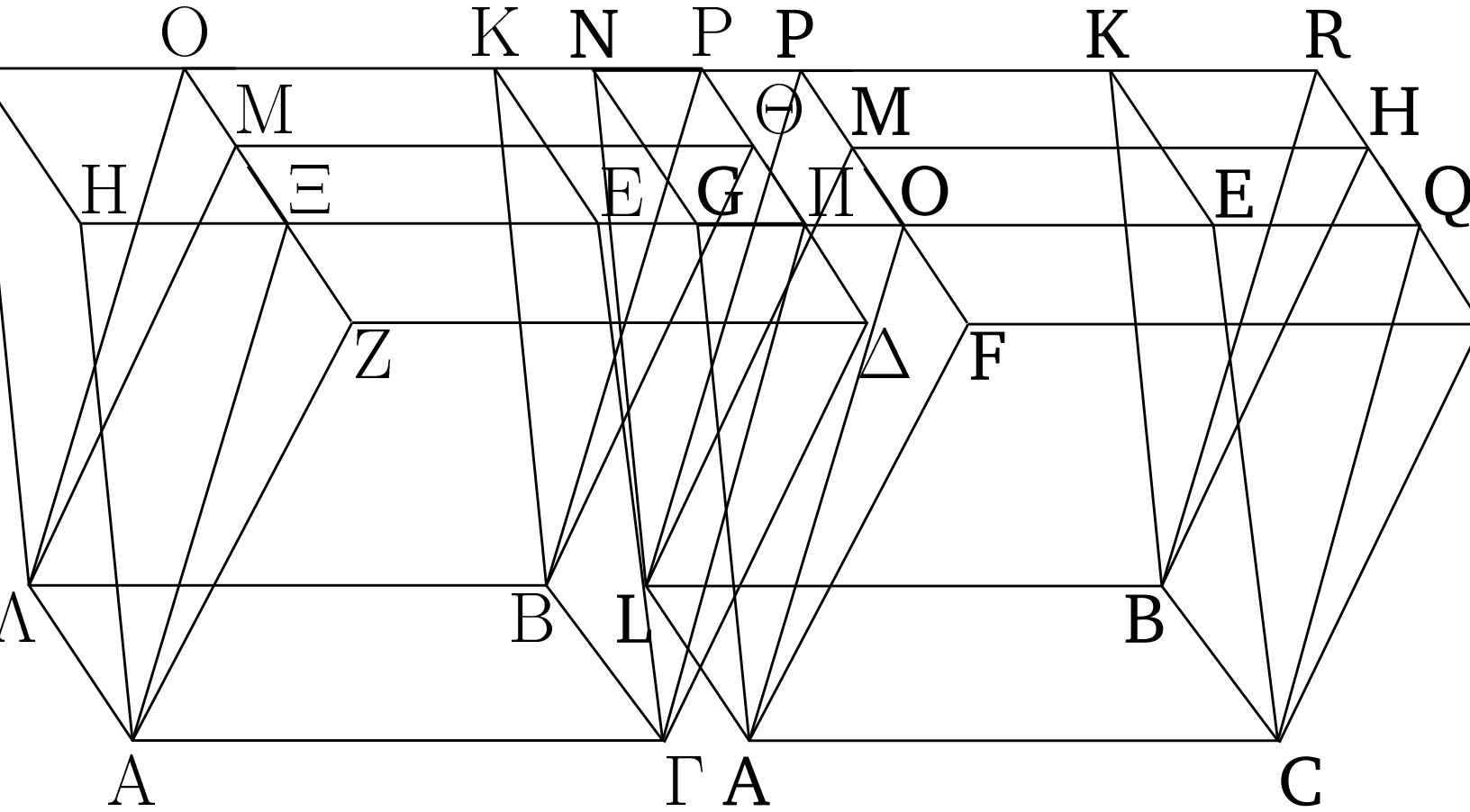
straight-lines) standing up (are) on the same straight-lines, are equal to one another. (Which is) the very thing it was required to show.

30.

Τὰ ἐπὶ τῆς αὐτῆς βάσεως ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι οὐκ εἰσὶν ἐπὶ τῶν αὐτῶν εὐθειῶν, ἴσα ἀλλήλοις ἐστίν.

Proposition 30

Parallelepiped solids which are on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up are not on the same straight-lines, are equal to one another.



Ἔστω ἐπὶ τῆς αὐτῆς βάσεως τῆς ΑΒ στερεὰ παραλληλεπίπεδα τὰ ΓΜ, ΓΝ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ ΑΖ, ΑΗ, ΛΜ, ΛΝ, ΓΔ, ΓΕ, ΒΘ, ΒΚ μὴ ὀέστωσαν ἐπὶ τῶν αὐτῶν εὐθειῶν· λέγω, ὅτι ἴσον ἐστὶ τὸ ΓΜ στερεὸν τῷ ΓΝ στερεῷ.

Ἐκβεβλήσθωσαν γὰρ αἱ ΝΚ, ΔΘ καὶ συμπιπτέτωσαν ἀλλήλαις κατὰ τὸ Ρ, καὶ ὀέτι ἐκβεβλήσθωσαν αἱ ΖΜ, ΗΕ ἐπὶ τὰ Ο, Π, καὶ ἐπεζεύθθωσαν αἱ ΑΧ, ΛΟ, ΓΠ, ΒΡ. ἴσον δή ἐστι τὸ ΓΜ στερεόν, οὗ βάσις μὲν τὸ ΑΓΒΛ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ ΖΔΘΜ, τῷ ΓΟ στερεῷ, οὗ βάσις μὲν τὸ ΑΓΒΛ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ ΧΠΡΟ· ἐπί τε γὰρ τῆς αὐτῆς βάσεώξ εἰσι τῆς ΑΓΒΛ καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ ΑΖ, ΑΧ, ΛΜ, ΛΟ, ΓΔ, ΓΠ, ΒΘ, ΒΡ ἐπὶ τῶν

Let the parallelepiped solids CM and CN be on the same base, AB, and (have) the same height, and let the (ends of the straight-lines) standing up in them, AF, AG, LM, LN, CD, CE, BH, and BK, not be on the same straight-lines. I say that the solid CM is equal to the solid CN.

For let NK and DH be produced, and let them have joined one another at R. And, further, let FM and GE be produced to P and Q (respectively). And let AO, LP, CQ, and BR be joined. So, solid CM, whose base (is) parallelogram $ACBL$, and opposite (face) $FDHM$, is equal to solid CP, whose base (is) parallelogram $ACBL$, and opposite (face) $OQRP$. For they are on the same base, $ACBL$, and (have) the same height, and the (ends of the straight-lines) standing up in them, AF, AO,

αὐτῶν εἰσιν εὐθειῶν τῶν ΖΟ, ΔΡ. ἀλλὰ τὸ ΓΟ στερεόν, οὗ βάσις μέν ἐστι τὸ ΑΓΒΛ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ ΧΠΡΟ, ἴσον ἐστὶ τῷ ΓΝ στερεῷ, οὗ βάσις μὲν τὸ ΑΓΒΛ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ ΗΕΚΝ· ἐπί τε γὰρ πάλιν τῆς αὐτῆς βάσεώξ εἰσι τῆς ΑΓΒΛ καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι αἱ ΑΗ, ΑΧ, ΓΕ, ΓΠ, ΛΝ, ΛΟ, ΒΚ, ΒΡ ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν τῶν ΗΠ, ΝΡ. ὥστε καὶ τὸ ΓΜ στερεὸν ἴσον ἐστὶ τῷ ΓΝ στερεῷ.

Τὰ ἄρα ἐπὶ τῆς αὐτῆς βάσεως στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος, ὧν αἱ ἐφεστῶσαι οὐκ εἰσὶν ἐπὶ τῶν αὐτῶν εὐθειῶν, ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖξαι.

LM, *LP*, *CD*, *CQ*, *BH*, and *BR*, are on the same straight-lines, *FP* and *DR* [Prop. 11.29]. But, solid *CP*, whose base is parallelogram *ACBL*, and opposite (face) *OQRP*, is equal to solid *CN*, whose base (is) parallelogram *ACBL*, and opposite (face) *GEKN*. For, again, they are on the same base, *ACBL*, and (have) the same height, and the (ends of the straight-lines) standing up in them, *AG*, *AO*, *CE*, *CQ*, *LN*, *LP*, *BK*, and *BR*, are on the same straight-lines, *GQ* and *NR* [Prop. 11.29]. Hence, solid *CM* is also equal to solid *CN*.

Thus, parallelepiped solids (which are) on the same base, and (have) the same height, and in which the (ends of the straight-lines) standing up are not on the same straight-lines, are equal to one another. (Which is) the very thing it was required to show.

31.

Τὰ ἐπὶ ὁίσων βάσεων ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοις ἐστίν.

Ἔστω ἐπὶ ὁίσων βάσεων τῶν ΑΒ, ΓΔ στερεὰ παραλληλεπίπεδα τὰ ΑΕ, ΓΖ ὑπὸ τὸ αὐτὸ ὕψος. λέγω, ὅτι ἴσον ἐστὶ τὸ ΑΕ στερεὸν τῷ ΓΖ στερεῷ.

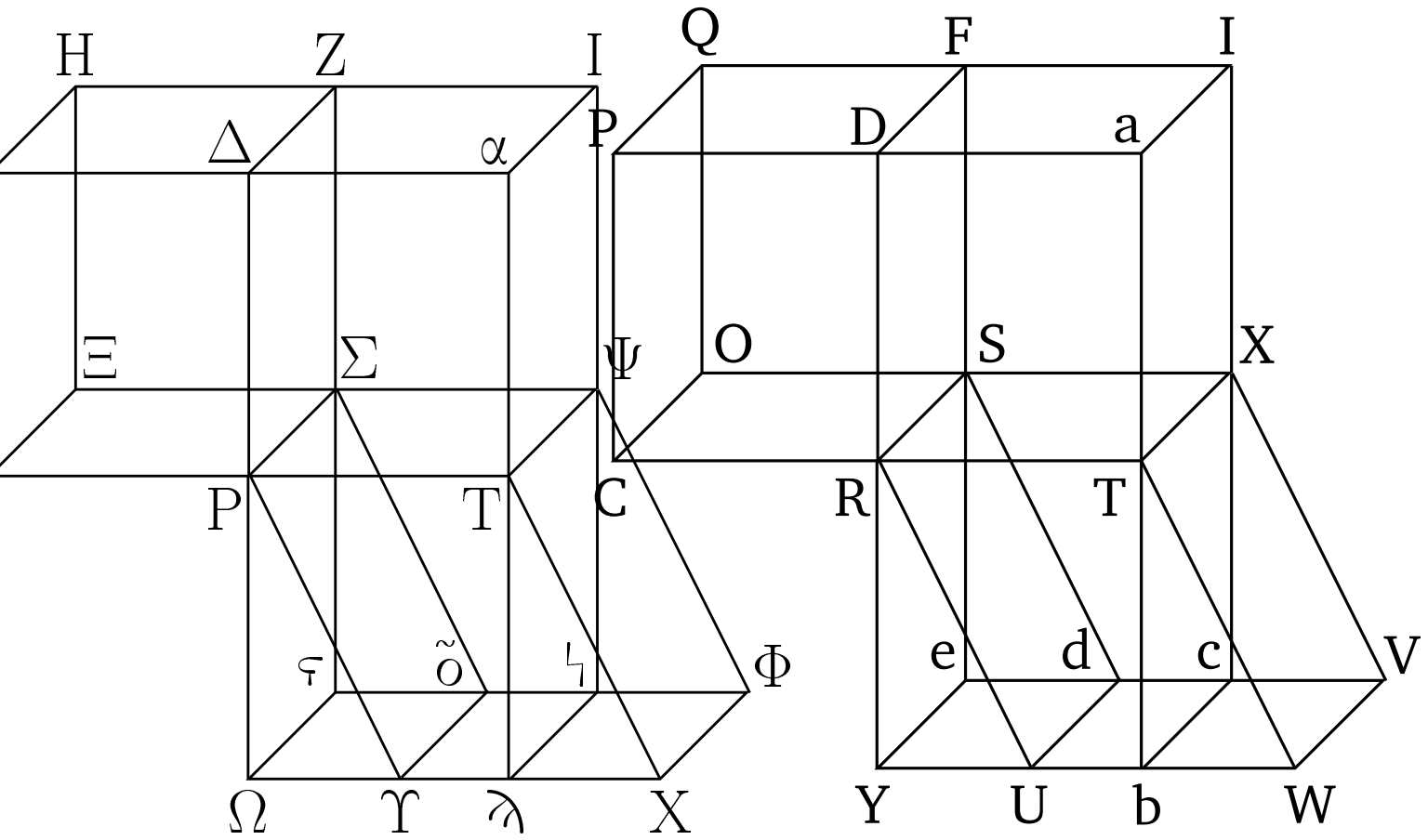
Ἔστωσαν δὴ πρότερον αἱ ἐφεστηκυῖαι αἱ ΘΚ, ΒΕ, ΑΗ, ΛΜ, ΟΠ, ΔΖ, ΓΧ, ΡΣ πρὸς ὀρθὰξ ταῖς ΑΒ, ΓΔ βάσεσιν, καὶ ἐκβεβλήσθω ἐπ΄ εὐθείας τῇ ΓΡ εὐθεῖα ἡ ΡΤ, καὶ συνεστάτω πρὸς τῇ ΡΤ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Ρ τῇ ὑπὸ ΑΛΒ γωνίᾳ ὁίση ἡ ὑπὸ ΤΡΥ, καὶ κείσθω τῇ μὲν ΑΛὁίση ἡ ΡΤ, τῇ δὲ ΛΒὁίση ἡ ΡΥ, καὶ συμπεπληρώσθω ἥ τε ΡΘ βάσις καὶ τὸ ΨΥ στερεόν.

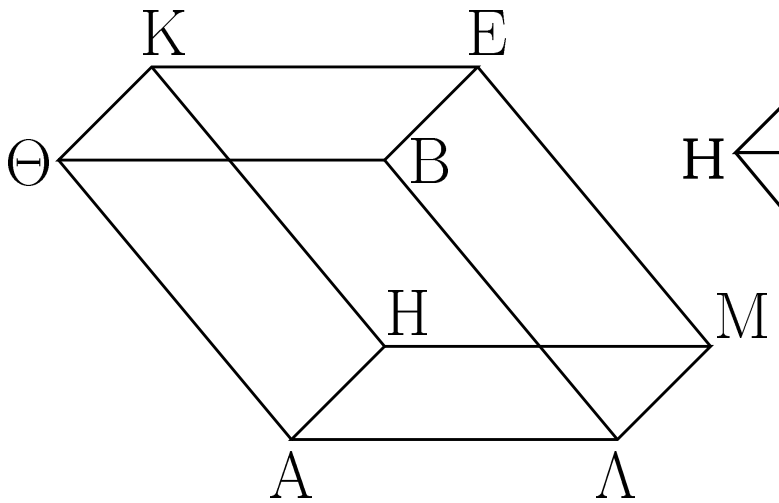
Proposition 31

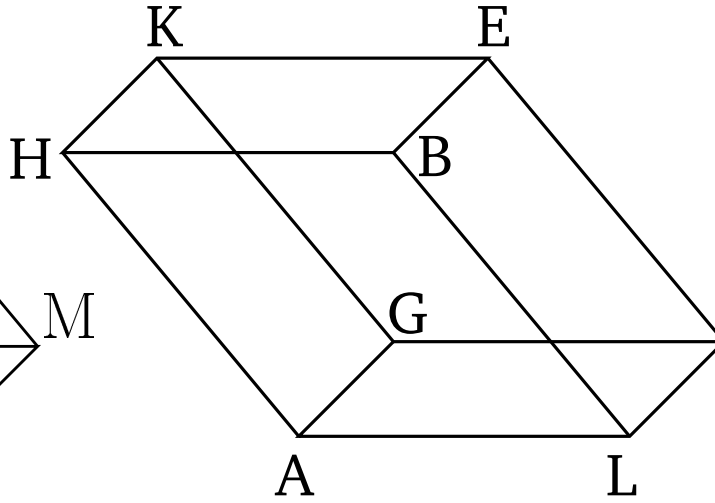
Parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another.

Let the parallelepiped solids *AE* and *CF* be on the equal bases *AB* and *CD* (respectively), and (have) the same height. I say that solid *AE* is equal to solid *CF*.

So, let the (straight-lines) standing up, *HK*, *BE*, *AG*, *LM*, *PQ*, *DF*, *CO*, and *RS*, first of all, be at right-angles to the bases *AB* and *CD*. And let *RT* be produced in a straight-line with *CR*. And let (angle) *TRU*, equal to angle *ALB*, be constructed on the straight-line *RT*, at the point *R* on it [Prop. 1.23]. And let *RT* be made equal to *AL*, and *RU* to *LB*. And let the base *RW*, and the solid *XU*, be completed.





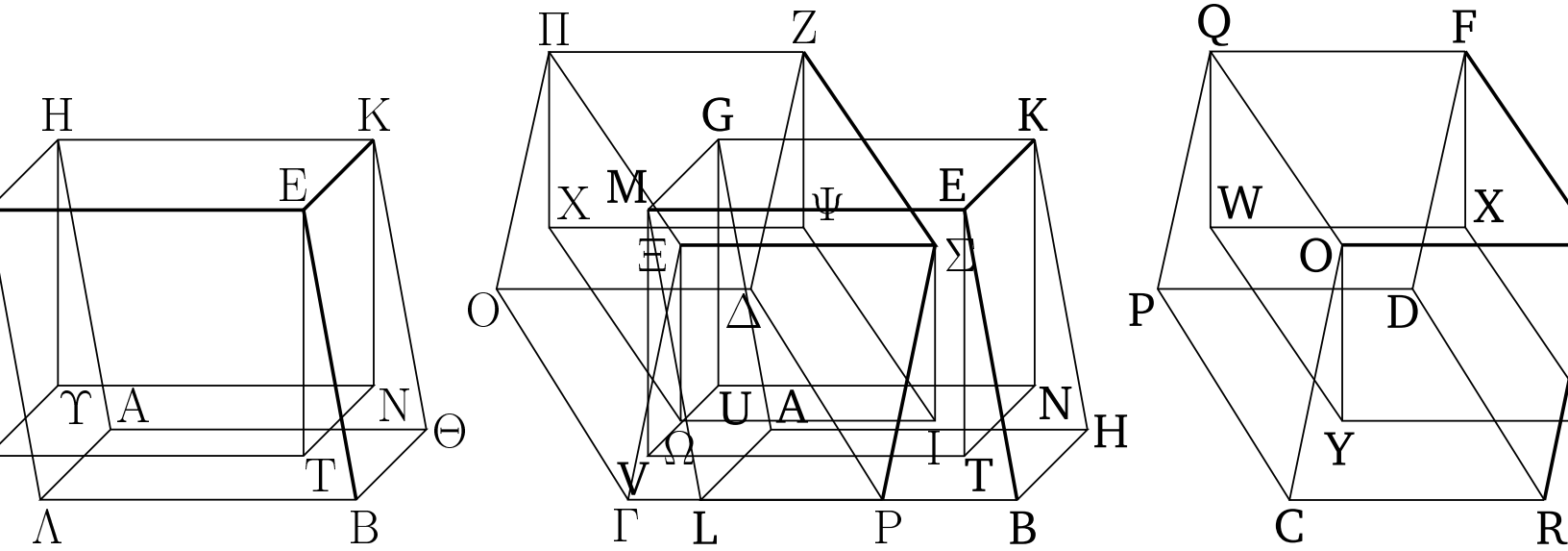


Καὶ ἐπεὶ δύο αἱ ΤΡ, ΡΥ δυσὶ ταῖς ΑΛ, ΛΒ ἴσαι εἰσίν, καὶ γωνίας ἴσας περιέθουσιν, ἴσον ἄρα καὶ ὅμοιον τὸ ΡΘ παραλληλόγραμμον τῷ ΘΛ παραλληλογράμμῳ. καὶ ἐπεὶ πάλινὁίση μὲν ἡ ΑΛ τῇ ΡΤ, ἡ δὲ ΛΜ τῇ ΡΣ, καὶ γωνίας ὀρθὰξ περιέθουσιν, ἴσον ἄρα καὶ ὅμοιόν ἐστι τὸ ΡΨ παραλληλόγραμμον

And since the two (straight-lines) *TR* and *RU* are equal to the two (straight-lines) *AL* and *LB* (respectively), and they contain equal angles, parallelogram *RW* is thus equal and similar to parallelogram *HL* [Prop. 6.14]. And, again, since *AL* is equal to *RT*, and *LM* to *RS*, and they contain right-angles,

τῷ AM παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΛE τῷ ΣY ἴσον τέ ἐστι καὶ ὅμοιον· τρία ἄρα παραλληλόγραμμα τοῦ AE στερεοῦ τρισὶ παραλληλογράμμοις τοῦ ΨY στερεοῦ ἴσα τέ ἐστι καὶ ὅμοια. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖς ἀπεναντίον ἴσα τέ ἐστι καὶ ὅμοια, τὰ δὲ τρία τρισὶ τοῖς ἀπεναντίον· ὅλον ἄρα τὸ AE στερεὸν παραλληλεπίπεδον ὅλῳ τῷ ΨY στερεῷ παραλληλεπιπέδῳ ἴσον ἐστίν. διήθθωσαν αἱ ΔP, ΘY καὶ συμπιπτέτωσαν ἀλλήλαις κατὰ τὸ Ω, καὶ διὰ τοῦ T τῇ $\Delta\Omega$ παράλληλος ἤθθω ἡ aT, καὶ ἐκβεβλήσθω ἡ $O\Delta$ κατὰ τὸ a, καὶ συμπεπληρώσθω τὰ $\Omega\Psi$, PI στερεά. ἴσον δή ἐστι τὸ $\Psi\Omega$ στερεόν, οὗ βάσις μέν ἐστι τὸ $P\Psi$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ Ω, τῷ ΨY στερεῷ, οὗ βάσις μὲν τὸ $P\Psi$ παραλληλόγραμμον, ἀπεναντίον δὲ τὸ $Y\Phi$· ἐπί τε γὰρ τῆς αὐτῆς βάσεώξ εἰσι τῆς $P\Psi$ καὶ ὑπὸ τὸ αὐτὸ ὑψος, ὧν αἱ ἐφεστῶσαι αἱ $P\Omega$, PY, T, $T\Theta$, Σ, Σo, $\Psi\rho$, $\Psi\Phi$ ἐπὶ τῶν αὐτῶν εἰσιν εὐθειῶν τῶν ΩX, $\varsigma\Phi$. ἀλλὰ τὸ ΨY στερεὸν τῷ AE ἐστιν ἴσον· καὶ τὸ $\Psi\Omega$ ἄρα στερεὸν τῷ AE στερεῷ ἐστιν ἴσον. καὶ ἐπεὶ ἴσον ἐστὶ τὸ $PYXT$ παραλληλόγραμμον τῷ ΩT παραλληλογράμμῳ· ἐπί τε γὰρ τῆς αὐτῆς βάσεώς εἰσι τῆς PT καὶ ἐν ταῖς αὐταῖς παραλλήλοις ταῖς PT, ΩX· ἀλλὰ τὸ $PYXT$ τῷ $\Gamma\Delta$ ἐστιν ἴσον, ἐπεὶ καὶ τῷ AB, καὶ τὸ ΩT ἄρα παραλληλόγραμμον τῷ $\Gamma\Delta$ ἐστιν ἴσον. ἄλλο δὲ τὸ ΔT· ἔστιν ἄρα ὡς ἡ $\Gamma\Delta$ βάσις πρὸς τὴν ΔT, οὕτως ἡ ΩT πρὸς τὴν ΔT. καὶ ἐπεὶ στερεὸν παραλληλεπίπεδον τὸ ΓI ἐπιπέδῳ τῷ PZ τέτμηται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἔστιν ὡς ἡ $\Gamma\Delta$ βάσις πρὸς τὴν ΔT βάσιν, οὕτως τὸ ΓZ στερεὸν πρὸς τὸ PI στερεόν. διὰ τὰ αὐτὰ δή, ἐπεὶ στερεὸν παραλληλεπίπεδον τὸ ΩI ἐπιπέδῳ τῷ $P\Psi$ τέτμηται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ἔστιν ὡς ἡ ΩT βάσις πρὸς τὴν $T\Lambda$ βάσιν, οὕτως τὸ $\Omega\Psi$ στερεὸν πρὸς τὸ PI. ἀλλ᾽ ὡς ἡ $\Gamma\Delta$ βάσις πρὸς τὴν ΔT, οὕτως ἡ ΩT πρὸς τὴν ΔT· καὶ ὡς ἄρα τὸ ΓZ στερεὸν πρὸς τὸ PI στερεόν, οὕτως τὸ $\Omega\Psi$ στερεὸν πρὸς τὸ PI. ἑκάτερον ἄρα τῶν ΓZ, $\Omega\Psi$ στερεῶν πρὸς τὸ PI τὸν αὐτὸν ἔχει λόγον· ἴσον ἄρα ἐστὶ τὸ ΓZ στερεὸν τῷ $\Omega\Psi$ στερεῷ. ἀλλὰ τὸ $\Omega\Psi$ τῷ AE ἐδείχθη ἴσον· καὶ τὸ AE ἄρα τῷ ΓZ ἐστιν ἴσον.

parallelogram RX is thus equal and similar to parallelogram AM [Prop. 6.14]. So, for the same (reasons), LE is also equal and similar to SU. Thus, three parallelograms of solid AE are equal and similar to three parallelograms of solid XU. But, the three (faces of the former solid) are equal and similar to the three opposite (faces), and the three (faces of the latter solid) to the three opposite (faces) [Prop. 11.24]. Thus, the whole parallelepiped solid AE is equal to the whole parallelepiped solid XU [Def. 11.10]. Let DR and WU be drawn across, and let them have met one another at Y. And let aTb be drawn through T parallel to DY. And let PD be produced to a. And let the solids YX and RI be completed. So, solid XY, whose base is parallelogram RX, and opposite (face) Yc, is equal to solid XU, whose base (is) parallelogram RX, and opposite (face) UV. For they are on the same base RX, and (have) the same height, and the (ends of the straight-lines) standing up in them, RY, RU, Tb, TW, Se, Sd, Xc and XV, are on the same straight-lines, YW and eV [Prop. 11.29]. But, solid XU is equal to AE. Thus, solid XY is also equal to solid AE. And since parallelogram $RUWT$ is equal to parallelogram YT. For they are on the same base RT, and between the same parallels RT and YW [Prop. 1.35]. But, $RUWT$ is equal to CD, since (it is) also (equal) to AB. Parallelogram YT is thus also equal to CD. And DT is another (parallelogram). Thus, as base CD is to DT, so YT (is) to DT [Prop. 5.7]. And since the parallelepiped solid CI has been cut by the plane RF, which is parallel to the opposite planes (of CI), as base CD is to base DT, so solid CF (is) to solid RI [Prop. 11.25]. So, for the same (reasons), since the parallelepiped solid YI has been cut by the plane RX, which is parallel to the opposite planes (of YI), as base YT is to base TD, so solid YX (is) to solid RI [Prop. 11.25]. But, as base CD (is) to DT, so YT (is) to DT. And, thus, as solid CF (is) to solid RI, so solid YX (is) to solid RI. Thus, solids CF and YX each have the same ratio to RI [Prop. 5.11]. Thus, solid CF is equal to solid YX [Prop. 5.9]. But, YX was show (to be) equal to AE. Thus, AE is also equal to CF.



μὴ ὀέστωσαν δὴ αἱ ἐφεστηκυῖαι αἱ ΑΗ, ΘΚ, ΒΕ, ΛΜ, ΓΧ, ΟΠ, ΔΖ, ΡΣ πρὸς ὀρθὰξ ταῖς ΑΒ, ΓΔ βάσεσιν· λέγω πάλιν, ὅτι ὅισον τὸ ΑΕ στερεὸν τῷ ΓΖ στερεῷ. ἤθθωσαν γὰρ ἀπὸ τῶν Κ, Ε, Η, Μ, Π, Ζ, Χ, Σ σημείων ἐπὶ τὸ ὑποκείμενον ἐπίπεδον κάθετοι αἱ ΚΝ, ΕΤ, ΗΥ, ΜΦ, ΠΘ, ΖΨ, ΧΩ, ΣΙ, καὶ συμβαλλέτωσαν τῷ ἐπιπέδῳ κατὰ τὰ Ν, Τ, Υ, Φ, Θ, Ψ, Ω, Ι σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΝΤ, ΝΥ, ΥΦ, ΤΦ, ΘΨ, ΘΩ, ΩΙ, ΙΨ. ἴσον δή ἐστι τὸ ΚΦ στερεὸν τῷ ΠΙ στερεῷ· ἐπί τε γὰρὸίσων βάσεών εἰσι τῶν ΚΜ, ΠΣ καὶ ὑπὸ τὸ αὐτὸ ὁύψος, ὁῶν αἱ ἐφεστῶσαι πρὸς ὀρθάξ εἰσι ταῖς βάσεσιν. ἀλλὰ τὸ μὲν ΚΦ στερεὸν τῷ ΑΕ στερεῷ ἐστιν ἴσον, τὸ δὲ ΠΙ τῷ ΓΖ· ἐπί τε γὰρ τῆς αὐτῆς βάσεώξ εἰσι καὶ ὑπὸ τὸ αὐτὸ ὁύψος, ὁῶν αἱ ἐφεστῶσαι οὐύκ εἰσιν ἐπὶ τῶν αὐτῶν εὐθειῶν. καὶ τὸ ΑΕ ἄρα στερεὸν τῷ ΓΖ στερεῷ ἐστιν ἴσον.

Τὰ ἄρα ἐπὶ ὁίσων βάσεων ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὁύψος ἴσα ἀλλήλοις ἐστίν· ὅπερ ἔδει δεῖχαι.

And so let the (straight-lines) standing up, AG, HK, BE, LM, CO, PQ, DF, and RS, not be at right-angles to the bases AB and CD. Again, I say that solid AE (is) equal to solid CF. For let KN, ET, GU, MV, QW, FX, OY, and SI be drawn from points K, E, G, M, Q, F, O, and S (respectively) perpendicular to the reference plane (i.e., the plane of the bases AB and CD), and let them have met the plane at points N, T, U, V, W, X, Y, and I (respectively). And let NT, NU, UV, TV, WX, WY, YI, and IX be joined. So solid KV is equal to solid QI. For they are on the equal bases KM and QS, and (have) the same height, and the (straight-lines) standing up in them are at right-angles to their bases (see first part of proposition). But, solid KV is equal to solid AE, and QI to CF. For they are on the same base, and (have) the same height, and the (straight-lines) standing up in them are not on the same straight-lines [Prop. 11.30]. Thus, solid AE is also equal to solid CF.

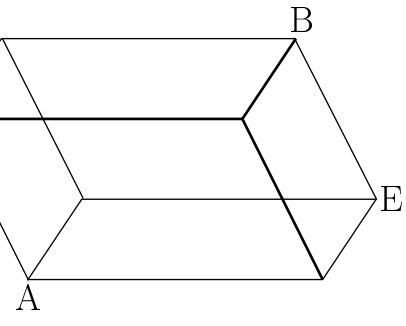
Thus, parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another. (Which is) the very thing it was required to show.

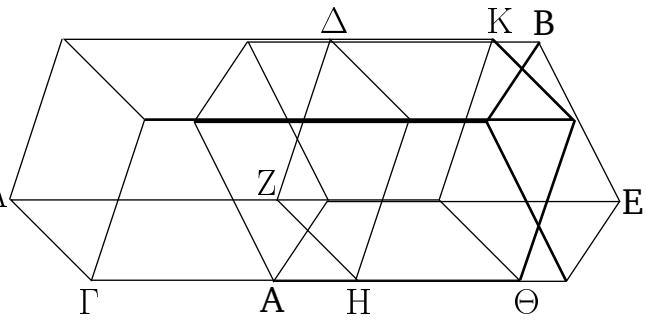
32.

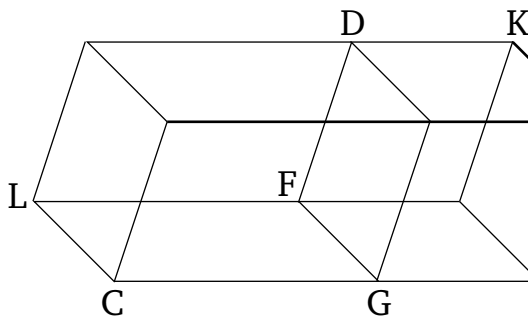
Τὰ ὑπὸ τὸ αὐτὸ ὁύψος ὄντα στερεὰ παραλληλεπίπεδα πρὸςὀάλληλά ἐστιν ὡξ αἱ βάσεις.

Proposition 32

Parallelepiped solids which (have) the same height are to one another as their bases.







Ἔστω ὑπὸ τὸ αὐτὸ ὕψος στερεὰ παραλληλεπίπεδα τὰ AB, ΓΔ· λέγω, ὅτι τὰ AB, ΓΔ στερεὰ παραλληλεπίπεδα πρὸς ἀλλήλα ἐστὶν ὡς αἱ βάσεις, τουτέστιν ὅτι ἐστὶν ὡς ἡ AE βάσις πρὸς τὴν ΓZ βάσιν, οὕτως τὸ AB στερεὸν πρὸς τὸ ΓΔ στερεόν.

Παραβεβλήσθω γὰρ παρὰ τὴν ZH τῷ AE ἴσον τὸ ZΘ, καὶ ἀπὸ βάσεως μὲν τῆς ZΘ, ὕψους δὲ τοῦ αὐτοῦ τῷ ΓΔ στερεὸν παραλληλεπίπεδον συμπεπληρώσθω τὸ HK. ἴσον δή ἐστι τὸ AB στερεὸν τῷ HK στερεῷ· ἐπί τε γὰρ ἴσων βάσεών εἰσι τῶν AE, ZΘ καὶ ὑπὸ τὸ αὐτὸ ὕψος. καὶ ἐπεὶ στερεὸν παραλληλεπίπεδον τὸ ΓK ἐπιπέδῳ τῷ ΔH τέτμηται παραλλήλῳ ὅντι τοῖς ἀπεναντίον ἐπιπέδοις, ὡς ἄρα ἡ βάσις ἡ ΓZ πρὸς τὴν ZΘ βάσιν, οὕτως τὸ ΓΔ στερεὸν πρὸς τὸ ΔΘ στερεόν. ἀλλ' ὡς μὲν ἡ ZΘ βάσις πρὸς τὴν AE βάσιν, οὕτως τὸ HK στερεὸν πρὸς τὸ AB στερεόν· ὡς ἄρα καὶ ἡ βάσις πρὸς τὴν ΓZ βάσιν, οὕτως τὸ AB στερεὸν πρὸς τὸ ΓΔ στερεόν.

Τὰ ἄρα ὑπὸ τὸ αὐτὸ ὕψος ὄντα στερεὰ παραλληλεπίπεδα πρὸς ἀλλήλα ἐστὶν ὡς αἱ βάσεις· ὅπερ ἔδει δεῖξαι.

Let *AB* and *CD* be parallelepiped solids (having) the same height. I say that the parallelepiped solids *AB* and *CD* are to one another as their bases. That is to say, as base *AE* is to base *CF*, so solid *AB* (is) to solid *CD*.

For let *FH*, equal to *AE*, be applied to *FG* (in the angle *FGH* equal to angle *LCG*) [Prop. 1.45]. And let the parallelepiped solid *GK*, (having) the same height as *CD*, be completed on the base *FH*. So solid *AB* is equal to solid *GK*. For they are on the equal bases *AE* and *FH*, and (have) the same height [Prop. 11.31]. And since the parallelepiped solid *CK* has been cut by the plane *DG*, which is parallel to the opposite planes (of *CK*), thus as the base *CF* is to the base *FH*, so the solid *CD* (is) to the solid *DH* [Prop. 11.25]. And base *FH* (is) equal to base *AE*, and solid *GK* to solid *AB*. And thus as base *AE* is to base *CF*, so solid *AB* (is) to solid *CD*.

Thus, parallelepiped solids which (have) the same height are to one another as their bases. (Which is) the very thing it was required to show.

33.

Τὰ ὅμοια στερεὰ παραλληλεπίπεδα πρὸς ἀλλήλα ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν.

Ἔστω ὅμοια στερεὰ παραλληλεπίπεδα τὰ AB, ΓΔ, ὁμόλογος δὲ ἐστω ἡ AE τῇ ΓZ· λέγω, ὅτι τὸ AB στερεὸν πρὸς τὸ ΓΔ στερεὸν τριπλασίονα λόγον ἔχει, ἤπερ ἡ AE πρὸς τὴν ΓZ.

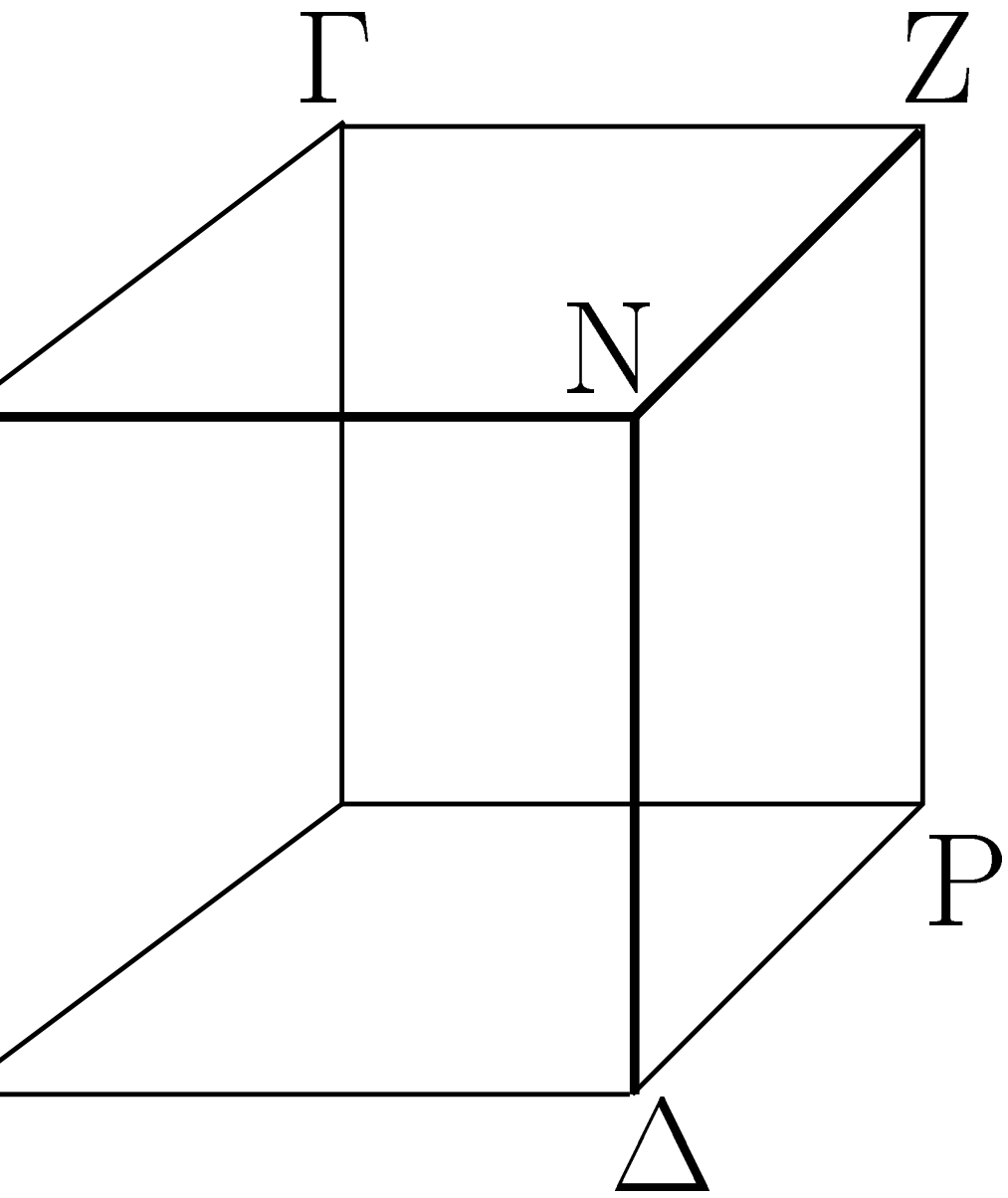
Ἐκβεβλήσθωσαν γὰρ ἐπ' εὐθείας ταῖς AE, HE, ΘE αἱ EK, EΛ, EM, καὶ κείσθω τῇ μὲν ΓZ ἴση ἡ EK, τῇ δὲ ZN ἴση ἡ EΛ, καὶ ἔτι τῇ ZP ἴση ἡ EM, καὶ συμπεπληρώσθω τὸ KΛ παραλληλόγραμμον καὶ τὸ KO στερεόν.

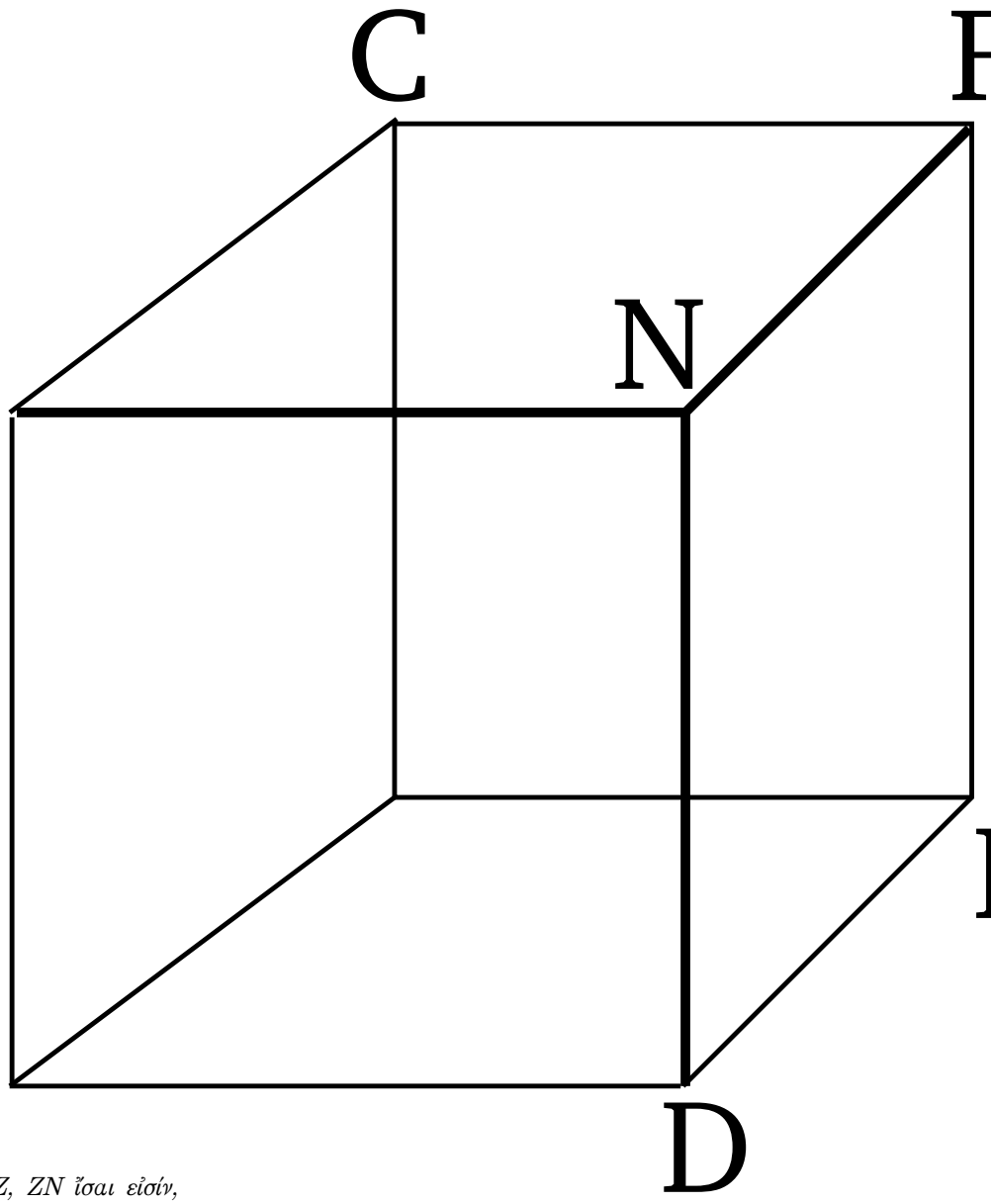
Proposition 33

Similar parallelepiped solids are to one another as the cubed ratio of their corresponding sides.

Let *AB* and *CD* be similar parallelepiped solids, and let *AE* correspond to *CF*. I say that solid *AB* has to solid *CD* the cubed ratio that *AE* (has) to *CF*.

For let *EK*, *EL*, and *EM* be produced in a straight-line with *AE*, *GE*, and *HE* (respectively). And let *EK* be made equal to *CF*, and *EL* equal to *FN*, and, further, *EM* equal to *FR*. And let the parallelogram *KL* be completed, and the solid *KP*.


Γ
Z
N
P
Δ

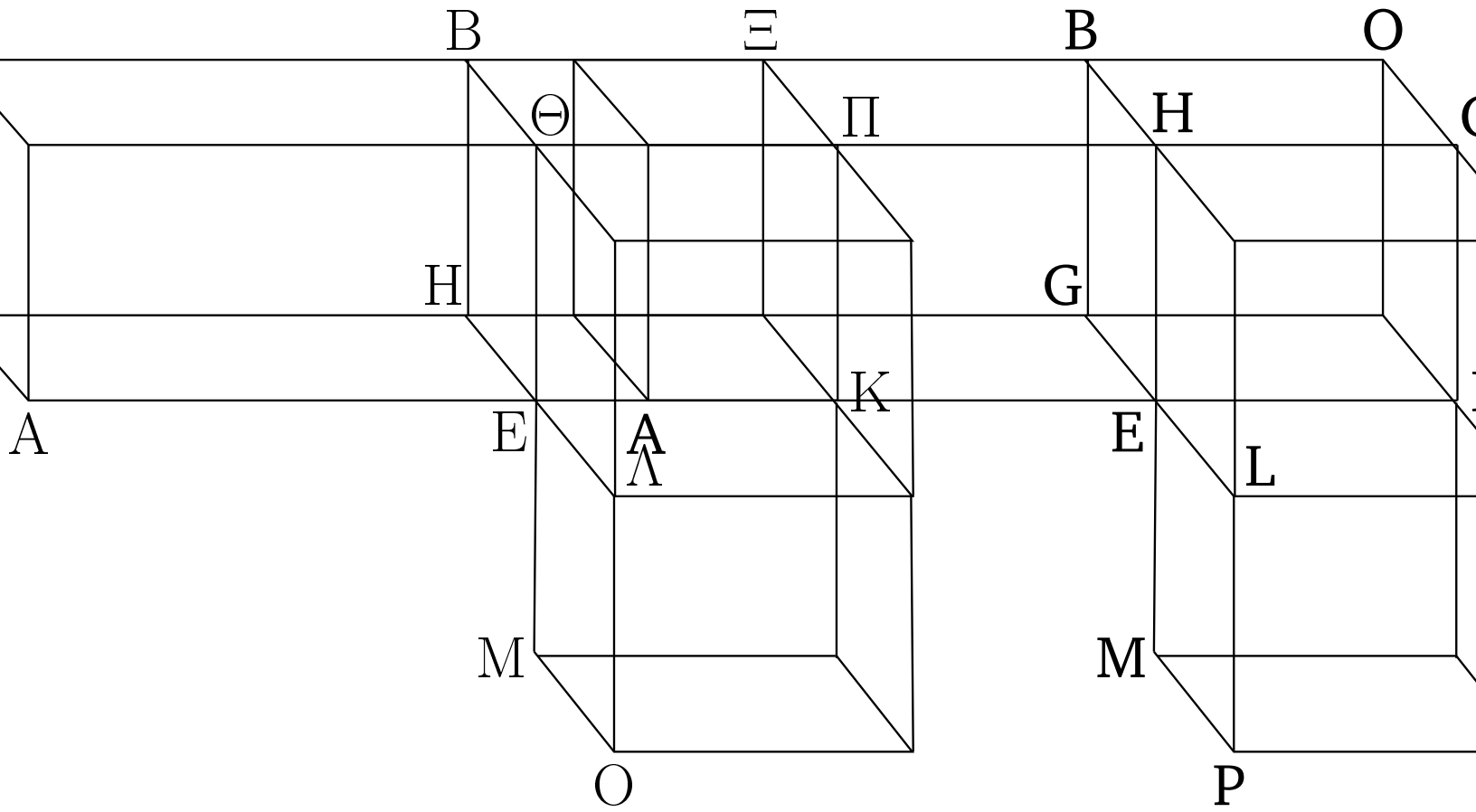


Καὶ ἐπεὶ δύο αἱ ΚΕ, ΕΛ δυσὶ ταῖς ΓΖ, ΖΝ ἴσαι εἰσίν, ἀλλὰ καὶ γωνία ἡ ὑπὸ ΚΕΛ γωνίᾳ τῇ ὑπὸ ΓΖΝ ἐστινὁίση, ἐπειδήπερ καὶ ἡ ὑπὸ ΑΕΗ τῇ ὑπὸ ΓΖΝ ἐστινὁίση διὰ τὴν ὁμοιότητα τῶν ΑΒ, ΓΔ στερεῶν, ἴσον ἄρα ἐστὶ [καὶ ὅμοιον] τὸ ΚΛ παραλληλόγραμμον τῷ ΓΝ παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν ΚΜ παραλληλόγραμμον ἴσον ἐστὶ καὶ ὅμοιον τῷ ΓΡ [παραλληλογράμμῳ] καὶ ὀέτι τὸ ΕΟ τῷ ΔΖ· τρία ἄρα παραλληλόγραμμα τοῦ ΚΟ στερεοῦ τρισὶ παραλληλογράμμοις τοῦ ΓΔ στερεοῦ ἴσα ἐστὶ καὶ ὅμοια. ἀλλὰ τὰ μὲν τρία τρισὶ τοῖς ἀπεναντίον ἴσα ἐστὶ καὶ ὅμοια, τὰ δὲ τρία τρισὶ τοῖς ἀπεναντίον ἴσα ἐστὶ καὶ ὅμοια· ὅλον ἄρα τὸ ΚΟ στερεὸν ὅλῳ τῷ ΓΔ στερεῷ ἴσον ἐστὶ καὶ ὅμοιον. συμπεπληρώσθω τὸ ΗΚ παραλληλόγραμμον, καὶ ἀπὸ βάσεων μὲν τῶν ΗΚ, ΚΛ παραλληλόγραμμων, ὑψους δὲ τοῦ αὐτοῦ τῷ ΑΒ στερεὰ

And since the two (straight-lines) KE and EL are equal to the two (straight-lines) CF and FN, but angle KEL is also equal to angle CFN, inasmuch as AEG is also equal to CFN, on account of the similarity of the solids AB and CD, parallelogram KL is thus equal [and similar] to parallelogram CN. So, for the same (reasons), parallelogram KM is also equal and similar to [parallelogram] CR, and, further, EP to DF. Thus, three parallelograms of solid KP are equal and similar to three parallelograms of solid CD. But the three (former parallelograms) are equal and similar to the three opposite (parallelograms), and the three (latter parallelograms) are equal and similar to the three opposite (parallelograms) [Prop. 11.24]. Thus, the whole of solid KP is equal and sim-

συμπεπληρώσθω τὰ ΕΧ, ΛΠ. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν ΑΒ, ΓΔ στερεῶν ἐστιν ὡς ἡ ΑΕ πρὸς τὴν ΓΖ, οὕτως ἡ ΕΗ πρὸς τὴν ΖΝ, καὶ ἡ ΕΘ πρὸς τὴν ΖΡ, ὁίση δὲ ἡ μὲν ΓΖ τῇ ΕΚ, ἡ δὲ ΖΝ τῇ ΕΛ, ἡ δὲ ΖΡ τῇ ΕΜ, ὀέστιν ἄρα ὡξ ἡ ΑΕ πρὸς τὴν ΕΚ, οὕτως ἡ ΗΕ πρὸς τὴν ΕΛ καὶ ἡ ΘΕ πρὸς τὴν ΕΜ. ἀλλὸ ὡξ μὲν ἡ ΑΕ πρὸς τὴν ΕΚ, οὕτως τὸ ΑΗ [παραλληλόγραμμον] πρὸς τὸ ΗΚ παραλληλόγραμμον, ὡξ δὲ ἡ ΗΕ πρὸς τὴν ΕΛ, οὕτως τὸ ΗΚ πρὸς τὸ ΚΛ, ὡξ δὲ ἡ ΘΕ πρὸς ΕΜ, οὕτως τὸ ΠΕ πρὸς τὸ ΚΜ· καὶ ὡξ ἄρα τὸ ΑΗ παραλληλόγραμμον πρὸς τὸ ΗΚ, οὕτως τὸ ΗΚ πρὸς τὸ ΚΛ καὶ τὸ ΠΕ πρὸς τὸ ΚΜ. ἀλλὸ ὡξ μὲν τὸ ΑΗ πρὸς τὸ ΗΚ, οὕτως τὸ ΑΒ στερεὸν πρὸς τὸ ΕΧ στερεόν, ὡξ δὲ τὸ ΗΚ πρὸς τὸ ΚΛ, οὕτως τὸ ΧΕ στερεὸν πρὸς τὸ ΠΛ στερεόν, ὡξ δὲ τὸ ΠΕ πρὸς τὸ ΚΜ, οὕτως τὸ ΠΛ στερεὸν πρὸς τὸ ΚΟ στερεόν· καὶ ὡξ ἄρα τὸ ΑΒ στερεὸν πρὸς τὸ ΕΧ, οὕτως τὸ ΕΧ πρὸς τὸ ΠΛ καὶ τὸ ΠΛ πρὸς τὸ ΚΟ. ἒαν δὲ τέσσαρα μεγέθη κατὰ τὸ συνεθὲξ ἀνάλογονὸῇ, τὸ πρῶτον πρὸς τὸ τέταρτον τριπλασίονα λόγον ἔχει ἤπερ πρὸς τὸ δεύτερον· τὸ ΑΒ ἄρα στερεὸν πρὸς τὸ ΚΟ τριπλασίονα λόγον ἔχει ἤπερ τὸ ΑΒ πρὸς τὸ ΕΧ. ἀλλὸ ὡξ τὸ ΑΒ πρὸς τὸ ΕΧ, οὕτως τὸ ΑΗ παραλληλόγραμμον πρὸς τὸ ΗΚ καὶ ἡ ΑΕ εὐθεῖα πρὸς τὴν ΕΚ· ὡστε καὶ τὸ ΑΒ στερεὸν πρὸς τὸ ΚΟ τριπλασίονα λόγον ἔχει ἤπερ ἡ ΑΕ πρὸς τὴν ΕΚ. ἴσον δὲ τὸ [μὲν] ΚΟ στερεὸν τῷ ΓΔ στερεῷ, ἡ δὲ ΕΚ εὐθεῖα τῇ ΓΖ· καὶ τὸ ΑΒ ἄρα στερεὸν πρὸς τὸ ΓΔ στερεὸν τριπλασίονα λόγον ἔχει ἤπερ ἡ ὁμόλογος αὐτοῦ πλευρὰ ἡ ΑΕ πρὸς τὴν ὁμόλογον πλευρὰν τὴν ΓΖ.

ilar to the whole of solid CD [Def. 11.10]. Let parallelogram GK be completed. And let the the solids EO and LQ, with bases the parallelograms GK and KL (respectively), and with the same height as AB, be completed. And since, on account of the similarity of solids AB and CD, as AE is to CF, so EG (is) to FN, and EH to FR [Defs. 6.1, 11.9], and CF (is) equal to EK, and FN to EL, and FR to EM, thus as AE is to EK, so GE (is) to EL, and HE to EM. But, as AE (is) to EK, so [parallelogram] AG (is) to parallelogram GK, and as GE (is) to EL, so GK (is) to KL, and as HE (is) to EM, so QE (is) to KM [Prop. 6.1]. And thus as parallelogram AG (is) to GK, so GK (is) to KL, and QE (is) to KM. But, as AG (is) to GK, so solid AB (is) to solid EO, and as GK (is) to KL, so solid OE (is) to solid QL, and as QE (is) to KM, so solid QL (is) to solid KP [Prop. 11.32]. And, thus, as solid AB is to EO, so EO (is) to QL, and QL to KP. And if four magnitudes are continuously proportional then the first has to the fourth the cubed ratio that (it has) to the second [Def. 5.10]. Thus, solid AB has to KP the cubed ratio which AB (has) to EO. But, as AB (is) to EO, so parallelogram AG (is) to GK, and the straight-line AE to EK [Prop. 6.1]. Hence, solid AB also has to KP the cubed ratio that AE (has) to EK. And solid KP (is) equal to solid CD, and straight-line EK to CF. Thus, solid AB also has to solid CD the cubed ratio which its corresponding side AE (has) to the corresponding side CF.



Τὰ ἄρα ὅμοια στερεὰ παραλληλεπίπεδα ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν· ὅπερ ἔδει δεῖξαι.

Thus, similar parallelepiped solids are to one another as the cubed ratio of their corresponding sides. (Which is) the very thing it was required to show.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον ὦσιν, ὡς ἡ πρώτη πρὸς τὴν τετάρτην, οὕτω τὸ ἀπὸ τῆς πρώτης στερεὸν παραλληλεπίπεδον πρὸς τὸ ἀπὸ τῆς δευτέρας τὸ ὅμοιον καὶ ὁμοίως ἀναγραφόμενον, ἐπείπερ καὶ ἡ πρώτη πρὸς τὴν τετάρτην τριπλασίονα λόγον ἔχει ἤπερ πρὸς τὴν δευτέραν.

Corollary

So, (it is) clear, from this, that if four straight-lines are (continuously) proportional then as the first is to the fourth, so the parallelepiped solid on the first will be to the similar, and similarly described, parallelepiped solid on the second, since the first also has to the fourth the cubed ratio that (it has) to the second.

34.

Τῶν ἴσων στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν· καὶ ὧν στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, ἴσα ἐστὶν ἐκεῖνα.

Ἔστω ἴσα στερεὰ παραλληλεπίπεδα τὰ AB, $ΓΔ$· λέγω, ὅτι τῶν AB, $ΓΔ$ στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, καί ἐστιν ὡς ἡ $EΘ$ βάσις πρὸς τὴν $NΠ$ βάσιν, οὕτως τὸ τοῦ $ΓΔ$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος.

Ἔστωσαν γὰρ πρότερον αἱ ἐφεστηκυῖαι αἱ AH, EZ, $ΛB$,

Proposition 34†

The bases of equal parallelepiped solids are reciprocally proportional to their heights. And those parallelepiped solids whose bases are reciprocally proportional to their heights are equal.

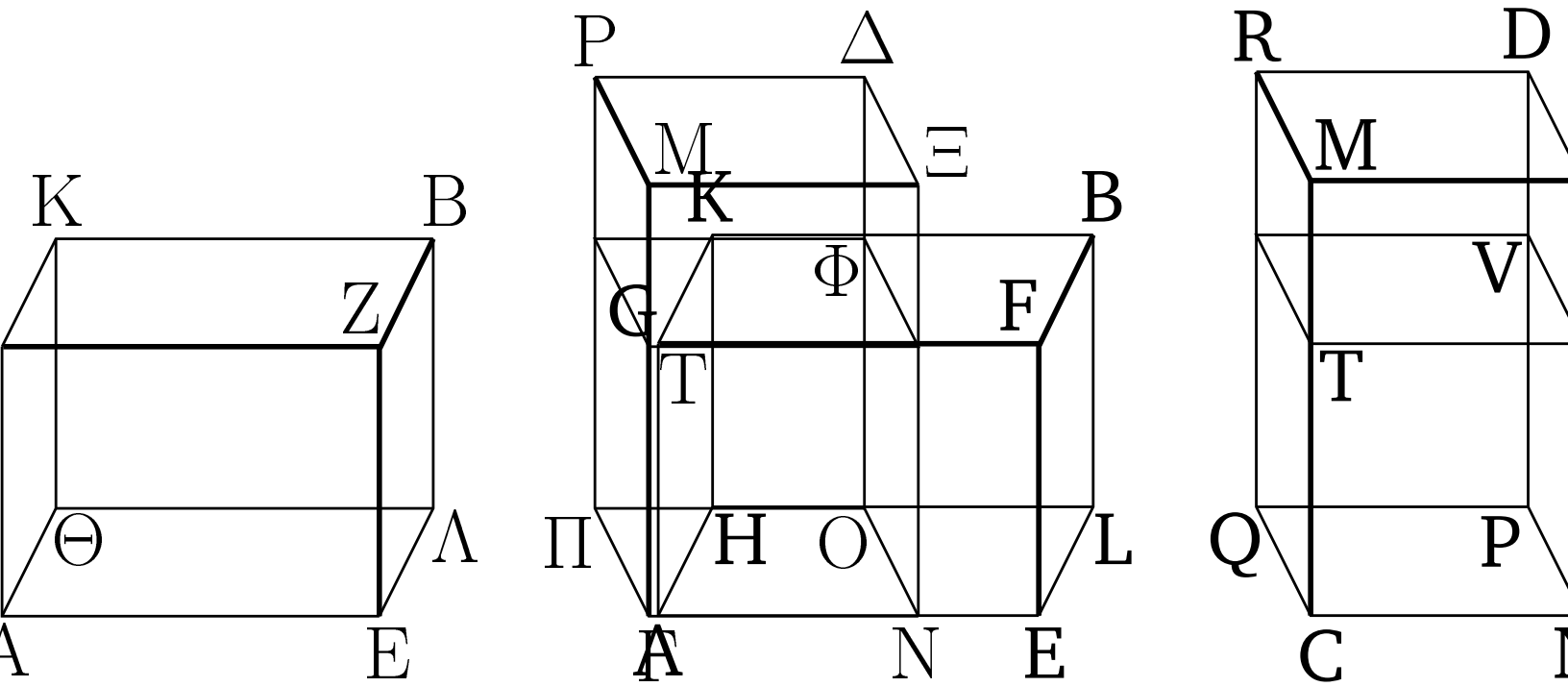
Let AB and CD be equal parallelepiped solids. I say that the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights, and (so) as base EH is to base NQ, so the height of solid CD (is) to the height of solid AB.

ΘΚ, ΓΜ, ΝΧ, ΟΔ, ΠΡ πρὸς ὀρθὰξ ταῖς βάσεσιν αὐτῶν· λέγω, ὅτι ἐστὶν ὡξ ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὕτως ἡ ΓΜ πρὸς τὴν ΑΗ.

Εἰ μὲν οὐῦνὀίση ἐστὶν ἡ ΕΘ βάσιν τῇ ΝΠ βάσει, ὀέστι δὲ καὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ ἴσον, ὀέσται καὶ ἡ ΓΜ τῇ ΑΗὀίση. τὰ γὰρ ὑπὸ τὸ αὐτὸ ὕψος στερεὰ παραλληλεπίπεδα πρὸςὀάλληλά ἐστιν ὡξ αἱ βάσεις. καὶ ὀέσται ὡξ ἡ ΕΘ βάσις πρὸς τὴν ΝΠ, οὕτως ἡ ΓΜ πρὸς τὴν ΑΗ, καὶ φανερόν, ὅτι τῶν ΑΒ, ΓΔ στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

For, first of all, let the (straight-lines) standing up, AG, EF, LB, HK, CM, NO, PD, and QR, be at right-angles to their bases. I say that as base EH is to base NQ, so CM (is) to AG.

Therefore, if base EH is equal to base NQ, and solid AB is also equal to solid CD, CM will also be equal to AG. For parallelepiped solids of the same height are to one another as their bases [Prop. 11.32]. And as base EH (is) to NQ, so CM will be to AG. And (so it is) clear that the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights.



μὴ ὀέστω δὴ ὀίση ἡ ΕΘ βάσις τῇ ΝΠ βάσει, ἀλλὸ ὀέστω μείζων ἡ ΕΘ. ὀέστι δὲ καὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ ἴσον· μείζων ἄρα ἐστὶ καὶ ἡ ΓΜ τῆς ΑΗ. κείσθω οὐῦν τῇ ΑΗὀίση ἡ ΓΤ, καὶ συμπεπληρώσθω ἀπὸ βάσεως μὲν τῆς ΝΠ, ὕψους δὲ τοῦ ΓΤ, στερεὸν παραλληλεπίπεδον τὸ ΦΓ. καὶ ἐπεὶ ἴσον ἐστὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ, ὀέχωθεν δὲ τὸ ΓΦ, τὰ δὲ ἴσα πρὸς τὸ αὐτὸ τὸν αὐτὸν ἔχει λόγον, ὀέστιν ἄρα ὡξ τὸ ΑΒ στερεὸν πρὸς τὸ ΓΦ στερεόν, οὕτως τὸ ΓΔ στερεὸν πρὸς τὸ ΓΦ στερεόν. ἀλλὸ ὡξ μὲν τὸ ΑΒ στερεὸν πρὸς τὸ ΓΦ στερεόν, οὕτως ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν· ἰσοϋψῆ γὰρ τὰ ΑΒ, ΓΦ στερεά· ὡξ δὲ τὸ ΓΔ στερεὸν πρὸς τὸ ΓΦ στερεόν, οὕτως ἡ ΜΠ βάσις πρὸς τὴν ΤΠ βάσιν καὶ ἡ ΓΜ πρὸς τὴν ΓΤ· καὶ ὡξ ἄρα ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὕτως ἡ ΜΓ πρὸς τὴν ΓΤ. ὀίση δὲ ἡ ΓΤ τῇ ΑΗ· καὶ ὡξ ἄρα ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὕτως ἡ ΜΓ πρὸς τὴν ΑΗ. τῶν ΑΒ, ΓΔ ἄρα στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

So let base EH not be equal to base NQ, but let EH be greater. And solid AB is also equal to solid CD. Thus, CM is also greater than AG. Therefore, let CT be made equal to AG. And let the parallelepiped solid VC be completed on the base NQ, with height CT. And since solid AB is equal to solid CD, and CV (is) extrinsic (to them), and equal (magnitudes) have the same ratio to the same (magnitude) [Prop. 5.7], thus as solid AB is to solid CV, so solid CD (is) to solid CV. But, as solid AB (is) to solid CV, so base EH (is) to base NQ. For the solids AB and CV (are) of equal height [Prop. 11.32]. And as solid CD (is) to solid CV, so base MQ (is) to base TQ [Prop. 11.25], and CM to CT [Prop. 6.1]. And, thus, as base EH is to base NQ, so MC (is) to AG. And CT (is) equal to AG. And thus as base EH (is) to base NQ, so MC (is) to AG. Thus, the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights.

Πάλιν δὴ τῶν ΑΒ, ΓΔ στερεῶν παραλληλεπιπέδων ἀντι-

So, again, let the bases of the parallelepipid solids AB and CD be reciprocally proportional to their heights, and let base

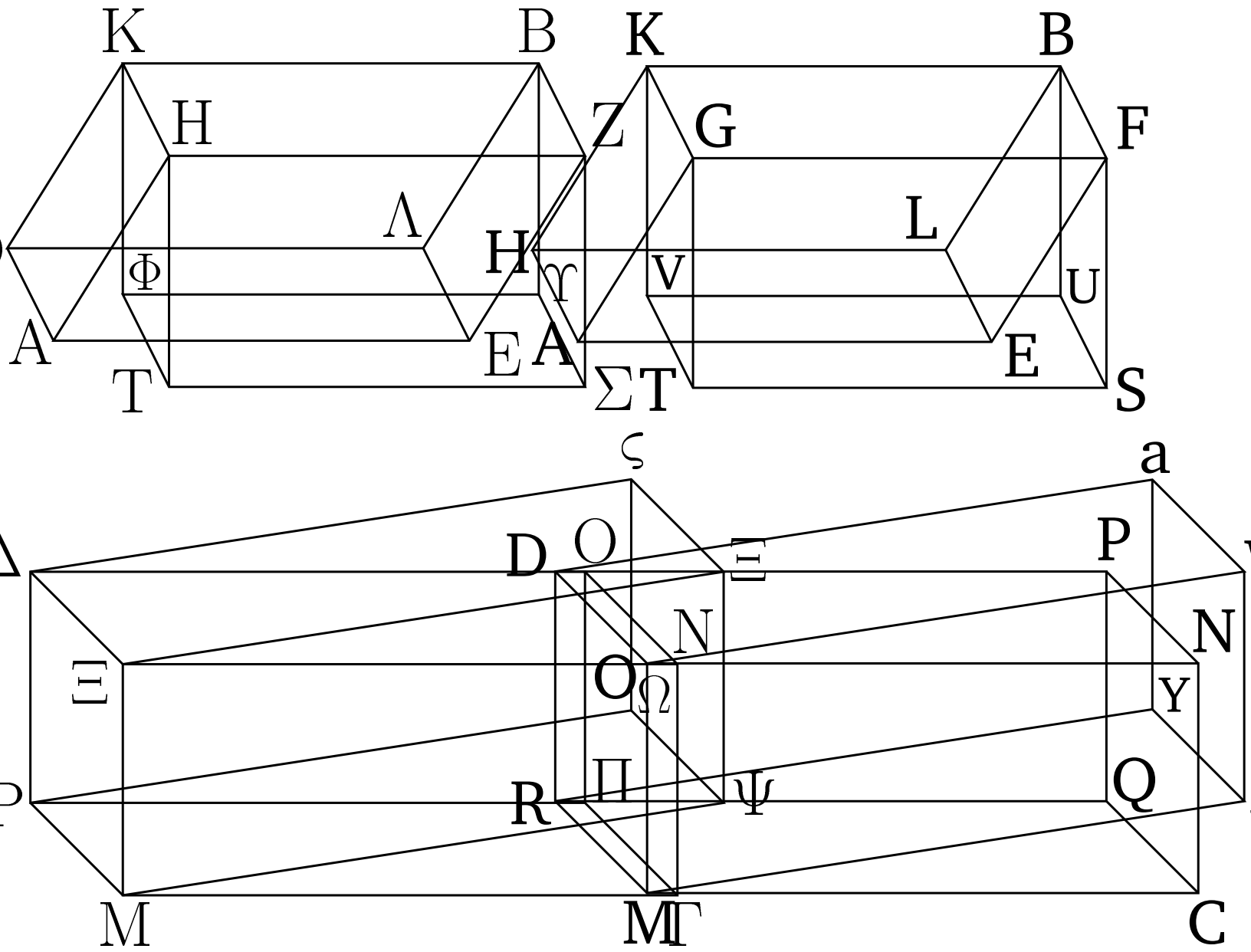
πεπονθέτωσαν αἱ βάσεις τοῖς ὑψεσιν, καὶ ὀέστω ὡς ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὑτως τὸ τοῦ ΓΔ στερεοῦ ὑψος πρὸς τὸ τοῦ ΑΒ στερεοῦ ὑψος· λέγω, ὅτι ἴσον ἐστὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ.

Ἔστωσαν [γὰρ] πάλιν αἱ ἐφεστηκυῖαι πρὸς ὀρθὰξ ταῖς βάσεσιν. καὶ εἰ μὲνὀίση ἐστὶν ἡ ΕΘ βάσις τῇ ΝΠ βάσει, καί ἐστιν ὡξ ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὑτως τὸ τοῦ ΓΔ στερεοῦ ὑψος πρὸς τὸ τοῦ ΑΒ στερεοῦ ὑψος, ἴσον ἄρα ἐστὶ καὶ τὸ τοῦ ΓΔ στερεοῦ ὑψος τῷ τοῦ ΑΒ στερεοῦ ὑψει. τὰ δὲ ἐπὶ ὁίσων βάσεων στερεά παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὑψος ἴσα ἀλλήλοις ἐστίν· ἴσον ἄρα ἐστὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ.

μὴ ὀέστω δὴ ἡ ΕΘ βάσις τῇ ΝΠ [βάσει] ὁίση, ἀλλὸ ὀέστω μείζων ἡ ΕΘ· μεῖζον ἄρα ἐστὶ καὶ τὸ τοῦ ΓΔ στερεοῦ ὑψος τοῦ τοῦ ΑΒ στερεοῦ ὑψους, τουτέστιν ἡ ΓΜ τῆς ΑΗ. κείσθω τῇ ΑΗὁίση πάλιν ἡ ΓΤ, καὶ συμπεπληρώσθω ὁμοίως τὸ ΓΦ στερεόν. ἐπεί ἐστιν ὡξ ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὑτως ἡ ΜΓ πρὸς τὴν ΑΗ, ὁίση δὲ ἡ ΑΗ τῇ ΓΤ, ὀέστιν ἄρα ὡξ ἡ ΕΘ βάσις πρὸς τὴν ΝΠ βάσιν, οὑτως ἡ ΓΜ πρὸς τὴν ΓΤ. ἀλλὸ ὡξ μὲν ἡ ΕΘ [βάσις] πρὸς τὴν ΝΠ βάσιν, οὑτως τὸ ΑΒ στερεὸν πρὸς τὸ ΓΦ στερεόν· ἰσοϋψῆ γάρ ἐστι τὰ ΑΒ, ΓΦ στερεά· ὡξ δὲ ἡ ΓΜ πρὸς τὴν ΓΤ, οὑτως ἥ τε ΜΠ βάσις πρὸς τὴν ΠΤ βάσιν καὶ τὸ ΓΔ στερεὸν πρὸς τὸ ΓΦ στερεόν. καὶ ὡξ ἄρα τὸ ΑΒ στερεὸν πρὸς τὸ ΓΦ στερεόν, οὑτως τὸ ΓΔ στερεὸν πρὸς τὸ ΓΦ στερεόν· ἑκάτερον ἄρα τῶν ΑΒ, ΓΔ πρὸς τὸ ΓΦ τὸν αὐτὸν ἔχει λόγον. ἴσον ἄρα ἐστὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ.

EH be to base NQ, as the height of solid CD (is) to the height of solid AB. I say that solid AB is equal to solid CD. [For] let the (straight-lines) standing up again be at right-angles to the bases. And if base EH is equal to base NQ, and as base EH is to base NQ, so the height of solid CD (is) to the height of solid AB, the height of solid CD is thus also equal to the height of solid AB. And parallelepiped solids on equal bases, and also with the same height, are equal to one another [Prop. 11.31]. Thus, solid AB is equal to solid CD.

So, let base EH not be equal to [base] NQ, but let EH be greater. Thus, the height of solid CD is also greater than the height of solid AB, that is to say CM (greater) than AG. Let CT again be made equal to AG, and let the solid CV be similarly completed. Since as base EH is to base NQ, so MC (is) to AG, and AG (is) equal to CT, thus as base EH (is) to base NQ, so CM (is) to CT. But, as [base] EH (is) to base NQ, so solid AB (is) to solid CV. For solids AB and CV are of equal heights [Prop. 11.32]. And as CM (is) to CT, so (is) base MQ to base QT [Prop. 6.1], and solid CD to solid CV [Prop. 11.25]. And thus as solid AB (is) to solid CV, so solid CD (is) to solid CV. Thus, AB and CD each have the same ratio to CV. Thus, solid AB is equal to solid CD [Prop. 5.9].



μὴ ὀέστωσαν δὴ αἱ ἐφεστηκυῖαι αἱ ΖΕ, ΒΛ, ΗΑ, ΚΘ, ΧΝ, ΔΟ, ΜΓ, ΡΠ πρὸς ὀρθὰξ ταῖς βάσεσιν αὐτῶν, καὶ ἤθθωσαν ἀπὸ τῶν Ζ, Η, Β, Κ, Χ, Μ, Ρ, Δ σημείων ἐπὶ τὰ διὰ τῶν ΕΘ, ΝΠ ἐπίπεδα κάθετοι καὶ συμβαλλέτωσαν τοῖς ἐπιπέδοις κατὰ τὰ Σ, Τ, Υ, Φ, Θ, Ψ, Ω, ξ, καὶ συμπεπληρώσθω τὰ ΖΦ, ΧΩ στερεά· λέγω, ὅτι καὶ οὑτωςὁίσων ὄντων τῶν ΑΒ, ΓΔ στερεῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν, καί ἐστιν ὡξ ἡ ΕΘ βᾶσιν πρὸς τὴν ΝΠ βάσιν, οὑτως τὸ τοῦ ΓΔ στερεοῦ ὑψος πρὸς τὸ τοῦ ΑΒ στερεοῦ ὑψος.

Ἐπεὶ ἴσον ἐστὶ τὸ ΑΒ στερεὸν τῷ ΓΔ στερεῷ, ἀλλὰ τὸ μὲν ΑΒ τῷ ΒΤ ἐστιν ἴσον· ἐπί τε γὰρ τῆς αὐτῆς βάσεώξ εἰσι τῆς ΖΚ καὶ ὑπὸ τὸ αὐτὸ ὑψος· τὸ δὲ ΓΔ στερεὸν τῷ ΔΨ ἐστιν

So, let the (straight-lines) standing up, FE, BL, GA, KH, ON, DP, MC, and RQ, not be at right-angles to their bases. And let perpendiculars be drawn to the planes through EH and NQ from points F, G, B, K, O, M, R, and D, and let them have joined the planes at (points) S, T, U, V, W, X, Y, and a (respectively). And let the solids FV and OY be completed. In this case, also, I say that the solids AB and CD being equal, their bases are reciprocally proportional to their heights, and (so) as base EH is to base NQ, so the height of solid CD (is) to the height of solid AB.

Since solid AB is equal to solid CD, but AB is equal to BT. For they are on the same base FK, and (have) the same

ἴσον· ἐπί τε γὰρ πάλιν τῆς αὐτῆς βάσεώξ εἰσι τῆς PX καὶ ὑπὸ τὸ αὐτὸ ὕψος· καὶ τὸ BT ἄρα στερεὸν τῷ $\Delta\Psi$ στερεῷ ἴσον ἐστίν. ὀέστιν ἄρα ὡξ ἡ ZK βάσις πρὸς τὴν XP βάσιν, οὕτως τὸ τοῦ $\Delta\Psi$ στερεοῦ ὕψος πρὸς τὸ τοῦ BT στερεοῦ ὕψος. ὁίση δὲ ἡ μὲν ZK βάσις τῇ $E\Theta$ βάσει, ἡ δὲ XP βάσις τῇ $N\Pi$ βάσει· ὀέστιν ἄρα ὡξ ἡ $E\Theta$ βάσις πρὸς τὴν $N\Pi$ βάσιν, οὕτως τὸ τοῦ $\Delta\Psi$ στερεοῦ ὕψος πρὸς τὸ τοῦ BT στερεοῦ ὕψος. τὰ δὸ αὐτὰ ὕψη ἐστὶ τῶν $\Delta\Psi$, BT στερεῶν καὶ τῶν $\Delta\Gamma$, BA· ὀέστιν ἄρα ὡξ ἡ $E\Theta$ βάσις πρὸς τὴν $N\Pi$ βάσιν, οὕτως τὸ τοῦ $\Delta\Gamma$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος. τῶν AB, $\Gamma\Delta$ ἄρα στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

Πάλιν δὴ τῶν AB, $\Gamma\Delta$ στερεῶν παραλληλεπιπέδων ἀντιπεπονθέτωσαν αἱ βάσεις τοῖς ὕψεσιν, καὶ ὀέστω ὡξ ἡ $E\Theta$ βάσις πρὸς τὴν $N\Pi$ βάσιν, οὕτως τὸ τοῦ $\Gamma\Delta$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος· λέγω, ὅτι ἴσον ἐστὶ τὸ AB στερεὸν τῷ $\Gamma\Delta$ στερεῷ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεί ἐστιν ὡξ ἡ $E\Theta$ βάσις πρὸς τὴν $N\Pi$ βάσιν, οὕτως τὸ τοῦ $\Gamma\Delta$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος, ὁίση δὲ ἡ μὲν $E\Theta$ βάσις τῇ ZK βάσει, ἡ δὲ $N\Pi$ τῇ XP, ὀέστιν ἄρα ὡξ ἡ ZK βάσις πρὸς τὴν XP βάσιν, οὕτως τὸ τοῦ $\Gamma\Delta$ στερεοῦ ὕψος πρὸς τὸ τοῦ AB στερεοῦ ὕψος. τὰ δὸ αὐτὰ ὕψη ἐστὶ τῶν AB, $\Gamma\Delta$ στερεῶν καὶ τῶν BT, $\Delta\Psi$· ὀέστιν ἄρα ὡξ ἡ ZK βάσις πρὸς τὴν XP βάσιν, οὕτως τὸ τοῦ $\Delta\Psi$ στερεοῦ ὕψος πρὸς τὸ τοῦ BT στερεοῦ ὕψος. τῶν BT, $\Delta\Psi$ ἄρα στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν· ἴσον ἄρα ἐστὶ τὸ BT στερεὸν τῷ $\Delta\Psi$ στερεῷ. ἀλλὰ τὸ μὲν BT τῷ BA ἴσον ἐστίν· ἐπί τε γὰρ τῆς αὐτῆς βάσεως [εἰσι] τῆς ZK καὶ ὑπὸ τὸ αὐτὸ ὕψος. τὸ δὲ $\Delta\Psi$ στερεὸν τῷ $\Delta\Gamma$ στερεῷ ἴσον ἐστίν. καὶ τὸ AB ἄρα στερεὸν τῷ $\Gamma\Delta$ στερεῷ ἐστιν ἴσον· ὅπερ ἔδει δεῖχαι.

height [Props. 11.29, 11.30]. And solid CD is equal is equal to DX. For, again, they are on the same base RO, and (have) the same height [Props. 11.29, 11.30]. Solid BT is thus also equal to solid DX. Thus, as base FK (is) to base OR, so the height of solid DX (is) to the height of solid BT (see first part of proposition). And base FK (is) equal to base EH, and base OR to NQ. Thus, as base EH is to base NQ, so the height of solid DX (is) to the height of solid BT. And solids DX, BT are the same height as (solids) DC, BA (respectively). Thus, as base EH is to base NQ, so the height of solid DC (is) to the height of solid AB. Thus, the bases of the parallelepiped solids AB and CD are reciprocally proportional to their heights.

So, again, let the bases of the parallelepiped solids AB and CD be reciprocally proportional to their heights, and (so) let base EH be to base NQ, as the height of solid CD (is) to the height of solid AB. I say that solid AB is equal to solid CD.

For, with the same construction (as before), since as base EH is to base NQ, so the height of solid CD (is) to the height of solid AB, and base EH (is) equal to base FK, and NQ to OR, thus as base FK is to base OR, so the height of solid CD (is) to the height of solid AB. And solids AB, CD are the same height as (solids) BT, DX (respectively). Thus, as base FK is to base OR, so the height of solid DX (is) to the height of solid BT. Thus, the bases of the parallelepiped solids BT and DX are reciprocally proportional to their heights. Thus, solid BT is equal to solid DX (see first part of proposition). But, BT is equal to BA. For [they are] on the same base FK, and (have) the same height [Props. 11.29, 11.30]. And solid DX is equal to solid DC [Props. 11.29, 11.30]. Thus, solid AB is also equal to solid CD. (Which is) the very thing it was required to show.

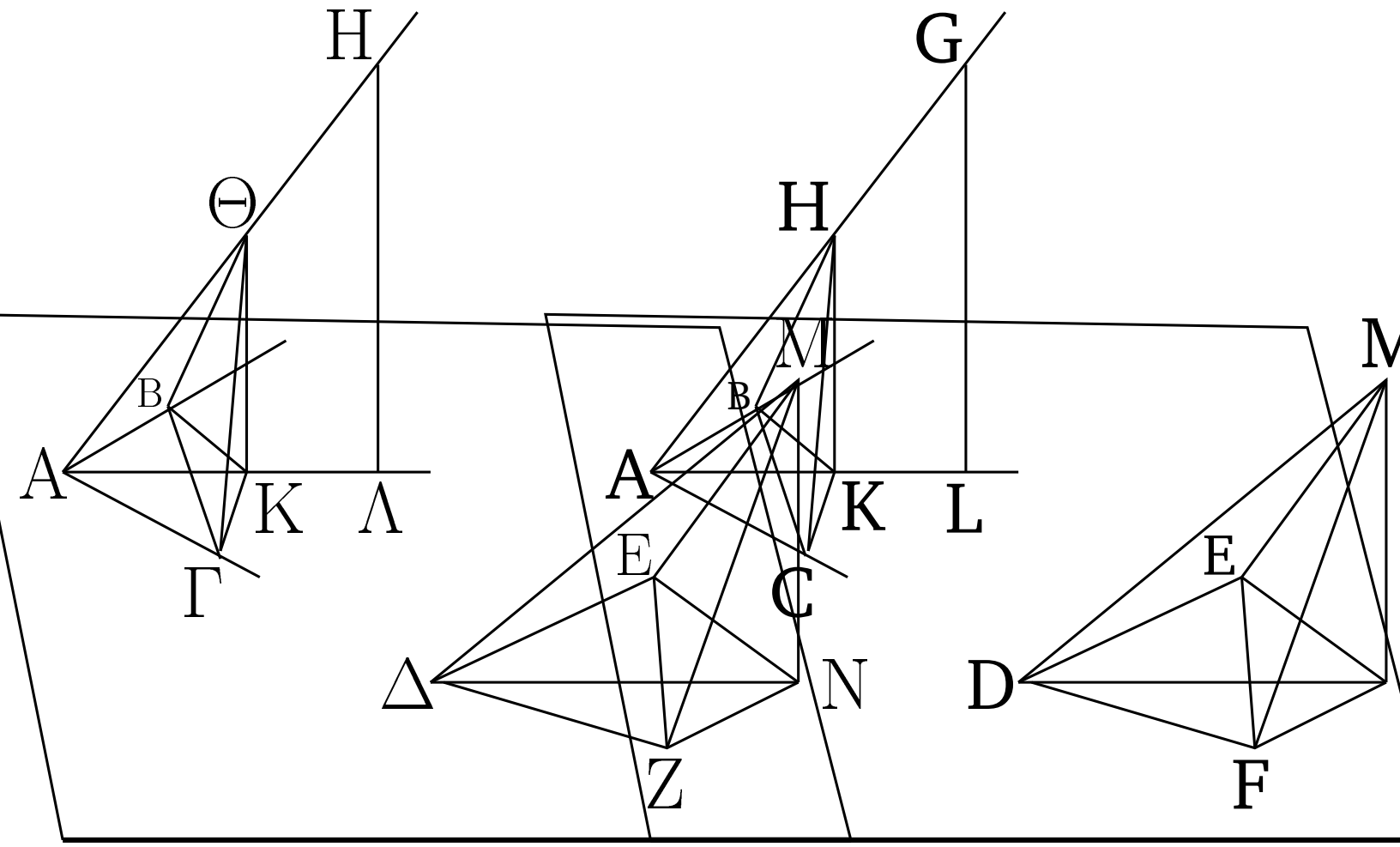
† This proposition assumes that (a) if two parallelepipeds are equal, and have equal bases, then their heights are equal, and (b) if the bases of two equal parallelepipeds are unequal, then that solid which has the lesser base has the greater height.

35.

Ἐὰνὸῶσι δύο γωνίαι ἐπίπεδοι ἴσαι, ἐπὶ δὲ τῶν κορυφῶν αὐτῶν μετέωροι εὐθεῖαι ἐπισταθῶσιν ἴσας γωνίας περιέθουσαι μετὰ τῶν ἐξ ἀρθῆς εὐθειῶν ἑκατέραν ἑκατέρᾳ, ἐπὶ δὲ τῶν μετεώρων ληφθῇ τυθόντα σημεῖα, καὶ ἀπὸ αὐτῶν ἐπὶ τὰ ἐπίπεδα, ἐν οὶῖξ εἰσιν αἱ ἐξ ἀρθῆς γωνίαι, κάθετοι ἀθθῶσιν, ἀπὸ δὲ τῶν γενομένων σημείων ἐν τοῖς ἐπιπέδοις ἐπὶ τὰξ ἐξ ἀρθῆς γωνίας ἐπιζευθθῶσιν εὐθεῖαι, ἴσας γωνίας περιέχουσι μετὰ τῶν μετεώρων.

Proposition 35

If there are two equal plane angles, and raised straight-lines are stood on the apexes of them, containing equal angles respectively with the original straight-lines (forming the angles), and random points are taken on the raised (straight-lines), and perpendiculars are drawn from them to the planes in which the original angles are, and straight-lines are joined from the points created in the planes to the (vertices of the) original angles, then they will enclose equal angles with the raised (straight-lines).



Ἔστωσαν δύο γωνίαι εὐθύγραμμοι ἴσαι αἱ ὑπὸ ΒΑΓ, ΕΔΖ, ἀπὸ δὲ τῶν Α, Δ σημείων μετέωροι εὐθεῖαι ἐφεστάτωσαν αἱ ΑΗ, ΔΜ ἴσας γωνίας περιέθουσιν μετὰ τῶν ἐξ ἀρθῆς εὐθειῶν ἑκατέραν ἑκατέρᾳ, τὴν μὲν ὑπὸ ΜΔΕ τῇ ὑπὸ ΗΑΒ, τὴν δὲ ὑπὸ ΜΔΖ τῇ ὑπὸ ΗΑΓ, καὶ εἰλήφθω ἐπὶ τῶν ΑΗ, ΔΜ τυθόντα σημεῖα τὰ Η, Μ, καὶ ἤχθωσαν ἀπὸ τῶν Η, Μ σημείων ἐπὶ τὰ διὰ τῶν ΒΑΓ, ΕΔΖ ἐπίπεδα κάθετοι αἱ ΗΛ, ΜΝ, καὶ συμβαλλέτωσαν τοῖς ἐπιπέδοις κατὰ τὰ Λ, Ν, καὶ ἐπεζεύθθωσαν αἱ ΛΑ, ΝΔ· λέγω, ὅτιὁίση ἐστὶν ἡ ὑπὸ ΗΑΛ γωνία τῇ ὑπὸ ΜΔΝ γωνίᾳ.

Κείσθω τῇ ΔΜὁίση ἡ ΑΘ, καὶ ἤχθω διὰ τοῦ Θ σημείου τῇ ΗΛ παράλληλος ἡ ΘΚ. ἡ δὲ ΗΛ κάθετόξ ἐστιν ἐπὶ τὸ διὰ τῶν ΒΑΓ ἐπίπεδον· καὶ ἡ ΘΚ ἄρα κάθετόξ ἐστιν ἐπὶ τὸ διὰ τῶν ΒΑΓ ἐπίπεδον. ἤχθωσαν ἀπὸ τῶν Κ, Ν σημείων ἐπὶ τὰξ ΑΓ, ΔΖ, ΑΒ, ΔΕ εὐθείας κάθετοι αἱ ΚΓ, ΝΖ, ΚΒ, ΝΕ, καὶ ἐπεζεύθθωσαν αἱ ΘΓ, ΓΒ, ΜΖ, ΖΕ. ἐπεὶ τὸ ἀπὸ τῆς ΘΑ ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΘΚ, ΚΑ, τῷ δὲ ἀπὸ τῆς ΚΑ ἴσα ἐστὶ τὰ ἀπὸ τῶν ΚΓ, ΓΑ, καὶ τὸ ἀπὸ τῆς ΘΑ ἄρα ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΘΚ, ΚΓ, ΓΑ. τοῖς δὲ ἀπὸ τῶν ΘΚ, ΚΓ ἴσον ἐστὶ τὸ ἀπὸ τῆς ΘΓ· τὸ ἄρα ἀπὸ τῆς ΘΑ ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΘΓ, ΓΑ.

Let *BAC* and *EDF* be two equal rectilinear angles. And let the raised straight-lines *AG* and *DM* be stood on points *A* and *D*, containing equal angles respectively with the original straight-lines. (That is) *MDE* (equal) to *GAB*, and *MDF* (to) *GAC*. And let the random points *G* and *M* be taken on *AG* and *DM* (respectively). And let the *GL* and *MN* be drawn from points *G* and *M* perpendicular to the planes through *BAC* and *EDF* (respectively). And let them have joined the planes at points *L* and *N* (respectively). And let *LA* and *ND* be joined. I say that angle *GAL* is equal to angle *MDN*.

Let *AH* be made equal to *DM*. And let *HK* be drawn through point *H* parallel to *GL*. And *GL* is perpendicular to the plane through *BAC*. Thus, *HK* is also perpendicular to the plane through *BAC* [Prop. 11.8]. And let *KC*, *NF*, *KB*, and *NE* be drawn from points *K* and *N* perpendicular to the straight-lines *AC*, *DF*, *AB*, and *DE*. And let *HC*, *CB*, *MF*, and *FE* be joined. Since the (square) on *HA* is equal to the (sum of the squares) on *HK* and *KA* [Prop. 1.47], and the (sum of the squares) on *KC* and *CA* is equal to the (square) on *KA* [Prop. 1.47], thus the (square) on *HA* is equal to the (sum of

ὀρθὴ ἄρα ἐστὶν ἡ ὑπὸ ΘΓΑ γωνία. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΔΖΜ γωνία ὀρθή ἐστιν. ὁίση ἄρα ἐστὶν ἡ ὑπὸ ΑΓΘ γωνία τῇ ὑπὸ ΔΖΜ. ὀέστι δὲ καὶ ἡ ὑπὸ ΘΑΓ τῇ ὑπὸ ΜΔΖὁίση. δύο δὴ τρίγωνά ἐστι τὰ ΜΔΖ, ΘΑΓ δύο γωνίας δυσὶ γωνίαις ἴσας ἔχοντα ἑκατέραν ἑκατέρᾳ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην τὴν ὑποτείνουσαν ὑπὸ μίαν τῶνὁίσων γωνιῶν τὴν ΘΑ τῇ ΜΔ· καὶ τὰξ λοιπὰξ ἄρα πλευρὰξ ταῖς λοιπαῖξ πλευραῖξ ἴσας ὁέχει ἑκατέραν ἑκαρέρᾳ. ὁίση ἄρα ἐστὶν ἡ ΑΓ τῇ ΔΖ. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ ΑΒ τῇ ΔΕ ἐστινὁίση. ἐπεὶ οὁῦνὁίση ἐστὶν ἡ μὲν ΑΓ τῇ ΔΖ, ἡ δὲ ΑΒ τῇ ΔΕ, δύο δὴ αἱ ΓΑ, ΑΒ δυσὶ ταῖς ΖΔ, ΔΕ ἴσαι εἰσίν. ἀλλὰ καὶ γωνία ἡ ὑπὸ ΓΑΒ γωνίᾳ τῇ ὑπὸ ΖΔΕ ἐστινὁίση· βάσις ἄρα ἡ ΒΓ βάσει τῇ ΕΖὁίση ἐστὶ καὶ τὸ τρίγωνον τῷ τριγώνῳ καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖξ γωνίαις· ὁίση ἄρα ἡ ὑπὸ ΑΓΒ γωνία τῇ ὑπὸ ΔΖΕ. ὀέστι δὲ καὶ ὀρθὴ ἡ ὑπὸ ΑΓΚ ὀρθῇ τῇ ὑπὸ ΔΖΝὁίση· καὶ λοιπὴ ἄρα ἡ ὑπὸ ΒΓΚ λοιπῇ τῇ ὑπὸ ΕΖΝ ἐστινὁίση. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΓΒΚ τῇ ὑπὸ ΖΕΝ ἐστινὁίση. δύο δὴ τρίγωνά ἐστι τὰ ΒΓΚ, ΕΖΝ [τὰξ] δύο γωνίας δυσὶ γωνίαις ἴσας ἔχοντα ἑκατέραν ἑκατέρᾳ καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην τὴν πρὸς ταῖς ἴσαις γωνίαις τὴν ΒΓ τῇ ΕΖ· καὶ τὰξ λοιπὰξ ἄρα πλευρὰξ ταῖς λοιπαῖξ πλευραῖξ ἴσας ὁέχουσιν. ὁίση ἄρα ἐστὶν ἡ ΓΚ τῇ ΖΝ. ὀέστι δὲ καὶ ἡ ΑΓ τῇ ΔΖὁίση· δύο δὴ αἱ ΑΓ, ΓΚ δυσὶ ταῖς ΔΖ, ΖΝ ἴσαι εἰσίν· καὶ ὀρθὰξ γωνίας περιέθουσιν. βάσις ἄρα ἡ ΑΚ βάσει τῇ ΔΝὁίση ἐστίν. καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΑΘ τῇ ΔΜ, ἴσον ἐστὶ καὶ τὸ ἀπὸ τῆς ΑΘ τῷ ἀπὸ τῆς ΔΜ. ἀλλὰ τῷ μὲν ἀπὸ τῆς ΑΘ ἴσα ἐστὶ τὰ ἀπὸ τῶν ΑΚ, ΚΘ· ὀρθὴ γὰρ ἡ ὑπὸ ΑΚΘ· τῷ δὲ ἀπὸ τῆς ΔΜ ἴσα τὰ ἀπὸ τῶν ΔΝ, ΝΜ· ὀρθὴ γὰρ ἡ ὑπὸ ΔΝΜ· τὰ ἄρα ἀπὸ τῶν ΑΚ, ΚΘ ἴσα ἐστὶ τοῖς ἀπὸ τῶν ΔΝ, ΝΜ, ὁῶν τὸ ἀπὸ τῆς ΑΚ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΔΝ· λοιπὸν ἄρα τὸ ἀπὸ τῆς ΚΘ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΝΜ· ὁίση ἄρα ἡ ΘΚ τῇ ΜΝ. καὶ ἐπεὶ δύο αἱ ΘΑ, ΑΚ δυσὶ ταῖς ΜΔ, ΔΝ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, καὶ βάσις ἡ ΘΚ βάσει τῇ ΜΝ ἐδείθηὁίση, γωνία ἄρα ἡ ὑπὸ ΘΑΚ γωνίᾳ τῇ ὑπὸ ΜΔΝ ἐστινὁίση.

Ἐὰν ἄραὁῶσι δύο γωνίαι ἐπίπεδοι ἴσαι καὶ τὰ ἑχῆς τῆς προτάσεως [ὅπερ ἔδει δεῖχαι].

the squares) on HK, KC, and CA. And the (square) on HC is equal to the (sum of the squares) on HK and KC [Prop. 1.47]. Thus, the (square) on HA is equal to the (sum of the squares) on HC and CA. Thus, angle HCA is a right-angle [Prop. 1.48]. So, for the same (reasons), angle DFM is also a right-angle. Thus, angle ACH is equal to (angle) DFM. And HAC is also equal to MDF. So, MDF and HAC are two triangles having two angles equal to two angles, respectively, and one side equal to one side—(namely), that subtending one of the equal angles —(that is), HA (equal) to MD. Thus, they will also have the remaining sides equal to the remaining sides, respectively [Prop. 1.26]. Thus, AC is equal to DF. So, similarly, we can show that AB is also equal to DE. Therefore, since AC is equal to DF, and AB to DE, so the two (straight-lines) CA and AB are equal to the two (straight-lines) FD and DE (respectively). But, angle CAB is also equal to angle FDE. Thus, base BC is equal to base EF, and triangle (ACB) to triangle (DFE), and the remaining angles to the remaining angles (respectively) [Prop. 1.4]. Thus, angle ACB (is) equal to DFE. And the right-angle ACK is also equal to the right-angle DFN. Thus, the remainder BCK is equal to the remainder EFN. So, for the same (reasons), CBK is also equal to FEN. So, BCK and EFN are two triangles having two angles equal to two angles, respectively, and one side equal to one side—(namely), that by the equal angles—(that is), BC (equal) to EF. Thus, they will also have the remaining sides equal to the remaining sides (respectively) [Prop. 1.26]. Thus, CK is equal to FN. And AC (is) also equal to DF. So, the two (straight-lines) AC and CK are equal to the two (straight-lines) DF and FN (respectively). And they enclose right-angles. Thus, base AK is equal to base DN [Prop. 1.4]. And since AH is equal to DM, the (square) on AH is also equal to the (square) on DM. But, the the (sum of the squares) on AK and KH is equal to the (square) on AH. For angle AKH (is) a right-angle [Prop. 1.47]. And the (sum of the squares) on DN and NM (is) equal to the square on DM. For angle DNM (is) a right-angle [Prop. 1.47]. Thus, the (sum of the squares) on AK and KH is equal to the (sum of the squares) on DN and NM, of which the (square) on AK is equal to the (square) on DN. Thus, the remaining (square) on KH is equal to the (square) on NM. Thus, HK (is) equal to MN. And since the two (straight-lines) HA and AK are equal to the two (straight-lines) MD and DN, respectively, and base HK was shown (to be) equal to base MN, angle HAK is thus equal to angle MDN [Prop. 1.8].

Thus, if there are two equal plane angles, and so on of the proposition. [(Which is) the very thing it was required to show].

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι, ἐανὁῶσι δύο γωνίαι ἐπίπε-

Corollary

So, it is clear, from this, that if there are two equal plane an-

δοι ἴσαι, ἐπισταθῶσι δὲ ἐπ᾽ αὐτῶν μετέωροι εὐθεῖαι ἴσαι ἴσας γωνίας περιέθουσαι μετὰ τῶν ἐξ ἀρθῆς εὐθειῶν ἑκατέραν ἑκατέρᾳ, αἱ ἀπὸ αὐτῶν κάθετοι ἀγόμεναι ἐπὶ τὰ ἐπίπεδα, ἐν οἷς εἰσιν αἱ ἐξ ἀρθῆς γωνίαι, ἴσαι ἀλλήλαις εἰσίν. ὅπερ ἔδει δεῖξαι.

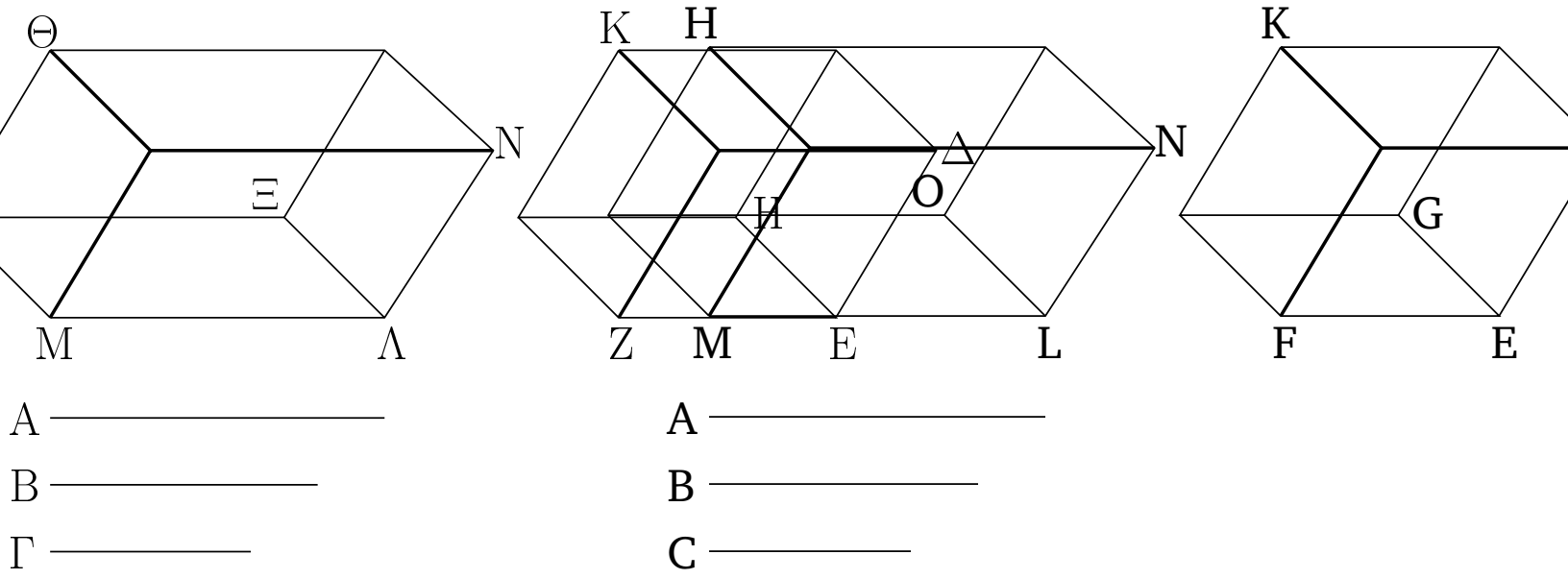
gles, and equal raised straight-lines are stood on them (at their apexes), containing equal angles respectively with the original straight-lines (forming the angles), then the perpendiculars drawn from (the raised ends of) them to the planes in which the original angles lie are equal to one another. (Which is) the very thing it was required to show.

36.

Ἐὰν τρεῖς εὐθεῖαι ἀνάλογον ὦσιν, τὸ ἐκ τῶν τριῶν στερεὸν παραλληλεπίπεδον ἴσον ἐστὶ τῷ ἀπὸ τῆς μέσης στερεῷ παραλληλεπιπέδῳ ἰσοπλεύρῳ μέν, ἰσογωνίῳ δὲ τῷ προειρημένῳ.

Proposition 36

If three straight-lines are (continuously) proportional then the parallelepiped solid (formed) from the three (straight-lines) is equal to the equilateral parallelepiped solid on the middle (straight-line which is) equiangular to the aforementioned (parallelepiped solid).



Ἔστωσαν τρεῖς εὐθεῖαι ἀνάλογον αἱ Α, Β, Γ, ὡς ἡ Α πρὸς τὴν Β, οὕτως ἡ Β πρὸς τὴν Γ· λέγω, ὅτι τὸ ἐκ τῶν Α, Β, Γ στερεὸν ἴσον ἐστὶ τῷ ἀπὸ τῆς Β στερεῷ ἰσοπλεύρῳ μέν, ἰσογωνίῳ δὲ τῷ προειρημένῳ.

Ἐκκείσθω στερεὰ γωνία ἡ πρὸς τῷ Ε περιεθομένη ὑπὸ τῶν ὑπὸ ΔΕΗ, ΗΕΖ, ΖΕΔ, καὶ κείσθω τῇ μὲν Βὁίση ἑκάστη τῶν ΔΕ, ΗΕ, ΕΖ, καὶ συμπεπληρώσθω τὸ ΕΚ στερεὸν παραλληλεπίπεδον, τῇ δὲ Αὁίση ἡ ΛΜ, καὶ συνεστάτω πρὸς τῇ ΛΜ εὐθείᾳ καὶ τῷ πρὸς αὐτῇ σημείῳ τῷ Λ τῇ πρὸς τῷ Ε στερεᾷ γωνίᾳ ὁίση στερεὰ γωνία ἡ περιεθομένη ὑπὸ τῶν ΝΛΧ, ΧΛΜ, ΜΛΝ, καὶ κείσθω τῇ μὲν Βὁίση ἡ ΛΧ, τῇ δὲ Γὁίση ἡ ΛΝ. καὶ ἐπεί ἐστιν ὡς ἡ Α πρὸς τὴν Β, οὕτως ἡ Β πρὸς τὴν Γ, ὁίση δὲ ἡ μὲν Α τῇ ΛΜ, ἡ δὲ Β ἑκατέρᾳ τῶν ΛΧ, ΕΔ, ἡ δὲ Γ τῇ ΛΝ, ὲστιν ἄρα ὡς ἡ ΛΜ πρὸς τὴν ΕΖ, οὕτως ἡ ΔΕ πρὸς τὴν ΛΝ. καὶ περὶ ἴσας γωνίας τὰξ ὑπὸ ΝΛΜ, ΔΕΖ αἱ πλευραὶ ἀντιπεπόνθασιν· ἴσον ἄρα ἐστὶ τὸ ΜΝ παραλληλόγραμμον τῷ ΔΖ παραλληλογραμμῳ. καὶ ἐπεὶ δύο γωνίαι ἐπίπεδοι εὐθύγραμμοι ἴσαι εἰσὶν αἱ ὑπὸ ΔΕΖ, ΝΛΜ,

Let *A*, *B*, and *C* be three (continuously) proportional straight-lines, (such that) as *A* (is) to *B*, so *B* (is) to *C*. I say that the (parallelepiped) solid (formed) from *A*, *B*, and *C* is equal to the equilateral solid on *B* (which is) equiangular with the aforementioned (solid).

Let the solid angle at *E*, contained by *DEG*, *GEF*, and *FED*, be set out. And let *DE*, *GE*, and *EF* each be made equal to *B*. And let the parallelepiped solid *EK* be completed. And (let) *LM* (be made) equal to *A*. And let the solid angle contained by *NLO*, *OLM*, and *MLN* be constructed on the straight-line *LM*, and at the point *L* on it, (so as to be) equal to the solid angle *E* [Prop. 11.23]. And let *LO* be made equal to *B*, and *LN* equal to *C*. And since as *A* (is) to *B*, so *B* (is) to *C*, and *A* (is) equal to *LM*, and *B* to each of *LO* and *ED*, and *C* to *LN*, thus as *LM* (is) to *EF*, so *DE* (is) to *LN*. And (so) the sides around the equal angles *NLM* and *DEF* are reciprocally proportional. Thus, parallelogram *MN* is equal to parallelogram *DF* [Prop. 6.14]. And since the two plane rectilinear

καὶ ἐπ΄ αὐτῶν μετέωροι εὐθεῖαι ἐφεστᾶσιν αἱ ΛΧ, ΕΗ ἴσαι τε ἀλλήλαις καὶ ἴσας γωνίας περιέθουσαι μετὰ τῶν ἐξ ἀρθῆς εὐθειῶν ἑκατέραν ἑκατέρᾳ, αἱ ἄρα ἀπὸ τῶν Η, Χ σημείων κάθετοι ἀγόμεναι ἐπὶ τὰ διὰ τῶν ΝΛΜ, ΔΕΖ ἐπίπεδα ἴσαι ἀλλήλαις εἰσίν· ὥστε τὰ ΛΘ, ΕΚ στερεὰ ὑπὸ τὸ αὐτὸ ὕψος ἐστίν. τὰ δὲ ἐπὶ ὁίσων βάσεων στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὕψος ἴσα ἀλλήλοις ἐστίν· ἴσον ἄρα ἐστὶ τὸ ΘΛ στερεὸν τῷ ΕΚ στερεῷ. καί ἐστι τὸ μὲν ΛΘ τὸ ἐκ τῶν Α, Β, Γ στερεόν, τὸ δὲ ΕΚ τὸ ἀπὸ τῆς Β στερεόν· τὸ ἄρα ἐκ τῶν Α, Β, Γ στερεὸν παραλληλεπίπεδον ἴσον ἐστὶ τῷ ἀπὸ τῆς Β στερεῷ ἰσοπλεύρῳ μέν, ἰσογωνίῳ δὲ τῷ προειρημένῳ· ὅπερ ἔδει δεῖξαι.

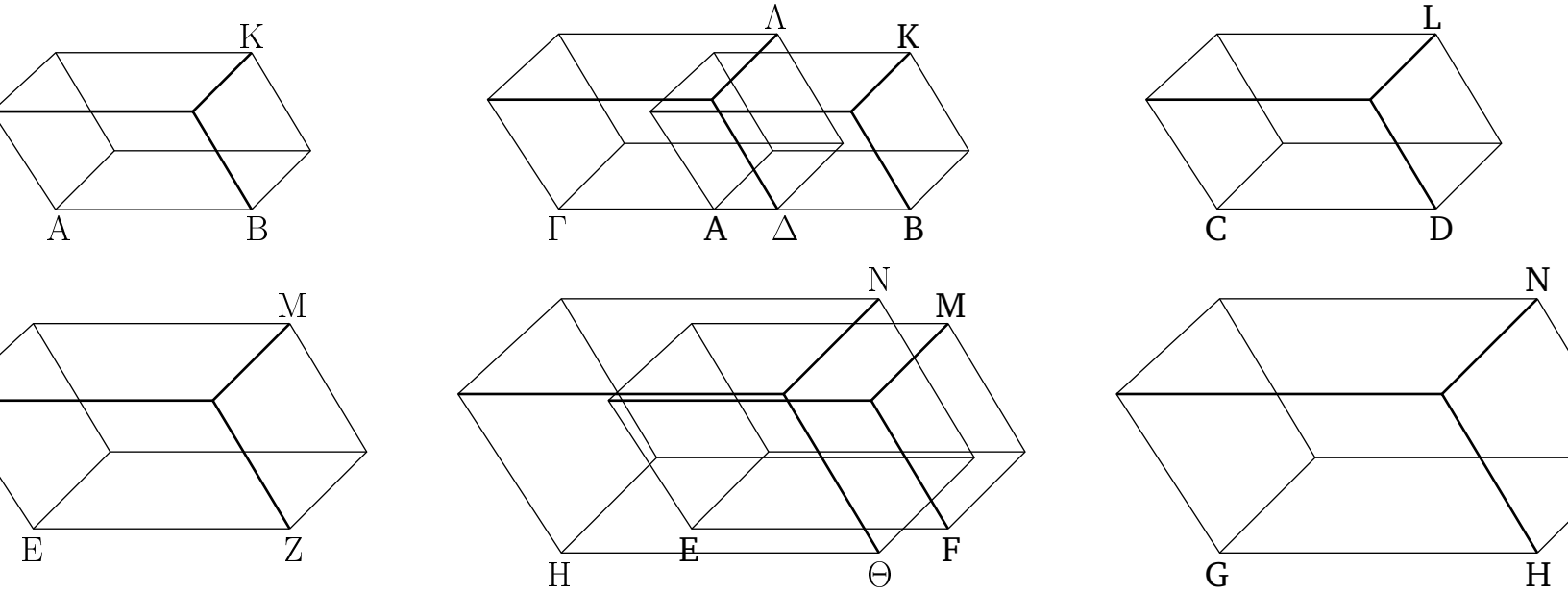
angles *DEF* and *NLM* are equal, and the raised straight-lines stood on them (at their apexes), *LO* and *EG*, are equal to one another, and contain equal angles respectively with the original straight-lines (forming the angles), the perpendiculars drawn from points *G* and *O* to the planes through *NLM* and *DEF* (respectively) are thus equal to one another [Prop. 11.35 corr.]. Thus, the solids *LH* and *EK* (have) the same height. And parallelepiped solids on equal bases, and with the same height, are equal to one another [Prop. 11.31]. Thus, solid *HL* is equal to solid *EK*. And *LH* is the solid (formed) from *A*, *B*, and *C*, and *EK* the solid on *B*. Thus, the parallelepiped solid (formed) from *A*, *B*, and *C* is equal to the equilateral solid on *B* (which is) equiangular with the aforementioned (solid). (Which is) the very thing it was required to show.

37.

Ἐὰν τέσσαρες εὐθεῖαι ἀνάλογον ὦσιν, καὶ τὰ ἀπὸ αὐτῶν στερεὰ παραλληλεπίπεδα ὅμοιά τε καὶ ὁμοίως ἀναγραφόμενα ἀνάλογονὸέσται· καὶ ἐαν τὰ ἀπὸ αὐτῶν στερεὰ παραλληλεπίπεδα ὅμοιά τε καὶ ὁμοίως ἀναγραφόμενα ἀνάλογονὸῇ, καὶ αὐταὶ αἱ εὐθεῖαι ἀνάλογονὸέσονται.

Proposition 37†

If four straight-lines are proportional then the similar, and similarly described, parallelepiped solids on them will also be proportional. And if the similar, and similarly described, parallelepiped solids on them are proportional then the straight-lines themselves will be proportional.



Ἔστωσαν τέσσαρες εὐθεῖαι ἀνάλογον αἱ ΑΒ, ΓΔ, ΕΖ, ΗΘ, ὡς ἡ ΑΒ πρὸς τὴν ΓΔ, οὕτως ἡ ΕΖ πρὸς τὴν ΗΘ, καὶ ἀναγεγράφθωσαν ἀπὸ τῶν ΑΒ, ΓΔ, ΕΖ, ΗΘ ὅμοιά τε καὶ ὁμοίως κείμενα στερεὰ παραλληλεπίπεδα τὰ ΚΑ, ΛΓ, ΜΕ, ΝΗ· λέγω, ὅτι ἐστὶν ὡς τὸ ΚΑ πρὸς τὸ ΛΓ, οὕτως τὸ ΜΕ πρὸς τὸ ΝΗ.

Ἐπεὶ γὰρ ὅμοιόν ἐστι τὸ ΚΑ στερεὸν παραλληλεπίπεδον τῷ ΛΓ, τὸ ΚΑ ἄρα πρὸς τὸ ΛΓ τριπλασίονα λόγον ἔχει ἤπερ ἡ ΑΒ πρὸς τὴν ΓΔ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΜΕ πρὸς τὸ ΝΗ

Let *AB*, *CD*, *EF*, and *GH*, be four proportional straight-lines, (such that) as *AB* (is) to *CD*, so *EF* (is) to *GH*. And let the similar, and similarly laid out, parallelepiped solids *KA*, *LC*, *ME* and *NG* be described on *AB*, *CD*, *EF*, and *GH* (respectively). I say that as *KA* is to *LC*, so *ME* (is) to *NG*.

For since the parallelepiped solid *KA* is similar to *LC*, *KA* thus has to *LC* the cubed ratio that *AB* (has) to *CD* [Prop. 11.33]. So, for the same (reasons), *ME* also has to *NG* the cubed ratio that *EF* (has) to *GH* [Prop. 11.33]. And since

τριπλασίονα λόγον ἔχει ἥπερ ἡ EZ πρὸς τὴν HΘ. καί ἐστιν ὡς ἡ AB πρὸς τὴν ΓΔ, οὕτως ἡ EZ πρὸς τὴν HΘ. καὶ ὡς ἄρα τὸ AK πρὸς τὸ ΛΓ, οὕτως τὸ ME πρὸς τὸ NH.

Ἀλλὰ δὴ ὀέστω ὡς τὸ AK στερεὸν πρὸς τὸ ΛΓ στερεόν, οὕτως τὸ ME στερεὸν πρὸς τὸ NH· λέγω, ὅτι ἐστὶν ὡς ἡ AB εὐθεῖα πρὸς τὴν ΓΔ, οὕτως ἡ EZ πρὸς τὴν HΘ.

Ἐπεὶ γὰρ πάλιν τὸ KA πρὸς τὸ ΛΓ τριπλασίονα λόγον ἔχει ἥπερ ἡ AB πρὸς τὴν ΓΔ, ἔχει δὲ καὶ τὸ ME πρὸς τὸ NH τριπλασίονα λόγον ἥπερ ἡ EZ πρὸς τὴν HΘ, καί ἐστιν ὡς τὸ KA πρὸς τὸ ΛΓ, οὕτως τὸ ME πρὸς τὸ NH, καὶ ὡς ἄρα ἡ AB πρὸς τὴν ΓΔ, οὕτως ἡ EZ πρὸς τὴν HΘ.

Ἐὰν ἄρα τέσσαρες εὐθεῖαι ἀνάλογονὸῶσι καὶ τὰ ἐχῆς τῆς προτάσεως· ὅπερ ἔδει δεῖχαι.

as *AB* is to *CD*, so *EF* (is) to *GH*, thus, also, as *AK* (is) to *LC*, so *ME* (is) to *NG*.

And so let solid *AK* be to solid *LC*, as solid *ME* (is) to *NG*. I say that as straight-line *AB* is to *CD*, so *EF* (is) to *GH*.

For, again, since *KA* has to *LC* the cubed ratio that *AB* (has) to *CD* [Prop. 11.33], and *ME* also has to *NG* the cubed ratio that *EF* (has) to *GH* [Prop. 11.33], and as *KA* is to *LC*, so *ME* (is) to *NG*, thus, also, as *AB* (is) to *CD*, so *EF* (is) to *GH*.

Thus, if four straight-lines are proportional, and so on of the proposition. (Which is) the very thing it was required to show.

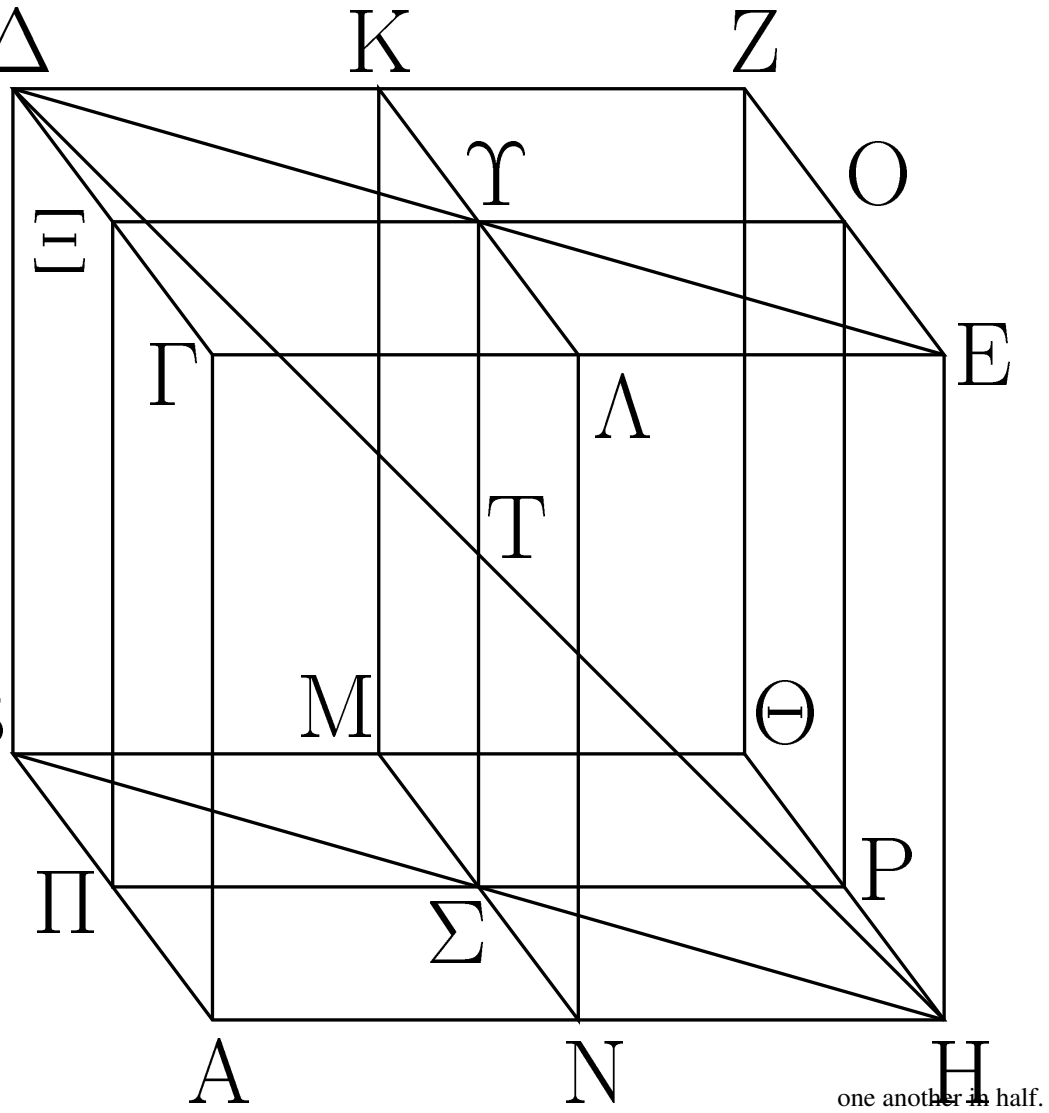
† This proposition assumes that if two ratios are equal then the cube of the former is also equal to the cube of the latter, and vice versa.

38.

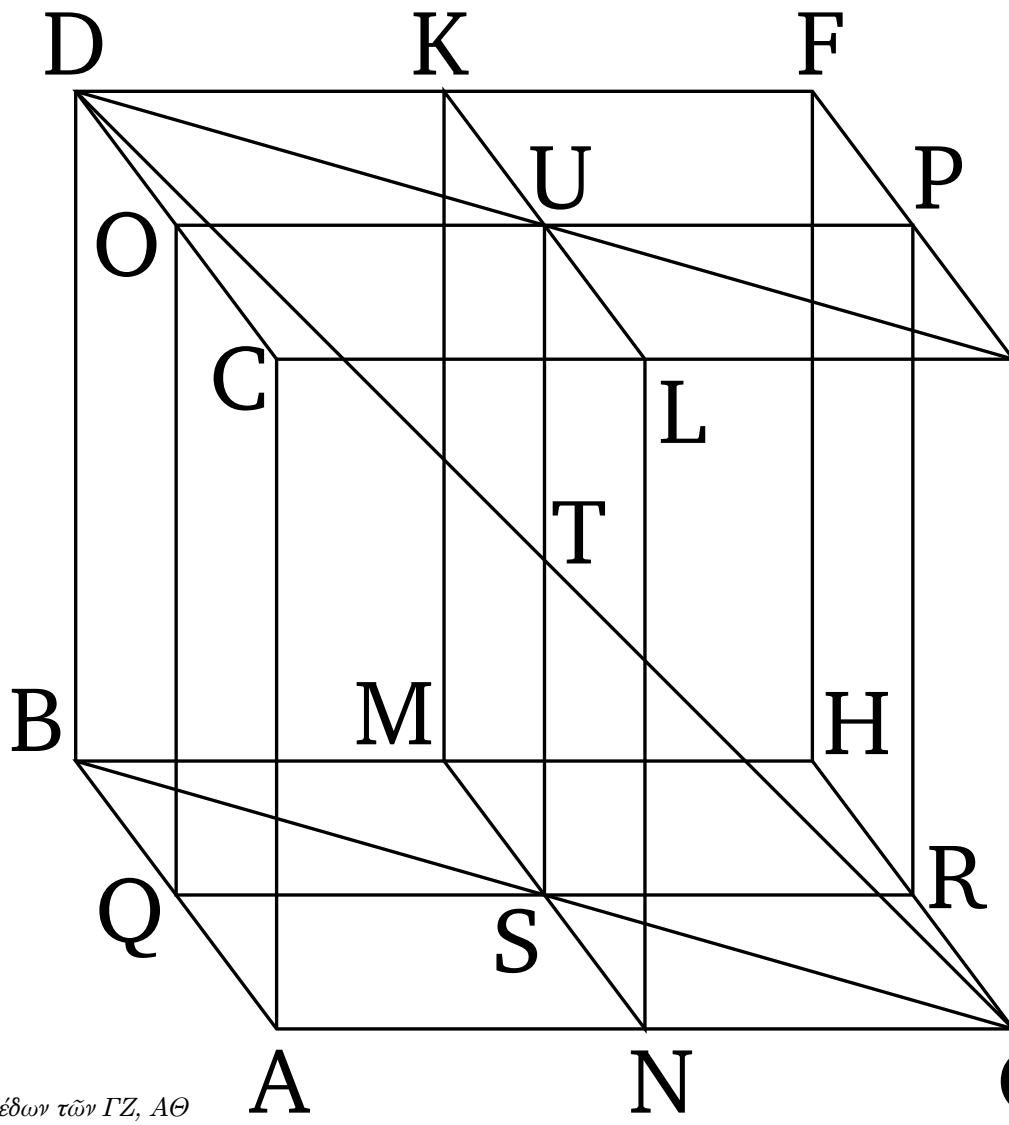
Ἐὰν κύβου τῶν ἀπεναντίον ἐπιπέδων αἱ πλευραὶ δίθα τμ ηθῶσιν, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβληθῇ, ἡ κοινὴ τομὴ τῶν ἐπιπέδων καὶ ἡ τοῦ κύβου διάμετρος δίθα τέμνουσιν ἀλλήλας.

Proposition 38

If the sides of the opposite planes of a cube are cut in half, and planes are produced through the pieces, then the common section of the (latter) planes and the diameter of the cube cut



one another in half.



Κύβου γὰρ τοῦ ΑΖ τῶν ἀπεναντίον ἐπιπέδων τῶν ΓΖ, ΑΘ αἱ πλευραὶ δίθα τετμήσθωσαν κατὰ τὰ Κ, Λ, Μ, Ν, Χ, Π, Ο, Ρ σημεῖα, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβεβλήσθω τὰ ΚΝ, ΧΡ, κοινὴ δὲ τομὴ τῶν ἐπιπέδωνὁέστω ἡ ΥΣ, τοῦ δὲ ΑΖ κύβου διαγώνιος ἡ ΔΗ. λέγω, ὅτιὁίση ἐστὶν ἡ μὲν ΥΤ τῇ ΤΣ, ἡ δὲ ΔΤ τῇ ΤΗ.

Ἐπεζεύθθωσαν γὰρ αἱ ΔΥ, ΥΕ, ΒΣ, ΣΗ. καὶ ἐπεὶ παράλληλόξ ἐστιν ἡ ΔΧ τῇ ΟΕ, αἱ ἐναλλὰχ γωνίαι αἱ ὑπὸ ΔΧΥ, ΥΟΕ ἴσαι ἀλλήλαις εἰσίν. καὶ ἐπεὶ ὁίση ἐστὶν ἡ μὲν ΔΧ τῇ ΟΕ, ἡ δὲ ΧΥ τῇ ΥΟ, καὶ γωνίας ἴσας περιέθουσιν, βάσις ἄρα ἡ ΔΥ τῇ ΥΕ ἐστινὁίση, καὶ τὸ ΔΧΥ τρίγωνον τῷ ΟΥΕ τριγώνῳ ἐστὶν ἴσον καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖξ γωνίαις ἴσαι· ὁίση ἄρα ἡ ὑπὸ ΧΥΔ γωνία τῇ ὑπὸ ΟΥΕ γωνίᾳ. διὰ δὴ τοῦτο εὐθεῖά ἐστιν ἡ ΔΥΕ. διὰ τὰ αὐτὰ δὴ καὶ ΒΣΗ εὐθεῖά ἐστιν, καὶ ὁίση ἡ ΒΣ τῇ ΣΗ. καὶ ἐπεὶ ἡ ΓΑ τῇ ΔΒὁίση ἐστὶ καὶ παράλληλος, ἀλλὰ ἡ ΓΑ καὶ τῇ ΕΗὁίση τέ ἐστι καὶ παράλληλος, καὶ ἡ ΔΒ ἄρα τῇ ΕΗὁίση τέ ἐστι καὶ παράλληλος.

For let the opposite planes CF and AH of the cube AF be cut in half at the points K, L, M, N, O, Q, P, and R. And let the planes KN and OR be produced through the pieces. And let US be the common section of the planes, and DG the diameter of cube AF. I say that UT is equal to TS, and DT to TG.

For let DU, UE, BS, and SG be joined. And since DO is parallel to PE, the alternate angles DOU and UPE are equal to one another [Prop. 1.29]. And since DO is equal to PE, and OU to UP, and they contain equal angles, base DU is thus equal to base UE, and triangle DOU is equal to triangle PUE, and the remaining angles (are) equal to the remaining angles [Prop. 1.4]. Thus, angle OUD (is) equal to angle PUE. So, for this (reason), DUE is a straight-line [Prop. 1.14]. So, for the same (reason), BSG is also a straight-line, and BS equal to SG. And since CA is equal and parallel to DB, but CA is also equal and parallel to EG, DB is thus also equal and parallel

καὶ ἐπιζευγνύουσιν αὐτὰξ εὐθεῖαι αἱ ΔΕ, ΒΗ· παράλληλος ἄρα ἐστὶν ἡ ΔΕ τῇ ΒΗ. ὁίση ἄρα ἡ μὲν ὑπὸ ΕΔΤ γωνία τῇ ὑπὸ ΒΗΤ· ἐναλλὰχ γάρ· ἡ δὲ ὑπὸ ΔΤΥ τῇ ὑπὸ ΗΤΣ. δύο δὴ τρίγωνά ἐστι τὰ ΔΤΥ, ΗΤΣ τὰξ δύο γωνίας ταῖς δυσὶ γωνίαις ἴσας ἔχοντα καὶ μίαν πλευρὰν μιᾷ πλευρᾷ ἴσην τὴν ὑποτείνουσαν ὑπὸ μίαν τῶνὁίσων γωνιῶν τὴν ΔΥ τῇ ΗΣ· ἡμίσειαι γάρ εἰσι τῶν ΔΕ, ΒΗ· καὶ τὰξ λοιπὰξ πλευρὰξ ταῖς λοιπαῖξ πλευραῖξ ἴσας ὁέχει. ὁίση ἄρα ἡ μὲν ΔΤ τῇ ΤΗ, ἡ δὲ ΥΤ τῇ ΤΣ.

Ἐὰν ἄρα κύβου τῶν ἀπεναντίον ἐπιπέδων αἱ πλευραὶ δίθα τμ ηθῶσιν, διὰ δὲ τῶν τομῶν ἐπίπεδα ἐκβληθῇ, ἡ κοινὴ τομὴ τῶν ἐπιπέδων καὶ ἡ τοῦ κύβου διάμετρος δίθα τέμνουσιν ἀλλήλας· ὅπερ ἔδει δεῖχαι.

to *EG* [Prop. 11.9]. And the straight-lines *DE* and *BG* join them. *DE* is thus parallel to *BG* [Prop. 1.33]. Thus, angle *EDT* (is) equal to *BGT*. For (they are) alternate [Prop. 1.29]. And (angle) *DTU* (is equal) to *GTS* [Prop. 1.15]. So, *DTU* and *GTS* are two triangles having two angles equal to two angles, and one side equal to one side—(namely), that subtended by one of the equal angles—(that is), *DU* (equal) to *GS*. For they are halves of *DE* and *BG* (respectively). (Thus), they will also have the remaining sides equal to the remaining sides [Prop. 1.26]. Thus, *DT* (is) equal to *TG*, and *UT* to *TS*.

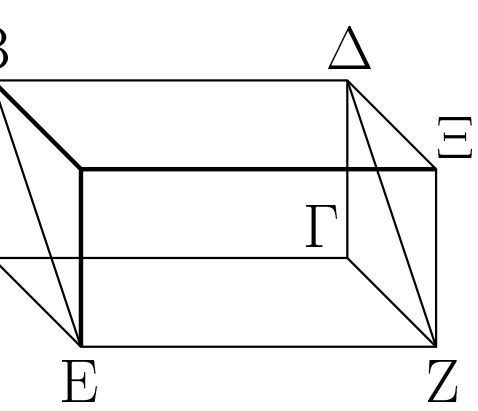
Thus, if the sides of the opposite planes of a cube are cut in half, and planes are produced through the pieces, then the common section of the (latter) planes and the diameter of the cube cut one another in half. (Which is) the very thing it was required to show.

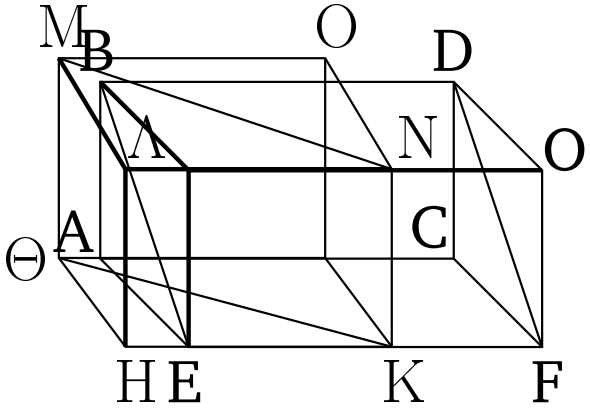
39.

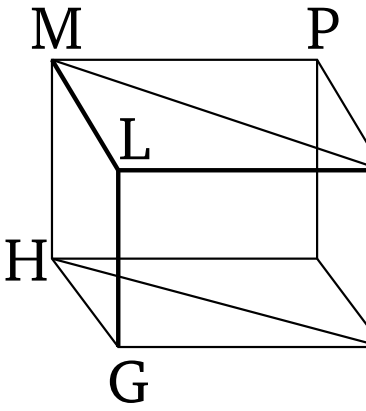
Proposition 39

Ἐὰνὸῇ δύο πρίσματα ἰσοϋψῆ, καὶ τὸ μὲνὸέθῇ βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ ὁῇ τὸ παραλληλόγραμμον τοῦ τριγώνου, ἴσαὸέσται τὰ πρίσματα.

If there are two equal height prisms, and one has a parallelogram, and the other a triangle, (as a) base, and the parallelogram is double the triangle, then the prisms will be equal.







Ἔστω δύο πρίσματα ἰσοϋψῆ τὰ ΑΒΓΔΕΖ, ΗΘΚΛΜΝ, καὶ τὸ μὲν ἐθέτω βάσιν τὸ ΑΖ παραλληλόγραμμον, τὸ δὲ τὸ ΗΘΚ τρίγωνον, διπλάσιον δὲ ὁέστω τὸ ΑΖ παραλληλόγραμμον τοῦ ΗΘΚ τριγώνου· λέγω, ὅτι ἴσον ἐστὶ τὸ ΑΒΓΔΕΖ πρίσμα τῷ ΗΘΚΛΜΝ πρίσματι.

Συμπεπληρώσθω γὰρ τὰ ΑΧ, ΗΟ στερεά. ἐπεὶ διπλάσιόν ἐστι τὸ ΑΖ παραλληλόγραμμον τοῦ ΗΘΚ τριγώνου, ὁέστι δὲ καὶ τὸ ΘΚ παραλληλόγραμμον διπλάσιον τοῦ ΗΘΚ τριγώνου, ἴσον ἄρα ἐστὶ τὸ ΑΖ παραλληλόγραμμον τῷ ΘΚ παραλληλογράμμῳ. τὰ δὲ ἐπὶ ὁίσων βάσεων ὄντα στερεὰ παραλληλεπίπεδα καὶ ὑπὸ τὸ αὐτὸ ὁύψος ἴσα ἀλλήλοις ἐστίν· ἴσον ἄρα ἐστὶ τὸ ΑΧ στερεὸν τῷ ΗΟ στερεῷ. καί ἐστι τοῦ μὲν ΑΧ στερεοῦ ἥμισυ τὸ ΑΒΓΔΕΖ πρίσμα, τοῦ δὲ ΗΟ στερεοῦ ἥμισυ τὸ ΗΘΚΛΜΝ πρίσμα· ἴσον ἄρα ἐστὶ τὸ ΑΒΓΔΕΖ πρίσμα τῷ ΗΘΚΛΜΝ πρίσματι.

Ἐὰν ἄραὸῇ δύο πρίσματα ἰσοϋψῆ, καὶ τὸ μὲνὸέθῇ βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ ὁῇ τὸ πα-

Let *ABCDEF* and *GHKLMN* be two equal height prisms, and let the former have the parallelogram *AF*, and the latter the triangle *GHK*, as a base. And let parallelogram *AF* be twice triangle *GHK*. I say that prism *ABCDEF* is equal to prism *GHKLMN*.

For let the solids *AO* and *GP* be completed. Since parallelogram *AF* is double triangle *GHK*, and parallelogram *HK* is also double triangle *GHK* [Prop. 1.34], parallelogram *AF* is thus equal to parallelogram *HK*. And parallelepiped solids which are on equal bases, and (have) the same height, are equal to one another [Prop. 11.31]. Thus, solid *AO* is equal to solid *GP*. And prism *ABCDEF* is half of solid *AO*, and prism *GHKLMN* half of solid *GP* [Prop. 11.28]. Prism *ABCDEF* is thus equal to prism *GHKLMN*.

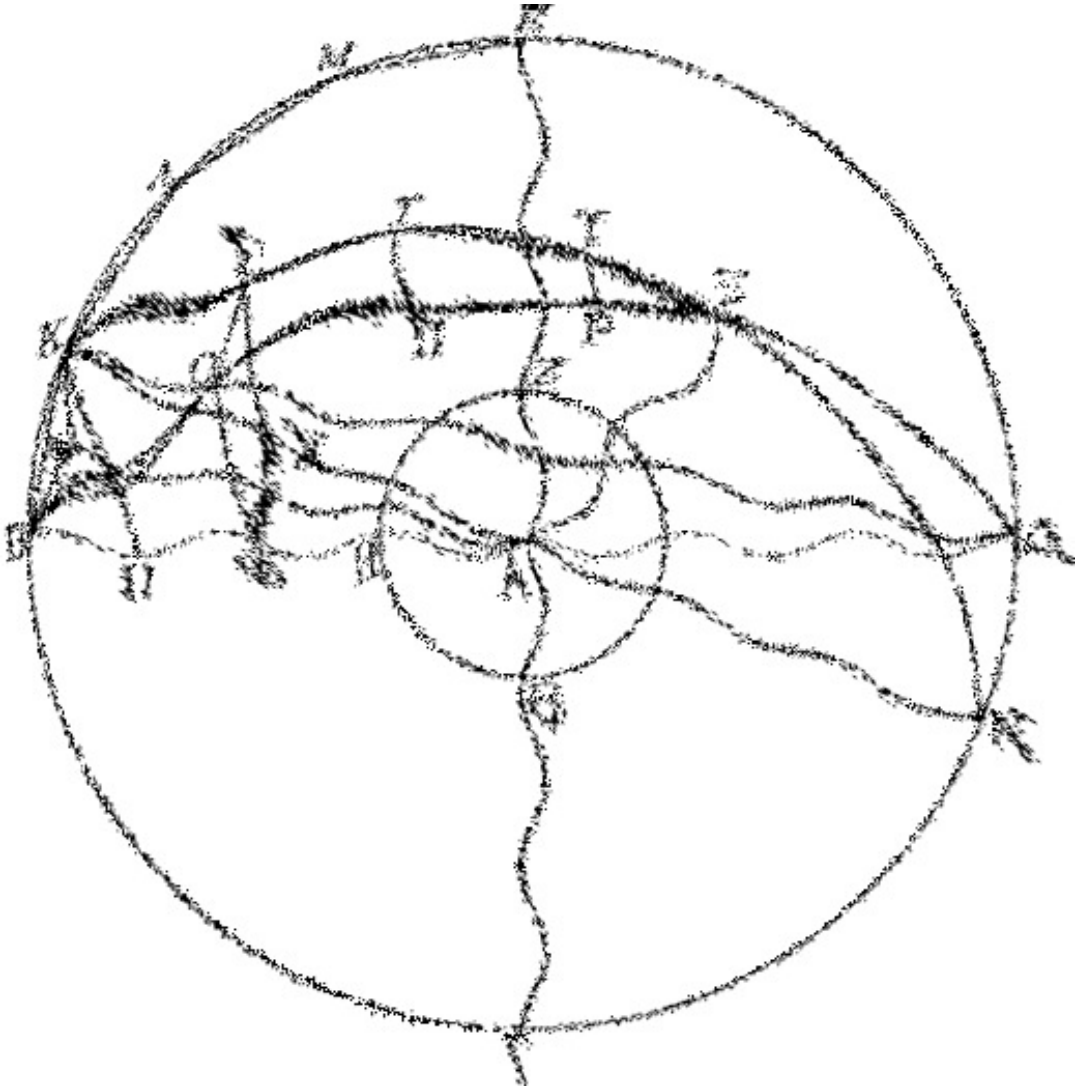
Thus, if there are two equal height prisms, and one has a parallelogram, and the other a triangle, (as a) base, and the parallelogram is double the triangle, then the prisms are equal.

ραλληλόγραμμον τοῦ τριγώνου, ἴσα δ' ἐστὶ τὰ πρίσματα· ὅπερ ἔδει δεῖξαι.

(Which is) the very thing it was required to show.

ELEMENTS BOOK 12

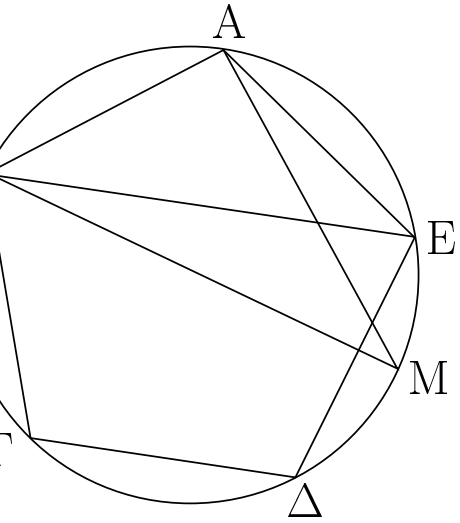
Proportional Stereometry[†]

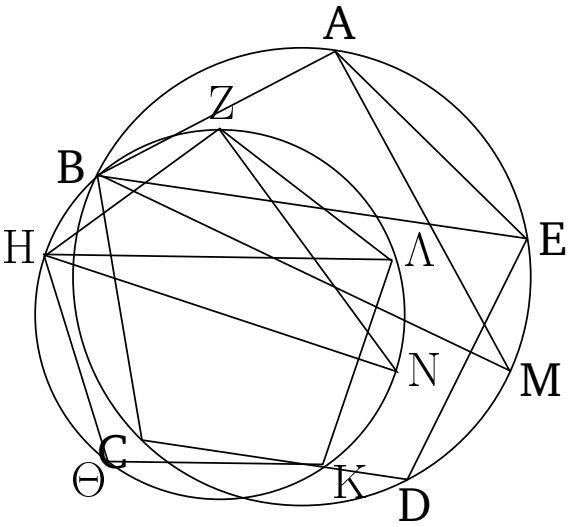


[†]The novel feature of this book is the use of the so-called method of exhaustion (see Prop. 10.1), a precursor to integration which is generally attributed to Eudoxus of Cnidus.

1.

Τὰ ἐν τοῖς κύκλοις ὅμοια πολύγωνα πρὸς ἀλλήλα ἐστὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα.





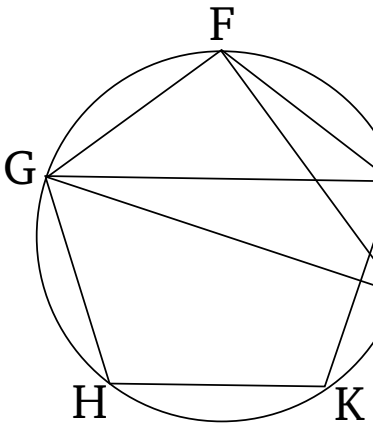
Ἔστωσαν κύκλοι οἱ ΑΒΓ, ΖΗΘ, καὶ ἐν αὐτοῖς ὅμοια πολύγωναὁέστω τὰ ΑΒΓΔΕ, ΖΗΘΚΛ, διάμετροι δὲ τῶν κύκλωνὁέστωσαν ΒΜ, ΗΝ· λέγω, ὅτι ἐστὶν ὡς τὸ ἀπὸ τῆς ΒΜ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΗΝ τετράγωνον, οὕτως τὸ ΑΒΓΔΕ πολύγωνον πρὸς τὸ ΖΗΘΚΛ πολύγωνον.

Ἐπεζεύθθωσαν γὰρ αἱ ΒΕ, ΑΜ, ΗΛ, ΖΝ. καὶ ἐπεὶ ὅμοιον τὸ ΑΒΓΔΕ πολύγωνον τῷ ΖΗΘΚΛ πολυγώνῳ, ὁίση ἐστὶ καὶ ἡ ὑπὸ ΒΑΕ γωνία τῇ ὑπὸ ΗΖΛ, καί ἐστιν ὡς ἡ ΒΑ πρὸς τὴν ΑΕ, οὕτως ἡ ΗΖ πρὸς τὴν ΖΛ. δύο δὴ τρίγωνά ἐστι τὰ ΒΑΕ, ΗΖΛ μίαν γωνίαν μιᾷ γωνίᾳ ἴσην ἔχοντα τὴν ὑπὸ ΒΑΕ τῇ ὑπὸ ΗΖΛ, περὶ δὲ τὰξ ἴσας γωνίας τὰξ πλευρὰξ ἀνάλογον· ἰσογώνιον ἄρα ἐστὶ τὸ ΑΒΕ τρίγωνον τῷ ΖΗΛ τριγώνῳ. ὁίση ἄρα ἐστὶν ἡ ὑπὸ ΑΕΒ γωνία τῇ ὑπὸ ΖΛΗ. ἀλλὰ ἡ μὲν ὑπὸ ΑΕΒ τῇ ὑπὸ ΑΜΒ ἐστινὁίση· ἐπὶ γὰρ τῆς αὐτῆς περιφερείας βεβήκασιν· ἡ δὲ ὑπὸ ΖΛΗ τῇ ὑπὸ ΖΝΗ· καὶ ἡ ὑπὸ ΑΜΒ ἄρα τῇ ὑπὸ ΖΝΗ ἐστινὁίση. ὁέστι δὲ καὶ ὀρθὴ ἡ ὑπὸ ΒΑΜ ὀρθῇ τῇ ὑπὸ ΗΖΝὁίση· καὶ ἡ λοιπὴ ἄρα τῇ λοιπῇ ἐστινὁίση. ἰσογώνιον ἄρα ἐστὶ τὸ ΑΒΜ τρίγωνον τῷ ΖΗΝ τρίγωνῳ. ἀνάλογον ἄρα ἐστὶν ὡς ἡ ΒΜ πρὸς τὴν ΗΝ, οὕτως ἡ ΒΑ πρὸς τὴν ΗΖ. ἀλλὰ τοῦ μὲν τῆς ΒΜ πρὸς τὴν ΗΝ λόγον διπλασίων ἐστὶν ὁ τοῦ ἀπὸ τῆς ΒΜ τετραγώνου πρὸς τὸ ἀπὸ τῆς ΗΝ τετράγωνον, τοῦ δὲ τῆς ΒΑ πρὸς τὴν ΗΖ διπλασίων ἐστὶν ὁ τοῦ ΑΒΓΔΕ πολυγώνου πρὸς τὸ ΖΗΘΚΛ πολύγωνον· καὶ ὡς ἄρα τὸ ἀπὸ τῆς ΒΜ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΗΝ τετράγωνον, οὕτως τὸ ΑΒΓΔΕ πολύγωνον πρὸς τὸ ΖΗΘΚΛ πολύγωνον.

Τὰ ἄρα ἐν τοῖς κύκλοις ὅμοια πολύγωνα πρὸς ἀλλήλα ἐστὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα· ὅπερ ἔδει δεῖξαι.

Proposition 1

Similar polygons (inscribed) in circles are to one another as the squares on the diameters (of the circles).



Let *ABC* and *FGH* be circles, and let *ABCDE* and *FGHKL* be similar polygons (inscribed) in them (respectively), and let *BM* and *GN* be the diameters of the circles (respectively). I say that as the square on *BM* is to the square on *GN*, so polygon *ABCDE* (is) to polygon *FGHKL*.

For let *BE*, *AM*, *GL*, and *FN* be joined. And since polygon *ABCDE* (is) similar to polygon *FGHKL*, angle *BAE* is also equal to (angle) *GFL*, and as *BA* is to *AE*, so *GF* (is) to *FL* [Def. 6.1]. So, *BAE* and *GFL* are two triangles having one angle equal to one angle, (namely), *BAE* (equal) to *GFL*, and the sides around the equal angles proportional. Triangle *ABE* is thus equiangular with triangle *FGL* [Prop. 6.6]. Thus, angle *AEB* is equal to (angle) *FLG*. But, *AEB* is equal to *AMB*, and *FLG* to *FNG*, for they stand on the same circumference [Prop. 3.27]. Thus, *AMB* is also equal to *FNG*. And the right-angle *BAM* is also equal to the right-angle *GFN* [Prop. 3.31]. Thus, the remaining (angle) is also equal to the remaining (angle) [Prop. 1.32]. Thus, triangle *ABM* is equiangular with triangle *FGN*. Thus, proportionally, as *BM* is to *GN*, so *BA* (is) to *GF* [Prop. 6.4]. But, the (ratio) of the square on *BM* to the square on *GN* is the square of the ratio of *BM* to *GN*, and the (ratio) of polygon *ABCDE* to polygon *FGHKL* is the square of the (ratio) of *BA* to *GF* [Prop. 6.20]. And, thus, as the square on *BM* (is) to the square on *GN*, so polygon *ABCDE* (is) to polygon *FGHKL*.

Thus, similar polygons (inscribed) in circles are to one another as the squares on the diameters (of the circles). (Which is) the very thing it was required to show.

2.

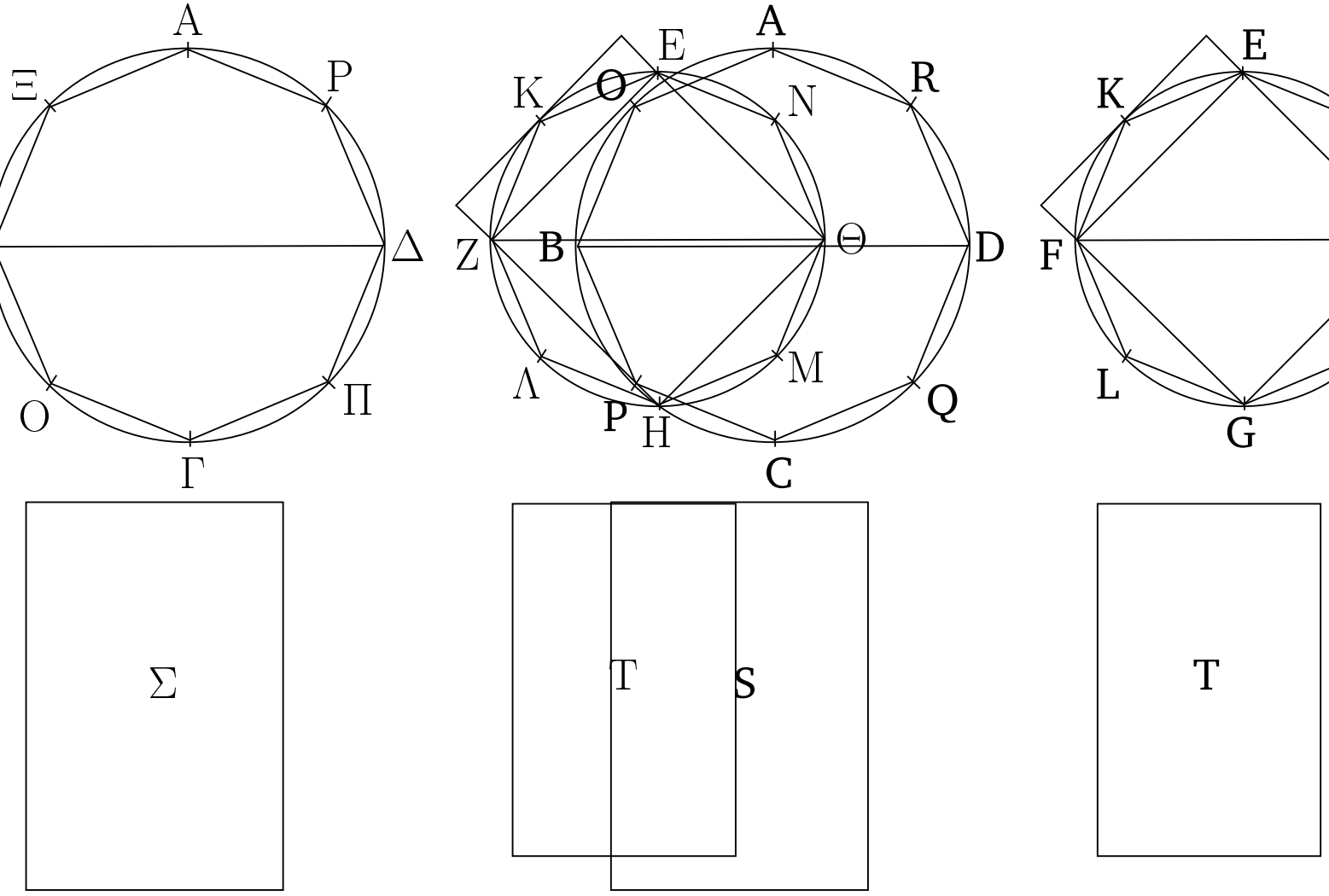
Proposition 2

Οἱ κύκλοι πρὸς ἀλλήλους εἰσὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα.

Ἔστωσαν κύκλοι οἱ ΑΒΓΔ, ΕΖΗΘ, διάμετροι δὲ αὐτῶν [δέστωσαν] αἱ ΒΔ, ΖΘ· λέγω, ὅτι ἐστὶν ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὕτως τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ τετράγωνον.

Circles are to one another as the squares on (their) diameters.

Let $ABCD$ and $EFGH$ be circles, and [let] BD and FH [be] their diameters. I say that as circle $ABCD$ is to circle $EFGH$, so the square on BD (is) to the square on FH.



Εἰ γὰρ μή ἐστιν ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ, οὕτως τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ, ἔσται ὡς τὸ ἀπὸ τῆς ΒΔ πρὸς τὸ ἀπὸ τῆς ΖΘ, οὕτως ὁ ΑΒΓΔ κύκλος ἤτοι πρὸς ἔλασσόν τι τοῦ ΕΖΗΘ κύκλου θωρίονὸή πρὸς μεῖζον. ἐστω πρότερον πρὸς ἔλασσον τὸ Σ. καὶ ἐγγεγράφθω εἰς τὸν ΕΖΗΘ κύκλον τετράγωνον τὸ ΕΖΗΘ. τὸ δὴ ἐγγεγραμμένον τετράγωνον μεῖζόν ἐστινὸή τὸ ἥμισυ τοῦ ΕΖΗΘ κύκλου, ἐπειδήπερ ἐὰν διὰ τῶν Ε, Ζ, Η, Θ σημείων ἐφαπτομένας [εὐθείας] τοῦ κύκλου ἀγάγωμεν, τοῦ περιγραφομένου περὶ τὸν κύκλον τετραγώνου ἥμισύ ἐστι τὸ ΕΖΗΘ τετράγωνον, τοῦ δὲ περιγραφέντος τετραγώνου ἐλάττων ἐστὶν

For if the circle $ABCD$ is not to the (circle) $EFGH$, as the square on BD (is) to the (square) on FH, then as the (square) on BD (is) to the (square) on FH, so circle $ABCD$ will be to some area either less than, or greater than, circle $EFGH$. Let it, first of all, be (in that ratio) to (some) lesser (area), S. And let the square $EFGH$ be inscribed in circle $EFGH$ [Prop. 4.6]. So the inscribed square is greater than half of circle $EFGH$, inasmuch as if we draw tangents to the circle through the points E, F, G, and H, then square $EFGH$ is half of the square circumscribed about the circle [Prop. 1.47], and the circle is less than the circumscribed square. Hence, the

ὁ κύκλος· ὥστε τὸ ΕΖΗΘ ἐγγεγραμμένομῆν τετράγωνον μεῖζόν ἐστι τοῦ ἡμίσεως τοῦ ΕΖΗΘ κύκλου. τετμήσθωσαν δίθα αἱ ΕΖ, ΖΗ, ΗΘ, ΘΕ περιφέρειαι κατὰ τὰ Κ, Λ, Μ, Ν σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΕΚ, ΚΖ, ΖΛ, ΛΗ, ΗΜ, ΜΘ, ΘΝ, ΝΕ· καὶ ὁέκαστον ἄρα τῶν ΕΚΖ, ΖΛΗ, ΗΜΘ, ΘΝΕ τριγώνων μεῖζόν ἐστινὸὴ τὸ ἥμισυ τοῦ καθὸ ἑαυτὸ τμήματος τοῦ κύκλου, ἐπειδήπερ ἔαν διὰ τῶν Κ, Λ, Μ, Ν σημείων ἐφαπτομένας τοῦ κύκλου ἀγάγωμεν καὶ ἀναπληρώσωμεν τὰ ἐπὶ τῶν ΕΖ, ΖΗ, ΗΘ, ΘΕ εὐθειῶν παραλληλόγραμμα, ὁέκαστον τῶν ΕΚΖ, ΖΛΗ, ΗΜΘ, ΘΝΕ τριγώνων ἥμισυὸέσται τοῦ καθὸ ἑαυτὸ παραλληλογράμμου, ἀλλὰ τὸ καθὸ ἑαυτὸ τμῆμαὸέλαττόν ἐστι τοῦ παραλληλογράμμου· ὥστε ὁέκαστον τῶν ΕΚΖ, ΖΛΗ, ΗΜΘ, ΘΝΕ τριγώνων μεῖζόν ἐστι τοῦ ἡμίσεως τοῦ καθὸ ἑαυτὸ τμήματος τοῦ κύκλου. τέμνοντες δὴ τὰξ ὑπολειπομένας περιφερείας δίθα καὶ ἐπιζευγνύντες εὐθείας καὶ τοῦτο ἀεὶ ποιοῦντες καταλείψομέν τινα ἀποτμήματα τοῦ κύκλου, ὁὰ ὀέσται ἐλάσσονα τῆς ὑπεροθῆς, ᾗ ὑπερέθει ὁ ΕΖΗΘ κύκλος τοῦ Σ θωρίου. ἐδείθθη γὰρ ἐν τῷ πρώτῳ θεωρήματι τοῦ δεκάτου βιβλίου, ὅτι δύο μεγεθῶν ἀνίσων ἐκκειμένων, ἔαν ἀπὸ τοῦ μείζονος ἀφαιρεθῇ μεῖζονὸὴ τὸ ἥμισυ καὶ τοῦ καταλειπομένου μεῖζονὸὴ τὸ ἥμισυ, καὶ τοῦτο ἀεὶ γίγνηται, λειφθήσεταί τι μέγεθος, ὃ ὀέσταιὸέλασσον τοῦ ἐκκειμένου ἐλάσσονος μεγέθους. λελείφθω οὐῦν, καὶ ὀέστω τὰ ἐπὶ τῶν ΕΚ, ΚΖ, ΖΛ, ΛΗ, ΗΜ, ΜΘ, ΘΝ, ΝΕ τμήματα τοῦ ΕΖΗΘ κύκλου ἐλάττονα τῆς ὑπεροθῆς, ᾗ ὑπερέθει ὁ ΕΖΗΘ κύκλος τοῦ Σ θωρίου. λοιπὸν ἄρα τὸ ΕΚΖΛΗΜΘΝ πολύγωνον μεῖζόν ἐστι τοῦ Σ θωρίου. ἐγγεγράφθω καὶ εἰξ τὸν ΑΒΓΔ κύκλον τῷ ΕΚΖΛΗΜΘΝ πολυγώνῳ ὅμοιον πολύγωνον τὸ ΑΧΒΟΓΠΔΡ· ὀέστιν ἄρα ὡξ τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ τετράγωνον, οὑτως τὸ ΑΧΒΟΓΠΔΡ πολύγωνον πρὸς τὸ ΕΚΖΛΗΜΘΝ πολύγωνον. ἀλλὰ καὶ ὡξ τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ, οὑτως ὁ ΑΒΓΔ κύκλος πρὸς τὸ Σ θωρίον· καὶ ὡξ ἄρα ὁ ΑΒΓΔ κύκλος πρὸς τὸ Σ θωρίον, οὑτως τὸ ΑΧΒΟΓΠΔΡ πολύγωνον πρὸς τὸ ΕΚΖΛΗΜΘΝ πολύγωνον· ἐναλλὰχ ἄρα ὡξ ὁ ΑΒΓΔ κύκλος πρὸς τὸ ἐν αὐτῷ πολύγωνον, οὑτως τὸ Σ θωρίον πρὸς τὸ ΕΚΖΛΗΜΘΝ πολύγωνον. μείζων δὲ ὁ ΑΒΓΔ κύκλος τοῦ ἐν αὐτῷ πολυγώνου· μεῖζον ἄρα καὶ τὸ Σ θωρίον τοῦ ΕΚΖΛΗΜΘΝ πολυγώνου. ἀλλὰ καὶ ὀέλαττον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐστὶν ὡξ τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ, οὑτως ὁ ΑΒΓΔ κύκλος πρὸςὀέλασσόν τι τοῦ ΕΖΗΘ κύκλου θωρίον. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ὡξ τὸ ἀπὸ ΖΘ πρὸς τὸ ἀπὸ ΒΔ, οὑτως ὁ ΕΖΗΘ κύκλος πρὸςὀέλασσόν τι τοῦ ΑΒΓΔ κύκλου θωρίον.

Λέγω δή, ὅτι οὐδὲ ὡξ τὸ ἀπὸ τῆς ΒΔ πρὸς τὸ ἀπὸ τῆς ΖΘ, οὑτως ὁ ΑΒΓΔ κύκλος πρὸς μεῖζόν τι τοῦ ΕΖΗΘ κύκλου θωρίον.

Εἰ γὰρ δυνατόν, ὀέστω πρὸς μεῖζον τὸ Σ. ἀνάπαλιν ἄρα [ἐστὶν] ὡξ τὸ ἀπὸ τῆς ΖΘ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΔΒ, οὑτως τὸ Σ θωρίον πρὸς τὸν ΑΒΓΔ κύκλον. ἀλλὰ ὡξ τὸ Σ θωρίον πρὸς τὸν ΑΒΓΔ κύκλον, οὑτως ὁ ΕΖΗΘ κύκλος

inscribed square $EFGH$ is greater than half of circle $EFGH$. Let the circumferences EF, FG, GH, and HE be cut in half at points K, L, M, and N (respectively), and let EK, KF, FL, LG, GM, MH, HN, and NE be joined. And, thus, each of the triangles EKF, FLG, GMH, and HNE is greater than half of the segment of the circle about it, inasmuch as if we draw tangents to the circle through points K, L, M, and N, and complete the parallelograms on the straight-lines EF, FG, GH, and HE, then each of the triangles EKF, FLG, GMH, and HNE will be half of the parallelogram about it, but the segment about it is less than the parallelogram. Hence, each of the triangles EKF, FLG, GMH, and HNE is greater than half of the segment of the circle about it. So, by cutting the circumferences remaining behind in half, and joining straight-lines, and doing this continually, we will (eventually) leave behind some segments of the circle whose (sum) will be less than the excess by which circle $EFGH$ exceeds the area S. For we showed in the first theorem of the tenth book that if two unequal magnitudes are laid out, and if (a part) greater than a half is subtracted from the greater, and (if from) the remainder (a part) greater than a half (is subtracted), and this happens continually, then some magnitude will (eventually) be left which will be less than the lesser laid out magnitude [Prop. 10.1]. Therefore, let the (segments) be left, and let the (sum of the) segments of the circle $EFGH$ on EK, KF, FL, LG, GM, MH, HN, and NE be less than the excess by which circle $EFGH$ exceeds area S. Thus, the remaining polygon $EKFLGMHN$ is greater than area S. And let the polygon $AOBPCQDR$, similar to the polygon $EKFLGMHN$, be inscribed in circle $ABCD$. Thus, as the square on BD is to the square on FH, so polygon $AOBPCQDR$ (is) to polygon $EKFLGMHN$ [Prop. 12.1]. But, also, as the square on BD (is) to the square on FH, so circle $ABCD$ (is) to area S. And, thus, as circle $ABCD$ (is) to area S, so polygon $AOBPGQDR$ (is) to polygon $EKFLGMHN$ [Prop. 5.11]. Thus, alternately, as circle $ABCD$ (is) to the polygon (inscribed) within it, so area S (is) to polygon $EKFLGMHN$ [Prop. 5.16]. And circle $ABCD$ (is) greater than the polygon (inscribed) within it. Thus, area S is also greater than polygon $EKFLGMHN$. But, (it is) also less. The very thing is impossible. Thus, the square on BD is not to the (square) on FH, as circle $ABCD$ (is) to some area less than circle $EFGH$. So, similarly, we can show that the (square) on FH (is) not to the (square) on BD as circle $EFGH$ (is) to some area less than circle $ABCD$ either.

So, I say that neither (is) the (square) on BD to the (square) on FH, as circle $ABCD$ (is) to some area greater than circle $EFGH$.

For, if possible, let it be (in that ratio) to (some) greater (area), S. Thus, inversely, as the square on FH [is] to the (square) on DB, so area S (is) to circle $ABCD$ [Prop. 5.7 corr.]. But, as area S (is) to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$ (see lemma). And, thus, as

πρὸς ὀλαττόν τι τοῦ ΑΒΓΔ κύκλου θωρίον· καὶ ὡς ἄρα τὸ ἀπὸ τῆς ΖΘ πρὸς τὸ ἀπὸ τῆς ΒΔ, οὕτως ὁ ΕΖΗΘ κύκλος πρὸς ὀλάσσόν τι τοῦ ΑΒΓΔ κύκλου θωρίον· ὅπερ ἀδύνατον ἐδείθθη. οὐκ ἄρα ἐστὶν ὡς τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ, οὕτως ὁ ΑΒΓΔ κύκλος πρὸς μεῖζόν τι τοῦ ΕΖΗΘ κύκλου θωρίον. ἐδείθθη δέ, ὅτι οὐδὲ πρὸς ὀλασσον· ὀέστιν ἄρα ὡς τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ, οὕτως ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον.

Οἱ ἄρα κύκλοι πρὸς ἀλλήλους εἰσὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα· ὅπερ ἔδει δεῖχαι.

Λῆμμα.

Λέγω δή, ὅτι τοῦ Σ θωρίου μείζονος ὄντος τοῦ ΕΖΗΘ κύκλου ἐστὶν ὡς τὸ Σ θωρίον πρὸς τὸν ΑΒΓΔ κύκλον, οὕτως ὁ ΕΖΗΘ κύκλος πρὸς ὀλαττόν τι τοῦ ΑΒΓΔ κύκλου θωρίον.

Γεγονέτω γὰρ ὡς τὸ Σ θωρίον πρὸς τὸν ΑΒΓΔ κύκλον, οὕτως ὁ ΕΖΗΘ κύκλος πρὸς τὸ Τ θωρίον. λέγω, ὅτιὸλαττόν ἐστι τὸ Τ θωρίον τοῦ ΑΒΓΔ κύκλου. ἐπεὶ γάρ ἐστιν ὡς τὸ Σ θωρίον πρὸς τὸν ΑΒΓΔ κύκλον, οὕτως ὁ ΕΖΗΘ κύκλος πρὸς τὸ Τ θωρίον, ἐναλλάχ ἐστιν ὡς τὸ Σ θωρίον πρὸς τὸν ΕΖΗΘ κύκλον, οὕτως ὁ ΑΒΓΔ κύκλος πρὸς τὸ Τ θωρίον. μεῖζον δὲ τὸ Σ θωρίον τοῦ ΕΖΗΘ κύκλου· μείζων ἄρα καὶ ὁ ΑΒΓΔ κύκλος τοῦ Τ θωρίου. ὅὡστε ἐστὶν ὡς τὸ Σ θωρίον πρὸς τὸν ΑΒΓΔ κύκλον, οὕτως ὁ ΕΖΗΘ κύκλος πρὸς ὀλαττόν τι τοῦ ΑΒΓΔ κύκλου θωρίον· ὅπερ ἔδει δεῖχαι.

3.

Πᾶσα πυραμὶξ τρίγωνονὸέθουσα βάσιν διαιρεῖται εἰξ δύο πυραμίδας ἴσας τε καὶ ὁμοίας ἀλλήλαις καὶ [ὁμοίας] τῇ ὅλῃ τριγώνους ἐθουσας βάσεις καὶ εἰξ δύο πρίσματα ἴσα· καὶ τὰ δύο πρίσματα μείζονά ἐστινὸὴ τὸ ἥμισυ τῆς ὅλης πυραμίδος.

the (square) on FH (is) to the (square) on BD, so circle $EFGH$ (is) to some area less than circle $ABCD$ [Prop. 5.11]. The very thing was shown (to be) impossible. Thus, as the square on BD is to the (square) on FH, so circle $ABCD$ (is) not to some area greater than circle $EFGH$. And it was shown that neither (is it in that ratio) to (some) lesser (area). Thus, as the square on BD is to the (square) on FH, so circle $ABCD$ (is) to circle $EFGH$.

Thus, circles are to one another as the squares on (their) diameters. (Which is) the very thing it was required to show.

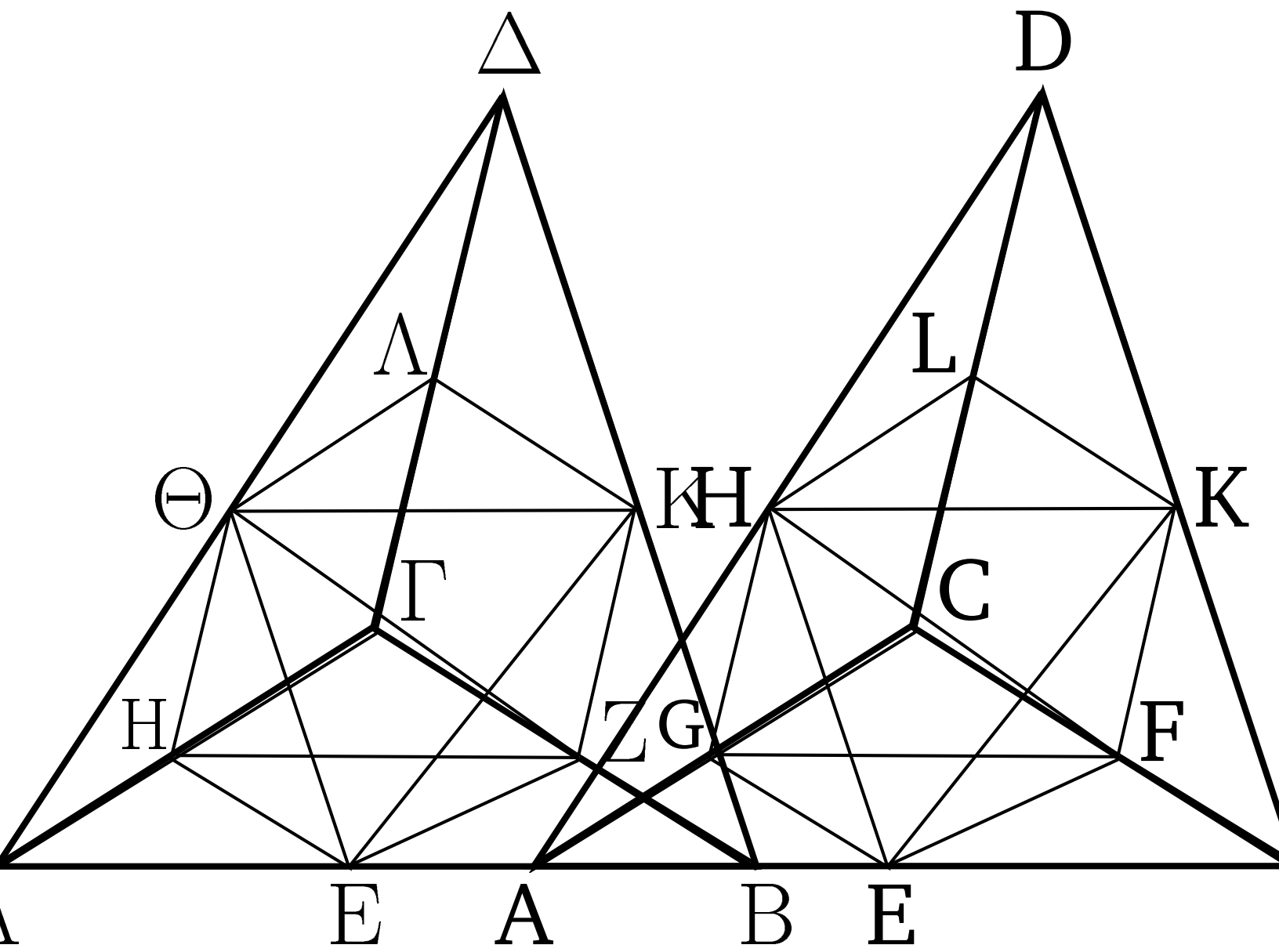
Lemma

So, I say that, area S being greater than circle $EFGH$, as area S is to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$.

For let it be contrived that as area S (is) to circle $ABCD$, so circle $EFGH$ (is) to area T. I say that area T is less than circle $ABCD$. For since as area S is to circle $ABCD$, so circle $EFGH$ (is) to area T, alternately, as area S is to circle $EFGH$, so circle $ABCD$ (is) to area T [Prop. 5.16]. And area S (is) greater than circle $EFGH$. Thus, circle $ABCD$ (is) also greater than area T [Prop. 5.14]. Hence, as area S is to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$. (Which is) the very thing it was required to show.

Proposition 3

Any pyramid having a triangular base is divided into two pyramids having triangular bases (which are) equal, similar to one another, and [similar] to the whole, and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid.



Ἔστω πυραμίξ, ἧς βάσις μέν ἐστι τὸ ΑΒΓ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον· λέγω, ὅτι ἡ ΑΒΓΔ πυραμὶξ διαιρεῖται εἰξ δύο πυραμίδας ἴσας ἀλλήλαις τριγώνους βάσεις ἐθούσας καὶ ὁμοίας τῇ ὅλῃ καὶ εἰξ δύο πρίσματα ἴσα· καὶ τὰ δύο πρίσματα μείζονά ἐστινὸὴ τὸ ἥμισυ τῆς ὅλης πυραμίδος.

Τετμήσθωσαν γὰρ αἱ ΑΒ, ΒΓ, ΓΑ, ΑΔ, ΔΒ, ΔΓ δίθα κατὰ τὰ Ε, Ζ, Η, Θ, Κ, Λ σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΘΕ, ΕΗ, ΗΘ, ΘΚ, ΚΛ, ΛΘ, ΚΖ, ΖΗ. ἐπεὶ ὁίση ἐστὶν ἡ μὲν ΑΕ τῇ ΕΒ, ἡ δὲ ΑΘ τῇ ΔΘ, παράλληλος ἄρα ἐστὶν ἡ ΕΘ τῇ ΔΒ. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΘΚ τῇ ΑΒ παράλληλός ἐστιν. παραλληλόγραμμον ἄρα ἐστὶ τὸ ΘΕΒΚ· ὁίση ἄρα ἐστὶν ἡ ΘΚ τῇ ΕΒ. ἀλλὰ ἡ ΕΒ τῇ ΕΑ ἐστινὸίση· καὶ ἡ ΑΕ ἄρα τῇ ΘΚ ἐστινὸίση. ὁέστι δὲ καὶ ἡ ΑΘ τῇ ΘΔὸίση· δύο δὴ

Let there be a pyramid whose base is triangle ABC, and (whose) apex (is) point D. I say that pyramid $ABCD$ is divided into two pyramids having triangular bases (which are) equal to one another, and similar to the whole, and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid.

For let AB, BC, CA, AD, DB, and DC be cut in half at points E, F, G, H, K, and L (respectively). And let HE, EG, GH, HK, KL, LH, KF, and FG be joined. Since AE is equal to EB, and AH to DH, EH is thus parallel to DB [Prop. 6.2]. So, for the same (reasons), HK is also parallel to AB. Thus, $HEBK$ is a parallelogram. Thus, HK is equal to EB [Prop. 1.34]. But, EB is equal to EA. Thus, AE is

αἱ ΕΑ, ΑΘ δυσὶ ταῖς ΚΘ, ΘΔ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ γωνία ἡ ὑπὸ ΕΑΘ γωνίᾳ τῇ ὑπὸ ΚΘΔ ὁίση· βάσις ἄρα ἡ ΕΘ βάσει τῇ ΚΔ ἐστινὁίση. ἴσον ἄρα καὶ ὅμοιόν ἐστι τὸ ΑΕΘ τρίγωνον τῷ ΘΚΔ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΑΘΗ τρίγωνον τῷ ΘΛΔ τριγώνῳ ἴσον τέ ἐστι καὶ ὅμοιον. καὶ ἐπεὶ δύο εὐθεῖαι ἁπτόμεναι ἀλλήλων αἱ ΕΘ, ΘΗ παρὰ δύο εὐθείας ἁπτομένας ἀλλήλων τὰξ ΚΔ, ΔΛ εἰσιν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ οὐσαι, ἴσας γωνίας περιέχουσιν. ὁίση ἄρα ἐστὶν ἡ ὑπὸ ΕΘΗ γωνία τῇ ὑπὸ ΚΔΛ γωνίᾳ. καὶ ἐπεὶ δύο εὐθεῖαι αἱ ΕΘ, ΘΗ δυσὶ ταῖς ΚΔ, ΔΛ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, καὶ γωνία ἡ ὑπὸ ΕΘΗ γωνίᾳ τῇ ὑπὸ ΚΔΛ ἐστινὁίση, βάσις ἄρα ἡ ΕΗ βάσει τῇ ΚΛ [ἐστιν] ὁίση· ἴσον ἄρα καὶ ὅμοιόν ἐστι τὸ ΕΘΗ τρίγωνον τῷ ΚΔΛ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΑΕΗ τρίγωνον τῷ ΘΚΛ τριγώνῳ ἴσον τε καὶ ὅμοιόν ἐστιν. ἡ ἄρα πυραμίξ, ἧς βάσις μέν ἐστι τὸ ΑΕΗ τρίγωνον, κορυφὴ δὲ τὸ Θ σημεῖον, ὁίση καὶ ὁμοία ἐστὶ πυραμίδι, ἧς βάσις μέν ἐστι τὸ ΘΚΛ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον. καὶ ἐπεὶ τριγώνου τοῦ ΑΔΒ παρὰ μίαν τῶν πλευρῶν τὴν ΑΒὀῆκται ἡ ΘΚ, ἰσογώνιόν ἐστι τὸ ΑΔΒ τρίγωνον τῷ ΔΘΚ τριγώνῳ, καὶ τὰξ πλευρὰξ ἀνάλογονὀέθουσιν· ὅμοιον ἄρα ἐστὶ τὸ ΑΔΒ τρίγωνον τῷ ΔΘΚ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν ΔΒΓ τρίγωνον τῷ ΔΚΛ τριγώνῳ ὅμοιόν ἐστιν, τὸ δὲ ΑΔΓ τῷ ΔΛΘ. καὶ ἐπεὶ δύο εὐθεῖαι ἁπτόμεναι ἀλλήλων αἱ ΒΑ, ΑΓ παρὰ δύο εὐθείας ἁπτομένας ἀλλήλων τὰξ ΚΘ, ΘΛ εἰσιν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ, ἴσας γωνίας περιέχουσιν. ὁίση ἄρα ἐστὶν ἡ ὑπὸ ΒΑΓ γωνία τῇ ὑπὸ ΚΘΛ. καί ἐστιν ὡξ ἡ ΒΑ πρὸς τὴν ΑΓ, οὁύτως ἡ ΚΘ πρὸς τὴν ΘΛ· ὅμοιον ἄρα ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΘΚΛ τριγώνῳ. καὶ πυραμὶξ ἄρα, ἧς βάσις μέν ἐστι τὸ ΑΒΓ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, ὁμοία ἐστὶ πυραμίδι, ἧς βάσις μέν ἐστι τὸ ΘΚΛ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον. ἀλλὰ πυραμίξ, ἧς βάσις μέν [ἐστι] τὸ ΘΚΛ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, ὁμοία ἐδείθθη πυραμίδι, ἧς βάσις μέν ἐστι τὸ ΑΕΗ τρίγωνον, κορυφὴ δὲ τὸ Θ σημεῖον. ἑκατέρα ἄρα τῶν ΑΕΗΘ, ΘΚΛΔ πυραμίδων ὁμοία ἐστὶ τῇ ὅλῃ τῇ ΑΒΓΔ πυραμίδι.

Καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΒΖ τῇ ΖΓ, διπλάσιόν ἐστι τὸ ΕΒΖΗ παραλληλόγραμμον τοῦ ΗΖΓ τριγώνου. καὶ ἐπεί, ἐανὸῇ δύο πρίσματα ἰσοϋψῆ, καὶ τὸ μὲνὸέθη βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ ὁῇ τὸ παραλληλόγραμμον τοῦ τριγώνου, ἴσα ἐστὶ τὰ πρίσματα, ἴσον ἄρα ἐστὶ τὸ πρίσμα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν ΒΚΖ, ΕΘΗ, τριῶν δὲ παραλληλογράμμων τῶν ΕΒΖΗ, ΕΒΚΘ, ΘΚΖΗ τῷ πρισματι τῷ περιεθομένῳ ὑπὸ δύο μὲν τριγώνων τῶν ΗΖΓ, ΘΚΛ, τριῶν δὲ παραλληλογράμμων τῶν ΚΖΓΛ, ΛΓΗΘ, ΘΚΖΗ. καὶ φανερόν, ὅτι ἑκάτρον τῶν πρισμάτων, οὗ τε βάσις τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΘΚ εὐθεῖα, καὶ οὗ βάσις τὸ ΗΖΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΘΚΛ τρίγωνον, μεῖζόν ἐστιν ἑκατέρας τῶν πυραμίδων, ὧν βάσεις μὲν τὰ ΑΕΗ, ΘΚΛ τρίγωνα, κορυφαί, δὲ τὰ Θ, Δ σημεῖα, ἐπειδήπερ [καί] ἐαν ἐπιζεύχωμεν τὰξ ΕΖ, ΕΚ εὐθείας, τὸ μὲν πρίσμα, οὗ βάσις τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΘΚ

also equal to HK. And AH is also equal to HD. So the two (straight-lines) EA and AH are equal to the two (straight-lines) KH and HD, respectively. And angle EAH (is) equal to angle KHD [Prop. 1.29]. Thus, base EH is equal to base KD [Prop. 1.4]. Thus, triangle AEH is equal and similar to triangle HKD [Prop. 1.4]. So, for the same (reasons), triangle AHG is also equal and similar to triangle HLD. And since EH and HG are two straight-lines joining one another (which are respectively) parallel to two straight-lines joining one another, KD and DL, not being in the same plane, they will contain equal angles [Prop. 11.10]. Thus, angle EHG is equal to angle KDL. And since the two straight-lines EH and HG are equal to the two straight-lines KD and DL, respectively, and angle EHG is equal to angle KDL, base EG [is] thus equal to base KL [Prop. 1.4]. Thus, triangle EHG is equal and similar to triangle KDL. So, for the same (reasons), triangle AEG is also equal and similar to triangle HKL. Thus, the pyramid whose base is triangle AEG, and apex the point H, is equal and similar to the pyramid whose base is triangle HKL, and apex the point D [Def. 11.10]. And since HK has been drawn parallel to one of the sides, AB, of triangle ADB, triangle ADB is equiangular to triangle DHK [Prop. 1.29], and they have proportional sides. Thus, triangle ADB is similar to triangle DHK [Def. 6.1]. So, for the same (reasons), triangle DBC is also similar to triangle DKL, and ADC to DLH. And since two straight-lines joining one another, BA and AC, are parallel to two straight-lines joining one another, KH and HL, not in the same plane, they will contain equal angles [Prop. 11.10]. Thus, angle BAC is equal to (angle) KHL. And as BA is to AC, so KH (is) to HL. Thus, triangle ABC is similar to triangle HKL [Prop. 6.6]. And, thus, the pyramid whose base is triangle ABC, and apex the point D, is similar to the pyramid whose base is triangle HKL, and apex the point D [Def. 11.9]. But, the pyramid whose base [is] triangle HKL, and apex the point D, was shown (to be) similar to the pyramid whose base is triangle AEG, and apex the point H. Thus, each of the pyramids $AEGH$ and $HKLD$ is similar to the whole pyramid $ABCD$.

And since BF is equal to FC, parallelogram $EBFG$ is double triangle GFC [Prop. 1.41]. And since, if two prisms (have) equal heights, and the former has a parallelogram as a base, and the latter a triangle, and the parallelogram (is) double the triangle, then the prisms are equal [Prop. 11.39], the prism contained by the two triangles BKF and EHG, and the three parallelograms $EBFG$, $EBKH$, and $HKFG$, is thus equal to the prism contained by the two triangles GFC and HKL, and the three parallelograms $KFCL$, $LCGH$, and $HKFG$. And (it is) clear that each of the prisms whose base (is) parallelogram $EBFG$, and opposite (side) straight-line HK, and whose base (is) triangle GFC, and opposite (plane) triangle HKL, is greater than each of the pyramids whose bases are triangles AEG and HKL, and apexes the points H and D (respec-

εὐθεῖα, μεῖζόν ἐστι τῆς πυραμίδος, ἧξ βάσις τὸ EBZ τρίγωνον, κορυφὴ δὲ τὸ K σημεῖον. ἀλλὰ ἡ πυραμίς, ἧξ βάσις τὸ EBZ τρίγωνον, κορυφὴ δὲ τὸ K σημεῖον, ὁίση ἐστὶ πυραμίδι, ἧξ βάσις τὸ AEH τρίγωνον, κορυφὴ δὲ τὸ Θ σημεῖον· ὑπὸ γὰρ ὁίσων καὶ ὁμοίων ἐπιπέδων περιέθονται. ὁώστε καὶ τὸ πρίσμα, οὗ βάσις μὲν τὸ $EBZH$ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΘK εὐθεῖα, μεῖζόν ἐστι πυραμίδος, ἧξ βάσις μὲν τὸ AEH τρίγωνον, κορυφὴ δὲ τὸ Θ σημεῖον. ἴσον δὲ τὸ μὲν πρίσμα, οὗ βάσις τὸ $EBZH$ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΘK εὐθεῖα, τῷ πρίσματι, οὗ βάσις μὲν τὸ $HZ\Gamma$ τρίγωνον, ἀπεναντίον δὲ τὸ $\Theta K\Lambda$ τρίγωνον· ἡ δὲ πυραμίς, ἧξ βάσις τὸ AEH τρίγωνον, κορυφὴ δὲ τὸ Θ σημεῖον, ὁίση ἐστὶ πυραμίδι, ἧξ βάσις τὸ $\Theta K\Lambda$ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον. τὰ ἄρα εἰρημένα δύο πρίσματα μείζονά ἐστι τῶν εἰρημένων δύο πυραμίδων, ὁῶν βάσεις μὲν τὰ AEH, $\Theta K\Lambda$ τρίγωνα, κορυφαὶ δὲ τὰ Θ, Δ σημεῖα.

Ἡ ἄρα ὅλη πυραμίς, ἧξ βάσις τὸ $AB\Gamma$ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, διῄρηται εἰίξ τε δύο πυραμίδας ἴσας ἀλλήλαις [καὶ ὁμοίας τῇ ὅλῃ] καὶ εἰξ δύο πρίσματα ἴσα, καὶ τὰ δύο πρίσματα μείζονά ἐστινὴ τὸ ἥμισυ τῆς ὅλης πυραμίδος· ὅπερ ἔδει δεῖχαι.

tively), inasmuch as, if we [also] join the straight-lines EF and EK then the prism whose base (is) parallelogram $EBFG$, and opposite (side) straight-line HK, is greater than the pyramid whose base (is) triangle EBF, and apex the point K. But the pyramid whose base (is) triangle EBF, and apex the point K, is equal to the pyramid whose base is triangle AEG, and apex point H. For they are contained by equal and similar planes. And, hence, the prism whose base (is) parallelogram $EBFG$, and opposite (side) straight-line HK, is greater than the pyramid whose base (is) triangle AEG, and apex the point H. And the prism whose base is parallelogram $EBFG$, and opposite (side) straight-line HK, (is) equal to the prism whose base (is) triangle GFC, and opposite (plane) triangle HKL. And the pyramid whose base (is) triangle AEG, and apex the point H, is equal to the pyramid whose base (is) triangle HKL, and apex the point D. Thus, the (sum of the) aforementioned two prisms is greater than the (sum of the) aforementioned two pyramids, whose bases (are) triangles AEG and HKL, and apexes the points H and D (respectively).

Thus, the whole pyramid, whose base (is) triangle ABC, and apex the point D, has been divided into two pyramids (which are) equal to one another [and similar to the whole], and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid. (Which is) the very thing it was required to show.

4.

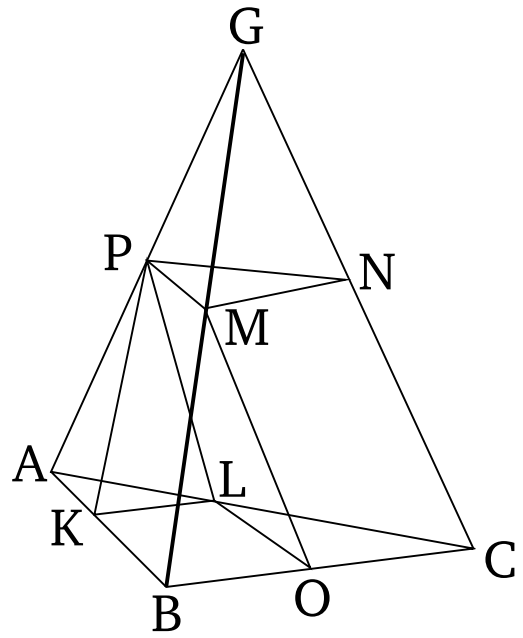
Ἐὰνὸῶσι δύο πυραμίδες ὑπὸ τὸ αὐτὸ ὁύψος τριγώνουςὸέθουσαι βάσεις, διαιρεθῇ δὲ ἑκατέρα αὐτῶν εὶίξ τε δύο πυραμίδας ἴσας ἀλλήλαις καὶ ὁμοίας τῇ ὅλῃ καὶ εἰξ δύο πρίσματα ἴσα, ὁέσται ὡξ ἡ τῆς μιᾶς πυραμίδος βάσις πρὸς τὴν τῆς ἑτέρας πυραμίδος βάσιν, οὁύτως τὰ ἐν τῇ μιᾷ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ ἑτάρᾳ πυραμίδι πρίσματα πάντα ἰσοπληθῆ.

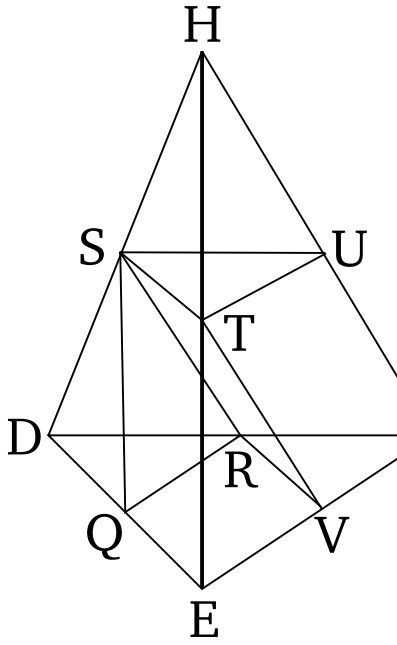
Ἔστωσαν δύο πυραμίδες ὑπὸ τὸ αὐτὸ ὁύψος τριγώνουςὸέθουσαι βάσεις τὰξ $AB\Gamma$, ΔEZ, κορυφὰξ δὲ τὰ H, Θ σημεῖα, καὶ διῃρήσθω ἑκατέρα αὐτῶν εὶίξ τε δύο πυραμίδας ἴσας ἀλλήλαις καὶ ὁμοίας τῇ ὅλῃ καὶ εἰξ δύο πρίσματα ἴσα· λέγω, ὅτι ἐστὶν ὡξ ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὁύτως τὰ ἐν τῇ $AB\Gamma H$ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ $\Delta EZ\Theta$ πυραμίδι πρίσματα ἰσοπληθῆ.

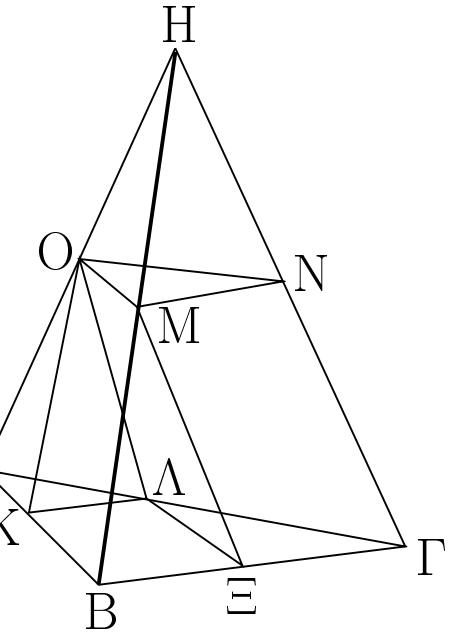
Proposition 4

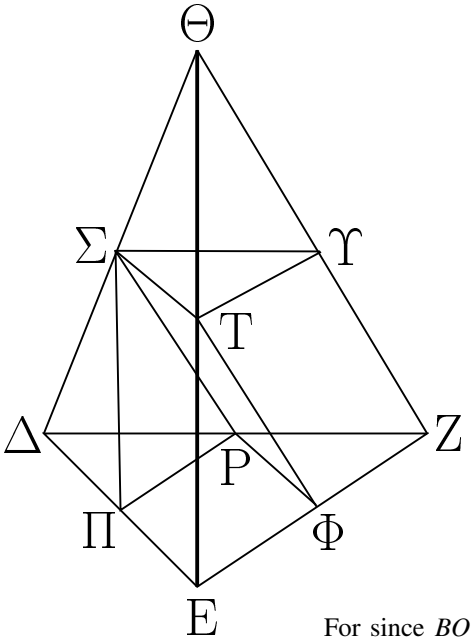
If there are two pyramids with the same height, having trianglular bases, and each of them is divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms then as the base of one pyramid (is) to the base of the other pyramid, so (the sum of) all the prisms in one pyramid will be to (the sum of all) the equal number of prisms in the other pyramid.

Let there be two pyramids with the same height, having the triangular bases ABC and DEF, (with) apexes the points G and H (respectively). And let each of them be divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms [Prop. 12.3]. I say that as base ABC is to base DEF, so (the sum of) all the prisms in pyramid $ABCG$ (is) to (the sum of) all the equal number of prisms in pyramid $DEFH$.









Ἐπεὶ γὰρ ὁίση ἐστὶν ἡ μὲν ΒΧ τῇ ΧΓ, ἡ δὲ ΑΛ τῇ ΛΓ, παράλληλος ἄρα ἐστὶν ἡ ΛΧ τῇ ΑΒ καὶ ὅμοιον τὸ ΑΒΓ τρίγωνον τῷ ΛΧΓ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΔΕΖ τρίγωνον τῷ ΡΦΖ τριγώνῳ ὅμοιόν ἐστιν. καὶ ἐπεὶ διπλασίων ἐστὶν ἡ μὲν ΒΓ τῆς ΓΧ, ἡ δὲ ΕΖ τῆς ΖΦ, ὐέστιν ἄρα ὡξ ἡ ΒΓ πρὸς τὴν ΓΧ, οὑτως ἡ ΕΖ πρὸς τὴν ΖΦ. καὶ ἀναγέγραπται ἀπὸ μὲν τῶν ΒΓ, ΓΧ ὅμοιά τε καὶ ὁμοίως κείμενα εὐθύγραμμα τὰ ΑΒΓ, ΛΧΓ, ἀπὸ δὲ τῶν ΕΖ, ΖΦ ὅμοιά τε καὶ ὁμοίως κείμενα [εὐθύγραμμα] τὰ ΔΕΖ, ΡΦΖ· ὐέστιν ἄρα ὡξ τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΛΧΓ τρίγωνον, οὑτως τὸ ΔΕΖ τρίγω-

For since *BO* is equal to *OC*, and *AL* to *LC*, *LO* is thus parallel to *AB*, and triangle *ABC* similar to triangle *LOC* [Prop. 12.3]. So, for the same (reasons), triangle *DEF* is also similar to triangle *RVF*. And since *BC* is double *CO*, and *EF* (double) *FV*, thus as *BC* (is) to *CO*, so *EF* (is) to *FV*. And the similar, and similarly laid out, rectilinear (figures) *ABC* and *LOC* be described on *BC* and *CO* (respectively), and the similar, and similarly laid out, [rectilinear] (figures) *DEF* and *RVF* on *EF* and *FV* (respectively). Thus, as triangle *ABC* is to triangle *LOC*, so triangle *DEF* (is) to triangle *RVF* [Prop. 6.22]. Thus, alternately, as triangle *ABC* is to [triangle] *DEF*, so [tri-

νον πρὸς τὸ ΡΦΖ τρίγωνον· ἐναλλὰχ ἄρα ἐστὶν ὡς τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΔΕΖ [τρίγωνον], οὕτως τὸ ΛΧΓ [τρίγωνον] πρὸς τὸ ΡΦΖ τρίγωνον. ἀλλὰ ὡς τὸ ΛΧΓ τρίγωνον πρὸς τὸ ΡΦΖ τρίγωνον, οὕτως τὸ πρίσμα, οὗ βάσις μέν [ἐστι] τὸ ΛΧΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΟΜΝ, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ ΡΦΖ τρίγωνον, ἀπεναντίον δὲ τὸ ΣΤΥ· καὶ ὡς ἄρα τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΔΕΖ τρίγωνον, οὕτως τὸ πρίσμα, οὗ βάσις μὲν τὸ ΛΧΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΟΜΝ, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ ΡΦΖ τρίγωνον, ἀπεναντίον δὲ τὸ ΣΤΥ. ὡς δὲ τὰ εἰρημένα πρίσματα πρὸς ἄλληλα, οὕτως τὸ πρίσμα, οὗ βάσις μὲν τὸ ΚΒΧΛ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΟΜ εὐθεῖα, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ ΠΕΦΡ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΣΤ εὐθεῖα. καὶ τὰ δύο ἄρα πρίσματα, οὗ τε βάσις μὲν τὸ ΚΒΧΛ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΟΜ, καὶ οὗ βάσις μὲν τὸ ΛΧΓ, ἀπεναντίον δὲ τὸ ΟΜΝ, πρὸς τὰ πρίσματα, οὗ τε βάσις μὲν τὸ ΠΕΦΡ, ἀπεναντίον δὲ ἡ ΣΤ εὐθεῖα, καὶ οὗ βάσις μὲν τὸ ΡΦΖ τρίγωνον, ἀπεναντίον δὲ τὸ ΣΤΥ. καὶ ὡς ἄρα ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὰ εἰρημένα δύο πρίσματα πρὸς τὰ εἰρημένα δύο πρίσματα.

Καὶ ὁμοίως, ἐὰν διαιρεθῶσιν αἱ ΟΜΝΗ, ΣΤΥΘ πυραμίδες εὐίξ τε δύο πρίσματα καὶ δύο πυραμίδας, ὀέσται ὡς ἡ ΟΜΝ βάσις πρὸς τὴν ΣΤΥ βάσιν, οὕτως τὰ ἐν τῇ ΟΜΝΗ πυραμίδι δύο πρίσματα πρὸς τὰ ἐν τῇ ΣΤΥΘ πυραμίδι δύο πρίσματα. ἀλλὰ ὡς ἡ ΟΜΝ βάσις πρὸς τὴν ΣΤΥ βάσιν, οὕτως ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν· ἴσον γὰρ ἑκάτερον τῶν ΟΜΝ, ΣΤΥ τριγώνων ἑκατέρῳ τῶν ΛΧΓ, ΡΦΖ. καὶ ὡς ἄρα ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὰ τέσσαρα πρίσματα πρὸς τὰ τέσσαρα πρίσματα. ὁμοίως δὲ κἂν τὰξ ὑπολειπομένας πυραμίδας διέλωμεν εὐίξ τε δύο πυραμίδας καὶ εἰξ δύο πρίσματα, ὀέσται ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὰ ἐν τῇ ΑΒΓΗ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα πάντα ἰσοπληθῆ· ὅπερ ἔδει δεῖξαι.

Λῆμμα.

Ὅτι δέ ἐστιν ὡς τὸ ΛΧΓ τρίγωνον πρὸς τὸ ΡΦΖ τρίγωνον, οὕτως τὸ πρίσμα, οὗ βάσις τὸ ΛΧΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΟΜΝ, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ ΡΦΖ [τρίγωνον], ἀπεναντίον δὲ τὸ ΣΤΥ, οὕτω δεικτέον.

Ἐπὶ γὰρ τῆς αὐτῆς καταγραφῆς νενοήσθωσαν ἀπὸ τῶν Η, Θ κάθετοι ἐπὶ τὰ ΑΒΓ, ΔΕΖ ἐπίπεδα, ἴσαι δηλαδὴ τυγθάνουσαι διὰ τὸ ἰσοϋψεῖς ὑποκεῖσθαι τὰξ πυραμίδας. καὶ ἐπεὶ δύο εὐθεῖαι ᾗ τε ΗΓ καὶ ἡ ἀπὸ τοῦ Η κάθετος ὑπὸ παραλλήλων ἐπιπέδων τῶν ΑΒΓ, ΟΜΝ τέμνονται, εἰξ τοὐξ αὐτοὐξ λόγους τμ ηθήσονται. καὶ τέτμ ηται ἡ ΗΓ δίθα ὑπὸ τοῦ ΟΜΝ ἐπιπέδου κατὰ τὸ Ν· καὶ ἡ ἀπὸ τοῦ Η ἄρα κάθετος ἐπὶ τὸ ΑΒΓ ἐπίπεδον δίθα τμ ηθήσεται ὑπὸ τοῦ ΟΜΝ ἐπιπέδου. διὰ τὰ αὐτὰ δὴ καὶ ἡ ἀπὸ τοῦ Θ κάθετος ἐπὶ τὸ ΔΕΖ ἐπίπεδον δίθα τμ ηθήσεται ὑπὸ τοῦ ΣΤΥ ἐπιπέδου. καί εἰσιν ἴσαι αἱ ἀπὸ τῶν angle] *LOC* (is) to triangle *RVF* [Prop. 5.16]. But, as triangle *LOC* (is) to triangle *RVF*, so the prism whose base [is] triangle *LOC*, and opposite (plane) *PMN*, (is) to the prism whose base (is) triangle *RVF*, and opposite (plane) *STU* (see lemma). And, thus, as triangle *ABC* (is) to triangle *DEF*, so the prism whose base (is) triangle *LOC*, and opposite (plane) *PMN*, (is) to the prism whose base (is) triangle *RVF*, and opposite (plane) *STU*. And as the aforementioned prisms (are) to one another, so the prism whose base (is) parallelogram *KBOL*, and opposite (side) straight-line *PM*, (is) to the prism whose base (is) parallelogram *QEVR*, and opposite (side) straight-line *ST* [Props. 11.39, 12.3]. Thus, also, (is) the (sum of the) two prisms—that whose base (is) parallelogram *KBOL*, and opposite (side) *PM*, and that whose base (is) *LOC*, and opposite (plane) *PMN*—to (the sum of) the (two) prisms—that whose base (is) *QEVR*, and opposite (side) straight-line *ST*, and that whose base (is) triangle *RVF*, and opposite (plane) *STU* [Prop. 5.12]. And, thus, as base *ABC* (is) to base *DEF*, so the (sum of the first) aforementioned two prisms (is) to the (sum of the second) aforementioned two prisms.

And, similarly, if pyramids *PMNG* and *STUH* are divided into two prisms, and two pyramids, as base *PMN* (is) to base *STU*, so (the sum of) the two prisms in pyramid *PMNG* will be to (the sum of) the two prisms in pyramid *STUH*. But, as base *PMN* (is) to base *STU*, so base *ABC* (is) to base *DEF*. For the triangles *PMN* and *STU* (are) equal to *LOC* and *RVF*, respectively. And, thus, as base *ABC* (is) to base *DEF*, so (the sum of) the four prisms (is) to (the sum of) the four prisms [Prop. 5.12]. So, similarly, even if we divide the pyramids left behind into two pyramids and into two prisms, as base *ABC* (is) to base *DEF*, so (the sum of) all the prisms in pyramid *ABCG* will be to (the sum of) all the equal number of prisms in pyramid *DEFH*. (Which is) the very thing it was required to show.

Lemma

And one may show, as follows, that as triangle *LOC* is to triangle *RVF*, so the prism whose base (is) triangle *LOC*, and opposite (plane) *PMN*, (is) to the prism whose base (is) [triangle] *RVF*, and opposite (plane) *STU*.

For, in the same figure, let perpendiculars be conceived (drawn) from (points) *G* and *H* to the planes *ABC* and *DEF* (respectively). These clearly turn out to be equal, on account of the pyramids being assumed (to be) of equal height. And since two straight-lines, *GC* and the perpendicular from *G*, are cut by the parallel planes *ABC* and *PMN* they will be cut in the same ratios [Prop. 11.17]. And *GC* was cut in half by the plane *PMN* at *N*. Thus, the perpendicular from *G* to the plane *ABC* will also be cut in half by the plane *PMN*. So, for the same (reasons), the perpendicular from *H* to the plane *DEF* will

Η, Θ κάθετοι ἐπὶ τὰ ΑΒΓ, ΔΕΖ ἐπίπεδα· ἴσαι ἄρα καὶ αἱ ἀπὸ τῶν ΟΜΝ, ΣΤΥ τριγώνων ἐπὶ τὰ ΑΒΓ, ΔΕΖ κάθετοι. ἰσοϋψῆ ἄρα [ἐστὶ] τὰ πρίσματα, ὧν βάσεις μέν εἰσι τὰ ΛΧΓ, ΡΦΖ τρίγωνα, ἀπεναντίον δὲ τὰ ΟΜΝ, ΣΤΥ. ὥστε καὶ τὰ στερεὰ παραλληλεπίπεδα τὰ ἀπὸ τῶν εἰρημένων πρισμάτων ἀναγραφόμενα ἰσοϋψῆ καὶ πρὸς ἄλληλά [εἰσιν] ὡς αἱ βάσεις· καὶ τὰ ἥμίση ἄρα ἐστὶν ὡς ἡ ΛΧΓ βάσις πρὸς τὴν ΡΦΖ βάσιν, οὕτως τὰ εἰρημένα πρίσματα πρὸς ἄλληλα· ὅπερ ἔδει δεῖχαι.

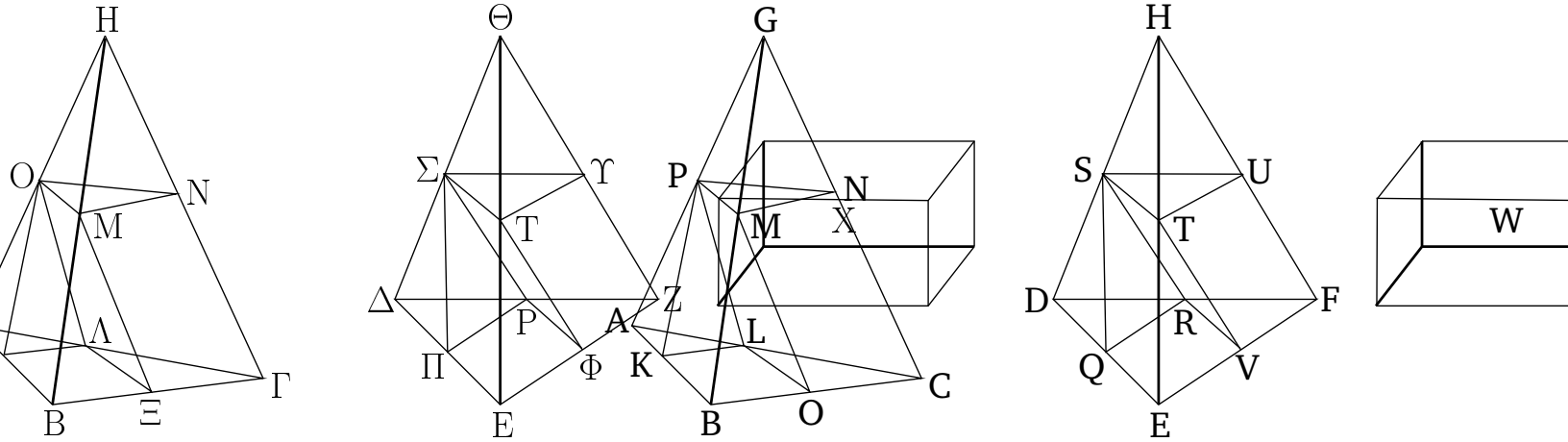
also be cut in half by the plane STU. And the perpendiculars from G and H to the planes ABC and DEF (respectively) are equal. Thus, the perpendiculars from the triangles PMN and STU to ABC and DEF (respectively, are) also equal. Thus, the prisms whose bases are triangles LOC and RVF, and opposite (sides) PMN and STU (respectively), [are] of equal height. And, hence, the parallelepiped solids described on the aforementioned prisms [are] of equal height and (are) to one another as their bases [Prop. 11.32]. Likewise, the halves (of the solids) [Prop. 11.28]. Thus, as base LOC is to base RVF, so the aforementioned prisms (are) to one another. (Which is) the very thing it was required to show.

5.

Αἱ ὑπὸ τὸ αὐτὸ ὕψος οὖσαι πυραμίδες καὶ τριγώνους ἐχούσας βάσεις πρὸς ἀλλήλας εἰσὶν ὡς αἱ βάσεις.

Proposition 5

Pyramids which are of the same height, and have triangular bases, are to one another as their bases.



Ἔστωσαν ὑπὸ τὸ αὐτὸ ὕψος πυραμίδες, ὧν βάσεις μὲν τὰ ΑΒΓ, ΔΕΖ τρίγωνα, κορυφαὶ δὲ τὰ Η, Θ σημεῖα· λέγω, ὅτι ἐστὶν ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς πρὸς τὴν ΔΕΖΘ πυραμίδα.

Εἰ γὰρ μή ἐστιν ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς πρὸς τὴν ΔΕΖΘ πυραμίδα, ὀέσται ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς ἤτοι πρὸς ἔλασσόν τι τῆς ΔΕΖΘ πυραμίδος στερεὸν ἢ πρὸς μεῖζον. ὀέστω πρότερον πρὸς ἔλασσον τὸ Θ, καὶ διῃρήσθω ἡ ΔΕΖΘ πυραμὶς εἰς τε δύο πυραμίδας ἴσας ἀλλήλαις καὶ ὁμοίας τῇ ὅλῃ καὶ εἰς δύο πρίσματα ἴσα· τὰ δὴ δύο πρίσαμτα μείζονά ἐστιν ἢ τὸ ἥμισυ τῆς ὅλης πυραμίδος. καὶ πάλιν αἱ ἐκ τῆς διαιρέσεως γινόμεναι πυραμίδες ὁμοίως διῃρήσθωσαν, καὶ τοῦτο ἀεὶ γινέσθω, ὀέως οὗ λειφθῶσί τινες πυραμίδες ἀπὸ τῆς ΔΕΖΘ πυραμίδος, αἱ εἰσιν ἐλάττονες τῆς ὑπεροθῆς, ᾗ ὑπερέθει ἡ ΔΕΖΘ πυραμὶς τοῦ Θ στερεοῦ. λελείφθωσαν καὶ ὀέστωσαν λόγου ὁένεκεν αἱ ΔΠΡΣ, ΣΤΥΘ· λοιπὰ ἄρα τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα μείζονά ἐστι τοῦ Θ στερεοῦ. διῃρήσθω καὶ ἡ ΑΒΓΗ πυραμὶς ὁμοίως καὶ ἰσοπληθῶς τῇ ΔΕΖΘ πυραμίδι· ὀέστιν ἄρα ὡς ἡ ΑΒΓ βάσις πρὸς

Let there be pyramids of the same height whose bases (are) the triangles ABC and DEF, and apexes the points G and H (respectively). I say that as base ABC is to base DEF, so pyramid $ABCG$ (is) to pyramid $DEFH$.

For if base ABC is not to base DEF, as pyramid $ABCG$ (is) to pyramid $DEFH$, then base ABC will be to base DEF, as pyramid $ABCG$ (is) to some solid either less than, or greater than, pyramid $DEFH$. Let it, first of all, be (in this ratio) to (some) lesser (solid), W. And let pyramid $DEFH$ be divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms. So, the (sum of the) two prisms is greater than half of the whole pyramid [Prop. 12.3]. And, again, let the pyramids generated by the division be similarly divided, and let this be done continually until some pyramids are left from pyramid $DEFH$ which (when added together) are less than the excess by which pyramid $DEFH$ exceeds the solid W [Prop. 10.1]. Let them be left, and, for the sake of argument, let them be $DQRS$ and $STUH$. Thus, the (sum of the) remaining prisms within pyramid $DEFH$ is greater than solid W. Let pyramid $ABCG$ also be divided simi-

τὴν ΔΕΖ βάσιν, οὕτως τὰ ἐν τῇ ΑΒΓΗ πυραμίδι πρίσματα πρὸς τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα, ἀλλὰ καὶ ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶξ πρὸς τὸ Θ στερεόν· καὶ ὡς ἄρα ἡ ΑΒΓΗ πυραμὶξ πρὸς τὸ Θ στερεόν, οὕτως τὰ ἐν τῇ ΑΒΓΗ πυραμίδι πρίσματα πρὸς τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα· ἐναλλὰχ ἄρα ὡς ἡ ΑΒΓΗ πυραμὶξ πρὸς τὰ ἐν αὐτῇ πρίσματα, οὕτως τὸ Θ στερεὸν πρὸς τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα. μείζων δὲ ἡ ΑΒΓΗ πυραμὶξ τῶν ἐν αὐτῇ πρισμάτων· μεῖζον ἄρα καὶ τὸ Θ στερεὸν τῶν ἐν τῇ ΔΕΖΘ πυραμίδι πρισμάτων. ἀλλὰ καὶ ἐλαττον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐστὶν ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶξ πρὸς ἔλασσόν τι τῆς ΔΕΖΘ πυραμίδος στερεόν. ὁμοίως δὴ δειθθήσεται, ὅτι οὐδὲ ὡς ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, οὕτως ἡ ΔΕΖΘ πυραμὶξ πρὸς ἔλαττόν τι τῆς ΑΒΓΗ πυραμίδος στερεόν.

Λέγω δή, ὅτι οὐκ ἔστιν οὐδὲ ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶξ πρὸς μεῖζόν τι τῆς ΔΕΖΘ πυραμίδος στερεόν.

Εἰ γὰρ δυνατόν, ἔστω πρὸς μεῖζον τὸ Θ· ἀνάπαλιν ἄρα ἐστὶν ὡς ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, οὕτως τὸ Θ στερεὸν πρὸς τὴν ΑΒΓΗ πυραμίδα. ὡς δὲ τὸ Θ στερεὸν πρὸς τὴν ΑΒΓΗ πυραμίδα, οὕτως ἡ ΔΕΖΘ πυραμὶξ πρὸς ἔλασσόν τι τῆς ΑΒΓΗ πυραμίδος, ὡξἐμπροσθεν ἐδείθθη· καὶ ὡς ἄρα ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, οὕτως ἡ ΔΕΖΘ πυραμὶξ πρὸς ἔλασσόν τι τῆς ΑΒΓΗ πυραμίδος· ὅπερ ἀτοπον ἐδείθθη. οὐκ ἄρα ἐστὶν ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶξ πρὸς μεῖζόν τι τῆς ΔΕΖΘ πυραμίδος στερεόν. ἐδείθθη δέ, ὅτι οὐδὲ πρὸς ἔλασσον. ἐστιν ἄρα ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶξ πρὸς τὴν ΔΕΖΘ πυραμίδα· ὅπερ ἔδει δεῖχαι.

larly, and a similar number of times, as pyramid $DEFH$. Thus, as base ABC is to base DEF, so the (sum of the) prisms within pyramid $ABCG$ (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 12.4]. But, also, as base ABC (is) to base DEF, so pyramid $ABCG$ (is) to solid W. And, thus, as pyramid $ABCG$ (is) to solid W, so the (sum of the) prisms within pyramid $ABCG$ (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.11]. Thus, alternately, as pyramid $ABCG$ (is) to the (sum of the) prisms within it, so solid W (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.16]. And pyramid $ABCG$ (is) greater than the (sum of the) prisms within it. Thus, solid W (is) also greater than the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.14]. But, (it is) also less. This very thing is impossible. Thus, as base ABC is to base DEF, so pyramid $ABCG$ (is) not to some solid less than pyramid $DEFH$. So, similarly, we can show that base DEF is not to base ABC, as pyramid $DEFH$ (is) to some solid less than pyramid $ABCG$ either.

So, I say that neither is base ABC to base DEF, as pyramid $ABCG$ (is) to some solid greater than pyramid $DEFH$.

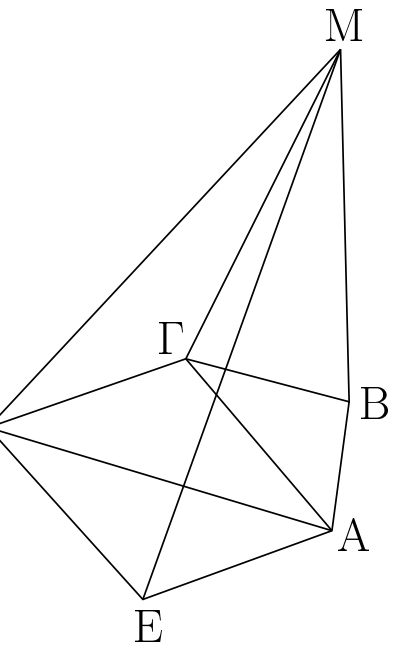
For, if possible, let it be (in this ratio) to some greater (solid), W. Thus, inversely, as base DEF (is) to base ABC, so solid W (is) to pyramid $ABCG$ [Prop. 5.7. corr.]. And as solid W (is) to pyramid $ABCG$, so pyramid $DEFH$ (is) to some (solid) less than pyramid $ABCG$, as shown before [Prop. 12.2 lem.]. And, thus, as base DEF (is) to base ABC, so pyramid $DEFH$ (is) to some (solid) less than pyramid $ABCG$ [Prop. 5.11]. The very thing was shown (to be) absurd. Thus, base ABC is not to base DEF, as pyramid $ABCG$ (is) to some solid greater than pyramid $DEFH$. And, it was shown that neither (is it in this ratio) to a lesser (solid). Thus, as base ABC is to base DEF, so pyramid $ABCG$ (is) to pyramid $DEFH$. (Which is) the very thing it was required to show.

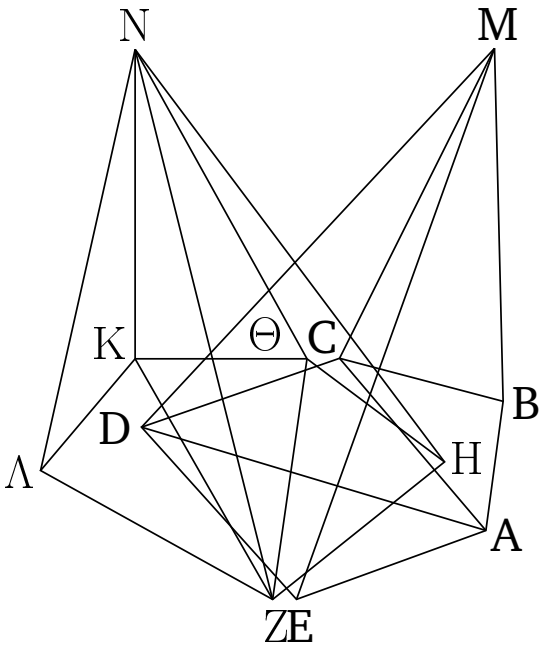
6.

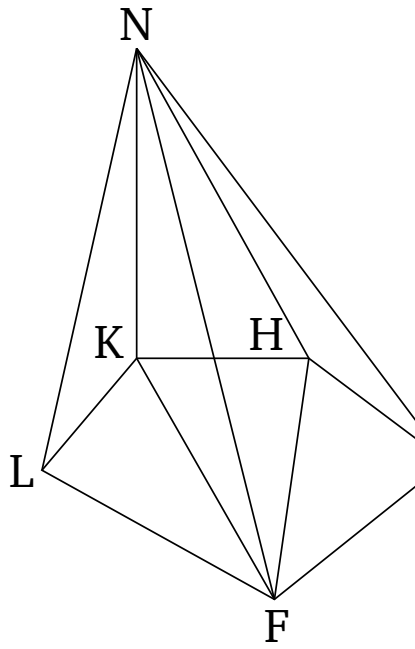
Αἱ ὑπὸ τὸ αὐτὸ ὕψος οὐσαι πυραμίδες καὶ πολυγώνουςὸέθουσαι βάσεις πρὸς ἀλλήλας εἰσὶν ὡς αἱ βάσεις.

Proposition 6

Pyramids which are of the same height, and have polygonal bases, are to one another as their bases.







Ἔστωσαν ὑπὸ τὸ αὐτὸ ὕψος πυραμίδες, ὧν [αἱ] βάσεις μὲν τὰ ΑΒΓΔΕ, ΖΗΘΚΛ πολύγωνα, κορυφαὶ δὲ τὰ Μ, Ν σημεῖα· λέγω, ὅτι ἐστὶν ὡς ἡ ΑΒΓΔΕ βάσις πρὸς τὴν ΖΗΘΚΛ βάσιν, οὕτως ἡ ΑΒΓΔΕΜ πυραμὶξ πρὸς τὴν ΖΗΘΚΛΝ πυραμίδα.

Ἐπεζεύθθωσαν γὰρ αἱ ΑΓ, ΑΔ, ΖΘ, ΖΚ. ἐπεὶ οὖν δύο πυραμίδες εἰσὶν αἱ ΑΒΓΜ, ΑΓΔΜ τριγώνους ὀέθουσαι βάσεις καὶ ὕψος ἴσον, πρὸς ἀλλήλας εἰσὶν ὡς αἱ βάσεις· ὀέστιν ἄρα ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΑΓΔ βάσιν, οὕτως ἡ ΑΒΓΜ πυραμὶξ πρὸς τὴν ΑΓΔΜ πυραμίδα. καὶ συνθέντι ὡς ἡ ΑΒΓΔ βάσις πρὸς τὴν ΑΓΔ βάσιν, οὕτως ἡ ΑΒΓΔΜ πυραμὶξ πρὸς τὴν ΑΓΔΜ πυραμίδα. ἀλλὰ καὶ ὡς ἡ ΑΓΔ βάσις πρὸς τὴν ΑΔΕ βάσιν, οὕτως ἡ ΑΓΔΜ πυραμὶξ πρὸς τὴν ΑΔΕΜ πυραμίδα. διὸ ὀίσου ἄρα ὡς ἡ ΑΒΓΔ βάσις πρὸς τὴν ΑΔΕ βάσιν, οὕτως ἡ ΑΒΓΔΜ πυραμὶξ πρὸς τὴν ΑΔΕΜ πυραμίδα. καὶ συνθέντι πάλιν, ὡς ἡ ΑΒΓΔΕ βάσις πρὸς τὴν ΑΔΕ βάσιν, οὕτως ἡ ΑΒΓΔΕΜ πυραμὶξ πρὸς τὴν ΑΔΕΜ πυραμίδα. ὁμοίως δὴ δειθθήσεται, ὅτι καὶ ὡς ἡ ΖΗΘΚΛ βάσις πρὸς τὴν ΖΗΘ βάσιν, οὕτως καὶ ἡ ΖΗΘΚΛΝ πυραμὶξ πρὸς τὴν ΖΗΘΝ πυραμίδα. καὶ ἐπεὶ δύο πυραμίδες εἰσὶν αἱ ΑΔΕΜ, ΖΗΘΝ τριγώνους ὀέθουσαι βάσεις καὶ ὕψος ἴσον, ὀέστιν ἄρα ὡς ἡ ΑΔΕ βάσις πρὸς τὴν ΖΗΘ βάσιν, οὕτως ἡ ΑΔΕΜ πυραμὶξ πρὸς τὴν ΖΗΘΝ πυραμίδα. ἀλλὰ ὡς ἡ ΑΔΕ βάσις πρὸς τὴν ΑΒΓΔΕ βάσιν, οὕτως ὂν ἡ ΑΔΕΜ πυραμὶξ πρὸς τὴν ΑΒΓΔΕΜ πυραμίδα. καὶ διὸ ὀίσου ἄρα ὡς ἡ ΑΒΓΔΕ βάσις πρὸς τὴν ΖΗΘ βάσιν, οὕτως ἡ ΑΒΓΔΕΜ πυραμὶξ πρὸς τὴν ΖΗΘΝ πυραμίδα. ἀλλὰ μὴν καὶ ὡς ἡ ΖΗΘ βάσις πρὸς τὴν ΖΗΘΚΛ βάσιν, οὕτως ὂν καὶ ἡ ΖΗΘΝ πυραμὶξ πρὸς τὴν ΖΗΘΚΛΝ πυραμίδα, καὶ διὸ ὀίσου ἄρα ὡς ἡ ΑΒΓΔΕ βάσις πρὸς τὴν ΖΗΘΚΛ βάσιν, οὕτως ἡ ΑΒΓΔΕΜ πυραμὶξ πρὸς τὴν ΖΗΘΚΛΝ πυραμίδα· ὅπερ

Let there be pyramids of the same height whose bases (are) the polygons *ABCDE* and *FGHKL*, and apexes the points *M* and *N* (respectively). I say that as base *ABCDE* is to base *FGHKL*, so pyramid *ABCDEM* (is) to pyramid *FGHKLN*.

For let *AC*, *AD*, *FH*, and *FK* be joined. Therefore, since *ABCM* and *ACDM* are two pyramids having triangular bases and equal height, they are to one another as their bases [Prop. 12.5]. Thus, as base *ABC* is to base *ACD*, so pyramid *ABCM* (is) to pyramid *ACDM*. And, via composition, as base *ABCD* (is) to base *ACD*, so pyramid *ABCDM* (is) to pyramid *ACDM* [Prop. 5.18]. But, as base *ACD* (is) to base *ADE*, so pyramid *ACDM* (is) also to pyramid *ADEM* [Prop. 12.5]. Thus, via equality, as base *ABCD* (is) to base *ADE*, so pyramid *ABCDM* (is) to pyramid *ADEM* [Prop. 5.22]. And, again, via composition, as base *ABCDE* (is) to base *ADE*, so pyramid *ABCDEM* (is) to pyramid *ADEM* [Prop. 5.18]. So, similarly, it can also be shown that as base *FGHKL* (is) to base *FGH*, so pyramid *FGHKLN* (is) also to pyramid *FGHN*. And since *ADEM* and *FGHN* are two pyramids having triangular bases and equal height, thus as base *ADE* (is) to base *FGH*, so pyramid *ADEM* (is) to pyramid *FGHN* [Prop. 12.5]. But, as base *ADE* (is) to base *ABCDE*, so pyramid *ADEM* (was) to pyramid *ABCDEM*. Thus, via equality, as base *ABCDE* (is) to base *FGH*, so pyramid *ABCDEM* (is) also to pyramid *FGHN* [Prop. 5.22]. But, furthermore, as base *FGH* (is) to base *FGHKL*, so pyramid *FGHN* was also to pyramid *FGHKLN*. Thus, via equality, as base *ABCDE* (is) to base *FGHKL*, so pyramid *ABCDEM* (is) also to pyramid *FGHKLN* [Prop. 5.22]. (Which is) the very thing it was required to show.

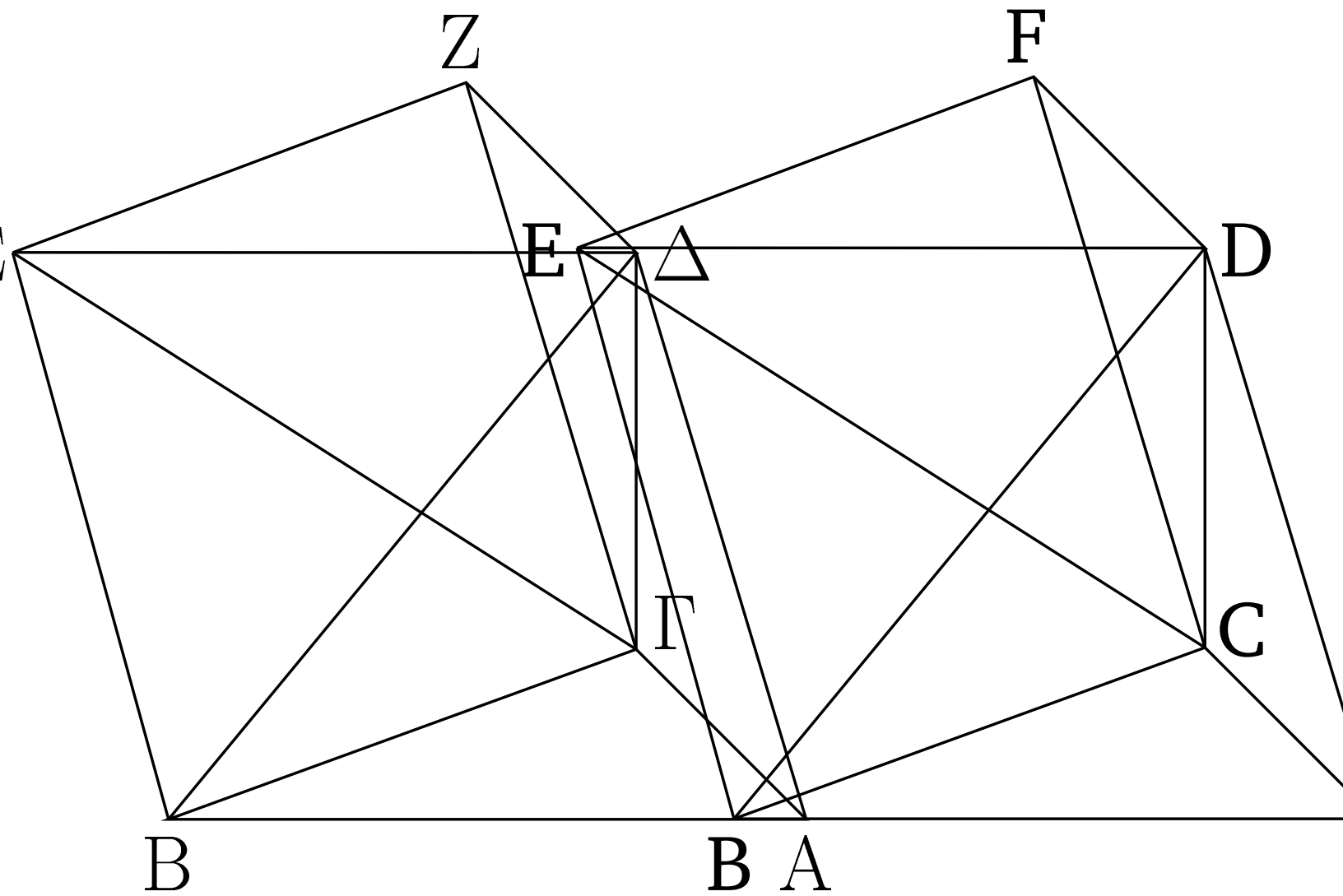
ἔδει δεῖχαι.

7.

Πᾶν πρίσμα τρίγωνον ἔχον βάσιν διαιρεῖται εἰξ τρεῖς πυραμίδας ἴσας ἀλλήλαις τριγώνους βάσεις ἐθούσας.

Proposition 7

Any prism having a triangular base is divided into three pyramids having triangular bases (which are) equal to one another.



Ἔστω πρίσμα, οὗ βάσις μὲν τὸ ΑΒΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΔΕΖ· λέγω, ὅτι τὸ ΑΒΓΔΕΖ πρίσμα διαιρεῖται εἰξ τρεῖς πυραμίδας ἴσας ἀλλήλαις τριγώνους ἐθούσας βάσεις.

Ἐπεζεύθθωσαν γὰρ αἱ ΒΔ, ΕΓ, ΓΔ. ἐπεὶ παραλληλόγραμμόν ἐστι τὸ ΑΒΕΔ, διάμετρος δὲ αὐτοῦ ἐστιν ἡ ΒΔ, ἴσον ἄρα ἐστι τὸ ΑΒΔ τρίγωνον τῷ ΕΒΔ τρίγωνῳ· καὶ ἡ πυραμὶξ ἄρα, ἧξ βάσις μὲν τὸ ΑΒΔ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ὁίση ἐστὶ πυραμίδι, ἧξ βάσις μέν ἐστι τὸ ΔΕΒ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον. ἀλλὰ ἡ πυραμίξ, ἧξ βάσις μέν ἐστι τὸ ΔΕΒ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ἡ αὐτή

Let there be a prism whose base (is) triangle ABC, and opposite (plane) DEF. I say that prism $ABCDEF$ is divided into three pyramids having triangular bases (which are) equal to one another.

For let BD, EC, and CD be joined. Since $ABED$ is a parallelogram, and BD is its diagonal, triangle ABD is thus equal to triangle EBD [Prop. 1.34]. And, thus, the pyramid whose base (is) triangle ABD, and apex the point C, is equal to the pyramid whose base is triangle DEB, and apex the point C [Prop. 12.5]. But, the pyramid whose base is triangle DEB,

ἐστι πυραμίδι, ἧξ βάσις μέν ἐστι τὸ ΕΒΓ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον· ὑπὸ γὰρ τῶν αὐτῶν ἐπιπέδων περιέθεται. καὶ πυραμὶξ ἄρα, ἧξ βάσις μέν ἐστι τὸ ΑΒΔ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ὁίση ἐστὶ πυραμίδι, ἧξ βάσις μέν ἐστι τὸ ΕΒΓ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον. πάλιν, ἐπεὶ παραλληλόγραμμόν ἐστι τὸ ΖΓΒΕ, διάμετρος δέ ἐστιν αὐτοῦ ἡ ΓΕ, ἴσον ἐστὶ τὸ ΓΕΖ τρίγωνον τῷ ΓΒΕ τριγώνῳ. καὶ πυραμὶξ ἄρα, ἧξ βάσις μέν ἐστι τὸ ΒΓΕ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, ὁίση ἐστὶ πυραμίδι, ἧξ βάσις μέν ἐστι τὸ ΕΓΖ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον. ἡ δὲ πυραμίξ, ἧξ βάσις μέν ἐστι τὸ ΒΓΕ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, ὁίση ἐδείχθη πυραμίδι, ἧξ βάσις μέν ἐστι τὸ ΑΒΔ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον· καὶ πυραμὶξ ἄρα, ἧξ βάσις μέν ἐστι τὸ ΓΕΖ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, ὁίση ἐστὶ πυραμίδι, ἧξ βάσις μέν [ἐστι] τὸ ΑΒΔ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον· διῄρηται ἄρα τὸ ΑΒΓΔΕΖ πρίσμα εἰξ τρεῖς πυραμίδας ἴσας ἀλλήλαις τριγώνους ἐθούσας βάσεις.

Καὶ ἐπεὶ πυραμίξ, ἧξ βάσις μέν ἐστι τὸ ΑΒΔ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ἡ αὐτή ἐστι πυραμίδι, ἧξ βάσις τὸ ΓΑΒ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον· ὑπὸ γὰρ τῶν αὐτῶν ἐπιπέδων περιέθονται· ἡ δὲ πυραμίξ, ἧξ βάσις τὸ ΑΒΔ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, τρίτον ἐδείθθη τοῦ πρίσματος, οὗ βάσις τὸ ΑΒΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΔΕΖ, καὶ ἡ πυραμὶξ ἄρα, ἧξ βάσις τὸ ΑΒΓ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, τρίτον ἐστὶ τοῦ πρίσματος τοῦ ἔχοντος βάσις τὴν αὐτὴν τὸ ΑΒΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΔΕΖ.

and apex the point C, is the same as the pyramid whose base is triangle EBC, and apex the point D. For they are contained by the same planes. And, thus, the pyramid whose base is ABD, and apex the point C, is equal to the pyramid whose base is EBC and apex the point D. Again, since $FCBE$ is a parallelogram, and CE is its diagonal, triangle CEF is equal to triangle CBE [Prop. 1.34]. And, thus, the pyramid whose base is triangle BCE, and apex the point D, is equal to the pyramid whose base is triangle ECF, and apex the point D [Prop. 12.5]. And the pyramid whose base is triangle BCE, and apex the point D, was shown (to be) equal to the pyramid whose base is triangle ABD, and apex the point C. Thus, the pyramid whose base is triangle CEF, and apex the point D, is also equal to the pyramid whose base [is] triangle ABD, and apex the point C. Thus, the prism $ABCDEF$ has been divided into three pyramids having triangular bases (which are) equal to one another.

And since the pyramid whose base is triangle ABD, and apex the point C, is the same as the pyramid whose base is triangle CAB, and apex the point D. For they are contained by the same planes. And the pyramid whose base (is) triangle ABD, and apex the point C, was shown (to be) a third of the prism whose base is triangle ABC, and opposite (plane) DEF, thus the pyramid whose base is triangle ABC, and apex the point D, is also a third of the pyramid having the same base, triangle ABC, and opposite (plane) DEF.

Πόρισμα.

Ἔκ δὴ τούτου φανερόν, ὅτι πᾶσα πυραμὶξ τρίτον μέρος ἐστὶ τοῦ πρίσματος τοῦ τὴν αὐτὴν βάσιν ἔχοντος αὐτῇ καὶ ὕψος ἴσον· ὅπερ ἔδει δεῖξαι.

Corollary

And, from this, (it is) clear that any pyramid is the third part of the prism having the same base as it, and an equal height. (Which is) the very thing it was required to show.

8.

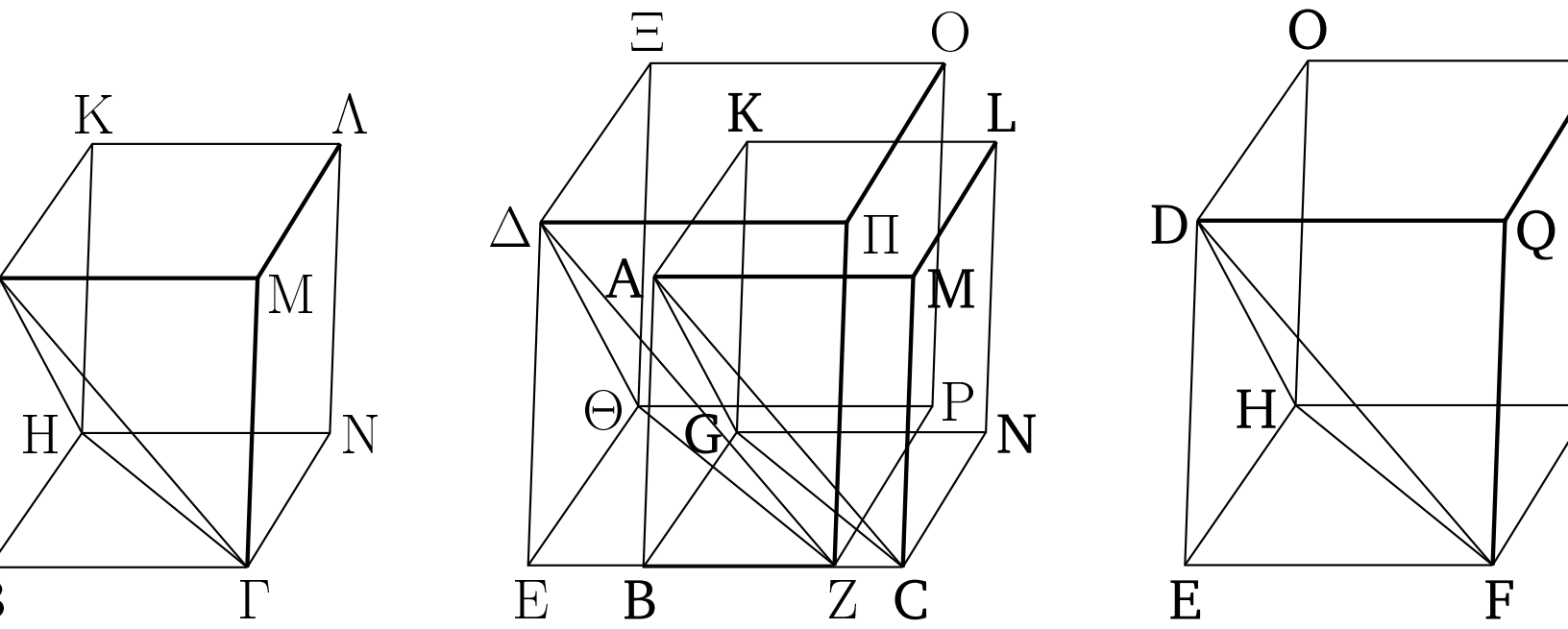
Αἱ ὅμοιαι πυραμίδες καὶ τριγώνουςὀέθουσαι βάσεις ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν.

Ἔστωσαν ὅμοιαι καὶ ὁμοίως κείμεναι πυραμίδες, ὧν βάσεις μέν εἰσι τὰ ΑΒΓ, ΔΕΖ τρίγωνα, κορυφαὶ δὲ τὰ Η, Θ σημεῖα· λέγω, ὅτι ἡ ΑΒΓΗ πυραμὶξ πρὸς τὴν ΔΕΖΘ πυραμίδα τριπλασίονα λόγον ἔχει ἤπερ ἡ ΒΓ πρὸς τὴν ΕΖ.

Proposition 8

Similar pyramids which also have triangular bases are in the cubed ratio of their corresponding sides.

Let there be similar, and similarly laid out, pyramids whose bases are triangles ABC and DEF, and apexes the points G and H (respectively). I say that pyramid $ABCG$ has to pyramid $DEFH$ the cubed ratio of that BC (has) to EF.



Συμπεπληρώσθω γὰρ τὰ ΒΗΜΛ, ΕΘΠΟ στερεὰ παραλληλεπίπεδα. καὶ ἐπεὶ ὅμοιά ἐστὶν ἡ ΑΒΓΗ πυραμὶς τῇ ΔΕΖΘ πυραμίδι, ὅίση ἄρα ἐστὶν ἡ μὲν ὑπὸ ΑΒΓ γωνία τῇ ὑπὸ ΔΕΖ γωνίᾳ, ἡ δὲ ὑπὸ ΗΒΓ τῇ ὑπὸ ΘΕΖ, ἡ δὲ ὑπὸ ΑΒΗ τῇ ὑπὸ ΔΕΘ, καί ἐστιν ὡξ ἡ ΑΒ πρὸς τὴν ΔΕ, οὕτως ἡ ΒΓ πρὸς τὴν ΕΖ, καὶ ἡ ΒΗ πρὸς τὴν ΕΘ. καὶ ἐπεί ἐστιν ὡξ ἡ ΑΒ πρὸς τὴν ΔΕ, οὕτως ἡ ΒΓ πρὸς τὴν ΕΖ, καὶ περὶ ἴσας γωνίας αἱ πλευραὶ ἀνάλογόν εἰσιν, ὅμοιον ἄρα ἐστὶ τὸ ΒΜ παραλληλόγραμμον τῷ ΕΠ παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν ΒΝ τῷ ΕΡ ὅμοιόν ἐστι, τὸ δὲ ΒΚ τῷ ΕΧ· τὰ τρία ἄρα τὰ ΜΒ, ΒΚ, ΒΝ τρισὶ τοῖς ΕΠ, ΕΧ, ΕΡ ὅμοιά ἐστιν. ἀλλὰ τὰ μὲν τρία τὰ ΜΒ, ΒΚ, ΒΝ τρισὶ τοῖς ἀπεναντίον ἴσα τε καὶ ὅμοιά ἐστιν, τὰ δὲ τρία τὰ ΕΠ, ΕΧ, ΕΡ τρισὶ τοῖς ἀπεναντίον ἴσα τε καὶ ὅμοιά ἐστιν. τὰ ΒΗΜΛ, ΕΘΠΟ ἄρα στερεὰ ὑπὸ ὁμοίων ἐπιπέδων ὑίσων τὸ πλῆθος περιέθεται. ὅμοιον ἄρα ἐστὶ τὸ ΒΗΜΛ στερεὸν τῷ ΕΘΠΟ στερεῷ. τὰ δὲ ὅμοια στερεὰ παραλληλεπίπεδα ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν. τὸ ΒΗΜΛ ἄρα στερεὸν πρὸς τὸ ΕΘΠΟ στερεὸν τριπλασίονα λόγον ἔχει ἤπερ ἡ ὁμόλογος πλευρὰ ἡ ΒΓ πρὸς τὴν ὁμόλογον πλευρὰν τὴν ΕΖ. ὡξ δὲ τὸ ΒΗΜΛ στερεὸν πρὸς τὸ ΕΘΠΟ στερεόν, οὕτως ἡ ΑΒΓΗ πυραμὶξ πρὸς τὴν ΔΕΖΘ πυραμίδα, ἐπειδήπερ ἡ πυραμὶξ ὲκτον μέρος ἐστὶ τοῦ στερεοῦ διὰ τὸ καὶ τὸ πρίσμα ἥμισυ ὂν τοῦ στερεοῦ παραλληλεπιπέδου τριπλάσιον εὸῖναι τῆς πυραμίδος. καὶ ἡ ΑΒΓΗ ἄρα πυραμὶξ πρὸς τὴν ΔΕΖΘ πυραμίδα τριπλασίονα λόγον ἔχει ἤπερ ἡ ΒΓ πρὸς τὴν ΕΖ· ὅπερ ἔδει δεῖχαι.

For let the parallelepiped solids $BGML$ and $EHQP$ be completed. And since pyramid $ABCG$ is similar to pyramid $DEFH$, angle ABC is thus equal to angle DEF, and GBC to HEF, and ABG to DEH. And as AB is to DE, so BC (is) to EF, and BG to EH [Def. 11.9]. And since as AB is to DE, so BC (is) to EF, and (so) the sides around equal angles are proportional, parallelogram BM is thus similar to parallelogram EQ. So, for the same (reasons), BN is also similar to ER, and BK to EO. Thus, the three (parallelograms) MB, BK, and BN are similar to the three (parallelograms) EQ, EO, ER (respectively). But, the three (parallelograms) MB, BK, and BN are (both) equal and similar to the three opposite (parallelograms), and the three (parallelograms) EQ, EO, and ER are (both) equal and similar to the three opposite (parallelograms) [Prop. 11.24]. Thus, the solids $BGML$ and $EHQP$ are contained by equal numbers of similar (and similarly laid out) planes. Thus, solid $BGML$ is similar to solid $EHQP$ [Def. 11.9]. And similar parallelepiped solids are in the cubed ratio of corresponding sides [Prop. 11.33]. Thus, solid $BGML$ has to solid $EHQP$ the cubed ratio that the corresponding side BC (has) to the corresponding side EF. And as solid $BGML$ (is) to solid $EHQP$, so pyramid $ABCG$ (is) to pyramid $DEFH$, inasmuch as the pyramid is the sixth part of the solid, on account of the prism, being half of the parallelepiped solid [Prop. 11.28], also being three times the pyramid [Prop. 12.7]. Thus, pyramid $ABCG$ also has to pyramid $DEFH$ the cubed ratio that BC (has) to EF. (Which is) the very thing it was required to show.

Πόρισμα.

Corollary

Ἐκ δὴ τούτου φανερόν, ὅτι καὶ αἱ πολυγώνουςὸέθουσαι βάσεις ὅμοιαι πυραμίδες πρὸς ἀλλήλας ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν. διαιρεθεισῶν γὰρ αὐτῶν εἰξ τὰξ ἐν αὐταῖς πυραμίδας τριγώνους βάσεις ἐθούσας τῷ καὶ τὰ ὅμοια πολύγωνα τῶν βάσεων εἰξ ὅμοια τρίγωνα διαιρεῖσθαι καὶ ἴσα τῷ πλήθει καὶ ὁμόλογα τοῖς ὅλοιςὸέσται ὡξ [ἡ] ἐν τῇ ἑτέρᾳ μία πυραμὶξ τρίγωνονὸέθουσα βάσιν πρὸς τὴν ἐν τῇ ἑτέρᾳ μίαν πυραμίδα τρίγωνονὸέθουσαν βάσιν, οὕτως καὶ ὁάπασαι αἱ ἐν τῇ ἑτέρᾳ πυραμίδι πυραμίδες τριγώνουςὸέθουσαι βάσεις πρὸς τὰξ ἐν τῇ ἑτέρᾳ πυραμίδι πυραμίδας τριγώνους βάσεις ἐθούσας, τουτέστιν αὐτὴ ἡ πολύγωνον βάσινὸέθουσα πυραμὶξ πρὸς τὴν πολύγωνον βάσινὸέθουσαν πυραμίδα. ἡ δὲ τρίγωνον βάσινὸέθουσα πυραμὶξ πρὸς τὴν τρίγωνον βάσινὸέθουσαν ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγον πλευρῶν· καὶ ἡ πολύγωνον ἄρα βάσινὸέθουσα πρὸς τὴν ὁμοίαν βάσινὸέθουσαν τριπλασίονα λόγον ἔχει ἥπερ ἡ πλευρὰ πρὸς τὴν πλευράν.

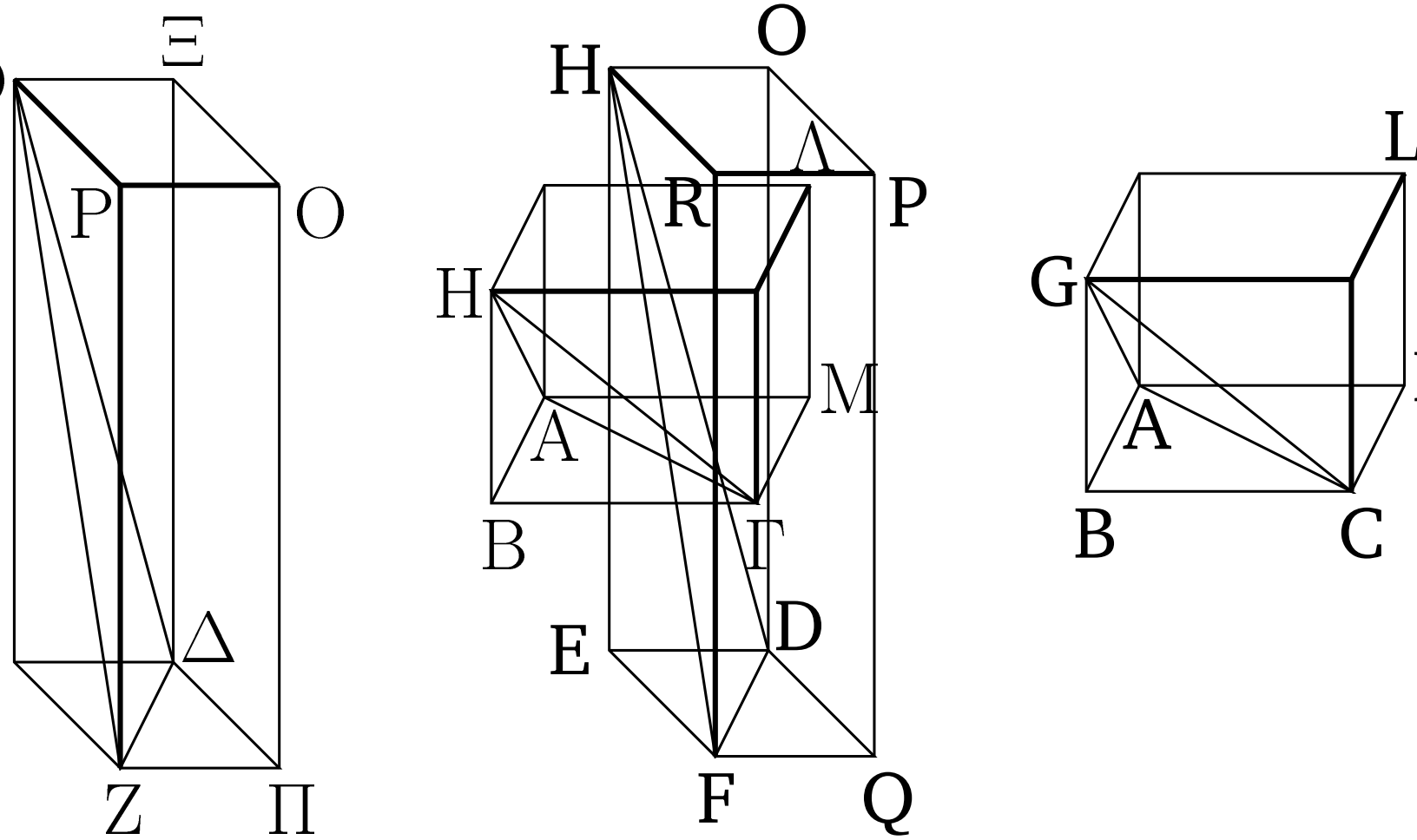
So, from this, (it is) also clear that similar pyramids having polygonal bases (are) to one another as the cubed ratio of their corresponding sides. For, dividing them into the pyramids (contained) within them which have triangular bases, with the similar polygons of the bases also being divided into similar triangles (which are) both equal in number, and corresponding, to the wholes [Prop. 6.20]. As one pyramid having a triangular base in the former (pyramid having a polygonal base is) to one pyramid having a triangular base in the latter (pyramid having a polygonal base), so (the sum of) all the pyramids having triangular bases in the former pyramid will also be to (the sum of) all the pyramids having triangular bases in the latter pyramid [Prop. 5.12]—that is to say, the (former) pyramid itself having a polygonal base to the (latter) pyramid having a polygonal base. And a pyramid having a triangular base is to a (pyramid) having a triangular base in the cubed ratio of corresponding sides [Prop. 12.8]. Thus, a (pyramid) having a polygonal base also has to to a (pyramid) having a similar base the cubed ratio of a (corresponding) side to a (corresponding) side.

9.

Τῶνὁίσων πυραμίδων καὶ τριγώνους βάσεις ἐθουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν· καὶ ὧν πυραμίδων τριγώνους βάσεις ἐθουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν, ἴσαι εἰσὶν ἐκεῖναι.

Proposition 9

The bases of equal pyramids which also have triangular bases are reciprocally proportional to their heights. And those pyramids which have triangular bases whose bases are reciprocally proportional to their heights are equal.



Ἔστωσαν γὰρ ἴσαι πυραμίδες τριγώνους βάσεις ὑπὸ ἐθουσαι τὰξ ΑΒΓ, ΔΕΖ, κορυφὰξ δὲ τὰ Η, Θ σημεῖα· λέγω, ὅτι τῶν ΑΒΓΗ, ΔΕΖΘ πυραμίδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν, καί ἐστιν ὡξ ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὑτως τὸ τῆς ΔΕΖΘ πυραμίδος ὑψος πρὸς τὸ τῆς ΑΒΓΗ πυραμίδος ὑψος.

Συμπεπληρώσθω γὰρ τὰ ΒΗΜΛ, ΕΘΠΟ στερεὰ παραλληλεπίπεδα. καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΑΒΓΗ πυραμὶξ τῇ ΔΕΖΘ πυραμίδι, καί ἐστι τῆς μὲν ΑΒΓΗ πυραμίδος ἑξαπλάσιον τὸ ΒΗΜΛ στερεόν, τῆς δὲ ΔΕΖΘ πυραμίδος ἑξαπλάσιον τὸ ΕΘΠΟ στερεόν, ἴσον ἄρα ἐστὶ τὸ ΒΗΜΛ στερεὸν τῷ ΕΘΠΟ στερεῷ. τῶν δὲ ὁίσων στερεῶν παραλληλεπιπώδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν· ὲστιν ἄρα ὡξ ἡ ΒΜ βάσις πρὸς τὴν ΕΠ βάσιν, οὑτως τὸ τοῦ ΕΘΠΟ στερεοῦ ὑψος πρὸς τὸ τοῦ ΒΗΜΛ στερεοῦ ὑψος. ἀλλὸ ὡξ ἡ ΒΜ βάσις πρὸς τὴν ΕΠ, οὑτως τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΔΕΖ τρίγωνον. καὶ ὡξ ἄρα τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΔΕΖ τρίγωνον, οὑτως τὸ τοῦ ΕΘΠΟ στερεοῦ ὑψος πρὸς τὸ τοῦ ΒΗΜΛ στερεοῦ ὑψος. ἀλλὰ τὸ μὲν τοῦ ΕΘΠΟ στερεοῦ ὑψος τὸ αὐτὸ ἐστι τῷ τῆς ΔΕΖΘ πυραμίδος ὑψει, τὸ δὲ τοῦ ΒΗΜΛ

For let there be (two) equal pyramids having the triangular bases *ABC* and *DEF*, and apexes the points *G* and *H* (respectively). I say that the bases of the pyramids *ABCG* and *DEFH* are reciprocally proportional to their heights, and (so) that as base *ABC* is to base *DEF*, so the height of pyramid *DEFH* (is) to the height of pyramid *ABCG*.

For let the parallelepiped solids *BGML* and *EHQP* be completed. And since pyramid *ABCG* is equal to pyramid *DEFH*, and solid *BGML* is six times pyramid *ABCG* (see previous proposition), and solid *EHQP* (is) six times pyramid *DEFH*, solid *BGML* is thus equal to solid *EHQP*. And the bases of equal parallelepiped solids are reciprocally proportional to their heights [Prop. 11.34]. Thus, as base *BM* is to base *EQ*, so the height of solid *EHQP* (is) to the height of solid *BGML*. But, as base *BM* (is) to base *EQ*, so triangle *ABC* (is) to triangle *DEF* [Prop. 1.34]. And, thus, as triangle *ABC* (is) to triangle *DEF*, so the height of solid *EHQP* (is) to the height of solid *BGML* [Prop. 5.11]. But, the height of solid *EHQP* is the same as the height of pyramid *DEFH*, and the height of solid *BGML* is the same as the height of pyramid

στερεοῦ ὕψος τὸ αὐτό ἐστι τῷ τῆς ΑΒΓΗ πυραμίδος ὕψει· ὀέστιν ἄρα ὡξ ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὸ τῆς ΔΕΖΘ πυραμίδος ὕψος πρὸς τὸ τῆς ΑΒΓΗ πυραμίδος ὕψος. τῶν ΑΒΓΗ, ΔΕΖΘ ἄρα πυραμίδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

Ἀλλὰ δὴ τῶν ΑΒΓΗ, ΔΕΖΘ πυραμίδων ἀντιπεπονθέτωσαν αἱ βάσεις τοῖς ὕψεσιν, καὶ ὀέστω ὡξ ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὸ τῆς ΔΕΖΘ πυραμίδος ὕψος πρὸς τὸ τῆς ΑΒΓΗ πυραμίδος ὕψος· λέγω, ὅτιὀίση ἐστὶν ἡ ΑΒΓΗ πυραμὶξ τῇ ΔΕΖΘ πυραμίδι.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεί ἐστιν ὡξ ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὸ τῆς ΔΕΖΘ πυραμίδος ὕψος πρὸς τὸ τῆς ΑΒΓΗ πυραμίδος ὕψος, ἀλλὰ ὡξ ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὸ ΒΜ παραλληλόγραμμον πρὸς τὸ ΕΠ παραλληλόγραμμον, καὶ ὡξ ἄρα τὸ ΒΜ παραλληλόγραμμον πρὸς τὸ ΕΠ παραλληλόγραμμον, οὕτως τὸ τῆς ΔΕΖΘ πυραμίδος ὕψος πρὸς τὸ τῆς ΑΒΓΗ πυραμίδος ὕψος. ἀλλὰ τὸ [μὲν] τῆς ΔΕΖΘ πυραμίδος ὕψος τὸ αὐτό ἐστι τῷ τοῦ ΕΘΠΟ παραλληλεπιπέδου ὕψει, τὸ δὲ τῆς ΑΒΓΗ πυραμίδος ὕψος τὸ αὐτό ἐστι τῷ τοῦ ΒΗΜΛ παραλληλεπιπέδου ὕψει· ὀέστιν ἄρα ὡξ ἡ ΒΜ βάσις πρὸς τὴν ΕΠ βάσιν, οὕτως τὸ τοῦ ΕΘΠΟ παραλληλεπιπέδου ὕψος πρὸς τὸ τοῦ ΒΗΜΛ παραλληλεπιπέδου ὕψος. ὧν δὲ στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, ἴσα ἐστὶν ἐκεῖνα· ἴσον ἄρα ἐστὶ τὸ ΒΗΜΛ στερεὸν παραλληλεπίπεδον τῷ ΕΘΠΟ στερεῷ παραλληλεπιπέδῳ. καί ἐστι τοῦ μὲν ΒΗΜΛ ἕκτον μέρος ἡ ΑΒΓΗ πυραμίξ, τοῦ δὲ ΕΘΠΟ παραλληλεπιπέδου ἕκτον μέρος ἡ ΔΕΖΘ πυραμίξ· ὀίση ἄρα ἡ ΑΒΓΗ πυραμὶξ τῇ ΔΕΖΘ πυραμίδι.

Τῶν ἄρα ἰσῶν πυραμίδων καὶ τριγώνους βάσεις ἐθουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν· καὶ ὧν πυραμίδων τριγώνους βάσεις ἐθουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, ἴσαι εἰσὶν ἐκεῖναι· ὅπερ ἔδει δεῖξαι.

10.

Πᾶξ κῶνος κυλίνδρου τρίτον μέρος ἐστὶ τοῦ τὴν αὐτὴν βάσιν ἔχοντος αὐτῷ καὶ ὕψος ἴσον.

Ἐθέτω γὰρ κῶνος κυλίνδρῳ βάσιν τε τὴν αὐτὴν τὸν ΑΒΓΔ κύκλον καὶ ὕψος ἴσον· λέγω, ὅτι ὁ κῶνος τοῦ κυλίνδρου τρίτον ἐστὶ μέρος, τουτέστιν ὅτι ὁ κύλινδρος τοῦ κώνου τριπλασίων ἐστίν.

ABCG. Thus, as base *ABC* is to base *DEF*, so the height of pyramid *DEFH* (is) to the height of pyramid *ABCG*. Thus, the bases of pyramids *ABCG* and *DEFH* are reciprocally proportional to their heights.

And so, let the bases of pyramids *ABCG* and *DEFH* be reciprocally proportional to their heights, and (thus) let base *ABC* be to base *DEF*, as the height of pyramid *DEFH* (is) to the height of pyramid *ABCG*. I say that pyramid *ABCG* is equal to pyramid *DEFH*.

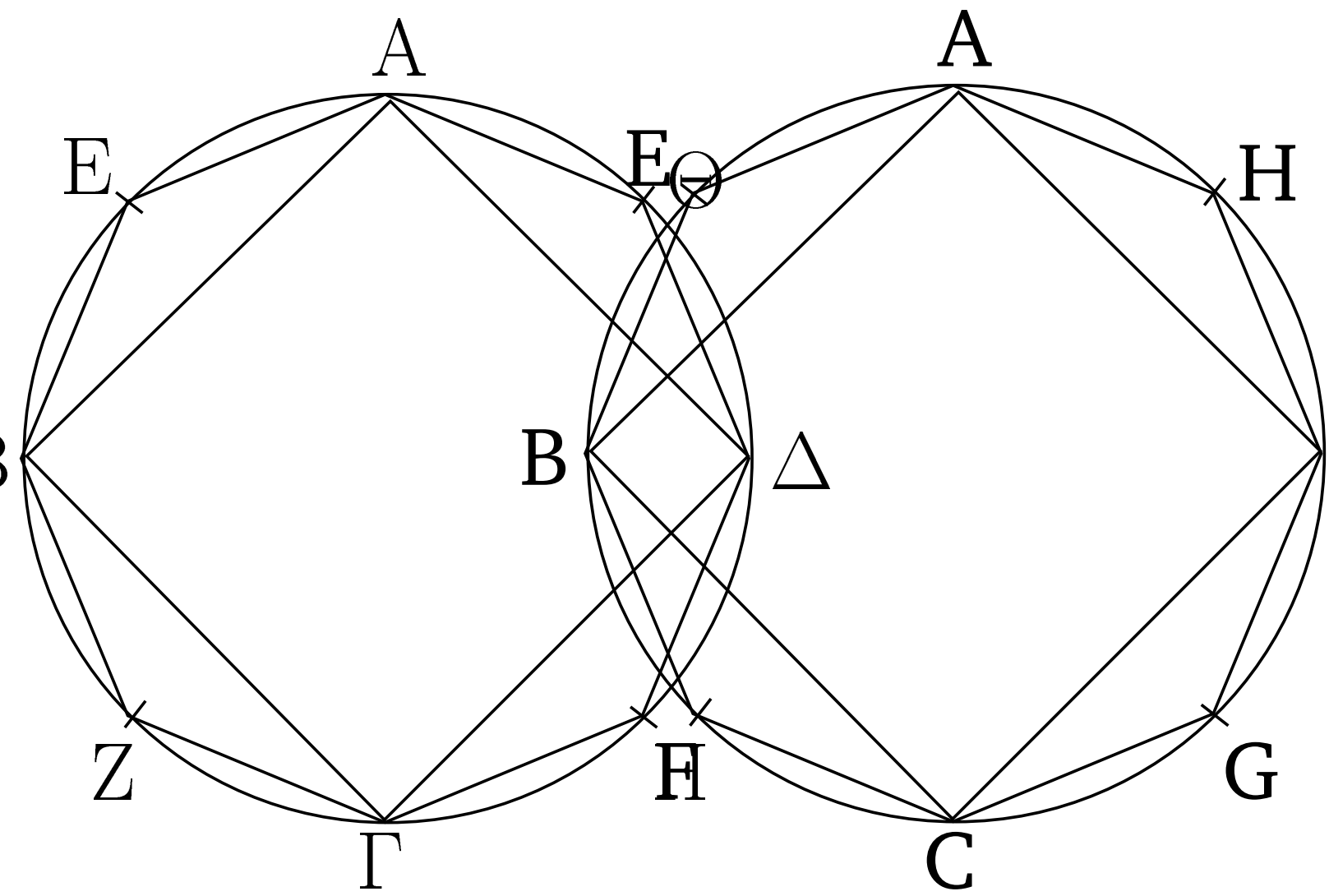
For, with the same construction, since as base *ABC* is to base *DEF*, so the height of pyramid *DEFH* (is) to the height of pyramid *ABCG*, but as base *ABC* (is) to base *DEF*, so parallelogram *BM* (is) to parallelogram *EQ* [Prop. 1.34], thus as parallelogram *BM* (is) to parallelogram *EQ*, so the height of pyramid *DEFH* (is) also to the height of pyramid *ABCG* [Prop. 5.11]. But, the height of pyramid *DEFH* is the same as the height of parallelepiped *EHQP*, and the height of pyramid *ABCG* is the same as the height of parallelepiped *BGML*. Thus, as base *BM* is to base *EQ*, so the height of parallelepiped *EHQP* (is) to the height of parallelepiped *BGML*. And those parallelepiped solids whose bases are reciprocally proportional to their heights are equal [Prop. 11.34]. Thus, the parallelepiped solid *BGML* is equal to the parallelepiped solid *EHQP*. And pyramid *ABCG* is a sixth part of *BGML*, and pyramid *DEFH* a sixth part of parallelepiped *EHQP*. Thus, pyramid *ABCG* is equal to pyramid *DEFH*.

Thus, the bases of equal pyramids which also have triangular bases are reciprocally proportional to their heights. And those pyramids having triangular bases whose bases are reciprocally proportional to their heights are equal. (Which is) the very thing it was required to show.

Proposition 10

Every cone is the third part of the cylinder which has the same base as it, and an equal height.

For let there be a cone (with) the same base as a cylinder, (namely) the circle *ABCD*, and an equal height. I say that the cone is the third part of the cylinder—that is to say, that the cylinder is three times the cone.



Εἰ γὰρ μή ἐστιν ὁ κύλινδρος τοῦ κώνου τριπλασίων, ὀέσται ὁ κύλινδρος τοῦ κώνου ἤτοι μείζωνόἠ τριπλασιωνόἠ ἐλάσσωνόἠ τριπλασίων. ὀέστω πρότερον μείζωνόἠ τριπλασίων, καὶ ἐγγεγράφθω εἰξ τὸν ΑΒΓΔ κύκλον τετράγωνον τὸ ΑΒΓΔ· τὸ δὴ ΑΒΓΔ τετράγωνον μείζόν ἐστινόἠ τὸ ἥμισυ τοῦ ΑΒΓΔ κύκλου. καὶ ἀνεστάτω ἀπὸ τοῦ ΑΒΓΔ τετραγώνου πρίσμα ἰσοϋψὲξ τῷ κυλίνδρῳ. τὸ δὴ ἀνιστάμενον πρίσμα μεῖζόν ἐστινόἠ τὸ ἥμισυ τοῦ κυλίνδου, ἐπειδήπερ κὸἀν περὶ τὸν ΑΒΓΔ κύκλον τετράγωνον περιγράψωμεν, τὸ ἐγγεγραμμένον εἰξ τὸν ΑΒΓΔ κύκλον τετράγωνον ἥμισύ ἐστι τοῦ περιγεγραμμένου· καί ἐστι τὰ ἀπὸ αὐτῶν ἀνιστάμενα στερεὰ παραλληλεπίπεδα πρίσματα ἰσοϋψῆ· τὰ δὲ ὑπὸ τὸ αὐτὸ ὁύψος ὄντα στερεὰ παραλληλεπίπεδα πρὸςὀάλληλά ἐστιν ὡξ αἱ βάσεις· καὶ τὸ ἐπὶ τοῦ ΑΒΓΔ ἄρα τετραγώνου ἀνασταθὲν πρίσμα ἥμισύ ἐστι τοῦ ἀνασταθέντος πρίσματος ἀπὸ τοῦ περὶ τὸν ΑΒΓΔ κύκλον περιγραφέντος τετραγώνου· καί ἐστιν ὁ κύλινδρος ἐλάττων τοῦ πρίσματος τοῦ ἀνατραθέντος ἀπὸ τοῦ περὶ τὸν ΑΒΓΔ κύκλον

For if the cylinder is not three times the cone then the cylinder will be either more than three times, or less than three times, (the cone). Let it, first of all, be more than three times (the cone). And let the square *ABCD* be inscribed in circle *ABCD* [Prop. 4.6]. So, square *ABCD* is more than half of circle *ABCD* [Prop. 12.2]. And let a prism of equal height to the cylinder be set up on square *ABCD*. So, the prism set up is more than half of the cylinder, inasmuch as if we also circumscribe a square around circle *ABCD* [Prop. 4.7] then the square inscribed in circle *ABCD* is half of the circumscribed (square). And the solids set up on them are parallelepiped prisms of equal height. And parallelepiped solids having the same height are to one another as their bases [Prop. 11.32]. And, thus, the prism set up on square *ABCD* is half of the prism set up on the square circumscribed about circle *ABCD*. And the cylinder is less than the prism set up on the square circumscribed about circle *ABCD*. Thus, the prism set up on

περιγραφέντος τετραγώνου· τὸ ἄρα πρίσμα τὸ ἀνασταθὲν ἀπὸ τοῦ ΑΒΓΔ τετραγώνου ἰσοϋψὲξ τῷ κυλίνδρῳ μεῖζόν ἐστι τοῦ ἡμίσεως τοῦ κυλίνδρου. τετμήσθωσαν αἱ ΑΒ, ΒΓ, ΓΔ, ΔΑ περιφέρειαι δίθα κατὰ τὰ Ε, Ζ, Η, Θ σημεῖα, καὶ ἐπεζεύχθωσαν αἱ ΑΕ, ΕΒ, ΒΖ, ΖΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ· καὶ ἑκάστον ἄρα τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων μειζόν ἐστινὸὴ τὸ ἥμισυ τοῦ καθὸ ἑαυτὸ τηήματος τοῦ ΑΒΓΔ κύκλου, ὡξὸέμπροσθεν ἐδείκνυμεν. ἀνεστάτω ἐφ᾽ ἑκάστου τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων πρίσματα ἰσοϋψῆ τῷ κυλίνδρῳ· καὶ ἑκάστον ἄρα τῶν ἀνασταθέντων πρισμάτων μεῖζόν ἐστινὸὴ τὸ ἥμισυ μέρος τοῦ καθὸ ἑαυτὸ τμήματος τοῦ κυλίνδρου, ἐπειδήπερ ἔαν διὰ τῶν Ε, Ζ, Η, Θ σημείων παραλλήλους ταῖς ΑΒ, ΒΓ, ΓΔ, ΔΑ ἀγάγωμεν, καὶ συμπληρώσωμεν τὰ ἐπὶ τῶν ΑΒ, ΒΓ, ΓΔ, ΔΑ παραλληλόγραμμα, καὶ ἀπὸ αὐτῶν ἀναστήσωμεν στερεὰ παραλληλεπίπεδα ἰσοϋψῆ τῷ κυλίνδρῳ, ἑκάσου τῶν ἀνασταθέντων ἡμίση ἐστὶ τὰ πρίσματα τὰ ἐπὶ τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων· καί ἐστι τὰ τοῦ κυλίνδρου τμήματα ἐλάττονα τῶν ἀνασταθέντων στερεῶν παραλληλεπιπέδων· ὁώστε καὶ τὰ ἐπὶ τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων πρίσματα μείζονά ἐστινὸὴ τὸ ἥμισυ τῶν καθὸ ἑαυτὰ τοῦ κυλίνδρου τμ ημάτων. τέμνοντες δὴ τὰξ ὑπολειπομένας περιφερείας δίθα καὶ ἐπιζευγνύντες εὐθείας καὶ ἀνιστάντες ἐφ᾽ ἑκάσου τῶν τριγώνων πρίσματα ἰσοϋψῆ τῷ κυλίνδρῳ καὶ τοῦτο ἀεὶ ποιοῦντες καταλείψομέν τινα ἀποτμήματα τοῦ κυλίνδρου, ὰ ὀέσται ἐλάττονα τῆς ὑπεροθῆς, ᾗ ὑπερέθει ὁ κυλίνδρος τοῦ τριπλασίου τοῦ κώνου. λελείφθω, καὶ ὀέστω τὰ ΑΕ, ΕΒ, ΒΖ, ΖΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ· λοιπὸν ἄρα τὸ πρίσμα, οὗ βάσις μὲν τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ, μεῖζόν ἐστινὸὴ τριπλάσιον τοῦ κώνου. ἀλλὰ τὸ πρίσμα, οὗ βάσις μὲν ἐστὶ τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ, τριπλάσιόν ἐστι τῆς πυραμίδος, ἧξ βάσις μέν ἐστι τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ· καὶ ἡ πυραμὶξ ἄρα, ἧξ βάσις μέν [ἐστι] τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, μείζων ἐστὶ τοῦ κώνου τοῦ βάσιν ἔχοντες τὸν ΑΒΓΔ κύκλον. ἀλλὰ καὶ ἐλάττων· ἐμπεριέθεται γὰρ ὑπὸ αὐτοῦ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐστὶν ὁ κύλινδρος τοῦ κώνου μεῖζωνὸὴ τριπλάσιος.

Λέγω δή, ὅτι οὐδὲ ἐλάττων ἐστινὸὴ τριπλάσιος ὁ κύλινδρος τοῦ κώνου.

Εἰ γὰρ δυνατόν, ὀέστω ἐλάττωνὸὴ τριπλάσιος ὁ κύλινδρος τοῦ κώνου· ἀνάπαλιν ἄρα ὁ κῶνος τοῦ κυλίνδρου μεῖζων ἐστινὸὴ τρίτον μέρος. ἐγγεγράφθω δὴ εἰξ τὸν ΑΒΓΔ κύκλον τετράγωνον τὸ ΑΒΓΔ· τὸ ΑΒΓΔ ἄρα τετράγωνον μεῖζόν ἐστινὸὴ τὸ ἥμισυ τοῦ ΑΒΓΔ κύκλου. καὶ ἀνεστάτω ἀπὸ τοῦ ΑΒΓΔ τετραγώνου πυραμὶξ τὴν αὐτὴν κορυφὴνὸέθουσα τῷ κώνῳ· ἡ ἄρα ἀναστα θεῖσα πυραμὶξ μείζων ἐστινὸὴ τὸ ἥμισυ μέρος τοῦ κώνου, ἐπειδήπερ, ὡξ ὁέμπροσθεν ἐδείκνυμεν, ὅτι ἔαν περὶ τὸν κύκλον τετράγωνον περιγράψωμεν, ὀέσται τὸ ΑΒΓΔ τετράγωνον ἥμισυ τοῦ περὶ τὸν κύκλον περιγεγραμμένου τετραγώνου· καὶ ἔαν ἀπὸ τῶν τετραγώνων στερεὰ παραλληλεπίπεδα ἀναστήσωμεν ἰσοϋψῆ τῷ κώνῳ, ὰ καὶ καλε-

square *ABCD* of the same height as the cylinder is more than half of the cylinder. Let the circumferences *AB*, *BC*, *CD*, and *DA* be cut in half at points *E*, *F*, *G*, and *H*. And let *AE*, *EB*, *BF*, *FC*, *CG*, *GD*, *DH*, and *HA* be joined. And thus each of the triangles *AEB*, *BFC*, *CGD*, and *DHA* is more than half of the segment of circle *ABCD* about it, as was shown previously [Prop. 12.2]. Let prisms of equal height to the cylinder be set up on each of the triangles *AEB*, *BFC*, *CGD*, and *DHA*. And each of the prisms set up is greater than the half part of the segment of the cylinder about it—inasmuch as if we draw (straight-lines) parallel to *AB*, *BC*, *CD*, and *DA* through points *E*, *F*, *G*, and *H* (respectively), and complete the parallelograms on *AB*, *BC*, *CD*, and *DA*, and set up parallelepiped solids of equal height to the cylinder on them, then the prisms on triangles *AEB*, *BFC*, *CGD*, and *DHA* are each half of the set up (parallelepipeds). And the segments of the cylinder are less than the set up parallelepiped solids. Hence, the prisms on triangles *AEB*, *BFC*, *CGD*, and *DHA* are also greater than half of the segments of the cylinder about them. So (if) the remaining circumferences are cut in half, and straight-lines are joined, and prisms of equal height to the cylinder are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cylinder whose (sum) is less than the excess by which the cylinder exceeds three times the cone [Prop. 10.1]. Let them be left, and let them be *AE*, *EB*, *BF*, *FC*, *CG*, *GD*, *DH*, and *HA*. Thus, the remaining prism whose base (is) polygon *AEBFCGDH*, and height the same as the cylinder, is greater than three times the cone. But, the prism whose base is polygon *AEBFCGDH*, and height the same as the cylinder, is three times the pyramid whose base is polygon *AEBFCGDH*, and apex the same as the cone [Prop. 12.7 corr.]. And thus the pyramid whose base [is] polygon *AEBFCGDH*, and apex the same as the cone, is greater than the cone having (as) base circle *ABCD*. But (it is) also less. For it is encompassed by it. The very thing (is) impossible. Thus, the cylinder is not more than three times the cone.

So, I say that neither (is) the cylinder less than three times the cone.

For, if possible, let the cylinder be less than three times the cone. Thus, inversely, the cone is greater than the third part of the cylinder. So, let the square *ABCD* be inscribed in circle *ABCD* [Prop. 4.6]. Thus, square *ABCD* is greater than half of circle *ABCD*. And let a pyramid having the same apex as the cone be set up on square *ABCD*. Thus, the pyramid set up is greater than the half part of the cone, inasmuch as we showed previously that if we circumscribe a square about the circle [Prop. 4.7] then the square *ABCD* will be half of the square circumscribed about the circle [Prop. 12.2]. And if we set up on the squares parallelepiped solids—which are also called prisms—of the same height as the cone, then the (prism)

ῖται πρίσματα, ὀέσται τὸ ἀνασταθὲν ἀπὸ τοῦ ΑΒΓΔ τετραγώνου ἥμισυ τοῦ ἀνασταθέντος ἀπὸ τοῦ περὶ τὸν κύκλον περιγραφέντος τετραγώνου· πρὸς ἀλλήλα γάρ εἰσιν ὡς αἱ βάσεις. ὥστε καὶ τὰ τρίτα· καὶ πυραμὶξ ἄρα, ἣξ βάσις τὸ ΑΒΓΔ τετράγωνον, ἥμισύ ἐστι τῆς πυραμίδος τῆς ἀνασταθείσης ἀπὸ τοῦ περὶ τὸν κύκλον περιγραφέντος τετραγώνου. καί ἐστι μείζων ἡ πυραμὶξ ἡ ἀνασταθεῖσα ἀπὸ τοῦ περὶ τὸν κύκλον τετραγώνου τοῦ κώνου· ἐμπεριέθει γὰρ αὐτόν. ἡ ἄρα πυραμίξ, ἣξ βάσις τὸ ΑΒΓΔ τετράγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, μείζων ἐστὶνὸὴ τὸ ἥμισυ τοῦ κώνου. τετμήσθωσαν αἱ ΑΒ, ΒΓ, ΓΔ, ΔΑ περιφέρειαι δίθα κατὰ τὰ Ε, Ζ, Η, Θ σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΑΕ, ΕΒ, ΒΖ, ΖΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ· καὶ ὁἑκαστον ἄρα τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων μεῖζόν ἐστινὸὴ τὸ ἥμισυ μέρος του καθὸ ἑαυτὸ τμήματος τοῦ ΑΒΓΔ κύκλου. καὶ ἀνεστάτωσαν ἐφ΄ ἑκάστου τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων πυραμίδες τὴν αὐτὴν κορυφὴνὸέθουσαι τῷ κώνῳ· καὶ ἑκάστη ἄρα τῶν ἀνασταθεισῶν πυραμίδων κατὰ τὸν αὐτὸν τρόπον μείζων ἐστὶνὸὴ τὸ ἥμισυ μέρος τοῦ καθὸ ἑαυτὴν τμήματος τοῦ κώνου. τέμνοντες δὴ τὰξ ὑπολειπομένας περιφερείας δίθα καὶ ἐπιζευγνύντες εὐθείας καὶ ἀνιστάντες ἐφ΄ ἑκάστου τῶν τριγώνων πυραμίδα τὴν αὐτὴν κορυφὴνὸέθουσαν τῷ κώνῳ καὶ τοῦτο ἀεὶ ποιοῦτες καταλείψομέν τινα ἀποτμήματα τοῦ κώνου, ὰ ὀέσται ἐλάττονα τῆς ὑπεροθῆς, ἣ ὑπερέθει ὁ κῶνος τοῦ τρίτου μέρους τοῦ κυλίνδρου. λελείφθω, καὶ ὀέστω τὰ ἐπὶ τῶν ΑΕ, ΕΒ, ΒΖ, ΖΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ· λοιπὴ ἄρα ἡ πυραμίξ, ἣξ βάσις μέν ἐστι τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, μείζων ἐστὶνὸὴ τρίτον μέρος τοῦ κυλίνδρου. ἀλλὸ ἡ πυραμίξ, ἣξ βάσις μέν ἐστι τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, κορυφὴ δὲ ἡ αυτὴ τῷ κώνῳ, τρίτον ἐστὶ μέρος τοῦ πρίσματος, οὗ βάσις μέν ἐστι τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ· τὸ ἄρα πρίσμα, οὗ βάσις μέν ἐστι τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ, μεῖζόν ἐστι τοῦ κυλίνδρου, οὗ βάσις ἐστὶν ὁ ΑΒΓΔ κύκλος. ἀλλὰ καὶ ὀέλαττον· ἐμπεριέθεται γὰρ ὑπὸ αὐτοῦ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ὁ κύλινδρος τοῦ κώνου ἐλάττων ἐστὶνὸὴ τριπλάσιος. ἐδείθθη δέ, ὅτι οὐδὲ μείζωνὸὴ τριπλάσιος· τριπλάσιος ἄρα ὁ κύλινδρος τοῦ κώνου· ὥστε ὁ κῶνος τρίτον ἐστὶ μέρος τοῦ κυλίνδρου.

Πᾶξ ἄρα κῶνος κυλίνδρου τρίτον μέρος ἐστὶ τοῦ τὴν αὐτὴν βάσιν ἔχοντος αὐτῷ καὶ ὕψος ἴσον· ὅπερ ἔδει δεῖχαι.

set up on square $ABCD$ will be half of the (prism) set up on the square circumscribed about the circle. For they are to one another as their bases [Prop. 11.32]. Hence, (the same) also (goes for) the thirds. Thus, the pyramid whose base is square $ABCD$ is half of the pyramid set up on the square circumscribed about the circle [Prop. 12.7 corr.]. And the pyramid set up on the square circumscribed about the circle is greater than the cone. For it encompasses it. Thus, the pyramid whose base is square $ABCD$, and apex the same as the cone, is greater than half of the cone. Let the circumferences AB, BC, CD, and DA be cut in half at points E, F, G, and H (respectively). And let AE, EB, BF, FC, CG, GD, DH, and HA be joined. And, thus, each of the triangles AEB, BFC, CGD, and DHA is greater than the half part of the segment of circle $ABCD$ about it [Prop. 12.2]. And let pyramids having the same apex as the cone be set up on each of the triangles AEB, BFC, CGD, and DHA. And, thus, in the same way, each of the pyramids set up is more than the half part of the segment of the cone about it. So, (if) the remaining circumferences are cut in half, and straight-lines are joined, and pyramids having the same apex as the cone are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone whose (sum) is less than the excess by which the cone exceeds the third part of the cylinder [Prop. 10.1]. Let them be left, and let them be the (segments) on AE, EB, BF, FC, CG, GD, DH, and HA. Thus, the remaining pyramid whose base is polygon $AEBFCGDH$, and apex the same as the cone, is greater than the third part of the cylinder. But, the pyramid whose base is polygon $AEBFCGDH$, and apex the same as the cone, is the third part of the prism whose base is polygon $AEBFCGDH$, and height the same as the cylinder [Prop. 12.7 corr.]. Thus, the prism whose base is polygon $AEBFCGDH$, and height the same as the cylinder, is greater than the cylinder whose base is circle $ABCD$. But, (it is) also less. For it is encompassed by it. The very thing is impossible. Thus, the cylinder is not less than three times the cone. And it was shown that neither (is it) greater than three times (the cone). Thus, the cylinder (is) three times the cone. Hence, the cone is the third part of the cylinder.

Thus, every cone is the third part of the cylinder which has the same base as it, and an equal height. (Which is) the very thing it was required to show.

11.

Οἱ ὑπο τὸ αὐτὸ ὕψος ὄντες κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡξ αἱ βάσεις.

Ἔστωσαν ὑπὸ τὸ αὐτὸ ὕψος κῶνοι καὶ κύλινδροι, ὧν βάσεις μὲν [εἰσιν] οἱ ΑΒΓΔ, ΕΖΗΘ κύκλοι, ὀάχονες δὲ οἱ ΚΛ, ΜΝ, διάμετροι δὲ τῶν βάσεων αἱ ΑΓ, ΕΗ· λέγω, ὅτι ἐστὶν ὡξ ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὕτως ὁ

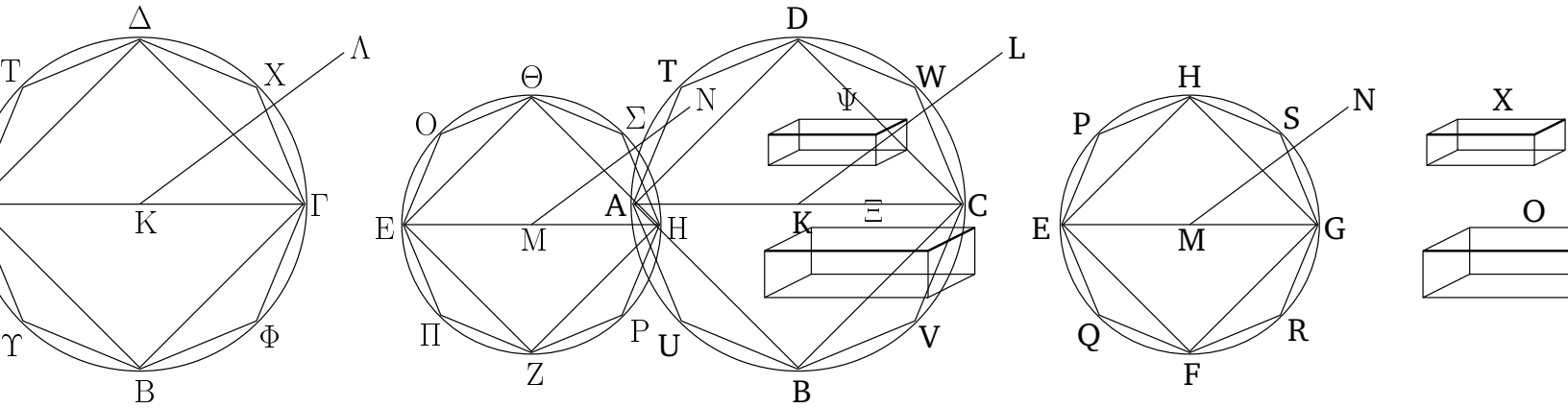
Proposition 11

Cones and cylinders having the same height are to one another as their bases.

Let there be cones and cylinders of the same height whose bases [are] the circles $ABCD$ and $EFGH$, axes KL and MN, and diameters of the bases AC and EG (respectively). I say that as circle $ABCD$ is to circle $EFGH$, so cone AL (is) to

ΑΛ κῶνος πρὸς τὸν ΕΝ κῶνον.

cone EN.



Εἰ γὰρ μή, ὀέσται ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὐτως ὁ ΑΛ κῶνος ἤτοι πρὸς ἔλασσόν τι τοῦ ΕΝ κώνου στερεὸν ὴ πρὸς μεῖζον. ὀέστω πρότερον πρὸς ἔλασσον τὸ Χ, καὶ ὁῷ ὀέλασσόν ἐστι τὸ Χ στερεὸν τοῦ ΕΝ κώνου, ἐκείνῳ ἴσον ὀέστω τὸ Ψ στερεόν· ὁ ΕΝ κῶνος ἄραὀίσος ἐστὶ τοῖς Χ, Ψ στερεοῖξ. ἐγγεγράφθω εἰξ τὸν ΕΖΗΘ κύκλον τετράγωνον τὸ ΕΖΗΘ· τὸ ἄρα τετράγωνον μεῖζόν ἐστινὸὴ τὸ ἥμισυ τοῦ κύκλου. ἀνεστάτω ἀπὸ τοῦ ΕΖΗΘ τετραγώνου πυραμὶξ ἰσοϋψὴξ τῷ κώνῳ· ἡ ἄρα ἀνασταθεῖσα πυραμὶξ μείζων ἐστὶνὸὴ τὸ ἥμισυ τοῦ κώνου, ἐπειδήπερ ἔαν περιγράψωμεν περὶ τὸν κύκλον τετράγωνον, καὶ ἀπὸ αὐτοῦ ἀναστήσωμεν πυραμίδα ἰσοϋψῆ τῷ κώνῳ, ἡ ἐγγραφεῖσα πυραμὶξ ἥμισύ ἐστι τῆς περιγραφείσης· πρὸς ἀλλήλας γάρ εἰσιν ὡς αἱ βάσεις· ἐλάττων δὲ ὁ κῶνος τῆς περιγραφείσης πυραμίδος. τετμήσθωσαν αἱ ΕΖ, ΖΗ, ΗΘ, ΘΕ περιφέρειαι δίθα κατὰ τὰ Ο, Π, Ρ, Σ σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΘΟ, ΟΕ, ΕΠ, ΠΖ, ΖΡ, ΡΗ, ΗΣ, ΣΘ. ὁέκαστον ἄρα τῶν ΘΟΕ, ΕΠΖ, ΖΡΗ, ΗΖΘ τριγώνων μεῖζόν ἐστινὸὴ τὸ ἥμισυ τοῦ καθὸ ἑαυτὸ τμήματος τοῦ κύκλου. ἀνεστάτω ἐφ᾽ ἑκάστου τῶν ΘΟΕ, ΕΠΖ, ΖΡΗ, ΗΣΘ τριγώνων πυραμὶξ ἰσοϋψὴξ τῷ κώνῳ· καὶ ἑκάστη ἄρα τῶν ἀνασταθεισῶν πυραμίδων μείζων ἐστὶνὸὴ τὸ ἥμισυ τοῦ καθὸ ἑαυτὴν τμήματος τοῦ κώνου. τέμνοντες δὴ τὰξ ὑπολειπομένας περιφερείας δίθα καὶ ἐπιζευγνύντες εὐθείας καὶ ἀνιστάντες ἐπὶ ἑκάστου τῶν τριγώνων πυραμίδας ἰσοϋψεῖς τῷ κώνῳ καὶ ἀεὶ τοῦτο ποιοῦντες καταλείψομέν τινα ἀποτμήματα τοῦ κώνου, ὁὰ ὀέσται ἐλάσσονα τοῦ Ψ στερεοῦ. λελείφθω, καὶ ὀέστω τὰ ἐπὶ τῶν ΘΟΕ, ΕΠΖ, ΖΡΗ, ΗΣΘ· λοιπὴ ἄρα ἡ πυραμίξ, ἧξ βάσις τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κώνῳ, μείζων ἐστὶ τοῦ Χ στερεοῦ. ἐγγεγράφθω καὶ εἰξ τὸν ΑΒΓΔ κύκλον τῷ ΘΟΕΠΖΡΗΣ πολυγώνῳ ὅμοιόν τε καὶ ὁμοίως κείμενον πολύγωνον τὸ ΔΤΑΥΒΦΓΘ, καὶ ἀνεστάτω ἐπ᾽ αὐτοῦ πυραμὶξ ἰσοϋψὴξ τῷ ΑΛ κώνῳ. ἐπεὶ οὐῦν ἐστιν ὡξ τὸ ἀπὸ τῆς ΑΓ πρὸς τὸ ἀπὸ τῆς ΕΗ, οὐτως τὸ ΔΤΑΥΒΦΓΘ πολύγωνον πρὸς τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, ὡξ δὲ τὸ ἀπὸ τῆς ΑΓ πρὸς τὸ ἀπὸ τῆς ΕΗ, οὐτως ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, καὶ ὡξ ἄρα ὁ ΑΒΓΔ κύκλος πρὸς τὸν

For if not, then as circle $ABCD$ (is) to circle $EFGH$, so cone AL will be to some solid either less than, or greater than, cone EN. Let it, first of all, be (in this ratio) to (some) lesser (solid), O. And let solid X be equal to that (magnitude) by which solid O is less than cone EN. Thus, cone EN is equal to (the sum of) solids O and X. Let the square $EFGH$ be inscribed in circle $EFGH$ [Prop. 4.6]. Thus, the square is greater than half of the circle [Prop. 12.2]. Let a pyramid of the same height as the cone be set up on square $EFGH$. Thus, the pyramid set up is greater than half of the cone, inasmuch as, if we circumscribe a square about the circle [Prop. 4.7], and set up on it a pyramid of the same height as the cone, then the inscribed pyramid is half of the circumscribed pyramid. For they are to one another as their bases [Prop. 12.6]. And the cone (is) less than the circumscribed pyramid. Let the circumferences EF, FG, GH, and HE be cut in half at points P, Q, R, and S. And let HP, PE, EQ, QF, FR, RG, GS, and SH be joined. Thus, each of the triangles HPE, EQF, FRG, and GSH is greater than half of the segment of the circle about it [Prop. 12.2]. Let pyramids of the same height as the cone be set up on each of the triangles HPE, EQF, FRG, and GSH. And, thus, each of the pyramids set up is greater than half of the segment of the cone about it [Prop. 12.10]. So, (if) the remaining circumferences are cut in half, and straight-lines are joined, and pyramids of equal height to the cone are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone (the sum of) which is less than solid X [Prop. 10.1]. Let them be left, and let them be the (segments) on HPE, EQF, FRG, and GSH. Thus, the remaining pyramid whose base is polygon $HPEQFRGS$, and height the same as the cone, is greater than solid O [Prop. 6.18]. And let the polygon $DTAUBVCW$, similar, and similarly laid out, to polygon $HPEQFRGS$, be inscribed in circle $ABCD$. And on it let a pyramid of the same height as cone AL be set up. Therefore, since as the (square) on AC is to the (square) on EG, so polygon $DTAUBVCW$

ΕΖΗΘ κύκλον, οὑτως τὸ ΔΤΑΥΒΦΓΘ πολύγωνον πρὸς τὸ ΘΟΕΠΖΡΗΣ πολύγωνον. ὡς δὲ ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὑτως ὁ ΑΛ κῶνος πρὸς τὸ Χ στερεόν, ὡς δὲ τὸ ΔΤΑΥΒΦΓΘ πολύγωνον πρὸς τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, οὑτως ἡ πυραμίξ, ἧξ βάσις μὲν τὸ ΔΤΑΥΒΦΓΘ πολύγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, πρὸς τὴν πυραμίδα, ἧξ βάσις μὲν τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, κορυφὴ δὲ τὸ Ν σημεῖον. καὶ ὡς ἄρα ὁ ΑΛ κῶνος πρὸς τὸ Χ στερεόν, οὑτως ἡ πυραμίξ, ἧξ βάσις μὲν τὸ ΔΤΑΥΒΦΓΘ πολύγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, πρὸς τὴν πυραμίδα, ἧξ βάσις μὲν τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, κορυφὴ δὲ τὸ Ν σημεῖον· ἐναλλὰχ ἄρα ἐστὶν ὡς ὁ ΑΛ κῶνος πρὸς τὴν ἐν αὐτῷ πυραμίδα, οὑτως τὸ Χ στερεὸν πρὸς τὴν ἐν τῷ ΕΝ κώνῳ πυραμίδα. μείζων δὲ ὁ ΑΛ κῶνος τῆς ἐν αὐτῷ πυραμίδος· μεῖζον ἄρα καὶ τὸ Χ στερεὸν τῆς ἐν τῷ ΕΝ κώνῳ πυραμίδος. ἀλλὰ καὶ ὐέλασσον· ὅπερὐάτοπον. οὐκ ἄρα ἐστὶν ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὑτως ὁ ΑΛ κῶνος πρὸςὐέλασσόν τι τοῦ ΕΝ κώνου στερεόν. ὁμοίως δὲ δείχομεν, ὅτι οὐδέ ἐστιν ὡς ὁ ΕΖΗΘ κύκλος πρὸς τὸν ΑΒΓΔ κύκλον, οὑτως ὁ ΕΝ κῶνος πρὸςὐέλασσόν τι τοῦ ΑΛ κώνου στερεόν.

Λέγω δή, ὅτι οὐδέ ἐστιν ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὑτως ὁ ΑΛ κῶνος πρὸς μεῖζόν τι τοῦ ΕΝ κώνου στερεόν.

Εἰ γὰρ δυνατόν, ὑέστω πρὸς μεῖζον τὸ Χ· ἀνάπαλιν ἄρα ἐστὶν ὡς ὁ ΕΖΗΘ κύκλος πρὸς τὸν ΑΒΓΔ κύκλον, οὑτως τὸ Χ στερεὸν πρὸς τὸν ΑΛ κῶνον. ἀλλὰ ὡς τὸ Χ στερεὸν πρὸς τὸν ΑΛ κῶνον, οὑτως ὁ ΕΝ κῶνος πρὸςὐέλασσόν τι τοῦ ΑΛ κώνου στερεόν· καὶ ὡς ἄρα ὁ ΕΖΗΘ κύκλος πρὸς τὸν ΑΒΓΔ κύκλον, οὑτως ὁ ΕΝ κῶνος πρὸςὐέλασσόν τι τοῦ ΑΛ κώνου στερεόν· ὅπερ ἀδύνατον ἐδείθθη. οὐκ ἄρα ἐστὶν ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὑτως ὁ ΑΛ κῶνος πρὸς μεῖζόν τι τοῦ ΕΝ κώνου στερεόν. ἐδείθθη δέ, ὅτι οὐδὲ πρὸςὐέλασσον· ὐέστιν ἄρα ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὑτως ὁ ΑΛ κῶνος πρὸς τὸν ΕΝ κῶνον.

Ἀλλὰ ὡς ὁ κῶνος πρὸς τὸν κῶνον, ὁ κύλινδρος πρὸς τὸν κύλινδρον· τριπλασίων γὰρ ἑκάτερος ἑκατέρου. καὶ ὡς ἄρα ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὑτως οἱ ἐπ᾽ αὐτῶν ἰσοϋψεῖς.

Οἱ ἄρα ὑπὸ τὸ αὐτὸ ὕψος ὄντες κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡς αἱ βάσεις· ὅπερ ἔδει δεῖχαι.

12.

Οἱ ὅμοιοι κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ἐν ταῖς βάσεσι διαμέτρων.

Ἔστωσαν ὅμοιοι κῶνοι καὶ κύλινδροι, ὧν βάσεις μὲν οἱ ΑΒΓΔ, ΕΖΗΘ κύκλοι, διάμετροι δὲ τῶν βάσεων αἱ ΒΔ, ΖΘ, ὐάχονες δὲ τῶν κώνων καὶ κυλίνδρων οἱ ΚΛ, ΜΝ· λέγω, ὅτι ὁ κῶνος, οὗ βάσις μέν [ἐστιν] ὁ ΑΒΓΔ κύκλος, κορυφὴ δὲ τὸ Λ σημεῖον, πρὸς τὸν κῶνον, οὗ βάσις μέν [ἐστιν] ὁ ΕΖΗΘ (is) to polygon $HPEQFRGS$ [Prop. 12.1], and as the (square) on AC (is) to the (square) on EG, so circle $ABCD$ (is) to circle $EFGH$ [Prop. 12.2], thus as circle $ABCD$ (is) to circle $EFGH$, so polygon $DTAUBVCW$ also (is) to polygon $HPEQFRGS$. And as circle $ABCD$ (is) to circle $EFGH$, so cone AL (is) to solid O. And as polygon $DTAUBVCW$ (is) to polygon $HPEQFRGS$, so the pyramid whose base is polygon $DTAUBVCW$, and apex the point L, (is) to the pyramid whose base is polygon $HPEQFRGS$, and apex the point N [Prop. 12.6]. And, thus, as cone AL (is) to solid O, so the pyramid whose base is $DTAUBVCW$, and apex the point L, (is) to the pyramid whose base is polygon $HPEQFRGS$, and apex the point N [Prop. 5.11]. Thus, alternately, as cone AL is to the pyramid within it, so solid O (is) to the pyramid within cone EN [Prop. 5.16]. But, cone AL (is) greater than the pyramid within it. Thus, solid O (is) also greater than the pyramid within cone EN [Prop. 5.14]. But, (it is) also less. The very thing (is) absurd. Thus, circle $ABCD$ is not to circle $EFGH$, as cone AL (is) to some solid less than cone EN. So, similarly, we can show that neither is circle $EFGH$ to circle $ABCD$, as cone EN (is) to some solid less than cone AL.

So, I say that neither is circle $ABCD$ to circle $EFGH$, as cone AL (is) to some solid greater than cone EN.

For, if possible, let it be (in this ratio) to (some) greater (solid), O. Thus, inversely, as circle $EFGH$ is to circle $ABCD$, so solid O (is) to cone AL [Prop. 5.7 corr.]. But, as solid O (is) to cone AL, so cone EN (is) to some solid less than cone AL [Prop. 12.2 lem.]. And, thus, as circle $EFGH$ (is) to circle $ABCD$, so cone EN (is) to some solid less than cone AL. The very thing was shown (to be) impossible. Thus, circle $ABCD$ is not to circle $EFGH$, as cone AL (is) to some solid greater than cone EN. And, it was shown that neither (is it in this ratio) to (some) lesser (solid). Thus, as circle $ABCD$ is to circle $EFGH$, so cone AL (is) to cone EN.

But, as the cone (is) to the cone, (so) the cylinder (is) to the cylinder. For each (is) three times each [Prop. 12.10]. Thus, circle $ABCD$ (is) also to circle $EFGH$, as (the ratio of the cylinders) on them (having) the same height.

Thus, cones and cylinders having the same height are to one another as their bases. (Which is) the very thing it was required to show.

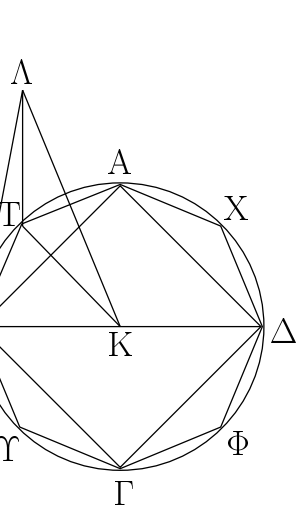
Proposition 12

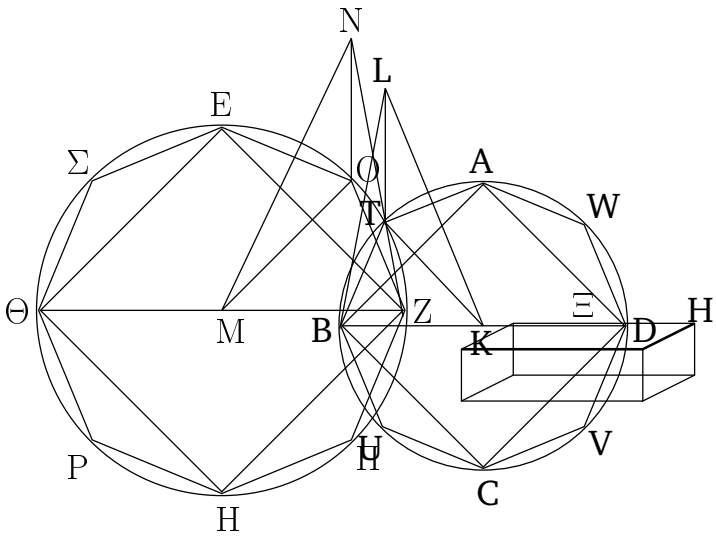
Similar cones and cylinders are to one another in the cubed ratio of the diameters of their bases.

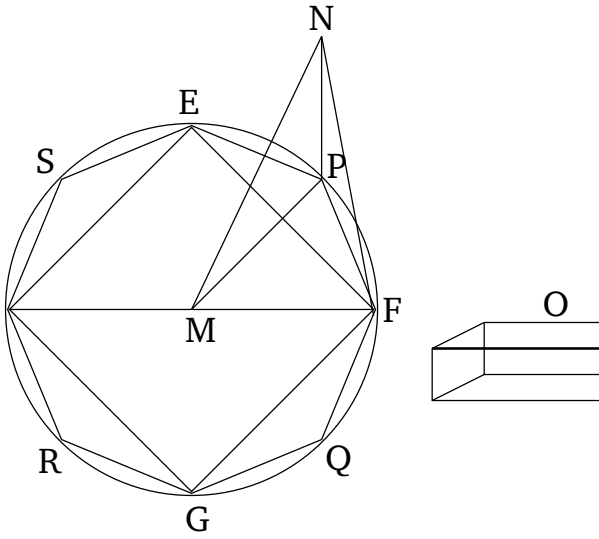
Let there be similar cones and cylinders of which the bases (are) the circles $ABCD$ and $EFGH$, the diameters of the bases (are) BD and FH, and the axes of the cones and cylinders (are) KL and MN (respectively). I say that the cone whose base [is] circle $ABCD$, and apex the point L, has to the cone whose base

κύκλος, κορυφὴ δὲ τὸ N σημεῖον, τριπλασίονα λόγον ἔχει ἤπερ ἡ ΒΔ πρὸς τὴν ΖΘ.

[is] circle $EFGH$, and apex the point N, the cubed ratio that BD (has) to FH.







Εἰ γὰρ μὴ ἔχει ὁ ΑΒΓΔΛ κῶνος πρὸς τὸν ΕΖΗΘΝ κῶνον πριπλασίονα λόγον ἤπερ ἡ ΒΔ πρὸς τὴν ΖΘ, ὁέχει ὁ ΑΒΓΔΛ κῶνοςὸὴ πρὸςὸέλασσόν τι τοῦ ΕΖΗΘΝ κώνου στερεὸν τριπλασίονα λόγονὸὴ πρὸς μεῖζον. ἐθέτω πρότερον πρὸςὸέλασσον τὸ Χ, καὶ ἐγγεγράφθω εἰξ τὸν ΕΖΗΘ κύκλον τετράγωνον τὸ ΕΖΗΘ· τὸ ἄρα ΕΖΗΘ τετράγωνον μεῖζόν ἐστινὸὴ τὸ ἥμισυ τοῦ ΕΖΗΘ κύκλου. καὶ ἀνεστάτω ἐπὶ τοῦ ΕΖΗΘ τετραγώνου πυραμὶξ τὴν αὐτὴν κορυφὴνὸέθουσα τῷ κώνῳ· ἡ ἄρα ἀναστα θεῖσα πυραμὶξ μείζων ἐστινὸὴ τὸ ἥμισυ μέρος τοῦ κώνου. τετμήσθωσαν δὴ αἱ ΕΖ, ΖΗ, ΗΘ, ΘΕ περιφέρειαι δίθα κατὰ τὰ Ο, Π, Ρ, Σ σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΕΟ, ΟΖ, ΖΠ, ΠΗ, ΗΡ, ΡΘ, ΘΣ, ΣΕ. καὶ ὁέκαστον ἄρα τῶν ΕΟΖ, ΖΠΗ, ΗΡΘ, ΘΣΕ τριγώνων μεῖζόν ἐστινὸὴ τὸ ἥμισυ μέρος τοῦ καθὸ ἑαυτὸ τμήματος τοῦ ΕΖΗΘ κύκλου. καὶ ἀνεστάτω ἐφ᾽ ἑκάστου τῶν ΕΟΖ, ΖΠΗ, ΗΡΘ, ΘΣΕ τριγώνων πυραμὶξ τὴν αὐτὴν κορυφὴνὸέθουσα τῷ κώνῳ· καὶ ἑκάστη ἄρα τῶν ἀνασταθεισῶν πυραμίδων μείζων ἐστινὸὴ τὸ ἥμισυ μέρος τοῦ καθὸ ἑαυτὴν τμήματος τοῦ κώνου. τέμνοντες δὴ τὰξ ὑπολειπομένας περιφερείας δίθα καὶ ἐπιζευγνύντες εὐθείας καὶ ἀνιστάντες ἐφ᾽ ἑκάστου τῶν τριγώνων πυραμίδας τὴν αὐτὴν κορυφὴν ἐθούσας τῷ κώνῳ καὶ τοῦτο ἀεὶ ποιοῦντες καταλείψομέν τινα ἀποτμήματα τοῦ κώνου, ὰ ὀέσται ἐλάσσονα τῆς ὑπεροθῆς, ᾗ ὑπερέθει ὁ ΕΖΗΘΝ κῶνος τοῦ Χ στερεοῦ. λελείφθω, καὶ ὀέστω τὰ ἐπὶ τῶν ΕΟ, ΟΖ, ΖΠ, ΠΗ, ΗΡ, ΡΘ, ΘΣ, ΣΕ· λοιπὴ ἄρα ἡ πυραμίξ, ἧξ βάσις μέν ἐστι τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν σημεῖον, μείζων ἐστὶ τοῦ Χ στερεοῦ. ἐγγεγράφθω καὶ εἰξ τὸν ΑΒΓΔ κύκλον τῷ ΕΟΖΠΗΡΘΣ πολυγώνῳ ὅμοιόν τε καὶ ὁμοίως κείμενον πολύγωνον τὸ ΑΤΒΥΓΦΔΘ, καὶ ἀνεστάτω ἐπὶ τοῦ ΑΤΒΥΓΦΔΘ πολυγώνου πυραμὶξ τὴν αὐτὴν κορυφὴνὸέθουσα τῷ κώνῳ, καὶ τῶν μὲν περιεθόντων τὴν πυραμίδα, ἧξ βάσις μέν ἐστι τὸ ΑΤΒΥΓΦΔΘ πολύγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, ὁὲν τρίγωνονὸέστω τὸ

For if cone $ABCDL$ does not have to cone $EFGHN$ the cubed ratio that BD (has) to FH then cone $ABCDL$ will have the cubed ratio to some solid either less than, or greater than, cone $EFGHN$. Let it, first of all, have (such a ratio) to (some) lesser (solid), O. And let the square $EFGH$ be inscribed in circle $EFGH$ [Prop. 4.6]. Thus, square $EFGH$ is greater than half of circle $EFGH$ [Prop. 12.2]. And let a pyramid having the same apex as the cone be set up on square $EFGH$. Thus, the pyramid set up is greater than the half part of the cone [Prop. 12.10]. So, let the circumferences EF, FG, GH, and HE be cut in half at points P, Q, R, and S (respectively). And let EP, PF, FQ, QG, GR, RH, HS, and SE be joined. And, thus, each of the triangles EPF, FQG, GRH, and HSE is greater than the half part of the segment of circle $EFGH$ about it [Prop. 12.2]. And let a pyramid having the same apex as the cone be set up on each of the triangles EPF, FQG, GRH, and HSE. And thus each of the pyramids set up is greater than the half part of the segment of the cone about it [Prop. 12.10]. So, (if) the the remaining circumferences are cut in half, and straight-lines are joined, and pyramids having the same apex as the cone are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone whose (sum) is less than the excess by which cone $EFGHN$ exceeds solid O [Prop. 10.1]. Let them be left, and let them be the (segments) on EP, PF, FQ, QG, GR, RH, HS, and SE. Thus, the remaining pyramid whose base is polygon $EPFQGRHS$, and apex the point N, is greater than solid O. And let the polygon $ATBUCVDW$, similar, and similarly laid out, to polygon $EPFQGRHS$, be inscribed in circle $ABCD$ [Prop. 6.18]. And let a pyramid having the same apex as the cone be set up on polygon $ATBUCVDW$. And let LBT be one of the triangles containing the pyramid whose base is polygon

ΛΒΤ, τῶν δὲ περειθόντων τὴν πυραμίδα, ἧξ βάσις μέν ἐστι τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν σημεῖον, δέν τρίγωνονὸέστω τὸ ΝΖΟ, καὶ ἐπεζεύθθωσαν αἱ ΚΤ, ΜΟ. καὶ ἐπεὶ ὅμοιόξ ἐστιν ὁ ΑΒΓΔΛ κῶνος τῷ ΕΖΗΘΝ κώνῳ, ὀέστιν ἄρα ὡξ ἡ ΒΔ πρὸς τὴν ΖΘ, οὑτως ὁ ΚΛὀάχων πρὸς τὸν ΜΝὀάχονα. ὡξ δὲ ἡ ΒΔ πρὸς τὴν ΖΘ, οὑτως ἡ ΒΚ πρὸς τὴν ΖΜ· καὶ ὡξ ἄρα ἡ ΒΚ πρὸς τὴν ΖΜ, οὑτως ἡ ΚΛ πρὸς τὴν ΜΝ. καὶ ἐναλλὰχ ὡξ ἡ ΒΚ πρὸς τὴν ΚΛ, οὑτως ἡ ΖΜ πρὸς τὴν ΜΝ. καὶ περὶ ἴσας γωνίας τὰξ ὑπὸ ΒΚΛ, ΖΜΝ αἱ πλευραὶ ἀνάλογόν εἰσιν· ὅμοιον ἄρα ἐστὶ τὸ ΒΚΛ τρίγωνον τῷ ΖΜΝ τριγώνῳ. πάλιν, ἐπεί ἐστιν ὡξ ἡ ΒΚ πρὸς τὴν ΚΤ, οὑτως ἡ ΖΜ πρὸς τὴν ΜΟ, καὶ περὶ ἴσας γωνίας τὰξ ὑπὸ ΒΚΤ, ΖΜΟ, ἐπειδήπερ, ὃ μέρος ἐστὶν ἡ ὑπὸ ΒΚΤ γωνία τῶν πρὸς τῷ Κ κέντρῳ τεσσάρων ὀρθῶν, τὸ αὐτὸ μέρος ἐστὶ καὶ ἡ ὑπὸ ΖΜΟ γωνία τῶν πρὸς τῷ Μ κέντρῳ τεσσάρων ὀρθῶν· ἐπεὶ οˇν περὶ ἴσας γωνίας αἱ πλευραὶ ἀνάλογόν εἰσιν, ὅμοιον ἄρα ἐστι τὸ ΒΚΤ τρίγωνον τῷ ΖΜΟ τριγώνῳ. πάλιν, ἐπεὶ ἐδείθθη ὡξ ἡ ΒΚ πρὸς τὴν ΚΛ, οὑτως ἡ ΖΜ πρὸς τὴν ΜΝ, ὀίση δὲ ἡ μὲν ΒΚ τῇ ΚΤ, ἡ δὲ ΖΜ τῇ ΟΜ, ὀέστιν ἄρα ὡξ ἡ ΤΚ πρὸς τὴν ΚΛ, οὑτως ἡ ΟΜ πρὸς τὴν ΜΝ. καὶ περὶ ἴσας γωνίας τὰξ ὑπὸ ΤΚΛ, ΟΜΝ· ὀρθαὶ γάρ· αἱ πλευραὶ ἀνάλογόν εἰσιν· ὅμοιον ἄρα ἐστὶ τὸ ΛΚΤ τρίγωνον τῷ ΝΜΟ τριγώνῳ. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν ΛΚΒ, ΝΜΖ τριγώνων ἐστὶν ὡξ ἡ ΛΒ πρὸς τὴν ΒΚ, οὑτως ἡ ΝΖ πρὸς τὴν ΖΜ, διὰ δὲ τὴν ὁμοιότητα τῶν ΒΚΤ, ΖΜΟ τριγώνων ἐστὶν ὡξ ἡ ΚΒ πρὸς τὴν ΒΤ, οὑτως ἡ ΜΖ πρὸς τὴν ΖΟ, διὸ ὀίσου ἄρα ὡξ ἡ ΛΒ πρὸς τὴν ΒΤ, οὑτως ἡ ΝΖ πρὸς τὴν ΖΟ. πάλιν, ἐπεὶ διὰ τὴν ομοιότητα τῶν ΛΤΚ, ΝΟΜ τριγώνων ἐστὶν ὡξ ἡ ΛΤ πρὸς τὴν ΤΚ, οὑτως ἡ ΝΟ πρὸς τὴν ΟΜ, διὰ δὲ τὴν ὁμοιότητα τῶν ΤΚΒ, ΟΜΖ τριγώνων ἐστὶν ὡξ ἡ ΚΤ πρὸς τὴν ΤΒ, οὑτως ἡ ΜΟ πρὸς τὴν ΟΖ, διὸ ὀίσου ἄρα ὡξ ἡ ΛΤ πρὸς τὴν ΤΒ, οὑτως ἡ ΝΟ πρὸς τὴν ΟΖ. ἐδείθθη δὲ καὶ ὡξ ἡ ΤΒ πρὸς τὴν ΒΛ, οὑτως ἡ ΟΖ πρὸς τὴν ΖΝ. διὸ ὀίσου ἄρα ὡξ ἡ ΤΛ πρὸς τὴν ΛΒ, οὑτως ἡ ΟΝ πρὸς τὴν ΝΖ. τῶν ΛΤΒ, ΝΟΖ ἄρα τριγώνων ἀνάλογόν εἰσιν αἱ πλευραί· ἰσογώνια ἄρα ἐστὶ τὰ ΛΤΒ, ΝΟΖ τρίγωνα· ὡστε καὶ ὅμοια. καὶ πυραμὶξ ἄρα, ἧξ βάσις μὲν τὸ ΒΚΤ τρίγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, ὁμοία ἐστὶ πυραμίδι, ἧξ βάσις μὲν τὸ ΖΜΟ τρίγωνον, κορυφὴ δὲ τὸ Ν σημεῖον· ὑπὸ γὰρ ὅμοίων ἐπιπέδων περιέθονταιὀίσων τὸ πλῆθος. αἱ δὲ ὅμοιαι πυραμίδες καὶ τριγώνουςὀέθουσαι βάσεις ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν. ἡ ἄρα ΒΚΤΛ πυραμὶξ πρὸς τὴν ΖΜΟΝ πυραμίδα τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΚ πρὸς τὴν ΖΜ. ὁμοίως δὴ ἐπιζευγνύντες ἀπὸ τῶν Α, Θ, Δ, Φ, Γ, Υ ἐπὶ τὸ Κ εὐθείας καὶ ἀπὸ τῶν Ε, Σ, Θ, Ρ, Η, Π ἐπὶ τὸ Μ καὶ ἀνιστάντες ἐφ΄ ἑκάστου τῶν τριγώνων πυραμίδας τὴν αὐτὴν κορυφὴν ἐθούσας τοῖς κώνοις δείχομεν, ὅτι καὶ ἑκάστη τῶν ὁμοταγῶν πυραμίδων πρὸς ἑκάστην ὁμοταγῆ πυραμίδα τριπλασίονα λόγον ὁέχει ἥπερ ἡ ΒΚ ὁμόλογος πλευρὰ πρὸς τὴν ΖΜ ὁμόλογον πλευράν, τουτέστιν ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ. καὶ ὡξ ὁὲν τῶν ἡγουμένων πρὸς ὁὲν τῶν ἑπομένων, o-

ATBUCVDW, and apex the point *L*. And let *NFP* be one of the triangles containing the pyramid whose base is triangle *EPFQGRHS*, and apex the point *N*. And let *KT* and *MP* be joined. And since cone *ABCDL* is similar to cone *EFGHN*, thus as *BD* is to *FH*, so axis *KL* (is) to axis *MN* [Def. 11.24]. And as *BD* (is) to *FH*, so *BK* (is) to *FM*. And, thus, as *BK* (is) to *FM*, so *KL* (is) to *MN*. And, alternately, as *BK* (is) to *KL*, so *FM* (is) to *MN* [Prop. 5.16]. And the sides around the equal angles *BKL* and *FMN* are proportional. Thus, triangle *BKL* is similar to triangle *FMN* [Prop. 6.6]. Again, since as *BK* (is) to *KT*, so *FM* (is) to *MP*, and (they are) about the equal angles *BKT* and *FMP*, inasmuch as whatever part angle *BKT* is of the four right-angles at the center *K*, angle *FMP* is also the same part of the four right-angles at the center *M*. Therefore, since the sides about equal angles are proportional, triangle *BKT* is thus similar to traingle *FMP* [Prop. 6.6]. Again, since it was shown that as *BK* (is) to *KL*, so *FM* (is) to *MN*, and *BK* (is) equal to *KT*, and *FM* to *PM*, thus as *TK* (is) to *KL*, so *PM* (is) to *MN*. And the sides about the equal angles *TKL* and *PMN*—for (they are both) right-angles—are proportional. Thus, triangle *LKT* (is) similar to triangle *NMP* [Prop. 6.6]. And since, on account of the similarity of triangles *LKB* and *NMF*, as *LB* (is) to *BK*, so *NF* (is) to *FM*, and, on account of the similarity of triangles *BKT* and *FMP*, as *KB* (is) to *BT*, so *MF* (is) to *FP* [Def. 6.1], thus, via equality, as *LB* (is) to *BT*, so *NF* (is) to *FP* [Prop. 5.22]. Again, since, on account of the similarity of triangles *LTK* and *NPM*, as *LT* (is) to *TK*, so *NP* (is) to *PM*, and, on account of the similarity of triangles *TKB* and *PMF*, as *KT* (is) to *TB*, so *MP* (is) to *PF*, thus, via equality, as *LT* (is) to *TB*, so *NP* (is) to *PF* [Prop. 5.22]. And it was shown that as *TB* (is) to *BL*, so *PF* (is) to *FN*. Thus, via equality, as *TL* (is) to *LB*, so *PN* (is) to *NF* [Prop. 5.22]. Thus, the sides of triangles *LTB* and *NPF* are proportional. Thus, triangles *LTB* and *NPF* are equiangular [Prop. 6.5]. And, hence, (they are) similar [Def. 6.1]. And, thus, the pyramid whose base is triangle *BKT*, and apex the point *L*, is similar to the pyramid whose base is triangle *FMP*, and apex the point *N*. For they are contained by equal numbers of similar planes [Def. 11.9]. And similar pyramids which also have triangular bases are in the cubed ratio of corresponding sides [Prop. 12.8]. Thus, pyramid *BKTL* has to pyramid *FMPN* the cubed ratio that *BK* (has) to *FM*. So, similarly, joining straight-lines from (points) *A*, *W*, *D*, *V*, *C*, and *U* to (center) *K*, and from (points) *E*, *S*, *H*, *R*, *G*, and *Q* to (center) *M*, and setting up pyramids having the same apexes as the cones on each of the triangles (so formed), we can also show that each of the pyramids (on base *ABCD* taken) in order will have to each of the pyramids (on base *EFGH* taken) in order the cubed ratio that the corresponding side *BK* (has) to the corresponding side *FM*—that is to say, that *BD* (has) to *FH*. And (for two sets of proportional magnitudes) as one of the

οὕτως ὁπάντα τὰ ἡγούμενα πρὸς ὁπάντα τὰ ἑπόμενα· ὐέστιν ἄρα καὶ ὡξ ἡ ΒΚΤΛ πυραμὶξ πρὸς τὴν ΖΜΟΝ πυραμίδα, οὑτως ἡ ὅλη πυραμίξ, ἧξ βάσις τὸ ΑΤΒΥΓΦΔΘ πολύγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, πρὸς τὴν ὅλην πυραμίδα, ἧξ βάσις μὲν τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν σημεῖον· ὁώστε καὶ πυραμίξ, ἧξ βάσις μὲν τὸ ΑΤΒΥΓΦΔΘ, κορυφὴ δὲ τὸ Λ, πρὸς τὴν πυραμίδα, ἧξ βάσις [μὲν] τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν σημεῖον, τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ. ὑπόκειται δὲ καὶ ὁ κῶνος, οὗ βάσις [μὲν] ὁ ΑΒΓΔ κύκλος, κορυφὴ δὲ τὸ Λ σημεῖον, πρὸς τὸ Χ στερεὸν τριπλασίονα λόγονὸέθων ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ· ὐέστιν ἄρα ὡξ ὁ κῶνος, οὗ βάσις μέν ἐστιν ὁ ΑΒΓΔ κύκλος, κορυφὴ δὲ τὸ Λ, πρὸς τὸ Χ στερεόν, οὑτως ἡ πυραμίξ, ἧξ βάσις μὲν τὸ ΑΤΒΥΓΦΔΘ [πολύγωνον], κορυφὴ δὲ τὸ Λ, πρὸς τὴν πυραμίδα, ἧξ βάσις μέν ἐστι τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν· ἐναλλὰχ ἄρα, ὡξ ὁ κῶνος, οὗ βάσις μὲν ὁ ΑΒΓΔ κύκλος, κορυφὴ δὲ τὸ Λ, πρὸς τὴν ἐν αὐτῷ πυραμίδα, ἧξ βάσις μὲν τὸ ΑΤΒΥΓΦΔΘ πολύγωνον, κορυφὴ δὲ τὸ Λ, οὑτως τὸ Χ [στερεὸν] πρὸς τὴν πυραμίδα, ἧξ βάσις μέν ἐστι τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν. μείζων δὲ ὁ εἰρημένος κῶνος τῆς ἐν αὐτῷ πυραμίδος· ἐμπεριέθει γὰρ αὐτὴν. μεῖζον ἄρα καὶ τὸ Χ στερεὸν τῆς πυραμίδος, ἧξ βάσις μέν ἐστι τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν. ἀλλὰ καὶ ὐέλαττον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ὁ κῶνος, οὗ βάσις ὁ ΑΒΓΔ κύκλος, κορυφὴ δὲ τὸ Λ [σημεῖον], πρὸςὐέλαττόν τι τοῦ κώνου στερεόν, οὗ βάσις μὲν ὁ ΕΖΗΘ κύκλος, κορυφὴ δὲ τὸ Ν σημεῖον, τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ὁ ΕΖΗΘΝ κῶνος πρὸςὐέλαττόν τι τοῦ ΑΒΓΔΛ κώνου στερεὸν τριπλασίονα λόγον ἔχει ἥπερ ἡ ΖΘ πρὸς τὴν ΒΔ.

Λέγω δή, ὅτι οὐδὲ ὁ ΑΒΓΔΛ κῶνος πρὸς μεῖζόν τι τοῦ ΕΖΗΘΝ κώνου στερεὸν τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ.

Εἰ γὰρ δυνατόν, ἐθέτω πρὸς μεῖζον τὸ Χ. ἀνάπαλιν ἄρα τὸ Χ στερεὸν πρὸς τὸν ΑΒΓΔΛ κῶνον τριπλασίονα λόγον ἔχει ἥπερ ἡ ΖΘ πρὸς τὴν ΒΔ. ὡξ δὲ τὸ Χ στερεὸν πρὸς τὸν ΑΒΓΔΛ κῶνον, οὑτως ὁ ΕΖΗΘΝ κῶνος πρὸςὐέλαττόν τι τοῦ ΑΒΓΔΛ κώνου στερεόν. καὶ ὁ ΕΖΗΘΝ ἄρα κῶνος πρὸςὐέλαττόν τι τοῦ ΑΒΓΔΛ κώνου στερεὸν τριπλασίονα λόγον ἔχει ἥπερ ἡ ΖΘ πρὸς τὴν ΒΔ· ὅπερ ἀδύνατον ἐδείθθη. οὐκ ἄρα ὁ ΑΒΓΔΛ κῶνος πρὸς μεῖζόν τι τοῦ ΕΖΗΘΝ κώνου στερεὸν τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ. ἐδείθθη δέ, ὅτι οὐδὲ πρὸςὐέλαττον. ὁ ΑΒΓΔΛ ἄρα κῶνος πρὸς τὸν ΕΖΗΘΝ κῶνον τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ.

Ὡξ δὲ ὁ κῶνος πρὸς τὸν κῶνον, ὁ κύλινδρος πρὸς τὸν κύλινδρον· τριπλάσιος γὰρ ὁ κύλινδρος τοῦ κώνου ὁ ἐπὶ τῆς αὐτῆς βάσεως τῷ κώνῳ καὶ ἰσοϋψὴς αὐτῷ. καὶ ὁ κύλινδρος ἄρα πρὸς τὸν κύλινδρον τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ.

Οἱ ἄρα ὅμοιοι κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους ἐν τριleading (magnitudes is) to one of the following, so (the sum of) all of the leading (magnitudes is) to (the sum of) all of the following (magnitudes) [Prop. 5.12]. And, thus, as pyramid $BKTL$ (is) to pyramid $FMPN$, so the whole pyramid whose base is polygon $ATBUCVDW$, and apex the point L, (is) to the whole pyramid whose base is polygon $EPFQGRHS$, and apex the point N. And, hence, the pyramid whose base is polygon $ATBUCVDW$, and apex the point L, has to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N, the cubed ratio that BD (has) to FH. And it was also assumed that the cone whose base is circle $ABCD$, and apex the point L, has to solid O the cubed ratio that BD (has) to FH. Thus, as the cone whose base is circle $ABCD$, and apex the point L, is to solid O, so the pyramid whose base (is) [polygon] $ATBUCVDW$, and apex the point L, (is) to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N. Thus, alternately, as the cone whose base (is) circle $ABCD$, and apex the point L, (is) to the pyramid within it whose base (is) the polygon $ATBUCVDW$, and apex the point L, so the [solid] O (is) to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N [Prop. 5.16]. And the aforementioned cone (is) greater than the pyramid within it. For it encompasses it. Thus, solid O (is) also greater than the pyramid whose base is polygon $EPFQGRHS$, and apex the point N. But, (it is) also less. The very thing is impossible. Thus, the cone whose base (is) circle $ABCD$, and apex the [point] L, does not have to some solid less than the cone whose base (is) circle $EFGH$, and apex the point N, the cubed ratio that BD (has) to EH. So, similarly, we can show that neither does cone $EFGHN$ have to some solid less than cone $ABCDL$ the cubed ratio that FH (has) to BD.

So, I say that neither does cone $ABCDL$ have to some solid greater than cone $EFGHN$ the cubed ratio that BD (has) to FH.

For, if possible, let it have (such a ratio) to a greater (solid), O. Thus, inversely, solid O has to cone $ABCDL$ the cubed ratio that FH (has) to BD [Prop. 5.7 corr.]. And as solid O (is) to cone $ABCDL$, so cone $EFGHN$ (is) to some solid less than cone $ABCDL$ [12.2 lem.]. Thus, cone $EFGHN$ also has to some solid less than cone $ABCDL$ the cubed ratio that FH (has) to BD. The very thing was shown (to be) impossible. Thus, cone $ABCDL$ does not have to some solid greater than cone $EFGHN$ the cubed ratio than BD (has) to FH. And it was shown that neither (does it have such a ratio) to a lesser (solid). Thus, cone $ABCDL$ has to cone $EFGHN$ the cubed ratio that BD (has) to FG.

And as the cone (is) to the cone, so the cylinder (is) to the cylinder. For a cylinder is three times a cone on the same base as the cone, and of the same height as it [Prop. 12.10]. Thus, the cylinder also has to the cylinder the cubed ratio that BD (has) to FH.

πλασίονι λόγῳ εἰσὶ τῶν ἐν ταῖς βάσεσι διαμέτρων· ὅπερ ἔδει δεῖξαι.

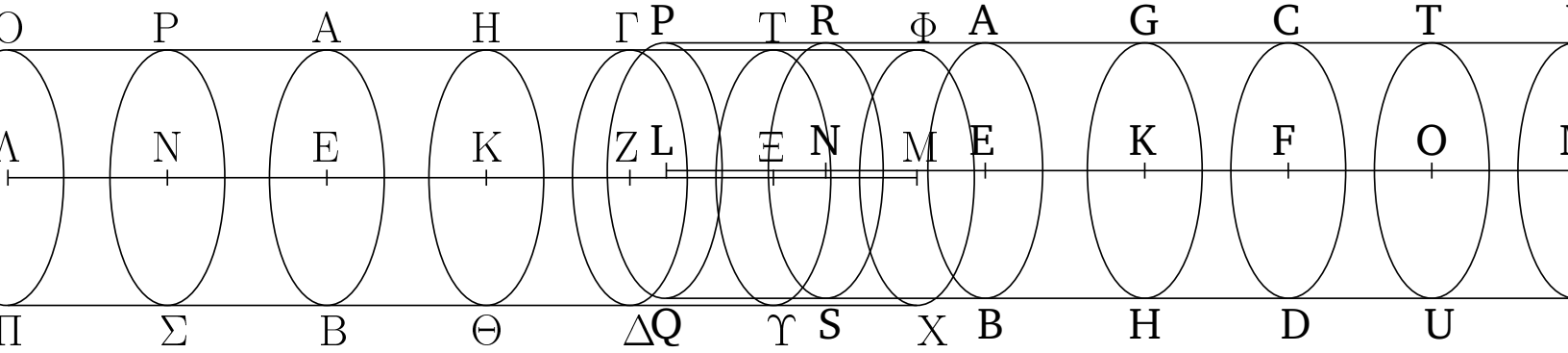
Thus, similar cones and cylinders are in the cubed ratio of the diameters of their bases. (Which is) the very thing it was required to show.

13.

Ἐὰν κύλινδρος ἐπιπέδῳ τμ ηθῇ παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ὀέσται ὡξ ὁ κύλινδρος πρὸς τὸν κύλινδρον, οὑτως ὁ ὀάχων πρὸς τὸνὀάχονα.

Proposition 13

If a cylinder is cut by a plane which is parallel to the opposite planes (of the cylinder) then as the cylinder (is) to the cylinder, so the axis will be to the axis.



Κύλινδρος γὰρ ὁ ΑΔ ἐπιπέδῳ τῷ ΗΘ τετμήσθω παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις τοῖς ΑΒ, ΓΔ, καὶ συμβαλλέτω τῷ ὀάχονι τὸ ΗΘ ἐπίπεδον κατὰ τὸ Κ σημεῖον· λέγω, ὅτι ἐστὶν ὡξ ὁ ΒΗ κύλινδρος πρὸς τὸν ΗΔ κύλινδρον, οὑτως ὁ ΕΚὀάχων πρὸς τὸν ΚΖὀάχονα.

Ἐκβεβλήσθω γὰρ ὁ ΕΖὀάχων ἐφ' ἑκάτερα τὰ μέρη ἐπὶ τὰ Λ, Μ σημεῖα, καὶ ἐκκείσθωσαν τῷ ΕΚὀάχονιὀίσοι ὁσοιδηποτοῦν οἱ ΕΝ, ΝΛ, τῷ δὲ ΖΚὀίσοι ὁσοιδηποτοῦν οἱ ΖΧ, ΧΜ, καὶ νοείσθω ὁ ἐπὶ τοῦ ΛΜὀάχονος κύλινδρος ὁ ΟΘ, οὗ βάσεις οἱ ΟΠ, ΦΘ κύκλοι. καὶ ἐκβεβλήσθω διὰ τῶν Ν, Χ σημείων ἐπίπεδα παράλληλα τοῖς ΑΒ, ΓΔ καὶ ταῖς βάσεσι τοῦ ΟΘ κυλίνδρου καὶ ποιείτωσαν τοὐξ ΡΣ, ΤΥ κύκλους περὶ τὰ Ν, Χ κέντρα. καὶ ἐπεὶ οἱ ΛΝ, ΝΕ, ΕΚὀάχονεςὀίσοι εἰσὶν ἀλλήλοις, οἱ ἄρα ΠΡ, ΡΒ, ΒΗ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡξ αἱ βάσεις. ἴσαι δέ εἰσιν αἱ βάσεις· ὀίσοι ἄρα καὶ οἱ ΠΡ, ΡΒ, ΒΗ κύλινδροι ἀλλήλοις. ἐπεὶ οὐῦν οἱ ΛΝ, ΝΕ, ΕΚὀάχονεςὀίσοι εἰσὶν ἀλλήλοις, εἰσὶ δὲ καὶ οἱ ΠΡ, ΡΒ, ΒΗ κύλινδροιὀίσοι ἀλλήλοις, καί ἐστιν ἴσον τὸ πλῆθος τῷ πλήθει, ὁσαπλασίων ἄρα ὁ ΚΛὀάχων τοῦ ΕΚὀάχονος, τοσαυταπλασίωνὀέσται καὶ ὁ ΠΗ κύλινδρος τοῦ ΗΒ κυλίνδρου. διὰ τὰ αὐτὰ δὴ καὶ ὁσαπλασίων ἐστὶν ὁ ΜΚὀάχων τοῦ ΚΖὀάχονος, τοσαυταπλασίων ἐστὶ καὶ ὁ ΘΗ κύλινδρος τοῦ ΗΔ κυλίνδρου. καὶ εἰ μὲνὀίσος ἐστὶν ὁ ΚΛὀάχων τῷ ΚΜὀάχονι, ὀίσοςὀέσται καὶ ὁ ΠΗ κύλινδρος τῷ ΗΘ κυλίνδρῳ, εἰ δὲ μείζων ὁ ὀάχων τοῦ ὀάχονος, μείζων καὶ ὁ κύλινδρος τοῦ κυλίνδρου, καὶ εἰ ἐλάσσων, ἐλάσσων. τεσσάρων δὴ μεγεθῶν ὄντων, ἀχόνων μὲν τῶν ΕΚ, ΚΖ, κυλίνδρων δὲ τῶν ΒΗ, ΗΔ, εὐίληπται ἰσάκις πολλαπλάσια, τοῦ μὲν ΕΚὀάχονος καὶ τοῦ ΒΗ κυλίνδρου ὅ τε ΛΚὀάχων καὶ ὁ ΠΗ κύλινδρος, τοῦ δὲ ΚΖὀάχονες καὶ τοῦ ΗΔ κυλίνδρου ὅ τε ΚΜὀάχων καὶ ὁ ΗΘ κύλινδρος, καὶ δέδεικται, ὅτι εἰ ὑπερέθει ὁ ΚΛὀάχων τοῦ ΚΜὀάχονος, ὑπερέθει καὶ ὁ ΠΗ κύλινδρος τοῦ ΗΘ κυλίνδρου,

For let the cylinder AD be cut by the plane GH which is parallel to the opposite planes (of the cylinder), AB and CD. And let the plane GH have met the axis at point K. I say that as cylinder BG is to cylinder GD, so axis EK (is) to axis KF.

For let axis EF be produced in each direction to points L and M. And let any number whatsoever (of lengths), EN and NL, equal to axis EK, be set out (on the axis EL), and any number whatsoever (of lengths), FO and OM, equal to (axis) FK, (on the axis KM). And let the cylinder PW, whose bases (are) the circles PQ and VW, be conceived on axis LM. And let planes parallel to AB, CD, and the bases of cylinder PW, be produced through points N and O, and let them have made the circles RS and TU around the centers N and O (respectively). And since axes LN, NE, and EK are equal to one another, the cylinders QR, RB, and BG are to one another as their bases [Prop. 12.11]. But the bases are equal. Thus, the cylinders QR, RB, and BG (are) also equal to one another. Therefore, since the axes LN, NE, and EK are equal to one another, and the cylinders QR, RB, and BG are also equal to one another, and the number (of the former) is equal to the number (of the latter), thus as many multiples as axis KL is of axis EK, so many multiples is cylinder QG also of cylinder GB. And so, for the same (reasons), as many multiples as axis MK is of axis KF, so many multiples is cylinder WG also of cylinder GD. And if axis KL is equal to axis KM then cylinder QG will also be equal to cylinder GW, and if the axis (is) greater than the axis then the cylinder (will also be) greater than the cylinder, and if (the axis is) less then (the cylinder will also be) less. So, there are four magnitudes—the axes EK and KF, and the cylinders BG and GD—and equal multiples be taken of axis EK and cylinder BG—(namely), axis LK and cylinder

καὶ εἰ ὁίσος, ὁίσος, καὶ εἰ ἐλάσσων, ἐλάσσων. ὁέστιν ἄρα ὡξ ὁ ΕΚὁάχων πρὸς τὸν ΚΖὁάχονα, οὁύτως ὁ ΒΗ κύλινδρος πρὸς τὸν ΗΔ κύλινδρον· ὅπερ ἔδει δεῖχαι.

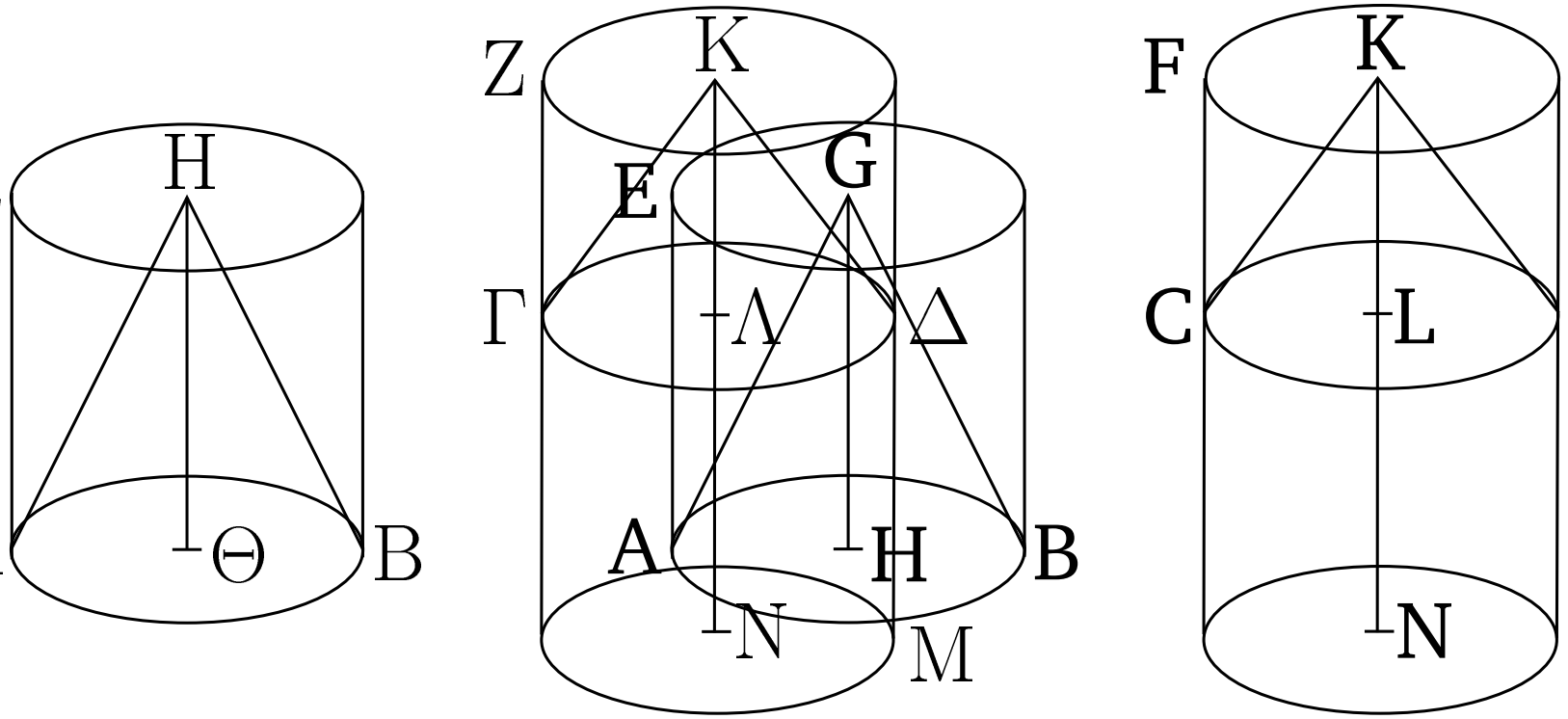
QG—and of axis *KF* and cylinder *GD*—(namely), axis *KM* and cylinder *GW*. And it has been shown that if axis *KL* exceeds axis *KM* then cylinder *QG* also exceeds cylinder *GW*, and if (the axes are) equal then (the cylinders are) equal, and if (*KL* is) less then (*QG* is) less. Thus, as axis *EK* is to axis *KF*, so cylinder *BG* (is) to cylinder *GD* [Def. 5.5]. (Which is) the very thing it was required to show.

14.

Οἱ ἐπὶ ὁίσων βάσεων ὄντες κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡξ τὰ ὁύψη.

Proposition 14

Cones and cylinders which are on equal bases are to one another as their heights.



Ἔστωσαν γὰρ ἐπὶ ὁίσων βάσεων τῶν ΑΒ, ΓΔ κύκλων κύλινδροι οἱ ΕΒ, ΖΔ· λέγω, ὅτι ἐστὶν ὡξ ὁ ΕΒ κύλινδρος πρὸς τὸν ΖΔ κύλινδρον, οὁύτως ὁ ΗΘὁάχων πρὸς τὸν ΚΛὁάχονα.

Ἐκβεβλήσθω γὰρ ὁ ΚΛὁάχων ἐπὶ τὸ Ν σημεῖον, καὶ κείσθω τῷ ΗΘὁάχονιὁίσος ὁ ΛΝ, καὶ περὶ ὁάχονα τὸν ΛΝ κύλινδρος νενοήσθω ὁ ΓΜ. ἐπεὶ οὁῦν οἱ ΕΒ, ΓΜ κύλινδροι ὑπὸ τὸ αὐτὸ ὁύψος εἰσίν, πρὸς ἀλλήλους εἰσὶν ὡξ αἱ βάσεις. ἴσαι δέ εἰσίν αἱ βάσεις ἀλλήλαις· ὁίσοι ἄρα εἰσὶ καὶ οἱ ΕΒ, ΓΜ κύλινδροι. καὶ ἐπεὶ κύλινδρος ὁ ΖΜ ἐπιπέδῳ τέτμ ηται τῷ ΓΔ παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ὁέστιν ἄρα ὡξ ὁ ΓΜ κύλινδρος πρὸς τὸν ΖΔ κύλινδρον, οὁύτως ὁ ΛΝὁάχων πρὸς τὸν ΚΛὁάχονα. ὁίσος δέ ἐστιν ὁ μὲν ΓΜ κύλινδρος τῷ ΕΒ κυλίνδρῳ, ὁ δὲ ΛΝὁάχων τῷ ΗΘὁάχονι· ὁέστιν ἄρα ὡξ ὁ ΕΒ κύλινδρος πρὸς τὸν ΖΔ κύλινδρον, οὁύτως ὁ ΗΘὁάχων πρὸς τὸν ΚΛὁάχονα. ὡξ δὲ ὁ ΕΒ κύλινδρος πρὸς τὸν ΖΔ

For let *EB* and *FD* be cylinders on equal bases, (namely) the circles *AB* and *CD* (respectively). I say that as cylinder *EB* is to cylinder *FD*, so axis *GH* (is) to axis *KL*.

For let the axis *KL* be produced to point *N*. And let *LN* be made equal to axis *GH*. And let the cylinder *CM* be conceived about axis *LN*. Therefore, since cylinders *EB* and *CM* have the same height they are to one another as their bases [Prop. 12.11]. And the bases are equal to one another. Thus, cylinders *EB* and *CM* are also equal to one another. And since cylinder *FM* has been cut by the plane *CD*, which is parallel to its opposite planes, thus as cylinder *CM* is to cylinder *FD*, so axis *LN* (is) to axis *KL* [Prop. 12.13]. And cylinder *CM* is equal to cylinder *EB*, and axis *LN* to axis *GH*. Thus, as cylinder *EB* is to cylinder *FD*, so axis *GH* (is) to axis *KL*. And as cylinder *EB* (is) to cylinder *FD*, so cone *ABG* (is) to cone

κύλινδρον, οὕτως ὁ ΑΒΗ κῶνος πρὸς τὸν ΓΔΚ κῶνον. καὶ ὡξ ἄρα ὁ ΗΘὀάχων πρὸς τὸν ΚΛὀάχονα, οὕτως ὁ ΑΒΗ κῶνος πρὸς τὸν ΓΔΚ κῶνον καὶ ὁ ΕΒ κύλινδρος πρὸς τὸν ΖΔ κύλινδρον· ὅπερ ἔδει δεῖχαι.

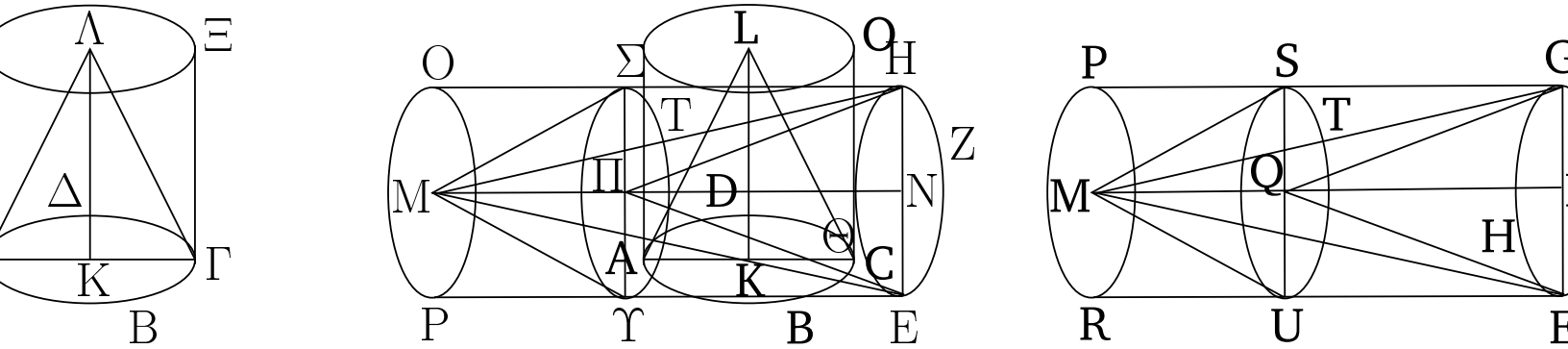
CDK [Prop. 12.10]. Thus, also, as axis *GH* (is) to axis *KL*, so cone *ABG* (is) to cone *CDK*, and cylinder *EB* to cylinder *FD*. (Which is) the very thing it was required to show.

15.

Τῶνὀίσων κώνων καὶ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν· καὶ ὧν κώνων καὶ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν, ὁίσοι εἰσὶν ἐκεῖνοι.

Proposition 15

The bases of equal cones and cylinders are reciprocally proportional to their heights. And, those cones and cylinders whose bases (are) reciprocally proportional to their heights are equal.



Ἔστωσανὁίσοι κῶνοι καὶ κύλινδροι, ὧν βάσεις μὲν οἱ ΑΒΓΔ, ΕΖΗΘ κύκλοι, διάμετροι δὲ αὐτῶν αἱ ΑΓ, ΕΗ, ὁάχονες δὲ οἱ ΚΛ, ΜΝ, οὁίτινες καὶ ὑψη εἰσὶ τῶν κώνωνὁή κυλίνδρων, καὶ συμπεπληρώσθωσαν οἱ ΑΧ, ΕΟ κύλινδροι. λέγω, ὅτι τῶν ΑΧ, ΕΟ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν, καί ἐστιν ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὕτως τὸ ΜΝ ὕψος πρὸς τὸ ΚΛ ὕψος.

Τὸ γὰρ ΛΚ ὕψος τῷ ΜΝ ὕψει ἤτοι ἴσον ἐστὶνὁή οὐ. ὁέστω πρότερον ἴσον. ὁέστι δὲ καὶ ὁ ΑΧ κύλινδρος τῷ ΕΟ κυλίνδρῳ ὁίσος. οἱ δὲ ὑπὸ τὸ αὐτὸ ὕψος ὄντες κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡξ αἱ βάσεις· ὁίση ἄρα καὶ ἡ ΑΒΓΔ βάσις τῇ ΕΖΗΘ βάσει. ὥστε καὶ ἀντιπέπονθεν, ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὕτως τὸ ΜΝ ὕψος πρὸς τὸ ΚΛ ὕψος. ἀλλὰ δὴ μὴ ὁέστω τὸ ΛΚ ὕψος τῷ ΜΝ ἴσον, ἀλλὸ ὁέστω μεῖζον τὸ ΜΝ, καὶ ἀφῃρήσθω ἀπὸ τοῦ ΜΝ ὕψους τῷ ΚΛ ἴσον τὸ ΠΝ, καὶ διὰ τοῦ Π σημείου τετμήσθω ὁ ΕΟ κύλινδρος ἐπιπέδῳ τῷ ΤΥΣ παραλλήλῳ τοῖς τῶν ΕΖΗΘ, ΡΟ κύκλων ἐπιπέδοις, καὶ ἀπὸ βάσεως μὲν τοῦ ΕΖΗΘ κύκλου, ὕψους δὲ τοῦ ΝΠ κύλινδρος νενοήσθω ὁ ΕΣ. καί ἐπεὶ ὁίσος ἐστὶν ὁ ΑΧ κύλινδρος τῷ ΕΟ κυλίνδρῳ, ὁέστιν ἄρα ὡξ ὁ ΑΧ κύλινδρος πρὸς τὸν ΕΣ κυλίνδρον, οὕτως ὁ ΕΟ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον. ἀλλὸ ὡξ μὲν ὁ ΑΧ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον, οὕτως ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ· ὑπὸ γὰρ τὸ αὐτὸ ὕψος εἰσὶν οἱ ΑΧ, ΕΣ κύλινδροι· ὡξ δὲ ὁ ΕΟ κύλινδρος πρὸς τὸν ΕΣ, οὕτως τὸ ΜΝ ὕψος πρὸς τὸ ΠΝ ὕψος· ὁ γὰρ ΕΟ κύλινδρος ἐπιπέδῳ τέτμ ηται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις. ὁέστιν ἄρα καὶ ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὕτως τὸ ΜΝ ὕψος πρὸς τὸ ΠΝ ὕψος. ἴσον δὲ τὸ ΠΝ ὕψος τῷ ΚΛ

Let there be equal cones and cylinders whose bases are the circles *ABCD* and *EFGH*, and the diameters of (the bases) *AC* and *EG*, and (whose) axes (are) *KL* and *MN*, which are also the heights of the cones and cylinders (respectively). And let the cylinders *AO* and *EP* be completed. I say that the bases of cylinders *AO* and *EP* are reciprocally proportional to their heights, and (so) as base *ABCD* is to base *EFGH*, so height *MN* (is) to height *KL*.

For height *LK* is either equal to height *MN*, or not. Let it, first of all, be equal. And cylinder *AO* is also equal to cylinder *EP*. And cones and cylinders having the same height are to one another as their bases [Prop. 12.11]. Thus, base *ABCD* (is) also equal to base *EFGH*. And, hence, reciprocally, as base *ABCD* (is) to base *EFGH*, so height *MN* (is) to height *KL*. And so, let height *LK* not be equal to *MN*, but let *MN* be greater. And let *QN*, equal to *KL*, be cut off from height *MN*. And let the cylinder *EP* be cut, through point *Q*, by the plane *TUS* (which is) parallel to the planes of the circles *EFGH* and *RP*. And let cylinder *ES* be conceived, with base the circle *EFGH*, and height *NQ*. And since cylinder *AO* is equal to cylinder *EP*, thus, as cylinder *AO* (is) to cylinder *ES*, so cylinder *EP* (is) to cylinder *ES* [Prop. 5.7]. But, as cylinder *AO* (is) to cylinder *ES*, so base *ABCD* (is) to base *EFGH*. For cylinders *AO* and *ES* (have) the same height [Prop. 12.11]. And as cylinder *EP* (is) to (cylinder) *ES*, so height *MN* (is) to height *QN*. For cylinder *EP* has been cut by a plane which is parallel to its opposite planes [Prop. 12.13]. And, thus, as base *ABCD* is to base *EFGH*, so height *MN* (is) to height *QN* [Prop. 5.11]. And height *QN* (is) equal to height *KL*. Thus,

ὑψει· ὐέστιν ἄρα ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὑτως τὸ ΜΝ ὑψος πρὸς τὸ ΚΛ ὑψος. τῶν ἄρα ΑΧ, ΕΟ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν.

Ἀλλὰ δὴ τῶν ΑΧ, ΕΟ κυλίνδρων ἀντιπεπονθέτωσαν αἱ βάσεις τοῖς ὑψεσιν, καὶ ὐέστω ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὑτως τὸ ΜΝ ὑψος πρὸς τὸ ΚΛ ὑψος· λέγω, ὅτιὐίσος ἐστὶν ὁ ΑΧ κύλινδρος τῷ ΕΟ κυλίνδρῳ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ἐπεί ἐστιν ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὑτως τὸ ΜΝ ὑψος πρὸς τὸ ΚΛ ὑψος, ἴσον δὲ τὸ ΚΛ ὑψος τῷ ΠΝ ὑψει, ὐέσται ἄρα ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὑτως τὸ ΜΝ ὑψος πρὸς τὸ ΠΝ ὑψος. ἀλλὰ ὡξ μὲν ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὑτως ὁ ΑΧ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον· ὑπὸ γὰρ τὸ αὐτὸ ὑψος εἰσίν· ὡξ δὲ τὸ ΜΝ ὑψος πρὸς τὸ ΠΝ [ὑψος], οὑτως ὁ ΕΟ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον· ὐέστιν ἄρα ὡξ ὁ ΑΧ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον, οὑτως ὁ ΕΟ κύλινδρος πρὸς τὸν ΕΣ. ὐίσος ἄρα ὁ ΑΧ κύλινδρος τῷ ΕΟ κυλίνδρῳ. ὡσαύτως δὲ καὶ ἐπὶ τῶν κώνων· ὅπερ ἔδει δεῖχαι.

as base $ABCD$ is to base $EFGH$, so height MN (is) to height KL. Thus, the bases of cylinders AO and EP are reciprocally proportional to their heights.

And, so, let the bases of cylinders AO and EP be reciprocally proportional to their heights, and (thus) let base $ABCD$ be to base $EFGH$, as height MN (is) to height KL. I say that cylinder AO is equal to cylinder EP.

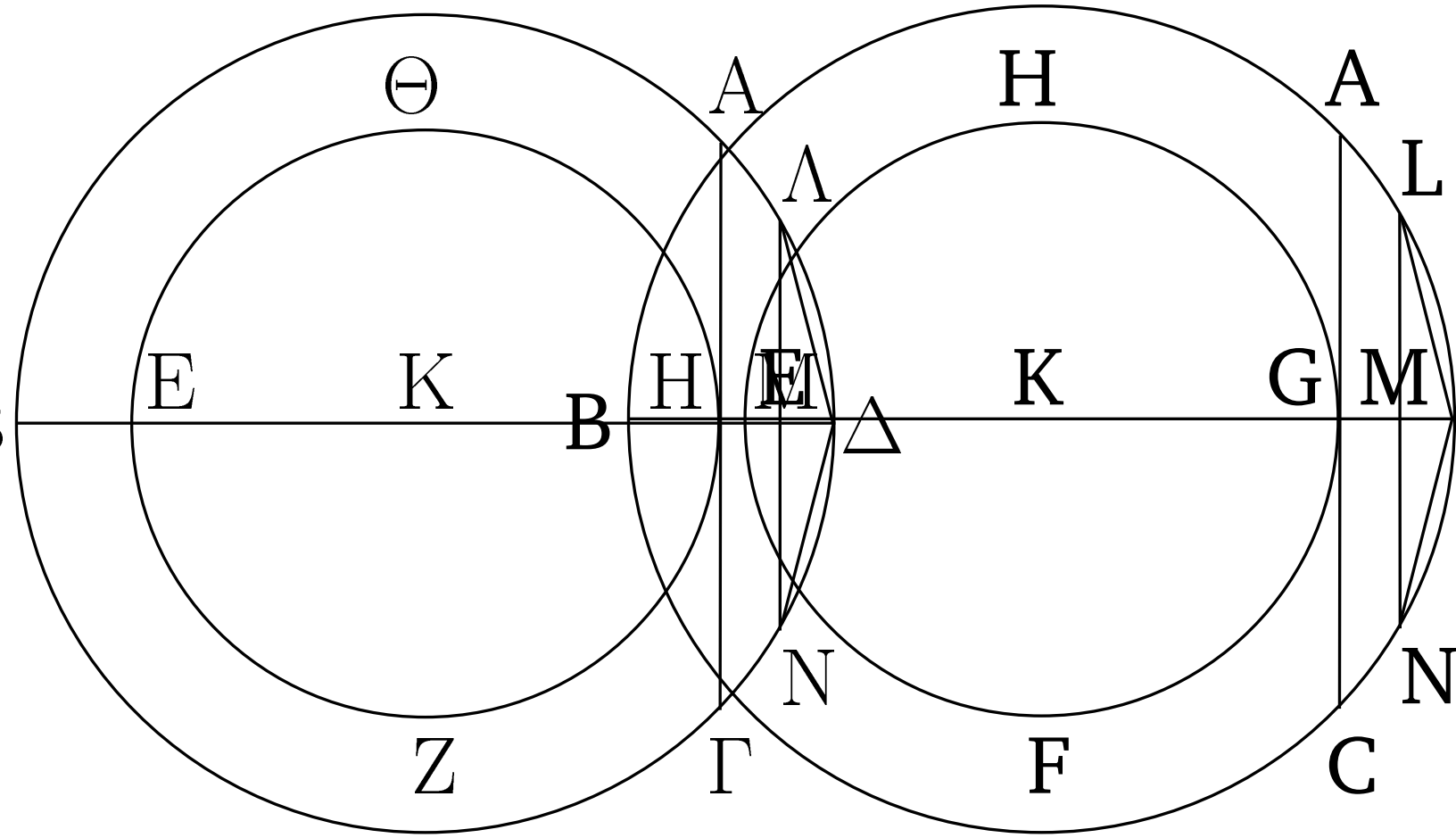
For, with the same construction, since as base $ABCD$ is to base $EFGH$, so height MN (is) to height KL, and height KL (is) equal to height QN, thus, as base $ABCD$ (is) to base $EFGH$, so height MN will be to height QN. But, as base $ABCD$ (is) to base $EFGH$, so cylinder AO (is) to cylinder ES. For they are the same height [Prop. 12.11]. And as height MN (is) to [height] QN, so cylinder EP (is) to cylinder ES [Prop. 12.13]. Thus, as cylinder AO is to cylinder ES, so cylinder EP (is) to (cylinder) ES [Prop. 5.11]. Thus, cylinder AO (is) equal to cylinder EP [Prop. 5.9]. In the same manner, (the proposition can) also (be demonstrated) for the cones. (Which is) the very thing it was required to show.

16.

Δύο κύκλων περὶ τὸ αὐτὸ κέντρον ὄντων εἰξ τὸν μείζονα κύκλον πολύγωνον ἰσόπλευρόν τε καὶ ἀρτιόπλευρον ἐγγράψαι μὴ ψαῦον τοῦ ἐλάσσονος κύκλου.

Proposition 16

There being two circles about the same center, to inscribe an equilateral and even-sided polygon in the greater circle, not touching the lesser circle.



Ἔστωσαν οἱ δοθέντες δύο κύκλοι οἱ ΑΒΓΔ, ΕΖΗΘ περὶ τὸ αὐτὸ κέντρον τὸ Κ· δεῖ δὴ εἰς τὸν μείζονα κύκλον τὸν ΑΒΓΔ πολύγωνον ἰσόπλευρόν τε καὶ ἀρτιόπλευρον ἐγγράψαι μὴ ψαῦον τοῦ ΕΖΗΘ κύκλου.

ὈΗθθω γὰρ διὰ τοῦ Κ κέντρου εὐθεῖα ἡ ΒΚΔ, καὶ ἀπὸ τοῦ Η σημείου τῇ ΒΔ εὐθείᾳ πρὸς ὀρθὰξ ἤθθω ἡ ΗΑ καὶ διήθθω ἐπὶ τὸ Γ· ἡ ΑΓ ἄρα ἐφάπτεται τοῦ ΕΖΗΘ κύκλου. τέμνοντες δὴ τὴν ΒΑΔ περιφέρειαν δίθα καὶ τὴν ἡμίσειαν αὐτῆς δίθα καὶ τοῦτο ἀεὶ ποιοῦντες καταλείψομεν περιφέρειαν ἐλάσσονα τῆς ΑΔ. λελείφθω, καὶ ὀέστω ἡ ΛΔ, καὶ ἀπὸ τοῦ Λ ἐπὶ τὴν ΒΔ κάθετος ἤθθω ἡ ΛΜ καὶ διήθθω ἐπὶ τὸ Ν, καὶ ἐπεζεύθθωσαν αἱ ΛΔ, ΔΝ· ὁίση ἄρα ἐστὶν ἡ ΛΔ τῇ ΔΝ. καὶ ἐπεὶ παράλληλόξ ἐστιν ἡ ΛΝ τῇ ΑΓ, ἡ δὲ ΑΓ ἐφάπτεται τοῦ ΕΖΗΘ κύκλου, ἡ ΛΝ ἄρα οὐκ ἐφάπτεται τοῦ ΕΖΗΘ κύκλου· πολλῷ ἄρα αἱ ΛΔ, ΔΝ οὐκ ἐφάπτονται τοῦ ΕΖΗΘ κύκλου. ἐαν δὴ τῇ ΛΔ εὐθείᾳ ἴσας κατὰ τὸ συνεθὲξ ἐναρμόσωμεν εἰξ τὸν ΑΒΓΔ κύκλον, ἐγγραφήσεται εἰξ τὸν ΑΒΓΔ κύκλον πολύγωνον ἰσόπλευρόν τε καὶ ἀρτιόπλευρον μὴ ψαῦον τοῦ ἐλάσσονος κύκλου τοῦ ΕΖΗΘ· ὅπερ ἔδει ποιῆσαι.

Let *ABCD* and *EFGH* be the given two circles, about the same center, *K*. So, it is necessary to inscribe an equilateral and even-sided polygon in the greater circle *ABCD*, not touching circle *EFGH*.

Let the straight-line *BKD* be drawn through the center *K*. And let *GA* be drawn, at right-angles to the straight-line *BD*, through point *G*, and let it be drawn through to *C*. Thus, *AC* touches circle *EFGH* [Prop. 3.16 corr.]. So, (by) cutting circumference *BAD* in half, and the half of it in half, and doing this continually, we will (eventually) leave a circumference less than *AD* [Prop. 10.1]. Let it be left, and let it be *LD*. And let *LM* be drawn, from *L*, perpendicular to *BD*, and let it be drawn through to *N*. And let *LD* and *DN* be joined. Thus, *LD* is equal to *DN* [Props. 3.3, 1.4]. And since *LN* is parallel to *AC* [Prop. 1.28], and *AC* touches circle *EFGH*, *LN* thus does not touch circle *EFGH*. Thus, even more so, *LD* and *DN* do not touch circle *EFGH*. And if we continuously insert (straight-lines) equal to straight-line *LD* into circle *ABCD* [Prop. 4.1] then an equilateral and even-sided polygon, not touching the lesser circle *EFGH*, will be inscribed in circle *ABCD*.† (Which is) the very thing it was required to do.

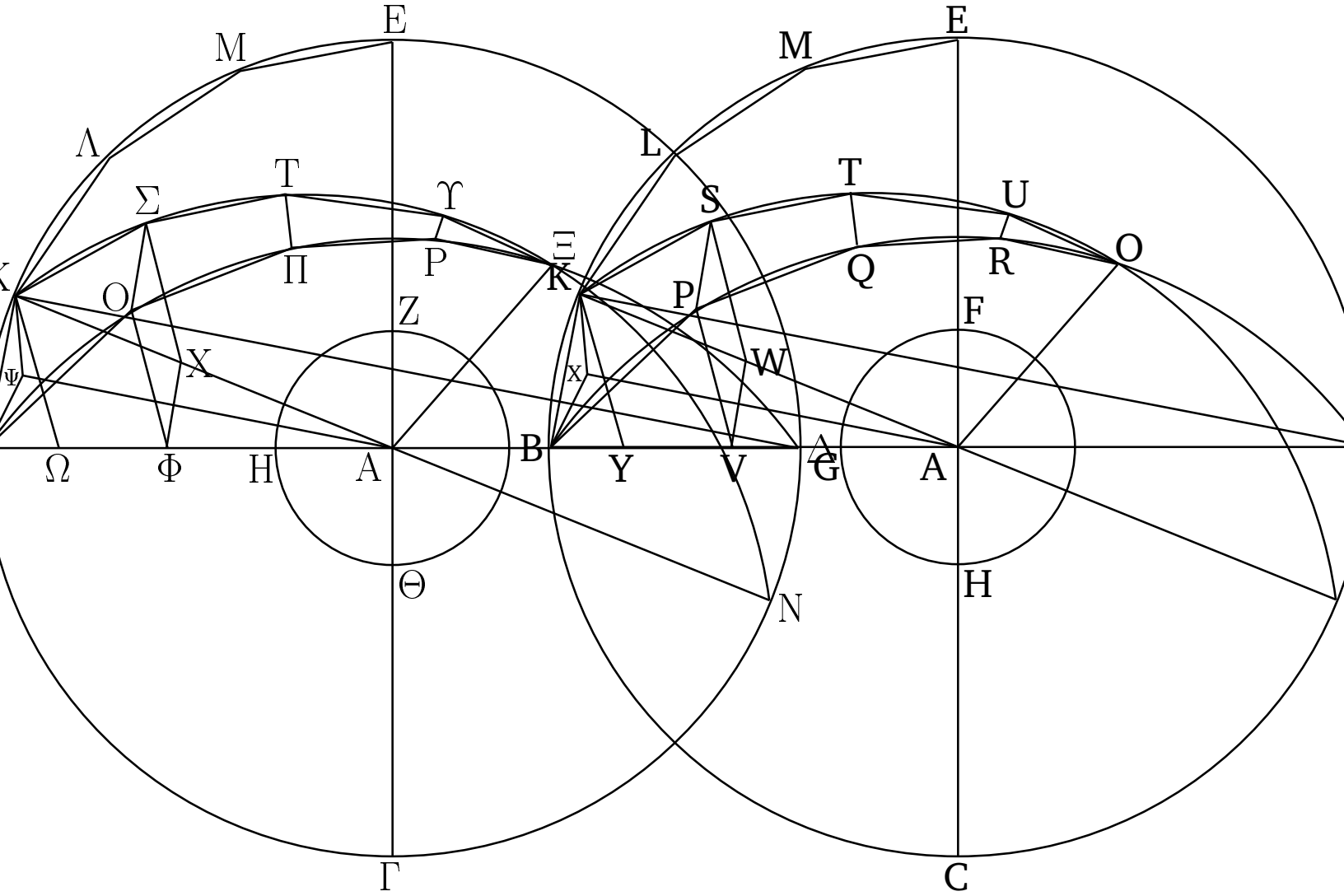
† Note that the chord of the polygon, *LN*, does not touch the inner circle either.

17.

Δύο σφαιρῶν περὶ τὸ αὐτὸ κέντρον οὐσῶν εἰξ τὴν μείζονα σφαῖραν στερεὸν πολύεδρον ἐγγράψαι μὴ ψαῦον τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν.

Proposition 17

There being two spheres about the same center, to inscribe a polyhedral solid in the greater sphere, not touching the lesser sphere on its surface.



Νενοήσθωσαν δύο σφαῖραι περὶ τὸ αὐτὸ κέντρον τὸ A· δεῖ δὴ εἰξ τὴν μείζονα σφαῖραν στερεὸν πολύεδρον ἐγγράψαι μὴ ψαῦον τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν.

Τετμήσθωσαν αἱ σφαῖραι ἐπιπέδῳ τινὶ διὰ τοῦ κέντρου· ὀέσονται δὴ αἱ τομαὶ κύκλοι, ἐπειδήπερ μενούσης τῆς διαμέτρου καὶ περιφερομένου τοῦ ἡμικυκλίου ἐγιγνετο ἡ σφαῖρα· ὥστε καὶ καθὸ οὐίαςὸὰν θέσεως ἐπινοήσωμεν τὸ ἡμικύκλιον, τὸ διὰ αὐτοῦ ἐκβαλλόμενον ἐπίπεδον ποιήσει ἐπὶ τῆς ἐπιφανείας τῆς σφαίρας κύκλον. καὶ φανερόν, ὅτι καὶ μέγιστον, ἐπειδήπερ ἡ διάμετρος τῆς σφαίρας, ἥτις ἐστὶ καὶ τοῦ ἡμι-

Let two spheres be conceived about the same center, *A*. So, it is necessary to inscribe a polyhedral solid in the greater sphere, not touching the lesser sphere on its surface.

Let the spheres be cut by some plane through the center. So, the sections will be circles, inasmuch as a sphere is generated by the diameter remaining behind, and a semi-circle being carried around [Def. 11.14]. And, hence, whatever position we conceive (of for) the semi-circle, the plane produced through it will make a circle on the surface of the sphere. And (it is) clear that (it is) also a great (circle), inasmuch as the diameter of

κυκλίου διάμετρος δηλαδὴ καὶ τοῦ κύκλου, μείζων ἐστὶ πασῶν τῶν εἰξ τὸν κύκλονὁὴ τὴν σφαῖραν διαγομένων [εὐθειῶν]. ὀέστω οὸῦν ἐν μὲν τῇ μείζονι σφαίρᾳ κύκλος ὁ ΒΓΔΕ, ἐν δὲ τῇ ἐλάσσονι σφαίρᾳ κύκλος ὁ ΖΗΘ, καὶ ἤθθωσαν αὐτῶν δύο διάμετροι πρὸς ὀρθὰξ ἀλλήλαις αἱ ΒΔ, ΓΕ, καὶ δύο κύκλων περὶ τὸ αὐτὸ κέντρον ὄντων τῶν ΒΓΔΕ, ΖΗΘ εἰξ τὸν μείζονα κύκλον τὸν ΒΓΔΕ πολύγωνον ἰσόπλευρον καὶ ἀρτιόπλευρον ἐγγεγράφθω μὴ ψαῦον τοῦ ἐλάσσονος κύκλου τοῦ ΖΗΘ, οὗ πλευραὶ ὀέστωσαν ἐν τῷ ΒΕ τεταρτημορίῳ αἱ ΒΚ, ΚΛ, ΛΜ, ΜΕ, καὶ ἐπιζευθθεῖσα ἡ ΚΑ διήθθω ἐπὶ τὸ Ν, καὶ ἀνεστάτω ἀπὸ τοῦ Α σημείου τῷ τοῦ ΒΓΔΕ κύκλου ἐπιπέδῳ πρὸς ὀρθὰξ ἡ ΑΧ καὶ συμβαλλέτω τῇ ἐπιφανείᾳ τῆς σφαίρας κατὰ τὸ Χ, καὶ διὰ τῆς ΑΧ καὶ ἑκατέρας τῶν ΒΔ, ΚΝ ἐπίπεδα ἐκβεβλήσθω· ποιήσουσι δὴ διὰ τὰ εἰρημένα ἐπὶ τῆς ἐπιφανείας τῆς σφαίρας μεγίστους κύκλους. ποιείτωσαν, ὁῶν ἡμικύκλιαὀέστω ἐπὶ τῶν ΒΔ, ΚΝ διαμέτρων τὰ ΒΧΔ, ΚΧΝ. καὶ ἐπεὶ ἡ ΧΑ ὀρθή ἐστι πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον, καὶ πάντα ἄρα τὰ διὰ τῆς ΧΑ ἐπίπεδά ἐστιν ὀρθὰ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον· ὁώστε καὶ τὰ ΒΧΔ, ΚΧΝ ἡμικύκλια ὀρθά ἐστι πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον. καὶ ἐπεὶ ἴσα ἐστὶ τὰ ΒΕΔ, ΒΧΔ, ΚΧΝ ἡμικύκλια· ἐπὶ γὰρὸίσων εἰσὶ διαμέτρων τῶν ΒΔ, ΚΝ· ἴσα ἐστὶ καὶ τὰ ΒΕ, ΒΧ, ΚΧ τεταρτημόρια ἀλλήλοις. ὅσαι ἄρα εἰσὶν ἐν τῷ ΒΕ τεταρτημορίῳ πλευραὶ τοῦ πολυγώνου, τοσαῦταί εἰσι καὶ ἐν τοῖς ΒΧ, ΚΧ τεταρτημορίοις ἴσαι ταῖς ΒΚ, ΚΛ, ΛΜ, ΜΕ εὐθείαις. ἐγγεγράφθωσαν καὶ ὀέστωσαν αἱ ΒΟ, ΟΠ, ΠΡ, ΡΧ, ΚΣ, ΣΤ, ΤΥ, ΥΧ, καὶ ἐπεζεύθθωσαν αἱ ΣΟ, ΤΠ, ΥΡ, καὶ ἀπὸ τῶν Ο, Σ ἐπὶ τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον κάθετοι ἤθθωσαν· πεσοῦνται δὴ ἐπὶ τὰξ κοινὰξ τομὰξ τῶν ἐπιπέδων τὰξ ΒΔ, ΚΝ, ἐπειδήπερ καὶ τὰ τῶν ΒΧΔ, ΚΧΝ ἐπίπεδα ὀρθά ἐστι πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον. πιπτέτωσαν, καὶ ὀέστωσαν αἱ ΟΦ, ΣΘ, καὶ ἐπεζεύθθω ἡ ΘΦ. καὶ ἐπεὶ ἐνὸίσοις ἡμικυκλίοις τοῖς ΒΧΔ, ΚΧΝ ἴσαι ἀπειλημμέναι εἰσὶν αἱ ΒΟ, ΚΣ, καὶ κάθετοι ἠγμέναι εἰσὶν αἱ ΟΦ, ΣΘ, ὁίση [ἄρα] ἐστὶν ἡ μὲν ΟΦ τῇ ΣΘ, ἡ δὲ ΒΦ τῇ ΚΘ. ὀέστι δὲ καὶ ὅλη ἡ ΒΑ ὅλῃ τῇ ΚΑὁίση· καὶ λοιπὴ ἄρα ἡ ΦΑ λοιπῇ τῇ ΘΑ ἐστινὁίση· ὀέστιν ἄρα ὡξ ἡ ΒΦ πρὸς τὴν ΦΑ, οὁύτως ἡ ΚΘ πρὸς τὴν ΘΑ· παράλληλος ἄρα ἐστὶν ἡ ΘΦ τῇ ΚΒ. καὶ ἐπεὶ ἑκατέρα τῶν ΟΦ, ΣΘ ὀρθή ἐστι πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον, παράλληλος ἄρα ἐστὶν ἡ ΟΦ τῇ ΣΘ. ἐδείθθη δὲ αὐτῇ καὶ ὁίση· καὶ αἱ ΘΦ, ΣΟ ἄρα ἴσαι εἰσὶ καὶ παράλληλοι. καὶ ἐπεὶ παράλληλόξ ἐστιν ἡ ΘΦ τῇ ΣΟ, ἀλλὰ ἡ ΘΦ τῇ ΚΒ ἐστι παράλληλος, καὶ ἡ ΣΟ ἄρα τῇ ΚΒ ἐστι παράλληλος. καὶ ἐπιζευγνύουσιν αὐτὰξ αἱ ΒΟ, ΚΣ· τὸ ΚΒΟΣ ἄρα τετράπλευρον ἐν ἑνί ἐστιν ἐπιπέδῳ, ἐπειδήπερ, ἔανὸῶσι δύο εὐθεῖαι παράλληλοι, καὶ ἐφ᾽ ἑκατέρας αὐτῶν ληφθῇ τυθόντα σημεῖα, ἡ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις. διὰ τὰ αὐτὰ δὴ καὶ ἑκάτερον τῶν ΣΟΠΤ, ΤΠΡΥ τετραπλεύρων ἐν ἑνί ἐστιν ἐπιπέδῳ. ὀέστι δὲ καὶ τὸ ΥΡΧ τρίγωνον ἐν ἑνὶ ἐπιπέδῳ. ἔαν δὴ νοήσωμεν ἀπὸ τῶν Ο, Σ, Π, Τ, Ρ, Υ σημείων ἐπὶ τὸ Α ἐπιζευγνυμένας εὐθείας, συσταθήσεταί τι σχῆμα στερεὸν πολύεδρον

the sphere, which is also manifestly the diameter of the semicircle and the circle, is greater than all of the (other) [straight-lines] drawn across in the circle or the sphere [Prop. 3.15]. Therefore, let *BCDE* be the circle in the greater sphere, and *FGH* the circle in the lesser sphere. And let two diameters of them be drawn at right-angles to one another, (namely), *BD* and *CE*. And there being two circles about the same center—(namely), *BCDE* and *FGH*—let an equilateral and even-sided polygon be inscribed in the greater circle, *BCDE*, not touching the lesser circle, *FGH* [Prop. 12.16], of which let the sides in the quadrant *BE* be *BK*, *KL*, *LM*, and *ME*. And, *KA* being joined, let it be drawn across to *N*. And let *AO* be set up at point *A*, at right-angles to the plane of circle *BCDE*. And let it meet the surface of the (greater) sphere at *O*. And let planes be produced through *AO* and each of *BD* and *KN*. So, according to the aforementioned (discussion), they will make great circles on the surface of the (greater) sphere. Let them make (great circles), of which let *BOD* and *KON* be semi-circles on the diameters *BD* and *KN* (respectively). And since *OA* is at right-angles to the plane of circle *BCDE*, all of the planes through *OA* are thus also at right-angles to the plane of circle *BCDE* [Prop. 11.18]. And, hence, the semi-circles *BOD* and *KON* are also at right-angles to the plane of circle *BCDE*. And since semi-circles *BED*, *BOD*, and *KON* are equal—for (they are) on the equal diameters *BD* and *KN* [Def. 3.1]—the quadrants *BE*, *BO*, and *KO* are also equal to one another. Thus, as many sides of the polygon as are in quadrant *BE*, so many are also in quadrants *BO* and *KO* equal to the straight-lines *BK*, *KL*, *LM*, and *ME*. Let them be inscribed, and let them be *BP*, *PQ*, *QR*, *RO*, *KS*, *ST*, *TU*, and *UO*. And let *SP*, *TQ*, and *UR* be joined. And let perpendiculars be drawn from *P* and *S* to the plane of circle *BCDE* [Prop. 11.11]. So, they will fall on the common sections of the planes *BD* and *KN* (with *BCDE*), inasmuch as the planes of *BOD* and *KON* are also at right-angles to the plane of circle *BCDE* [Def. 11.4]. Let them have fallen, and let them be *PV* and *SW*. And let *WV* be joined. And since *BP* and *KS* are equal (circumferences) having been cut off in the equal semi-circles *BOD* and *KON* [Def. 3.28], and *PV* and *SW* are perpendiculars having been drawn (from them), *PV* is [thus] equal to *SW*, and *BV* to *KW* [Props. 3.27, 1.26]. And the whole of *BA* is also equal to the whole of *KA*. And, thus, as *BV* is to *VA*, so *KW* (is) to *WA*. *WV* is thus parallel to *KB* [Prop. 6.2]. And since *PV* and *SW* are each at right-angles to the plane of circle *BCDE*, *PV* is thus parallel to *SW* [Prop. 11.6]. And it was also shown (to be) equal to it. And, thus, *WV* and *SP* are equal and parallel [Prop. 1.33]. And since *WV* is parallel to *SP*, but *WV* is parallel to *KB*, *SP* is thus also parallel to *KB* [Prop. 11.1]. And *BP* and *KS* join them. Thus, the quadrilateral *KBPS* is in one plane, inasmuch as if there are two parallel straight-lines, and a random point is taken on each of them, then the straight-line joining the points

ματαχὺ τῶν BX, KX περιφερειῶν ἐκ πυραμίδων συγκείμενον, ὧν βάσεις μὲν τὰ KBOΣ, ΣOΠT, TΠPY τετράπλευρα καὶ τὸ YPX τρίγωνον, κορυφὴ δὲ τὸ A σημεῖον. ἔαν δὲ καὶ ἐπὶ ἑκάστης τῶν KΛ, ΛM, ME πλευρῶν καθάπερ ἐπὶ τῆς BK τὰ αὐτὰ κατασκευάσωμεν καὶ ὁέτι τῶν λοιπῶν τριῶν τεταρτημορίων, συσταθήσεταί τι σχῆμα πολύεδρον ἐγγεγραμμένον εἰξ τὴν σφαῖραν πυραμίσι περιεχόμενον, ὧν βάσιες [μὲν] τὰ εἰρημένα τετράπλευρα καὶ τὸ YPX τρίγωνον καὶ τὰ ὁμοταγῆ αὐτοῖς, κορυφὴ δὲ τὸ A σημεῖον.

Λέγω ὅτι τὸ εἰρημένον πολύεδρον οὐκ ἐφάψεται τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν, ἐφ' ἣξ ἐστιν ὁ ZHΘ κύκλος.

ὀΗθθω ἀπὸ τοῦ A σημείου ἐπὶ τὸ τοῦ KBOΣ τετραπλεύρου ἐπίπεδον κάθετος ἡ AΨ καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ Ψ σημεῖον, καὶ ἐπεζεύθθωσαν αἱ ΨB, ΨK. καὶ ἐπεὶ ἡ AΨ ὀρθή ἐστι πρὸς τὸ τοῦ KBOΣ τετραπλεύρου ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ τοῦ τετραπλεύρου ἐπιπέδῳ ὀρθή ἐστιν. ἡ AΨ ἄρα ὀρθή ἐστι πρὸς ἑκατέραν τῶν BΨ, ΨK. καὶ ἐπεὶ ὀίση ἐστὶν ἡ AB τῇ AK, ὁίσον ἐστὶ καὶ τὸ ἀπὸ τῆς AB τῷ ἀπὸ τῆς AK. καί ἐστι τῷ μὲν ἀπὸ τῆς AB ἴσα τὰ ἀπὸ τῶν AΨ, ΨB· ὀρθὴ γὰρ ἡ πρὸς τῷ Ψ· τῷ δὲ ἀπὸ τῆς AK ἴσα τὰ ἀπὸ τῶν AΨ, ΨK. τὰ ἄρα ἀπὸ τῶν AΨ, ΨB ἴσα ἐστὶ τοῖς ἀπὸ τῶν AΨ, ΨK. κοινὸν ἀφῃρήσθω τὸ ἀπὸ τῆς AΨ· λοιπὸν ἄρα τὸ ἀπὸ τῆς BΨ λοιπῷ τῷ ἀπὸ τῆς ΨK ἴσον ἐστίν· ὀίση ἄρα ἡ BΨ τῇ ΨK. ὁμοίως δὴ δείχομεν, ὅτι καὶ αἱ ἀπὸ τοῦ Ψ ἐπὶ τὰ O, Σ ἐπιζευγνύμεναι εὐθεῖαι ἴσαι εἰσὶν ἑκατέρᾳ τῶν BΨ, ΨK. ὁ ἄρα κέντρῳ τῷ Ψ καὶ διαστήματι ἑνὶ τῶν ΨB, ΨK γραφόμενος κύκλος ἥξει καὶ διὰ τῶν O, Σ, καὶ ὀέσται ἐν κύκλῳ τὸ KBOΣ τετράπλευρον.

Καὶ ἐπεὶ μείζων ἐστὶν ἡ KB τῆς ΘΦ, ὀίση δὲ ἡ ΘΦ τῇ ΣO, μείζων ἄρα ἡ KB τῆς ΣO. ὀίση δὲ ἡ KB ἑκατέρᾳ τῶν KΣ, BO· καὶ ἑκατέρα ἄρα τῶν KΣ, BO τῆς ΣO μείζων ἐστίν. καὶ ἐπεὶ ἐν κύκλῳ τετράπλευρόν ἐστι τὸ KBOΣ, καὶ ἴσαι αἱ KB, BO, KΣ, καὶ ἐλάττων ἡ OΣ, καὶ ἐκ τοῦ κέντρου τοῦ κύκλου ἐστὶν ἡ BΨ, τὸ ἄρα ἀπὸ τῆς KB τοῦ ἀπὸ τῆς BΨ μεῖζόν ἐστινὸὴ διπλάσιον. ἤθθω ἀπὸ τοῦ K ἐπὶ τὴν BΦ κάθετος ἡ KΩ. καὶ ἐπεὶ ἡ BΔ τῆς ΔΩ ἐλάττων ἐστὶνὸὴ διπλῆ, καί ἐστιν ὡξ ἡ BΔ πρὸς τὴν ΔΩ, οὕτως τὸ ὑπὸ τῶν ΔB, BΩ πρὸς τὸ ὑπὸ [τῶν] ΔΩ, ΩB, ἀναγραφομένου ἀπὸ τῆς BΩ τετραγώνου καὶ συμπληρουμένου τοῦ ἐπὶ τῆς ΩΔ παραλληλογράμμου καὶ τὸ ὑπὸ ΔB, BΩ ἄρα τοῦ ὑπὸ ΔΩ, ΩBὀέλαττόν ἐστινὸὴ διπλάσιον. καί ἐστι τῆς KΔ ἐπιζευγνυμένης τὸ μὲν ὑπὸ ΔB, BΩ ἴσον τῷ ἀπὸ τῆς BK, τὸ δὲ ὑπὸ τῶν ΔΩ, ΩB ἴσον τῷ ἀπὸ τῆς KΩ· τὸ ἄρα ἀπὸ τῆς KB τοῦ ἀπὸ τῆς KΩὀέλασσόν ἐστινὸὴ διπλάσιον. ἀλλὰ τὸ ἀπὸ τῆς KB τοῦ ἀπὸ τῆς BΨ μεῖζόν ἐστινὸὴ διπλάσιον· μεῖζον ἄρα τὸ ἀπὸ τῆς KΩ τοῦ ἀπὸ τῆς BΨ. καὶ ἐπεὶ ὀίση ἐστὶν ἡ BA τῇ KA, ἴσον ἐστὶ τὸ ἀπὸ τῆς BA τῷ ἀπὸ τῆς AK. καί ἐστι τῷ μὲν ἀπὸ τῆς BA ἴσα τὰ ἀπὸ τῶν BΨ, ΨA, τῷ δὲ ἀπὸ τῆς KA ἴσα τὰ ἀπὸ τῶν KΩ, ΩA· τὰ ἄρα ἀπὸ τῶν BΨ, ΨA ἴσα ἐστὶ τοῖς ἀπὸ τῶν KΩ, ΩA, ὧν τὸ ἀπὸ τῆς KΩ μεῖζον τοῦ ἀπὸ τῆς BΨ· λοιπὸν ἄρα

is in the same plane as the parallel (straight-lines) [Prop. 11.7]. So, for the same (reasons), each of the quadrilaterals $SPQT$ and $TQRU$ is also in one plane. And triangle URO is also in one plane [Prop. 11.2]. So, if we conceive straight-lines joining points P, S, Q, T, R, and U to A then some solid polyhedral figure will be constructed between the circumferences BO and KO, being composed of pyramids whose bases (are) the quadrilaterals $KBPS$, $SPQT$, $TQRU$, and the triangle URO, and apex the point A. And if we also make the same construction on each of the sides KL, LM, and ME, just as on BK, and, further, (repeat the construction) in the remaining three quadrants, then some polyhedral figure which has been inscribed in the sphere will be constructed, being contained by pyramids whose bases (are) the aforementioned quadrilaterals, and triangle URO, and the (quadrilaterals and triangles) similarly arranged to them, and apex the point A.

So, I say that the aforementioned polyhedron will not touch the lesser sphere on the surface on which the circle FGH is (situated).

Let the perpendicular (straight-line) AX be drawn from point A to the plane $KBPS$, and let it meet the plane at point X [Prop. 11.11]. And let XB and XK be joined. And since AX is at right-angles to the plane of quadrilateral $KBPS$, it is thus also at right-angles to all of the straight-lines joined to it which are also in the plane of the quadrilateral [Def. 11.3]. Thus, AX is at right-angles to each of BX and XK. And since AB is equal to AK, the (square) on AB is also equal to the (square) on AK. And the (sum of the squares) on AX and XB is equal to the (square) on AB. For the angle at X (is) a right-angle [Prop. 1.47]. And the (sum of the squares) on AX and XK is equal to the (square) on AK [Prop. 1.47]. Thus, the (sum of the squares) on AX and XB is equal to the (sum of the squares) on AX and XK. Let the (square) on AX be subtracted from both. Thus, the remaining (square) on BX is equal to the remaining (square) on XK. Thus, BX (is) equal to XK. So, similarly, we can show that the straight-lines joined from X to P and S are equal to each of BX and XK. Thus, a circle drawn (in the plane of the quadrilateral) with center X, and radius one of XB or XK, will also pass through P and S, and the quadrilateral $KBPS$ will be inside the circle.

And since KB is greater than WV, and WV (is) equal to SP, KB (is) thus greater than SP. And KB (is) equal to each of KS and BP. Thus, KS and BP are each greater than SP. And since quadrilateral $KBPS$ is in a circle, and KB, BP, and KS are equal (to one another), and PS (is) less (than them), and BX is the radius of the circle, the (square) on KB is thus greater than double the (square) on BX.[†] Let the perpendicular KY be drawn from K to BV.[‡] And since BD is less than double DY, and as BD is to DY, so the (rectangle contained) by DB and BY (is) to the (rectangle contained) by DY and YB—a square being described on BY, and a (rectangular) parallel-

τὸ ἀπὸ τῆς ΩΑ ὲλασσόν ἐστι τοῦ ἀπὸ τῆς ΨΑ. μείζων ἄρα ἡ ΑΨ τῆς ΑΩ· πολλῷ ἄρα ἡ ΑΨ μείζων ἐστὶ τῆς ΑΗ. καί ἐστιν ἡ μὲν ΑΨ ἐπὶ μίαν τοῦ πολυέδρου βάσιν, ἡ δὲ ΑΗ ἐπὶ τὴν τῆς ἐλάσσονος σφαίρας ἐπιφάνειαν· ὥστε τὸ πολύεδρον οὐ ψαύσει τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν.

Δύο ἄρα σφαιρῶν περὶ τὸ αὐτὸ κέντρον οὐσῶν εἰξ τὴν μείζονα σφαῖραν στερεὸν πολύεδρον ἐγγέγραπται μὴ ψαῦον τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν· ὅπερ ἔδει ποιῆσαι.

ogram (with short side equal to BY) completed on YD—the (rectangle contained) by DB and BY is thus also less than double the (rectangle contained) by DY and YB. And, KD being joined, the (rectangle contained) by DB and BY is equal to the (square) on BK, and the (rectangle contained) by DY and YB equal to the (square) on KY [Props. 3.31, 6.8 corr.]. Thus, the (square) on KB is less than double the (square) on KY. But, the (square) on KB is greater than double the (square) on BX. Thus, the (square) on KY (is) greater than the (square) on BX. And since BA is equal to KA, the (square) on BA is equal to the (square) on AK. And the (sum of the squares) on BX and XA is equal to the (square) on BA, and the (sum of the squares) on KY and YA (is) equal to the (square) on KA [Prop. 1.47]. Thus, the (sum of the squares) on BX and XA is equal to the (sum of the squares) on KY and YA, of which the (square) on KY (is) greater than the (square) on BX. Thus, the remaining (square) on YA is less than the (square) on XA. Thus, AX (is) greater than AY. Thus, AX is much greater than AG.[§] And AX is (a perpendicular) on one of the bases of the polyhedron, and AG (is a perpendicular) on the surface of the lesser sphere. Hence, the polyhedron will not touch the lesser sphere on its surface.

Thus, there being two spheres about the same center, a polyhedral solid has been inscribed in the greater sphere which does not touch the lesser sphere on its surface. (Which is) the very thing it was required to do.

[†] Since KB, BP, and KS are greater than the sides of an inscribed square, which are each of length $\sqrt{2}\,BX$.

[‡] Note that points Y and V are actually identical.

[§] This conclusion depends on the fact that the chord of the polygon in proposition 12.16 does not touch the inner circle.

Πόρισμα.

Ἐὰν δὲ καὶ εἰξ ἑτέραν σφαῖραν τῷ ἐν τῇ ΒΓΔΕ σφαίρᾳ στερεῷ πολυέδρῳ ὅμοιον στερεὸν πολύεδρον ἐγγραφῇ, τὸ ἐν τῇ ΒΓΔΕ σφαίρᾳ στερεὸν πολύεδρον πρὸς τὸ ἐν τῇ ἑτέρᾳ σφαίρᾳ στερεὸν πολύεδρον τριπλασίονα λόγον ἔχει, ἥπερ ἡ τῆς ΒΓΔΕ σφαίρας διάμετρος πρὸς τὴν τῆς ἑτέρας σφαίρας διάμετρον. διαιρεθέντων γὰρ τῶν στερεῶν εἰξ τὰξ ὁμοιοπληθεῖς καὶ ὁμοιοταγεῖς πυραμίδας ὁέσονται αἱ πυραμίδες ὅμοιαι. αἱ δὲ ὅμοιαι πυραμίδες πρὸς ἀλλήλας ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν· ἡ ἄρα πυραμίξ, ἧξ βάσις μέν ἐστι τὸ ΚΒΟΣ τετράπλευρον, κορυφὴ δὲ τὸ Α σημεῖον, πρὸς τὴν ἐν τῇ ἑτέρᾳ σφαίρᾳ ὁμοιοταγῆ πυραμίδα τριπλασίονα λόγον ἔχει, ἥπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν ἥπερ ἡ ΑΒ ἐκ τοῦ κέντρου τῆς σφαίρας τῆς περὶ κέντρον τὸ Α πρὸς τὴν ἐκ τοῦ κέντρου τῆς ἑτέρας σφαίρας. ὁμοίως καὶ ἑκάστη πυραμὶξ τῶν ἐν τῇ περὶ κέντρον τὸ Α σφαίρᾳ πρὸς ἑκάστην ὁμοταγῆ πυραμίδα τῶν ἐν τῇ ἑτέρᾳ σφαίρᾳ τριπλασίονα λόγον ὁέχει, ἥπερ ἡ ΑΒ πρὸς τὴν ἐκ τοῦ κέντρου τῆς ἑτέρας σφαίρας. καὶ ὡξ ὁὲν τῶν ἡγουμένων πρὸς ὁὲν τῶν

Corollary

And, also, if a similar polyhedral solid to that in sphere $BCDE$ is inscribed in another sphere then the polyhedral solid in sphere $BCDE$ has to the polyhedral solid in the other sphere the cubed ratio that the diameter of sphere $BCDE$ has to the diameter of the other sphere. For if the solids are divided into similarly numbered, and similarly situated, pyramids, then the pyramids will be similar. And similar pyramids are in the cubed ratio of corresponding sides [Prop. 12.8 corr.]. Thus, the pyramid whose base is quadrilateral $KBPS$, and apex the point A, will have to the similarly situated pyramid in the other sphere the cubed ratio that a corresponding side (has) to a corresponding side. That is to say, that of radius AB of the sphere about center A to the radius of the other sphere. And, similarly, each pyramid in the sphere about center A will have to each similarly situated pyramid in the other sphere the cubed ratio that AB (has) to the radius of the other sphere. And as one of the leading (magnitudes is) to one of the following (in two sets of proportional magnitudes), so (the sum of) all the

ἑπομένων, οὕτως ὅλον τὸ ἐν τῇ περὶ κέντρον τὸ A σφαίρᾳ στερεὸν πολύεδρον πρὸς ὅλον τὸ ἐν τῇ ἑτέρᾳ [σφαίρᾳ] στερεὸν πολύεδρον τριπλασίονα λόγον ἕξει, ἥπερ ἡ AB πρὸς τὴν ἐκ τοῦ κέντρου τῆς ἑτέρας σφαίρας, τουτέστιν ἥπερ ἡ $BΔ$ διάμετρος πρὸς τὴν τῆς ἑτέρας σφαίρας διάμετρον· ὅπερ ἔδει δεῖξαι.

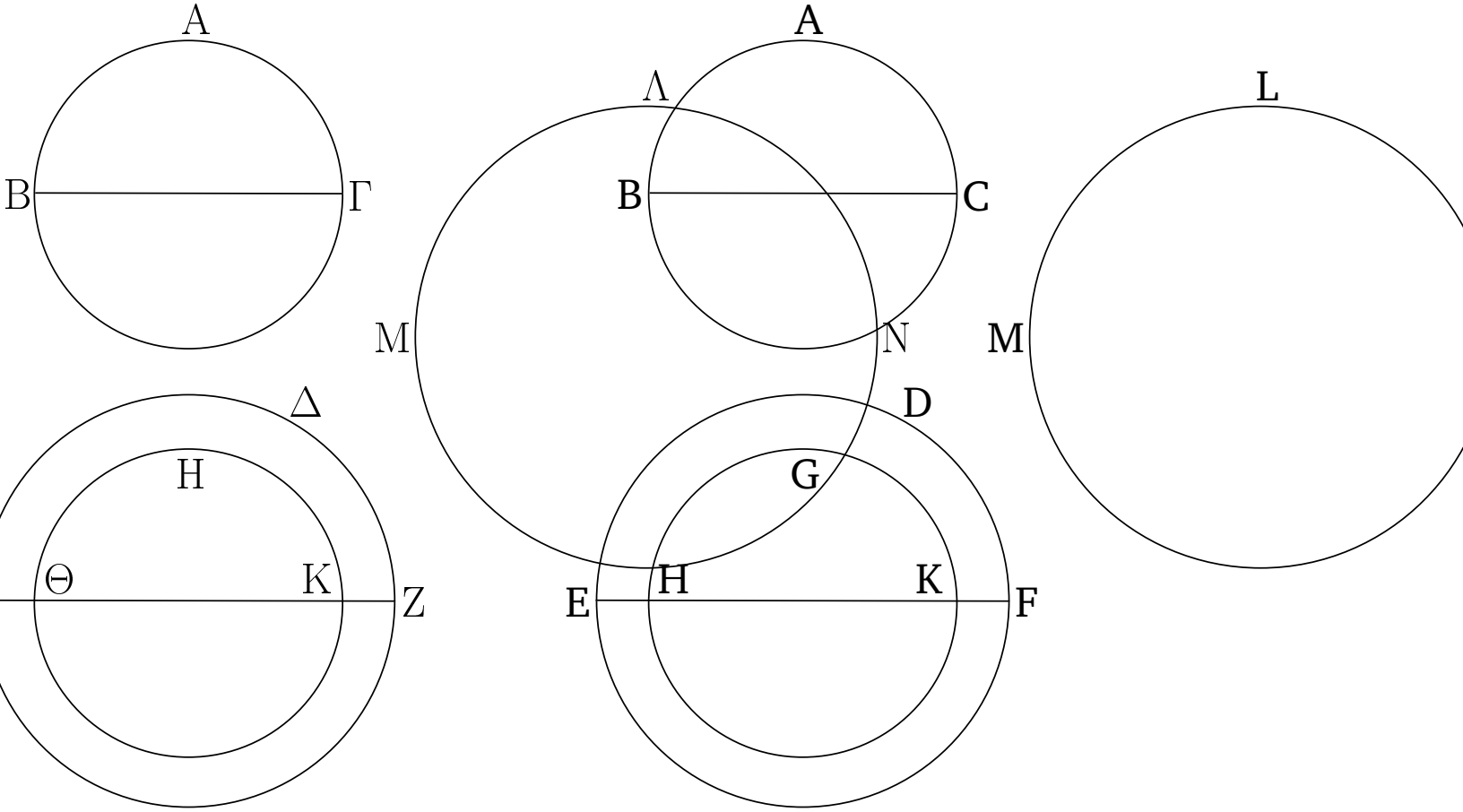
leading (magnitudes is) to (the sum of) all of the following (magnitudes) [Prop. 5.12]. Hence, the whole polyhedral solid in the sphere about center A will have to the whole polyhedral solid in the other [sphere] the cubed ratio that (radius) AB (has) to the radius of the other sphere. That is to say, that diameter BD (has) to the diameter of the other sphere. (Which is) the very thing it was required to show.

18.

Proposition 18

Αἱ σφαῖραι πρὸς ἀλλήλας ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ἰδίων διαμέτρων.

Spheres are to one another in the cubed ratio of their respective diameters.



Νενοήσθωσαν σφαῖραι αἱ $ABΓ$, $ΔEZ$, διάμετροι δὲ αὐτῶν αἱ $BΓ$, EZ· λέγω, ὅτι ἡ $ABΓ$ σφαῖρα πρὸς τὴν $ΔEZ$ σφαῖραν τριπλασίονα λόγον ἔχει ἥπερ ἡ $BΓ$ πρὸς τὴν EZ.

Εἰ γὰρ μὴ ἡ $ABΓ$ σφαῖρα πρὸς τὴν $ΔEZ$ σφαῖραν τριπλασίονα λόγον ἔχει ἥπερ ἡ $BΓ$ πρὸς τὴν EZ, ἕξει ἄρα ἡ $ABΓ$ σφαῖρα πρὸς ἐλάσσονά τινα τῆς $ΔEZ$ σφαίρας τριπλασίονα λόγονὁἡ πρὸς μείζονα ἥπερ ἡ $BΓ$ πρὸς τὴν EZ. ἐνθέτω πρότερον πρὸς ἐλάσσονα τὴν $HΘK$, καὶ νενοήσθω ἡ $ΔEZ$ τῇ $HΘK$ περὶ τὸ αὐτὸ κέντρον, καὶ ἐγγεγράφθω εἰς τὴν μείζονα σφαῖραν τὴν $ΔEZ$ στερεὸν πολύεδρον μὴ ψαῦον τῆς ἐλάσσονος σφαίρας τῆς $HΘK$ κατὰ τὴν ἐπιφάνειαν, ἐγγεγράφθω δὲ καὶ

Let the spheres ABC and DEF be conceived, and (let) their diameters (be) BC and EF (respectively). I say that sphere ABC has to sphere DEF the cubed ratio that BC (has) to EF.

For if sphere ABC does not have to sphere DEF the cubed ratio that BC (has) to EF then sphere ABC will have to some (sphere) either less than, or greater than, sphere DEF the cubed ratio that BC (has) to EF. Let it, first of all, have (such a ratio) to a lesser (sphere), GHK. And let DEF be conceived about the same center as GHK. And let a polyhedral solid be inscribed in the greater sphere DEF, not touching the lesser sphere GHK on its surface [Prop. 12.17]. And let a poly-

εἰξ τὴν ΑΒΓ σφαῖραν τῷ ἐν τῇ ΔΕΖ σφαίρᾳ στερεῷ πολυέδρῳ ὅμοιον στερεὸν πολύεδρον· τὸ ἄρα ἐν τῇ ΑΒΓ στερεὸν πολύεδρον πρὸς τὸ ἐν τῇ ΔΕΖ στερεὸν πολύεδρον τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΓ πρὸς τὴν ΕΖ. ἔχει δὲ καὶ ἡ ΑΒΓ σφαῖρα πρὸς τὴν ΗΘΚ σφαῖραν τριπλασίονα λόγον ἥπερ ἡ ΒΓ πρὸς τὴν ΕΖ· ὀέστιν ἄρα ὡξ ἡ ΑΒΓ σφαῖρα πρὸς τὴν ΗΘΚ σφαῖραν, οὑτως τὸ ἐν τῇ ΑΒΓ σφαίρᾳ στερεὸν πολύεδρον πρὸς τὸ ἐν τῇ ΔΕΖ σφαίρᾳ στερεὸν πολύεδρον· ἐναλλὰχ [ἄρα] ὡξ ἡ ΑΒΓ σφαῖρα πρὸς τὸ ἐν αὐτῇ πολύεδρον, οὑτως ἡ ΗΘΚ σφαῖρα πρὸς τὸ ἐν τῇ ΔΕΖ σφαίρᾳ στερεὸν πολύεδρον. μείζων δὲ ἡ ΑΒΓ σφαῖρα τοῦ ἐν αὐτῇ πολυέδρου· μείζων ἄρα καὶ ἡ ΗΘΚ σφαῖρα τοῦ ἐν τῇ ΔΕΖ σφαίρᾳ πολυέδρου. ἀλλὰ καὶ ἐλάττων· ἐμπεριέθεται γὰρ ὑπὸ αὐτοῦ. οὐκ ἄρα ἡ ΑΒΓ σφαῖρα πρὸς ἐλάσσονα τῆς ΔΕΖ σφαίρας τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΓ διάμετρος πρὸς τὴν ΕΖ. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ἡ ΔΕΖ σφαῖρα πρὸς ἐλάσσονα τῆς ΑΒΓ σφαίρας τριπλασίονα λόγον ἔχει ἥπερ ἡ ΕΖ πρὸς τὴν ΒΓ.

Λέγω δή, ὅτι οὐδὲ ἡ ΑΒΓ σφαῖρα πρὸς μείζονά τινα τῆς ΔΕΖ σφαίρας τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΓ πρὸς τὴν ΕΖ.

Εἰ γὰρ δυνατόν, ἐθέτω πρὸς μείζονα τὴν ΛΜΝ· ἀνάπαλιν ἄρα ἡ ΛΜΝ σφαῖρα πρὸς τὴν ΑΒΓ σφαῖραν τριπλασίονα λόγον ἔχει ἥπερ ἡ ΕΖ διάμετρος πρὸς τὴν ΒΓ διάμετρον. ὡξ δὲ ἡ ΛΜΝ σφαῖρα πρὸς τὴν ΑΒΓ σφαῖραν, οὑτως ἡ ΔΕΖ σφαῖρα πρὸς ἐλάσσονά τινα τῆς ΑΒΓ σφαίρας, ἐπειδήπερ μείζων ἐστὶν ἡ ΛΜΝ τῆς ΔΕΖ, ὡξὸέμπροσθεν ἐδείθθη. καὶ ἡ ΔΕΖ ἄρα σφαῖρα πρὸς ἐλάσσονά τινα τῆς ΑΒΓ σφαίρας τριπλασίονα λόγον ἔχει ἥπερ ἡ ΕΖ πρὸς τὴν ΒΓ· ὅπερ ἀδύνατον ἐδείθθη. οὐκ ἄρα ἡ ΑΒΓ σφαῖρα πρὸς μείζονά τινα τῆς ΔΕΖ σφαίρας τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΓ πρὸς τὴν ΕΖ. ἐδείθθη δέ, ὅτι οὐδὲ πρὸς ἐλάσσονα. ἡ ἄρα ΑΒΓ σφαῖρα πρὸς τὴν ΔΕΖ σφαῖραν τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΓ πρὸς τὴν ΕΖ· ὅπερ ἔδει δεῖχαι.

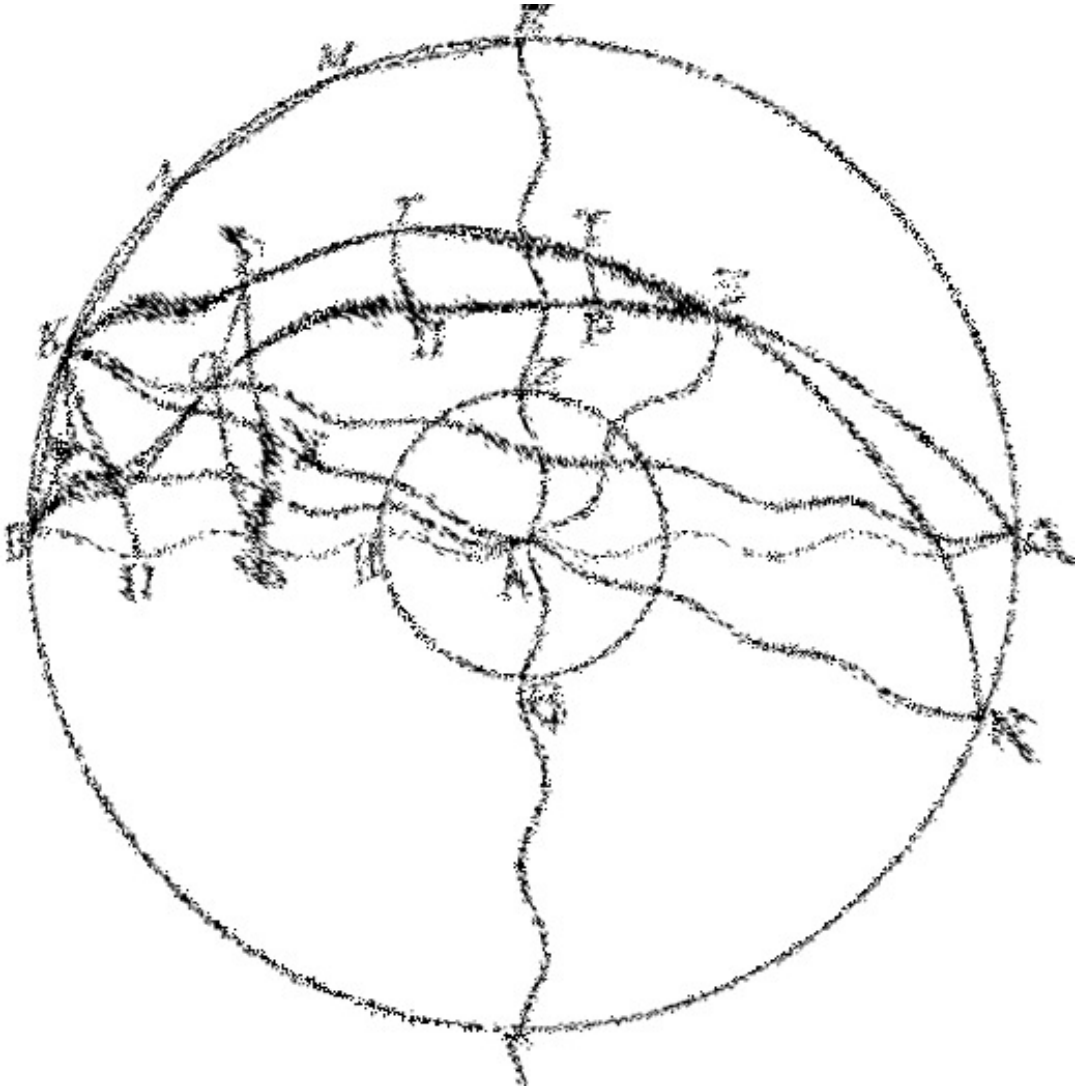
hedral solid, similar to the polyhedral solid in sphere DEF, have also been inscribed in sphere ABC. Thus, the polyhedral solid in sphere ABC has to the polyhedral solid in sphere DEF the cubed ratio that BC (has) to EF [Prop. 12.17 corr.]. And sphere ABC also has to sphere GHK the cubed ratio that BC (has) to EF. Thus, as sphere ABC is to sphere GHK, so the polyhedral solid in sphere ABC (is) to the polyhedral solid is sphere DEF. [Thus], alternately, as sphere ABC (is) to the polygon within it, so sphere GHK (is) to the polyhedral solid within sphere DEF [Prop. 5.16]. And sphere ABC (is) greater than the polyhedron within it. Thus, sphere GHK (is) also greater than the polyhedron within sphere DEF [Prop. 5.14]. But, (it is) also less. For it is encompassed by it. Thus, sphere ABC does not have to (a sphere) less than sphere DEF the cubed ratio that diameter BC (has) to EF. So, similarly, we can show that sphere DEF does not have to (a sphere) less than sphere ABC the cubed ratio that EF (has) to BC either.

So, I say that sphere ABC does not have to some (sphere) greater than sphere DEF the cubed ratio that BC (has) to EF either.

For, if possible, let it have (the cubed ratio) to a greater (sphere), LMN. Thus, inversely, sphere LMN (has) to sphere ABC the cubed ratio that diameter EF (has) to diameter BC [Prop. 5.7 corr.]. And as sphere LMN (is) to sphere ABC, so sphere DEF (is) to some (sphere) less than sphere ABC, inasmuch as LMN is greater than DEF, as was shown before [Prop. 12.2 lem.]. And, thus, sphere DEF has to some (sphere) less than sphere ABC the cubed ratio that EF (has) to BC. The very thing was shown (to be) impossible. Thus, sphere ABC does not have to some (sphere) greater than sphere DEF the cubed ratio that BC (has) to EF. And it was shown that neither (does it have such a ratio) to a lesser (sphere). Thus, sphere ABC has to sphere DEF the cubed ratio that BC (has) to EF. (Which is) the very thing it was required to show.

ELEMENTS BOOK 12

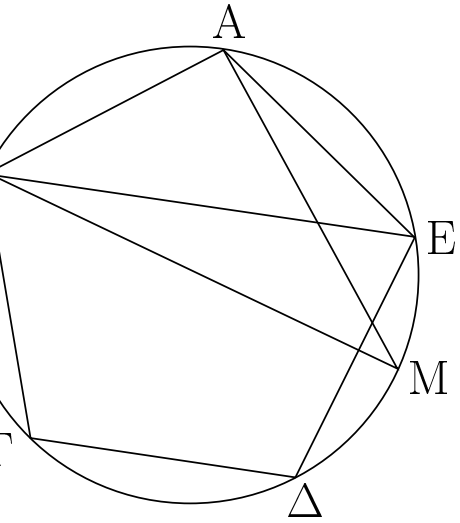
Proportional Stereometry[†]

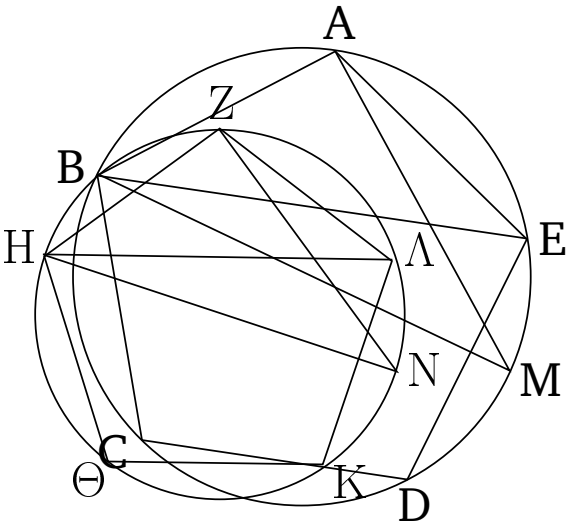


[†]The novel feature of this book is the use of the so-called method of exhaustion (see Prop. 10.1), a precursor to integration which is generally attributed to Eudoxus of Cnidus.

1.

Τὰ ἐν τοῖς κύκλοις ὅμοια πολύγωνα πρὸς ἀλλήλα ἐστὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα.





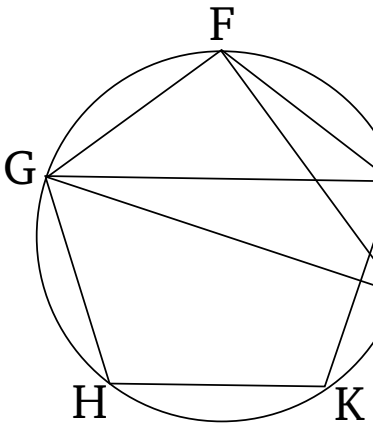
Ἔστωσαν κύκλοι οἱ ΑΒΓ, ΖΗΘ, καὶ ἐν αὐτοῖς ὅμοια πολύγωναὁέστω τὰ ΑΒΓΔΕ, ΖΗΘΚΛ, διάμετροι δὲ τῶν κύκλωνὁέστωσαν ΒΜ, ΗΝ· λέγω, ὅτι ἐστὶν ὡς τὸ ἀπὸ τῆς ΒΜ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΗΝ τετράγωνον, οὕτως τὸ ΑΒΓΔΕ πολύγωνον πρὸς τὸ ΖΗΘΚΛ πολύγωνον.

Ἐπεζεύθθωσαν γὰρ αἱ ΒΕ, ΑΜ, ΗΛ, ΖΝ. καὶ ἐπεὶ ὅμοιον τὸ ΑΒΓΔΕ πολύγωνον τῷ ΖΗΘΚΛ πολυγώνῳ, ὁίση ἐστὶ καὶ ἡ ὑπὸ ΒΑΕ γωνία τῇ ὑπὸ ΗΖΛ, καί ἐστιν ὡς ἡ ΒΑ πρὸς τὴν ΑΕ, οὕτως ἡ ΗΖ πρὸς τὴν ΖΛ. δύο δὴ τρίγωνά ἐστι τὰ ΒΑΕ, ΗΖΛ μίαν γωνίαν μιᾷ γωνίᾳ ἴσην ἔχοντα τὴν ὑπὸ ΒΑΕ τῇ ὑπὸ ΗΖΛ, περὶ δὲ τὰξ ἴσας γωνίας τὰξ πλευρὰξ ἀνάλογον· ἰσογώνιον ἄρα ἐστὶ τὸ ΑΒΕ τρίγωνον τῷ ΖΗΛ τριγώνῳ. ὁίση ἄρα ἐστὶν ἡ ὑπὸ ΑΕΒ γωνία τῇ ὑπὸ ΖΛΗ. ἀλλὰ ἡ μὲν ὑπὸ ΑΕΒ τῇ ὑπὸ ΑΜΒ ἐστινὁίση· ἐπὶ γὰρ τῆς αὐτῆς περιφερείας βεβήκασιν· ἡ δὲ ὑπὸ ΖΛΗ τῇ ὑπὸ ΖΝΗ· καὶ ἡ ὑπὸ ΑΜΒ ἄρα τῇ ὑπὸ ΖΝΗ ἐστινὁίση. ὁέστι δὲ καὶ ὀρθὴ ἡ ὑπὸ ΒΑΜ ὀρθῇ τῇ ὑπὸ ΗΖΝὁίση· καὶ ἡ λοιπὴ ἄρα τῇ λοιπῇ ἐστινὁίση. ἰσογώνιον ἄρα ἐστὶ τὸ ΑΒΜ τρίγωνον τῷ ΖΗΝ τρίγωνῳ. ἀνάλογον ἄρα ἐστὶν ὡς ἡ ΒΜ πρὸς τὴν ΗΝ, οὕτως ἡ ΒΑ πρὸς τὴν ΗΖ. ἀλλὰ τοῦ μὲν τῆς ΒΜ πρὸς τὴν ΗΝ λόγον διπλασίων ἐστὶν ὁ τοῦ ἀπὸ τῆς ΒΜ τετραγώνου πρὸς τὸ ἀπὸ τῆς ΗΝ τετράγωνον, τοῦ δὲ τῆς ΒΑ πρὸς τὴν ΗΖ διπλασίων ἐστὶν ὁ τοῦ ΑΒΓΔΕ πολυγώνου πρὸς τὸ ΖΗΘΚΛ πολύγωνον· καὶ ὡς ἄρα τὸ ἀπὸ τῆς ΒΜ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΗΝ τετράγωνον, οὕτως τὸ ΑΒΓΔΕ πολύγωνον πρὸς τὸ ΖΗΘΚΛ πολύγωνον.

Τὰ ἄρα ἐν τοῖς κύκλοις ὅμοια πολύγωνα πρὸς ἀλλήλα ἐστὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα· ὅπερ ἔδει δεῖξαι.

Proposition 1

Similar polygons (inscribed) in circles are to one another as the squares on the diameters (of the circles).



Let *ABC* and *FGH* be circles, and let *ABCDE* and *FGHKL* be similar polygons (inscribed) in them (respectively), and let *BM* and *GN* be the diameters of the circles (respectively). I say that as the square on *BM* is to the square on *GN*, so polygon *ABCDE* (is) to polygon *FGHKL*.

For let *BE*, *AM*, *GL*, and *FN* be joined. And since polygon *ABCDE* (is) similar to polygon *FGHKL*, angle *BAE* is also equal to (angle) *GFL*, and as *BA* is to *AE*, so *GF* (is) to *FL* [Def. 6.1]. So, *BAE* and *GFL* are two triangles having one angle equal to one angle, (namely), *BAE* (equal) to *GFL*, and the sides around the equal angles proportional. Triangle *ABE* is thus equiangular with triangle *FGL* [Prop. 6.6]. Thus, angle *AEB* is equal to (angle) *FLG*. But, *AEB* is equal to *AMB*, and *FLG* to *FNG*, for they stand on the same circumference [Prop. 3.27]. Thus, *AMB* is also equal to *FNG*. And the right-angle *BAM* is also equal to the right-angle *GFN* [Prop. 3.31]. Thus, the remaining (angle) is also equal to the remaining (angle) [Prop. 1.32]. Thus, triangle *ABM* is equiangular with triangle *FGN*. Thus, proportionally, as *BM* is to *GN*, so *BA* (is) to *GF* [Prop. 6.4]. But, the (ratio) of the square on *BM* to the square on *GN* is the square of the ratio of *BM* to *GN*, and the (ratio) of polygon *ABCDE* to polygon *FGHKL* is the square of the (ratio) of *BA* to *GF* [Prop. 6.20]. And, thus, as the square on *BM* (is) to the square on *GN*, so polygon *ABCDE* (is) to polygon *FGHKL*.

Thus, similar polygons (inscribed) in circles are to one another as the squares on the diameters (of the circles). (Which is) the very thing it was required to show.

2.

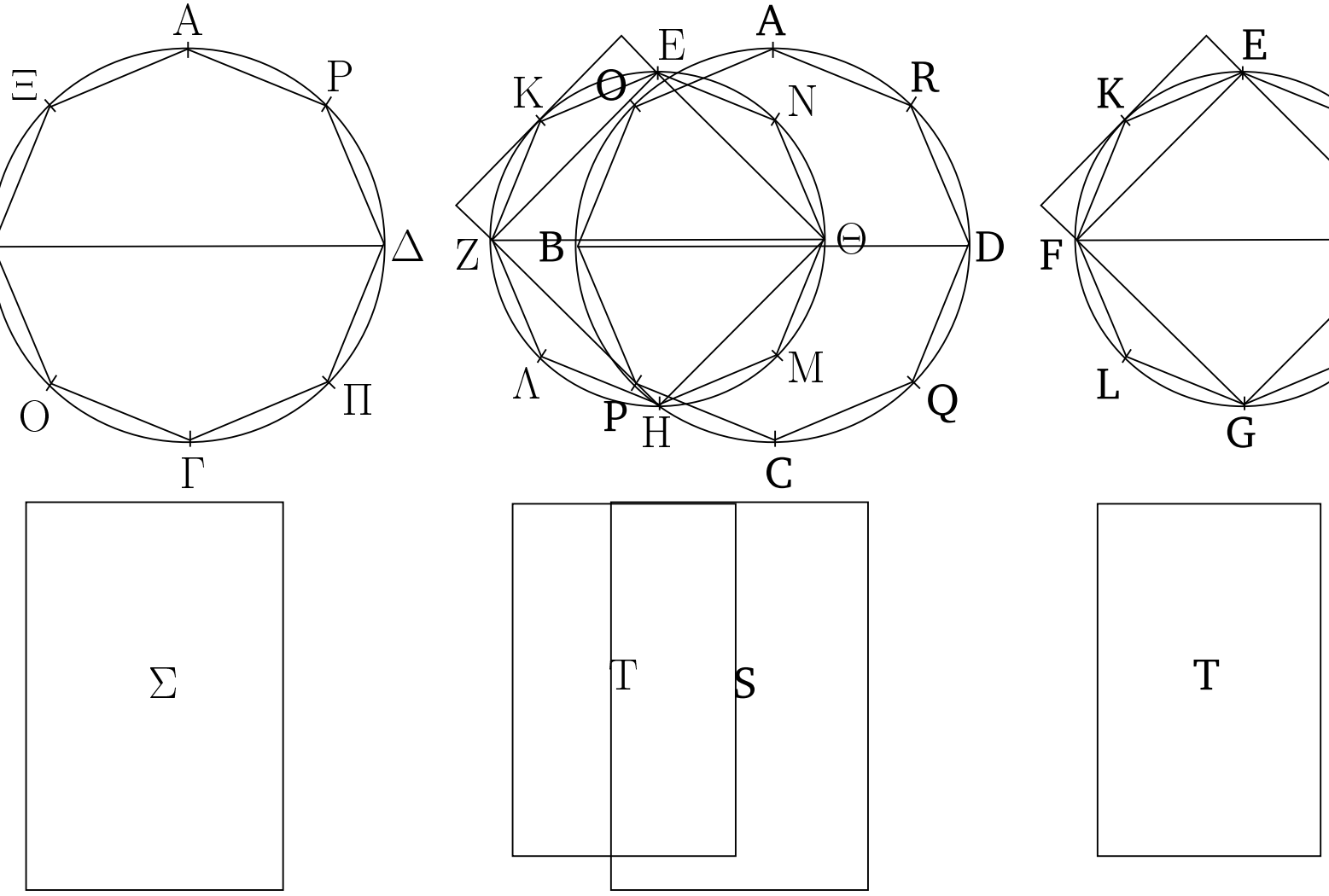
Proposition 2

Οἱ κύκλοι πρὸς ἀλλήλους εἰσὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα.

Ἔστωσαν κύκλοι οἱ ΑΒΓΔ, ΕΖΗΘ, διάμετροι δὲ αὐτῶν [δέστωσαν] αἱ ΒΔ, ΖΘ· λέγω, ὅτι ἐστὶν ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὕτως τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ τετράγωνον.

Circles are to one another as the squares on (their) diameters.

Let $ABCD$ and $EFGH$ be circles, and [let] BD and FH [be] their diameters. I say that as circle $ABCD$ is to circle $EFGH$, so the square on BD (is) to the square on FH.



Εἰ γὰρ μή ἐστιν ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ, οὕτως τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ, ἔσται ὡς τὸ ἀπὸ τῆς ΒΔ πρὸς τὸ ἀπὸ τῆς ΖΘ, οὕτως ὁ ΑΒΓΔ κύκλος ἤτοι πρὸς ἔλασσόν τι τοῦ ΕΖΗΘ κύκλου θωρίονὸή πρὸς μεῖζον. ἐστω πρότερον πρὸς ἔλασσον τὸ Σ. καὶ ἐγγεγράφθω εἰς τὸν ΕΖΗΘ κύκλον τετράγωνον τὸ ΕΖΗΘ. τὸ δὴ ἐγγεγραμμένον τετράγωνον μεῖζόν ἐστινὸή τὸ ἥμισυ τοῦ ΕΖΗΘ κύκλου, ἐπειδήπερ ἐὰν διὰ τῶν Ε, Ζ, Η, Θ σημείων ἐφαπτομένας [εὐθείας] τοῦ κύκλου ἀγάγωμεν, τοῦ περιγραφομένου περὶ τὸν κύκλον τετραγώνου ἥμισύ ἐστι τὸ ΕΖΗΘ τετράγωνον, τοῦ δὲ περιγραφέντος τετραγώνου ἐλάττων ἐστὶν

For if the circle $ABCD$ is not to the (circle) $EFGH$, as the square on BD (is) to the (square) on FH, then as the (square) on BD (is) to the (square) on FH, so circle $ABCD$ will be to some area either less than, or greater than, circle $EFGH$. Let it, first of all, be (in that ratio) to (some) lesser (area), S. And let the square $EFGH$ be inscribed in circle $EFGH$ [Prop. 4.6]. So the inscribed square is greater than half of circle $EFGH$, inasmuch as if we draw tangents to the circle through the points E, F, G, and H, then square $EFGH$ is half of the square circumscribed about the circle [Prop. 1.47], and the circle is less than the circumscribed square. Hence, the

ὁ κύκλος· ὥστε τὸ ΕΖΗΘ ἐγγεγραμμένομῆν τετράγωνον μεῖζόν ἐστι τοῦ ἡμίσεως τοῦ ΕΖΗΘ κύκλου. τετμήσθωσαν δίθα αἱ ΕΖ, ΖΗ, ΗΘ, ΘΕ περιφέρειαι κατὰ τὰ Κ, Λ, Μ, Ν σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΕΚ, ΚΖ, ΖΛ, ΛΗ, ΗΜ, ΜΘ, ΘΝ, ΝΕ· καὶ ὁέκαστον ἄρα τῶν ΕΚΖ, ΖΛΗ, ΗΜΘ, ΘΝΕ τριγώνων μεῖζόν ἐστινὸὴ τὸ ἥμισυ τοῦ καθὸ ἑαυτὸ τμήματος τοῦ κύκλου, ἐπειδήπερ ἔαν διὰ τῶν Κ, Λ, Μ, Ν σημείων ἐφαπτομένας τοῦ κύκλου ἀγάγωμεν καὶ ἀναπληρώσωμεν τὰ ἐπὶ τῶν ΕΖ, ΖΗ, ΗΘ, ΘΕ εὐθειῶν παραλληλόγραμμα, ὁέκαστον τῶν ΕΚΖ, ΖΛΗ, ΗΜΘ, ΘΝΕ τριγώνων ἥμισυὸέσται τοῦ καθὸ ἑαυτὸ παραλληλογράμμου, ἀλλὰ τὸ καθὸ ἑαυτὸ τμῆμαὸέλαττόν ἐστι τοῦ παραλληλογράμμου· ὥστε ὁέκαστον τῶν ΕΚΖ, ΖΛΗ, ΗΜΘ, ΘΝΕ τριγώνων μεῖζόν ἐστι τοῦ ἡμίσεως τοῦ καθὸ ἑαυτὸ τμήματος τοῦ κύκλου. τέμνοντες δὴ τὰξ ὑπολειπομένας περιφερείας δίθα καὶ ἐπιζευγνύντες εὐθείας καὶ τοῦτο ἀεὶ ποιοῦντες καταλείψομέν τινα ἀποτμήματα τοῦ κύκλου, ὁὰ ὀέσται ἐλάσσονα τῆς ὑπεροθῆς, ᾗ ὑπερέθει ὁ ΕΖΗΘ κύκλος τοῦ Σ θωρίου. ἐδείθθη γὰρ ἐν τῷ πρώτῳ θεωρήματι τοῦ δεκάτου βιβλίου, ὅτι δύο μεγεθῶν ἀνίσων ἐκκειμένων, ἔαν ἀπὸ τοῦ μείζονος ἀφαιρεθῇ μεῖζονὸὴ τὸ ἥμισυ καὶ τοῦ καταλειπομένου μεῖζονὸὴ τὸ ἥμισυ, καὶ τοῦτο ἀεὶ γίγνηται, λειφθήσεταί τι μέγεθος, ὃ ὀέσταιὸέλασσον τοῦ ἐκκειμένου ἐλάσσονος μεγέθους. λελείφθω οὐῦν, καὶ ὀέστω τὰ ἐπὶ τῶν ΕΚ, ΚΖ, ΖΛ, ΛΗ, ΗΜ, ΜΘ, ΘΝ, ΝΕ τμήματα τοῦ ΕΖΗΘ κύκλου ἐλάττονα τῆς ὑπεροθῆς, ᾗ ὑπερέθει ὁ ΕΖΗΘ κύκλος τοῦ Σ θωρίου. λοιπὸν ἄρα τὸ ΕΚΖΛΗΜΘΝ πολύγωνον μεῖζόν ἐστι τοῦ Σ θωρίου. ἐγγεγράφθω καὶ εἰξ τὸν ΑΒΓΔ κύκλον τῷ ΕΚΖΛΗΜΘΝ πολυγώνῳ ὅμοιον πολύγωνον τὸ ΑΧΒΟΓΠΔΡ· ὀέστιν ἄρα ὡξ τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ τετράγωνον, οὑτως τὸ ΑΧΒΟΓΠΔΡ πολύγωνον πρὸς τὸ ΕΚΖΛΗΜΘΝ πολύγωνον. ἀλλὰ καὶ ὡξ τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ, οὑτως ὁ ΑΒΓΔ κύκλος πρὸς τὸ Σ θωρίον· καὶ ὡξ ἄρα ὁ ΑΒΓΔ κύκλος πρὸς τὸ Σ θωρίον, οὑτως τὸ ΑΧΒΟΓΠΔΡ πολύγωνον πρὸς τὸ ΕΚΖΛΗΜΘΝ πολύγωνον· ἐναλλὰχ ἄρα ὡξ ὁ ΑΒΓΔ κύκλος πρὸς τὸ ἐν αὐτῷ πολύγωνον, οὑτως τὸ Σ θωρίον πρὸς τὸ ΕΚΖΛΗΜΘΝ πολύγωνον. μείζων δὲ ὁ ΑΒΓΔ κύκλος τοῦ ἐν αὐτῷ πολυγώνου· μεῖζον ἄρα καὶ τὸ Σ θωρίον τοῦ ΕΚΖΛΗΜΘΝ πολυγώνου. ἀλλὰ καὶ ὀέλαττον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐστὶν ὡξ τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ, οὑτως ὁ ΑΒΓΔ κύκλος πρὸςὀέλασσόν τι τοῦ ΕΖΗΘ κύκλου θωρίον. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ὡξ τὸ ἀπὸ ΖΘ πρὸς τὸ ἀπὸ ΒΔ, οὑτως ὁ ΕΖΗΘ κύκλος πρὸςὀέλασσόν τι τοῦ ΑΒΓΔ κύκλου θωρίον.

Λέγω δή, ὅτι οὐδὲ ὡξ τὸ ἀπὸ τῆς ΒΔ πρὸς τὸ ἀπὸ τῆς ΖΘ, οὑτως ὁ ΑΒΓΔ κύκλος πρὸς μεῖζόν τι τοῦ ΕΖΗΘ κύκλου θωρίον.

Εἰ γὰρ δυνατόν, ὀέστω πρὸς μεῖζον τὸ Σ. ἀνάπαλιν ἄρα [ἐστὶν] ὡξ τὸ ἀπὸ τῆς ΖΘ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΔΒ, οὑτως τὸ Σ θωρίον πρὸς τὸν ΑΒΓΔ κύκλον. ἀλλὰ ὡξ τὸ Σ θωρίον πρὸς τὸν ΑΒΓΔ κύκλον, οὑτως ὁ ΕΖΗΘ κύκλος

inscribed square $EFGH$ is greater than half of circle $EFGH$. Let the circumferences EF, FG, GH, and HE be cut in half at points K, L, M, and N (respectively), and let EK, KF, FL, LG, GM, MH, HN, and NE be joined. And, thus, each of the triangles EKF, FLG, GMH, and HNE is greater than half of the segment of the circle about it, inasmuch as if we draw tangents to the circle through points K, L, M, and N, and complete the parallelograms on the straight-lines EF, FG, GH, and HE, then each of the triangles EKF, FLG, GMH, and HNE will be half of the parallelogram about it, but the segment about it is less than the parallelogram. Hence, each of the triangles EKF, FLG, GMH, and HNE is greater than half of the segment of the circle about it. So, by cutting the circumferences remaining behind in half, and joining straight-lines, and doing this continually, we will (eventually) leave behind some segments of the circle whose (sum) will be less than the excess by which circle $EFGH$ exceeds the area S. For we showed in the first theorem of the tenth book that if two unequal magnitudes are laid out, and if (a part) greater than a half is subtracted from the greater, and (if from) the remainder (a part) greater than a half (is subtracted), and this happens continually, then some magnitude will (eventually) be left which will be less than the lesser laid out magnitude [Prop. 10.1]. Therefore, let the (segments) be left, and let the (sum of the) segments of the circle $EFGH$ on EK, KF, FL, LG, GM, MH, HN, and NE be less than the excess by which circle $EFGH$ exceeds area S. Thus, the remaining polygon $EKFLGMHN$ is greater than area S. And let the polygon $AOBPCQDR$, similar to the polygon $EKFLGMHN$, be inscribed in circle $ABCD$. Thus, as the square on BD is to the square on FH, so polygon $AOBPCQDR$ (is) to polygon $EKFLGMHN$ [Prop. 12.1]. But, also, as the square on BD (is) to the square on FH, so circle $ABCD$ (is) to area S. And, thus, as circle $ABCD$ (is) to area S, so polygon $AOBPGQDR$ (is) to polygon $EKFLGMHN$ [Prop. 5.11]. Thus, alternately, as circle $ABCD$ (is) to the polygon (inscribed) within it, so area S (is) to polygon $EKFLGMHN$ [Prop. 5.16]. And circle $ABCD$ (is) greater than the polygon (inscribed) within it. Thus, area S is also greater than polygon $EKFLGMHN$. But, (it is) also less. The very thing is impossible. Thus, the square on BD is not to the (square) on FH, as circle $ABCD$ (is) to some area less than circle $EFGH$. So, similarly, we can show that the (square) on FH (is) not to the (square) on BD as circle $EFGH$ (is) to some area less than circle $ABCD$ either.

So, I say that neither (is) the (square) on BD to the (square) on FH, as circle $ABCD$ (is) to some area greater than circle $EFGH$.

For, if possible, let it be (in that ratio) to (some) greater (area), S. Thus, inversely, as the square on FH [is] to the (square) on DB, so area S (is) to circle $ABCD$ [Prop. 5.7 corr.]. But, as area S (is) to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$ (see lemma). And, thus, as

πρὸς ὀλαττόν τι τοῦ ΑΒΓΔ κύκλου θωρίον· καὶ ὡς ἄρα τὸ ἀπὸ τῆς ΖΘ πρὸς τὸ ἀπὸ τῆς ΒΔ, οὕτως ὁ ΕΖΗΘ κύκλος πρὸς ὀλάσσόν τι τοῦ ΑΒΓΔ κύκλου θωρίον· ὅπερ ἀδύνατον ἐδείθθη. οὐκ ἄρα ἐστὶν ὡς τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ, οὕτως ὁ ΑΒΓΔ κύκλος πρὸς μεῖζόν τι τοῦ ΕΖΗΘ κύκλου θωρίον. ἐδείθθη δέ, ὅτι οὐδὲ πρὸς ὀλασσον· ὀέστιν ἄρα ὡς τὸ ἀπὸ τῆς ΒΔ τετράγωνον πρὸς τὸ ἀπὸ τῆς ΖΘ, οὕτως ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον.

Οἱ ἄρα κύκλοι πρὸς ἀλλήλους εἰσὶν ὡς τὰ ἀπὸ τῶν διαμέτρων τετράγωνα· ὅπερ ἔδει δεῖχαι.

Λῆμμα.

Λέγω δή, ὅτι τοῦ Σ θωρίου μείζονος ὄντος τοῦ ΕΖΗΘ κύκλου ἐστὶν ὡς τὸ Σ θωρίον πρὸς τὸν ΑΒΓΔ κύκλον, οὕτως ὁ ΕΖΗΘ κύκλος πρὸς ὀλαττόν τι τοῦ ΑΒΓΔ κύκλου θωρίον.

Γεγονέτω γὰρ ὡς τὸ Σ θωρίον πρὸς τὸν ΑΒΓΔ κύκλον, οὕτως ὁ ΕΖΗΘ κύκλος πρὸς τὸ Τ θωρίον. λέγω, ὅτιὸλαττόν ἐστι τὸ Τ θωρίον τοῦ ΑΒΓΔ κύκλου. ἐπεὶ γάρ ἐστιν ὡς τὸ Σ θωρίον πρὸς τὸν ΑΒΓΔ κύκλον, οὕτως ὁ ΕΖΗΘ κύκλος πρὸς τὸ Τ θωρίον, ἐναλλάχ ἐστιν ὡς τὸ Σ θωρίον πρὸς τὸν ΕΖΗΘ κύκλον, οὕτως ὁ ΑΒΓΔ κύκλος πρὸς τὸ Τ θωρίον. μεῖζον δὲ τὸ Σ θωρίον τοῦ ΕΖΗΘ κύκλου· μείζων ἄρα καὶ ὁ ΑΒΓΔ κύκλος τοῦ Τ θωρίου. ὅὡστε ἐστὶν ὡς τὸ Σ θωρίον πρὸς τὸν ΑΒΓΔ κύκλον, οὕτως ὁ ΕΖΗΘ κύκλος πρὸς ὀλαττόν τι τοῦ ΑΒΓΔ κύκλου θωρίον· ὅπερ ἔδει δεῖχαι.

3.

Πᾶσα πυραμὶξ τρίγωνονὸέθουσα βάσιν διαιρεῖται εἰξ δύο πυραμίδας ἴσας τε καὶ ὁμοίας ἀλλήλαις καὶ [ὁμοίας] τῇ ὅλῃ τριγώνους ἐθουσας βάσεις καὶ εἰξ δύο πρίσματα ἴσα· καὶ τὰ δύο πρίσματα μείζονά ἐστινὸὴ τὸ ἥμισυ τῆς ὅλης πυραμίδος.

the (square) on FH (is) to the (square) on BD, so circle $EFGH$ (is) to some area less than circle $ABCD$ [Prop. 5.11]. The very thing was shown (to be) impossible. Thus, as the square on BD is to the (square) on FH, so circle $ABCD$ (is) not to some area greater than circle $EFGH$. And it was shown that neither (is it in that ratio) to (some) lesser (area). Thus, as the square on BD is to the (square) on FH, so circle $ABCD$ (is) to circle $EFGH$.

Thus, circles are to one another as the squares on (their) diameters. (Which is) the very thing it was required to show.

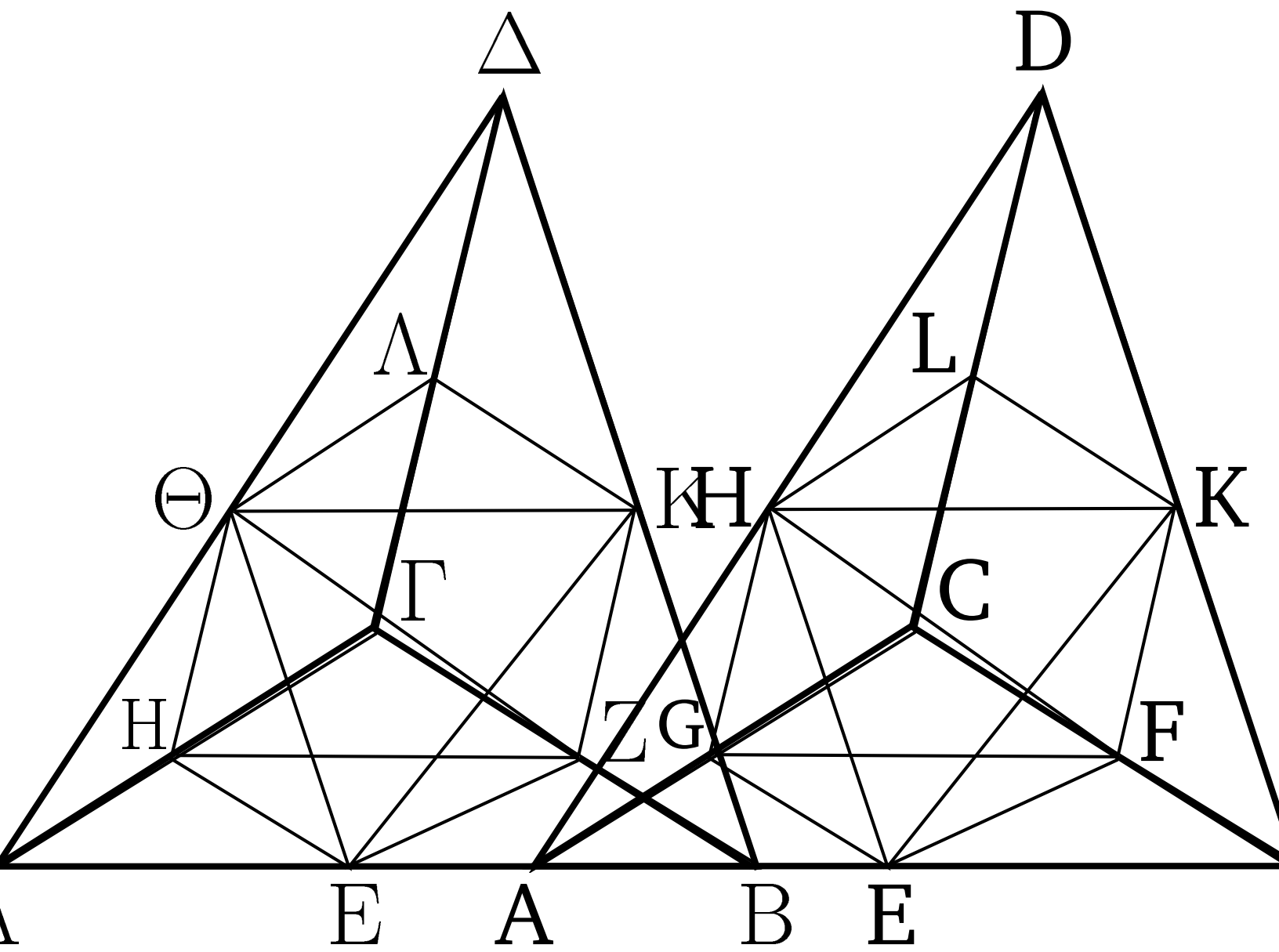
Lemma

So, I say that, area S being greater than circle $EFGH$, as area S is to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$.

For let it be contrived that as area S (is) to circle $ABCD$, so circle $EFGH$ (is) to area T. I say that area T is less than circle $ABCD$. For since as area S is to circle $ABCD$, so circle $EFGH$ (is) to area T, alternately, as area S is to circle $EFGH$, so circle $ABCD$ (is) to area T [Prop. 5.16]. And area S (is) greater than circle $EFGH$. Thus, circle $ABCD$ (is) also greater than area T [Prop. 5.14]. Hence, as area S is to circle $ABCD$, so circle $EFGH$ (is) to some area less than circle $ABCD$. (Which is) the very thing it was required to show.

Proposition 3

Any pyramid having a triangular base is divided into two pyramids having triangular bases (which are) equal, similar to one another, and [similar] to the whole, and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid.



Ἔστω πυραμίξ, ἧς βάσις μέν ἐστι τὸ ΑΒΓ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον· λέγω, ὅτι ἡ ΑΒΓΔ πυραμὶξ διαιρεῖται εἰξ δύο πυραμίδας ἴσας ἀλλήλαις τριγώνους βάσεις ἐθούσας καὶ ὁμοίας τῇ ὅλῃ καὶ εἰξ δύο πρίσματα ἴσα· καὶ τὰ δύο πρίσματα μείζονά ἐστινὸὴ τὸ ἥμισυ τῆς ὅλης πυραμίδος.

Τετμήσθωσαν γὰρ αἱ ΑΒ, ΒΓ, ΓΑ, ΑΔ, ΔΒ, ΔΓ δίθα κατὰ τὰ Ε, Ζ, Η, Θ, Κ, Λ σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΘΕ, ΕΗ, ΗΘ, ΘΚ, ΚΛ, ΛΘ, ΚΖ, ΖΗ. ἐπεὶ ὁίση ἐστὶν ἡ μὲν ΑΕ τῇ ΕΒ, ἡ δὲ ΑΘ τῇ ΔΘ, παράλληλος ἄρα ἐστὶν ἡ ΕΘ τῇ ΔΒ. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΘΚ τῇ ΑΒ παράλληλός ἐστιν. παραλληλόγραμμον ἄρα ἐστὶ τὸ ΘΕΒΚ· ὁίση ἄρα ἐστὶν ἡ ΘΚ τῇ ΕΒ. ἀλλὰ ἡ ΕΒ τῇ ΕΑ ἐστινὸίση· καὶ ἡ ΑΕ ἄρα τῇ ΘΚ ἐστινὸίση. ὁέστι δὲ καὶ ἡ ΑΘ τῇ ΘΔὸίση· δύο δὴ

Let there be a pyramid whose base is triangle ABC, and (whose) apex (is) point D. I say that pyramid $ABCD$ is divided into two pyramids having triangular bases (which are) equal to one another, and similar to the whole, and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid.

For let AB, BC, CA, AD, DB, and DC be cut in half at points E, F, G, H, K, and L (respectively). And let HE, EG, GH, HK, KL, LH, KF, and FG be joined. Since AE is equal to EB, and AH to DH, EH is thus parallel to DB [Prop. 6.2]. So, for the same (reasons), HK is also parallel to AB. Thus, $HEBK$ is a parallelogram. Thus, HK is equal to EB [Prop. 1.34]. But, EB is equal to EA. Thus, AE is

αἱ ΕΑ, ΑΘ δυσὶ ταῖς ΚΘ, ΘΔ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ· καὶ γωνία ἡ ὑπὸ ΕΑΘ γωνίᾳ τῇ ὑπὸ ΚΘΔ ὁίση· βάσις ἄρα ἡ ΕΘ βάσει τῇ ΚΔ ἐστινὁίση. ἴσον ἄρα καὶ ὅμοιόν ἐστι τὸ ΑΕΘ τρίγωνον τῷ ΘΚΔ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΑΘΗ τρίγωνον τῷ ΘΛΔ τριγώνῳ ἴσον τέ ἐστι καὶ ὅμοιον. καὶ ἐπεὶ δύο εὐθεῖαι ἁπτόμεναι ἀλλήλων αἱ ΕΘ, ΘΗ παρὰ δύο εὐθείας ἁπτομένας ἀλλήλων τὰξ ΚΔ, ΔΛ εἰσιν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ οὐσαι, ἴσας γωνίας περιέχουσιν. ὁίση ἄρα ἐστὶν ἡ ὑπὸ ΕΘΗ γωνία τῇ ὑπὸ ΚΔΛ γωνίᾳ. καὶ ἐπεὶ δύο εὐθεῖαι αἱ ΕΘ, ΘΗ δυσὶ ταῖς ΚΔ, ΔΛ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, καὶ γωνία ἡ ὑπὸ ΕΘΗ γωνίᾳ τῇ ὑπὸ ΚΔΛ ἐστινὁίση, βάσις ἄρα ἡ ΕΗ βάσει τῇ ΚΛ [ἐστιν] ὁίση· ἴσον ἄρα καὶ ὅμοιόν ἐστι τὸ ΕΘΗ τρίγωνον τῷ ΚΔΛ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΑΕΗ τρίγωνον τῷ ΘΚΛ τριγώνῳ ἴσον τε καὶ ὅμοιόν ἐστιν. ἡ ἄρα πυραμίξ, ἧς βάσις μέν ἐστι τὸ ΑΕΗ τρίγωνον, κορυφὴ δὲ τὸ Θ σημεῖον, ὁίση καὶ ὁμοία ἐστὶ πυραμίδι, ἧς βάσις μέν ἐστι τὸ ΘΚΛ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον. καὶ ἐπεὶ τριγώνου τοῦ ΑΔΒ παρὰ μίαν τῶν πλευρῶν τὴν ΑΒὀῆκται ἡ ΘΚ, ἰσογώνιόν ἐστι τὸ ΑΔΒ τρίγωνον τῷ ΔΘΚ τριγώνῳ, καὶ τὰξ πλευρὰξ ἀνάλογονὀέθουσιν· ὅμοιον ἄρα ἐστὶ τὸ ΑΔΒ τρίγωνον τῷ ΔΘΚ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν ΔΒΓ τρίγωνον τῷ ΔΚΛ τριγώνῳ ὅμοιόν ἐστιν, τὸ δὲ ΑΔΓ τῷ ΔΛΘ. καὶ ἐπεὶ δύο εὐθεῖαι ἁπτόμεναι ἀλλήλων αἱ ΒΑ, ΑΓ παρὰ δύο εὐθείας ἁπτομένας ἀλλήλων τὰξ ΚΘ, ΘΛ εἰσιν οὐκ ἐν τῷ αὐτῷ ἐπιπέδῳ, ἴσας γωνίας περιέχουσιν. ὁίση ἄρα ἐστὶν ἡ ὑπὸ ΒΑΓ γωνία τῇ ὑπὸ ΚΘΛ. καί ἐστιν ὡξ ἡ ΒΑ πρὸς τὴν ΑΓ, οὁύτως ἡ ΚΘ πρὸς τὴν ΘΛ· ὅμοιον ἄρα ἐστὶ τὸ ΑΒΓ τρίγωνον τῷ ΘΚΛ τριγώνῳ. καὶ πυραμὶξ ἄρα, ἧς βάσις μέν ἐστι τὸ ΑΒΓ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, ὁμοία ἐστὶ πυραμίδι, ἧς βάσις μέν ἐστι τὸ ΘΚΛ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον. ἀλλὰ πυραμίξ, ἧς βάσις μέν [ἐστι] τὸ ΘΚΛ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, ὁμοία ἐδείθθη πυραμίδι, ἧς βάσις μέν ἐστι τὸ ΑΕΗ τρίγωνον, κορυφὴ δὲ τὸ Θ σημεῖον. ἑκατέρα ἄρα τῶν ΑΕΗΘ, ΘΚΛΔ πυραμίδων ὁμοία ἐστὶ τῇ ὅλῃ τῇ ΑΒΓΔ πυραμίδι.

Καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΒΖ τῇ ΖΓ, διπλάσιόν ἐστι τὸ ΕΒΖΗ παραλληλόγραμμον τοῦ ΗΖΓ τριγώνου. καὶ ἐπεί, ἐανὸῇ δύο πρίσματα ἰσοϋψῆ, καὶ τὸ μὲνὸέθη βάσιν παραλληλόγραμμον, τὸ δὲ τρίγωνον, διπλάσιον δὲ ὁῇ τὸ παραλληλόγραμμον τοῦ τριγώνου, ἴσα ἐστὶ τὰ πρίσματα, ἴσον ἄρα ἐστὶ τὸ πρίσμα τὸ περιεχόμενον ὑπὸ δύο μὲν τριγώνων τῶν ΒΚΖ, ΕΘΗ, τριῶν δὲ παραλληλογράμμων τῶν ΕΒΖΗ, ΕΒΚΘ, ΘΚΖΗ τῷ πρισματι τῷ περιεθομένῳ ὑπὸ δύο μὲν τριγώνων τῶν ΗΖΓ, ΘΚΛ, τριῶν δὲ παραλληλογράμμων τῶν ΚΖΓΛ, ΛΓΗΘ, ΘΚΖΗ. καὶ φανερόν, ὅτι ἑκάτρον τῶν πρισμάτων, οὗ τε βάσις τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΘΚ εὐθεῖα, καὶ οὗ βάσις τὸ ΗΖΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΘΚΛ τρίγωνον, μεῖζόν ἐστιν ἑκατέρας τῶν πυραμίδων, ὧν βάσεις μὲν τὰ ΑΕΗ, ΘΚΛ τρίγωνα, κορυφαί, δὲ τὰ Θ, Δ σημεῖα, ἐπειδήπερ [καί] ἐαν ἐπιζεύχωμεν τὰξ ΕΖ, ΕΚ εὐθείας, τὸ μὲν πρίσμα, οὗ βάσις τὸ ΕΒΖΗ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΘΚ

also equal to HK. And AH is also equal to HD. So the two (straight-lines) EA and AH are equal to the two (straight-lines) KH and HD, respectively. And angle EAH (is) equal to angle KHD [Prop. 1.29]. Thus, base EH is equal to base KD [Prop. 1.4]. Thus, triangle AEH is equal and similar to triangle HKD [Prop. 1.4]. So, for the same (reasons), triangle AHG is also equal and similar to triangle HLD. And since EH and HG are two straight-lines joining one another (which are respectively) parallel to two straight-lines joining one another, KD and DL, not being in the same plane, they will contain equal angles [Prop. 11.10]. Thus, angle EHG is equal to angle KDL. And since the two straight-lines EH and HG are equal to the two straight-lines KD and DL, respectively, and angle EHG is equal to angle KDL, base EG [is] thus equal to base KL [Prop. 1.4]. Thus, triangle EHG is equal and similar to triangle KDL. So, for the same (reasons), triangle AEG is also equal and similar to triangle HKL. Thus, the pyramid whose base is triangle AEG, and apex the point H, is equal and similar to the pyramid whose base is triangle HKL, and apex the point D [Def. 11.10]. And since HK has been drawn parallel to one of the sides, AB, of triangle ADB, triangle ADB is equiangular to triangle DHK [Prop. 1.29], and they have proportional sides. Thus, triangle ADB is similar to triangle DHK [Def. 6.1]. So, for the same (reasons), triangle DBC is also similar to triangle DKL, and ADC to DLH. And since two straight-lines joining one another, BA and AC, are parallel to two straight-lines joining one another, KH and HL, not in the same plane, they will contain equal angles [Prop. 11.10]. Thus, angle BAC is equal to (angle) KHL. And as BA is to AC, so KH (is) to HL. Thus, triangle ABC is similar to triangle HKL [Prop. 6.6]. And, thus, the pyramid whose base is triangle ABC, and apex the point D, is similar to the pyramid whose base is triangle HKL, and apex the point D [Def. 11.9]. But, the pyramid whose base [is] triangle HKL, and apex the point D, was shown (to be) similar to the pyramid whose base is triangle AEG, and apex the point H. Thus, each of the pyramids $AEGH$ and $HKLD$ is similar to the whole pyramid $ABCD$.

And since BF is equal to FC, parallelogram $EBFG$ is double triangle GFC [Prop. 1.41]. And since, if two prisms (have) equal heights, and the former has a parallelogram as a base, and the latter a triangle, and the parallelogram (is) double the triangle, then the prisms are equal [Prop. 11.39], the prism contained by the two triangles BKF and EHG, and the three parallelograms $EBFG$, $EBKH$, and $HKFG$, is thus equal to the prism contained by the two triangles GFC and HKL, and the three parallelograms $KFCL$, $LCGH$, and $HKFG$. And (it is) clear that each of the prisms whose base (is) parallelogram $EBFG$, and opposite (side) straight-line HK, and whose base (is) triangle GFC, and opposite (plane) triangle HKL, is greater than each of the pyramids whose bases are triangles AEG and HKL, and apexes the points H and D (respec-

εὐθεῖα, μεῖζόν ἐστι τῆς πυραμίδος, ἧξ βάσις τὸ EBZ τρίγωνον, κορυφὴ δὲ τὸ K σημεῖον. ἀλλὰ ἡ πυραμίς, ἧξ βάσις τὸ EBZ τρίγωνον, κορυφὴ δὲ τὸ K σημεῖον, ὁίση ἐστὶ πυραμίδι, ἧξ βάσις τὸ AEH τρίγωνον, κορυφὴ δὲ τὸ Θ σημεῖον· ὑπὸ γὰρ ὁίσων καὶ ὁμοίων ἐπιπέδων περιέθονται. ὁώστε καὶ τὸ πρίσμα, οὗ βάσις μὲν τὸ $EBZH$ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΘK εὐθεῖα, μεῖζόν ἐστι πυραμίδος, ἧξ βάσις μὲν τὸ AEH τρίγωνον, κορυφὴ δὲ τὸ Θ σημεῖον. ἴσον δὲ τὸ μὲν πρίσμα, οὗ βάσις τὸ $EBZH$ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΘK εὐθεῖα, τῷ πρίσματι, οὗ βάσις μὲν τὸ $HZ\Gamma$ τρίγωνον, ἀπεναντίον δὲ τὸ $\Theta K\Lambda$ τρίγωνον· ἡ δὲ πυραμίς, ἧξ βάσις τὸ AEH τρίγωνον, κορυφὴ δὲ τὸ Θ σημεῖον, ὁίση ἐστὶ πυραμίδι, ἧξ βάσις τὸ $\Theta K\Lambda$ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον. τὰ ἄρα εἰρημένα δύο πρίσματα μείζονά ἐστι τῶν εἰρημένων δύο πυραμίδων, ὁῶν βάσεις μὲν τὰ AEH, $\Theta K\Lambda$ τρίγωνα, κορυφαὶ δὲ τὰ Θ, Δ σημεῖα.

Ἡ ἄρα ὅλη πυραμίς, ἧξ βάσις τὸ $AB\Gamma$ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, διῄρηται εἰίξ τε δύο πυραμίδας ἴσας ἀλλήλαις [καὶ ὁμοίας τῇ ὅλῃ] καὶ εἰξ δύο πρίσματα ἴσα, καὶ τὰ δύο πρίσματα μείζονά ἐστινὴ τὸ ἥμισυ τῆς ὅλης πυραμίδος· ὅπερ ἔδει δεῖχαι.

tively), inasmuch as, if we [also] join the straight-lines EF and EK then the prism whose base (is) parallelogram $EBFG$, and opposite (side) straight-line HK, is greater than the pyramid whose base (is) triangle EBF, and apex the point K. But the pyramid whose base (is) triangle EBF, and apex the point K, is equal to the pyramid whose base is triangle AEG, and apex point H. For they are contained by equal and similar planes. And, hence, the prism whose base (is) parallelogram $EBFG$, and opposite (side) straight-line HK, is greater than the pyramid whose base (is) triangle AEG, and apex the point H. And the prism whose base is parallelogram $EBFG$, and opposite (side) straight-line HK, (is) equal to the prism whose base (is) triangle GFC, and opposite (plane) triangle HKL. And the pyramid whose base (is) triangle AEG, and apex the point H, is equal to the pyramid whose base (is) triangle HKL, and apex the point D. Thus, the (sum of the) aforementioned two prisms is greater than the (sum of the) aforementioned two pyramids, whose bases (are) triangles AEG and HKL, and apexes the points H and D (respectively).

Thus, the whole pyramid, whose base (is) triangle ABC, and apex the point D, has been divided into two pyramids (which are) equal to one another [and similar to the whole], and into two equal prisms. And the (sum of the) two prisms is greater than half of the whole pyramid. (Which is) the very thing it was required to show.

4.

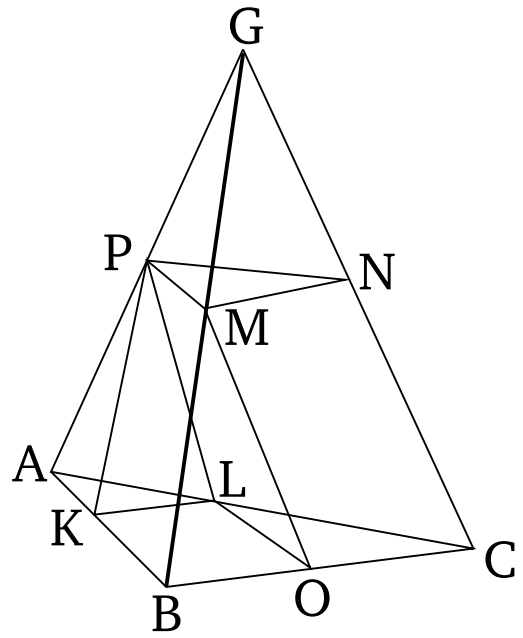
Ἐὰνὸῶσι δύο πυραμίδες ὑπὸ τὸ αὐτὸ ὁύψος τριγώνουςὸέθουσαι βάσεις, διαιρεθῇ δὲ ἑκατέρα αὐτῶν εὶίξ τε δύο πυραμίδας ἴσας ἀλλήλαις καὶ ὁμοίας τῇ ὅλῃ καὶ εἰξ δύο πρίσματα ἴσα, ὁέσται ὡξ ἡ τῆς μιᾶς πυραμίδος βάσις πρὸς τὴν τῆς ἑτέρας πυραμίδος βάσιν, οὁύτως τὰ ἐν τῇ μιᾷ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ ἑτάρᾳ πυραμίδι πρίσματα πάντα ἰσοπληθῆ.

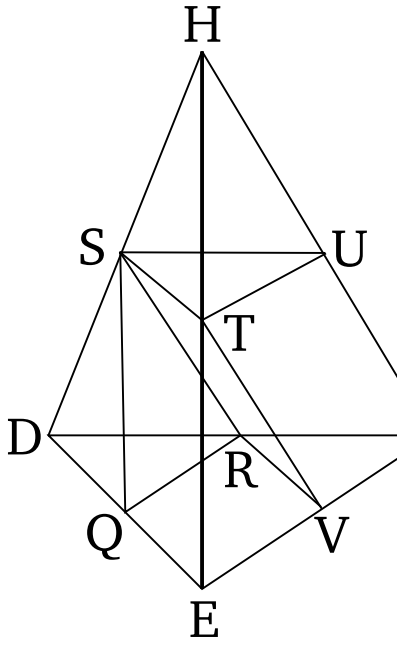
Ἔστωσαν δύο πυραμίδες ὑπὸ τὸ αὐτὸ ὁύψος τριγώνουςὸέθουσαι βάσεις τὰξ $AB\Gamma$, ΔEZ, κορυφὰξ δὲ τὰ H, Θ σημεῖα, καὶ διῃρήσθω ἑκατέρα αὐτῶν εὶίξ τε δύο πυραμίδας ἴσας ἀλλήλαις καὶ ὁμοίας τῇ ὅλῃ καὶ εἰξ δύο πρίσματα ἴσα· λέγω, ὅτι ἐστὶν ὡξ ἡ $AB\Gamma$ βάσις πρὸς τὴν ΔEZ βάσιν, οὁύτως τὰ ἐν τῇ $AB\Gamma H$ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ $\Delta EZ\Theta$ πυραμίδι πρίσματα ἰσοπληθῆ.

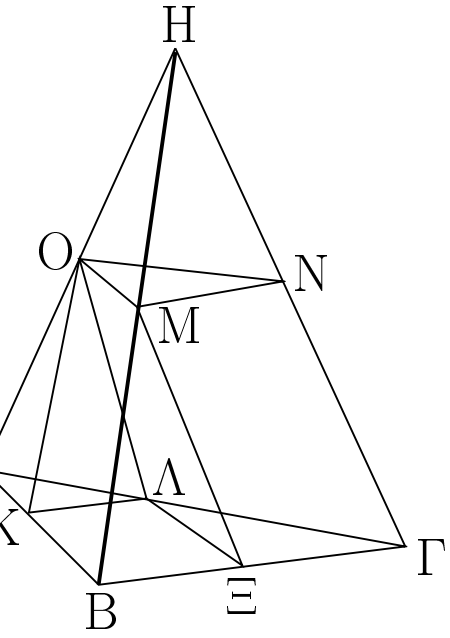
Proposition 4

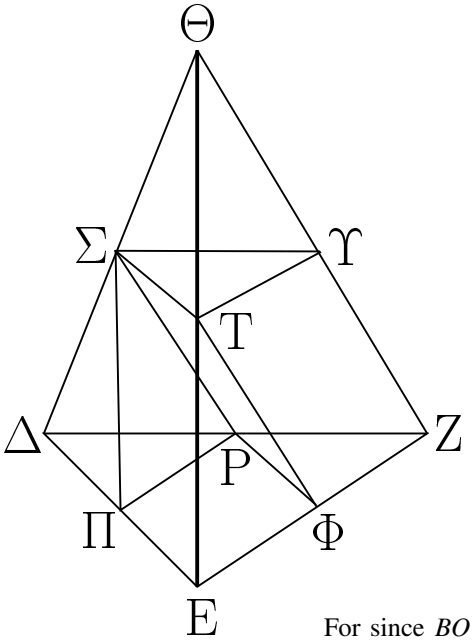
If there are two pyramids with the same height, having trianglular bases, and each of them is divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms then as the base of one pyramid (is) to the base of the other pyramid, so (the sum of) all the prisms in one pyramid will be to (the sum of all) the equal number of prisms in the other pyramid.

Let there be two pyramids with the same height, having the triangular bases ABC and DEF, (with) apexes the points G and H (respectively). And let each of them be divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms [Prop. 12.3]. I say that as base ABC is to base DEF, so (the sum of) all the prisms in pyramid $ABCG$ (is) to (the sum of) all the equal number of prisms in pyramid $DEFH$.









Ἐπεὶ γὰρ ὁίση ἐστὶν ἡ μὲν ΒΧ τῇ ΧΓ, ἡ δὲ ΑΛ τῇ ΛΓ, παράλληλος ἄρα ἐστὶν ἡ ΛΧ τῇ ΑΒ καὶ ὅμοιον τὸ ΑΒΓ τρίγωνον τῷ ΛΧΓ τριγώνῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ ΔΕΖ τρίγωνον τῷ ΡΦΖ τριγώνῳ ὅμοιόν ἐστιν. καὶ ἐπεὶ διπλασίων ἐστὶν ἡ μὲν ΒΓ τῆς ΓΧ, ἡ δὲ ΕΖ τῆς ΖΦ, ὐέστιν ἄρα ὡξ ἡ ΒΓ πρὸς τὴν ΓΧ, οὑτως ἡ ΕΖ πρὸς τὴν ΖΦ. καὶ ἀναγέγραπται ἀπὸ μὲν τῶν ΒΓ, ΓΧ ὅμοιά τε καὶ ὁμοίως κείμενα εὐθύγραμμα τὰ ΑΒΓ, ΛΧΓ, ἀπὸ δὲ τῶν ΕΖ, ΖΦ ὅμοιά τε καὶ ὁμοίως κείμενα [εὐθύγραμμα] τὰ ΔΕΖ, ΡΦΖ· ὐέστιν ἄρα ὡξ τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΛΧΓ τρίγωνον, οὑτως τὸ ΔΕΖ τρίγω-

For since *BO* is equal to *OC*, and *AL* to *LC*, *LO* is thus parallel to *AB*, and triangle *ABC* similar to triangle *LOC* [Prop. 12.3]. So, for the same (reasons), triangle *DEF* is also similar to triangle *RVF*. And since *BC* is double *CO*, and *EF* (double) *FV*, thus as *BC* (is) to *CO*, so *EF* (is) to *FV*. And the similar, and similarly laid out, rectilinear (figures) *ABC* and *LOC* be described on *BC* and *CO* (respectively), and the similar, and similarly laid out, [rectilinear] (figures) *DEF* and *RVF* on *EF* and *FV* (respectively). Thus, as triangle *ABC* is to triangle *LOC*, so triangle *DEF* (is) to triangle *RVF* [Prop. 6.22]. Thus, alternately, as triangle *ABC* is to [triangle] *DEF*, so [tri-

νον πρὸς τὸ ΡΦΖ τρίγωνον· ἐναλλὰχ ἄρα ἐστὶν ὡς τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΔΕΖ [τρίγωνον], οὕτως τὸ ΛΧΓ [τρίγωνον] πρὸς τὸ ΡΦΖ τρίγωνον. ἀλλὰ ὡς τὸ ΛΧΓ τρίγωνον πρὸς τὸ ΡΦΖ τρίγωνον, οὕτως τὸ πρίσμα, οὗ βάσις μέν [ἐστι] τὸ ΛΧΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΟΜΝ, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ ΡΦΖ τρίγωνον, ἀπεναντίον δὲ τὸ ΣΤΥ· καὶ ὡς ἄρα τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΔΕΖ τρίγωνον, οὕτως τὸ πρίσμα, οὗ βάσις μὲν τὸ ΛΧΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΟΜΝ, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ ΡΦΖ τρίγωνον, ἀπεναντίον δὲ τὸ ΣΤΥ. ὡς δὲ τὰ εἰρημένα πρίσματα πρὸς ἄλληλα, οὕτως τὸ πρίσμα, οὗ βάσις μὲν τὸ ΚΒΧΛ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΟΜ εὐθεῖα, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ ΠΕΦΡ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΣΤ εὐθεῖα. καὶ τὰ δύο ἄρα πρίσματα, οὗ τε βάσις μὲν τὸ ΚΒΧΛ παραλληλόγραμμον, ἀπεναντίον δὲ ἡ ΟΜ, καὶ οὗ βάσις μὲν τὸ ΛΧΓ, ἀπεναντίον δὲ τὸ ΟΜΝ, πρὸς τὰ πρίσματα, οὗ τε βάσις μὲν τὸ ΠΕΦΡ, ἀπεναντίον δὲ ἡ ΣΤ εὐθεῖα, καὶ οὗ βάσις μὲν τὸ ΡΦΖ τρίγωνον, ἀπεναντίον δὲ τὸ ΣΤΥ. καὶ ὡς ἄρα ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὰ εἰρημένα δύο πρίσματα πρὸς τὰ εἰρημένα δύο πρίσματα.

Καὶ ὁμοίως, ἐὰν διαιρεθῶσιν αἱ ΟΜΝΗ, ΣΤΥΘ πυραμίδες εὐίξ τε δύο πρίσματα καὶ δύο πυραμίδας, ὀέσται ὡς ἡ ΟΜΝ βάσις πρὸς τὴν ΣΤΥ βάσιν, οὕτως τὰ ἐν τῇ ΟΜΝΗ πυραμίδι δύο πρίσματα πρὸς τὰ ἐν τῇ ΣΤΥΘ πυραμίδι δύο πρίσματα. ἀλλὰ ὡς ἡ ΟΜΝ βάσις πρὸς τὴν ΣΤΥ βάσιν, οὕτως ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν· ἴσον γὰρ ἑκάτερον τῶν ΟΜΝ, ΣΤΥ τριγώνων ἑκατέρῳ τῶν ΛΧΓ, ΡΦΖ. καὶ ὡς ἄρα ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὰ τέσσαρα πρίσματα πρὸς τὰ τέσσαρα πρίσματα. ὁμοίως δὲ κἂν τὰξ ὑπολειπομένας πυραμίδας διέλωμεν εὐίξ τε δύο πυραμίδας καὶ εἰξ δύο πρίσματα, ὀέσται ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὰ ἐν τῇ ΑΒΓΗ πυραμίδι πρίσματα πάντα πρὸς τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα πάντα ἰσοπληθῆ· ὅπερ ἔδει δεῖξαι.

Λῆμμα.

Ὅτι δέ ἐστιν ὡς τὸ ΛΧΓ τρίγωνον πρὸς τὸ ΡΦΖ τρίγωνον, οὕτως τὸ πρίσμα, οὗ βάσις τὸ ΛΧΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΟΜΝ, πρὸς τὸ πρίσμα, οὗ βάσις μὲν τὸ ΡΦΖ [τρίγωνον], ἀπεναντίον δὲ τὸ ΣΤΥ, οὕτω δεικτέον.

Ἐπὶ γὰρ τῆς αὐτῆς καταγραφῆς νενοήσθωσαν ἀπὸ τῶν Η, Θ κάθετοι ἐπὶ τὰ ΑΒΓ, ΔΕΖ ἐπίπεδα, ἴσαι δηλαδὴ τυγθάνουσαι διὰ τὸ ἰσοϋψεῖς ὑποκεῖσθαι τὰξ πυραμίδας. καὶ ἐπεὶ δύο εὐθεῖαι ᾗ τε ΗΓ καὶ ἡ ἀπὸ τοῦ Η κάθετος ὑπὸ παραλλήλων ἐπιπέδων τῶν ΑΒΓ, ΟΜΝ τέμνονται, εἰξ τοὐξ αὐτοὐξ λόγους τμ ηθήσονται. καὶ τέτμ ηται ἡ ΗΓ δίθα ὑπὸ τοῦ ΟΜΝ ἐπιπέδου κατὰ τὸ Ν· καὶ ἡ ἀπὸ τοῦ Η ἄρα κάθετος ἐπὶ τὸ ΑΒΓ ἐπίπεδον δίθα τμ ηθήσεται ὑπὸ τοῦ ΟΜΝ ἐπιπέδου. διὰ τὰ αὐτὰ δὴ καὶ ἡ ἀπὸ τοῦ Θ κάθετος ἐπὶ τὸ ΔΕΖ ἐπίπεδον δίθα τμ ηθήσεται ὑπὸ τοῦ ΣΤΥ ἐπιπέδου. καί εἰσιν ἴσαι αἱ ἀπὸ τῶν angle] *LOC* (is) to triangle *RVF* [Prop. 5.16]. But, as triangle *LOC* (is) to triangle *RVF*, so the prism whose base [is] triangle *LOC*, and opposite (plane) *PMN*, (is) to the prism whose base (is) triangle *RVF*, and opposite (plane) *STU* (see lemma). And, thus, as triangle *ABC* (is) to triangle *DEF*, so the prism whose base (is) triangle *LOC*, and opposite (plane) *PMN*, (is) to the prism whose base (is) triangle *RVF*, and opposite (plane) *STU*. And as the aforementioned prisms (are) to one another, so the prism whose base (is) parallelogram *KBOL*, and opposite (side) straight-line *PM*, (is) to the prism whose base (is) parallelogram *QEVR*, and opposite (side) straight-line *ST* [Props. 11.39, 12.3]. Thus, also, (is) the (sum of the) two prisms—that whose base (is) parallelogram *KBOL*, and opposite (side) *PM*, and that whose base (is) *LOC*, and opposite (plane) *PMN*—to (the sum of) the (two) prisms—that whose base (is) *QEVR*, and opposite (side) straight-line *ST*, and that whose base (is) triangle *RVF*, and opposite (plane) *STU* [Prop. 5.12]. And, thus, as base *ABC* (is) to base *DEF*, so the (sum of the first) aforementioned two prisms (is) to the (sum of the second) aforementioned two prisms.

And, similarly, if pyramids *PMNG* and *STUH* are divided into two prisms, and two pyramids, as base *PMN* (is) to base *STU*, so (the sum of) the two prisms in pyramid *PMNG* will be to (the sum of) the two prisms in pyramid *STUH*. But, as base *PMN* (is) to base *STU*, so base *ABC* (is) to base *DEF*. For the triangles *PMN* and *STU* (are) equal to *LOC* and *RVF*, respectively. And, thus, as base *ABC* (is) to base *DEF*, so (the sum of) the four prisms (is) to (the sum of) the four prisms [Prop. 5.12]. So, similarly, even if we divide the pyramids left behind into two pyramids and into two prisms, as base *ABC* (is) to base *DEF*, so (the sum of) all the prisms in pyramid *ABCG* will be to (the sum of) all the equal number of prisms in pyramid *DEFH*. (Which is) the very thing it was required to show.

Lemma

And one may show, as follows, that as triangle *LOC* is to triangle *RVF*, so the prism whose base (is) triangle *LOC*, and opposite (plane) *PMN*, (is) to the prism whose base (is) [triangle] *RVF*, and opposite (plane) *STU*.

For, in the same figure, let perpendiculars be conceived (drawn) from (points) *G* and *H* to the planes *ABC* and *DEF* (respectively). These clearly turn out to be equal, on account of the pyramids being assumed (to be) of equal height. And since two straight-lines, *GC* and the perpendicular from *G*, are cut by the parallel planes *ABC* and *PMN* they will be cut in the same ratios [Prop. 11.17]. And *GC* was cut in half by the plane *PMN* at *N*. Thus, the perpendicular from *G* to the plane *ABC* will also be cut in half by the plane *PMN*. So, for the same (reasons), the perpendicular from *H* to the plane *DEF* will

Η, Θ κάθετοι ἐπὶ τὰ ΑΒΓ, ΔΕΖ ἐπίπεδα· ἴσαι ἄρα καὶ αἱ ἀπὸ τῶν ΟΜΝ, ΣΤΥ τριγώνων ἐπὶ τὰ ΑΒΓ, ΔΕΖ κάθετοι. ἰσοϋψῆ ἄρα [ἐστὶ] τὰ πρίσματα, ὧν βάσεις μέν εἰσι τὰ ΛΧΓ, ΡΦΖ τρίγωνα, ἀπεναντίον δὲ τὰ ΟΜΝ, ΣΤΥ. ὥστε καὶ τὰ στερεὰ παραλληλεπίπεδα τὰ ἀπὸ τῶν εἰρημένων πρισμάτων ἀναγραφόμενα ἰσοϋψῆ καὶ πρὸς ἄλληλά [εἰσιν] ὡς αἱ βάσεις· καὶ τὰ ἥμίση ἄρα ἐστὶν ὡς ἡ ΛΧΓ βάσις πρὸς τὴν ΡΦΖ βάσιν, οὕτως τὰ εἰρημένα πρίσματα πρὸς ἄλληλα· ὅπερ ἔδει δεῖχαι.

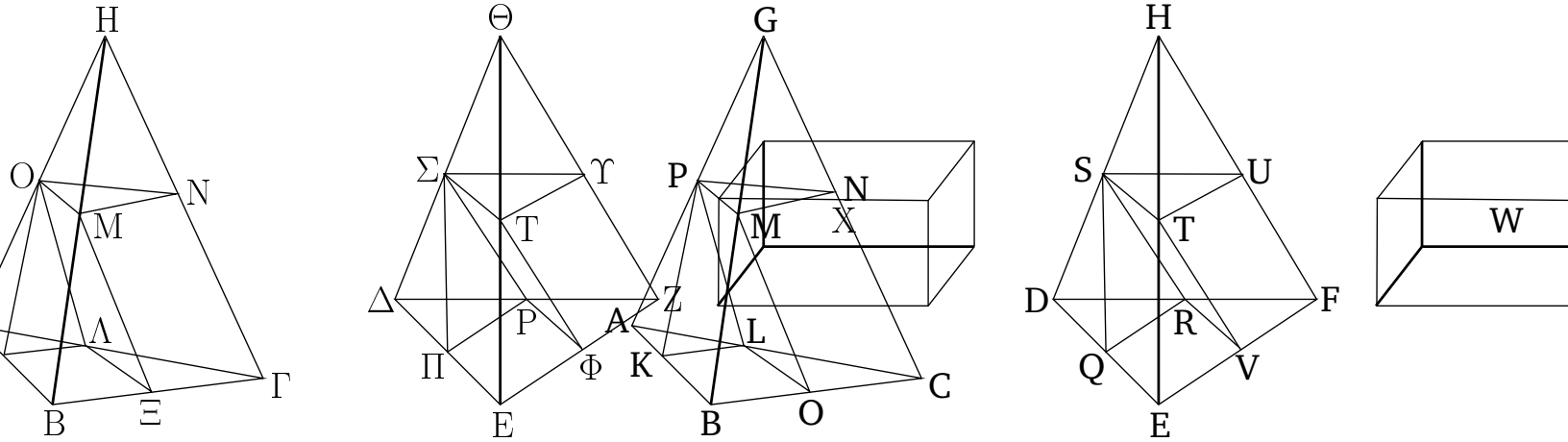
also be cut in half by the plane STU. And the perpendiculars from G and H to the planes ABC and DEF (respectively) are equal. Thus, the perpendiculars from the triangles PMN and STU to ABC and DEF (respectively, are) also equal. Thus, the prisms whose bases are triangles LOC and RVF, and opposite (sides) PMN and STU (respectively), [are] of equal height. And, hence, the parallelepiped solids described on the aforementioned prisms [are] of equal height and (are) to one another as their bases [Prop. 11.32]. Likewise, the halves (of the solids) [Prop. 11.28]. Thus, as base LOC is to base RVF, so the aforementioned prisms (are) to one another. (Which is) the very thing it was required to show.

5.

Αἱ ὑπὸ τὸ αὐτὸ ὕψος οὖσαι πυραμίδες καὶ τριγώνους ἐχούσας βάσεις πρὸς ἀλλήλας εἰσὶν ὡς αἱ βάσεις.

Proposition 5

Pyramids which are of the same height, and have triangular bases, are to one another as their bases.



Ἔστωσαν ὑπὸ τὸ αὐτὸ ὕψος πυραμίδες, ὧν βάσεις μὲν τὰ ΑΒΓ, ΔΕΖ τρίγωνα, κορυφαὶ δὲ τὰ Η, Θ σημεῖα· λέγω, ὅτι ἐστὶν ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς πρὸς τὴν ΔΕΖΘ πυραμίδα.

Εἰ γὰρ μή ἐστιν ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς πρὸς τὴν ΔΕΖΘ πυραμίδα, ὀέσται ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶς ἤτοι πρὸς ἔλασσόν τι τῆς ΔΕΖΘ πυραμίδος στερεὸν ἢ πρὸς μεῖζον. ὀέστω πρότερον πρὸς ἔλασσον τὸ Θ, καὶ διῃρήσθω ἡ ΔΕΖΘ πυραμὶς εἰς τε δύο πυραμίδας ἴσας ἀλλήλαις καὶ ὁμοίας τῇ ὅλῃ καὶ εἰς δύο πρίσματα ἴσα· τὰ δὴ δύο πρίσαμτα μείζονά ἐστιν ἢ τὸ ἥμισυ τῆς ὅλης πυραμίδος. καὶ πάλιν αἱ ἐκ τῆς διαιρέσεως γινόμεναι πυραμίδες ὁμοίως διῃρήσθωσαν, καὶ τοῦτο ἀεὶ γινέσθω, ὀέως οὗ λειφθῶσί τινες πυραμίδες ἀπὸ τῆς ΔΕΖΘ πυραμίδος, αἱ εἰσιν ἐλάττονες τῆς ὑπεροθῆς, ᾗ ὑπερέθει ἡ ΔΕΖΘ πυραμὶς τοῦ Θ στερεοῦ. λελείφθωσαν καὶ ὀέστωσαν λόγου ὁένεκεν αἱ ΔΠΡΣ, ΣΤΥΘ· λοιπὰ ἄρα τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα μείζονά ἐστι τοῦ Θ στερεοῦ. διῃρήσθω καὶ ἡ ΑΒΓΗ πυραμὶς ὁμοίως καὶ ἰσοπληθῶς τῇ ΔΕΖΘ πυραμίδι· ὀέστιν ἄρα ὡς ἡ ΑΒΓ βάσις πρὸς

Let there be pyramids of the same height whose bases (are) the triangles ABC and DEF, and apexes the points G and H (respectively). I say that as base ABC is to base DEF, so pyramid $ABCG$ (is) to pyramid $DEFH$.

For if base ABC is not to base DEF, as pyramid $ABCG$ (is) to pyramid $DEFH$, then base ABC will be to base DEF, as pyramid $ABCG$ (is) to some solid either less than, or greater than, pyramid $DEFH$. Let it, first of all, be (in this ratio) to (some) lesser (solid), W. And let pyramid $DEFH$ be divided into two pyramids equal to one another, and similar to the whole, and into two equal prisms. So, the (sum of the) two prisms is greater than half of the whole pyramid [Prop. 12.3]. And, again, let the pyramids generated by the division be similarly divided, and let this be done continually until some pyramids are left from pyramid $DEFH$ which (when added together) are less than the excess by which pyramid $DEFH$ exceeds the solid W [Prop. 10.1]. Let them be left, and, for the sake of argument, let them be $DQRS$ and $STUH$. Thus, the (sum of the) remaining prisms within pyramid $DEFH$ is greater than solid W. Let pyramid $ABCG$ also be divided simi-

τὴν ΔΕΖ βάσιν, οὕτως τὰ ἐν τῇ ΑΒΓΗ πυραμίδι πρίσματα πρὸς τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα, ἀλλὰ καὶ ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶξ πρὸς τὸ Θ στερεόν· καὶ ὡς ἄρα ἡ ΑΒΓΗ πυραμὶξ πρὸς τὸ Θ στερεόν, οὕτως τὰ ἐν τῇ ΑΒΓΗ πυραμίδι πρίσματα πρὸς τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα· ἐναλλὰχ ἄρα ὡς ἡ ΑΒΓΗ πυραμὶξ πρὸς τὰ ἐν αὐτῇ πρίσματα, οὕτως τὸ Θ στερεὸν πρὸς τὰ ἐν τῇ ΔΕΖΘ πυραμίδι πρίσματα. μείζων δὲ ἡ ΑΒΓΗ πυραμὶξ τῶν ἐν αὐτῇ πρισμάτων· μεῖζον ἄρα καὶ τὸ Θ στερεὸν τῶν ἐν τῇ ΔΕΖΘ πυραμίδι πρισμάτων. ἀλλὰ καὶ ἐλαττον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐστὶν ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶξ πρὸς ἔλασσόν τι τῆς ΔΕΖΘ πυραμίδος στερεόν. ὁμοίως δὴ δειθθήσεται, ὅτι οὐδὲ ὡς ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, οὕτως ἡ ΔΕΖΘ πυραμὶξ πρὸς ἔλαττόν τι τῆς ΑΒΓΗ πυραμίδος στερεόν.

Λέγω δή, ὅτι οὐκ ἔστιν οὐδὲ ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶξ πρὸς μεῖζόν τι τῆς ΔΕΖΘ πυραμίδος στερεόν.

Εἰ γὰρ δυνατόν, ἔστω πρὸς μεῖζον τὸ Θ· ἀνάπαλιν ἄρα ἐστὶν ὡς ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, οὕτως τὸ Θ στερεὸν πρὸς τὴν ΑΒΓΗ πυραμίδα. ὡς δὲ τὸ Θ στερεὸν πρὸς τὴν ΑΒΓΗ πυραμίδα, οὕτως ἡ ΔΕΖΘ πυραμὶξ πρὸς ἔλασσόν τι τῆς ΑΒΓΗ πυραμίδος, ὡξἐμπροσθεν ἐδείθθη· καὶ ὡς ἄρα ἡ ΔΕΖ βάσις πρὸς τὴν ΑΒΓ βάσιν, οὕτως ἡ ΔΕΖΘ πυραμὶξ πρὸς ἔλασσόν τι τῆς ΑΒΓΗ πυραμίδος· ὅπερ ἀτοπον ἐδείθθη. οὐκ ἄρα ἐστὶν ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶξ πρὸς μεῖζόν τι τῆς ΔΕΖΘ πυραμίδος στερεόν. ἐδείθθη δέ, ὅτι οὐδὲ πρὸς ἔλασσον. ἐστιν ἄρα ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως ἡ ΑΒΓΗ πυραμὶξ πρὸς τὴν ΔΕΖΘ πυραμίδα· ὅπερ ἔδει δεῖχαι.

larly, and a similar number of times, as pyramid $DEFH$. Thus, as base ABC is to base DEF, so the (sum of the) prisms within pyramid $ABCG$ (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 12.4]. But, also, as base ABC (is) to base DEF, so pyramid $ABCG$ (is) to solid W. And, thus, as pyramid $ABCG$ (is) to solid W, so the (sum of the) prisms within pyramid $ABCG$ (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.11]. Thus, alternately, as pyramid $ABCG$ (is) to the (sum of the) prisms within it, so solid W (is) to the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.16]. And pyramid $ABCG$ (is) greater than the (sum of the) prisms within it. Thus, solid W (is) also greater than the (sum of the) prisms within pyramid $DEFH$ [Prop. 5.14]. But, (it is) also less. This very thing is impossible. Thus, as base ABC is to base DEF, so pyramid $ABCG$ (is) not to some solid less than pyramid $DEFH$. So, similarly, we can show that base DEF is not to base ABC, as pyramid $DEFH$ (is) to some solid less than pyramid $ABCG$ either.

So, I say that neither is base ABC to base DEF, as pyramid $ABCG$ (is) to some solid greater than pyramid $DEFH$.

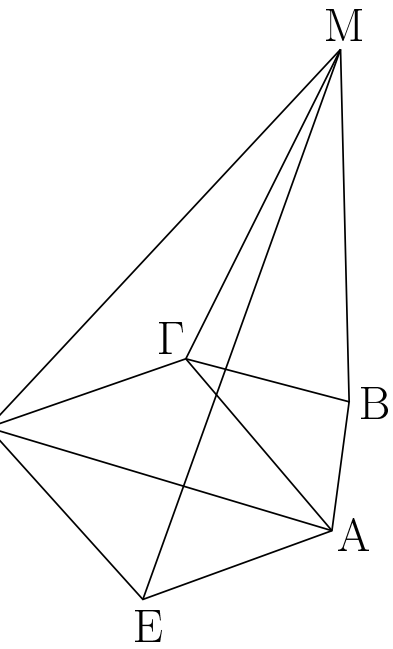
For, if possible, let it be (in this ratio) to some greater (solid), W. Thus, inversely, as base DEF (is) to base ABC, so solid W (is) to pyramid $ABCG$ [Prop. 5.7. corr.]. And as solid W (is) to pyramid $ABCG$, so pyramid $DEFH$ (is) to some (solid) less than pyramid $ABCG$, as shown before [Prop. 12.2 lem.]. And, thus, as base DEF (is) to base ABC, so pyramid $DEFH$ (is) to some (solid) less than pyramid $ABCG$ [Prop. 5.11]. The very thing was shown (to be) absurd. Thus, base ABC is not to base DEF, as pyramid $ABCG$ (is) to some solid greater than pyramid $DEFH$. And, it was shown that neither (is it in this ratio) to a lesser (solid). Thus, as base ABC is to base DEF, so pyramid $ABCG$ (is) to pyramid $DEFH$. (Which is) the very thing it was required to show.

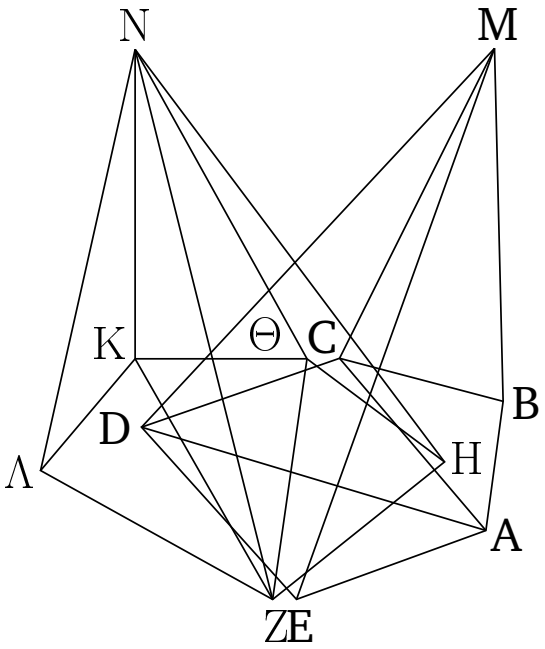
6.

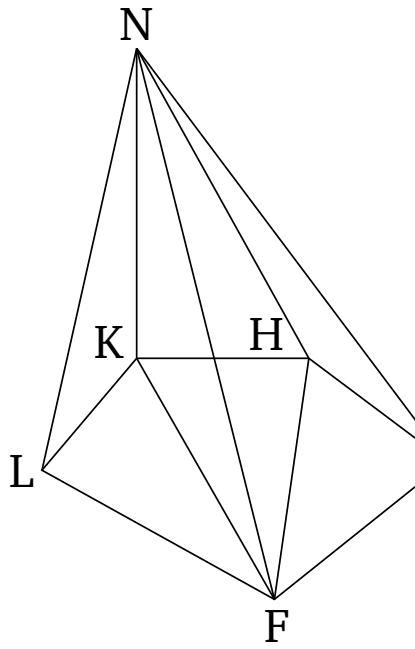
Αἱ ὑπὸ τὸ αὐτὸ ὕψος οὐσαι πυραμίδες καὶ πολυγώνουςὸέθουσαι βάσεις πρὸς ἀλλήλας εἰσὶν ὡς αἱ βάσεις.

Proposition 6

Pyramids which are of the same height, and have polygonal bases, are to one another as their bases.







Ἔστωσαν ὑπὸ τὸ αὐτὸ ὕψος πυραμίδες, ὧν [αἱ] βάσεις μὲν τὰ ΑΒΓΔΕ, ΖΗΘΚΛ πολύγωνα, κορυφαὶ δὲ τὰ Μ, Ν σημεῖα· λέγω, ὅτι ἐστὶν ὡς ἡ ΑΒΓΔΕ βάσις πρὸς τὴν ΖΗΘΚΛ βάσιν, οὕτως ἡ ΑΒΓΔΕΜ πυραμὶξ πρὸς τὴν ΖΗΘΚΛΝ πυραμίδα.

Ἐπεζεύθθωσαν γὰρ αἱ ΑΓ, ΑΔ, ΖΘ, ΖΚ. ἐπεὶ οὖν δύο πυραμίδες εἰσὶν αἱ ΑΒΓΜ, ΑΓΔΜ τριγώνους ὀέθουσαι βάσεις καὶ ὕψος ἴσον, πρὸς ἀλλήλας εἰσὶν ὡς αἱ βάσεις· ὀέστιν ἄρα ὡς ἡ ΑΒΓ βάσις πρὸς τὴν ΑΓΔ βάσιν, οὕτως ἡ ΑΒΓΜ πυραμὶξ πρὸς τὴν ΑΓΔΜ πυραμίδα. καὶ συνθέντι ὡς ἡ ΑΒΓΔ βάσις πρὸς τὴν ΑΓΔ βάσιν, οὕτως ἡ ΑΒΓΔΜ πυραμὶξ πρὸς τὴν ΑΓΔΜ πυραμίδα. ἀλλὰ καὶ ὡς ἡ ΑΓΔ βάσις πρὸς τὴν ΑΔΕ βάσιν, οὕτως ἡ ΑΓΔΜ πυραμὶξ πρὸς τὴν ΑΔΕΜ πυραμίδα. διὸ ὀίσου ἄρα ὡς ἡ ΑΒΓΔ βάσις πρὸς τὴν ΑΔΕ βάσιν, οὕτως ἡ ΑΒΓΔΜ πυραμὶξ πρὸς τὴν ΑΔΕΜ πυραμίδα. καὶ συνθέντι πάλιν, ὡς ἡ ΑΒΓΔΕ βάσις πρὸς τὴν ΑΔΕ βάσιν, οὕτως ἡ ΑΒΓΔΕΜ πυραμὶξ πρὸς τὴν ΑΔΕΜ πυραμίδα. ὁμοίως δὴ δειθθήσεται, ὅτι καὶ ὡς ἡ ΖΗΘΚΛ βάσις πρὸς τὴν ΖΗΘ βάσιν, οὕτως καὶ ἡ ΖΗΘΚΛΝ πυραμὶξ πρὸς τὴν ΖΗΘΝ πυραμίδα. καὶ ἐπεὶ δύο πυραμίδες εἰσὶν αἱ ΑΔΕΜ, ΖΗΘΝ τριγώνους ὀέθουσαι βάσεις καὶ ὕψος ἴσον, ὀέστιν ἄρα ὡς ἡ ΑΔΕ βάσις πρὸς τὴν ΖΗΘ βάσιν, οὕτως ἡ ΑΔΕΜ πυραμὶξ πρὸς τὴν ΖΗΘΝ πυραμίδα. ἀλλὰ ὡς ἡ ΑΔΕ βάσις πρὸς τὴν ΑΒΓΔΕ βάσιν, οὕτως ὂν ἡ ΑΔΕΜ πυραμὶξ πρὸς τὴν ΑΒΓΔΕΜ πυραμίδα. καὶ διὸ ὀίσου ἄρα ὡς ἡ ΑΒΓΔΕ βάσις πρὸς τὴν ΖΗΘ βάσιν, οὕτως ἡ ΑΒΓΔΕΜ πυραμὶξ πρὸς τὴν ΖΗΘΝ πυραμίδα. ἀλλὰ μὴν καὶ ὡς ἡ ΖΗΘ βάσις πρὸς τὴν ΖΗΘΚΛ βάσιν, οὕτως ὂν καὶ ἡ ΖΗΘΝ πυραμὶξ πρὸς τὴν ΖΗΘΚΛΝ πυραμίδα, καὶ διὸ ὀίσου ἄρα ὡς ἡ ΑΒΓΔΕ βάσις πρὸς τὴν ΖΗΘΚΛ βάσιν, οὕτως ἡ ΑΒΓΔΕΜ πυραμὶξ πρὸς τὴν ΖΗΘΚΛΝ πυραμίδα· ὅπερ

Let there be pyramids of the same height whose bases (are) the polygons *ABCDE* and *FGHKL*, and apexes the points *M* and *N* (respectively). I say that as base *ABCDE* is to base *FGHKL*, so pyramid *ABCDEM* (is) to pyramid *FGHKLN*.

For let *AC*, *AD*, *FH*, and *FK* be joined. Therefore, since *ABCM* and *ACDM* are two pyramids having triangular bases and equal height, they are to one another as their bases [Prop. 12.5]. Thus, as base *ABC* is to base *ACD*, so pyramid *ABCM* (is) to pyramid *ACDM*. And, via composition, as base *ABCD* (is) to base *ACD*, so pyramid *ABCDM* (is) to pyramid *ACDM* [Prop. 5.18]. But, as base *ACD* (is) to base *ADE*, so pyramid *ACDM* (is) also to pyramid *ADEM* [Prop. 12.5]. Thus, via equality, as base *ABCD* (is) to base *ADE*, so pyramid *ABCDM* (is) to pyramid *ADEM* [Prop. 5.22]. And, again, via composition, as base *ABCDE* (is) to base *ADE*, so pyramid *ABCDEM* (is) to pyramid *ADEM* [Prop. 5.18]. So, similarly, it can also be shown that as base *FGHKL* (is) to base *FGH*, so pyramid *FGHKLN* (is) also to pyramid *FGHN*. And since *ADEM* and *FGHN* are two pyramids having triangular bases and equal height, thus as base *ADE* (is) to base *FGH*, so pyramid *ADEM* (is) to pyramid *FGHN* [Prop. 12.5]. But, as base *ADE* (is) to base *ABCDE*, so pyramid *ADEM* (was) to pyramid *ABCDEM*. Thus, via equality, as base *ABCDE* (is) to base *FGH*, so pyramid *ABCDEM* (is) also to pyramid *FGHN* [Prop. 5.22]. But, furthermore, as base *FGH* (is) to base *FGHKL*, so pyramid *FGHN* was also to pyramid *FGHKLN*. Thus, via equality, as base *ABCDE* (is) to base *FGHKL*, so pyramid *ABCDEM* (is) also to pyramid *FGHKLN* [Prop. 5.22]. (Which is) the very thing it was required to show.

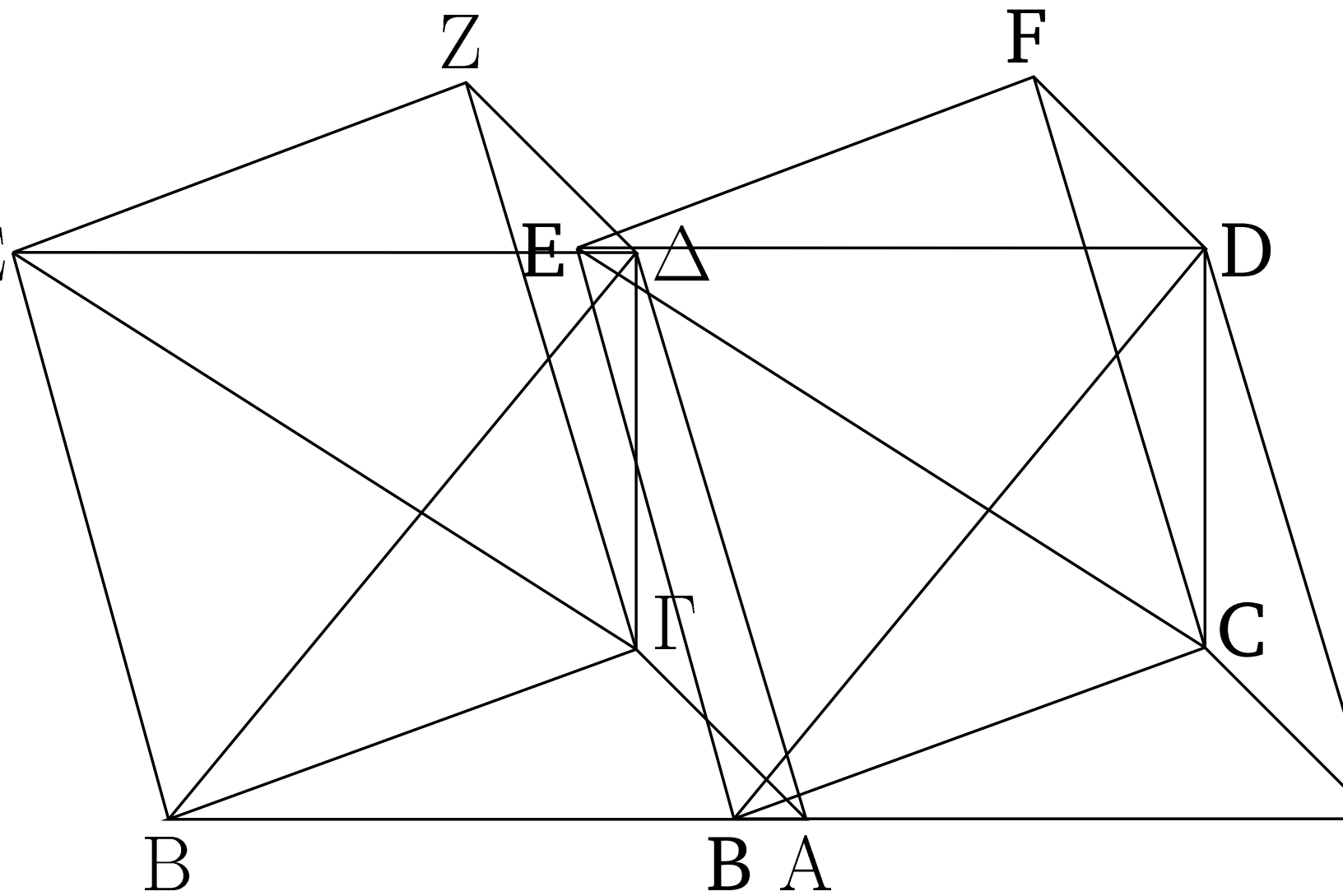
ἔδει δεῖχαι.

7.

Πᾶν πρίσμα τρίγωνον ἔχον βάσιν διαιρεῖται εἰξ τρεῖς πυραμίδας ἴσας ἀλλήλαις τριγώνους βάσεις ἐθούσας.

Proposition 7

Any prism having a triangular base is divided into three pyramids having triangular bases (which are) equal to one another.



Ἔστω πρίσμα, οὗ βάσις μὲν τὸ ΑΒΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΔΕΖ· λέγω, ὅτι τὸ ΑΒΓΔΕΖ πρίσμα διαιρεῖται εἰξ τρεῖς πυραμίδας ἴσας ἀλλήλαις τριγώνους ἐθούσας βάσεις.

Ἐπεζεύθθωσαν γὰρ αἱ ΒΔ, ΕΓ, ΓΔ. ἐπεὶ παραλληλόγραμμόν ἐστι τὸ ΑΒΕΔ, διάμετρος δὲ αὐτοῦ ἐστιν ἡ ΒΔ, ἴσον ἄρα ἐστι τὸ ΑΒΔ τρίγωνον τῷ ΕΒΔ τρίγωνῳ· καὶ ἡ πυραμὶξ ἄρα, ἧξ βάσις μὲν τὸ ΑΒΔ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ὁίση ἐστὶ πυραμίδι, ἧξ βάσις μέν ἐστι τὸ ΔΕΒ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον. ἀλλὰ ἡ πυραμίξ, ἧξ βάσις μέν ἐστι τὸ ΔΕΒ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ἡ αὐτή

Let there be a prism whose base (is) triangle ABC, and opposite (plane) DEF. I say that prism $ABCDEF$ is divided into three pyramids having triangular bases (which are) equal to one another.

For let BD, EC, and CD be joined. Since $ABED$ is a parallelogram, and BD is its diagonal, triangle ABD is thus equal to triangle EBD [Prop. 1.34]. And, thus, the pyramid whose base (is) triangle ABD, and apex the point C, is equal to the pyramid whose base is triangle DEB, and apex the point C [Prop. 12.5]. But, the pyramid whose base is triangle DEB,

ἐστι πυραμίδι, ἧξ βάσις μέν ἐστι τὸ ΕΒΓ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον· ὑπὸ γὰρ τῶν αὐτῶν ἐπιπέδων περιέθεται. καὶ πυραμὶξ ἄρα, ἧξ βάσις μέν ἐστι τὸ ΑΒΔ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ὁίση ἐστὶ πυραμίδι, ἧξ βάσις μέν ἐστι τὸ ΕΒΓ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον. πάλιν, ἐπεὶ παραλληλόγραμμόν ἐστι τὸ ΖΓΒΕ, διάμετρος δέ ἐστιν αὐτοῦ ἡ ΓΕ, ἴσον ἐστὶ τὸ ΓΕΖ τρίγωνον τῷ ΓΒΕ τριγώνῳ. καὶ πυραμὶξ ἄρα, ἧξ βάσις μέν ἐστι τὸ ΒΓΕ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, ὁίση ἐστὶ πυραμίδι, ἧξ βάσις μέν ἐστι τὸ ΕΓΖ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον. ἡ δὲ πυραμίξ, ἧξ βάσις μέν ἐστι τὸ ΒΓΕ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, ὁίση ἐδείχθη πυραμίδι, ἧξ βάσις μέν ἐστι τὸ ΑΒΔ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον· καὶ πυραμὶξ ἄρα, ἧξ βάσις μέν ἐστι τὸ ΓΕΖ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, ὁίση ἐστὶ πυραμίδι, ἧξ βάσις μέν [ἐστι] τὸ ΑΒΔ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον· διῄρηται ἄρα τὸ ΑΒΓΔΕΖ πρίσμα εἰξ τρεῖς πυραμίδας ἴσας ἀλλήλαις τριγώνους ἐθούσας βάσεις.

Καὶ ἐπεὶ πυραμίξ, ἧξ βάσις μέν ἐστι τὸ ΑΒΔ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, ἡ αὐτή ἐστι πυραμίδι, ἧξ βάσις τὸ ΓΑΒ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον· ὑπὸ γὰρ τῶν αὐτῶν ἐπιπέδων περιέθονται· ἡ δὲ πυραμίξ, ἧξ βάσις τὸ ΑΒΔ τρίγωνον, κορυφὴ δὲ τὸ Γ σημεῖον, τρίτον ἐδείθθη τοῦ πρίσματος, οὗ βάσις τὸ ΑΒΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΔΕΖ, καὶ ἡ πυραμὶξ ἄρα, ἧξ βάσις τὸ ΑΒΓ τρίγωνον, κορυφὴ δὲ τὸ Δ σημεῖον, τρίτον ἐστὶ τοῦ πρίσματος τοῦ ἔχοντος βάσις τὴν αὐτὴν τὸ ΑΒΓ τρίγωνον, ἀπεναντίον δὲ τὸ ΔΕΖ.

and apex the point C, is the same as the pyramid whose base is triangle EBC, and apex the point D. For they are contained by the same planes. And, thus, the pyramid whose base is ABD, and apex the point C, is equal to the pyramid whose base is EBC and apex the point D. Again, since $FCBE$ is a parallelogram, and CE is its diagonal, triangle CEF is equal to triangle CBE [Prop. 1.34]. And, thus, the pyramid whose base is triangle BCE, and apex the point D, is equal to the pyramid whose base is triangle ECF, and apex the point D [Prop. 12.5]. And the pyramid whose base is triangle BCE, and apex the point D, was shown (to be) equal to the pyramid whose base is triangle ABD, and apex the point C. Thus, the pyramid whose base is triangle CEF, and apex the point D, is also equal to the pyramid whose base [is] triangle ABD, and apex the point C. Thus, the prism $ABCDEF$ has been divided into three pyramids having triangular bases (which are) equal to one another.

And since the pyramid whose base is triangle ABD, and apex the point C, is the same as the pyramid whose base is triangle CAB, and apex the point D. For they are contained by the same planes. And the pyramid whose base (is) triangle ABD, and apex the point C, was shown (to be) a third of the prism whose base is triangle ABC, and opposite (plane) DEF, thus the pyramid whose base is triangle ABC, and apex the point D, is also a third of the pyramid having the same base, triangle ABC, and opposite (plane) DEF.

Πόρισμα.

Ἔκ δὴ τούτου φανερόν, ὅτι πᾶσα πυραμὶξ τρίτον μέρος ἐστὶ τοῦ πρίσματος τοῦ τὴν αὐτὴν βάσιν ἔχοντος αὐτῇ καὶ ὕψος ἴσον· ὅπερ ἔδει δεῖξαι.

Corollary

And, from this, (it is) clear that any pyramid is the third part of the prism having the same base as it, and an equal height. (Which is) the very thing it was required to show.

8.

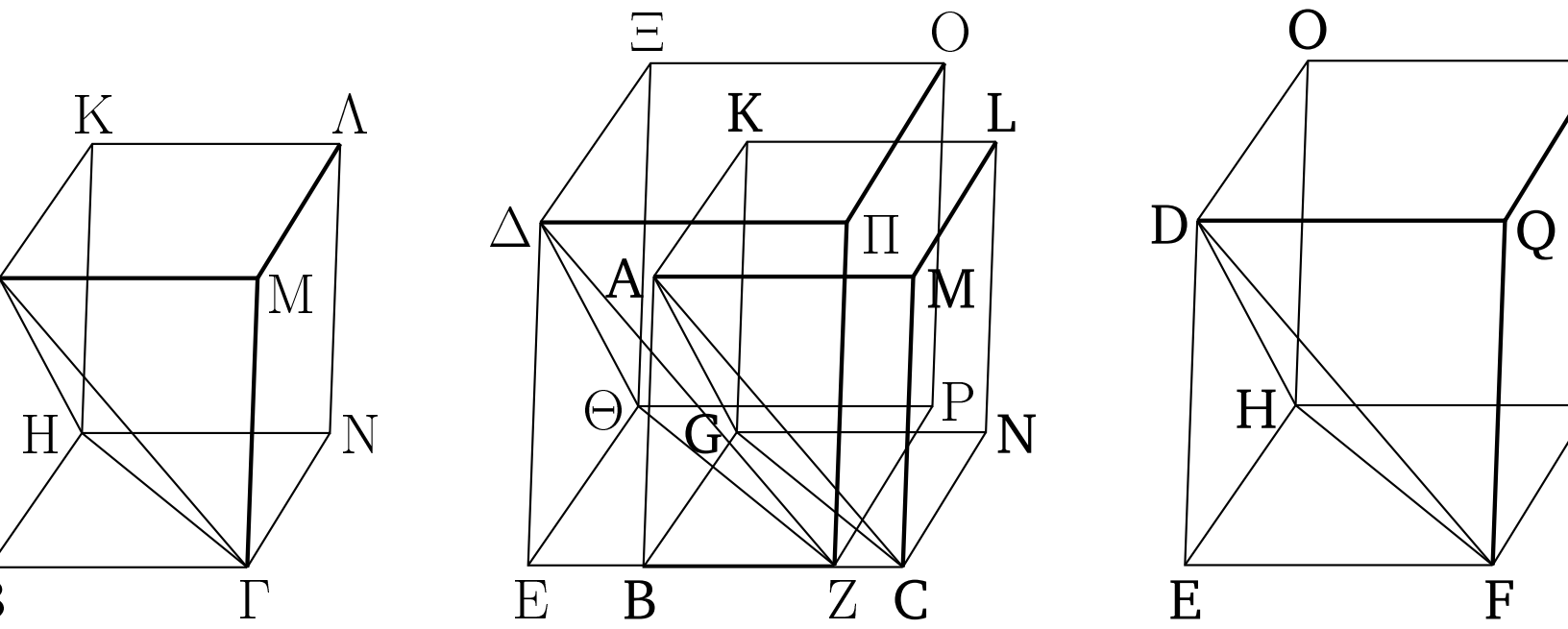
Αἱ ὅμοιαι πυραμίδες καὶ τριγώνουςὀέθουσαι βάσεις ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν.

Ἔστωσαν ὅμοιαι καὶ ὁμοίως κείμεναι πυραμίδες, ὧν βάσεις μέν εἰσι τὰ ΑΒΓ, ΔΕΖ τρίγωνα, κορυφαὶ δὲ τὰ Η, Θ σημεῖα· λέγω, ὅτι ἡ ΑΒΓΗ πυραμὶξ πρὸς τὴν ΔΕΖΘ πυραμίδα τριπλασίονα λόγον ἔχει ἤπερ ἡ ΒΓ πρὸς τὴν ΕΖ.

Proposition 8

Similar pyramids which also have triangular bases are in the cubed ratio of their corresponding sides.

Let there be similar, and similarly laid out, pyramids whose bases are triangles ABC and DEF, and apexes the points G and H (respectively). I say that pyramid $ABCG$ has to pyramid $DEFH$ the cubed ratio of that BC (has) to EF.



Συμπεπληρώσθω γὰρ τὰ ΒΗΜΛ, ΕΘΠΟ στερεὰ παραλληλεπίπεδα. καὶ ἐπεὶ ὅμοιά ἐστὶν ἡ ΑΒΓΗ πυραμὶς τῇ ΔΕΖΘ πυραμίδι, ὅίση ἄρα ἐστὶν ἡ μὲν ὑπὸ ΑΒΓ γωνία τῇ ὑπὸ ΔΕΖ γωνίᾳ, ἡ δὲ ὑπὸ ΗΒΓ τῇ ὑπὸ ΘΕΖ, ἡ δὲ ὑπὸ ΑΒΗ τῇ ὑπὸ ΔΕΘ, καί ἐστιν ὡξ ἡ ΑΒ πρὸς τὴν ΔΕ, οὕτως ἡ ΒΓ πρὸς τὴν ΕΖ, καὶ ἡ ΒΗ πρὸς τὴν ΕΘ. καὶ ἐπεί ἐστιν ὡξ ἡ ΑΒ πρὸς τὴν ΔΕ, οὕτως ἡ ΒΓ πρὸς τὴν ΕΖ, καὶ περὶ ἴσας γωνίας αἱ πλευραὶ ἀνάλογόν εἰσιν, ὅμοιον ἄρα ἐστὶ τὸ ΒΜ παραλληλόγραμμον τῷ ΕΠ παραλληλογράμμῳ. διὰ τὰ αὐτὰ δὴ καὶ τὸ μὲν ΒΝ τῷ ΕΡ ὅμοιόν ἐστι, τὸ δὲ ΒΚ τῷ ΕΧ· τὰ τρία ἄρα τὰ ΜΒ, ΒΚ, ΒΝ τρισὶ τοῖς ΕΠ, ΕΧ, ΕΡ ὅμοιά ἐστιν. ἀλλὰ τὰ μὲν τρία τὰ ΜΒ, ΒΚ, ΒΝ τρισὶ τοῖς ἀπεναντίον ἴσα τε καὶ ὅμοιά ἐστιν, τὰ δὲ τρία τὰ ΕΠ, ΕΧ, ΕΡ τρισὶ τοῖς ἀπεναντίον ἴσα τε καὶ ὅμοιά ἐστιν. τὰ ΒΗΜΛ, ΕΘΠΟ ἄρα στερεὰ ὑπὸ ὁμοίων ἐπιπέδων ὑίσων τὸ πλῆθος περιέθεται. ὅμοιον ἄρα ἐστὶ τὸ ΒΗΜΛ στερεὸν τῷ ΕΘΠΟ στερεῷ. τὰ δὲ ὅμοια στερεὰ παραλληλεπίπεδα ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγων πλευρῶν. τὸ ΒΗΜΛ ἄρα στερεὸν πρὸς τὸ ΕΘΠΟ στερεὸν τριπλασίονα λόγον ἔχει ἤπερ ἡ ὁμόλογος πλευρὰ ἡ ΒΓ πρὸς τὴν ὁμόλογον πλευρὰν τὴν ΕΖ. ὡξ δὲ τὸ ΒΗΜΛ στερεὸν πρὸς τὸ ΕΘΠΟ στερεόν, οὕτως ἡ ΑΒΓΗ πυραμὶξ πρὸς τὴν ΔΕΖΘ πυραμίδα, ἐπειδήπερ ἡ πυραμὶξ ὲκτον μέρος ἐστὶ τοῦ στερεοῦ διὰ τὸ καὶ τὸ πρίσμα ἥμισυ ὂν τοῦ στερεοῦ παραλληλεπιπέδου τριπλάσιον εὸῖναι τῆς πυραμίδος. καὶ ἡ ΑΒΓΗ ἄρα πυραμὶξ πρὸς τὴν ΔΕΖΘ πυραμίδα τριπλασίονα λόγον ἔχει ἤπερ ἡ ΒΓ πρὸς τὴν ΕΖ· ὅπερ ἔδει δεῖχαι.

For let the parallelepiped solids $BGML$ and $EHQP$ be completed. And since pyramid $ABCG$ is similar to pyramid $DEFH$, angle ABC is thus equal to angle DEF, and GBC to HEF, and ABG to DEH. And as AB is to DE, so BC (is) to EF, and BG to EH [Def. 11.9]. And since as AB is to DE, so BC (is) to EF, and (so) the sides around equal angles are proportional, parallelogram BM is thus similar to parallelogram EQ. So, for the same (reasons), BN is also similar to ER, and BK to EO. Thus, the three (parallelograms) MB, BK, and BN are similar to the three (parallelograms) EQ, EO, ER (respectively). But, the three (parallelograms) MB, BK, and BN are (both) equal and similar to the three opposite (parallelograms), and the three (parallelograms) EQ, EO, and ER are (both) equal and similar to the three opposite (parallelograms) [Prop. 11.24]. Thus, the solids $BGML$ and $EHQP$ are contained by equal numbers of similar (and similarly laid out) planes. Thus, solid $BGML$ is similar to solid $EHQP$ [Def. 11.9]. And similar parallelepiped solids are in the cubed ratio of corresponding sides [Prop. 11.33]. Thus, solid $BGML$ has to solid $EHQP$ the cubed ratio that the corresponding side BC (has) to the corresponding side EF. And as solid $BGML$ (is) to solid $EHQP$, so pyramid $ABCG$ (is) to pyramid $DEFH$, inasmuch as the pyramid is the sixth part of the solid, on account of the prism, being half of the parallelepiped solid [Prop. 11.28], also being three times the pyramid [Prop. 12.7]. Thus, pyramid $ABCG$ also has to pyramid $DEFH$ the cubed ratio that BC (has) to EF. (Which is) the very thing it was required to show.

Πόρισμα.

Corollary

Ἐκ δὴ τούτου φανερόν, ὅτι καὶ αἱ πολυγώνουςὸέθουσαι βάσεις ὅμοιαι πυραμίδες πρὸς ἀλλήλας ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν. διαιρεθεισῶν γὰρ αὐτῶν εἰξ τὰξ ἐν αὐταῖς πυραμίδας τριγώνους βάσεις ἐθούσας τῷ καὶ τὰ ὅμοια πολύγωνα τῶν βάσεων εἰξ ὅμοια τρίγωνα διαιρεῖσθαι καὶ ἴσα τῷ πλήθει καὶ ὁμόλογα τοῖς ὅλοιςὸέσται ὡξ [ἡ] ἐν τῇ ἑτέρᾳ μία πυραμὶξ τρίγωνονὸέθουσα βάσιν πρὸς τὴν ἐν τῇ ἑτέρᾳ μίαν πυραμίδα τρίγωνονὸέθουσαν βάσιν, οὕτως καὶ ὁάπασαι αἱ ἐν τῇ ἑτέρᾳ πυραμίδι πυραμίδες τριγώνουςὸέθουσαι βάσεις πρὸς τὰξ ἐν τῇ ἑτέρᾳ πυραμίδι πυραμίδας τριγώνους βάσεις ἐθούσας, τουτέστιν αὐτὴ ἡ πολύγωνον βάσινὸέθουσα πυραμὶξ πρὸς τὴν πολύγωνον βάσινὸέθουσαν πυραμίδα. ἡ δὲ τρίγωνον βάσινὸέθουσα πυραμὶξ πρὸς τὴν τρίγωνον βάσινὸέθουσαν ἐν τριπλασίονι λόγῳ ἐστὶ τῶν ὁμολόγον πλευρῶν· καὶ ἡ πολύγωνον ἄρα βάσινὸέθουσα πρὸς τὴν ὁμοίαν βάσινὸέθουσαν τριπλασίονα λόγον ἔχει ἥπερ ἡ πλευρὰ πρὸς τὴν πλευράν.

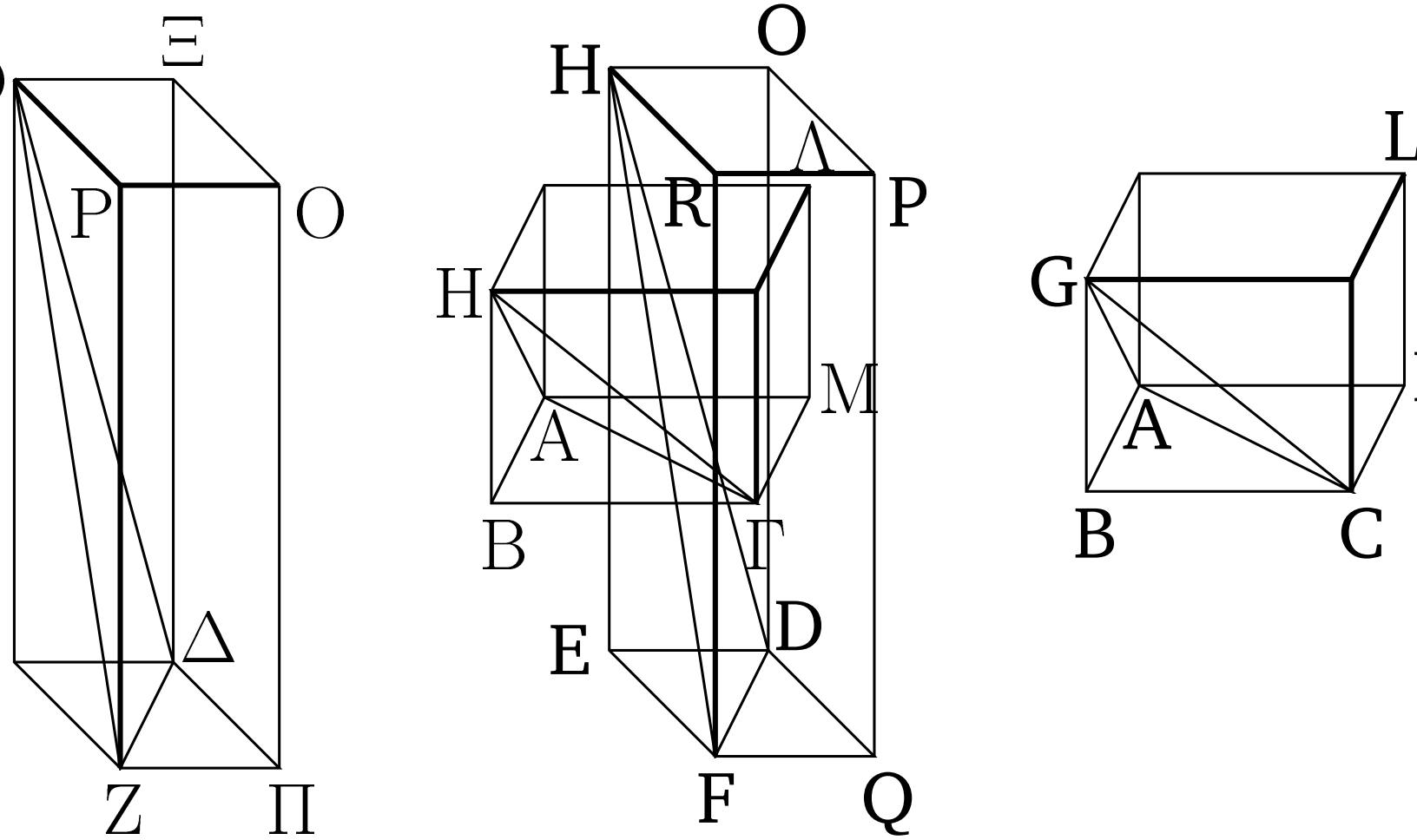
So, from this, (it is) also clear that similar pyramids having polygonal bases (are) to one another as the cubed ratio of their corresponding sides. For, dividing them into the pyramids (contained) within them which have triangular bases, with the similar polygons of the bases also being divided into similar triangles (which are) both equal in number, and corresponding, to the wholes [Prop. 6.20]. As one pyramid having a triangular base in the former (pyramid having a polygonal base is) to one pyramid having a triangular base in the latter (pyramid having a polygonal base), so (the sum of) all the pyramids having triangular bases in the former pyramid will also be to (the sum of) all the pyramids having triangular bases in the latter pyramid [Prop. 5.12]—that is to say, the (former) pyramid itself having a polygonal base to the (latter) pyramid having a polygonal base. And a pyramid having a triangular base is to a (pyramid) having a triangular base in the cubed ratio of corresponding sides [Prop. 12.8]. Thus, a (pyramid) having a polygonal base also has to to a (pyramid) having a similar base the cubed ratio of a (corresponding) side to a (corresponding) side.

9.

Τῶνὁίσων πυραμίδων καὶ τριγώνους βάσεις ἐθουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν· καὶ ὧν πυραμίδων τριγώνους βάσεις ἐθουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν, ἴσαι εἰσὶν ἐκεῖναι.

Proposition 9

The bases of equal pyramids which also have triangular bases are reciprocally proportional to their heights. And those pyramids which have triangular bases whose bases are reciprocally proportional to their heights are equal.



Ἔστωσαν γὰρ ἴσαι πυραμίδες τριγώνους βάσεις ὑπὸ ἐθουσαι τὰξ ΑΒΓ, ΔΕΖ, κορυφὰξ δὲ τὰ Η, Θ σημεῖα· λέγω, ὅτι τῶν ΑΒΓΗ, ΔΕΖΘ πυραμίδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν, καί ἐστιν ὡξ ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὑτως τὸ τῆς ΔΕΖΘ πυραμίδος ὑψος πρὸς τὸ τῆς ΑΒΓΗ πυραμίδος ὑψος.

Συμπεπληρώσθω γὰρ τὰ ΒΗΜΛ, ΕΘΠΟ στερεὰ παραλληλεπίπεδα. καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΑΒΓΗ πυραμὶξ τῇ ΔΕΖΘ πυραμίδι, καί ἐστι τῆς μὲν ΑΒΓΗ πυραμίδος ἑξαπλάσιον τὸ ΒΗΜΛ στερεόν, τῆς δὲ ΔΕΖΘ πυραμίδος ἑξαπλάσιον τὸ ΕΘΠΟ στερεόν, ἴσον ἄρα ἐστὶ τὸ ΒΗΜΛ στερεὸν τῷ ΕΘΠΟ στερεῷ. τῶν δὲ ὁίσων στερεῶν παραλληλεπιπώδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν· ὲστιν ἄρα ὡξ ἡ ΒΜ βάσις πρὸς τὴν ΕΠ βάσιν, οὑτως τὸ τοῦ ΕΘΠΟ στερεοῦ ὑψος πρὸς τὸ τοῦ ΒΗΜΛ στερεοῦ ὑψος. ἀλλὸ ὡξ ἡ ΒΜ βάσις πρὸς τὴν ΕΠ, οὑτως τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΔΕΖ τρίγωνον. καὶ ὡξ ἄρα τὸ ΑΒΓ τρίγωνον πρὸς τὸ ΔΕΖ τρίγωνον, οὑτως τὸ τοῦ ΕΘΠΟ στερεοῦ ὑψος πρὸς τὸ τοῦ ΒΗΜΛ στερεοῦ ὑψος. ἀλλὰ τὸ μὲν τοῦ ΕΘΠΟ στερεοῦ ὑψος τὸ αὐτὸ ἐστι τῷ τῆς ΔΕΖΘ πυραμίδος ὑψει, τὸ δὲ τοῦ ΒΗΜΛ

For let there be (two) equal pyramids having the triangular bases *ABC* and *DEF*, and apexes the points *G* and *H* (respectively). I say that the bases of the pyramids *ABCG* and *DEFH* are reciprocally proportional to their heights, and (so) that as base *ABC* is to base *DEF*, so the height of pyramid *DEFH* (is) to the height of pyramid *ABCG*.

For let the parallelepiped solids *BGML* and *EHQP* be completed. And since pyramid *ABCG* is equal to pyramid *DEFH*, and solid *BGML* is six times pyramid *ABCG* (see previous proposition), and solid *EHQP* (is) six times pyramid *DEFH*, solid *BGML* is thus equal to solid *EHQP*. And the bases of equal parallelepiped solids are reciprocally proportional to their heights [Prop. 11.34]. Thus, as base *BM* is to base *EQ*, so the height of solid *EHQP* (is) to the height of solid *BGML*. But, as base *BM* (is) to base *EQ*, so triangle *ABC* (is) to triangle *DEF* [Prop. 1.34]. And, thus, as triangle *ABC* (is) to triangle *DEF*, so the height of solid *EHQP* (is) to the height of solid *BGML* [Prop. 5.11]. But, the height of solid *EHQP* is the same as the height of pyramid *DEFH*, and the height of solid *BGML* is the same as the height of pyramid

στερεοῦ ὕψος τὸ αὐτό ἐστι τῷ τῆς ΑΒΓΗ πυραμίδος ὕψει· ὀέστιν ἄρα ὡξ ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὸ τῆς ΔΕΖΘ πυραμίδος ὕψος πρὸς τὸ τῆς ΑΒΓΗ πυραμίδος ὕψος. τῶν ΑΒΓΗ, ΔΕΖΘ ἄρα πυραμίδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν.

Ἀλλὰ δὴ τῶν ΑΒΓΗ, ΔΕΖΘ πυραμίδων ἀντιπεπονθέτωσαν αἱ βάσεις τοῖς ὕψεσιν, καὶ ὀέστω ὡξ ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὸ τῆς ΔΕΖΘ πυραμίδος ὕψος πρὸς τὸ τῆς ΑΒΓΗ πυραμίδος ὕψος· λέγω, ὅτιὀίση ἐστὶν ἡ ΑΒΓΗ πυραμὶξ τῇ ΔΕΖΘ πυραμίδι.

Τῶν γὰρ αὐτῶν κατασκευασθέντων, ἐπεί ἐστιν ὡξ ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὸ τῆς ΔΕΖΘ πυραμίδος ὕψος πρὸς τὸ τῆς ΑΒΓΗ πυραμίδος ὕψος, ἀλλὰ ὡξ ἡ ΑΒΓ βάσις πρὸς τὴν ΔΕΖ βάσιν, οὕτως τὸ ΒΜ παραλληλόγραμμον πρὸς τὸ ΕΠ παραλληλόγραμμον, καὶ ὡξ ἄρα τὸ ΒΜ παραλληλόγραμμον πρὸς τὸ ΕΠ παραλληλόγραμμον, οὕτως τὸ τῆς ΔΕΖΘ πυραμίδος ὕψος πρὸς τὸ τῆς ΑΒΓΗ πυραμίδος ὕψος. ἀλλὰ τὸ [μὲν] τῆς ΔΕΖΘ πυραμίδος ὕψος τὸ αὐτό ἐστι τῷ τοῦ ΕΘΠΟ παραλληλεπιπέδου ὕψει, τὸ δὲ τῆς ΑΒΓΗ πυραμίδος ὕψος τὸ αὐτό ἐστι τῷ τοῦ ΒΗΜΛ παραλληλεπιπέδου ὕψει· ὀέστιν ἄρα ὡξ ἡ ΒΜ βάσις πρὸς τὴν ΕΠ βάσιν, οὕτως τὸ τοῦ ΕΘΠΟ παραλληλεπιπέδου ὕψος πρὸς τὸ τοῦ ΒΗΜΛ παραλληλεπιπέδου ὕψος. ὧν δὲ στερεῶν παραλληλεπιπέδων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, ἴσα ἐστὶν ἐκεῖνα· ἴσον ἄρα ἐστὶ τὸ ΒΗΜΛ στερεὸν παραλληλεπίπεδον τῷ ΕΘΠΟ στερεῷ παραλληλεπιπέδῳ. καί ἐστι τοῦ μὲν ΒΗΜΛ ἕκτον μέρος ἡ ΑΒΓΗ πυραμίξ, τοῦ δὲ ΕΘΠΟ παραλληλεπιπέδου ἕκτον μέρος ἡ ΔΕΖΘ πυραμίξ· ὀίση ἄρα ἡ ΑΒΓΗ πυραμὶξ τῇ ΔΕΖΘ πυραμίδι.

Τῶν ἄρα ἰσῶν πυραμίδων καὶ τριγώνους βάσεις ἐθουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν· καὶ ὧν πυραμίδων τριγώνους βάσεις ἐθουσῶν ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὕψεσιν, ἴσαι εἰσὶν ἐκεῖναι· ὅπερ ἔδει δεῖξαι.

10.

Πᾶξ κῶνος κυλίνδρου τρίτον μέρος ἐστὶ τοῦ τὴν αὐτὴν βάσιν ἔχοντος αὐτῷ καὶ ὕψος ἴσον.

Ἐθέτω γὰρ κῶνος κυλίνδρῳ βάσιν τε τὴν αὐτὴν τὸν ΑΒΓΔ κύκλον καὶ ὕψος ἴσον· λέγω, ὅτι ὁ κῶνος τοῦ κυλίνδρου τρίτον ἐστὶ μέρος, τουτέστιν ὅτι ὁ κύλινδρος τοῦ κώνου τριπλασίων ἐστίν.

ABCG. Thus, as base *ABC* is to base *DEF*, so the height of pyramid *DEFH* (is) to the height of pyramid *ABCG*. Thus, the bases of pyramids *ABCG* and *DEFH* are reciprocally proportional to their heights.

And so, let the bases of pyramids *ABCG* and *DEFH* be reciprocally proportional to their heights, and (thus) let base *ABC* be to base *DEF*, as the height of pyramid *DEFH* (is) to the height of pyramid *ABCG*. I say that pyramid *ABCG* is equal to pyramid *DEFH*.

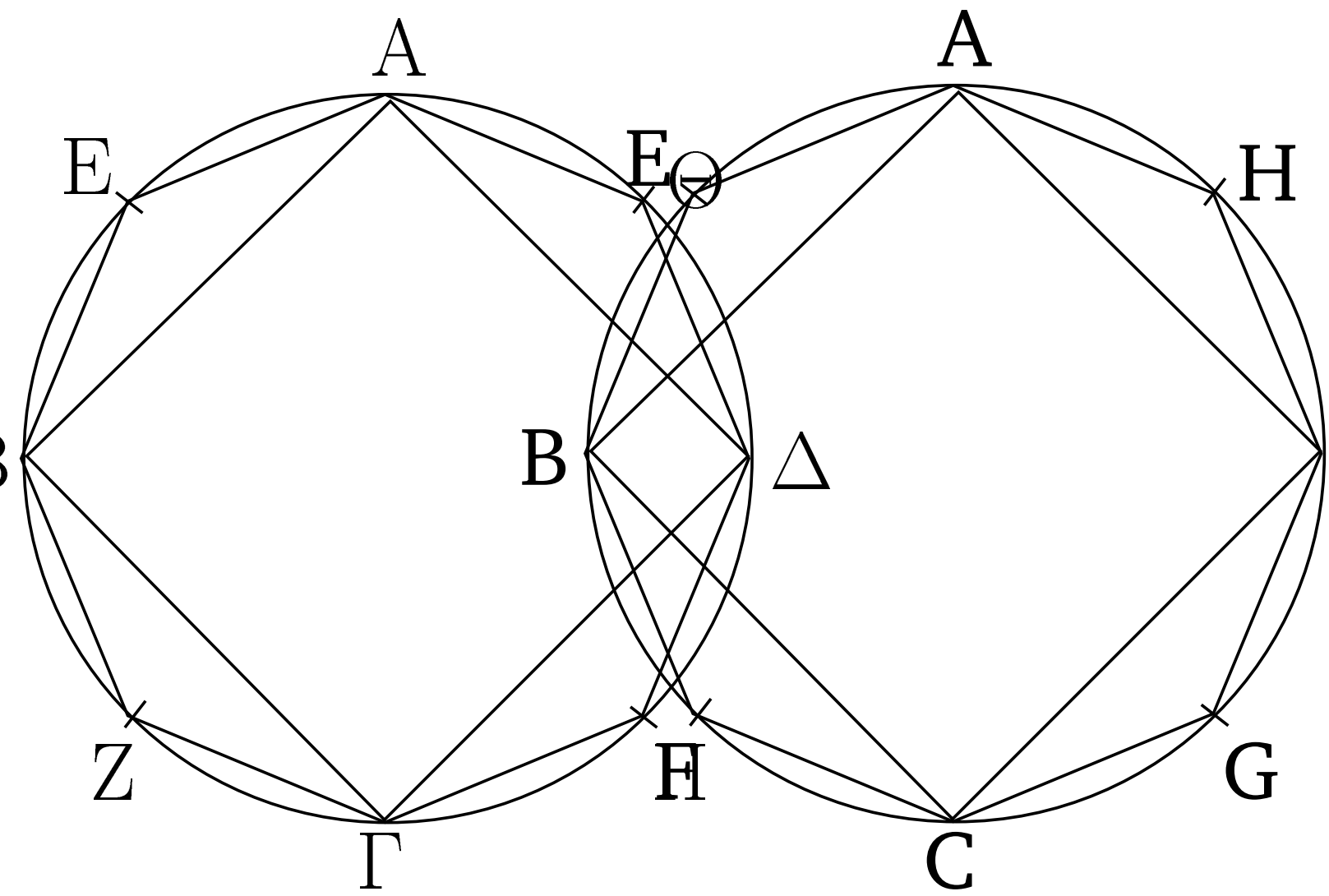
For, with the same construction, since as base *ABC* is to base *DEF*, so the height of pyramid *DEFH* (is) to the height of pyramid *ABCG*, but as base *ABC* (is) to base *DEF*, so parallelogram *BM* (is) to parallelogram *EQ* [Prop. 1.34], thus as parallelogram *BM* (is) to parallelogram *EQ*, so the height of pyramid *DEFH* (is) also to the height of pyramid *ABCG* [Prop. 5.11]. But, the height of pyramid *DEFH* is the same as the height of parallelepiped *EHQP*, and the height of pyramid *ABCG* is the same as the height of parallelepiped *BGML*. Thus, as base *BM* is to base *EQ*, so the height of parallelepiped *EHQP* (is) to the height of parallelepiped *BGML*. And those parallelepiped solids whose bases are reciprocally proportional to their heights are equal [Prop. 11.34]. Thus, the parallelepiped solid *BGML* is equal to the parallelepiped solid *EHQP*. And pyramid *ABCG* is a sixth part of *BGML*, and pyramid *DEFH* a sixth part of parallelepiped *EHQP*. Thus, pyramid *ABCG* is equal to pyramid *DEFH*.

Thus, the bases of equal pyramids which also have triangular bases are reciprocally proportional to their heights. And those pyramids having triangular bases whose bases are reciprocally proportional to their heights are equal. (Which is) the very thing it was required to show.

Proposition 10

Every cone is the third part of the cylinder which has the same base as it, and an equal height.

For let there be a cone (with) the same base as a cylinder, (namely) the circle *ABCD*, and an equal height. I say that the cone is the third part of the cylinder—that is to say, that the cylinder is three times the cone.



Εἰ γὰρ μή ἐστιν ὁ κύλινδρος τοῦ κώνου τριπλασίων, ὀέσται ὁ κύλινδρος τοῦ κώνου ἤτοι μείζωνόἠ τριπλασιωνόἠ ἐλάσσωνόἠ τριπλασίων. ὀέστω πρότερον μείζωνόἠ τριπλασίων, καὶ ἐγγεγράφθω εἰξ τὸν ΑΒΓΔ κύκλον τετράγωνον τὸ ΑΒΓΔ· τὸ δὴ ΑΒΓΔ τετράγωνον μείζόν ἐστινόἠ τὸ ἥμισυ τοῦ ΑΒΓΔ κύκλου. καὶ ἀνεστάτω ἀπὸ τοῦ ΑΒΓΔ τετραγώνου πρίσμα ἰσοϋψὲξ τῷ κυλίνδρῳ. τὸ δὴ ἀνιστάμενον πρίσμα μεῖζόν ἐστινόἠ τὸ ἥμισυ τοῦ κυλίνδου, ἐπειδήπερ κὸἀν περὶ τὸν ΑΒΓΔ κύκλον τετράγωνον περιγράψωμεν, τὸ ἐγγεγραμμένον εἰξ τὸν ΑΒΓΔ κύκλον τετράγωνον ἥμισύ ἐστι τοῦ περιγεγραμμένου· καί ἐστι τὰ ἀπὸ αὐτῶν ἀνιστάμενα στερεὰ παραλληλεπίπεδα πρίσματα ἰσοϋψῆ· τὰ δὲ ὑπὸ τὸ αὐτὸ ὁύψος ὄντα στερεὰ παραλληλεπίπεδα πρὸςὀάλληλά ἐστιν ὡξ αἱ βάσεις· καὶ τὸ ἐπὶ τοῦ ΑΒΓΔ ἄρα τετραγώνου ἀνασταθὲν πρίσμα ἥμισύ ἐστι τοῦ ἀνασταθέντος πρίσματος ἀπὸ τοῦ περὶ τὸν ΑΒΓΔ κύκλον περιγραφέντος τετραγώνου· καί ἐστιν ὁ κύλινδρος ἐλάττων τοῦ πρίσματος τοῦ ἀνατραθέντος ἀπὸ τοῦ περὶ τὸν ΑΒΓΔ κύκλον

For if the cylinder is not three times the cone then the cylinder will be either more than three times, or less than three times, (the cone). Let it, first of all, be more than three times (the cone). And let the square *ABCD* be inscribed in circle *ABCD* [Prop. 4.6]. So, square *ABCD* is more than half of circle *ABCD* [Prop. 12.2]. And let a prism of equal height to the cylinder be set up on square *ABCD*. So, the prism set up is more than half of the cylinder, inasmuch as if we also circumscribe a square around circle *ABCD* [Prop. 4.7] then the square inscribed in circle *ABCD* is half of the circumscribed (square). And the solids set up on them are parallelepiped prisms of equal height. And parallelepiped solids having the same height are to one another as their bases [Prop. 11.32]. And, thus, the prism set up on square *ABCD* is half of the prism set up on the square circumscribed about circle *ABCD*. And the cylinder is less than the prism set up on the square circumscribed about circle *ABCD*. Thus, the prism set up on

περιγραφέντος τετραγώνου· τὸ ἄρα πρίσμα τὸ ἀνασταθὲν ἀπὸ τοῦ ΑΒΓΔ τετραγώνου ἰσοϋψὲξ τῷ κυλίνδρῳ μεῖζόν ἐστι τοῦ ἡμίσεως τοῦ κυλίνδρου. τετμήσθωσαν αἱ ΑΒ, ΒΓ, ΓΔ, ΔΑ περιφέρειαι δίθα κατὰ τὰ Ε, Ζ, Η, Θ σημεῖα, καὶ ἐπεζεύχθωσαν αἱ ΑΕ, ΕΒ, ΒΖ, ΖΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ· καὶ ἑκάστον ἄρα τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων μειζόν ἐστινὸὴ τὸ ἥμισυ τοῦ καθὸ ἑαυτὸ τηήματος τοῦ ΑΒΓΔ κύκλου, ὡξὸέμπροσθεν ἐδείκνυμεν. ἀνεστάτω ἐφ᾽ ἑκάστου τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων πρίσματα ἰσοϋψῆ τῷ κυλίνδρῳ· καὶ ἑκάστον ἄρα τῶν ἀνασταθέντων πρισμάτων μεῖζόν ἐστινὸὴ τὸ ἥμισυ μέρος τοῦ καθὸ ἑαυτὸ τμήματος τοῦ κυλίνδρου, ἐπειδήπερ ἔαν διὰ τῶν Ε, Ζ, Η, Θ σημείων παραλλήλους ταῖς ΑΒ, ΒΓ, ΓΔ, ΔΑ ἀγάγωμεν, καὶ συμπληρώσωμεν τὰ ἐπὶ τῶν ΑΒ, ΒΓ, ΓΔ, ΔΑ παραλληλόγραμμα, καὶ ἀπὸ αὐτῶν ἀναστήσωμεν στερεὰ παραλληλεπίπεδα ἰσοϋψῆ τῷ κυλίνδρῳ, ἑκάσου τῶν ἀνασταθέντων ἡμίση ἐστὶ τὰ πρίσματα τὰ ἐπὶ τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων· καί ἐστι τὰ τοῦ κυλίνδρου τμήματα ἐλάττονα τῶν ἀνασταθέντων στερεῶν παραλληλεπιπέδων· ὁώστε καὶ τὰ ἐπὶ τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων πρίσματα μείζονά ἐστινὸὴ τὸ ἥμισυ τῶν καθὸ ἑαυτὰ τοῦ κυλίνδρου τμ ημάτων. τέμνοντες δὴ τὰξ ὑπολειπομένας περιφερείας δίθα καὶ ἐπιζευγνύντες εὐθείας καὶ ἀνιστάντες ἐφ᾽ ἑκάσου τῶν τριγώνων πρίσματα ἰσοϋψῆ τῷ κυλίνδρῳ καὶ τοῦτο ἀεὶ ποιοῦντες καταλείψομέν τινα ἀποτμήματα τοῦ κυλίνδρου, ὰ ὀέσται ἐλάττονα τῆς ὑπεροθῆς, ᾗ ὑπερέθει ὁ κυλίνδρος τοῦ τριπλασίου τοῦ κώνου. λελείφθω, καὶ ὀέστω τὰ ΑΕ, ΕΒ, ΒΖ, ΖΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ· λοιπὸν ἄρα τὸ πρίσμα, οὗ βάσις μὲν τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ, μεῖζόν ἐστινὸὴ τριπλάσιον τοῦ κώνου. ἀλλὰ τὸ πρίσμα, οὗ βάσις μὲν ἐστὶ τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ, τριπλάσιόν ἐστι τῆς πυραμίδος, ἧξ βάσις μέν ἐστι τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ· καὶ ἡ πυραμὶξ ἄρα, ἧξ βάσις μέν [ἐστι] τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, μείζων ἐστὶ τοῦ κώνου τοῦ βάσιν ἔχοντες τὸν ΑΒΓΔ κύκλον. ἀλλὰ καὶ ἐλάττων· ἐμπεριέθεται γὰρ ὑπὸ αὐτοῦ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ἐστὶν ὁ κύλινδρος τοῦ κώνου μεῖζωνὸὴ τριπλάσιος.

Λέγω δή, ὅτι οὐδὲ ἐλάττων ἐστινὸὴ τριπλάσιος ὁ κύλινδρος τοῦ κώνου.

Εἰ γὰρ δυνατόν, ὀέστω ἐλάττωνὸὴ τριπλάσιος ὁ κύλινδρος τοῦ κώνου· ἀνάπαλιν ἄρα ὁ κῶνος τοῦ κυλίνδρου μεῖζων ἐστινὸὴ τρίτον μέρος. ἐγγεγράφθω δὴ εἰξ τὸν ΑΒΓΔ κύκλον τετράγωνον τὸ ΑΒΓΔ· τὸ ΑΒΓΔ ἄρα τετράγωνον μεῖζόν ἐστινὸὴ τὸ ἥμισυ τοῦ ΑΒΓΔ κύκλου. καὶ ἀνεστάτω ἀπὸ τοῦ ΑΒΓΔ τετραγώνου πυραμὶξ τὴν αὐτὴν κορυφὴνὸέθουσα τῷ κώνῳ· ἡ ἄρα ἀναστα θεῖσα πυραμὶξ μείζων ἐστινὸὴ τὸ ἥμισυ μέρος τοῦ κώνου, ἐπειδήπερ, ὡξ ὁέμπροσθεν ἐδείκνυμεν, ὅτι ἔαν περὶ τὸν κύκλον τετράγωνον περιγράψωμεν, ὀέσται τὸ ΑΒΓΔ τετράγωνον ἥμισυ τοῦ περὶ τὸν κύκλον περιγεγραμμένου τετραγώνου· καὶ ἔαν ἀπὸ τῶν τετραγώνων στερεὰ παραλληλεπίπεδα ἀναστήσωμεν ἰσοϋψῆ τῷ κώνῳ, ὰ καὶ καλε-

square *ABCD* of the same height as the cylinder is more than half of the cylinder. Let the circumferences *AB*, *BC*, *CD*, and *DA* be cut in half at points *E*, *F*, *G*, and *H*. And let *AE*, *EB*, *BF*, *FC*, *CG*, *GD*, *DH*, and *HA* be joined. And thus each of the triangles *AEB*, *BFC*, *CGD*, and *DHA* is more than half of the segment of circle *ABCD* about it, as was shown previously [Prop. 12.2]. Let prisms of equal height to the cylinder be set up on each of the triangles *AEB*, *BFC*, *CGD*, and *DHA*. And each of the prisms set up is greater than the half part of the segment of the cylinder about it—inasmuch as if we draw (straight-lines) parallel to *AB*, *BC*, *CD*, and *DA* through points *E*, *F*, *G*, and *H* (respectively), and complete the parallelograms on *AB*, *BC*, *CD*, and *DA*, and set up parallelepiped solids of equal height to the cylinder on them, then the prisms on triangles *AEB*, *BFC*, *CGD*, and *DHA* are each half of the set up (parallelepipeds). And the segments of the cylinder are less than the set up parallelepiped solids. Hence, the prisms on triangles *AEB*, *BFC*, *CGD*, and *DHA* are also greater than half of the segments of the cylinder about them. So (if) the remaining circumferences are cut in half, and straight-lines are joined, and prisms of equal height to the cylinder are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cylinder whose (sum) is less than the excess by which the cylinder exceeds three times the cone [Prop. 10.1]. Let them be left, and let them be *AE*, *EB*, *BF*, *FC*, *CG*, *GD*, *DH*, and *HA*. Thus, the remaining prism whose base (is) polygon *AEBFCGDH*, and height the same as the cylinder, is greater than three times the cone. But, the prism whose base is polygon *AEBFCGDH*, and height the same as the cylinder, is three times the pyramid whose base is polygon *AEBFCGDH*, and apex the same as the cone [Prop. 12.7 corr.]. And thus the pyramid whose base [is] polygon *AEBFCGDH*, and apex the same as the cone, is greater than the cone having (as) base circle *ABCD*. But (it is) also less. For it is encompassed by it. The very thing (is) impossible. Thus, the cylinder is not more than three times the cone.

So, I say that neither (is) the cylinder less than three times the cone.

For, if possible, let the cylinder be less than three times the cone. Thus, inversely, the cone is greater than the third part of the cylinder. So, let the square *ABCD* be inscribed in circle *ABCD* [Prop. 4.6]. Thus, square *ABCD* is greater than half of circle *ABCD*. And let a pyramid having the same apex as the cone be set up on square *ABCD*. Thus, the pyramid set up is greater than the half part of the cone, inasmuch as we showed previously that if we circumscribe a square about the circle [Prop. 4.7] then the square *ABCD* will be half of the square circumscribed about the circle [Prop. 12.2]. And if we set up on the squares parallelepiped solids—which are also called prisms—of the same height as the cone, then the (prism)

ῖται πρίσματα, ὀέσται τὸ ἀνασταθὲν ἀπὸ τοῦ ΑΒΓΔ τετραγώνου ἥμισυ τοῦ ἀνασταθέντος ἀπὸ τοῦ περὶ τὸν κύκλον περιγραφέντος τετραγώνου· πρὸς ἀλλήλα γάρ εἰσιν ὡς αἱ βάσεις. ὥστε καὶ τὰ τρίτα· καὶ πυραμὶξ ἄρα, ἣξ βάσις τὸ ΑΒΓΔ τετράγωνον, ἥμισύ ἐστι τῆς πυραμίδος τῆς ἀνασταθείσης ἀπὸ τοῦ περὶ τὸν κύκλον περιγραφέντος τετραγώνου. καί ἐστι μείζων ἡ πυραμὶξ ἡ ἀνασταθεῖσα ἀπὸ τοῦ περὶ τὸν κύκλον τετραγώνου τοῦ κώνου· ἐμπεριέθει γὰρ αὐτόν. ἡ ἄρα πυραμίξ, ἣξ βάσις τὸ ΑΒΓΔ τετράγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, μείζων ἐστὶνὸὴ τὸ ἥμισυ τοῦ κώνου. τετμήσθωσαν αἱ ΑΒ, ΒΓ, ΓΔ, ΔΑ περιφέρειαι δίθα κατὰ τὰ Ε, Ζ, Η, Θ σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΑΕ, ΕΒ, ΒΖ, ΖΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ· καὶ ὁἑκαστον ἄρα τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων μεῖζόν ἐστινὸὴ τὸ ἥμισυ μέρος του καθὸ ἑαυτὸ τμήματος τοῦ ΑΒΓΔ κύκλου. καὶ ἀνεστάτωσαν ἐφ΄ ἑκάστου τῶν ΑΕΒ, ΒΖΓ, ΓΗΔ, ΔΘΑ τριγώνων πυραμίδες τὴν αὐτὴν κορυφὴνὸέθουσαι τῷ κώνῳ· καὶ ἑκάστη ἄρα τῶν ἀνασταθεισῶν πυραμίδων κατὰ τὸν αὐτὸν τρόπον μείζων ἐστὶνὸὴ τὸ ἥμισυ μέρος τοῦ καθὸ ἑαυτὴν τμήματος τοῦ κώνου. τέμνοντες δὴ τὰξ ὑπολειπομένας περιφερείας δίθα καὶ ἐπιζευγνύντες εὐθείας καὶ ἀνιστάντες ἐφ΄ ἑκάστου τῶν τριγώνων πυραμίδα τὴν αὐτὴν κορυφὴνὸέθουσαν τῷ κώνῳ καὶ τοῦτο ἀεὶ ποιοῦτες καταλείψομέν τινα ἀποτμήματα τοῦ κώνου, ὰ ὀέσται ἐλάττονα τῆς ὑπεροθῆς, ἣ ὑπερέθει ὁ κῶνος τοῦ τρίτου μέρους τοῦ κυλίνδρου. λελείφθω, καὶ ὀέστω τὰ ἐπὶ τῶν ΑΕ, ΕΒ, ΒΖ, ΖΓ, ΓΗ, ΗΔ, ΔΘ, ΘΑ· λοιπὴ ἄρα ἡ πυραμίξ, ἣξ βάσις μέν ἐστι τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, κορυφὴ δὲ ἡ αὐτὴ τῷ κώνῳ, μείζων ἐστὶνὸὴ τρίτον μέρος τοῦ κυλίνδρου. ἀλλὸ ἡ πυραμίξ, ἣξ βάσις μέν ἐστι τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, κορυφὴ δὲ ἡ αυτὴ τῷ κώνῳ, τρίτον ἐστὶ μέρος τοῦ πρίσματος, οὗ βάσις μέν ἐστι τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ· τὸ ἄρα πρίσμα, οὗ βάσις μέν ἐστι τὸ ΑΕΒΖΓΗΔΘ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κυλίνδρῳ, μεῖζόν ἐστι τοῦ κυλίνδρου, οὗ βάσις ἐστὶν ὁ ΑΒΓΔ κύκλος. ἀλλὰ καὶ ὀέλαττον· ἐμπεριέθεται γὰρ ὑπὸ αὐτοῦ· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ὁ κύλινδρος τοῦ κώνου ἐλάττων ἐστὶνὸὴ τριπλάσιος. ἐδείθθη δέ, ὅτι οὐδὲ μείζωνὸὴ τριπλάσιος· τριπλάσιος ἄρα ὁ κύλινδρος τοῦ κώνου· ὥστε ὁ κῶνος τρίτον ἐστὶ μέρος τοῦ κυλίνδρου.

Πᾶξ ἄρα κῶνος κυλίνδρου τρίτον μέρος ἐστὶ τοῦ τὴν αὐτὴν βάσιν ἔχοντος αὐτῷ καὶ ὕψος ἴσον· ὅπερ ἔδει δεῖχαι.

set up on square $ABCD$ will be half of the (prism) set up on the square circumscribed about the circle. For they are to one another as their bases [Prop. 11.32]. Hence, (the same) also (goes for) the thirds. Thus, the pyramid whose base is square $ABCD$ is half of the pyramid set up on the square circumscribed about the circle [Prop. 12.7 corr.]. And the pyramid set up on the square circumscribed about the circle is greater than the cone. For it encompasses it. Thus, the pyramid whose base is square $ABCD$, and apex the same as the cone, is greater than half of the cone. Let the circumferences AB, BC, CD, and DA be cut in half at points E, F, G, and H (respectively). And let AE, EB, BF, FC, CG, GD, DH, and HA be joined. And, thus, each of the triangles AEB, BFC, CGD, and DHA is greater than the half part of the segment of circle $ABCD$ about it [Prop. 12.2]. And let pyramids having the same apex as the cone be set up on each of the triangles AEB, BFC, CGD, and DHA. And, thus, in the same way, each of the pyramids set up is more than the half part of the segment of the cone about it. So, (if) the remaining circumferences are cut in half, and straight-lines are joined, and pyramids having the same apex as the cone are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone whose (sum) is less than the excess by which the cone exceeds the third part of the cylinder [Prop. 10.1]. Let them be left, and let them be the (segments) on AE, EB, BF, FC, CG, GD, DH, and HA. Thus, the remaining pyramid whose base is polygon $AEBFCGDH$, and apex the same as the cone, is greater than the third part of the cylinder. But, the pyramid whose base is polygon $AEBFCGDH$, and apex the same as the cone, is the third part of the prism whose base is polygon $AEBFCGDH$, and height the same as the cylinder [Prop. 12.7 corr.]. Thus, the prism whose base is polygon $AEBFCGDH$, and height the same as the cylinder, is greater than the cylinder whose base is circle $ABCD$. But, (it is) also less. For it is encompassed by it. The very thing is impossible. Thus, the cylinder is not less than three times the cone. And it was shown that neither (is it) greater than three times (the cone). Thus, the cylinder (is) three times the cone. Hence, the cone is the third part of the cylinder.

Thus, every cone is the third part of the cylinder which has the same base as it, and an equal height. (Which is) the very thing it was required to show.

11.

Οἱ ὑπο τὸ αὐτὸ ὕψος ὄντες κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡξ αἱ βάσεις.

Ἔστωσαν ὑπὸ τὸ αὐτὸ ὕψος κῶνοι καὶ κύλινδροι, ὧν βάσεις μὲν [εἰσιν] οἱ ΑΒΓΔ, ΕΖΗΘ κύκλοι, ὀάχονες δὲ οἱ ΚΛ, ΜΝ, διάμετροι δὲ τῶν βάσεων αἱ ΑΓ, ΕΗ· λέγω, ὅτι ἐστὶν ὡξ ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὕτως ὁ

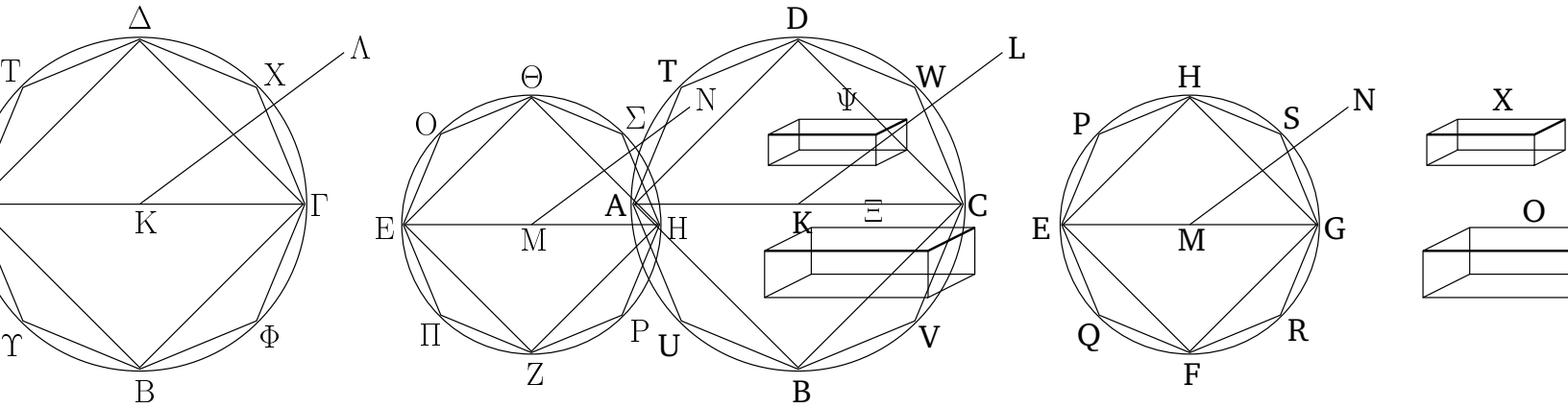
Proposition 11

Cones and cylinders having the same height are to one another as their bases.

Let there be cones and cylinders of the same height whose bases [are] the circles $ABCD$ and $EFGH$, axes KL and MN, and diameters of the bases AC and EG (respectively). I say that as circle $ABCD$ is to circle $EFGH$, so cone AL (is) to

ΑΛ κῶνος πρὸς τὸν ΕΝ κῶνον.

cone EN.



Εἰ γὰρ μή, ὀέσται ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὐτως ὁ ΑΛ κῶνος ἤτοι πρὸς ἔλασσόν τι τοῦ ΕΝ κώνου στερεὸν ὴ πρὸς μεῖζον. ὀέστω πρότερον πρὸς ἔλασσον τὸ Χ, καὶ ὁῷ ὀέλασσόν ἐστι τὸ Χ στερεὸν τοῦ ΕΝ κώνου, ἐκείνῳ ἴσον ὀέστω τὸ Ψ στερεόν· ὁ ΕΝ κῶνος ἄραὀίσος ἐστὶ τοῖς Χ, Ψ στερεοῖξ. ἐγγεγράφθω εἰξ τὸν ΕΖΗΘ κύκλον τετράγωνον τὸ ΕΖΗΘ· τὸ ἄρα τετράγωνον μεῖζόν ἐστινὸὴ τὸ ἥμισυ τοῦ κύκλου. ἀνεστάτω ἀπὸ τοῦ ΕΖΗΘ τετραγώνου πυραμὶξ ἰσοϋψὴξ τῷ κώνῳ· ἡ ἄρα ἀνασταθεῖσα πυραμὶξ μείζων ἐστὶνὸὴ τὸ ἥμισυ τοῦ κώνου, ἐπειδήπερ ἔαν περιγράψωμεν περὶ τὸν κύκλον τετράγωνον, καὶ ἀπὸ αὐτοῦ ἀναστήσωμεν πυραμίδα ἰσοϋψῆ τῷ κώνῳ, ἡ ἐγγραφεῖσα πυραμὶξ ἥμισύ ἐστι τῆς περιγραφείσης· πρὸς ἀλλήλας γάρ εἰσιν ὡς αἱ βάσεις· ἐλάττων δὲ ὁ κῶνος τῆς περιγραφείσης πυραμίδος. τετμήσθωσαν αἱ ΕΖ, ΖΗ, ΗΘ, ΘΕ περιφέρειαι δίθα κατὰ τὰ Ο, Π, Ρ, Σ σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΘΟ, ΟΕ, ΕΠ, ΠΖ, ΖΡ, ΡΗ, ΗΣ, ΣΘ. ὁέκαστον ἄρα τῶν ΘΟΕ, ΕΠΖ, ΖΡΗ, ΗΖΘ τριγώνων μεῖζόν ἐστινὸὴ τὸ ἥμισυ τοῦ καθὸ ἑαυτὸ τμήματος τοῦ κύκλου. ἀνεστάτω ἐφ᾽ ἑκάστου τῶν ΘΟΕ, ΕΠΖ, ΖΡΗ, ΗΣΘ τριγώνων πυραμὶξ ἰσοϋψὴξ τῷ κώνῳ· καὶ ἑκάστη ἄρα τῶν ἀνασταθεισῶν πυραμίδων μείζων ἐστὶνὸὴ τὸ ἥμισυ τοῦ καθὸ ἑαυτὴν τμήματος τοῦ κώνου. τέμνοντες δὴ τὰξ ὑπολειπομένας περιφερείας δίθα καὶ ἐπιζευγνύντες εὐθείας καὶ ἀνιστάντες ἐπὶ ἑκάστου τῶν τριγώνων πυραμίδας ἰσοϋψεῖς τῷ κώνῳ καὶ ἀεὶ τοῦτο ποιοῦντες καταλείψομέν τινα ἀποτμήματα τοῦ κώνου, ὁὰ ὀέσται ἐλάσσονα τοῦ Ψ στερεοῦ. λελείφθω, καὶ ὀέστω τὰ ἐπὶ τῶν ΘΟΕ, ΕΠΖ, ΖΡΗ, ΗΣΘ· λοιπὴ ἄρα ἡ πυραμίξ, ἧξ βάσις τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, ὕψος δὲ τὸ αὐτὸ τῷ κώνῳ, μείζων ἐστὶ τοῦ Χ στερεοῦ. ἐγγεγράφθω καὶ εἰξ τὸν ΑΒΓΔ κύκλον τῷ ΘΟΕΠΖΡΗΣ πολυγώνῳ ὅμοιόν τε καὶ ὁμοίως κείμενον πολύγωνον τὸ ΔΤΑΥΒΦΓΘ, καὶ ἀνεστάτω ἐπ᾽ αὐτοῦ πυραμὶξ ἰσοϋψὴξ τῷ ΑΛ κώνῳ. ἐπεὶ οὐῦν ἐστιν ὡξ τὸ ἀπὸ τῆς ΑΓ πρὸς τὸ ἀπὸ τῆς ΕΗ, οὐτως τὸ ΔΤΑΥΒΦΓΘ πολύγωνον πρὸς τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, ὡξ δὲ τὸ ἀπὸ τῆς ΑΓ πρὸς τὸ ἀπὸ τῆς ΕΗ, οὐτως ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, καὶ ὡξ ἄρα ὁ ΑΒΓΔ κύκλος πρὸς τὸν

For if not, then as circle $ABCD$ (is) to circle $EFGH$, so cone AL will be to some solid either less than, or greater than, cone EN. Let it, first of all, be (in this ratio) to (some) lesser (solid), O. And let solid X be equal to that (magnitude) by which solid O is less than cone EN. Thus, cone EN is equal to (the sum of) solids O and X. Let the square $EFGH$ be inscribed in circle $EFGH$ [Prop. 4.6]. Thus, the square is greater than half of the circle [Prop. 12.2]. Let a pyramid of the same height as the cone be set up on square $EFGH$. Thus, the pyramid set up is greater than half of the cone, inasmuch as, if we circumscribe a square about the circle [Prop. 4.7], and set up on it a pyramid of the same height as the cone, then the inscribed pyramid is half of the circumscribed pyramid. For they are to one another as their bases [Prop. 12.6]. And the cone (is) less than the circumscribed pyramid. Let the circumferences EF, FG, GH, and HE be cut in half at points P, Q, R, and S. And let HP, PE, EQ, QF, FR, RG, GS, and SH be joined. Thus, each of the triangles HPE, EQF, FRG, and GSH is greater than half of the segment of the circle about it [Prop. 12.2]. Let pyramids of the same height as the cone be set up on each of the triangles HPE, EQF, FRG, and GSH. And, thus, each of the pyramids set up is greater than half of the segment of the cone about it [Prop. 12.10]. So, (if) the remaining circumferences are cut in half, and straight-lines are joined, and pyramids of equal height to the cone are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone (the sum of) which is less than solid X [Prop. 10.1]. Let them be left, and let them be the (segments) on HPE, EQF, FRG, and GSH. Thus, the remaining pyramid whose base is polygon $HPEQFRGS$, and height the same as the cone, is greater than solid O [Prop. 6.18]. And let the polygon $DTAUBVCW$, similar, and similarly laid out, to polygon $HPEQFRGS$, be inscribed in circle $ABCD$. And on it let a pyramid of the same height as cone AL be set up. Therefore, since as the (square) on AC is to the (square) on EG, so polygon $DTAUBVCW$

ΕΖΗΘ κύκλον, οὑτως τὸ ΔΤΑΥΒΦΓΘ πολύγωνον πρὸς τὸ ΘΟΕΠΖΡΗΣ πολύγωνον. ὡς δὲ ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὑτως ὁ ΑΛ κῶνος πρὸς τὸ Χ στερεόν, ὡς δὲ τὸ ΔΤΑΥΒΦΓΘ πολύγωνον πρὸς τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, οὑτως ἡ πυραμίξ, ἧξ βάσις μὲν τὸ ΔΤΑΥΒΦΓΘ πολύγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, πρὸς τὴν πυραμίδα, ἧξ βάσις μὲν τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, κορυφὴ δὲ τὸ Ν σημεῖον. καὶ ὡς ἄρα ὁ ΑΛ κῶνος πρὸς τὸ Χ στερεόν, οὑτως ἡ πυραμίξ, ἧξ βάσις μὲν τὸ ΔΤΑΥΒΦΓΘ πολύγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, πρὸς τὴν πυραμίδα, ἧξ βάσις μὲν τὸ ΘΟΕΠΖΡΗΣ πολύγωνον, κορυφὴ δὲ τὸ Ν σημεῖον· ἐναλλὰχ ἄρα ἐστὶν ὡς ὁ ΑΛ κῶνος πρὸς τὴν ἐν αὐτῷ πυραμίδα, οὑτως τὸ Χ στερεὸν πρὸς τὴν ἐν τῷ ΕΝ κώνῳ πυραμίδα. μείζων δὲ ὁ ΑΛ κῶνος τῆς ἐν αὐτῷ πυραμίδος· μεῖζον ἄρα καὶ τὸ Χ στερεὸν τῆς ἐν τῷ ΕΝ κώνῳ πυραμίδος. ἀλλὰ καὶ ὐέλασσον· ὅπερὐάτοπον. οὐκ ἄρα ἐστὶν ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὑτως ὁ ΑΛ κῶνος πρὸςὐέλασσόν τι τοῦ ΕΝ κώνου στερεόν. ὁμοίως δὲ δείχομεν, ὅτι οὐδέ ἐστιν ὡς ὁ ΕΖΗΘ κύκλος πρὸς τὸν ΑΒΓΔ κύκλον, οὑτως ὁ ΕΝ κῶνος πρὸςὐέλασσόν τι τοῦ ΑΛ κώνου στερεόν.

Λέγω δή, ὅτι οὐδέ ἐστιν ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὑτως ὁ ΑΛ κῶνος πρὸς μεῖζόν τι τοῦ ΕΝ κώνου στερεόν.

Εἰ γὰρ δυνατόν, ὑέστω πρὸς μεῖζον τὸ Χ· ἀνάπαλιν ἄρα ἐστὶν ὡς ὁ ΕΖΗΘ κύκλος πρὸς τὸν ΑΒΓΔ κύκλον, οὑτως τὸ Χ στερεὸν πρὸς τὸν ΑΛ κῶνον. ἀλλὰ ὡς τὸ Χ στερεὸν πρὸς τὸν ΑΛ κῶνον, οὑτως ὁ ΕΝ κῶνος πρὸςὐέλασσόν τι τοῦ ΑΛ κώνου στερεόν· καὶ ὡς ἄρα ὁ ΕΖΗΘ κύκλος πρὸς τὸν ΑΒΓΔ κύκλον, οὑτως ὁ ΕΝ κῶνος πρὸςὐέλασσόν τι τοῦ ΑΛ κώνου στερεόν· ὅπερ ἀδύνατον ἐδείθθη. οὐκ ἄρα ἐστὶν ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὑτως ὁ ΑΛ κῶνος πρὸς μεῖζόν τι τοῦ ΕΝ κώνου στερεόν. ἐδείθθη δέ, ὅτι οὐδὲ πρὸςὐέλασσον· ὐέστιν ἄρα ὡς ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὑτως ὁ ΑΛ κῶνος πρὸς τὸν ΕΝ κῶνον.

Ἀλλὰ ὡς ὁ κῶνος πρὸς τὸν κῶνον, ὁ κύλινδρος πρὸς τὸν κύλινδρον· τριπλασίων γὰρ ἑκάτερος ἑκατέρου. καὶ ὡς ἄρα ὁ ΑΒΓΔ κύκλος πρὸς τὸν ΕΖΗΘ κύκλον, οὑτως οἱ ἐπ᾽ αὐτῶν ἰσοϋψεῖς.

Οἱ ἄρα ὑπὸ τὸ αὐτὸ ὕψος ὄντες κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡς αἱ βάσεις· ὅπερ ἔδει δεῖχαι.

12.

Οἱ ὅμοιοι κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ἐν ταῖς βάσεσι διαμέτρων.

Ἔστωσαν ὅμοιοι κῶνοι καὶ κύλινδροι, ὧν βάσεις μὲν οἱ ΑΒΓΔ, ΕΖΗΘ κύκλοι, διάμετροι δὲ τῶν βάσεων αἱ ΒΔ, ΖΘ, ὐάχονες δὲ τῶν κώνων καὶ κυλίνδρων οἱ ΚΛ, ΜΝ· λέγω, ὅτι ὁ κῶνος, οὗ βάσις μέν [ἐστιν] ὁ ΑΒΓΔ κύκλος, κορυφὴ δὲ τὸ Λ σημεῖον, πρὸς τὸν κῶνον, οὗ βάσις μέν [ἐστιν] ὁ ΕΖΗΘ (is) to polygon $HPEQFRGS$ [Prop. 12.1], and as the (square) on AC (is) to the (square) on EG, so circle $ABCD$ (is) to circle $EFGH$ [Prop. 12.2], thus as circle $ABCD$ (is) to circle $EFGH$, so polygon $DTAUBVCW$ also (is) to polygon $HPEQFRGS$. And as circle $ABCD$ (is) to circle $EFGH$, so cone AL (is) to solid O. And as polygon $DTAUBVCW$ (is) to polygon $HPEQFRGS$, so the pyramid whose base is polygon $DTAUBVCW$, and apex the point L, (is) to the pyramid whose base is polygon $HPEQFRGS$, and apex the point N [Prop. 12.6]. And, thus, as cone AL (is) to solid O, so the pyramid whose base is $DTAUBVCW$, and apex the point L, (is) to the pyramid whose base is polygon $HPEQFRGS$, and apex the point N [Prop. 5.11]. Thus, alternately, as cone AL is to the pyramid within it, so solid O (is) to the pyramid within cone EN [Prop. 5.16]. But, cone AL (is) greater than the pyramid within it. Thus, solid O (is) also greater than the pyramid within cone EN [Prop. 5.14]. But, (it is) also less. The very thing (is) absurd. Thus, circle $ABCD$ is not to circle $EFGH$, as cone AL (is) to some solid less than cone EN. So, similarly, we can show that neither is circle $EFGH$ to circle $ABCD$, as cone EN (is) to some solid less than cone AL.

So, I say that neither is circle $ABCD$ to circle $EFGH$, as cone AL (is) to some solid greater than cone EN.

For, if possible, let it be (in this ratio) to (some) greater (solid), O. Thus, inversely, as circle $EFGH$ is to circle $ABCD$, so solid O (is) to cone AL [Prop. 5.7 corr.]. But, as solid O (is) to cone AL, so cone EN (is) to some solid less than cone AL [Prop. 12.2 lem.]. And, thus, as circle $EFGH$ (is) to circle $ABCD$, so cone EN (is) to some solid less than cone AL. The very thing was shown (to be) impossible. Thus, circle $ABCD$ is not to circle $EFGH$, as cone AL (is) to some solid greater than cone EN. And, it was shown that neither (is it in this ratio) to (some) lesser (solid). Thus, as circle $ABCD$ is to circle $EFGH$, so cone AL (is) to cone EN.

But, as the cone (is) to the cone, (so) the cylinder (is) to the cylinder. For each (is) three times each [Prop. 12.10]. Thus, circle $ABCD$ (is) also to circle $EFGH$, as (the ratio of the cylinders) on them (having) the same height.

Thus, cones and cylinders having the same height are to one another as their bases. (Which is) the very thing it was required to show.

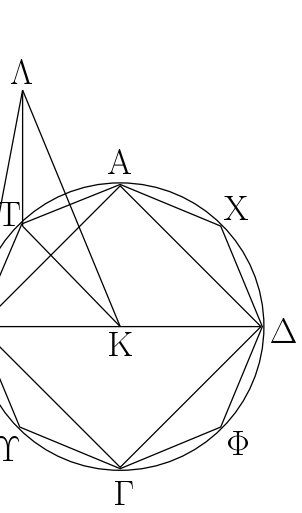
Proposition 12

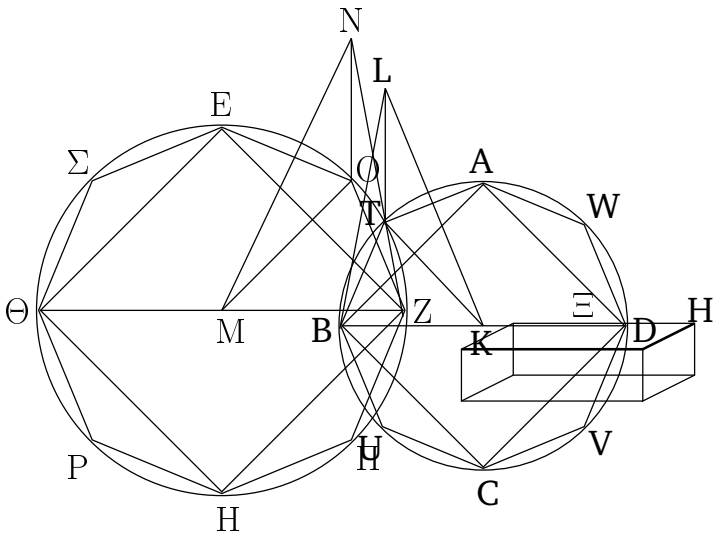
Similar cones and cylinders are to one another in the cubed ratio of the diameters of their bases.

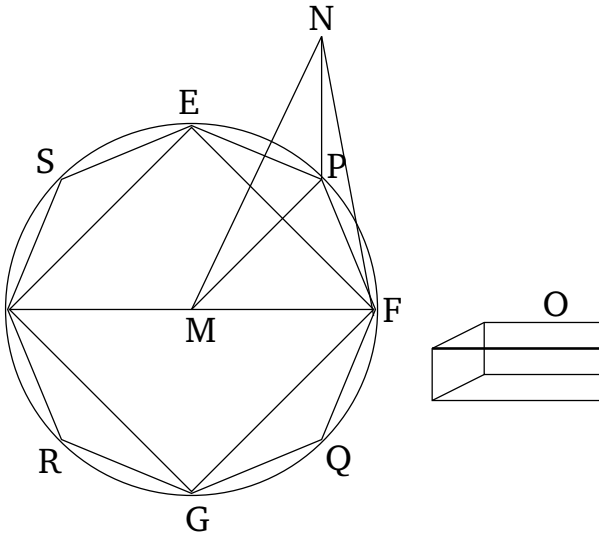
Let there be similar cones and cylinders of which the bases (are) the circles $ABCD$ and $EFGH$, the diameters of the bases (are) BD and FH, and the axes of the cones and cylinders (are) KL and MN (respectively). I say that the cone whose base [is] circle $ABCD$, and apex the point L, has to the cone whose base

κύκλος, κορυφὴ δὲ τὸ N σημεῖον, τριπλασίονα λόγον ἔχει ἤπερ ἡ ΒΔ πρὸς τὴν ΖΘ.

[is] circle $EFGH$, and apex the point N, the cubed ratio that BD (has) to FH.







Εἰ γὰρ μὴ ἔχει ὁ ΑΒΓΔΛ κῶνος πρὸς τὸν ΕΖΗΘΝ κῶνον πριπλασίονα λόγον ἤπερ ἡ ΒΔ πρὸς τὴν ΖΘ, ὁέχει ὁ ΑΒΓΔΛ κῶνοςὸὴ πρὸςὸέλασσόν τι τοῦ ΕΖΗΘΝ κώνου στερεὸν τριπλασίονα λόγονὸὴ πρὸς μεῖζον. ἐθέτω πρότερον πρὸςὸέλασσον τὸ Χ, καὶ ἐγγεγράφθω εἰξ τὸν ΕΖΗΘ κύκλον τετράγωνον τὸ ΕΖΗΘ· τὸ ἄρα ΕΖΗΘ τετράγωνον μεῖζόν ἐστινὸὴ τὸ ἥμισυ τοῦ ΕΖΗΘ κύκλου. καὶ ἀνεστάτω ἐπὶ τοῦ ΕΖΗΘ τετραγώνου πυραμὶξ τὴν αὐτὴν κορυφὴνὸέθουσα τῷ κώνῳ· ἡ ἄρα ἀναστα θεῖσα πυραμὶξ μείζων ἐστινὸὴ τὸ ἥμισυ μέρος τοῦ κώνου. τετμήσθωσαν δὴ αἱ ΕΖ, ΖΗ, ΗΘ, ΘΕ περιφέρειαι δίθα κατὰ τὰ Ο, Π, Ρ, Σ σημεῖα, καὶ ἐπεζεύθθωσαν αἱ ΕΟ, ΟΖ, ΖΠ, ΠΗ, ΗΡ, ΡΘ, ΘΣ, ΣΕ. καὶ ὁέκαστον ἄρα τῶν ΕΟΖ, ΖΠΗ, ΗΡΘ, ΘΣΕ τριγώνων μεῖζόν ἐστινὸὴ τὸ ἥμισυ μέρος τοῦ καθὸ ἑαυτὸ τμήματος τοῦ ΕΖΗΘ κύκλου. καὶ ἀνεστάτω ἐφ᾽ ἑκάστου τῶν ΕΟΖ, ΖΠΗ, ΗΡΘ, ΘΣΕ τριγώνων πυραμὶξ τὴν αὐτὴν κορυφὴνὸέθουσα τῷ κώνῳ· καὶ ἑκάστη ἄρα τῶν ἀνασταθεισῶν πυραμίδων μείζων ἐστινὸὴ τὸ ἥμισυ μέρος τοῦ καθὸ ἑαυτὴν τμήματος τοῦ κώνου. τέμνοντες δὴ τὰξ ὑπολειπομένας περιφερείας δίθα καὶ ἐπιζευγνύντες εὐθείας καὶ ἀνιστάντες ἐφ᾽ ἑκάστου τῶν τριγώνων πυραμίδας τὴν αὐτὴν κορυφὴν ἐθούσας τῷ κώνῳ καὶ τοῦτο ἀεὶ ποιοῦντες καταλείψομέν τινα ἀποτμήματα τοῦ κώνου, ὰ ὀέσται ἐλάσσονα τῆς ὑπεροθῆς, ᾗ ὑπερέθει ὁ ΕΖΗΘΝ κῶνος τοῦ Χ στερεοῦ. λελείφθω, καὶ ὀέστω τὰ ἐπὶ τῶν ΕΟ, ΟΖ, ΖΠ, ΠΗ, ΗΡ, ΡΘ, ΘΣ, ΣΕ· λοιπὴ ἄρα ἡ πυραμίξ, ἧξ βάσις μέν ἐστι τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν σημεῖον, μείζων ἐστὶ τοῦ Χ στερεοῦ. ἐγγεγράφθω καὶ εἰξ τὸν ΑΒΓΔ κύκλον τῷ ΕΟΖΠΗΡΘΣ πολυγώνῳ ὅμοιόν τε καὶ ὁμοίως κείμενον πολύγωνον τὸ ΑΤΒΥΓΦΔΘ, καὶ ἀνεστάτω ἐπὶ τοῦ ΑΤΒΥΓΦΔΘ πολυγώνου πυραμὶξ τὴν αὐτὴν κορυφὴνὸέθουσα τῷ κώνῳ, καὶ τῶν μὲν περιεθόντων τὴν πυραμίδα, ἧξ βάσις μέν ἐστι τὸ ΑΤΒΥΓΦΔΘ πολύγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, ὁὲν τρίγωνονὸέστω τὸ

For if cone $ABCDL$ does not have to cone $EFGHN$ the cubed ratio that BD (has) to FH then cone $ABCDL$ will have the cubed ratio to some solid either less than, or greater than, cone $EFGHN$. Let it, first of all, have (such a ratio) to (some) lesser (solid), O. And let the square $EFGH$ be inscribed in circle $EFGH$ [Prop. 4.6]. Thus, square $EFGH$ is greater than half of circle $EFGH$ [Prop. 12.2]. And let a pyramid having the same apex as the cone be set up on square $EFGH$. Thus, the pyramid set up is greater than the half part of the cone [Prop. 12.10]. So, let the circumferences EF, FG, GH, and HE be cut in half at points P, Q, R, and S (respectively). And let EP, PF, FQ, QG, GR, RH, HS, and SE be joined. And, thus, each of the triangles EPF, FQG, GRH, and HSE is greater than the half part of the segment of circle $EFGH$ about it [Prop. 12.2]. And let a pyramid having the same apex as the cone be set up on each of the triangles EPF, FQG, GRH, and HSE. And thus each of the pyramids set up is greater than the half part of the segment of the cone about it [Prop. 12.10]. So, (if) the the remaining circumferences are cut in half, and straight-lines are joined, and pyramids having the same apex as the cone are set up on each of the triangles, and this is done continually, then we will (eventually) leave some segments of the cone whose (sum) is less than the excess by which cone $EFGHN$ exceeds solid O [Prop. 10.1]. Let them be left, and let them be the (segments) on EP, PF, FQ, QG, GR, RH, HS, and SE. Thus, the remaining pyramid whose base is polygon $EPFQGRHS$, and apex the point N, is greater than solid O. And let the polygon $ATBUCVDW$, similar, and similarly laid out, to polygon $EPFQGRHS$, be inscribed in circle $ABCD$ [Prop. 6.18]. And let a pyramid having the same apex as the cone be set up on polygon $ATBUCVDW$. And let LBT be one of the triangles containing the pyramid whose base is polygon

ΛΒΤ, τῶν δὲ περειθόντων τὴν πυραμίδα, ἧξ βάσις μέν ἐστι τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν σημεῖον, δέν τρίγωνονὸέστω τὸ ΝΖΟ, καὶ ἐπεζεύθθωσαν αἱ ΚΤ, ΜΟ. καὶ ἐπεὶ ὅμοιόξ ἐστιν ὁ ΑΒΓΔΛ κῶνος τῷ ΕΖΗΘΝ κώνῳ, ὀέστιν ἄρα ὡξ ἡ ΒΔ πρὸς τὴν ΖΘ, οὑτως ὁ ΚΛὀάχων πρὸς τὸν ΜΝὀάχονα. ὡξ δὲ ἡ ΒΔ πρὸς τὴν ΖΘ, οὑτως ἡ ΒΚ πρὸς τὴν ΖΜ· καὶ ὡξ ἄρα ἡ ΒΚ πρὸς τὴν ΖΜ, οὑτως ἡ ΚΛ πρὸς τὴν ΜΝ. καὶ ἐναλλὰχ ὡξ ἡ ΒΚ πρὸς τὴν ΚΛ, οὑτως ἡ ΖΜ πρὸς τὴν ΜΝ. καὶ περὶ ἴσας γωνίας τὰξ ὑπὸ ΒΚΛ, ΖΜΝ αἱ πλευραὶ ἀνάλογόν εἰσιν· ὅμοιον ἄρα ἐστὶ τὸ ΒΚΛ τρίγωνον τῷ ΖΜΝ τριγώνῳ. πάλιν, ἐπεί ἐστιν ὡξ ἡ ΒΚ πρὸς τὴν ΚΤ, οὑτως ἡ ΖΜ πρὸς τὴν ΜΟ, καὶ περὶ ἴσας γωνίας τὰξ ὑπὸ ΒΚΤ, ΖΜΟ, ἐπειδήπερ, ὃ μέρος ἐστὶν ἡ ὑπὸ ΒΚΤ γωνία τῶν πρὸς τῷ Κ κέντρῳ τεσσάρων ὀρθῶν, τὸ αὐτὸ μέρος ἐστὶ καὶ ἡ ὑπὸ ΖΜΟ γωνία τῶν πρὸς τῷ Μ κέντρῳ τεσσάρων ὀρθῶν· ἐπεὶ οˇν περὶ ἴσας γωνίας αἱ πλευραὶ ἀνάλογόν εἰσιν, ὅμοιον ἄρα ἐστι τὸ ΒΚΤ τρίγωνον τῷ ΖΜΟ τριγώνῳ. πάλιν, ἐπεὶ ἐδείθθη ὡξ ἡ ΒΚ πρὸς τὴν ΚΛ, οὑτως ἡ ΖΜ πρὸς τὴν ΜΝ, ὀίση δὲ ἡ μὲν ΒΚ τῇ ΚΤ, ἡ δὲ ΖΜ τῇ ΟΜ, ὀέστιν ἄρα ὡξ ἡ ΤΚ πρὸς τὴν ΚΛ, οὑτως ἡ ΟΜ πρὸς τὴν ΜΝ. καὶ περὶ ἴσας γωνίας τὰξ ὑπὸ ΤΚΛ, ΟΜΝ· ὀρθαὶ γάρ· αἱ πλευραὶ ἀνάλογόν εἰσιν· ὅμοιον ἄρα ἐστὶ τὸ ΛΚΤ τρίγωνον τῷ ΝΜΟ τριγώνῳ. καὶ ἐπεὶ διὰ τὴν ὁμοιότητα τῶν ΛΚΒ, ΝΜΖ τριγώνων ἐστὶν ὡξ ἡ ΛΒ πρὸς τὴν ΒΚ, οὑτως ἡ ΝΖ πρὸς τὴν ΖΜ, διὰ δὲ τὴν ὁμοιότητα τῶν ΒΚΤ, ΖΜΟ τριγώνων ἐστὶν ὡξ ἡ ΚΒ πρὸς τὴν ΒΤ, οὑτως ἡ ΜΖ πρὸς τὴν ΖΟ, διὸ ὀίσου ἄρα ὡξ ἡ ΛΒ πρὸς τὴν ΒΤ, οὑτως ἡ ΝΖ πρὸς τὴν ΖΟ. πάλιν, ἐπεὶ διὰ τὴν ομοιότητα τῶν ΛΤΚ, ΝΟΜ τριγώνων ἐστὶν ὡξ ἡ ΛΤ πρὸς τὴν ΤΚ, οὑτως ἡ ΝΟ πρὸς τὴν ΟΜ, διὰ δὲ τὴν ὁμοιότητα τῶν ΤΚΒ, ΟΜΖ τριγώνων ἐστὶν ὡξ ἡ ΚΤ πρὸς τὴν ΤΒ, οὑτως ἡ ΜΟ πρὸς τὴν ΟΖ, διὸ ὀίσου ἄρα ὡξ ἡ ΛΤ πρὸς τὴν ΤΒ, οὑτως ἡ ΝΟ πρὸς τὴν ΟΖ. ἐδείθθη δὲ καὶ ὡξ ἡ ΤΒ πρὸς τὴν ΒΛ, οὑτως ἡ ΟΖ πρὸς τὴν ΖΝ. διὸ ὀίσου ἄρα ὡξ ἡ ΤΛ πρὸς τὴν ΛΒ, οὑτως ἡ ΟΝ πρὸς τὴν ΝΖ. τῶν ΛΤΒ, ΝΟΖ ἄρα τριγώνων ἀνάλογόν εἰσιν αἱ πλευραί· ἰσογώνια ἄρα ἐστὶ τὰ ΛΤΒ, ΝΟΖ τρίγωνα· ὡστε καὶ ὅμοια. καὶ πυραμὶξ ἄρα, ἧξ βάσις μὲν τὸ ΒΚΤ τρίγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, ὁμοία ἐστὶ πυραμίδι, ἧξ βάσις μὲν τὸ ΖΜΟ τρίγωνον, κορυφὴ δὲ τὸ Ν σημεῖον· ὑπὸ γὰρ ὅμοίων ἐπιπέδων περιέθονταιὀίσων τὸ πλῆθος. αἱ δὲ ὅμοιαι πυραμίδες καὶ τριγώνουςὀέθουσαι βάσεις ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν. ἡ ἄρα ΒΚΤΛ πυραμὶξ πρὸς τὴν ΖΜΟΝ πυραμίδα τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΚ πρὸς τὴν ΖΜ. ὁμοίως δὴ ἐπιζευγνύντες ἀπὸ τῶν Α, Θ, Δ, Φ, Γ, Υ ἐπὶ τὸ Κ εὐθείας καὶ ἀπὸ τῶν Ε, Σ, Θ, Ρ, Η, Π ἐπὶ τὸ Μ καὶ ἀνιστάντες ἐφ΄ ἑκάστου τῶν τριγώνων πυραμίδας τὴν αὐτὴν κορυφὴν ἐθούσας τοῖς κώνοις δείχομεν, ὅτι καὶ ἑκάστη τῶν ὁμοταγῶν πυραμίδων πρὸς ἑκάστην ὁμοταγῆ πυραμίδα τριπλασίονα λόγον ὁέχει ἥπερ ἡ ΒΚ ὁμόλογος πλευρὰ πρὸς τὴν ΖΜ ὁμόλογον πλευράν, τουτέστιν ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ. καὶ ὡξ ὁὲν τῶν ἡγουμένων πρὸς ὁὲν τῶν ἑπομένων, o-

ATBUCVDW, and apex the point *L*. And let *NFP* be one of the triangles containing the pyramid whose base is triangle *EPFQGRHS*, and apex the point *N*. And let *KT* and *MP* be joined. And since cone *ABCDL* is similar to cone *EFGHN*, thus as *BD* is to *FH*, so axis *KL* (is) to axis *MN* [Def. 11.24]. And as *BD* (is) to *FH*, so *BK* (is) to *FM*. And, thus, as *BK* (is) to *FM*, so *KL* (is) to *MN*. And, alternately, as *BK* (is) to *KL*, so *FM* (is) to *MN* [Prop. 5.16]. And the sides around the equal angles *BKL* and *FMN* are proportional. Thus, triangle *BKL* is similar to triangle *FMN* [Prop. 6.6]. Again, since as *BK* (is) to *KT*, so *FM* (is) to *MP*, and (they are) about the equal angles *BKT* and *FMP*, inasmuch as whatever part angle *BKT* is of the four right-angles at the center *K*, angle *FMP* is also the same part of the four right-angles at the center *M*. Therefore, since the sides about equal angles are proportional, triangle *BKT* is thus similar to traingle *FMP* [Prop. 6.6]. Again, since it was shown that as *BK* (is) to *KL*, so *FM* (is) to *MN*, and *BK* (is) equal to *KT*, and *FM* to *PM*, thus as *TK* (is) to *KL*, so *PM* (is) to *MN*. And the sides about the equal angles *TKL* and *PMN*—for (they are both) right-angles—are proportional. Thus, triangle *LKT* (is) similar to triangle *NMP* [Prop. 6.6]. And since, on account of the similarity of triangles *LKB* and *NMF*, as *LB* (is) to *BK*, so *NF* (is) to *FM*, and, on account of the similarity of triangles *BKT* and *FMP*, as *KB* (is) to *BT*, so *MF* (is) to *FP* [Def. 6.1], thus, via equality, as *LB* (is) to *BT*, so *NF* (is) to *FP* [Prop. 5.22]. Again, since, on account of the similarity of triangles *LTK* and *NPM*, as *LT* (is) to *TK*, so *NP* (is) to *PM*, and, on account of the similarity of triangles *TKB* and *PMF*, as *KT* (is) to *TB*, so *MP* (is) to *PF*, thus, via equality, as *LT* (is) to *TB*, so *NP* (is) to *PF* [Prop. 5.22]. And it was shown that as *TB* (is) to *BL*, so *PF* (is) to *FN*. Thus, via equality, as *TL* (is) to *LB*, so *PN* (is) to *NF* [Prop. 5.22]. Thus, the sides of triangles *LTB* and *NPF* are proportional. Thus, triangles *LTB* and *NPF* are equiangular [Prop. 6.5]. And, hence, (they are) similar [Def. 6.1]. And, thus, the pyramid whose base is triangle *BKT*, and apex the point *L*, is similar to the pyramid whose base is triangle *FMP*, and apex the point *N*. For they are contained by equal numbers of similar planes [Def. 11.9]. And similar pyramids which also have triangular bases are in the cubed ratio of corresponding sides [Prop. 12.8]. Thus, pyramid *BKTL* has to pyramid *FMPN* the cubed ratio that *BK* (has) to *FM*. So, similarly, joining straight-lines from (points) *A*, *W*, *D*, *V*, *C*, and *U* to (center) *K*, and from (points) *E*, *S*, *H*, *R*, *G*, and *Q* to (center) *M*, and setting up pyramids having the same apexes as the cones on each of the triangles (so formed), we can also show that each of the pyramids (on base *ABCD* taken) in order will have to each of the pyramids (on base *EFGH* taken) in order the cubed ratio that the corresponding side *BK* (has) to the corresponding side *FM*—that is to say, that *BD* (has) to *FH*. And (for two sets of proportional magnitudes) as one of the

οὕτως ὁπάντα τὰ ἡγούμενα πρὸς ὁπάντα τὰ ἑπόμενα· ὐέστιν ἄρα καὶ ὡξ ἡ ΒΚΤΛ πυραμὶξ πρὸς τὴν ΖΜΟΝ πυραμίδα, οὑτως ἡ ὅλη πυραμίξ, ἧξ βάσις τὸ ΑΤΒΥΓΦΔΘ πολύγωνον, κορυφὴ δὲ τὸ Λ σημεῖον, πρὸς τὴν ὅλην πυραμίδα, ἧξ βάσις μὲν τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν σημεῖον· ὁώστε καὶ πυραμίξ, ἧξ βάσις μὲν τὸ ΑΤΒΥΓΦΔΘ, κορυφὴ δὲ τὸ Λ, πρὸς τὴν πυραμίδα, ἧξ βάσις [μὲν] τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν σημεῖον, τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ. ὑπόκειται δὲ καὶ ὁ κῶνος, οὗ βάσις [μὲν] ὁ ΑΒΓΔ κύκλος, κορυφὴ δὲ τὸ Λ σημεῖον, πρὸς τὸ Χ στερεὸν τριπλασίονα λόγονὸέθων ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ· ὐέστιν ἄρα ὡξ ὁ κῶνος, οὗ βάσις μέν ἐστιν ὁ ΑΒΓΔ κύκλος, κορυφὴ δὲ τὸ Λ, πρὸς τὸ Χ στερεόν, οὑτως ἡ πυραμίξ, ἧξ βάσις μὲν τὸ ΑΤΒΥΓΦΔΘ [πολύγωνον], κορυφὴ δὲ τὸ Λ, πρὸς τὴν πυραμίδα, ἧξ βάσις μέν ἐστι τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν· ἐναλλὰχ ἄρα, ὡξ ὁ κῶνος, οὗ βάσις μὲν ὁ ΑΒΓΔ κύκλος, κορυφὴ δὲ τὸ Λ, πρὸς τὴν ἐν αὐτῷ πυραμίδα, ἧξ βάσις μὲν τὸ ΑΤΒΥΓΦΔΘ πολύγωνον, κορυφὴ δὲ τὸ Λ, οὑτως τὸ Χ [στερεὸν] πρὸς τὴν πυραμίδα, ἧξ βάσις μέν ἐστι τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν. μείζων δὲ ὁ εἰρημένος κῶνος τῆς ἐν αὐτῷ πυραμίδος· ἐμπεριέθει γὰρ αὐτὴν. μεῖζον ἄρα καὶ τὸ Χ στερεὸν τῆς πυραμίδος, ἧξ βάσις μέν ἐστι τὸ ΕΟΖΠΗΡΘΣ πολύγωνον, κορυφὴ δὲ τὸ Ν. ἀλλὰ καὶ ὐέλαττον· ὅπερ ἐστὶν ἀδύνατον. οὐκ ἄρα ὁ κῶνος, οὗ βάσις ὁ ΑΒΓΔ κύκλος, κορυφὴ δὲ τὸ Λ [σημεῖον], πρὸςὐέλαττόν τι τοῦ κώνου στερεόν, οὗ βάσις μὲν ὁ ΕΖΗΘ κύκλος, κορυφὴ δὲ τὸ Ν σημεῖον, τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ὁ ΕΖΗΘΝ κῶνος πρὸςὐέλαττόν τι τοῦ ΑΒΓΔΛ κώνου στερεὸν τριπλασίονα λόγον ἔχει ἥπερ ἡ ΖΘ πρὸς τὴν ΒΔ.

Λέγω δή, ὅτι οὐδὲ ὁ ΑΒΓΔΛ κῶνος πρὸς μεῖζόν τι τοῦ ΕΖΗΘΝ κώνου στερεὸν τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ.

Εἰ γὰρ δυνατόν, ἐθέτω πρὸς μεῖζον τὸ Χ. ἀνάπαλιν ἄρα τὸ Χ στερεὸν πρὸς τὸν ΑΒΓΔΛ κῶνον τριπλασίονα λόγον ἔχει ἥπερ ἡ ΖΘ πρὸς τὴν ΒΔ. ὡξ δὲ τὸ Χ στερεὸν πρὸς τὸν ΑΒΓΔΛ κῶνον, οὑτως ὁ ΕΖΗΘΝ κῶνος πρὸςὐέλαττόν τι τοῦ ΑΒΓΔΛ κώνου στερεόν. καὶ ὁ ΕΖΗΘΝ ἄρα κῶνος πρὸςὐέλαττόν τι τοῦ ΑΒΓΔΛ κώνου στερεὸν τριπλασίονα λόγον ἔχει ἥπερ ἡ ΖΘ πρὸς τὴν ΒΔ· ὅπερ ἀδύνατον ἐδείθθη. οὐκ ἄρα ὁ ΑΒΓΔΛ κῶνος πρὸς μεῖζόν τι τοῦ ΕΖΗΘΝ κώνου στερεὸν τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ. ἐδείθθη δέ, ὅτι οὐδὲ πρὸςὐέλαττον. ὁ ΑΒΓΔΛ ἄρα κῶνος πρὸς τὸν ΕΖΗΘΝ κῶνον τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ.

Ὡξ δὲ ὁ κῶνος πρὸς τὸν κῶνον, ὁ κύλινδρος πρὸς τὸν κύλινδρον· τριπλάσιος γὰρ ὁ κύλινδρος τοῦ κώνου ὁ ἐπὶ τῆς αὐτῆς βάσεως τῷ κώνῳ καὶ ἰσοϋψὴς αὐτῷ. καὶ ὁ κύλινδρος ἄρα πρὸς τὸν κύλινδρον τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΔ πρὸς τὴν ΖΘ.

Οἱ ἄρα ὅμοιοι κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους ἐν τριleading (magnitudes is) to one of the following, so (the sum of) all of the leading (magnitudes is) to (the sum of) all of the following (magnitudes) [Prop. 5.12]. And, thus, as pyramid $BKTL$ (is) to pyramid $FMPN$, so the whole pyramid whose base is polygon $ATBUCVDW$, and apex the point L, (is) to the whole pyramid whose base is polygon $EPFQGRHS$, and apex the point N. And, hence, the pyramid whose base is polygon $ATBUCVDW$, and apex the point L, has to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N, the cubed ratio that BD (has) to FH. And it was also assumed that the cone whose base is circle $ABCD$, and apex the point L, has to solid O the cubed ratio that BD (has) to FH. Thus, as the cone whose base is circle $ABCD$, and apex the point L, is to solid O, so the pyramid whose base (is) [polygon] $ATBUCVDW$, and apex the point L, (is) to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N. Thus, alternately, as the cone whose base (is) circle $ABCD$, and apex the point L, (is) to the pyramid within it whose base (is) the polygon $ATBUCVDW$, and apex the point L, so the [solid] O (is) to the pyramid whose base is polygon $EPFQGRHS$, and apex the point N [Prop. 5.16]. And the aforementioned cone (is) greater than the pyramid within it. For it encompasses it. Thus, solid O (is) also greater than the pyramid whose base is polygon $EPFQGRHS$, and apex the point N. But, (it is) also less. The very thing is impossible. Thus, the cone whose base (is) circle $ABCD$, and apex the [point] L, does not have to some solid less than the cone whose base (is) circle $EFGH$, and apex the point N, the cubed ratio that BD (has) to EH. So, similarly, we can show that neither does cone $EFGHN$ have to some solid less than cone $ABCDL$ the cubed ratio that FH (has) to BD.

So, I say that neither does cone $ABCDL$ have to some solid greater than cone $EFGHN$ the cubed ratio that BD (has) to FH.

For, if possible, let it have (such a ratio) to a greater (solid), O. Thus, inversely, solid O has to cone $ABCDL$ the cubed ratio that FH (has) to BD [Prop. 5.7 corr.]. And as solid O (is) to cone $ABCDL$, so cone $EFGHN$ (is) to some solid less than cone $ABCDL$ [12.2 lem.]. Thus, cone $EFGHN$ also has to some solid less than cone $ABCDL$ the cubed ratio that FH (has) to BD. The very thing was shown (to be) impossible. Thus, cone $ABCDL$ does not have to some solid greater than cone $EFGHN$ the cubed ratio than BD (has) to FH. And it was shown that neither (does it have such a ratio) to a lesser (solid). Thus, cone $ABCDL$ has to cone $EFGHN$ the cubed ratio that BD (has) to FG.

And as the cone (is) to the cone, so the cylinder (is) to the cylinder. For a cylinder is three times a cone on the same base as the cone, and of the same height as it [Prop. 12.10]. Thus, the cylinder also has to the cylinder the cubed ratio that BD (has) to FH.

πλασίονι λόγῳ εἰσὶ τῶν ἐν ταῖς βάσεσι διαμέτρων· ὅπερ ἔδει δεῖξαι.

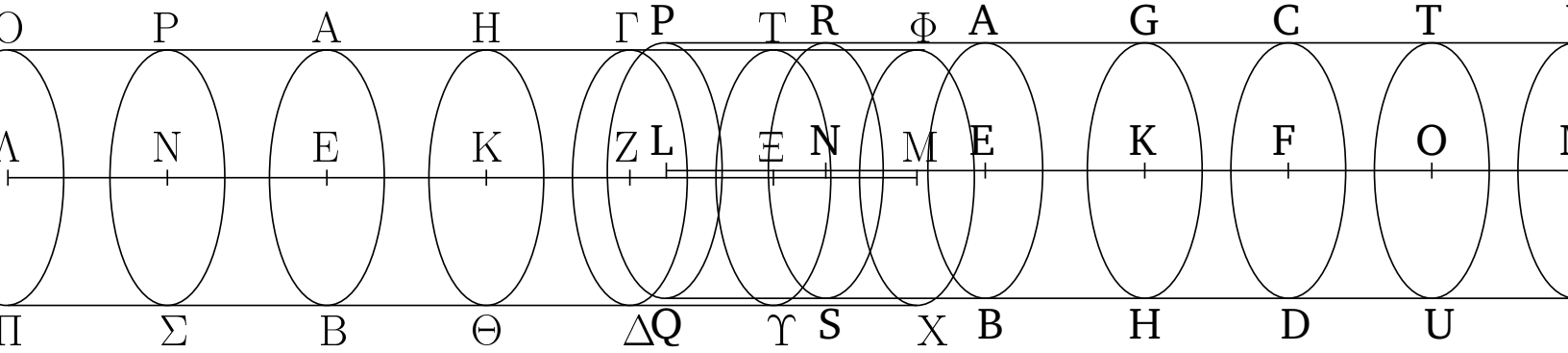
Thus, similar cones and cylinders are in the cubed ratio of the diameters of their bases. (Which is) the very thing it was required to show.

13.

Ἐὰν κύλινδρος ἐπιπέδῳ τμ ηθῇ παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ὀέσται ὡξ ὁ κύλινδρος πρὸς τὸν κύλινδρον, οὑτως ὁ ὀάχων πρὸς τὸνὀάχονα.

Proposition 13

If a cylinder is cut by a plane which is parallel to the opposite planes (of the cylinder) then as the cylinder (is) to the cylinder, so the axis will be to the axis.



Κύλινδρος γὰρ ὁ ΑΔ ἐπιπέδῳ τῷ ΗΘ τετμήσθω παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις τοῖς ΑΒ, ΓΔ, καὶ συμβαλλέτω τῷ ὀάχονι τὸ ΗΘ ἐπίπεδον κατὰ τὸ Κ σημεῖον· λέγω, ὅτι ἐστὶν ὡξ ὁ ΒΗ κύλινδρος πρὸς τὸν ΗΔ κύλινδρον, οὑτως ὁ ΕΚὀάχων πρὸς τὸν ΚΖὀάχονα.

Ἐκβεβλήσθω γὰρ ὁ ΕΖὀάχων ἐφ' ἑκάτερα τὰ μέρη ἐπὶ τὰ Λ, Μ σημεῖα, καὶ ἐκκείσθωσαν τῷ ΕΚὀάχονιὀίσοι ὁσοιδηποτοῦν οἱ ΕΝ, ΝΛ, τῷ δὲ ΖΚὀίσοι ὁσοιδηποτοῦν οἱ ΖΧ, ΧΜ, καὶ νοείσθω ὁ ἐπὶ τοῦ ΛΜὀάχονος κύλινδρος ὁ ΟΘ, οὗ βάσεις οἱ ΟΠ, ΦΘ κύκλοι. καὶ ἐκβεβλήσθω διὰ τῶν Ν, Χ σημείων ἐπίπεδα παράλληλα τοῖς ΑΒ, ΓΔ καὶ ταῖς βάσεσι τοῦ ΟΘ κυλίνδρου καὶ ποιείτωσαν τοὐξ ΡΣ, ΤΥ κύκλους περὶ τὰ Ν, Χ κέντρα. καὶ ἐπεὶ οἱ ΛΝ, ΝΕ, ΕΚὀάχονεςὀίσοι εἰσὶν ἀλλήλοις, οἱ ἄρα ΠΡ, ΡΒ, ΒΗ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡξ αἱ βάσεις. ἴσαι δέ εἰσιν αἱ βάσεις· ὀίσοι ἄρα καὶ οἱ ΠΡ, ΡΒ, ΒΗ κύλινδροι ἀλλήλοις. ἐπεὶ οὐῦν οἱ ΛΝ, ΝΕ, ΕΚὀάχονεςὀίσοι εἰσὶν ἀλλήλοις, εἰσὶ δὲ καὶ οἱ ΠΡ, ΡΒ, ΒΗ κύλινδροιὀίσοι ἀλλήλοις, καί ἐστιν ἴσον τὸ πλῆθος τῷ πλήθει, ὁσαπλασίων ἄρα ὁ ΚΛὀάχων τοῦ ΕΚὀάχονος, τοσαυταπλασίωνὀέσται καὶ ὁ ΠΗ κύλινδρος τοῦ ΗΒ κυλίνδρου. διὰ τὰ αὐτὰ δὴ καὶ ὁσαπλασίων ἐστὶν ὁ ΜΚὀάχων τοῦ ΚΖὀάχονος, τοσαυταπλασίων ἐστὶ καὶ ὁ ΘΗ κύλινδρος τοῦ ΗΔ κυλίνδρου. καὶ εἰ μὲνὀίσος ἐστὶν ὁ ΚΛὀάχων τῷ ΚΜὀάχονι, ὀίσοςὀέσται καὶ ὁ ΠΗ κύλινδρος τῷ ΗΘ κυλίνδρῳ, εἰ δὲ μείζων ὁ ὀάχων τοῦ ὀάχονος, μείζων καὶ ὁ κύλινδρος τοῦ κυλίνδρου, καὶ εἰ ἐλάσσων, ἐλάσσων. τεσσάρων δὴ μεγεθῶν ὄντων, ἀχόνων μὲν τῶν ΕΚ, ΚΖ, κυλίνδρων δὲ τῶν ΒΗ, ΗΔ, εὐίληπται ἰσάκις πολλαπλάσια, τοῦ μὲν ΕΚὀάχονος καὶ τοῦ ΒΗ κυλίνδρου ὅ τε ΛΚὀάχων καὶ ὁ ΠΗ κύλινδρος, τοῦ δὲ ΚΖὀάχονες καὶ τοῦ ΗΔ κυλίνδρου ὅ τε ΚΜὀάχων καὶ ὁ ΗΘ κύλινδρος, καὶ δέδεικται, ὅτι εἰ ὑπερέθει ὁ ΚΛὀάχων τοῦ ΚΜὀάχονος, ὑπερέθει καὶ ὁ ΠΗ κύλινδρος τοῦ ΗΘ κυλίνδρου,

For let the cylinder AD be cut by the plane GH which is parallel to the opposite planes (of the cylinder), AB and CD. And let the plane GH have met the axis at point K. I say that as cylinder BG is to cylinder GD, so axis EK (is) to axis KF.

For let axis EF be produced in each direction to points L and M. And let any number whatsoever (of lengths), EN and NL, equal to axis EK, be set out (on the axis EL), and any number whatsoever (of lengths), FO and OM, equal to (axis) FK, (on the axis KM). And let the cylinder PW, whose bases (are) the circles PQ and VW, be conceived on axis LM. And let planes parallel to AB, CD, and the bases of cylinder PW, be produced through points N and O, and let them have made the circles RS and TU around the centers N and O (respectively). And since axes LN, NE, and EK are equal to one another, the cylinders QR, RB, and BG are to one another as their bases [Prop. 12.11]. But the bases are equal. Thus, the cylinders QR, RB, and BG (are) also equal to one another. Therefore, since the axes LN, NE, and EK are equal to one another, and the cylinders QR, RB, and BG are also equal to one another, and the number (of the former) is equal to the number (of the latter), thus as many multiples as axis KL is of axis EK, so many multiples is cylinder QG also of cylinder GB. And so, for the same (reasons), as many multiples as axis MK is of axis KF, so many multiples is cylinder WG also of cylinder GD. And if axis KL is equal to axis KM then cylinder QG will also be equal to cylinder GW, and if the axis (is) greater than the axis then the cylinder (will also be) greater than the cylinder, and if (the axis is) less then (the cylinder will also be) less. So, there are four magnitudes—the axes EK and KF, and the cylinders BG and GD—and equal multiples be taken of axis EK and cylinder BG—(namely), axis LK and cylinder

καὶ εἰ ὁίσος, ὁίσος, καὶ εἰ ἐλάσσων, ἐλάσσων. ὁέστιν ἄρα ὡξ ὁ ΕΚὁάχων πρὸς τὸν ΚΖὁάχονα, οὁύτως ὁ ΒΗ κύλινδρος πρὸς τὸν ΗΔ κύλινδρον· ὅπερ ἔδει δεῖχαι.

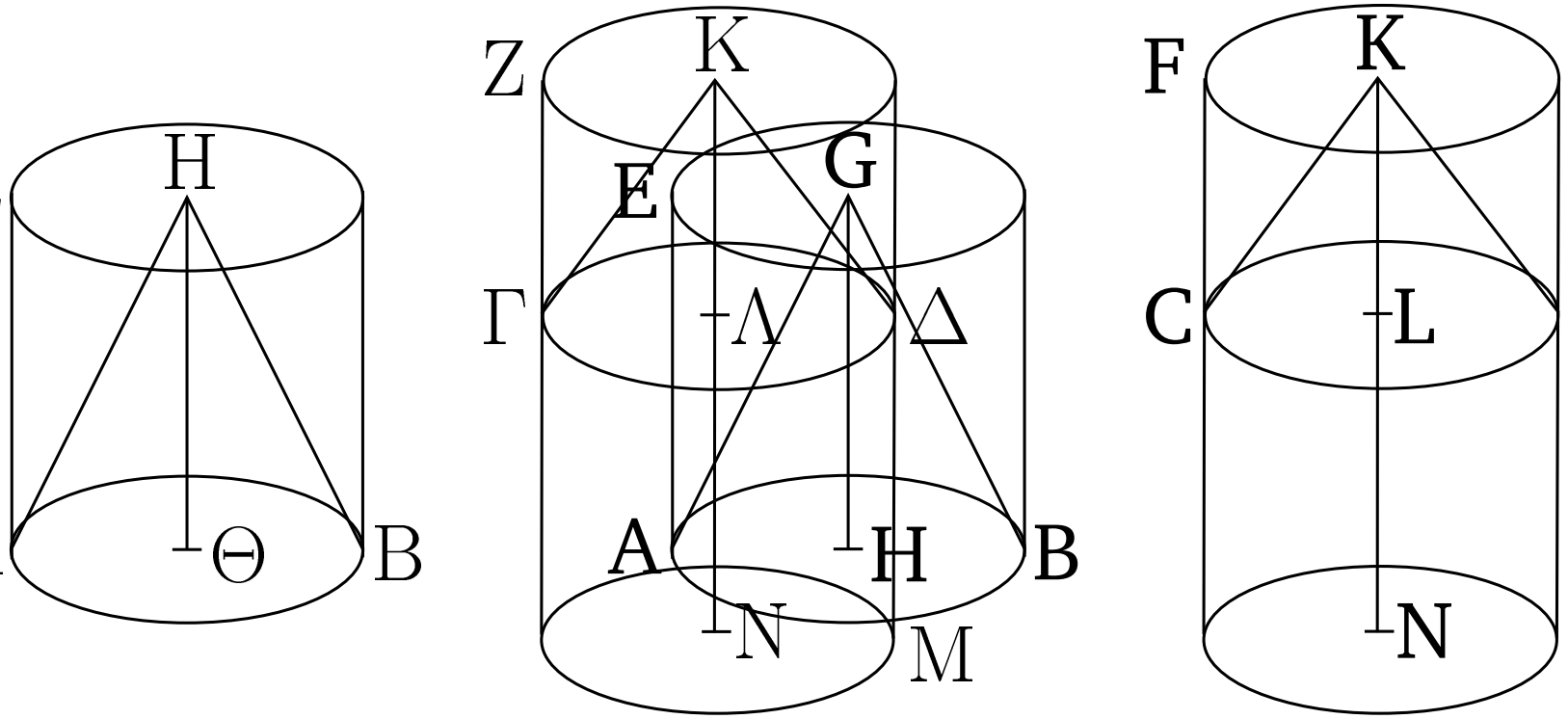
QG—and of axis *KF* and cylinder *GD*—(namely), axis *KM* and cylinder *GW*. And it has been shown that if axis *KL* exceeds axis *KM* then cylinder *QG* also exceeds cylinder *GW*, and if (the axes are) equal then (the cylinders are) equal, and if (*KL* is) less then (*QG* is) less. Thus, as axis *EK* is to axis *KF*, so cylinder *BG* (is) to cylinder *GD* [Def. 5.5]. (Which is) the very thing it was required to show.

14.

Οἱ ἐπὶ ὁίσων βάσεων ὄντες κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡξ τὰ ὁύψη.

Proposition 14

Cones and cylinders which are on equal bases are to one another as their heights.



Ἔστωσαν γὰρ ἐπὶ ὁίσων βάσεων τῶν ΑΒ, ΓΔ κύκλων κύλινδροι οἱ ΕΒ, ΖΔ· λέγω, ὅτι ἐστὶν ὡξ ὁ ΕΒ κύλινδρος πρὸς τὸν ΖΔ κύλινδρον, οὁύτως ὁ ΗΘὁάχων πρὸς τὸν ΚΛὁάχονα.

Ἐκβεβλήσθω γὰρ ὁ ΚΛὁάχων ἐπὶ τὸ Ν σημεῖον, καὶ κείσθω τῷ ΗΘὁάχονιὁίσος ὁ ΛΝ, καὶ περὶ ὁάχονα τὸν ΛΝ κύλινδρος νενοήσθω ὁ ΓΜ. ἐπεὶ οὁῦν οἱ ΕΒ, ΓΜ κύλινδροι ὑπὸ τὸ αὐτὸ ὁύψος εἰσίν, πρὸς ἀλλήλους εἰσὶν ὡξ αἱ βάσεις. ἴσαι δέ εἰσίν αἱ βάσεις ἀλλήλαις· ὁίσοι ἄρα εἰσὶ καὶ οἱ ΕΒ, ΓΜ κύλινδροι. καὶ ἐπεὶ κύλινδρος ὁ ΖΜ ἐπιπέδῳ τέτμ ηται τῷ ΓΔ παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις, ὁέστιν ἄρα ὡξ ὁ ΓΜ κύλινδρος πρὸς τὸν ΖΔ κύλινδρον, οὁύτως ὁ ΛΝὁάχων πρὸς τὸν ΚΛὁάχονα. ὁίσος δέ ἐστιν ὁ μὲν ΓΜ κύλινδρος τῷ ΕΒ κυλίνδρῳ, ὁ δὲ ΛΝὁάχων τῷ ΗΘὁάχονι· ὁέστιν ἄρα ὡξ ὁ ΕΒ κύλινδρος πρὸς τὸν ΖΔ κύλινδρον, οὁύτως ὁ ΗΘὁάχων πρὸς τὸν ΚΛὁάχονα. ὡξ δὲ ὁ ΕΒ κύλινδρος πρὸς τὸν ΖΔ

For let *EB* and *FD* be cylinders on equal bases, (namely) the circles *AB* and *CD* (respectively). I say that as cylinder *EB* is to cylinder *FD*, so axis *GH* (is) to axis *KL*.

For let the axis *KL* be produced to point *N*. And let *LN* be made equal to axis *GH*. And let the cylinder *CM* be conceived about axis *LN*. Therefore, since cylinders *EB* and *CM* have the same height they are to one another as their bases [Prop. 12.11]. And the bases are equal to one another. Thus, cylinders *EB* and *CM* are also equal to one another. And since cylinder *FM* has been cut by the plane *CD*, which is parallel to its opposite planes, thus as cylinder *CM* is to cylinder *FD*, so axis *LN* (is) to axis *KL* [Prop. 12.13]. And cylinder *CM* is equal to cylinder *EB*, and axis *LN* to axis *GH*. Thus, as cylinder *EB* is to cylinder *FD*, so axis *GH* (is) to axis *KL*. And as cylinder *EB* (is) to cylinder *FD*, so cone *ABG* (is) to cone

κύλινδρον, οὕτως ὁ ΑΒΗ κῶνος πρὸς τὸν ΓΔΚ κῶνον. καὶ ὡξ ἄρα ὁ ΗΘὀάχων πρὸς τὸν ΚΛὀάχονα, οὕτως ὁ ΑΒΗ κῶνος πρὸς τὸν ΓΔΚ κῶνον καὶ ὁ ΕΒ κύλινδρος πρὸς τὸν ΖΔ κύλινδρον· ὅπερ ἔδει δεῖχαι.

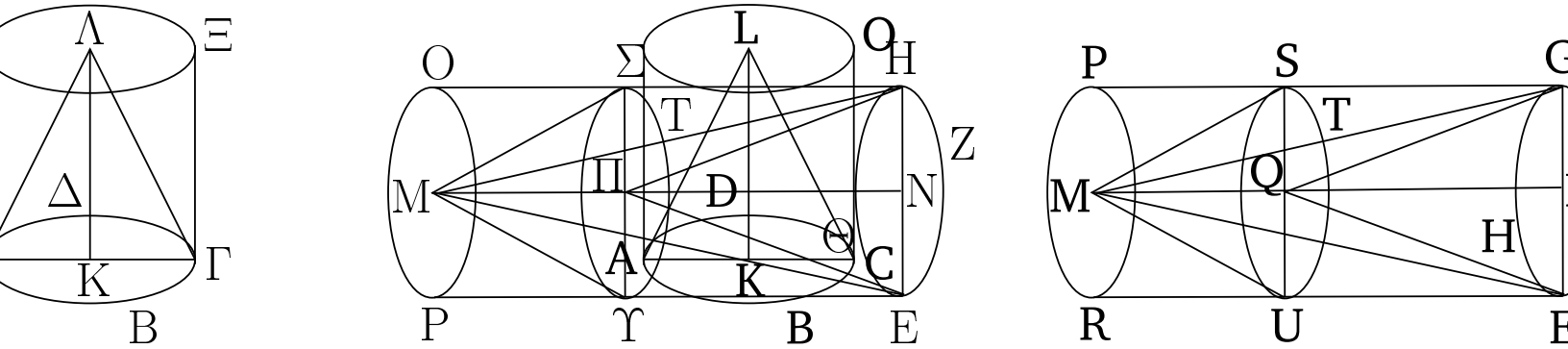
CDK [Prop. 12.10]. Thus, also, as axis *GH* (is) to axis *KL*, so cone *ABG* (is) to cone *CDK*, and cylinder *EB* to cylinder *FD*. (Which is) the very thing it was required to show.

15.

Τῶνὀίσων κώνων καὶ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν· καὶ ὧν κώνων καὶ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν, ὁίσοι εἰσὶν ἐκεῖνοι.

Proposition 15

The bases of equal cones and cylinders are reciprocally proportional to their heights. And, those cones and cylinders whose bases (are) reciprocally proportional to their heights are equal.



Ἔστωσανὁίσοι κῶνοι καὶ κύλινδροι, ὧν βάσεις μὲν οἱ ΑΒΓΔ, ΕΖΗΘ κύκλοι, διάμετροι δὲ αὐτῶν αἱ ΑΓ, ΕΗ, ὁάχονες δὲ οἱ ΚΛ, ΜΝ, οὁίτινες καὶ ὑψη εἰσὶ τῶν κώνωνὁή κυλίνδρων, καὶ συμπεπληρώσθωσαν οἱ ΑΧ, ΕΟ κύλινδροι. λέγω, ὅτι τῶν ΑΧ, ΕΟ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν, καί ἐστιν ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὕτως τὸ ΜΝ ὕψος πρὸς τὸ ΚΛ ὕψος.

Τὸ γὰρ ΛΚ ὕψος τῷ ΜΝ ὕψει ἤτοι ἴσον ἐστὶνὁή οὐ. ὁέστω πρότερον ἴσον. ὁέστι δὲ καὶ ὁ ΑΧ κύλινδρος τῷ ΕΟ κυλίνδρῳ ὁίσος. οἱ δὲ ὑπὸ τὸ αὐτὸ ὕψος ὄντες κῶνοι καὶ κύλινδροι πρὸς ἀλλήλους εἰσὶν ὡξ αἱ βάσεις· ὁίση ἄρα καὶ ἡ ΑΒΓΔ βάσις τῇ ΕΖΗΘ βάσει. ὥστε καὶ ἀντιπέπονθεν, ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὕτως τὸ ΜΝ ὕψος πρὸς τὸ ΚΛ ὕψος. ἀλλὰ δὴ μὴ ὁέστω τὸ ΛΚ ὕψος τῷ ΜΝ ἴσον, ἀλλὸ ὁέστω μεῖζον τὸ ΜΝ, καὶ ἀφῃρήσθω ἀπὸ τοῦ ΜΝ ὕψους τῷ ΚΛ ἴσον τὸ ΠΝ, καὶ διὰ τοῦ Π σημείου τετμήσθω ὁ ΕΟ κύλινδρος ἐπιπέδῳ τῷ ΤΥΣ παραλλήλῳ τοῖς τῶν ΕΖΗΘ, ΡΟ κύκλων ἐπιπέδοις, καὶ ἀπὸ βάσεως μὲν τοῦ ΕΖΗΘ κύκλου, ὕψους δὲ τοῦ ΝΠ κύλινδρος νενοήσθω ὁ ΕΣ. καί ἐπεὶ ὁίσος ἐστὶν ὁ ΑΧ κύλινδρος τῷ ΕΟ κυλίνδρῳ, ὁέστιν ἄρα ὡξ ὁ ΑΧ κύλινδρος πρὸς τὸν ΕΣ κυλίνδρον, οὕτως ὁ ΕΟ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον. ἀλλὸ ὡξ μὲν ὁ ΑΧ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον, οὕτως ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ· ὑπὸ γὰρ τὸ αὐτὸ ὕψος εἰσὶν οἱ ΑΧ, ΕΣ κύλινδροι· ὡξ δὲ ὁ ΕΟ κύλινδρος πρὸς τὸν ΕΣ, οὕτως τὸ ΜΝ ὕψος πρὸς τὸ ΠΝ ὕψος· ὁ γὰρ ΕΟ κύλινδρος ἐπιπέδῳ τέτμ ηται παραλλήλῳ ὄντι τοῖς ἀπεναντίον ἐπιπέδοις. ὁέστιν ἄρα καὶ ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὕτως τὸ ΜΝ ὕψος πρὸς τὸ ΠΝ ὕψος. ἴσον δὲ τὸ ΠΝ ὕψος τῷ ΚΛ

Let there be equal cones and cylinders whose bases are the circles *ABCD* and *EFGH*, and the diameters of (the bases) *AC* and *EG*, and (whose) axes (are) *KL* and *MN*, which are also the heights of the cones and cylinders (respectively). And let the cylinders *AO* and *EP* be completed. I say that the bases of cylinders *AO* and *EP* are reciprocally proportional to their heights, and (so) as base *ABCD* is to base *EFGH*, so height *MN* (is) to height *KL*.

For height *LK* is either equal to height *MN*, or not. Let it, first of all, be equal. And cylinder *AO* is also equal to cylinder *EP*. And cones and cylinders having the same height are to one another as their bases [Prop. 12.11]. Thus, base *ABCD* (is) also equal to base *EFGH*. And, hence, reciprocally, as base *ABCD* (is) to base *EFGH*, so height *MN* (is) to height *KL*. And so, let height *LK* not be equal to *MN*, but let *MN* be greater. And let *QN*, equal to *KL*, be cut off from height *MN*. And let the cylinder *EP* be cut, through point *Q*, by the plane *TUS* (which is) parallel to the planes of the circles *EFGH* and *RP*. And let cylinder *ES* be conceived, with base the circle *EFGH*, and height *NQ*. And since cylinder *AO* is equal to cylinder *EP*, thus, as cylinder *AO* (is) to cylinder *ES*, so cylinder *EP* (is) to cylinder *ES* [Prop. 5.7]. But, as cylinder *AO* (is) to cylinder *ES*, so base *ABCD* (is) to base *EFGH*. For cylinders *AO* and *ES* (have) the same height [Prop. 12.11]. And as cylinder *EP* (is) to (cylinder) *ES*, so height *MN* (is) to height *QN*. For cylinder *EP* has been cut by a plane which is parallel to its opposite planes [Prop. 12.13]. And, thus, as base *ABCD* is to base *EFGH*, so height *MN* (is) to height *QN* [Prop. 5.11]. And height *QN* (is) equal to height *KL*. Thus,

ὑψει· ὐέστιν ἄρα ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὑτως τὸ ΜΝ ὑψος πρὸς τὸ ΚΛ ὑψος. τῶν ἄρα ΑΧ, ΕΟ κυλίνδρων ἀντιπεπόνθασιν αἱ βάσεις τοῖς ὑψεσιν.

Ἀλλὰ δὴ τῶν ΑΧ, ΕΟ κυλίνδρων ἀντιπεπονθέτωσαν αἱ βάσεις τοῖς ὑψεσιν, καὶ ὐέστω ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὑτως τὸ ΜΝ ὑψος πρὸς τὸ ΚΛ ὑψος· λέγω, ὅτιὐίσος ἐστὶν ὁ ΑΧ κύλινδρος τῷ ΕΟ κυλίνδρῳ.

Τῶν γὰρ αὐτῶν κατασκευασθέντων ἐπεί ἐστιν ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὑτως τὸ ΜΝ ὑψος πρὸς τὸ ΚΛ ὑψος, ἴσον δὲ τὸ ΚΛ ὑψος τῷ ΠΝ ὑψει, ὐέσται ἄρα ὡξ ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὑτως τὸ ΜΝ ὑψος πρὸς τὸ ΠΝ ὑψος. ἀλλὰ ὡξ μὲν ἡ ΑΒΓΔ βάσις πρὸς τὴν ΕΖΗΘ βάσιν, οὑτως ὁ ΑΧ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον· ὑπὸ γὰρ τὸ αὐτὸ ὑψος εἰσίν· ὡξ δὲ τὸ ΜΝ ὑψος πρὸς τὸ ΠΝ [ὑψος], οὑτως ὁ ΕΟ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον· ὐέστιν ἄρα ὡξ ὁ ΑΧ κύλινδρος πρὸς τὸν ΕΣ κύλινδρον, οὑτως ὁ ΕΟ κύλινδρος πρὸς τὸν ΕΣ. ὐίσος ἄρα ὁ ΑΧ κύλινδρος τῷ ΕΟ κυλίνδρῳ. ὡσαύτως δὲ καὶ ἐπὶ τῶν κώνων· ὅπερ ἔδει δεῖχαι.

as base $ABCD$ is to base $EFGH$, so height MN (is) to height KL. Thus, the bases of cylinders AO and EP are reciprocally proportional to their heights.

And, so, let the bases of cylinders AO and EP be reciprocally proportional to their heights, and (thus) let base $ABCD$ be to base $EFGH$, as height MN (is) to height KL. I say that cylinder AO is equal to cylinder EP.

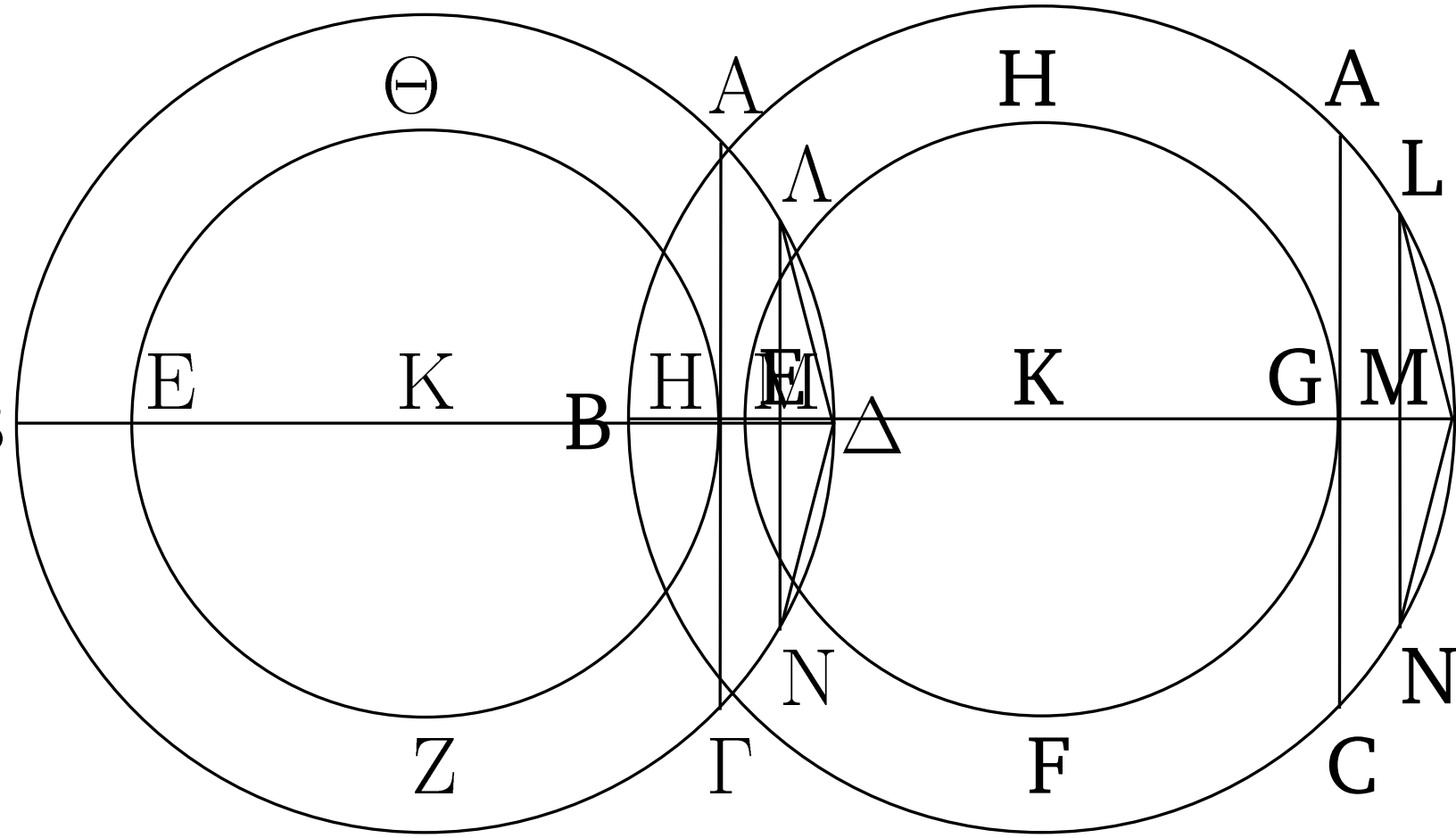
For, with the same construction, since as base $ABCD$ is to base $EFGH$, so height MN (is) to height KL, and height KL (is) equal to height QN, thus, as base $ABCD$ (is) to base $EFGH$, so height MN will be to height QN. But, as base $ABCD$ (is) to base $EFGH$, so cylinder AO (is) to cylinder ES. For they are the same height [Prop. 12.11]. And as height MN (is) to [height] QN, so cylinder EP (is) to cylinder ES [Prop. 12.13]. Thus, as cylinder AO is to cylinder ES, so cylinder EP (is) to (cylinder) ES [Prop. 5.11]. Thus, cylinder AO (is) equal to cylinder EP [Prop. 5.9]. In the same manner, (the proposition can) also (be demonstrated) for the cones. (Which is) the very thing it was required to show.

16.

Δύο κύκλων περὶ τὸ αὐτὸ κέντρον ὄντων εἰξ τὸν μείζονα κύκλον πολύγωνον ἰσόπλευρόν τε καὶ ἀρτιόπλευρον ἐγγράψαι μὴ ψαῦον τοῦ ἐλάσσονος κύκλου.

Proposition 16

There being two circles about the same center, to inscribe an equilateral and even-sided polygon in the greater circle, not touching the lesser circle.



Ἔστωσαν οἱ δοθέντες δύο κύκλοι οἱ ΑΒΓΔ, ΕΖΗΘ περὶ τὸ αὐτὸ κέντρον τὸ Κ· δεῖ δὴ εἰς τὸν μείζονα κύκλον τὸν ΑΒΓΔ πολύγωνον ἰσόπλευρόν τε καὶ ἀρτιόπλευρον ἐγγράψαι μὴ ψαῦον τοῦ ΕΖΗΘ κύκλου.

ὈΗθθω γὰρ διὰ τοῦ Κ κέντρου εὐθεῖα ἡ ΒΚΔ, καὶ ἀπὸ τοῦ Η σημείου τῇ ΒΔ εὐθείᾳ πρὸς ὀρθὰξ ἤθθω ἡ ΗΑ καὶ διήθθω ἐπὶ τὸ Γ· ἡ ΑΓ ἄρα ἐφάπτεται τοῦ ΕΖΗΘ κύκλου. τέμνοντες δὴ τὴν ΒΑΔ περιφέρειαν δίθα καὶ τὴν ἡμίσειαν αὐτῆς δίθα καὶ τοῦτο ἀεὶ ποιοῦντες καταλείψομεν περιφέρειαν ἐλάσσονα τῆς ΑΔ. λελείφθω, καὶ ὀέστω ἡ ΛΔ, καὶ ἀπὸ τοῦ Λ ἐπὶ τὴν ΒΔ κάθετος ἤθθω ἡ ΛΜ καὶ διήθθω ἐπὶ τὸ Ν, καὶ ἐπεζεύθθωσαν αἱ ΛΔ, ΔΝ· ὁίση ἄρα ἐστὶν ἡ ΛΔ τῇ ΔΝ. καὶ ἐπεὶ παράλληλόξ ἐστιν ἡ ΛΝ τῇ ΑΓ, ἡ δὲ ΑΓ ἐφάπτεται τοῦ ΕΖΗΘ κύκλου, ἡ ΛΝ ἄρα οὐκ ἐφάπτεται τοῦ ΕΖΗΘ κύκλου· πολλῷ ἄρα αἱ ΛΔ, ΔΝ οὐκ ἐφάπτονται τοῦ ΕΖΗΘ κύκλου. ἐαν δὴ τῇ ΛΔ εὐθείᾳ ἴσας κατὰ τὸ συνεθὲξ ἐναρμόσωμεν εἰξ τὸν ΑΒΓΔ κύκλον, ἐγγραφήσεται εἰξ τὸν ΑΒΓΔ κύκλον πολύγωνον ἰσόπλευρόν τε καὶ ἀρτιόπλευρον μὴ ψαῦον τοῦ ἐλάσσονος κύκλου τοῦ ΕΖΗΘ· ὅπερ ἔδει ποιῆσαι.

Let *ABCD* and *EFGH* be the given two circles, about the same center, *K*. So, it is necessary to inscribe an equilateral and even-sided polygon in the greater circle *ABCD*, not touching circle *EFGH*.

Let the straight-line *BKD* be drawn through the center *K*. And let *GA* be drawn, at right-angles to the straight-line *BD*, through point *G*, and let it be drawn through to *C*. Thus, *AC* touches circle *EFGH* [Prop. 3.16 corr.]. So, (by) cutting circumference *BAD* in half, and the half of it in half, and doing this continually, we will (eventually) leave a circumference less than *AD* [Prop. 10.1]. Let it be left, and let it be *LD*. And let *LM* be drawn, from *L*, perpendicular to *BD*, and let it be drawn through to *N*. And let *LD* and *DN* be joined. Thus, *LD* is equal to *DN* [Props. 3.3, 1.4]. And since *LN* is parallel to *AC* [Prop. 1.28], and *AC* touches circle *EFGH*, *LN* thus does not touch circle *EFGH*. Thus, even more so, *LD* and *DN* do not touch circle *EFGH*. And if we continuously insert (straight-lines) equal to straight-line *LD* into circle *ABCD* [Prop. 4.1] then an equilateral and even-sided polygon, not touching the lesser circle *EFGH*, will be inscribed in circle *ABCD*.† (Which is) the very thing it was required to do.

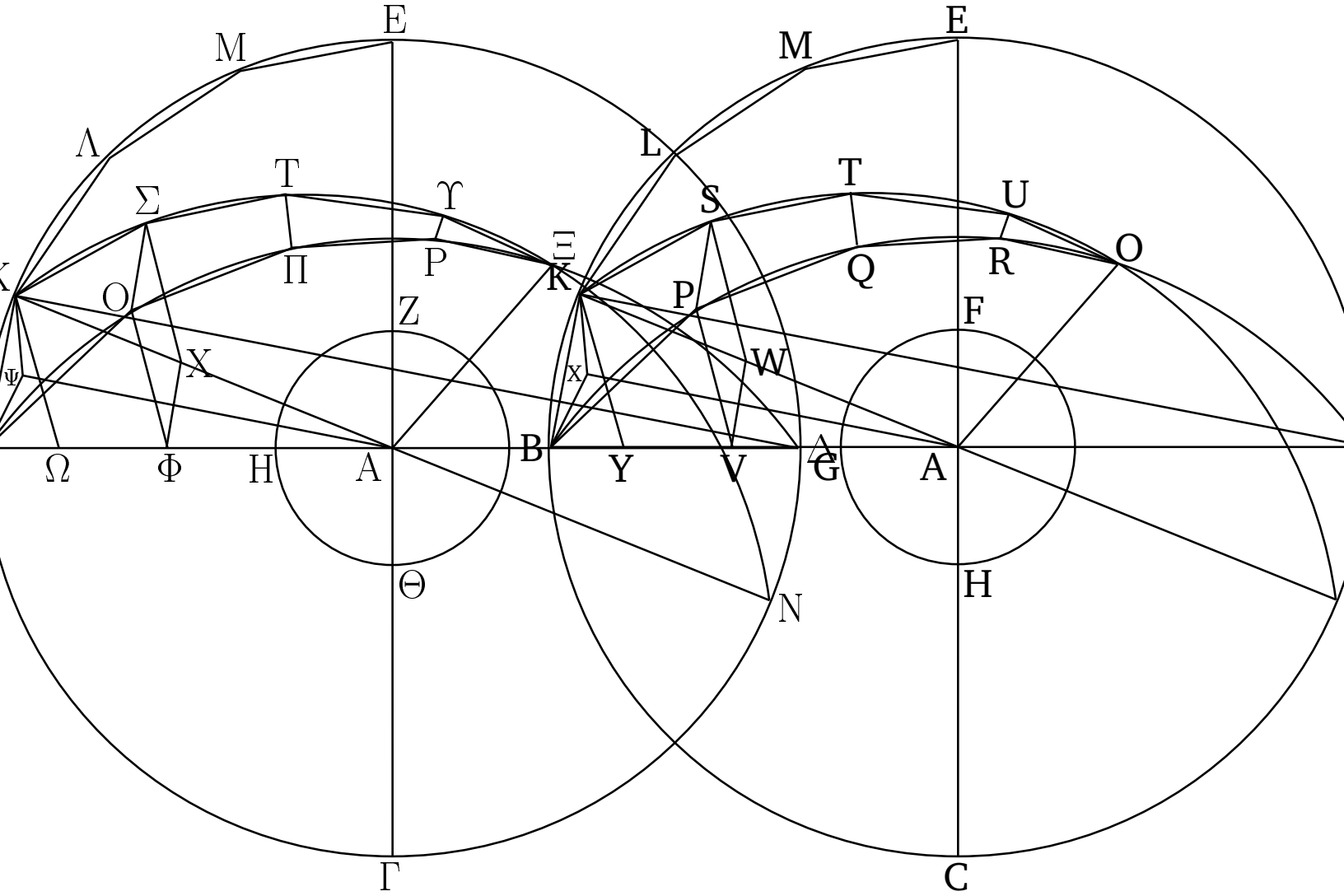
† Note that the chord of the polygon, *LN*, does not touch the inner circle either.

17.

Δύο σφαιρῶν περὶ τὸ αὐτὸ κέντρον οὐσῶν εἰξ τὴν μείζονα σφαῖραν στερεὸν πολύεδρον ἐγγράψαι μὴ ψαῦον τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν.

Proposition 17

There being two spheres about the same center, to inscribe a polyhedral solid in the greater sphere, not touching the lesser sphere on its surface.



Νενοήσθωσαν δύο σφαῖραι περὶ τὸ αὐτὸ κέντρον τὸ A· δεῖ δὴ εἰξ τὴν μείζονα σφαῖραν στερεὸν πολύεδρον ἐγγράψαι μὴ ψαῦον τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν.

Τετμήσθωσαν αἱ σφαῖραι ἐπιπέδῳ τινὶ διὰ τοῦ κέντρου· ὀέσονται δὴ αἱ τομαὶ κύκλοι, ἐπειδήπερ μενούσης τῆς διαμέτρου καὶ περιφερομένου τοῦ ἡμικυκλίου ἐγιγνετο ἡ σφαῖρα· ὥστε καὶ καθὸ οὐίαςὸὰν θέσεως ἐπινοήσωμεν τὸ ἡμικύκλιον, τὸ διὰ αὐτοῦ ἐκβαλλόμενον ἐπίπεδον ποιήσει ἐπὶ τῆς ἐπιφανείας τῆς σφαίρας κύκλον. καὶ φανερόν, ὅτι καὶ μέγιστον, ἐπειδήπερ ἡ διάμετρος τῆς σφαίρας, ἥτις ἐστὶ καὶ τοῦ ἡμι-

Let two spheres be conceived about the same center, *A*. So, it is necessary to inscribe a polyhedral solid in the greater sphere, not touching the lesser sphere on its surface.

Let the spheres be cut by some plane through the center. So, the sections will be circles, inasmuch as a sphere is generated by the diameter remaining behind, and a semi-circle being carried around [Def. 11.14]. And, hence, whatever position we conceive (of for) the semi-circle, the plane produced through it will make a circle on the surface of the sphere. And (it is) clear that (it is) also a great (circle), inasmuch as the diameter of

κυκλίου διάμετρος δηλαδὴ καὶ τοῦ κύκλου, μείζων ἐστὶ πασῶν τῶν εἰξ τὸν κύκλονὁὴ τὴν σφαῖραν διαγομένων [εὐθειῶν]. ὀέστω οὸῦν ἐν μὲν τῇ μείζονι σφαίρᾳ κύκλος ὁ ΒΓΔΕ, ἐν δὲ τῇ ἐλάσσονι σφαίρᾳ κύκλος ὁ ΖΗΘ, καὶ ἤθθωσαν αὐτῶν δύο διάμετροι πρὸς ὀρθὰξ ἀλλήλαις αἱ ΒΔ, ΓΕ, καὶ δύο κύκλων περὶ τὸ αὐτὸ κέντρον ὄντων τῶν ΒΓΔΕ, ΖΗΘ εἰξ τὸν μείζονα κύκλον τὸν ΒΓΔΕ πολύγωνον ἰσόπλευρον καὶ ἀρτιόπλευρον ἐγγεγράφθω μὴ ψαῦον τοῦ ἐλάσσονος κύκλου τοῦ ΖΗΘ, οὗ πλευραὶ ὀέστωσαν ἐν τῷ ΒΕ τεταρτημορίῳ αἱ ΒΚ, ΚΛ, ΛΜ, ΜΕ, καὶ ἐπιζευθθεῖσα ἡ ΚΑ διήθθω ἐπὶ τὸ Ν, καὶ ἀνεστάτω ἀπὸ τοῦ Α σημείου τῷ τοῦ ΒΓΔΕ κύκλου ἐπιπέδῳ πρὸς ὀρθὰξ ἡ ΑΧ καὶ συμβαλλέτω τῇ ἐπιφανείᾳ τῆς σφαίρας κατὰ τὸ Χ, καὶ διὰ τῆς ΑΧ καὶ ἑκατέρας τῶν ΒΔ, ΚΝ ἐπίπεδα ἐκβεβλήσθω· ποιήσουσι δὴ διὰ τὰ εἰρημένα ἐπὶ τῆς ἐπιφανείας τῆς σφαίρας μεγίστους κύκλους. ποιείτωσαν, ὁῶν ἡμικύκλιαὀέστω ἐπὶ τῶν ΒΔ, ΚΝ διαμέτρων τὰ ΒΧΔ, ΚΧΝ. καὶ ἐπεὶ ἡ ΧΑ ὀρθή ἐστι πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον, καὶ πάντα ἄρα τὰ διὰ τῆς ΧΑ ἐπίπεδά ἐστιν ὀρθὰ πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον· ὁώστε καὶ τὰ ΒΧΔ, ΚΧΝ ἡμικύκλια ὀρθά ἐστι πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον. καὶ ἐπεὶ ἴσα ἐστὶ τὰ ΒΕΔ, ΒΧΔ, ΚΧΝ ἡμικύκλια· ἐπὶ γὰρὸίσων εἰσὶ διαμέτρων τῶν ΒΔ, ΚΝ· ἴσα ἐστὶ καὶ τὰ ΒΕ, ΒΧ, ΚΧ τεταρτημόρια ἀλλήλοις. ὅσαι ἄρα εἰσὶν ἐν τῷ ΒΕ τεταρτημορίῳ πλευραὶ τοῦ πολυγώνου, τοσαῦταί εἰσι καὶ ἐν τοῖς ΒΧ, ΚΧ τεταρτημορίοις ἴσαι ταῖς ΒΚ, ΚΛ, ΛΜ, ΜΕ εὐθείαις. ἐγγεγράφθωσαν καὶ ὀέστωσαν αἱ ΒΟ, ΟΠ, ΠΡ, ΡΧ, ΚΣ, ΣΤ, ΤΥ, ΥΧ, καὶ ἐπεζεύθθωσαν αἱ ΣΟ, ΤΠ, ΥΡ, καὶ ἀπὸ τῶν Ο, Σ ἐπὶ τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον κάθετοι ἤθθωσαν· πεσοῦνται δὴ ἐπὶ τὰξ κοινὰξ τομὰξ τῶν ἐπιπέδων τὰξ ΒΔ, ΚΝ, ἐπειδήπερ καὶ τὰ τῶν ΒΧΔ, ΚΧΝ ἐπίπεδα ὀρθά ἐστι πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον. πιπτέτωσαν, καὶ ὀέστωσαν αἱ ΟΦ, ΣΘ, καὶ ἐπεζεύθθω ἡ ΘΦ. καὶ ἐπεὶ ἐνὸίσοις ἡμικυκλίοις τοῖς ΒΧΔ, ΚΧΝ ἴσαι ἀπειλημμέναι εἰσὶν αἱ ΒΟ, ΚΣ, καὶ κάθετοι ἠγμέναι εἰσὶν αἱ ΟΦ, ΣΘ, ὁίση [ἄρα] ἐστὶν ἡ μὲν ΟΦ τῇ ΣΘ, ἡ δὲ ΒΦ τῇ ΚΘ. ὀέστι δὲ καὶ ὅλη ἡ ΒΑ ὅλῃ τῇ ΚΑὁίση· καὶ λοιπὴ ἄρα ἡ ΦΑ λοιπῇ τῇ ΘΑ ἐστινὁίση· ὀέστιν ἄρα ὡξ ἡ ΒΦ πρὸς τὴν ΦΑ, οὁύτως ἡ ΚΘ πρὸς τὴν ΘΑ· παράλληλος ἄρα ἐστὶν ἡ ΘΦ τῇ ΚΒ. καὶ ἐπεὶ ἑκατέρα τῶν ΟΦ, ΣΘ ὀρθή ἐστι πρὸς τὸ τοῦ ΒΓΔΕ κύκλου ἐπίπεδον, παράλληλος ἄρα ἐστὶν ἡ ΟΦ τῇ ΣΘ. ἐδείθθη δὲ αὐτῇ καὶ ὁίση· καὶ αἱ ΘΦ, ΣΟ ἄρα ἴσαι εἰσὶ καὶ παράλληλοι. καὶ ἐπεὶ παράλληλόξ ἐστιν ἡ ΘΦ τῇ ΣΟ, ἀλλὰ ἡ ΘΦ τῇ ΚΒ ἐστι παράλληλος, καὶ ἡ ΣΟ ἄρα τῇ ΚΒ ἐστι παράλληλος. καὶ ἐπιζευγνύουσιν αὐτὰξ αἱ ΒΟ, ΚΣ· τὸ ΚΒΟΣ ἄρα τετράπλευρον ἐν ἑνί ἐστιν ἐπιπέδῳ, ἐπειδήπερ, ἔανὸῶσι δύο εὐθεῖαι παράλληλοι, καὶ ἐφ᾽ ἑκατέρας αὐτῶν ληφθῇ τυθόντα σημεῖα, ἡ ἐπὶ τὰ σημεῖα ἐπιζευγνυμένη εὐθεῖα ἐν τῷ αὐτῷ ἐπιπέδῳ ἐστὶ ταῖς παραλλήλοις. διὰ τὰ αὐτὰ δὴ καὶ ἑκάτερον τῶν ΣΟΠΤ, ΤΠΡΥ τετραπλεύρων ἐν ἑνί ἐστιν ἐπιπέδῳ. ὀέστι δὲ καὶ τὸ ΥΡΧ τρίγωνον ἐν ἑνὶ ἐπιπέδῳ. ἔαν δὴ νοήσωμεν ἀπὸ τῶν Ο, Σ, Π, Τ, Ρ, Υ σημείων ἐπὶ τὸ Α ἐπιζευγνυμένας εὐθείας, συσταθήσεταί τι σχῆμα στερεὸν πολύεδρον

the sphere, which is also manifestly the diameter of the semicircle and the circle, is greater than all of the (other) [straight-lines] drawn across in the circle or the sphere [Prop. 3.15]. Therefore, let *BCDE* be the circle in the greater sphere, and *FGH* the circle in the lesser sphere. And let two diameters of them be drawn at right-angles to one another, (namely), *BD* and *CE*. And there being two circles about the same center—(namely), *BCDE* and *FGH*—let an equilateral and even-sided polygon be inscribed in the greater circle, *BCDE*, not touching the lesser circle, *FGH* [Prop. 12.16], of which let the sides in the quadrant *BE* be *BK*, *KL*, *LM*, and *ME*. And, *KA* being joined, let it be drawn across to *N*. And let *AO* be set up at point *A*, at right-angles to the plane of circle *BCDE*. And let it meet the surface of the (greater) sphere at *O*. And let planes be produced through *AO* and each of *BD* and *KN*. So, according to the aforementioned (discussion), they will make great circles on the surface of the (greater) sphere. Let them make (great circles), of which let *BOD* and *KON* be semi-circles on the diameters *BD* and *KN* (respectively). And since *OA* is at right-angles to the plane of circle *BCDE*, all of the planes through *OA* are thus also at right-angles to the plane of circle *BCDE* [Prop. 11.18]. And, hence, the semi-circles *BOD* and *KON* are also at right-angles to the plane of circle *BCDE*. And since semi-circles *BED*, *BOD*, and *KON* are equal—for (they are) on the equal diameters *BD* and *KN* [Def. 3.1]—the quadrants *BE*, *BO*, and *KO* are also equal to one another. Thus, as many sides of the polygon as are in quadrant *BE*, so many are also in quadrants *BO* and *KO* equal to the straight-lines *BK*, *KL*, *LM*, and *ME*. Let them be inscribed, and let them be *BP*, *PQ*, *QR*, *RO*, *KS*, *ST*, *TU*, and *UO*. And let *SP*, *TQ*, and *UR* be joined. And let perpendiculars be drawn from *P* and *S* to the plane of circle *BCDE* [Prop. 11.11]. So, they will fall on the common sections of the planes *BD* and *KN* (with *BCDE*), inasmuch as the planes of *BOD* and *KON* are also at right-angles to the plane of circle *BCDE* [Def. 11.4]. Let them have fallen, and let them be *PV* and *SW*. And let *WV* be joined. And since *BP* and *KS* are equal (circumferences) having been cut off in the equal semi-circles *BOD* and *KON* [Def. 3.28], and *PV* and *SW* are perpendiculars having been drawn (from them), *PV* is [thus] equal to *SW*, and *BV* to *KW* [Props. 3.27, 1.26]. And the whole of *BA* is also equal to the whole of *KA*. And, thus, as *BV* is to *VA*, so *KW* (is) to *WA*. *WV* is thus parallel to *KB* [Prop. 6.2]. And since *PV* and *SW* are each at right-angles to the plane of circle *BCDE*, *PV* is thus parallel to *SW* [Prop. 11.6]. And it was also shown (to be) equal to it. And, thus, *WV* and *SP* are equal and parallel [Prop. 1.33]. And since *WV* is parallel to *SP*, but *WV* is parallel to *KB*, *SP* is thus also parallel to *KB* [Prop. 11.1]. And *BP* and *KS* join them. Thus, the quadrilateral *KBPS* is in one plane, inasmuch as if there are two parallel straight-lines, and a random point is taken on each of them, then the straight-line joining the points

ματαχὺ τῶν BX, KX περιφερειῶν ἐκ πυραμίδων συγκείμενον, ὧν βάσεις μὲν τὰ KBOΣ, ΣOΠT, TΠPY τετράπλευρα καὶ τὸ YPX τρίγωνον, κορυφὴ δὲ τὸ A σημεῖον. ἔαν δὲ καὶ ἐπὶ ἑκάστης τῶν KΛ, ΛM, ME πλευρῶν καθάπερ ἐπὶ τῆς BK τὰ αὐτὰ κατασκευάσωμεν καὶ ὁέτι τῶν λοιπῶν τριῶν τεταρτημορίων, συσταθήσεταί τι σχῆμα πολύεδρον ἐγγεγραμμένον εἰξ τὴν σφαῖραν πυραμίσι περιεχόμενον, ὧν βάσιες [μὲν] τὰ εἰρημένα τετράπλευρα καὶ τὸ YPX τρίγωνον καὶ τὰ ὁμοταγῆ αὐτοῖς, κορυφὴ δὲ τὸ A σημεῖον.

Λέγω ὅτι τὸ εἰρημένον πολύεδρον οὐκ ἐφάψεται τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν, ἐφ' ἣξ ἐστιν ὁ ZHΘ κύκλος.

ὀΗθθω ἀπὸ τοῦ A σημείου ἐπὶ τὸ τοῦ KBOΣ τετραπλεύρου ἐπίπεδον κάθετος ἡ AΨ καὶ συμβαλλέτω τῷ ἐπιπέδῳ κατὰ τὸ Ψ σημεῖον, καὶ ἐπεζεύθθωσαν αἱ ΨB, ΨK. καὶ ἐπεὶ ἡ AΨ ὀρθή ἐστι πρὸς τὸ τοῦ KBOΣ τετραπλεύρου ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ τοῦ τετραπλεύρου ἐπιπέδῳ ὀρθή ἐστιν. ἡ AΨ ἄρα ὀρθή ἐστι πρὸς ἑκατέραν τῶν BΨ, ΨK. καὶ ἐπεὶ ὀίση ἐστὶν ἡ AB τῇ AK, ὁίσον ἐστὶ καὶ τὸ ἀπὸ τῆς AB τῷ ἀπὸ τῆς AK. καί ἐστι τῷ μὲν ἀπὸ τῆς AB ἴσα τὰ ἀπὸ τῶν AΨ, ΨB· ὀρθὴ γὰρ ἡ πρὸς τῷ Ψ· τῷ δὲ ἀπὸ τῆς AK ἴσα τὰ ἀπὸ τῶν AΨ, ΨK. τὰ ἄρα ἀπὸ τῶν AΨ, ΨB ἴσα ἐστὶ τοῖς ἀπὸ τῶν AΨ, ΨK. κοινὸν ἀφῃρήσθω τὸ ἀπὸ τῆς AΨ· λοιπὸν ἄρα τὸ ἀπὸ τῆς BΨ λοιπῷ τῷ ἀπὸ τῆς ΨK ἴσον ἐστίν· ὀίση ἄρα ἡ BΨ τῇ ΨK. ὁμοίως δὴ δείχομεν, ὅτι καὶ αἱ ἀπὸ τοῦ Ψ ἐπὶ τὰ O, Σ ἐπιζευγνύμεναι εὐθεῖαι ἴσαι εἰσὶν ἑκατέρᾳ τῶν BΨ, ΨK. ὁ ἄρα κέντρῳ τῷ Ψ καὶ διαστήματι ἑνὶ τῶν ΨB, ΨK γραφόμενος κύκλος ἥξει καὶ διὰ τῶν O, Σ, καὶ ὀέσται ἐν κύκλῳ τὸ KBOΣ τετράπλευρον.

Καὶ ἐπεὶ μείζων ἐστὶν ἡ KB τῆς ΘΦ, ὀίση δὲ ἡ ΘΦ τῇ ΣO, μείζων ἄρα ἡ KB τῆς ΣO. ὀίση δὲ ἡ KB ἑκατέρᾳ τῶν KΣ, BO· καὶ ἑκατέρα ἄρα τῶν KΣ, BO τῆς ΣO μείζων ἐστίν. καὶ ἐπεὶ ἐν κύκλῳ τετράπλευρόν ἐστι τὸ KBOΣ, καὶ ἴσαι αἱ KB, BO, KΣ, καὶ ἐλάττων ἡ OΣ, καὶ ἐκ τοῦ κέντρου τοῦ κύκλου ἐστὶν ἡ BΨ, τὸ ἄρα ἀπὸ τῆς KB τοῦ ἀπὸ τῆς BΨ μεῖζόν ἐστινὸὴ διπλάσιον. ἤθθω ἀπὸ τοῦ K ἐπὶ τὴν BΦ κάθετος ἡ KΩ. καὶ ἐπεὶ ἡ BΔ τῆς ΔΩ ἐλάττων ἐστὶνὸὴ διπλῆ, καί ἐστιν ὡξ ἡ BΔ πρὸς τὴν ΔΩ, οὕτως τὸ ὑπὸ τῶν ΔB, BΩ πρὸς τὸ ὑπὸ [τῶν] ΔΩ, ΩB, ἀναγραφομένου ἀπὸ τῆς BΩ τετραγώνου καὶ συμπληρουμένου τοῦ ἐπὶ τῆς ΩΔ παραλληλογράμμου καὶ τὸ ὑπὸ ΔB, BΩ ἄρα τοῦ ὑπὸ ΔΩ, ΩBὀέλαττόν ἐστινὸὴ διπλάσιον. καί ἐστι τῆς KΔ ἐπιζευγνυμένης τὸ μὲν ὑπὸ ΔB, BΩ ἴσον τῷ ἀπὸ τῆς BK, τὸ δὲ ὑπὸ τῶν ΔΩ, ΩB ἴσον τῷ ἀπὸ τῆς KΩ· τὸ ἄρα ἀπὸ τῆς KB τοῦ ἀπὸ τῆς KΩὀέλασσόν ἐστινὸὴ διπλάσιον. ἀλλὰ τὸ ἀπὸ τῆς KB τοῦ ἀπὸ τῆς BΨ μεῖζόν ἐστινὸὴ διπλάσιον· μεῖζον ἄρα τὸ ἀπὸ τῆς KΩ τοῦ ἀπὸ τῆς BΨ. καὶ ἐπεὶ ὀίση ἐστὶν ἡ BA τῇ KA, ἴσον ἐστὶ τὸ ἀπὸ τῆς BA τῷ ἀπὸ τῆς AK. καί ἐστι τῷ μὲν ἀπὸ τῆς BA ἴσα τὰ ἀπὸ τῶν BΨ, ΨA, τῷ δὲ ἀπὸ τῆς KA ἴσα τὰ ἀπὸ τῶν KΩ, ΩA· τὰ ἄρα ἀπὸ τῶν BΨ, ΨA ἴσα ἐστὶ τοῖς ἀπὸ τῶν KΩ, ΩA, ὧν τὸ ἀπὸ τῆς KΩ μεῖζον τοῦ ἀπὸ τῆς BΨ· λοιπὸν ἄρα

is in the same plane as the parallel (straight-lines) [Prop. 11.7]. So, for the same (reasons), each of the quadrilaterals $SPQT$ and $TQRU$ is also in one plane. And triangle URO is also in one plane [Prop. 11.2]. So, if we conceive straight-lines joining points P, S, Q, T, R, and U to A then some solid polyhedral figure will be constructed between the circumferences BO and KO, being composed of pyramids whose bases (are) the quadrilaterals $KBPS$, $SPQT$, $TQRU$, and the triangle URO, and apex the point A. And if we also make the same construction on each of the sides KL, LM, and ME, just as on BK, and, further, (repeat the construction) in the remaining three quadrants, then some polyhedral figure which has been inscribed in the sphere will be constructed, being contained by pyramids whose bases (are) the aforementioned quadrilaterals, and triangle URO, and the (quadrilaterals and triangles) similarly arranged to them, and apex the point A.

So, I say that the aforementioned polyhedron will not touch the lesser sphere on the surface on which the circle FGH is (situated).

Let the perpendicular (straight-line) AX be drawn from point A to the plane $KBPS$, and let it meet the plane at point X [Prop. 11.11]. And let XB and XK be joined. And since AX is at right-angles to the plane of quadrilateral $KBPS$, it is thus also at right-angles to all of the straight-lines joined to it which are also in the plane of the quadrilateral [Def. 11.3]. Thus, AX is at right-angles to each of BX and XK. And since AB is equal to AK, the (square) on AB is also equal to the (square) on AK. And the (sum of the squares) on AX and XB is equal to the (square) on AB. For the angle at X (is) a right-angle [Prop. 1.47]. And the (sum of the squares) on AX and XK is equal to the (square) on AK [Prop. 1.47]. Thus, the (sum of the squares) on AX and XB is equal to the (sum of the squares) on AX and XK. Let the (square) on AX be subtracted from both. Thus, the remaining (square) on BX is equal to the remaining (square) on XK. Thus, BX (is) equal to XK. So, similarly, we can show that the straight-lines joined from X to P and S are equal to each of BX and XK. Thus, a circle drawn (in the plane of the quadrilateral) with center X, and radius one of XB or XK, will also pass through P and S, and the quadrilateral $KBPS$ will be inside the circle.

And since KB is greater than WV, and WV (is) equal to SP, KB (is) thus greater than SP. And KB (is) equal to each of KS and BP. Thus, KS and BP are each greater than SP. And since quadrilateral $KBPS$ is in a circle, and KB, BP, and KS are equal (to one another), and PS (is) less (than them), and BX is the radius of the circle, the (square) on KB is thus greater than double the (square) on BX.[†] Let the perpendicular KY be drawn from K to BV.[‡] And since BD is less than double DY, and as BD is to DY, so the (rectangle contained) by DB and BY (is) to the (rectangle contained) by DY and YB—a square being described on BY, and a (rectangular) parallel-

τὸ ἀπὸ τῆς ΩΑ ὲλασσόν ἐστι τοῦ ἀπὸ τῆς ΨΑ. μείζων ἄρα ἡ ΑΨ τῆς ΑΩ· πολλῷ ἄρα ἡ ΑΨ μείζων ἐστὶ τῆς ΑΗ. καί ἐστιν ἡ μὲν ΑΨ ἐπὶ μίαν τοῦ πολυέδρου βάσιν, ἡ δὲ ΑΗ ἐπὶ τὴν τῆς ἐλάσσονος σφαίρας ἐπιφάνειαν· ὥστε τὸ πολύεδρον οὐ ψαύσει τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν.

Δύο ἄρα σφαιρῶν περὶ τὸ αὐτὸ κέντρον οὐσῶν εἰξ τὴν μείζονα σφαῖραν στερεὸν πολύεδρον ἐγγέγραπται μὴ ψαῦον τῆς ἐλάσσονος σφαίρας κατὰ τὴν ἐπιφάνειαν· ὅπερ ἔδει ποιῆσαι.

ogram (with short side equal to BY) completed on YD—the (rectangle contained) by DB and BY is thus also less than double the (rectangle contained) by DY and YB. And, KD being joined, the (rectangle contained) by DB and BY is equal to the (square) on BK, and the (rectangle contained) by DY and YB equal to the (square) on KY [Props. 3.31, 6.8 corr.]. Thus, the (square) on KB is less than double the (square) on KY. But, the (square) on KB is greater than double the (square) on BX. Thus, the (square) on KY (is) greater than the (square) on BX. And since BA is equal to KA, the (square) on BA is equal to the (square) on AK. And the (sum of the squares) on BX and XA is equal to the (square) on BA, and the (sum of the squares) on KY and YA (is) equal to the (square) on KA [Prop. 1.47]. Thus, the (sum of the squares) on BX and XA is equal to the (sum of the squares) on KY and YA, of which the (square) on KY (is) greater than the (square) on BX. Thus, the remaining (square) on YA is less than the (square) on XA. Thus, AX (is) greater than AY. Thus, AX is much greater than AG.[§] And AX is (a perpendicular) on one of the bases of the polyhedron, and AG (is a perpendicular) on the surface of the lesser sphere. Hence, the polyhedron will not touch the lesser sphere on its surface.

Thus, there being two spheres about the same center, a polyhedral solid has been inscribed in the greater sphere which does not touch the lesser sphere on its surface. (Which is) the very thing it was required to do.

[†] Since KB, BP, and KS are greater than the sides of an inscribed square, which are each of length $\sqrt{2}\,BX$.

[‡] Note that points Y and V are actually identical.

[§] This conclusion depends on the fact that the chord of the polygon in proposition 12.16 does not touch the inner circle.

Πόρισμα.

Ἐὰν δὲ καὶ εἰξ ἑτέραν σφαῖραν τῷ ἐν τῇ ΒΓΔΕ σφαίρᾳ στερεῷ πολυέδρῳ ὅμοιον στερεὸν πολύεδρον ἐγγραφῇ, τὸ ἐν τῇ ΒΓΔΕ σφαίρᾳ στερεὸν πολύεδρον πρὸς τὸ ἐν τῇ ἑτέρᾳ σφαίρᾳ στερεὸν πολύεδρον τριπλασίονα λόγον ἔχει, ἥπερ ἡ τῆς ΒΓΔΕ σφαίρας διάμετρος πρὸς τὴν τῆς ἑτέρας σφαίρας διάμετρον. διαιρεθέντων γὰρ τῶν στερεῶν εἰξ τὰξ ὁμοιοπληθεῖς καὶ ὁμοιοταγεῖς πυραμίδας ὁέσονται αἱ πυραμίδες ὅμοιαι. αἱ δὲ ὅμοιαι πυραμίδες πρὸς ἀλλήλας ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ὁμολόγων πλευρῶν· ἡ ἄρα πυραμίξ, ἧξ βάσις μέν ἐστι τὸ ΚΒΟΣ τετράπλευρον, κορυφὴ δὲ τὸ Α σημεῖον, πρὸς τὴν ἐν τῇ ἑτέρᾳ σφαίρᾳ ὁμοιοταγῆ πυραμίδα τριπλασίονα λόγον ἔχει, ἥπερ ἡ ὁμόλογος πλευρὰ πρὸς τὴν ὁμόλογον πλευράν, τουτέστιν ἥπερ ἡ ΑΒ ἐκ τοῦ κέντρου τῆς σφαίρας τῆς περὶ κέντρον τὸ Α πρὸς τὴν ἐκ τοῦ κέντρου τῆς ἑτέρας σφαίρας. ὁμοίως καὶ ἑκάστη πυραμὶξ τῶν ἐν τῇ περὶ κέντρον τὸ Α σφαίρᾳ πρὸς ἑκάστην ὁμοταγῆ πυραμίδα τῶν ἐν τῇ ἑτέρᾳ σφαίρᾳ τριπλασίονα λόγον ὁέχει, ἥπερ ἡ ΑΒ πρὸς τὴν ἐκ τοῦ κέντρου τῆς ἑτέρας σφαίρας. καὶ ὡξ ὁὲν τῶν ἡγουμένων πρὸς ὁὲν τῶν

Corollary

And, also, if a similar polyhedral solid to that in sphere $BCDE$ is inscribed in another sphere then the polyhedral solid in sphere $BCDE$ has to the polyhedral solid in the other sphere the cubed ratio that the diameter of sphere $BCDE$ has to the diameter of the other sphere. For if the solids are divided into similarly numbered, and similarly situated, pyramids, then the pyramids will be similar. And similar pyramids are in the cubed ratio of corresponding sides [Prop. 12.8 corr.]. Thus, the pyramid whose base is quadrilateral $KBPS$, and apex the point A, will have to the similarly situated pyramid in the other sphere the cubed ratio that a corresponding side (has) to a corresponding side. That is to say, that of radius AB of the sphere about center A to the radius of the other sphere. And, similarly, each pyramid in the sphere about center A will have to each similarly situated pyramid in the other sphere the cubed ratio that AB (has) to the radius of the other sphere. And as one of the leading (magnitudes is) to one of the following (in two sets of proportional magnitudes), so (the sum of) all the

ἑπομένων, οὕτως ὅλον τὸ ἐν τῇ περὶ κέντρον τὸ A σφαίρᾳ στερεὸν πολύεδρον πρὸς ὅλον τὸ ἐν τῇ ἑτέρᾳ [σφαίρᾳ] στερεὸν πολύεδρον τριπλασίονα λόγον ἕξει, ἥπερ ἡ AB πρὸς τὴν ἐκ τοῦ κέντρου τῆς ἑτέρας σφαίρας, τουτέστιν ἥπερ ἡ $BΔ$ διάμετρος πρὸς τὴν τῆς ἑτέρας σφαίρας διάμετρον· ὅπερ ἔδει δεῖξαι.

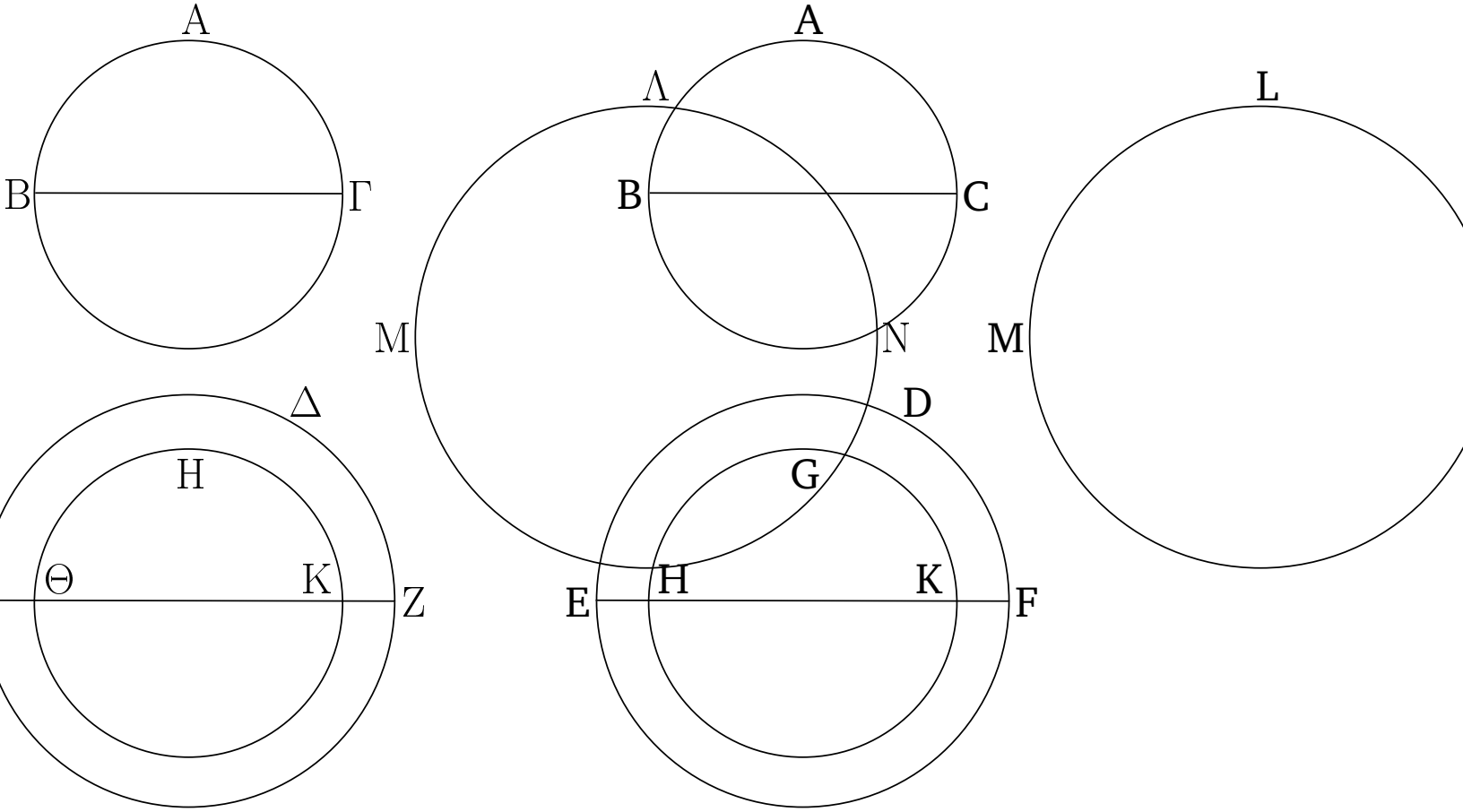
leading (magnitudes is) to (the sum of) all of the following (magnitudes) [Prop. 5.12]. Hence, the whole polyhedral solid in the sphere about center A will have to the whole polyhedral solid in the other [sphere] the cubed ratio that (radius) AB (has) to the radius of the other sphere. That is to say, that diameter BD (has) to the diameter of the other sphere. (Which is) the very thing it was required to show.

18.

Proposition 18

Αἱ σφαῖραι πρὸς ἀλλήλας ἐν τριπλασίονι λόγῳ εἰσὶ τῶν ἰδίων διαμέτρων.

Spheres are to one another in the cubed ratio of their respective diameters.



Νενοήσθωσαν σφαῖραι αἱ $ABΓ$, $ΔEZ$, διάμετροι δὲ αὐτῶν αἱ $BΓ$, EZ· λέγω, ὅτι ἡ $ABΓ$ σφαῖρα πρὸς τὴν $ΔEZ$ σφαῖραν τριπλασίονα λόγον ἔχει ἥπερ ἡ $BΓ$ πρὸς τὴν EZ.

Εἰ γὰρ μὴ ἡ $ABΓ$ σφαῖρα πρὸς τὴν $ΔEZ$ σφαῖραν τριπλασίονα λόγον ἔχει ἥπερ ἡ $BΓ$ πρὸς τὴν EZ, ἕξει ἄρα ἡ $ABΓ$ σφαῖρα πρὸς ἐλάσσονά τινα τῆς $ΔEZ$ σφαίρας τριπλασίονα λόγονὁἡ πρὸς μείζονα ἥπερ ἡ $BΓ$ πρὸς τὴν EZ. ἐνθέτω πρότερον πρὸς ἐλάσσονα τὴν $HΘK$, καὶ νενοήσθω ἡ $ΔEZ$ τῇ $HΘK$ περὶ τὸ αὐτὸ κέντρον, καὶ ἐγγεγράφθω εἰς τὴν μείζονα σφαῖραν τὴν $ΔEZ$ στερεὸν πολύεδρον μὴ ψαῦον τῆς ἐλάσσονος σφαίρας τῆς $HΘK$ κατὰ τὴν ἐπιφάνειαν, ἐγγεγράφθω δὲ καὶ

Let the spheres ABC and DEF be conceived, and (let) their diameters (be) BC and EF (respectively). I say that sphere ABC has to sphere DEF the cubed ratio that BC (has) to EF.

For if sphere ABC does not have to sphere DEF the cubed ratio that BC (has) to EF then sphere ABC will have to some (sphere) either less than, or greater than, sphere DEF the cubed ratio that BC (has) to EF. Let it, first of all, have (such a ratio) to a lesser (sphere), GHK. And let DEF be conceived about the same center as GHK. And let a polyhedral solid be inscribed in the greater sphere DEF, not touching the lesser sphere GHK on its surface [Prop. 12.17]. And let a poly-

εἰξ τὴν ΑΒΓ σφαῖραν τῷ ἐν τῇ ΔΕΖ σφαίρᾳ στερεῷ πολυέδρῳ ὅμοιον στερεὸν πολύεδρον· τὸ ἄρα ἐν τῇ ΑΒΓ στερεὸν πολύεδρον πρὸς τὸ ἐν τῇ ΔΕΖ στερεὸν πολύεδρον τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΓ πρὸς τὴν ΕΖ. ἔχει δὲ καὶ ἡ ΑΒΓ σφαῖρα πρὸς τὴν ΗΘΚ σφαῖραν τριπλασίονα λόγον ἥπερ ἡ ΒΓ πρὸς τὴν ΕΖ· ὀέστιν ἄρα ὡξ ἡ ΑΒΓ σφαῖρα πρὸς τὴν ΗΘΚ σφαῖραν, οὑτως τὸ ἐν τῇ ΑΒΓ σφαίρᾳ στερεὸν πολύεδρον πρὸς τὸ ἐν τῇ ΔΕΖ σφαίρᾳ στερεὸν πολύεδρον· ἐναλλὰχ [ἄρα] ὡξ ἡ ΑΒΓ σφαῖρα πρὸς τὸ ἐν αὐτῇ πολύεδρον, οὑτως ἡ ΗΘΚ σφαῖρα πρὸς τὸ ἐν τῇ ΔΕΖ σφαίρᾳ στερεὸν πολύεδρον. μείζων δὲ ἡ ΑΒΓ σφαῖρα τοῦ ἐν αὐτῇ πολυέδρου· μείζων ἄρα καὶ ἡ ΗΘΚ σφαῖρα τοῦ ἐν τῇ ΔΕΖ σφαίρᾳ πολυέδρου. ἀλλὰ καὶ ἐλάττων· ἐμπεριέθεται γὰρ ὑπὸ αὐτοῦ. οὐκ ἄρα ἡ ΑΒΓ σφαῖρα πρὸς ἐλάσσονα τῆς ΔΕΖ σφαίρας τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΓ διάμετρος πρὸς τὴν ΕΖ. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ἡ ΔΕΖ σφαῖρα πρὸς ἐλάσσονα τῆς ΑΒΓ σφαίρας τριπλασίονα λόγον ἔχει ἥπερ ἡ ΕΖ πρὸς τὴν ΒΓ.

Λέγω δή, ὅτι οὐδὲ ἡ ΑΒΓ σφαῖρα πρὸς μείζονά τινα τῆς ΔΕΖ σφαίρας τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΓ πρὸς τὴν ΕΖ.

Εἰ γὰρ δυνατόν, ἐθέτω πρὸς μείζονα τὴν ΛΜΝ· ἀνάπαλιν ἄρα ἡ ΛΜΝ σφαῖρα πρὸς τὴν ΑΒΓ σφαῖραν τριπλασίονα λόγον ἔχει ἥπερ ἡ ΕΖ διάμετρος πρὸς τὴν ΒΓ διάμετρον. ὡξ δὲ ἡ ΛΜΝ σφαῖρα πρὸς τὴν ΑΒΓ σφαῖραν, οὑτως ἡ ΔΕΖ σφαῖρα πρὸς ἐλάσσονά τινα τῆς ΑΒΓ σφαίρας, ἐπειδήπερ μείζων ἐστὶν ἡ ΛΜΝ τῆς ΔΕΖ, ὡξὸέμπροσθεν ἐδείθθη. καὶ ἡ ΔΕΖ ἄρα σφαῖρα πρὸς ἐλάσσονά τινα τῆς ΑΒΓ σφαίρας τριπλασίονα λόγον ἔχει ἥπερ ἡ ΕΖ πρὸς τὴν ΒΓ· ὅπερ ἀδύνατον ἐδείθθη. οὐκ ἄρα ἡ ΑΒΓ σφαῖρα πρὸς μείζονά τινα τῆς ΔΕΖ σφαίρας τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΓ πρὸς τὴν ΕΖ. ἐδείθθη δέ, ὅτι οὐδὲ πρὸς ἐλάσσονα. ἡ ἄρα ΑΒΓ σφαῖρα πρὸς τὴν ΔΕΖ σφαῖραν τριπλασίονα λόγον ἔχει ἥπερ ἡ ΒΓ πρὸς τὴν ΕΖ· ὅπερ ἔδει δεῖχαι.

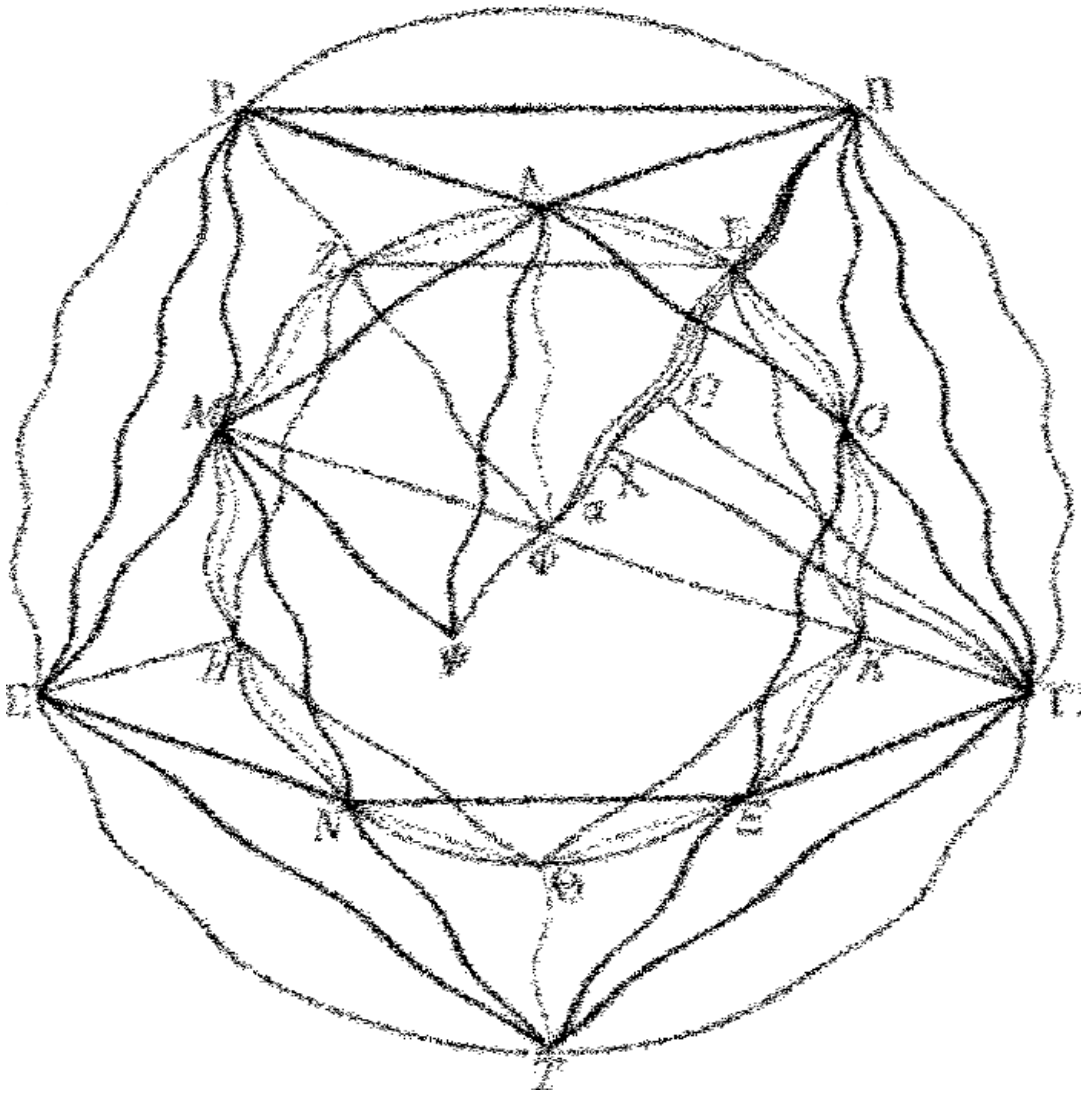
hedral solid, similar to the polyhedral solid in sphere DEF, have also been inscribed in sphere ABC. Thus, the polyhedral solid in sphere ABC has to the polyhedral solid in sphere DEF the cubed ratio that BC (has) to EF [Prop. 12.17 corr.]. And sphere ABC also has to sphere GHK the cubed ratio that BC (has) to EF. Thus, as sphere ABC is to sphere GHK, so the polyhedral solid in sphere ABC (is) to the polyhedral solid is sphere DEF. [Thus], alternately, as sphere ABC (is) to the polygon within it, so sphere GHK (is) to the polyhedral solid within sphere DEF [Prop. 5.16]. And sphere ABC (is) greater than the polyhedron within it. Thus, sphere GHK (is) also greater than the polyhedron within sphere DEF [Prop. 5.14]. But, (it is) also less. For it is encompassed by it. Thus, sphere ABC does not have to (a sphere) less than sphere DEF the cubed ratio that diameter BC (has) to EF. So, similarly, we can show that sphere DEF does not have to (a sphere) less than sphere ABC the cubed ratio that EF (has) to BC either.

So, I say that sphere ABC does not have to some (sphere) greater than sphere DEF the cubed ratio that BC (has) to EF either.

For, if possible, let it have (the cubed ratio) to a greater (sphere), LMN. Thus, inversely, sphere LMN (has) to sphere ABC the cubed ratio that diameter EF (has) to diameter BC [Prop. 5.7 corr.]. And as sphere LMN (is) to sphere ABC, so sphere DEF (is) to some (sphere) less than sphere ABC, inasmuch as LMN is greater than DEF, as was shown before [Prop. 12.2 lem.]. And, thus, sphere DEF has to some (sphere) less than sphere ABC the cubed ratio that EF (has) to BC. The very thing was shown (to be) impossible. Thus, sphere ABC does not have to some (sphere) greater than sphere DEF the cubed ratio that BC (has) to EF. And it was shown that neither (does it have such a ratio) to a lesser (sphere). Thus, sphere ABC has to sphere DEF the cubed ratio that BC (has) to EF. (Which is) the very thing it was required to show.

ELEMENTS BOOK 13

The Platonic Solids[†]



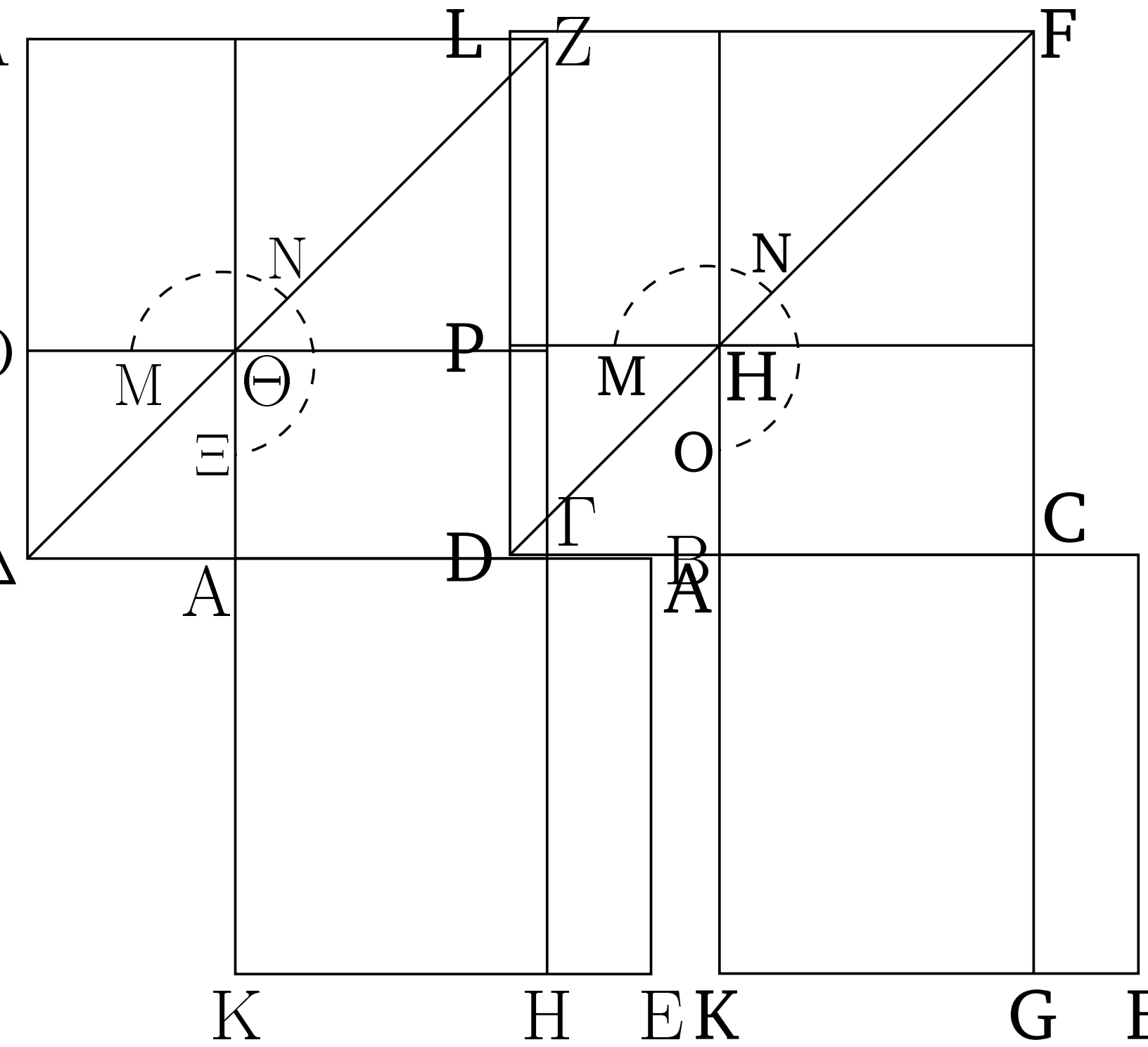
[†]The five regular solids—the cube, tetrahedron (i.e., pyramid), octahedron, icosahedron, and dodecahedron—were problably discovered by the school of Pythagoras. They are generally termed "Platonic" solids because they feature prominently in Plato's famous dialogue *Timaeus*. Many of the theorems contained in this book—particularly those which pertain to the last two solids—are ascribed to Theaetetus of Athens.

1.

Ἐὰν εὐθεῖα γραμμὴ ὀάκρον καὶ μέσον λόγον τμ ηθῇ, τὸ μεῖζον τμῆμα προσλαβὸν τὴν ἡμίσειαν τῆς ὅλης πενταπλάσιον δύναται τοῦ ἀπὸ τῆς ἡμισείας τετραγώνου.

Proposition 1

If a straight-line is cut in extreme and mean ratio then the square on the greater piece, added to half of the whole, is five times the square on the half.



Εὐθεῖα γὰρ γραμμὴ ἡ ΑΒ ὁάκρον καὶ μέσον λόγον τετμήσθω κατὰ τὸ Γ σημεῖον, καὶ ὁέστω μεῖζον τμῆμα τὸ ΑΓ, καὶ ἐκβεβλήσθω ἐπ᾽ εὐθείας τῇ ΓΑ εὐθεῖα ἡ ΑΔ, καὶ κείσθω τῆς ΑΒ ἡμίσεια ἡ ΑΔ· λέγω, ὅτι πενταπλάσιόν ἐστι τὸ ἀπὸ

For let the straight-line *AB* be cut in extreme and mean ratio at point *C*, and let *AC* be the greater piece. And let the straight-line *AD* be produced in a straight-line with *CA*. And let *AD* be made (equal to) half of *AB*. I say that the (square)

τῆς ΓΔ τοῦ ἀπὸ τῆς ΔΑ.

Ἀναγεγράφθωσαν γὰρ ἀπὸ τῶν ΑΒ, ΔΓ τετράγωνα τὰ ΑΕ, ΔΖ, καὶ καταγεγράφθω ἐν τῷ ΔΖ τὸ σχῆμα, καὶ διήχθω ἡ ΖΓ ἐπὶ τὸ Η. καὶ ἐπεὶ ἡ ΑΒ ὀάκρον καὶ μέσον λόγον τέτμηται κατὰ τὸ Γ, τὸ ἄρα ὑπὸ τῶν ΑΒΓ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΓ. καί ἐστι τὸ μὲν ὑπὸ τῶν ΑΒΓ τὸ ΓΕ, τὸ δὲ ἀπὸ τῆς ΑΓ τὸ ΖΘ· ἴσον ἄρα τὸ ΓΕ τῷ ΖΘ. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΒΑ τῆς ΑΔ, ὁίση δὲ ἡ μὲν ΒΑ τῇ ΚΑ, ἡ δὲ ΑΔ τῇ ΑΘ, διπλῆ ἄρα καὶ ἡ ΚΑ τῆς ΑΘ. ὡς δὲ ἡ ΚΑ πρὸς τὴν ΑΘ, οὕτως τὸ ΓΚ πρὸς τὸ ΓΘ· διπλάσιον ἄρα τὸ ΓΚ τοῦ ΓΘ. εἰσὶ δὲ καὶ τὰ ΛΘ, ΘΓ διπλάσια τοῦ ΓΘ. ἴσον ἄρα τὸ ΚΓ τοῖς ΛΘ, ΘΓ. ἐδείχθη δὲ καὶ τὸ ΓΕ τῷ ΘΖ ἴσον· ὅλον ἄρα τὸ ΑΕ τετράγωνον ἴσον ἐστὶ τῷ ΜΝΧ γνώμονι. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΒΑ τῆς ΑΔ, τετραπλάσιόν ἐστι τὸ ἀπὸ τῆς ΒΑ τοῦ ἀπὸ τῆς ΑΔ, τουτέστι τὸ ΑΕ τοῦ ΔΘ. ἴσον δὲ τὸ ΑΕ τῷ ΜΝΧ γνώμονι· καὶ ὁ ΜΝΧ ἄρα γνώμων τετραπλάσιόξ ἐστι τοῦ ΑΟ· ὅλον ἄρα τὸ ΔΖ πενταπλάσιόν ἐστι τοῦ ΑΟ. καί ἐστι τὸ μὲν ΔΖ τὸ ἀπὸ τῆς ΔΓ, τὸ δὲ ΑΟ τὸ ἀπὸ τῆς ΔΑ· τὸ ἄρα ἀπὸ τῆς ΓΔ πενταπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΔΑ.

Ἐὰν ἄρα εὐθεῖα ὀάκρον καὶ μέσον λόγον τμ ηθῇ, τὸ μεῖζον τμῆμα προσλαβὸν τὴν ἡμίσειαν τῆς ὅλης πενταπλάσιον δύναται τοῦ ἀπὸ τῆς ἡμισείας τετραγώνου· ὅπερ ἔδει δεῖχαι.

on CD is five times the (square) on DA.

For let the squares AE and DF be described on AB and DC (respectively). And let the figure in DF be drawn. And let FC be drawn across to G. And since AB has been cut in extreme and mean ratio at C, the (rectangle contained) by ABC is thus equal to the (square) on AC [Def. 6.3, Prop. 6.17]. And CE is the (rectangle contained) by ABC, and FH the (square) on AC. Thus, CE (is) equal to FH. And since BA is double AD, and BA (is) equal to KA, and AD to AH, KA (is) thus also double AH. And as KA (is) to AH, so CK (is) to CH [Prop. 6.1]. Thus, CK (is) double CH. And LH plus HC is also double CH [Prop. 1.43]. Thus, KC (is) equal to LH plus HC. And CE was also shown (to be) equal to HF. Thus, the whole square AE is equal to the gnomon MNO. And since BA is double AD, the (square) on BA is four times the (square) on AD—that is to say, AE (is four times) DH. And AE (is) equal to gnomon MNO. And, thus, gnomon MNO is also four times AP. Thus, the whole of DF is five times AP. And DF is the (square) on DC, and AP the (square) on DA. Thus, the (square) on CD is five times the (square) on DA.

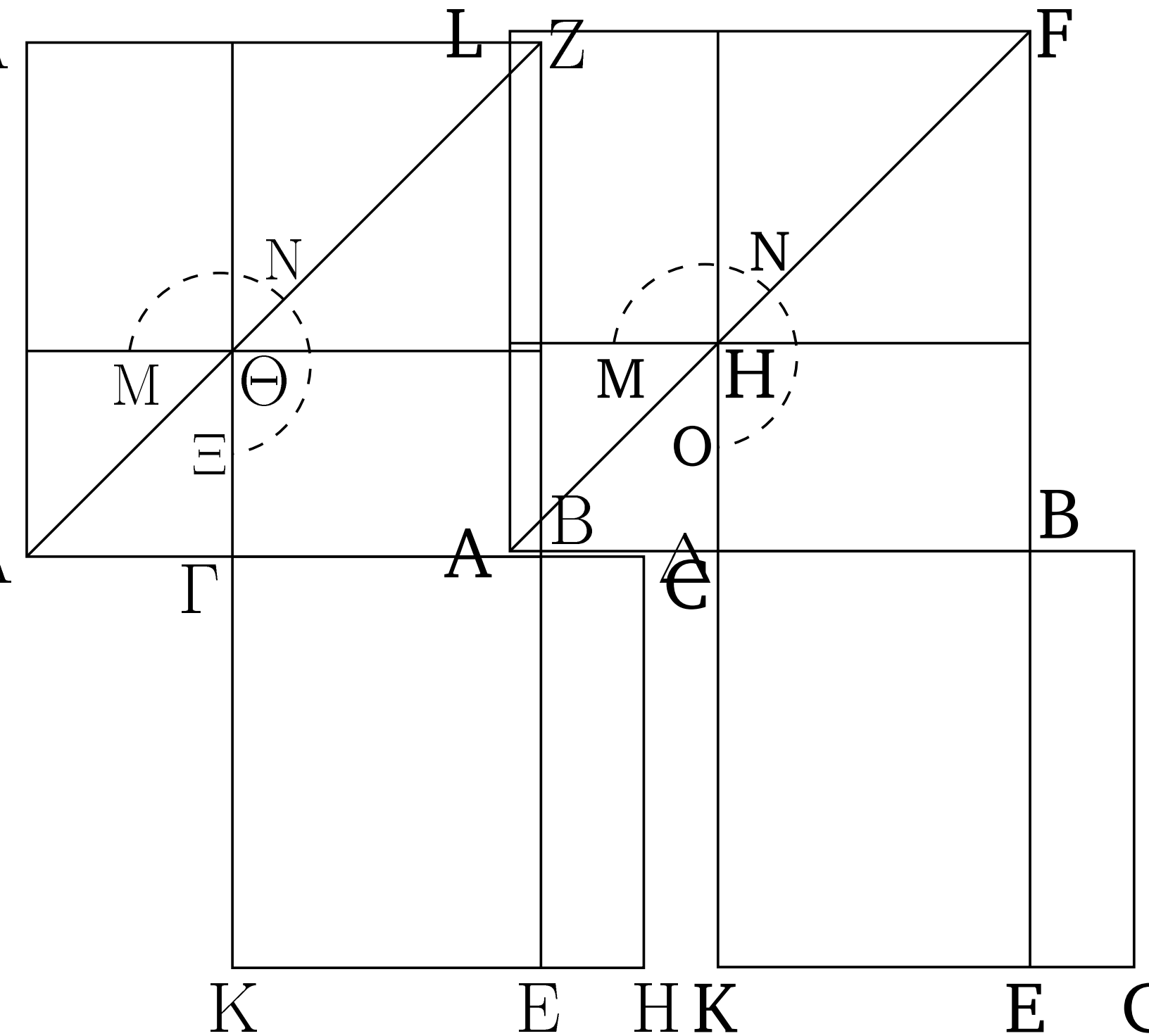
Thus, if a straight-line is cut in extreme and mean ratio then the square on the greater piece, added to half of the whole, is five times the square on the half. (Which is) the very thing it was required to show.

2.

Ἐὰν εὐθεῖα γραμμὴ τμήματος ἑαυτῆς πενταπλάσιον δύνηται, τῆς διπλασίας τοῦ εἰρημένου τμήματος ὀάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμα τὸ λοιπόν μέρος ἐστὶ τῆς ἐξ ἀρθῆς εὐθείας.

Proposition 2

If the square on a straight-line is five times the (square) on a piece of it, and double the aforementioned piece is cut in extreme and mean ratio, then the greater piece is the remaining part of the original straight-line.



Εὐθεῖα γὰρ γραμμὴ ἡ AB τμήματος ἑαυτῆς τοῦ AΓ πενταπλάσιον δυνάσθω, τῆς δὲ AΓ διπλῆ ὑπόστω ἡ ΓΔ. λέγω, ὅτι τῆς ΓΔ ὑάκρον καὶ μέσον λόγον τεμνομένος τὸ μεῖζον τμῆμά ἐστιν ἡ ΓB.

Ἀναγεγράφθω γὰρ ἀφ' ἑκατέρας τῶν AB, ΓΔ τετράγωνα

For let the square on the straight-line AB be five times the (square) on the piece of it, AC. And let CD be double AC. I say that if CD is cut in extreme and mean ratio then the greater piece is CB.

For let the squares AF and CG be described on each of AB

τὰ ΑΖ, ΓΗ, καὶ καταγεγράφθω ἐν τῷ ΑΖ τὸ σχῆμα, καὶ διήχθω ἡ ΒΕ. καὶ ἐπεὶ πενταπλάσιόν ἐστι τὸ ἀπὸ τῆς ΒΑ τοῦ ἀπὸ τῆς ΑΓ, πενταπλάσιόν ἐστι τὸ ΑΖ τοῦ ΑΘ. τετραπλάσιος ἄρα ὁ ΜΝΧ γνώμων τοῦ ΑΘ. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΔΓ τῆς ΓΑ, τετραπλάσιον ἄρα ἐστὶ τὸ ἀπὸ ΔΓ τοῦ ἀπὸ ΓΑ, τουτέστι τὸ ΓΗ τοῦ ΑΘ. ἐδείχθη δὲ καὶ ὁ ΜΝΧ γνώμων τετραπλάσιος τοῦ ΑΘ· ὁ ἄρα ὁ ΜΝΧ γνώμων τῷ ΓΗ. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΔΓ τῆς ΓΑ, ἴση δὲ ἡ μὲν ΔΓ τῇ ΓΚ, ἡ δὲ ΑΓ τῇ ΓΘ, [διπλῆ ἄρα καὶ ἡ ΚΓ τῆς ΓΘ], διπλάσιον ἄρα καὶ τὸ ΚΒ τοῦ ΒΘ. εἰσὶ δὲ καὶ τὰ ΛΘ, ΘΒ τοῦ ΘΒ διπλάσια· ἴσον ἄρα τὸ ΚΒ τοῖς ΛΘ, ΘΒ. ἐδείχθη δὲ καὶ ὅλος ὁ ΜΝΧ γνώμων ὅλῳ τῷ ΓΗ ἴσος· καὶ λοιπὸν ἄρα τὸ ΘΖ τῷ ΒΗ ἐστιν ἴσον. καί ἐστι τὸ μὲν ΒΗ τὸ ὑπὸ τῶν ΓΔΒ· ἴση γὰρ ἡ ΓΔ τῇ ΔΗ· τὸ δὲ ΘΖ τὸ ἀπὸ τῆς ΓΒ· τὸ ἄρα ὑπὸ τῶν ΓΔΒ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΓΒ. ὡέστιν ἄρα ὡξ ἡ ΔΓ πρὸς τὴν ΓΒ, οὕτως ἡ ΓΒ πρὸς τὴν ΒΔ. μείζων δὲ ἡ ΔΓ τῆς ΓΒ· μείζων ἄρα καὶ ἡ ΓΒ τῆς ΒΔ. τῆς ΓΔ ἄρα εὐθείαςὁάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμά ἐστιν ἡ ΓΒ.

Ἐὰν ἄρα εὐθεῖα γραμμὴ τμήματος ἑαυτῆς πενταπλάσιον δύνηται, τῆς διπλασίας τοῦ εἰρημένου τμήματοςὁάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμα τὸ λοιπὸν μέρος ἐστὶ τῆς ἐξ ἀρθῆς εὐθείας· ὅπερ ἔδει δεῖξαι.

and *CD* (respectively). And let the figure in *AF* be drawn. And let *BE* be drawn across. And since the (square) on *BA* is five times the (square) on *AC*, *AF* is five times *AH*. Thus, gnomon *MNO* (is) four times *AH*. And since *DC* is double *CA*, the (square) on *DC* is thus four times the (square) on *CA*—that is to say, *CG* (is four times) *AH*. And the gnomon *MNO* was also shown (to be) four times *AH*. Thus, gnomon *MNO* (is) equal to *CG*. And since *DC* is double *CA*, and *DC* (is) equal to *CK*, and *AC* to *CH*, [*KC* (is) thus also double *CH*], (and) *KB* (is) also double *BH* [Prop. 6.1]. And *LH* plus *HB* is also double *HB* [Prop. 1.43]. Thus, *KB* (is) equal to *LH* plus *HB*. And the whole gnomon *MNO* was also shown (to be) equal to the whole of *CG*. Thus, the remainder *HF* is also equal to (the remainder) *BG*. And *BG* is the (rectangle contained by) *CDB*. For *CD* (is) equal to *DG*. And *HF* (is) the square on *CB*. Thus, the (rectangle contained by) *CDB* is equal to the (square) on *CB*. Thus, as *DC* is to *CB*, so *CB* (is) to *BD* [Prop. 6.17]. And *DC* (is) greater than *CB* (see lemma). Thus, *CB* (is) also greater than *BD* [Prop. 5.14]. Thus, if the straight-line *CD* is cut in extreme and mean ratio then the greater piece is *CB*.

Thus, if the square on a straight-line is five times the (square) on a piece of itself, and double the aforementioned piece is cut in extreme and mean ratio, then the greater piece is the remaining part of the original straight-line. (Which is) the very thing it was required to show.

Λῆμμα.

Ὅτι δὲ ἡ διπλῆ τῆς ΑΓ μείζων ἐστὶ τῆς ΒΓ, οὕτως δεικτέον.

Εἰ γὰρ μή, ὡέστω, εἰ δυνατόν, ἡ ΒΓ διπλῆ τῆς ΓΑ. τετραπλάσιον ἄρα τὸ ἀπὸ τῆς ΒΓ τοῦ ἀπὸ τῆς ΓΑ· πενταπλάσια ἄρα τὰ ἀπὸ τῶν ΒΓ, ΓΑ τοῦ ἀπὸ τῆς ΓΑ. ὑπόκειται δὲ καὶ τὸ ἀπὸ τῆς ΒΑ πενταπλάσιον τοῦ ἀπὸ τῆς ΓΑ· τὸ ἄρα ἀπὸ τῆς ΒΑ ἴσον ἐστὶ τοῖς ἀπὸ τῶν ΒΓ, ΓΑ· ὅπερ ἀδύνατον. οὐκ ἄρα ἡ ΓΒ διπλασία ἐστὶ τῆς ΑΓ. ὁμοίως δὴ δείχομεν, ὅτι οὐδὲ ἡ ἐλάττων τῆς ΓΒ διπλασίων ἐστὶ τῆς ΓΑ· πολλῷ γὰρ [μεῖζον] τὸ ὀάτοπον.

Ἡ ἄρα τῆς ΑΓ διπλῆ μείζων ἐστὶ τῆς ΓΒ· ὅπερ ἔδει δεῖξαι.

Lemma

And it can be shown that double *AC* (*i.e.*, *DC*) is greater than *BC*, as follows.

For if (double *AC* is) not (greater than *BC*), if possible, let *BC* be double *CA*. Thus, the (square) on *BC* (is) four times the (square) on *CA*. Thus, the (sum of) the (squares) on *BC* and *CA* (is) five times the (square) on *CA*. And the (square) on *BA* was assumed (to be) five times the (square) on *CA*. Thus, the (square) on *BA* is equal to the (sum of) the (squares) on *BC* and *CA*. The very thing (is) impossible [Prop. 2.4]. Thus, *CB* is not double *AC*. So, similarly, we can show that a (straight-line) less than *CB* is not double *AC* either. For (in this case) the absurdity is much [greater].

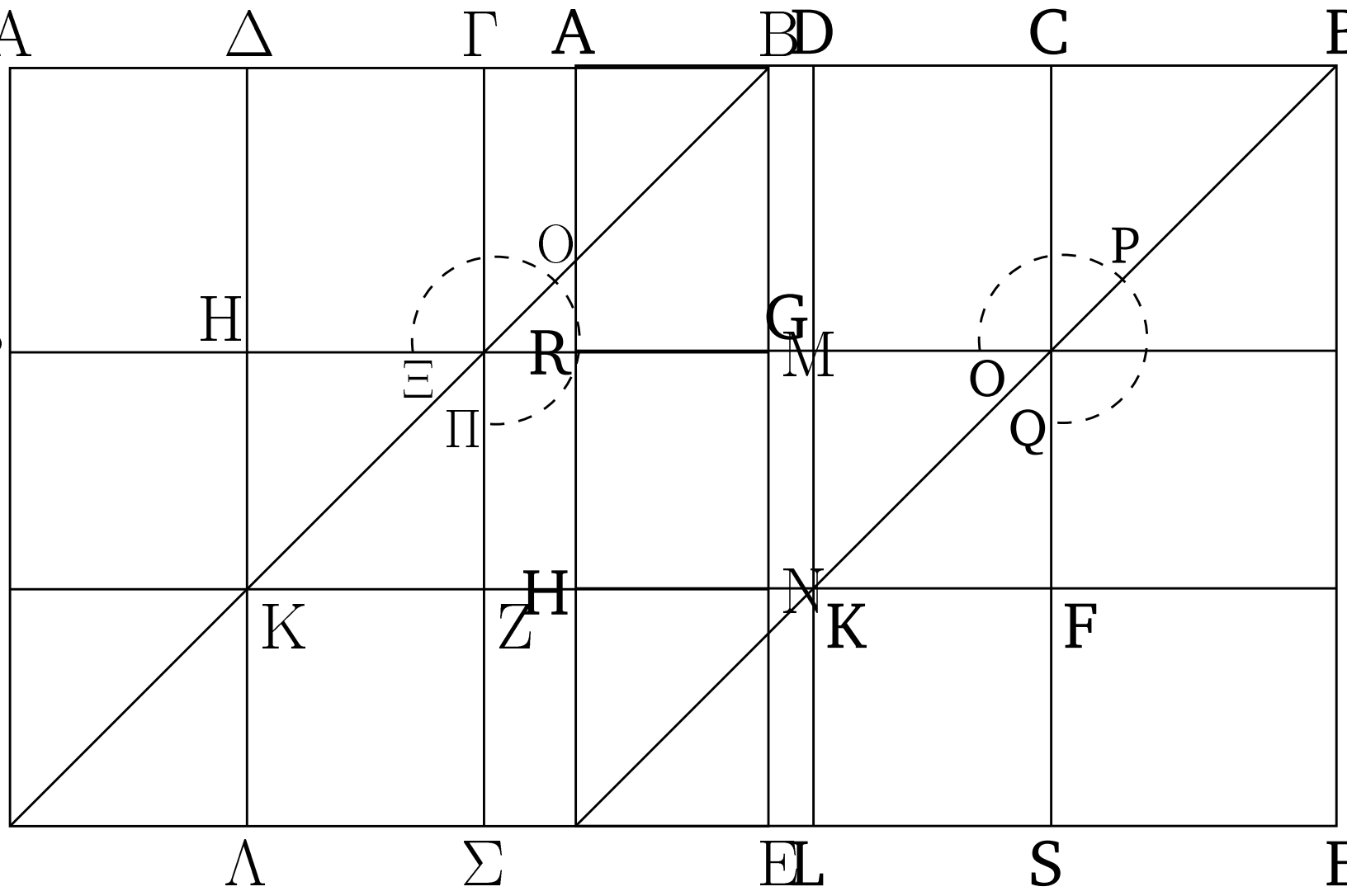
Thus, double *AC* is greater than *CB*. (Which is) the very thing it was required to show.

3.

Ἐὰν εὐθεῖα γραμμὴ ὁάκρον καὶ μέσον λόγον τμ ηθῇ, τὸ ὁέλασσον τμῆμα προσλαβὸν τὴν ἡμίσειαν τοῦ μείζονος τμήματος πενταπλάσιον δύναται τοῦ ἀπὸ τῆς ἡμισείας τοῦ μείζονος τμήματος τετραγώνου.

Proposition 3

If a straight-line is cut in extreme and mean ratio then the square on the lesser piece added to half of the greater piece is five times the square on half of the greater piece.



Εὐθεῖα γάρ τις ἡ ΑΒ ὀάκρον καὶ μέσον λόγον τετμήσθω κατὰ τὸ Γ σημεῖον, καὶ ὀέστω μεῖζον τμῆμα τὸ ΑΓ, καὶ τετμήσθω ἡ ΑΓ δίθα κατὰ τὸ Δ· λέγω, ὅτι πενταπλάσιόν ἐστι τὸ ἀπὸ τῆς ΒΔ τοῦ ἀπὸ τῆς ΔΓ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΑΒ τετράγωνον τὸ ΑΕ, καὶ καταγεγράφθω διπλοῦν τὸ σχῆμα. ἐπεὶ διπλῆ ἐστιν ἡ ΑΓ τῆς ΔΓ, τετραπλάσιον ἄρα τὸ ἀπὸ τῆς ΑΓ τοῦ ἀπὸ τῆς ΔΓ, τουτέστι τὸ ΡΣ τοῦ ΖΗ. καὶ ἐπεὶ τὸ ὑπὸ τῶν ΑΒΓ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΓ, καί ἐστι τὸ ὑπὸ τῶν ΑΒΓ τὸ ΓΕ, τὸ ἄρα ΓΕ ἴσον ἐστὶ τῷ ΡΣ. τετραπλάσιον δὲ τὸ ΡΣ τοῦ ΖΗ· τετραπλάσιον ἄρα καὶ τὸ ΓΕ τοῦ ΖΗ. πάλιν ἐπεὶ ὀίση ἐστὶν ἡ ΑΔ τῇ ΔΓ, ὀίση ἐστὶ καὶ ἡ ΘΚ τῇ ΚΖ. ὥστε καὶ τὸ ΗΖ τετράγωνον ἴσον ἐστὶ τῷ ΘΛ τετραγώνῳ. ὀίση ἄρα ἡ ΗΚ τῇ ΚΛ, τουτέστιν ἡ ΜΝ τῇ ΝΕ· ὥστε καὶ τὸ ΜΖ τῷ ΖΕ ἐστιν ἴσον. ἀλλὰ τὸ ΜΖ τῷ ΓΗ ἐστιν ἴσον· καὶ τὸ ΓΗ ἄρα τῷ ΖΕ ἐστιν ἴσον. κοινὸν προσκείσθω τὸ ΓΝ· ὁ ἄρα ΧΟΠ γνώμων ὀίσος ἐστὶ τῷ ΓΕ. ἀλλὰ τὸ ΓΕ τετραπλάσιον ἐδείχθη τοῦ ΗΖ· καὶ ὁ ΧΟΠ ἄρα

For let some straight-line AB be cut in extreme and mean ratio at point C. And let AC be the greater piece. And let AC be cut in half at D. I say that the (square) on BD is five times the (square) on DC.

For let the square AE be described on AB. And let the figure be drawn double. Since AC is double DC, the (square) on AC (is) thus four times the (square) on DC—that is to say, RS (is four times) FG. And since the (rectangle contained) by ABC is equal to the (square) on AC [Def. 6.3, Prop. 6.17], and CE is the (rectangle contained) by ABC, CE is thus equal to RS. And RS (is) four times FG. Thus, CE (is) also four times FG. Again, since AD is equal to DC, HK is also equal to KF. Hence, square GF is also equal to square HL. Thus, GK (is) equal to KL—that is to say, MN to NE. Hence, MF is also equal to FE. But, MF is equal to CG. Thus, CG is also equal to FE. Let CN be added to both. Thus, gnomon OPQ is equal to CE. But, CE was shown (to be) equal to four times

γνώμων τετραπλάσιόξ ἐστι τοῦ ΖΗ τετραγώνου. ὁ ΧΟΠ ἄρα γνώμων καὶ τὸ ΖΗ τετράγωνον πενταπλάσιόξ ἐστι τοῦ ΖΗ. ἀλλὰ ὁ ΧΟΠ γνώμων καὶ τὸ ΖΗ τετράγωνόν ἐστι τὸ ΔΝ. καί ἐστι τὸ μὲν ΔΝ τὸ ἀπὸ τῆς ΔΒ, τὸ δὲ ΗΖ τὸ ἀπὸ τῆς ΔΓ. τὸ ἄρα ἀπὸ τῆς ΔΒ πενταπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΔΓ· ὅπερ ἔδει δεῖχαι.

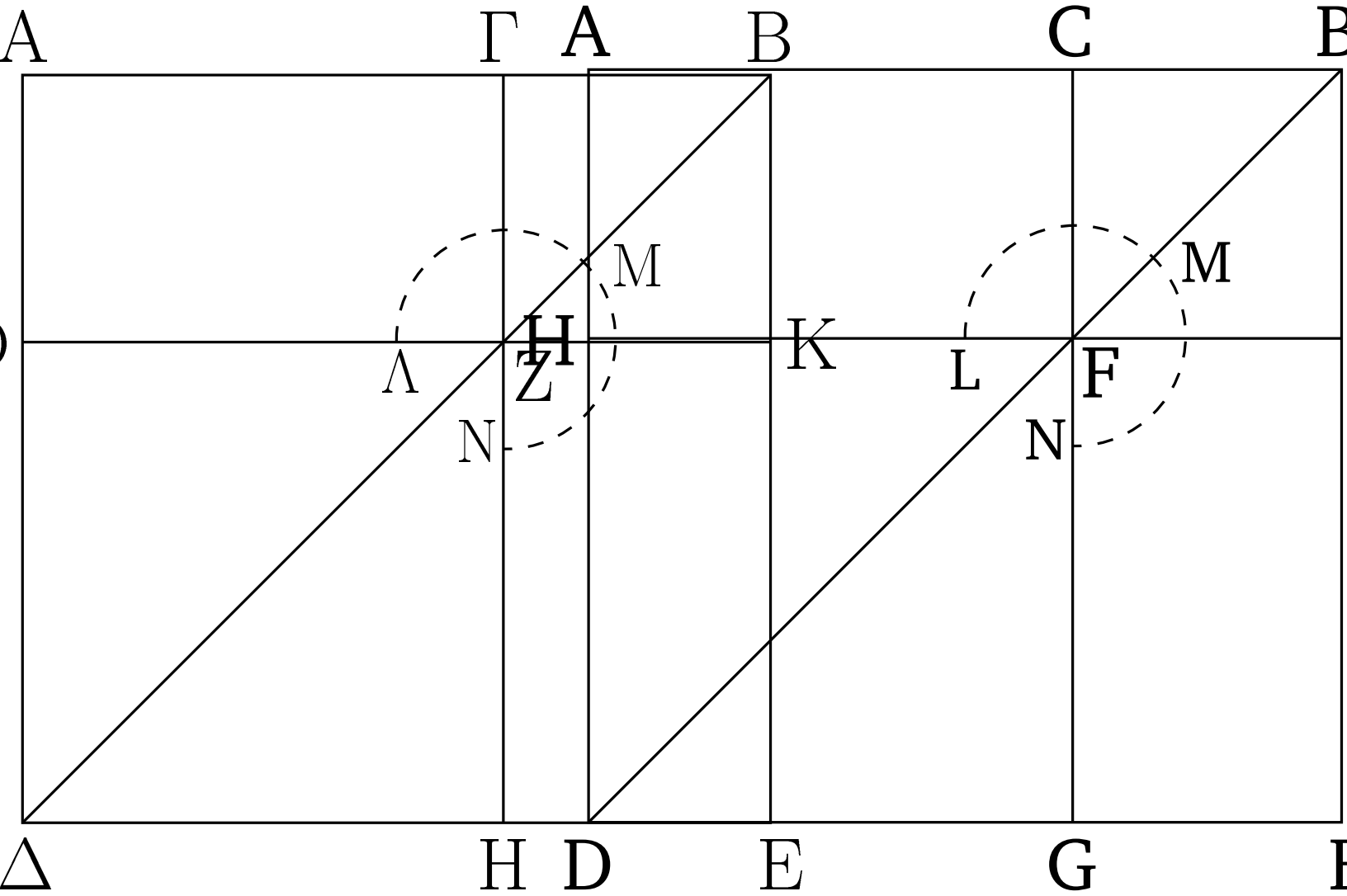
GF. Thus, gnomon *OPQ* is also four times square *FG*. Thus, gnomon *OPQ* plus square *FG* is five times *FG*. But, gnomon *OPQ* plus square *FG* is (square) *DN*. And *DN* is the (square) on *DB*, and *GF* the (square) on *DC*. Thus, the (square) on *DB* is five times the (square) on *DC*. (Which is) the very thing it was required to show.

4.

Ἐὰν εὐθεῖα γραμμὴ ὀάκρον καὶ μέσον λόγον τμ ηθῇ, τὸ ἀπὸ τῆς ὅλης καὶ τοῦ ἐλάσσονος τμήματος, τὰ συναμφότερα τετράγωνα, τριπλάσιά ἐστι τοῦ ἀπὸ τοῦ μείζονος τμήματος τετραγώνου.

Proposition 4

If a straight-line is cut in extreme and mean ratio then the sum of the squares on the whole and the lesser piece is three times the square on the greater piece.



Ἔστω εὐθεῖα ἡ ΑΒ, καὶ τετμήσθω ὀάκρον καὶ μέσον λόγον κατὰ τὸ Γ, καὶ ὀέστω μεῖζον τμῆμα τὸ ΑΓ· λέγω, ὅτι τὰ ἀπὸ τῶν ΑΒ, ΒΓ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΓΑ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΑΒ τετράγωνον τὸ ΑΔΕΒ,

Let *AB* be a straight-line, and let it be cut in extreme and mean ratio at *C*, and let *AC* be the greater piece. I say that the (sum of the squares) on *AB* and *BC* is three times the (square) on *CA*.

καὶ καταγεγράφθω τὸ σχῆμα. ἐπεὶ οὐῦν ἡ ΑΒὀάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Γ, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ΑΓ, τὸ ἄρα ὑπὸ τῶν ΑΒΓ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΓ. καί ἐστι τὸ μὲν ὑπὸ τῶν ΑΒΓ τὸ ΑΚ, τὸ δὲ ἀπὸ τῆς ΑΓ τὸ ΘΗ· ἴσον ἄρα ἐστὶ τὸ ΑΚ τῷ ΘΗ. καὶ ἐπεὶ ἴσον ἐστὶ τὸ ΑΖ τῷ ΖΕ, κοινὸν προσκείσθω τὸ ΓΚ· ὅλον ἄρα τὸ ΑΚ ὅλῳ τῷ ΓΕ ἐστιν ἴσον· τὰ ἄρα ΑΚ, ΓΕ τοῦ ΑΚ ἐστι διπλάσια. ἀλλὰ τὰ ΑΚ, ΓΕ ὁ ΛΜΝ γνώμων ἐστὶ καὶ τὸ ΓΚ τετράγωνον· ὁ ἄρα ΛΜΝ γνώμων καὶ τὸ ΓΚ τετράγωνον διπλάσιά ἐστι τοῦ ΑΚ. ἀλλὰ μὴν καὶ τὸ ΑΚ τῷ ΘΗ ἐδείθθη ἴσον· ὁ ἄρα ΛΜΝ γνώμων καὶ [τὸ ΓΚ τετράγωνον διπλάσιά ἐστι τοῦ ΘΗ· ὥστε ὁ ΛΜΝ γνώμων καὶ] τὰ ΓΚ, ΘΗ τετράγωνα τριπλάσιά ἐστι τοῦ ΘΗ τετραγώνου. καί ἐστιν ὁ [μὲν] ΛΜΝ γνώμων καὶ τὰ ΓΚ, ΘΗ τετράγωνα ὅλον τὸ ΑΕ καὶ τὸ ΓΚ, ὁάπερ ἐστὶ τὰ ἀπὸ τῶν ΑΒ, ΒΓ τετράγωνα, τὸ δὲ ΗΘ τὸ ἀπὸ τῆς ΑΓ τετράγωνον. τὰ ἄρα ἀπὸ τῶν ΑΒ, ΒΓ τετράγωνα τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΑΓ τετραγώνου· ὅπερ ἔδει δεῖχαι.

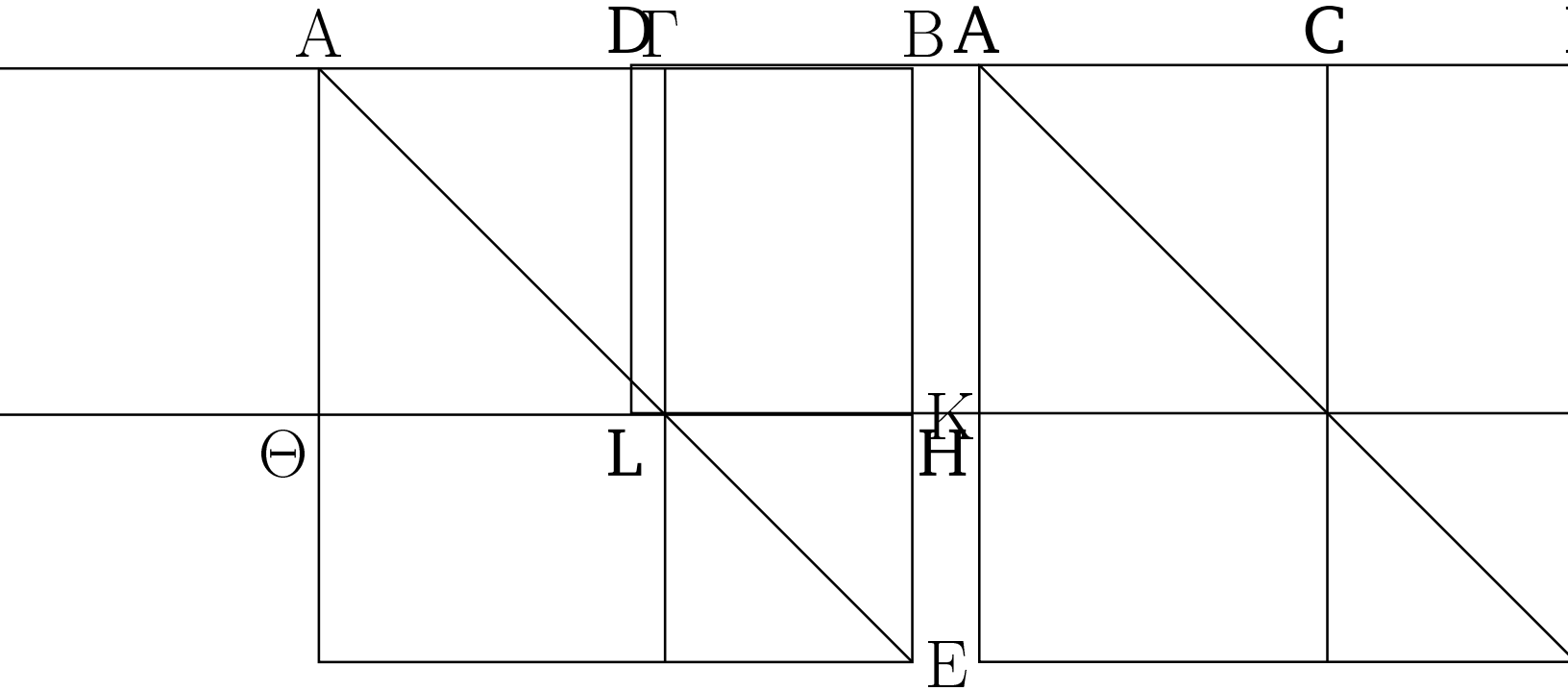
For let the square $ADEB$ be described on AB, and let the (remainder of the) figure be drawn. Therefore, since AB has been cut in extreme and mean ratio at C, and AC is the greater piece, the (rectangle contained) by ABC is thus equal to the (square) on AC [Def. 6.3, Prop. 6.17]. And AK is the (rectangle contained) by ABC, and HG the (square) on AC. Thus, AK is equal to HG. And since AF is equal to FE [Prop. 1.43], let CK be added to both. Thus, the whole of AK is equal to the whole of CE. Thus, AK plus CE is double AK. But, AK plus CE is the gnomon LMN plus the square CK. Thus, gnomon LMN plus square CK is double AK. But, indeed, AK was also shown (to be) equal to HG. Thus, gnomon LMN plus [square CK is double HG. Hence, gnomon LMN plus] the squares CK and HG is three times the square HG. And gnomon LMN plus the squares CK and HG is the whole of AE plus CK—which are the squares on AB and BC (respectively)—and GH (is) the square on AC. Thus, the (sum of the) squares on AB and BC is three times the square on AC. (Which is) the very thing it was required to show.

5.

Proposition 5

Ἐὰν εὐθεῖα γραμμὴ ὀάκρον καὶ μέσον λόγον τμ ηθῇ, καὶ προστεθῇ αὐτῇ ὀίση τῷ μείζονι τμήματι, ἡ ὅλη εὐθεῖαὀάκρον καὶ μέσον λόγον τέτμ ηται, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ἐξ ἀρθῆς εὐθεῖα.

If a straight-line is cut in extreme and mean ratio, and a (straight-line) equal to the greater piece is added to it, then the whole straight-line has been cut in extreme and mean ratio, and the original straight-line is the greater piece.



Εὐθεῖα γὰρ γραμμὴ ἡ ΑΒὀάκρον καὶ μέσον λόγον τετμήσθω κατὰ τὸ Γ σημεῖον, καὶ ὀέστω μεῖζον τμῆμα ἡ ΑΓ,

For let the straight-line AB be cut in extreme and mean ratio at point C. And let AC be the greater piece. And let

καὶ τῇ ΑΓ ὁίση [κείσθω] ἡ ΑΔ. λέγω, ὅτι ἡ ΔΒ εὐθεῖα ὁάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Α, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ἐξ ἀρθῆς εὐθεῖα ἡ ΑΒ.

Ἀναγεγράφθω γὰρ ἀπὸ τῆς ΑΒ τετράγωνον τὸ ΑΕ, καὶ καταγεγράφθω τὸ σχῆμα. ἐπεὶ ἡ ΑΒ ὁάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Γ, τὸ ἄρα ὑπὸ ΑΒΓ ἴσον ἐστὶ τῷ ἀπὸ ΑΓ. καί ἐστι τὸ μὲν ὑπὸ ΑΒΓ τὸ ΓΕ, τὸ δὲ ἀπὸ τῆς ΑΓ τὸ ΓΘ· ἴσον ἄρα τὸ ΓΕ τῷ ΘΓ. ἀλλὰ τῷ μὲν ΓΕ ἴσον ἐστὶ τὸ ΘΕ, τῷ δὲ ΘΓ ἴσον τὸ ΔΘ· καὶ τὸ ΔΘ ἄρα ἴσον ἐστὶ τῷ ΘΕ [κοινὸν προσκείσθω τὸ ΘΒ]. ὅλον ἄρα τὸ ΔΚ ὅλῳ τῷ ΑΕ ἐστιν ἴσον. καί ἐστι τὸ μὲν ΔΚ τὸ ὑπὸ τῶν ΒΔ, ΔΑ· ὁίση γὰρ ἡ ΑΔ τῇ ΔΛ· τὸ δὲ ΑΕ τὸ ἀπὸ τῆς ΑΒ· τὸ ἄρα ὑπὸ τῶν ΒΔΑ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΒ. ὁέστιν ἄρα ὡς ἡ ΔΒ πρὸς τὴν ΒΑ, οὁύτως ἡ ΒΑ πρὸς τὴν ΑΔ. μείζων δὲ ἡ ΔΒ τῆς ΒΑ· μείζων ἄρα καὶ ἡ ΒΑ τῆς ΑΔ.

Ἡ ἄρα ΔΒ ὁάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Α, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ΑΒ· ὅπερ ἔδει δεῖχαι.

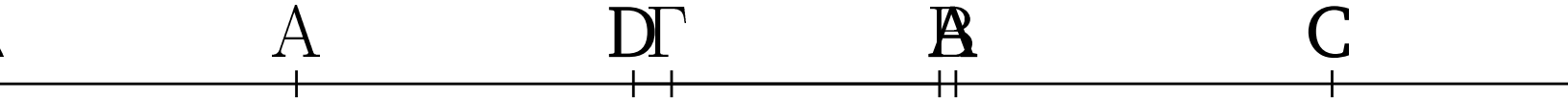
AD be [made] equal to AC. I say that the straight-line DB has been cut in extreme and mean ratio at A, and that the original straight-line AB is the greater piece.

For let the square AE be described on AB, and let the (remainder of the) figure be drawn. And since AB has been cut in extreme and mean ratio at C, the (rectangle contained) by ABC is thus equal to the (square) on AC [Def. 6.3, Prop. 6.17]. And CE is the (rectangle contained) by ABC, and CH the (square) on AC. But, HE is equal to CE [Prop. 1.43], and DH equal to HC. Thus, DH is also equal to HE. [Let HB be added to both.] Thus, the whole of DK is equal to the whole of AE. And DK is the (rectangle contained) by BD and DA. For AD (is) equal to DL. And AE (is) the (square) on AB. Thus, the (rectangle contained) by BDA is equal to the (square) on AB. Thus, as DB (is) to BA, so BA (is) to AD [Prop. 6.17]. And DB (is) greater than BA. Thus, BA (is) also greater than AD [Prop. 5.14].

Thus, DB has been cut in extreme and mean ratio at A, and the greater piece is AB. (Which is) the very thing it was required to show.

6.

Ἐὰν εὐθεῖα ὁρητὴ ὁάκρον καὶ μέσον λόγον τμ ηθῇ, ἑκάτερον τῶν τμ ημάτων ὁάλογόξ ἐστιν ἡ καλουμένη ἀποτομή.



Ἔστω εὐθεῖα ὁρητὴ ἡ ΑΒ καὶ τετμήσθω ὁάκρον καὶ μέσον λόγον κατὰ τὸ Γ, καὶ ὁέστω μεῖζον τμῆμα ἡ ΑΓ· λέγω, ὅτι ἑκατέρα τῶν ΑΓ, ΓΒ ὁάλογόξ ἐστιν ἡ καλουμένη ἀποτομή.

Ἐκβεβλήσθω γὰρ ἡ ΒΑ, καὶ κείσθω τῆς ΒΑ ἡμίσεια ἡ ΑΔ. ἐπεὶ οὁῦν εὐθεῖα ἡ ΑΒ τέτμ ηταιὁάκρον καὶ μέσον λόγον κατὰ τὸ Γ, καὶ τῷ μείζονι τμήματι τῷ ΑΓ πρόσκειται ἡ ΑΔ ἡμίσεια οὁῦσα τῆς ΑΒ, τὸ ἄρα ἀπὸ ΓΔ τοῦ ἀπὸ ΔΑ πενταπλάσιόν ἐστιν. τὸ ἄρα ἀπὸ ΓΔ πρὸς τὸ ἀπὸ ΔΑ λόγον ἔχει, ὃν ἀριθμὸξ πρὸς ἀριθμόν· σύμμετρον ἄρα τὸ ἀπὸ ΓΔ τῷ ἀπὸ ΔΑ. ὁρητὸν δὲ τὸ ἀπὸ ΔΑ· ὁρητὴ γάρ [ἐστιν] ἡ ΔΑ ἡμίσεια οὁῦσα τῆς ΑΒ ὁρητῆς οὁύσης· ὁρητὸν ἄρα καὶ τὸ ἀπὸ ΓΔ· ὁρητὴ ἄρα ἐστὶ καὶ ἡ ΓΔ. καὶ ἐπεὶ τὸ ἀπὸ ΓΔ πρὸς τὸ ἀπὸ ΔΑ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν, ἀσύμμετρος ἄρα μήκει ἡ ΓΔ τῇ ΔΑ· αἱ ΓΔ, ΔΑ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι· ἀποτομὴ ἄρα ἐστὶν ἡ ΑΓ. πάλιν, ἐπεὶ ἡ ΑΒ ὁάκρον καὶ μέσον λόγον τέτμ ηται, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ΑΓ, τὸ ἄρα ὑπὸ ΑΒ, ΒΓ τῷ ἀπὸ ΑΓ ἴσον ἐστίν. τὸ ἄρα ἀπὸ τῆς ΑΓ ἀποτομῆς παρὰ τὴν ΑΒ ὁρητὴν παραβληθὲν πλάτος ποιεῖ τὴν ΒΓ. τὸ δὲ ἀπὸ ἀποτομῆς παρὰ ὁρητὴν παραβαλλόμενον πλάτος ποιεῖ ἀποτομὴν πρώτην· ἀποτομὴ ἄρα πρώτη ἐστὶν ἡ ΓΒ. ἐδείθθη δὲ καὶ ἡ ΓΑ ἀποτομή.

Proposition 6

If a rational straight-line is cut in extreme and mean ratio then each of the pieces is that irrational (straight-line) called an apotome.

Let AB be a rational straight-line cut in extreme and mean ratio at C, and let AC be the greater piece. I say that AC and CB is each that irrational (straight-line) called an apotome.

For let BA be produced, and let AD be made (equal) to half of BA. Therefore, since the straight-line AB has been cut in extreme and mean ratio at C, and AD, which is half of AB, has been added to the greater piece AC, the (square) on CD is thus five times the (square) on DA [Prop. 13.1]. Thus, the (square) on CD has to the (square) on DA the ratio which a number (has) to a number. The (square) on CD (is) thus commensurable with the (square) on DA [Prop. 10.6]. And the (square) on DA (is) rational. For DA [is] rational, being half of AB, which is rational. Thus, the (square) on CD (is) also rational [Def. 10.4]. Thus, CD is also rational. And since the (square) on CD does not have to the (square) on DA the ratio which a square number (has) to a square number, CD (is) thus incommensurable in length with DA [Prop. 10.9]. Thus, CD and DA are rational (straight-lines which are) commensurable in square only. Thus, AC is an apotome [Prop. 10.73]. Again, since AB has been cut in extreme and mean ratio, and AC is the greater piece, the (rectangle contained) by AB and BC is thus equal to the (square) on AC [Def. 6.3, Prop. 6.17].

Ἐὰν ἄρα εὐθεῖα ὁρητὴ ὁάκρον καὶ μέσον λόγον τμ ηθῇ, ἑκάτερον τῶν τμ ημάτωνὁάλογόξ ἐστιν ἡ καλουμένη ἀποτομή· ὅπερ ἔδει δεῖχαι.

Thus, the (square) on the apotome AC, applied to the rational (straight-line) AB, makes BC as width. And the (square) on an apotome, applied to a rational (straight-line), makes a first apotome as width [Prop. 10.97]. Thus, CB is a first apotome. And CA was also shown (to be) an apotome.

Thus, if a rational straight-line is cut in extreme and mean ratio then each of the pieces is that irrational (straight-line) called an apotome.

7.

Ἐὰν πενταγώνου ἰσοπλεύρου αἱ τρεῖς γωνίαι ἤτοι αἱ κατὰ τὸ ἑχῆςὸἡ αἱ μὴ κατὰ τὸ ἑχῆς ἴσαι ὦσιν, ἰσογώνιονὸέσται τὸ πεντάγωνον.

Πενταγώνου γὰρ ἰσοπλεύρον τοῦ ΑΒΓΔΕ αἱ τρεῖς γωνίαι πρότερον αἱ κατὰ τὸ ἑχῆς αἱ πρὸς τοῖς Α, Β, Γ ἴσαι ἀλλήλαιςὸέστωσαν· λέγω, ὅτι ἰσογώνιόν ἐστι τὸ ΑΒΓΔΕ πεντάγωνον.

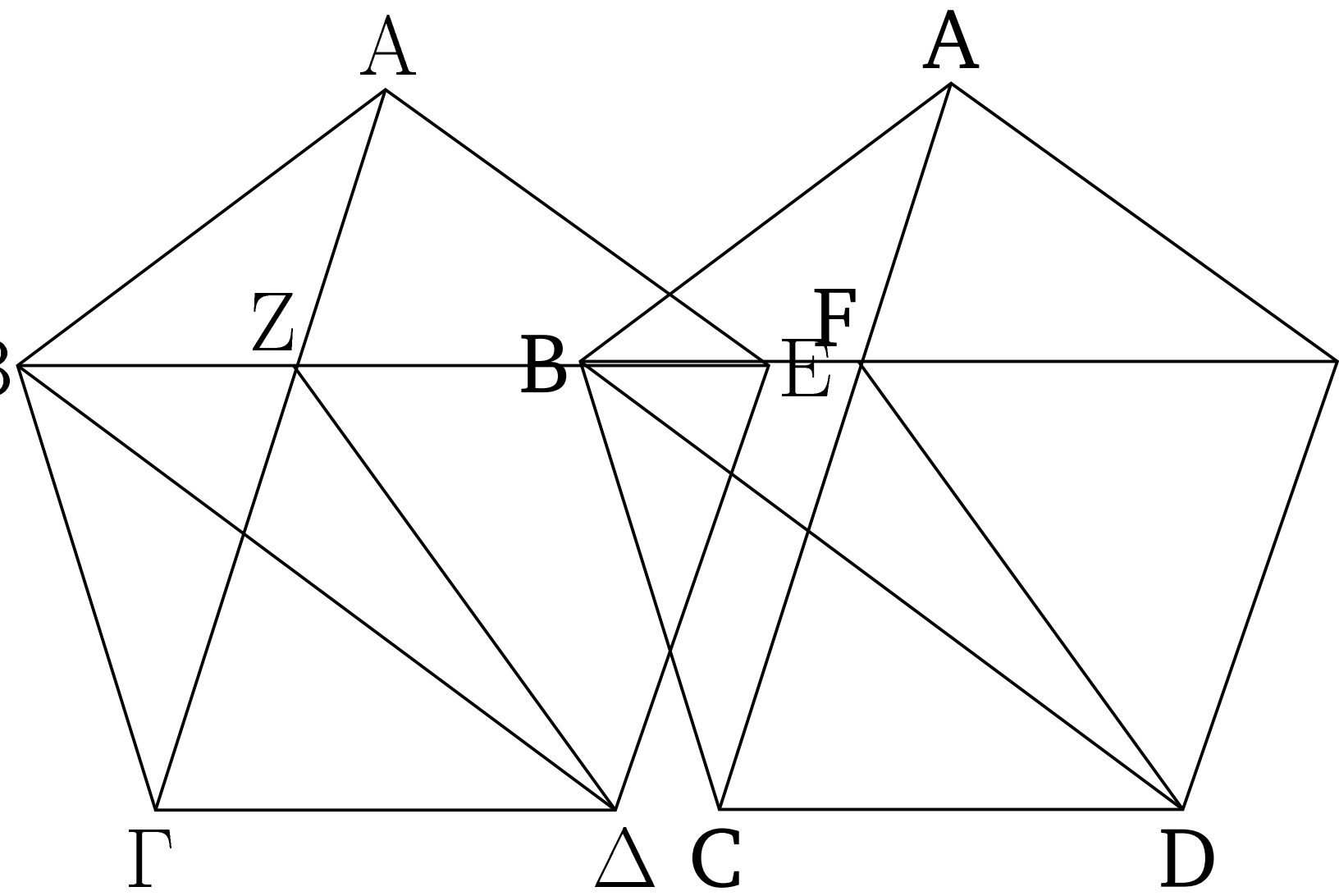
Ἐπεζεύθθωσαν γὰρ αἱ ΑΓ, ΒΕ, ΖΔ. καὶ ἐπεὶ δύο αἱ ΓΒ, ΒΑ δυσὶ ταῖς ΒΑ, ΑΕ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, καὶ γωνία ἡ ὑπὸ ΓΒΑ γωνίᾳ τῇ ὑπὸ ΒΑΕ ἐστινὸίση, βάσις ἄρα ἡ ΑΓ βάσει τῇ ΒΕ ἐστινὸίση, καὶ τὸ ΑΒΓ τρίγωνον τῷ ΑΒΕ τριγώνῳ ἴσον, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖξ γωνίαις ἴσαιὸέσονται, ὑφὸ ὰὰξ αἱ ἴσαι πλευραὶ ὑποτείνουσιν, ἡ μὲν ὑπὸ ΒΓΑ τῇ ὑπὸ ΒΕΑ, ἡ δὲ ὑπὸ ΑΒΕ τῇ ὑπὸ ΓΑΒ· ὁώστε καὶ πλευρὰ ἡ ΑΖ πλευρᾷ τῇ ΒΖ ἐστινὸίση. ἐδείθθη δὲ καὶ ὅλη ἡ ΑΓ ὅλῃ τῇ ΒΕὸίση· καὶ λοιπὴ ἄρα ἡ ΖΓ λοιπῇ τῇ ΖΕ ἐστινὸίση. ὸέστι δὲ καὶ ἡ ΓΔ τῇ ΔΕὸίση. δύο δὴ αἱ ΖΓ, ΓΔ δυσὶ ταῖς ΖΕ, ΕΔ ἴσαι εἰσίν· καὶ βάσις αὐτῶν κοινὴ ἡ ΖΔ· γωνία ἄρα ἡ ὑπὸ ΖΓΔ γωνίᾳ τῇ ὑπὸ ΖΕΔ ἐστινὸίση. ἐδείθθη δὲ καὶ ἡ ὑπὸ ΒΓΑ τῇ ὑπὸ ΑΕΒὸίση· καὶ ὅλη ἄρα ἡ ὑπὸ ΒΓΔ ὅλῃ τῇ ὑπὸ ΑΕΔὸίση. ἀλλὸ ἡ ὑπὸ ΒΓΔὸίση ὑπόκειται ταῖς πρὸς τοῖς Α, Β γωνίαις· καὶ ἡ ὑπὸ ΑΕΔ ἄρα ταῖς πρὸς τοῖς Α, Β γωνίαιςὸίση ἐστίν. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ ὑπὸ ΓΔΕ γωνίαὸίση ἐστὶ ταῖς πρὸς τοῖς Α, Β, Γ γωνίαις· ἰσογώνιον ἄρα ἐστὶ τὸ ΑΒΓΔΕ πεντάγωνον.

Proposition 7

If three angles, either consecutive or not consecutive, of an equilateral pentagon are equal then the pentagon will be equiangular.

For let three angles of the equilateral pentagon $ABCDE$—first of all, the consecutive (angles) at A, B, and C—-be equal to one another. I say that pentagon $ABCDE$ is equiangular.

For let AC, BE, and FD be joined. And since the two (straight-lines) CB and BA are equal to the two (straight-lines) BA and AE, respectively, and angle CBA is equal to angle BAE, base AC is thus equal to base BE, and triangle ABC equal to triangle ABE, and the remaining angles will be equal to the remaining angles which the equal sides subtend [Prop. 1.4], (that is), BCA (equal) to BEA, and ABE to CAB. And hence side AF is also equal to side BF [Prop. 1.6]. And the whole of AC was also shown (to be) equal to the whole of BE. Thus, the remainder FC is also equal to the remainder FE. And CD is also equal to DE. So, the two (straight-lines) FC and CD are equal to the two (straight-lines) FE and ED (respectively). And FD is their common base. Thus, angle FCD is equal to angle FED [Prop. 1.8]. And BCA was also shown (to be) equal to AEB. And thus the whole of BCD (is) equal to the whole of AED. But, (angle) BCD was assumed (to be) equal to the angles at A and B. Thus, (angle) AED is also equal to the angles at A and B. So, similarly, we can show that angle CDE is also equal to the angles at A, B, C. Thus, pentagon $ABCDE$ is equiangular.



Ἀλλὰ δὴ μὴ ὀέστωσαν ἴσαι αἱ κατὰ τὸ ἑξῆς γωνίαι, ἀλλὸ ὀέστωσαν ἴσαι αἱ πρὸς τοῖς Α, Γ, Δ σημείοις· λέγω, ὅτι καὶ οὑτως ἰσογώνιόν ἐστι τὸ ΑΒΓΔΕ πεντάγωνον.

Ἐπεζεύθθω γὰρ ἡ ΒΔ. καὶ ἐπεὶ δύο αἱ ΒΑ, ΑΕ δυσὶ ταῖς ΒΓ, ΓΔ ἴσαι εἰσὶ καὶ γωνίας ἴσας περιέθουσιν, βάσις ἄρα ἡ ΒΕ βάσει τῇ ΒΔὀίση ἐστίν, καὶ τὸ ΑΒΕ τρίγωνον τῷ ΒΓΔ τριγώνῳ ἴσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖξ γωνίαις ἴσαιὀέσονται, ὑφὸ ὁὰξ αἱ ἴσαι πλευραὶ ὑποτείνουσιν· ὀίση ἄρα ἐστὶν ἡ ὑπὸ ΑΕΒ γωνία τῇ ὑπὸ ΓΔΒ. ὀέστι δὲ καὶ ἡ ὑπὸ ΒΕΔ γωνία τῇ ὑπὸ ΒΔΕὀίση, ἐπεὶ καὶ πλευρὰ ἡ ΒΕ πλευρᾷ τῇ ΒΔ ἐστινὀίση. καὶ ὅλη ἄρα ἡ ὑπὸ ΑΕΔ γωνία ὅλῃ τῇ ὑπὸ ΓΔΕ ἐστινὀίση. ἀλλὰ ἡ ὑπὸ ΓΔΕ ταῖς πρὸς τοῖς Α, Γ γωνίαις ὑπόκειταιὀίση· καὶ ἡ ὑπὸ ΑΕΔ ἄρα γωνία ταῖς πρὸς τοῖς Α, Γὀίση ἐστίν. διὰ τὰ αὐτὰ δὴ καὶ ἡ ὑπὸ ΑΒΓὀίση ἐστὶ ταῖς πρὸς τοῖς Α, Γ, Δ γωνίαις. ἰσογώνιον ἄρα ἐστὶ τὸ ΑΒΓΔΕ πεντάγωνον· ὅπερ ἔδει δεῖχαι.

And so let consecutive angles not be equal, but let the (angles) at points *A*, *C*, and *D* be equal. I say that pentagon *ABCDE* is also equiangular in this case.

For let *BD* be joined. And since the two (straight-lines) *BA* and *AE* are equal to the (straight-lines) *BC* and *CD*, and they contain equal angles, base *BE* is thus equal to base *BD*, and triangle *ABE* is equal to triangle *BCD*, and the remaining angles will be equal to the remaining angles which the equal sides subtend [Prop. 1.4]. Thus, angle *AEB* is equal to (angle) *CDB*. And angle *BED* is also equal to (angle) *BDE*, since side *BE* is also equal to side *BD* [Prop. 1.5]. Thus, the whole angle *AED* is also equal to the whole (angle) *CDE*. But, (angle) *CDE* was assumed (to be) equal to the angles at *A* and *C*. Thus, angle *AED* is also equal to the (angles) at *A* and *C*. So, for the same (reasons), (angle) *ABC* is also equal to the angles at *A*, *C*, and *D*. Thus, pentagon *ABCDE* is equiangular. (Which is)

8.

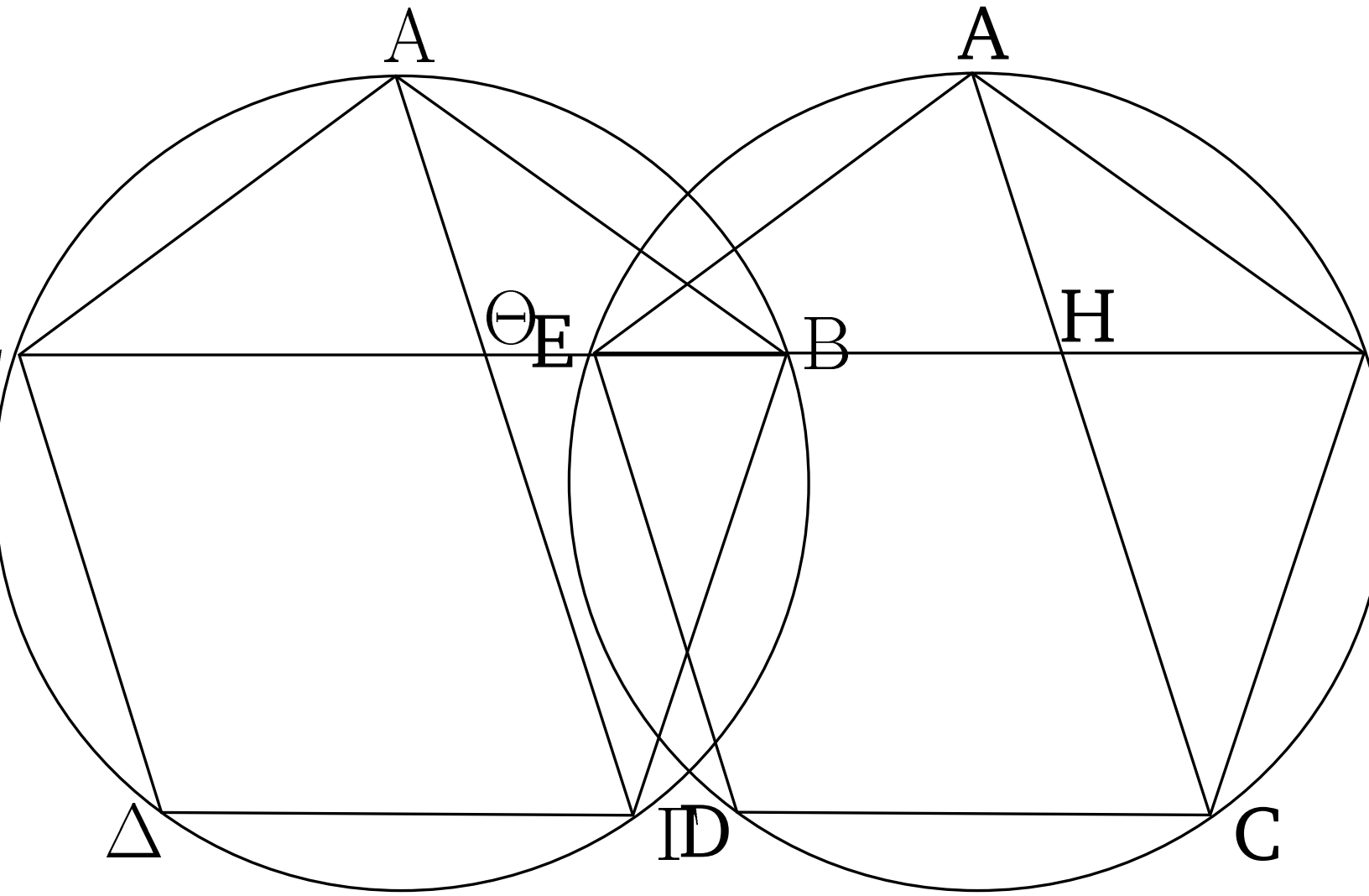
Ἐὰν πενταγώνου ἰσοπλεύρου καὶ ἰσογωνίου τὰξ κατὰ τὸ ἑξῆς δύο γωνίας ὑποτείνωσιν εὐθεῖαι, ὀάκρον καὶ μέσον λόγον τέμνουσιν ἀλλήλας, καὶ τὰ μείζονα αὐτῶν τμήματα ἴσα ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ.

Πενταγώνου γὰρ ἰσοπλεύρον καὶ ἰσογωνίου τοῦ ΑΒΓΔΕ δύο γωνίας τὰξ κατὰ τὸ ἑξῆς τὰξ πρὸς τοῖς Α, Β ὑποτεινέτωσαν εὐθεῖαι αἱ ΑΓ, ΒΕ τέμνουσαι ἀλλήλας κατὰ τὸ Θ σημεῖον· λέγω, ὅτι ἑκατέρα αὐτῶνὀάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Θ σημεῖον, καὶ τὰ μείζονα αὐτῶν τμήματα ἴσα ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ.

Proposition 8

If straight-lines subtend two consecutive angles of an equilateral and equiangular pentagon then they cut one another in extreme and mean ratio, and their greater pieces are equal to the sides of the pentagon.

For let the two straight-lines, *AC* and *BE*, cutting one another at point *H*, have subtended two consecutive angles, at *A* and *B* (respectively), of the equilateral and equiangular pentagon *ABCDE*. I say that each of them has been cut in extreme and mean ratio at point *H*, and that their greater pieces are equal to the sides of the pentagon.



Περιγεγράφθω γὰρ περὶ τὸ ΑΒΓΔΕ πεντάγωνον κύκλος ὁ ΑΒΓΔΕ. καὶ ἐπεὶ δύο εὐθεῖαι αἱ ΕΑ, ΑΒ δυσὶ ταῖς ΑΒ, ΒΓ ἴσαι εἰσὶ καὶ γωνίας ἴσας περιέθουσιν, βάσις ἄρα ἡ ΒΕ βάσει

For let the circle *ABCDE* be circumscribed about pentagon *ABCDE* [Prop. 4.14]. And since the two straight-lines *EA* and *AB* are equal to the two (straight-lines) *AB* and *BC*

τῇ ΑΓὁίση ἐστίν, καὶ τὸ ΑΒΕ τρίγωνον τῷ ΑΒΓ τριγώνῳ ἴσον ἐστίν, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖξ γωνίαις ἴσαιὁέσονται ἑκατέρα ἑκατέρᾳ, ὑφὸ ἃὰξ αἱ ἴσαι πλευραὶ ὑποτείνουσιν. ὁίση ἄρα ἐστὶν ἡ ὑπὸ ΒΑΓ γωνία τῇ ὑπὸ ΑΒΕ· διπλῆ ἄρα ἡ ὑπὸ ΑΘΕ τῆς ὑπὸ ΒΑΘ. ὁέστι δὲ καὶ ἡ ὑπὸ ΕΑΓ τῆς ὑπὸ ΒΑΓ διπλῆ, ἐπειδήπερ καὶ περιφέρεια ἡ ΕΔΓ περιφερείας τῆς ΓΒ ἐστι διπλῆ· ὁίση ἄρα ἡ ὑπὸ ΘΑΕ γωνία τῇ ὑπὸ ΑΘΕ· ὡστε καὶ ἡ ΘΕ εὐθεῖα τῇ ΕΑ, τουτέστι τῇ ΑΒ ἐστινὁίση. καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΒΑ εὐθεῖα τῇ ΑΕ, ὁίση ἐστὶ καὶ γωνία ἡ ὑπὸ ΑΒΕ τῇ ὑπὸ ΑΕΒ. ἀλλὰ ἡ ὑπὸ ΑΒΕ τῇ ὑπὸ ΒΑΘ ἐδείθηὁίση· καὶ ἡ ὑπὸ ΒΕΑ ἄρα τῇ ὑπὸ ΒΑΘ ἐστινὁίση. καὶ κοινὴ τῶν δύο τριγώνων τοῦ τε ΑΒΕ καὶ τοῦ ΑΒΘ ἐστιν ἡ ὑπὸ ΑΒΕ· λοιπὴ ἄρα ἡ ὑπὸ ΒΑΕ γωνία λοιπῇ τῇ ὑπὸ ΑΘΒ ἐστινὁίση· ἰσογώνιον ἄρα ἐστὶ τὸ ΑΒΕ τρίγωνον τῷ ΑΒΘ τριγώνῳ· ἀνάλογον ἄρα ἐστὶν ὡξ ἡ ΕΒ πρὸς τὴν ΒΑ, οὕτως ἡ ΑΒ πρὸς τὴν ΒΘ. ὁίση δὲ ἡ ΒΑ τῇ ΕΘ· ὡξ ἄρα ἡ ΒΕ πρὸς τὴν ΕΘ, οὕτως ἡ ΕΘ πρὸς τὴν ΘΒ. μείζων δὲ ἡ ΒΕ τῆς ΕΘ· μείζων ἄρα καὶ ἡ ΕΘ τῆς ΘΒ. ἡ ΒΕ ὁάραὁάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Θ, καὶ τὸ μεῖζον τμῆμα τὸ ΘΕ ἴσον ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ ΑΓὁάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Θ, καὶ τὸ μεῖζον αὐτῆς τμῆμα ἡ ΓΘ ἴσον ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ· ὅπερ ἔδει δεῖχαι.

(respectively), and they contain equal angles, the base BE is thus equal to the base AC, and triangle ABE is equal to triangle ABC, and the remaining angles will be equal to the remaining angles, respectively, which the equal sides subtend [Prop. 1.4]. Thus, angle BAC is equal to (angle) ABE. Thus, (angle) AHE (is) double (angle) BAH [Prop. 1.32]. And EAC is also double BAC, inasmuch as circumference EDC is also double circumference CB [Props. 3.28, 6.33]. Thus, angle HAE (is) equal to (angle) AHE. Hence, straight-line HE is also equal to (straight-line) EA—that is to say, to (straight-line) AB [Prop. 1.6]. And since straight-line BA is equal to AE, angle ABE is also equal to AEB [Prop. 1.5]. But, ABE was shown (to be) equal to BAH. Thus, BEA is also equal to BAH. And (angle) ABE is common to the two triangles ABE and ABH. Thus, the remaining angle BAE is equal to the remaining (angle) AHB [Prop. 1.32]. Thus, triangle ABE is equiangular to triangle ABH. Thus, proportionally, as EB is to BA, so AB (is) to BH [Prop. 6.4]. And BA (is) equal to EH. Thus, as BE (is) to EH, so EH (is) to HB. And BE (is) greater than EH. EH (is) thus also greater than HB [Prop. 5.14]. Thus, BE has been cut in extreme and mean ratio at H, and the greater piece HE is equal to the side of the pentagon. So, similarly, we can show that AC has also been cut in extreme and mean ratio at H, and that its greater piece CH is equal to the side of the pentagon. (Which is) the very thing it was required to show.

9.

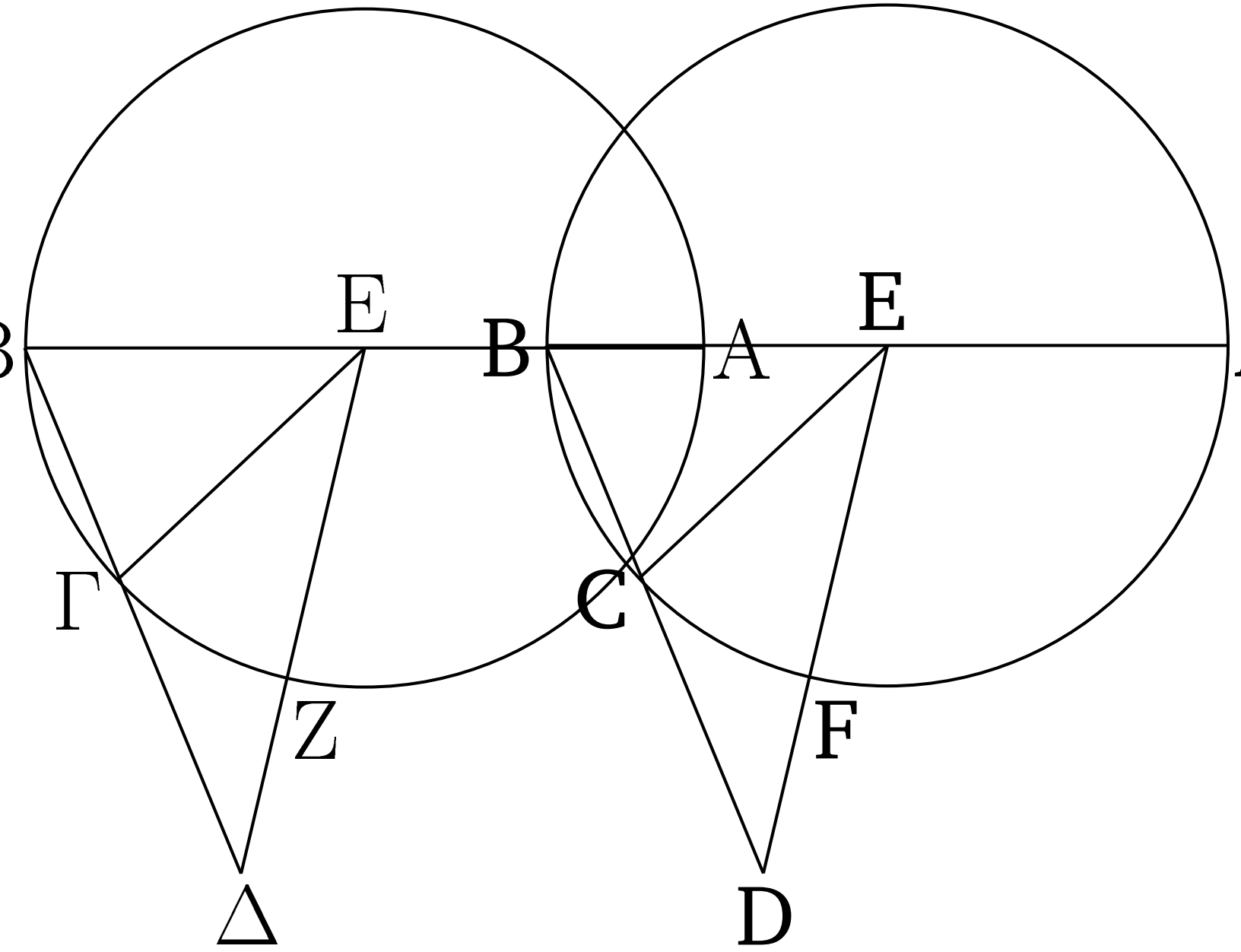
Ἐὰν ἡ τοῦ ἑχαγώνου πλευρὰ καὶ ἡ τοῦ δεκαγώνου τῶν εἰξ τὸν αὐτὸν κύκλον ἐγγραφομένων συντεθῶσιν, ἡ ὅλη εὐθεῖαὁάκρον καὶ μέσον λόγον τέτμ ηται, καὶ τὸ μεῖχον αὐτῆς τμῆμά ἐστιν ἡ τοῦ ἑχαγώνου πλευρά.

Ἔστω κύκλος ὁ ΑΒΓ, καὶ τῶν εἰξ τὸν ΑΒΓ κύκλον ἐγγραφομένων σχημάτων, δεκαγώνου μὲνὁέστω πλευρὰ ἡ ΒΓ, ἑχαγώνου δὲ ἡ ΓΔ, καὶ ὁέστωσαν ἐπ΄ εὐθείας· λέγω, ὅτι ἡ ὅλη εὐθεῖα ἡ ΒΔὁάκρον καὶ μέσον λόγον τέτμ ηται, καὶ τὸ μεῖζον αὐτῆς τμῆμά ἐστιν ἡ ΓΔ.

Proposition 9

If the side of a hexagon and of a decagon inscribed in the same circle are added together then the whole straight-line has been cut in extreme and mean ratio (at the junction point), and its greater piece is the side of the hexagon.[†]

Let ABC be a circle. And of the figures inscribed in circle ABC, let BC be the side of a decagon, and CD (the side) of a hexagon. And let them be (laid down) straight-on (to one another). I say that the whole straight-line BD has been cut in extreme and mean ratio (at C), and that CD is its greater piece.



Εἰλήφθω γὰρ τὸ κέντρον τοῦ κύκλου τὸ Ε σημεῖον, καὶ ἐπεζεύθθωσαν αἱ ΕΒ, ΕΓ, ΕΔ, καὶ διήθθω ἡ ΒΕ ἐπὶ τὸ Α. ἐπεὶ δεκαγώνου ἰσοπλεύρου πλευρά ἐστιν ἡ ΒΓ, πενταπλασίων ἄρα ἡ ΑΓΒ περιφέρεια τῆς ΒΓ περιφερείας· τετραπλασίων ἄρα ἡ ΑΓ περιφέρεια τῆς ΓΒ. ὡξ δὲ ἡ ΑΓ περιφέρεια πρὸς τὴν ΓΒ, οὑτως ἡ ὑπὸ ΑΕΓ γωνία πρὸς τὴν ὑπὸ ΓΕΒ· τετραπλασίων ἄρα ἡ ὑπὸ ΑΕΓ τῆς ὑπὸ ΓΕΒ. καὶ ἐπεὶ ὁίση ἡ ὑπὸ ΕΒΓ γωνία τῇ ὑπὸ ΕΓΒ, ἡ ἄρα ὑπὸ ΑΕΓ γωνία διπλασία ἐστὶ τῆς ὑπὸ ΕΓΒ. καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΕΓ εὐθεῖα τῇ ΓΔ· ἑκατέρα γὰρ αὐτῶνὁίση ἐστὶ τῇ τοῦ ἑχαγώνου πλευρᾷ τοῦ εἰξ τὸν ΑΒΓ κύκλον [ἐγγραφομένου]· ὁίση ἐστὶ καὶ ἡ ὑπὸ ΓΕΔ

For let the center of the circle, point E, be found [Prop. 3.1], and let EB, EC, and ED be joined, and let BE be drawn across to A. Since BC is a side on an equilateral decagon, circumference ACB (is) thus five times circumference BC. Thus, circumference AC (is) four times CB. And as circumference AC (is) to CB, so angle AEC (is) to CEB [Prop. 6.33]. Thus, (angle) AEC (is) four times CEB. And since angle EBC (is) equal to ECB [Prop. 1.5], angle AEC is thus double ECB [Prop. 1.32]. And since straight-line EC is equal to CD—for each of them is equal to the side of the hexagon [inscribed] in circle ABC [Prop. 4.15 corr.]— angle CED is also equal to

γωνία τῇ ὑπὸ ΓΔΕ γωνίᾳ· διπλασία ἄρα ἡ ὑπὸ ΕΓΒ γωνία τῆς ὑπὸ ΕΔΓ. ἀλλὰ τῆς ὑπὸ ΕΓΒ διπλασία ἐδείχθη ἡ ὑπὸ ΑΕΓ· τετραπλασία ἄρα ἡ ὑπὸ ΑΕΓ τῆς ὑπὸ ΕΔΓ. ἐδείχθη δὲ καὶ τῆς ὑπὸ ΒΕΓ τετραπλασία ἡ ὑπὸ ΑΕΓ· ὁίση ἄρα ἡ ὑπὸ ΕΔΓ τῇ ὑπὸ ΒΕΓ. κοινὴ δὲ τῶν δύο τριγώνων, τοῦ τε ΒΕΓ καὶ τοῦ ΒΕΔ, ἡ ὑπὸ ΕΒΔ γωνία· καὶ λοιπὴ ἄρα ἡ ὑπὸ ΒΕΔ τῇ ὑπὸ ΕΓΒ ἐστινὁίση· ἰσογώνιον ἄρα ἐστὶ τὸ ΕΒΔ τρίγωνον τῷ ΕΒΓ τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὡς ἡ ΔΒ πρὸς τὴν ΒΕ, οὑτως ἡ ΕΒ πρὸς τὴν ΒΓ. ὁίση δὲ ἡ ΕΒ τῇ ΓΔ. ὁέστιν ἄρα ὡξ ἡ ΒΔ πρὸς τὴν ΔΓ, οὑτως ἡ ΔΓ πρὸς τὴν ΓΒ. μείζων δὲ ἡ ΒΔ τῆς ΔΓ· μείζων ἄρα καὶ ἡ ΔΓ τῆς ΓΒ. ἡ ΒΔ ἄρα εὐθεῖαὁάκρον καὶ μέσον λόγον τέτμ ηται [κατὰ τὸ Γ], καὶ τὸ μεῖζον τμῆμα αὐτῆς ἐστιν ἡ ΔΓ· ὅπερ ἔδει δεῖχαι.

angle *CDE* [Prop. 1.5]. Thus, angle *ECB* (is) double *EDC* [Prop. 1.32]. But, *AEC* was shown (to be) double *ECB*. Thus, *AEC* (is) four times *EDC*. And *AEC* was also shown (to be) four times *BEC*. Thus, *EDC* (is) equal to *BEC*. And angle *EBD* (is) common to the two triangles *BEC* and *BED*. Thus, the remaining (angle) *BED* is equal to the (remaining angle) *ECB* [Prop. 1.32]. Thus, triangle *EBD* is equiangular to triangle *EBC*. Thus, proportionally, as *DB* is to *BE*, so *EB* (is) to *BC* [Prop. 6.4]. And *EB* (is) equal to *CD*. Thus, as *BD* is to *DC*, so *DC* (is) to *CB*. And *BD* (is) greater than *DC*. Thus, *DC* (is) also greater than *CB* [Prop. 5.14]. Thus, the straight-line *BD* has been cut in extreme and mean ratio [at *C*], and *DC* is its greater piece. (Which is), the very thing it was required to show.

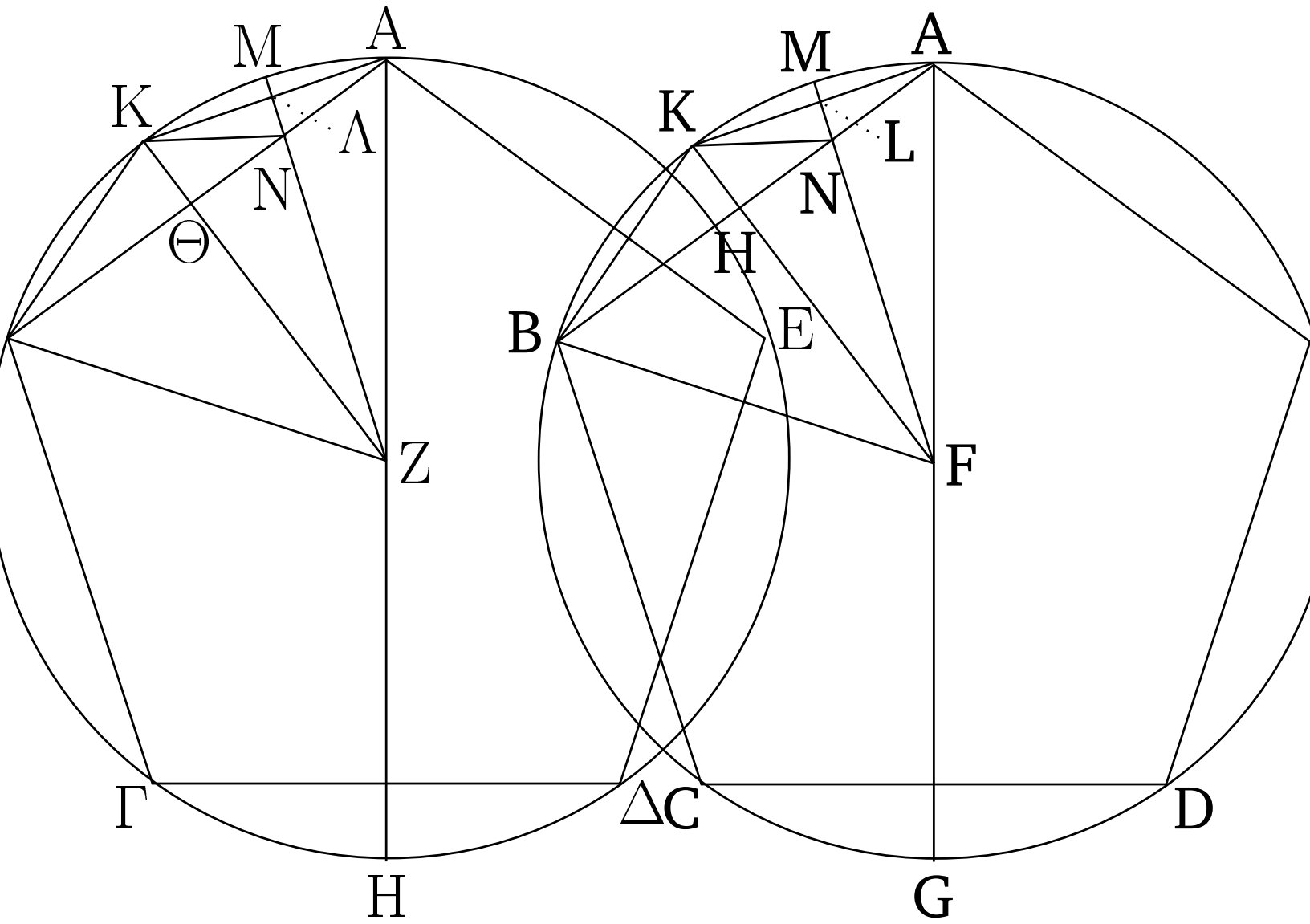
† If the circle is of unit radius then the side of the hexagon is 1, whereas the side of the decagon is $(1/2)(\sqrt{5}-1)$.

10.

Ἐὰν εἰξ κύκλον πεντάγωνον ἰσόπλευρον ἐγγραφῇ, ἡ τοῦ πενταγώνου πλευρὰ δύναται τήν τε τοῦ ἑξαγώνου καὶ τὴν τοῦ δεκαγώνου τῶν εἰξ τὸν αὐτὸν κύκλον ἐγγραφομένων.

Proposition 10

If an equilateral pentagon is inscribed in a circle then the square on the side of the pentagon is (equal to) the (sum of the squares) on the (sides) of the hexagon and of the decagon inscribed in the same circle.†



Ἔστω κύκλος ὁ ΑΒΓΔΕ, καὶ εἰξ τὸ ΑΒΓΔΕ κύκλον πεντάγωνον ἰσόπλευρον ἐγγεγράφθω τὸ ΑΒΓΔΕ. λέγω, ὅτι ἡ τοῦ ΑΒΓΔΕ πενταγώνου πλευρὰ δύναται τήν τε τοῦ ἑξαγώνου καὶ τὴν τοῦ δεκαγώνου πλευρὰν τῶν εἰξ τὸν ΑΒΓΔΕ κύκλον ἐγγραφομένων.

Εἰλήφθω γὰρ τὸ κέντρον τοῦ κύκλου τὸ Ζ σημεῖον, καὶ ἐπιζευθθεῖσα ἡ ΑΖ διήθθω ἐπὶ τὸ Η σημεῖον, καὶ ἐπεζεύθθω ἡ ΖΒ, καὶ ἀπὸ τοῦ Ζ ἐπὶ τὴν ΑΒ κάθετος ἤθθω ἡ ΖΘ, καὶ διήθθω ἐπὶ τὸ Κ, καὶ ἐπεζεύθθωσαν αἱ ΑΚ, ΚΒ, καὶ πάλιν ἀπὸ τοῦ Ζ ἐπὶ τὴν ΑΚ κάθετος ἤθθω ἡ ΖΛ, καὶ διήθθω ἐπὶ τὸ Μ, καὶ ἐπεζεύθθω ἡ ΚΝ.

Ἐπεὶ ὁίση ἐστὶν ἡ ΑΒΓΗ περιφέρεια τῇ ΑΕΔΗ περιφερείᾳ, ὁῶν ἡ ΑΒΓ τῇ ΑΕΔ ἐστινὸίση, λοιπὴ ἄρα ἡ ΓΗ περιφέρεια λοιπῇ τῇ ΗΔ ἐστινὸίση. πενταγώνου δὲ ἡ ΓΔ· δεκαγώνου ἄρα ἡ ΓΗ. καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΖΑ τῇ ΖΒ, καὶ

Let *ABCDE* be a circle. And let the equilateral pentagon *ABCDE* be inscribed in circle *ABCDE*. I say that the square on the side of pentagon *ABCDE* is the (sum of the squares) on the sides of the hexagon and of the decagon inscribed in circle *ABCDE*.

For let the center of the circle, point *F*, be found [Prop. 3.1]. And, *AF* being joined, let it be drawn across to point *G*. And let *FB* be joined. And let *FH* be drawn from *F* perpendicular to *AB*. And let it be drawn across to *K*. And let *AK* and *KB* be joined. And, again, let *FL* have been drawn from *F* perpendicular to *AK*. And let it be drawn across to *M*. And let *KN* be joined.

Since circumference *ABCG* is equal to circumference *AEDG*, of which *ABC* is equal to *AED*, the remaining circumference *CG* is thus equal to the remaining (circumference)

κάθετος ἡ ΖΘ, ὁίση ἄρα καὶ ἡ ὑπὸ ΑΖΚ γωνία τῇ ὑπὸ ΚΖΒ. ὁώστε καὶ περιφέρεια ἡ ΑΚ τῇ ΚΒ ἐστινὁίση· διπλῆ ἄρα ἡ ΑΒ περιφέρεια τῆς ΒΚ περιφερείας· δεκαγώνου ἄρα πλευρά ἐστιν ἡ ΑΚ εὐθεῖα. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΑΚ τῆς ΚΜ ἐστι διπλῆ. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΑΒ περιφέρεια τῆς ΒΚ περιφερείας, ὁίση δὲ ἡ ΓΔ περιφέρεια τῇ ΑΒ περιφερείᾳ, διπλῆ ἄρα καὶ ἡ ΓΔ περιφέρεια τῆς ΒΚ περιφερείας. ὁέστι δὲ ἡ ΓΔ περιφέρεια καὶ τῆς ΓΗ διπλῆ· ὁίση ἄρα ἡ ΓΗ περιφέρεια τῇ ΒΚ περιφερείᾳ. ἀλλὰ ἡ ΒΚ τῆς ΚΜ ἐστι διπλῆ, ἐπεὶ καὶ ἡ ΚΑ· καὶ ἡ ΓΗ ἄρα τῆς ΚΜ ἐστι διπλῆ. ἀλλὰ μὴν καὶ ἡ ΓΒ περιφέρεια τῆς ΒΚ περιφερείας ἐστὶ διπλῆ· ὁίση γὰρ ἡ ΓΒ περιφέρεια τῇ ΒΑ. καὶ ὅλη ἄρα ἡ ΗΒ περιφέρεια τῆς ΒΜ ἐστι διπλῆ· ὁώστε καὶ γωνία ἡ ὑπὸ ΗΖΒ γωνίας τῆς ὑπὸ ΒΖΜ [ἐστι] διπλῆ. ὁέστι δὲ ἡ ὑπὸ ΗΖΒ καὶ τῆς ὑπὸ ΖΑΒ διπλῆ· ὁίση γὰρ ἡ ὑπὸ ΖΑΒ τῇ ὑπὸ ΑΒΖ. καὶ ἡ ὑπὸ ΒΖΝ ἄρα τῇ ὑπὸ ΖΑΒ ἐστινὁίση. κοινὴ δὲ τῶν δύο τριγώνων, τοῦ τε ΑΒΖ καὶ τοῦ ΒΖΝ, ἡ ὑπὸ ΑΒΖ γωνία· λοιπὴ ἄρα ἡ ὑπὸ ΑΖΒ λοιπῇ τῇ ὑπὸ ΒΝΖ ἐστινὁίση· ὁισογώνιον ἄρα ἐστὶ τὸ ΑΒΖ τρίγωνον τῷ ΒΖΝ τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὡς ἡ ΑΒ εὐθεῖα πρὸς τὴν ΒΖ, ὁούτως ἡ ΖΒ πρὸς τὴν ΒΝ· τὸ ἄρα ὑπὸ τῶν ΑΒΝ ἴσον ἐστὶ τῷ ἀπὸ ΒΖ. πάλιν ἐπεὶ ὁίση ἐστὶν ἡ ΑΛ τῇ ΛΚ, κοινὴ δὲ καὶ πρὸς ὀρθὰξ ἡ ΛΝ, βάσις ἄρα ἡ ΚΝ βάσει τῇ ΑΝ ἐστινὁίση· καὶ γωνία ἄρα ἡ ὑπὸ ΛΚΝ γωνίᾳ τῇ ὑπὸ ΛΑΝ ἐστινὁίση. ἀλλὰ ἡ ὑπὸ ΛΑΝ τῇ ὑπὸ ΚΒΝ ἐστινὁίση· καὶ ἡ ὑπὸ ΛΚΝ ἄρα τῇ ὑπὸ ΚΒΝ ἐστινὁίση. καὶ κοινὴ τῶν δύο τριγώνων τοῦ τε ΑΚΒ καὶ τοῦ ΑΚΝ ἡ πρὸς τῷ Α. λοιπὴ ἄρα ἡ ὑπὸ ΑΚΒ λοιπῇ τῇ ὑπὸ ΚΝΑ ἐστινὁίση· ὁισογώνιον ἄρα ἐστὶ τὸ ΚΒΑ τρίγωνον τῷ ΚΝΑ τριγώνῳ. ἀνάλογον ἄρα ἐστὶν ὡς ἡ ΒΑ εὐθεῖα πρὸς τὴν ΑΚ, ὁούτως ἡ ΚΑ πρὸς τὴν ΑΝ· τὸ ἄρα ὑπὸ τῶν ΒΑΝ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΚ. ἐδείθθη δὲ καὶ τὸ ὑπὸ τῶν ΑΒΝ ἴσον τῷ ἀπὸ τῆς ΒΖ· τὸ ἄρα ὑπὸ τῶν ΑΒΝ μετὰ τοῦ ὑπὸ ΒΑΝ, ὅπερ ἐστὶ τὸ ἀπὸ τῆς ΒΑ, ἴσον ἐστὶ τῷ ἀπὸ τῆς ΒΖ μετὰ τοῦ ἀπὸ τῆς ΑΚ. καί ἐστιν ἡ μὲν ΒΑ πενταγώνου πλευρά, ἡ δὲ ΒΖ ἑξαγώνου, ἡ δὲ ΑΚ δεκαγώνου.

Ἡ ἄρα τοῦ πενταγώνου πλευρὰ δύναται τήν τε τοῦ ἑξαγώνου καὶ τὴν τοῦ δεκαγώνου τῶν εἰξ τὸν αὐτὸν κύκλον ἐγγραφομένων· ὅπερ ἔδει δεῖχαι.

GD. And CD (is the side) of the pentagon. CG (is) thus (the side) of the decagon. And since FA is equal to FB, and FH is perpendicular (to AB), angle AFK (is) thus also equal to KFB [Props. 1.5, 1.26]. Hence, circumference AK is also equal to KB [Prop. 3.26]. Thus, circumference AB (is) double circumference BK. Thus, straight-line AK is the side of the decagon. So, for the same (reasons, circumference) AK is also double KM. And since circumference AB is double circumference BK, and circumference CD (is) equal to circumference AB, circumference CD (is) thus also double circumference BK. And circumference CD is also double CG. Thus, circumference CG (is) equal to circumference BK. But, BK is double KM, since KA (is) also (double KM). Thus, (circumference) CG is also double KM. But, indeed, circumference CB is also double circumference BK. For circumference CB (is) equal to BA. Thus, the whole circumference GB is also double BM. Hence, angle GFB [is] also double angle BFM [Prop. 6.33]. And GFB (is) also double FAB. For FAB (is) equal to ABF. Thus, BFN is also equal to FAB. And angle ABF (is) common to the two triangles ABF and BFN. Thus, the remaining (angle) AFB is equal to the remaining (angle) BNF [Prop. 1.32]. Thus, triangle ABF is equiangular to triangle BFN. Thus, proportionally, as straight-line AB (is) to BF, so FB (is) to BN [Prop. 6.4]. Thus, the (rectangle contained) by ABN is equal to the (square) on BF [Prop. 6.17]. Again, since AL is equal to LK, and LN is common and at right-angles (to KA), base KN is thus equal to base AN [Prop. 1.4]. And, thus, angle LKN is equal to angle LAN. But, LAN is equal to KBN [Props. 3.29, 1.5]. Thus, LKN is also equal to KBN. And the (angle) at A (is) common to the two triangles AKB and AKN. Thus, the remaining (angle) AKB is equal to the remaining (angle) KNA [Prop. 1.32]. Thus, triangle KBA is equiangular to triangle KNA. Thus, proportionally, as straight-line BA is to AK, so KA (is) to AN [Prop. 6.4]. Thus, the (rectangle contained) by BAN is equal to the (square) on AK [Prop. 6.17]. And the (rectangle contained) by ABN was also shown (to be) equal to the (square) on BF. Thus, the (rectangle contained) by ABN plus the (rectangle contained) by BAN, which is the (square) on BA [Prop. 2.2], is equal to the (square) on BF plus the (square) on AK. And BA is the side of the pentagon, and BF (the side) of the hexagon [Prop. 4.15 corr.], and AK (the side) of the decagon.

Thus, the square on the side of the pentagon (inscribed in a circle) is (equal to) the (sum of the squares) on the (sides) of the hexagon and of the decagon inscribed in the same circle.

† If the circle is of unit radius then the side of the pentagon is $(1/2)\sqrt{10-2\sqrt{5}}$.

11.

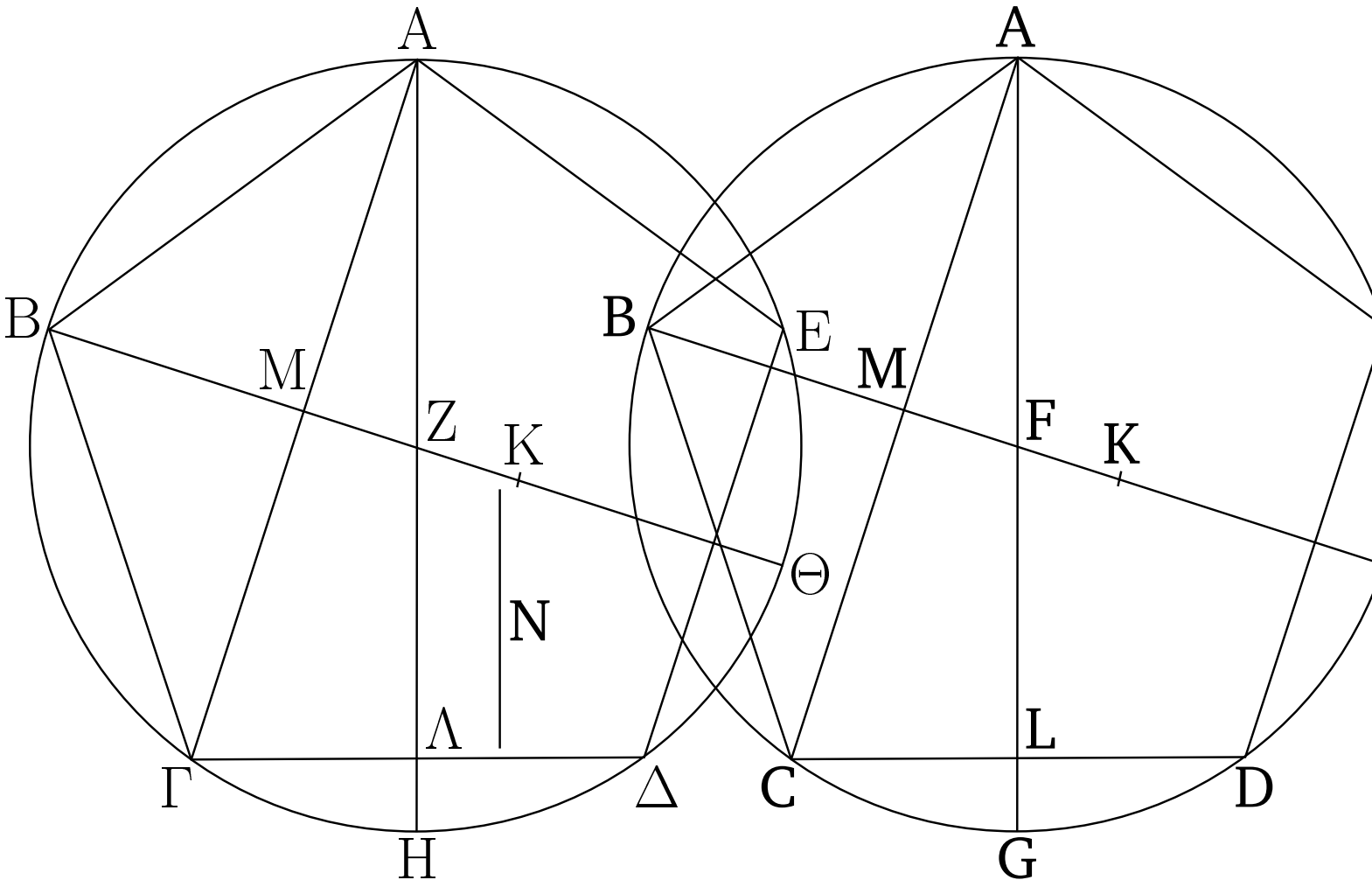
Ἐὰν εἰξ κύκλον ὁρητὴν ἔχοντα τὴν διάμετρον πεντάγωνον

Proposition 11

If an equilateral pentagon is inscribed in a circle which

ἰσόπλευρον ἐγγραφῇ, ἡ τοῦ πενταγώνου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἐλάσσων.

has a rational diameter then the side of the pentagon is that irrational (straight-line) called minor.



Εἰξ γὰρ κύκλον τὸν ΑΒΓΔΕ ὀρητὴν ἔχοντα τὴν διάμετρον πεντάγωνον ἰσόπλευρον ἐγγεγράφθω τὸ ΑΒΓΔΕ· λέγω, ὅτι ἡ τοῦ [ΑΒΓΔΕ] πενταγώνου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἐλάσσων.

Εἰλήφθω γὰρ τὸ κέντρον τοῦ κύκλου τὸ Ζ σημεῖον, καὶ ἐπεζεύθθωσαν αἱ ΑΖ, ΖΒ καὶ διήθθωσαν ἐπὶ τὰ Η, Θ σημεῖα, καὶ ἐπεζεύθθω ἡ ΑΓ, καὶ κείσθω τῆς ΑΖ τέταρτον μέρος ἡ ΖΚ. ὀρητὴ δὲ ἡ ΑΖ· ὀρητὴ ἄρα καὶ ἡ ΖΚ. ὀέστι δὲ καὶ ἡ ΒΖ ὀρητή· ὅλη ἄρα ἡ ΒΚ ὀρητή ἐστιν. καὶ ἐπεὶ ὀίση ἐστὶν ἡ ΑΓΗ περιφέρεια τῇ ΑΔΗ περιφερείᾳ, ὁῶν ἡ ΑΒΓ τῇ ΑΕΔ ἐστινὀίση, λοιπὴ ἄρα ἡ ΓΗ λοιπῇ τῇ ΗΔ ἐστινὀίση. καὶ ἐαν ἐπιζεύχωμεν τὴν ΑΔ, συνάγονται ὀρθαὶ αἱ πρὸς τῷ Λ γωνίαι, καὶ διπλῆ ἡ ΓΔ τῆς ΓΛ. διὰ τὰ αὐτὰ δὴ καὶ αἱ πρὸς τῷ Μ ὀρθαί εἰσιν, καὶ διπλῆ ἡ ΑΓ τῆς ΓΜ. ἐπεὶ οὀῦνὀίση ἐστὶν ἡ ὑπὸ ΑΛΓ γωνία τῇ ὑπὸ ΑΜΖ, κοινὴ δὲ τῶν δύο τριγώνων τοῦ τε ΑΓΛ καὶ τοῦ ΑΜΖ ἡ ὑπὸ ΛΑΓ, λοιπὴ ἄρα ἡ ὑπὸ ΑΓΛ λοιπῇ τῇ ὑπὸ ΜΖΑ ἐστινὀίση· ἰσογώνιον ἄρα ἐστὶ τὸ

For let the equilateral pentagon *ABCDE* be inscribed in the circle *ABCDE* which has a rational diameter. I say that the side of pentagon [*ABCDE*] is that irrational (straight-line) called minor.

For let the center of the circle, point *F*, be found [Prop. 3.1]. And let *AF* and *FB* be joined. And let them be drawn across to points *G* and *H* (respectively). And let *AC* be joined. And let *FK* made (equal) to the fourth part of *AF*. And *AF* (is) rational. *FK* (is) thus also rational. And *BF* is also rational. Thus, the whole of *BK* is rational. And since circumference *ACG* is equal to circumference *ADG*, of which *ABC* is equal to *AED*, the remainder *CG* is thus equal to the remainder *GD*. And if we join *AD* then the angles at *L* are inferred (to be) right-angles, and *CD* (is inferred to be) double *CL* [Prop. 1.4]. So, for the same (reasons), the (angles) at *M* are also right-angles, and *AC* (is) double *CM*. Therefore, since angle *ALC* (is) equal to *AMF*, and (angle) *LAC* (is) common to the two

ΑΓΛ τρίγωνον τῷ ΑΜΖ τριγώνῳ· ἀνάλογον ἄρα ἐστὶν ὡς ἡ ΛΓ πρὸς ΓΑ, οὕτως ἡ ΜΖ πρὸς ΖΑ· καὶ τῶν ἡγουμένων τὰ διπλάσια· ὡς ἄρα ἡ τῆς ΛΓ διπλῆ πρὸς τὴν ΓΑ, οὕτως ἡ τῆς ΜΖ διπλῆ πρὸς τὴν ΖΑ. ὡς δὲ ἡ τῆς ΜΖ διπλῆ πρὸς τὴν ΖΑ, οὕτως ἡ ΜΖ πρὸς τὴν ἡμίσειαν τῆς ΖΑ· καὶ ὡς ἄρα ἡ τῆς ΛΓ διπλῆ πρὸς τὴν ΓΑ, οὕτως ἡ ΜΖ πρὸς τὴν ἡμίσειαν τῆς ΖΑ· καὶ τῶν ἑπομένων τὰ ἡμίσεα· ὡς ἄρα ἡ τῆς ΛΓ διπλῆ πρὸς τὴν ἡμίσειαν τῆς ΓΑ, οὕτως ἡ ΜΖ πρὸς τὸ τέτατρον τῆς ΖΑ. καί ἐστι τῆς μὲν ΛΓ διπλῆ ἡ ΔΓ, τῆς δὲ ΓΑ ἡμίσεια ἡ ΓΜ, τῆς δὲ ΖΑ τέτατρον μέρος ἡ ΖΚ· ὡς ἄρα ἡ ΔΓ πρὸς τὴν ΓΜ, οὕτως ἡ ΜΖ πρὸς τὴν ΖΚ. συνθέντι καὶ ὡς συναμφότερος ἡ ΔΓΜ πρὸς τὴν ΓΜ, οὕτως ἡ ΜΚ πρὸς ΚΖ· καὶ ὡς ἄρα τὸ ἀπὸ συναμφοτέρου τῆς ΔΓΜ πρὸς τὸ ἀπὸ ΓΜ, οὕτως τὸ ἀπὸ ΜΚ πρὸς τὸ ἀπὸ ΚΖ. καὶ ἐπεὶ τῆς ὑπὸ δύο πλευρὰξ τοῦ πενταγώνου ὑποτεινούσης, οὗτον τῆς ΑΓ, ὀάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμα ἴσον ἐστὶ τῇ τοῦ πενταγώνου πλευρᾷ, τουτέστι τῇ ΔΓ, τὸ δὲ μεῖζον τμῆμα προσλαβὸν τὴν ἡμίσειαν τῆς ὅλης πενταπλάσιον δύναται τοῦ ἀπὸ τῆς ἡμισείας τῆς ὅλης, καί ἐστιν ὅλης τῆς ΑΓ ἡμίσεια ἡ ΓΜ, τὸ ἄρα ἀπὸ τῆς ΔΓΜ ὡξ μιᾶς πενταπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΓΜ. ὡξ δὲ τὸ ἀπὸ τῆς ΔΓΜ ὡξ μιᾶς πρὸς τὸ ἀπὸ τῆς ΓΜ, οὕτως ἐδείθθη τὸ ἀπὸ τῆς ΜΚ πρὸς τὸ ἀπὸ τῆς ΚΖ· πενταπλάσιον ἄρα τὸ ἀπὸ τῆς ΜΚ τοῦ ἀπὸ τῆς ΚΖ. ὁρητὸν δὲ τὸ ἀπὸ τῆς ΚΖ· ὁρητὴ γὰρ ἡ διάμετρος· ὁρητὸν ἄρα καὶ τὸ ἀπὸ τῆς ΜΚ· ὁρητὴ ἄρα ἐστὶν ἡ ΜΚ [δυνάμει μόνον]. καὶ ἐπεὶ τετραπλασία ἐστὶν ἡ ΒΖ τῆς ΖΚ, πενταπλασία ἄρα ἐστὶν ἡ ΒΚ τῆς ΚΖ· εἰκοσιπενταπλάσιον ἄρα τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΖ. πενταπλάσιον δὲ τὸ ἀπὸ τῆς ΜΚ τοῦ ἀπὸ τῆς ΚΖ· πενταπλάσιον ἄρα τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΜ· τὸ ἄρα ἀπὸ τῆς ΒΚ πρὸς τὸ ἀπὸ ΚΜ λόγον οὐκ ἔχει, ὃν τετράγωνος ἀριθμὸξ πρὸς τετράγωνον ἀριθμόν· ἀσύμμετρος ἄρα ἐστὶν ἡ ΒΚ τῇ ΚΜ μήκει. καί ἐστι ὁρητὴ ἑκατέρα αὐτῶν. αἱ ΒΚ, ΚΜ ἄρα ὁρηταί εἰσι δυνάμει μόνον σύμμετροι. ἐαν δὲ ἀπὸ ὁρητῆς ὁρητὴ ἀφαιρεθῇ δυνάμει μόνον σύμμετρος οὐὖσα τῇ ὅλῃ, ἡ λοιπὴ ὀάλογόξ ἐστιν ἀποτομή· ἀποτομὴ ἄρα ἐστὶν ἡ ΜΒ, προσαρμόζουσα δὲ αὐτῇ ἡ ΜΚ. λέγω δή, ὅτι καὶ τετάρτη. ὁῷ δὴ μεῖζόν ἐστι τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΜ, ἐκείνῳ ἴσονὸέστω τὸ ἀπὸ τῆς Ν· ἡ ΒΚ ἄρα τῆς ΚΜ μεῖζον δύναται τῇ Ν. καὶ ἐπεὶ σύμμετρόξ ἐστιν ἡ ΚΖ τῇ ΖΒ, καὶ συνθέντι σύμμετρόξ ἐστι ἡ ΚΒ τῇ ΖΒ. ἀλλὰ ἡ ΒΖ τῇ ΒΘ σύμμετρόξ ἐστιν· καὶ ἡ ΒΚ ἄρα τῇ ΒΘ σύμμετρόξ ἐστιν. καὶ ἐπεὶ πενταπλάσιόν ἐστι τὸ ἀπὸ τῆς ΒΚ τοῦ ἀπὸ τῆς ΚΜ, τὸ ἄρα ἀπὸ τῆς ΒΚ πρὸς τὸ ἀπὸ τῆς ΚΜ λόγον ἔχει, ὃν ε πρὸς ἕν. ἀναστρέψαντι ἄρα τὸ ἀπὸ τῆς ΒΚ πρὸς τὸ ἀπὸ τῆς Ν λόγον ἔχει, ὃν $\overline{\varepsilon}$ πρὸς $\overline{\delta}$, οὐχ ὃν τετράγωνος πρὸς τετράγωνον· ἀσύμμετρος ἄρα ἐστὶν ἡ ΒΚ τῇ Ν· ἡ ΒΚ ἄρα τῆς ΚΜ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ. ἐπεὶ οὖν ὅλη ἡ ΒΚ τῆς προσαρμοζούσης τῆς ΚΜ μεῖζον δύναται τῷ ἀπὸ ἀσυμμέτρου ἑαυτῇ, καὶ ὅλη ἡ ΒΚ σύμμετρός ἐστι τῇ ἐκκειμένῃ ῥητῇ τῇ ΒΘ, ἀποτομὴ ἄρα τετάρτη ἐστὶν ἡ ΜΒ. τὸ δὲ ὑπὸ ῥητῆς καὶ ἀποτομῆς τετάρτης περιεχόμενον ὀρθογώνιον ἄλογόν ἐστιν, καὶ

triangles ACL and AMF, the remaining (angle) ACL is thus equal to the remaining (angle) MFA [Prop. 1.32]. Thus, triangle ACL is equiangular to triangle AMF. Thus, proportionally, as LC (is) to CA, so MF (is) to FA [Prop. 6.4]. And (we can take) the doubles of the leading (magnitudes). Thus, as double LC (is) to CA, so double MF (is) to FA. And as double MF (is) to FA, so MF (is) to half of FA. And, thus, as double LC (is) to CA, so MF (is) to half of FA. And (we can take) the halves of the following (magnitudes). Thus, as double LC (is) to half of CA, so MF (is) to the fourth of FA. And DC is double LC, and CM half of CA, and FK the fourth part of FA. Thus, as DC is to CM, so MF (is) to FK. Via composition, as the sum of DCM (*i.e.*, DC and CM) (is) to CM, so MK (is) to KF [Prop. 5.18]. And, thus, as the (square) on the sum of DCM (is) to the (square) on CM, so the (square) on MK (is) to the (square) on KF. And since the greater piece of a (straight-line) subtending two sides of a pentagon, such as AC, (which is) cut in extreme and mean ratio is equal to the side of the pentagon [Prop. 13.8]—that is to say, to DC—-and the square on the greater piece added to half of the whole is five times the (square) on half of the whole [Prop. 13.1], and CM (is) half of the whole, AC, thus the (square) on DCM, (taken) as one, is five times the (square) on CM. And the (square) on DCM, (taken) as one, (is) to the (square) on CM, so the (square) on MK was shown (to be) to the (square) on KF. Thus, the (square) on MK (is) five times the (square) on KF. And the square on KF (is) rational. For the diameter (is) rational. Thus, the (square) on MK (is) also rational. Thus, MK is rational [in square only]. And since BF is four times FK, BK is thus five times KF. Thus, the (square) on BK (is) twenty-five times the (square) on KF. And the (square) on MK (is) five times the square on KF. Thus, the (square) on BK (is) five times the (square) on KM. Thus, the (square) on BK does not have to the (square) on KM the ratio which a square number (has) to a square number. Thus, BK is incommensurable in length with KM [Prop. 10.9]. And each of them is a rational (straight-line). Thus, BK and KM are rational (straight-lines which are) commensurable in square only. And if from a rational (straight-line) a rational (straight-line) is subtracted, which is commensurable in square only with the whole, then the remainder is that irrational (straight-line called) an apotome [Prop. 10.73]. Thus, MB is an apotome, and MK its attachment. So, I say that (it is) also a fourth (apotome). So, let the (square) on N be (made) equal to that (magnitude) by which the (square) on BK is greater than the (square) on KM. Thus, the square on BK is greater than the (square) on KM by the (square) on N. And since KF is commensurable (in length) with FB then, via composition, KB is also commensurable (in length) with FB [Prop. 10.15]. But, BF is commensurable (in length) with BH. Thus, BK is also commensurable (in length) with BH [Prop. 10.12]. And since the (square)

ἡ δυναμένη αὐτὸ ἄλογός ἐστιν, καλεῖται δὲ ἐλάττων. δύναται δὲ τὸ ὑπὸ τῶν ΘΒΜ ἡ ΑΒ διὰ τὸ ἐπιζευγνυμένης τῆς ΑΘ ἰσογώνιον γίνεσθαι τὸ ΑΒΘ τρίγωνον τῷ ΑΒΜ τριγώνῳ καὶ εἶναι ὡς τὴν ΘΒ πρὸς τὴν ΒΑ, οὕτως τὴν ΑΒ πρὸς τὴν ΒΜ.

Ἡ ἄρα ΑΒ τοῦ πενταγώνου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἐλάττων· ὅπερ ἔδει δεῖξαι.

on BK is five times the (square) on KM, the (square) on BK thus has to the (square) on KM the ratio which 5 (has) to one. Thus, via conversion, the (square) on BK has to the (square) on N the ratio which 5 (has) to 4 [Prop. 5.19 corr.], which is not (that) of a square (number) to a square (number). BK is thus incommensurable (in length) with N [Prop. 10.9]. Thus, the square on BK is greater than the (square) on KM by the (square) on (some straight-line which is) incommensurable (in length) with (BK). Therefore, since the square on the whole, BK, is greater than the (square) on the attachment, KM, by the (square) on (some straight-line which is) incommensurable (in length) with (BK), and the whole, BK, is commensurable (in length) with the (previously) laid down rational (straight-line) BH, MB is thus a fourth apotome [Def. 10.14]. And the rectangle contained by a rational (straight-line) and a fourth apotome is irrational, and its square-root is that irrational (straight-line) called minor [Prop. 10.94]. And the square on AB is the rectangle contained by HBM, on account of joining AH, (so that) triangle ABH becomes equiangular with triangle ABM [Prop. 6.8], and (proportionally) as HB is to BA, so AB (is) to BM.

Thus, the side AB of the pentagon is that irrational (straight-line) called minor.[†] (Which is) the very thing it was required to show.

[†] If the circle has unit radius then the side of the pentagon is $(1/2)\sqrt{10 - 2\sqrt{5}}$. However, this length can be written in the "minor" form (see Prop. 10.94) $(\rho/\sqrt{2})\sqrt{1 + k/\sqrt{1+k^2}} - (\rho/\sqrt{2})\sqrt{1 - k/\sqrt{1+k^2}}$, with $\rho = \sqrt{5/2}$ and $k = 2$.

12.

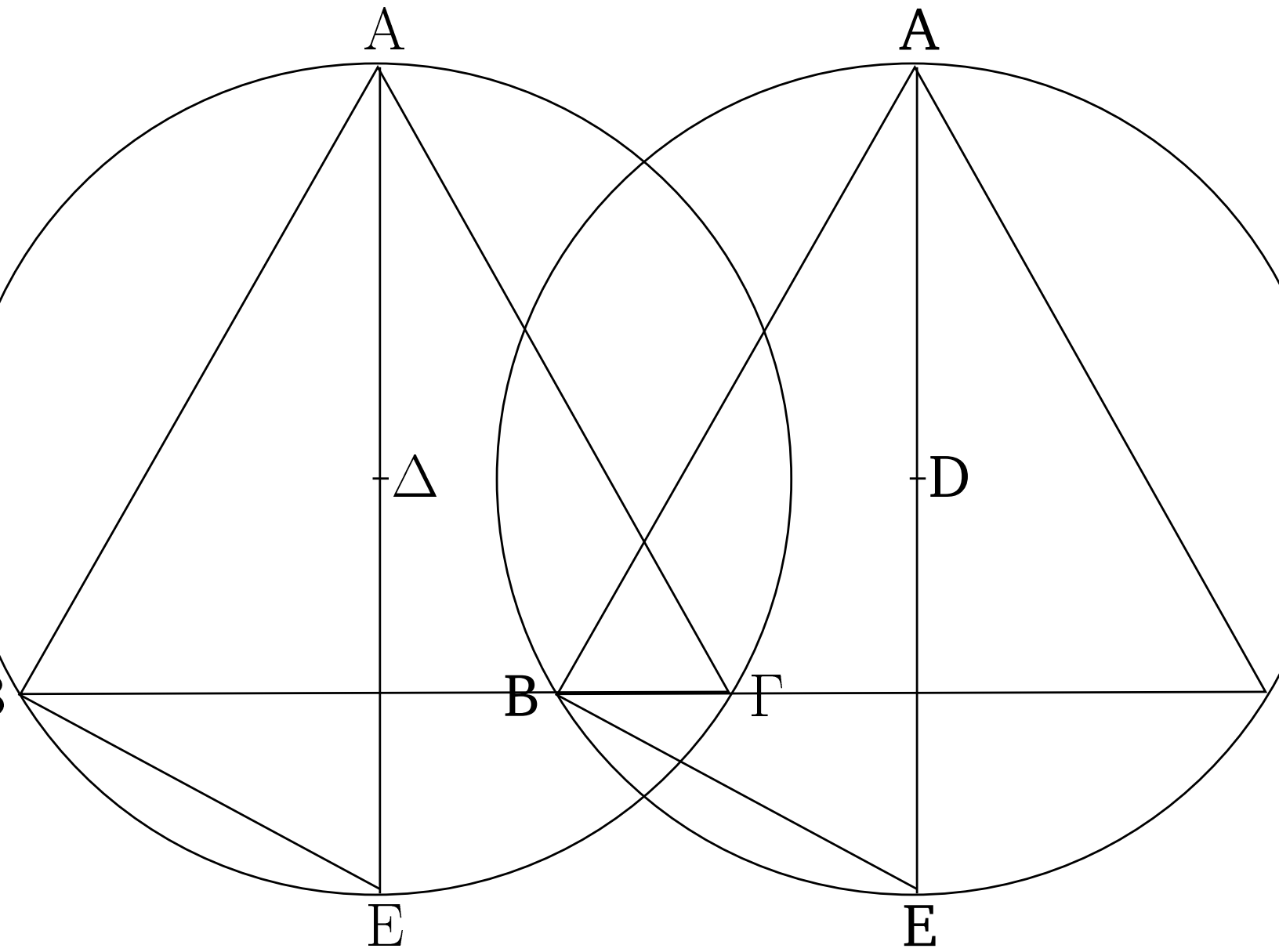
Ἐὰν εἰξ κύκλον τρίγωνον ἰσόπλευρον ἐγγραφῇ, ἡ τοῦ τριγώνου πλευρὰ δυνάμει τριπλασίων ἐστὶ τῆς ἐκ τοῦ κέντρου τοῦ κύκλου.

Ἔστω κύκλος ὁ ΑΒΓ, καὶ εἰξ αὐτὸν τρίγωνον ἰσόπλευρον ἐγγεγράφθω τὸ ΑΒΓ· λέγω, ὅτι τοῦ ΑΒΓ τριγώνου μία πλευρὰ δυνάμει τριπλασίων ἐστὶ τῆς ἐκ τοῦ κέντρου τοῦ ΑΒΓ κύκλου.

Proposition 12

If an equilateral triangle is inscribed in a circle then the square on the side of the triangle is three times the (square) on the radius of the circle.

Let there be a circle ABC, and let the equilateral triangle ABC have been inscribed in it [Prop. 4.2]. I say that the square on one side of triangle ABC is three times the (square) on the radius of circle ABC.



Εἰλήφθω γὰρ τὸ κέντρον τοῦ ΑΒΓ κύκλου τὸ Δ, καὶ ἐπιζευθθεῖσα ἡ ΑΔ διήθθω ἐπὶ τὸ Ε, καὶ ἐπεζεύθθω ἡ ΒΕ.

Καὶ ἐπεὶ ἰσόπλευρόν ἐστι τὸ ΑΒΓ τρίγωνον, ἡ ΒΕΓ ἄρα περιφέρεια τρίτον μέρος ἐστὶ τῆς τοῦ ΑΒΓ κύκλου περιφερείας. ἡ ἄρα ΒΕ περιφέρεια ἕκτον ἐστὶ μέρος τῆς τοῦ κύκλου περιφερείας· ἐχαγώνου ἄρα ἐστὶν ἡ ΒΕ εὐθεῖα· ὁίση ἄρα ἐστὶ τῇ ἐκ τοῦ κέντρου τῇ ΔΕ. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΑΕ τῆς ΔΕ, τετραπλάσιον ἐστι τὸ ἀπὸ τῆς ΑΕ τοῦ ἀπὸ τῆς ΕΔ, τουτέστι τοῦ ἀπὸ τῆς ΒΕ. ἴσον δὲ τὸ ἀπὸ τῆς ΑΕ τοῖς ἀπὸ τῶν ΑΒ, ΒΕ· τὰ ἄρα ἀπὸ τῶν ΑΒ, ΒΕ τετραπλάσιά ἐστι τοῦ ἀπὸ τῆς ΒΕ. διελόντι ἄρα τὸ ἀπὸ τῆς ΑΒ τριπλάσιόν ἐστι τοῦ ἀπὸ ΒΕ. ὁίση δὲ ἡ ΒΕ τῇ ΔΕ· τὸ ἄρα ἀπὸ τῆς ΑΒ τριπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΔΕ.

For let the center, D, of circle ABC be found [Prop. 3.1]. And AD (being) joined, let it be drawn across to E. And let BE be joined.

And since triangle ABC is equilateral, circumference BEC is thus the third part of the circumference of circle ABC. Thus, circumference BE is the sixth part of the circumference of the circle. Thus, straight-line BE is (the side) of a hexagon. Thus, it is equal to the radius DE [Prop. 4.15 corr.]. And since AE is double DE, the (square) on AE is four times the (square) on ED—that is to say, of the (square) on BE. And the (square) on AE (is) equal to the (sum of the squares) on AB and BE [Props. 3.31, 1.47]. Thus, the (sum of the squares) on AB and BE is four times the (square) on BE. Thus, via separation, the

Ἡ ἄρα τοῦ τριγώνου πλευρὰ δυνάμει τριπλασία ἐστὶ τῆς ἐκ τοῦ κέντρου [τοῦ κύκλου]· ὅπερ ἔδει δεῖξαι.

(square) on AB is three times the (square) on BE. And BE (is) equal to DE. Thus, the (square) on AB is three times the (square) on DE.

Thus, the square on the side of the triangle is three times the (square) on the radius [of the circle]. (Which is) the very thing it was required to show.

13.

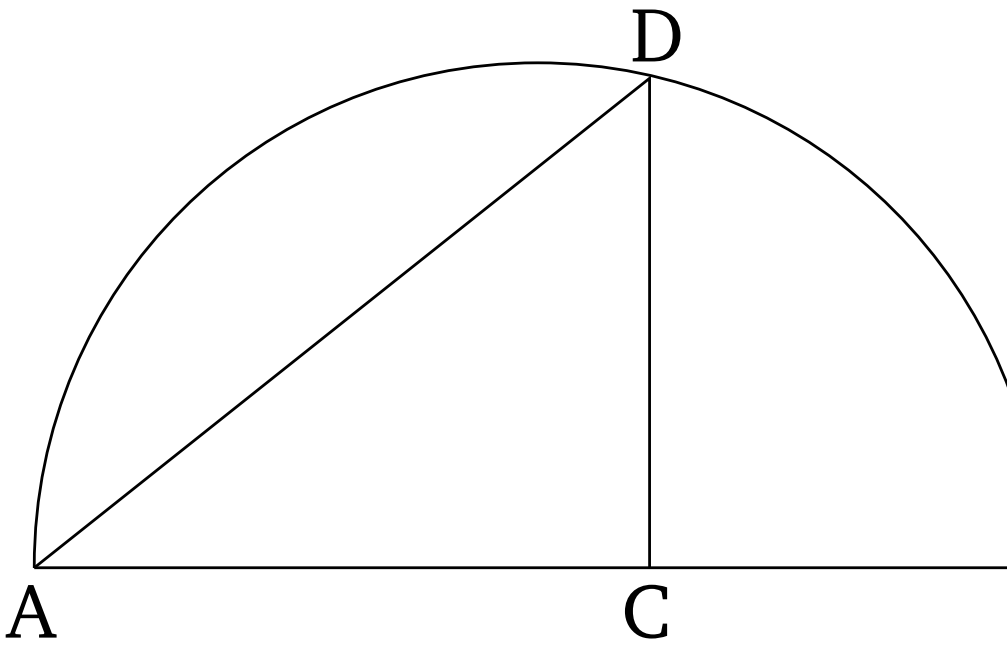
Πυραμίδα συστήσασθαι καὶ σφαίρᾳ περιλαβεῖν τῇ δοθείσῃ καὶ δεῖξαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει ἡμιολία ἐστὶ τῆς πλευρᾶς τῆς πυραμίδος.

Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας δίαμετρος ἡ AB, καὶ τετμήσθω κατὰ τὸ $Γ$ σημεῖον, ὥστε διπλασίαν εἶναι τὴν $AΓ$ τῆς $ΓB$· καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ $AΔB$, καὶ ἤχθω ἀπὸ τοῦ $Γ$ σημείου τῇ AB πρὸς ὀρθὰξ ἡ $ΓΔ$, καὶ ἐπεζεύχθω ἡ $ΔA$· καὶ ἐκκείσθω κύκλος ὁ EZH ἴσην ὀέθων τὴν ἐκ τοῦ κέντρου τῇ $ΔΓ$, καὶ ἐγγεγράφθω εἰς τὸν EZH κύκλον τρίγωνον ἰσόπλευρον τὸ EZH· καὶ εἰλήφθω τὸ κέντρον τοῦ κύκλου τὸ $Θ$ σημεῖον, καὶ ἐπεζεύχθωσαν αἱ $EΘ$, $ΘZ$, $ΘH$· καὶ ἀνεστάτω ἀπὸ τοῦ $Θ$ σημείου τῷ τοῦ EZH κύκλου ἐπιπέδῳ πρὸς ὀρθὰξ ἡ $ΘK$, καὶ ἀφῃρήσθω ἀπὸ τῆς $ΘK$ τῇ $AΓ$ εὐθείᾳ ὁίση ἡ $ΘK$, καὶ ἐπεζεύχθωσαν αἱ KE, KZ, KH. καὶ ἐπεὶ ἡ $KΘ$ ὀρθή ἐστι πρὸς τὸ τοῦ EZH κύκλου ἐπίπεδον, καὶ πρὸς πάσας ἄρα τὰξ ἁπτομένας αὐτῆς εὐθείας καὶ οὐούσας ἐν τῷ τοῦ EZH κύκλου ἐπιπέδῳ ὀρθὰξ ποιήσει γωνίας. ὁάπτεται δὲ αὐτῆς ἑκάστη τῶν $ΘE$, $ΘZ$, $ΘH$· ἡ $ΘK$ ἄρα πρὸς ἑκάστη τῶν $ΘE$, $ΘZ$, $ΘH$ ὀρθή ἐστιν. καὶ ἐπεὶ ὁίση ἐστὶν ἡ μὲν $AΓ$ τῇ $ΘK$, ἡ δὲ $ΓΔ$ τῇ $ΘE$, καὶ ὀρθὰξ γωνίας περιέθουσιν, βάσις ἄρα ἡ $ΔA$ βάσει τῇ KE ἐστινὸίση. διὰ τὰ αὐτὰ δὴ καὶ ἑκατέρα τῶν KZ, KH τῇ $ΔA$ ἐστινὸίση· αἱ τρεῖς ἄρα αἱ KE, KZ, KH ἴσαι ἀλλήλαις εἰσίν. καὶ ἐπεὶ διπλῆ ἐστιν ἡ $AΓ$ τῆς $ΓB$, τριπλῆ ἄρα ἡ AB τῆς $BΓ$. ὡς δὲ ἡ AB πρὸς τὴν $BΓ$, οὕτως τὸ ἀπὸ τῆς $AΔ$ πρὸς τὸ ἀπὸ τῆς $ΔΓ$, ὡξ ἔχῃς δειθθήσεται. τριπλάσιον ἄρα τὸ ἀπὸ τῆς $AΔ$ τοῦ ἀπὸ τῆς $ΔΓ$. ὀέστι δὲ καὶ τὸ ἀπὸ τῆς ZE τοῦ ἀπὸ τῆς $EΘ$ τριπλάσιον, καὶ ἐστινὸίση ἡ $ΔΓ$ τῇ $EΘ$· ὁίση ἄρα καὶ ἡ $ΔA$ τῇ EZ. ἀλλὰ ἡ $ΔA$ ἑκάστῃ τῶν KE, KZ, KH ἐδείθηὸίση· καὶ ἑκάστη ἄρα τῶν EZ, ZH, HE ἑκάστῃ τῶν KE, KZ, KH ἐστινὸίση· ἰσόπλευρα ἄρα ἐστὶ τὰ τέσσαρα τρίγωνα τὰ EZH, KEZ, KZH, KEH. πυραμὶξ ἄρα συνέσταται ἐκ τεσσάρων τριγώνων ἰσοπλέυρων, ἧξ βάσις μέν ἐστι τὸ EZH τρίγωνον, κορυφὴ δὲ τὸ K σημεῖον.

Proposition 13

To construct a (regular) pyramid (*i.e.*, a tetrahedron), and to enclose (it) in a given sphere, and to show that the square on the diameter of the sphere is one and a half times the (square) on the side of the pyramid.

Let the diameter AB of the given sphere be laid out, and let it be cut at point C such that AC is double CB [Prop. 6.10]. And let the semi-circle ADB be drawn on AB. And let CD be drawn from point C at right-angles to AB. And let DA be joined. And let the circle EFG be laid down having a radius equal to DC, and let the equilateral triangle EFG be inscribed in circle EFG [Prop. 4.2]. And let the center of the circle, point H, be found [Prop. 3.1]. And let EH, HF, and HG be joined. And let HK be set up, at point H, at right-angles to the plane of circle EFG [Prop. 11.12]. And let HK, equal to the straight-line AC, be cut off from HK. And let KE, KF, and KG be joined. And since KH is at right-angles to the plane of circle EFG, it will thus also make right-angles with all of the straight-lines joining it (which are) also in the plane of circle EFG [Def. 11.3]. And HE, HF, and HG each join it. Thus, HK is at right-angles to each of HE, HF, and HG. And since AC is equal to HK, and CD to HE, and they contain right-angles, the base DA is thus equal to the base KE [Prop. 1.4]. So, for the same (reasons), KF and KG is each equal to DA. Thus, the three (straight-lines) KE, KF, and KG are equal to one another. And since AC is double CB, AB (is) thus triple BC. And as AB (is) to BC, so the (square) on AD (is) to the (square) on DC, as will be shown later [see lemma]. Thus, the (square) on AD (is) three times the (square) on DC. And the (square) on FE is also three times the (square) on EH [Prop. 13.12], and DC is equal to EH. Thus, DA (is) also equal to EF. But, DA was shown (to be) equal to each of KE, KF, and KG. Thus, EF, FG, and GE are equal to KE, KF, and KG, respectively. Thus, the four triangles EFG, KEF, KFG, and KEG are equilateral. Thus, a pyramid, whose base is triangle EFG, and apex the point K, has been constructed from four equilateral triangles.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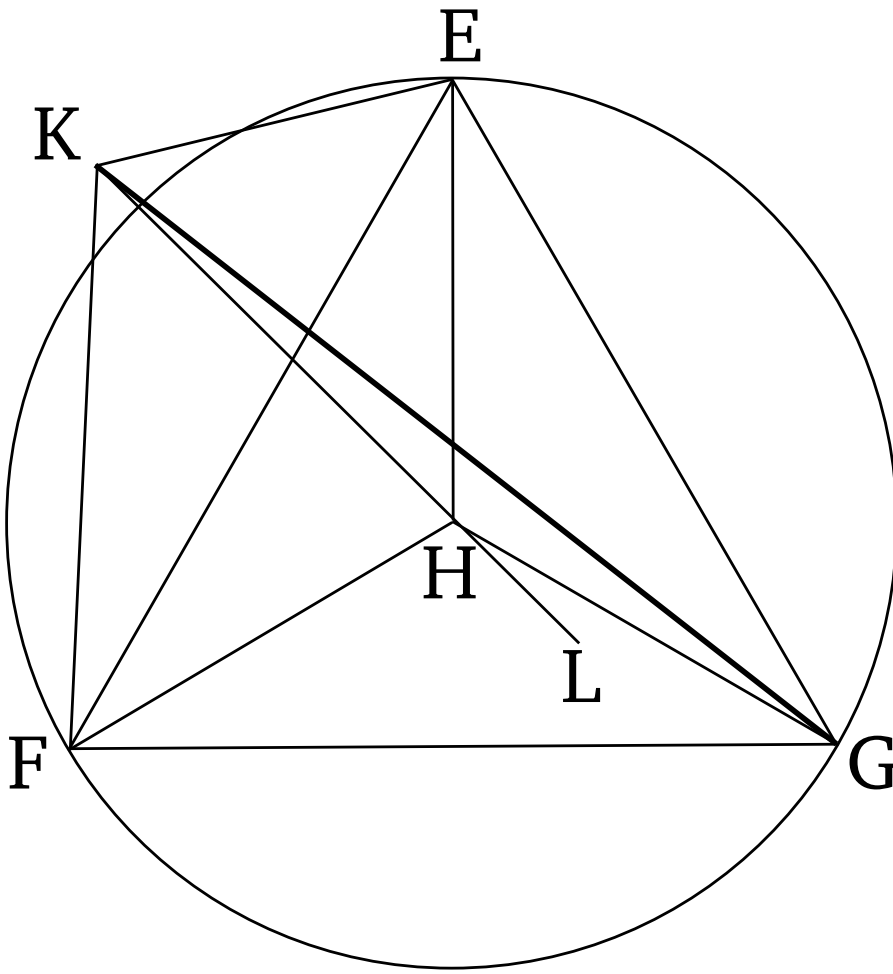
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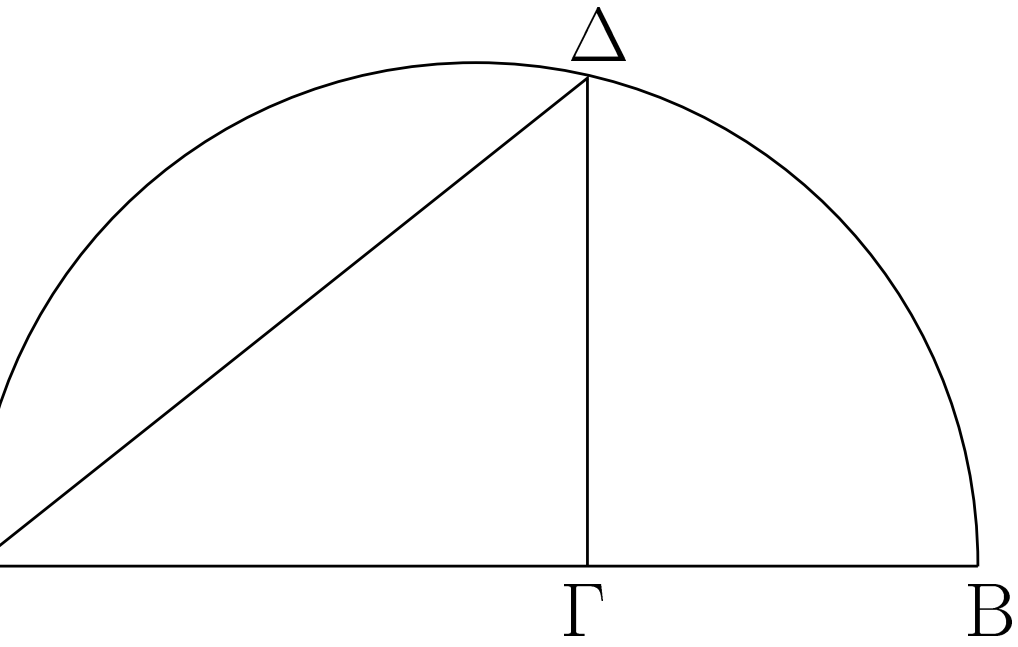
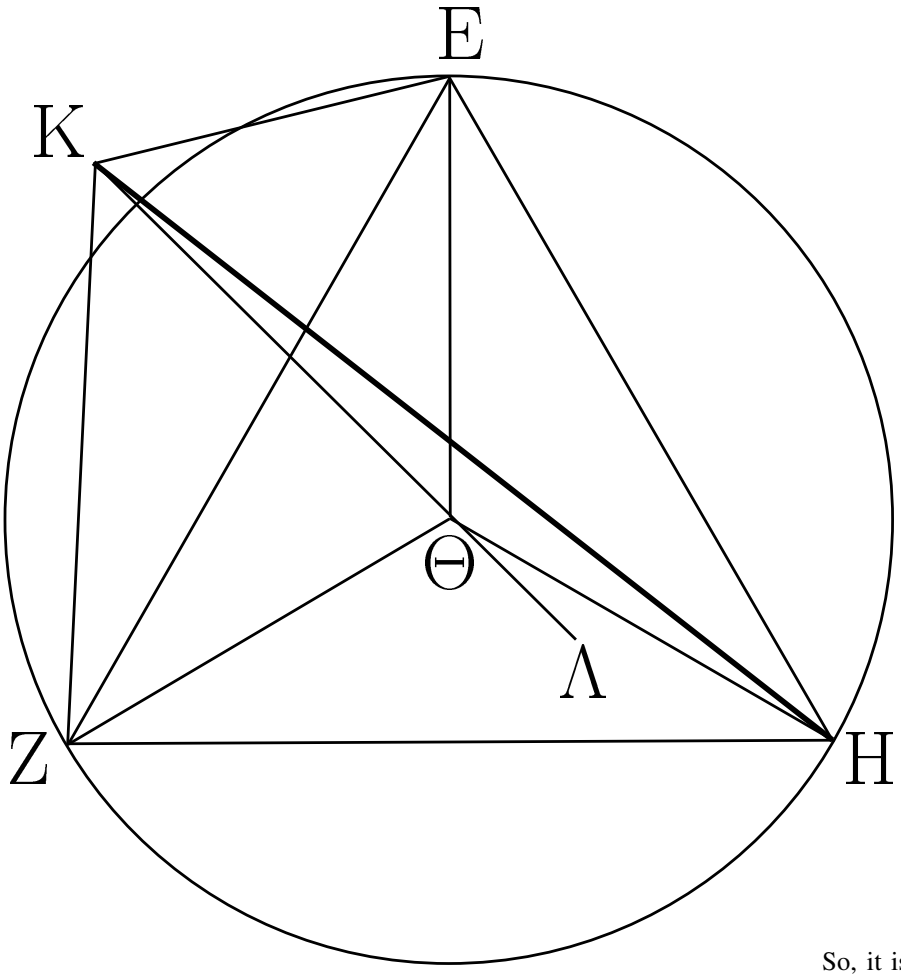
G





So, it is also necessary to enclose it in the given sphere,

Δεῖ δὴ αὐτὴν καὶ σφαίρᾳ περιλαβεῖν τῇ δοθείσῃ καὶ δεῖξαι, ὅτι ἡ τῆς σφαίρας διάμετρος ἡμιολία ἐστὶ δυνάμει τῆς πλευρᾶξ τῆς πυραμίδος.

Ἐκβεβλήσθω γὰρ ἐπ᾽ εὐθείας τῇ ΚΘ εὐθεῖα ἡ ΘΛ, καὶ κείσθω τῇ ΓΒὁίση ἡ ΘΛ. καὶ ἐπεί ἐστιν ὡξ ἡ ΑΓ πρὸς τὴν ΓΔ, οὕτως ἡ ΓΔ πρὸς τὴν ΓΒ, ὁίση δὲ ἡ μὲν ΑΓ τῇ ΚΘ, ἡ δὲ ΓΔ τῇ ΘΕ, ἡ δὲ ΓΒ τῇ ΘΛ, ὁέστιν ἄρα ὡξ ἡ ΚΘ πρὸς τὴν ΘΕ, οὕτως ἡ ΕΘ πρὸς τὴν ΘΛ· τὸ ἄρα ὑπὸ τῶν ΚΘ, ΘΛ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΘ. καί ἐστιν ὀρθὴ ἑκατέρα τῶν ὑπὸ ΚΘΕ, ΕΘΛ γωνιῶν· τὸ ἄρα ἐπὶ τῆς ΚΛ γραφόμενον ἡμικύκλιον ἥξει καὶ διὰ τοῦ Ε [ἐπειδήπερ ἐαν ἐπιζεύχωμεν τὴν ΕΛ, ὀρθὴ γίνεται ἡ ὑπὸ ΛΕΚ γωνία διὰ τὸ ἰσογώνιον γίνεσθαι τὸ ΕΛΚ τρίγωνον ἑκατέρῳ τῶν ΕΛΘ, ΕΘΚ τριγώνων]. ἐαν δὴ μενούσης τῆς ΚΛ περιενεθθὲν τὸ ἡμικύκλιον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, ἥξει καὶ διὰ τῶν Ζ, Η σημείων ἐπιζευγνυμένων τῶν ΖΛ, ΛΗ καὶ ὀρθῶν ὁμοίως γινομένων τῶν πρὸς τοῖς Ζ, Η γωνιῶν· καὶ ὀέσται ἡ πυραμὶξ σφαίρᾳ περιειλημμένη τῇ δοθείσῃ. ἡ γὰρ ΚΛ τῆς σφαίρας διάμετροςὁίση ἐστὶ τῇ τῆς δοθείσης σφαίρας διαμετρῳ τῇ ΑΒ, ἐπειδήπερ τῇ μὲν ΑΓὁίση κεῖται ἡ ΚΘ, τῇ δὲ ΓΒ ἡ ΘΛ.

Λέγω δή, ὅτι ἡ τῆς σφαίρας διάμετρος ἡμιολία ἐστὶ δυνάμει τῆς πλευρᾶξ τῆς πυραμίδος.

Ἐπεὶ γὰρ διπλῆ ἐστιν ἡ ΑΓ τῆς ΓΒ, τριπλῆ ἄρα ἐστὶν ἡ ΑΒ τῆς ΒΓ· ἀναστρέψαντι ἡμιολία ἄρα ἐστὶν ἡ ΒΑ τῆς ΑΓ. ὡξ δὲ ἡ ΒΑ πρὸς τὴν ΑΓ, οὕτως τὸ ἀπὸ τῆς ΒΑ πρὸς τὸ ἀπὸ τῆς ΑΔ [ἐπειδήπερ ἐπιζευγνμένης τῆς ΔΒ ἐστιν ὡξ ἡ ΒΑ πρὸς τὴν ΑΔ, οὕτως ἡ ΔΑ πρὸς τὴν ΑΓ διὰ τὴν ὁμοιότητα τῶν ΔΑΒ, ΔΑΓ τριγώνων, καὶ εὐῖναι ὡξ τὴν πρώτην πρὸς τὴν τρίτην, οὕτως τὸ ἀπὸ τῆς τρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας]. ἡμιόλιον ἄρα καὶ τὸ ἀπὸ τῆς ΒΑ τοῦ ἀπὸ τῆς ΑΔ. καί ἐστιν ἡ μὲν ΒΑ ἡ τῆς δοθείσης σφαίρας διάμετρος, ἡ δὲ ΑΔὁίση τῇ πλευρᾷ τῆς πυραμίδος.

Ἡ ἄρα τῆς σφαίρας διάμετρος ἡμιολία ἐστὶ τῆς πλευρᾶξ τῆς πυραμίδος· ὅπερ ἔδει δεῖξαι.

and to show that the square on the diameter of the sphere is one and a half times the (square) on the side of the pyramid.

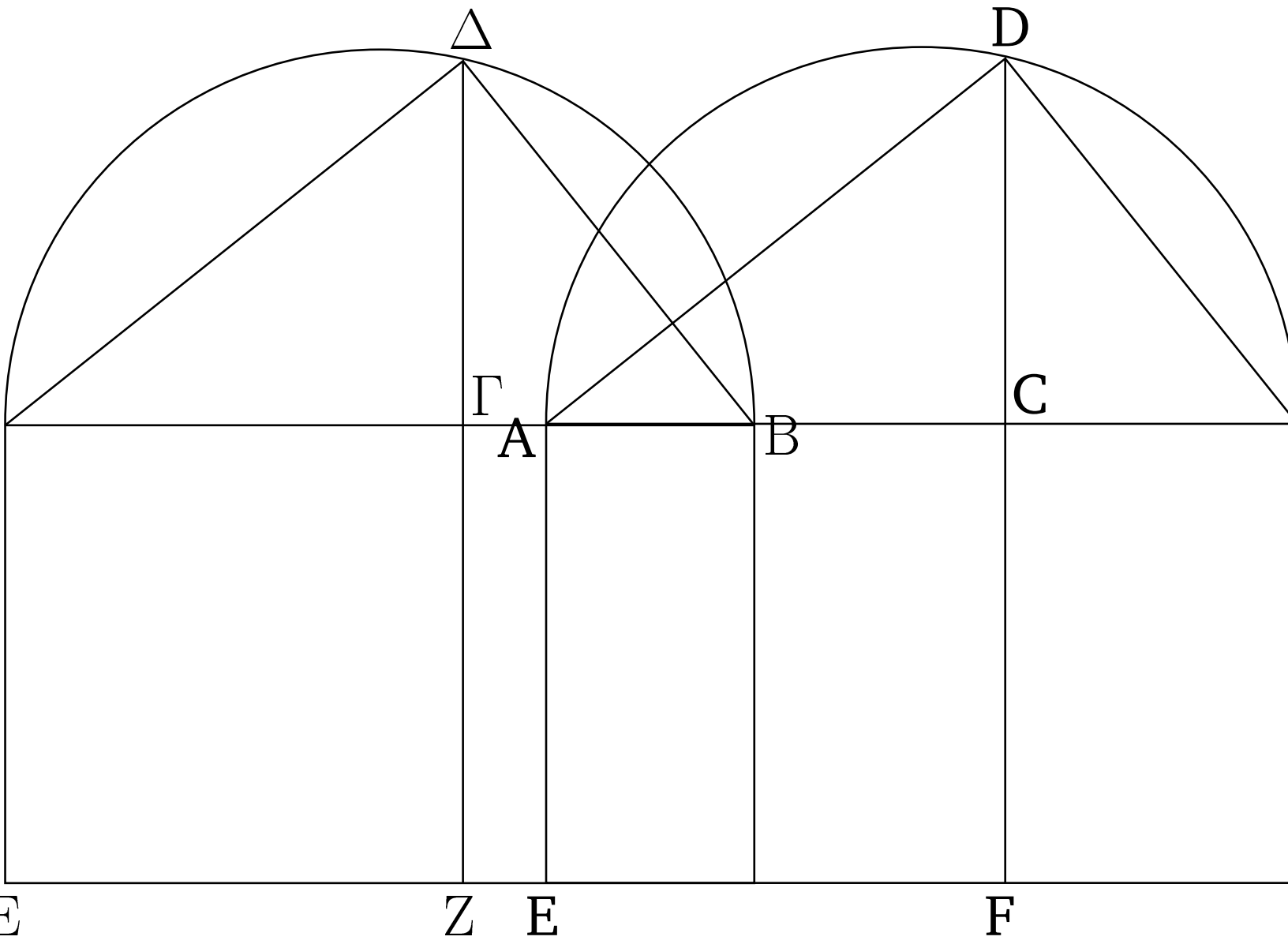
For let the straight-line HL be produced in a straight-line with KH, and let HL be made equal to CB. And since as AC (is) to CD, so CD (is) to CB [Prop. 6.8 corr.], and AC (is) equal to KH, and CD to HE, and CB to HL, thus as KH is to HE, so EH (is) to HL. Thus, the (rectangle contained) by KH and HL is equal to the (square) on EH [Prop. 6.17]. And each of the angles KHE and EHL is a right-angle. Thus, the semi-circle drawn on KL will also pass through E [inasmuch as if we join EL then the angle LEK becomes a right-angle, on account of triangle ELK becoming equiangular to each of the triangles ELH and EHK [Props. 6.8, 3.31]]. So, if KL remains (fixed), and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, it will also pass through points F and G, (because) if FL and LG are joined, the angles at F and G will similarly become right-angles. And the pyramid will be enclosed by the given sphere. For the diameter, KL, of the sphere is equal to the diameter, AB, of the given sphere—inasmuch as KH was made equal to AC, and HL to CB.

So, I say that the square on the diameter of the sphere is one and a half times the (square) on the side of the pyramid.

For since AC is double CB, AB is thus triple BC. Thus, via conversion, BA is one and a half times AC. And as BA (is) to AC, so the (square) on BA (is) to the (square) on AD [inasmuch as if DB is joined then as BA is to AD, so DA (is) to AC, on account of the similarity of triangles DAB and DAC. And as the first is to the third (of four proportional magnitudes), so the (square) on the first (is) to the (square) on the second.] Thus, the (square) on BA (is) also one and a half times the (square) on AD. And BA is the diameter of the given sphere, and AD (is) equal to the side of the pyramid.

Thus, the square on the diameter of the sphere is one and a half times the (square) on the side of the pyramid.† (Which is) the very thing it was required to show.

† If the radius of the sphere is unity then the side of the pyramid (*i.e.*, tetrahedron) is $\sqrt{8/3}$.



Λῆμμα.

Δεικτέον, ὅτι ἐστὶν ὡς ἡ ΑΒ πρὸς τὴν ΒΓ, οὕτως τὸ ἀπὸ τῆς ΑΔ πρὸς τὸ ἀπὸ τῆς ΔΓ.

Ἐκκείσθω γὰρ ἡ τοῦ ἡμικυκλίου καταγραφή, καὶ ἐπεζεύθθω ἡ ΔΒ, καὶ ἀναγεγράφθω ἀπὸ τῆς ΑΓ τετράγωνον τὸ ΕΓ, καὶ συμπεπληρώσθω τὸ ΖΒ παραλληλόγραμμον. ἐπεὶ οὖν διὰ τὸ ἰσογώνιον εἶναι τὸ ΔΑΒ τρίγωνον τῷ ΔΑΓ τριγώνῳ ἐστὶν ὡς ἡ ΒΑ πρὸς τὴν ΑΔ, οὕτως ἡ ΔΑ πρὸς τὴν ΑΓ, τὸ ἄρα ὑπὸ τῶν ΒΑ, ΑΓ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΑΔ. καὶ ἐπεί ἐστιν ὡς ἡ ΑΒ πρὸς τὴν ΒΓ, οὕτως τὸ ΕΒ πρὸς τὸ ΒΖ, καί ἐστι τὸ μὲν ΕΒ τὸ ὑπὸ τῶν ΒΑ, ΑΓ· ὁίση γὰρ ἡ ΕΑ τῇ ΑΓ· τὸ δὲ ΒΖ τὸ ὑπὸ τῶν ΑΓ, ΓΒ, ὡς ἄρα ἡ ΑΒ πρὸς τὴν ΒΓ, οὕτως τὸ

Lemma

It must be shown that as AB is to BC, so the (square) on AD (is) to the (square) on DC.

For, let the figure of the semi-circle be set out, and let DB have been joined. And let the square EC be described on AC. And let the parallelogram FB be completed. Therefore, since, on account of triangle DAB being equiangular to triangle DAC [Props. 6.8, 6.4], (proportionally) as BA is to AD, so DA (is) to AC, the (rectangle contained) by BA and AC is thus equal to the (square) on AD [Prop. 6.17]. And since as AB is to BC, so EB (is) to BF [Prop. 6.1]. And EB is the (rectangle contained) by BA and AC—for EA (is) equal to AC. And BF the (rect-

ὑπὸ τῶν BA, $AΓ$ πρὸν τὸ ὑπὸ τῶν $AΓ$, $ΓB$. καί ἐστι τὸ μὲν ὑπὸ τῶν BA, $AΓ$ ἴσον τῷ ἀπὸ τῆς $AΔ$, τὸ δὲ ὑπὸ τῶν $AΓB$ ἴσον τῷ ἀπὸ τῆς $ΔΓ$· ἡ γὰρ $ΔΓ$ κάθετος τῶν τῆς βάσεως τμημάτων τῶν $AΓ$, $ΓB$ μέση ἀνάλογόν ἐστι διὰ τὸ ὀρθὴν εἶναι τὴν ὑπὸ $AΔB$. ὡς ἄρα ἡ AB πρὸς τὴν $BΓ$, οὕτως τὸ ἀπὸ τῆς $AΔ$ πρὸς τὸ ἀπὸ τῆς $ΔΓ$· ὅπερ ἔδει δεῖξαι.

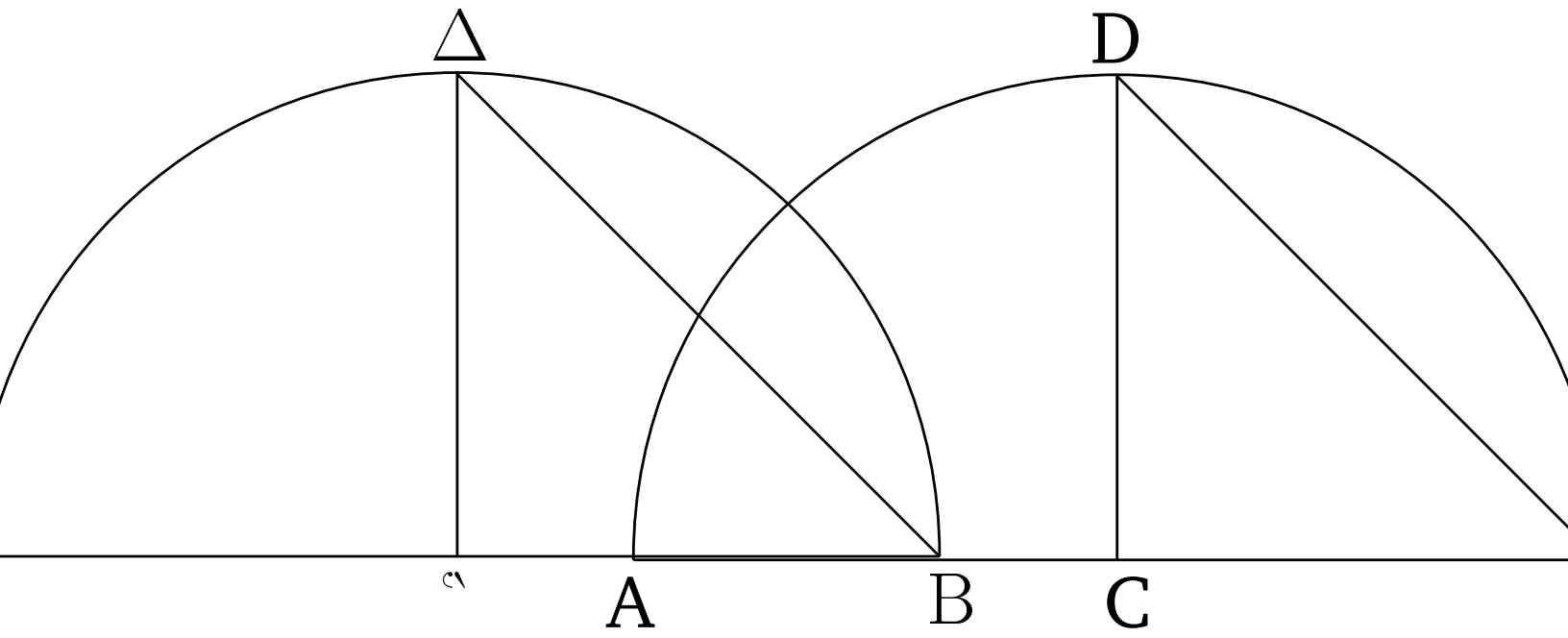
angle contained) by AC and CB. Thus, as AB (is) to BC, so the (rectangle contained) by BA and AC (is) to the (rectangle contained) by AC and CB. And the (rectangle contained) by BA and AC is equal to the (square) on AD, and the (rectangle contained) by ACB (is) equal to the (square) on DC. For the perpendicular DC is the mean proportional to the pieces of the base, AC and CB, on account of ADB being a right-angle [Prop. 6.8 corr.]. Thus, as AB (is) to BC, so the (square) on AD (is) to the (square) on DC. (Which is) the very thing it was required to show.

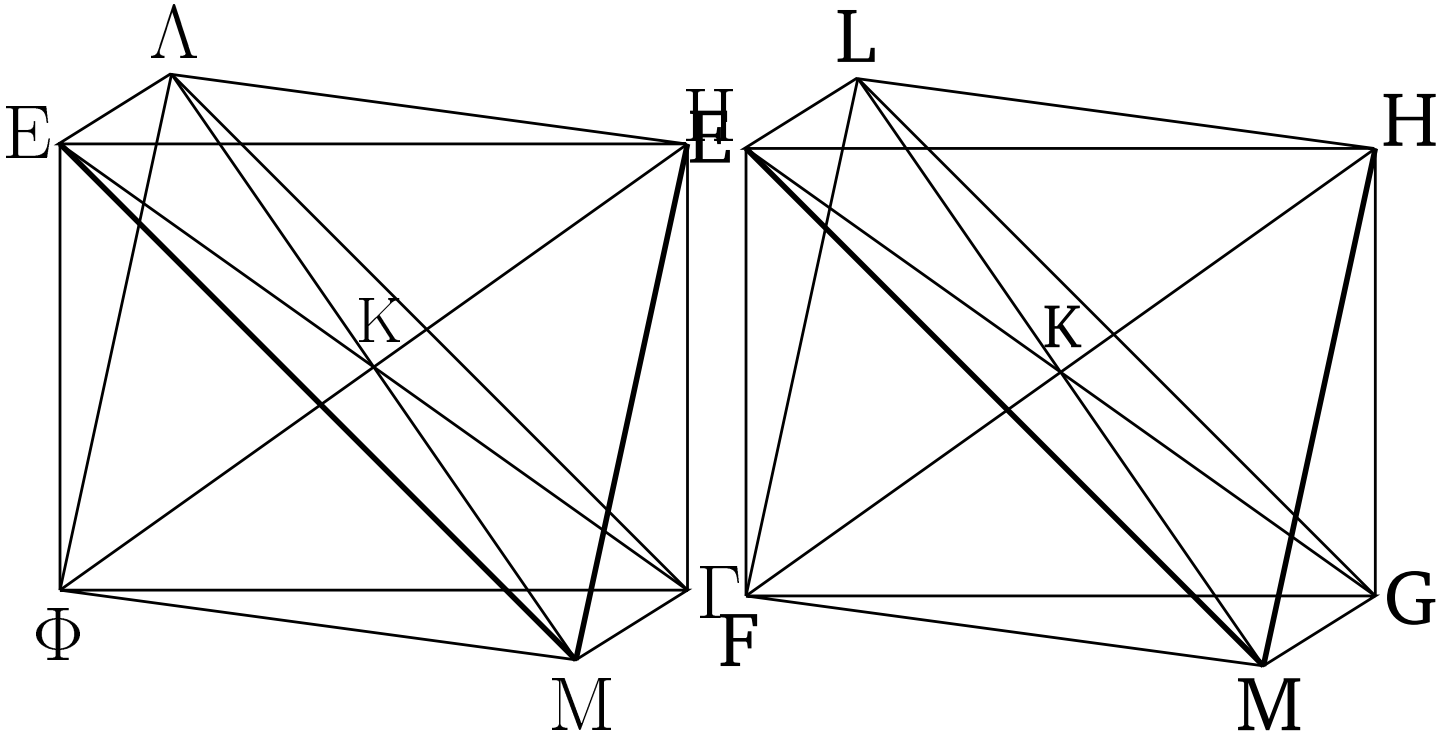
14.

Ὀκτάεδρον συστήσασθαι καὶ σφαίρᾳ περιλαβεῖν, ᾗ καὶ τὰ πρότερα, καὶ δεῖξαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασία ἐστὶ τῆς πλευρᾶξ τοῦ ὀκταέδρου.

Proposition 14

To construct an octahedron, and to enclose (it) in a (given) sphere, like in the preceding (proposition), and to show that the square on the diameter of the sphere is double the (square) on the side of the octahedron.





Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB, καὶ τετμήσθω δίθα κατὰ τὸ Γ, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ AΔB, καὶ ἤχθω ἀπὸ τοῦ Γ τῇ AB πρὸς ὀρθὰξ ἡ ΓΔ, καὶ ἐπεζεύχθω ἡ ΔB, καὶ ἐκκείσθω τετράγωνον τὸ EZHΘ ἴσην ἔχον ἑκάστην τῶν πλευρῶν τῇ ΔB, καὶ ἐπεζε-

Let the diameter *AB* of the given sphere be laid out, and let it have been cut in half at *C*. And let the semi-circle *ADB* be drawn on *AB*. And let *CD* be drawn from *C* at right-angles to *AB*. And let *DB* be joined. And let the square *EFGH*, having each of its sides equal to *DB*, be laid out. And let *HF* and *EG*

ὑθθωσαν αἱ ΘΖ, ΕΗ, καὶ ἀνεστάτω ἀπὸ τοῦ Κ σημείου τῷ τοῦ ΕΖΗΘ τετραγώνου ἐπιπέδῳ πρὸς ὀρθὰξ εὐθεῖα ἡ ΚΛ καὶ διήθθω ἐπὶ τὰ ὁέτερα μέρη τοῦ ἐπιπέδου ὡξ ἡ ΚΜ, καὶ ἀφῃρήσθω ἀφ' ἑκατέρας τῶν ΚΛ, ΚΜ μιᾷ τῶν ΕΚ, ΖΚ, ΗΚ, ΘΚὁίση ἑκατέρα τῶν ΚΛ, ΚΜ, καὶ ἐπεζεύθθωσαν αἱ ΛΕ, ΛΖ, ΛΗ, ΛΘ, ΜΕ, ΜΖ, ΜΗ, ΜΘ.

Καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΚΕ τῇ ΚΘ, καί ἐστιν ὀρθὴ ἡ ὑπὸ ΕΚΘ γωνία, τὸ ἄρα ἀπὸ τῆς ΘΕ διπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΕΚ. πάλιν, ἐπεὶ ὁίση ἐστὶν ἡ ΛΚ τῇ ΚΕ, καί ἐστιν ὀρθὴ ἡ ὑπὸ ΛΚΕ γωνία, τὸ ἄρα ἀπὸ τῆς ΕΛ διπλάσιόν ἐστι τοῦ ἀπὸ ΕΚ. ἐδείθθη δὲ καὶ τὸ ἀπὸ τῆς ΘΕ διπλάσιον τοῦ ἀπὸ τῆς ΕΚ· τὸ ἄρα ἀπὸ τῆς ΛΕ ἴσον ἐστὶ τῷ ἀπὸ τῆς ΕΘ· ὁίση ἄρα ἐστὶν ἡ ΛΕ τῇ ΕΘ. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΛΘ τῇ ΘΕ ἐστινὁίση· ἰσόπλευρον ἄρα ἐστὶ τὸ ΛΕΘ τρίγωνον. ὁμοίως δὴ δείχομεν, ὅτι καὶ ὁέκαστον τῶν λοιπῶν τριγώνων, ὁῶν βάσεις μέν εἰσιν αἱ τοῦ ΕΖΗΘ τετραγώνου πλευραί, κορυφαὶ δὲ τὰ Λ, Μ σημεῖα, ἰσόπλευρόν ἐστιν· ὀκτάεδρον ἄρα συνέσταται ὑπὸ ὀκτὼ τριγώνων ἰσοπλεύρων περιεχόμενον.

Δεῖ δὴ αὐτὸ καὶ σφαίρᾳ περιλαβεῖν τῇ δοθείσῃ καὶ δεῖχαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασίων ἐστὶ τῆς τοῦ ὀκταέδρου πλευρᾶξ.

Ἐπεὶ γὰρ αἱ τρεῖς αἱ ΛΚ, ΚΜ, ΚΕ ἴσαι ἀλλήλαις εἰσίν, τὸ ἄρα ἐπὶ τῆς ΛΜ γραφόμενον ἡμικύκλιον ἥξει καὶ διὰ τοῦ Ε. καὶ διὰ τὰ αὐτά, ἐαν μενούσης τῆς ΛΜ περιενεθθὲν τὸ ἡμικύκλιον εἰξ τὸ αὐτὸ ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, ἥξει καὶ διὰ τῶν Ζ, Η, Θ σημείων, καὶ ὀέσται σφαίρᾳ περιειλημμένον τὸ ὀκτάεδρον. λέγω δή, ὅτι καὶ τῇ δοθείσῃ. ἐπεὶ γὰρὁίση ἐστὶν ἡ ΛΚ τῇ ΚΜ, κοινὴ δὲ ἡ ΚΕ, καὶ γωνίας ὀρθὰξ περιέθουσιν, βάσις ἄρα ἡ ΛΕ βάσει τῇ ΕΜ ἐστινὁίση. καὶ ἐπεὶ ὀρθή ἐστιν ἡ ὑπὸ ΛΕΜ γωνία· ἐν ἡμικυκλίῳ γάρ· τὸ ἄρα ἀπὸ τῆς ΛΜ διπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΛΕ. πάλιν, ἐπεὶ ὁίση ἐστὶν ἡ ΑΓ τῇ ΓΒ, διπλασία ἐστὶν ἡ ΑΒ τῆς ΒΓ. ὡξ δὲ ἡ ΑΒ πρὸς τὴν ΒΓ, οὁύτως τὸ ἀπὸ τῆς ΑΒ πρὸς τὸ ἀπὸ τῆς ΒΔ· διπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΒ τοῦ ἀπὸ τῆς ΒΔ. ἐδείθθη δὲ καὶ τὸ ἀπὸ τῆς ΛΜ διπλάσιον τοῦ ἀπὸ τῆς ΛΕ. καί ἐστιν ἴσον τὸ ἀπὸ τῆς ΔΒ τῷ ἀπὸ τῆς ΛΕ· ὁίση γὰρ κεῖται ἡ ΕΘ τῇ ΔΒ. ἴσον ἄρα καὶ τὸ ἀπὸ τῆς ΑΒ τῷ ἀπὸ τῆς ΛΜ· ὁίση ἄρα ἡ ΑΒ τῇ ΛΜ. καί ἐστιν ἡ ΑΒ ἡ τῆς δοθείσης σφαίρας διάμετρος· ἡ ΛΜ ἄραὁίση ἐστὶ τῇ τῆς δοθείσης σφαίρας διαμέτρῳ.

Περιείληπται ἄρα τὸ ὀκτάεδρον τῇ δοθείσῃ σφαίρᾳ. καὶ συναποδέδεικται, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασίων ἐστὶ τῆς τοῦ ὀκταέδρου πλευρᾶξ· ὅπερ ἔδει δεῖχαι.

be joined. And let the straight-line KL be set up, at point K, at right-angles to the plane of square $EFGH$ [Prop. 11.12]. And let it be drawn across on the other side of the plane, like KM. And let KL and KM, equal to one of EK, FK, GK, and HK, be cut off from KL and KM, respectively. And let LE, LF, LG, LH, ME, MF, MG, and MH be joined.

And since KE is equal to KH, and angle EKH is a right-angle, the (square) on the HE is thus double the (square) on EK [Prop. 1.47]. Again, since LK is equal to KE, and angle LKE is a right-angle, the (square) on EL is thus double the (square) on EK [Prop. 1.47]. And the (square) on HE was also shown (to be) double the (square) on EK. Thus, the (square) on LE is equal to the (square) on EH. Thus, LE is equal to EH. So, for the same (reasons), LH is also equal to HE. Triangle LEH is thus equilateral. So, similarly, we can show that each of the remaining triangles, whose bases are the sides of the square $EFGH$, and apexes the points L and M, are equilateral. Thus, an octahedron contained by eight equilateral triangles has been constructed.

So, it is also necessary to enclose it by the given sphere, and to show that the square on the diameter of the sphere is double the (square) on the side of the octahedron.

For since the three (straight-lines) LK, KM, and KE are equal to one another, the semi-circle drawn on LM will thus also pass through E. And, for the same (reasons), if LM remains (fixed), and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, then it will also pass through points F, G, and H, and the octahedron will be enclosed by a sphere. So, I say that (it is) also (enclosed) by the given (sphere). For since LK is equal to KM, and KE (is) common, and they contain right-angles, the base LE is thus equal to the base EM [Prop. 1.4]. And since angle LEM is a right-angle—for (it is) in a semi-circle [Prop. 3.31]—the (square) on LM is thus double the (square) on LE [Prop. 1.47]. Again, since AC is equal to CB, AB is double BC. And as AB (is) to BC, so the (square) on AB (is) to the (square) on BD [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is double the (square) on BD. And the (square) on LM was also shown (to be) double the (square) on LE. And the (square) on DB is equal to the (square) on LE. For EH was made equal to DB. Thus, the (square) on AB (is) also equal to the (square) on LM. Thus, AB (is) equal to LM. And AB is the diameter of the given sphere. Thus, LM is equal to the diameter of the given sphere.

Thus, the octahedron has been enclosed by the given sphere, and it has been simultaneously proved that the square on the diameter of the sphere is double the (square) on the side of the octahedron.† (Which is) the very thing it was required to show.

† If the radius of the sphere is unity then the side of octahedron is $\sqrt{2}$.

15.

Κύβον συστήσασθαι καὶ σφαίρᾳ περιλαβεῖν, ᾗ καὶ τὴν πυραμίδα, καὶ δεῖξαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίων ἐστὶ τῆς τοῦ κύβου πλευρᾶξ.

Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB καὶ τετμήσθω κατὰ τὸ $Γ$ ὥστε διπλῆν εἶναι τὴν $AΓ$ τῆς $ΓB$, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ $AΔB$, καὶ ἀπὸ τοῦ $Γ$ τῇ AB πρὸς ὀρθὰξ ἤχθω ἡ $ΓΔ$, καὶ ἐπεζεύχθω ἡ $ΔB$, καὶ ἐκκείσθω τετράγωνον τὸ $EZHΘ$ ἴσην ἔχον τὴν πλευρὰν τῇ $ΔB$, καὶ ἀπὸ τῶν E, Z, H, $Θ$ τῷ τοῦ $EZHΘ$ τετραγώνου ἐπιπέδῳ πρὸς ὀρθὰξ ἤχθωσαν αἱ EK, $ZΛ$, HM, $ΘN$, καὶ ἀφῃρήσθω ἀπὸ ἑκάστης τῶν EK, $ZΛ$, HM, $ΘN$ μιᾷ τῶν EZ, ZH, $HΘ$, $ΘE$ὁίση ἑκάστη τῶν EK, $ZΛ$, HM, $ΘN$, καὶ ἐπεζεύχθωσαν αἱ $KΛ$, $ΛM$, MN, NK· κύβος ἄρα συνέσταται ὁ ZN ὑπὸ ὁἐχ τετραγώνωνὁίσων περιεθόμενος.

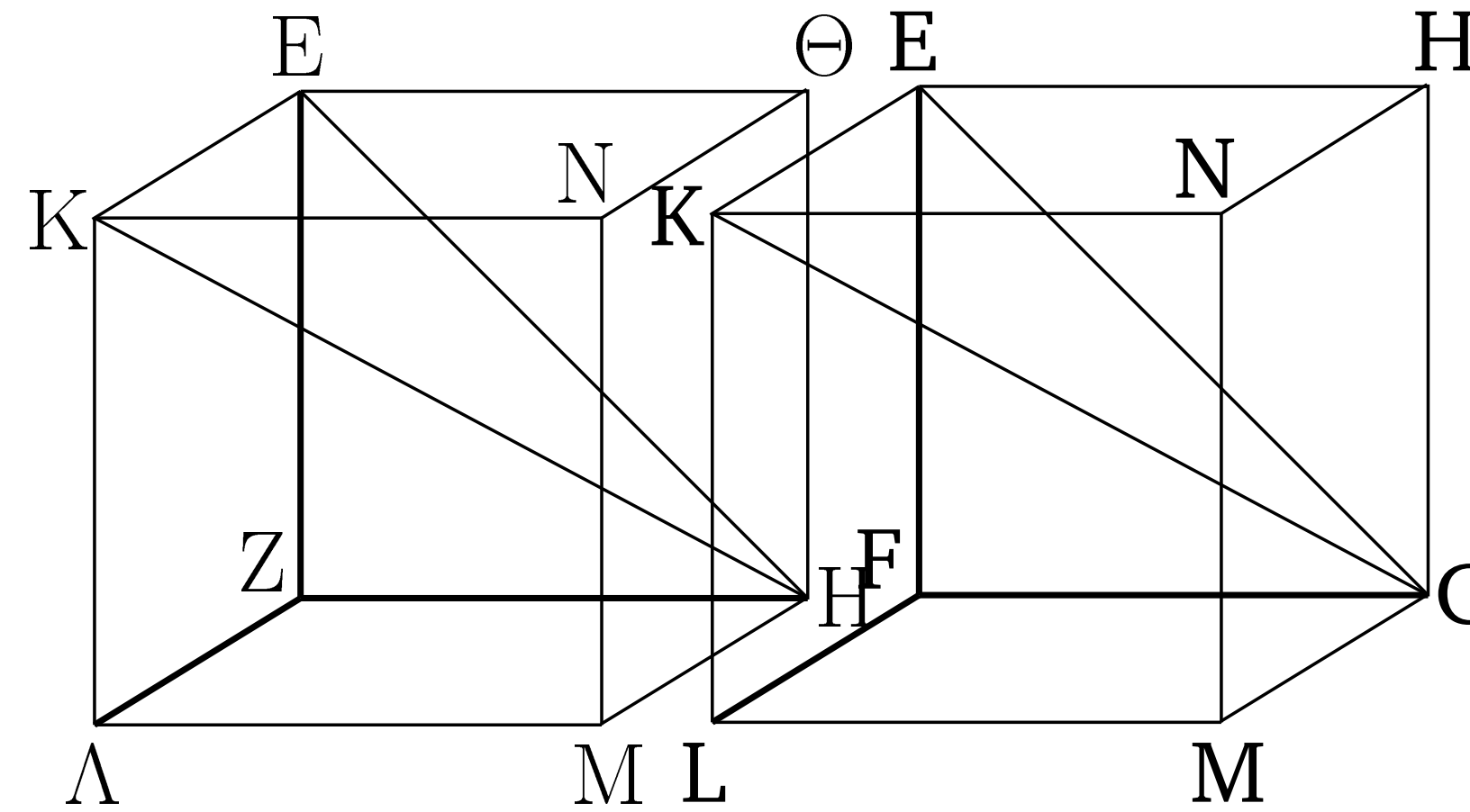
Δεῖ δὴ αὐτὸν καὶ σφαίρᾳ περιλαβεῖν τῇ δοθείσῃ καὶ δεῖξαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασία ἐστὶ τῆς πλευρᾶξ τοῦ κύβου.

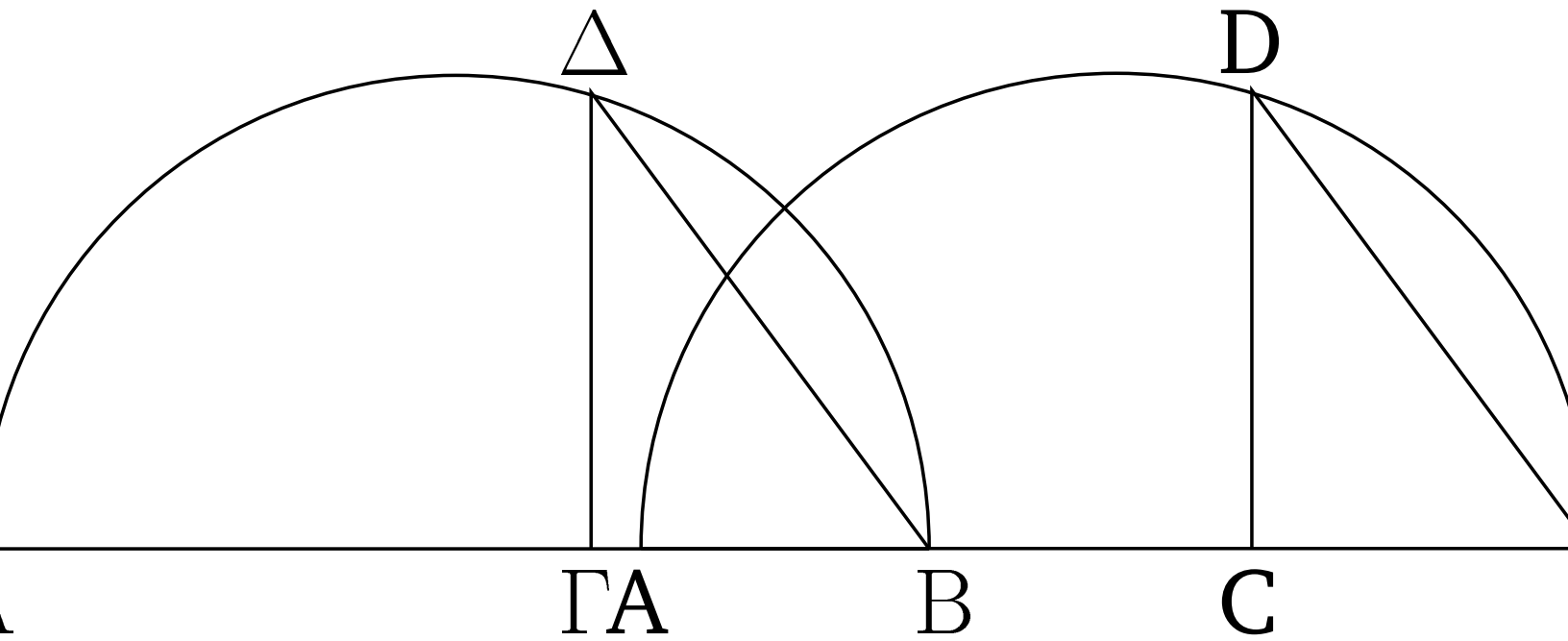
Proposition 15

To construct a cube, and to enclose (it) in a sphere, like in the (case of the) pyramid, and to show that the square on the diameter of the sphere is three times the (square) on the side of the cube.

Let the diameter AB of the given sphere be laid out, and let it be cut at C such that AC is double CB. And let the semicircle ADB be drawn on AB. And let CD be drawn from C at right-angles to AB. And let DB be joined. And let the square $EFGH$, having (its) side equal to DB, be laid out. And let EK, FL, GM, and HN be drawn from (points) E, F, G, and H, (respectively), at right-angles to the plane of square $EFGH$. And let EK, FL, GM, and HN, equal to one of EF, FG, GH, and HE, be cut off from EK, FL, GM, and HN, respectively. And let KL, LM, MN, and NK be joined. Thus, a cube contained by six equal squares has been constructed.

So, it is also necessary to enclose it by the given sphere, and to show that the square on the diameter of the sphere is three times the (square) on the side of the cube.





Ἐπεζεύθθωσαν γὰρ αἱ ΚΗ, ΕΗ. καὶ ἐπεὶ ὀρθή ἐστιν ἡ

For let *KG* and *EG* be joined. And since angle *KEG* is

ὑπὸ ΚΕΗ γωνία διὰ τὸ καὶ τὴν ΚΕ ὀρθὴν εἶναι πρὸς τὸ ΕΗ ἐπίπεδον δηλαδὴ καὶ πρὸς τὴν ΕΗ εὐθεῖαν, τὸ ἄρα ἐπὶ τῆς ΚΗ γραφόμενον ἡμικύκλιον ἥξει καὶ διὰ τοῦ Ε σημείου. πάλιν, ἐπεὶ ἡ ΗΖ ὀρθή ἐστι πρὸς ἑκατέραν τῶν ΖΛ, ΖΕ, καὶ πρὸς τὸ ΖΚ ἄρα ἐπίπεδον ὀρθή ἐστιν ἡ ΗΖ· ὥστε καὶ ἐαν ἐπιζεύχωμεν τὴν ΖΚ, ἡ ΗΖ ὀρθὴ ὀέσται καὶ πρὸς τὴν ΖΚ· καὶ διὰ τοῦτο πάλιν τὸ ἐπὶ τῆς ΗΚ γραφόμενον ἡμικύκλιον ἥξει καὶ διὰ τοῦ Ζ. ὁμοίως καὶ διὰ τῶν λοιπῶν τοῦ κύβου σημείων ἥξει. ἐαν δὴ μενούσης τῆς ΚΗ περιενεθθὲν τὸ ἡμικύκλιον εἰξ τὸ αὐτὸ ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, ὀέσται σφαίρᾳ περιειλημμένος ὁ κύβος. λέγω δή, ὅτι καὶ τῇ δοθείσῃ. ἐπεὶ γὰρὀίση ἐστὶν ἡ ΗΖ τῇ ΖΕ, καί ἐστιν ὀρθὴ ἡ πρὸς τῷ Ζ γωνία, τὸ ἄρα ἀπὸ τῆς ΕΗ διπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΕΖ. ὀίση δὲ ἡ ΕΖ τῇ ΕΚ· τὸ ἄρα ἀπὸ τῆς ΕΗ διπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΕΚ· ὥστε τὰ ἀπὸ τῶν ΗΕ, ΕΚ, τουτέστι τὸ ἀπὸ τῆς ΗΚ, τριπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΕΚ. καὶ ἐπεὶ τριπλασίων ἐστὶν ἡ ΑΒ τῆς ΒΓ, ὡς δὲ ἡ ΑΒ πρὸς τὴν ΒΓ, οὕτως τὸ ἀπὸ τῆς ΑΒ πρὸς τὸ ἀπὸ τῆς ΒΔ, τριπλάσιον ἄρα τὸ ἀπὸ τῆς ΑΒ τοῦ ἀπὸ τῆς ΒΔ. ἐδείθθη δὲ καὶ τὸ ἀπὸ τῆς ΗΚ τοῦ ἀπὸ τῆς ΚΕ τριπλάσιον. καὶ κεῖταιὀίση ἡ ΚΕ τῇ ΔΒ· ὀίση ἄρα καὶ ἡ ΚΗ τῇ ΑΒ. καί ἐστιν ἡ ΑΒ τῆς δοθείσης σφαίρας διάμετρος· καὶ ἡ ΚΗ ἄραὀίση ἐστὶ τῇ τῆς δοθείσης σφαίρας διαμέτρῳ.

Τῇ δοθείσῃ ἄρα σφαίρα περιείληπται ὁ κύβος· καὶ συναποδέδεικται, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίων ἐστὶ τῆς τοῦ κύβου πλευρᾶξ· ὅπερ ἔδει δεῖχαι.

a right-angle—on account of KE also being at right-angles to the plane EG, and manifestly also to the straight-line EG [Def. 11.3]—the semi-circle drawn on KG will thus also pass through point E. Again, since GF is at right-angles to each of FL and FE, GF is thus also at right-angles to the plane FK [Prop. 11.4]. Hence, if we also join FK then GF will also be at right-angles to FK. And, again, on account of this, the semi-circle drawn on GK will also pass through point F. Similarly, it will also pass through the remaining (angular) points of the cube. So, if KG remains (fixed), and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, then the cube will be enclosed by a sphere. So, I say that (it is) also (enclosed) by the given (sphere). For since GF is equal to FE, and the angle at F is a right-angle, the (square) on EG is thus double the (square) on EF [Prop. 1.47]. And EF (is) equal to EK. Thus, the (square) on EG is double the (square) on EK. Hence, the (sum of the squares) on GE and EK—that is to say, the (square) on GK [Prop. 1.47]—is three times the (square) on EK. And since AB is three times BC, and as AB (is) to BC, so the (square) on AB (is) to the (square) on BD [Prop. 6.8, Def. 5.9], the (square) on AB (is) thus three times the (square) on BD. And the (square) on GK was also shown (to be) three times the (square) on KE. And KE was made equal to DB. Thus, KG (is) also equal to AB. And AB is the radius of the given sphere. Thus, KG is also equal to the diameter of the given sphere.

Thus, the cube has been enclosed by the given sphere. And it has simultaneously been shown that the square on the diameter of the sphere is three times the (square) on the side of the cube.[†] (Which is) the very thing it was required to show.

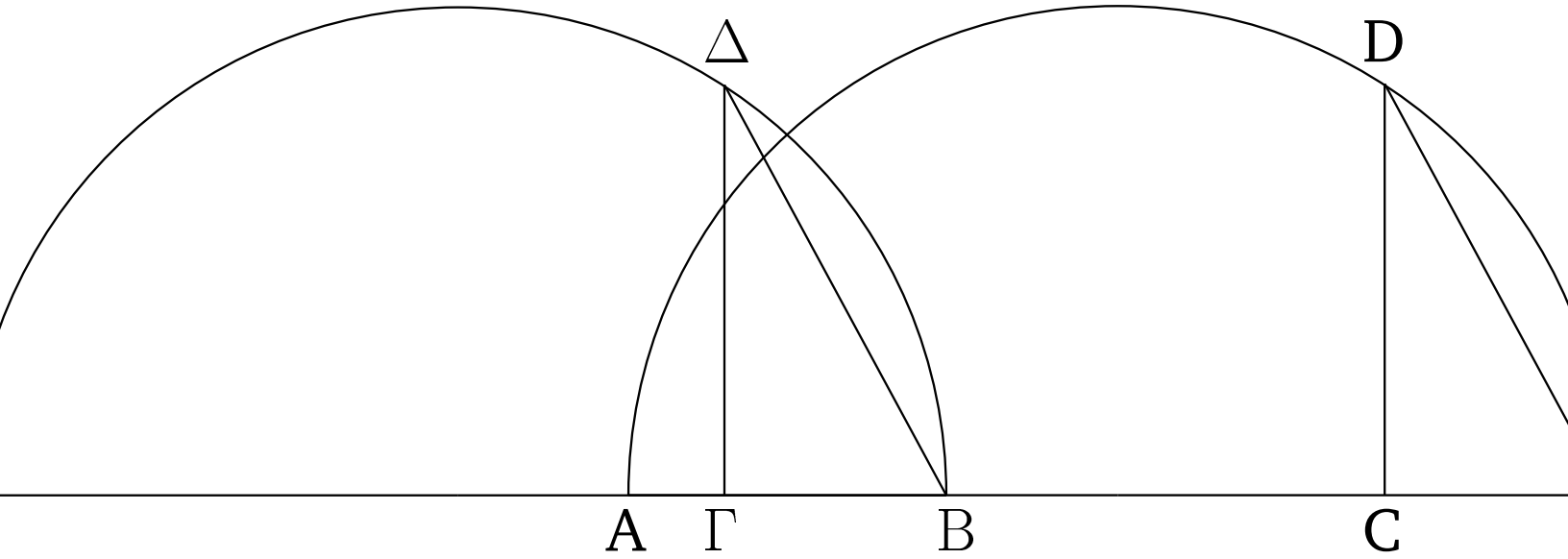
[†] If the radius of the sphere is unity then the side of the cube is $\sqrt{4/3}$.

16.

Εἰκοσάεδρον συστήσασθαι καὶ σφαίρᾳ περιλαβεῖν, ᾗ καὶ τὰ προειρημένα σθήματα, καὶ δεῖχαι, ὅτι ἡ τοῦ εἰκοσαέδρου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἐλάττων.

Proposition 16

To construct an icosahedron, and to enclose (it) in a sphere, like the aforementioned figures, and to show that the side of the icosahedron is that irrational (straight-line) called minor.



Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB καὶ τετμήσθω κατὰ τὸ Γ ὥστε τετραπλῆν εἶναι τὴν $A\Gamma$ τῆς ΓB, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ $A\Delta B$, καὶ ἤχθω ἀπὸ τοῦ Γ τῇ AB πρὸς ὀρθὰξ γωνίας εὐθεῖα γραμμὴ ἡ $\Gamma\Delta$, καὶ ἐπεζεύχθω ἡ ΔB, καὶ ἐκκείσθω κύκλος ὁ $EZH\Theta K$, οὗ ἡ ἐν τοῦ κέντρου ὁ ἡ ΔB, καὶ ἐγγεγράφθω εἰξ τὸν $EZH\Theta K$ κύκλον πεντάγωνον ἰσόπλευρόν τε καὶ ἰσογώνιον τὸ $EZH\Theta K$, καὶ τετμήσθωσαν αἱ EZ, ZH, $H\Theta$, ΘK, KE περιφέρειαι δίθα κατὰ τὸ Λ, M, N, X, O σημεῖα, καὶ ἐπεζεύχθωσαν αἱ ΛM, MN, NX, XO, $O\Lambda$, $E O$. ὁἰσόπλευρον ἄρα ἐστὶ καὶ τὸ $\Lambda MNXO$ πεντάγωνον, καὶ δεκαγώνου ἡ EO εὐθεῖα. καὶ ἀνεστάτωσανὁἀπὸ τῶν E, Z, H, Θ, K σημείων τῷ τοῦ κύκλου ἐπιπέδῳ πρὸς ὀρθὰξ γωνίας εὐθεῖαι αἱ $E\Pi$, ZP, $H\Sigma$, ΘT, KY ἴσαι οὐσαι τῇ ἐκ τοῦ κέντρου τοῦ $EZH\Theta K$ κύκλου, καὶ ἐπεζεύχθωσαν αἱ ΠP, $P\Sigma$, ΣT, TY, $Y\Pi$, $\Pi\Lambda$, ΛP, PM, $M\Sigma$, ΣN, NT, TX, XY, YO, $O\Pi$.

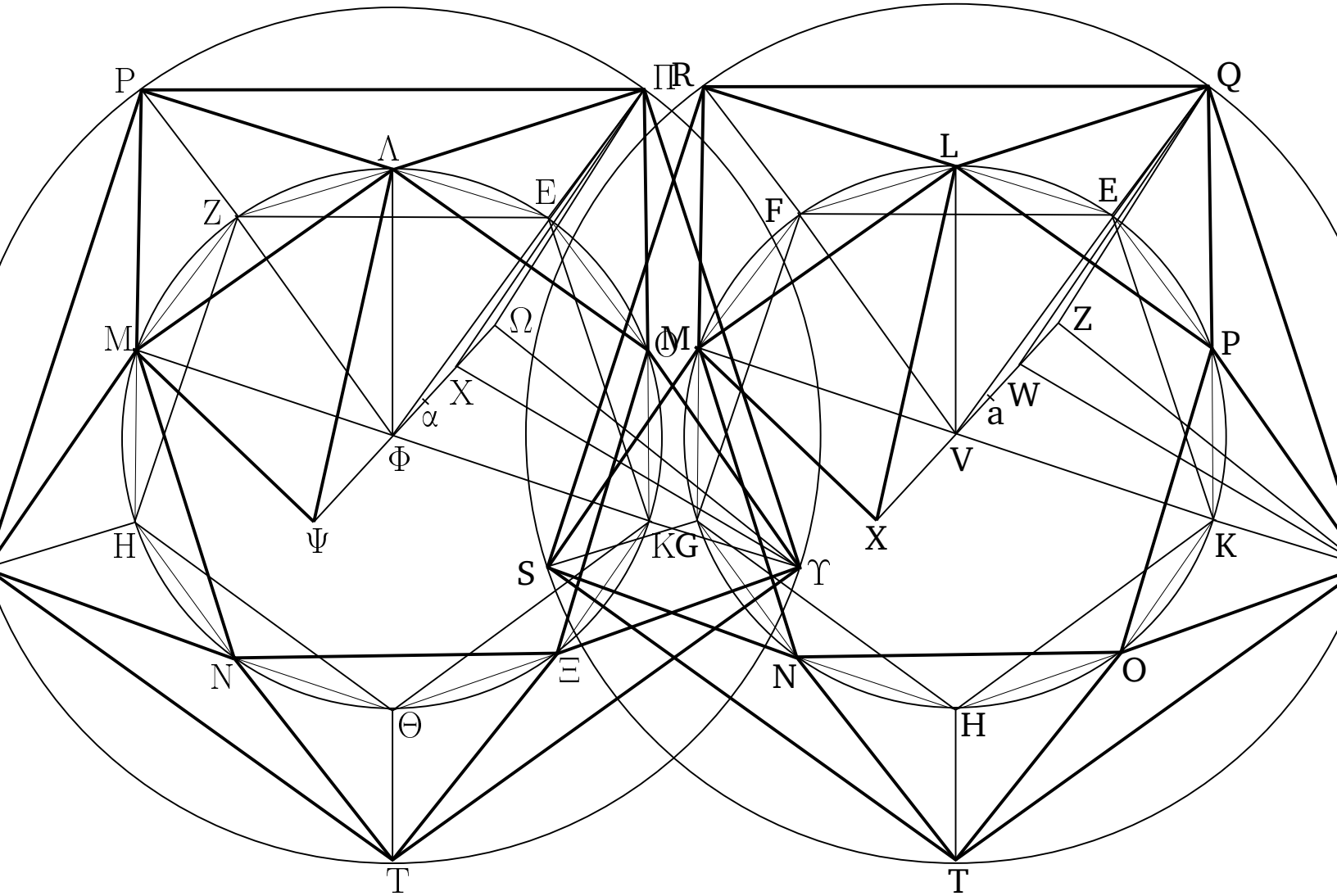
Καὶ ἐπεὶ ἑκατέρα τῶν $E\Pi$, KY τῷ αὐτῷ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν, παράλληλος ἄρα ἐστὶν ἡ $E\Pi$ τῇ KY. ὁέστι δὲ αὐτῇ καὶ ὁίση· αἱ δὲ τὰξ ἴσας τε καὶ παραλλήλους ἐπιζευγνύουσαι ἐπὶ τὰ αὐτὰ μέρη εὐθεῖαι ἴσαι τε καὶ παράλληλοί εἰσιν. ἡ ΠY ἄρα τῇ EKὁίση τε καὶ παράλληλόξ ἐστιν. πενταγώνου δὲ ἰσοπλεύρου ἡ EK· πενταγώνου ἄρα ἰσοπλεύρου καὶ ἡ ΠY τοῦ εἰξ τὸν $EZH\Theta K$ κύκλον ἐγγραφομένου. διὰ τὰ αὐτὰ δὴ καὶ ἑκάστη τῶν ΠP, $P\Sigma$, ΣT, TY πενταγώνου ἐστὶν ἰσοπλεύρου τοῦ εἰξ τὸν $EZH\Theta K$ κύκλον ἐγγραφομένου· ἰσόπλευρον ἄρα τὸ $\Pi P\Sigma TY$ πεντάγωνον. καὶ ἐπεὶ ἑχαγώνου μέν ἐστιν ἡ ΠE, δεκαγώνου δὲ ἡ EO, καὶ ἐστιν ὀρθὴ ἡ ὑπὸ ΠEO, πενταγώνου ἄρα ἐστὶν ἡ ΠO· ἡ γὰρ τοῦ πενταγώνου πλευρὰ δύναται τήν τε τοῦ ἑχαγώνου καὶ τὴν τοῦ δεκαγώνου τῶν εἰξ τὸν αὐτὸν κύκλον ἐγγραφομένων. διὰ τὰ αὐτὰ δὴ καὶ ἡ OY πενταγώνου ἐστὶ πλευρά. ὁέστι δὲ καὶ ἡ ΠY πενταγώνου· ἰσόπλευρον ἄρα ἐστὶ τὸ ΠOY τρίγωνον. διὰ τὰ αὐτὰ δὴ καὶ ὁέκαστον τῶν $\Pi\Lambda P$, $PM\Sigma$, ΣNT, TXY ἰσόπλευρόν

Let the diameter AB of the given sphere be laid out, and let it be cut at C such that AC is four times CB [Prop. 6.10]. And let the semi-circle ADB be drawn on AB. And let the straight-line CD be drawn from C at right-angles to AB. And let DB be joined. And let the circle $EFGHK$ be set down, and let its radius be equal to DB. And let the equilateral and equiangular pentagon $EFGHK$ be inscribed in circle $EFGHK$ [Prop. 4.11]. And let the circumferences EF, FG, GH, HK, and KE be cut in half at points L, M, N, O, and P (respectively). And let LM, MN, NO, OP, PL, and EP be joined. Thus, pentagon $LMNOP$ is also equilateral, and EP (is) the side of the decagon (inscribed in the circle). And let the straight-lines EQ, FR, GS, HT, and KU, which are equal to the radius of circle $EFGHK$, have been set up at right-angles to the plane of the circle, at points E, F, G, H, and K (respectively). And let QR, RS, ST, TU, UQ, QL, LR, RM, MS, SN, NT, TO, OU, UP, and PQ be joined.

And since EQ and KU are each at right-angles to the same plane, EQ is thus parallel to KU [Prop. 11.6]. And it is also equal to it. And straight-lines joining equal and parallel (straight-lines) on the same side are (themselves) equal and parallel [Prop. 1.33]. Thus, QU is equal and parallel to EK. And EK (is the side) of an equilateral pentagon (inscribed in circle $EFGHK$). Thus, QU (is) also the side of an equilateral pentagon inscribed in circle $EFGHK$. So, for the same (reasons), QR, RS, ST, and TU are also the sides of an equilateral pentagon inscribed in circle $EFGHK$. Pentagon $QRSTU$ (is) thus equilateral. And side QE is (the side) of a hexagon (inscribed in circle $EFGHK$), and EP (the side) of a decagon, and (angle) QEP is a right-angle, thus QP is (the side) of a pentagon (inscribed in the same circle). For the square on the side of a pentagon is (equal to the sum of) the (squares) on (the sides of) a hexagon and a decagon inscribed in the same circle

ἐστιν. καὶ ἐπεὶ πενταγώνου ἐδείχθη ἑκατέρα τῶν ΠΛ, ΠΟ, ὀέστι δὲ καὶ ἡ ΛΟ πενταγώνου, ἰσόπλευρον ἄρα ἐστὶ τὸ ΠΛΟ τρίγωνον. διὰ τὰ αὐτὰ δὴ καὶ ὁέκαστον τῶν ΛΡΜ, ΜΣΝ, ΝΤΧ, ΧΥΟ τριγώνων ἰσόπλευρόν ἐστιν.

[Prop. 13.10]. So, for the same (reasons), PU is also the side of a pentagon. And QU is also (the side) of a pentagon. Thus, triangle QPU is equilateral. So, for the same (reasons), (triangles) QLR, RMS, SNT, and TOU are each also equilateral. And since QL and QP were each shown (to be the sides) of a pentagon, and LP is also (the side) of a pentagon, triangle QLP is thus equilateral. So, for the same (reasons), triangles LRM, MSN, NTO, and OUP are each also equilateral.



Εἰλήφθω τὸ κέντρον τοῦ ΕΖΗΘΚ κύκλου τὸ Φ σημεῖον· καὶ ἀπὸ τοῦ Φ τῷ τοῦ κύκλου ἐπιπέδῳ πρὸς ὀρθὰξ ἀνεστάτω ἡ ΦΩ, καὶ ἐκβεβλήσθω ἐπὶ τὰ ὁέτερα μέρη ὡξ ἡ ΦΨ, καὶ ἀφῃρήσθω ἑχαγώνου μὲν ἡ ΦΘ, δεκαγώνου δὲ ἑκατέρα τῶν ΦΨ, ΘΩ, καὶ ἐπεζεύθθωσαν αἱ ΠΩ, ΠΘ, ΥΩ, ΕΦ, ΛΦ, ΛΨ, ΨΜ.

Καὶ ἐπεὶ ἑκατέρα τῶν ΦΘ, ΠΕ τῷ τοῦ κύκλου ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν, παράλληλος ἄρα ἐστὶν ἡ ΦΘ τῇ ΠΕ. εἰσὶ

Let the center, point V, of circle $EFGHK$ be found [Prop. 3.1]. And let VZ be set up, at (point) V, at right-angles to the plane of the circle. And let it be produced on the other side (of the circle), like VX. And let VW be cut off (from XZ so as to be equal to the side) of a hexagon, and each of VX and WZ (so as to be equal to the side) of a decagon. And let QZ, QW, UZ, EV, LV, LX, and XM be joined.

And since VW and QE are each at right-angles to the plane

δὲ καὶ ἴσαι· καὶ αἱ ΕΦ, ΠΘ ἄρα ἴσαι τε καὶ παράλληλοί εἰσιν. ἐχαγώνου δὲ ἡ ΕΦ· ἐχαγώνου ἄρα καὶ ἡ ΠΘ. καὶ ἐπεὶ ἐχαγώνου μέν ἐστιν ἡ ΠΘ, δεκαγώνου δὲ ἡ ΘΩ, καὶ ὀρθή ἐστιν ἡ ὑπὸ ΠΘΩ γωνία, πενταγώνου ἄρα ἐστὶν ἡ ΠΩ. διὰ τὰ αὐτὰ δὴ καὶ ἡ ΥΩ πενταγώνου ἐστίν, ἐπειδήπερ, ἂν ἐπιζεύχωμεν τὰξ ΦΚ, ΘΥ, ἴσαι καὶ ἀπεναντίον ὲσονται, καί ἐστιν ἡ ΦΚ ἐκ τοῦ κέντρου οὐῦσα ἑχαγώνου. ἑχαγώνου ἄρα καὶ ἡ ΘΥ. δεκαγώνου δὲ ἡ ΘΩ, καὶ ὀρθὴ ἡ ὑπὸ ΥΘΩ· πενταγώνου ἄρα ἡ ΥΩ. ὀέστι δὲ καὶ ἡ ΠΥ πενταγώνου· ἰσόπλευρον ἄρα ἐστὶ τὸ ΠΥΩ τρίγωνον. διὰ τὰ αὐτὰ δὴ καὶ ὁέκαστον τῶν λοιπῶν τριγώνων, ὁῶν βάσεις μέν εἰσιν αἱ ΠΡ, ΡΣ, ΣΤ, ΤΥ εὐθεῖαι, κορυφὴ δὲ τὸ Ω σημεῖον, ἰσόπλευρόν ἐστιν. πάλιν, ἐπεὶ ἑχαγώνου μὲν ἡ ΦΛ, δεκαγώνου δὲ ἡ ΦΨ, καὶ ὀρθή ἐστιν ἡ ὑπὸ ΛΦΨ γωνία, πενταγώνου ἄρα ἐστὶν ἡ ΛΨ. διὰ τὰ αὐτὰ δὴ ἐαν ἐπιζεύχωμεν τὴν ΜΦ οὐῦσαν ἑχαγώνου, συνάγεται καὶ ἡ ΜΨ πενταγώνου. ὀέστι δὲ καὶ ἡ ΛΜ πενταγώνου· ἰσόπλευρον ἄρα ἐστὶ τὸ ΛΜΨ τρίγωνον. ὁμοίως δὴ δειθθήσεται, ὅτι καὶ ὁέκαστον τῶν λοιπῶν τριγώνων, ὁῶν βάσεις μέν εἰσιν αἱ ΜΝ, ΝΧ, ΧΟ, ΟΛ, κορυφὴ δὲ τὸ Ψ σημεῖον, ἰσόπλευρόν ἐστιν. συνέσταται ἄρα εἰκοσάεδρον ὑπὸ εὀίκοσι τριγώνων ἰσοπλεύρων περιεχόμενον.

Δεῖ δὴ αὐτὸ καὶ σφαίρᾳ περιλαβεῖν τῇ δοθείσῃ καὶ δεῖχαι, ὅτι ἡ τοῦ εἰκοσαέδρου πλευρὰ ὀάλογός ἐστιν ἡ καλουμένη ἐλάσσων.

Ἐπεὶ γὰρ ἑχαγώνου ἐστὶν ἡ ΦΘ, δεκαγώνου δὲ ἡ ΘΩ, ἡ ΦΩ ἄραὀάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Θ, καὶ τὸ μεῖζον αὐτῆς τμῆμά ἐστιν ἡ ΦΘ· ὀέστιν ἄρα ὡξ ἡ ΩΦ πρὸς τὴν ΦΘ, οὑτως ἡ ΦΘ πρὸς τὴν ΘΩ. ὀίση δὲ ἡ μὲν ΦΘ τῇ ΦΕ, ἡ δὲ ΘΩ τῇ ΦΨ· ὀέστιν ἄρα ὡξ ἡ ΩΦ πρὸς τὴν ΦΕ, οὑτως ἡ ΕΦ πρὸς τὴν ΦΨ. καί εἰσιν ὀρθαὶ αἱ ὑπὸ ΩΦΕ, ΕΦΨ γωνίαι· ἐαν ἄρα ἐπιζεύχωμεν τὴν ΕΩ εὐθεῖαν, ὀρθὴ ὀέσται ἡ ὑπὸ ΨΕΩ γωνία διὰ τὴν ὁμοιότητα τῶν ΨΕΩ, ΦΕΩ τριγώνων. διὰ τὰ αὐτὰ δὴ ἐπεί ἐστιν ὡξ ἡ ΩΦ πρὸς τὴν ΦΘ, οὑτως ἡ ΦΘ πρὸς τὴν ΘΩ, ὀίση δὲ ἡ μὲν ΩΦ τῇ ΨΘ, ἡ δὲ ΦΘ τῇ ΘΠ, ὀέστιν ἄρα ὡξ ἡ ΨΘ πρὸς τὴν ΘΠ, οὑτως ἡ ΠΘ πρὸς τὴν ΘΩ. καὶ διὰ τοῦτο πάλιν ἐαν ἐπιζεύχωμεν τὴν ΠΨ, ὀρθὴ ὀέσται ἡ πρὸς τῷ Π γωνία· τὸ ἄρα ἐπὶ τῆς ΨΩ γραφόμενον ἡμικύκλιον ἥχει καὶ διὰ τοῦ Π. καὶ ἐαν μενούσης τῆς ΨΩ περιενεθθὲν τὸ ἡμικύκλιον εἰξ τὸ αὐτὸ πάλιν ἀποκατασταθῇ, ὅθεν ἤρχατο φέρεσθαι, ἥχει καὶ διὰ τοῦ Π καὶ τῶν λοιπῶν σημείων τοῦ εἰκοσαέδρου, καὶ ὀέσται σφαίρᾳ περιειλημμένον τὸ εἰκοσάεδρον. λέγω δή, ὅτι καὶ τῇ δοθείσῃ. τετμήσθω γὰρ ἡ ΦΘ δίθα κατὰ τὸ α. καὶ ἐπεὶ εὐθεῖα γραμμὴ ἡ ΦΩὀάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Θ, καὶ τὸ ὀέλασσον αὐτῆς τμῆμά ἐστιν ἡ ΩΘ, ἡ ἄρα ΩΘ προσλαβοῦσα τὴν ἡμίσειαν τοῦ μείζονος τμήματος τὴν Θα πενταπλάσιον δύναται τοῦ ἀπὸ τῆς ἡμισείας τοῦ μείζονος τμήματος· πενταπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς Ωα τοῦ ἀπὸ τῆς αΘ. καί ἐστι τῆς μὲν Ωα διπλῆ ἡ ΩΨ, τῆς δὲ αΘ διπλῆ ἡ ΦΘ· πενταπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΩΨ τοῦ ἀπὸ τῆς ΘΦ. καὶ ἐπεί τετραπλῆ ἐστιν ἡ ΑΓ τῆς ΓΒ, πενταπλῆ ἄρα ἐστὶν ἡ AB τῆς ΒΓ. ὡξ δὲ ἡ AB πρὸς τὴν ΒΓ, οὑτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς $B\Delta$· πενταπλάσιον ἄρα ἐστὶ

of the circle, VW is thus parallel to QE [Prop. 11.6]. And they are also equal. EV and QW are thus equal and parallel (to one another) [Prop. 1.33]. And EV (is the side) of a hexagon. Thus, QW (is) also (the side) of a hexagon. And since QW is (the side) of a hexagon, and WZ (the side) of a decagon, and angle QWZ is a right-angle [Def. 11.3, Prop. 1.29], QZ is thus (the side) of a pentagon [Prop. 13.10]. So, for the same (reasons), UZ is also (the side) of a pentagon—inasmuch as, if we join VK and WU then they will be equal and opposite. And VK, being (equal) to the radius (of the circle), is (the side) of a hexagon [Prop. 4.15 corr.]. Thus, WU (is) also the side of a hexagon. And WZ (is the side) of a decagon, and (angle) UWZ (is) a right-angle. Thus, UZ (is the side) of a pentagon [Prop. 13.10]. And QU is also (the side) of a pentagon. Triangle QUZ is thus equilateral. So, for the same (reasons), each of the remaining triangles, whose bases are the straight-lines QR, RS, ST, and TU, and apexes the point Z, are also equilateral. Again, since VL (is the side) of a hexagon, and VX (the side) of a decagon, and angle LVX is a right-angle, LX is thus (the side) of a pentagon [Prop. 13.10]. So, for the same (reasons), if we join MV, which is (the side) of a hexagon, MX is also inferred (to be the side) of a pentagon. And LM is also (the side) of a pentagon. Thus, triangle LMX is equilateral. So, similarly, it can be shown that each of the remaining triangles, whose bases are the (straight-lines) MN, NO, OP, and PL, and apexes the point X, are also equilateral. Thus, an icosahedron contained by twenty equilateral triangles has been constructed.

So, it is also necessary to enclose it in the given sphere, and to show that the side of the icosahedron is that irrational (straight-line) called minor.

For, since VW is (the side) of a hexagon, and WZ (the side) of a decagon, VZ has thus been cut in extreme and mean ratio at W, and VW is its greater piece [Prop. 13.9]. Thus, as ZV is to VW, so VW (is) to WZ. And VW (is) equal to VE, and WZ to VX. Thus, as ZV is to VE, so EV (is) to VX. And angles ZVE and EVX are right-angles. Thus, if we join straight-line EZ then angle XEZ will be a right-angle, on account of the similarity of triangles XEZ and VEZ. [Prop. 6.8]. So, for the same (reasons), since as ZV is to VW, so VW (is) to WZ, and ZV (is) equal to XW, and VW to WQ, thus as XW is to WQ, so QW (is) to WZ. And, again, on account of this, if we join QX then the angle at Q will be a right-angle [Prop. 6.8]. Thus, the semi-circle drawn on XZ will also pass through Q [Prop. 3.31]. And if XZ remains fixed, and the semi-circle is carried around, and again established at the same (position) from which it began to be moved, then it will also pass through (point) Q, and (through) the remaining (angular) points of the icosahedron. And the icosahedron will be enclosed by a sphere. So, I say that (it is) also (enclosed) by the given (sphere). For let VW be cut in half at a. And since the straight-line VZ has been cut in extreme and mean ratio at W, and ZW is its lesser piece, then

τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς $BΔ$. ἐδείθη δὲ καὶ τὸ ἀπὸ τῆς $ΩΨ$ πενταπλάσιον τοῦ ἀπὸ τῆς $ΦΘ$. καί ἐστινὁίση ἡ $ΔB$ τῇ $ΦΘ$· ἑκατέρα γὰρ αὐτῶνὁίση ἐστὶ τῇ ἐκ τοῦ κέντρου τοῦ $EZHΘK$ κύκλου· ὁίση ἄρα καὶ ἡ AB τῇ $ΨΩ$. καί ἐστιν ἡ AB ἡ τῆς δοθείσης σφαίρας διάμετρος· καὶ ἡ $ΨΩ$ ἄραὁίση ἐστὶ τῇ τῆς δοθείσης σφαίρας διαμέτρῳ· τῇ ἄρα δοθείσῃ σφαίρᾳ περιείληπται τὸ εἰκοσάεδρον.

Λέγω δή, ὅτι ἡ τοῦ εἰκοσαέδρου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἐλάττων. ἐπεὶ γὰρ ὁρητή ἐστιν ἡ τῆς σφαίρας διάμετρος, καί ἐστι δυνάμει πενταπλασίων τῆς ἐκ τοῦ κέντρου τοῦ $EZHΘK$ κύκλου, ὁρητὴ ἄρα ἐστὶ καὶ ἡ ἐκ τοῦ κέντρου τοῦ $EZHΘK$ κύκλου· ὥστε καὶ ἡ διάμετρος αὐτοῦ ὁρητή ἐστιν. ἔαν δὲ εἰξ κύκλον ὁρητὴν ἔχοντα τὴν διάμετρον πεντάγωνον ἰσόπλευρον ἐγγραφῃ, ἡ τοῦ πενταγώνου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἐλάττων. ἡ δὲ τοῦ $EZHΘK$ πενταγώνου πλευρὰ ἡ τοῦ εἰκοσαέδρου ἐστίν. ἡ ἄρα τοῦ εἰκοσαέδρου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἐλάττων.

the square on ZW added to half of the greater piece, Wa, is five times the (square) on half of the greater piece [Prop. 13.3]. Thus, the (square) on Za is five times the (square) on aW. And ZX is double Za, and VW double aW. Thus, the (square) on ZX is five times the (square) on WV. And since AC is four times CB, AB is thus five times BC. And as AB (is) to BC, so the (square) on AB (is) to the (square) on BD [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is five times the (square) on BD. And the (square) on ZX was also shown (to be) five times the (square) on VW. And DB is equal to VW. For each of them is equal to the radius of circle $EFGHK$. Thus, AB (is) also equal to XZ. And AB is the diameter of the given sphere. Thus, XZ is equal to the diameter of the given sphere. Thus, the icosahedron has been enclosed by the given sphere.

So, I say that the side of the icosahedron is that irrational (straight-line) called minor. For since the diameter of the sphere is rational, and the square on it is five times the (square) on the radius of circle $EFGHK$, the radius of circle $EFGHK$ is thus also rational. Hence, its diameter is also rational. And if an equilateral pentagon is inscribed in a circle having a rational diameter then the side of the pentagon is that irrational (straight-line) called minor [Prop. 13.11]. And the side of pentagon $EFGHK$ is (the side) of the icosahedron. Thus, the side of the icosahedron is that irrational (straight-line) called minor.

Πόρισμα.

Ἐκ δὴ τούτου φανερόν, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει πενταπλασίων ἐστὶ τῆς ἐκ τοῦ κέντρου τοῦ κύκλου, ἀφ΄ οὗ τὸ εἰκοσάεδρον ἀναγέγραπται, καὶ ὅτι ἡ τῆς σφαίρας διάμετρος σύγκειταιὁέκ τε τῆς τοῦ ἑξαγώνου καὶ δύο τῶν τοῦ δεκαγώνου τῶν εἰξ τὸν αὐτὸν κύκλον ἐγγραφομένων. ὅπερ ἔδει δεῖξαι.

Corollary

So, (it is) clear, from this, that the square on the diameter of the sphere is five times the square on the radius of the circle from which the icosahedron has been described, and that the the diameter of the sphere is the sum of (the side) of the hexagon, and two of (the sides) of the decagon, inscribed in the same circle.†

† If the radius of the sphere is unity then the radius of the circle is $2/\sqrt{5}$, and the sides of the hexagon, decagon, and pentagon/icosahedron are $2/\sqrt{5}$, $1 - 1/\sqrt{5}$, and $(1/\sqrt{5})\sqrt{10 - 2\sqrt{5}}$, respectively.

17.

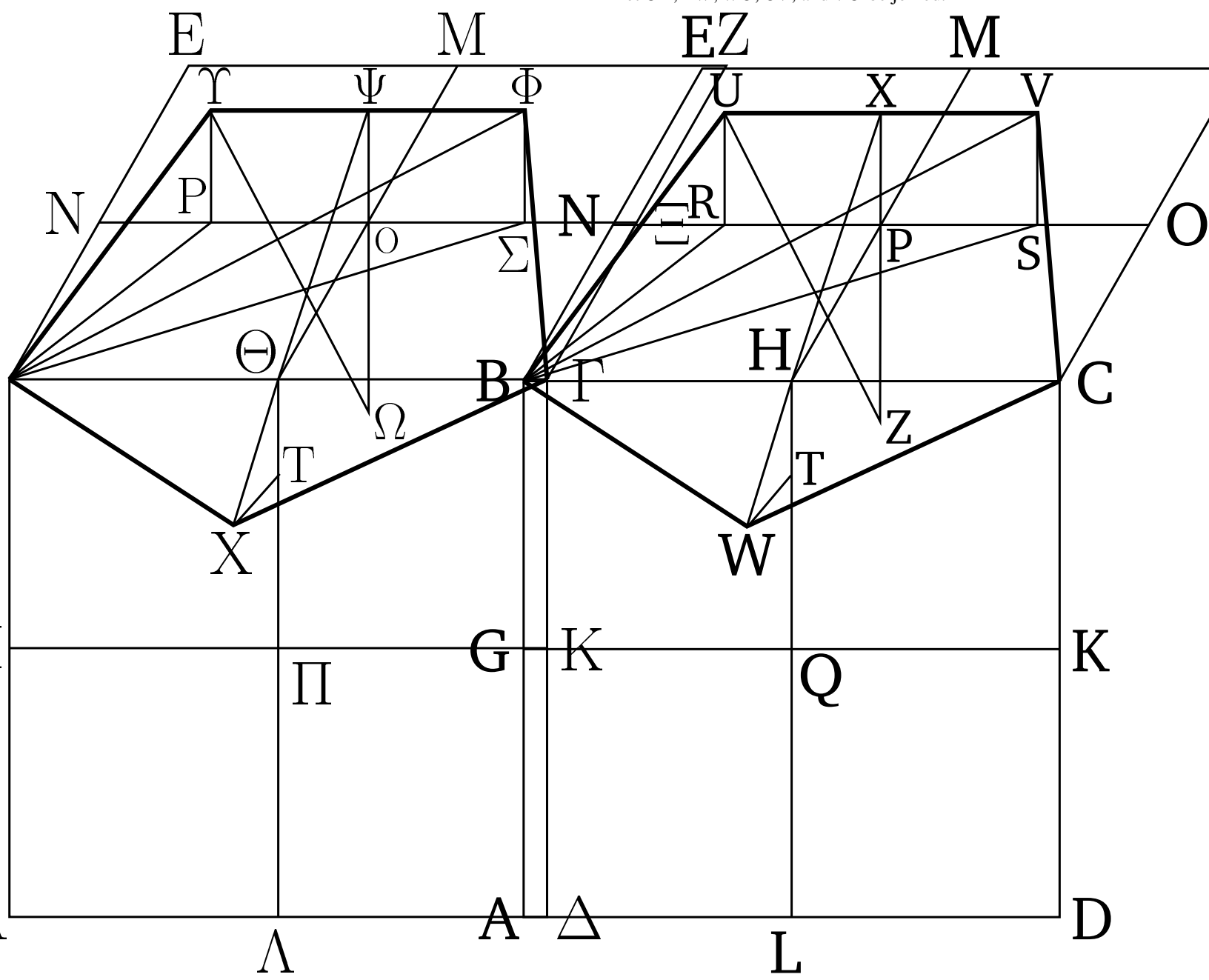
Δωδεκάεδρον συστήσασθαι καὶ σφαίρᾳ περιλαβεῖν, ᾗ καὶ τὰ προειρημένα σθήματα, καὶ δεῖξαι, ὅτι ἡ τοῦ δωδεκαέδρου πλευρὰ ὀάλογόξ ἐστιν ἡ καλουμένη ἀποτομή.

Ἐκκείσθωσαν τοῦ προειρημένου κύβου δύο ἐπίπεδα πρὸς ὀρθὰξ ἀλλήλοις τὰ $ABΓΔ$, $ΓBEZ$, καὶ τετμήσθω ἑκάστη τῶν AB, $BΓ$, $ΓΔ$, $ΔA$, EZ, EB, $ZΓ$ πλευρῶν δίθα κατὰ τὰ H, $Θ$, K, $Λ$, M, N, X, καὶ ἐπεζεύθθωσαν αἱ HK, $ΘΛ$, $MΘ$, NX, καὶ τετηήσθω ἑκάστη τῶν NO, OX, $ΘΠ$ὀάκρον καὶ μέσον λόγον κατὰ τὰ P, $Σ$, T σημεῖα, καὶ ὀέστω αὐτῶν μείζονα τμήματα τὰ PO, $OΣ$, $TΠ$, καὶ ἀνεστάτωσαν ἀπὸ τῶν P, $Σ$, T σημείων τοῖς τοῦ κύβου ἐπιπέδοις πρὸς ὀρθὰξ ἐπὶ τὰ ἐκτὸξ μέρη τοῦ κύβου αἱ PY, $ΣΦ$, $TΘ$, καὶ κείσθωσαν ἴσαι ταῖς PO, $OΣ$, $TΠ$, καὶ ἐπεζεύθθωσαν αἱ YB, $BΘ$, $ΘΓ$, $ΓΦ$, $ΦY$.

Proposition 17

To construct a dodecahedron, and to enclose (it) in a sphere, like the aforementioned figures, and to show that the side of the dodecahedron is that irrational (straight-line) called an apotome.

Let two planes of the aforementioned cube [Prop. 13.15], $ABCD$ and $CBEF$, (which are) at right-angles to one another, be laid out. And let the sides AB, BC, CD, DA, EF, EB, and FC have each been cut in half at points G, H, K, L, M, N, and O (respectively). And let GK, HL, MH, and NO be joined. And let NP, PO, and HQ have each been cut in extreme and mean ratio at points R, S, and T (respectively). And let their greater pieces be RP, PS, and TQ (respectively). And let RU, SV, and TW be set up on the exterior side of the cube, at points



Λέγω, ὅτι τὸ ΥΒΘΓΦ πεντάγωνον ἰσόπλευρόν τε καὶ ἐν ἑνὶ ἐπιπέδῳ καὶ ὀέτι ἰσογώνιόν ἐστιν. ἐπεζεύθθωσαν γὰρ αἱ ΡΒ, ΣΒ, ΦΒ. καὶ ἐπεὶ εὐθεῖα ἡ ΝΟὀάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Ρ, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ ΡΟ, τὰ ἄρα ἀπὸ τῶν ΟΝ, ΝΡ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΡΟ. ὁίση δὲ ἡ μὲν ΟΝ τῇ ΝΒ, ἡ δὲ ΟΡ τῇ ΡΥ· τὰ ἄρα ἀπὸ τῶν ΒΝ, ΝΡ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΡΥ. τοῖς δὲ ἀπὸ τῶν ΒΝ, ΝΡ τὸ

I say that the pentagon $UBWCV$ is equilateral, and in one plane, and, further, equiangular. For let RB, SB, and VB be joined. And since the straight-line NP has been cut in extreme and mean ratio at R, and RP is the greater piece, the (sum of the squares) on PN and NR is thus three times the (square) on RP [Prop. 13.4]. And PN (is) equal to NB, and PR to RU. Thus, the (sum of the squares) on BN and NR is three times the

R, S, and T (respectively), at right-angles to the planes of the cube. And let them be made equal to RP, PS, and TQ. And let UB, BW, WC, CV, and VU be joined.

ἀπὸ τῆς BP ἐστιν ἴσον· τὸ ἄρα ἀπὸ τῆς BP τριπλάσιόν ἐστι τοῦ ἀπὸ τῆς PY· ὥστε τὰ ἀπὸ τῶν BP, PY τετραπλάσιά ἐστι τοῦ ἀπὸ τῆς PY. τοῖς δὲ ἀπὸ τῶν BP, PY ἴσον ἐστι τὸ ἀπὸ τῆς BY· τὸ ἄραἀπὸ τῆς BY τετραπλάσιόν ἐστι τοῦ ἀπὸ τῆς YP· διπλῆ ἄρα ἐστὶν ἡ BY τῆς PY. ὀέστι δὲ καὶ ἡ ΦY τῆς YP διπλῆ, ἐπειδήπερ καὶ ἡ ΣP τῆς OP, τουτέστι τῆς PY, ἐστι διπλῆ· ὁίση ἄρα ἡ BY τῇ YΦ. ὁμοίως δὴ δειθθήσεται, ὅτι καὶ ἑκάστη τῶν BΘ, ΘΓ, ΓΦ ἑκατέρᾳ τῶν BY, YΦ ἐστινὁίση. ἰσόπλευρον ἄρα ἐστὶ τὸ BYΦΓΘ πεντάγωνον. λέγω δή, ὅτι καὶ ἐν ἑνί ἐστιν ἐπιπέδῳ. ἤθθω γὰρ ἀπὸ τοῦ O ἑκατέρᾳ τῶν PY, ΣΦ παράλληλος ἐπὶ τὰ ἐκτὸξ τοῦ κύβου μέρη ἡ OΨ, καὶ ἐπεζεύθθωσαν αἱ ΨΘ, ΘΘ· λέγω, ὅτι ἡ ΨΘΘ εὐθεῖά ἐστιν. ἐπεὶ γὰρ ἡ ΘΠὁάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ T, καὶ τὸ μεῖζον αὐτῆς τμῆμά ἐστιν ἡ ΠT, ὀέστιν ἄρα ὡξ ἡ ΘΠ πρὸς τὴν ΠT, οὕτως ἡ ΠT πρὸς τὴν TΘ. ὁίση δὲ ἡ μὲν ΘΠ τῇ ΘO, ἡ δὲ ΠT ἑκατέρᾳ τῶν TΘ, OΨ· ὀέστιν ἄρα ὡξ ἡ ΘO πρὸς τὴν OΨ, οὕτως ἡ ΘT πρὸς τὴν TΘ. καί ἐστι παράλληλος ἡ μὲν ΘO τῇ TΘ· ἑκατέρα γὰρ αὐτῶν τῷ BΔ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν· ἡ δὲ TΘ τῇ OΨ· ἑκατέρα γὰρ αὐτῶν τῷ BZ ἐπιπέδῳ πρὸς ὀρθάξ ἐστιν. ἔαν δὲ δύο τρίγωνα συντεθῇ κατὰ μίαν γωνίαν, ὡξ τὰ ΨOΘ, ΘTΘ, τὰξ δύο πλευρὰξ ταῖς δυνὶν ἀνάλογον ἔχοντα, ὥστε τὰξ ὁμολόγους αὐτῶν πλευρὰξ καὶ παραλλήλους εὸῖναι, αἱ λοιπαὶ εὐθεῖαι ἐπ΄ εὐθείαςὀέσονται· ἐπ΄ εὐθείας ἄρα ἐστὶν ἡ ΨΘ τῇ ΘΘ. πᾶσα δὲ εὐθεῖα ἐν ἑνί ἐστιν ἐπιπέδῳ· ἐν ἑνὶ ἄρα ἐπιπέδῳ ἐστὶ τὸ YBΘΓΦ πεντάγωνον.

Λέγω δή, ὅτι καὶ ἰσογώνιόν ἐστιν.

Ἐπεὶ γὰρ εὐθεῖα γραμμὴ ἡ NOὁάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ P, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ OP [ὀέστιν ἄρα ὡξ συναμφότερος ἡ NO, OP πρὸς τὴν ON, οὕτως ἡ NO πρὸς τὴν OP], ὁίση δὲ ἡ OP τῇ OΣ [ὀέστιν ἄρα ὡξ ἡ ΣN πρὸς τὴν NO, οὕτως ἡ NO πρὸς τὴν OΣ], ἡ NΣ ἄραὁάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ O, καὶ τὸ μεῖζον τμῆμά ἐστιν ἡ NO· τὰ ἄρα ἀπὸ τῶν NΣ, ΣO τριπλάσιά ἐστι τοῦ ἀπὸ τῆς NO. ὁίση δὲ ἡ μὲν NO τῇ NB, ἡ δὲ OΣ τῇ ΣΦ· τὰ ἄρα ἀπὸ τῶν NΣ, ΣΦ τετράγωνα τριπλάσιά ἐστι τοῦ ἀπὸ τῆς NB· ὥστε τὰ ἀπὸ τῶν ΦΣ, ΣN, NB τετραπλάσιά ἐστι τοῦ ἀπὸ τῆς NB. τοῖς δὲ ἀπὸ τῶν ΣN, NB ἴσον ἐστὶ τὸ ἀπὸ τῆς ΣB· τὰ ἄρα ἀπὸ τῶν BΣ, ΣΦ, τουτέστι τὸ ἀπὸ τῆς BΦ [ὀρθὴ γὰρ ἡ ὑπὸ ΦΣB γωνία], τετραπλάσιόν ἐστι τοῦ ἀπὸ τῆς NB· διπλῆ ἄρα ἐστὶν ἡ ΦB τῆς BN. ὀέστι δὲ καὶ ἡ BΓ τῆς BN διπλῆ· ὁίση ἄρα ἐστὶν ἡ BΦ τῇ BΓ. καὶ ἐπεὶ δύο αἱ BY, YΦ δυσὶ ταῖς BΘ, ΘΓ ἴσαι εἰσίν, καὶ βάσις ἡ BΦ βάσει τῇ BΓὁίση, γωνία ἄρα ἡ ὑπὸ BYΦ γωνίᾳ τῇ ὑπὸ BΘΓ ἐστινὁίση. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἡ ὑπὸ YΦΓ γωνίαὁίση ἐστὶ τῇ ὑπὸ BΘΓ· αἱ ἄρα ὑπὸ BΘΓ, BYΦ, YΦΓ τρεῖς γωνίαι ἴσαι ἀλλήλαις εἰσίν. ἔαν δὲ πενταγώνου ἰσοπλεύρου αἱ τρεῖς γωνίαι ἴσαι ἀλλήλαις ὦσιν, ἰσογώνιονὀέσται τὸ πεντάγωνον· ἰσογώνιον ἄρα ἐστὶ τὸ BYΦΓΘ πεντάγωνον. ἐδείθθη δὲ καὶ ἰσόπλευρον· τὸ ἄρα BYΦΓΘ πεντάγωνον ἰσόπλευρόν ἐστι καὶ ἰσογώνιον, καί ἐστιν ἐπὶ μιᾶς τοῦ κύβου πλευρᾶξ τῆς BΓ. ἔαν

(square) on *RU*. And the (square) on *BR* is equal to the (sum of the squares) on *BN* and *NR* [Prop. 1.47]. Thus, the (square) on *BR* is three times the (square) on *RU*. Hence, the (sum of the squares) on *BR* and *RU* is four times the (square) on *RU*. And the (square) on *BU* is equal to the (sum of the squares) on *BR* and *RU* [Prop. 1.47]. Thus, the (square) on *BU* is four times the (square) on *UR*. Thus, *BU* is double *RU*. And *VU* is also double *UR*, inasmuch as *SR* is also double *PR*—that is to say, *RU*. Thus, *BU* (is) equal to *UV*. So, similarly, it can be shown that each of *BW*, *WC*, *CV* is equal to each of *BU* and *UV*. Thus, pentagon *BUVCW* is equilateral. So, I say that it is also in one plane. For let *PX* be drawn from *P*, parallel to each of *RU* and *SV*, on the exterior side of the cube. And let *XH* and *HW* be joined. I say that *XHW* is a straight-line. For since *HQ* has been cut in extreme and mean ratio at *T*, and *QT* is its greater piece, thus as *HQ* is to *QT*, so *QT* (is) to *TH*. And *HQ* (is) equal to *HP*, and *QT* to each of *TW* and *PX*. Thus, as *HP* is to *PX*, so *WT* (is) to *TH*. And *HP* is parallel to *TW*. For of each of them is at right-angles to the plane *BD* [Prop. 11.6]. And *TH* (is parallel) to *PX*. For each of them is at right-angles to the plane *BF* [Prop. 11.6]. And if two triangles, like *XPH* and *HTW*, having two sides proportional to two sides, are placed together at a single angle such that their corresponding sides are also parallel then the remaining sides will be straight-on (to one another) [Prop. 6.32]. Thus, *XH* is straight-on to *HW*. And every straight-line is in one plane [Prop. 11.1]. Thus, pentagon *UBWCV* is in one plane.

So, I say that it is also equiangular.

For since the straight-line *NP* has been cut in extreme and mean ratio at *R*, and *PR* is the greater piece [thus as the sum of *NP* and *PR* is to *PN*, so *NP* (is) to *PR*], and *PR* (is) equal to *PS* [thus as *SN* is to *NP*, so *NP* (is) to *PS*], *NS* has thus also been cut in extreme and mean ratio at *P*, and *NP* is the greater piece [Prop. 13.5]. Thus, the (sum of the squares) on *NS* and *SP* is three times the (square) on *NP* [Prop. 13.4]. And *NP* (is) equal to *NB*, and *PS* to *SV*. Thus, the (sum of the) squares on *NS* and *SV* is three times the (square) on *NB*. Hence, the (sum of the squares) on *VS*, *SN*, and *NB* is four times the (square) on *NB*. And the (square) on *SB* is equal to the (sum of the squares) on *SN* and *NB* [Prop. 1.47]. Thus, the (sum of the squares) on *BS* and *SV*—that is to say, the (square) on *BV* [for angle *VSB* (is) a right-angle]—is four times the (square) on *NB* [Def. 11.3, Prop. 1.47]. Thus, *VB* is double *BN*. And *BC* (is) also double *BN*. Thus, *BV* is equal to *BC*. And since the two (straight-lines) *BU* and *UV* are equal to the two (straight-lines) *BW* and *WC* (respectively), and the base *BV* (is) equal to the base *BC*, angle *BUV* is thus equal to angle *BWC* [Prop. 1.8]. So, similarly, we can show that angle *UVC* is equal to angle *BWC*. Thus, the three angles *BWC*, *BUV*, and *UVC* are equal to one another. And if three angles of an equilateral pentagon are equal to one another then the pen-

ἄρα ἐφ' ἑκάστης τῶν τοῦ κύβου δώδεκα πλευρῶν τὰ αὐτὰ κατασκευάσωμεν, συσταθήσεταί τι σχῆμα στερεὸν ὑπὸ δώδεκα πενταγώνων ἰσοπλεύρων τε καὶ ἰσογωνίων περιεχόμενον, ὃ καλεῖται δωδεκάεδρον.

Δεῖ δὴ αὐτὸ καὶ σφαίρᾳ περιλαβεῖν τῇ δοθείσῃ καὶ δεῖξαι, ὅτι ἡ τοῦ δωδεκαέδρου πλευρὰ ὁάλογός ἐστιν ἡ καλουμένη ἀποτομή.

Ἐκβεβλήσθω γὰρ ἡ ΨΟ, καὶ ὁέστω ἡ ΨΩ· συμβάλλει ἄρα ἡ ΟΩ τῇ τοῦ κύβου διαμέτρῳ, καὶ δίθα τέμνουσιν ἀλλήλας· τοῦτο γὰρ δέδεικται ἐν τῷ παρατελεύτῳ θεωρήματι τοῦ ἑνδεκάτου βιβλίου. τεμνέτωσαν κατὰ τὸ Ω· τὸ Ω ἄρα κέντρον ἐστὶ τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον, καὶ ἡ ΩΟ ἡμίσεια τῆς πλευρᾶξ τοῦ κύβου. ἐπεζεύθθω δὴ ἡ ΥΩ. καὶ ἐπεὶ εὐθεῖα γραμμὴ ἡ ΝΣὸάκρον καὶ μέσον λόγον τέτμ ηται κατὰ τὸ Ο, καὶ τὸ μεῖζον αὐτῆς τμῆμά ἐστιν ἡ ΝΟ, τὰ ἄρα ἀπὸ τῶν ΝΣ, ΣΟ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΝΟ. ὁίση δὲ ἡ μὲν ΝΣ τῇ ΨΩ, ἐπειδήπερ καὶ ἡ μὲν ΝΟ τῇ ΟΩ ἐστινὁίση, ἡ δὲ ΨΟ τῇ ΟΣ. ἀλλὰ μὴν καὶ ἡ ΟΣ τῇ ΨΥ, ἐπεὶ καὶ τῇ ΡΟ· τὰ ἄρα ἀπὸ τῶν ΩΨ, ΨΥ τριπλάσιά ἐστι τοῦ ἀπὸ τῆς ΝΟ. τοῖς δὲ ἀπὸ τῶν ΩΨ, ΨΥ ἴσον ἐστὶ τὸ ἀπὸ τῆς ΥΩ· τὸ ἄρα ἀπὸ τῆς ΥΩ τριπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΝΟ. ὁέστι δὲ καὶ ἡ ἐκ τοῦ κέντρου τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον δυνάμει τριπλασίων τῆς ἡμισείας τῆς τοῦ κύβου πλευρᾶξ· προδέδεικται γὰρ κύβον συστήσασθαι καὶ σφαίρᾳ περιλαβεῖν καὶ δεῖχαι, ὅτι ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίων ἐστὶ τῆς πλευρᾶξ τοῦ κύβου. εἰ δὲ ὅλη τῆς ὅλης, καὶ [ἡ] ἡμίσεια τῆς ἡμισείας· καί ἐστιν ἡ ΝΟ ἡμίσεια τῆς τοῦ κύβου πλευρᾶξ· ἡ ἄρα ΥΩὁίση ἐστὶ τῇ ἐκ τοῦ κέντρου τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον. καί ἐστι τὸ Ω κέντρον τῆς σφαίρας τῆς περιλαμβανούσης τὸν κύβον· τὸ Υ ἄρα σημεῖον πρὸς τῇ ἐπιφανείᾳ ἐστι τῆς σφαίρας. ὁμοίως δὴ δείχομεν, ὅτι καὶ ἑκάστη τῶν λοιπῶν γωνιῶν τοῦ δωδεκαέδρου πρὸς τῇ ἐπιφανείᾳ ἐστὶ τῆς σφαίρας· περιείληπται ἄρα τὸ δωδεκαέδρον τῇ δοθείσῃ σφαίρᾳ.

Λέγω δή, ὅτι ἡ τοῦ δωδεκαέδρου πλευρὰ ὁάλογός ἐστιν ἡ καλουμένη ἀποτομή.

Ἐπεὶ γὰρ τῆς ΝΟὁάκρον καὶ μέσον λόγον τετμ ημένης τὸ μεῖζον τμῆμά ἐστιν ὁ ΡΟ, τῆς δὲ ΟΧὁάκρον καὶ μέσον λόγον τετμ ημένης τὸ μεῖζον τμῆμά ἐστιν ἡ ΟΣ, ὅλης ἄρα τῆς ΝΧὁάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμά ἐστιν ἡ ΡΣ. [οὗτον ἐπεί ἐστιν ὡξ ἡ ΝΟ πρὸς τὴν ΟΡ, ἡ ΟΡ πρὸς τὴν ΡΝ, καὶ τὰ διπλάσια· τὰ γὰρ μέρη τοῖς ἰσάκις πολλαπλασίοις τὸν αὐτὸν ἔχει λόγον· ὡξ ἄρα ἡ ΝΧ πρὸς τὴν ΡΣ, οὑτως ἡ ΡΣ πρὸς συναμφότερον τὴν ΝΡ, ΣΧ. μείζων δὲ ἡ ΝΧ τῆς ΡΣ· μείζων ἄρα καὶ ἡ ΡΣ συναμφοτέρου τῆς ΝΡ, ΣΧ· ἡ ΝΧ ἄραὁάκρον καὶ μέσον λόγον τέτμ ηται, καὶ τὸ μεῖζον αὐτῆς τμῆμά ἐστιν ἡ ΡΣ.] ὁίση δὲ ἡ ΡΣ τῇ ΥΦ· τῆς ἄρα ΝΧὁάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμά ἐστιν ἡ ΥΦ. καὶ ἐπεὶ ὁρητή ἐστιν τῆς σφαίρας διάμετρος καί ἐστι δυνάμει τριπλασίων τῆς τοῦ κύβου πλευρᾶξ, ὁρητὴ ἄρα ἐστὶν ἡ ΝΧ πλευρὰ οὑσα τοῦ κύβου. ἐαν δὲ ὁρητὴ γραμμὴ ὁάκρον καὶ

tagon is equiangular [Prop. 13.7]. Thus, pentagon $BUVCW$ is equiangular. And it was also shown (to be) equilateral. Thus, pentagon $BUVCW$ is equilateral and equiangular, and it is on one of the sides, BC, of the cube. Thus, if we make the same construction on each of the twelve sides of the cube then some solid figure contained by twelve equilateral and equiangular pentagons will be constructed, which is called a dodecahedron.

So, it is necessary to enclose it in the given sphere, and to show that the side of the dodecahedron is that irrational (straight-line) called an apotome.

For let XP be produced, and let (the produced straight-line) be XZ. Thus, PZ meets the diameter of the cube, and they cut one another in half. For, this has been proved in the penultimate theorem of the eleventh book [Prop. 11.38]. Let them cut (one another) at Z. Thus, Z is the center of the sphere enclosing the cube, and ZP (is) half the side of the cube. So, let UZ be joined. And since the straight-line NS has been cut in extreme and mean ratio at P, and its greater piece is NP, the (sum of the squares) on NS and SP is thus three times the (square) on NP [Prop. 13.4]. And NS (is) equal to XZ, inasmuch as NP is also equal to PZ, and XP to PS. But, indeed, PS (is) also (equal) to XU, since (it is) also (equal) to RP. Thus, the (sum of the squares) on ZX and XU is three times the (square) on NP. And the (square) on UZ is equal to the (sum of the squares) on ZX and XU [Prop. 1.47]. Thus, the (square) on UZ is three times the (square) on NP. And the square on the radius of the sphere enclosing the cube is also three times the (square) on half the side of the cube. For it has previously been demonstrated (how to) construct the cube, and to enclose (it) in a sphere, and to show that the square on the diameter of the sphere is three times the (square) on the side of the cube [Prop. 13.15]. And if the (square on the) whole (is three times) the (square on the) whole, then the (square on the) half (is) also (three times) the (square on the) half. And NP is half of the side of the cube. Thus, UZ is equal to the radius of the sphere enclosing the cube. And Z is the center of the sphere enclosing the cube. Thus, point U is on the surface of the sphere. So, similarly, we can show that each of the remaining angles of the dodecahedron is also on the surface of the sphere. Thus, the dodecahedron has been enclosed by the given sphere.

So, I say that the side of the dodecahedron is that irrational straight-line called an apotome.

For since RP is the greater piece of NP, which has been cut in extreme and mean ratio, and PS is the greater piece of PO, which has been cut in extreme and mean ratio, RS is thus the greater piece of the whole of NO, which has been cut in extreme and mean ratio. [Thus, since as NP is to PR, (so) PR (is) to RN, and (the same is also true) of the doubles. For parts have the same ratio as similar multiples (taken in corresponding order) [Prop. 5.15]. Thus, as NO (is) to RS, so RS (is) to the sum of NR and SO. And NO (is) greater

μέσον λόγον τμ ηθῇ, ἑκάτερον τῶν τμ ημάτωνὁάλογόξ ἐστιν ἀποτομή.

Ἡ ΥΦ ἄρα πλευρὰ οὐῦσα τοῦ δωδεκαέδρουὁάλογόξ ἐστιν ἀποτομή.

than RS. Thus, RS (is) also greater than the sum of NR and SO [Prop. 5.14]. Thus, NO has been cut in extreme and mean ratio, and RS is its greater piece.] And RS (is) equal to UV. Thus, UV is the greater piece of NO, which has been cut in extreme and mean ratio. And since the diameter of the sphere is rational, and the square on it is three times the (square) on the side of the cube, NO, which is the side of the cube, is thus rational. And if a rational (straight)-line is cut in extreme and mean ratio then each of the pieces is the irrational (straight-line called) an apotome.

Thus, UV, which is the side of the dodecahedron, is the irrational (straight-line called) an apotome [Prop. 13.6].

Πόρισμα.

Ἔκ δὴ τούτου φανερόν, ὅτι τῆς τοῦ κύβου πλευρᾶξὁάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμά ἐστιν ἡ τοῦ δωδεκαέδρου πλευρά. ὅπερ ἔδει δεῖχαι.

Corollary

So, (it is) clear, from this, that the side of the dodecahedron is the greater piece of the side of the cube, when it is cut in extreme and mean ratio.† (Which is) the very thing it was required to show.

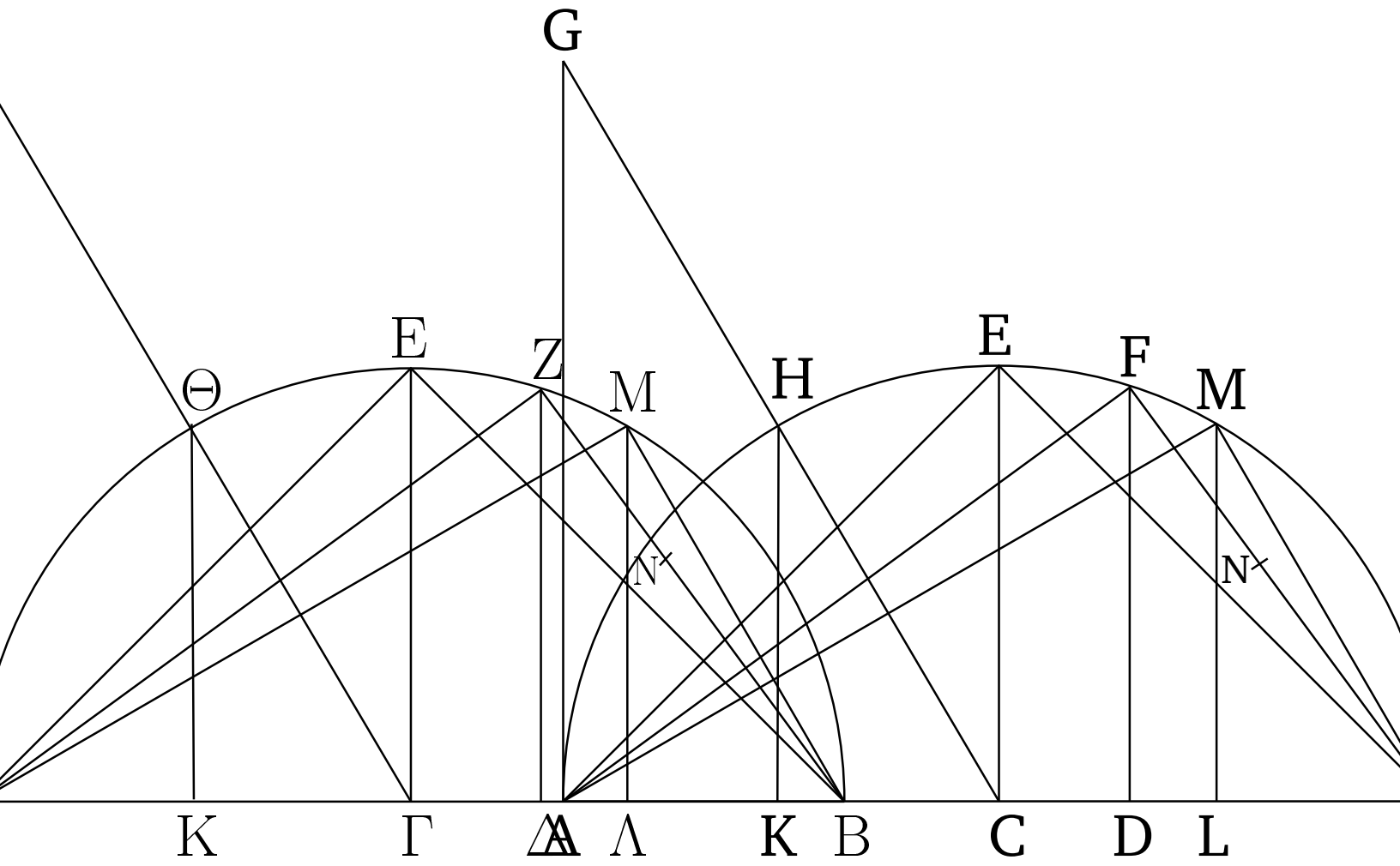
† If the radius of the circumscribed sphere is unity then the side of the cube is $\sqrt{4/3}$, and the side of the dodecahedron is $(1/3)\,(\sqrt{15}-\sqrt{3})$.

18.

Τὰξ πλευρὰξ τῶν πέντε σχημάτων ἐκθέσθαι καὶ συγκρῖναι πρὸς ἀλλήλας.

Proposition 18

To set out the sides of the five (aforementioned) figures, and to compare (them) with one another.†



Ἐκκείσθω ἡ τῆς δοθείσης σφαίρας διάμετρος ἡ AB, καὶ τετμήσθω κατὰ τὸ Γ ὥστε ἴσην εἶναι τὴν AΓ τῇ ΓB, κατὰ δὲ τὸ Δ ὥστε διπλασίονα εἶναι τὴν AΔ τῆς ΔB, καὶ γεγράφθω ἐπὶ τῆς AB ἡμικύκλιον τὸ AEB, καὶ ἀπὸ τῶν Γ, Δ τῇ AB πρὸς ὀρθὰξ ἤχθωσαν αἱ ΓE, ΔZ, καὶ ἐπεζεύχθωσαν αἱ AZ, ZB, EB. καὶ ἐπεὶ διπλῆ ἐστιν ἡ AΔ τῆς ΔB, τριπλῆ ἄρα ἐστὶν ἡ AB τῆς BΔ. ἀναστρέψαντι ἡμιολία ἄρα ἐστὶν ἡ BA τῆς AΔ. ὡξ δὲ ἡ BA πρὸς τὴν AΔ, οὕτως τὸ ἀπὸ τῆς BA πρὸς τὸ ἀπὸ τῆς AZ· ἰσογώνιον γάρ ἐστι τὸ AZB τρίγωνον τῷ AZΔ τριγώνῳ· ἡμιόλιον ἄρα ἐστὶ τὸ ἀπὸ τῆς BA τοῦ ἀπὸ τῆς AZ. ὀέστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει ἡμιολία τῆς πλευρᾶξ τῆς πυραμίδος. καί ἐστιν ἡ AB ἡ τῆς σφαίρας διάμετρος· ἡ AZ ἄραὀίση ἐστὶ τῇ πλευρᾷ τῆς πυραμίδος.

Πάλιν, ἐπεὶ διπλασίων ἐστὶν ἡ AΔ τῆς ΔB, τριπλῆ ἄρα ἐστὶν ἡ AB τῆς BΔ. ὡξ δὲ ἡ AB πρὸς τὴν BΔ, οὕτως τὸ ἀπὸ τῆς AB πρὸς τὸ ἀπὸ τῆς BZ· τριπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς AB τοῦ ἀπὸ τῆς BZ. ὀέστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει τριπλασίων τῆς τοῦ κύβου πλευρᾶξ. καί ἐστιν ἡ AB ἡ τῆς σφαίρας διάμετρος· ἡ BZ ἄρα τοῦ κύβου ἐστὶ πλευρά.

Let the diameter, *AB*, of the given sphere be laid out. And let it be cut at *C*, such that *AC* is equal to *CB*, and at *D*, such that *AD* is double *DB*. And let the semi-circle *AEB* be drawn on *AB*. And let *CE* and *DF* be drawn from *C* and *D* (respectively), at right-angles to *AB*. And let *AF*, *FB*, and *EB* be joined. And since *AD* is double *DB*, *AB* is thus triple *BD*. Thus, via conversion, *BA* is one and a half times *AD*. And as *BA* (is) to *AD*, so the (square) on *BA* (is) to the (square) on *AF* [Def. 5.9]. For triangle *AFB* is equiangular to triangle *AFD* [Prop. 6.8]. Thus, the (square) on *BA* is one and a half times the (square) on *AF*. And the square on the diameter of the sphere is also one and a half times the (square) on the side of the pyramid [Prop. 13.13]. And *AB* is the diameter of the sphere. Thus, *AF* is equal to the side of the pyramid.

Again, since *AD* is double *DB*, *AB* is thus triple *BD*. And as *AB* (is) to *BD*, so the (square) on *AB* (is) to the (square) on *BF* [Prop. 6.8, Def. 5.9]. Thus, the (square) on *AB* is three times the (square) on *BF*. And the square on the diameter of the sphere is also three times the (square) on the side of the cube [Prop. 13.15]. And *AB* is the diameter of the sphere.

Καὶ ἐπεὶ ὁίση ἐστὶν ἡ ΑΓ τῇ ΓΒ, διπλῆ ἄρα ἐστὶν ἡ ΑΒ τῆς ΒΓ. ὡξ δὲ ἡ ΑΒ πρὸς τὴν ΒΓ, οὑτως τὸ ἀπὸ τῆς ΑΒ πρὸς τὸ ἀπὸ τῆς ΒΕ· διπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΒ τοῦ ἀπὸ τῆς ΒΕ. ὀέστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει διπλασίων τῆς τοῦ ὀκταέδρου πλευρᾶξ. καὶ ἐστιν ἡ ΑΒ ἡ τῆς δοθείσης σφαίρας διάμετρος· ἡ ΒΕ ἄρα τοῦ ὀκταέδρου ἐστὶ πλευρά.

ὀΗθθω δὴ ἀπὸ τοῦ Α σημείου τῇ ΑΒ εὐθείᾳ πρὸς ὀρθὰξ ἡ ΑΗ, καὶ κείσθω ἡ ΑΗὁίση τῇ ΑΒ, καὶ ἐπεζεύθθω ἡ ΗΓ, καὶ ἀπὸ τοῦ Θ ἐπὶ τὴν ΑΒ κάθετος ἦθθω ἡ ΘΚ. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΗΑ τῆς ΑΓ· ὁίση γὰρ ἡ ΗΑ τῇ ΑΒ· ὡξ δὲ ἡ ΗΑ πρὸς τὴν ΑΓ, οὑτως ἡ ΘΚ πρὸς τὴν ΚΓ, διπλῆ ἄρα καὶ ἡ ΘΚ τῆς ΚΓ. τετραπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΘΚ τοῦ ἀπὸ τῆς ΚΓ· τὰ ἄρα ἀπὸ τῶν ΘΚ, ΚΓ, ὅπερ ἐστὶ τὸ ἀπὸ τῆς ΘΓ, πενταπλάσιόν ἐστι τοῦ ἀπὸ τῆς ΚΓ. ὁίση δὲ ἡ ΘΓ τῇ ΓΒ· πενταπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΒΓ τοῦ ἀπὸ τῆς ΓΚ. καὶ ἐπεὶ διπλῆ ἐστιν ἡ ΑΒ τῆς ΓΒ, ὧν ἡ ΑΔ τῆς ΔΒ ἐστι διπλῆ, λοιπὴ ἄρα ἡ ΒΔ λοιπῆς τῆς ΔΓ ἐστι διπλῆ. τριπλῆ ἄρα ἡ ΒΓ τῆς ΓΔ· ἐνναπλάσιον ἄρα τὸ ἀπὸ τῆς ΒΓ τοῦ ἀπὸ τῆς ΓΔ. πενταπλάσιον δὲ τὸ ἀπὸ τῆς ΒΓ τοῦ ἀπὸ τῆς ΓΚ· μεῖζον ἄρα τὸ ἀπὸ τῆς ΓΚ τοῦ ἀπὸ τῆς ΓΔ. μείζων ἄρα ἐστὶν ἡ ΓΚ τῆς ΓΔ. κείσθω τῇ ΓΚὁίση ἡ ΓΛ, καὶ ἀπὸ τοῦ Λ τῇ ΑΒ πρὸς ὀρθὰξ ἦθθω ἡ ΛΜ, καὶ ἐπεζεύθθω ἡ ΜΒ. καὶ ἐπεὶ πενταπλάσιόν ἐστι τὸ ἀπὸ τῆς ΒΓ τοῦ ἀπὸ τῆς ΓΚ, καί ἐστι τῆς μὲν ΒΓ διπλῆ ἡ ΑΒ, τῆς δὲ ΓΚ διπλῆ ἡ ΚΛ, πενταπλάσιον ἄρα ἐστὶ τὸ ἀπὸ τῆς ΑΒ τοῦ ἀπὸ τῆς ΚΛ. ὀέστι δὲ καὶ ἡ τῆς σφαίρας διάμετρος δυνάμει πενταπλασίων τῆς ἐκ τοῦ κέντρου τοῦ κύκλου, ἀφ' οὗ τὸ εἰκοσάεδρον ἀναγέγραπται. καί ἐστιν ἡ ΑΒ ἡ τῆς σφαίρας διάμετρος· ἡ ΚΛ ἄρα ἐκ τοῦ κέντρου ἐστὶ τοῦ κύκλου, ἀφ' οὗ τὸ εἰκοσάεδρον ἀναγέγραπται· ἡ ΚΛ ἄρα ἑξαγώνου ἐστὶ πλευρὰ τοῦ εἰρημένου κύκλου. καὶ ἐπεὶ ἡ τῆς σφαίρας διάμετρος σύγκειταιὁέκ τε τῆς τοῦ ἑξαγώνου καὶ δύο τῶν τοῦ δεκαγώνου τῶν εἰξ τὸν εἰρημένον κύκλον ἐγγραφομένων, καί ἐστιν ἡ μὲν ΑΒ ἡ τῆς σφαίρας διάμετρος, ἡ δὲ ΚΛ ἑξαγώνου πλευρά, καὶ ὁίση ἡ ΑΚ τῇ ΛΒ, ἑκατέρα ἄρα τῶν ΑΚ, ΛΒ δεκαγώνου ἐστὶ πλευρὰ τοῦ ἐγγραφομένου εἰξ τὸν κύκλον, ἀφ' οὗ τὸ εἰκοσάεδρον ἀναγέγραπται. καὶ ἐπεὶ δεκαγώνου μὲν ἡ ΛΒ, ἑξαγώνου δὲ ἡ ΜΛ· ὁίση γάρ ἐστι τῇ ΚΛ, ἐπεὶ καὶ τῇ ΘΚ· ἴσον γὰρ ἀπέθουσιν ἀπὸ τοῦ κέντρου· καί ἐστιν ἑκατέρα τῶν ΘΚ, ΚΛ διπλασίων τῆς ΚΓ· πενταγώνου ἄρα ἐστὶν ἡ ΜΒ. ἡ δὲ τοῦ πενταγώνου ἐστὶν ἡ τοῦ εἰκοσαέδρου· εἰκοσαέδρου ἄρα ἐστὶν ἡ ΜΒ.

Καὶ ἐπεὶ ἡ ΖΒ κύβου ἐστὶ πλευρά, τετμήσθωὁάκρον καὶ μέσον λόγον κατὰ τὸ Ν, καὶ ὁέστω μεῖζον τμῆμα τὸ ΝΒ· ἡ ΝΒ ἄρα δωδεκαέδρου ἐστὶ πλευρά.

Καὶ ἐπεὶ ἡ τῆς σφαίρας διάμετρος ἐδείθθη τῆς μὲν ΑΖ πλευρᾶξ τῆς πυραμίδος δυνάμει ἡμιολία, τῆς δὲ τοῦ ὀκταέδρου τῆς ΒΕ δυνάμει διπλασίων, τῆς δὲ τοῦ κύβου τῆς ΖΒ δυνάμει τριπλασίων, οὁίων ἄρα ἡ τῆς σφαίρας διάμετρος δυνάμει ὁέχ, τοιούτων ἡ μὲν τῆς πυραμίδος τεσσάρων, ἡ δὲ τοῦ ὀκταέδρου τριῶν, ἡ δὲ τοῦ κύβου δύο. ἡ μὲν ἄρα τῆς πυρα-

Thus, BF is the side of the cube.

And since AC is equal to CB, AB is thus double BC. And as AB (is) to BC, so the (square) on AB (is) to the (square) on BE [Prop. 6.8, Def. 5.9]. Thus, the (square) on AB is double the (square) on BE. And the square on the diameter of the sphere is also double the (square) on the side of the octagon [Prop. 13.14]. And AB is the diameter of the given sphere. Thus, BE is the side of the octagon.

So let AG be drawn from point A at right-angles to the straight-line AB. And let AG be made equal to AB. And let GC be joined. And let HK be drawn from H, perpendicular to AB. And since GA is double AC. For GA (is) equal to AB. And as GA (is) to AC, so HK (is) to KC [Prop. 6.4]. HK (is) thus also double KC. Thus, the (square) on HK is four times the (square) on KC. Thus, the (sum of the squares) on HK and KC, which is the (square) on HC [Prop. 1.47], is five times the (square) on KC. And HC (is) equal to CB. Thus, the (square) on BC (is) five times the (square) on CK. And since AB is double CB, of which AD is double DB, the remainder BD is thus double the remainder DC. BC (is) thus triple CD. The (square) on BC (is) thus nine times the (square) on CD. And the (square) on BC (is) five times the (square) on CK. Thus, the (square) on CK (is) greater than the (square) on CD. CK is thus greater than CD. Let CL be made equal to CK. And let LM be drawn from L at right-angles to AB. And let MB be joined. And since the (square) on BC is five times the (square) on CK, and AB is double BC, and KL double CK, the (square) on AB is thus five times the (square) on KL. And the square on the diameter of the sphere is also five times the (square) on the radius of the circle from which the icosahedron has been described [Prop. 13.16 corr.]. And AB is the diameter of the sphere. Thus, KL is the radius of the circle from which the icosahedron has been described. Thus, KL is (the side) of the hexagon (inscribed) in the aforementioned circle [Prop. 4.15 corr.]. And since the diameter of the sphere is composed of (the side) of the hexagon, and two of (the sides) of the decagon, inscribed in the aforementioned circle, and AB is the diameter of the sphere, and KL the side of the hexagon, and AK (is) equal to LB, thus AK and LB are each sides of the decagon inscribed in the circle from which the icosahedron has been described. And since LB is (the side) of the decagon. And ML (is the side) of the hexagon—for (it is) equal to KL, since (it is) also (equal) to HK, for they are equally far from the center. And HK and KL are each double KC. MB is thus (the side) of the pentagon (inscribed in the circle) [Props. 13.10, 1.47]. And (the side) of the pentagon is (the side) of the icosahedron [Prop. 13.16]. Thus, MB is (the side) of the icosahedron.

And since FB is the side of the cube, let it be cut in extreme and mean ratio at N, and let NB be the greater piece. Thus, NB is the side of the dodecahedron [Prop. 13.17 corr.].

μίδος πλευρὰ τῆς μὲν τοῦ ὀκταέδρου πλευρᾶξ δυνάμει ἐστὶν ἐπίτριτος, τῆς δὲ τοῦ κύβου δυνάμει διπλῆ, ἡ δὲ τοῦ ὀκταέδρου τῆς τοῦ κύβου δυνάμει ἡμιολία. αἱ μὲν οὖν εἰρημέναι τῶν τριῶν σχημάτων πλευραί, λέγω δὴ πυραμίδος καὶ ὀκταέδρου καὶ κύβου, πρὸς ἀλλήλας εἰσὶν ἐν λόγοις ὀρητοῖς. αἱ δὲ λοιπαὶ δύο, λέγω δὴ ἥ τε τοῦ εἰκοσαέδρου καὶ ἡ τοῦ δωδεκαέδρου, οὐδέτε πρὸς ἀλλήλας οὐδέτε πρὸς τὰξ προειρημένας εἰσὶν ἐν λόγοις ὀρητοῖς· ἀλογοι γάρ εἰσιν, ἡ μὲν ἐλάττων, ἡ δὲ ἀποτομή.

Ὅτι μείζων ἐστὶν ἡ τοῦ εἰκοσαέδρου πλευρὰ ἡ MB τῆς τοῦ δωδεκαέδρου τῆς NB, δείχομεν οὕτως.

Ἐπεὶ γὰρ ἰσογώνιόν ἐστι τὸ $Z\Delta B$ τρίγωνον τῷ ZAB τριγώνῳ, ἀνάλογόν ἐστιν ὡς ἡ ΔB πρὸς τὴν BZ, οὕτως ἡ BZ πρὸς τὴν BA. καὶ ἐπεὶ τρεῖς εὐθεῖαι ἀνάλογόν εἰσιν, ἐστιν ὡς ἡ πρώτη πρὸς τὴν τρίτην, οὕτως τὸ ἀπὸ τῆς πρώτης πρὸς τὸ ἀπὸ τῆς δευτέρας· ἐστιν ἄρα ὡς ἡ ΔB πρὸς τὴν BA, οὕτως τὸ ἀπὸ τῆς ΔB πρὸς τὸ ἀπὸ τῆς BZ· ἀνάπαλιν ἄρα ὡς ἡ AB πρὸς τὴν $B\Delta$, οὕτως τὸ ἀπὸ τῆς ZB πρὸς τὸ ἀπὸ τῆς $B\Delta$. τριπλῆ δὲ ἡ AB τῆς $B\Delta$· τριπλάσιον ἄρα τὸ ἀπὸ τῆς ZB τοῦ ἀπὸ τῆς $B\Delta$. ἐστι δὲ καὶ τὸ ἀπὸ τῆς $A\Delta$ τοῦ ἀπὸ τῆς ΔB τετραπλάσιον· διπλῆ γὰρ ἡ $A\Delta$ τῆς ΔB· μεῖζον ἄρα τὸ ἀπὸ τῆς $A\Delta$ τοῦ ἀπὸ τῆς ZB· μείζων ἄρα ἡ $A\Delta$ τῆς ZB· πολλῷ ἄρα ἡ $A\Lambda$ τῆς ZB μείζων ἐστίν. καὶ τῆς μὲν $A\Lambda$ὀάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμά ἐστιν ἡ $K\Lambda$, ἐπειδήπερ ἡ μὲν ΛK ἑξαγώνου ἐστίν, ἡ δὲ KA δεκαγώνου· τῆς δὲ ZBὀάκρον καὶ μέσον λόγον τεμνομένης τὸ μεῖζον τμῆμά ἐστιν ἡ NB· μείζων ἄρα ἡ $K\Lambda$ τῆς NB. ὁίση δὲ ἡ $K\Lambda$ τῇ ΛM· μείζων ἄρα ἡ ΛM τῆς NB [τῆς δὲ ΛM μείζων ἐστὶν ἡ MB]. πολλῷ ἄρα ἡ MB πλευρὰ οὐδῦσα τοῦ εἰκοσαέδρου μείζων ἐστὶ τῆς NB πλευρᾶξ οὐδύσης τοῦ δωδεκαέδρου· ὅπερ ἔδει δεῖχαι.

And since the (square) on the diameter of the sphere was shown (to be) one and a half times the square on the side, AF, of the pyramid, and twice the square on (the side), BE, of the octagon, and three times the square on (the side), FB, of the cube, thus, of whatever (parts) the (square) on the diameter of the sphere (makes) six, of such (parts) the (square) on (the side) of the pyramid (makes) four, and (the square) on (the side) of the octagon three, and (the square) on (the side) of the cube two. Thus, the (square) on the side of the pyramid is one and a third times the square on the side of the octagon, and double the square on (the side) of the cube. And the (square) on (the side) of the octahedron is one and a half times the square on (the side) of the cube. Therefore, the aforementioned sides of the three figures—I mean, of the pyramid, and of the octahedron, and of the cube—are in rational ratios to one another. And (the sides of) the remaining two (figures)—I mean, of the icosahedron, and of the dodecahedron—are neither in rational ratios to one another, nor to the (sides) of the aforementioned (three figures). For they are irrational (straight-lines): (namely), a minor [Prop. 13.16], and an apotome [Prop. 13.17].

(And), we can show that the side, MB, of the icosahedron is greater that the (side), NB, or the dodecahedron, as follows.

For, since triangle FDB is equiangular to triangle FAB [Prop. 6.8], proportionally, as DB is to BF, so BF (is) to BA [Prop. 6.4]. And since three straight-lines are (continually) proportional, as the first (is) to the third, so the (square) on the first (is) to the (square) on the second [Def. 5.9, Prop. 6.20 corr.]. Thus, as DB is to BA, so the (square) on DB (is) to the (square) on BF. Thus, inversely, as AB (is) to BD, so the (square) on FB (is) to the (square) on BD. And AB (is) triple BD. Thus, the (square) on FB (is) three times the (square) on BD. And the (square) on AD is also four times the (square) on DB. For AD (is) double DB. Thus, the (square) on AD (is) greater than the (square) on FB. Thus, AD (is) greater than FB. Thus, AL is much greater than FB. And KL is the greater piece of AL, which is cut in extreme and mean ratio—inasmuch as LK is (the side) of the hexagon, and KA (the side) of the decagon [Prop. 13.9]. And NB is the greater piece of FB, which is cut in extreme and mean ratio. Thus, KL (is) greater than NB. And KL (is) equal to LM. Thus, LM (is) greater than NB [and MB is greater than LM]. Thus, MB, which is (the side) of the icosahedron, is much greater than NB, which is (the side) of the dodecahedron. (Which is) the very thing it was required to show.

† If the radius of the given sphere is unity then the sides of the pyramid (*i.e.*, tetrahedron), octahedron, cube, icosahedron, and dodecahedron, respectively, satisfy the following inequality: $\sqrt{8/3} > \sqrt{2} > \sqrt{4/3} > (1/\sqrt{5})\sqrt{10 - 2\sqrt{5}} > (1/3)(\sqrt{15} - \sqrt{3})$.

Λέγω δή, ὅτι παρὰ τὰ εἰρημένα πέντε σθήματα οὐ συστα-

So, I say that, beside the five aforementioned figures, no

θήσεται ὅτερον σχῆμα περιεχόμενον ὑπὸ ἰσοπλεύρων τε καὶ ἰσογωνίωνὅσων ἀλλήλοις.

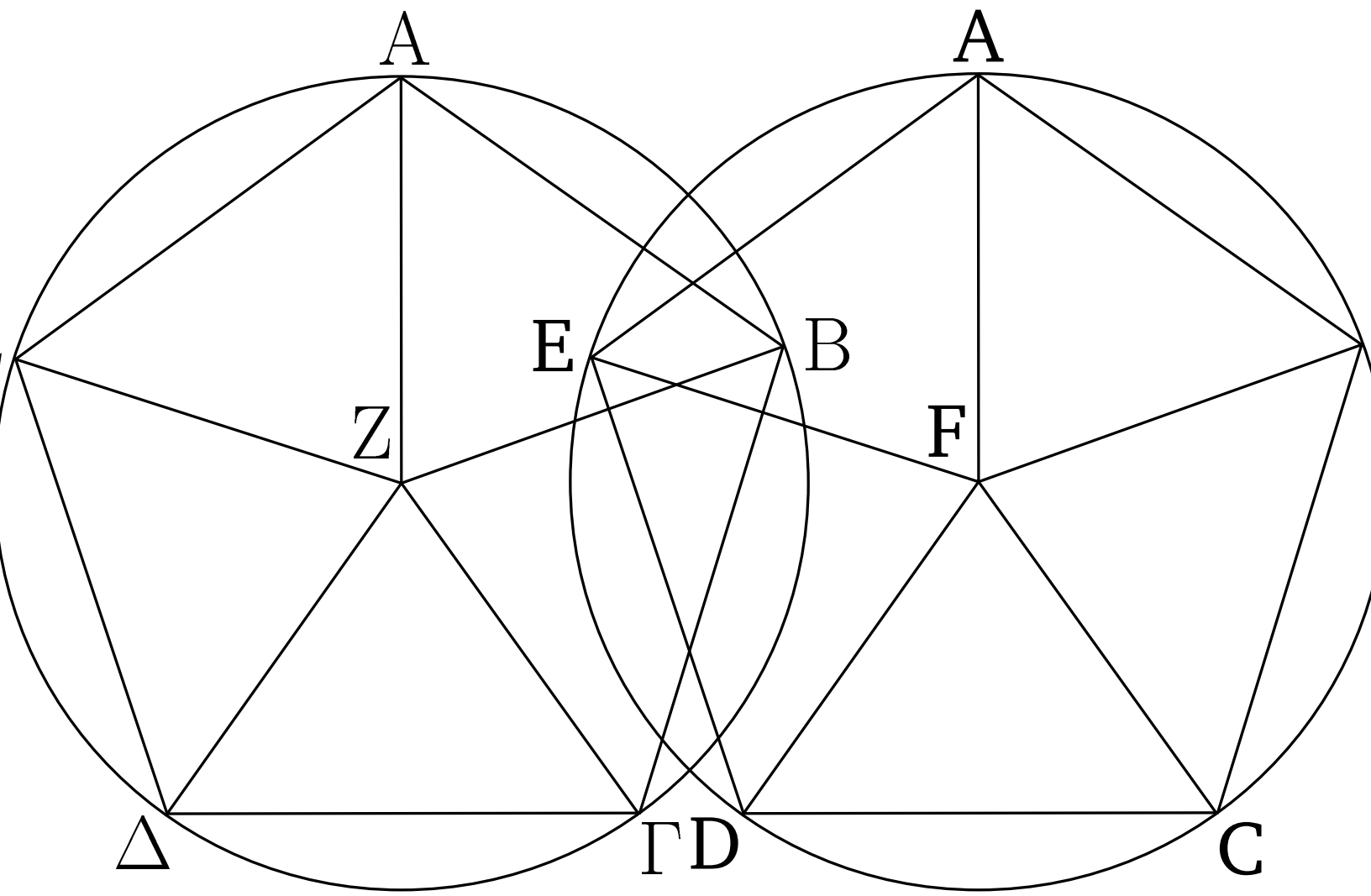
Ὑπὸ μὲν γὰρ δύο τριγώνωνὃ ὅλως ἐπιπέδων στερεὰ γωνία οὐ συνίσταται. ὑπὸ δὲ τριῶν τριγώνων ἡ τῆς πυραμίδος, ὑπὸ δὲ τεσσάρων ἡ τοῦ ὀκταέδρου, ὑπὸ δὲ πέντε ἡ τοῦ εἰκοσαέδρου· ὑπὸ δὲ ὁὲχ τριγώνων ἰσοπλεύρων τε καὶ ἰσογωνίων πρὸς ἑνὶ σημείῳ συνισταμένων οὐκὸέσται στερεὰ γωνία· οὐὁύσης γὰρ τῆς τοῦ ἰσοπλεύρου τριγώνου γωνίας διμοίρου ὀρθῆςὁέσονται αἱ ὁὲχ τέσσαρσιν ὀρθαῖξ ἴσαι· ὅπερ ἀδύνατον· ὁάπασα γὰρ στερεὰ γωνία ὑπὸ ἐλασσόνωνὃῆ τεσσάρων ὀρθῶν περέθεται. διὰ τὰ αὐτὰ δὴ οὐδὲ ὑπὸ πλειόνωνὃῆ ὁὲχ γωνιῶν ἐπιπέδων στερεὰ γωνία συνίσταται. ὑπὸ δὲ τετραγώνων τριῶν ἡ τοῦ κύβου γωνία περιέθεται· ὑπὸ δὲ τεσσάρων ἀδύνατον· ὁέσονται γὰρ πάλιν τέσσαρες ὀρθαί. ὑπὸ δὲ πενταγώνων ἰσοπλεύρων καὶ ἰσογωνίων, ὑπὸ μὲν τριῶν ἡ τοῦ δωδεκαέδρου· ὑπὸ δὲ τεσσάρων ἀδύνατον· οὐὁύσης γὰρ τῆς τοῦ πενταγώνου ἰσοπλεύρου γωνίας ὀρθῆς καὶ πέμπτου, ὁέσονται αἱ τέσσαρες γωνίαι τεσσάρων ὀρθῶν μείζους· ὅπερ ἀδύνατον. οὐδὲ μὴν ὑπὸ πολυγώνων ἑτέρων σχημάτων περισθεθήσεται στερεὰ γωνία διὰ τὸ αὐτὸ ὁάτοπον.

Οὐκ ἄρα παρὰ τὰ εἰρημένα πέντε σθήματα ὁέτερον σχῆμα στερεὸν συσταθήσεται ὑπὸ ἰσοπλεύρων τε καὶ ἰσογωνίων περιεχόμενον· ὅπερ ἔδει δεῖξαι.

other (solid) figure can be constructed (which is) contained by equilateral and equiangular (planes), equal to one another.

For a solid angle cannot be constructed from two triangles, or indeed (two) planes (of any sort) [Def. 11.11]. And (the solid angle) of the pyramid (is constructed) from three (equiangular) triangles, and (that) of the octahedron from four (triangles), and (that) of the icosahedron from (five) triangles. And a solid angle cannot be (made) from six equilateral and equiangular triangles set up together at one point. For, since the angles of a equilateral triangle are (each) two-thirds of a right-angle, the (sum of the) six (plane) angles (containing the solid angle) will be four right-angles. The very thing (is) impossible. For every solid angle is contained by (plane angles whose sum is) less than four right-angles [Prop. 11.21]. So, for the same (reasons), a solid angle cannot be constructed from more than six plane angles (equal to two-thirds of a right-angle) either. And the (solid) angle of a cube is contained by three squares. And (a solid angle contained) by four (squares is) impossible. For, again, the (sum of the plane angles containing the solid angle) will be four right-angles. And (the solid angle) of a dodecahedron (is contained) by three equilateral and equiangular pentagons. And (a solid angle contained) by four (equiangular pentagons is) impossible. For, the angle of an equilateral pentagon being one and one-fifth of right-angle, four (such) angles will be greater (in sum) than four right-angles. The very thing (is) impossible. And, on account of the same absurdity, a solid angle cannot be constructed from any other (equiangular) polygonal figures either.

Thus, beside the five aforementioned figures, no other solid figure can be constructed (which is) contained by equilateral and equiangular (planes). (Which is) the very thing it was required to show.



Λῆμμα.

Ὅτι δὲ ἡ τοῦ ἰσοπλεύρου καὶ ἰσογωνίου πενταγώνου γωνία ὀρθή ἐστι καὶ πέμπτου, οὕτω δεικτέον.

Ἔστω γὰρ πεντάγωνον ἰσόπλευρον καὶ ἰσογώνιον τὸ ΑΒΓΔΕ, καὶ περιγεγράφθω περὶ αὐτὸ κύκλος ὁ ΑΒΓΔΕ, καὶ εἰλήφθω αὐτοῦ τὸ κέντρον τὸ Ζ, καὶ ἐπεζεύχθωσαν αἱ ΖΑ, ΖΒ, ΖΓ, ΖΔ, ΖΕ. δίχα ἄρα τέμνουσι τὰξ πρὸς τοῖς Α, Β, Γ, Δ, Ε τοῦ πενταγώνου γωνίας. καὶ ἐπεὶ αἱ πρὸς τῷ Ζ πέντε γωνίαι τέσσαρσιν ὀρθαῖξ ἴσαι εἰσὶ καὶ εἰσὶν ἴσαι, μία ἄρα αὐτῶν, ὡξ ἡ ὑπὸ ΑΖΒ, μιᾶς ὀρθῆς ἐστι παρὰ πέμπτον· λοιπαὶ ἄρα αἱ ὑπὸ ΖΑΒ, ΑΒΖ μιᾶς εἰσιν ὀρθῆς καὶ πέμπτου. ὁίση δὲ ἡ ὑπὸ ΖΑΒ τῇ ὑπὸ ΖΒΓ· καὶ ὅλη ἄρα ἡ ὑπὸ ΑΒΓ τοῦ πενταγώνου γωνία μιᾶς ἐστιν ὀρθῆς καὶ πέμπτου· ὅπερ ἔδει δεῖχαι.

Lemma

It can be shown that the angle of an equilateral and equiangular pentagon is one and one-fifth of a right-angle, as follows.

For let $ABCDE$ be an equilateral and equiangular pentagon, and let the circle $ABCDE$ be circumscribed about it [Prop. 4.14]. And let its center, F, be found [Prop. 3.1]. And let FA, FB, FC, FD, and FE be joined. Thus, they cut the angles of the pentagon in half at (points) A, B, C, D, and E [Prop. 1.4]. And since the five angles at F are equal (in sum) to four right-angles, and are also equal (to one another), (any) one of them, like AFB, is thus one less a fifth of a right-angle. Thus, the (sum of the) remaining (angles in triangle ABF), FAB and ABF, is one plus a fifth of a right-angle [Prop. 1.32]. And FAB (is) equal to FBC. Thus, the whole angle, ABC, of the pentagon is also one and one-fifth of a right-angle. (Which is) the very thing it was required to show.

GREEK-ENGLISH LEXICON

ABBREVIATIONS: *act* - active; *adj* - adjective; *adv* - adverb; *conj* - conjunction; *fut* - future; *gen* - genitive; *imperat* - imperative; *impf* - imperfect; *ind* - indeclinable; *indic* - indicative; *intr* - intransitive; *mid* - middle; *neut* - neuter; *no* - noun; *par* - particle; *part* - participle; *pass* - passive; *perf* - perfect; *pre* - preposition; *pres* - present; *pro* - pronoun; *sg* - singular; *tr* - transitive; *vb* - verb.

ἄγω, *ἄχω*, *ἤγαγον*, *-ἦθα*, *ἦγμαι*, *ἤχθην* : *vb*, lead, draw (a line).

ἀδύνατος -ον : *adj*, impossible.

ἀεί : *adv*, always, for ever.

αἱρέω, *αἱρήσω*, *ε[˘]λον*, *ἥρηκα*, *ἥρημαι*, *ἡρέθην* : *vb*, grasp.

ἀιτέω, *αἰτήσω*, *ἤτησα*, *ἤτηκα*, *ἤτημαι*, *ἠτήθη* : *vb*, postulate.

αἴτημα -ατος, τό : *no*, postulate.

ἀκόλουθος -ον : *adj*, analogous, consequent on, in conformity with.

ἄκρος -α -ον : *adj*, outermost, end, extreme.

ἀλλά : *conj*, but, otherwise.

ἄλογος -ον : *adj*, irrational.

ἅμα : *adv*, at once, at the same time, together.

ἀμβλυγώνιος -ον : *adj*, obtuse-angled; *τὸ ἀμβλυγώνιον*, *no*, obtuse angle.

ἀμβλύς -εῖα -ύ : *adj*, obtuse.

ἀμφότερος -α -ον : *pro*, both.

ἀναγράφω : *vb*, describe (a figure); see *γράφω*.

ἀναλογία, ἡ : *no*, proportion, (geometric) progression.

ἀνάλογος -ον : *adj*, proportional.

ἀνάπαλιν : *adv*, inverse(ly).

ἀναπληρόω : *vb*, fill up.

ἀναστρέφω : *vb*, turn upside down, convert (ratio); see *στρέφω*.

ἀναστροφή, ἡ : *no*, turning upside down, conversion (of ratio).

ἀνθυφαιρέω : *vb*, take away in turn; see *αἱρέω*.

ἀνίστημι : *vb*, set up; see *ἵστημι*.

ἄνισος -ον : *adj*, unequal, uneven.

ἀντιπάσθω : *vb*, be reciprocally proportional; see *πάσθω*.

ἄχων -ονος, ὁ : *vb*, axis.

ἅπαξ : *adv*, once.

ἅπας, ἅπασα, ἅπαν : *adj*, quite all, the whole.

ἄπειρος -ον : *adj*, infinite.

ἀπεναντίον : *ind*, opposite.

ἀπέθω : *vb*, be far from, be away from; see *ἔθω*.

ἀπλατής -ές : *adj*, without breadth.

ἀπόδειχις -εως, ἡ : *no*, proof.

ἀποκαθίστημι : *vb*, re-establish, restore; see *ἵστημι*.

ἀπολαμβάνω : *vb*, take from, subtract from, cut off from; see *λαμβάνω*.

ἀποτέμνω : *vb*, cut off, subtend.

ἀπότμημα -ατος, τό : *no*, piece cut off, segment.

ἀποτομή, ἡ : *vb*, piece cut off, apotome.

ἅπτω, *ἅψω*, *ἧψα*, —, *ἧμμαι*, — : *vb*, touch, join, meet.

ἀπώτερος -α -ον : *adj*, further off.

ἄρα : *par*, thus, as it seems (inferential).

ἀριθμός, ὁ : *no*, number.

ἀρτιάκις : *adv*, an even number of times.

αρτιόπλευρος -ον : *adj*, having a even number of sides.

ἄρθω, *ἄρχω*, *ἦρχα*, *ἦρθα*, *ἦργμαι*, *ἤρθθην* : *vb*, rule; *mid.*, begin.

ἀσύμμετρος -ον : *adj*, incommensurable.

ἀσύμπτωτος -ον : *adj*, not touching, not meeting.

ἄρτιος -α -ον : *adj*, even, perfect.

ἄτμητος -ον : *adj*, uncut.

ἀτόπος -ον : *adj*, absurd, paradoxical.

αὐτόθεν : *adv*, immediately, obviously.

ἀφαίρεω : *vb*, take from, subtract from, cut off from; see *αἱρέω*.

ἁφή, ἡ : *no*, point of contact.

βάθος -εος, τό : *no*, depth, height.

βαίνω, *-βήσομαι*, *-έβην*, *βέβηκα*, —, — : *vb*, walk; *perf*, stand (of angle).

βάλλω, *βαλῶ*, *ἔβαλον*, *βέβληκα*, *βέβλημαι*, *ἐβλήθην* : *vb*, throw.

βάσις -εως, ἡ : *no*, base (of a triangle).

γάρ : *conj*, for (explanatory).

γι[γ]νομαι, *γενήσομαι*, *ἐγενόμην*, *γέγονα*, *γεγένημαι*, — : *vb*, happen, become.

γνώμων -ονος, ἡ : *no*, gnomon.

γραμμή, ἡ : *no*, line.

γράφω, *γράψω*, *ἔγρα[ψφ]α*, *γέγραφα*, *γέγραμμαι*, *ἐραψάμην* : *vb*, draw (a figure).

γωνία, ἡ : *no*, angle.

δεῖ : *vb*, be necessary; *δεῖ*, it is necessary; *ἔδει*, it was necassary; *δέον*, being necessary.

δείκνυμι, *δείχω*, *ἔδειχα*, *δέδειθα*, *δέδειγμαι*, *ἐδείθθην* : *vb*, show, demonstrate.

δεικτέον : *ind*, one must show.

δεῖχις -εως, ἡ : *no*, proof.

δεκαγώνος -ον : *adj*, ten-sided; *τὸ δεκαγώνον*, *no*, decagon.

δέθομαι, *δέχομαι*, *ἐδεχάμην*, —, *δέδεγμαι*, *ἐδέθθην* : *vb*, receive, accept.

δή : *conj*, so (explanatory).

δηλαδή : *ind*, quite clear, manifest.

δῆλος -η -ον : *adj*, clear.

δηλονότι : *adv*, manifestly.

διάγω : *vb*, carry over, draw through, draw across; see *ἄγω*.

διαγώνιος -ον : *adj*, diagonal.

διαλείπω : *vb*, leave an interval between.

διάμετρος -ον : *adj*, diametrical; *ἡ διάμετρος*, *no*, diameter, diagonal.

διαίρεσις -εως, ἡ : *no*, division, separation.

διαιρέω : *vb*, divide (in two); *διαρεθέντος -η -ον*, *adj*, separated (ratio); see *αἱρέω*.

διάστημα -ατος, τό : *no*, radius.

διαφέρω : *vb*, differ; see *φέρω*.

δίδωμι, *δώσω*, *ἔδωκα*, *δέδωκα*, *δέδομαι*, *ἐδόθην* : *vb*, give.

διμοίρος -ον : *adj*, two-thirds.

διπλασιάζω : *vb*, double.

διπλάσιος -α -ον : *adj*, double, twofold.

διπλασίων -ον : *adj*, double, twofold.

διπλοῦς -ῆ -οῦν : *adj*, double.

δίς : *adv*, twice.

δίχα : *adv*, in two, in half.

διχορομία, ἡ : *no*, point of bisection.

δυάς -άδος, ἡ : *no*, the number two, dyad.

δύναμαι : *vb*, be able, be capable, generate, square, be when squared; *δυναμένη, ἡ, no*, square-root (of area)—*i.e.*, straight-line whose square is equal to a given area.

δύναμις -εως, ἡ : *no*, power (usually 2nd power when used in mathematical sence, hence), square.

δυνατός -ή -όν : *adj*, possible.

δωδεκάεδρος -ον : *adj*, twelve-sided.

ἑαυτοῦ -ῆς -οῦ : *adj*, of him/her/it/self, his/her/its/own.

ἐγγίων -ον : *adj*, nearer, nearest.

ἐγγράφω : *vb*, inscribe; see *γράφω*.

εἶδος -εος, τό : *no*, figure, form, shape.

εἰκοσάεδρος -ον : *adj*, twenty-sided.

εἶρω/λέγω, ἐρῶ/ερέω, εἶπον, εἴρηκα, εἴρημαι, ἐρρήθην : *vb*, say, speak; *per pass part*, *εἰρημένος -η -ον*, *adj*, said, aforementioned.

εἴτε . . . εἴτε : *ind*, either . . . or.

ἕκαστος -η -ον : *pro*, each, every one.

ἑκατέρος -α -ον : *pro*, each (of two).

ἐκβάλλω : *vb*, produce (a line); see *βάλλω*.

ἐκθέω : *vb*, set out.

ἔκκειμαι : *vb*, be set out, be taken; see *κεῖμαι*.

ἐκτίθημι : *vb*, set out; see *τίθημι*.

ἐκτός : *pre + gen*, outside, external.

ἐλάσσων/ἐλάττων -ον : *adj*, less, lesser.

ἐλάθιστος -η -ον : *adj*, least.

ἐλλείπω : *vb*, be less than, fall short of.

ἐμπίπτω : *vb*, meet (of lines), fall on; see *πίπτω*.

ἔμπροσθεν : *adv*, in front.

ἐναλλάξ : *adv*, alternate(ly).

ἐναρμόζω : *vb*, insert; *perf indic pass 3rd sg*, *ἐνήρμοσται*.

ἐνδέθομαι : *vb*, admit, allow.

ἕνεκεν : *ind*, on account of, for the sake of.

ἐννεαπλάσιος -α -ον : *adj*, nine-fold, nine-times.

ἔννοια, ἡ : *no*, notion.

ἐνπεριέθω : *vb*, encompass; see *ἔθω*.

ἐνπίπτω : see *ἐμπίπτω*.

ἐντός : *pre + gen*, inside, interior, within, internal.

ἑξάγωνος -ον : *adj*, hexagonal; *τὸ ἑξάγωνον*, *no*, hexagon.

ἑξαπλάσιος -α -ον : *adj*, sixfold.

ἑξῆς : *adv*, in order, successively, consecutively.

ἔξωθεν : *adv*, outside, extrinsic.

ἐπάνω : *adv*, above.

ἐπαφή, ἡ : *no*, point of contact.

ἐπεί : *conj*, since (causal).

ἐπειδήπερ : *ind*, inasmuch as, seeing that.

ἐπιζεύγνῡμι, ἐπιζεύχω, ἐπέζευχα, —, ἐπέζευγμαι, ἐπεζεύθθην : *vb*, join (by a line).

ἐπιλογίζομαι : *vb*, conclude.

ἐπινοέω : *vb*, think of, contrive.

ἐπιπέδος -ον : *adj*, level, flat, plane; *τὸ ἐπιπέδον*, *no*, plane.

ἐπισκέπτομαι : *vb*, investigate.

ἐπίσκεψις -εως, ἡ : *no*, inspection, investigation.

ἐπιτάσσω : *vb*, put upon, enjoin; *τὸ ἐπιταθθέν*, *no*, the (thing) prescribed; see *τάσσω*.

ἐπίτριτος -ον : *adj*, one and a third times.

ἐπιφάνεια, ἡ : *no*, surface.

ἕπομαι : *vb*, follow.

ἔρθομαι, ἐλεύσομαι, ἦλθον, ἐλήλυθα, —, — : *vb*, come, go.

ἔσθατος -η -ον : *adj*, outermost, uttermost, last.

ἑτερόμηκης -εξ : *adj*, oblong; *τὸ ἑτερόμηκεξ*, *no*, rectangle.

ἕτερος -α -ον : *adj*, other (of two).

ἔτι : *par*, yet, still, besides.

εὐθύγραμμος -ον : *adj*, rectilinear; *τὸ εὐθύγραμμον*, *no*, rectilinear figure.

εὐθύς -εῖα -ύ : *adj*, straight; *ἡ εὐθεῖα*, *no*, straight-line; *ἐπ᾽ εὐθεῖαξ*, in a straight-line, straight-on.

εὑρίσκω, εὑρήσκω, ηὗρον, εὕρεκα, εὕρηκα, εὕρημαι, εὑρέθην : *vb*, find.

ἐφάπτω : *vb*, bind to; *mid*, touch; *ἡ ἐφαπτομένη*, *no*, tangent; see *ἅπτω*.

ἐφαρμόζω, ἐφαρμόσω, ἐφήρμοσα, ἐφήμοκα, ἐφήμοσμαι, ἐφημόσθην : *vb*, coincide; *pass*, be applied.

ἐφεχῆς : *adv*, in order, adjacent.

ἐφίστημι : *vb*, set, stand, place upon; see *ἵστημι*.

ἔθω, ἕχω, ἔσθον, ἔσθηκα, -έσθημαι, — : *vb*, have.

ἡγέομαι, ἡγήσομαι, ἡγησάμην, ἥγημαι, —, ἡγήθην : *vb*, lead.

ἤδη : *ind*, already, now.

ἥκω, ἵκω, —, —, —, — : *vb*, have come, be present.

ἡμικύκλιον, τό : *no*, semi-circle.

ἡμιόλιος -α -ον : *adj*, containing one and a half, one and a half times.

ἥμισυς -εια -υ : *adj*, half.

ἤπερ = ἤ + περ : *conj*, than, than indeed.

ἤτοι . . . ἤ : *par*, surely, either . . . or; in fact, either . . . or.

θέσις -εως, ἡ : *no*, placing, setting, position.

θεωρημα -ατος, τό : *no*, theorem.

ἴδιος -α -ον : *adj*, one's own.

ἰσάκις : *adv*, the same number of times; *ἰσάκις πολλαπλάσια*, the same multiples, equal multiples.

ἰσογώνιος -ον : *adj*, equiangular.

ἰσόπλευρος -ον : *adj*, equilateral.

ἰσοπληθής -ές : *adj*, equal in number.

ἴσος -η -ον : *adj*, equal; *ἐχ ἴσου*, equally, evenly.

ἰσοσκελής -ές : *adj*, isosceles.

ἵστημι, στήσω, ἔστησα, —, —, *ἑστάθην* : *vb tr*, stand (something).

ἵστημι, στήσω, ἔστην, ἕστηκα, ἕσταμαι, ἑσταθην : *vb intr*, stand up (oneself); Note: perfect *I have stood up* can be taken to mean present *I am standing*.

ἰσοϋψής -ές : *adj*, of equal height.

καθάπερ : *ind*, according as, just as.

κάθετος -ον : *adj*, perpendicular.

καθόλου : *adv*, on the whole, in general.

καλέω : *vb*, call.

κἄκεινος = *καὶ ἐκεῖνος* .

κἄν = *καὶ ἄν* : *ind*, even if, and if.

καταγραφή, ἡ : *no*, diagram, figure.

καταγράφω : *vb*, describe/draw, inscribe (a figure); see *γράφω*.

κατακολουθέω : *vb*, follow after.

καταλείπω : *vb*, leave behind; see *λείπω*; *τὰ καταλειπόμενα*, *no*, remainder.

κατάλληλος -ον : *adj*, in succession, in corresponding order.

καταμετρέω : *vb*, measure (exactly).

καταντάω : *vb*, come to, arrive at.

κατασκενάζω : *vb*, furnish, construct.

κεῖμαι, κεῖσομαι, —, —, —, — : *vb*, have been placed, lie, be made; see *τίθημι*.

κέντρον, τό : *no*, center.

κλάω : *vb*, break off, inflect.

κλίνω, κλίνω, ἔκλινα, κέκλικα, κέκλιμαι, ἐκλίθην : *vb*, lean, incline.

κλίσις -εως, ἡ : *no*, inclination, bending.

κοῖλος -η -ον : *adj*, hollow, concave.

κορυφή, ἡ : *no*, top, summit, apex; *κατὰ κορυφήν*, vertically opposite (of angles).

κρίνω, κρινῶ, ἔκρῖνα, κέκρικα, κέκριμαι, ἐκρίθην : *vb*, judge.

κύβος, ὁ : *no*, cube.

κύκλος, ὁ : *no*, circle.

κύλινδρος, ὁ : *no*, cylinder.

κυρτός -ή -όν : *adj*, convex.

κῶνος, ὁ : *no*, cone.

λαμβάνω, λήψομαι, ἔλαβον, εἴληφα εἴλημμαι, ἐλήφθην : *vb*, take.

λέγω : *vb*, say; *pres pass part*, *λεγόμενος -η -ον*, *adj*, so-called; see *εἴρω*.

λείπω, λείψω, ἔλιπον, λέλοιπα, λέλειμμαι, ἐλείφθην : *vb*, leave, leave behind.

λημμάτιον, τό : *no*, diminutive of *λῆμμα*.

λῆμμα -ατος, τό : *no*, lemma.

λῆψις -εως, ἡ : *no*, taking, catching.

λόγος, ὁ : *no*, ratio, proportion, argument.

λοιπός -ή -όν : *adj*, remaining.

μανθάνω, μαθήσομαι, ἔμαθον, μεμάθηκα, —, — : *vb*, learn.

μέγεθος -εος, τό : *no*, magnitude, size.

μείζων -ον : *adj*, greater.

μένω, μενῶ, ἔμεινα, μεμένηκα, —, — : *vb*, stay, remain.

μέρος -ους, τό : *no*, part, direction, side.

μέσος -η -ον : *adj*, middle, mean, medial; *ἐκ δύο μέσων*, bimedial.

μεταλαμβάνω : *vb*, take up.

μεταχύ : *adv*, between.

μετέωρος -ον : *adj*, raised off the ground.

μετρέω : *vb*, measure.

μέτρον, τό : *no*, measure.

μηδείς, μηδεμία, μηδέν : *adj*, not even one, (neut.) nothing.

μηδέποτε : *adv*, never.

μηδέτερος -α -ον : *pro*, neither (of two).

μῆκος -εος, τό : *no*, length.

μήν : *par*, truely, indeed.

μονάς -άδος, ἡ : *no*, unit, unity.

μοναθός -ή -όν : *adj*, unique.

μοναθῶς : *adv*, uniquely.

μόνος -η -ον : *adj*, alone.

νοέω, —, *νόησα, νενόηκα, νενόημαι, ἐνοήθην* : *vb*, apprehend, conceive.

οἷος -α -ον : *pre*, such as, of what sort.

ὀκτάεδρος -ον : *adj*, eight-sided.

ὅλος -η -ον : *adj*, whole.

ὁμογενής -ές : *adj*, of the same kind.

ὅμοιος -α -ον : *adj*, similar.

ὁμοιοπληθής -ές : *adj*, similar in number.

ὁμοιοταγής -ές : *adj*, similarly arranged.

ὁμοιότης -ητος, ἡ : *no* similarity.

ὁμοίως : *adv*, similarly.

ὁμόλογος -ον : *adj*, corresponding, homologous.

ὁμοταγής -ές : *adj*, ranged in the same row or line.

ὁμώνυμος -ον : *adj*, having the same name.

ὄνομα -ατος, τό : *no*, name; *ἐκ δύο ὀνομάτων*, binomial.

ὀχυγώνιος -ον : *adj*, acute-angled; *τὸ ὀχυγώνιον*, *no*, acute angle.

ὀχύς -εῖα -ύ : *adj*, acute.

ὁποιοσοῦν = *ὁποῖος -α -ον* + *οὖν* : *adj*, of whatever kind, any kind whatsoever.

ὁπόσος -η -ον : *pro*, as many, as many as.

ὁποσοσδηποτοῦν = *ὁπόσος -η -ον* + *δή* + *ποτέ* + *οὖν* : *adj*, of whatever number, any number whatsoever.

ὁποσοσοῦν = *ὁπόσος -η -ον* + *οὖν* : *adj*, of whatever number, any number whatsoever.

ὁπότερος -α -ον : *pro*, either (of two), which (of two).

ὀρθογώνιον, τό : *no*, rectangle, right-angle.

ὀρθός -ή -όν : *adj*, straight, right-angled, perpendicular; *πρὸς ὀρθὰς γωνίας*, at right-angles.

ὅρος, ὁ : *no*, boundary, definition, term (of a ratio).

ὁσαδηποτοῦν = *ὅσα* + *δή* + *ποτέ* + *οὖν* : *ind*, any number whatsoever.

ὁσάκις : *ind*, as many times as, as often as.

ὁσαπλάσιος -ον : *pro*, as many times as.

ὅσος -η -ον : *pro*, as many as.

ὁσπερ, ἥπερ, ὅπερ : *pro*, the very man who, the very thing which.

ὁστις, ἥτις, ὅ τι : *pro*, anyone who, anything which.

ὅταν : *adv*, when, whenever.

ὁτιοῦν : *ind*, whatsoever.

οὐδείς, οὐδεμία, οὐδέν : *pro*, not one, nothing.

οὐδέτερος -α -ον : *pro*, not either.

οὐθέτερος : see *οὐδέτερος*.

οὐθέν : *ind*, nothing.

οὖν : *adv*, therefore, in fact.

οὕτως : *adv*, thusly, in this case.

πάλιν : *adv*, back, again.

πάντως : *adv*, in all ways.

παρά : *prep + acc*, parallel to.

παραβάλλω : *vb*, apply (a figure); see *βάλλω*.

παραβολή, ἡ : *no*, application.

παράκειμαι : *vb*, lie beside, apply (a figure); see *κεῖμαι*.

παραλλάσσω, παραλλάχω, **—,** *παρήλλαθα,* **—, —** : *vb*, miss, fall awry.

παραλληλεπίπεδος, -ον : *adj*, with parallel surfaces; *τὸ παραλληλεπίπεδον, no*, parallelepiped.

παραλληλόγραμμος -ον : *adj*, bounded by parallel lines; *τὸ παραλληλόγραμμον, no*, parallelogram.

παράλληλος -ον : *adj*, parallel; *τὸ παράλληλον, no*, parallel, parallel-line.

παραπλήρωμα -ατος, τό : *no*, complement (of a parallelogram).

παρατέλυετος -ον : *adj*, penultimate.

παρέκ : *prep + gen*, except.

παρεμπίπτω : *vb*, insert; see *πίπτω*.

πάσθω, πείσομαι, ἔπαθον, πέπονθα, **—, —** : *vb*, suffer.

πεντάγωνος -ον : *adj*, pentagonal; *τὸ πεντάγωνον, no*, pentagon.

πενταπλάσιος -α -ον : *adj*, five-fold, five-times.

πεντεκαιδεκάγωνον, τό : *no*, fifteen-sided figure.

πεπερασμένος -η -ον : *adj*, finite, limited; see *περαίνω*.

περαίνω, περανῶ, ἐπέρανα, **—,** *πεπεράσμαι, ἐπερανάνθην* : *vb*, bring to end, finish, complete; *pass*, be finite.

πέραξ -ατος, τό : *no*, end, extremity.

περατόω, **—, —, —, —, —** : *vb*, bring to an end.

περιγράφω : *vb*, circumscribe; see *γράφω*.

περιέθω : *vb*, encompass, surround, contain, comprise; see *ἔθω*.

περιλαμβάνω : *vb*, enclose; see *λαμβάνω*.

περισσάκις : *adv*, an odd number of times.

περισσός -ή -όν : *adj*, odd.

περιφέρεια, ἡ : *no*, circumference.

περιφέρω : *vb*, carry round; see *φέρω*.

πηλικότης -ητος, ἡ : *no*, magnitude, size.

πίπτω, πεσοῦμαι, ἔπεσον, πέπτωκα, **—, —** : *vb*, fall.

πλάτος -εος, τό : *no*, breadth, width.

πλείων -ον : *adj*, more, several.

πλευρά, ἡ : *no*, side.

πλῆθος -εος, τό : *no*, great number, multitude, number.

πλήν : *adv & prep + gen*, more than.

ποιός -ά -όν : *adj*, of a certain nature, kind, quality, type.

πολλαπλασιάζω : *vb*, multiply.

πολλαπλασιασμός, ὁ : *no*, multiplication.

πολλαπλάσιον, τό : *no*, multiple.

πολύεδρος -ον : *adj*, polyhedral; *τό πολύεδρον, no*, polyhedron.

πολύγωνος -ον : *adj*, polygonal; *τό πολύγωνον, no*, polygon.

πολύπλευρος -ον : *adj*, multilateral.

πόρισμα -ατος, τό : *no*, corollary.

ποτέ : *ind*, at some time.

πρῖσμα -ατος, τό : *no*, prism.

προβαίνω : *vb*, step forward, advance.

προδείκνυμι : *vb*, show previously; see *δείκνυμι*.

προεκτίθημι : *vb*, set forth beforehand; see *τίθημι*.

προερέω : *vb*, say beforehand; *perf pass part*, *προειρημένος -η -ον, adj*, aforementioned; see *ἐῖρω*.

προσαναπληρόω : *vb*, fill up, complete.

προσαναγράφω : *vb*, complete (tracing of); see *γράφω*.

προσαρμόζω : *vb*, fit to, attach to.

προσεκβάλλω : *vb*, produce (a line); see *ἐκβάλλω*.

προσευρίσκω : *vb*, find besides, find; see *εὑρίσκω*.

προσλαμβάμω : *vb*, add.

πρόκειμαι : *vb*, set before, prescribe; see *κεῖμαι*.

πρόσκειμαι : *vb*, be laid on, have been added to; see *κεῖμαι*.

προσπίπτω : *vb*, fall on, fall toward, meet; see *πίπτω*.

προτασις -εως, ἡ : *no*, proposition.

προστάσσω : *vb*, prescribe, enjoin; *τὸ τροσταθθέν, no*, the thing prescribed; see *τάσσω*.

προστίθημι : *vb*, add; see *τίθημι*.

πρότερος -α -ον : *adj*, first (comparative), before, former.

προτίθημι : *vb*, assign; see *τίθημι*.

προθωρέω : *vb*, go/come forward, advance.

πρῶτος -α -ον : *adj*, first, prime.

πυραμίξ -ίδος, ἡ : *no*, pyramid.

ῥητός -ή -όν : *adj*, expressible, rational.

ῥομβοειδής -ές : *adj*, rhomboidal; *τὸ ῥομβοειδές, no*, romboid.

ῥόμβος, ὁ no, rhombus.

σημεῖον, τό : *no*, point.

σκαληνός -ή -όν : *adj*, scalene.

στερεός -ά -όν : *adj*, solid; *τὸ στερεόν, no*, solid, solid body.

στοιθεῖον, τό : *no*, element.

στρέφω, -στρέψω, ἔστρεψα, **—,** *ἐσταμμαι, ἐστάφην* : *vb*, turn.

σύγκειμαι : *vb*, lie together, be the sum of, be composed; *συγκείμενος -η -ον, adj*, composed (ratio), compounded; see *κεῖμαι*.

σύγκρίνω : *vb*, compare; see *κρίνω*.

συμβαίνω : *vb*, come to pass, happen, follow; see *βαίνω*.

συμβάλλω : *vb*, throw together, meet; see *βάλλω*.

σύμμετρος -ον : *adj*, commensurable.

σύμπαξ -αντος, ὁ : *no*, sum, whole.

συμπίπτω : *vb*, meet together (of lines); see *πίπτω.*

συμπληρόω : *vb*, complete (a figure), fill in.

συνάγω : *vb*, conclude, infer; see *ἄγω.*

συναμφότεροι -αι -α : *adj*, both together; *ὁ συναμφότεροξ*, *no*, sum (of two things).

συναποδείκνυμι : *no*, demonstrate together; see *δείκνυμι.*

συναφή, ἡ : *no*, point of junction.

σύνδυο, οἱ, αἱ, τά : *no*, two together, in pairs.

συνεθής -έξ : *adj*, continuous; *κατὰ τὸ συνεθέξ*, continuously.

σύνθεσις -εως, ἡ : *no*, putting together, composition.

σύνθετος -ον : *adj*, composite.

συ[ν]ίστημι : *vb*, construct (a figure), set up together; *perf imperat pass 3rd sg*, *συνεστάτω*; see *ἵστημι.*

συντίθημι : *vb*, put together, add together, compound (ratio); see *τίθημι.*

σθέσις -εως, ἡ : *no*, state, condition.

σθῆμα -ατος, τό : *no*, figure.

σφαῖρα -ας, ἡ : *no*, sphere.

τάχις -εως, ἡ : *no*, arrangement, order.

ταράσσω, ταράχω, **—, —,** *τετάραγμαι, ἐταράθθην* : *vb*, stir, trouble, disturbe; *τεταραγμένος -η -ον*, *adj*, disturbed, perturbed.

τάσσω, τάχω, ἔταχα, τέταθα, τέταγμαι, ἐτάθθην : *vb*, arrange, draw up.

τέλειος -α -ον : *adj*, perfect.

τέμνω, τεμνῶ, ἔτεμον, -τέτμηκα, τέτμημαι, ἐτμήθην : *vb*, cut; *pres/fut indic act 3rd sg*, *τέμει.*

τεταρτημοριον, τό : *no*, quadrant.

τετράγωνος -ον : *adj*, square; *τὸ τετράγωνον*, *no*, square.

τετράκις : *adv*, four times.

τετραπλάσιος -α -ον : *adj*, quadruple.

τετράπλευρος -ον : *adj*, quadrilateral.

τετραπλόος -η -ον : *adj*, fourfold.

τίθημι, θήσω, ἔθηκα, τέθηκα, κεῖμαι, ἐτέθην : *vb*, place, put.

τμῆμα -ατος, τό : *no*, part cut off, piece, segment.

τοίνυν : *par*, accordingly.

τοιοῦτος -αύτη -οῦτο : *pro*, such as this.

τομεύς -έως, ὁ : *no*, sector (of circle).

τομή, ἡ : *no*, cutting, stump, piece.

τόπος, ὁ : *no*, place, space.

τοσαυτάκις : *adv*, so many times.

τοσαυταπλάσιος -α -ον : *pro*, so many times.

τοσοῦτος -αύτη -οῦτο : *pro*, so many.

τουτέστι = *τοῦτ᾽ ἔστι* : *par*, that is to say.

τραπέζιον, τό : *no*, trapezium.

τρίγωνος -ον : *adj*, triangular; *τὸ τρίγωνον*, *no*, triangle.

τριπλάσιος -α -ον : *adj*, triple, threefold.

τρίπλευρος -ον : *adj*, trilateral.

τριπλ -όος -η -ον : *adj*, triple.

τρόπος, ὁ : *no*, way.

τυγθάνω, τεύχομαι, ἔτυθον, τετύθηκα, τέτευγμαι, ἐτεύθθην : *vb*, hit, happen to be at (a place).

ὑπάρθω : *vb*, begin, be, exist; see *ἄρθω.*

ὑπεχαίρεσις -εως, ἡ : *no*, removal.

ὑπερβάλλω : *vb*, overshoot, exceed; see *βάλλω.*

ὑπεροθή, ἡ : *no*, excess, difference.

ὑπερέθω : *vb*, exceed; see *ἔθω.*

ὑπόθεσις -εως, ἡ : *no*, hypothesis.

ὑπόκειμαι : *vb*, underlie, be assumed (as hypothesis); see *κεῖμαι.*

ὑπολείπω : *vb*, leave remaining.

ὑποτείνω, ὑποτενῶ, ὑπέτεινα, ὑποτέτακα, ὑποτέταμαι, ὑπετάθην : *vb*, subtend.

ὕψος -εος, τό : *no*, height.

φανερός -ά -όν : *adj*, visible, manifest.

φημί, φήσω, ἔφην, **—, —, —** : *vb*, say; *ἔφαμεν*, we said.

φέρω, οἴσω, ἤνεγκον, ἐνήνοθα, ἐνήνεγμαι, ἠνέθθην : *vb*, carry.

θώριον, τό : *no*, place, spot, area, figure.

θωρίξ : *pre* + *gen*, apart from.

ψαύω : *vb*, touch.

ὡξ : *par*, as, like, for instance.

ὡξ ἔτυθεν : *par*, at random.

ὡσαύτως : *adv*, in the same manner, just so.

ὥστε : *conj*, so that (causal), hence.

GREEK-ENGLISH LEXICON

ABBREVIATIONS: *act* - active; *adj* - adjective; *adv* - adverb; *conj* - conjunction; *fut* - future; *gen* - genitive; *imperat* - imperative; *impf* - imperfect; *ind* - indeclinable; *indic* - indicative; *intr* - intransitive; *mid* - middle; *neut* - neuter; *no* - noun; *par* - particle; *part* - participle; *pass* - passive; *perf* - perfect; *pre* - preposition; *pres* - present; *pro* - pronoun; *sg* - singular; *tr* - transitive; *vb* - verb.

ἄγω, *ἄχω*, *ἤγαγον*, *-ἦθα*, *ἦγμαι*, *ἤχθην* : *vb*, lead, draw (a line).

ἀδύνατος -ον : *adj*, impossible.

ἀεί : *adv*, always, for ever.

αἱρέω, *αἱρήσω*, *ε[˘]λον*, *ἥρηκα*, *ἥρημαι*, *ἡρέθην* : *vb*, grasp.

ἀιτέω, *αἰτήσω*, *ἤτησα*, *ἤτηκα*, *ἤτημαι*, *ἠτήθη* : *vb*, postulate.

αἴτημα -ατος, τό : *no*, postulate.

ἀκόλουθος -ον : *adj*, analogous, consequent on, in conformity with.

ἄκρος -α -ον : *adj*, outermost, end, extreme.

ἀλλά : *conj*, but, otherwise.

ἄλογος -ον : *adj*, irrational.

ἅμα : *adv*, at once, at the same time, together.

ἀμβλυγώνιος -ον : *adj*, obtuse-angled; *τὸ ἀμβλυγώνιον*, *no*, obtuse angle.

ἀμβλύς -εῖα -ύ : *adj*, obtuse.

ἀμφότερος -α -ον : *pro*, both.

ἀναγράφω : *vb*, describe (a figure); see *γράφω*.

ἀναλογία, ἡ : *no*, proportion, (geometric) progression.

ἀνάλογος -ον : *adj*, proportional.

ἀνάπαλιν : *adv*, inverse(ly).

ἀναπληρόω : *vb*, fill up.

ἀναστρέφω : *vb*, turn upside down, convert (ratio); see *στρέφω*.

ἀναστροφή, ἡ : *no*, turning upside down, conversion (of ratio).

ἀνθυφαιρέω : *vb*, take away in turn; see *αἱρέω*.

ἀνίστημι : *vb*, set up; see *ἵστημι*.

ἄνισος -ον : *adj*, unequal, uneven.

ἀντιπάσθω : *vb*, be reciprocally proportional; see *πάσθω*.

ἄχων -ονος, ὁ : *vb*, axis.

ἅπαξ : *adv*, once.

ἅπας, ἅπασα, ἅπαν : *adj*, quite all, the whole.

ἄπειρος -ον : *adj*, infinite.

ἀπεναντίον : *ind*, opposite.

ἀπέθω : *vb*, be far from, be away from; see *ἔθω*.

ἀπλατής -ές : *adj*, without breadth.

ἀπόδειχις -εως, ἡ : *no*, proof.

ἀποκαθίστημι : *vb*, re-establish, restore; see *ἵστημι*.

ἀπολαμβάνω : *vb*, take from, subtract from, cut off from; see *λαμβάνω*.

ἀποτέμνω : *vb*, cut off, subtend.

ἀπότμημα -ατος, τό : *no*, piece cut off, segment.

ἀποτομή, ἡ : *vb*, piece cut off, apotome.

ἅπτω, *ἅψω*, *ἧψα*, —, *ἧμμαι*, — : *vb*, touch, join, meet.

ἀπώτερος -α -ον : *adj*, further off.

ἄρα : *par*, thus, as it seems (inferential).

ἀριθμός, ὁ : *no*, number.

ἀρτιάκις : *adv*, an even number of times.

αρτιόπλευρος -ον : *adj*, having a even number of sides.

ἄρθω, *ἄρχω*, *ἦρχα*, *ἦρθα*, *ἦργμαι*, *ἤρθθην* : *vb*, rule; *mid.*, begin.

ἀσύμμετρος -ον : *adj*, incommensurable.

ἀσύμπτωτος -ον : *adj*, not touching, not meeting.

ἄρτιος -α -ον : *adj*, even, perfect.

ἄτμητος -ον : *adj*, uncut.

ἀτόπος -ον : *adj*, absurd, paradoxical.

αὐτόθεν : *adv*, immediately, obviously.

ἀφαίρεω : *vb*, take from, subtract from, cut off from; see *αἱρέω*.

ἁφή, ἡ : *no*, point of contact.

βάθος -εος, τό : *no*, depth, height.

βαίνω, *-βήσομαι*, *-έβην*, *βέβηκα*, —, — : *vb*, walk; *perf*, stand (of angle).

βάλλω, *βαλῶ*, *ἔβαλον*, *βέβληκα*, *βέβλημαι*, *ἐβλήθην* : *vb*, throw.

βάσις -εως, ἡ : *no*, base (of a triangle).

γάρ : *conj*, for (explanatory).

γι[γ]νομαι, *γενήσομαι*, *ἐγενόμην*, *γέγονα*, *γεγένημαι*, — : *vb*, happen, become.

γνώμων -ονος, ἡ : *no*, gnomon.

γραμμή, ἡ : *no*, line.

γράφω, *γράψω*, *ἔγρα[ψφ]α*, *γέγραφα*, *γέγραμμαι*, *ἐραψάμην* : *vb*, draw (a figure).

γωνία, ἡ : *no*, angle.

δεῖ : *vb*, be necessary; *δεῖ*, it is necessary; *ἔδει*, it was necassary; *δέον*, being necessary.

δείκνυμι, *δείχω*, *ἔδειχα*, *δέδειθα*, *δέδειγμαι*, *ἐδείθθην* : *vb*, show, demonstrate.

δεικτέον : *ind*, one must show.

δεῖχις -εως, ἡ : *no*, proof.

δεκαγώνος -ον : *adj*, ten-sided; *τὸ δεκαγώνον*, *no*, decagon.

δέθομαι, *δέχομαι*, *ἐδεχάμην*, —, *δέδεγμαι*, *ἐδέθθην* : *vb*, receive, accept.

δή : *conj*, so (explanatory).

δηλαδή : *ind*, quite clear, manifest.

δῆλος -η -ον : *adj*, clear.

δηλονότι : *adv*, manifestly.

διάγω : *vb*, carry over, draw through, draw across; see *ἄγω*.

διαγώνιος -ον : *adj*, diagonal.

διαλείπω : *vb*, leave an interval between.

διάμετρος -ον : *adj*, diametrical; *ἡ διάμετρος*, *no*, diameter, diagonal.

διαίρεσις -εως, ἡ : *no*, division, separation.

διαιρέω : *vb*, divide (in two); *διαρεθέντος -η -ον*, *adj*, separated (ratio); see *αἱρέω*.

διάστημα -ατος, τό : *no*, radius.

διαφέρω : *vb*, differ; see *φέρω*.

δίδωμι, *δώσω*, *ἔδωκα*, *δέδωκα*, *δέδομαι*, *ἐδόθην* : *vb*, give.

διμοίρος -ον : *adj*, two-thirds.

διπλασιάζω : *vb*, double.

διπλάσιος -α -ον : *adj*, double, twofold.

διπλασίων -ον : *adj*, double, twofold.

διπλοῦς -ῆ -οῦν : *adj*, double.

δίς : *adv*, twice.

δίχα : *adv*, in two, in half.

διχορομία, ἡ : *no*, point of bisection.

δυάς -άδος, ἡ : *no*, the number two, dyad.

δύναμαι : *vb*, be able, be capable, generate, square, be when squared; *δυναμένη, ἡ, no*, square-root (of area)—*i.e.*, straight-line whose square is equal to a given area.

δύναμις -εως, ἡ : *no*, power (usually 2nd power when used in mathematical sence, hence), square.

δυνατός -ή -όν : *adj*, possible.

δωδεκάεδρος -ον : *adj*, twelve-sided.

ἑαυτοῦ -ῆς -οῦ : *adj*, of him/her/it/self, his/her/its/own.

ἐγγίων -ον : *adj*, nearer, nearest.

ἐγγράφω : *vb*, inscribe; see *γράφω*.

εἶδος -εος, τό : *no*, figure, form, shape.

εἰκοσάεδρος -ον : *adj*, twenty-sided.

εἶρω/λέγω, ἐρῶ/ερέω, εἶπον, εἴρηκα, εἴρημαι, ἐρρήθην : *vb*, say, speak; *per pass part*, *εἰρημένος -η -ον*, *adj*, said, aforementioned.

εἴτε . . . εἴτε : *ind*, either . . . or.

ἕκαστος -η -ον : *pro*, each, every one.

ἑκατέρος -α -ον : *pro*, each (of two).

ἐκβάλλω : *vb*, produce (a line); see *βάλλω*.

ἐκθέω : *vb*, set out.

ἔκκειμαι : *vb*, be set out, be taken; see *κεῖμαι*.

ἐκτίθημι : *vb*, set out; see *τίθημι*.

ἐκτός : *pre + gen*, outside, external.

ἐλάσσων/ἐλάττων -ον : *adj*, less, lesser.

ἐλάθιστος -η -ον : *adj*, least.

ἐλλείπω : *vb*, be less than, fall short of.

ἐμπίπτω : *vb*, meet (of lines), fall on; see *πίπτω*.

ἔμπροσθεν : *adv*, in front.

ἐναλλάξ : *adv*, alternate(ly).

ἐναρμόζω : *vb*, insert; *perf indic pass 3rd sg*, *ἐνήρμοσται*.

ἐνδέθομαι : *vb*, admit, allow.

ἕνεκεν : *ind*, on account of, for the sake of.

ἐννεαπλάσιος -α -ον : *adj*, nine-fold, nine-times.

ἔννοια, ἡ : *no*, notion.

ἐνπεριέθω : *vb*, encompass; see *ἔθω*.

ἐνπίπτω : see *ἐμπίπτω*.

ἐντός : *pre + gen*, inside, interior, within, internal.

ἑξάγωνος -ον : *adj*, hexagonal; *τὸ ἑξάγωνον*, *no*, hexagon.

ἑξαπλάσιος -α -ον : *adj*, sixfold.

ἑξῆς : *adv*, in order, successively, consecutively.

ἔξωθεν : *adv*, outside, extrinsic.

ἐπάνω : *adv*, above.

ἐπαφή, ἡ : *no*, point of contact.

ἐπεί : *conj*, since (causal).

ἐπειδήπερ : *ind*, inasmuch as, seeing that.

ἐπιζεύγνῡμι, ἐπιζεύχω, ἐπέζευχα, —, ἐπέζευγμαι, ἐπεζεύθθην : *vb*, join (by a line).

ἐπιλογίζομαι : *vb*, conclude.

ἐπινοέω : *vb*, think of, contrive.

ἐπιπέδος -ον : *adj*, level, flat, plane; *τὸ ἐπιπέδον*, *no*, plane.

ἐπισκέπτομαι : *vb*, investigate.

ἐπίσκεψις -εως, ἡ : *no*, inspection, investigation.

ἐπιτάσσω : *vb*, put upon, enjoin; *τὸ ἐπιταθθέν*, *no*, the (thing) prescribed; see *τάσσω*.

ἐπίτριτος -ον : *adj*, one and a third times.

ἐπιφάνεια, ἡ : *no*, surface.

ἕπομαι : *vb*, follow.

ἔρθομαι, ἐλεύσομαι, ἦλθον, ἐλήλυθα, —, — : *vb*, come, go.

ἔσθατος -η -ον : *adj*, outermost, uttermost, last.

ἑτερόμηκης -εξ : *adj*, oblong; *τὸ ἑτερόμηκεξ*, *no*, rectangle.

ἕτερος -α -ον : *adj*, other (of two).

ἔτι : *par*, yet, still, besides.

εὐθύγραμμος -ον : *adj*, rectilinear; *τὸ εὐθύγραμμον*, *no*, rectilinear figure.

εὐθύς -εῖα -ύ : *adj*, straight; *ἡ εὐθεῖα*, *no*, straight-line; *ἐπ᾽ εὐθεῖαξ*, in a straight-line, straight-on.

εὑρίσκω, εὑρήσκω, ηὗρον, εὕρεκα, εὕρηκα, εὕρημαι, εὑρέθην : *vb*, find.

ἐφάπτω : *vb*, bind to; *mid*, touch; *ἡ ἐφαπτομένη*, *no*, tangent; see *ἅπτω*.

ἐφαρμόζω, ἐφαρμόσω, ἐφήρμοσα, ἐφήμοκα, ἐφήμοσμαι, ἐφημόσθην : *vb*, coincide; *pass*, be applied.

ἐφεχῆς : *adv*, in order, adjacent.

ἐφίστημι : *vb*, set, stand, place upon; see *ἵστημι*.

ἔθω, ἕχω, ἔσθον, ἔσθηκα, -έσθημαι, — : *vb*, have.

ἡγέομαι, ἡγήσομαι, ἡγησάμην, ἥγημαι, —, ἡγήθην : *vb*, lead.

ἤδη : *ind*, already, now.

ἥκω, ἵκω, —, —, —, — : *vb*, have come, be present.

ἡμικύκλιον, τό : *no*, semi-circle.

ἡμιόλιος -α -ον : *adj*, containing one and a half, one and a half times.

ἥμισυς -εια -υ : *adj*, half.

ἤπερ = ἤ + περ : *conj*, than, than indeed.

ἤτοι . . . ἤ : *par*, surely, either . . . or; in fact, either . . . or.

θέσις -εως, ἡ : *no*, placing, setting, position.

θεωρημα -ατος, τό : *no*, theorem.

ἴδιος -α -ον : *adj*, one's own.

ἰσάκις : *adv*, the same number of times; *ἰσάκις πολλαπλάσια*, the same multiples, equal multiples.

ἰσογώνιος -ον : *adj*, equiangular.

ἰσόπλευρος -ον : *adj*, equilateral.

ἰσοπληθής -ές : *adj*, equal in number.

ἴσος -η -ον : *adj*, equal; *ἐχ ἴσου*, equally, evenly.

ἰσοσκελής -ές : *adj*, isosceles.

ἵστημι, στήσω, ἔστησα, —, —, *ἑστάθην* : *vb tr*, stand (something).

ἵστημι, στήσω, ἔστην, ἕστηκα, ἕσταμαι, ἑσταθην : *vb intr*, stand up (oneself); Note: perfect *I have stood up* can be taken to mean present *I am standing*.

ἰσοϋψής -ές : *adj*, of equal height.

καθάπερ : *ind*, according as, just as.

κάθετος -ον : *adj*, perpendicular.

καθόλου : *adv*, on the whole, in general.

καλέω : *vb*, call.

κἄκεινος = *καὶ ἐκεῖνος* .

κἄν = *καὶ ἄν* : *ind*, even if, and if.

καταγραφή, ἡ : *no*, diagram, figure.

καταγράφω : *vb*, describe/draw, inscribe (a figure); see *γράφω*.

κατακολουθέω : *vb*, follow after.

καταλείπω : *vb*, leave behind; see *λείπω*; *τὰ καταλειπόμενα*, *no*, remainder.

κατάλληλος -ον : *adj*, in succession, in corresponding order.

καταμετρέω : *vb*, measure (exactly).

καταντάω : *vb*, come to, arrive at.

κατασκενάζω : *vb*, furnish, construct.

κεῖμαι, κεῖσομαι, —, —, —, — : *vb*, have been placed, lie, be made; see *τίθημι*.

κέντρον, τό : *no*, center.

κλάω : *vb*, break off, inflect.

κλίνω, κλίνω, ἔκλινα, κέκλικα, κέκλιμαι, ἐκλίθην : *vb*, lean, incline.

κλίσις -εως, ἡ : *no*, inclination, bending.

κοῖλος -η -ον : *adj*, hollow, concave.

κορυφή, ἡ : *no*, top, summit, apex; *κατὰ κορυφήν*, vertically opposite (of angles).

κρίνω, κρινῶ, ἔκρῖνα, κέκρικα, κέκριμαι, ἐκρίθην : *vb*, judge.

κύβος, ὁ : *no*, cube.

κύκλος, ὁ : *no*, circle.

κύλινδρος, ὁ : *no*, cylinder.

κυρτός -ή -όν : *adj*, convex.

κῶνος, ὁ : *no*, cone.

λαμβάνω, λήψομαι, ἔλαβον, εἴληφα εἴλημμαι, ἐλήφθην : *vb*, take.

λέγω : *vb*, say; *pres pass part*, *λεγόμενος -η -ον*, *adj*, so-called; see *εἴρω*.

λείπω, λείψω, ἔλιπον, λέλοιπα, λέλειμμαι, ἐλείφθην : *vb*, leave, leave behind.

λημμάτιον, τό : *no*, diminutive of *λῆμμα*.

λῆμμα -ατος, τό : *no*, lemma.

λῆψις -εως, ἡ : *no*, taking, catching.

λόγος, ὁ : *no*, ratio, proportion, argument.

λοιπός -ή -όν : *adj*, remaining.

μανθάνω, μαθήσομαι, ἔμαθον, μεμάθηκα, —, — : *vb*, learn.

μέγεθος -εος, τό : *no*, magnitude, size.

μείζων -ον : *adj*, greater.

μένω, μενῶ, ἔμεινα, μεμένηκα, —, — : *vb*, stay, remain.

μέρος -ους, τό : *no*, part, direction, side.

μέσος -η -ον : *adj*, middle, mean, medial; *ἐκ δύο μέσων*, bimedial.

μεταλαμβάνω : *vb*, take up.

μεταχύ : *adv*, between.

μετέωρος -ον : *adj*, raised off the ground.

μετρέω : *vb*, measure.

μέτρον, τό : *no*, measure.

μηδείς, μηδεμία, μηδέν : *adj*, not even one, (neut.) nothing.

μηδέποτε : *adv*, never.

μηδέτερος -α -ον : *pro*, neither (of two).

μῆκος -εος, τό : *no*, length.

μήν : *par*, truely, indeed.

μονάς -άδος, ἡ : *no*, unit, unity.

μοναθός -ή -όν : *adj*, unique.

μοναθῶς : *adv*, uniquely.

μόνος -η -ον : *adj*, alone.

νοέω, —, *νόησα, νενόηκα, νενόημαι, ἐνοήθην* : *vb*, apprehend, conceive.

οἷος -α -ον : *pre*, such as, of what sort.

ὀκτάεδρος -ον : *adj*, eight-sided.

ὅλος -η -ον : *adj*, whole.

ὁμογενής -ές : *adj*, of the same kind.

ὅμοιος -α -ον : *adj*, similar.

ὁμοιοπληθής -ές : *adj*, similar in number.

ὁμοιοταγής -ές : *adj*, similarly arranged.

ὁμοιότης -ητος, ἡ : *no* similarity.

ὁμοίως : *adv*, similarly.

ὁμόλογος -ον : *adj*, corresponding, homologous.

ὁμοταγής -ές : *adj*, ranged in the same row or line.

ὁμώνυμος -ον : *adj*, having the same name.

ὄνομα -ατος, τό : *no*, name; *ἐκ δύο ὀνομάτων*, binomial.

ὀχυγώνιος -ον : *adj*, acute-angled; *τὸ ὀχυγώνιον*, *no*, acute angle.

ὀχύς -εῖα -ύ : *adj*, acute.

ὁποιοσοῦν = *ὁποῖος -α -ον* + *οὖν* : *adj*, of whatever kind, any kind whatsoever.

ὁπόσος -η -ον : *pro*, as many, as many as.

ὁποσοσδηποτοῦν = *ὁπόσος -η -ον* + *δή* + *ποτέ* + *οὖν* : *adj*, of whatever number, any number whatsoever.

ὁποσοσοῦν = *ὁπόσος -η -ον* + *οὖν* : *adj*, of whatever number, any number whatsoever.

ὁπότερος -α -ον : *pro*, either (of two), which (of two).

ὀρθογώνιον, τό : *no*, rectangle, right-angle.

ὀρθός -ή -όν : *adj*, straight, right-angled, perpendicular; *πρὸς ὀρθὰς γωνίας*, at right-angles.

ὅρος, ὁ : *no*, boundary, definition, term (of a ratio).

ὁσαδηποτοῦν = *ὅσα* + *δή* + *ποτέ* + *οὖν* : *ind*, any number whatsoever.

ὁσάκις : *ind*, as many times as, as often as.

ὁσαπλάσιος -ον : *pro*, as many times as.

ὅσος -η -ον : *pro*, as many as.

ὁσπερ, ἥπερ, ὅπερ : *pro*, the very man who, the very thing which.

ὁστις, ἥτις, ὅ τι : *pro*, anyone who, anything which.

ὅταν : *adv*, when, whenever.

ὁτιοῦν : *ind*, whatsoever.

οὐδείς, οὐδεμία, οὐδέν : *pro*, not one, nothing.

οὐδέτερος -α -ον : *pro*, not either.

οὐθέτερος : see *οὐδέτερος*.

οὐθέν : *ind*, nothing.

οὖν : *adv*, therefore, in fact.

οὕτως : *adv*, thusly, in this case.

πάλιν : *adv*, back, again.

πάντως : *adv*, in all ways.

παρά : *prep + acc*, parallel to.

παραβάλλω : *vb*, apply (a figure); see *βάλλω*.

παραβολή, ἡ : *no*, application.

παράκειμαι : *vb*, lie beside, apply (a figure); see *κεῖμαι*.

παραλλάσσω, παραλλάχω, **—,** *παρήλλαθα,* **—, —** : *vb*, miss, fall awry.

παραλληλεπίπεδος, -ον : *adj*, with parallel surfaces; *τὸ παραλληλεπίπεδον, no*, parallelepiped.

παραλληλόγραμμος -ον : *adj*, bounded by parallel lines; *τὸ παραλληλόγραμμον, no*, parallelogram.

παράλληλος -ον : *adj*, parallel; *τὸ παράλληλον, no*, parallel, parallel-line.

παραπλήρωμα -ατος, τό : *no*, complement (of a parallelogram).

παρατέλυετος -ον : *adj*, penultimate.

παρέκ : *prep + gen*, except.

παρεμπίπτω : *vb*, insert; see *πίπτω*.

πάσθω, πείσομαι, ἔπαθον, πέπονθα, **—, —** : *vb*, suffer.

πεντάγωνος -ον : *adj*, pentagonal; *τὸ πεντάγωνον, no*, pentagon.

πενταπλάσιος -α -ον : *adj*, five-fold, five-times.

πεντεκαιδεκάγωνον, τό : *no*, fifteen-sided figure.

πεπερασμένος -η -ον : *adj*, finite, limited; see *περαίνω*.

περαίνω, περανῶ, ἐπέρανα, **—,** *πεπεράσμαι, ἐπερανάνθην* : *vb*, bring to end, finish, complete; *pass*, be finite.

πέραξ -ατος, τό : *no*, end, extremity.

περατόω, **—, —, —, —, —** : *vb*, bring to an end.

περιγράφω : *vb*, circumscribe; see *γράφω*.

περιέθω : *vb*, encompass, surround, contain, comprise; see *ἔθω*.

περιλαμβάνω : *vb*, enclose; see *λαμβάνω*.

περισσάκις : *adv*, an odd number of times.

περισσός -ή -όν : *adj*, odd.

περιφέρεια, ἡ : *no*, circumference.

περιφέρω : *vb*, carry round; see *φέρω*.

πηλικότης -ητος, ἡ : *no*, magnitude, size.

πίπτω, πεσοῦμαι, ἔπεσον, πέπτωκα, **—, —** : *vb*, fall.

πλάτος -εος, τό : *no*, breadth, width.

πλείων -ον : *adj*, more, several.

πλευρά, ἡ : *no*, side.

πλῆθος -εος, τό : *no*, great number, multitude, number.

πλήν : *adv & prep + gen*, more than.

ποιός -ά -όν : *adj*, of a certain nature, kind, quality, type.

πολλαπλασιάζω : *vb*, multiply.

πολλαπλασιασμός, ὁ : *no*, multiplication.

πολλαπλάσιον, τό : *no*, multiple.

πολύεδρος -ον : *adj*, polyhedral; *τό πολύεδρον, no*, polyhedron.

πολύγωνος -ον : *adj*, polygonal; *τό πολύγωνον, no*, polygon.

πολύπλευρος -ον : *adj*, multilateral.

πόρισμα -ατος, τό : *no*, corollary.

ποτέ : *ind*, at some time.

πρῖσμα -ατος, τό : *no*, prism.

προβαίνω : *vb*, step forward, advance.

προδείκνυμι : *vb*, show previously; see *δείκνυμι*.

προεκτίθημι : *vb*, set forth beforehand; see *τίθημι*.

προερέω : *vb*, say beforehand; *perf pass part*, *προειρημένος -η -ον, adj*, aforementioned; see *ἐῖρω*.

προσαναπληρόω : *vb*, fill up, complete.

προσαναγράφω : *vb*, complete (tracing of); see *γράφω*.

προσαρμόζω : *vb*, fit to, attach to.

προσεκβάλλω : *vb*, produce (a line); see *ἐκβάλλω*.

προσευρίσκω : *vb*, find besides, find; see *εὑρίσκω*.

προσλαμβάμω : *vb*, add.

πρόκειμαι : *vb*, set before, prescribe; see *κεῖμαι*.

πρόσκειμαι : *vb*, be laid on, have been added to; see *κεῖμαι*.

προσπίπτω : *vb*, fall on, fall toward, meet; see *πίπτω*.

προτασις -εως, ἡ : *no*, proposition.

προστάσσω : *vb*, prescribe, enjoin; *τὸ τροσταθθέν, no*, the thing prescribed; see *τάσσω*.

προστίθημι : *vb*, add; see *τίθημι*.

πρότερος -α -ον : *adj*, first (comparative), before, former.

προτίθημι : *vb*, assign; see *τίθημι*.

προθωρέω : *vb*, go/come forward, advance.

πρῶτος -α -ον : *adj*, first, prime.

πυραμίξ -ίδος, ἡ : *no*, pyramid.

ῥητός -ή -όν : *adj*, expressible, rational.

ῥομβοειδής -ές : *adj*, rhomboidal; *τὸ ῥομβοειδές, no*, romboid.

ῥόμβος, ὁ no, rhombus.

σημεῖον, τό : *no*, point.

σκαληνός -ή -όν : *adj*, scalene.

στερεός -ά -όν : *adj*, solid; *τὸ στερεόν, no*, solid, solid body.

στοιθεῖον, τό : *no*, element.

στρέφω, -στρέψω, ἔστρεψα, **—,** *ἐσταμμαι, ἐστάφην* : *vb*, turn.

σύγκειμαι : *vb*, lie together, be the sum of, be composed; *συγκείμενος -η -ον, adj*, composed (ratio), compounded; see *κεῖμαι*.

σύγκρίνω : *vb*, compare; see *κρίνω*.

συμβαίνω : *vb*, come to pass, happen, follow; see *βαίνω*.

συμβάλλω : *vb*, throw together, meet; see *βάλλω*.

σύμμετρος -ον : *adj*, commensurable.

σύμπαξ -αντος, ὁ : *no*, sum, whole.

συμπίπτω : *vb*, meet together (of lines); see *πίπτω.*

συμπληρόω : *vb*, complete (a figure), fill in.

συνάγω : *vb*, conclude, infer; see *ἄγω.*

συναμφότεροι -αι -α : *adj*, both together; *ὁ συναμφότεροξ*, *no*, sum (of two things).

συναποδείκνυμι : *no*, demonstrate together; see *δείκνυμι.*

συναφή, ἡ : *no*, point of junction.

σύνδυο, οἱ, αἱ, τά : *no*, two together, in pairs.

συνεθής -έξ : *adj*, continuous; *κατὰ τὸ συνεθέξ*, continuously.

σύνθεσις -εως, ἡ : *no*, putting together, composition.

σύνθετος -ον : *adj*, composite.

συ[ν]ίστημι : *vb*, construct (a figure), set up together; *perf imperat pass 3rd sg*, *συνεστάτω*; see *ἵστημι.*

συντίθημι : *vb*, put together, add together, compound (ratio); see *τίθημι.*

σθέσις -εως, ἡ : *no*, state, condition.

σθῆμα -ατος, τό : *no*, figure.

σφαῖρα -ας, ἡ : *no*, sphere.

τάχις -εως, ἡ : *no*, arrangement, order.

ταράσσω, ταράχω, **—, —,** *τετάραγμαι, ἐταράθθην* : *vb*, stir, trouble, disturbe; *τεταραγμένος -η -ον*, *adj*, disturbed, perturbed.

τάσσω, τάχω, ἔταχα, τέταθα, τέταγμαι, ἐτάθθην : *vb*, arrange, draw up.

τέλειος -α -ον : *adj*, perfect.

τέμνω, τεμνῶ, ἔτεμον, -τέτμηκα, τέτμημαι, ἐτμήθην : *vb*, cut; *pres/fut indic act 3rd sg*, *τέμει.*

τεταρτημοριον, τό : *no*, quadrant.

τετράγωνος -ον : *adj*, square; *τὸ τετράγωνον*, *no*, square.

τετράκις : *adv*, four times.

τετραπλάσιος -α -ον : *adj*, quadruple.

τετράπλευρος -ον : *adj*, quadrilateral.

τετραπλόος -η -ον : *adj*, fourfold.

τίθημι, θήσω, ἔθηκα, τέθηκα, κεῖμαι, ἐτέθην : *vb*, place, put.

τμῆμα -ατος, τό : *no*, part cut off, piece, segment.

τοίνυν : *par*, accordingly.

τοιοῦτος -αύτη -οῦτο : *pro*, such as this.

τομεύς -έως, ὁ : *no*, sector (of circle).

τομή, ἡ : *no*, cutting, stump, piece.

τόπος, ὁ : *no*, place, space.

τοσαυτάκις : *adv*, so many times.

τοσαυταπλάσιος -α -ον : *pro*, so many times.

τοσοῦτος -αύτη -οῦτο : *pro*, so many.

τουτέστι = *τοῦτ᾽ ἔστι* : *par*, that is to say.

τραπέζιον, τό : *no*, trapezium.

τρίγωνος -ον : *adj*, triangular; *τὸ τρίγωνον*, *no*, triangle.

τριπλάσιος -α -ον : *adj*, triple, threefold.

τρίπλευρος -ον : *adj*, trilateral.

τριπλ -όος -η -ον : *adj*, triple.

τρόπος, ὁ : *no*, way.

τυγθάνω, τεύχομαι, ἔτυθον, τετύθηκα, τέτευγμαι, ἐτεύθθην : *vb*, hit, happen to be at (a place).

ὑπάρθω : *vb*, begin, be, exist; see *ἄρθω.*

ὑπεχαίρεσις -εως, ἡ : *no*, removal.

ὑπερβάλλω : *vb*, overshoot, exceed; see *βάλλω.*

ὑπεροθή, ἡ : *no*, excess, difference.

ὑπερέθω : *vb*, exceed; see *ἔθω.*

ὑπόθεσις -εως, ἡ : *no*, hypothesis.

ὑπόκειμαι : *vb*, underlie, be assumed (as hypothesis); see *κεῖμαι.*

ὑπολείπω : *vb*, leave remaining.

ὑποτείνω, ὑποτενῶ, ὑπέτεινα, ὑποτέτακα, ὑποτέταμαι, ὑπετάθην : *vb*, subtend.

ὕψος -εος, τό : *no*, height.

φανερός -ά -όν : *adj*, visible, manifest.

φημί, φήσω, ἔφην, **—, —, —** : *vb*, say; *ἔφαμεν*, we said.

φέρω, οἴσω, ἤνεγκον, ἐνήνοθα, ἐνήνεγμαι, ἠνέθθην : *vb*, carry.

θώριον, τό : *no*, place, spot, area, figure.

θωρίξ : *pre* + *gen*, apart from.

ψαύω : *vb*, touch.

ὡξ : *par*, as, like, for instance.

ὡξ ἔτυθεν : *par*, at random.

ὡσαύτως : *adv*, in the same manner, just so.

ὥστε : *conj*, so that (causal), hence.